Computing the Riemann Hypothesis

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Thorin Measures, Pólya Densities and Hitting Times

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Cover: the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ at eight resolutions, computed by the authors.

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Preface

The Riemann Hypothesis can be stated as the positivity of a measure. The reciprocal Xi clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution exactly when every nontrivial zero lies on the critical line, and then its Thorin measure is the discrete measure $\sum_{\gamma>0}\delta_{\gamma^2}$ on the squared ordinates. This book asks what computation can say about that positivity condition.

Its answer has three parts. First, the Thorin measure is computable at every positive resolution, with explicit error bounds, from the Pólya kernel alone, with no primes and no list of zeros. Second, the distance between the computed measure and the measure demanded by RH can be measured exactly, and the signature an off line zero would leave can be calibrated. Third, a finite computation could refute RH by certifying a zero off the line, but neither a computation of the zeros up to a finite height nor a finite set of positivity tests on the heat trace can prove it. A last question is how much of the positivity can be proved. Moved from the centre of the critical strip to the basepoint one, the clock has a positive Thorin measure with no hypothesis. Chapters 9 to 11 follow each construction of the book back across the strip as far as it holds unconditionally; below the basepoint one, zero density estimates still show that the zeros to the right of the basepoint, the only possible source of negativity, carry a vanishing fraction of the mass of the density.

The book grew out of the first author's programme on generalised gamma convolutions and the Pólya density, collected in the monograph *On Hilbert's 8th Problem* [252], and out of a joint paper computing the Thorin measure directly [263]. The first author's arXiv paper of the same title is a separate work, which we cite by version: [249] is its version 24, of 2026, and [250] its version 5, of 2017. Chapters 6 and 7 incorporate Part IX of that monograph [252], the first author's work on the reciprocal zeta function, written as the manuscript [258], and Chapters 9 to 11 a paper of the authors on the reciprocal Xi function at the basepoint one and across the critical strip, together with notes on Riesz's criterion and on almost generalised gamma convolutions. Sections 8.5 and 11.3 draw on a manuscript of the first author on the Thorin measures of the Dedekind eta function [255], Chapter 13 on a note of the first author on ferromagnetic random variables, Chapter 14 on a manuscript of the first author on the Pólya density as a maximum order statistic [257], Chapter 15 on a working report of the authors

[264], Chapter 16 on a chapter of the first author on Pólya and the zeros of entire functions, and Chapters 17 to 21 on five further chapters of the first author, on Yoshida's Hermitian forms, on the compact windows of Zhu [350], on three families of generalised gamma convolutions, on deconvolution as a damped Weil criterion, and on Dirichlet L functions and the Linnik constant. The core evaluator runs in ball arithmetic, and Appendix A says where the scripts and data files behind the tables and figures are, and `code/README.md` of the repository lists them and what the archive lacks. No proof of RH is claimed.

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16.4 Heights beyond which Theorem 16.12.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width d of the window and the number k of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8} \le \beta \le \frac{7}{8}$; "on line" uses $\beta = \frac{1}{2}$, valid for $r + d \le H$. Computed by `polya_qrh.py` with the bound of Lemma 16.12.6. 637

17.1 The tent value $D(x)$ from the primes (Proposition 17.6.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x = \log 2$ and $x = \log 5$ differ by Landau's term, see the text. 653

17.2 The least eigenvalue $\lambda_{\min}(\alpha, a)$ of the damped Yoshida form on the window $[-a, a]$, from the matrices of the form on the modes χ_n, $|n| \le N$. For $\alpha > \frac{1}{2}$ the entries are converged values, from Richardson extrapolation of $N = 96$, 192, 384, with an uncertainty of at most 3×10^{-7} at $a = 0.35$ and 1.1×10^{-8} for $a \ge 0.8$ (an independent Legendre basis agrees to 1.5×10^{-7} at most). For $\alpha = \frac{1}{2}$ the entries are the finite section values at $N = \max(24, 24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a = 4$ ($N = 96$) was computed with 640 digits. The last row is the floor $2b_\alpha = 2\xi'(\alpha)/\xi(\alpha)$. Computed by `yo_damped.py` with `yo_matrix.py`, data in `yo_damped.json`. 661

18.1 Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order N shown (`code/zh_window_eigs.py`). The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling. . . 673

18.2 The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number K of atoms killed and the even exponent m, with the optimal K and m, the last ordinate γ_K killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 18.4, and T_* itself. 675

18.3 Smallest window half length a at which Yoshida's form on $[-a, a]$ detects a hypothetical off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05 (script `code/zh_detect.py`, record `data/zh_detect.json`). The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\leq$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90. 678

18.4 The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem. 680

18.5 The comb constant $A_a(\alpha)$ of (18.5.1) and the barrier height $T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha = \frac{1}{2}$ is Zhu's [350], who prints $A_{0.8}(\frac{1}{2}) = 2.9420$ and $T_1 = 119$ at $a = 0.8$, 1.0×10^7 at $a = 1.6$ and 2.4×10^{11} at $a = 2$; the digits here are 2.94197, 119, 1.04×10^7 and 2.44×10^{11}. For $\alpha > 1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha) = 18.88$, 3.01, 1.14 at $\alpha = 1.1$, 1.5, 2. . . 681

18.6 The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 18.5.2 for several values of $\lambda = \lambda_{\min}^{\text{even}}(\frac{1}{2}, a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices (script `code/zh_window_eigs.py`, record `data/zh_window_eigs.json`), not bounds; Zhu's reference values of his Table 1, 9.34×10^{-7} and 1.65×10^{-17}, give 1.01×10^{-5} and 1.79×10^{-16}. 684

19.1 Mean modulus of $E(N, s)$ of (19.2.2) for χ_4 over $s = \alpha + 10i$ and $s = \alpha + 25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10} N$ fitted by least squares on the four values of N. Four values of N and two values of b illustrate the GRH statement of the proposition and do not verify it. Script `tf_primes.py`. 693

19.2 The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate γ_{n_0} of the n_0th zero. Replacing a zero: the 20 zeros γ_n, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of γ_n and γ_{n+1}, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros γ_n and γ_{n+1} are removed as well. Median and least are over the 20 values of δ_c, and the ratio is the median over the 20 points of δ_c divided by the local gap. Script `tf_deltac.py`, data `tf_deltac.json`. 695

19.3 Levels ε at which ρ_ε is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)). Script `tf_window.py`, data `tf_window.json`. 696

19.4 The number K of greedy levels of Corollary 19.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$. Script `tf_visibility.py`. . . 700

20.1 The ladder $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, by power law extrapolation from $N = 384$, 768, 1536 (`code/dc_ladder.py`, parts `grid` for $\varepsilon > 0$ and `y0` for $\varepsilon = 0$, `data/dc_ladder.json`). The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2} + \varepsilon)$ of Theorem 20.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed by `code/dc_weil.py`, part W2 (`data/dc_weil.json`). At $\varepsilon = 0$ the entries for $a \ge 0.75$ are omitted: in double precision they are below the rounding level of about 10^{-13}, and the upper bounds from extended precision are given in the text. 712

20.2 The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is 1.3293×10^{-3}. Script `code/dc_ladder.py`, part `y0`, data `data/dc_ladder.json`. 712

20.3 The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of 10^{-4}, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 20.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times a^2, is at most 4×10^{-5} in these units, an error estimate and not a bound. Script `code/dc_rate.py`, parts `const` and `table`, data `data/dc_rate.json`. 715

20.4 Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of 10^{-4}: the lower bound of the proof of Theorem 20.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1-\ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 20.3, extended to $a = 6$ in `data/dc_rate.json`), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound. Script `code/dc_rate.py`, parts `lower` and `trial`. 715

21.1 The moments $\mathbb{E}[T^k]$ from the zeros (800 ordinates and the smooth tail from γ_{800}) and from the function (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`). 726

21.2 Relative errors of p_1, p_2, p_3 for the smooth and the anchored tail with the first N ordinates; the cuts are $T = 1183.1475$ ($S(T) = 0.083$) for $N = 799$, $T = 397.1503$ ($S(T) = 0.244$) for $N = 200$ and $T = 680.8186$ ($S(T) = -0.211$) for $N = 400$ (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`). 727

21.3 Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ of the density of Y (`gl_density_pf.py`, 70 digits). 732

21.4 The minimum of $v_\alpha(b)$ over $0 \le b \le 150$ for χ_4, always at $b = 0$, against the bound $2(\alpha - \frac{1}{2})\, p_1(\chi_4) = (\alpha - \frac{1}{2})V_f$ of Theorem 9.1.18(i), with $V_f = 2p_1(\chi_4) = 0.1560296499$. Computed by `code/gl_sinsq_chi4.py` (`data/gl_sinsq_chi4.json`). 746

21.5 The power sums $p_j(\chi)$ of the ordinates for the characters modulo 4 and 5: the zero side, the function side, the relative error, and the difference of the two sides times 300^{2j}. 748

21.6 The zero y of the mixture $M_\lambda = \lambda A + (1-\lambda)B$ modulo 5 in each of the two intervals between a pole of A and a pole of B, for five weights; each interval contains exactly one zero. 750

B.1 Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [81, Chapters 6 to 8]. 761

Chapter 1

Introduction and foundations

The book rests on one equivalence, between the Riemann Hypothesis and the positivity of the Thorin measure of a function built from the completed zeta function.

1.1 The Thorin condition and what this book shows

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function, whose zeros are the nontrivial zeros ρ of ζ. The Riemann Hypothesis (RH) asserts that every ρ has real part $\frac{1}{2}$.

This book works with a reformulation. Put

$$G(u) = \frac{\xi(\frac{1}{2} + \sqrt{u})}{\xi(\frac{1}{2})}.$$

Because $\xi(\frac{1}{2} + s)$ is even in s, G is an entire function of u with no branch point, of order one half, and Hadamard's theorem gives

$$G(u) = \prod_k \Big(1 + \frac{u}{a_k}\Big), \qquad a_k = -(\rho_k - \tfrac{1}{2})^2,$$

one factor for each pair $\{\rho, 1-\rho\}$ (Section 1.2 derives the product as (1.2.1)). A zero on the line has $a_k = \gamma_k^2 > 0$; a zero off the line has a_k complex. The reciprocal $1/G$ is the *clock*. The classical theory of generalised gamma convolutions (GGC) of Thorin and Bondesson [319, 320, 39] says that a function of the form $\exp\{-\int \log(1 + u/z)\, U(dz)\}$ with a positive measure U is the Laplace transform of a positive random variable, and that U, the Thorin measure, is unique. The rigidity theorem of [252], which Section 1.2 restates with its source as Proposition 1.2.1, says that the clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the Laplace transform of a generalised gamma convolution if and only if RH holds, and that its Thorin measure is then

$$U_{\mathrm{RH}} = \sum_{\gamma > 0} \delta_{\gamma^2}, \tag{1.1.1}$$

while the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ.

We call the requirement that the Thorin measure of the clock be a positive measure the *Thorin condition*. It has several equivalent faces, each of which is a sign condition. The logarithmic derivative $G'/G(u) = \sum_k (u + a_k)^{-1}$ must be a Stieltjes transform. The heat trace $W(t) = \sum_k e^{-a_k t}$ must be completely monotone. The Thorin density $\rho_\varepsilon(\theta) = \pi^{-1} \operatorname{Re}(\xi'/\xi)(\frac{1}{2}+\varepsilon+i\theta)$ at every resolution $\varepsilon > 0$ must be nonnegative, which is Lagarias's criterion [160]; for $\varepsilon \geq \frac{1}{2}$, where the basepoint $\frac{1}{2}+\varepsilon$ is at least one, it is positive with no hypothesis (Section 9.1). Weil's explicit formula functional must be positive on positive definite test functions [338]. The clock must be hyperbolically completely monotone (Corollary 4.1.3). All of them are computable at a finite resolution and on a finite window (Section 1.3.3).

1.1.1 What computation can and cannot say

Platt and Trudgian verified RH below height 3×10^{12} [247]; Appendix B tells the history of such computations, with the landmarks in Table B.1. None of it proves RH, and the reason is structural rather than technological: every computation sees finitely many zeros, and there are functions with the order, type and symmetry of ξ that pass a given finite set of heat trace tests and yet have zeros off the line just above any sufficiently large height R for which RH has been verified up to $R + 50$ (Theorem 5.2.1 [249, Theorem D]).

The Thorin condition gives computation a different role. Instead of locating zeros one at a time, it asks for the sign of a measure, and the measure can be computed at every positive resolution without knowing where a single zero is.

Three ways of evaluating ξ appear throughout. The first is Pólya's integral, $\Xi(\theta) = \int \Phi(u) \cos(\theta u)\, du$, with the Pólya kernel Φ, a positive, doubly exponentially decaying function built from the Jacobi theta function. The integral is exact and needs no primes and no zeros, and since Φ is analytic in a strip the trapezoid rule applied to it is spectrally accurate; the cost is the cancellation $e^{-\pi\theta/4}$, which the tilted contour of Section 2.1 largely removes. The second is the Riemann Siegel formula and its fast multi point descendant, the Odlyzko Schönhage algorithm [220], the tools of choice at large height. The third is the explicit formula, which evaluates the same objects from the primes and the gamma factor; the sine squared lemma is its Thorin form.

The principle that ties them together is that the Thorin measure at resolution ε is the Poisson smoothing of the zeros, and simultaneously the sine squared image of the primes. The first description makes it computable from any evaluator of ξ off the line. The second explains why the primes alone cannot compute it near the line. In Selberg's smoothed explicit formula the prime cutoff x plays the part of the resolution: if no zero has real part greater than $\frac{1}{2}$, the error at resolution ε is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that k correct digits need $x \approx 10^{k/\varepsilon}$, with primes up to x^2 (Section 3.5.1). Figure 1.1 shows the first description at work.

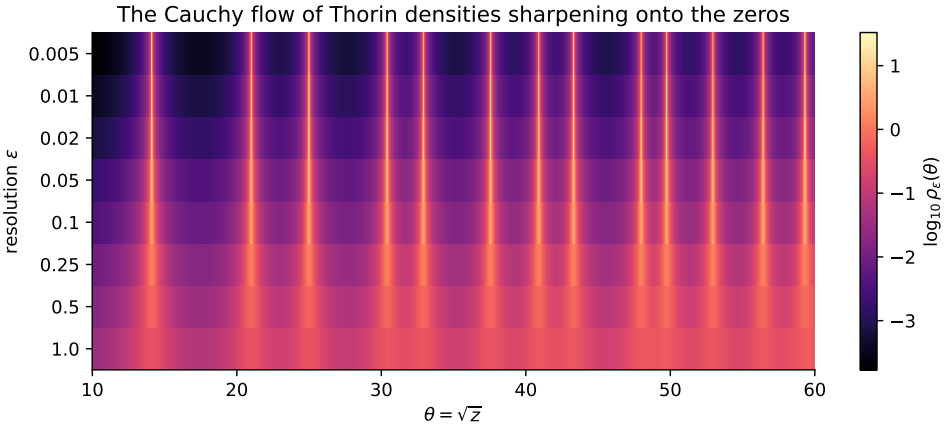


Figure 1.1: The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$.

1.1.2 Plan of the book and its principal results

Two ideas set out in Sections 1.2 and 1.3 carry the rest of the book. The first, stated above, is that the Thorin measure at resolution ε is the Poisson smoothing of the zeros; as $\varepsilon \downarrow 0$ it converges to the atoms on the line. The second is that the hierarchy of infinitely divisible laws becomes a hierarchy of positivity conditions on one function, the heat trace, of which the lower rungs are theorems and the top rung is RH.

Chapter 2 evaluates ξ and ξ' in the complex plane by the Pólya route. Chapters 3 to 5 are the core of the book. Chapter 3 measures the distance from the computed Thorin measure to U_{RH} and computes the signature of an off line zero. Chapter 4 treats weak convergence and hyperbolic complete monotonicity. Chapter 5 climbs the hierarchy of positivity on the heat trace.

Chapters 6 and 7 study the reciprocal zeta function in the variable u of the clock. Chapter 8, with the Pólya route as its evaluator, treats the clock as a random variable.

Chapters 9 to 11 move the basepoint. The shifted clock T_α has Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{u})$; at $\alpha = \frac{1}{2}$ it is the clock. These chapters carry both ideas from the basepoint one, the resolution $\varepsilon = \frac{1}{2}$, into the critical strip. Chapter 9 works out the basepoint one and constructs T_α as a hitting time at every basepoint $\alpha > \frac{1}{2}$. Chapter 10 extends the hierarchy to symmetric laws on the line and reads zero free half planes as completeness statements after Wiener, Salem and Riesz.

Chapter 11 reads the hierarchy through the books of Sato and Bondesson, uses zero density estimates to bound how far each T_α can be from a GGC, and closes with the work of Anthony Pakes on size biasing, the Lambert law and ladders of self decomposable laws.

Chapter 12 assembles the results into a path towards RH on which every step is marked proved, computed, announced or open. Chapter 13 reads RH as the statement that the Pólya variable is ferromagnetic, in the sense that its moment generating function vanishes only on the imaginary axis, and builds provably ferromagnetic laws within total variation distance 1.3×10^{-4} of it with no hypothesis and 2×10^{-6} by a computer assisted proof. Chapter 14 reads the Pólya density as the density of a maximum order statistic of independent exponential variables and follows the clock through that representation. Chapter 15 places the computations of the book on the Thorin phase diagram of the de Bruijn Newman flow, the set of points (t, α) at which the shifted clock of ξ_t is a generalised gamma convolution, and certifies the finite part of one of its points below height 160. Chapter 16 reads Pólya's papers on the zeros of entire functions through the reciprocal Xi law and the basepoint laws, and ends with what the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [226], which has not been refereed, would give. Chapter 17 reads Yoshida's Hermitian forms attached to the explicit formula against the programme. Chapter 18 reads the compact windows of Zhu [350] in the same language. Chapter 19 compares the three one parameter families of generalised gamma convolutions that contain the clock of ξ, the Esscher tilt, the basepoint family and the heat flow, and proves the deconvolution theorem. Chapter 20 reads the deconvolution as a damped Weil criterion. Chapter 21 carries the tilt and the Thorin representation to Dedekind zeta functions and Dirichlet L functions and asks what the programme says about the Linnik constant.

Tables 1.1, 1.2 and 1.3 collect the principal results of Chapters 2 to 11, each with the section where it is developed. A result is marked proved when that section proves it or cites a proof, and computed when it rests on a computation; the section states how far the computation is certified.

Result	Status	Where
Pólya route: ξ and ξ' from two fixed polynomials in ball arithmetic, about 60 correct digits at height 200 at 420 bits, 0.7 ms per point	computed	Section 2.1
Tilted contour: 25 digits of $\rho_{0.05}$ at height 5000 against `mpmath`, in 28 ms per point; to leading exponential order the loss to cancellation falls from about $0.341\,\theta$ digits to about $\delta\theta/\log 10$	proved and computed	Section 2.1
The Lehmer pair near height 7005 resolved by the tilted Pólya route at 34 ms per point	computed	Section 2.2.3.1
$N(1000) = 649$ from the Thorin mass at resolution $\frac{1}{2}$ and the argument along a horizontal segment	computed	Section 2.2.3.2

Result	Status	Where
Raw total variation to U_{RH} maximal; matched total variation at the floor $3.7 \times 10^{-9}\varepsilon$ of the tail model; cell distance $5.44\,\varepsilon$	computed	Section 3.2
Mass defect $-(\varepsilon/\pi)\Xi'(\Theta)/\Xi(\Theta)$ to first order; at $\Theta = 200$ the observed slope agrees with it to five digits at $\varepsilon = 0.005$	proved and computed	Section 3.2
Signatures of an off line pair in total variation and negative mass, exact for an omitted pair and for the negative mass, to leading order for a displaced pair	proved	Section 3.3
Zero density $\nu(\frac{1}{2} + \varepsilon, 200) = 0$ at eight resolutions, from the argument principle identity of Proposition 3.4.2	computed	Section 3.4
Prime side of the sine squared lemma: when no zero has real part greater than $\frac{1}{2}$, Selberg's smoothed sum computes ρ_ε with relative error at most $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon \log x)$, so with error of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at the peaks, the single zero law; tested at $\varepsilon = \frac{1}{4}$	proved and computed	Section 3.5
The Thorin measures converge vaguely to the measure of the zeros on the line, whether or not RH holds	proved	Section 4.1
The clock is HCM if and only if RH holds, and the shifted clock at any basepoint $\alpha \geq 1$ is HCM unconditionally; the 216 normalised derivatives of a finite HCM test are all positive, certified in ball arithmetic	proved and computed	Section 4.1
Bosch type threshold: $1/G(x^{1/\beta})$ is HCM if and only if $\lvert\beta\rvert \geq 1/(1-\alpha_\xi)$, with α_ξ set by the largest angle of the zeros; unconditionally $1/G(x^p)$ is HCM for $\lvert p\rvert \leq 1 - 1/(\pi T_0)$, where $T_0 = 3 \times 10^{12}$ is the verified height, in particular for $\lvert p\rvert \leq 1 - 1.1 \times 10^{-13}$	proved	Section 4.3
The clock is infinitely divisible and self decomposable with no hypothesis, and $(-1)^k W^{(k)} > 0$ for every $k \leq 1.16 \times 10^{25}$, that is, the Williamson measures, inverse gamma smoothings of the Thorin measure, are positive to that order; the two proofs use the verification of RH below heights 1000 and 3×10^{12} respectively [247]	proved	Section 5.1

Table 1.1: Principal results of Chapters 2 to 5.

Result	Status	Where
The normalised reciprocal zeta function is infinitely divisible with no hypothesis, and it is not self decomposable, not a GGC and not in the Bondesson class	proved and computed	Section 7.1
Conjecture S, that the symmetrised truncations of the Biane, Pitman and Yor lattice have their zeros in the strip on the line, holds for the sizes $n \leq 24$ and $n = 32, 36$ above explicit heights and numerically below them; for infinitely many n it would imply RH	proved and computed	Section 7.2
Law of the clock by Talbot inversion, with no zero as input; its hazard tends to γ_1^2, reached to eleven digits at $t = 0.25$	computed	Section 8.1

Result	Status	Where
Kent approximants with 1700 zeros and a moment matched Weyl tail: relative error 3.5×10^{-23} at $u = 1$ and 7.0×10^{-13} at $u = 10^4$	computed	Section 8.1
The clock is not the first passage time of Brownian motion to an independent random level; the boundary from the Volterra equation, solved in three zones, reproduces its law to Monte Carlo accuracy (Kolmogorov distance 0.0020, against 0.0059 for the tangent boundary)	proved and computed	Section 8.2
Keiper Li coefficients $\lambda_1, \ldots, \lambda_{100}$ from the Pólya density, λ_1 to 25 digits	computed	Section 8.3
Closest pair below height 200 collides at de Bruijn Newman time -0.288; the Lehmer pair near 7005 at -7.11×10^{-4}	computed	Section 8.4
For $L(s, \chi_{12})$ and $L(s, \Delta)$ the sine squared lemma has no pole piece but a signed prime piece	proved	Section 8.5
The eta kernel of $L(s, \chi_{12})$: GRH below height 100 from the zero density identity and the 67 sign changes	computed	Section 8.5
For an imaginary quadratic field the Pólya kernel of ζ_K is positive exactly when an explicit theta sum is positive on a finite interval, which gives $L(\frac{1}{2}, \chi_d) > 0$ for $d = -3, -4, -7, -8$; it takes negative values for every $d \equiv 5 \pmod{24}$	proved	Section 8.5
For the Dirichlet beta function $L(s, \chi_4)$, whose kernel is the cube of the eta function: GRH below height 100; a positive Thorin density at every basepoint $\alpha \geq 1$, which the absolute values of the prime weights certify only for $\alpha > 2.2823$; the density as the difference of those of $\xi_{\mathbb{Q}(i)}$ and ξ, and as the derivative of the argument of the completed function on the line	proved and computed	Section 8.5.7
Nonnegative deconvolution of $\rho_{0.1}$ on $[15, 55]$, with the zeros outside $[5, 65]$ as background, recovers every zero in the window to within 10^{-5}	computed	Section 8.6
For Pólya's Bessel function $K_{i\theta/2}(2\pi)$ the index side clock is the hitting time of an explicit diffusion, its Thorin atoms are the squared zeros and its trace is completely monotone; its de Bruijn Newman constant is finite, by Newman's theorem, although none of its real zeros below height 90 collide down to $t = -60$	proved and computed	Section 8.7

Table 1.2: Principal results of Chapters 6 to 8.

Result	Status	Where
T_1, with Laplace transform $\xi(1)/\xi(1+\sqrt{u})$, is a GGC with positive Thorin density, smallest at the origin, by a computation of the zeros below height 2000 in ball arithmetic and explicit zero counts above it, so that T_1 is a positive stable one half variable plus an independent GGC; its law has infinite mean and a stable one half tail, and the Lévy form of $\xi(1+s)$, which converges by the prime number theorem, gives its Thorin density from the primes	proved and computed	Section 9.1
Below the basepoint one, $v_\alpha(\theta)/(\alpha-\frac{1}{2})$ is nonincreasing in α away from the zeros off the line; the Thorin density is smallest at the origin, and only there, for every $\alpha \in (\frac{1}{2}, 7.35]$ on $\lvert\theta\rvert \le 3\times 10^{12} - 1.21$, for $L(s,\chi_4)$ and $L(s,\chi_{12})$ at every height for every $\alpha \ge 1$, and at every height for every $\alpha \in (\frac{1}{2}, 1)$ exactly when RH holds; for every $\alpha > \frac{1}{2}$ it is positive on $\lvert\theta\rvert \le 3\times 10^{12} - \frac{1}{2}$	proved and computed	Sections 9.1.6, 11.4
Every T_α, $\alpha > \frac{1}{2}$, exists as the tilted centre clock run through the inverse local time of a sticky Brownian motion, and it is a GGC if and only if no zero has real part greater than α	proved	Section 9.2
For real u and $\alpha \ge \frac{1}{2}$, $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ is the characteristic function of an infinitely divisible normal scale mixture; for $\frac{1}{2} < \alpha < 1$ it is an extended GGC, and the translates of every tilted Pólya kernel $\Phi_{\alpha'}$ with $\alpha' > \alpha$ are complete in L^1, exactly when no zero has real part greater than α	proved	Section 10.1
The complete monotonicity conditions of the Poisson trace $K_{1/2}$, which together are RH, hold for every order $n < T_0(\pi - y)$ at the point y; $K_{1/2}$ and the trace of Pólya's Bessel function differ by $\frac{9}{8} + O(y \log \frac{1}{y})$ as $y \to 0$	proved and computed	Section 10.1.5
$1/\zeta(2k)$ is, up to an elementary factor, a value of the Laplace transform of the centre clock and of T_1; for $\frac{1}{2} \le \alpha < 1$ the bound $O(x^{\alpha/2+\delta})$ on Riesz's function, for every $\delta > 0$, holds exactly when every $T_{\alpha'}$ with $\alpha' > \alpha$ is a GGC	proved	Section 10.2
The constant of $\sum_{n\le x} \mu(n)^2/n$ is an elementary term plus $\mathbb{E}V_1 - 2\,\mathbb{E}V_2$, two tilted Pólya means; RH holds exactly when the squarefree count smoothed by an exponential or a Gamma(2) law, or the count in mean square, has error $O(x^{1/4+\varepsilon})$; the error is $\Omega_\pm(x^{1/4})$; Gamma(2) smoothing writes it as an average of the Mertens function under a positive law, and two is the largest shape for which this holds	proved	Section 10.3
The centre clock is driven by a subordinator with Lévy density $-W'$ and is unimodal, as is T_α for $\alpha \ge 1$; the tilted Pólya variable is quasi infinitely divisible for every $\alpha > 1$ but not at $\alpha = 1$	proved	Section 11.1
For $L(s,\chi_{12})$ the Thorin measure at the basepoint one is positive, and its Lévy form there has no pole to compensate and prime atoms of both signs	proved	Section 11.3

Result	Status	Where
For $\frac{1}{2} < \alpha < 1$, in the density $\rho_{\alpha-1/2}$ the Cauchy kernels of the zeros to the right of α carry at heights $\theta \le T$ at most $T^{e(\alpha)-1+o(1)}$ times the mass of the kernels of the other zeros, with $e(\alpha) < 1$, by the zero density estimates of Ingham, of Guth and Maynard and of Huxley [124, 104, 119]	proved	Section 11.4
With no hypothesis, the size biased centre clock is the clock plus an independent variable with density $W/\mathbb{E}T$, and RH holds exactly when this variable is a mixture of exponential laws, with weight $\gamma^{-2}/\mathbb{E}T$ at the rate γ^2, about a fifth at the lowest zero; the Lambert law, with Laplace transform $\mathcal{W}(u)/u$, is a GGC whose Thorin measure is the image of the uniform law on $(0,\pi)$ under $b \mapsto b\,e^{-b\cot b}/\sin b$, by a proof of which only one step fails for the clock, and that step is RH	proved and computed	Sections 11.5.2, 11.5.3
The centre clock lies in Urbanik's class L_2 of multiply self decomposable laws and not in L_3; it is not the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter	proved and computed	Section 11.5.4
A property of even kernels that also holds for the damped Pólya kernel $e^{tv^2/4}\Phi(v)$ for some $t < 0$ cannot by itself imply RH, by the theorem of Rodgers and Tao; the GGC property of the variable of Biane, Pitman and Yor fails for the damped kernel at every $t < 0$ tested and, provably, for small $\lvert t\rvert$, so the filter does not exclude it there	proved and computed	Section 11.5.5

Table 1.3: Principal results of Chapters 9 to 11. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 5.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic.

1.1.3 Conventions and how to read this book

The clock is a function of $u = s^2$; earlier papers of the programme used $s^2/2$, which rescales Thorin measures by $z \mapsto z/2$. Thorin measures are measures in the variable z, and we write $\theta = \sqrt{z}$ for the chart in which the Thorin measure at the basepoint $\frac{1}{2} + \varepsilon$ has density $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$; ε is the resolution. The Pólya kernel Φ is taken in Titchmarsh's normalisation, $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u)\cos(\theta u)\,du$. $N(T)$ counts zeros with $0 < \operatorname{Im}\rho \le T$ and $\nu(\sigma, T)$ those with, in addition, $\operatorname{Re}\rho > \sigma$. Total variation of a signed measure is the total variation norm $\lvert\mu\rvert(E)$ on the stated window. Chapters 6 and 7, and Sections 4.2 and 4.3, write s for u, and the Kent approximants of Section 8.1 write w; each declares its variables where it starts, and Appendix C lists the notation of the book with the place where each symbol is defined. Numerical values are quoted to the accuracy at which they have been checked.

Readers mainly interested in computation can read Sections 1.2, 2.1 and 2.3 and then go straight to Chapters 3 to 5. Readers interested in the probabilistic

side can start with Sections 1.3, 8.1 and 8.2 and continue with Chapters 9 to 11; these chapters build on Sections 1.2 and 1.3 and draw on Section 3.4 for the zero count. Chapters 6 and 7 can be read after Chapter 1 and the statements of Theorems 5.1.1 and 5.1.6 in Section 5.1; they form the most technical part of the book. Readers looking for the conclusions can go directly to Sections 12.1 and 12.2.6. Throughout, statements are marked as proved, computed, announced or open.

1.2 The Xi function, the Pólya density and the clock

Most of the material is classical or taken from [252].

1.2.1 The completed zeta function and the clock

The completed zeta function ξ of Section 1.1 is entire of order one and maximal type, satisfies $\xi(s) = \xi(1-s)$ and $\overline{\xi(\bar{s})} = \xi(s)$, and is positive on the real axis. On the critical line $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ is real and even. The zeros $\rho = \beta + i\gamma$ of ξ lie in the open strip $0 < \operatorname{Re} s < 1$ and are symmetric under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar{\rho}$.

Hadamard's theorem applied to G, which is entire of order $\frac{1}{2}$, gives, with $u = s^2$,

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \prod_k \Big(1 - \frac{s^2}{(\rho_k - \frac{1}{2})^2}\Big) = \prod_k \Big(1 + \frac{s^2}{a_k}\Big), \tag{1.2.1}$$

with $\sum |a_k|^{-1} < \infty$. Logarithmic differentiation gives the partial fraction expansion

$$\frac{\xi'}{\xi}(s) = \sum_\rho \frac{1}{s-\rho}, \tag{1.2.2}$$

in which each ρ is grouped with $1-\rho$, as the differentiation of (1.2.1) produces, or with $\bar{\rho}$. Grouped either way the series converges absolutely, since each group is $O(\gamma^{-2})$ for fixed s and $\sum_\rho \gamma^{-2} < \infty$; without grouping it converges only when summed symmetrically in $\operatorname{Im}\rho$.

Passing to $u = s^2$ folds each pair $\pm(\rho_k - \frac{1}{2})$ of zeros of $\xi(\frac{1}{2}+s)$ into the single zero $-a_k$ of G, so the product has genus zero and needs no exponential factors. Since ξ has no real zeros, a_k is real only when ρ_k lies on the critical line, and then $a_k = \gamma_k^2 > 0$. RH is therefore the statement that every zero of G lies on the negative real axis. The reciprocal of a product of genus zero with negative zeros is a product of gamma Laplace transforms, and this is how generalised gamma convolutions enter.

A random variable $H \geq 0$ is a generalised gamma convolution (GGC) if $\mathbb{E}e^{-uH} = \exp\{-du - \int_{(0,\infty)} \log(1+u/z)\, U(dz)\}$ for $u > 0$, with a drift $d \geq 0$ and a positive measure U, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$. The drift and the Thorin measure are unique, and the

Laplace transform of a GGC extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$.

Unconditionally, $1/G$ is the Laplace transform of a probability law on $[0, \infty)$ [252, Theorem A]; Section 5.1 states this as Theorem 5.1.1 and proves it. A random variable T with $\mathbb{E}e^{-uT} = 1/G(u)$ for $u \geq 0$ is called the clock variable. Its mean and variance are $\mathbb{E}T = \sum_k a_k^{-1} = G'(0) = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ and $\operatorname{Var} T = \sum_k a_k^{-2} = 3.71726 \times 10^{-5}$; these constants, and $\xi(\frac{1}{2}) = 0.4971207781883141$, recur throughout the book.

Proposition 1.2.1 (Rigidity, [252, Proposition 3.4]). *The clock $1/G$ is the Laplace transform of a GGC if and only if RH holds, and then its Thorin measure is U_{RH} of (1.1.1) and the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ.*

If $1/G$ is the Laplace transform of a GGC with drift d and Thorin measure U, then $d + \int U(dz)/(u+z) = G'/G(u) = \sum_k (u + a_k)^{-1}$ for $u > 0$. The left side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, so every pole $-a_k$ of the right side lies on the negative real axis. Hence every a_k is positive, which is RH, and uniqueness of the Stieltjes representation forces $d = 0$ and $U = \sum_k \delta_{a_k}$. Conversely, under RH, (1.2.1) writes $1/G$ as a product of gamma Laplace transforms.

1.2.2 The Pólya kernel and the Pólya density

Riemann's continuation of ζ writes ξ as a Mellin transform of a combination of $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and its derivatives, and uses the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$, $x > 0$. The substitution $x = e^{2u}$ turns the Mellin transform into a Fourier transform, and in Titchmarsh's normalisation [324, Section 10.1] the result is

$$\begin{aligned} \Phi(u) &= 2 \sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right) \exp(-\pi n^2 e^{2u}), \\ \Xi(\theta) &= \int_{\mathbb{R}} \Phi(u) \cos(\theta u)\, du. \end{aligned} \tag{1.2.3}$$

Here u is the variable of the kernel, not the argument of the clock. Pólya studied this representation [273], and Φ is called the Pólya kernel.

The kernel is even and strictly positive, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2} \exp(-\pi e^{2|u|})$ as $|u| \to \infty$. Evenness is the functional equation read through the theta transformation (Exercise 1.2). For $u \geq 0$ every term of (1.2.3) is positive, because $2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2} = \pi n^2 e^{5u/2}(2\pi n^2 e^{2u} - 3)$ and $2\pi n^2 e^{2u} \geq 2\pi > 3$; positivity for $u < 0$ follows by evenness. Evenness decides how Φ is computed: term n is $n^{-1/2}$ times the first term translated to the left by $\log n$, so for $u \geq 0$ the first term dominates, while for $u < 0$ the terms are individually of order $n^{-1/2}$ and cancel almost completely, and computations therefore evaluate $\Phi(|u|)$.

Each term of (1.2.3) is entire in u, and since $\operatorname{Re} e^{2u} = e^{2x}\cos 2y$ for $u = x+iy$, the series converges locally uniformly in the strip $|\operatorname{Im} u| < \pi/4$, so Φ is analytic there; the boundary lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary inherited from the theta function [284, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The strip of analyticity makes the trapezoid rule spectrally accurate and permits the tilted contour of Section 2.1.3. Figure 1.2 shows $|\Phi|$ on lines inside the strip.

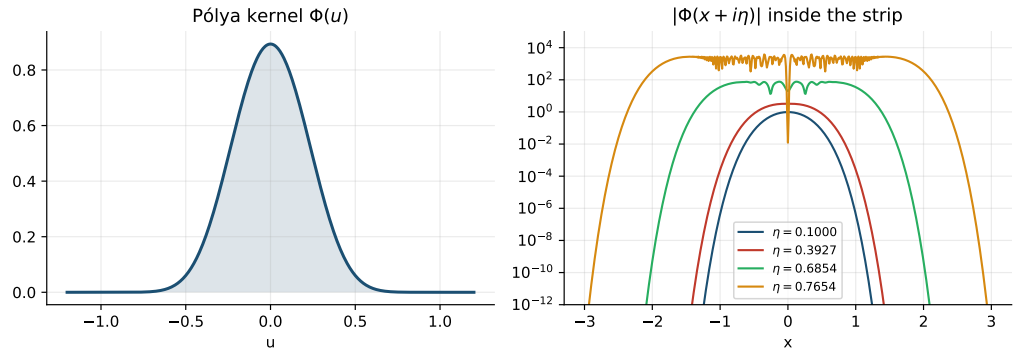


Figure 1.2: Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$.

Since $\Xi(0) = \xi(\frac{1}{2}) = \int_{\mathbb{R}} \Phi$, the Pólya density $p = \Phi/\Xi(0)$ is the law of a symmetric random variable X with all exponential moments, and

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad s \in \mathbb{C}. \tag{1.2.4}$$

The clock is $1/\mathbb{E}\cosh(\sqrt{u}\,X)$, and at $u = \theta^2$ it forms with the characteristic function of X a van Dantzig pair, that is, a pair (φ_1, φ_2) of characteristic functions with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$; the name goes back to a problem of van Dantzig, which Lukacs studied [195]. Here $\varphi_1(\theta) = \Xi(\theta)/\Xi(0)$ is the characteristic function of X, and $1/\varphi_1(i\theta) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta) = \mathbb{E}e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T [252, Theorem A]. G. Letac asked whether φ_1 is the first member of a van Dantzig pair (private communication), and Theorem 5.1.1 shows that it is, unconditionally.

Two classical pairs show the shape to expect. The first is $(\cos\theta, 1/\cosh\theta)$ and the second is $(\sin\theta/\theta, \theta/\sinh\theta)$. In both the first member has only real zeros and the second is the product of the reciprocal factors: $1/\cosh\theta = \prod_{k\geq 1}(1+\theta^2/((k-\frac{1}{2})\pi)^2)^{-1}$ is the characteristic function of $\sum_{k\geq 1} L_k/((k-\frac{1}{2})\pi)$, where the L_k are

independent standard Laplace variables, with density $\frac{1}{2}e^{-|x|}$ and characteristic function $1/(1+\theta^2)$, and likewise $\theta/\sinh\theta$ is that of $\sum_{k\geq 1} L_k/(k\pi)$.

The Xi pair has the classical shape exactly when RH holds. If RH holds, $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), so $Z\sqrt{2T}$ has the law of $\sum_{\gamma>0} L_\gamma/\gamma$, the sum that Section 4.2 calls $\log W$. Conversely, if $Z\sqrt{2T}$ has the law of $\sum_k L_k/c_k$ for real c_k with $\sum_k c_k^{-2} < \infty$, then $\xi(\frac{1}{2}+\theta)/\xi(\frac{1}{2}) = \prod_k (1+\theta^2/c_k^2)$ for real θ, hence for complex θ, and every zero of ξ lies on the critical line. Konstantopoulos, Patie and Sarkar state the same equivalence in the language of the Laguerre Pólya class and of Pólya frequency functions [153, Theorem 2.3 and Corollary 2.4].

The Pólya variable also has a probabilistic construction. Let $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponentials e_k, e'_k; it has the law of the sum of two independent hitting times of level 1 by a three dimensional Bessel process started at 0. By the theorem of Biane, Pitman and Yor [32], going back to Williams [344], $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$. With $Y = \frac{1}{2}\log(\pi S_2/2)$ this reads $\mathbb{E}e^{sY} = 2\xi(s)$, so X is Y tilted by $e^{Y/2}$: $\mathbb{E}f(X) = \mathbb{E}[f(Y)e^{Y/2}]/\mathbb{E}e^{Y/2}$. Tilting Y by $e^{\alpha Y}$ in place of $e^{Y/2}$ gives, at every real basepoint α and with no hypothesis on the zeros, a variable V_α with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$ and $V_{1/2} = X$ (Section 9.1.1).

1.2.3 The heat trace: spectral and arithmetic forms

The heat trace of the zeros is

$$W(t) = \sum_k e^{-a_k t} = \tfrac{1}{2}\sum_\rho e^{(\rho-\frac{1}{2})^2 t}, \qquad t > 0,$$

and the series converges absolutely for every $t > 0$ wherever the zeros lie, since $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > \gamma_k^2 - \frac{1}{4}$ for $\rho_k = \beta_k + i\gamma_k$. Integrating $e^{-(u+a_k)t}$ term by term, and using Frullani's integral $\log(1+u/a) = \int_0^\infty (1-e^{-ut})e^{-at}t^{-1}dt$ for $\operatorname{Re} a > 0$, gives, unconditionally for $u > 0$ [252, Theorem 4.3],

$$\frac{G'}{G}(u) = \int_0^\infty e^{-ut} W(t)\, dt, \qquad \log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\frac{W(t)}{t}\, dt.$$

The second formula shows that $W(t)/t$ is the only possible Lévy density of the clock variable T.

The classes of infinitely divisible laws of Section 1.3.2 become sign conditions on W [252, Theorem 4.6]: T is infinitely divisible if and only if $W \geq 0$, self decomposable if and only if W is nonincreasing, and a GGC if and only if W is completely monotone, which by Proposition 1.2.1 happens if and only if RH holds. The first two conditions hold unconditionally, in the strict forms $W > 0$ and $W' < 0$ on $(0,\infty)$ (Theorem 5.1.1, [252, Theorem A]), by a proof that uses the verification of RH below height 1000.

The explicit formula expresses W through the primes. For even test functions h holomorphic and rapidly decreasing in a strip $|\operatorname{Im} r| \leq \frac{1}{2} + c$ with $c > 0$, Weil's

explicit formula [338], [128, Theorem 5.12], reads

$$\sum_{\rho} h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\big(\operatorname{Re}\psi(\tfrac{1}{4}+\tfrac{ir}{2}) - \log\pi\big)dr \\ - 2\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\, g(\log n),$$

where $\rho = \frac{1}{2}+i\gamma_\rho$, $g(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)e^{-irx}dr$ and $\psi = \Gamma'/\Gamma$ is the digamma function. The Gaussian $h(r) = e^{-r^2t}$ has $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$ and $\sum_\rho h(\gamma_\rho) = 2W(t)$, which gives the arithmetic form [252, Theorem 4.4]

$$W(t) = e^{t/4} + \frac{1}{4\pi}\int_{\mathbb{R}} e^{-r^2t} m(r)\, dr - \frac{1}{2\sqrt{\pi t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/4t}, \\ m(r) = \operatorname{Re}\psi\big(\tfrac{1}{4}+\tfrac{ir}{2}\big) - \log\pi. \tag{1.2.5}$$

The three terms of (1.2.5) are the polar, archimedean and prime contributions. The polar term $e^{t/4}$ comes from $h(\pm\frac{i}{2})$, that is, from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s=1$ and $s=0$, which the factor $s(s-1)$ of ξ removes. The prime term is the series $\sum\Lambda(n)n^{-1/2}$ smoothed by a Gaussian of standard deviation $\sqrt{2t}$ in $\log n$.

Table 1.4 reproduces the comparison of [252, Table 4.1], and the two sides agree to fifteen digits. The polar and archimedean terms are of order one while $W(t)$ decays like $e^{-\gamma_1^2 t}$ with $\gamma_1^2 = 199.79$. An estimate of W from (1.2.5) at moderate t must therefore survive a cancellation whose depth grows linearly in t. Section 5.3.2 extends the comparison to smaller and larger t and gives that depth explicitly.

t	$W(t)$ from zeros	polar + archimedean − prime	archimedean	prime
0.01	0.149698011253780	0.149698011253780	−0.8527967	8.3995×10^{-6}
0.02	$1.85412917313195\times10^{-2}$	$1.85412917313195\times10^{-2}$	−0.9840612	0.0024101
0.05	$4.58783504857357\times10^{-5}$	$4.58783504857357\times10^{-5}$	−0.9546186	0.0579140
0.10	$2.10479979582235\times10^{-9}$	$2.10479979582235\times10^{-9}$	−0.8625096	0.1628055
0.20	$4.43018218022364\times10^{-18}$	$4.43018218022364\times10^{-18}$	−0.7497538	0.3015173

Table 1.4: The heat trace from the zeros and from the explicit formula, after [252, Table 4.1]. At $t=1$ the cancellation between the arithmetic terms reaches about eighty seven digits.

1.2.4 The Thorin density at a shifted basepoint

The formula for the Thorin density in the theorem below defines a density at every basepoint $\alpha > \frac{1}{2}$, whatever the zeros do, and it is the starting point of the computations in Chapters 3 to 5. For a real basepoint α write $v_\alpha(r) = \operatorname{Re}(\xi'/\xi)(\alpha+ir)$, and let U_α be the signed measure on $(0,\infty)$ given by

$$U_\alpha(dz) = \frac{v_\alpha(\sqrt{z})}{2\pi\sqrt{z}}\, dz. \tag{1.2.6}$$

At $\alpha = \frac{1}{2} + \varepsilon$, where $\varepsilon > 0$ is the resolution, the chart $\theta = \sqrt{z}$ gives U_α the density

$$\rho_\varepsilon(\theta) = \frac{1}{\pi}\, v_{1/2+\varepsilon}(\theta) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}\Big(\frac{1}{2} + \varepsilon + i\theta\Big), \tag{1.2.7}$$

so that $U_\alpha(dz) = \rho_\varepsilon(\theta)\, d\theta$. The family $\varepsilon \mapsto \rho_\varepsilon$ is the Cauchy flow of [252, Chapter 8].

THEOREM 1.2.2 ([252, Theorem 3.7 and Remark 3.10]). *If $\alpha > \operatorname{Re}\rho$ for every zero, in particular for every $\alpha \geq 1$, then $u \mapsto \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a GGC T_α with Thorin density* (1.2.6)*, and $v_\alpha > 0$. It is the Laplace transform of a GGC if and only if $\alpha \geq \sup_\rho \operatorname{Re}\rho$.*

Until 2026 no basepoint $\alpha < 1$ was known to satisfy the hypothesis of the theorem; the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would give every $\alpha > \frac{7}{8}$ (Section 12.1.2). At every basepoint $\alpha > \frac{1}{2}$ the shifted clock is still the Laplace transform of a probability law, with no hypothesis on the zeros (Theorem 9.2.5); what the zeros decide is whether that law is a GGC. Let $P_a(x) = a/(\pi(a^2 + x^2))$ denote the Poisson kernel, with $P_{-a} = -P_a$. Write μ_ε for the signed measure $U_{1/2+\varepsilon}$, and use the same symbol for its image in the chart $\theta = \sqrt{z}$, where it has the density (1.2.7). By Theorem 1.2.2, μ_ε is the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2} + \varepsilon > \operatorname{Re}\rho$ for every zero, in particular for every $\varepsilon \geq \frac{1}{2}$, and we call it the Thorin measure at resolution ε in general.

PROPOSITION 1.2.3 (Poisson structure). *For $\varepsilon > 0$ and $\theta \geq 0$ such that $\frac{1}{2} + \varepsilon + i\theta$ is not a zero of ξ,*

$$\begin{aligned}\rho_\varepsilon(\theta) = &\sum_{\gamma>0,\ \operatorname{Re}\rho=1/2} \big[P_\varepsilon(\theta-\gamma) + P_\varepsilon(\theta+\gamma)\big] \\ &+ \sum_{\text{off line } \rho=\frac{1}{2}\pm\delta+i\gamma,\ \gamma>0} \Big(\big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta-\gamma) + \big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta+\gamma)\Big),\end{aligned} \tag{1.2.8}$$

where the second sum counts each off line pair $\{\frac{1}{2} + \delta + i\gamma, \frac{1}{2} - \delta + i\gamma\}$, $0 < \delta < \frac{1}{2}$, once. Equivalently, with $\alpha = \frac{1}{2} + \varepsilon$, $\rho_\varepsilon(\theta)$ is the sum of $P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)$ over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$. In particular, under RH, $\mu_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ restricted to $\theta > 0$, where $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma + \delta_{-\gamma})$.

Proof. Take real parts in (1.2.2) at $s = \alpha + i\theta$ with $\alpha = \frac{1}{2} + \varepsilon$: a zero $\rho = \beta + i\gamma$ contributes $\pi P_{\alpha-\beta}(\theta-\gamma)$ to $v_\alpha(\theta)$, and $\alpha - \beta = \varepsilon \mp \delta$ when $\beta = \frac{1}{2} \pm \delta$. Dividing by π and grouping each zero ρ with $\bar\rho$ gives the second form, and collecting the zeros ρ, $\bar\rho$, $1-\rho$ and $1-\bar\rho$ gives (1.2.8). The series converges absolutely, and uniformly for θ in compact sets avoiding the zeros, since each term is $O(\gamma^{-2})$ there and $\sum_\rho \gamma^{-2} < \infty$. □

An off line pair contributes kernels of widths $\varepsilon + \delta$ and $\varepsilon - \delta$, and the second is negative once $\varepsilon < \delta$, that is, once the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ lies to the left of the zero

$\frac{1}{2} + \delta + i\gamma$. If every zero lies to the left of the line, every kernel is positive and so is v_α; a negative value of v_α therefore proves a zero to the right of α.

Remark 1.2.4 (Positivity on one line). The criterion of Theorem 1.2.2 concerns the analytic continuation of the shifted clock, not the sign of v_α on the single line $\operatorname{Re} s = \alpha$. Once a zero ρ lies to the right of the line, the shifted clock ceases to be a GGC, since $\xi(\alpha + \sqrt{u})$ vanishes at $u = (\rho - \alpha)^2 \in \mathbb{C} \setminus (-\infty, 0]$, but the density need not change sign, because the positive kernels of the other zeros can outweigh the negative one. Exercise 1.7 gives an entire function with the order, symmetry and reality of ξ whose density is positive on a line with a zero to its right; for ξ itself it is not known, without assuming RH, whether positivity on one line excludes zeros to its right. Positivity on every line to the right of $\operatorname{Re} s = \sigma_0$ does exclude zeros there: if $v_\alpha \geq 0$ for every $\alpha > \sigma_0$, then no zero has $\operatorname{Re}\rho > \sigma_0$, because $\operatorname{Re}(\xi'/\xi)(\rho - b) \sim -m/b$ as $b \downarrow 0$ at a zero ρ of multiplicity m. With $\sigma_0 = \frac{1}{2}$ this is Lagarias's criterion [160]: RH holds if and only if $v_\alpha \geq 0$ for every $\alpha > \frac{1}{2}$.

Integrated over $\theta > 0$, a line zero contributes mass one, which concentrates at $\theta = \gamma$ as $\varepsilon \downarrow 0$. An off line pair contributes mass two when $\varepsilon > \delta$, and when $\varepsilon < \delta$ a signed measure of mass zero whose total variation tends to zero as $\varepsilon \downarrow 0$. These limits can be summed over the zeros: if f is continuous with support in $[a, b] \subset (0, \infty)$, then for $0 < \varepsilon \leq 1$ each term of (1.2.8) with $\gamma > b + 1$ integrates f to a number of modulus at most $C_f \gamma^{-2}$, where C_f depends only on f, because the widths c in (1.2.8) satisfy $|c| < \varepsilon + \frac{1}{2} \leq \frac{3}{2}$, $|P_c(x)| \leq |c|/(\pi x^2)$, and $|\theta \pm \gamma| \geq \gamma - b \geq \gamma/(b+1)$ for $\theta \in [a, b]$; as $\sum_\rho \gamma^{-2} < \infty$, dominated convergence applies to the sum over the zeros. Hence $\mu_\varepsilon \to U_{\text{line}} = \sum_{\text{line zeros}} \delta_{\gamma^2}$ vaguely, whatever the truth of RH [249, Proposition 9.5].

1.2.5 The sine squared lemma: the Thorin density from the primes

The sine squared lemma produces the same Thorin density from the Lévy side, where the primes appear as atoms. For $x > 0$ and $s \geq 0$,

$$e^{-sx} + sx - 1 = x \int_0^\infty \log\Big(1 + \frac{s^2}{z}\Big) \frac{\sin^2(x\sqrt{z}/2)}{\pi\sqrt{z}}\, dz. \tag{1.2.9}$$

To prove it, substitute $z = \theta^2$, which turns the right side into $\frac{x}{\pi} \int_0^\infty \log(1 + s^2/\theta^2)(1 - \cos x\theta)\, d\theta$. Both sides vanish at $s = 0$, and their derivatives in s agree because $\int_0^\infty (1 - \cos x\theta)\, 2s(\theta^2 + s^2)^{-1} d\theta = \pi(1 - e^{-sx})$.

The identity (1.2.9) is [249, Lemma 8.1].

Integrating (1.2.9) against a measure turns Lévy representations into Thorin representations. A Lévy representation $\log f(\alpha + s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1) L(dx)$ for $s \geq 0$, with c real and L a real measure on $(0, \infty)$ such that $\int \min(x, x^2)\, |L|(dx) < \infty$, becomes a Thorin representation with density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\, xL(dx)]$, and the bracket equals $\operatorname{Re}(f'/f)(\alpha + i\sqrt{z})$ [249, Theorem 8.2]. For $f = \xi$ at $\alpha > 1$ the input is the signed canonical measure

α	θ	sine squared, primes $\le 2\times10^6$	Pólya route	error of prime side
3.0	14.1347	0.561447238789	0.561447238789	1.1×10^{-13}
3.0	50	1.120967014740	1.120967015289	5.5×10^{-10}
2.0	14.1347	0.766398694727	0.766399169607	4.7×10^{-7}
2.0	50	1.224475714058	1.224476216908	5.0×10^{-7}
1.5	14.1347	1.065762504228	1.067140080666	1.4×10^{-3}
1.5	50	1.385288214458	1.386706213218	1.4×10^{-3}

Table 1.5: The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure and from the Pólya density; $\theta = 14.1347$ is the first ordinate γ_1, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through Γ and ζ to about 10^{-15}.

$xL(dx)$, with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density [252, Theorem 8.1].

Table 1.5 compares the two routes: the bracket computed from the canonical measure with the primes truncated at $N = 2\times10^6$, and $v_\alpha(\theta)$ computed by the Pólya route of Section 2.1. The difference is the prime tail $\sum_{n>N}\Lambda(n)n^{-\alpha}(1-\cos(\theta\log n))$, of order $N^{1-\alpha}/(\alpha-1)$, except at $\alpha=3$, $\theta=50$, where it comes from the double precision quadrature of the archimedean integral.

1.3 Generalised gamma convolutions and the Thorin condition

The Thorin condition on the clock $1/G$, equivalent to RH by Proposition 1.2.1, is that it be the Laplace transform of a generalised gamma convolution.

1.3.1 The definition and its three readings

As in Section 1.2, a random variable $H \ge 0$, or its law, is a generalised gamma convolution (GGC) if

$$\mathbb{E}e^{-uH} = \exp\Big\{ -du - \int_{(0,\infty)} \log\Big(1+\frac{u}{z}\Big)U(dz)\Big\}, \qquad u>0, \tag{1.3.1}$$

with drift $d \ge 0$ and a positive Radon measure U on $(0,\infty)$, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1}U(dz) < \infty$ [319, 320], [39, p. 29]. The drift and the Thorin measure are unique [39, p. 30]. Since $\log(1+u/z)$ is the Laplace exponent of the exponential law with rate z, a discrete measure $U = \sum_j c_j\delta_{z_j}$ makes H the sum of d and independent gamma variables with shapes c_j and rates z_j. The class is the closure of such sums under weak limits (Thorin's closure theorem, Section 4.1.1).

The first reading is through the Lévy measure. Frullani's integral $\log(1+u/z)=\int_0^\infty(1-e^{-ut})e^{-zt}t^{-1}dt$ turns (1.3.1) into

$$-\log \mathbb{E}e^{-uH} = du + \int_0^\infty \bigl(1-e^{-ut}\bigr)\frac{k(t)}{t}\,dt, \qquad k(t)=\int e^{-zt}\,U(dz),$$

so H is infinitely divisible with Lévy density $k(t)/t$, and its canonical function k, t times the Lévy density, is completely monotone. Conversely, by Bernstein's theorem, an infinitely divisible law on $[0,\infty)$ with drift d and Lévy density $k(t)/t$, k completely monotone, is a GGC whose Thorin measure is the measure representing k [129, Section 1.2]. For the clock, the unconditional formula for $\log G$ in Section 1.2 gives $k=W$, the heat trace, so the Thorin condition is the complete monotonicity of W.

The second reading is through the Stieltjes transform of the Thorin measure. Differentiating the logarithm of (1.3.1) gives

$$f_H(u) = -\frac{d}{du}\log \mathbb{E}e^{-uH} = d + \int \frac{U(dz)}{u+z},$$

a nonnegative constant plus the Stieltjes transform of a positive measure, holomorphic on $\mathbb{C}\setminus(-\infty,0]$. At distance $\eta>0$ below the cut, $\pi^{-1}\operatorname{Im} f_H(-x-i\eta)=\int P_\eta(x-z)\,U(dz)$ is the Poisson smoothing of U, with P_η the Poisson kernel of Section 1.2, and letting $\eta\downarrow 0$ recovers U by Stieltjes inversion: for continuity points $a<b$ of U,

$$U\bigl((a,b]\bigr) = \lim_{\eta\downarrow 0}\frac{1}{\pi}\int_a^b \operatorname{Im} f_H(-x-i\eta)\,dx. \tag{1.3.2}$$

For the clock, $G'/G=-\frac{d}{du}\log(1/G)$ takes the place of f_H whether or not RH holds. The chain rule applied to $G(s^2)=\xi(\frac12+s)/\xi(\frac12)$ gives $(\xi'/\xi)(\frac12+s)=2s\,(G'/G)(s^2)$. At $s=\varepsilon+i\theta$ we have $\bar s^2=-x-i\eta$ with $x=\theta^2-\varepsilon^2$ and $\eta=2\varepsilon\theta$, and since G is real on the real axis, (1.2.7) becomes

$$\rho_\varepsilon(\theta) = 2\theta\cdot\frac{1}{\pi}\operatorname{Im}\frac{G'}{G}(-x-i\eta) + \frac{2\varepsilon}{\pi}\operatorname{Re}\frac{G'}{G}(s^2).$$

The first term is the integrand of (1.3.2) at distance $\eta=2\varepsilon\theta$ from the cut, multiplied by the Jacobian $dx/d\theta=2\theta$; the second carries the factor ε. The Thorin density at resolution ε is therefore a regularised Stieltjes inversion, taken along the parabola $u=(\varepsilon-i\theta)^2$ rather than along a horizontal line.

Under RH the logarithm gives the counting function directly: for $y>0$ not of the form γ^2, $U_{\mathrm{RH}}((0,y])=-\pi^{-1}\operatorname{Im}\log G(-y-i0)=N(\sqrt{y})$, with $\log G$ continued from $(0,\infty)$ through the lower half plane, since the factor $1+u/\gamma^2$ of G has argument $-\pi$ at $u=-y-i0$ when $\gamma^2<y$ and 0 otherwise.

The third reading is as a product. If $U=\sum_j c_j\delta_{z_j}$ is discrete, (1.3.1) is the product $e^{-du}\prod_j(1+u/z_j)^{-c_j}$ of gamma Laplace transforms. Conversely, if F is

entire of genus zero with $F(0) = 1$ and all its zeros real and negative, then $1/F$ is the Laplace transform of a GGC with $d = 0$ and a unit atom at $-u_j$ for each zero u_j of F, counted with multiplicity. For the clock $F = G$, the product is Hadamard's (1.2.1), and the Thorin condition says that every zero $-a_k$ of G is real and negative, that is, $a_k > 0$.

1.3.2 The Bondesson hierarchy and the clock

Between the infinitely divisible laws on $[0, \infty)$ (ID) and the GGCs lie two classes defined by conditions on the Lévy density $k(t)/t$: the self decomposable laws (SD), for which k is nonincreasing, and the Bondesson class (BO), for which $k(t)/t$ is completely monotone. The GGCs are the laws for which k itself is completely monotone, and inside them lie the laws with a hyperbolically completely monotone density (HCM, defined below). The inclusions are

$$\text{HCM} \subset \text{GGC} \subset \text{SD} \subset \text{ID}, \qquad \text{GGC} \subset \text{BO} \subset \text{ID},$$

and SD and BO are not nested. The Lévy density $e^{-t}(1 + \frac{1}{2t})$ is completely monotone, but $k(t) = e^{-t}(t + \frac{1}{2})$ increases on $(0, \frac{1}{2})$, so the law is in BO and not in SD [249, Section 2]; the Lévy density $t^{-1}\mathbf{1}_{t<1}$ gives a law in SD and not in BO, since a completely monotone function that vanishes on $(1, \infty)$ vanishes identically.

A positive continuous function φ on $(0, \infty)$ is hyperbolically completely monotone if for every $u > 0$ the function $\varphi(uv)\varphi(u/v)$ is completely monotone in $w = v + v^{-1}$, and a law is HCM if it has an HCM density. By Bondesson's theorem [39, Theorem 6.1.1], a function φ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM, so the Thorin condition becomes a family of sign conditions on real derivatives. An HCM law is a GGC, so the HCM property of a law is stronger than that of its Laplace transform, and strictly so: the GGC of Theorem 4.3.20 has an HCM Laplace transform and a density that is not HCM.

The extended generalised gamma convolutions (EGGC), introduced by Thorin [321], are the weak limits of convolutions of gamma laws and their reflections [39, Section 7.1]. A symmetric law is an EGGC if and only if it is the law of $Z\sqrt{2H}$ with Z standard normal and H an independent GGC [39, Theorem 7.3.1]. Since $\mathbb{E}e^{i\theta Z\sqrt{2T}} = \mathbb{E}e^{-\theta^2 T}$ determines the law of T, the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2 is an EGGC if and only if T is a GGC, that is, by Proposition 1.2.1, if and only if RH holds. With no hypothesis $Z\sqrt{2T}$ is self decomposable: subordinating the Lévy measure $W(v)\,dv/v$ of T to the heat kernel gives the Lévy density $k(|x|)/|x|$ with $k(r) = \int_0^\infty e^{-1/4v}(4\pi v)^{-1/2}W(r^2 v)\,dv/v = \sum_j e^{-r\sqrt{a_j}}$ (the first passage identity for each term, with the principal root), and k decreases because W does (Theorem 5.1.1). The z distributions of Barndorff-Nielsen, Kent and Sørensen [18] are the lattice analogue of this family of variance mixtures.

Table 1.6 records what is known about the sign conditions on W that the classes become for the clock. Section 5.2 explains why no finite number of derivative conditions can reach the last row.

class or property	condition on W	status for the clock
ID	$W \geq 0$	proved, with the zeros verified to height 1000 (Theorem 5.1.1)
SD	W nonincreasing	proved, with the zeros verified to height 1000 (Theorem 5.1.1)
strict log convexity	$WW'' > W'^2$ and $W'W''' > W''^2$	proved, with the zeros verified to height 1000 (Section 5.1)
derivative signs to order 1.16×10^{25}	$(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$	proved, with the zeros verified to height 3×10^{12} (Theorem 5.1.13)
complete monotonicity near the origin	$(-1)^n W^{(n)}(t) > 0$ for every n and $t \leq 1.29$	proved, with the zeros verified to height 10^{12} (Theorem 5.1.9)
BO	$W(t)/t$ completely monotone	equivalent to RH (Section 8.5.4); open
GGC	W completely monotone	equivalent to RH; open

Table 1.6: The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$.

Urbanik's classes of multiply self decomposable laws give conditions on W of another kind: with L_0 the class of self decomposable laws and L_n that of the self decomposable laws whose driving process has its law in L_{n-1}, the clock lies in L_n exactly when $(-t\frac{d}{dt})^j W \geq 0$ for $j \leq n+1$ (Section 11.5.4), and for $n \geq 1$ this does not follow from the GGC property, since the exponential law is a GGC whose canonical function e^{-t} has $(-t\frac{d}{dt})^2 e^{-t} = -t(1-t)e^{-t} < 0$ for $t < 1$. For the clock these conditions hold for $n = 2$ and fail for $n = 3$, with no hypothesis (Theorem 11.5.8), while membership of every L_n would imply RH, since $\bigcap_n L_n$ lies inside the GGCs [27].

1.3.3 Diagnostics for the Thorin condition

RH can be stated as a GGC identity on a real interval. The right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$ (Exercise 1.14), so RH holds if and only if there is a GGC H with $\xi(\frac{1}{2})/\xi(\frac{1}{2}+s) = \mathbb{E}e^{-s^2 H}$ for real $0 < s < 1$ alone [249, Proposition 3.3].

As it stands the identity is of no use to a computation, because analytic continuation from an interval is unstable. A unit Thorin atom at z_0 multiplies the transform by $(1 + u/z_0)^{-1}$, which differs from one by less than $1/z_0$ on $(0, 1)$, so transforms that agree to high accuracy on $(0, 1)$ can have very different Thorin

measures at large z. Riesz's criterion meets the same obstruction: its numbers $1/\zeta(2k)$ are, up to explicit factors, values of the clock at the real points $(2k-\frac{1}{2})^2$, and they bear on the zeros only through the analytic interpolation of $k \mapsto 1/\zeta(2k)$ into the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$ (Section 10.2.6). The information about the zeros enters through the analyticity that the representation forces, and a computation has to supply it in another way: through values of ξ and ξ' at complex arguments, which Chapter 2 computes, or through sign conditions on derivatives of every order. Each characterisation of Sections 1.3.1 and 1.3.2 yields such a test, and Table 1.7 lists them with what a failure proves.

diagnostic	computes	a failure proves
pole test	poles of G'/G off $(-\infty, 0]$	not GGC; for the clock, a zero off the line
Thorin density at resolution ε	$\rho_\varepsilon = \pi^{-1}\operatorname{Re}(\xi'/\xi)$ at $\frac{1}{2}+\varepsilon+i\theta$	a zero with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ (Proposition 1.2.3, [160])
zero density scan	$M_\varepsilon(T)+\tau_\varepsilon(T)$ against $N(T)$ (Proposition 3.4.2)	$\frac{1}{2}[N(T)-M_\varepsilon(T)-\tau_\varepsilon(T)]$ zeros with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ below height T
heat trace signs	$(-1)^n W^{(n)}(t)$	not GGC; finitely many orders are consistent with off line zeros (Section 5.2)
Hankel blocks	$\det[(-1)^{i+j}W^{(i+j)}(t)]$	not GGC (Weil's criterion at one scale, Section 5.1)
HCM test	$(-1)^n\partial_w^n[\varphi(uv)\varphi(u/v)]$	not GGC (Section 4.1)
Bernstein test	$(-1)^{n-1}L^{(n)}$, $L=\log G$	not ID, which Theorem 5.1.1 excludes for the clock
Stieltjes moments	Hankel matrices $[m_{i+j}]$ and $[m_{i+j+1}]$, $m_n=\sum_k a_k^{-n-1}$	not GGC

Table 1.7: Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally.

The diagnostics differ in how strongly they weight the higher zeros. Those built from W see a zero at height γ through $e^{-\gamma^2 t}$: in $(-1)^n W^{(n)}(t) = \sum_j a_j^n e^{-a_j t}$ it enters with weight $(\gamma/\gamma_1)^{2n} e^{-(\gamma^2-\gamma_1^2)t}$ relative to the first zero. These tests certify easily and discriminate poorly. The Bernstein test on $L(u) = \log G(u)$, computed for $u > \frac{1}{4}$, where the Dirichlet series of $\log\zeta(\frac{1}{2}+\sqrt{u})$ converges, is better conditioned [252, Section 18.2.2]: since $(-1)^{n-1}L^{(n)}(u) = (n-1)!\sum_k (u+a_k)^{-n}$, a zero at height γ enters with relative weight of order $(\gamma_1/\gamma)^{2n}$. It too sees the higher zeros only through subdominant terms, and for the clock it cannot fail: by the formula for $\log G$ in Section 1.2, $(-1)^{n-1}L^{(n)}(u) = \int_0^\infty t^{n-1}e^{-ut}W(t)\,dt$, which is positive because $W > 0$ (Theorem 5.1.1). Read at $u = 0$ through (1.2.4), which gives $\log \mathbb{E}e^{hX} = \log G(h^2)$, the same formula makes the even cumulants

of the Pólya variable X alternate in sign with no hypothesis: $(-1)^{n-1}\kappa_{2n}(X) = \frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt > 0$ for every $n \geq 1$. Under RH this is the sign rule that Newman proved for laws of Lee Yang type [209]. Complete monotonicity of L' is weaker than the Thorin condition, which asks that L' be the Stieltjes transform $\int U(dz)/(u+z)$ of a positive measure U.

The Thorin density and the zero density scan built from it are local in the ordinate without a search for zeros: by Proposition 1.2.3, a zero at height γ contributes about $1/(\pi\varepsilon)$ to ρ_ε at $\theta = \gamma$ and about $\varepsilon/(\pi d^2)$ at distance $d \gg \varepsilon$.

1.3.4 Explicit Thorin measures and the sinh benchmark

Table 1.8 lists GGCs whose Thorin measures are explicit, in the variable u of the clock. They are models for the clock, test cases for code, and the building blocks of the Vere-Jones representation of Section 6.1 (Theorem 6.1.8 and Table 6.1).

Laplace transform	Thorin measure	law
$(1+u/z_0)^{-a}$	$a\,\delta_{z_0}$	Gamma(a, z_0)
$\pi\sqrt{u}/\sinh(\pi\sqrt{u})$	$\sum_{k\geq 1}\delta_{k^2}$	$2K^2$, K Kolmogorov
$(\sqrt{u}/\sinh\sqrt{u})^2$	$2\sum_{k\geq 1}\delta_{\pi^2k^2}$	$S_2/2$ (Williams)
$\operatorname{sech}\sqrt{u}$	$\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	exit time of $\sqrt{2}B$ from $(-1,1)$
$(\operatorname{sech}\sqrt{u})^{1/2}$	$\frac{1}{2}\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	Meixner time
$e^{-c\sqrt{u}}$	$c\,(2\pi\sqrt{z})^{-1}dz$	positive stable, index $\frac{1}{2}$
$(1+\sqrt{u}/c)^{-1}$	$c\,\{2\pi\sqrt{z}(z+c^2)\}^{-1}dz$	E^2S/c^2
$e^{x(\alpha-\sqrt{\alpha^2+2u})}$	$x\{\pi\sqrt{2z-\alpha^2}\}^{-1}\mathbf{1}_{z>\alpha^2/2}dz$	inverse Gaussian
$\exp\{1-x\sqrt{u}-e^{-x\sqrt{u}}\}$	$x\sin^2(x\sqrt{z}/2)(\pi\sqrt{z})^{-1}dz$	one compensated jump, by (1.2.9)
$\mathcal{W}(u)/u$	image of db/π on $(0,\pi)$ under $r(b) = b\,e^{-b\cot b}/\sin b$	Lambert law (Theorem 11.5.3)
$K_0(2\pi)/K_{\sqrt{u}}(2\pi)$	$\sum_{n\geq 1}\delta_{\nu_n^2}$, ν_n the zeros of $\nu \mapsto K_{i\nu}(2\pi)$	hitting time of a diffusion (Theorem 8.7.10)
$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, $\alpha \geq 1$	$v_\alpha(\sqrt{z})(2\pi\sqrt{z})^{-1}dz$	shifted clock (Theorem 1.2.2)
$\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$	$\sum_{\gamma>0}\delta_{\gamma^2}$ if and only if RH holds	the clock (Proposition 1.2.1)

Table 1.8: GGC Laplace transforms in the variable u, their Thorin measures in z, and the corresponding laws.

In the table, K is the Kolmogorov variable; S_2 is as in Section 1.2; B is a standard Brownian motion; the Meixner time H has $\mathbb{E}e^{-uH} = (\operatorname{sech}\sqrt{u})^{1/2}$, so that $Z\sqrt{2H}$, with Z standard normal and independent of H, has the Meixner law of Section 6.2, whose time in Proposition 6.2.10 is $2H$; and S is the positive stable

variable with $\mathbb{E}e^{-uS} = e^{-\sqrt{u}}$, independent of the standard exponential variable E. Rows two to five are the classical hyperbolic laws [32]. At $u = -t^2$ the reciprocals of rows two and four become $\sin(\pi t)/(\pi t)$ and $\cos t$, the characteristic functions of the uniform law on $(-\pi, \pi)$ and of a symmetric sign. Like Ξ, both have infinitely many real zeros, so real zeros do not obstruct the Thorin property; the property requires them. For the clock, unlike these models, the positivity of the points a_k is the conclusion to be proved [249, Section 10].

The sinh clock $1/G_0$, with $G_0(u) = \sinh(\pi\sqrt{u})/(\pi\sqrt{u}) = \prod_{k\geq 1}(1 + u/k^2)$, is the second row of the table. It has the structure that the clock has under RH, a GGC with unit Thorin atoms, but its ordinates are the integers, equally spaced, and no arithmetic enters. Its Thorin density at resolution ε, the density in $\theta = \sqrt{z}$ of a measure $\mu_\varepsilon^{(0)}$, is

$$\rho_\varepsilon^{(0)}(\theta) = \frac{1}{\pi}\operatorname{Re}\Big[\pi\coth\big(\pi(\varepsilon + i\theta)\big) - \frac{1}{\varepsilon + i\theta}\Big] = \sum_{k\neq 0} P_\varepsilon(\theta - k),$$

exactly the Poisson smoothing of the atoms and their reflections, so the matched total variation of Section 3.2 vanishes identically. The cumulative mass is explicit. As for ξ (Section 3.2), it is π^{-1} times the change in the argument of $\sinh(\pi s)/(\pi s)$ along $s = \varepsilon + i\theta$, and at a half integer $\Theta = m + \frac{1}{2}$ the continuous argument of $\sinh(\pi(\varepsilon + i\theta))$ has grown from 0 to $\pi\Theta$ and that of $\varepsilon + i\theta$ to $\frac{\pi}{2} - \arctan(\varepsilon/\Theta)$, so that

$$\mu_\varepsilon^{(0)}\big((0, \Theta^2]\big) = m + \frac{1}{\pi}\arctan\frac{\varepsilon}{\Theta}.$$

The mass defect $\mu_\varepsilon^{(0)}((0, \Theta^2]) - m$ is therefore $\varepsilon/(\pi\Theta) + O(\varepsilon^3)$. On the window $0 < \theta \leq 50.5$, which holds 50 atoms, the defect is $0.00630\,\varepsilon$ to first order and 3.151×10^{-3} at $\varepsilon = 0.5$.

The cell distance, the sum over the cells C of $|\mu_\varepsilon^{(0)}(C) - 1|$ as in Section 3.2, is explicit as well. We use the cells $(k - \frac{1}{2}, k + \frac{1}{2}]$, $k = 1, \ldots, 50$, and add the mass of the strip $(0, \frac{1}{2}]$, which lies in no cell, as unmatched. For the full lattice $\mathbb{Z}$ each cell would have mass exactly one. The sinh clock lacks the atom at 0, so each of its cells is short by the mass that P_ε puts there, and the strip carries $\pi^{-1}\arctan 2\varepsilon$; Section 5.4 computes these masses. Summing,

$$d_{\mathrm{cell}}^{(0)}(\varepsilon) = \frac{1}{\pi}\Big[2\arctan 2\varepsilon - \arctan\frac{\varepsilon}{50.5}\Big] = \frac{4 - 1/50.5}{\pi}\,\varepsilon + O(\varepsilon^3),$$

which is 0.4968, 0.1250, 0.0631, 0.01267 and 0.00633 at $\varepsilon = 0.5$, 0.1, 0.05, 0.01 and 0.005, and $1.267\,\varepsilon$ in the limit. All of it is due to the atom missing at 0.

For the 79 zeros below height 200 the constant is 5.44 (Section 3.2), and it has a different origin. The zeros are irregularly spaced, so the mass that leaks out of a cell is no longer replaced by the mass that leaks in from its neighbours, and every cell contributes; Section 5.4 reads the constant as a statistic of the spacings. Figure 1.3 divides both distances by the number of atoms, which gives

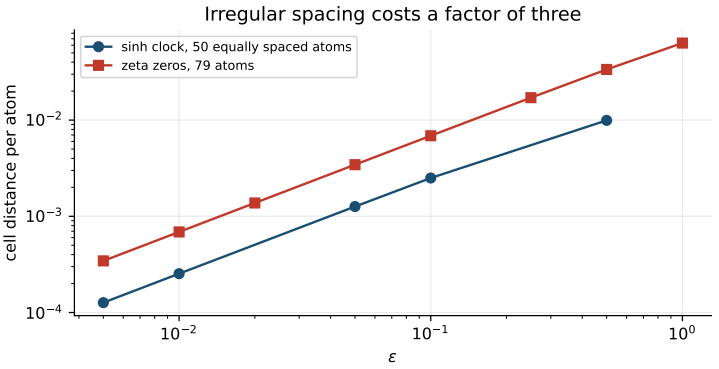


Figure 1.3: Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.2). Both vanish linearly in ε. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only.

$1.267\,\varepsilon/50 = 0.0253\,\varepsilon$ for the sinh clock and $5.44\,\varepsilon/79 = 0.069\,\varepsilon$ for the zeros. Their ratio, nearly three, depends on the window, since for the lattice the distance per atom falls like $1/N$ as the window grows to N atoms. Williams' Thorin measure $2\sum_k \delta_{\pi^2 k^2}$ of Table 1.8 needs no separate benchmark: it is the same lattice rescaled by π, with atoms of mass two, so its Thorin density at resolution ε is $\frac{2}{\pi}\rho^{(0)}_{\varepsilon/\pi}(\theta/\pi)$ and its distances follow by scaling from the closed forms for the sinh clock.

1.4 Exercises

On Section 1.2.

1.1. Show from the Hadamard product (1.2.1) that $\sum_k a_k^{-1} = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$, and compute it to ten digits from the Pólya density by differentiating (1.2.4) twice under the integral.

1.2. Prove that Φ is even by applying the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$ to $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and differentiating twice.

1.3. Show that if X has the Pólya density then $\mathbb{E}X^2 = 2\sum_k a_k^{-1}$, and deduce that the Pólya law has variance 0.0462.

1.4. Verify the first row of Table 1.4 by summing $e^{-\gamma^2 t}$ over the first forty zeros, and estimate the error of truncation using $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$.

1.5. Show that an off line quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ contributes $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, and find the first time at which that contribution is negative.

1.6. Show that at a basepoint $\alpha > 1$ the Thorin density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ is bounded below by a positive multiple of $z^{-1/2}\log(2+z)$ for large z. Use Proposition 1.2.3 with the Riemann von Mangoldt formula, or, more simply, $v_\alpha(r) = \mathrm{Re}[\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)]$ at $s = \alpha + ir$ with Stirling's formula.

1.7. Let $\delta = 0.49$, $\gamma_0 = 10.5$ and

$$F(s) = \frac{\sinh(\pi(s-\frac{1}{2}))}{\pi(s-\frac{1}{2})}\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta+i\gamma_0)^2}\Big)\Big(1 - \frac{(s-\frac{1}{2})^2}{(\delta-i\gamma_0)^2}\Big).$$

Show that F is entire of order one, positive on the real axis and symmetric under $s \mapsto 1-s$, with zeros $\frac{1}{2} \pm ik$ $(k \geq 1)$ and $\frac{1}{2} \pm \delta \pm i\gamma_0$. Show that $\mathrm{Re}(F'/F)(0.6+ir) > 0$ for every real r, although F vanishes at $0.99+10.5i$. (Put $\varepsilon = 0.1$. The zeros $\frac{1}{2}\pm\delta+i\gamma_0$ together contribute $2\varepsilon(t^2-\delta^2+\varepsilon^2)/((t^2+(\delta-\varepsilon)^2)(t^2+(\delta+\varepsilon)^2))$ at $t = r-\gamma_0$, and their conjugates the same expression at $t = r+\gamma_0$; it is at least $-2\varepsilon/(\delta^2-\varepsilon^2)$ and is nonnegative for $t^2 \geq \delta^2-\varepsilon^2$; the zeros on the line contribute $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) - \varepsilon/(\varepsilon^2+r^2) > 0$, and $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) \geq \pi\tanh(\pi\varepsilon)$.)

On Section 1.3.

1.8. Show that the Gamma(a,b) law, with shape a and rate b, has Thorin measure $a\delta_b$, and that the sum of independent Gamma(a_j,b_j) variables has Thorin measure $\sum_j a_j\delta_{b_j}$.

1.9. Derive the Thorin measure of sech $\sqrt{u}$ from the product $\cosh x = \prod_{k\geq 1}(1+4x^2/((2k-1)^2\pi^2))$.

1.10. Prove that $\int_0^\infty \log(1+u/z)\,(2\pi\sqrt{z})^{-1}dz = \sqrt{u}$ (integrate by parts), and deduce the entry for the positive stable law of index $\frac{1}{2}$ in Table 1.8.

1.11. Show that $e^{-t}(1+\frac{1}{2t})$ is completely monotone while $k(t) = e^{-t}(t+\frac{1}{2})$ is not monotone, and conclude that the infinitely divisible law with Lévy density $e^{-t}(1+\frac{1}{2t})$ and no drift is in BO but not in SD.

1.12. For the sinh clock, derive $\mu_\varepsilon^{(0)}((0,\Theta^2]) = m + \pi^{-1}\arctan(\varepsilon/\Theta)$ at $\Theta = m+\frac{1}{2}$ from the periodicity of $\sum_{k\in\mathbb{Z}} P_\varepsilon(\theta-k)$, and check the first order value $0.00630\,\varepsilon$ of the mass defect at $\Theta = 50.5$.

1.13. Show that the HCM property of $\varphi(u) = a/(a+u)$ reduces to complete monotonicity of $a^2/(a^2+auw+u^2)$ in w, and deduce that finite products of such functions are HCM.

1.14. Show that the right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$. Deduce that if an entire function F with $F(0) = 1$ satisfies $1/F(u) = \mathbb{E}e^{-uH}$ for $0 < u < 1$, with H a GGC, then F has no zeros in $\mathbb{C} \setminus (-\infty, 0]$.

Chapter 2

Evaluating the Xi function

2.1 The Pólya route

Pólya's integral (1.2.4) evaluates ξ in the complex plane from the kernel alone, without primes or zeros. In this chapter, δ always denotes the distance from the line of integration to the edge of the strip of analyticity of Φ, not the displacement of an off line zero.

2.1.1 The trapezoid rule on Pólya's integral

Recall the Pólya kernel (1.2.3) in Titchmarsh's normalisation [324, Section 10.1],

$$\Phi(u) = 2\sum_{n\ge 1} \bigl(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\bigr)\exp(-\pi n^2 e^{2u}), \tag{2.1.1}$$

even and positive on the real line, analytic in the strip $|\operatorname{Im} u| < \pi/4$ (Section 1.2) and doubly exponentially decaying in every closed substrip (proof of Proposition 2.1.1). With the Pólya variable X of density $p = \Phi/\Xi(0)$, the moment generating function (1.2.4) and its derivative under the integral sign give, for every complex s,

$$\frac{\xi(\frac12 + s)}{\xi(\frac12)} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad \frac{\xi'(\frac12 + s)}{\xi(\frac12)} = \mathbb{E}\, X e^{sX}, \qquad X \sim p, \tag{2.1.2}$$

so the Thorin density (1.2.7) is $\pi^{-1}\operatorname{Re}\{\mathbb{E}Xe^{sX}/\mathbb{E}e^{sX}\}$ at $s = \varepsilon + i\theta$. Throughout, log is the natural logarithm.

Analyticity and decay in the strip make the trapezoid rule with nodes $u_j = jh$ converge geometrically in $1/h$ [326, Theorem 5.1] (Exercise 2.1): for $s = \varepsilon + i\theta$,

$$\Bigl|\int_{\mathbb{R}} \Phi(u)e^{su}\,du - h\sum_{j\in\mathbb{Z}} \Phi(jh)e^{sjh}\Bigr| \le \frac{2K_\eta(\varepsilon)\, e^{\eta|\theta|}}{e^{2\pi\eta/h} - 1}, \qquad 0 < \eta < \pi/4,$$

where $K_\eta(\varepsilon) = \int_{\mathbb{R}} |\Phi(x \pm i\eta)|\, e^{\varepsilon|x|}\,dx$, the same for both signs because Φ is real on the real axis. We write $K_\eta = K_\eta(0)$. Since $|\xi(\frac12 + \varepsilon + i\theta)|$ decays like $e^{-\pi\theta/4}$,

the relative error is at most of order $\exp\{(\eta+\pi/4)\theta - 2\pi\eta/h\}$. With $h = 1/200$ the bound is best for η close to $\pi/4$, and at $\theta = 200$ it is about 10^{-289}; halving h changes the computed values by less than 10^{-64}.

The limit in practice is the cancellation in (2.1.2): the integrand is of order one and the integral of order $e^{-\pi\theta/4}$, so at b bits the rounding error is of order $2^{-b}e^{\pi\theta/4}$ relative to $|\xi|$, a loss of $\pi\theta/(4\log 10) = 0.341\,\theta$ decimal digits. Discretisation, rounding and the truncation of the u axis at $|u| \le L$ can be budgeted separately. To reach D correct digits at height θ one needs

$$b \;\ge\; \Big(D + \frac{\pi\theta}{4\log 10}\Big)\log_2 10, \qquad h \;\le\; \frac{2\pi\eta}{(D + 0.682\,\theta)\log 10},$$

where the condition on h uses $\eta + \pi/4 \le \pi/2$. The truncation needs $\pi e^{2L} \ge (D + 0.341\,\theta)\log 10 + |\varepsilon|L$, up to lower order terms.

We take $h = 1/200$ and $L = 3$, beyond which Φ is below 10^{-500}, and work at 420 bits (126 decimal digits): at height 200, where $|\xi|$ is about 10^{-63}, the cancellation consumes about half of them and leaves about 60. Truncating the sum at $|j| \le J = L/h$ and folding it with the evenness of Φ gives, with $E = e^{sh}$,

$$\xi(\tfrac12 + s) \approx P(E) + P(E^{-1}), \qquad \xi'(\tfrac12 + s) \approx Q(E) - Q(E^{-1}),$$

$$P(x) = \tfrac{h}{2}\Phi_0 + h\sum_{j=1}^{J}\Phi_j x^j, \quad Q(x) = h^2\sum_{j=1}^{J} j\Phi_j x^j,$$

with $\Phi_j = \Phi(jh)$ and $J = 600$. The node values, the only place where the theta series is summed, are computed once for a given step, truncation and precision and serve every resolution and every height; one evaluation of ξ and ξ' then costs four polynomial evaluations in ball arithmetic [130], about 0.7 ms at 420 bits. The ball radii certify the rounding error only, and they overestimate it: at height 200 they certify about 38 digits of ξ and, at $\varepsilon = 0.005$, about 36 of $\mathrm{Re}(\xi'/\xi)$, while comparison with `mpmath` at 90 digits shows that the values themselves carry about 63.

2.1.2 Zeros and Thorin masses below height 200

With the parameters of Section 2.1.1, the zeros and the Thorin masses below height $\Theta = 200$ are computed at the eight resolutions $\varepsilon = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, 0.005$ in five steps.

1. Compute Φ_j, $0 \le j \le J$, from (2.1.1) at 420 bits.

2. For each resolution ε, evaluate $\rho_\varepsilon(\theta_k)$ and $M_\varepsilon(\theta_k) = \pi^{-1}\arg\xi(\frac12 + \varepsilon + i\theta_k)$, with the branch of the argument that vanishes at $\theta = 0$, on the grid $\theta_k = k\varepsilon/8$ ($k\varepsilon/4$ at $\varepsilon = 0.005$), $0 \le \theta_k \le \Theta$. Since $\partial_\theta \arg\xi(\alpha + i\theta) = v_\alpha(\theta)$ with $\alpha = \frac12 + \varepsilon$, $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2])$ is exactly the cumulative Thorin mass.

3. Locate the line zeros as sign changes of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid of step 0.05, reading each sign from a ball that excludes zero, and refine them by Newton's method using $\Xi'(\theta) = -\operatorname{Im} \xi'(\frac{1}{2} + i\theta)$.

4. Count all zeros with $0 < \operatorname{Im} \rho \le \Theta$ by the argument principle, $N(\Theta) = \pi^{-1} \Delta \arg \xi(s)$ along $2 \to 2 + i\Theta \to \frac{1}{2} + i\Theta$ [324, Section 9.3].

5. Compare with U_{RH} in the metrics of Section 3.2.

Steps 3 and 4, which use the same two polynomials, together bound the number of off line zeros. Each sign change of Ξ on the grid encloses at least one zero on the line, so $N(\Theta)$ minus the number of sign changes is an upper bound for the number of off line zeros with $0 < \operatorname{Im} \rho \le \Theta$. When the two numbers agree, every zero below height Θ is on the line and simple.

In step 2 the argument is accumulated from the increments $\arg(\xi(s_{k+1})/\xi(s_k))$ along $\operatorname{Re} s = \frac{1}{2} + \varepsilon$. The principal value is the true increment only if $\arg \xi$ changes by less than π between consecutive points, which the samples cannot show; the grids are chosen so that every observed increment is below $\pi/2$ in modulus, a margin checked after the fact rather than a guarantee, and the largest observed was 0.29. At large height ξ underflows double precision ($|\xi|$ is about 10^{-335} at height 1000), so the quotient must be formed in ball arithmetic before the argument is taken.

With $\Theta = 200$ there are 79 sign changes, and at none of the 4001 grid points does the ball for Ξ contain zero. The argument principle gives $N(200) = 79.0000000000004$, with a largest increment of 0.02 radians along the contour, and $\Theta = 200$ lies between $\gamma_{79} = 198.0153$ and $\gamma_{80} = 201.2648$, so it is not an ordinate. The Newton refined roots agree to 5.0×10^{-48} with the ordinates computed independently by `mpmath` [205] at 50 digits. The computation thus recovers, with no input but (2.1.1), the classical fact that the 79 zeros below height 200 lie on the line and are simple [11, 247]. It is not yet a proof: the argument increments are summed in double precision from samples, and the analytic error bounds are not added to the radii (Section 12.2.2 lists what a certified version requires). Figure 2.1 shows the computed density.

2.1.3 Tilting the contour into the strip

Since Φ is analytic in $|\operatorname{Im} u| < \pi/4$, the integrals behind (2.1.2) can be taken along any horizontal line inside the strip.

Proposition 2.1.1. *Let $0 < \delta < \pi/4$ and $\eta = \pi/4 - \delta$. For every complex s,*

$$\xi\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} \Phi(x + i\eta)\, e^{sx}\, dx,$$
$$\xi'\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} (x + i\eta)\, \Phi(x + i\eta)\, e^{sx}\, dx.$$

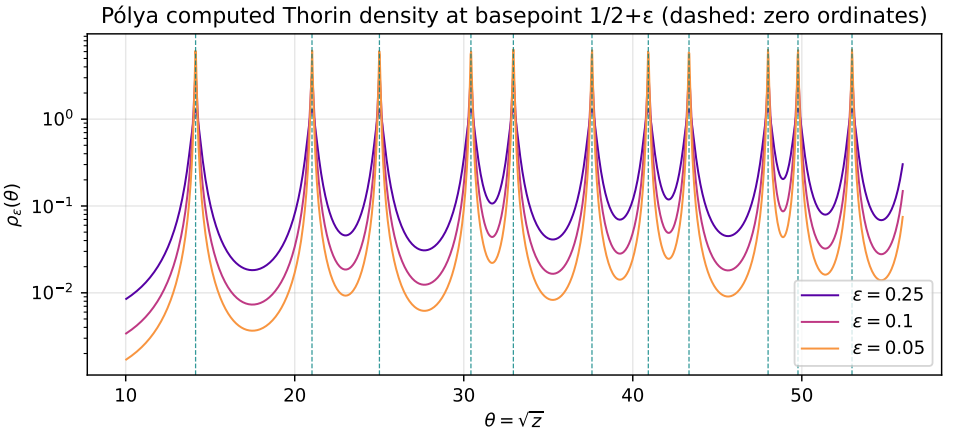


Figure 2.1: The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2}+\varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 1.2.3).

For real x, $|\Phi(x+i\eta)| \le C_\delta$ and $|(x+i\eta)\Phi(x+i\eta)| \le C_\delta$, with $C_\delta = O(\delta^{-5/2})$ as $\delta \to 0$.

Proof. By Section 1.2, Φ is analytic in $|\operatorname{Im} u| < \pi/4$, and by the identity theorem its evenness $\Phi(u) = \Phi(-u)$ extends from the real line to the strip. For $u = x + iy$ with $|y| \le \eta$ and $x \ge 0$ we have $\operatorname{Re} e^{2u} = e^{2x}\cos 2y \ge e^{2x}\sin 2\delta$, so

$$|\Phi(x+iy)| \le 2\sum_{n\ge 1}\left(2\pi^2 n^4 e^{9x/2} + 3\pi n^2 e^{5x/2}\right)\exp(-\pi n^2 e^{2x}\sin 2\delta), \tag{2.1.3}$$

and Φ decays doubly exponentially as $x \to +\infty$, uniformly in the closed strip $|y| \le \eta$; by evenness it does so as $x \to -\infty$ as well. Cauchy's theorem therefore moves the line of integration in $\xi(\frac{1}{2}+s) = \int_{\mathbb{R}} \Phi(u)e^{su}\,du$, which is (2.1.2), from $\operatorname{Im} u = 0$ to $\operatorname{Im} u = \eta$, and the second identity follows in the same way from $\xi'(\frac{1}{2}+s) = \int_{\mathbb{R}} u\Phi(u)e^{su}\,du$. Finally, $\sum_{n\ge1} n^k e^{-bn^2} = O(b^{-(k+1)/2})$ uniformly in $b > 0$, so the majorant (2.1.3) is $O(e^{-x/2}\delta^{-5/2})$ for $x \ge 0$, of order $\delta^{-5/2}$ at $x = 0$; with evenness this gives both bounds. □

On the real axis the integrand $\Phi(u)e^{su}$, with $s = \varepsilon + i\theta$, oscillates with frequency θ, and its integral, of order $\theta^{7/4+\varepsilon/2}e^{-\pi\theta/4}|\zeta(\frac{1}{2}+\varepsilon+i\theta)|$ by Stirling's formula, is exponentially smaller than the integrand. On the line $\operatorname{Im} u = \eta$ the factor $e^{i\theta u}$ has constant modulus $e^{-\theta\eta}$ and leaves the integral as the prefactor $e^{i\eta s}$, which carries almost all of the decay of ξ; the integral that remains is of order $e^{-\delta\theta}$ up to the same factors, against an integrand of size at most $C_\delta e^{\varepsilon x}$. The cancellation thus

drops from $e^{-\pi\theta/4}$ to $e^{-\delta\theta}$. The price is a larger kernel, by the factor K_η/K_0 with $K_0 = \xi(\frac{1}{2}) = 0.497$, and a finer step, because the natural boundary at $\operatorname{Im} u = \pi/4$ [284, Chapter 11, Section 1.4, and Chapter 5, Section 3.2] is now only δ away.

On the tilted line the kernel constant measures the size of the integrand and costs $\log_{10} K_\eta$ digits of precision. By Table 2.1 it grows like $\delta^{-5/2}$, but $\log_{10} K_\eta$ stays under five even at $\delta = 0.01$, which is negligible against the saving $(\pi/4 - \delta)\theta/\log 10$ at any height where tilting is worthwhile.

$\delta = \pi/4 - \eta$	K_η	$\max_x \lvert\Phi(x + i\eta)\rvert$
0.6854	0.554	0.981
0.3927	2.52	3.24
0.2	19.0	15.4
0.1	126	76.5
0.05	774	415
0.02	7836	3938
0.01	43503	22207

Table 2.1: Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts.

The trapezoid rule on the tilted line converges geometrically under two conditions (Exercise 2.2). By Poisson summation its error is $\sum_{k\neq 0} I(\theta + 2\pi k/h)$, where $I(t) = \int_{\mathbb{R}} \Phi(x + i\eta) e^{\varepsilon x} e^{itx}\, dx = e^{\eta t - i\eta\varepsilon}\xi(\frac{1}{2} + \varepsilon + it)$. The integrand is analytic in a strip that extends δ above the line and $\pi/2 - \delta$ below it, and correspondingly $|I(t)|$ is of order $e^{-\delta t}$ as $t \to +\infty$ and $e^{-(\pi/2-\delta)|t|}$ as $t \to -\infty$, up to the factors of Stirling's formula. Relative to $I(\theta)$ the terms with $k \ge 1$ are of order $e^{-2\pi\delta k/h}$, and those with $k \le -1$ are small only when the step resolves the oscillation $e^{i\theta x}$ with a margin. For D correct digits this gives

$$\frac{1}{h} \;\ge\; \frac{D\log 10}{2\pi\delta}, \qquad \frac{2\pi}{h} \;\ge\; \theta + \frac{\delta\theta + D\log 10}{\pi/2 - \delta}, \tag{2.1.4}$$

up to terms of order $\log\theta + \log(1/h)$. The first condition asks for about $D\log 10/(2\pi\delta)$ nodes per unit length, 458 at $D = 25$ and $\delta = 0.02$, that is $h \le 2.2 \times 10^{-3}$. The second is the binding one at large height: at $\delta = 0.02$ and 25 digits it is the stronger from about $\theta = 2800$ on, and it asks for $h \le 1.23 \times 10^{-3}$ at $\theta = 5000$ and $h \le 8.8 \times 10^{-4}$ at $\theta = 7005$.

Taking δ proportional to $1/\theta$ holds the loss $\delta\theta/\log 10$ fixed, so the working precision grows only through the $\log_{10} K_\eta$ digits of the kernel constant, about $\frac{5}{2}\log_{10}(1/\delta)$, while both conditions (2.1.4) make the number of nodes, and with it the cost per point, grow linearly in θ up to a logarithmic factor from the truncation.

δ	bits	nodes	θ	error of ξ'/ξ	ms per point	untilted digits needed
0.10	200	2001	200	$< 2 \times 10^{-45}$	7	68
0.10	200	2001	500	9.8×10^{-42}	7	171
0.10	200	2001	1000	1.8×10^{-20}	7	341
0.02	220	8401	1000	$< 6 \times 10^{-45}$	28	341
0.02	220	8401	2000	$< 4 \times 10^{-45}$	28	682
0.02	220	8401	5000	2.9×10^{-25}	28	1705

Table 2.2: The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of ξ'/ξ, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height.

2.1.4 Precision and cost of the tilted route

The tilt makes height 5000 accessible (Table 2.2).

On the tilted line the rounding term of Section 2.1.1 becomes $2^{-b}e^{\delta\theta}$ times the kernel constant, and the step obeys (2.1.4). Table 2.3 compares the working precision of the two routes. The formula of the table for the tilted bits is conservative: it ignores the factor $\theta^{7/4+\varepsilon/2}$ in $|\xi(\frac{1}{2}+\varepsilon+i\theta)|$, worth about six digits at $\theta = 5000$, and it adds five digits for the kernel constant. The run at $\theta = 5000$ in Table 2.2 used 220 bits, fewer than the 244 of Table 2.3, and returned 25 digits; the Lehmer pair of Section 2.2.3.1 was computed with 256 bits at $\delta = 0.01$, where the formula asks for 201.

height θ	target digits	bits (untilted)	bits (tilted, $\delta = 0.02$)	nodes (tilted)
100	30	213	119	8400
200	30	326	122	8400
1000	30	1233	145	8400
5000	25	5748	244	8400
7005	25	8020	302	8400

Table 2.3: Working precision needed by the untilted and the tilted Pólya route for D target digits. The untilted bits are $(D + \pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D + \delta\theta/\log 10 + 5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8 \times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices.

Because the tilt reduces the cancellation to $e^{-\delta\theta}$, ordinary double precision suffices at moderate height. With $\delta = 0.15$ and the node values rounded to double, a vectorised evaluation agrees with the ball arithmetic values of $\rho_{0.05}$ to within an

absolute error of 2.8×10^{-12} for $\theta \le 50$, 1.3×10^{-9} for $\theta \le 100$ and 7.6×10^{-7} for $\theta \le 150$, at 93 μs per point. The untilted route fails in double precision above $\theta \approx 40$ (Exercise 2.5).

2.1.5 Ball arithmetic on the tilted line

Horner's rule, which serves the untilted evaluator well, fails in ball arithmetic on the tilted line. The tilted evaluator writes the trapezoid sum as one polynomial of degree $2J$ in $E = e^{sh}$, with coefficients $h\Phi(x_k + i\eta)$ for x_k running from $-L$ to L, times the prefactor $e^{i\eta s}E^{-J}$. With 2001 nodes, $\delta = 0.1$ and 200 bits, the radius of the Horner recursion grew by a factor of about 1.3 per step, and the final ball at $\theta = 100$ had radius 10^{109} although its midpoint was correct to many digits. Evaluating each exponential e^{sx_k} directly and summing gives balls of radius about 10^{-55} at the same point.

The cause is the wrapping effect of rectangular complex balls. Arb stores a complex number as a real ball and an imaginary ball [130], so a multiplication by $E = e^{\varepsilon h}e^{i\theta h}$ rotates the midpoint but multiplies the radii by $e^{\varepsilon h}(|\cos\theta h| + |\sin\theta h|)$, up to rounding; at $\theta h = 0.4$ this factor is 1.31. In Horner's rule the coefficient of E^k passes through k such multiplications. On the untilted line the effect is contained. The polynomials are folded at $x = 0$, so the coefficient at $x = jh$ passes through at most 600 multiplications, and the coefficients $h\Phi(jh)$ decay like $\exp(-\pi e^{2jh})$, so those that pass through many multiplications are negligible and the radii stay far below $|\xi|$ up to height 200. On the tilted line the coefficients decay only like $\exp(-\pi e^{2x}\sin 2\delta)$, and the coefficient at x passes through $(x+L)/h$ multiplications. With $h = 0.004$ and $L = 4$, which give 2001 nodes, the coefficients near $x = 2$ are still of order 10^{-12} at $\delta = 0.1$, and they pass through about 1500 multiplications, which enlarge their radii by a factor of about 10^{176}.

2.2 Classical evaluators, Turing's method and the Pólya route

The classical evaluators on the critical line pay for height differently and, at large height, far less than the Pólya route, tilted or not.

2.2.1 The Riemann Siegel formula, Gram points and certified counts

On the critical line $\zeta(\frac{1}{2}+it) = e^{-i\vartheta(t)}Z(t)$, where $\vartheta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann Siegel theta function and $Z(t)$ is real for real t. Write $m = \lfloor\sqrt{t/2\pi}\rfloor$ and $q = \sqrt{t/2\pi} - m$. For every $K \ge 0$,

$$Z(t) = 2\sum_{n\le m}\frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + (-1)^{m-1}\Big(\frac{t}{2\pi}\Big)^{-1/4}\sum_{j=0}^{K}C_j(q)\Big(\frac{t}{2\pi}\Big)^{-j/2} + R_K(t),$$

where $R_K(t) = O(t^{-(2K+3)/4})$ as $t \to \infty$, $C_0(q) = \cos(2\pi(q^2 - q - \frac{1}{16}))/\cos(2\pi q)$, and the higher C_j are given in [81, Section 7.5] and [89]. This asymptotic expansion was found in Riemann's unpublished notes and published by Siegel [304].

The main sum is the approximate functional equation $\zeta(s) = \sum_{n \le x} n^{-s} + \chi(s) \sum_{n \le y} n^{s-1} + \cdots$, where $\zeta(s) = \chi(s)\zeta(1-s)$ and $2\pi xy = t$ [324, Chapter 4], taken at $x = y = \sqrt{t/2\pi}$ and written for Z; the C_j correct for the abrupt truncation at $n = m$. The terms are bounded by 2 and there is no exponential cancellation, so double precision suffices at every height computed in this book; near height 10^{12} and above, the phases $\vartheta(t) - t \log n$ must be reduced modulo 2π in higher precision. Gabcke's thesis gives explicit bounds for the remainders R_K with $K \le 10$ [89], and with them the formula can be used in rigorous computation.

To evaluate Z at many points near height T, Odlyzko and Schönhage [220] compute the main sum on a grid by fast Fourier transforms and rational function evaluation, and interpolate between grid points by band limited interpolation. About $T^{1/2}$ values then cost $T^{1/2+o(1)}$ operations, that is $T^{o(1)}$ per point. Odlyzko compared the spacings of the zeros near zero number 10^{12} with the Gaussian unitary ensemble [219], and with the algorithm he later took such computations to zero numbers 10^{20} [221] and 10^{22} [223].

The classical searches for zeros on the line sample Z at the Gram points g_n, defined by $\vartheta(g_n) = n\pi$ [98, pp. 301 and 302]. There $Z(g_n)$ usually has the sign of $(-1)^n$, and each Gram interval $[g_{n-1}, g_n)$ usually contains exactly one zero. This is Gram's law, and it fails with positive frequency [328, Corollary 5.4]. Call g_n good when $(-1)^n Z(g_n) > 0$, and call an interval $[g_j, g_{j+k})$ between consecutive good Gram points a Gram block. Rosser's rule asserts that such a block contains exactly k zeros [292], [328, Section 7.3]. It holds far longer than Gram's law, since its first exception is the block that begins at $g_{13\,999\,525}$ [47, Table 3], but it too fails with positive frequency [328, Corollary 7.8]. Counting sign changes of Z gives a lower bound for the number of zeros on the line. The difficulty is to show that none is missing, whether off the line or in a close pair that produces no sign change.

Turing's answer [332] uses the counting function $N(T) = \vartheta(T)/\pi + 1 + S(T)$, where, for T not an ordinate, $\pi S(T) = \arg \zeta(\frac{1}{2} + iT)$, the argument being defined by continuous variation along $2 \to 2 + iT \to \frac{1}{2} + iT$. The function S is small on average: Littlewood proved $\int_0^T S(t)\,dt = O(\log T)$ [324, Theorem 9.9(A)], and Turing gave explicit constants for $|\int_{t_1}^{t_2} S(t)\,dt|$, later sharpened by Trudgian [327]. If the sign changes found near height T were fewer than $N(T)$, then S would be displaced by at least 2 over an interval long enough to violate the integral bound. The count is therefore certified by locating the sign changes in a range slightly beyond T and checking them against Turing's bound for $\int S$. The method never evaluates ζ off the line. Turing's published argument contained errors, corrected by Lehman [172], and the corrected method underlies every large verification since, including Platt and Trudgian's to height 3×10^{12} [247].

	Riemann Siegel and Turing	Odlyzko Schönhage	Pólya route
object	sign changes of $Z(t)$	zeros in blocks, statistics	Thorin density ρ_ε, mass, count
where evaluated	critical line	critical line	line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ and contour
count certificate	Gram points and $\int S$	Turing's method	argument principle count equal to the number of sign changes; computed, not yet certified
cost per point	$\sqrt{t/2\pi}$ terms, double	amortised, $t^{o(1)}$	$0.341\,t$ extra digits and step below π/t (untilted); $\delta t/\log 10$ digits and step below $2\pi/t$ (tilted)
accuracy	asymptotic remainder	asymptotic remainder	spectral; rounding certified by balls
reach	$t \approx 1540$ in 1953	$t \approx 10^{20}$ and beyond [221, 223]	$t \approx 7005$ (Section 2.2.3.1); 25 digits at $t = 5000$ (Table 2.2)

Table 2.4: The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached.

Before Turing, counts were certified by Backlund's method. Since $\operatorname{Re}\zeta(2+it) \geq 2 - \zeta(2) > 0$ for every t, it suffices to check that $\operatorname{Re}\zeta(\sigma + iT) \neq 0$ for $\frac{1}{2} \leq \sigma \leq 2$: then $\arg\zeta$ stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$ along the whole path, so $|S(T)| < \frac{1}{2}$ and $N(T)$ is the integer nearest to $\vartheta(T)/\pi + 1$. Backlund carried this out for $T = 200$ [11] and Hutchinson for $T = 300$ [118, pp. 54 to 57], [327].

2.2.2 The Pólya route beside the classical evaluators

The Pólya route evaluates ξ and ξ' on and off the line, at $\frac{1}{2} + \varepsilon + i\theta$, from the Pólya kernel, and so delivers the Thorin density ρ_ε, its mass and the zero count from one evaluator. Its count has the logic of Turing's certificate: when the sign changes of Ξ and the argument principle count $N(T)$ agree, every zero up to height T is on the line and simple (Section 2.1.2).

The methods differ most in how their cost grows with height (Table 2.4). The untilted Pólya route needs about $0.341\,t$ extra digits; and by the budget of Section 2.1.1 its step must be below π/t, so that its number of nodes grows linearly with t as well. The cause is aliasing. By Poisson summation the trapezoid sum at $s = \varepsilon + it$ equals $\sum_{k\in\mathbb{Z}} \xi(\frac{1}{2} + s - 2\pi ik/h)$, and the term $k = 1$, at height $t - 2\pi/h$, is exponentially larger than $\xi(\frac{1}{2} + s)$ once $t > \pi/h$. The tilted contour of Section 2.1.3 cuts the loss of digits to about $\delta t/\log 10$, but aliasing still requires $2\pi/h > t$, by the second condition in (2.1.4), so there too the number of nodes grows linearly with t.

The distances of Section 3.2 do not depend on how ξ is evaluated, and this

suggests a hybrid at large height. The approximate functional equation holds off the critical line [324, Chapter 4], so Riemann Siegel type expansions of ζ [5, Theorems 3.1 and 4.2] and of ζ', and the Odlyzko Schönhage algorithm for blocks of points, apply at $\frac{1}{2} + \varepsilon + i\theta$ as well. With them as the evaluator and Odlyzko's zeros [224] as the RH reference, one could compute ρ_ε, the matched total variation and the positivity margin at the heights of Odlyzko's computations, from about 10^{11} to beyond 10^{20}, while the tilted Pólya route supplies values with explicitly bounded errors on low windows.

2.2.3 Two computations beyond the untilted route

2.2.3.1 A Lehmer pair near height 7005

Lehmer found two zeros that are unusually close, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ [173, pp. 297 and 298], at distance 0.0377, the classical example of a Lehmer pair; such pairs give lower bounds for the de Bruijn Newman constant [70] (Section 8.4.6). The untilted Pólya route cannot reach this height: it would need about 2400 digits and a step below 4.5×10^{-4}. The tilted route with $\delta = 0.01$, 256 bits, 13001 nodes, step 6×10^{-4} and truncation $|x| \le 3.9$ computes $\Xi(\theta)e^{\pi\theta/4}$ at 34 milliseconds per point. On a grid of step 0.005 it finds sign changes in $(7005.060, 7005.065)$ and $(7005.100, 7005.105)$, the two members of the pair (Figure 2.2).

The Thorin density shows what resolution separates them. A single zero on the line produces a Cauchy peak of height $1/(\pi\varepsilon)$, that is 6.4, 15.9 and 31.8 at $\varepsilon = 0.05$, 0.02 and 0.01. The largest values of ρ_ε on the grid are 11.18, 19.58 and 33.89. At $\varepsilon = 0.05$ the two peaks have merged into one of nearly double height; at $\varepsilon = 0.02$ they are partly separated; at $\varepsilon = 0.01$ each carries its own peak, lifted slightly by its neighbour. The Cauchy kernel predicts this: two kernels with half width at half maximum ε at distance g have a single maximum when $g \le 2\varepsilon/\sqrt{3}$ and two maxima when $g > 2\varepsilon/\sqrt{3}$. With $g = 0.0377$ the pair is resolved as two peaks for $\varepsilon < \sqrt{3}\, g/2 \approx 0.033$. The criterion concerns two isolated kernels; Exercise 2.11 proves it and asks how much the other zeros move the threshold.

2.2.3.2 The zero count at height 1000

Proposition 3.4.2 of Section 3.4 gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, where $M_{1/2}(T) = \pi^{-1} \arg \xi(1 + iT)$ is the cumulative Thorin mass at resolution $\frac{1}{2}$, that is at the basepoint one of Section 9.1, and $\tau_{1/2}(T)$ is π^{-1} times the change of $\arg \xi$ along the segment from $1 + iT$ to $\frac{1}{2} + iT$. With the tilted route ($\delta = 0.1$, 200 bits, step 0.004 and truncation $|x| \le 4.2$, that is 2101 nodes), 10000 steps of 0.1 along $\operatorname{Re} s = 1$ and 2000 steps along the segment, we find

$$M_{1/2}(1000) = 648.7562315041018, \qquad \tau_{1/2}(1000) = 0.2437684958980,$$

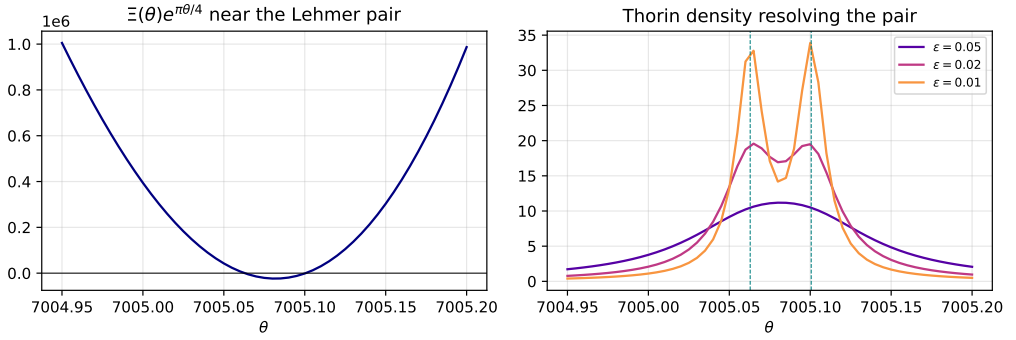


Figure 2.2: The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed.

$$M_{1/2}(1000) + \tau_{1/2}(1000) = 648.9999999999999,$$

so $N(1000) = 649$, with a largest argument increment of 0.43 radians. The computation took 77.5 seconds; `mpmath`'s `nzeros` returns 649 in 0.017 seconds. Where Backlund bounds the argument, this computation accumulates it from samples, and ball arithmetic certifies the rounding in each sampled value but not the absence of winding between samples. The count is therefore computed rather than certified. Certifying it would need, besides the analytic error bounds of Section 2.1, an enclosure of ξ between consecutive sample points s_k and s_{k+1}, for instance a check that $\mathrm{Re}\, \xi(s)/\xi(s_k) > 0$ on a ball containing the segment between them.

For counting and locating zeros at height, the Riemann Siegel family with Turing's method is faster by orders of magnitude, and the Pólya route is not a competitor there. It reads the count as an integer from a smooth quantity, the Thorin mass, with no Gram point bookkeeping; it gives the zero density strip by strip (Section 3.4); and it evaluates objects that the classical methods, used on the critical line, do not compute: Thorin densities and their distances to U_{RH}.

2.3 A worked computation from first principles

Where Section 2.2 took the tilted route to large height, we return to the untilted route and follow one small computation with it at low height, from the theta series to a Thorin mass and a zero. Every value can be reproduced with `code/polya_flint.py` (Section A.1).

The parameters are those of Section 2.1.1, step $h = 1/200$ and truncation $|u| \le 3$, and the working precision is 240 bits, about 72 decimal digits, the default of the code. As there, s is the displacement from $\frac{1}{2}$: with $E = e^{sh}$ and $\Phi_j = \Phi(jh)$, the evaluator returns $\xi(\frac{1}{2}+s) \approx P(E) + P(E^{-1})$ and $\xi'(\frac{1}{2}+s) \approx Q(E) - Q(E^{-1})$,

from which $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta)$.

Step one: the kernel at a single point. At $u=0$ the kernel (1.2.3) is

$$\Phi(0) = 2\sum_{n\ge1}\bigl(2\pi^2n^4-3\pi n^2\bigr)e^{-\pi n^2}.$$

The nth term is of order $n^4e^{-\pi n^2}$, and the terms with $n\ge8$ add up to less than 10^{-82}, so seven terms already give more than eighty digits:

$$\Phi(0) = 0.893393800934246888173969334109418226440848734189 65\ldots.$$

The ball returned by the arithmetic has radius 3×10^{-71}.

Step two: the coefficients. The truncation leaves 601 nodes $u_j = jh$, $0\le j\le600$. The coefficients of P are $\frac{h}{2}\Phi_0$ and $h\Phi_j$, $1\le j\le600$, and those of Q are $h^2j\Phi_j$; the first is

$$\tfrac{h}{2}\Phi(0) = 0.00223348450233561722043492333527354556610\ldots.$$

The coefficients decay doubly exponentially. By the asymptotic formula of Section 1.2, $\Phi(2.7)\approx4\pi^2e^{12.15}\exp(-\pi e^{5.4})\approx6\times10^{-296}$, so the coefficients beyond $j\approx540$ are below 10^{-297}.

Step three: $\xi(\frac12)$ as a trapezoid sum. At $s=0$, $E=e^{sh}=1$, and $\xi(\frac12)=2P(1)$ is the trapezoid sum of the kernel:

$$\xi(\tfrac12) = 0.49712077818831410991277373968539771980729360955 77\ldots,$$

with ball radius 5×10^{-71}. `mpmath`'s evaluation from the definition, $\frac12z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ at $z=\frac12$ in thirty digit arithmetic, gives

$$0.497120778188314109912773739685,$$

agreeing to every digit. By the bound of Section 2.1.1 the quadrature error is below 10^{-400}, so the seventy digits are set by the working precision, not by the rule.

Step four: the Thorin density at the first ordinate. At

$$s = 0.1 + 14.134725141734693790i,$$

that is at resolution $\varepsilon=0.1$ and the first ordinate γ_1, the evaluator returns $\xi(\frac12+s)$ and $\xi'(\frac12+s)$, and

$$\rho_{0.1}(\gamma_1) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}\Bigl(\frac12+s\Bigr) = 3.18525309376337225967538\ldots.$$

A single Cauchy kernel of width 0.1 contributes exactly $1/(\pi\varepsilon) = 3.18309886183791$ at its centre. By Proposition 1.2.3 the excess, 0.00215423, is the sum of the contributions of every other zero, $\sum_{j\neq 1}[P_{0.1}(\gamma_1 - \gamma_j) + P_{0.1}(\gamma_1 + \gamma_j)]$, plus the reflected kernel of γ_1 itself. The nearest neighbour, at distance 6.887, contributes 6.7×10^{-4}, the reflected kernel 4×10^{-5}, and the many distant zeros, whose contributions decay like the inverse square of the distance, the remaining 1.4×10^{-3}. Figure 2.3 shows the same decomposition along the whole stretch between the first two zeros.

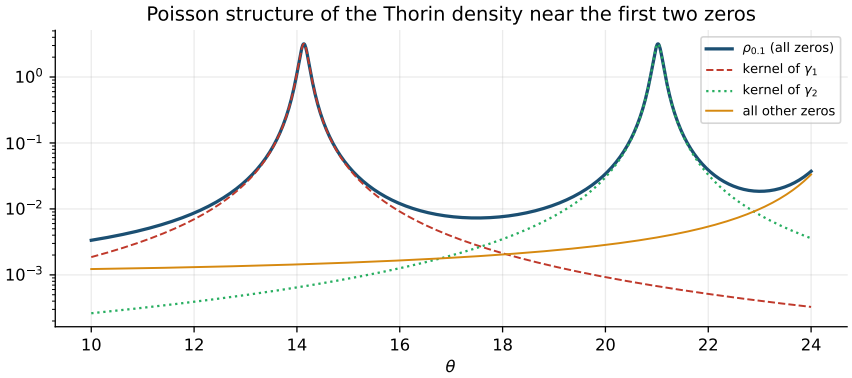


Figure 2.3: The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of γ_1, the kernel of γ_2 and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction.

Between zeros the density is small but strictly positive. At $\theta = 17$, between γ_1 and $\gamma_2 = 21.022$,

$$\rho_{0.1}(17) = 0.0076857707896205949783\ldots,$$

close to the minimum of $\rho_{0.1}$ between the two zeros (Figure 2.3). Lagarias's criterion [160] asks for this positivity at every θ and every resolution $\varepsilon > 0$.

Step five: the Thorin mass up to height 20. The cumulative Thorin mass of Section 2.1.2 is $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2} + \varepsilon + i\theta)$, with the argument continued from the real axis. Stepping along the line $\frac{1}{2} + 0.1 + i\theta$ from $\theta = 0$ to $\theta = 20$ in 1600 steps of $\varepsilon/8 = 0.0125$, as in that section, and accumulating the arguments of the quotients $\xi(\frac{1}{2} + s_{k+1})/\xi(\frac{1}{2} + s_k)$, where $s_k = 0.1 + i\theta_k$ and $\theta_k = 0.0125\,k$, gives

$$\mu_{0.1}\big((0,400]\big) = M_{0.1}(20) = 1.0509304\ldots,$$

in 0.77 seconds. There is exactly one zero below height 20. By Proposition 3.2.2 the excess 0.0509 over the count agrees to $O(\varepsilon^2)$ with the first order mass defect

$-(\varepsilon/\pi)\Xi'(20)/\Xi(20)$ (Exercise 2.16): it is the mass carried across the edge $\theta = 20$ by the Cauchy tails of the first zero and of the zeros above 20, while what crosses $\theta = 0$ is returned exactly by the reflected kernels. Adding $\tau_{0.1}(20)$, which is π^{-1} times the change of $\arg\xi$ along the segment from $\frac{1}{2} + 0.1 + 20i$ to $\frac{1}{2} + 20i$, returns the integer 1, as Proposition 3.4.2 requires.

Step six: locating the first zero. The ordinate γ_1 in step four was only a convenient point, and the density would have revealed it, up to the small shift caused by the other zeros, as the location of its first peak. To locate it from the Pólya route alone, one evaluates $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ on a grid, finds the sign change in $(14.10, 14.15)$, and refines by Newton's method with $\Xi'(\theta) = -\operatorname{Im}\xi'(\frac{1}{2} + i\theta)$; six iterations give γ_1 to forty digits. Section 2.1.2 refines all 79 zeros below height 200 in the same way, at 420 bits.

2.4 Exercises

On Section 2.1.

2.1. Let f be analytic in a strip containing $|\operatorname{Im} u| \le \eta$, with $\int_{\mathbb{R}} |f(x \pm i\eta)|\,dx \le M$ and $f(u) \to 0$ uniformly as $|\operatorname{Re} u| \to \infty$ in $|\operatorname{Im} u| \le \eta$. Prove that $|\int_{\mathbb{R}} f(u)\,du - h\sum_j f(jh)| \le 2M/(e^{2\pi\eta/h} - 1)$ by writing the error as the sum of the integrals, from left to right, of $f(u)(e^{-2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = \eta$ and of $f(u)(e^{2\pi iu/h} - 1)^{-1}$ along $\operatorname{Im} u = -\eta$ [326, Theorem 5.1]. Deduce the bound of Section 2.1.1.

2.2. Let $g(x) = \Phi(x + i\eta)e^{\varepsilon x}$ with $\eta = \pi/4 - \delta$ and $\varepsilon \ge 0$, and $I(t) = \int_{\mathbb{R}} g(x)e^{itx}\,dx$. Show by Poisson summation that $h\sum_{j\in\mathbb{Z}} g(jh)e^{i\theta jh} - I(\theta) = \sum_{k\neq 0} I(\theta + 2\pi k/h)$. For $0 < a < \delta$, $0 < b < \pi/2 - \delta$ and $0 < \theta < 2\pi/h$, shift the line of integration up by a when $k \ge 1$ and down by b when $k \le -1$ to bound the error by $Ae^{-a\theta}/(e^{2\pi a/h} - 1) + Be^{b\theta}/(e^{2\pi b/h} - 1)$, where A and B are the suprema of $\int_{\mathbb{R}} |g(x + iy)|\,dx$ over $0 \le y \le a$ and over $-b \le y \le 0$. Derive (2.1.4) from $|I(t)| = e^{\eta t}|\xi(\frac{1}{2} + \varepsilon + it)|$ and Stirling's formula.

2.3. Using the formulas in the caption of Table 2.3, estimate the number of bits the untilted route needs at height 2000 for twenty digits, and compare with the tilted route at $\delta = 0.02$.

2.4. Show that on the tilted line $\operatorname{Im} u = \eta$ the factor e^{su} with $s = \varepsilon + i\theta$ has modulus $e^{\varepsilon x - \theta\eta}$. Explain why the prefactor $e^{i\eta s}$ carries all of the exponential decay of ξ except the factor $e^{-\delta\theta}$, and where that factor comes from.

2.5. Implement the two polynomial evaluator in double precision and find the height at which it first returns the wrong sign of Ξ.

2.6. Show that in midpoint radius arithmetic, with a complex number stored as a real and an imaginary ball, multiplication by an exact $w = |w|e^{i\varphi}$ rotates the midpoint and maps real and imaginary radii r to $|w|(|\cos\varphi| + |\sin\varphi|)\,r$, up to rounding. Use this to explain why Horner's rule loses ball accuracy on the tilted line but not on the untilted line.

2.7. Use the tilted route to compute $\Xi(1000)$ and check its sign against $Z(1000)$ from the Riemann Siegel formula, with Z as in Section 2.2 and $\Xi(t) = -\frac{1}{2}\big(t^2 + \frac{1}{4}\big)\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{1}{2}it)|\,Z(t)$.

On Section 2.2.

2.8. Compute the Gram points $g_0, \dots, g_{20}$ and check Gram's law for them; find the first Gram interval that contains no zero (it lies beyond g_{100}).

2.9. Show that $N(T) = \vartheta(T)/\pi + 1 + S(T)$ follows from the argument principle applied to ξ on a rectangle.

2.10. Explain why a close pair of zeros that falls inside one Gram interval produces no sign change there, and how Turing's method detects that zeros are missing.

2.11. Prove that $P_\varepsilon(x - g/2) + P_\varepsilon(x + g/2)$ has a single maximum if $g \le 2\varepsilon/\sqrt{3}$ and two maxima if $g > 2\varepsilon/\sqrt{3}$, and estimate how the other zeros near height 7005 shift the threshold 0.033 for the Lehmer pair.

2.12. Estimate the cost of counting zeros to height 10^4 by the Thorin mass at resolution $\frac{1}{2}$ with the tilted route, including the growth of the number of nodes with height, and compare with the Riemann Siegel count.

On Section 2.3.

2.13. Compute $\Phi(0)$ with two, three, four and five terms of the series and explain the number of correct digits in each case.

2.14. Repeat step three with $h = 1/5$, $1/10$ and $1/20$ and plot $\log_{10}$ of the error against $1/h$. Show by Poisson summation that the error of the untruncated sum is exactly $2\sum_{k\ge1}\Xi(2\pi k/h)$, and confirm that it falls like $e^{-\pi^2/(2h)}$, the rate set by the strip half width $\pi/4$, up to slowly varying factors.

2.15. Compute $\rho_{0.1}(\gamma_1) - 1/(0.1\pi)$ directly as a sum over the first 1700 zeros with the smooth tail model of Section 3.2, and compare with step four.

2.16. Predict $M_{0.1}(20) - 1$ from the first order formula of Proposition 3.2.2, with $\Xi'(20)/\Xi(20)$ computed by the evaluator of step four, and compare with step five.

Chapter 3

The Thorin measure at finite resolution

The Thorin condition is RH (Proposition 1.2.1), and by Lagarias's criterion RH is the positivity of ρ_ε at every resolution (Remark 1.2.4); the question is what finite computations of ρ_ε can establish about it. The answer, which Chapters 4 and 5 complete, has two sides. On every window where the zeros are verified, the computed Thorin measure μ_ε converges to U_{RH}, the discrete measure demanded by RH, at explicit rates as $\varepsilon \downarrow 0$. Above the computed range every finite computation of the kinds in this book remains consistent with a failure of the Thorin condition, and the step that remains, positivity at every resolution and every height, is not a computational one (Section 5.2).

3.1 What computation can reach, and what the sine squared lemma adds

Two facts govern what computation can establish about the Thorin condition and which evaluator it should use (Sections 3.1.1 and 3.1.2). The Pólya density is the evaluator in the rest of this chapter and in Chapters 4 and 5.

3.1.1 What a finite computation can measure

Below height 200 the Pólya route finds 79 sign changes of Ξ and an argument principle count $N(200) = 79.0000000000004$ (Section 2.1.2). So, numerically and in agreement with [11, 247], the Thorin functional $U = \sum_k \delta_{a_k}$ of Section 5.3 coincides with U_{RH} on $(0, 200^2]$. The matched total variation distance of Section 3.2, computed against a model of the zero list, is $3.7 \times 10^{-9}\,\varepsilon$ at every resolution computed, the numerical floor of the comparison.

None of this constrains the zeros above the window. Platt and Trudgian [247] have verified every zero below height 3×10^{12}, yet the perturbation of ξ in Theorem 5.2.1 [252, Chapter 11] keeps the order, type and symmetry of $\xi(\frac{1}{2} + s)$, puts zeros off the line just above any large enough height R for which RH has been verified up to $R + 50$, and still passes the finite tests of the heat trace, from

Hankel blocks and derivative signs (to order 10^{17}) to the small time expansion. So a computation from the zeros verified up to a given height, or from finitely many tests of the heat trace, cannot by itself measure a scalar distance to RH, although a finite computation could refute RH by certifying a zero off the line. What it can measure is a margin at a stated resolution and height.

The margin is the minimum of ρ_ε, and positivity on one line is weaker than a zero free half plane (Remark 1.2.4).

Resolution is the first limitation. On $0 \le \theta \le 200$ the computed minimum is $\min \rho_\varepsilon = 0.0147\,\varepsilon$ at each of the eight resolutions of Section 2.1.2, attained at $\theta = 0$, where $\rho_\varepsilon(0) = \pi^{-1}\xi'(\frac12+\varepsilon)/\xi(\frac12+\varepsilon) = 2\varepsilon\pi^{-1}\sum_k a_k^{-1} + O(\varepsilon^3)$, because $(\xi'/\xi)(\frac12+\varepsilon) = \sum_k 2\varepsilon/(\varepsilon^2+a_k)$ and $\sum_k a_k^{-1} = 0.0231050$ (Section 1.2). So $\inf_\theta \rho_\varepsilon(\theta) \le \rho_\varepsilon(0) \to 0$ as $\varepsilon \downarrow 0$: near the critical line positivity is never won by a fixed margin. On the verified window the origin is in fact the minimum: for every $\varepsilon \in (0, 6.85]$ the minimum of ρ_ε over $0 \le \theta \le 3\times 10^{12} - 1.21$ is $\rho_\varepsilon(0)$, attained only at the origin, and that this holds at every height for every $\varepsilon \in (0, \frac12)$ is equivalent to RH (Theorem 9.1.18 and Remark 9.1.19).

Height is the second limitation. Even on the tilted contour of Section 2.1.3 the number of nodes, and with it the cost per point, grows linearly in the height, so the Pólya route is not a competitor for zero verification (Section 2.2.3.2). Its value is accuracy at small ε on moderate windows, where the Thorin measure is sharp and the prime side is ill conditioned.

3.1.2 The sine squared decomposition of the Thorin bracket

The sine squared lemma (1.2.9) converts the Lévy description of $\log\xi$, in which the primes are atoms, into the Thorin description of the reciprocal. At a basepoint $\alpha > 1$, applying the kernel $2\sin^2(x\theta/2) = 1-\cos(x\theta)$ to the canonical measure of [252, Theorem 8.1] writes the bracket $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta)$ as a sum of four explicit pieces:

$$v_\alpha(\theta) = \underbrace{\frac{\xi'}{\xi}(\alpha)}_{\text{drift}} + \underbrace{\int_0^\infty (1-\cos\theta x)\,\frac{e^{-\alpha x}}{e^{2x}-1}\,dx}_{\text{archimedean } \ge 0} - \underbrace{\frac{\theta^2}{(\alpha-1)\big((\alpha-1)^2+\theta^2\big)}}_{\text{pole } \ge 0} + \underbrace{\sum_{n\ge 2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big)}_{\text{primes } \ge 0}. \quad (3.1.1)$$

The pole piece is the closed form of $\int_0^\infty (1-\cos\theta x)e^{-(\alpha-1)x}dx$, and the prime piece equals $-\frac{\zeta'}{\zeta}(\alpha) + \mathrm{Re}\,\frac{\zeta'}{\zeta}(\alpha+i\theta)$.

The drift is the linear coefficient $c = \xi'/\xi(\alpha)$ of the Lévy representation of Section 1.2, not the drift d of a GGC. The archimedean piece is the contribution of $\Gamma(s/2)$ together with the factor s of $\frac12 s(s-1)$; it is smooth in θ and grows like

α	θ	drift	archimedean	pole	primes	$v_\alpha(\theta)$	$\kappa_\alpha(\theta)$
3.0	2	0.1144	0.0533	−0.2500	0.1988	0.1165	3.85
3.0	50	0.1144	1.2594	−0.4992	0.2464	1.1210	0.67
2.0	2	0.0691	0.0859	−0.8000	0.7154	0.0704	21.5
2.0	50	0.0691	1.3989	−0.9996	0.7561	1.2245	1.43
1.5	2	0.0461	0.1137	−1.8824	1.7683	0.0470	77.6
1.5	50	0.0461	1.4863	−1.9998	1.8527	1.3867	2.78

Table 3.1: The four pieces of the sine squared decomposition (3.1.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond 2×10^6 (1.3×10^{-3} at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece.

$\frac{1}{2}\log\theta$, which accounts for the average density of zeros. The prime piece is a sum of oscillations with frequencies $\log n$ and amplitudes $\Lambda(n)n^{-\alpha}$, one for each prime power. The drift is positive.

Near $\alpha = 1$ the pole piece is almost cancelled by the prime piece (Table 3.1). Both pieces grow like $1/(\alpha - 1)$, the pole explicitly and the primes through the pole of $-\zeta'/\zeta$ at one. At each fixed $\theta \neq 0$ their difference stays bounded as $\alpha \downarrow 1$ because ζ has no zero on the line $\operatorname{Re} s = 1$, so the analytic input of the prime number theorem appears in the Thorin description as this cancellation, which [252, Section 8.2] identifies with the collapse onto the zeros. Measured by the condition number

$$\kappa_\alpha(\theta) = \frac{|\text{pole}| + |\text{primes}|}{|v_\alpha(\theta)|},$$

the prime side amplifies relative errors by 77.6 at $\alpha = 1.5$ and $\theta = 2$.

The lemma also prices the prime side. By the prime number theorem the truncation tail satisfies, for fixed $\theta \neq 0$,

$$\sum_{n>N} \frac{\Lambda(n)}{n^\alpha}\big(1 - \cos(\theta \log n)\big) \asymp \frac{N^{1-\alpha}}{\alpha - 1},$$

which is 1.4×10^{-3} at $\alpha = 1.5$ with $N = 2 \times 10^6$, the size of the errors in Table 1.5; an absolute error of 10^{-10} at $\alpha = 1.5$ would need a cutoff $N \approx 4 \times 10^{20}$. For $\alpha \le 1$ the series diverges, and the lemma must be read through the regularised pairing of [252, Chapter 7] and the explicit formula. At the basepoint one the regularisation is explicit: the pole and prime pieces can be compensated against each other, and Section 9.1 converts the resulting Lévy form of $\xi(1 + s)$ term by term into the Thorin density. Near the critical line Selberg's smoothed sum over the primes up to x^2 takes the place of the series, and at resolution ε, if no zero lies to the right of the critical line, its error at an ordinate is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that k correct digits need $x \approx 10^{k/\varepsilon}$ (Section 3.5.1). The Pólya route has none

of these costs: its cost per point is polynomial in the height and independent of ε, and at height 200 it reaches $\varepsilon = 0.005$ with about 60 digits (Section 2.1.1).

3.1.3 Consequences for computation

The bracket is the derivative of an argument, $v_\alpha(\theta) = \partial_\theta \arg \xi(\alpha + i\theta)$, because $\partial_\theta \log \xi(\alpha + i\theta) = i(\xi'/\xi)(\alpha + i\theta)$. So the cumulative Thorin mass is an argument,

$$\mu_\varepsilon\big((0, \theta^2]\big) = \frac{1}{\pi} \arg \xi\Big(\frac{1}{2} + \varepsilon + i\theta\Big),$$

continued along the line from the real axis, where $\xi > 0$. It is computed without quadrature, by accumulating the arguments of quotients of consecutive values of ξ, with steps chosen so that each increment can be read from its principal value, a condition checked after the fact rather than guaranteed (Section 2.1.2). Section 3.2 uses it for masses, cell masses, cumulative distances and the mass defect of Proposition 3.2.2, and Section 3.4 for the zero count.

The lemma is linear with a nonnegative kernel, so a nonnegative Lévy input gives a nonnegative Thorin output, and positivity certificates can be posed as linear programs in the sine squared basis [252, Sections 18.2.8 and 18.2.9]. For ξ the only negative part of the input $xL(dx)$ is the pole density $-e^{-(\alpha-1)x}dx$, so a certificate must show that the positive pieces dominate the pole piece at every θ. For $\alpha > 1$ they do (Theorem 1.2.2), although the proof goes through the zeros and not through the input. For $\frac{1}{2} < \alpha \le 1$, where the lemma is read through the regularised pairing, domination at every basepoint is Lagarias's criterion, that is RH, and Section 3.5.2 states RH as a domination of the same kind at the centre, Weil's criterion on variograms. Beurling generalised prime systems, in which the sign of the input is controlled by construction, are among the open computations of Section 12.2.3.

The lemma also supplies an exact consistency check at $\alpha = \frac{1}{2}$, where the constant parts of the pole and prime pieces diverge and are continued analytically. The constant part $1/(\alpha - 1)$ of the pole piece continues to -2 and enters with a minus sign; the constant part $-\zeta'/\zeta(\alpha)$ of the prime piece continues to $-\zeta'/\zeta(\frac{1}{2})$; and the gamma factor contributes the archimedean constant of [252, Section 6.8], $C_\Gamma = \frac{1}{2}\log\pi - \frac{1}{2}\psi(\frac{1}{4}) - 2 = 0.6860917096$, which is minus the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ at $s = \frac{1}{2}$. They sum to

$$C_\Gamma + 2 - \frac{\zeta'}{\zeta}\Big(\frac{1}{2}\Big) = 0,$$

which is equivalent to $\xi'(\frac{1}{2}) = 0$, because $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$ and $1/s + 1/(s-1)$ vanishes at $s = \frac{1}{2}$.

3.2 Distances to the discrete measure implied by RH

On a window where the zeros are known, the Thorin measure μ_ε computed by the Pólya route can be compared with U_{RH} directly (Section 3.1.1), and how close the two are depends on the metric.

Fix $\Theta = 200$ and the window $E = (0, \Theta^2]$ in z, equivalently $(0, \Theta]$ in θ. The window contains 79 zeros, all simple and on the line, and Θ is not an ordinate, since $\gamma_{80} = 201.2648$ (Section 2.1.2). For a signed measure ν on E write $\|\nu\|_{\mathrm{TV}} = |\nu|(E)$, and write $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ for the cumulative mass.

3.2.1 Total variation at matched resolution

A smooth approximation to a sum of point masses never converges to it in total variation.

Proposition 3.2.1. *If $\nu \geq 0$ is absolutely continuous and U is purely atomic with N unit atoms in E, then $\|\nu - U\|_{\mathrm{TV}} = \nu(E) + N$.*

Proof. Let A be the finite set of atoms. Then $\nu(A) = 0$, so $|\nu - U|(A) = U(A) = N$ and $|\nu - U|(E \setminus A) = \nu(E \setminus A) = \nu(E)$. □

The Thorin measure is positive on E, where its density has minimum $0.0147\,\varepsilon$ (Section 3.1.1), so $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}} = \mu_\varepsilon(E) + 79 \approx 158$ for every ε. Total variation becomes informative only after both measures have been brought to the same resolution, or summed over cells.

By Proposition 1.2.3 the object to compare with μ_ε is the Poisson smoothing of the RH measure at the same resolution, and we write $d_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\|_{\mathrm{TV}}$ for the matched distance. Under RH the two measures coincide and $d_{\mathrm{match}}(\varepsilon) = 0$, so a computation of the matched distance measures its own error unless some zero is off the line.

A computation replaces $U^{\pm}_{\mathrm{RH}}$ by a model $\widetilde{U}$, which we build from the first 1700 ordinates, computed with `mpmath` [205], and from the smooth density $\pi^{-1}\vartheta'(t)\,dt$ of the Riemann Siegel theta function beyond the Gram point $T_* = g_{1699} = 2197.8796$ of Section 2.2.1, at which $\vartheta(T_*)/\pi + 1 = 1700$, each with its reflection. We write $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$ for the matched distance against the model. Since $\gamma_{1700} = 2197.2587 < T_* < \gamma_{1701} = 2198.6875$, we have $N(T_*) = \vartheta(T_*)/\pi + 1$, so the boundary term vanishes when $\int_{T_*}^{\infty} P_\varepsilon(\theta - t)\, d\big(N(t) - \vartheta(t)/\pi - 1\big)$ is integrated by parts, and the error of the tail model is at most of order ε/T_*^2 per unit length. The observed floor is far smaller: $\tilde{d}_{\mathrm{match}}(\varepsilon) = 3.66 \times 10^{-9}\,\varepsilon$ at every resolution from $\varepsilon = 1$ to $\varepsilon = 0.005$ (Table 3.2), that is $1.8 \times 10^{-11}\,\varepsilon$ per unit length. The exact proportionality to ε identifies the floor as the residual of the tail model; it is an observed value, not a bound for $d_{\mathrm{match}}(\varepsilon)$: such a bound would need certified bounds on the error of the tail model and on the evaluation of μ_ε. The densities that enter the comparison are stored in double precision.

ε	mass	raw TV	matched TV	sup diff	cell TV	L^1 in θ	$W_\varepsilon(0.01)$
1	79.36738	158.37	3.66×10^{-9}	1.9×10^{-11}	4.992	59.23	0.29504
0.5	79.19417	158.19	1.83×10^{-9}	9.6×10^{-12}	2.655	41.86	0.22362
0.25	79.09887	158.10	9.14×10^{-10}	4.8×10^{-12}	1.351	28.77	0.18698
0.1	79.03977	158.04	3.66×10^{-10}	1.9×10^{-12}	0.5431	16.01	0.16469
0.05	79.01990	158.02	1.83×10^{-10}	9.6×10^{-13}	0.2718	9.737	0.15721
0.02	79.00796	158.01	7.31×10^{-11}	3.8×10^{-13}	0.1087	4.816	0.15270
0.01	79.00398	158.00	3.66×10^{-11}	2.0×10^{-13}	0.05437	2.756	0.15120
0.005	79.00199	158.00	1.83×10^{-11}	1.5×10^{-13}	0.02718	1.553	0.15045
RH	79	0	0	0	0	0	0.14970

Table 3.2: Distances between the Pólya computed Thorin measure μ_ε and U_{RH} on $0 < z \le 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.2.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \le 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, L^1 in θ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row.

3.2.2 Cumulative distances and extrapolation in the resolution

The total mass is the simplest quantity to compare, since $U_{\mathrm{RH}}(E) = N(\Theta) = 79$. By Section 3.1.3 it is an argument, $\mu_\varepsilon(E) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\Theta)$, continued from $\arg \xi(\frac{1}{2} + \varepsilon) = 0$.

Proposition 3.2.2. *Suppose Θ is not an ordinate and every zero with $0 < \operatorname{Im} \rho \le \Theta$ lies on the line. Then*

$$\mu_\varepsilon\big((0, \Theta^2]\big) = N(\Theta) - \frac{\varepsilon}{\pi} \frac{\Xi'(\Theta)}{\Xi(\Theta)} + O(\varepsilon^2), \qquad \varepsilon \downarrow 0.$$

Proof. Let $\sigma_1 > 1$. By hypothesis, and because ξ is positive on the real axis, ξ has no zero in the closed rectangle with vertices $\frac{1}{2}+\varepsilon$, σ_1, $\sigma_1 + i\Theta$ and $\frac{1}{2}+\varepsilon+i\Theta$. So the change of $\arg \xi$ up its left side, which is $\pi\mu_\varepsilon((0, \Theta^2])$, equals the change along the path $\frac{1}{2} + \varepsilon \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + \varepsilon + i\Theta$. The classical formula $\pi N(\Theta) = \Delta \arg \xi$ along $\frac{1}{2} \to \sigma_1 \to \sigma_1 + i\Theta \to \frac{1}{2} + i\Theta$ [324, Section 9.3] differs from it only by the segment from $\frac{1}{2} + \varepsilon + i\Theta$ to $\frac{1}{2} + i\Theta$, since $\arg \xi$ is constant on the real axis. Put $h(\varepsilon) = \xi(\frac{1}{2} + \varepsilon + i\Theta)$. Since $h(0) = \Xi(\Theta) \neq 0$, a branch of $\arg h$ is real analytic near $\varepsilon = 0$, and

$$\pi\mu_\varepsilon\big((0, \Theta^2]\big) = \pi N(\Theta) + \arg h(\varepsilon) - \arg h(0) = \pi N(\Theta) + \varepsilon \operatorname{Im} \frac{\xi'}{\xi}\Big(\frac{1}{2} + i\Theta\Big) + O(\varepsilon^2).$$

From $\frac{d}{d\theta}\Xi(\theta) = i\xi'(\frac{1}{2} + i\theta)$ we get $(\xi'/\xi)(\frac{1}{2} + i\Theta) = -i\,\Xi'(\Theta)/\Xi(\Theta)$, and Ξ'/Ξ is real, so the coefficient of ε is $-\Xi'(\Theta)/\Xi(\Theta)$. □

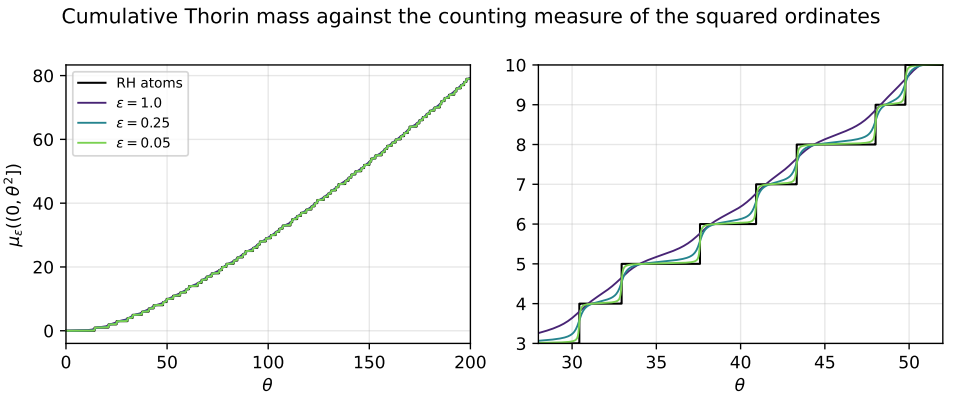


Figure 3.1: Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of U_{RH}, on the whole window (left) and on $28 \le \theta \le 52$ (right).

Remark 3.2.3. The remainder in Proposition 3.2.2 is in fact $O(\varepsilon^3)$: $\xi(s) = \xi(1-s)$ and the reality of ξ on the real axis give $h(-\varepsilon) = \overline{h(\varepsilon)}$, so $\arg h(\varepsilon) - \arg h(0)$ is odd in ε and its expansion has no term in ε^2.

Choosing Θ midway between zeros makes the correction small, and adding it recovers $N(\Theta)$ up to $O(\varepsilon^3)$; Proposition 3.4.2 makes the recovery exact. At $\Theta = 200$ the Pólya route gives $\Xi'(200)/\Xi(200) = -1.2506879$, so the proposition predicts $\mu_\varepsilon(E) - 79 = -\varepsilon\,\Xi'(200)/(\pi\,\Xi(200)) + O(\varepsilon^3)$, with $-\Xi'(200)/(\pi\,\Xi(200)) = 0.3981063$; the observed $(\mu_\varepsilon(E) - 79)/\varepsilon$ is 0.398105 at $\varepsilon = 0.005$ and 0.398102 at $\varepsilon = 0.01$. Figure 3.1 compares M_ε with the counting function of U_{RH} across the window.

Summing over cells gives a distance that resolves the individual zeros. Let $b_0 = 0$, $b_k = (\gamma_k + \gamma_{k+1})/2$ for $1 \le k \le 78$ and $b_{79} = \Theta$, and let $C_k = (b_{k-1}, b_k]$. The cell masses $\mu_\varepsilon(C_k) = M_\varepsilon(b_k) - M_\varepsilon(b_{k-1})$ are differences of the argument, and the coarse grained distance

$$d_{\mathrm{cell}}(\varepsilon) = \sum_k |\mu_\varepsilon(C_k) - 1|$$

is the total variation distance between the images of μ_ε and U_{RH} under the map sending each cell to its atom. No b_k is an ordinate, so Proposition 3.2.2 applies at both ends of every cell, with $\Xi'(0) = 0$ at b_0, and with the remainder of Remark 3.2.3 gives $\mu_\varepsilon(C_k) - 1 = -(\varepsilon/\pi)\big[\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\big] + O(\varepsilon^3)$. Hence $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$, where $\kappa_{\mathrm{cell}} = \pi^{-1}\sum_k |\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})|$, the cell constant of this window, is 5.43665, and every cell mass lies in $[0.99888, 1.00126]$ at $\varepsilon = 0.005$. Section 5.4 reads this constant as a statistic of the irregularity of the spacings.

The L^1 distance $\int_0^\Theta |M_\varepsilon(\theta) - N(\theta)|\,d\theta$ between the cumulative mass and the counting function is a transport distance, and because of the Cauchy tails it behaves like $\varepsilon\log(1/\varepsilon)$: it is 1.55 in θ and 360 in z at $\varepsilon = 0.005$, and Section 3.4.1 derives the rate. The heat trace $W(t) = \int e^{-tz}U_{\mathrm{RH}}(dz)$ is approximated by $W_\varepsilon(t) = \int_0^\infty e^{-t\theta^2}\rho_\varepsilon(\theta)\,d\theta = \int e^{-tz}\mu_\varepsilon(dz)$. At $t = 0.01$ the zeros give $W = 0.1496980112538$ (Table 1.4), while $W_\varepsilon(0.01) = 0.1504500280599$, 0.1512017825455 and 0.1527045050533 at $\varepsilon = 0.005$, 0.01 and 0.02, again with error linear in ε.

The mass and the heat trace are smooth in ε at $\varepsilon = 0$, so Richardson extrapolation applies to them: the mass by Proposition 3.2.2, and the heat trace because by Proposition 1.2.3 it is a sum over the zeros of Poisson integrals of the Gaussian $e^{-t\theta^2}$. The combination $2\mu_{0.005}(E) - \mu_{0.01}(E)$ equals $79 + 3.2\times10^{-8}$, against a raw defect of 2.0×10^{-3}. For the heat trace the error 7.5×10^{-4} of $W_{0.005}(0.01)$ falls to 2.6×10^{-7} after one Richardson step ($\varepsilon = 0.005$ and 0.01) and to 1.7×10^{-10} after two (adding $\varepsilon = 0.02$). Without extrapolation the same accuracy would need $\varepsilon \approx 8\times10^{-8}$ for the mass, and $\varepsilon \approx 2\times10^{-6}$ and 10^{-9} for the heat trace.

3.3 What an off line zero would do

How each distance of Section 3.2 would change if a zero left the line can be computed in closed form, because by Proposition 1.2.3 an off line zero at distance δ from the line enters the Thorin density, together with its reflection in the line, only through the kernels $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at its ordinate, the second of which is negative once $\varepsilon < \delta$. The three signatures of Proposition 3.3.1 show that an off line zero is most visible at resolutions comparable to its distance from the line, a theme that returns in Section 5.2.

3.3.1 Three signatures of an off line zero

Off line zeros come in pairs $\frac12 \pm \delta + i\gamma$ with $0 < \delta < \frac12$, together with their conjugates, and near γ a pair contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ to ρ_ε where two line zeros would contribute $2P_\varepsilon$.

Proposition 3.3.1. *Let $\varepsilon > 0$ and $0 < \delta < \frac12$; when $\delta = \varepsilon$, P_0 denotes the unit point mass at the origin.*

(i) (Displaced zeros, same count.) If two line zeros at height γ are replaced by the quadruple $\frac12 \pm \delta \pm i\gamma$, the density near γ changes by $2g$, where $g = \frac12(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$ is the change per displaced zero, and

$$\|g\|_{\mathrm{TV}} = \frac{3\sqrt3}{4\pi}\Big(\frac{\delta}{\varepsilon}\Big)^2 + O\Big(\frac{\delta^4}{\varepsilon^4}\Big), \qquad \delta < \varepsilon.$$

(ii) (Pair omitted from the list.) If a pair at $\frac{1}{2} \pm \delta + i\gamma$ *exists but a list of line zeros omits it, the difference is* $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ *and*

$$\|P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\|_{\mathrm{TV}} = \begin{cases} 2, & \delta \le \varepsilon, \\ \dfrac{8}{\pi} \arctan \sqrt{\dfrac{\delta+\varepsilon}{\delta-\varepsilon}} - 2, & \delta > \varepsilon. \end{cases}$$

(iii) (Negative mass.) For $\delta > \varepsilon$ *the negative part of* $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$*, the density of one displaced zero, has mass*

$$\frac{2}{\pi} \arctan \sqrt{\frac{\delta+\varepsilon}{\delta-\varepsilon}} - \frac{1}{2} = \frac{\varepsilon}{\pi\delta} + O\Big(\frac{\varepsilon^3}{\delta^3}\Big).$$

Proof. (i) Taylor expansion in δ gives $g = \frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$ [252, Proposition 19.7], with a remainder of L^1 norm $O(\delta^4/\varepsilon^4)$. Since $\partial_a^2 P_a(x) = (2a/\pi)(a^2 - 3x^2)/(a^2+x^2)^3$, the substitution $x = \varepsilon v$ gives $\partial_a^2 P_a|_{a=\varepsilon} = 2(1-3v^2)/(\pi\varepsilon^3(1+v^2)^3)$ and $dx = \varepsilon\, dv$. The function $(1-3v^2)/(1+v^2)^3$ is the derivative of $v/(1+v^2)^2$, whose total variation is four times its maximum $9/(16\sqrt{3})$, that is $3\sqrt{3}/4$, so the L^1 norm of $\partial_a^2 P_a$ at $a = \varepsilon$ is $(2/(\pi\varepsilon^2)) \cdot 3\sqrt{3}/4 = \frac{3\sqrt{3}}{2\pi\varepsilon^2}$, and multiplying by $\delta^2/2$ gives (i). (ii) For $\delta \le \varepsilon$ both kernels are nonnegative with mass one. For $\delta > \varepsilon$ put $a = \delta + \varepsilon$, $b = \delta - \varepsilon$, so the difference is $P_a - P_b$, which is negative exactly on $|x| < \sqrt{ab}$. Integrating, $\|P_a - P_b\|_{\mathrm{TV}} = \frac{4}{\pi}[\arctan\sqrt{a/b} - \arctan\sqrt{b/a}]$, and $\arctan y + \arctan(1/y) = \pi/2$. (iii) $P_a - P_b$ has total mass zero, so its negative part is half its norm; halve again for the factor $\frac{1}{2}$. □

With $\varepsilon/\delta = \sin\phi$ one has $\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} = \pi/4 + \phi/2$, so for $\delta > \varepsilon$ the norm in (ii) is $(4/\pi)\arcsin(\varepsilon/\delta)$ and the mass in (iii) is $(1/\pi)\arcsin(\varepsilon/\delta)$. Figure 3.2 plots the three signatures.

3.3.2 Calibration of the matched distance

Proposition 3.3.1 converts the matched distance of Section 3.2 into detection thresholds. By part (ii), a computed matched distance $\tilde{d}_{\mathrm{match}}(\varepsilon)$ of at most $3.7 \times 10^{-9}\,\varepsilon$ excludes any omitted pair with ordinate in $(0, 200)$: such a pair contributes total variation $(4/\pi)\arcsin(\varepsilon/\delta)$, or 2 if $\delta \le \varepsilon$, less only the part of its kernels outside the window. By part (i), a displaced pair that keeps the count adds $2g$ to a difference whose norm is the floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, so by the triangle inequality the matched distance rises above the floor as soon as $\|g\|_{\mathrm{TV}} > \tilde{d}_{\mathrm{match}}(\varepsilon)$, that is, to leading order, once δ exceeds

$$\delta_{\min}(\varepsilon) = \varepsilon\Big(\frac{4\pi\,\tilde{d}_{\mathrm{match}}(\varepsilon)}{3\sqrt{3}}\Big)^{1/2}. \tag{3.3.1}$$

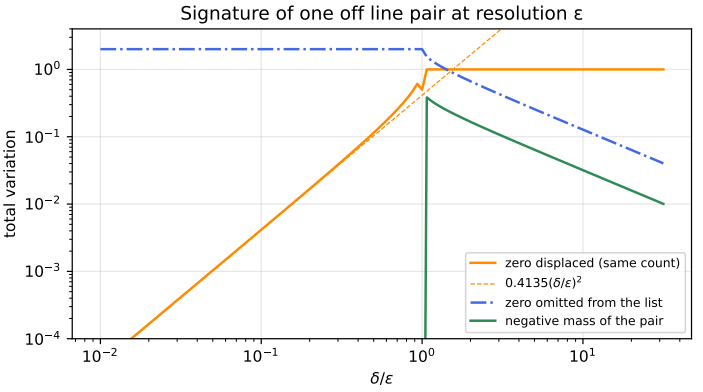


Figure 3.2: The three signatures of Proposition 3.3.1 as functions of δ/ε: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1.

With the floors of Table 3.2 this threshold is 3.3×10^{-5} at $\varepsilon = 0.5$, 1.1×10^{-6} at $\varepsilon = 0.05$ and 3.3×10^{-8} at $\varepsilon = 0.005$.

The threshold (3.3.1) is the square root law of [252, Proposition 19.7]: the signature grows like the square of the displacement, so the smallest detectable displacement falls like the square root of the achievable distance. The prime side calibration of [252, Section 19.3.6] obeys the same law and reaches $\delta \approx 0.05$ at $\varepsilon = 2$. What changes is the reach: the Pólya route evaluates ρ_ε at small ε to about 60 digits below height 200 (Section 2.1.1), so the floor is set by the tail model for the zero list (Section 3.2.1) rather than by the evaluator, whereas the prime side is limited by the truncation of Table 1.5. A positivity test sees less than the matched distance: only the negative part of the whole density, which by part (iii) is at most of order ε/δ per displaced zero and which the kernels of the neighbouring zeros can cancel entirely (Section 5.2).

These thresholds concern zeros below height 200 and calibrate the method above the verified window.

3.4 Counting zeros with the Thorin measure

Turing's method certifies the count $N(T)$ without evaluating ζ off the critical line, from the sign changes of Z and an explicit bound on the integral of S, where $\pi S(t) = \arg \zeta(\frac{1}{2} + it)$; Backlund's method, which it replaced, bounds the argument of ζ along a horizontal segment off the line (Section 2.2.1). Without the assumption of Proposition 3.2.2 that the zeros below the window lie on the line,

the argument principle gives at every resolution an exact identity for $N(T)$ and for the number $\nu(\frac{1}{2}+\varepsilon, T)$ of zeros to the right of the line of evaluation, so that off line zeros are counted and not only detected. At $\varepsilon = \frac{1}{2}$ the identity is Backlund's count with the vertical side on $\operatorname{Re} s = 1$.

3.4.1 The cumulative Thorin mass as a smoothed counting function

Recall that $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$, with the argument continued along the line $\operatorname{Re} s = \frac{1}{2}+\varepsilon$ from $\theta = 0$, where $\xi > 0$ (Sections 3.1.3 and 3.2). Since $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the mass splits as

$$M_\varepsilon(\theta) = N_\varepsilon^{\mathrm{sm}}(\theta) + S_\varepsilon(\theta), \qquad S_\varepsilon(\theta) = \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+\varepsilon+i\theta\Big),$$

where, with $s = \frac{1}{2}+\varepsilon+i\theta$ and $\theta > 0$, the smooth part $N_\varepsilon^{\mathrm{sm}}(\theta) = \pi^{-1}\big[\arg s + \arg(s-1) + \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{\varepsilon}{2}+\frac{i\theta}{2}) - \frac{\theta}{2}\log\pi\big]$ comes from $s(s-1)\pi^{-s/2}\Gamma(s/2)$, with the principal arguments of s and $s-1$, which lie in $(0,\pi)$, and the branch of $\log\Gamma$ that is real on the positive axis. Thus S_ε is $\pi^{-1}\arg\zeta$ continued along the vertical line from $S_\varepsilon(0+) = -1$ if $0 < \varepsilon < \frac{1}{2}$, when $\arg(s-1) \to \pi$ as $\theta \downarrow 0$, and from $S_\varepsilon(0+) = 0$ if $\varepsilon > \frac{1}{2}$.

As $\varepsilon \downarrow 0$, $N_\varepsilon^{\mathrm{sm}} \to \vartheta/\pi + 1$, where $\vartheta(\theta) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{i\theta}{2}) - \frac{\theta}{2}\log\pi$ is the Riemann Siegel theta function, and for θ not an ordinate Proposition 3.4.2 below gives $M_\varepsilon(\theta) \to N(\theta) - 2\nu(\frac{1}{2},\theta)$, where $\nu(\sigma,\theta)$ counts the zeros with real part greater than σ and ordinate in $(0,\theta]$ (Section 1.1). When no zero below height θ lies to the right of the critical line, the limit is therefore the Riemann von Mangoldt formula $N(\theta) = \vartheta(\theta)/\pi + 1 + S(\theta)$ [324, Section 9.3].

Integrating Proposition 1.2.3 in θ writes M_ε as a sum of Cauchy steps. Under RH, for $\theta \geq 0$,

$$M_\varepsilon(\theta) = \sum_{\gamma>0}\big[F_\varepsilon(\theta-\gamma) - F_\varepsilon(-\gamma) + F_\varepsilon(\theta+\gamma) - F_\varepsilon(\gamma)\big],$$

$$F_\varepsilon(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x}{\varepsilon},$$

with the sum over all positive ordinates. Since $F_\varepsilon(-\gamma) + F_\varepsilon(\gamma) = 1$, this says that $M_\varepsilon = P_\varepsilon * \tilde{N}$, where $\tilde{N}$ is the odd extension of N. As ϑ is odd, S_ε is then the Poisson average of the odd extension of S up to an explicit smooth function. If only the zeros below height $T+B$ are known to lie on the line, the display holds on $[0,T]$ up to an error of order $\varepsilon\delta^2/B^3$ for each off line pair above $T+B$ at distance δ from the line, counted as a double ordinate, because $P_{\varepsilon+\delta}(x) + P_{\varepsilon-\delta}(x) - 2P_\varepsilon(x) = -6\varepsilon\delta^2/(\pi x^4) + O(x^{-6})$.

For a single step,

$$\int_{-L}^{L} \bigl|F_\varepsilon(x) - \mathbf{1}_{x\ge 0}\bigr|\,dx = \frac{2}{\pi}\Bigl[L\arctan\frac{\varepsilon}{L} + \frac{\varepsilon}{2}\log\Bigl(1+\frac{L^2}{\varepsilon^2}\Bigr)\Bigr]$$
$$= \frac{2\varepsilon}{\pi}\Bigl(\log\frac{L}{\varepsilon}+1\Bigr) + O\Bigl(\frac{\varepsilon^3}{L^2}\Bigr),$$

which explains the $\varepsilon\log(1/\varepsilon)$ rate of the L^1 distance in Table 3.2. Taking half of it on each side of each of the 79 zeros below height 200, with L the half gap on that side, gives 1.61 at $\varepsilon = 0.005$, against the computed 1.55; the difference comes from the tails of the other steps.

3.4.2 A Littlewood identity and the zero density

Proposition 3.4.1. *Let $T > 0$. For every $\varepsilon > 0$ such that ξ has no zero on the segment from $\frac12+\varepsilon$ to $\frac12+\varepsilon+iT$,*

$$\pi M_\varepsilon(T) = \frac{\partial}{\partial\varepsilon}\int_0^T \log\Bigl|\xi\Bigl(\frac12+\varepsilon+it\Bigr)\Bigr|\,dt,$$

and for every $\varepsilon > 0$

$$\int_0^T \log\Bigl|\xi\Bigl(\frac12+\varepsilon+it\Bigr)\Bigr|\,dt = \int_0^T \log\Bigl|\xi\Bigl(\frac12+it\Bigr)\Bigr|\,dt + \pi\int_0^\varepsilon M_{\varepsilon'}(T)\,d\varepsilon'.$$

Proof. Since $\partial_\sigma \log|\xi(\sigma+it)| = \mathrm{Re}(\xi'/\xi)(\sigma+it) = v_\sigma(t)$ and $v_\sigma(t) = \partial_t \arg\xi(\sigma+it)$, we have $M_\varepsilon(T) = \pi^{-1}\int_0^T v_{1/2+\varepsilon}(t)\,dt$, and when no zero lies on the segment, differentiation under the integral gives the first identity. The integral $I(\varepsilon)$ on the left of the second is continuous on $[0,\infty)$, because the logarithmic singularities at the zeros are integrable uniformly in ε, and continuously differentiable except at the finitely many ε' at which the segment meets a zero. There I' jumps by 2π for each zero met at an interior point of the segment, since $\int_{-c}^{c} a\,du/(a^2+u^2) \to \pm\pi$ as $a \to 0\pm$. So I is absolutely continuous, and $I(\varepsilon) - I(0) = \int_0^\varepsilon I'(\varepsilon')\,d\varepsilon'$ is the second identity. □

The jump of 2π in the slope $\pi M_\varepsilon(T)$ when the line crosses a zero is the term 2ν of Proposition 3.4.2 below. Littlewood's lemma [324, Section 9.9] is the same Cauchy Riemann identity with the argument taken along a path through $\mathrm{Re}\,s = 2$; the two arguments differ by $2\pi\nu(\frac12+\varepsilon, T)$, and Exercise 3.14 derives the lemma from the two propositions. Let $\tau_\varepsilon(T) = \pi^{-1}\Delta\arg\xi$ along the horizontal segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$.

Proposition 3.4.2. *Let $\varepsilon > 0$ and $T > 0$, with T not an ordinate and no zero of ξ on the segment from $\frac12+\varepsilon$ to $\frac12+\varepsilon+iT$. Then*

$$M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\,\nu\Bigl(\frac12+\varepsilon, T\Bigr).$$

The segment condition holds at $\varepsilon = \frac{1}{2}$, and $\nu(1,T) = 0$, which gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, so that $\nu(\frac{1}{2}+\varepsilon,T) = \frac{1}{2}\big[N(T) - M_\varepsilon(T) - \tau_\varepsilon(T)\big]$ for every ε satisfying the hypotheses.

Proof. Let $\sigma_1 > 1$. Write Δ_{right} for the change of $\arg\xi$ along the segment from σ_1 to $\sigma_1 + iT$, and $\Delta_{\text{top}}(a \to b)$ for its change along the segment from $a + iT$ to $b+iT$. The argument principle on the rectangle $[\frac{1}{2}+\varepsilon, \sigma_1]\times[0,T]$, on whose bottom edge ξ is positive, gives $\Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2}+\varepsilon) - \pi M_\varepsilon(T) = 2\pi\nu(\frac{1}{2}+\varepsilon,T)$. Subtracting this from the classical formula $\pi N(T) = \Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2})$ [324, Section 9.3] leaves $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) + \pi M_\varepsilon(T) = \pi N(T) - 2\pi\nu(\frac{1}{2}+\varepsilon,T)$, and $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) = \pi\tau_\varepsilon(T)$. At $\varepsilon = \frac{1}{2}$ the segment lies on $\operatorname{Re} s = 1$, where ζ has no zeros, and $\nu(1,T) = 0$ because the Euler product has no zeros in $\operatorname{Re} s > 1$. □

Half the difference of $M_\varepsilon(T) + \tau_\varepsilon(T)$ at two resolutions counts the zeros in the strip between the two lines. The factor two comes from the functional equation: a zero $\frac{1}{2} + \delta + i\gamma$ with $\delta > \varepsilon$ and its partner $\frac{1}{2} - \delta + i\gamma$ add 2 to $N(T)$ and 1 to $\nu(\frac{1}{2}+\varepsilon,T)$, hence nothing to $M_\varepsilon(T) + \tau_\varepsilon(T)$, whereas two zeros on the line at the same ordinate add 2. In the density the pair enters through the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of Proposition 1.2.3, whose masses cancel, as in the proof of Proposition 3.3.1(iii) (Exercise 3.12).

Zero density estimates such as Ingham's [124] bound the number $N(\sigma,T)$ of zeros with real part at least σ and ordinate in $(0,T]$, which is at least $\nu(\sigma,T)$, by $T^{e(\sigma)+o(1)}$, with an exponent $e(\sigma) < 1$ for every $\sigma > \frac{1}{2}$, and Section 11.4 turns these bounds, with no hypothesis, into a bound on the mass of the Cauchy kernels of the zeros to the right of the basepoint σ, the negative part of the split (11.4.1) (Theorem 11.4.2). Proposition 3.4.2 evaluates ν itself, as an integer, but only up to a finite height.

3.4.3 The count and the zero density below height 200

The right side of the identity is an integer, so a computation of the left side with error below one half determines it, provided no winding of the argument is missed. We follow the argument on a grid, summing the principal values of the arguments of quotients of successive values of ξ, with steps $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) up the vertical side from $\theta = 0$ and 4000 steps along the horizontal segment. A certificate would need, besides ball arithmetic at the grid points, a bound on the change of the argument between them. We do not supply that bound, and the conclusions below are computed, not certified.

Table 3.3 gives the identity at $T = 200$, evaluated by the Pólya route. Every row returns 79 to within 6×10^{-13}. The row $\varepsilon = \frac{1}{2}$ gives $N(200) = 79$, and with the 79 sign changes of Ξ on $(0,200)$ (Section 2.1.2) this places every zero below height 200 on the line. Independently of the sign changes, the rows $\varepsilon < \frac{1}{2}$ give

$\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every $\varepsilon \geq 0.005$, since ν is nonincreasing in σ, so that by the functional equation no zero below height 200 lies farther than 0.005 from the line.

ε	$M_\varepsilon(200)$	$\tau_\varepsilon(200)$	$M_\varepsilon + \tau_\varepsilon = N - 2\nu$
1	79.36737810665	−0.36737810665	79.00000000000003
0.5	79.19416875933	−0.19416875933	78.99999999999993
0.25	79.09886991193	−0.09886991193	79.00000000000024
0.1	79.03976768776	−0.03976768776	79.00000000000006
0.05	79.01989993142	−0.01989993142	78.99999999999967
0.02	79.00796178162	−0.00796178162	78.99999999999970
0.01	79.00398102016	−0.00398102016	78.99999999999993
0.005	79.00199052625	−0.00199052625	78.99999999999947

Table 3.3: Proposition 3.4.2 at $T = 200$ by the Pólya route (computed, not certified).

The identity also gives the mass defect of Proposition 3.2.2 exactly. If T is not an ordinate and every zero with ordinate in $(0, T]$ lies on the line, then $\nu(\frac{1}{2}+\varepsilon, T) = 0$ and $M_\varepsilon(T) = N(T) - \tau_\varepsilon(T)$ for every $\varepsilon > 0$. Comparing with that proposition and Remark 3.2.3 gives $\tau_\varepsilon(T) = (\varepsilon/\pi)\,\Xi'(T)/\Xi(T) + O(\varepsilon^3)$. At $T = 200$ the first order term is $-0.3981063\,\varepsilon$, minus the mass defect found in Section 3.2.2, and for $\varepsilon \leq 0.1$ the column $\tau_\varepsilon(200)$ of Table 3.3 exceeds it by about $0.043\,\varepsilon^3$.

3.5 Further readings of the sine squared lemma

This section follows the sine squared lemma (1.2.9) of Section 1.2 in two directions.

3.5.1 The prime side near the critical line

For $\alpha > 1$ the integrated lemma is the argument of ζ. In the chart $\theta = \sqrt{z}$ a unit of $xL(dx)$ at x contributes $\pi^{-1}(1-\cos\theta x)$ to the density, and $\pi^{-1}\int_0^\Theta (1-\cos\theta x)\,d\theta = \pi^{-1}(\Theta - \sin(\Theta x)/x)$, so the prime atoms $\Lambda(n)n^{-\alpha}$ at $x = \log n$ return a term linear in Θ plus π^{-1} times

$$-\sum_{n\geq 2} \frac{\Lambda(n)}{n^\alpha}\,\frac{\sin(\Theta \log n)}{\log n} = \arg\zeta(\alpha + i\Theta),$$

on the branch with $\arg\zeta(\alpha) = 0$, since $\log\zeta(s) = \sum_n \Lambda(n)n^{-s}/\log n$ for $\operatorname{Re} s > 1$. Divided by π, this is $S_{\alpha-1/2}(\Theta)$ in the notation of Section 3.4.1. The series converges absolutely for $\alpha > 1$ and diverges for $\alpha < 1$, so the function S of Turing's method (Section 2.2.1), at $\alpha = \frac{1}{2}$, is reached only through a smoothing.

Selberg [303] truncates the Dirichlet series of ζ'/ζ smoothly and accounts for the truncation exactly by a sum over the zeros [324, Section 14.20]. With $\Lambda_x(n) =$

$\Lambda(n)$ for $n \le x$ and $\Lambda_x(n) = \Lambda(n)\log(x^2/n)/\log x$ for $x < n \le x^2$, and for $x > 1$ and s different from 1, from every zero ρ and from every $-2q$,

$$\begin{aligned}\frac{\zeta'}{\zeta}(s) = &-\sum_{n\le x^2}\frac{\Lambda_x(n)}{n^s} + \frac{x^{2(1-s)}-x^{1-s}}{(1-s)^2\log x}\\ &+\frac{1}{\log x}\sum_{q\ge1}\frac{x^{-2q-s}-x^{-2(2q+s)}}{(2q+s)^2} + \frac{1}{\log x}\sum_{\rho}\frac{x^{\rho-s}-x^{2(\rho-s)}}{(s-\rho)^2}.\end{aligned} \tag{3.5.1}$$

At $s = \alpha + i\theta$ with $\alpha = \frac12 + \varepsilon$, a zero $\rho = \beta + i\gamma$ with $\beta < \alpha$ contributes to the last sum a term of modulus about $x^{\beta-\alpha}/(|s-\rho|^2\log x)$. For a line zero at its own ordinate the term is $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2\log x)$, a fraction $x^{-\varepsilon}(1-x^{-\varepsilon})/(\varepsilon\log x)$ of the contribution $1/\varepsilon$ of that zero to v_α; we call this the single zero law. The same computation bounds the whole sum. If no zero has real part greater than $\frac12$, then $|x^{\rho-s}| = x^{-\varepsilon}$ and $|x^{2(\rho-s)}| = x^{-2\varepsilon}$ for every zero, and every zero contributes $\operatorname{Re}(1/(s-\rho)) = \varepsilon/|s-\rho|^2$ to $v_\alpha(\theta)$, so that $\sum_\rho |s-\rho|^{-2} = v_\alpha(\theta)/\varepsilon$ by Proposition 1.2.3. Hence the bracket v_α computed from Selberg's smoothed prime sum has error at most

$$\frac{x^{-\varepsilon}+x^{-2\varepsilon}}{\varepsilon\log x}\,v_\alpha(\theta),$$

which is of order $x^{-\varepsilon}/(\varepsilon^2\log x)$ at a peak, where v_α is about $1/\varepsilon$, and k correct digits at an ordinate need $x^\varepsilon\,\varepsilon\log x \approx 10^k$, that is

$$x \approx 10^{k/\varepsilon}$$

to leading order, with primes up to x^2. A zero with $\frac12 < \beta < \alpha$ gives a term that decays like $x^{\beta-\alpha}$, more slowly than $x^{-\varepsilon}$, and one with $\beta > \alpha$ a term that grows like $x^{2(\beta-\alpha)}$, since then $x^{2(\rho-s)}$ dominates. With $B = \sup_\rho \operatorname{Re}\rho$ the error is $O(x^{B-\alpha}/\log x)$ when $B < \alpha$, and when $B > \alpha$ it reaches $x^{2(B-\alpha)+o(1)}$ along a sequence of cutoffs, so that the prime side diverges as $x \to \infty$ if a zero lies to the right of the line $\operatorname{Re} s = \alpha$. Selberg's abscissa $\sigma_{x,t} = \frac12 + 2\max(\beta - \frac12, 2/\log x)$, the maximum over the zeros near height t, takes this into account; when none of these zeros lies to the right of $\frac12 + 2/\log x$ it equals $\frac12 + 4/\log x$, where $x^{-\varepsilon} = e^{-4}$ [303]. Section 9.1 meets the same dichotomy without smoothing.

The terms of the zeros to the right of the line $\operatorname{Re} s = \alpha$ cannot cancel one another. Suppose that $B > \alpha$, write $E_x(s)$ for the last term of (3.5.1), the sum over the zeros divided by $\log x$, and put $y = \log x$, $\lambda_\rho = \rho - s$ and $R(y) = y\operatorname{Re} E_{e^y}(s)$. Bounding each term by its modulus gives $|R(y)| \le \sum_\rho |s-\rho|^{-2}\big(e^{(B-\alpha)y} + e^{2(B-\alpha)y}\big)$, so for $\operatorname{Re} w > 2(B-\alpha)$ the Laplace transform $\int_0^\infty e^{-wy}R(y)\,dy$ may be taken term by term, and it is

$$\frac12\sum_\rho\Big[\frac{1}{\lambda_\rho^2}\Big(\frac{1}{w-\lambda_\rho} - \frac{1}{w-2\lambda_\rho}\Big) + \frac{1}{\bar\lambda_\rho^2}\Big(\frac{1}{w-\bar\lambda_\rho} - \frac{1}{w-2\bar\lambda_\rho}\Big)\Big].$$

The series converges locally uniformly away from its poles, since $\sum_\rho |\lambda_\rho|^{-2} < \infty$ and $|\lambda_\rho| \to \infty$, and so defines a meromorphic function in the whole plane. A zero ρ_0 with $\operatorname{Re}\rho_0 > (B+\alpha)/2$ gives a pole at $2\lambda_{\rho_0}$, to the right of every pole λ_ρ and $\bar\lambda_\rho$; each fraction with a pole there contributes $-\frac12\lambda_{\rho_0}^{-2}$ to the residue, which is therefore not zero. If $R(y) = O(e^{\eta y})$, the Laplace integral converges on $\operatorname{Re} w > \eta$ and continues the series holomorphically there, so $\eta \geq 2(\operatorname{Re}\rho_0 - \alpha)$. Letting $\operatorname{Re}\rho_0$ tend to B, and using the bound on $|R|$, gives $\limsup_{x\to\infty} \log|\operatorname{Re} E_x(s)|/\log x = 2(B-\alpha)$.

The cutoff interpolates between the zeros and the primes. For $x < \sqrt{2}$ the prime sum is empty, and as $x \downarrow 1$ each remaining term tends to its partial fraction, $1/(1-s)$, $1/(s+2q)$ or $1/(s-\rho)$. The limits cannot be taken inside the sums separately: the sum over q grows like $\frac12\log(1/\log x)$ and the real part of the sum over the zeros falls like $-\frac12\log(1/\log x)$, and only their total, $\zeta'/\zeta(s) - (x^{2(1-s)} - x^{1-s})/((1-s)^2\log x)$, has a limit, $\zeta'/\zeta(s) + 1/(s-1)$. Adding the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ turns this limit into $\xi'/\xi(s)$, whose real part divided by π is the density $\rho_\varepsilon(\theta)$, the sum of the kernels of Proposition 1.2.3.

Figure 3.3 tests the formula at $\varepsilon = 0.25$, that is $\alpha = \frac34$, on 501 points of $10 \leq \theta \leq 60$, against the Pólya values of $v_{3/4}$. With the primes alone the maximal relative error is 66%, 20% and 7% for $x = 30$, 300 and 3000, attained where $v_{3/4}$ is small, and the maximal absolute errors 1.14, 0.53 and 0.23, attained at peaks over zeros, follow the single zero law, which predicts 1.15, 0.51 and 0.23; the relative errors stay below the bound $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon\log x)$, which is 0.72, 0.21 and 0.077. Both errors decrease faster than $x^{-\varepsilon}$. Adding the sum over the first 1700 zeros reduces the error below 1.6×10^{-7}, the residual being the zeros above height 2197. At $\varepsilon = 0.005$ ten digits would need $x \approx 10^{2000}$, about e^{4600}, to leading order (about e^{4000} by the single zero law), while the cost of one Pólya evaluation does not depend on ε (Section 3.1.2). So the sine squared lemma identifies the Thorin density, but near the line the primes compute it only at a cost exponential in $1/\varepsilon$.

3.5.2 Positivity: Brownian subordination and Weil's criterion

The lemma cannot produce a Brownian first passage time to a random level from a positive input. Brownian motion with variance two per unit time first reaches a level $x > 0$ at a time with Laplace transform $e^{-x\sqrt{u}}$, so its first passage time T_Y to an independent level $Y \geq 0$ has $\mathbb{E}e^{-uT_Y} = \mathbb{E}e^{-Y\sqrt{u}}$. The exponent $\log f(\alpha+s)/f(\alpha) = cs + \int(e^{-sx} + sx - 1)L(dx)$ to which the lemma applies (Section 1.2) would have to equal $-\log\mathbb{E}e^{-sY}$, which is concave in s by Hölder's inequality, whereas $L \geq 0$, $L \neq 0$ makes it strictly convex, with second derivative $\int x^2 e^{-sx}L(dx) > 0$. With $c \geq 0$ the lemma gives a GGC with Thorin density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\,xL(dx)]$, but not a first passage time. Conversely, an infinitely divisible level with $\mathbb{E}e^{-sY} = \exp(-bs - \int(1-e^{-sx})\Pi(dx))$ enters the lemma with $c = b + \int x\,\Pi(dx)$ and the negative input $L = -\Pi$, and T_Y receives the signed density $(2\pi\sqrt{z})^{-1}[b + \int x\cos(x\sqrt{z})\,\Pi(dx)]$. For a Poisson level ($\Pi = \delta_1$,

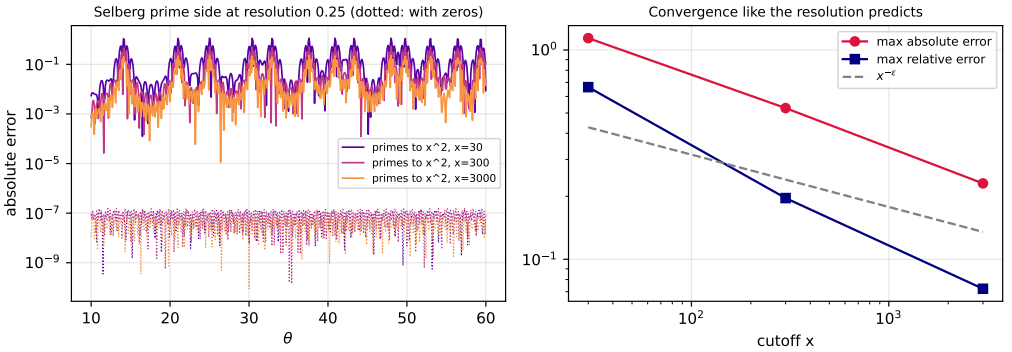


Figure 3.3: Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison.

$b = 0$) this is $\cos(\sqrt{z})/(2\pi\sqrt{z})$, and T_Y is not a GGC: it has an atom e^{-1} at 0. A positive input and a random level coexist only for pure drift, with transform $e^{-c\sqrt{u}}$ and Thorin density $c/(2\pi\sqrt{z})$.

The reciprocal direction does work: for $Y \geq 0$ infinitely divisible with finite mean m, $T_m \stackrel{d}{=} T_Y + R$, with T_m the first passage time to the level m and R a GGC independent of T_Y, which for the clock is Proposition 9.1.7 and for a Barnes beta law is Proposition 9.1.9.

For ξ there is no random level at any basepoint. By Stirling's formula and the polynomial growth of ζ in vertical strips, $|\xi(\alpha)/\xi(\alpha + it)|$ is at least $e^{\pi|t|/4}$ divided by a power of $|t|$, while $|\mathbb{E}e^{-itY}| \leq 1$; so $s \mapsto \xi(\alpha)/\xi(\alpha + s)$ is never a Laplace transform in s, although for $\alpha \geq 1$ it is one in $u = s^2$ (Theorem 1.2.2). For $\alpha > 1$ the input of ξ is signed, and the positivity of U_α is created by the zeros through the Poisson kernels of Proposition 1.2.3, not inherited from a positive input (Section 3.1.3); compare [252, Remark 3.8]. At $\alpha = \frac{1}{2}$ the obstruction is the drift zero case of Proposition 8.2.1, whose Brownian motion has variance one, and the curved boundaries of Section 8.2 take the place of the straight boundary with a random intercept. The obstruction concerns first passage alone (Section 9.2).

The same kernel puts Weil's criterion in variogram form. Section 5.3 defines the Thorin functional U on the class $\mathcal{T}$ of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic on a strip $|\operatorname{Im} r| \leq \frac{1}{2} + c$ for some $c > 0$, and $O((1 + |r|)^{-2-c})$ there. With $g(x) = \frac{1}{2\pi} \int h(r) e^{-irx}\, dr$,

$$\int_0^\infty \varphi(z)\, \frac{2\sin^2(x\sqrt{z}/2)}{2\pi\sqrt{z}}\, dz = g(0) - g(x), \tag{3.5.2}$$

both sides being $\pi^{-1} \int_0^\infty h(r)(1 - \cos xr)\, dr$. If $h \geq 0$ on the real line, g is positive

definite and $g(0) - g(x) \geq 0$ is a variogram. Weil's criterion [338] states that RH holds if and only if $\langle U, \varphi \rangle \geq 0$ for every such φ, and by the explicit formula U evaluates φ at every zero. Formally, the lemma writes $\langle U, \varphi \rangle$ as the pairing of variograms with the canonical measure of (3.1.1) taken at the centre, the measure ν of Section 5.3, whose archimedean density and prime atoms are nonnegative and whose only negative input is the pole density $-e^{x/2}dx$. So RH says that on every variogram of a positive definite function the pole mass is dominated by the prime and archimedean mass. Both masses are infinite, and the absolutely convergent form of Section 5.3 makes the statement exact, the divergent constants cancelling through $\xi'(\frac{1}{2}) = 0$ (Section 3.1.3).

The linear programming formulation of [252, Sections 18.2.8 and 18.2.9] works on this form, with the variograms as feasible directions. In a space of test functions of finite dimension the regularised domination is a feasibility problem with dual certificates, which Section 12.2.4 proposes to solve at increasing dimension. Weil's criterion concerns U, which sees every zero, whereas the limit $\alpha \downarrow \frac{1}{2}$ of the Thorin measures is U_{line}, which pairs nonnegatively with every such φ whatever the zeros do (Section 5.3).

3.6 Exercises

On Section 3.2.

3.1. Prove Proposition 3.2.1 for a positive measure ν on a general measurable space that does not charge the atoms of U.

3.2. Show that the total variation distance between the unit point masses at a and b is 2 for $a \neq b$, and deduce that total variation cannot compare two atomic measures with slightly displaced atoms.

3.3. Compute the L^1 distance between the cumulative functions of a Cauchy kernel of width ε and a unit step on $[-L, L]$, derive the $\varepsilon \log(1/\varepsilon)$ rate, and compare with the formula for a single step in Section 3.4.1.

3.4. Derive the first order formula of Proposition 3.2.2 for the sinh clock of Section 1.3, where it reads $-(\varepsilon/\pi) f'/f$ with $f(\theta) = \sin(\pi\theta)/(\pi\theta)$, and compare it at $\Theta = m + \frac{1}{2}$ with the exact mass $m + \pi^{-1} \arctan(\varepsilon/\Theta)$ of Section 1.3.4; explain why the difference is of order ε^3.

3.5. Explain why the two point Richardson combination removes the linear term in ε and the three point combination the quadratic term, and reproduce the heat trace errors 2.6×10^{-7} and 1.7×10^{-10} from the values of $W_\varepsilon(0.01)$ given in Section 3.2.2.

On Section 3.3.

3.6. Verify the expansion $\frac{1}{2}(P_{\varepsilon+\delta}+P_{\varepsilon-\delta})-P_\varepsilon=\frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon}+O(\delta^4)$, show that the remainder has L^1 norm $O(\delta^4/\varepsilon^4)$, and compute the L^1 norm of $\partial_a^2 P_a$ at $a=\varepsilon$.

3.7. Show that for $\delta>\varepsilon$ the kernel $P_{\varepsilon+\delta}-P_{\delta-\varepsilon}$ is negative exactly on $|x|<\sqrt{\delta^2-\varepsilon^2}$.

3.8. Compute the depth $2\varepsilon/(\pi(\delta^2-\varepsilon^2))$ of the negative dip of an off line pair at its centre, that is $\varepsilon/(\pi(\delta^2-\varepsilon^2))$ per displaced zero, compare it with the contribution $\varepsilon/(\pi(\varepsilon^2+\delta_j^2))\approx\varepsilon/(\pi\delta_j^2)$ of a line zero at distance δ_j, and derive the local visibility condition of Section 5.2.

3.9. Given the matched floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, derive the smallest displacement δ of a pair that keeps the count and would be detected, and evaluate it at $\varepsilon=0.05$ using Table 3.2.

On Section 3.4.

3.10. Show that $I(\varepsilon)=\int_0^T\log|\xi(\frac{1}{2}+\varepsilon+it)|\,dt$ is continuous for $\varepsilon\geq 0$, that its derivative jumps by 2π when the line $\operatorname{Re}s=\frac{1}{2}+\varepsilon$ crosses a zero, and deduce the second identity of Proposition 3.4.1.

3.11. Prove Proposition 3.4.2 in detail, paying attention to the orientation of each side of the rectangle.

3.12. Show that the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of an off line pair at distance $\delta>\varepsilon$ from the line have total mass zero, and compute to leading order the mass that they and their mirror images at $-\gamma$ put on a window $(0,T]$ containing the ordinate γ. Explain how the factor two in Proposition 3.4.2 accounts for the pair.

3.13. Using the first order formula for $\tau_\varepsilon(T)$ in Section 3.4.3, predict $\tau_{0.05}(200)$ from $\Xi'(200)/\Xi(200)=-1.2506879$ and compare with Table 3.3; estimate the coefficient of the ε^3 term from the rows $\varepsilon=0.01$ and 0.02.

3.14. Show that Littlewood's lemma follows from Proposition 3.4.1 and Proposition 3.4.2 by integrating in ε.

On Section 3.5.

3.15. Show that $\frac{1}{\pi}\int_0^\Theta(1-\cos\theta x)\,d\theta=\frac{1}{\pi}(\Theta-\sin(\Theta x)/x)$ and, for $\alpha>1$, derive the Dirichlet series for $\arg\zeta(\alpha+i\Theta)$ from the integrated sine squared lemma.

3.16. Implement Selberg's formula at $s = \frac{3}{4} + i\theta$ with $x = 100$ and check that adding the zero sum over the first hundred zeros reduces the error to the size of the omitted zeros.

3.17. Estimate, from the single zero law, the cutoff x needed for six correct digits of $v_{0.55}(\theta)$ at an ordinate $\theta = \gamma$ from the primes, and compare with the cost of the Pólya route at the same point.

3.18. Show that $g(0) - g(x) \geq 0$ for every real positive definite function g, and that $g(0) - g(x)$ is conditionally negative definite.

Chapter 4

Weak limits, hyperbolic complete monotonicity and a Bosch type theorem

Sections 4.2 and 4.3 bring in Bosch's theorem on powers of the half Cauchy law. The first explains why his result cannot be transferred as an input, and the second transfers his method instead and proves a threshold theorem for powers of the reciprocal Xi function, in which the angle of the zeros replaces Bosch's drift parameter and RH is the extreme case.

4.1 The limit of vanishing resolution

Neither closure under weak limits nor Bondesson's theorem reaches RH.

4.1.1 Weak limits of Thorin measures

Theorem 4.1.1 (Thorin [319, 320]; Bondesson [39, Theorem 3.1.5]; see also [252, Theorem 18.9]). *The class of GGC laws is closed under convergence in law.*

Corollary 4.1.2. *RH holds if and only if there are GGC variables X_N with $\mathbb{E}e^{-uX_N} \to 1/G(u)$ for all u in a set with an accumulation point in $(0,\infty)$ [252, Propositions 18.10 and 18.14].*

Proof. The transforms are analytic and bounded by one on $\operatorname{Re} u > 0$, so by Vitali's theorem they converge locally uniformly there, and the limit is $1/G$ on $(0,\infty)$. Since $1/G(u) \to 1$ as $u \downarrow 0$, the X_N converge in law to a proper limit with transform $1/G$, which is a GGC by the closure theorem, and Proposition 1.2.1 gives RH. Conversely, under RH the clock variable T is a GGC with transform $1/G$; take $X_N = T$. □

The Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2 is the candidate at hand. Its measures μ_ε are the Thorin measures of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ at $\alpha = \frac{1}{2}+\varepsilon$ whenever these are GGC transforms (Theorem 1.2.2), the transforms converge to $1/G$ pointwise as $\varepsilon \downarrow 0$, and the measures converge vaguely. The flow fails for two reasons.

First, the limit is the wrong measure. As shown after Proposition 1.2.3, $\mu_\varepsilon \to U_{\mathrm{line}}$ vaguely whatever the truth of RH [249, Proposition 9.5], because the contribution of an off line pair tends to zero in total variation (Proposition 3.3.1(ii)); the limit omits the off line zeros. It is the Thorin measure of the clock only under RH, since the Thorin functional U of Section 5.3 counts every zero and $U = U_{\mathrm{line}}$ exactly when RH holds (Section 5.3.1 and [252, Proposition 7.6]). Positivity of the vague limit therefore carries no information.

Second, the closure theorem needs GGC laws along the sequence. The flow supplies a law at every resolution with no hypothesis, that of the shifted clock T_α of Theorem 9.2.5 at $\alpha = \frac{1}{2} + \varepsilon$, but at small ε it supplies a GGC law only under the hypothesis to be proved: the transform at basepoint $\frac{1}{2} + \varepsilon$ is a GGC transform if and only if no zero lies to the right of the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ (Theorem 1.2.2). Positivity of ρ_ε at that one ε is necessary for this but is not known to imply it, since near a cluster of line zeros the negative kernel of an off line pair can be outweighed by the kernels of its neighbours (Exercise 1.7 and Section 5.2.2). Positivity of $\rho_{\varepsilon'}$ for every $\varepsilon' > \varepsilon$ does imply it, as shown in Remark 1.2.4; for $\varepsilon = 0$ this is Lagarias's criterion [160], which is RH. Weak closure adds the endpoint $\varepsilon = 0$ to a family already known to consist of GGCs; it cannot show that the family consists of GGCs [252, Remark 3.10].

On the window $0 \le \theta \le 200$ the Pólya route computes $\min \rho_\varepsilon = 0.0147\,\varepsilon > 0$ at each of the eight resolutions of Table 3.2, down to $\varepsilon = 0.005$ (Section 3.1.1); with the verification of RH to height 3×10^{12} this is a theorem, since for every $\varepsilon \le 6.85$ the minimum of ρ_ε over $[0, 3 \times 10^{12} - 1.21]$ is $\rho_\varepsilon(0)$ (Theorem 9.1.18 and Remark 9.1.19), and Section 12.2.2 lists what would make the Pólya computation itself a certificate.

Weak convergence becomes useful only with approximants whose positivity is proved independently of ξ, such as Beurling generalised prime systems whose zeta functions provably have only real ordinates, finite Krein strings, and the partial products over verified zeros completed by the Weyl tail; the last two are the Kent approximants of Section 8.1.3, without and with the Weyl tail. For such approximants the distances of Section 3.2 are the right measure of the approximation. Verification also removes factors from the question: if the first N zeros are verified to lie on the line, then $1/G_N$, with $G_N(u) = \prod_{k \le N}(1 + u/\gamma_k^2)$, is a GGC transform with no hypothesis, and RH holds if and only if the tail factor G_N/G is a GGC transform as well [252, Corollary 18.11].

4.1.2 The HCM form of the Thorin condition

Bondesson's theorem (Section 1.3.2; [39, Theorem 6.1.1], see also [129]) and the results of Section 1.2 give the following.

Corollary 4.1.3. *(i) For every $\alpha \ge 1$ the function $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is HCM, unconditionally. (ii) The reciprocal Xi function $\varphi_{1/2}(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$,*

which is the clock $1/G$, is HCM if and only if RH holds.

Proof. (i) is Theorem 1.2.2 with Bondesson's theorem, and (ii) is Proposition 1.2.1 with Bondesson's theorem. □

By the Hadamard product (1.2.1), with $a_k = -(\rho_k - \frac{1}{2})^2$ and $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_k \frac{a_k^2}{a_k^2 + a_k\,u\,w + u^2} = \prod_k \frac{a_k/u}{w + c_k(u)}, \qquad c_k(u) = \frac{a_k}{u} + \frac{u}{a_k}. \tag{4.1.1}$$

The numbers $c_k(u)$ are local to this subsection, and are not the derivatives $c_k(t)$ of the heat trace in Section 5.1. An off line pair, with values a and $\bar{a}$ of a_k, contributes $(|a|^2/u^2)/((w+c)(w+\bar{c}))$ with $c = c(u)$ complex, and $1/((w+c)(w+\bar{c}))$ is the Laplace transform of the signed function $e^{-t\,\mathrm{Re}\,c}\sin(t\,\mathrm{Im}\,c)/\,\mathrm{Im}\,c$. At a single u the remaining factors may smooth this to a positive function; failure is guaranteed only across all u, by Bondesson's theorem. At $u = |a|$ both values of c equal $2\cos\arg a$, and the pair contributes $(w + 2\cos\arg a)^{-2}$, which is completely monotone on $[2,\infty)$. Complete monotonicity in w at finitely many u is therefore necessary for HCM but not sufficient.

We test this necessary condition. From 60 digit values of ξ at real arguments, computed by (2.1.2) without cancellation, numerical differentiation gives the normalised derivatives $d_k = (-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\,\varphi_\alpha(u/v)$ for $k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{2.05, 3, 5, 10, 20, 50\}$, at $\alpha = \frac{1}{2}$ and at the control $\alpha = 1$. All 216 values are positive (Figure 4.1 shows $w = 3$ and $w = 20$), and the signs are certified: from the Taylor coefficients of H_u in w, computed in ball arithmetic [130] from $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, and so independently of the Pólya route, every d_k is enclosed in a ball of relative radius below 10^{-70} that lies in $(0,\infty)$. Under RH, (4.1.1) gives $H_u(w+t)/H_u(w) = \mathbb{E}e^{-tY}$ with $Y = \sum_k e_k/(w + c_k(u))$ and independent standard exponentials e_k, so $d_k = \mathbb{E}Y^k$ is of order d_1^k. The smallest value, 2.5×10^{-10} at $\alpha = \frac{1}{2}$, $u = 1$, $w = 50$, $k = 6$, is $2.54\,d_1^6$ with $d_1 = 0.0215$: it reflects the scale of Y, to which many zeros contribute, not closeness to a sign change. The test is consistent with (ii) and cannot decide it (Section 5.2, Table 5.5, and [252, Chapter 11]).

Under RH the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), and if moreover the zeros are simple and satisfy two further hypotheses, its density is not HCM (Theorem 4.3.22); for the Cauchy analogue of the clock variable this holds unconditionally (Theorem 4.3.20).

4.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function

By Corollary 4.1.3(ii) of Section 4.1 the clock $\varphi = 1/G$ is HCM if and only if RH holds; this is the HCM form of the reciprocal Xi programme of [249]. In

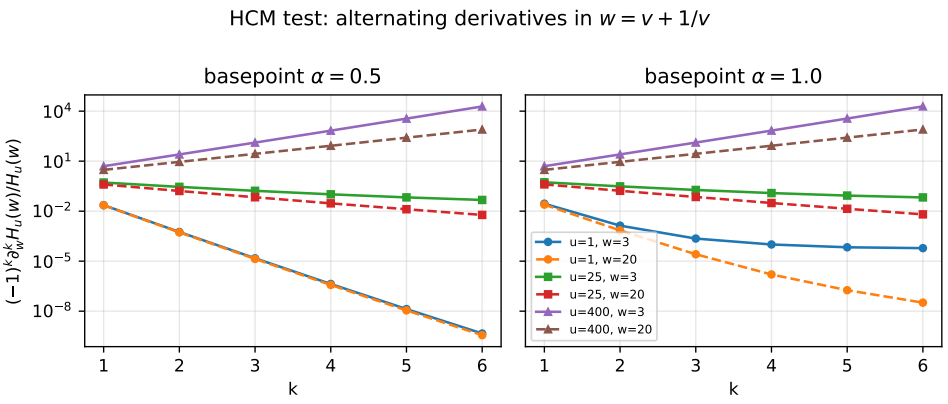


Figure 4.1: Normalised alternating derivatives $(-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$.

this section and in Section 4.3 the clock is a function of s, the variable u of Section 1.2: $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is the function $\varphi_{1/2}$ of Corollary 4.1.3. A sharp HCM theorem for a nearby, exactly solvable function might transfer to φ, and the candidate is Bosch's theorem on powers of the half Cauchy law [41].

4.2.1 HCM and Bosch's theorem at the half Cauchy point

Bosch's theorem concerns densities, so we recall HCM from Section 1.3.2 in the form that applies to them: a continuous function $f : (0, \infty) \to [0, \infty)$ is HCM if for every $u > 0$ the map $v \mapsto f(uv)f(u/v)$, the pair function of f at u, is a completely monotone function of $w = v + v^{-1}$ on $[2, \infty)$, and a positive random variable is HCM if it has an HCM density. In logarithmic coordinates $u = e^a$, $v = e^y$ the condition reads: $y \mapsto f(e^{a+y})f(e^{a-y})$ is completely monotone in $w = 2\cosh y$. We use the following facts, all due to Bondesson; (F1), (F2), (F5) and (F6) are also stated in Section 1.3.2.

(F1) If X_1, X_2 are independent and HCM then X_1X_2 and X_1/X_2 are HCM [39, Theorem 5.1.1].

(F2) If X is HCM then X^q is HCM for $|q| \ge 1$ [39, pp. 68 and 69]. The mechanism is that for $|p| \le 1$ the function $v^p + v^{-p}$ is a Bernstein function of $v + v^{-1}$, so if h is an HCM function then so is $x \mapsto h(x^p)$ for $|p| \le 1$; if X has density f, the density of X^q is $|q|^{-1}x^{1/q-1}f(x^{1/q})$, and the power factor is HCM by (F3).

(F3) Pointwise products of HCM functions are HCM, and powers x^r, $r \in \mathbb{R}$, are HCM [39, p. 68].

(F4) Pointwise limits of HCM functions, when finite and positive, are HCM [39, p. 68 and Theorem 5.1.3].

(F5) Every HCM law is a GGC [39, Theorem 5.1.2].

(F6) A probability law on $[0,\infty)$ is GGC if and only if its Laplace transform is an HCM function [39, Theorem 6.1.1].

(F7) A function f that is not identically zero is HCM if and only if

$$\begin{aligned} f(x) &= Kx^{r-1}h_1(x)h_2(1/x), \\ h_j(x) &= \exp\Big(-b_j x + \int_{(0,\infty)} \log\frac{1+t}{x+t}\,\Gamma_j(dt)\Big), \end{aligned} \tag{4.2.1}$$

with $K>0$, $r\in\mathbb{R}$, $b_j\ge 0$ and nonnegative measures Γ_j satisfying $\int(1+t)^{-1}\Gamma_j(dt)<\infty$ [39, (5.2.3) and Theorem 5.3.1]. In particular every HCM function that is not identically zero extends to an analytic and zero free function on $\mathbb{C}\setminus(-\infty,0]$ [39, p. 68].

THEOREM 4.2.1 (Bosch [41, Theorem 1.2]). *Let $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with $Z_\alpha, \tilde{Z}_\alpha$ independent positive α stable (here α is a stability index, not a basepoint, and T_α is not the variable of Theorem 1.2.2). Then T_α has density $\sin(\pi\alpha)/(\pi\alpha(t^2+2t\cos\pi\alpha+1))$, a half Cauchy law with drift, and the density of T_α^β is HCM if and only if $\alpha\le 1/2$ and $|\beta|\ge 1/(1-\alpha)$.*

At $\alpha=1/2$ we have $T_{1/2}\overset{d}{=}|C|$ with density $2/(\pi(1+t^2))$. The density of $|C|^\beta$ is

$$f_\beta(x) = \frac{2}{\pi|\beta|}\,\frac{x^{1/\beta-1}}{1+x^{2/\beta}}, \qquad x>0.$$

With $x=e^{a\pm y}$ and $q=2/|\beta|$ a short computation gives

$$f_\beta(e^{a+y})f_\beta(e^{a-y}) = \frac{4}{\pi^2\beta^2}\,e^{-2a}\,\frac{1}{2\cosh(qa)+2\cosh(qy)}. \tag{4.2.2}$$

So at $\alpha=1/2$ Bosch's theorem is the statement that $1/(c+\cosh(qy))$ is completely monotone in $\cosh y$ for every $c\ge 1$ exactly when $q\le 1$; here $c=\cosh(qa)$, which ranges over $[1,\infty)$.

PROPOSITION 4.2.2. *The half Cauchy density is not HCM.*

Proof. Take $\beta=1$, so $q=2$ in (4.2.2). Since $\cosh(2a)+\cosh(2y)=2\sinh^2 a+w^2/2$ with $w=2\cosh y$, the pair function is proportional to $h(w)=1/(c^2+w^2)$ with $c=2|\sinh a|>0$ for $a\ne 0$. For $x>0$,

$$h(2+x) = \int_0^\infty e^{-xt}\,e^{-2t}\,\frac{\sin(ct)}{c}\,dt.$$

If h were completely monotone on $(2,\infty)$, Bernstein's theorem would give $h(2+x)=\int e^{-xt}\mu(dt)$ with $\mu\ge 0$, and uniqueness of the Laplace transform would force $\mu(dt)=e^{-2t}c^{-1}\sin(ct)\,dt$, which is not nonnegative. □

Proposition 4.2.3. *The density of $|C|^\beta$ is HCM for $|\beta|\ge 2$.*

Proof. For $q=2/|\beta|\le 1$ the function $\cosh(qy)$ is a Bernstein function of $\cosh y$ by (F2), and $x\mapsto 1/(c+x)$ is completely monotone for $c>0$, so the composition in (4.2.2) is completely monotone. □

At $\alpha=1/2$ sufficiency is therefore elementary; the deep part of Bosch's paper is $\alpha<1/2$.

4.2.2 The half Cauchy law and the Riemann Xi function

On the Cauchy side everything is explicit. The logarithm of $|C|$ has density $g_C(y)=1/(\pi\cosh y)$ and characteristic function $\operatorname{sech}(\pi t/2)$. The Weierstrass product

$$\operatorname{sech}\Big(\frac{\pi t}{2}\Big)=\prod_{k\ge 1}\Big(1+\frac{t^2}{(2k-1)^2}\Big)^{-1}$$

says $\log|C|\overset{d}{=}\sum_{k\ge1}L_k/(2k-1)$ with L_k independent standard Laplace variables. The Mellin transform is $\mathbb{E}|C|^z=\sec(\pi z/2)$ for $|\operatorname{Re} z|<1$, the reciprocal of $\cos(\pi z/2)$, a function of the Laguerre Pólya class with real zeros $\pm(2k-1)$; these odd integers are the frequencies of the half Cauchy law. The Cauchy reciprocal Xi function

$$\varphi_C(s)=\mathbb{E}|C|^{i\sqrt{s}}=\operatorname{sech}\Big(\frac{\pi\sqrt{s}}{2}\Big)=\prod_{k\ge1}\Big(1+\frac{s}{(2k-1)^2}\Big)^{-1}$$

is the Laplace transform of the GGC $\sum_k e_k/(2k-1)^2$, with independent standard exponentials e_k and Thorin measure $\sum_k\delta_{(2k-1)^2}$, hence HCM by (F6). The checks below that name no source used mpmath [205] at 20 to 30 digits. Bourgade, Fujita and Yor [43] derive Euler's formulae for $\zeta(2n)$ from products of independent Cauchy variables, whose Mellin transforms are powers of $\sec(\pi z/2)$. Section 10.2 develops their construction.

On the Riemann side the frequencies are the zeros γ of $\Xi(z)=\xi(\frac12+iz)$. By (1.2.1), whose product converges absolutely because $\sum|\gamma|^{-2}<\infty$, unconditionally

$$\Xi(z)=\Xi(0)\prod_{\gamma}\Big(1-\frac{z^2}{\gamma^2}\Big),$$

the product over zeros γ of Ξ with $\operatorname{Re}\gamma > 0$, counted with multiplicity. No zero is purely imaginary because $\xi(\sigma) > 0$ for real σ. Hence the clock is

$$\varphi(s) = \frac{\Xi(0)}{\Xi(i\sqrt{s})} = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_{\gamma}\Big(1+\frac{s}{\gamma^2}\Big)^{-1}.$$

Under RH define, in the spirit of Biane, Pitman and Yor [32], $\log W = \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables L_γ. Then $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1 \approx 14.1347$, the characteristic function of $\log W$ is $\xi(1/2)/\xi(1/2+t)$, and $\varphi(s) = \mathbb{E}W^{i\sqrt{s}}$. The variable W, not to be confused with the heat trace $W(t)$, exists without RH: $\log W$ has the law of the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2, where Z is standard normal and independent of the clock variable T. Only the series for $\log W$ needs RH. Biane, Pitman and Yor represent 2ξ itself, not its reciprocal, as a Mellin transform (Sections 1.2 and 9.1).

	half Cauchy	Riemann
entire function	$\cos(\pi z/2)$	$\Xi(z)/\Xi(0)$
frequencies	$2k-1$	γ
log variable	$\sum_k L_k/(2k-1)$	$\sum_\gamma L_\gamma/\gamma$
Mellin transform	$\sec(\pi z/2)$	$\Xi(0)/\Xi(z)$
reciprocal Xi	$\operatorname{sech}(\pi\sqrt{s}/2)$	$\xi(1/2)/\xi(1/2+\sqrt{s})$
Thorin measure	$\sum_k \delta_{(2k-1)^2}$	$\sum_\gamma \delta_{\gamma^2}$
exit time	Brownian motion on $(-1,1)$	Krein string with spectrum $\{\gamma^2\}$
density of the power	HCM iff $\lvert\beta\rvert \geq 2$ (Bosch)	HCM only if $\lvert\beta\rvert \geq \gamma_2 - \gamma_1$

Table 4.1: The dictionary. In the Riemann column the Mellin transform (Section 1.2 and Theorem 4.3.25) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 8.1.3), presumes RH and simple zeros. The last entry is the necessary condition of Theorem 4.3.27, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 4.3.28).

On the Riemann side the analogue of g_C is the density of $\log W$. The function $t \mapsto \xi(1/2)/\xi(1/2+t)$ is defined for all real t without any hypothesis (ξ has no real zeros) and decays like $\exp(-\frac{1}{2}|t|\log|t|)$, so its Fourier inverse

$$g(y) = \frac{1}{\pi}\int_0^\infty \frac{\xi(1/2)}{\xi(1/2+t)}\cos(ty)\,dt$$

is an even function with $\int g = 1$, entire because the integrand decays faster than any exponential; $g > 0$ unconditionally (Theorem 4.3.25). Under RH, with simple zeros and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$, the residues at the poles $\pm i\gamma_n$ of the integrand give, for $|y| > u_0$ with the constant $u_0 \geq \pi/4$ of Theorem 4.3.27,

$$g(y) = \sum_n \frac{\gamma_n}{2} A_n e^{-\gamma_n|y|}, \qquad A_n = \prod_{m\neq n}\Big(1-\frac{\gamma_n^2}{\gamma_m^2}\Big)^{-1}, \tag{4.2.3}$$

the analogue of $g_C(y) = (2/\pi)\sum_k (-1)^{k-1} e^{-(2k-1)|y|}$, which converges for every $y \neq 0$ because its coefficients are bounded. The Riemann coefficients $A_1 \approx 50.9$, $A_2 \approx -2664.6$, $A_3 \approx 31420$ (`code/README.md`) alternate in sign, as the Cauchy coefficients do, but grow roughly like $e^{\pi\gamma_n/4}$ (Theorem 4.3.27), so (4.2.3) diverges for $|y| < \pi/4$.

4.2.3 The reciprocal Xi function is HCM if and only if RH holds

Bosch's method works with the singularities of a function and with its logarithmic derivative, so we state Corollary 4.1.3(ii) in four forms. Form (iii) is Proposition 1.2.1, and since $\psi = G'/G$, form (iv) is the Stieltjes form of the Thorin condition in Section 1.1. A Stieltjes function is a function $a/s + b + \int_{(0,\infty)} \sigma(dt)/(s+t)$ with $a, b \geq 0$ and σ a nonnegative measure satisfying $\int (1+t)^{-1}\sigma(dt) < \infty$ [299, Definition 2.1].

Theorem 4.2.4. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$ *is HCM on* $(0,\infty)$.

(iii) φ *is the Laplace transform of a GGC, namely* $\sum_\gamma e_\gamma/\gamma^2$.

(iv) $\psi(s) = -\varphi'(s)/\varphi(s) = \dfrac{1}{2\sqrt{s}}\dfrac{\xi'}{\xi}\big(\tfrac12 + \sqrt{s}\big)$ *is a Stieltjes function.*

Proof. (i) implies (iii): each factor $(1 + s/\gamma^2)^{-1}$ is the Laplace transform of an exponential variable with rate $\gamma^2 > 0$, and the product converges because $\sum \gamma^{-2} < \infty$; the sum $\sum_\gamma e_\gamma/\gamma^2$ converges almost surely. (iii) implies (ii) by (F6).

(ii) implies (i): by (F7), φ extends analytically to $\mathbb{C} \setminus (-\infty, 0]$. Suppose a zero γ has $\operatorname{Im}\gamma \neq 0$. Since $\operatorname{Re}\gamma > 0$ the point $s_0 = -\gamma^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re}\gamma\,\operatorname{Im}\gamma \neq 0$, and its principal square root is $\pm i\gamma$, so $\xi(1/2 + \sqrt{s_0}) = \Xi(\pm\gamma) = 0$ by evenness. Then φ has a pole at a nonreal point of $\mathbb{C} \setminus (-\infty, 0]$, a contradiction.

(i) is equivalent to (iv): under RH, $\psi(s) = \sum_\gamma 1/(s+\gamma^2)$, a Stieltjes function with measure $\sum_\gamma \delta_{\gamma^2}$. Conversely a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and an off line zero produces the nonreal pole s_0 of ψ found above. □

Condition (iv) is a companion of Lagarias's criterion [160], and it is the pole test of Table 1.7 in analytic form.

The de Bruijn Newman flow embeds φ in a family of the same kind. Let Φ_{dBN} and $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be as in Section 8.4, so that $H_0(z) = \frac18\Xi(z/2)$. Put $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$, the function R_t of Theorem 7.2.15, so that $\varphi_0 = \varphi$. The function H_t is even, real on $\mathbb{R}$ and entire of order one, as in the proof of Theorem 7.2.15, so its Hadamard product in z^2 converges. It is positive on the imaginary axis, $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, since

$\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u) > 0$; hence $\varphi_t > 0$ on $(0,\infty)$ and no zero of H_t is purely imaginary. With these facts the argument of Theorem 4.2.4 applies verbatim.

Corollary 4.2.5. *φ_t is HCM if and only if $t \geq \Lambda$, the de Bruijn Newman constant. Hence φ_t is HCM for $t \geq 0.22$ [282], φ_t is not HCM for $t < 0$ [289], and RH holds if and only if φ_t is HCM for every $t > 0$.*

Proof. The zeros of H_t are real if and only if $t \geq \Lambda$ (de Bruijn [73], Newman [210]). If RH holds, H_0 has only real zeros, so $\Lambda \leq 0$ and φ_t is HCM for every $t > 0$. Conversely, $\varphi_t \to \varphi_0$ pointwise as $t \downarrow 0$, so if every φ_t with $t > 0$ is HCM then φ is HCM by (F4), and Theorem 4.2.4 gives RH. □

The bound $\Lambda \leq 0.2$ of Platt and Trudgian [247] (Section 8.4) lowers the unconditional threshold from 0.22 to 0.2. The reciprocal Xi function, if HCM at all, is HCM with no room to spare, since the property fails for every $t < 0$. This is the Riemann counterpart of the boundary $|\beta| = 1/(1-\alpha)$ in Bosch's theorem, which is attained; the difference is that Bosch can compute on the boundary because his kernel is explicit.

4.2.4 What Bosch's theorem can and cannot supply

Bosch's theorem cannot be the input to a proof of Theorem 4.2.4. For products the reason is elementary.

Proposition 4.2.6. *For any positive numbers c_n with $\sum_n c_n^{-1} < \infty$ the function $\prod_n (1 + s/c_n)^{-1}$ is HCM. If one c_n is replaced by a pair $c, \bar{c}$ with c nonreal, the product is still positive on $(0,\infty)$ but is not HCM.*

Proof. The first claim is the argument for (i) implies (ii) in Theorem 4.2.4. For the second, $(1 + s/c)(1 + s/\bar{c}) = |1 + s/c|^2 > 0$ for real s, and the product has a pole at the nonreal point $-c$, contradicting (F7). □

Heuristically, every HCM closure operation carries real frequencies to real frequencies and none creates them: on Mellin transforms, products and quotients of independent variables multiply the transforms, and powers rescale the frequencies. Bosch's theorem lives in the class where the frequency set $\{2k-1\}$ is given, so any deduction of Theorem 4.2.4 from it must import the location of the γ from elsewhere.

Nor can the Riemann statement be proved factor by factor. Write $\xi(1/2 + \sqrt{s}) = \frac{1}{2}(s - \frac{1}{4})\,\pi^{-(1/2+\sqrt{s})/2}\,\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\,\zeta(\frac{1}{2} + \sqrt{s})$. The reciprocal factors are not individually HCM. The reciprocal of $\pi^{-(1/2+\sqrt{s})/2}$ is $\pi^{1/4}\pi^{\sqrt{s}/2}$, and the pair function of $\pi^{\sqrt{s}/2}$ is $\exp(\frac{1}{2}\log\pi\,\sqrt{u}\,\sqrt{w+2})$, which increases in w. The factor $1/\zeta(1/2+\sqrt{s})$ changes sign at $s = 1/4$, since $\zeta < 0$ on $(1/2, 1)$, and HCM functions are positive. The positive combination $1/((s - \frac{1}{4})\zeta(\frac{1}{2} + \sqrt{s}))$ is continuous at $s = 1/4$, and by the argument of Theorem 4.2.4 any HCM statement for it would

again imply RH. It is the function to which the sech expansion of $1/\zeta(\frac{1}{2}+\sqrt{s})$ in Theorem 6.1.5 should be applied.

Bosch's theorem is a density statement, and HCM of a density differs from HCM of a Laplace transform: the density of $|C|$ is not HCM, yet φ_C is. On the Cauchy side the two levels are linked only because sech is self dual, its own Fourier transform up to scaling, so that the density g_C of $\log|C|$ and the transform φ_C are the same function in different coordinates. On the Riemann side there is no self duality: the density level object is g in (4.2.3), the Fourier inverse of $\Xi(0)/\Xi(i\,\cdot)$, and the function level object is φ.

Question 4.2.7. The variable W, with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, exists unconditionally (Section 4.2.2 and Theorem 4.3.25). For which β is the density of W^β HCM? On the Cauchy side the answer is $|\beta| \geq 2$.

Question 4.2.8. Does HCM of the density of W^β for a single β imply RH?

Theorem 4.3.27 answers Question 4.2.7 in part: under RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$, the density of W^β can be HCM only if $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873$, and Question 4.3.28 asks whether this is also sufficient. At the Cauchy point the same rule gives $3 - 1 = 2$, Bosch's threshold.

One might hope that an HCM variable has a logarithm with an EGGC law (Section 1.3.2), which would force the singularities of $\mathbb{E}W^z$, that is the zeros of Ξ, onto the real axis. This implication is false in general. The generalised inverse Gaussian law with density proportional to $x^{-1}e^{-(x+1/x)}$ is HCM by (F7), yet its Mellin transform at $z = it$, $K_{it}(2)/K_0(2)$, has real zeros in t, the first at $t \approx 4.4255$ (by a theorem of Pólya [273], every zero of $t \mapsto K_{it}(2)$ is real). The characteristic function of its logarithm therefore vanishes, and the logarithm is not infinitely divisible. So density level HCM does not by itself give RH.

Two things do transfer. First, the dictionary of Table 4.1 is a solved model in which every quantity is explicit, and it tests any proposed argument: a proof that φ is HCM which does not use the values of the frequencies would also prove HCM for $\cos(\pi z/2)$ with frequencies moved off the axis, which is false by Proposition 4.2.6. Second, the boundary phenomenon of Corollary 4.2.5 gives a target: to show that φ_t is HCM for all $t > 0$, which by that corollary is equivalent to $\Lambda \leq 0$, that is, by Rodgers and Tao [289], to $\Lambda = 0$.

4.3 A Bosch type theorem for the reciprocal Xi function

The observation that closes Section 4.2 becomes a theorem through an elasticity criterion (Theorem 4.3.2). Applied to $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ it gives a Bosch type theorem for $\varphi(x^{1/\beta})$ (Theorem 4.3.5), and the same elasticity identifies the Thorin measure of $\varphi(s^p)$ (Theorem 4.3.12); the positivity of its density is where the Riemann problem now sits.

The notation is that of Section 4.2: the clock φ is a function of s, and W is the

Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace $W(t)$ of Section 5.1. The numerical checks are made by `code_bosch/verify_bosch_xi_v7.py` in the repository and documented in `code/README.md`; the checks confirm formulas and constants and prove nothing.

4.3.1 The elasticity criterion

We use HCM as defined in Section 4.2.1, with the facts (F1) to (F7) listed there, above all Bondesson's representation (4.2.1) [39, (5.2.3) and Theorem 5.3.1]. Throughout, $\Omega = \mathbb{C}\setminus(-\infty, 0]$, $\mathbb{C}^+ = \{\operatorname{Im} x > 0\}$, and a Pick function is an analytic map $\mathbb{C}^+ \to \overline{\mathbb{C}^+}$. For a GGC with Laplace transform h, drift $b \geq 0$ and Thorin measure U, in the sense of (1.3.1), $-h'/h = b + \int U(dt)/(x+t)$ is a Stieltjes function. We also need two facts about complete Bernstein functions [299, Theorem 6.2]: a function f on $(0,\infty)$ is complete Bernstein if and only if $f \geq 0$ on $(0,\infty)$ and f continues to a Pick function analytic on Ω, and then $f(x)/x$ is a Stieltjes function.

The elasticity criterion below restates (4.2.1) as a Pick function condition, of the kind through which Bosch verifies HCM [41, Section 2.2].

Definition 4.3.1. For h analytic and zero free on Ω and positive on $(0,\infty)$, let $\log h$ be the branch that is real on $(0,\infty)$ and define the elasticity $\varepsilon_h(x) = -xh'(x)/h(x)$. In logarithmic coordinates $x = e^z$ one has $\varepsilon_h(e^z) = -\frac{d}{dz}\log h(e^z)$.

Theorem 4.3.2. *Let $h > 0$ on $(0,\infty)$. Then h is HCM if and only if h continues to a zero free analytic function on Ω and ε_h is a Pick function, that is* $\operatorname{Im}\varepsilon_h(x) \geq 0$ *for* $\operatorname{Im} x > 0$.

Proof. Necessity. By (4.2.1), h is zero free and analytic on Ω, and its elasticity is

$$\varepsilon_h(x) = 1 - r + b_1 x + \int \frac{x}{x+t}\,\Gamma_1(dt) - \frac{b_2}{x} - \int \frac{\Gamma_2(dt)}{1+tx}.$$

For $\operatorname{Im} x > 0$ each term has nonnegative imaginary part, for instance $\operatorname{Im}\frac{x}{x+t} = \frac{t\operatorname{Im} x}{|x+t|^2}$.

Sufficiency. Fix $u > 0$. The principal branch $y(w) = \operatorname{arccosh}(w/2)$ maps $\mathbb{C}\setminus(-\infty, 2]$ conformally onto the half strip $S = \{\operatorname{Re} y > 0,\ |\operatorname{Im} y| < \pi\}$, and $w = 2\cosh y$. For $y \in S$ both ue^y and ue^{-y} lie in Ω, so

$$F(w) = -\log h\big(ue^{y(w)}\big) - \log h\big(ue^{-y(w)}\big)$$

is analytic on $\mathbb{C}\setminus(-\infty, 2]$ and real on $(2,\infty)$, and $e^{-F(w)}$ is the pair function. Let $\operatorname{Im} w > 0$, so $y = \varrho + i\theta$ with $\varrho > 0$ and $\theta \in (0,\pi)$. Put $r_1 = ue^{\varrho} > r_2 = ue^{-\varrho}$. Since $\log h(\bar{x}) = \overline{\log h(x)}$,

$$\operatorname{Im} F(w) = -\operatorname{Im}\log h(r_1 e^{i\theta}) + \operatorname{Im}\log h(r_2 e^{i\theta}).$$

Along the ray, $\partial_r \operatorname{Im} \log h(re^{i\theta}) = \operatorname{Im}\big(e^{i\theta}h'/h\big) = -r^{-1}\operatorname{Im}\varepsilon_h(re^{i\theta}) \le 0$, so $\operatorname{Im} F(w) \ge 0$. Thus F is a Pick function that continues analytically across $(2,\infty)$, where it is real. By the Nevanlinna representation $F(w) = a + bw + \int(\frac{1}{t-w} - \frac{t}{1+t^2})\sigma(dt)$ with $b \ge 0$ and $\sigma \ge 0$ carried by $(-\infty, 2]$. Hence for $w > 2$,

$$F'(w) = b + \int_{(-\infty,2]} \frac{\sigma(dt)}{(w-t)^2} = b + \int_{(-\infty,2]} \int_0^\infty \lambda e^{-\lambda(w-t)}\, d\lambda\, \sigma(dt),$$

which is completely monotone. A function whose derivative is completely monotone is a Bernstein function up to an additive constant, and e^{-F} is then completely monotone. So every pair function is completely monotone and h is HCM. □

The two conditions play different roles. Analyticity without zeros is global and rigid: a single zero or pole of h anywhere off the negative axis destroys HCM, however small its effect on the real line. The Pick condition is local and quantitative: it can fail along a ray without any singularity.

4.3.2 Bosch's theorem from the criterion

Bosch's variable $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$, $0 < \alpha < 1$, of Section 4.2.1 has the density $\sin(\pi\alpha)/(\pi\alpha(t^2 + 2t\cos\pi\alpha + 1))$ of a half Cauchy law with drift. Let $c = \cos\pi\alpha$ and $q = 1/|\beta|$, half the q of (4.2.2). Since $T_\alpha \overset{d}{=} 1/T_\alpha$ we may take $\beta > 0$. The density of T_α^β is proportional to

$$h(x) = \frac{x^{q-1}}{D(x)}, \qquad D(x) = x^{2q} + 2cx^q + 1 = (x^q + e^{i\pi\alpha})(x^q + e^{-i\pi\alpha}).$$

Theorem 4.3.3 (Bosch [41, Theorem 1.2]). *h is HCM if and only if $\alpha \le 1/2$ and $q \le 1 - \alpha$.*

Proof. Zeros. For $x \in \Omega$ the value x^q ranges over the sector $|\arg| < q\pi$. The zeros of D satisfy $x^q = e^{\pm i(1-\alpha)\pi}$, so D is zero free on Ω if and only if $q \le 1 - \alpha$.

Elasticity. For $x = re^{i\theta}$ with $0 < \theta < \pi$ write $x^q = Re^{i\omega}$ with $R = r^q$ and $\omega = q\theta \in (0, q\pi)$. Then $\varepsilon_h = 1 - q + xD'(x)/D(x)$ and $\operatorname{Im}\varepsilon_h = qR\,\partial_R \arg D$. A direct computation of $\operatorname{Im}(\partial_R D \cdot \overline{D})$ gives

$$\operatorname{Im}\varepsilon_h(x) = \frac{2qR\sin\omega\,\big(c(1+R^2) + 2R\cos\omega\big)}{|D(x)|^2}. \tag{4.3.1}$$

If $\alpha \le 1/2$ and $q \le 1 - \alpha$ then $c \ge 0$ and $\cos\omega \ge \cos((1-\alpha)\pi) = -c$, so, using $1 + R^2 \ge 2R$, the bracket is at least $2R(c + \cos\omega) \ge 0$. Theorem 4.3.2 gives HCM. If $\alpha > 1/2$ then $c < 0$ and the bracket tends to $c < 0$ as $R \to 0$, so ε_h is not Pick. If $q > 1 - \alpha$ then h has poles in Ω. In both cases Theorem 4.3.2 rules out HCM. □

In terms of β, the theorem says that the density of T_α^β is HCM if and only if $\alpha \le 1/2$ and $|\beta| \ge 1/(1-\alpha)$. For the half Cauchy variable, $\alpha = 1/2$, the threshold is $|\beta| \ge 2$; there $c = 0$ and (4.3.1) reduces to $\operatorname{Im}\varepsilon_h = 4qR^2 \sin\omega\cos\omega/|D|^2$, which is nonnegative for every $\omega \in (0, q\pi)$ exactly when $q \le 1/2$. So the half Cauchy threshold is the condition $\omega \le \pi/2$, equivalently that the zeros $x = e^{\pm i\pi/(2q)}$ of $1 + x^{2q}$ stay on or beyond the cut.

4.3.3 The Riemann analogue

As in Section 4.2.2, γ runs over the zeros of $\Xi(z) = \xi(1/2 + iz)$ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; the nonreal ones come in conjugate pairs $\gamma, \bar\gamma$. By (1.2.1), unconditionally,

$$\varphi(s) = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_\gamma \Big(1 + \frac{s}{\gamma^2}\Big)^{-1}, \qquad \varepsilon_\varphi(s) = \frac{\sqrt{s}}{2}\frac{\xi'}{\xi}\Big(\frac12 + \sqrt{s}\Big) = \sum_\gamma \frac{s}{s+\gamma^2},$$

both absolutely convergent since $\sum |\gamma|^{-2} < \infty$. Write $\gamma = \varrho e^{i\delta}$. Since the zeros of ξ lie in the open critical strip, $|\delta| < \arctan(1/(2\operatorname{Re}\gamma))$, and since every zero has $\operatorname{Re}\gamma > 14$, also $|\delta| < \pi/4$. Define

$$\delta_* = \max_\gamma |\arg\gamma|, \qquad \alpha_\xi = \frac{2\delta_*}{\pi}, \qquad p_* = 1 - \alpha_\xi.$$

The maximum is attained if some zero is off the line, because for every positive angle only finitely many γ have $|\arg\gamma|$ larger than it. RH is the statement $\delta_* = 0$.

Lemma 4.3.4. *Let $\gamma = \varrho e^{i\delta}$ with $0 \le \delta < \pi/4$ and $s = re^{i\theta}$ with $0 < \theta < \pi - 2\delta$. Then*

$$\operatorname{Im}\Big(\frac{s}{s+\gamma^2} + \frac{s}{s+\bar\gamma^2}\Big) \ge 0,$$

with strict inequality. For $\delta = 0$ the single term $\operatorname{Im}\frac{s}{s+\gamma^2} = \frac{\gamma^2 r\sin\theta}{|s+\gamma^2|^2}$ is positive for $0 < \theta < \pi$.

Proof. For $c \in \mathbb{C}$, $\operatorname{Im}\frac{s}{s+c} = \operatorname{Im}\big(-\frac{c}{s+c}\big) = \frac{\operatorname{Im}(\bar c s)}{|s+c|^2}$. With $c = \varrho^2 e^{\pm 2i\delta}$ this is $\varrho^2 r\sin(\theta \mp 2\delta)/|s+c|^2$. The sum is

$$\varrho^2 r\Big(\frac{\sin(\theta-2\delta)}{A_-} + \frac{\sin(\theta+2\delta)}{A_+}\Big), \qquad A_\mp = r^2 + \varrho^4 + 2r\varrho^2\cos(\theta \mp 2\delta).$$

Since $\theta + 2\delta < \pi$ the second sine is positive. If $\theta \ge 2\delta$ both terms are nonnegative. If $\theta < 2\delta$ then $|\theta - 2\delta| < \theta + 2\delta$, so $A_- > A_+$, and $\sin(\theta+2\delta) - \sin(2\delta - \theta) = 2\cos(2\delta)\sin\theta > 0$; the positive term has the larger numerator and the smaller denominator. □

Theorem 4.3.5. *For $p \in \mathbb{R}$ the function $x \mapsto \varphi(x^p)$ is HCM if and only if $|p| \le p_* = 1 - \alpha_\xi$. Equivalently, $\varphi(x^{1/\beta})$ is HCM if and only if $|\beta| \ge 1/(1-\alpha_\xi)$.*

Proof. Since $h(x)$ is HCM if and only if $h(1/x)$ is, take $p > 0$. Let $h_p(x) = \varphi(x^p)$, positive on $(0,\infty)$ because $\xi > 0$ on $\mathbb{R}$. Its singularities are the poles $x^p = -\gamma^2 = \varrho^2 e^{\pm i(\pi - 2|\delta|)}$, and x^p ranges over $|\arg| < p\pi$ for $x \in \Omega$. Hence h_p is zero free and analytic on Ω if and only if $p\pi \le \pi - 2|\delta|$ for every γ, that is $p \le p_*$. When $p \le p_*$,

$$\varepsilon_{h_p}(x) = p\,\varepsilon_\varphi(x^p) = p\sum_\gamma \frac{x^p}{x^p + \gamma^2},$$

and for $\operatorname{Im} x > 0$ the point $s = x^p$ has $0 < \arg s < p\pi \le \pi - 2\delta_*$. Grouping conjugate pairs, Lemma 4.3.4 makes every term of $\operatorname{Im}\varepsilon_{h_p}$ nonnegative. Theorem 4.3.2 gives HCM. When $p > p_*$ the pole coming from a zero with $|\arg\gamma| = \delta_*$ lies in Ω, and Theorem 4.3.2 rules out HCM. □

Corollary 4.3.6. *The following are equivalent: (i) RH; (ii) φ is HCM; (iii) $\varphi(x^p)$ is HCM for every $|p| < 1$; (iv) ε_φ is a Pick function; (v)* $\operatorname{Im}\big[z\,(\xi'/\xi)(\frac12+z)\big] \ge 0$ *for all z with* $\operatorname{Re} z > 0$ *and* $\operatorname{Im} z > 0$.

Proof. By Theorem 4.3.5, each of (i), (ii) and (iii) says that $p_* = 1$; the equivalence of (i) and (ii) is also Corollary 4.1.3(ii), restated as Theorem 4.2.4, so only (iii) to (v) are new. (iv) is the elasticity form of (ii) through Theorem 4.3.2. (v) is (iv) with $s = z^2$, which maps the open first quadrant onto $\mathbb{C}^+$. □

Criterion (v) is an angular companion to Lagarias's criterion $\operatorname{Re}(\xi'/\xi)(\frac12+z) > 0$ for $\operatorname{Re} z > 0$ [160].

Corollary 4.3.7 (unconditional). *RH has been verified for all zeros with* $0 < \operatorname{Re}\gamma \le T_0 = 3\times 10^{12}$ *[247]. Hence* $\delta_* < \arctan(1/(2T_0))$ *and*

$$\varphi(x^p) \text{ is HCM for all } |p| \le 1 - \frac{1}{\pi T_0}, \qquad 1 - \frac{1}{\pi T_0} > 1 - 1.1\times 10^{-13}.$$

Unconditionally, $\operatorname{Im}\varepsilon_\varphi \ge 0$ *on the sector* $0 < \arg s < \pi - 1/T_0$.

Proof. Any off line zero has $\operatorname{Re}\gamma > T_0$ and $|\operatorname{Im}\gamma| < 1/2$, so $|\arg\gamma| < 1/(2T_0)$ and $2\delta_*/\pi < 1/(\pi T_0)$. The sector statement is Lemma 4.3.4 applied termwise. □

Corollary 4.3.7 is a statement about the elasticity $\varepsilon_\varphi(s) = s\psi(s)$, where $\psi = -\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$ is the function of the Stieltjes form (iv) of Theorem 4.2.4.

Lemma 4.3.8. *Let $c = c_1 + ic_2$ with $c_2 > 0$, and let $s = x + iy$ with $y > 0$ and $s \ne c$. Then*

$$\operatorname{Im}\Big(\frac{1}{s-c} + \frac{1}{s-\bar c}\Big) = -\frac{2y\,\big(|s-c_1|^2 - c_2^2\big)}{|s-c|^2\,|s-\bar c|^2},$$

so the imaginary part is positive on the open disc $|s - c_1| < c_2$, whose vertical diameter is the segment from $\bar c$ to c, zero on its boundary and negative outside it.

Proof. With $d = (x - c_1)^2$, $\operatorname{Im} \frac{1}{s-c} = -\frac{y-c_2}{d+(y-c_2)^2}$ and $\operatorname{Im} \frac{1}{s-\bar{c}} = -\frac{y+c_2}{d+(y+c_2)^2}$. Over the common denominator the numerator of the sum is

$$-(y-c_2)\big(d+(y+c_2)^2\big) - (y+c_2)\big(d+(y-c_2)^2\big) = -2y\,(d+y^2-c_2^2),$$

and $d + y^2 = |s - c_1|^2$. □

A nonreal zero $\gamma = a + ib$ of Ξ, with $a > 0$ and, after conjugation, $b > 0$, contributes to ψ the pair of terms $1/(s-c)+1/(s-\bar{c})$ with $c = -\bar{\gamma}^2 = b^2-a^2+2iab$. Let

$$D_\gamma = \big\{ s : \ |s - (b^2 - a^2)| \le 2ab \big\}$$

be the closed disc with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$. By Lemma 4.3.8 the pair has positive imaginary part only at interior points of D_γ, and negative imaginary part at every point of the upper half plane outside D_γ.

THEOREM 4.3.9 (unconditional). *Let $T_0 = 3 \times 10^{12}$, so that every zero γ with $0 < \operatorname{Re}\gamma \le T_0$ is real [247], and let*

$$\mathcal{E}_{T_0} = \Big\{ x + iy : \ x < -T_0^2 + T_0 + \tfrac{1}{4}, \ y^2 < \tfrac{1}{2} - x \Big\}.$$

Split $\psi = S_{\text{on}} + R_{\text{off}}$, where $S_{\text{on}}(s) = \sum_{\gamma \text{ real}} (s+\gamma^2)^{-1}$ runs over the zeros on the critical line and R_{off} is the same sum over the nonreal γ.

(i) S_{on} is a Stieltjes function, with measure $U_{\text{line}} = \sum_{\gamma \text{ real}} \delta_{\gamma^2}$.

(ii) R_{off} is meromorphic on $\mathbb{C}$ and nonnegative on $[0,\infty)$. For every nonreal γ the disc D_γ, and with it the poles $-\gamma^2$ and $-\bar{\gamma}^2$ of R_{off} on its boundary, lies in $\mathcal{E}_{T_0}$, and $\operatorname{Im} R_{\text{off}} \le 0$ at every point of the upper half plane outside these discs.

(iii) $\psi > 0$ on $[0,\infty)$, and $\operatorname{Im}\psi(s) < 0$ at every $s = x + iy$ with $y > 0$ outside $\mathcal{E}_{T_0}$.

Proof. Since $\sum |\gamma|^{-2} < \infty$, the series for ψ, S_{on} and R_{off} converge absolutely and locally uniformly off the points $-\gamma^2$, so R_{off} is meromorphic on $\mathbb{C}$, and (i) holds because $\sum_{\gamma \text{ real}} (1+\gamma^2)^{-1} < \infty$. A real γ contributes $-y/|s+\gamma^2|^2 < 0$ to $\operatorname{Im}\psi(x+iy)$ for $y > 0$, and $(x+\gamma^2)^{-1} > 0$ to $\psi(x)$ for $x \ge 0$.

A nonreal zero is $\gamma = a + ib$ with $a > T_0$ and, after conjugation, $0 < b < \frac{1}{2}$, because ξ vanishes at $\frac{1}{2} \pm b + ia$. With $c = b^2 - a^2 + 2iab$ as above, its pair of terms equals $2(x - c_1)/((x - c_1)^2 + c_2^2) > 0$ at $x \ge 0$, since $c_1 = b^2 - a^2 < 0$, and its imaginary part is negative at every point of the upper half plane outside D_γ. It remains to show that $D_\gamma \subset \mathcal{E}_{T_0}$. A point $x + iy$ of D_γ has

$$x \le b^2 - a^2 + 2ab = 2b^2 - (a-b)^2 < -T_0^2 + T_0 + \tfrac{1}{4},$$

because the middle term increases in b and, for $a > b$, decreases in a, and equals $-T_0^2 + T_0 + \frac{1}{4}$ at $a = T_0$, $b = \frac{1}{2}$; in particular $x < 0$. With $e = x - (b^2 - a^2)$, so that $a^2 = b^2 - x + e$, it also has

$$y^2 \le 4a^2b^2 - e^2 = 8b^4 - 4b^2x - (e - 2b^2)^2 \le 4b^2(2b^2 - x) < 2b^2 - x < \tfrac{1}{2} - x,$$

because $4b^2 < 1$ and $2b^2 - x > 0$. Hence $D_\gamma \subset \mathcal{E}_{T_0}$, and (ii) follows from the signs of the pairs. At a point of the upper half plane outside $\mathcal{E}_{T_0}$ every pair of terms has negative imaginary part, and so has the term of every real γ; with the signs on $[0, \infty)$, since γ_1 is real, this gives (iii). □

Both bounds that define $\mathcal{E}_{T_0}$ are sharp for zeros with $a > T_0$ and $0 < b < \frac{1}{2}$. The rightmost point $2b^2 - (a - b)^2$ of D_γ tends to $-T_0^2 + T_0 + \frac{1}{4}$ as $b \uparrow \frac{1}{2}$ and $a \downarrow T_0$. For $x < \frac{3}{4} - T_0^2$ the supremum of y^2 over the points $x + iy$ of the discs is $\frac{1}{2} - x$: the bound $8b^4 - 4b^2x$ of the proof is attained, at $e = 2b^2$, by the disc with $a^2 = 3b^2 - x$, which has $a > T_0$ once b is close to $\frac{1}{2}$, and it tends to $\frac{1}{2} - x$ as $b \uparrow \frac{1}{2}$. At the tip $x_0 = -T_0^2 + T_0 + \frac{1}{4}$ the height of $\mathcal{E}_{T_0}$ is exactly $T_0 - \frac{1}{2}$, since $\frac{1}{2} - x_0 = (T_0 - \frac{1}{2})^2$, so the ray from the origin to the corner $x_0 + i(T_0 - \frac{1}{2})$ makes the angle $\theta_0 = \arctan \frac{T_0 - 1/2}{T_0^2 - T_0 - 1/4} = 3.33333 \times 10^{-13}$ with the negative axis, which exceeds $1/T_0$ by the factor $1 + 1.7 \times 10^{-13}$. Further out the angle decreases like $|x|^{-1/2}$.

Corollary 4.3.7 gives the sign of the elasticity $s\psi(s)$ outside a sector of opening $1/T_0$ beside the cut, through Lemma 4.3.4; Theorem 4.3.9 gives the sign of ψ itself outside a parabola, through the exact region of Lemma 4.3.8, and the parabola is much thinner. The theorem does not make ψ a Stieltjes function off $\mathcal{E}_{T_0}$, since being a Stieltjes function is a global property; it says that if the Stieltjes form fails, it fails through poles of ψ in $\mathcal{E}_{T_0}$, which come from zeros above the verification height. Like the corollary, it brings no new information about the zeros, and nothing in it limits the number of poles in $\mathcal{E}_{T_0}$ (Section 11.4.7).

Remark 4.3.10 (the right end). The right end of $\mathcal{E}_{T_0}$ cannot be moved to $-(T_0 - \frac{1}{2})^2$, the value the argument would give if the rightmost point of D_γ were $-(a-b)^2$: it is $2b^2 - (a - b)^2$, and the extra $2b^2$ reaches into the strip $-(T_0 - \frac{1}{2})^2 \le x < -(T_0 - \frac{1}{2})^2 + \frac{1}{2}$. Whether $\operatorname{Im}\psi$ itself is positive at such points depends on zeros that nobody knows; the termwise argument cannot exclude them.

Figure 4.2 shows the sliver and the discs.

The threshold p_* of Theorem 4.3.5 can be followed along the de Bruijn Newman flow. With H_t and $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$ as in Section 4.2.3, so that $\varphi_0 = \varphi$, let $\delta_*(t)$ be the largest angle $|\arg z|$ over the zeros z of H_t with $\operatorname{Re} z > 0$, and $p_*(t) = 1 - 2\delta_*(t)/\pi$. The proof of Theorem 4.3.5 uses only the product over the zeros, positivity on $(0, \infty)$ and the bound $|\arg z| < \pi/4$ required by Lemma 4.3.4, so whenever the zeros of H_t satisfy that bound, $\varphi_t(x^p)$ is HCM if and only if $|p| \le p_*(t)$. By de Bruijn [73] and Newman [210] (Corollary 4.2.5), $p_*(t) = 1$ exactly for $t \ge \Lambda$; by Rodgers and Tao [289], $p_*(t) < 1$ for every $t < 0$.

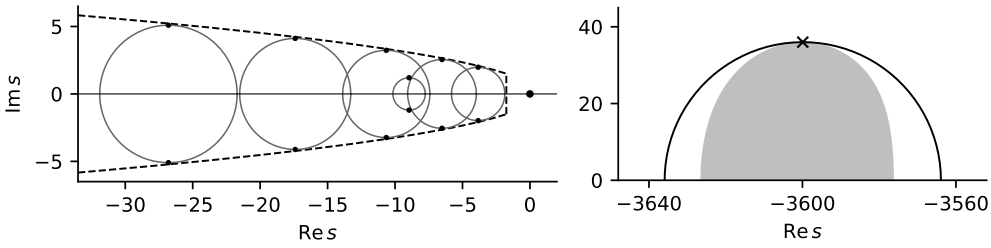


Figure 4.2: Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 4.3.9 for $T_0 = 2$ (dashed) and the discs D_γ of six hypothetical zeros $a + ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar{\gamma}^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of ζ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of 481×300 points, which lies inside the upper half of D_γ (solid); the cross is the pole $-\bar{\gamma}^2 = -3599.91 + 36i$. Drawn by `pakes_disc.py`.

Question 4.3.11. Describe $p_*(t) = 1 - \frac{2}{\pi}\delta_*(t)$ for $t < 0$. By Rodgers and Tao $p_*(t) < 1$ for every $t < 0$. Assume RH; how fast does $1 - p_*(t) \to 0$ as $t \uparrow 0$?

Heuristically, the off line zeros of H_t for small $|t|$ come from nearly coincident pairs of real zeros: by the isolated pair estimate of Section 8.4, a pair at distance D_0 collides near $t = -D_0^2/8$ and then has imaginary parts of order $\sqrt{|t|}$, at heights where the gap is of order $\sqrt{|t|}$. Since $1 - p_*(t)$ is governed by $\max \operatorname{Im} z / \operatorname{Re} z$, the answer depends on the lowest height at which such a pair has collided, and so on the small gap statistics of the zeros. Both ingredients of an estimate are missing: de Bruijn's bound on the imaginary parts of the zeros runs only forward in time, from H_0 to H_t with $t > 0$, and no lower bound is known for the heights of the pairs that have collided by time t.

4.3.4 The Thorin measure of $\varphi(s^p)$ and Bosch's variable

By Bondesson's theorem [39, Theorem 6.1.1], a function with value 1 at 0 is HCM exactly when it is the Laplace transform of a GGC, so Theorem 4.3.5 already makes $\varphi(s^p)$ a GGC transform for $0 < p \le p_*$.

Theorem 4.3.12. *Let $0 < p \le p_*$. Then $\varphi(s^p) = \mathbb{E}e^{-sX_p}$ for a GGC variable X_p with no drift. If $p < p_*$, its Thorin measure has the density*

$$u_p(\tau) = \frac{p}{\pi\tau} \operatorname{Im} \varepsilon_\varphi\big(\tau^p e^{ip\pi}\big) = \frac{p}{2\pi\tau} \operatorname{Im}\Big[z\,\frac{\xi'}{\xi}\Big(\frac{1}{2} + z\Big)\Big]_{z=\tau^{p/2}e^{ip\pi/2}} \ge 0, \qquad \tau > 0, \tag{4.3.2}$$

which in terms of the zeros is

$$u_p(\tau) = \frac{p}{\pi\tau} \sum_\gamma \operatorname{Im} \frac{\tau^p e^{ip\pi}}{\tau^p e^{ip\pi} + \gamma^2}. \tag{4.3.3}$$

If $p = p_*$, its Thorin measure is $u_{p_*}(\tau)\, d\tau$ plus a unit atom at $|\gamma|^{2/p_*}$ for every zero γ with $\arg\gamma = -\delta_*$, counted with multiplicity, that is, one atom for each conjugate pair of zeros at the extreme angle; when $p_* = 1$ there is an atom at γ^2 for every zero, and $u_1 = 0$.

Proof. Let $h = h_p$ and $\varepsilon_h(x) = p\,\varepsilon_\varphi(x^p)$. By the proof of Theorem 4.3.5, ε_h is analytic on Ω and Pick. On $(0,\infty)$ it is nonnegative: a real zero contributes $x^p/(x^p+\gamma^2) > 0$ and a pair contributes $2\operatorname{Re}\frac{s}{s+\varrho^2 e^{2i\delta}} = \frac{2s(s+\varrho^2\cos 2\delta)}{|s+\varrho^2 e^{2i\delta}|^2} > 0$. Hence ε_h is complete Bernstein and $\varepsilon_h(x)/x = -(\log h)'(x)$ is a Stieltjes function, $a/x + b + \int U(dt)/(x+t)$. Here $a = \varepsilon_h(0+) = 0$ and $b = \lim_{x\to\infty} \varepsilon_h(x)/x = 0$ because $\varepsilon_\varphi(s) = O(\sqrt{s}\log s)$. Integrating from 0 with $h(0) = 1$ gives $h(x) = \exp(-\int \log(1 + x/t)U(dt))$, a GGC transform. For $p < p_*$ the function ε_h is continuous up to the upper edge of the cut, and Stieltjes inversion at $x = \tau e^{i\pi}$ gives $U(d\tau) = u_p(\tau)d\tau$ with $u_p(\tau) = \frac{1}{\pi\tau}\operatorname{Im}\varepsilon_h(\tau e^{i\pi})$, which is (4.3.2); (4.3.3) follows from the Hadamard form of ε_φ. For $p = p_*$ the same inversion applies away from the points $x_0 = \tau_0 e^{i\pi}$ with $\tau_0 = |\gamma|^{2/p_*}$ and $\arg\gamma = -\delta_*$, at which $x_0^{p_*} = -\gamma^2$. Near such a point the term of γ in ε_h is $p_* x^{p_*}/(x^{p_*} + \gamma^2) = x_0/(x - x_0) + O(1)$, so $\varepsilon_h(x)/x = 1/(x + \tau_0) + O(1)$, and inversion gives a unit atom at τ_0. On the ray itself this term equals $p_*\tau^{p_*}/(\tau^{p_*} - |\gamma|^2)$, which is real, so u_{p_*} does not see the atom. □

Corollary 4.3.13. *Let $L_p = Z_p/\tilde{Z}_p$ be the ratio of independent positive p stable variables, whose density $\frac{\sin p\pi}{\pi}\frac{t^{p-1}}{t^{2p}+2t^p\cos p\pi+1}$ is Lamperti's [163]; its power L_p^p is Bosch's variable T_α with $\alpha = p$. Under RH the Thorin measure of X_p is*

$$U_p = p\sum_\gamma \mathcal{L}\big(\gamma^{2/p} L_p\big),$$

and $X_p \overset{d}{=} S_p\, X^{1/p}$, where $X = \sum_\gamma e_\gamma/\gamma^2$ is the clock variable of Proposition 1.2.1, with independent standard exponential variables e_γ, and S_p is a positive p stable variable independent of X with $\mathbb{E}e^{-\lambda S_p} = e^{-\lambda^p}$. At $p = 1/2$, $L_{1/2} \overset{d}{=} C^2$ for a standard Cauchy variable C, and $U_{1/2} = \frac{1}{2}\sum_\gamma \mathcal{L}(\gamma^4 C^2)$. As $p \uparrow 1$, $L_p \to 1$ and $U_p \to \sum_\gamma \delta_{\gamma^2}$ vaguely, the Thorin measure of φ itself.

Proof. For real γ the term of (4.3.3) equals $\frac{p}{\pi}\frac{\gamma^2\tau^{p-1}\sin p\pi}{\tau^{2p}+2\gamma^2\tau^p\cos p\pi+\gamma^4}$; substituting $\tau = \gamma^{2/p}t$ turns it into p times Lamperti's density in t. The subordination identity is $\mathbb{E}e^{-sS_pY} = \mathbb{E}e^{-s^pY^p}$ with $Y = X^{1/p}$. □

Bosch's variable is therefore the kernel of the Thorin measure of $\varphi(s^p)$. Formula (4.3.2) makes sense whatever the zeros are, and RH is the statement that it stays nonnegative as p increases to 1 (Question 4.3.16 and the argument after it).

COROLLARY 4.3.14. *RH holds if and only if X_p exists, that is $\varphi(s^p)$ is a GGC transform, for every $p \in (0,1)$. Unconditionally X_p exists for $0 < p \le 1 - 1/(\pi T_0)$, in particular for $0 < p \le 1 - 1.1 \times 10^{-13}$, with the Thorin density* (4.3.2).

Proof. If RH fails, $p_* < 1$, and for $p \in (p_*, 1)$ the function $\varphi(s^p)$ has a pole in Ω, which no GGC transform can have, since $\exp(-bs - \int \log(1+s/t)U(dt))$ is analytic and zero free on Ω. The rest is Theorem 4.3.12 with Corollary 4.3.7, which gives $1 - 1/(\pi T_0) < p_*$. □

QUESTION 4.3.15. Construct $X_{1/2}$ from Brownian motion, Bessel processes or the Biane, Pitman and Yor variables [32] without reference to the zeros. A construction that survived $p \uparrow 1$ would prove RH by Corollary 4.3.14.

4.3.5 Positivity of the Thorin density from the primes

The method reads the threshold off the zeros: the elasticity has nonnegative imaginary part termwise on the sector $0 < \arg s < \pi - 2\delta_*$, and no information about δ_* is produced. A proof of RH along these lines must show $u_p \ge 0$ for every $p < 1$ directly from (4.3.2), for instance from the Pólya kernel, from the prime sum of Remark 4.3.17 or from a probabilistic model; this is the Thorin measure form of the reciprocal Xi programme of [249]. It cannot be done factor by factor: only the combination $(s - \frac{1}{4})\zeta(\frac{1}{2} + \sqrt{s})$ can enter an HCM or Pick argument (Section 4.2.4).

QUESTION 4.3.16. Prove $u_p \ge 0$ for every $p \in (0,1)$ directly from (4.3.2), without locating the zeros.

The statement for all $p \in (0,1)$ is equivalent to RH. Under RH it holds by Theorem 4.3.12. Conversely, suppose RH fails, let $\gamma = \varrho e^{i\delta_*}$ be a zero of maximal angle, and let $p \in (p_*, 1)$; formula (4.3.2) still defines u_p, except at isolated poles, although it is no longer a Thorin density. At $\tau = \varrho^{2/p}$ the term of $\bar\gamma$ in (4.3.3) equals $-\frac{p}{2\pi\tau} \cot \frac{\pi(p-p_*)}{2}$, which tends to $-\infty$ as $p \downarrow p_*$. The sum of the other terms stays bounded, because the other points $-\gamma'^2$ keep a positive distance from $-\bar\gamma^2$ and the tail of the sum is $O(\sum |\gamma'|^{-2})$. So u_p takes negative values for p slightly above p_*.

Along the ray write $\vartheta = p\pi/2 \in (0, \pi/2)$, $t = \tau^{p/2}$, $z = te^{i\vartheta}$ and $w = \frac{1}{2} + z$, so that $u_p(\tau) = \frac{p}{2\pi\tau} \operatorname{Im}[z\, \frac{\xi'}{\xi}(w)]$ by (4.3.2), and split $\xi'/\xi = \Gamma_\xi + \zeta'/\zeta$, where

$$\Gamma_\xi(w) = \frac{1}{w} + \frac{1}{w-1} - \frac{\log \pi}{2} + \frac{1}{2}\psi\Big(\frac{w}{2}\Big)$$

is the logarithmic derivative of the factor $\frac12 w(w-1)\pi^{-w/2}\Gamma(w/2)$ of ξ, and $\psi = \Gamma'/\Gamma$ is the digamma function, not the function ψ of Theorem 4.2.4. The ray crosses the critical strip for $t < t_e = 1/(2\cos\vartheta)$ and then enters $\operatorname{Re} w > 1$.

REMARK 4.3.17 (primes). For $t > t_e$, that is $\tau > (2\cos(p\pi/2))^{-2/p}$, we have $\operatorname{Re} w > 1$ and

$$u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}\Big[z\Big(\frac1w + \frac{1}{w-1} - \frac{\log\pi}{2} + \frac12\psi\Big(\frac{w}{2}\Big)\Big)\Big] - \frac{p}{2\pi\tau}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt n}\operatorname{Im}\big(z\,n^{-z}\big),$$

with Λ the von Mangoldt function.

PROPOSITION 4.3.18. *Let* $\operatorname{Re} w > 1$ *and* $\sigma = \operatorname{Re} w$. *Then*

$$\frac{2\pi\tau}{p}\,u_p(\tau) \;\ge\; m_p(t) := \operatorname{Im}\Big[z\Big(\frac{1}{2w} + \frac{1}{w-1} + \frac12\log\frac{w}{2\pi}\Big)\Big] - \frac{t}{6\sigma^2} - t\Big(-\frac{\zeta'}{\zeta}\Big)(\sigma). \tag{4.3.4}$$

Every term of m_p *is explicit, no zero of* ζ *enters, and* $m_p(t) = \frac{t}{2}\sin\vartheta\log t\,(1+o(1))$ *as* $t\to\infty$. *Consequently for each* $p\in(0,1)$ *there is* $\tau_1(p)<\infty$ *with* $u_p(\tau)>0$ *for all* $\tau > \tau_1(p)$.

Proof. Binet's formula [225, (5.9.13)] gives, for $\operatorname{Re} x > 0$, $\psi(x) = \log x - \frac{1}{2x} - \int_0^\infty b(u)e^{-xu}du$ with $b(u) = \frac{1}{e^u-1} - \frac1u + \frac12 \in [0, u/12]$. Hence $\psi(x) = \log x - \frac{1}{2x} - R$ with $|R| \le \frac{1}{12(\operatorname{Re} x)^2}$. With $x = w/2$ this turns $\Gamma_\xi(w)$ into $\frac{1}{2w} + \frac{1}{w-1} + \frac12\log\frac{w}{2\pi} - \frac{R}{2}$ with $|R| \le \frac{1}{3\sigma^2}$, so $\operatorname{Im}[z\Gamma_\xi(w)]$ is at least the first term of m_p minus $t/(6\sigma^2)$. For the prime part, $\operatorname{Im}[z\,\zeta'/\zeta(w)] = -\sum\Lambda(n)\operatorname{Im}(zn^{-w}) \ge -t\sum\Lambda(n)n^{-\sigma}$. The asymptotic follows from $\operatorname{Im}[z\log z] = t(\sin\vartheta\log t + \vartheta\cos\vartheta)$ and $-\zeta'/\zeta(\sigma)\to 0$ as $\sigma\to\infty$. □

Table 4.2 gives the point beyond which $m_p > 0$, computed on a logarithmic grid of 50 points per decade, with $-\zeta'/\zeta(\sigma)$ evaluated directly. The entry t_1 is the last grid point at which $m_p \le 0$, so the sign change lies within one grid step, a factor $10^{1/50}$, above it. These values are computed, not proved: nothing controls m_p between grid points or beyond the last one.

The prime term of m_p can be made elementary, because $-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$ for every $\sigma > 1$. Delange [74, Théorème] proves the sharper bound $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) - 1/(2\sigma^2)$. Since $-\zeta'/\zeta(\sigma)$ enters m_p with the factor $-t < 0$, replacing it by $1/(\sigma-1)$ lowers m_p, and (4.3.4) still holds.

REMARK 4.3.19. The ratio t_1/t_e tends to 1. Near $\vartheta = \pi/2$ write $\sigma = 1+x$, so $t = t_e(1+2x)$. The dominant terms of m_p/t are $\frac12\log\frac{t}{2\pi}$ and $-\frac1x$, so positivity begins at $x \approx 2/\log(t_e/2\pi)$, that is $t_1(p) = t_e(p)\big(1 + \frac{4+o(1)}{\log(t_e/2\pi)}\big)$.

So Question 4.3.16 reduces, for each p, to the segment $|z| \le t_1(p)$ of the ray, which is asymptotically the part inside the critical strip. That part has length $t_e = 1/(2\cos(p\pi/2)) \approx 1/(\pi(1-p))$, so heights and exponents correspond through $1-p \approx 1/(\pi T)$, as in Corollary 4.3.7.

p	strip exit $t_e = \frac{1}{2\cos\vartheta}$	t_1	t_1/t_e	$\tau_1 = t_1^{2/p}$
0.5	0.707	4.07	5.75	274
0.9	3.20	11.1	3.47	210
0.99	31.8	69.6	2.19	5.29×10^3
0.999	318	553	1.74	3.10×10^5
0.9999	3183	4818	1.51	2.33×10^7

Table 4.2: Where the lower bound (4.3.4) becomes positive. On a logarithmic grid from t_e to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches t_1 at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved.

4.3.6 The density level and natural boundaries

A variable with an HCM density is a GGC and so has an HCM Laplace transform ((F5) and (F6) of Section 4.2.1). Under RH the clock variable $X = \sum_\gamma e_\gamma/\gamma^2$ is the GGC with transform φ. Its Cauchy counterpart is $X_C = \sum_k e_k/(2k-1)^2$, with transform $\operatorname{sech}(\pi\sqrt{s}/2)$.

Theorem 4.3.20. *The density of X_C is*

$$f_C(t) = \frac{4}{\pi}\sum_{k\ge1}(-1)^{k-1}(2k-1)e^{-(2k-1)^2t} = \frac{4}{\pi}\,\eta\Big(\frac{4it}{\pi}\Big)^3, \qquad t>0,$$

with η the Dedekind eta function. The line $\operatorname{Re} t = 0$ is a natural boundary of f_C. Hence X_C is a GGC whose Laplace transform is HCM, but the density of X_C is not HCM.

Proof. The residue of $\operatorname{sech}(\pi\sqrt{s}/2)$ at $s = -(2k-1)^2$ is $\frac{4}{\pi}(-1)^{k-1}(2k-1)$, which gives the series once the Bromwich contour is moved to the left across the circles $|s| = 4k^2$: for $s = 4k^2e^{i\theta}$, $|\cosh(\pi\sqrt{s}/2)|^2 = \sinh^2(\pi k\cos\frac{\theta}{2}) + \cos^2(\pi k\sin\frac{\theta}{2})$ is at least $\frac14$ for every θ and at least $\sinh^2(\pi k/2)$ for $|\theta| \le 2\pi/3$, while $|e^{st}| \le e^{-2k^2t}$ for $2\pi/3 \le |\theta| \le \pi$, so for $t > 0$ the integrals over the arcs of these circles to the left of the Bromwich line tend to 0. Jacobi's identity $\eta(z)^3 = \sum_{n\ge0}(-1)^n(2n+1)e^{\pi i(2n+1)^2z/4}$ [107, Theorem 357], at $z = 4it/\pi$, where $e^{\pi i(2k-1)^2z/4} = e^{-(2k-1)^2t}$, gives the eta form. As a modular form η has the real axis as natural boundary [284, Chapter 11, Section 1.4, and Chapter 1, Section 4.2], and the real axis in z is the line $\operatorname{Re} t = 0$. An HCM density continues analytically to Ω by (4.2.1), so f_C is not HCM. □

Question 4.3.21 (density level). Assume RH and that all zeros are simple. Does the density $f_X(t) = \sum_n A_n\gamma_n^2e^{-\gamma_n^2t}$ of X, with A_n as in (4.2.3), have $\operatorname{Re} t = 0$ as a natural boundary? If so, X is not HCM.

Pólya's gap theorem for Dirichlet series [271, 277] says that if the exponents μ_n of a Dirichlet series $\sum_n c_n e^{-\mu_n t}$ have density $d = \lim n/\mu_n$ and satisfy $\inf_n(\mu_{n+1} - \mu_n) > 0$, then every segment of length greater than $2\pi d$ on the line of convergence carries a singularity. When $\limsup \log n/\mu_n = 0$ the abscissa of convergence equals $\limsup_n \log|c_n|/\mu_n$ [196].

THEOREM 4.3.22. *Assume RH and that all zeros are simple, and assume*

(G) $\inf_n(\gamma_{n+1}^2 - \gamma_n^2) > 0$;

(D) $\log\bigl(1/|\zeta'(\frac12 + i\gamma_n)|\bigr) = o(\gamma_n^2)$.

Then for $t > 0$

$$f_X(t) = \sum_n c_n e^{-\gamma_n^2 t}, \qquad c_n = A_n\gamma_n^2 = -\frac{2\gamma_n\,\Xi(0)}{\Xi'(\gamma_n)},$$

the series has abscissa of convergence 0, *and* $\operatorname{Re} t = 0$ *is a natural boundary of* f_X. *In particular the density of* X *is not HCM.*

Proof. Residue expansion. The residue of $\varphi(s) = \Xi(0)/\Xi(i\sqrt{s})$ at $s = -\gamma_n^2$ is c_n as stated. Write f_X as a Bromwich integral and move the contour to the left across the arcs of the circles $|s| = T_k^2$ that lie to the left of the Bromwich line, where, for large k, $T_k \in [k, k+1]$ is chosen at distance at least $c/\log k$ from every ordinate, with a small absolute constant $c > 0$; this is possible because $[k-1, k+2]$ contains $O(\log k)$ ordinates [109]. Then $\zeta'/\zeta(\sigma + iy) = O(\log^2 k)$ uniformly in $-1 \le \sigma \le 2$ and $|y - T_k| \le c/(2\log k)$ [324, Theorem 9.6(A) and Section 14.4], and integration from $\sigma = 2$ gives $1/|\zeta(\sigma + iy)| \le \exp(O(\log^2 k))$ there. On the part of such an arc within angle $O(1/T_k)$ of the negative axis, $w = \frac12 + \sqrt{s}$ has $\frac12 \le \operatorname{Re} w \le 2$ and $|\operatorname{Im} w| = \sqrt{T_k^2 - (\operatorname{Re} w - \frac12)^2} = T_k + O(1/T_k)$, so for large k it lies in this band or in its reflection in the real axis, and Stirling's formula gives $1/|\Xi(i\sqrt{s})| \le \exp(O(T_k \log T_k))$, while $|e^{st}| \le e^{-tT_k^2/2}$. On the rest of the arc $\operatorname{Re} w \ge 2$, and Stirling's formula gives $|e^{st}\varphi(s)| \le \exp\bigl(tT_k^2\cos\theta - \frac{T_k}{2}\cos\frac{\theta}{2}\log T_k + O(T_k)\bigr)$ for $s = T_k^2 e^{i\theta}$. For $t > 0$ both bounds tend to zero faster than any power of T_k, so the arcs contribute nothing in the limit and f_X equals the residue series.

Abscissa. The exponents γ_n^2 satisfy $\log n/\gamma_n^2 \to 0$ and have density zero. By Stirling, $|\Xi'(\gamma_n)| = |\xi'(\rho_n)| = \gamma_n^{O(1)} e^{-\pi\gamma_n/4}|\zeta'(\rho_n)|$, so $\log|c_n| = \frac{\pi}{4}\gamma_n + O(\log\gamma_n) + \log(1/|\zeta'(\rho_n)|)$. Since $|\zeta'(\rho_n)| \le \gamma_n^{O(1)}$ always, (D) gives $\log|c_n|/\gamma_n^2 \to 0$, so the abscissa of convergence is 0.

Natural boundary. By (G) and density zero, Pólya's theorem places a singularity in every segment of the imaginary axis. An HCM density continues analytically to Ω, which contains the imaginary axis minus the origin. □

The same argument, with $\mu_k = (2k-1)^2$, gaps $8k$ and $c_k = \frac{4}{\pi}(-1)^{k-1}(2k-1)$, gives a second proof of Theorem 4.3.20 that does not use modularity: by Pólya's gap theorem [271, 277] the line $\operatorname{Re} t = 0$ is a natural boundary of f_C.

REMARK 4.3.23 (status of the hypotheses). Hypothesis (D) is far weaker than the conjectured polynomial lower bounds for $|\zeta'(\rho)|$. Hypothesis (G) asks $\gamma_{n+1}-\gamma_n \geq c_0/(\gamma_n+\gamma_{n+1})$ for some $c_0 > 0$, a gap of order $1/\gamma$ against a mean spacing of order $1/\log\gamma$. Under the GUE model the expected number of violations is $\sum_n(\log\gamma_n/\gamma_n)^3 < \infty$, so by the Borel Cantelli heuristic (G) fails at most finitely often, and then holds with a smaller c_0 if the zeros are simple.

Numerically, for the first 200 zeros the coefficients c_n computed from $\Xi'(\gamma_n)$ alternate in sign without exception; $\log|c_n|/\gamma_n^2$ equals 0.0462, 0.0144, 0.00524, 0.00322, 0.00195 at $n = 1, 10, 50, 100, 200$, tracking $\pi/(4\gamma_n)$; and the smallest gap $\gamma_{n+1}^2-\gamma_n^2$ is 159, at $n = 4$. The c_n are the residues of the spectral expansion of the clock density in Section 8.1, written R_k there, and Theorem 4.3.22 describes, under its hypotheses, why that expansion cannot be continued past $t = 0$.

For small p the density of X_p escapes this obstruction. The transform $\varphi(s^p)$ is analytic on the wider sector $|\arg s| < (\pi-2\delta_*)/p$ and, by Stirling's formula applied to $\xi(\frac{1}{2}+z)$ for $|\arg z| < \pi/2$, decays along every ray strictly inside it. The inverse Laplace transform then continues the density of X_p to the sector $|\arg t| < (\pi-2\delta_*)/p - \pi/2$, which contains Ω when $p \leq 2p_*/3$.

QUESTION 4.3.24. Determine $\{p:\ X_p \text{ is HCM}\}$, and the same set for the Cauchy family $X_{C,p}$ with transform $\operatorname{sech}(\pi s^{p/2}/2)$.

Two facts constrain the answer. Under RH, $X_p \overset{d}{=} S_p X^{1/p}$, and positive p stable variables are HCM exactly for $p \leq 1/2$, a conjecture of Bondesson [39, Sections 5.7, 5.8 and 7.2] proved by Bosch and Simon [42, p. 32]. Since HCM is preserved under products of independent variables and under powers of modulus at least one ((F1) and (F2) of Section 4.2.1), HCM of X would give HCM of X_p for $p \leq 1/2$; but Theorem 4.3.22 shows that X is not HCM under (G) and (D), so this route is closed unless those hypotheses fail. The elasticity criterion reduces Question 4.3.24 to the absence of zeros of the density f of X_p in Ω and the Pick property of $-tf'(t)/f(t)$; by the continuation argument before Question 4.3.24, f is at least analytic on Ω for small p. A numerical scan of this elasticity for $X_{C,1/2}$ would be the first test.

4.3.7 THE MELLIN VARIABLE W AND THE FIRST GAP

Question 4.2.7 asks for which β the density of W^β is HCM; on the Cauchy side the answer is $|\beta| \geq 2$. Let $\chi(t) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+t) = \prod_\gamma(1+t^2/\gamma^2)^{-1}$ and let $g(y) = \frac{1}{\pi}\int_0^\infty \chi(t)\cos(ty)\,dt$ be the function g of Section 4.2.2, which is entire because $\chi(t) = \exp(-\frac{1}{2}|t|\log|t|(1+o(1)))$. By the van Dantzig pair of Section 1.2, recalled in Section 4.2.2, χ is a characteristic function without RH: the clock variable T exists unconditionally [252, Theorem A], and if Z is a standard normal variable independent of T then $\mathbb{E}e^{itZ\sqrt{2T}} = \mathbb{E}e^{-t^2T} = \chi(t)$. So g is the density of the normal scale mixture $Z\sqrt{2T}$, and $W = \exp(Z\sqrt{2T})$ exists without any

hypothesis. By the elasticity criterion, HCM of W^β becomes a sign condition on $-g'/g$ in a strip (Proposition 4.3.26), and since g is entire no natural boundary stands in the way.

4.3.7.1 A lower bound for g

Theorem 4.3.25 proves the existence of W a second time, independently of [252, Theorem A].

THEOREM 4.3.25. *Unconditionally, by the verification of RH to height T_0 [247], $g(y) \geq 0.9998\, g_R(y) > 0$ for all real y, where g_R is the density of $\sum_{\gamma\ \mathrm{real}} L_\gamma/\gamma$. Hence χ is a characteristic function and there is a positive random variable W with $\mathbb{E}W^{it} = \chi(t)$ and $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1$.*

Proof. Split $\chi = \chi_R \chi_{\mathrm{off}}$ into the products over real zeros and over conjugate pairs of nonreal zeros. For a pair $\gamma = \varrho e^{i\delta} = a + ib$ with $a, b > 0$, the factor $(1 + t^2/\gamma^2)^{-1}(1 + t^2/\bar\gamma^2)^{-1}$ is the Fourier transform of the even kernel

$$k_\gamma(x) = \frac{\varrho}{2\sin 2\delta}\, e^{-a|x|} \sin(b|x| + \delta),$$

of total mass one, positive for $|x| < (\pi - \delta)/b$ and bounded by $\frac{\varrho}{2\sin 2\delta} e^{-a|x|}$. List the pairs as $\gamma^{(1)}, \gamma^{(2)}, \dots$ in order of $\operatorname{Re}\gamma$, write $k_j = k_{\gamma^{(j)}}$, and let χ_m be the product of the first m pair factors, $\nu_m = k_1 * \cdots * k_m$ and $g_m = g_R * \nu_m$, the function with Fourier transform $\chi_R \chi_m$. We bound g_m uniformly in m and let m grow at the end.

Comparison. $g_R = \ell * \tilde g_R$, where $\ell(z) = \frac{\gamma_1}{2} e^{-\gamma_1|z|}$ is the density of L_{γ_1}/γ_1 and $\tilde g_R$ is the density of the sum over the other real zeros, so $e^{-\gamma_1|x|} g_R(y) \leq g_R(y - x) \leq e^{\gamma_1|x|} g_R(y)$. Writing $\nu_m = \nu_m^+ - \nu_m^-$,

$$g_m(y) \geq g_R(y)\Big(\int e^{-\gamma_1|x|} \nu_m(x)dx - 2\int \cosh(\gamma_1 x)\, \nu_m^-(x)dx \Big) =: g_R(y)(A_m - 2B_m).$$

The term A_m. By Parseval, $A_m = \frac{1}{\pi}\int \chi_m(t) \frac{\gamma_1}{\gamma_1^2 + t^2} dt = \mathbb{E}\,\chi_m(\gamma_1 C)$ with C a standard Cauchy variable. For $s \geq 0$ each pair factor is $|1 + s/\gamma^2|^{-2} \geq (1 + s/\varrho^2)^{-2} \geq e^{-2s/\varrho^2}$, so $A_m \geq \mathbb{E}e^{-\kappa C^2} = e^\kappa \operatorname{erfc}(\sqrt\kappa)$ with $\kappa = 2\gamma_1^2 \Sigma_{\mathrm{off}}$ and $\Sigma_{\mathrm{off}} = \sum_{\mathrm{off}} |\gamma|^{-2}$. All nonreal zeros have $\operatorname{Re}\gamma > T_0 = 3 \times 10^{12}$ [247], so with $N(T) \leq T \log T$ for $T \geq T_0$, $\Sigma_{\mathrm{off}} \leq 2\int_{T_0}^\infty N(T) T^{-3} dT \leq 2(\log T_0 + 1)/T_0 < 2 \times 10^{-11}$, $\kappa < 8 \times 10^{-9}$ and $A_m > 0.99989$.

The term B_m. For even signed measures $\int \cosh(\gamma_1 x)\, d(|\mu| * |\nu|) = \int \cosh(\gamma_1 x) d|\mu| \cdot \int \cosh(\gamma_1 x) d|\nu|$ exactly, and $(\mu * \nu)^- \leq \mu^+ * \nu^- + \mu^- * \nu^+$. Hence, with $M(\cdot) = \int \cosh(\gamma_1 x)\, d(\cdot)$ and since $M(|k_\gamma|) \geq M(k_\gamma) = |1 - \gamma_1^2/\gamma^2|^{-2} \geq 1$,

$$B_m \leq \sum_{j \leq m} M(k_j^-) \prod_{i \leq m,\ i \neq j} M(|k_i|) \leq SP, \qquad S = \sum_\gamma M(k_\gamma^-), \quad P = \prod_\gamma M(|k_\gamma|),$$

with the sum and the product over all pairs. Here k_γ^- lives on $|x| \geq (\pi-\delta)/b$ and $M(k_\gamma^-) \leq \frac{\varrho}{\sin 2\delta\,(a-\gamma_1)} \exp\bigl(-(a-\gamma_1)\frac{\pi-\delta}{b}\bigr)$. As $\varrho/\sin 2\delta = \varrho^3/(2ab)$, this bound is $\frac{\varrho^3}{2a(a-\gamma_1)}\, b^{-1}e^{-K/b}$ with $K = (a-\gamma_1)(\pi-\delta)$. For fixed a it increases with b: ϱ increases, K decreases because $\delta = \arctan(b/a)$ increases, and $b \mapsto b^{-1}e^{-K/b}$ increases on $(0, K)$. So, as $b < \frac{1}{2}$, it is at most

$$F(a) = \frac{(a^2+\frac{1}{4})^{3/2}}{a(a-\gamma_1)} \exp\Bigl(-2(a-\gamma_1)\bigl(\pi - \arctan\tfrac{1}{2a}\bigr)\Bigr),$$

and $F(T_0) < \exp(-1.88\times 10^{13})$, while $-(\log F)' > 6$ on $[T_0, \infty)$. Every pair has $a > T_0$, so Stieltjes integration against $N(T) \leq T\log T$ gives

$$S \leq F(T_0)\Bigl(T_0\log T_0 + \frac{\log T_0 + 1}{6} + \frac{1}{36T_0}\Bigr) < \exp(-1.8\times 10^{13}).$$

Moreover $M(|k_\gamma|) = M(k_\gamma) + 2M(k_\gamma^-)$ and $M(k_\gamma) \leq \exp(3\gamma_1^2/\varrho^2)$, so $\log P \leq 3\gamma_1^2\Sigma_{\text{off}} + 2S < 1.2\times 10^{-8}$. So $A_m - 2B_m > 0.9998$ for every m.

Limit. For real t each pair factor lies in $(0,1]$, and χ_R is integrable because it contains the factor $(1+t^2/\gamma_1^2)^{-1}$. By dominated convergence in the Fourier inversion integral, $g_m \to g$ uniformly on $\mathbb{R}$, so $g \geq 0.9998\, g_R$. The comparison also gives $g_m(y) \leq g_R(y)\int e^{\gamma_1|x|}|\nu_m(x)|dx \leq 2P\, g_R(y)$, since $e^{\gamma_1|x|} \leq 2\cosh(\gamma_1 x)$ and $|\nu_m| \leq |k_1| * \cdots * |k_m|$, so $g \leq 2P\, g_R$. Hence g is integrable, its Fourier transform is χ, and $\chi(0) = 1$ makes g a probability density. Finally $\int e^{cy} g_R(y)dy = \prod_{\gamma\ \text{real}}(1 - c^2/\gamma^2)^{-1} < \infty$ for $|c| < \gamma_1$, so $\int e^{c|y|}g(y)dy < \infty$ there, $\mathbb{E}W^z = \int e^{zy}g(y)dy$ is analytic on $|\operatorname{Re} z| < \gamma_1$, and it equals $\Xi(0)/\Xi(z)$ there because the two agree on the imaginary axis. □

The proof uses only that the zeros with $0 < \operatorname{Re}\gamma \leq T_0$ are real. The pair kernel formula and the constants of the proof were also checked numerically (`code/README.md`).

4.3.7.2 Which powers of W are HCM

For real $\beta \neq 0$ the density of W^β is $x^{-1}g(\log x/\beta)/|\beta|$. With $y = \log x$, its elasticity is $1 - \frac{1}{\beta}\frac{g'}{g}(y/\beta)$, and $x \in \mathbb{C}^+$ corresponds to $0 < \operatorname{Im} y < \pi$; for $\beta < 0$ the point $u = y/\beta$ then lies in the lower half of the strip $|\operatorname{Im} u| < \pi/|\beta|$, where the factor $\frac{1}{\beta} < 0$ reverses the sign condition, and $g(\bar u) = \overline{g(u)}$, as g is real on $\mathbb{R}$, moves it to the upper half. Theorem 4.3.2 therefore gives the following.

Proposition 4.3.26. *Let $\beta \in \mathbb{R}\setminus\{0\}$. Then W^β is HCM if and only if g has no zeros in the strip $|\operatorname{Im} u| < \pi/|\beta|$ and $\operatorname{Im}\bigl(-g'(u)/g(u)\bigr) \geq 0$ for $0 < \operatorname{Im} u < \pi/|\beta|$.*

As $\operatorname{Im} u \downarrow 0$ the condition requires $(\log g)'' \leq 0$ on $\mathbb{R}$; numerically $(\log g)'' \leq -2.2$ on $[0,1]$. On the Cauchy side $g_C = 1/(\pi\cosh u)$, $-g_C'/g_C = \tanh u$, which

is Pick on $0 < \operatorname{Im} u < \pi/2$ with a pole at $i\pi/2$; Proposition 4.3.26 gives $|\beta| \geq 2$, Bosch's theorem once more.

THEOREM 4.3.27. *Assume RH, simple zeros, and* $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$. *Then* $u_0 = \frac{\pi}{4} + \limsup_n \gamma_n^{-1} \log(1/|\zeta'(\rho_n)|)$ *is finite, and for* $\operatorname{Re} u > u_0$

$$g(u) = -\Xi(0) \sum_n \frac{e^{-\gamma_n u}}{\Xi'(\gamma_n)} = \sum_n b_n e^{-\gamma_n u}, \qquad b_1 > 0 > b_2,$$

and if W^β *is HCM then* $|\beta| \geq \gamma_2 - \gamma_1 = 6.8873\ldots$.

Proof. The expansion is the residue computation for the Fourier integral $g(u) = \frac{1}{2\pi} \int \chi(t) e^{-itu}\, dt$ closed in the lower half plane. We move the line of integration to $\operatorname{Im} t = -T_k$, where $T_k \in [k, k+1]$ are heights at which $1/\zeta(\sigma \pm iT_k) = T_k^{O(1)}$ uniformly in $-1 \leq \sigma \leq 2$ [324, Theorem 9.7]; by Stirling's formula and the functional equation the moved integral is $O(e^{(\pi/4 - \operatorname{Re} u)T_k + O(\log T_k)})$, which tends to 0 for $\operatorname{Re} u > \pi/4$, so there $g(u)$ is the limit of the sums of the residue terms over $\gamma_n < T_k$. By Stirling, $\log|b_n| = \frac{\pi}{4}\gamma_n + \log(1/|\zeta'(\rho_n)|) + O(\log \gamma_n)$, so the series converges absolutely for $\operatorname{Re} u > u_0$, and $u_0 \geq \pi/4$ because $|\zeta'(\rho_n)| \leq \gamma_n^{O(1)}$, so its sum is $g(u)$ there (conjecturally $u_0 = \pi/4$); the signs come from $b_n = \frac{\gamma_n}{2} A_n$. Put $\Delta = \gamma_2 - \gamma_1$. As $\operatorname{Re} u \to \infty$ with $\operatorname{Im} u = v$ fixed,

$$\operatorname{Im}\Big(-\frac{g'}{g}\Big)(x + iv) = \Delta \Big|\frac{b_2}{b_1}\Big| e^{-\Delta x} \sin(\Delta v)\, \big(1 + O(e^{-cx})\big), \qquad c > 0.$$

For $\pi/\Delta < v < 2\pi/\Delta$ this is negative for large x. If $|\beta| < \Delta$ the strip $0 < \operatorname{Im} u < \pi/|\beta|$ contains such v, and Proposition 4.3.26 fails. □

QUESTION 4.3.28 (first gap conjecture). Is W^β HCM exactly when $|\beta| \geq \gamma_2 - \gamma_1$? On the Cauchy side the analogous statement, with frequencies $2k-1$, is Bosch's theorem at $\alpha = 1/2$: $3 - 1 = 2$.

With b_n computed from $\Xi'(\gamma_n)$ ($b_1 = 359.52$, $b_2 = -28007$, $b_3 = 392918$) and g computed by quadrature near the origin, a scan of 4840 points with $0 \leq x \leq 6$ and $0 < v < 0.99\, v_*$, where $v_* = \pi/\Delta = 0.45614$, found no value of $\operatorname{Im}(-g'/g)$ below -1.3×10^{-15}, which is rounding; in 40 digit arithmetic the minimum over $1.1 \leq x \leq 9.1$ at $v = 0.999\, v_*$ is $+1.0 \times 10^{-27}$. Just above the threshold the sign flips: the minimum is -6.5×10^{-8} at $v = 1.001\, v_*$, so numerically the threshold is sharp to 0.1 per cent.

Proving sufficiency requires the Pick property of $-g'/g$ in the full strip $0 < \operatorname{Im} u < \pi/\Delta$, which near the origin is a statement about the Pólya type kernel g that does not reduce to the zeros.

4.3.7.3 Solvable models

To see whether the first gap rule is a coincidence of the half Cauchy case, we test it on a family in which everything is explicit. The logistic variable $\log W$ has characteristic function $\pi t/\sinh(\pi t) = \prod_{k\geq 1}(1+t^2/k^2)^{-1}$. Removing a finite set R of frequencies multiplies the characteristic function by $\prod_{k\in R}(1+t^2/k^2)$, which gives $g = \prod_{k\in R}(1-D^2/k^2)L$ with $D = d/dy$ and $L(y) = \frac{1}{4}\operatorname{sech}^2(y/2)$. For each model we locate numerically the largest strip on which $\operatorname{Im}(-g'/g) \geq 0$; by Theorem 4.3.2, as in Proposition 4.3.26, its width h_* gives the HCM threshold $\beta_* = \pi/h_*$. Table 4.3 lists the results.

frequencies	gaps	first gap	largest gap	$\beta_* = \pi/h_*$
$1, 2, 3, \ldots$	$1, 1, 1, \ldots$	1	1	1.000
$1, 3, 5, \ldots$ (half Cauchy)	$2, 2, 2, \ldots$	2	2	2 (Bosch)
$1, 3, 4, 5, \ldots$	$2, 1, 1, \ldots$	2	2	2.000
$1, 3, 5, 6, \ldots$	$2, 2, 1, \ldots$	2	2	2.000
$1, 3, 4, 6, \ldots$	$2, 1, 2, 1, \ldots$	2	2	2.000
$1, 4, 5, 6, \ldots$	$3, 1, 1, \ldots$	3	3	3.000
$1, 4, 5, 7, \ldots$	$3, 1, 2, 1, \ldots$	3	3	3.000
$1, 4, 7, 8, \ldots$	$3, 3, 1, \ldots$	3	3	3.000
$1, 5, 6, 7, \ldots$	$4, 1, 1, \ldots$	4	4	4.000
$1, 2, 4, 5, \ldots$	$1, 2, 1, \ldots$	1	2	1.269
$1, 2, 3, 6, \ldots$	$1, 1, 3, 1, \ldots$	1	3	1.317
$1, 2, 5, 6, \ldots$	$1, 3, 1, \ldots$	1	3	1.483

Table 4.3: HCM threshold for powers of W in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width h_* of the largest strip on which $\operatorname{Im}(-g'/g) \geq 0$. The half Cauchy row is Bosch's theorem, not a computation.

Whenever the first gap is at least as large as every later gap, the computed threshold equals the first gap, to the resolution of the computation. When some later gap exceeds the first, the threshold lies strictly between the first gap and the largest gap, and depends on where the large gap sits. The asymptotic argument of Theorem 4.3.27 explains the lower bound in all cases. For the Riemann zeros the first gap $\gamma_2 - \gamma_1 = 6.8873$ exceeds every later gap among the first 1000 zeros (the next largest is $\gamma_4 - \gamma_3 = 5.414$). The explicit form of the Riemann von Mangoldt formula of Hasanalizade, Shen and Wong, $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac{7}{8}| \leq 2.5167 + \frac{1}{50T}$ for $e \leq T \leq X_0 = 30\,610\,046\,000$ and $\leq 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}$ for $T > X_0$ [109, (1.7), (2.7) and Corollary 1.4], shows that every interval $(T, T+6.887]$ with $T \geq 1000$ contains an ordinate: the lower bound it gives for the number of ordinates there exceeds 0.52 at $T = 1000$, increases with T while $T + 6.887 \leq X_0$, and exceeds 11 beyond. So every gap that starts above height 1000 is at most 6.887, less than the first, and every gap that starts below it is a gap between two

of the first 650 zeros, since $N(1000) = 649$. The first gap therefore exceeds every later gap. This refines Question 4.3.28.

Question 4.3.29 (first gap conjecture, refined). Let $Y = \sum_n L_n/\lambda_n$ with independent standard Laplace variables L_n and $0 < \lambda_1 < \lambda_2 < \cdots$, and let $W = e^Y$. If the characteristic function of Y decays faster than every exponential, or if the first gap $\lambda_2 - \lambda_1$ is at least every later gap, is the HCM threshold of W^β exactly $\lambda_2 - \lambda_1$? In particular, is it $\gamma_2 - \gamma_1$ for the Riemann variable?

4.3.8 Where the difficulty lies

Bosch's theorem cannot be the input to a proof that the reciprocal Xi function is HCM, because for a product over the zeros HCM is equivalent to the reality of the zeros (Proposition 4.2.6), and Section 11.1 finds that every GGC criterion in the books of Sato and Bondesson is, at the centre, equivalent to RH. Bosch's method does transfer: at each of three levels it isolates one concrete object on which the question turns.

At the level of transforms the object is the Thorin density u_p of (4.3.2). Its positivity for every $p < 1$ is equivalent to RH; the primes prove it on each ray beyond some point, that point is asymptotically where the ray leaves the critical strip, and what remains is a segment of length about $1/(\pi(1-p))$. The first direction is to prove positivity on that segment from the explicit formula for ξ'/ξ rather than from the zeros; computation alone extends the unconditional range $p \le 1 - 1/(\pi T_0)$ only as fast as the verification height T_0 grows.

At the level of the Mellin variable the object is the sign of $\operatorname{Im}(-g'/g)$ in the strip $0 < \operatorname{Im} u < \pi/(\gamma_2 - \gamma_1)$. The second direction is to prove the first gap conjecture, first in the logistic models of Table 4.3, where every quantity is elementary, and then for W, whose threshold the scan after Question 4.3.28 already locates to 0.1 per cent. Question 4.2.8 remains open.

At the level of densities the object is the natural boundary. The third direction is to establish it for the clock variable without the hypotheses (G) and (D) of Theorem 4.3.22, which would show what replaces, on the Riemann side, the modularity that produces the natural boundary on the Cauchy side.

Section 5.1 approaches the Thorin functional $\sum_\gamma \delta_{\gamma^2}$ of the clock, its Thorin measure under RH, from the other end of Bondesson's hierarchy, through its Laplace transform, the heat trace $W(t) = \sum_\gamma e^{-\gamma^2 t}$.

4.4 Exercises

On Section 4.1.

4.1. Show that if X_N are GGCs converging in law to a proper limit, the Thorin measures converge vaguely on $(0, \infty)$, and use the laws Gamma(n, n) of

Section 1.3.1, which converge to the point mass at one, to show that Thorin mass escaping to infinity can reappear in the limit as drift.

4.2. For the sinh clock of Section 1.3.4 with one atom k^2 replaced by the conjugate pair $a, \bar{a}$, $a = -(\delta + ik)^2$ with $0 < \delta < \frac{1}{2}$, as in Proposition 1.2.3, construct the Thorin densities at resolution ε and show that their vague limit is the sinh measure with the atom k^2 removed.

4.3. Verify that $\varphi_{1/2}(uv)\varphi_{1/2}(u/v) = \prod_k a_k^2/(a_k^2 + a_k uw + u^2)$ for the clock, and that at $u = |a|$ an off line pair contributes the completely monotone factor $(w + 2\cos\arg a)^{-2}$.

4.4. Compute $(-1)^k \partial_w^k H_u(w)/H_u(w)$ for $k \le 6$ at $u = 1$ and $w = 3$ from the Pólya density and compare with Figure 4.1.

Chapter 5

The heat trace, the arithmetic and the limits of finite information

Chapter 4 worked at the top of the Bondesson hierarchy; here we climb it from the bottom. Every rung is a sign condition on the heat trace $W(t)$ of Section 1.2, the pairing of the Thorin functional with e^{-tz} (not the Mellin variable W of Section 4.3). The lower rungs are unconditional theorems, proved with a finite verification of the zeros; the top rung is RH.

5.1 The heat trace and the hierarchy of positivity

The rungs of the hierarchy proved unconditionally are infinite divisibility and self decomposability of the clock, log convexity of W, positive definite Hankel blocks up to size 3×10^7, the signs of all derivatives, equivalently the positivity of the Williamson measures, up to order 1.16×10^{25}, and near the origin the signs of every derivative: W is completely monotone on $(0, 1.29]$. One scheme proves them all, with a reach in size and order set by the height to which RH has been verified; no verification reaches the top, which is where Section 5.2 begins. The results are from [252, Chapters 4 and 5] and [249], except those stated without a citation, which are proved here.

5.1.1 The heat trace and the first rungs

By Section 1.2, unconditionally,

$$\log G(u) = \int_0^\infty \bigl(1 - e^{-ut}\bigr) \frac{W(t)}{t}\, dt, \qquad W(t) = \sum_j e^{-a_j t},$$

where $a_j = -(\rho_j - \frac{1}{2})^2$ and the sum runs over one zero of each pair $\{\rho, 1-\rho\}$. So $W(t)/t$ is the only candidate for the Lévy density of the clock, and the classes of Section 1.3 become conditions on W: nonnegativity for infinite divisibility, monotonicity for self decomposability, complete monotonicity for the GGC property.

The function W is analytic in $\operatorname{Re} t > 0$, real on the positive axis, and can be differentiated term by term,

$$c_k(t) = (-1)^k W^{(k)}(t) = \sum_j a_j^k e^{-a_j t}.$$

Under RH every $c_k(t)$ is a sum of positive terms. A quadruple of zeros $\frac12 \pm \delta \pm i\gamma$, $0 < \delta < \frac12$, instead contributes the exponents $a = -(\delta + i\gamma)^2 = \gamma^2 - \delta^2 - 2i\delta\gamma$ and $\bar a$, hence the oscillating term $2|a|^k e^{-\operatorname{Re} a\, t}\cos(2\delta\gamma t - k\phi)$, with $|a| = \gamma^2 + \delta^2$ and $\tan\phi = 2\delta\gamma/(\gamma^2 - \delta^2)$. Its amplitude exceeds the contribution $2\gamma^{2k}e^{-\gamma^2 t}$ of the same two zeros placed at $\frac12 + i\gamma$ by the factor $(1+\delta^2/\gamma^2)^k e^{\delta^2 t} \le \exp\{k\delta^2/\gamma^2 + \delta^2 t\}$ [249, Proposition 11.1].

As $t \downarrow 0$ the archimedean term of (1.2.5) dominates. Write ϑ for the Riemann Siegel theta function. Since $\operatorname{Re}\psi(\frac14 + \frac{iu}{2}) - \log\pi = 2\vartheta'(u)$, the explicit formula gives, with no hypothesis,

$$W(t) = e^{t/4} + \frac{1}{\pi}\int_0^\infty e^{-u^2 t}\vartheta'(u)\,du - \frac{1}{\sqrt{4\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt n}\,e^{-(\log n)^2/4t},$$

and the prime terms are $O(t^{-1/2}e^{-(\log 2)^2/4t})$, smaller than every power of t. Mellin's method for the integral [225, Section 2.5] gives, for every N and as $t \downarrow 0$,

$$W(t) = \frac{\log\frac1t - \gamma_E - 4\log 2 - 2\log\pi}{8\sqrt{\pi t}} + \sum_{n=0}^{N} d_n t^n + \sum_{k=1}^{N} b_k t^{k-\frac12} + O\big(t^{N+\frac12}\big),$$

with $d_n = (1 - \frac{E_{2n}}{8})\frac{4^{-n}}{n!}$ and $b_k = \frac{(1-2k)\Gamma(\frac12 - k)}{2\pi}\beta_k$, where γ_E is Euler's constant, E_{2n} are the Euler numbers and $\beta_k = \frac{(1-2^{1-2k})|B_{2k}|}{4k(2k-1)}$ are the coefficients of Stirling's series $\vartheta(u) \sim \frac{u}{2}\log\frac{u}{2\pi} - \frac{u}{2} - \frac{\pi}{8} + \sum_k \beta_k u^{1-2k}$. The terms $b_k t^{k-\frac12}$ come from the expansion of ϑ' at infinity, and the terms $d_n t^n$ from the Taylor series of $e^{-u^2 t}$ through the Hurwitz value $\zeta(-2n, \frac14) = E_{2n}/4^{2n+1}$, together with the coefficients $4^{-n}/n!$ of $e^{t/4}$. After the leading term the series is $\frac78 + \frac{\sqrt t}{48\sqrt\pi} + \frac{9}{32}t - \frac{7t^{3/2}}{2880\sqrt\pi} + \frac{3}{256}t^2 + \frac{31t^{5/2}}{60480\sqrt\pi} + \frac{69}{3072}t^3 + \cdots$, and it diverges; $\frac78 = d_0$ is the constant of the Riemann von Mangoldt formula. The error of the two term expansion (the leading term and $\frac78$) is $b_1\sqrt t + d_1 t + O(t^{3/2})$, which is 6.5×10^{-4}, 1.5×10^{-4} and 4.0×10^{-5} at $t = 10^{-3}, 10^{-4}, 10^{-5}$ [249, Corollary 5.5]; with the terms through $t^{5/2}$ the remainder agrees with $\frac{69}{3072}t^3$ to the order of the next term, $t^{7/2}$, in a 40 digit computation for $10^{-6} \le t \le 10^{-3}$, and at $t = 10^{-2}$ it is -8.377×10^{-6}, the prime 2. Consequently $\int_\tau^\infty W(t)\,dt/t = \frac{\log\frac1\tau - \gamma_E - 4\log 2 - 2\log\pi - 2}{4\sqrt{\pi\tau}} + \frac78\log\frac1\tau + O(1)$ as $\tau \downarrow 0$, so the clock has infinitely many jumps. As $t \to \infty$ the first zero dominates, $W(t) = e^{-\gamma_1^2 t}(1 + o(1))$ with $\gamma_1^2 = 199.7904548$. In between, the arithmetic terms cancel to a depth that grows linearly in t (Table 1.4).

THEOREM 5.1.1 ([252, Theorem A]). *Unconditionally, that is, using only the verification of RH below height* 1000 *[247],* $W(t) > 0$ *and* $W'(t) < 0$ *for every* $t > 0$. *Consequently the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density* $W(t)/t$, *and* $(\Xi(\theta)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta))$ *is a van Dantzig pair in the sense of Section 1.2.*

Proof. The proof is that of [252, Theorem A], with the constants made explicit. Let $\tau = 1/20000$, a constant of this section only, $\ell = -\frac{1}{2}\log t$, and $s_k(t) = \Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}/(4\pi)$, which is $(-1)^k$ times the kth derivative of $s_0(t) = 1/(4\sqrt{\pi t})$. It suffices to show that c_0 and c_1 are positive, and we do so first for $t \le \tau$, from (1.2.5), and then for $t \ge \tau$, from the zeros.

Let A and P be the archimedean and prime terms of (1.2.5). The series $\psi(z) = -\gamma_E + \sum_{n\ge0}\big(\frac{1}{n+1} - \frac{1}{n+z}\big)$ [225, (5.7.6)] shows that m increases on $[0,\infty)$, from $m(0) = \psi(\frac{1}{4}) - \log\pi = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi = -5.372$. Binet's formula $\psi(z) = \log z - \frac{1}{2z} - 2\int_0^\infty v\,dv/\big((v^2+z^2)(e^{2\pi v}-1)\big)$ [225, (5.9.15)] at $z = \frac{1}{4} + \frac{ir}{2}$, with the integral split at $v = r/4$, gives $|m(r) - \log\frac{r}{2\pi}| \le \frac{9}{8r^2} + 10^{-10} < 0.003$ for $r \ge 20$. Hence $|m| \le 6$ on $[0,20]$ and $\int_0^{20}|m(r) - \log\frac{r}{2\pi}|\,dr \le 120 + 4\pi + 20\log\frac{10}{\pi} - 20 < 136$. Since $\frac{1}{2\pi}\int_0^\infty r^{2k}e^{-r^2t}\log\frac{r}{2\pi}\,dr = s_k(t)(\ell+\alpha_k)$, with $\alpha_0 = -\frac{1}{2}\gamma_E - \log 4\pi = -2.8196$ and $\alpha_1 = \alpha_0 + 1$, it follows that

$$\Big|\frac{(-1)^k A^{(k)}(t)}{s_k(t)} - \ell - \alpha_k\Big| \le \frac{272\cdot 20^{2k}\,t^{k+\frac{1}{2}}}{\Gamma(k+\frac{1}{2})} + 0.003 \qquad (k = 0, 1),$$

and the right side, which increases with t, is below 1.089 for $k = 0$ and 0.047 for $k = 1$ at $t = \tau$. For the prime term put $y_n = (\log n)^2/4t$ and $q = (\log 2)/8t$. Both e^{-y_n} and $|y_n - \frac{1}{2}|e^{-y_n}$ are at most $e^{-y_n/2}$, which is at most n^{-q} because $(\log n)^2 \ge (\log 2)\log n$, and $\Lambda(n) \le \log n < \sqrt{n}$; so $|P^{(k)}(t)| \le (2\sqrt{\pi})^{-1}t^{-k-\frac{1}{2}}\sum_{n\ge2} n^{-q} < t^{-k-\frac{1}{2}}2^{-q}$, that is $|P^{(k)}(t)| < t^{-k-\frac{1}{2}}e^{-\kappa/t}$ for $k = 0, 1$ and $t \le \tau$, where $\kappa = (\log 2)^2/8$ is another constant of this section. Since $\kappa/\tau > 1201$, this is below $10^{-500}s_k(t)$. The polar term adds $e^{t/4}$ to c_0 and subtracts $\frac{1}{4}e^{t/4} < 10^{-5}s_1(t)$ from c_1. So for $t \le \tau$

$$c_0(t) \ge s_0(t)\,(\ell - 3.92), \qquad c_1(t) \ge s_1(t)\,(\ell - 1.87),$$

and both are positive, because $\ell \ge \frac{1}{2}\log 20000 > 4.95$.

For $t \ge \tau$ we use the zeros. Those below height 1000 lie on the line [247] and contribute $\gamma^{2k}e^{-\gamma^2 t} > 0$ to c_k, and by the explicit bound for $N(T)$ of [109] one of them lies in $(20, 50]$ and contributes at least $400^k e^{-2500t}$. A zero above height 1000, wherever it lies, has $\operatorname{Re} a \ge \gamma^2 - \frac{1}{4}$ and $|a| \le 2\gamma^2$. The same bound gives $N(r) \le r^2$ for $r \ge 1000$, and integration by parts against dN bounds the total modulus of the contributions of these zeros to c_0 and c_1 by

$$H_0(t) = e^{-(10^6-\frac{1}{4})t}\big(10^6 + t^{-1}\big), \qquad H_1(t) = 2e^{-(10^6-\frac{1}{4})t}\big(10^{12} + 2\cdot 10^6 t^{-1} + 2t^{-2}\big).$$

At $t = \tau$ these are below 2×10^{-16} and 4.1×10^{-10}, while $e^{-2500\tau} > 0.88$, and $e^{2500t} H_k(t)$ decreases with t. So $c_k(t) \geq 400^k e^{-2500t} - H_k(t) > 0$ for $t \geq \tau$ and $k = 0, 1$.

Thus $W > 0$ and $W' < 0$ on $(0, \infty)$. Since $0 < W(t) \leq \sum_j e^{-t \operatorname{Re} a_j}$, the measure $W(t)\, dt/t$ satisfies $\int_0^\infty (1 \wedge t)\, W(t)\, dt/t \leq \sum_j 1/\operatorname{Re} a_j < \infty$, so it is a Lévy measure, and the formula for $\log G$ above makes the clock infinitely divisible with this Lévy measure. A law on the half line whose Lévy density is $W(t)/t$ with W nonincreasing is self decomposable [297, Corollary 15.11]. Finally, $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta) = \mathbb{E} e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T, and with the characteristic function $\Xi(\theta)/\Xi(0)$ of the variable X of Section 1.2 it forms a van Dantzig pair. The program `idsd_check.py` (`code/README.md`) evaluates every numerical inequality of this proof in ball arithmetic. □

Tilting the clock by e^{-T} gives $\xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+s}) = G(1)/G(1+s) = \mathbb{E} e^{-sT^{(1)}}$, with $T^{(1)} \geq 0$ infinitely divisible, so unconditionally $h_k = \xi(\frac{3}{2})/\xi(\frac{1}{2} + \sqrt{1+k}) = \mathbb{E}(e^{-T^{(1)}})^k$ is a Hausdorff moment sequence.

The proof needs far fewer zeros than those below height 1000. With the same bounds it goes through with $\tau = 1/8300$ and the zeros below height 400. If the archimedean error is bounded by computation instead, from $\int_0^5 r^{2k} |m(r) - \log \frac{r}{2\pi}|\, dr < 0.836$ and 0.404 for $k = 0, 1$ and $r^2 |m(r) - \log \frac{r}{2\pi}| < 0.05$ for $r \geq 5$, it goes through with $\tau = 1/500$ and height 100. If in addition the polar term $e^{t/4}$ is kept in c_0, the prime terms with $n \leq 30$ are taken exactly, and the lower bound for $t \geq \tau$ comes from the first zero, in $(14.1347, 14.1348)$, rather than from a zero in $(20, 50]$, it goes through with $\tau = 1/50$ and height 25, below which there are only two zeros. The program `idsd_height.py` evaluates every numerical inequality of the three versions in ball arithmetic.

The same scheme, which we call the two range method, gives more. With the prime term bounded by $100\, t^{-k-\frac{1}{2}} e^{-\kappa/t}$ for $k \leq 3$ and a second zero in $(100, 200]$, it shows that W and $-W'$ are strictly log convex [249, Theorem 7.3], that is $c_0 c_2 > c_1^2$ and $c_1 c_3 > c_2^2$ for every $t > 0$, and that $(-1)^k W^{(k)} > 0$ for $k \leq 3$.

For the first two rungs the explicit formula can be avoided altogether, because where the size of an off line quadruple cannot be bounded its phase fixes its sign, as the first author's note [261] observes. Suppose that the zeros with ordinate in $(0, H]$ lie on the line, with $H \geq 100$, and let $0 \leq k \leq H/8$. A quadruple $\frac{1}{2} \pm \delta \pm i\gamma$ then has $\gamma > H$, and since $0 < \phi < 2\delta/\gamma < 1/\gamma$ its phase satisfies $|2\delta\gamma t - k\phi| < \gamma t + 1$, so its term in $c_k(t)$ is positive unless $\gamma t > \frac{\pi}{2} - 1 > 0.57$. When $\gamma t > 0.57$ the term is at most $2(\gamma^2 + \frac{1}{4})^k e^{-(\gamma^2 - \frac{1}{4})t}$ in modulus, and since $(\gamma^2 + \frac{1}{4})/\gamma_1^2 \leq \gamma^2/99.9$, $\gamma^2 - \frac{1}{4} - \gamma_1^2 \geq \gamma^2/2$ and $\gamma^2 t/2 > 0.285\gamma$, it is at most $2(\gamma^2/99.9)^k e^{-0.285\gamma}$ times the term $\gamma_1^{2k} e^{-\gamma_1^2 t}$ of the first zero. A quadruple has two

zeros of ordinate γ, so at every $t > 0$

$$c_k(t) \geq \big(1 - S_k(H)\big)\gamma_1^{2k} e^{-\gamma_1^2 t}, \qquad S_k(H) = \sum_{\gamma > H} \Big(\frac{\gamma^2}{99.9}\Big)^k e^{-0.285\gamma},$$

the sum running over the zeros above height H, with multiplicity. The explicit bound for $N(T)$ of [109] gives $N(T+1) - N(T) < 2\log T$ for $T \geq 100$, and summing over unit intervals gives $S_0(100) < 10^{-10.5}$ and $S_1(100) < 10^{-8.5}$. Since Hutchinson placed the zeros below height 300 on the line in 1925 [118], this proves Theorem 5.1.1 without the explicit formula. For larger k the factor $(\gamma^2/99.9)^k$ must be absorbed by $e^{-0.285\gamma}$ at $\gamma = H$, so the reachable order, about $0.285H/\log(H^2/99.9)$, stays below the height: the verification below 3×10^{12} [247] gives every $k \leq 10^{10}$, with $S_k(H) < 10^{-1.4\times 10^{11}}$. The two range method reaches orders proportional to H^2 (Theorems 5.1.6 and 5.1.13). The program `ladder_bound.py` evaluates these bounds in ball arithmetic.

Positivity of W also limits how the clock factorises. By Mase's theorem [200, Theorems 1 and 2], an infinitely divisible law without Gaussian component whose Lévy measure has a density that is positive on the whole half line, or that has mass on both half lines, has an indecomposable factor, hence a factor that is not infinitely divisible, so it lies outside Linnik's class I_0 [184] of the infinitely divisible laws all of whose factors are infinitely divisible. Since $W > 0$, this applies unconditionally to the clock variable T, to $Z\sqrt{2T}$ and to every nondegenerate GGC [249, Proposition 12.1].

Proposition 5.1.2 (Truncation error). *Let the zeros be simple, so that the a_j are distinct, and let r_k and $r_k^{(K)}$ be the residues at $u = -a_k$, $k \leq K$, of $1/G(u) = \prod_j a_j/(a_j + u)$ and of its product over $j \leq K$. Then $r_k = a_k \prod_{j \neq k} a_j/(a_j - a_k)$ and, when $|a_j| > |a_k|$ for $j > K$,*

$$\log \frac{r_k^{(K)}}{r_k} = \sum_{j>K} \log\Big(1 - \frac{a_k}{a_j}\Big) = -a_k \sum_{j>K} a_j^{-1} - \frac{a_k^2}{2} \sum_{j>K} a_j^{-2} - \cdots .$$

Under RH $a_j = \gamma_j^2$ and $\sum_{j>K} \gamma_j^{-2} \sim \log\gamma_K/(2\pi\gamma_K)$, so the first term is $-\gamma_k^2 \log\gamma_K/(2\pi\gamma_K)$: the truncated coefficient is close to the residue for $\gamma_k \leq (\gamma_K/\log\gamma_K)^{1/2}$, and not beyond.

Proof. The product converges absolutely, and its pole at $-a_k$ is simple, with residue a_k times the other factors at $u = -a_k$. The ratio of the two residues is the tail product $\prod_{j>K}(a_j - a_k)/a_j$. For the sum, $\sum_{j>K} \gamma_j^{-2} = \int_{\gamma_K}^{\infty} T^{-2}\, dN(T)$ with $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, which is $(\log(\gamma_K/2\pi) + 1)/(2\pi\gamma_K)$ to leading order. □

5.1.2 Hankel blocks and derivative signs

Fix $t > 0$ and form the Hankel matrices $[c_{i+j}(t)]$ and $[c_{i+j+1}(t)]$, $i, j \geq 0$.

Theorem 5.1.3 ([249, Theorem 7.1]). *For any fixed $t > 0$ the following are equivalent: both infinite Hankel matrices are positive semidefinite; Weil's quadratic form is positive semidefinite on the span of $r^m e^{-r^2 t/2}$, $m = 0, 1, 2, \ldots$; RH.*

Under RH the derivatives at one time are the moments of the positive measure $e^{-tz}U(dz)$ on the half line, so both matrices are positive semidefinite. Conversely, if both Hankel matrices are positive semidefinite, Stieltjes' theorem makes the $c_k(t)$ the moments of a positive measure μ on the half line, and since $\sum_k c_k(t)(-s)^k/k! = W(t+s)$ for $|s| < t$, $W(t+s) = \int e^{-sx}\mu(dx)$ there. Both sides are analytic in $\operatorname{Re} s > -t$, the right side because $\int e^{|s|x}\mu(dx) = \sum_k c_k(t)|s|^k/k! = W(t-|s|) < \infty$ for $|s| < t$, so the identity theorem extends the equality to $\operatorname{Re} s > -t$, and W is completely monotone, which is RH. Through the explicit formula, Weil's form on the span of $r^m e^{-r^2 t/2}$ gives the same two matrices, even m the first and odd m the second.

Theorem 5.1.3 tests RH on the real axis. The next one tests it on the vertical line, where W is a sum of oscillating terms.

Theorem 5.1.4. *RH holds if and only if $v \mapsto W(\sigma + iv)$ is bounded on $\mathbb{R}$, for one, equivalently every, $\sigma > 0$.*

Proof. Write $s_j = -a_j = (\rho_j - \frac{1}{2})^2 = x_j + iy_j$, $\rho_j = \frac{1}{2} + \delta_j + i\gamma_j$, so that $x_j = \delta_j^2 - \gamma_j^2$, $y_j = 2\delta_j\gamma_j$, $|\delta_j| < \frac{1}{2}$, $|y_j| \leq |\gamma_j|$ and $\gamma_j \neq 0$. Only three facts on the zeros are used: $|\delta_j| < \frac{1}{2}$, $\gamma_j \neq 0$ (ξ has no zeros in $(0,1)$) and $N(T) = O(T \log T)$, so that $\sum_j e^{-c\gamma_j^2} < \infty$ for every $c > 0$ and only finitely many $|\gamma_j|$ lie below any height. Put $F(v) = W(\sigma + iv) = \sum_j c_j e^{is_j v}$, $c_j = e^{s_j \sigma}$, $C = \sum_j |c_j| \leq e^{\sigma/4} \sum_j e^{-\sigma\gamma_j^2} < \infty$. Since $|e^{is_j v}| = e^{-y_j v} \leq e^{|\gamma_j v|}$, the series converges absolutely and locally uniformly in v.

Under RH every $s_j = -\gamma_j^2$ is real, so $|F(v)| \leq \sum_j e^{-\sigma\gamma_j^2} = W(\sigma)$ for every $\sigma > 0$. Conversely let $|F| \leq M$ on $\mathbb{R}$ for one σ, and suppose RH fails. For $\omega \in \mathbb{R}$ and complex ε with $\operatorname{Re}\varepsilon > 0$ put $L(\varepsilon) = \int_{\mathbb{R}} F(v) e^{-\varepsilon v^2 - i\omega v}\, dv$.

Step 1. On $\operatorname{Re}\varepsilon \geq \eta > 0$ the integrand is entire in ε and bounded by $Me^{-\eta v^2}$, so L is continuous by dominated convergence and holomorphic by Morera's theorem, with Fubini over triangles justified by the same bound. For real $\varepsilon > 0$, $|L(\varepsilon)| \leq M\sqrt{\pi/\varepsilon}$.

Step 2. For $\varepsilon > 0$ and $b \in \mathbb{C}$, $\int_{\mathbb{R}} e^{-\varepsilon v^2 + bv}\, dv = \sqrt{\pi/\varepsilon}\, e^{b^2/4\varepsilon}$: both sides are entire in b and agree for real b by completing the square. With $b = i(s_j - \omega)$ this gives $\int_{\mathbb{R}} e^{is_j v - \varepsilon v^2 - i\omega v}\, dv = \sqrt{\pi/\varepsilon}\, e^{-(s_j-\omega)^2/4\varepsilon}$. The sum over j may be taken under the integral (Fubini and Tonelli) if $\sum_j |c_j| \int_{\mathbb{R}} e^{-y_j v - \varepsilon v^2}\, dv = \sqrt{\pi/\varepsilon} \sum_j e^{\sigma\delta_j^2 - \sigma\gamma_j^2 + \delta_j^2\gamma_j^2/\varepsilon} < \infty$. For real $\varepsilon \geq 1/(2\sigma)$ we

have $\delta_j^2/\varepsilon \le \sigma/2$, each term is at most $e^{\sigma/4}e^{-\sigma\gamma_j^2/2}$, and the sum is finite. Hence $L(\varepsilon) = \sqrt{\pi/\varepsilon}\,\Psi_\varepsilon(\omega)$ for $\varepsilon \ge 1/(2\sigma)$, where $\Psi_\varepsilon(\omega) = \sum_j c_j e^{-(s_j-\omega)^2/4\varepsilon}$ and the root is principal.

Step 3. The series $\Psi_\varepsilon(\omega)$ converges locally uniformly on $\operatorname{Re}\varepsilon > 0$. Let K be compact there. Then $u = 1/(4\varepsilon)$ stays in a compact set with $\operatorname{Re} u \ge u_0 > 0$ and $|\operatorname{Im} u| \le u_1$. Write $(s_j-\omega)^2 = X + iY$ with $X = (x_j-\omega)^2 - y_j^2$ and $Y = 2(x_j-\omega)y_j$. For $\gamma_j^2 \ge \frac14 + |\omega|$ we have $|x_j-\omega| \ge \gamma_j^2 - \frac14 - |\omega|$, so $X \ge (\gamma_j^2 - \frac14 - |\omega|)^2 - \gamma_j^2$ and $|Y| \le 2|\gamma_j|(\gamma_j^2 + \frac14 + |\omega|)$, of orders γ_j^4 and $|\gamma_j|^3$. Hence there is $G = G(K,\omega)$ such that $X > 0$ and $u_0X - u_1|Y| \ge 0$ for $|\gamma_j| \ge G$, and then $|e^{-(X+iY)u}| = e^{-(X\operatorname{Re}u - Y\operatorname{Im}u)} \le e^{-(u_0X-u_1|Y|)} \le 1$, so the jth term is at most $|c_j|$. Only finitely many j have $|\gamma_j| < G$, and $\sum_j |c_j| = C$, so the M-test applies and Ψ is holomorphic on $\operatorname{Re}\varepsilon > 0$. Now L and $\sqrt{\pi/\varepsilon}\,\Psi_\varepsilon(\omega)$ are holomorphic on the connected half plane and agree on $[1/(2\sigma),\infty)$, so by the identity theorem they agree throughout. With Step 1,

$$|\Psi_\varepsilon(\omega)| \le M \qquad (\varepsilon > 0,\ \omega \in \mathbb{R}). \tag{$*$}$$

Step 4. Put $t = 1/(4\varepsilon)$ and $q_j(\omega) = y_j^2 - (x_j-\omega)^2$. Then $|e^{-(s_j-\omega)^2 t}| = e^{q_j t}$ and its phase is $e^{-i\phi_j t}$, $\phi_j = 2(x_j-\omega)y_j$. For fixed ω and level l only finitely many j have $q_j(\omega) \ge l$, because $q_j \le \gamma_j^2 - (\gamma_j^2 - \frac14 - |\omega|)^2 \to -\infty$ as $|\gamma_j| \to \infty$.

Step 5. Take a zero off the line, with $s_0 = x_0 + iy_0$, $y_0 = 2\delta\gamma \ne 0$. On the open interval $I_0 = (x_0 - |y_0|, x_0 + |y_0|)$, $q_0 > 0$. For two values $s \ne s'$ in $\{s_j\}$ with $(x,|y|) \ne (x',|y'|)$, the difference $q - q' = y^2 - y'^2 - x^2 + x'^2 + 2\omega(x-x')$ is a nonconstant affine function of ω if $x \ne x'$ and a nonzero constant if $x = x'$, so it vanishes for at most one ω. Let E be the countable set of these points together with $\{x_j\}$, and fix $\omega \in I_0 \setminus E$. By Step 4, $Q = \max_j q_j(\omega)$ is attained, $Q \ge q_0 > 0$, and by the choice of ω every maximiser has the same $(x,|y|)$, so $s_j \in \{s^*, \bar s^*\}$ for a fixed $s^* = x^* + iy^*$. Now $Q > 0$ forces $y^* \ne 0$, and $\omega \ne x^*$ forces $\phi := 2(x^*-\omega)y^* \ne 0$. Only finitely many j have $q_j \ge 0$, so $Q' = \max\{0,\, q_j(\omega) : s_j \notin \{s^*, \bar s^*\}\}$ is a maximum over a finite set and $Q' < Q$.

Step 6. Let $m_1 \ge 1$ and $m_2 \ge 0$ be the numbers of j with $s_j = s^*$ and $s_j = \bar s^*$, and $A = e^{s^*\sigma} \ne 0$. The terms with $\bar s^*$ have the same q, the phase $e^{+i\phi t}$ and the coefficient $\bar A$. Hence $\Psi_\varepsilon(\omega) = e^{Qt}g(t) + r(t)$ with $g(t) = m_1 A e^{-i\phi t} + m_2 \bar A e^{i\phi t}$ and $|r(t)| \le \sum_j |c_j| e^{Q't} = Ce^{Q't}$ for $t > 0$. The function g is continuous and periodic with period $2\pi/|\phi|$, and not identically zero, since its Fourier coefficient at the frequency $-\phi \ne \phi$ is $m_1 A \ne 0$. So $\kappa = \max|g| > 0$ is attained in every period, at points $t_k \to \infty$, and with $\varepsilon_k = 1/(4t_k)$, using $Q > 0$ for the growth and $Q > Q'$ for the remainder,

$$|\Psi_{\varepsilon_k}(\omega)| \ge \kappa e^{Qt_k} - Ce^{Q't_k} \to \infty,$$

contradicting $(*)$. So RH holds. Since boundedness for one σ gives RH, and RH gives it for every σ, the two readings agree. □

THEOREM 5.1.5 ([249, Theorem 7.4]). *If all zeros below* $T_N = 20000N\sqrt{1+\log N}$ *are on the line, the leading* $N \times N$ *blocks are positive definite at every* $t > 0$. *The verification to height* 3×10^{12} *[247] gives this for every* $N \le 3 \times 10^7$.

Table 5.1 shows normalised leading minors recomputed in sixty digit arithmetic from the first 1700 zeros, whose ordinates are stored in double precision; they agree with [249, Numerical observation 7.2] to every digit printed there.

t	c_0c_2/c_1^2	$N=2$	$N=3$	$N=4$	$N=5$	block
0.01	1.1346548	0.119	2.45×10^{-3}	1.05×10^{-5}	1.04×10^{-8}	even
0.01		0.182	5.87×10^{-3}	3.30×10^{-5}	4.57×10^{-8}	odd
0.05	1.0000081	8.11×10^{-6}	1.77×10^{-14}	5.35×10^{-28}	2.20×10^{-44}	even
0.05		1.79×10^{-5}	1.22×10^{-13}	1.71×10^{-26}	3.81×10^{-42}	odd

Table 5.1: Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N}/\prod_{i<N} c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N}/\prod_{i<N} c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic.

At $t = 0.05$ (Figure 5.1, left) the first exponents dominate and the others enter only through $e^{-(a_j-a_1)t}$, so the matrix is nearly of rank one. Computed from the zeros, the minors are still sums of positive terms, by the Cauchy Binet formula $\det[c_{i+j}]_{i,j<N} = \sum_{|S|=N} \prod_{l\in S} e^{-a_l t} \prod_{l<l',\, l,l'\in S} (a_l - a_{l'})^2$ over the sets S of N zeros, so what is lost is conditioning, not sign. The derivative signs (Figure 5.1, right) reach much further, because a sign needs only that the verified terms dominate, not that a small determinant stay positive.

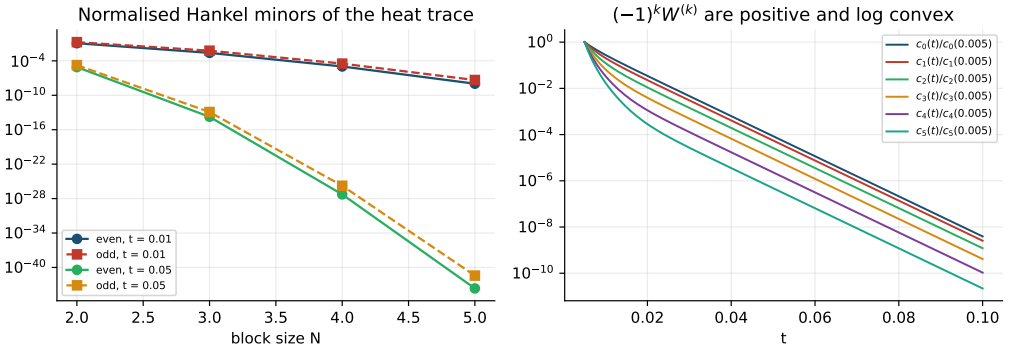


Figure 5.1: Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as t grows. Right: the first six $(-1)^k W^{(k)}(t)$, divided by their values at $t = 0.005$, from 1700 zeros.

THEOREM 5.1.6 ([249, Theorem 7.10]). *Unconditionally, that is, using only the verification of RH below height* 10^{12} *[247],* $(-1)^k W^{(k)}(t) > 0$ *for every* $t > 0$ *and*

every integer $0 \le k \le 10^{17}$.

The proof again splits at τ. Near the origin, Cauchy's estimate on a circle about t of radius close to t, such as $(1 - \frac{1}{k+1})t$, on which $\mathrm{Re}(1/z) \ge 1/(2t)$ and $|z| \ge t/(k+1)$, bounds the kth derivative of the prime term by a multiple of $(k+1)\Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}e^{-\kappa/t}$. Against $s_k(t)$ this loses a factor of order k and gains $e^{-\kappa/t} \le e^{-1201}$, and for $k \ge 1$ the archimedean term dominates by a margin of more than $2.9\, s_k(t)$, the case $k = 0$ being Theorem 5.1.1. Away from the origin a zero in $(20, 50]$ dominates all zeros above the verified height $H = 10^{12}$, wherever they lie, provided $tH^2 \ge 2(k+1)$ and a logarithmic condition hold; Section 5.1.6 explains why these allow orders up to about 10^{18}.

The Stieltjes moments of Table 1.7 give a Hankel test at the other end of the heat trace, and the test also returns the zeros [261]. They are the moments of W, $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_j a_j^{-n-1}$, and since $\log G(u) = \sum_j \log(1 + u/a_j) = \sum_{n\ge1}(-1)^{n+1}m_{n-1}u^n/n$ they follow from the Taylor coefficients of $G(u) = \Xi(i\sqrt{u})/\Xi(0) = \sum_k M_k u^k/((2k)!\,M_0)$, where $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$; no zero enters. Under RH the m_n are the moments of the positive measure $\nu = \sum_j a_j^{-1}\delta_{1/a_j}$, so every Hankel matrix $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is positive definite. Conversely, if all of them are, Hamburger's theorem makes the m_n the moments of a positive measure, of compact support because $|m_n| \le \sum_j |a_j|^{-n-1}$ grows at most geometrically, and then $\sum_n m_n w^n = \sum_j 1/(a_j - w)$ has all its poles on the real axis: every a_j is real, and positive because ξ has no zeros in $(0, 1)$, which is RH. The shifted matrices $\mathcal{H}'_N = [m_{i+j+1}]_{i,j<N}$ give the same test, and either suffices alone: if every $\mathcal{H}'_N$ is positive definite, Hamburger's theorem applied to (m_{n+1}) and $\sum_n m_n w^n = m_0 + w\sum_n m_{n+1}w^n$ give the same conclusion. Positivity is tested by the Cholesky factorisation $\mathcal{H}_N = LL^\top$, and the factor gives more: $J_N = L^{-1}\mathcal{H}'_N L^{-\top}$ is the Jacobi matrix of ν, tridiagonal, and its eigenvalues are the nodes of the N point Gauss rule for ν. They approximate the largest atoms $1/a_j$ first, so their inverse square roots return the lowest ordinates (Table 5.2). With $M_0, \dots, M_{100}$ computed at 1000 digits by the trapezoid rule, every $\mathcal{H}_N$ up to order 50 is positive definite.

This is Gauss quadrature, equivalently Padé approximation of G'/G at the origin, and Table 5.2 shows its two limits. The first is precision. The last Cholesky pivot $\det\mathcal{H}_N/\det\mathcal{H}_{N-1}$ falls from 10^{-61} at $N = 10$ to 10^{-398} at $N = 50$, and the computation must resolve it. The precision thus grows linearly, by about 17 digits for each further zero. The second is reach. An off line quadruple at height γ enters m_n with weight of order γ^{-2n-2}, against γ_1^{-2n-2} for the first zero, so a block of order N sees essentially only the zeros that its Gauss rule resolves; Section 8.4.3 shows this along the de Bruijn Newman flow, where the order 30 test registers the collision of the pair just above the zeros it resolves, and not a collision higher up. The program `jacobi_zeros.py` computes the table.

N	zeros	error of γ_1	last error	$\det \mathcal{H}_N / \det \mathcal{H}_{N-1}$	digits
10	2	4.4×10^{-18}	1.3×10^{-9}	2.0×10^{-61}	60
20	7	1.6×10^{-49}	3.1×10^{-7}	1.2×10^{-138}	140
30	12	8.6×10^{-87}	2.4×10^{-7}	1.0×10^{-221}	230
40	17	6.1×10^{-128}	5.4×10^{-8}	1.3×10^{-308}	310
50	22	5.3×10^{-172}	5.4×10^{-8}	1.4×10^{-398}	400

Table 5.2: The lowest ordinates from the Stieltjes moments $m_0, \dots, m_{2N-1}$, computed from the Taylor coefficients of Ξ at the centre: the number returned by the Jacobi matrix of order N to within 10^{-6}, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned.

The Hankel matrices count the zeros off the line as well as test RH. For a real polynomial this is a theorem attributed to Hermite [115]: the Hankel form of the power sums of the roots has as many negative squares as the polynomial has distinct pairs of nonreal roots. Let ι_N and ι'_N be the numbers of negative eigenvalues of $\mathcal{H}_N$ and $\mathcal{H}'_N$.

THEOREM 5.1.7. *Let $K \in \{0, 1, 2, \dots\} \cup \{\infty\}$ be the number of distinct zeros ρ of ξ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$. Then ι_N and ι'_N are nondecreasing in N, $\iota_N \le K$ and $\iota'_N \le K$ for every N, and $\iota_N \to K$ and $\iota'_N \to K$ as $N \to \infty$. In particular RH holds if and only if $\iota_N = 0$ for every N, and if it fails, $\iota_N \ge 1$ from some N on; the same holds for ι'_N.*

Proof. Put $b_k = 1/a_k$, listed with multiplicity, so that $m_n = \sum_k b_k^{n+1}$ and $\sum_k |b_k| < \infty$. For a real vector $x = (x_0, \dots, x_{N-1})$ and $P(w) = \sum_i x_i w^i$,

$$x^\top \mathcal{H}_N x = \sum_k b_k P(b_k)^2, \qquad x^\top \mathcal{H}'_N x = \sum_k b_k^2 P(b_k)^2,$$

the sums converging absolutely. A zero on the line gives $b_k = \gamma_k^{-2} > 0$ and nonnegative terms. A zero $\rho = \frac{1}{2} + \delta + i\gamma$ with $\delta > 0$ and $\gamma > 0$, of multiplicity μ, gives $a = -(\delta + i\gamma)^2$ with $\operatorname{Im} a = -2\delta\gamma < 0$, hence $b = 1/a$ with $\operatorname{Im} b > 0$; its mirror image $1 - \bar\rho$ gives $\bar a$ and $\bar b$ with the same multiplicity, and since P has real coefficients the two contribute $2\mu \operatorname{Re}\big(bP(b)^2\big)$ to the first form. With $u = \operatorname{Re} P(b)$ and $v = \operatorname{Im} P(b)$ this is $2\mu\big(\operatorname{Re} b\,(u^2 - v^2) - 2 \operatorname{Im} b\, uv\big)$, a quadratic form in (u, v) whose matrix has determinant $-4\mu^2|b|^2 < 0$, so it has exactly one negative square, and composed with the linear map $x \mapsto (u, v)$ it has at most one. The terms of the zeros on the line add up to a positive semidefinite form, so $\iota_N \le K$. In the second form the weight is b^2, and $2\mu \operatorname{Re}\big(b^2 P(b)^2\big)$ is again a form in (u, v) of determinant $-4\mu^2|b|^4 < 0$, so $\iota'_N \le K$. Since $\mathcal{H}_N$ is the leading principal submatrix of $\mathcal{H}_{N+1}$,

Cauchy's interlacing theorem shows that ι_N does not decrease, and the same holds for ι'_N.

For the lower bound let $\mathcal{B}$ be the closure of $\{b_k\}$. It is a countable compact set, symmetric under conjugation, whose only possible accumulation point is 0, and a countable compact set does not separate the plane, so $\mathbb{C}\setminus\mathcal{B}$ is connected. Fix a finite $m \le K$ and distinct atoms $\omega_1,\dots,\omega_m$ with $\operatorname{Im}\omega_j > 0$, of multiplicities μ_j; they are isolated points of $\mathcal{B}$. On $\mathbb{C}^m = \mathbb{R}^{2m}$ the form $q(y) = \sum_{j\le m} 2\mu_j \operatorname{Re}(\omega_j y_j^2)$ is a sum of m forms in the pairs $(\operatorname{Re} y_j, \operatorname{Im} y_j)$, each with one negative square, so there are an m dimensional real subspace E and $\lambda_0 > 0$ with $q(y) \le -\lambda_0|y|^2$ on E. Let $y^{(1)},\dots,y^{(m)}$ be a basis of E and $\lambda_1 > 0$ such that $|\sum_i \alpha_i y^{(i)}|^2 \ge \lambda_1|\alpha|^2$ for $\alpha\in\mathbb{R}^m$. Take open discs D_j about ω_j, and their mirror images D_j^* about $\bar\omega_j$, whose closures are disjoint and meet $\mathcal{B}$ only in their centres, and an open set $O \supset \mathcal{B}\setminus\{\omega_1,\bar\omega_1,\dots,\omega_m,\bar\omega_m\}$, symmetric under conjugation and disjoint from the closed discs, which exists because that set is compact. Let f_i be $y_j^{(i)}$ on D_j, $\overline{y_j^{(i)}}$ on D_j^* and 0 on O. Each f_i is holomorphic on a neighbourhood of $\mathcal{B}$, so by Runge's theorem there are polynomials $\tilde P_i$ with $|\tilde P_i - f_i| < \varepsilon$ on $\mathcal{B}$, and since $\mathcal{B}$ and f_i are symmetric under conjugation, $P_i(w) = \frac12\big(\tilde P_i(w) + \overline{\tilde P_i(\bar w)}\big)$ has real coefficients and satisfies the same bound. For $\alpha\in\mathbb{R}^m$ and $P = \sum_i \alpha_i P_i$ we have $|P(b_k)| \le \varepsilon\sqrt{m}\,|\alpha|$ at every atom other than the ω_j and $\bar\omega_j$, and $P(\omega_j) = \sum_i \alpha_i y_j^{(i)} + \eta_j$ with $|\eta_j| \le \varepsilon\sqrt{m}\,|\alpha|$. The atoms ω_j and $\bar\omega_j$ contribute $\sum_j 2\mu_j \operatorname{Re}\big(\omega_j P(\omega_j)^2\big)$ to the first form, so for $\varepsilon \le 1$ and a constant C that depends on neither ε nor α,

$$\sum_k b_k P(b_k)^2 \le q\Big(\sum_i \alpha_i y^{(i)}\Big) + C\varepsilon|\alpha|^2 \le -(\lambda_0\lambda_1 - C\varepsilon)|\alpha|^2.$$

For $\varepsilon < \lambda_0\lambda_1/C$ the right side is negative for $\alpha \ne 0$, so the coefficient vectors of $P_1,\dots,P_m$ span an m dimensional space on which the form of $\mathcal{H}_N$ is negative definite as soon as N exceeds their degrees, and $\iota_N \ge m$ for those N. For $\mathcal{H}'_N$ the same argument applies with ω_j^2 in place of ω_j in q and the weights b_k^2 in the sum. Since $m \le K$ is arbitrary, $\iota_N \to K$ and $\iota'_N \to K$. □

In determinant form the theorem is classical. For a real entire function of order one with zeros z_j, Grommer [102] proved that all its zeros are real if and only if the determinants $D_n = \det(s_{i+j})_{i,j<n}$ of the power sums $s_k = \sum_j z_j^{-k-2}$ are positive for every $n \ge 1$, and Chebotarev [331] that if the sequence $D_{n-1}D_n$, $n \ge 1$, with $D_0 = 1$, has exactly m negative terms, the function has m distinct pairs of complex conjugate zeros; Baricz and Štampach state both in this form [16, Theorem 10]. For Ξ, whose zeros are the square roots of the a_k, $s_{2l} = 2m_l$ and $s_{2l+1} = 0$, so after the even indices are put first $(s_{i+j})_{i,j<n}$ is block diagonal with the blocks $2\mathcal{H}_{\lceil n/2\rceil}$ and $2\mathcal{H}'_{\lfloor n/2\rfloor}$, whose negative indices add up to $\iota_{\lceil n/2\rceil} + \iota'_{\lfloor n/2\rfloor}$; this tends to $2K$, and each zero ρ counted by K gives two pairs of complex conjugate zeros of Ξ, as Chebotarev's count requires. Counting by eigenvalues rather than by the signs of determinants, Theorem 5.1.7 needs no determinant to be nonzero,

allows $K = \infty$ and holds for each family separately. Along the de Bruijn Newman flow the proof applies at every time to the zeros of H_t, none of which is purely imaginary (Section 8.4.3).

5.1.3 Every order near the origin

Let $s_k(t) = \Gamma(k+\frac12)t^{-k-\frac12}/(4\pi)$ as in the proof of Theorem 5.1.1, let $E_k(t)$ be the average of m against the weight $r^{2k}e^{-r^2t}$ on $[0,\infty)$, and let $F_k(x) = {}_1F_1(k+\frac12;\frac12;-x^2)$. Expanding the cosine in its power series gives $\frac{1}{2\pi}\int_{\mathbb{R}} r^{2k}e^{-r^2t}\cos(ry)\,dr = 2s_k(t)F_k(y/2\sqrt{t})$, whose case $k=0$ is the kernel $e^{-y^2/4t}/(2\sqrt{\pi t})$ of the prime term of (1.2.5). Differentiating (1.2.5) k times term by term, which the Gaussian factors in r and in $\log n$ allow, therefore gives, with no hypothesis on the zeros,

$$\frac{c_k(t)}{s_k(t)} = E_k(t) - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}F_k(x_n) + \frac{(-\frac14)^k e^{t/4}}{s_k(t)}, \qquad x_n = \frac{\log n}{2\sqrt{t}}. \tag{5.1.1}$$

In (5.1.1) the archimedean term $E_k(t)$ grows with k.

Lemma 5.1.8. (a) *For every integer $k \ge 0$ and every $x \ge 0$, $|F_k(x)| \le B(x)e^{-x^2/2}$, where $B(x) = \max\{2.62,\ (\pi(2ex^2+\frac12))^{1/4}\}$.*

(b) *Let $0<\varepsilon<1$, and let $k \ge 1$ and $x \ge 0$ satisfy $e^2x^2 \le 4\varepsilon k$. Then $|F_k(x)| \le C_\varepsilon(k)e^{-x^2/2}$, where $C_\varepsilon(k) = \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}$.*

Proof. Taking $t=1$ and $y=2x$ above, $F_k(x) = \Gamma(k+\frac12)^{-1}\int_{\mathbb{R}} r^{2k}e^{-r^2+2ixr}\,dr$. Put $\mu = x^2/4$. The integrand is entire and decays in every horizontal strip, so the path can be moved to $\operatorname{Im} r = x/2$, where $|r^{2k}| = (\sigma^2+\mu)^k$, $|e^{-r^2}| = e^{-\sigma^2+\mu}$ and $|e^{2ixr}| = e^{-4\mu}$, with $\sigma = \operatorname{Re} r$. Expanding $(\sigma^2+\mu)^k$ by the binomial theorem gives, for every k and x,

$$|F_k(x)| \le e^{-3\mu}\sum_{j=0}^{k}\frac{\mu^j}{j!}P_{k,j}, \qquad P_{k,j} = \frac{\Gamma(k+1)}{\Gamma(k+\frac12)}\,\frac{\Gamma(k-j+\frac12)}{\Gamma(k-j+1)}.$$

Kershaw's inequalities $\sqrt{y+\frac14} \le \Gamma(y+1)/\Gamma(y+\frac12) \le \sqrt{y+0.37}$, proved for $y>0$ in [147] and true at $y=0$, where the ratio is $1/\sqrt{\pi}$, give $P_{k,j} \le \sqrt{(k+0.37)/(k-j+\frac14)}$, which is less than $\sqrt{2}$ for $j \le k/2$ and at most $2\sqrt{k+0.37}$ for every j. For (a), suppose first that $k \ge 2ex^2$. Since $k/2 \ge 4e\mu$, Chernoff's bound $\sum_{j\ge a}\mu^j/j! \le (e\mu/a)^a$ for the Poisson law, valid for $a>\mu$, gives $\sum_{j\ge k/2}\mu^j/j! \le (2e\mu/k)^{k/2} \le 2^{-k}$, and $2\sqrt{k+0.37}\,2^{-k} \le \sqrt{1.37} < 1.2$ for $k \ge 1$. Hence $|F_k(x)| \le e^{-3\mu}(\sqrt{2}\,e^{\mu}+1.2) \le (\sqrt{2}+1.2)e^{-x^2/2}$, and $\sqrt{2}+1.2<2.62$; for $k=0$ the assumption forces $x=0$, where $F_0=1$.

Suppose now that $k < 2ex^2$. Kummer's transformation and the Hermite case of ${}_1F_1$ [225, (13.2.39), (13.6.16)] give $F_k(x) = e^{-x^2}\,{}_1F_1(-k;\frac12;x^2) = (-1)^k\frac{k!}{(2k)!}e^{-x^2}H_{2k}(x)$, and the bound $e^{-x^2/2}|H_n(x)| \le (2^n n!)^{1/2}$ [225, (18.14.9)] gives

$$|F_k(x)| \le \Big(\frac{4^k(k!)^2}{(2k)!}\Big)^{1/2} e^{-x^2/2} \le \big(\pi(k+\tfrac12)\big)^{1/4} e^{-x^2/2} < \big(\pi(2ex^2+\tfrac12)\big)^{1/4} e^{-x^2/2},$$

since $4^k(k!)^2/(2k)! = \sqrt{\pi}\,\Gamma(k+1)/\Gamma(k+\frac12) \le \sqrt{\pi(k+\frac12)}$ by Kershaw's upper bound.

For (b), split the sum at $j = \varepsilon k$ instead. For $j \le \varepsilon k$, $P_{k,j} \le \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}$, and these j contribute at most that times e^{μ}. For $j > \varepsilon k$, $P_{k,j} \le 2\sqrt{k+0.37}$, and since $\varepsilon k \ge e^2\mu > \mu$, Chernoff's bound gives $\sum_{j\ge\varepsilon k}\mu^j/j! \le (e\mu/\varepsilon k)^{\varepsilon k} \le e^{-\varepsilon k}$. Hence $|F_k(x)| \le e^{-3\mu}\big(\sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}\,e^{\mu} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}\big) \le C_\varepsilon(k)e^{-x^2/2}$. □

Theorem 5.1.9. (a) *For every integer $k \ge 1$ and every $t \in (0, 0.019]$, $(-1)^k W^{(k)}(t) > 0$. The proof uses no information on the zeros.*

(b) *Unconditionally, that is, using only the verification of RH below height 10^{12} [247], W is completely monotone on $(0, 1.29]$: $(-1)^k W^{(k)}(t) > 0$ for every $t \in (0, 1.29]$ and every integer $k \ge 0$.*

Proof. By (5.1.1) and Lemma 5.1.8(a), for every $k \ge 0$ and $t > 0$,

$$\frac{c_k(t)}{s_k(t)} \ge E_k(t) - S_B(t) - Q_k(t),$$

where

$$S_B(t) = 2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}B(x_n)e^{-(\log n)^2/8t}, \qquad Q_k(t) = \frac{4\pi e^{t/4}t^{k+\frac12}}{4^k\,\Gamma(k+\frac12)}.$$

Above the order $K = 10^{17}$ part (b) of the lemma does better. Take $\varepsilon = 10^{-6}$. The first term of $C_\varepsilon(k)$ decreases in k, because $\frac14 < 0.37(1-\varepsilon)$, and the second decreases for $k > 1/(2\varepsilon)$, so $C_\varepsilon(k) \le C = C_\varepsilon(K+1) < 1 + 6\times10^{-7}$ for $k > K$, and $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds 2.3×10^5. For $t \ge 0.019$ every prime power $n \le 2^{24}$ has $x_n < 61$. So for $k > K$ and $t \ge 0.019$ the bound holds with S_B replaced by S_C, the sum S_B with $B(x_n)$ replaced by C for $n \le 2^{24}$.

The three terms are monotone. Since m increases on $[0,\infty)$, and the weights $r^{2k}e^{-r^2t}$ have likelihood ratios r^2 from k to $k+1$ and $e^{r^2(t'-t)}$ from t' to $t < t'$, both increasing in r, $E_k(t)$ increases in k and decreases in t. $Q_k(t)$ increases in t, and for $t \le 2$ it decreases in k, the ratio of consecutive terms being $t/(4k+2)$. $B(x)e^{-x^2/2}$ decreases on $(0,\infty)$: the second branch of B increases and is 1.78 at

$x = 2^{-1/2}$, so $B = 2.62$ below that point, and above it the logarithmic derivative of the second branch times $e^{-x^2/2}$ is less than $1/(2x) - x < 0$; since x_n decreases in t, S_B and S_C increase in t. So for $k \geq k_0$ and $t \leq T \leq 2$ each bound is at least its value at (k_0, T), and it suffices to prove $E_1(0.019) - S_B(0.019) - Q_1(0.019) > 0$, which gives (a), and $E_{K+1}(1.29) - S_C(1.29) - Q_{K+1}(1.29) > 0$, which gives the orders $k > K$ on $[0.019, 1.29]$; with (a) for $t \leq 0.019$ and Theorem 5.1.6 for the orders $k \leq K$, this gives (b).

For the first, lower sums over $[0, 64]$, which the monotonicity of m allows, and the positivity of m beyond 64 give $E_1(0.019) > 0.15924$; the prime powers $n \leq 30000$ and the tail bound below give $S_B(0.019) < 0.13628$; and $Q_1(0.019) < 0.00933$. For the second, the weight $r^{2K+2}e^{-r^2t}$ makes $v = r^2t$ a gamma variable of shape $\lambda = K + \frac{3}{2}$, and Chernoff's bound for the gamma law, $\Pr(v < a) \leq (a/\lambda)^\lambda e^{\lambda - a}$ for $a < \lambda$, gives $\Pr(v < a) < e^{-1000}$ for $a = (1 - 2^{-13})^2\lambda$. As m increases, $E_{K+1}(t) \geq m(r_1) - (m(r_1) - m(0))e^{-1000}$ with $r_1 = \sqrt{a/t}$, which gives $E_{K+1}(1.29) > 17.6066$; the prime powers $n \leq 2^{24}$ and the tail bound below give $S_C(1.29) < 17.5848$; and $Q_{K+1}(1.29) < e^{-1000}$. In both cases the prime powers above the cutoff n_0, which is 30000 and 2^{24}, add less than $2\big(2.62 + 2.08(2\sqrt{t})^{-1/2}\big)n_0^{1-q}/(q-1)$ with $q = \log n_0/8t$, because $x_n \geq x_{n_0} \geq 1$ and $e^{-(\log n)^2/8t} \leq n^{-q}$ for $n > n_0$, $B(x) \leq 2.62 + 2.08\sqrt{x}$ for $x \geq 1$, and $\log n$ and $(\log n)^{3/2}$ are at most $\sqrt{n}$ for $n \geq e^{10}$; this is below 10^{-299} and 5×10^{-4}. The program `cm_orders.py` evaluates every numerical inequality of this proof and of the proof of Lemma 5.1.8 in ball arithmetic. □

Computed in floating point, the bound $E_1 - S_B - Q_1$ of (a) vanishes at $t = 0.0193$; E_1 itself vanishes at $t = 0.0261$, so no bound on the size of the prime terms carries (a) much further. Part (b) goes further because $E_k(t) = \frac{1}{2}(\psi(k + \frac{1}{2}) - \log t) - \log 2\pi + O(t/k)$ grows like $\frac{1}{2}\log k$, while at such orders part (b) of the lemma bounds the prime terms at almost no cost: the bound $E_{K+1} - S_C$, computed in the same way, vanishes at $t = 1.2912$, while $E_{K+1} - S_B$ vanishes at $t = 0.578$.

The ray of Section 5.1.5 shows what the proof uses. On the ray $t = k/u^2$, $e^{-f_0}c_k(t)$ is to leading order the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros (Proposition 5.1.10), $E_k(t)$ is close to $\log(u/2\pi)$, and for large k the factor $F_k(x_n)$ is close to $e^{-(\log n)^2/8t}\cos(u \log n)$. The proof is therefore the bound of Section 12.2.6 that ignores the phases of the primes, made uniform in the order; with C replaced by one, S_C is twice the sum S of that section, up to the prime powers above 2^{24}. Above the order 10^{17} uniformity costs a factor below $1 + 6 \times 10^{-7}$, so 1.2912, the root of $E_{K+1}(t) = 2S(t)$, is as far as this argument can carry (b) with $K = 10^{17}$: it bounds the prime terms by one envelope for all orders above K, and the archimedean term by its value at the order $K + 1$. A bound that ignores the phases but treats the orders separately is not covered by this limit. At low orders it costs the factor $B \geq 2.62$ of part (a), and part (a) is not sharp: on a grid of $0.5 \leq x \leq 20$ and $k \leq 1200$ the ratio $|F_k(x)|e^{x^2/2}/B(x)$ is at most 0.44 (computed). The ray also

says where (b) looks. To leading order, complete monotonicity on $(0,T]$ asks for positive Gaussian smoothings at resolutions $1/(2\sqrt{t})$ of at least $1/(2\sqrt{T})$, which is 0.44 for $T = 1.29$, finer than the 0.46 at which the quadruples of Table 5.4, at distance $\frac{1}{4}$ from the line, first make the smoothing negative, at $t = 1.18$. There is no conflict, because the orders above 10^{17} at $t \le 1.29$ lie on the rays at heights $u = \sqrt{k/t} > 2.78 \times 10^8$, where the time $t_0(u)$ of Section 12.2.6, below which the phase free bound makes the smoothing positive whatever the zeros, exceeds 1.29. The quadruples lie below height 2000, at heights where t_0 is at most 0.511, and there (b) rests on the verified zeros, through Theorem 5.1.6.

5.1.4 Williamson measures: the Thorin measure smoothed by gamma laws

Complete monotonicity is the limit of k monotonicity, which asks only that the first k derivatives alternate in sign, and by Williamson's theorem [345] a k monotone function that vanishes at infinity is a positive mixture of the truncated powers $(1 - tz/k)_+^{k-1}$. For the heat trace the mixing measure is explicit,

$$W(t) = \int_0^\infty \Big(1 - \frac{tz}{k}\Big)_+^{k-1} U_k(dz), \qquad U_k(dz) = \frac{k^k}{(k-1)!} z^{-k-1} c_k\Big(\frac{k}{z}\Big) dz,$$

and we call U_k the Williamson measure of order k. By Theorem 5.1.6 it is positive for every $k \le 10^{17}$ [249, Corollary 7.13], and by Theorem 5.1.13 for every $k \le 1.16 \times 10^{25}$. RH holds if and only if $U_k \ge 0$ for infinitely many k, since then W is k monotone for infinitely many k, hence completely monotone.

The summands of U_k are inverse gamma laws, by an elementary identity: for $k \ge 1$ and $c \ge 0$,

$$e^{-c} = \mathbb{E}\Big(1 - \frac{c}{\Gamma_k}\Big)_+^{k-1}, \qquad \Gamma_k \sim \mathrm{Gamma}(k,1), \tag{5.1.2}$$

since $\int_c^\infty (\omega - c)^{k-1} e^{-\omega} d\omega/\Gamma(k) = e^{-c}$. With $c = a_j t$ and $z = k a_j/\Gamma_k$, so that $1 - a_j t/\Gamma_k = 1 - tz/k$, the identity shows that U_k is the sum over the zeros of the densities $(k a_j)^k z^{-k-1} e^{-k a_j/z}/(k-1)!$. For real a_j this is the inverse gamma law of $k a_j/\Gamma_k$. Under RH, therefore, writing $\mathcal{L}(Y)$ for the law of Y,

$$U_k = \sum_\gamma \mathcal{L}\Big(\gamma^2 \frac{k}{\Gamma_k}\Big),$$

a sum of probability laws: the Thorin measure with each atom γ^2 spread over a relative width of about $1/\sqrt{k}$. As $k \to \infty$ the ratio Γ_k/k concentrates at one and the atoms reappear, which is the passage from Williamson's theorem to Bernstein's; in statistics, Polson, Scott and Windle used k monotone mixtures of this kind to build samplers for the Bayesian bridge [248, Section 2.2, Theorem 2.1] and noted the same limit [248, Section 3].

The Williamson measures form a second resolution family for the Thorin measure, beside the Poisson family ρ_ε, which is the Cauchy flow of Section 1.2. In the chart $\theta = \sqrt{z}$ the gamma smoothing spreads the atom at γ over a width of about $\gamma/(2\sqrt{k})$, proportional to the height, while the Poisson smoothing spreads every atom over the width ε, and the inverse gamma kernel has tails far lighter than the x^{-2} tails of the Cauchy kernel. In Figure 5.2, where the widths match at $\theta = 30$, the Williamson densities fall by many orders of magnitude between neighbouring zeros and the Poisson densities by only one to four.

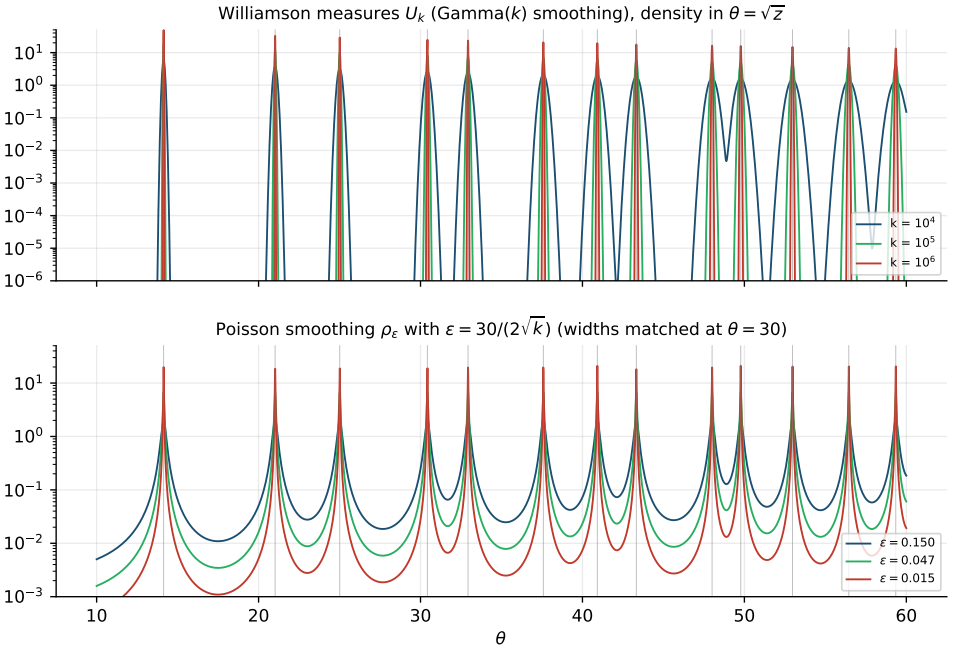


Figure 5.2: Top: the Williamson measures U_k of the heat trace for $k = 10^4$, 10^5, 10^6, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities ρ_ε with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$.

The Poisson smoothing of the zeros is $\pi^{-1}\operatorname{Re}(\xi'/\xi)$ off the line and can be computed from ξ alone at every resolution, which is why ρ_ε is the central object of Chapter 3 and Section 4.1; the price is the heavy tail, which makes positivity at one resolution a statement about all zeros at once. The gamma smoothing is local, and an off line pair contributes to U_k a signed, oscillating term near its height.

5.1.5 Along a ray: the Gaussian smoothing of the zeros

Fix a height u and move along the ray $t = k/u^2$ of the (k,t) plane, on which the weight $x^k e^{-xt}$ of $c_k(t)$ peaks at $x = u^2$ and the density of U_k is taken at $z = u^2$. Write each zero as $\rho = \frac12 + \delta + i\gamma$ and put $\tilde\gamma = \gamma - i\delta$, so that $a = \tilde\gamma^2$; as in every sum of this section, the zero is taken with $\gamma > 0$ from each pair $\{\rho, 1-\rho\}$. The Gaussian smoothing of the zeros at resolution $1/(2\sqrt{t})$ is

$$\mathcal{G}(u,t) = \sum e^{-2t(\tilde\gamma - u)^2}.$$

A zero on the line contributes a Gaussian bump at its ordinate. The zeros $\frac12 \pm \delta + i\gamma$ of a quadruple $\frac12 \pm \delta \pm i\gamma$ have conjugate values of $\tilde\gamma$, so $\mathcal{G}$ is real, and together they contribute $2e^{2t(\delta^2 - (\gamma-u)^2)}\cos(4t\delta(\gamma-u))$, a bump amplified by $e^{2t\delta^2}$ and modulated in u with period $\pi/(2t\delta)$. At $u = 0$, $\mathcal{G}(0,t) = W(2t)$.

Proposition 5.1.10. *Let $u \ge 2$, $t > 0$ and $k = u^2 t$, not necessarily an integer, with a^k the principal power in $c_k(t) = \sum a^k e^{-at}$, and let $f_0 = k(\log u^2 - 1)$. Then*

$$e^{-f_0} c_k(t) = \sum_{|\tilde\gamma - u| \le u/2} e^{-2t(\tilde\gamma - u)^2}\big(1 + \varepsilon_\rho\big) + R, \qquad |\varepsilon_\rho| \le \exp\Big\{\frac{4t}{3u}|\tilde\gamma - u|^3\Big\} - 1,$$

where, with $\ell_\pm = u \pm \frac12\sqrt{u^2-1}$, $H(r) = u^2 \log\frac{r^2+1/4}{u^2} - r^2 + \frac14 + u^2$ and $\eta = 1 - u^2/(\ell_+^2 + \frac14) > 0$,

$$|R| \le N(\ell_-) e^{tH(\ell_-)} + \Big(\ell_+^2 + \frac{1}{t\eta}\Big) e^{tH(\ell_+)}.$$

No hypothesis on the zeros is used.

Proof. Put $\tilde\gamma = u + v$ and $w = v/u$. Since $\operatorname{Re}\tilde\gamma > 0$ and $|\arg\tilde\gamma| < \frac{\pi}{4}$, the principal logarithm gives $e^{-f_0} a^k e^{-at} = e^E$ with

$$E = tu^2\big(2\log(1+w) - 2w - w^2\big) = -2tv^2 + 2tu^2 \sum_{m \ge 3} \frac{(-1)^{m+1} w^m}{m},$$

and for $|w| \le \frac12$ the series has modulus at most $|w|^3/(3(1-|w|)) \le \frac23 |w|^3$. This gives $\varepsilon_\rho = e^{E + 2tv^2} - 1$ and its bound, since $|e^z - 1| \le e^{|z|} - 1$. For a zero with $|v| > u/2$, $|v|^2 = (\gamma - u)^2 + \delta^2$ and $\delta^2 < \frac14$ give $|\gamma - u| > \frac12\sqrt{u^2-1}$, so that $\gamma < \ell_-$ or $\gamma > \ell_+$, and $|e^E| = \exp t\big(u^2 \log(|a|/u^2) - \operatorname{Re} a + u^2\big) \le e^{tH(\gamma)}$, because $|a| = \gamma^2 + \delta^2$ and $\operatorname{Re} a = \gamma^2 - \delta^2$. The function H increases up to $\sqrt{u^2 - \frac14}$, which lies between ℓ_- and ℓ_+, so each of the at most $N(\ell_-)$ zeros below ℓ_- contributes at most $e^{tH(\ell_-)}$. For $r \ge \ell_+$, $\log(1+x) \le x$ gives $H(r) \le H(\ell_+) - \eta(r^2 - \ell_+^2)$. The explicit Riemann von Mangoldt formula [109, Corollary 1.2] gives $N(r) \le r^2$ for every r, and partial summation then gives $\sum_{\gamma > \ell_+} e^{-q(\gamma^2 - \ell_+^2)} \le \ell_+^2 + 1/q$ for $q > 0$; take $q = t\eta$. □

For $u \geq 10$ the ratios $H(\ell_-)/u^2$ and $H(\ell_+)/u^2$ decrease in u, from -0.616 and -0.431 at $u = 10$, and $\eta > 0.55$, so that $|R| \leq (3u^2 + 2/t)e^{-0.43k}$. What remains is ε_ρ, whose term of order $1/u$ is $\frac{2t}{3u}(\tilde\gamma - u)^3$. Along the ray, therefore, $e^{-f_0}c_k(t)$ is $\mathcal{G}(u,t)$ up to an exponentially small remainder and, in each term, a relative error at most $e^{4t|\tilde\gamma-u|^3/3u} - 1$. That error is small, of order $t|\tilde\gamma - u|^3/u$, for the terms that carry the sum provided $t|\tilde\gamma - u|^3$ is small against u for them; for the zeros within a few multiples of $1/\sqrt{t}$ of u it is of order $1/(\sqrt{t}\,u)$. It is not small in general: at a fixed height, as t grows, the term of the nearest zero dominates, and its relative error grows with t. For an integer k the density of U_k in the chart $\theta = \sqrt{z}$ is exactly $\sqrt{2t/\pi}\,e^{-\lambda_k}e^{-f_0}c_k(t)$ at $\theta = u$, where $t = k/u^2$ and $\lambda_k \in (\frac{1}{12k+1}, \frac{1}{12k})$ is the remainder in Stirling's formula for $k!$ [286]. The density of U_k at θ is therefore about $\sqrt{2t/\pi}\,\mathcal{G}(\theta,t)$ with $t = k/\theta^2$: the zeros smoothed by normal densities of standard deviation $1/(2\sqrt{t}) = \theta/(2\sqrt{k})$, which is the width found above.

Proposition 5.1.11. *RH holds if and only if $\mathcal{G}(u,t) \geq 0$ for every real u and every $t > 0$.*

Proof. Under RH every term is positive. Suppose that $\frac12 + d + ic$ is a zero with $d > 0$, and let $|u - c| < \frac{d}{2}$. The term of a zero has modulus $e^{2t\phi}$ with $\phi = \delta^2 - (\gamma - u)^2$. For the zero $\frac12 + d + ic$, $\phi > \frac34 d^2$; for a zero with $|\gamma - c| > 1$, $\phi < \frac14 - \frac{9}{16} < 0$. Group the finitely many zeros with $|\gamma - c| \leq 1$ by their type $(\gamma, |\delta|)$. Two types at the same ordinate have exponents ϕ differing by a nonzero constant, and two at different ordinates have exponents differing by a nonconstant affine function of u, so outside finitely many points one type has the largest ϕ. Fix such a u_*, different from every ordinate, and let (γ_*, d_*) be the maximising type and ϕ_* its exponent. Then $\phi_* > \frac34 d^2$, so $d_* > 0$, and the type consists of $\nu \geq 2$ zeros counted with multiplicity, which contribute $\nu e^{2t\phi_*}\cos(4td_*(\gamma_* - u_*))$. Along $t_j = (2j+1)\pi/(4d_*|\gamma_* - u_*|)$ this is $-\nu e^{2t_j\phi_*}$. The other n zeros with $|\gamma - c| \leq 1$ contribute at most $ne^{2t_j(\phi_* - \Delta)}$, with $\Delta > 0$ the gap to the next exponent, and the zeros with $|\gamma - c| > 1$ at most $32\log(c+2)e^{-5t_j/8}$ once $t_j \geq 1$, since by the explicit Riemann von Mangoldt formula every interval of heights $[y, y+1]$ holds at most $4\log(y+3)$ zeros. Hence

$$e^{-2t_j\phi_*}\mathcal{G}(u_*, t_j) \leq -\nu + ne^{-2t_j\Delta} + 32\log(c+2)e^{-(2\phi_* + 5/8)t_j},$$

which is negative for all large j. □

Proposition 5.1.12. *For $\frac12 < \alpha < 1$ let $\eta = \alpha - \frac12$ and $\mathcal{G}_\alpha(u,t) = \operatorname{Re}\sum e^{-2t(\tilde\gamma + i\eta - u)^2}$, over the zeros with $\gamma > 0$ as for $\mathcal{G}$, which is $\mathcal{G}$ at $\eta = 0$. Under RH $\mathcal{G}_\alpha(u,t) < 0$ for some u and t: there is no Gaussian criterion on the line $\operatorname{Re} s = \alpha$.*

Proof. Under RH $\tilde\gamma = \gamma$, and the term of γ is $e^{2t\eta^2}e^{-2t(\gamma-u)^2}\cos\big(4t\eta(\gamma - u)\big)$. Take $u = \gamma_1 + \pi/(4t\eta)$, where γ_1 is the lowest ordinate and a simple zero, and let

$g > 0$ be the gap to the next ordinate. The term of γ_1 is $-e^{2t\eta^2-\pi^2/(8t\eta^2)}$. For $t \geq \pi/(2\eta g)$ every other γ has $\gamma - u \geq (\gamma-\gamma_1)/2$, so the other terms total at most $e^{2t\eta^2}\sum_{\gamma>\gamma_1} e^{-t(\gamma-\gamma_1)^2/2} = o(e^{2t\eta^2})$. Hence $\mathcal{G}_\alpha(u,t) = -e^{2t\eta^2}(1+o(1)) < 0$ as $t \to \infty$. □

Proposition 5.1.11 restricts Weil's criterion to the modulated Gaussians. For $f(x) = e^{-x^2/4t-iux}$ the transform $\hat f(r) = \int f(x)e^{irx}dx = \sqrt{4\pi t}\, e^{-t(r-u)^2}$ gives, summed over all zeros $\rho = \frac12 + i\gamma_\rho$,

$$\sum_\rho \hat f(\gamma_\rho)\overline{\hat f(\bar\gamma_\rho)} = 4\pi t\big(\mathcal{G}(u,t) + \mathcal{G}(-u,t)\big),$$

where $\mathcal{G}(-u,t)$ comes from the zeros $1-\rho$ and is at most $e^{t/2}((u+14)^2 + \frac{1}{2t})e^{-2t(u+14)^2}$ in modulus for $u \geq 0$, too little to change the sign in the proof. At $u = 0$ the family gives the first rung, $\mathcal{G}(0,t) = W(2t) > 0$ (Theorem 5.1.1). Since the Gaussian is entire, Weil's formula of Section 1.2 applies to $h(r) = e^{-2t(r-u)^2} + e^{-2t(r+u)^2}$, for which $\sum_\rho h(\gamma_\rho) = 2(\mathcal{G}(u,t) + \mathcal{G}(-u,t))$, and gives $\mathcal{G}$ from the primes:

$$\begin{aligned} \mathcal{G}(u,t) + \mathcal{G}(-u,t) &= 2e^{t/2-2tu^2}\cos 2tu + A(u,t) - P(u,t), \\ A(u,t) &= \frac{1}{4\pi}\int_{\mathbb{R}} \big(e^{-2t(r-u)^2} + e^{-2t(r+u)^2}\big)m(r)\,dr, \\ P(u,t) &= \frac{1}{\sqrt{2\pi t}}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt n}e^{-(\log n)^2/8t}\cos(u\log n), \end{aligned} \tag{5.1.3}$$

with m as in (1.2.5).

Table 5.3 computes both sides. The prime sum converges at every resolution, but more slowly the finer the resolution: since Chebyshev's function $\psi(x) = \sum_{n\leq x}\Lambda(n)$ grows like x, the weight $n^{-1/2}e^{-(\log n)^2/8t}$ acts, as a density in $\log n$, like $e^{t/2}$ times a Gaussian with centre $2t$ and standard deviation $2\sqrt t$. At $t = 3$ (last row) the rest of the prime sum beyond 10^9 is nearly all main term: with the integral $(2\pi t)^{-1/2}\int_N^\infty x^{-1/2}e^{-(\log x)^2/8t}\cos(u\log x)\,dx$ that the prime number theorem predicts for it, the difference at $N = 10^9$ falls to 3×10^{-10}. Where $\mathcal{G}$ is small the two arithmetic terms cancel, the more deeply the finer the resolution, as for W in Table 1.4; these are the points where an off line quadruple would first make $\mathcal{G}$ negative (Section 5.2).

5.1.6 What a verification height buys

Chapters 6 and 7 reach further rungs by the same method. Adding to the clock variable an independent positive stable variable of index $\frac12$, that is, multiplying the clock by $e^{-c\sqrt u}$ with $c = \zeta'(\frac12)/\zeta(\frac12) = 2.6860917$, gives the floor $e^{-c\sqrt u}\,\xi(\frac12)/\xi(\frac12 + \sqrt u)$ of Section 6.2, whose canonical function, y times its Lévy density, is $k(y) =$

u	t	$\mathcal{G}(u,t)$	$A(u,t)$	$P(u,t)$	$\lvert\mathcal{G}-A+P_N\rvert$		
					$N=10^4$	$N=10^6$	$N=10^9$
100	1	0.0960845325	0.5519917891	0.4559072567	2.3×10^{-6}	1.2×10^{-10}	9.6×10^{-19}
1977.22	1.75	1.9835406644	0.8672562847	−1.1162843797	1.2×10^{-3}	1.3×10^{-7}	2.2×10^{-13}
1000	2	0.8452691893	0.7150931047	−0.1301760846	5.8×10^{-4}	5.8×10^{-6}	1.9×10^{-11}
100	3	0.0003053958	0.3186935678	0.3183870527	2.0×10^{-3}	5.4×10^{-4}	1.1×10^{-6}

Table 5.3: The Gaussian smoothing of the zeros from the zeros and from the primes, by (5.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than 10^{-70000}; the polar term and $\mathcal{G}(-u,t)$ are below 10^{-8000}. P_N is P with the sum cut at $n\le N$, and the P column is P_{10^9}. The bound on the rest of the prime sum that ignores its phases, from $\psi(x)<1.03883x$ [291, Theorem 12], is 1.6×10^{-20}, 2.1×10^{-10}, 1.0×10^{-8} and 1.0×10^{-4} at $N=10^9$ for the four rows; in the first row the difference at $N=10^9$ is the rounding error of the sums, in extended precision.

$W(y)+c/(2\sqrt{\pi y})$. By Theorem 5.1.1, $k>W>0$, so the floor is infinitely divisible unconditionally; Section 6.2 proves this independently, and proves that the floor is a GGC if and only if RH holds (Theorem 6.2.15). Weil's functional is positive on the cone of test functions completely monotone in r^2 (Theorem 7.2.14); this is the first rung integrated against a positive weight, $\sum_\rho \hat h(\gamma_\rho)=2\int_0^\infty W(y)g(y)\,dy$ with $g\ge0$, and for it Section 7.2 proves $W>0$ afresh.

Every rung is reached by the same mechanism: near the origin the archimedean term dominates the primes, and away from it a known positive term, usually a verified low zero, dominates the unverified high zeros. Only the second limits the order, as the proof of Theorem 5.1.6 shows. The first limits the time instead: it gives every order $k\ge1$ up to $t=0.019$, and every order above 10^{17} up to $t=1.29$ (Theorem 5.1.9).

THEOREM 5.1.13. *Let $H=3\times10^{12}$, the height to which RH is verified [247]. Then $(-1)^kW^{(k)}(t)>0$ for every $t\ge1.29$ and every integer k with $0\le k\le t(H-23)^2$. Consequently $(-1)^kW^{(k)}(t)>0$ for every $t>0$ and every integer k with $0\le k\le 1.29(H-23)^2$, and $1.29(H-23)^2>1.16\times10^{25}$.*

Proof. Fix $t\ge1.29$ and k, and put $u=\sqrt{k/t}$, $D=H-u\ge23$ and $f(v)=v^{2k}e^{-tv^2}$. As in Section 5.1.5, write each zero as $\frac12+\delta+i\gamma$ and put $\tilde\gamma=\gamma-i\delta$, so that $c_k(t)=\sum f(\tilde\gamma)$ over one zero of each pair $\{\rho,1-\rho\}$, taken with $\gamma>0$. On $(0,\infty)$, $(\log f)''=-2k/v^2-2t$ is at most $-2t$, and at least $-4t$ on $[u,\infty)$, while $(\log f)'(u)=0$; hence $f(v)\le f(u)e^{-t(v-u)^2}$ for every $v>0$, and $f(v)\ge f(u)e^{-2t(v-u)^2}$ for $v\ge u$. The zeros below height H lie on the line and contribute $f(\gamma)>0$. One of them lies in $[u,u+15]$: for $u\le1000$ because $\gamma_1<15$ and consecutive ordinates below 1015 differ by less than 7, and for $u\ge1000$ because the explicit Riemann von Mangoldt formula of [109, (1.7) and Corollary 1.4]

gives $N(u+15)-N(u)>4$. That zero contributes at least $f(u)e^{-450t}$. A zero above H, wherever it lies, has $|\delta|<\frac12$ and $|\tilde\gamma|\ge\gamma>u$, so its term has modulus $f(|\tilde\gamma|)e^{2t\delta^2}\le e^{t/2}f(u)e^{-t(\gamma-u)^2}$. Partial summation with $N(r)\le r^2$, and $\int_D^\infty e^{-tv^2}dv\le e^{-tD^2}/(2tD)$, give

$$\sum_{\gamma>H} e^{-t(\gamma-u)^2}\le e^{-tD^2}\Big(H^2+\frac1t+\frac{H}{tD}\Big),$$

and therefore

$$\frac{c_k(t)}{f(u)}\ge e^{-450t}-e^{t/2-tD^2}\Big(H^2+\frac1t+\frac{H}{tD}\Big)>0,$$

since $t(D^2-450.5)\ge 1.29\cdot 78.5>101$ while $\log(H^2+1+H/23)<57.5$. For the second statement, Theorem 5.1.9(b) gives every order for $t\le 1.29$, and for $t>1.29$ every $k\le 1.29(H-23)^2$ satisfies $k\le t(H-23)^2$. The program `window_orders.py` (`code/README.md`) computes the zeros below height 1015 in ball arithmetic and evaluates every numerical inequality of the proof. □

For the floor, $c/(2\sqrt{\pi y})$ is completely monotone, so Theorem 5.1.13 gives $(-1)^n k^{(n)}>0$ for every $n\le 1.16\times 10^{25}$; Theorem 6.2.17 proves $n\le 10^{10}$ independently, with a Cauchy circle of radius $y/2$, which loses a factor 2^n and makes its reach grow only linearly in H.

5.1.7 A compound Poisson defect of the off line zeros

Group the exponents a_k into the real ones, $a_k=\gamma^2>0$, which belong to zeros on the line, and the conjugate pairs $\{a,\bar a\}$ of nonreal exponents, one for each off line quadruple $\frac12\pm\delta\pm i\gamma$, for which $a=A-iB$ with $A=\gamma^2-\delta^2$ and $B=2\delta\gamma$. Each pair contributes $2e^{-At}\cos(Bt)$ to W, and $\cos\theta=1-2\sin^2(\theta/2)$ splits this into a Laplace transform part and a nonnegative remainder.

Theorem 5.1.14. *Unconditionally, $W(t)=\int_{(0,\infty)}e^{-tz}\,U_+(dz)-R(t)$, where U_+ is a Thorin measure, $R\ge 0$, and*

$$U_+=\sum_{a_k\ real}\delta_{a_k}+2\sum_{pairs}\delta_A,\qquad R(t)=4\sum_{pairs}e^{-At}\sin^2\Big(\frac{Bt}{2}\Big).$$

Let T_+ be the GGC with Thorin measure U_+, and Y the infinitely divisible variable with Lévy density $R(t)/t$, independent of T. Then

$$T+Y\overset{d}{=}T_+,\qquad \mathbb{E}e^{-uY}=\prod_{pairs}\frac{(1+u/a)(1+u/\bar a)}{(1+u/A)^2}.$$

The defect Y is compound Poisson, with rate and jump density

$$\lambda = \int_0^\infty \frac{R(t)}{t}\,dt = \sum_{pairs} \log\Big(1+\frac{B^2}{A^2}\Big), \qquad \frac{R(t)}{\lambda t},$$

and $\lambda < 10^{-12}$, so $\Pr(Y=0) = e^{-\lambda} > 1-10^{-12}$. The following are equivalent: $R \equiv 0$; $Y = 0$; RH. In that case $U_+ = U_{\mathrm{RH}}$ and $T_+ = T$.

Proof. The decomposition is the identity applied pair by pair, and U_+ is a Thorin measure because $A \ge \frac12|a|$, since $|a| = \gamma^2+\delta^2$ and $\gamma > 14$, and $\sum_k |a_k|^{-1} < \infty$. By Tonelli and the Frullani type integral $\int_0^\infty e^{-At}\sin^2(Bt/2)\,t^{-1}dt = \frac14\log(1+B^2/A^2)$, the function $R(t)/t$ is integrable on $(0,\infty)$ with integral λ, which is finite by the bound below, and a Lévy measure of finite mass is compound Poisson. For the bound let $H = 3\times10^{12}$. Every pair has $\gamma > H$, and $\delta < \frac12$ gives $B^2/A^2 = 4\delta^2\gamma^2/(\gamma^2-\delta^2)^2 < \gamma^2/(\gamma^2-\frac14)^2 < (1+10^{-24})\gamma^{-2}$. The two zeros $\frac12 \pm \delta + i\gamma$ of a pair have the same ordinate, so $\lambda < \frac12(1+10^{-24})\sum_{\gamma>H}\gamma^{-2}$, the sum running over all zeros of ordinate above H. Partial summation with the explicit bound for $N(T)$ of [109, Corollary 1.2], exactly as for ε_H in the proof of Proposition 14.7.2, gives $\sum_{\gamma>H}\gamma^{-2} < 1.4798\times10^{-12}$, hence $\lambda < 7.4\times10^{-13}$. The program `xg_defect.py` (`code/README.md`) evaluates this bound in ball arithmetic.

By the formula for $\log G$ at the start of this section and Frullani's integral $\log(1+u/z) = \int_0^\infty (1-e^{-ut})e^{-zt}t^{-1}dt$,

$$\log G(u) = \int_0^\infty \big(1-e^{-ut}\big)\frac{W(t)}{t}\,dt = \int \log\Big(1+\frac{u}{z}\Big)U_+(dz) - \int_0^\infty \big(1-e^{-ut}\big)\frac{R(t)}{t}\,dt,$$

that is, $\mathbb{E}e^{-uT}\,\mathbb{E}e^{-uY} = \mathbb{E}e^{-uT_+}$, and the product formula follows from $\mathbb{E}e^{-uT_+} = \prod_{a_k\ \mathrm{real}}(1+u/a_k)^{-1}\prod_{\mathrm{pairs}}(1+u/A)^{-2}$ and (1.2.1). For the equivalences: every term of R is nonnegative, and a pair contributes a positive term outside the discrete set $\{t>0 : Bt \in 2\pi\mathbb{Z}\}$, so $R \equiv 0$ if and only if there is no pair, that is, every a_k is real, which happens only for zeros on the line; and $Y = 0$ if and only if $\lambda = 0$ if and only if $R \equiv 0$. Under RH, $U_+ = \sum_k \delta_{a_k} = U_{\mathrm{RH}}$ and $T_+ = T$ by Proposition 1.2.1. □

Remark 5.1.15. Unconditionally, therefore, the clock plus an independent compound Poisson variable of rate below 10^{-12} is a GGC, and the law of the clock is within total variation 10^{-12} of that GGC, since $T = T+Y$ on the event $\{Y=0\}$.

5.1.8 The prime side of the ladder

The proofs of Theorems 5.1.9 and 5.1.13 stop at 1.29 and 1.16×10^{25} because of the constant $K = 10^{17}$ with which the proof of Theorem 5.1.9 uses Lemma 5.1.8(b): as the discussion after Theorem 5.1.9 says, 1.2912 is the root of $E_{K+1}(t) = 2S(t)$ for that K. Theorem 5.1.13 supplies every order up to 1.16×10^{25}, and the lemma

can be used with that order in the place of 10^{17}. The root of $E_{K+1}(t) = 2S(t)$ then moves to about 1.719 in floating point, and the argument of both theorems runs at $t = 1.70$.

THEOREM 5.1.16. *W is completely monotone on $(0, 1.70]$, and $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every integer k with $0 \le k \le 1.70(H-23)^2$, where $H = 3\times10^{12}$; and $1.70(H-23)^2 > 1.52 \times 10^{25}$.*

Proof. Let $K = 1.16 \times 10^{25}$. By Theorem 5.1.13, $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every $k \le K$. For $k > K$ and $t \le 0.019$, Theorem 5.1.9(a) applies. For $k > K$ and $t \in [0.019, 1.70]$ the proof of Theorem 5.1.9(b) applies with K in the place of 10^{17} and 1.70 in the place of 1.29, and with the cutoff $n_0 = 10^9$ in the place of 2^{24}: Lemma 5.1.8(b) with $\varepsilon = 10^{-6}$ gives $C_\varepsilon(k) \le C_\varepsilon(K+1) < 1 + 6 \times 10^{-7}$ for $k > K$, and the condition $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds 2.5×10^9, which covers every prime power $n \le 10^9$ at $t \ge 0.019$ (there $x_n < 76$). The monotonicity in k and t is that of the proof, so it suffices to show $E_{K+1}(1.70) - S_C(1.70) - Q_{K+1}(1.70) > 0$, with S_C the sum S_B with $B(x_n)$ replaced by C for $n \le 10^9$. Chernoff's bound for the gamma law with $a = (1 - 2^{-13})^2\lambda$ gives $E_{K+1}(1.70) > 26.7532$; the prime powers $n \le 10^9$ give $S_C(1.70) < 26.2968$, with the tail bound of that proof, which is 2.9×10^{-4} at $q = \log n_0/8t = 1.52$; and $Q_{K+1}(1.70) < e^{-1000}$. The margin exceeds 0.4564. So the orders $k > K$ have the right signs on $(0, 1.70]$ and, with the orders $k \le K$, W is completely monotone there.

For $t > 1.70$ and $k \le 1.70(H-23)^2$ the proof of Theorem 5.1.13 applies with 1.70 in the place of 1.29, since $k \le t(H-23)^2$ and the only place where the lower bound on t enters is $t(D^2 - 450.5) \ge 1.70 \cdot 78.5 > 133$, against $\log(H^2 + 1 + H/23) < 57.5$. Finally $1.70(H-23)^2 > 1.52 \times 10^{25}$. The program `cm_orders_k116e25.py` (`code/README.md`) evaluates every numerical inequality of the argument for $k > K$ in ball arithmetic, with the prime powers $n \le 10^9$ (50 851 223 of them): all of them hold. □

In floating point the root of $E_{K+1}(t) = 2S(t)$ for $K = 1.52 \times 10^{25}$ is about 1.724, which is the time $t_0 = 1.72$ of the phase free bound at the verified height (Section 12.2.6); 1.70 is the safe round number.

By Proposition 5.1.10, on the ray $t = k/u^2$ the rung is, up to the errors of that proposition, $\mathcal{G}(u,t)$, and (5.1.3) writes $\mathcal{G}(u,t) + \mathcal{G}(-u,t)$ as a polar term plus $A(u,t) - P(u,t)$, where $A(u,t)$ is close to $\log(u/2\pi)/(2\sqrt{2\pi t})$ and $P(u,t) = (2\pi t)^{-1/2}\Pi(u;t)$, with $\Pi(T;x) = \sum_{n\ge2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/8x} \cos(T \log n)$ the smoothed prime sum. To leading order, therefore, the rung at height T and scale x is positive when $2\Pi(T;x) < \log(T/2\pi)$, and this is the comparison of the archimedean term with the prime term for which Section 12.2.6 gives the phase free time $t_0(T)$, which agrees to five digits with $x_c(T)$, the root of $M(x) = \log(T/2\pi)$ with $M(x) = 2\sum_{n\ge2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/8x}$ (twice the sum $S(x)$ of that section). Below $x_c(T)$ it holds whatever the phases are; above it, it

is a statement about cancellation among the primes, which Proposition 5.1.11 makes an exact criterion for RH, with A and $\mathcal{G}$ in place of the leading terms.

The comparison can be evaluated at heights far beyond verification, because the cost depends on T only through the precision of $T \log n$ modulo 2π. The table gives $2\Pi(T;x)$ against $\log(T/2\pi)$, with the prime powers $n \le N$, $N = \lfloor e^{\sqrt{8xD}} \rfloor$ for the tolerance $D = 22$ at $x \le 1$, $D = 20$ at $x = 1.7$, $D = 16$ at $x = 2.5$ and $D = 14$ at $x = 3$, and the phases in ball arithmetic at 300 bits. The weight of the omitted prime powers is about $e^{\sqrt{2xD}-D}$ times a factor of order one, not e^{-D}; the Prime Number Theorem estimate of the sum of their absolute values is below 1.5×10^{-2} in every row. The values are not certified, and the leading term $\log(T/2\pi)$ stands for the archimedean term $A(T,x)$, with a relative error $O(T^{-2})$. The condition holds in every row; the scales $x = 1.7$, 2.5 and 3 at $T = 10^{12}$ are above $x_c(10^{12}) = 1.679$ (computed), where the weights alone no longer decide.

T	x	N	$2\Pi(T;x)$	$\log(T/2\pi)$	$\pi/(4x)$
10^{12}	0.5	1.2×10^4	-0.187	25.79	1.57
10^{12}	1.0	5.8×10^5	-0.374	25.79	0.79
10^{12}	1.7	1.5×10^7	-0.142	25.79	0.46
10^{12}	2.5	5.9×10^7	0.630	25.79	0.31
10^{12}	3.0	9.1×10^7	1.279	25.79	0.26
3×10^{12}	1.7	1.5×10^7	-0.266	26.89	0.46
3×10^{12}	3.0	9.1×10^7	-2.713	26.89	0.26
10^{20}	1.0	5.8×10^5	-0.720	44.21	0.79
10^{30}	1.0	5.8×10^5	-0.077	67.24	0.79
10^{50}	1.0	5.8×10^5	0.881	113.29	0.79

The last column is the product $\delta\Delta$ of the displacement and the gap of an isolated off line pair at which, in the local model of Proposition 5.2.4 and to leading order only, a failure of the comparison at the scale x would first be visible, since $t^*_{\mathrm{loc}} = \omega^*/(2\delta\Delta)$ with ω^* just above $\pi/2$ (Section 5.2.4). The failure sits at one offset of the window in the height, so one value of T excludes nothing and a scan in T over about Δ is needed. In this reading the prime side resolves only zeros that are well off the line and isolated, and a pair with $\delta\Delta \ll 1$, such as a Lehmer type near miss, needs a scale $x \gg 1$ at which the cutoff N is out of reach. The program `prime_side_check.py` (`code/README.md`) computes the table and the values of M and x_c quoted above.

5.2 What finite information cannot decide

Every computation in this book is finite: a finite window, finitely many resolutions, finitely many derivatives or matrix blocks. Theorem 5.2.1 replaces two close zeros of ξ on the line by a quadruple off it [252, Chapter 11] and obtains an entire function whose heat trace passes the finite tests of Section 5.1.

5.2.1 A perturbation that passes the finite tests of the heat trace

Theorem 5.2.1 ([249, Theorem D]). (a) *There are ordinates* $1000 < A < B \le 2000$ *of zeros on the critical line, with* $B - A \le 2$, *such that, with* $g = (A+B)/2$ *and* $w = \frac{1}{4} + ig$,

$$F(s) = \frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} \, \frac{(1 - s^2/w^2)(1 - s^2/\bar{w}^2)}{(1 + s^2/A^2)(1 + s^2/B^2)}$$

is an even entire function with the order, type and reflection symmetry of $\xi(\frac{1}{2}+s)$ *and with four zeros off the critical line, and the following hold. The reciprocal clock* $1/F(\sqrt{u})$ *is the Laplace transform of an infinitely divisible, self decomposable law, and its heat trace*

$$W_F(t) = W(t) + 2\,\mathrm{Re}\, e^{-at} - e^{-A^2 t} - e^{-B^2 t}, \qquad a = -w^2 = g^2 - \tfrac{1}{16} - \tfrac{ig}{2},$$

which is t *times its Lévy density, is strictly log convex.* W_F *has the same small time expansion as* W *through the constant term, and* $1/F$ *has the same nearest poles as the reciprocal Xi function.*

(b) *Let* $N \ge 1$ *and* $R \ge T_N$, *where* $T_N = 20000N\sqrt{1+\log N}$ *as in Section 5.1, and suppose that every zero of* ξ *with ordinate in* $(0, R+50]$ *lies on the critical line. Then the pair can instead be chosen with* $R < A \le B \le R+50$ *and* $B - A < 2$ *(if* $A = B$, *a multiple zero supplies both), so that* F *keeps every property listed in (a) and the leading* $N \times N$ *Hankel blocks of* W_F *remain positive definite at every* $t > 0$. *If moreover* $R \ge 10^{12}$, *the derivative signs of Theorem 5.1.6 are also preserved:* $(-1)^k W_F^{(k)}(t) > 0$ *for every* $t > 0$ *and every integer* $0 \le k \le 10^{17}$. *For every* $N \le 3 \times 10^7$ *the verification to height* 3×10^{12} *[247] establishes the hypothesis on the zeros for* $R = \max(T_N, 10^{12})$, *so that such an* F *with both properties exists unconditionally.*

The construction moves the zeros at heights A and B, with their conjugates, to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$. The heat trace changes by the oscillating term $W_F - W$, whose envelope is bounded as in Section 5.1 [249, Proposition 11.1], and at the heights of (b) this change is smaller than the margin by which each of its tests holds. The pair of (b), with the properties of (a) and the Hankel blocks, comes from [249, Theorem 12.5], the derivative signs from [249, Proposition 12.7], which holds for any pair of zeros on the line with $A > 10^{12}$ and $B - A \le 2$, and the unconditional range from [249, Corollary 12.6], since $T_{3\times 10^7} + 50 < 2.6 \times 10^{12}$. Figure 5.3 shows the change for a pair in the range of (a); the pair it moves, near height 1977.22, is the closest pair among the first 1700 zeros, whether or not the spacings are normalised (Section 5.4.2).

Not every finite test is passed. Below any height $T > B$ the argument count of F equals that of ξ, since F loses the zeros at heights A and B and gains two at height g, but these two are off the line, so F has two zeros fewer on the line,

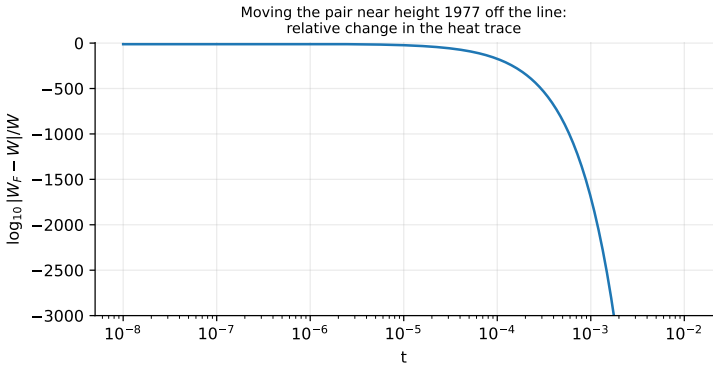


Figure 5.3: Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with W replaced by its sum over the first 1700 zeros. The change is at most 4×10^{-11}, near $t = 7 \times 10^{-7}$, and falls below 10^{-1000} by $t = 10^{-3}$. The sum over 1700 zeros is smaller than W, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with W from the explicit formula (1.2.5) the maximum is 3.66×10^{-11}, at $t = 7.2 \times 10^{-7}$.

counted with multiplicity; when $A < B$ and both zeros are simple, as for the pair of Figure 5.3, it has two sign changes fewer. Turing's comparison of the two counts, the first test of Table 5.5, then detects the perturbation once it passes height B, and the zero density identity at any height $T > g$ returns $\nu(\frac{1}{2}+\varepsilon, T) \geq 1$ for every $\varepsilon < \frac{1}{4}$, whatever the multiplicities.

The pair of Figure 5.3 lies far below the heights of (b), and at its height the derivative signs of F do fail at a finite order, which can be computed. On the ray $t = k/u^2$ the normalised value $e^{-f_0} c_k^F(t)$ of Proposition 5.1.10, whose proof applies to F because F has at most one zero more than ξ below any height, is a sum of positive terms, one for each zero on the line, the oscillating term of the quadruple and terms, negligible here, from any zeros off the line above 3×10^{12}, and only the zeros within a few units of u matter. A scan over u and over the orders, at every order up to 500, then on a grid of ratio 1.002 and at every order of the last grid interval, finds that the first order at which $c_k^F(t) < 0$ for some $t > 0$ is $k = 6\,683\,655$, at $t = 1.7113393$. The grid loses nothing, because $c_{k-1}^F(t) = \int_t^\infty c_k^F(s)\, ds$: if $c_k^F \geq 0$ on $(0, \infty)$ then so is c_{k-1}^F, and the orders that fail are all those from the first one on. The negative value is certified in ball arithmetic, with explicit bounds for the zeros below $g - 40$, which lie on the line, and for those above $g + 40$, wherever the ones above 3×10^{12} lie; the positivity at the lower orders is computed, not certified. Table 5.4 repeats the computation for the closest pair of consecutive zeros in five other windows of height.

window	zeros moved	g	δ	k^*	t^*	$c^F_{k^*}(t^*)$, normalised
$(100, 200)$	71, 72	185.24	$\frac{1}{4}$	40 018	1.18413	-6.37×10^{-7}
$(200, 400)$	186, 187	376.08	$\frac{1}{4}$	219 169	1.54072	-6.70×10^{-8}
$(400, 700)$	363, 364	630.64	$\frac{1}{4}$	624 921	1.576635	-1.68×10^{-7}
$(700, 1000)$	453, 454	750.81	$\frac{1}{4}$	792 746	1.410725	-9.26×10^{-8}
$(1000, 1500)$	922, 923	1329.12	$\frac{1}{4}$	4 433 220	2.5121193	-3.37×10^{-8}
$(1500, 2000)$	1496, 1497	1977.22	$\frac{1}{4}$	6 683 655	1.7113393	-7.12×10^{-9}
$(1500, 2000)$	1496, 1497	1977.22	0.1	15 722 219	4.0257254	-6.71×10^{-11}

Table 5.4: The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, g the midpoint of the pair. k^* is the least order at which $c^F_k(t) < 0$ for some $t > 0$, found by the scan above (computed, not certified); $c^F_{k^*}(t^*) < 0$ is certified in ball arithmetic. The last column is $c^F_{k^*}(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$.

The first failing order is $k^* = t^* u_*^2$, where $u_* = \sqrt{k^*/t^*}$ is the height of the failure, between 0.69 and 1.40 from g, so that k^* is within 1.6 per cent of $t^* g^2$, with t^* between 1.18 and 2.51 for $\delta = \frac{1}{4}$: it grows roughly with the square of the height. Proposition 5.1.10 explains this. Along the ray, $e^{-f_0} c^F_k(t)$ is, up to relative errors of order $t|\tilde{\gamma} - u|^3/u$ in the terms near u, small at these heights and times (Section 5.1.5), the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros of F, in which the quadruple contributes $2e^{2t(\delta^2 - (g-u)^2)} \cos(4t\delta(g - u))$. This term is negative only where $4t\delta|g - u|$ lies in $(\frac{\pi}{2}, \frac{3\pi}{2})$ modulo 2π, hence at distance at least $\pi/(8t\delta)$ from g, where its modulus is at most $2e^{2t\delta^2 - \pi^2/(32t\delta^2)}$, while a zero on the line at distance d from u contributes e^{-2td^2}. The first failure time is set by δ and by the gaps between the line zeros near the moved pair, which its removal widens; the mean spacing $2\pi/\log(g/2\pi)$ falls from 1.86 to 1.09 between the first and last rows, but the gaps near the moved pairs vary more, which is why t^* shows no trend. The first t at which $\min_{|u-g| \le 8} \mathcal{G}(u, t) < 0$, computed in floating point, predicts k^* to 0.21 per cent at height 185 and to 0.08 per cent or better at the other heights, and the minimum lies just beyond the distance $\pi/(8t\delta)$ at which the first negative lobe begins. These failure times of $\mathcal{G}$ within 8 of g are certified in ball arithmetic, with the zeros from Arb, for all seven rows of Table 5.4: $\mathcal{G}(u, t) > 0$ at every centre u within 8 of g for every t up to the failure time rounded down to two decimals, and $\mathcal{G} < 0$ at one centre at the failure time rounded up to four decimals. Far from g the bracket says nothing: near or above 3×10^{12}, zeros off the line could make $\mathcal{G}$ negative earlier. One time suffices for the first statement, since $\sqrt{2t/\pi}\, \mathcal{G}(\cdot, t)$ at a smaller t is its value at a larger one convolved with a centred normal density: the sum over the zeros below 3×10^{12} is positive on the whole line at $t = 1.71$, by an interval cover within 8 of g and by domination beyond, so it has a uniform positive lower bound at every smaller t within 8 of g, and the zeros above 3×10^{12},

wherever they lie, can lower $\mathcal{G}$ there by less than $10^{-2\times10^{12}}$ (`gsmooth_arb.py`); what they add may be large. Unlike the Hankel condition of Theorem 5.1.3 at one time, positivity of $\mathcal{G}$ at one resolution does not decide whether the zeros are on the line. Moving the pair near 1977.22 to distance 0.1 raises t^* from 1.71 to 4.03: the closer the quadruple to the line, the finer the resolution needed to see it.

The same mechanism puts the first failure for a quadruple just above the verified height far beyond the orders of (b). There $g^2 \approx 10^{25}$, so the first failing order is about $10^{25}t^*$, and t^* depends on the gaps around the quadruple, which a computation at height 3×10^{12} would have to supply. The mean spacing there is 0.23, and for $\delta = \frac{1}{4}$ a failure at $t \le 1$ would need a stretch of length 2.9, about twelve mean spacings, free of zeros on the line other than the moved pair. A model with the line zeros spread at their mean density gives $t^* \approx 10$ there, but the same model gives 5.9 at height 1977 against the computed 1.71, because it ignores the gaps, so it gives the order of magnitude only. Theorem 5.1.13 certifies the signs of ξ to every order up to $1.29(H-23)^2 > 1.16\times10^{25}$, with $H = 3\times10^{12}$, so a quadruple of ξ just above H could make a sign fail only at a time $t^* > 1.29$, at an order of about t^*H^2: the orders certified at the time t^* reach that size, and by the model the orders certified at every time fall short of it by a factor of about eight. This is the entry of Table 5.5 on derivative signs.

5.2.2 What finite tests establish, and the local visibility condition

Each entry of the middle column of Table 5.5, made rigorous, is an unconditional theorem on its window; the right column is the reason why no accumulation of such entries proves RH.

The second row rests on a local computation. Suppose that ξ has a pair of zeros $\frac{1}{2} \pm d + i\gamma$ with $d > \varepsilon$, and that the other zeros near height γ lie on the line at the ordinates $\gamma + \delta_j$. A zero ρ contributes $\mathrm{Re}(s-\rho)^{-1}$ to $\pi\rho_\varepsilon(\theta)$, where $s = \frac{1}{2} + \varepsilon + i\theta$, so at the ordinate, ignoring distant zeros and those at negative heights,

$$\pi\rho_\varepsilon(\gamma) = \varepsilon\Big(\sum_j \frac{1}{\delta_j^2+\varepsilon^2} - \frac{2}{d^2-\varepsilon^2}\Big).$$

To first order in ε, the off line zero therefore makes ρ_ε negative at its ordinate if and only if $2/d^2 > \sum_j \delta_j^{-2}$; this is the local visibility condition. For neighbours at the mean spacing $\Delta = 2\pi/\log(\gamma/2\pi)$, that is $\delta_j = j\Delta$ with $j \neq 0$, the sum is $\pi^2/(3\Delta^2)$, and the condition becomes $d < (\sqrt{6}/\pi)\Delta = \sqrt{24}/\log(\gamma/2\pi)$, about 0.78 of the mean spacing. A zero farther from the line leaves ρ_ε positive at its ordinate at every small resolution and makes it negative there only at resolutions comparable to d.

The condition concerns the ordinate itself. For evenly spaced neighbours the neighbour sum is smallest and the pair term most negative there, so the condition also decides whether ρ_ε is negative near γ; in general it does not, because away

test	establishes	cannot rule out
sign changes and argument count to height T (Sections 2.1 and 2.2)	all zeros below T on the line	anything above T
$\rho_\varepsilon \geq 0$ on $0 < \theta \leq \Theta$ (Section 3.1)	no zero in the window at distance $d > \varepsilon$ from the line with $2/d^2 > \sum_j \delta_j^{-2}$ (to first order in ε)	zeros within ε of the line; zeros farther from it than about 0.78 of the mean spacing at their height; zeros above the window
zero density scan at T (Section 3.4)	$\nu(\frac{1}{2}+\varepsilon, T) = 0$ for the ε computed	zeros within ε of the line; zeros above T
Hankel blocks of size N (Section 5.1)	Weil positivity on a finite space	off line zeros above T_N
derivative signs to order K (Section 5.1)	finite order Williamson positivity	off line zeros above a height of about $\sqrt{K/1.29}$
HCM derivative signs (Section 4.1)	finitely many CM conditions	failure at other u or orders
matched total variation at resolution ε (Section 3.2)	agreement with the Poisson smoothing of the list	zeros outside the window
Li coefficients λ_n, $n \leq N$ (Section 8.3)	finitely many of Li's inequalities	off line zeros $\frac{1}{2} \pm d + i\gamma$ with Nd/γ^2 small

Table 5.5: What finite tests establish and what they cannot exclude.

from the ordinate the neighbour sum can fall faster than the pair term grows. With $d = 0.3$, $\varepsilon = 0.005$ and neighbours at $\gamma + 0.2 + m$ and $\gamma - 4 - m$, $m = 0, 1, 2, \ldots$, the sum $\sum_j \delta_j^{-2} = 26.55$ exceeds $2/d^2 = 22.2$ and $\rho_\varepsilon(\gamma) = 0.0069$, yet $\rho_\varepsilon(\gamma - 0.079) = -0.0060$ (Exercise 5.14). A search for negative values of ρ_ε must therefore cover every point of the window, not only the ordinates, and every resolution.

A zero on the line at the ordinate of the pair, $\delta_j = 0$, makes the neighbour sum infinite, and the pair is then invisible to first order at every distance. The exact statement is short.

Proposition 5.2.2 (A pair at the height of a line zero). *Let $d > 0$, $\varepsilon > 0$ with $\varepsilon \neq d$, and $u = (\theta - \gamma)^2$. The zeros $\frac{1}{2} + i\gamma$ and $\frac{1}{2} \pm d + i\gamma$ contribute to $\pi\rho_\varepsilon(\theta)$*

$$\frac{\varepsilon\big(3u^2 + 6\varepsilon^2 u + (3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)\big)}{(\varepsilon^2 + u)\big((d-\varepsilon)^2 + u\big)\big((d+\varepsilon)^2 + u\big)},$$

which is positive at every θ if and only if $\varepsilon < d/\sqrt{3}$ or $\varepsilon > d$. Hence, if every zero off the line of an entire function with the order, symmetry and reality of ξ shares its ordinate with a zero on the line of at least its multiplicity, and the distances d have a positive infimum d_0, its density is positive at every height on every line

$\operatorname{Re} s = \frac{1}{2} + \varepsilon$ *with* $0 < \varepsilon < d_0/\sqrt{3}$*, although zeros lie to the right of these lines.*

Proof. The line zero contributes $\varepsilon/(\varepsilon^2 + u)$ and the pair $2\varepsilon(\varepsilon^2 - d^2 + u)/\big(((\varepsilon - d)^2 + u)((\varepsilon + d)^2 + u)\big)$ (proof of Corollary 11.4.7); over the common denominator the numerator is the one displayed. Its coefficients of u^2 and u are positive, so it is positive for every $u \geq 0$ exactly when its value $(3\varepsilon^2 - d^2)(\varepsilon^2 - d^2)$ at $u = 0$ is, that is when $\varepsilon^2 < d^2/3$ or $\varepsilon^2 > d^2$. For the second statement group each pair, and its conjugate at $-\gamma$, with its line zero; every group and every remaining line zero contributes positively, and the sum converges absolutely, as in the proof of Corollary 11.4.7. □

To first order in ε the contribution is $\varepsilon(3u^2 + d^4)/(u(u + d^2)^2) > 0$ for every d: the polynomial $P(x) = x(x^2 + d^2)$ satisfies the Laguerre inequality $P'^2 - PP'' = 3x^4 + d^4 > 0$ on the whole real line without having only real zeros. Exercise 1.7 gives one such function at one line; the proposition gives a band of lines whose width is fixed by the nearest off line pair, so the Hadamard structure and the functional equation alone cannot make positivity on one line detect zeros to its right.

The zero density identity of Proposition 3.4.2 has no blind spot of this kind: one evaluation at resolution ε returns $\nu(\frac{1}{2} + \varepsilon, T)$, the number of zeros with real part greater than $\frac{1}{2} + \varepsilon$ and ordinate in $(0, T]$, however far from the line they lie. The same count controls the negative part of ρ_ε: the negative kernels in its Poisson form, which come from the zeros to the right of $\frac{1}{2} + \varepsilon$, have total mass on $(0, T]$ at most the number of such zeros below height $2T$ plus a tail that tends to zero as $T \to \infty$ when that number is $O(T^c)$ for some $c < 1$ (Lemma 11.4.1). The zero density estimates of Ingham [124] and of Guth and Maynard [104] give such a bound with no hypothesis and make this mass a vanishing fraction of the positive mass (Theorem 11.4.2).

5.2.3 Refuting sufficient routes with one certificate

Not every route to RH through the Pólya density is a positivity question on the clock. Konstantopoulos, Patie and Sarkar [153] introduced a class $\mathcal{D}_P$ of solutions of the van Dantzig problem, the entire functions $\sum_{n\geq 0}(-1)^n t^{2n}/(\Psi(1)\cdots\Psi(n))$ in which Ψ is the Laplace exponent of a spectrally negative Lévy process whose largest zero q is at most $\frac{1}{2}$; then $\Psi(u) = (u - q)\varphi(u)$ with φ a Bernstein function [153, Proposition 5.2]. If $q = 0$ and φ is a Bernstein Pick function whose zeros and poles satisfy a separation condition, every zero of the entire function is real [153, Theorem 4.4], so a representation of $\Xi/\Xi(0)$ of this kind would prove RH; this is the Bernstein Pick route. Let X have the Pólya density and put $M(s) = \mathbb{E}|X|^{2s}$. Since $\Xi(t)/\Xi(0) = \sum_{n\geq 0}(-1)^n M(n)t^{2n}/(2n)!$, such a representation needs $\Psi(n) = n\varphi_I(n)$ for every $n \geq 1$, where $\varphi_I(s) = 4(s - \frac{1}{2})M(s - 1)/M(s)$ is the canonical interpolant of the moment ratios; that is, φ must interpolate $n\varphi_I(n)/(n - q)$ at

the integers. The function M is analytic for $\operatorname{Re} s > -\frac{1}{2}$, and for $-\frac{1}{2} < \operatorname{Re} s < 0$ the classical formula for negative absolute moments gives

$$M(s) = \frac{2\,\Gamma(2s+1)\sin(-\pi s)}{\pi\,\Xi(0)} \int_0^\infty \Xi(\theta)\,\theta^{-2s-1}\,d\theta,$$

whose right side continues M meromorphically to $\operatorname{Re} s < 0$.

THEOREM 5.2.3 ([249, Theorem E]). *M has exactly one zero in the disc $|s - s_1| < 10^{-8}$ with centre $s_1 = -8.494182256992559 + 2.937767068567584i$; the zero is simple, and $M(s-1)$ does not vanish in the disc. Hence φ_I has a nonreal pole, and every Bernstein interpolant of the moment ratios agrees with φ_I on $(1,\infty)$ and cannot be continued through that pole, so none is a complete Bernstein function. For every $0 < q < 1$ the sequence $n\varphi_I(n)/(n-q)$ has no Bernstein interpolant either.*

Since the Bernstein Pick functions are the complete Bernstein functions, the theorem closes the Bernstein Pick route, and its last statement excludes $\Xi/\Xi(0)$ from $\mathcal{D}_P$ with $0 < q \le \frac{1}{2}$, the case $q > 0$ of the question raised in [153, Section 4.1]. The theorem belongs to the arcsine route of [252, Chapter 12], where its last statement is proved by a beta multiplier obstruction, and at the single remaining value $q = 0$ membership in $\mathcal{D}_P$ reduces to a Hausdorff moment criterion.

5.2.4 A local model for the first failing order

Once the resolution $1/(2\sqrt{t})$ of the Gaussian smoothing (Section 5.1.5) is below the gaps between the line zeros near the moved pair, $c_k^F(t)$ near the quadruple is governed by the quadruple and its two neighbours on the line, and the first failing order becomes explicit.

PROPOSITION 5.2.4. *Let F be obtained as in Table 5.4 by replacing two consecutive zeros of ξ on the line by the quadruple $\frac{1}{2} \pm \delta \pm ig$, g their midpoint, let $g - \Delta_-$ and $g + \Delta_+$ be the ordinates of the nearest zeros of F on the line below and above g, and put $\Delta = \max(\Delta_-, \Delta_+)$. Assume $\delta \ll \Delta \ll g$ and write $\omega = 2k\delta\Delta/g^2$, $\kappa = 2k\Delta^2/g^2$. Restricted to the quadruple and these two zeros, $c_k^F(t) < 0$ for some $t > 0$ if and only if, to leading order,*

$$\min_{0 \le \eta \le 1} \Big[2\cos(\omega\eta) + e^{-\kappa(1-\eta)} \Big] < 0,$$

and the first failing order is

$$k^* = \frac{\omega^* g^2}{2\delta\Delta}, \qquad \omega^* = \frac{\pi}{2} + \frac{\delta}{\Delta}\Big(1 + \log\frac{\Delta}{2\delta}\Big) + O\Big(\frac{\delta^2}{\Delta^2} + \frac{\Delta}{g}\Big),$$

so that $k^* = \frac{\pi}{4}\, g^2/(\delta\Delta)\,(1 + o(1))$ as $\delta/\Delta \to 0$ and $\Delta/g \to 0$. The failure is at the time $t^* = k^*/g^2\,(1 + O(\Delta/g)) = \omega^*/(2\delta\Delta)\,(1 + O(\Delta/g))$, which involves δ and Δ only.

Proof. Put $t = k(1+v)/g^2$ and $a = (g - i\delta)^2$. The quadruple contributes $2\operatorname{Re} a^k e^{-at} = 2|a|^k e^{-t\operatorname{Re} a}\cos\big(2\delta g t - 2k\arctan(\delta/g)\big)$ to $c_k^F(t)$, with $|a|^k e^{-t\operatorname{Re} a} = g^{2k}e^{-tg^2}e^{O(k\delta^2/g^2)}$ and phase $2\delta g k(1+v)/g^2 - 2k\arctan(\delta/g) = 2k\delta v/g + O(k\delta^3/g^3)$. The zero at $g+\Delta_+$ contributes $(g+\Delta_+)^{2k}e^{-t(g+\Delta_+)^2}$, and the logarithm of the ratio of its weight to $g^{2k}e^{-tg^2}$ is $2k\log(1+\Delta_+/g) - t(2g\Delta_+ + \Delta_+^2) = -(2k\Delta_+/g)(\Delta_+/g + v) + O(k\Delta_+^3/g^3)$. Up to a common factor these are, to this order, the quadruple term and the term of a zero on the line of the Gaussian smoothing of Section 5.1.5, at $u = g(1+v)^{-1/2}$. With $v = -\eta\Delta_+/g$ the phase is $-\omega_+\eta$ and the ratio is $e^{-\kappa_+(1-\eta)}$, the subscripted parameters being built from Δ_+; the other neighbour is reached with $v > 0$ and the parameters built from Δ_-. For $\eta > 1$ the neighbour dominates exponentially, so the minimum is attained on $[0,1]$, and on the side of the larger gap, where a given phase is reached at a smaller neighbour weight. Since $k \asymp g^2/(\delta\Delta)$ in the relevant range, the neglected terms are a relative error $O(\delta/\Delta)$ in the weight of the quadruple, $O(\delta^2/(g\Delta))$ in its phase and $O(\kappa\Delta/g)$ in the exponent of the neighbour. With $\kappa/\omega = \Delta/\delta \gg 1$ the minimum is attained at $\eta = 1 - \sigma$ with ω just above $\pi/2$. Writing $\omega = \pi/2 + \theta$, the condition is $2(\theta - \omega\sigma) > e^{-\kappa\sigma}$; optimising $\sigma = \kappa^{-1}\log(\kappa/2\omega)$ gives $\theta > (\omega/\kappa)(1 + \log(\kappa/2\omega))$, the stated ω^*, with $\omega/\kappa = \delta/\Delta$. Because $e^{-\kappa\sigma} = 2\delta/\Delta$ at the optimum, a relative error $O(\delta/\Delta)$ in the weight of the quadruple moves θ by $O(\delta^2/\Delta^2)$ only, an error $O(\kappa\Delta/g)$ in the exponent of the neighbour by $O(\Delta/g)$, and an error in the phase by that error, $O(\delta^2/(g\Delta))$. Finally $t = k(1+v)/g^2$ with $|v| \le \Delta/g$, which gives the time. □

The table at the end of this section compares $t^*_{\mathrm{loc}} = \omega^*/(2\delta\Delta)$, the time of Proposition 5.2.4 without its remainder, with the certified failure times of Table 5.4, row by row, Δ being the larger of the gaps from g to the zeros next to the moved pair, read from the zeros; its last column is $100\,(t^*_{\mathrm{loc}}/t^* - 1)$. The local model is within 2.8 per cent of the certified time in every row, and the order $\omega^* g^2/(2\delta\Delta)$ within 2.9 per cent of k^*. Keeping only the quadruple and its two neighbours, with their exact weights, gives the first failing order of Table 5.4 exactly in four rows and to within 0.03 per cent in the other three (computed, not certified), so the differences of the table belong to the expansion of the proposition and not to the zeros it omits. They are positive because the expansion drops the factor $e^{-2t\delta^2}$ in the weight of the neighbour relative to the quadruple, the amplification $e^{2t\delta^2}$ of the quadruple in Section 5.1.5; with that factor kept, the seven times are within 0.25 per cent of the certified ones. The program `xg_local.py` (`code/README.md`) evaluates the model and the sum over the quadruple and its two neighbours from the zeros.

g	δ	Δ	t^*_{loc}	t^*	difference, per cent
185.24	$\frac{1}{4}$	3.03	1.1896	1.18413	0.46
376.08	$\frac{1}{4}$	2.36	1.5591	1.54072	1.19
630.64	$\frac{1}{4}$	2.31	1.5941	1.576635	1.11
750.81	$\frac{1}{4}$	2.57	1.4230	1.410725	0.87
1329.12	$\frac{1}{4}$	1.49	2.5815	2.5121193	2.76
1977.22	$\frac{1}{4}$	2.14	1.7341	1.7113393	1.33
1977.22	0.1	2.14	4.0339	4.0257254	0.20

5.3 The Thorin functional from the arithmetic

The explicit formula gives the Thorin measure from the gamma factor and the primes without reference to any zero [252, Chapter 7].

5.3.1 The Thorin functional and the explicit formula

The Thorin measure cannot be introduced as a measure without assuming the conclusion. If W were known to be completely monotone, Bernstein's theorem would supply a unique positive measure U with $W(t) = \int e^{-tz} U(dz)$, but complete monotonicity of W is RH. Nor is $\sum_k \delta_{a_k}$ a measure on $[0,\infty)$ unless every a_k is real and nonnegative, which is RH again; an atom at a nonreal a_k can be paired only with functions defined off the real axis. The remedy is to define U as a linear functional on a class of test functions large enough to determine it and small enough that the arithmetic side converges.

Let $\mathcal{T}$ be the set of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic in $|\operatorname{Im} r| \le \frac{1}{2} + c$ for some $c > 0$ and $O((1+|r|)^{-2-c})$ there, and let $g(x) = \frac{1}{2\pi}\int h(r) e^{-irx} dr$. Then φ is holomorphic on the image $\{r^2 : |\operatorname{Im} r| < \frac{1}{2} + c\}$, which contains $-\frac{1}{4}$ and every a_k. Define

$$\begin{aligned} \langle U, \varphi\rangle &= \varphi\Big(-\frac{1}{4}\Big) + \frac{1}{4\pi}\int_{\mathbb{R}} \varphi(r^2)\, m(r)\, dr - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \\ m(r) &= \operatorname{Re}\psi\Big(\frac{1}{4} + \frac{ir}{2}\Big) - \log\pi, \end{aligned} \tag{5.3.1}$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. No zero appears in this definition. Weil's explicit formula [338], [128, Theorem 5.12] identifies the functional: $\langle U, \varphi\rangle = \sum_k \varphi(a_k)$ for every $\varphi \in \mathcal{T}$, the sum running over the exponents $a_k = -(\rho_k - \frac{1}{2})^2$ of Section 1.2, one for each pair $\{\rho, 1-\rho\}$, and in particular $\langle U, e^{-t\cdot}\rangle = W(t)$ [249, Proposition 9.4].

The term $\varphi(-\frac{1}{4}) = h(\frac{i}{2})$ comes from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of ξ removes; it is the only evaluation of φ off $[0,\infty)$. The integral is the contribution of the gamma factor: with the Riemann Siegel theta function $\vartheta(r) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{ir}{2}) - \frac{r}{2}\log\pi$ we have $m = 2\vartheta'$,

so the integral is $\pi^{-1}\int_0^\infty h\,d\vartheta$, the integral of h against the smooth part of the zero counting function. The prime sum is the fluctuating part. The half width $\frac{1}{2}+c$ of the strip makes $g(x)=O(e^{-(1/2+c)|x|})$, so that the prime sum converges absolutely against the weight $n^{-1/2}$.

The sine squared lemma (1.2.9) rewrites the functional in terms of variograms. By (3.5.2) its kernel carries each $\varphi\in\mathcal{T}$ to $g(0)-g(x)$, a variogram when $h\geq 0$, and the functional takes the absolutely convergent form [249, Proposition 9.2]

$$\langle U,\varphi\rangle=\int_0^\infty \frac{e^{-x/2}}{e^{2x}-1}\big(g(0)-g(x)\big)\,dx - C_\Gamma g(0) + \int_0^\infty e^{x/2}g(x)\,dx - \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}g(\log n),$$

where $C_\Gamma=\frac{1}{2}\log\pi-\frac{1}{2}\psi(\frac{1}{4})-2=0.6860917096$ is the archimedean constant of Section 3.1.3. To derive this form, split $\varphi(-\frac{1}{4})=\int_0^\infty(e^{x/2}+e^{-x/2})g(x)\,dx$; the half with $e^{-x/2}$ combines with the gamma integral into the first two terms.

This form regularises a formal pairing. Formally $\langle U,\varphi\rangle=\int_0^\infty\big(g(0)-g(x)\big)\,\nu(dx)$, where ν is the canonical measure of (3.1.1) at $\alpha=\frac{1}{2}$,

$$\nu(dx)=\frac{e^{-x/2}\,dx}{e^{2x}-1}-e^{x/2}\,dx+\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx),$$

but the pole and prime parts of this pairing diverge through their $g(0)$ terms. Part by part, the absolutely convergent form exceeds the formal pairing by multiples of $g(0)$: by $-C_\Gamma$ for the gamma factor, by $\int_0^\infty e^{x/2}dx$ for the pole and by $-\sum_n\Lambda(n)n^{-1/2}$ for the primes. Continued analytically as in Section 3.1.3, the last two are -2 and $\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, and the three constants cancel by the identity $C_\Gamma+2-\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})=0$ of that section, which is equivalent to $\xi'(\frac{1}{2})=0$. The only negative part of ν is the pole density $-e^{x/2}dx$, and Section 3.5.2 reads Weil's criterion from this sign pattern. At the basepoint one the pole density is $-dx$, and no analytic continuation is needed there: the prime atoms compensated by the pole density have a convergent sine squared integral, and its convergence is the prime number theorem (Proposition 9.1.3).

The functional is not in general the limit of the Thorin measures as the basepoint approaches the line. As $\alpha\downarrow\frac{1}{2}$ the measures U_α of (1.2.6) converge vaguely to $U_{\text{line}}=\sum\delta_{\gamma^2}$, the sum running over the zeros on the line, whatever the truth of RH (Section 1.2 and [249, Proposition 9.5]), whereas unconditionally

$$U=U_{\text{line}}+\sum_{a_k\notin(0,\infty)}\delta_{a_k}\quad\text{on }\mathcal{T},$$

the sum running over the nonreal exponents. These come in conjugate pairs $(\gamma-id)^2$ and $(\gamma+id)^2$, one pair for each off line pair $\frac{1}{2}\pm d+i\gamma$ of Proposition 1.2.3

(written $\frac{1}{2} \pm \delta + i\gamma$ there), so $U = U_{\mathrm{line}}$ exactly when RH holds [249, Proposition 9.6]. The functional evaluates φ at both exponents of an off line pair, while in (1.2.8) the pair contributes kernels $P_{\varepsilon+d} + P_{\varepsilon-d}$ of total mass zero once $\varepsilon < d$, which vanish in the limit.

5.3.2 The heat trace from the primes

For $\varphi(z) = e^{-tz}$, that is $h(r) = e^{-tr^2}$ and $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$, the functional is the arithmetic form (1.2.5) of the heat trace, $W = \Pi + A - P$, with the polar term $\Pi = e^{t/4}$, the archimedean term A and the prime term P described in Section 1.2. Table 5.6 extends the comparison of Table 1.4 to smaller and larger t, with the three terms in sixty digit arithmetic (P over $n \le 2 \times 10^5$) and the sum over the first 1700 zeros (`data/zeros_1700.txt`); the last column is the number of decimal digits lost to cancellation, $\log_{10}\{(\Pi + |A| + P)/W\}$. The two columns agree to every digit shown.

t	polar Π	archimedean A	prime P
0.002	1.000500	0.782881	2.6×10^{-26}
0.005	1.001251	−0.464059	7.2×10^{-11}
0.01	1.002503	−0.852797	8.4×10^{-6}
0.02	1.005013	−0.984061	0.002410
0.05	1.012578	−0.954619	0.057914
0.1	1.025315	−0.862510	0.162806
0.15	1.038212	−0.797634	0.240578
0.2	1.051271	−0.749754	0.301517
0.3	1.077884	−0.681852	0.396032

t	$\Pi + A - P$	W from zeros	digits lost
0.002	1.78338120537913	1.78338120537913	0.0
0.005	0.537191964091041	0.537191964091041	0.4
0.01	0.14969801125378	0.14969801125378	1.1
0.02	0.0185412917313195	0.0185412917313195	2.0
0.05	$4.58783504857357 \times 10^{-5}$	$4.58783504857357 \times 10^{-5}$	4.6
0.1	$2.10479979582235 \times 10^{-9}$	$2.10479979582235 \times 10^{-9}$	9.0
0.15	$9.65642094572633 \times 10^{-14}$	$9.65642094572633 \times 10^{-14}$	13.3
0.2	$4.43018218022364 \times 10^{-18}$	$4.43018218022364 \times 10^{-18}$	17.7
0.3	$9.32464654810622 \times 10^{-27}$	$9.32464654810622 \times 10^{-27}$	26.4

Table 5.6: The heat trace from the explicit formula and from the zeros. For $t \ge 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$.

The cancellation follows a simple law. For $t \ge 0.005$ the archimedean term is negative (Exercise 5.8), so $\Pi + |A| + P = 2e^{t/4} - W$ and the digits lost are $\log_{10}(2e^{t/4}/W - 1)$. Since $W = e^{-\gamma_1^2 t}\big(1 + O(e^{-(\gamma_2^2 - \gamma_1^2)t})\big)$, this is

$$\log_{10} 2 + \frac{(\gamma_1^2 + \frac{1}{4})\,t}{\log 10} \approx 0.30 + 86.88\,t$$

up to an exponentially small error, below 2×10^{-5} for $t \geq 0.05$.

The explicit formula holds exactly, and the cancellation costs working precision and nothing more: 26.4 digits at $t = 0.3$, well within sixty digit arithmetic. The zeros give W cheaply at moderate t, the primes only near $t = 0$. Section 11.1 finds the same profile for the Lévy density $-W'$ of the subordinator that drives the clock.

The arithmetic side is cheap where g is concentrated near $x = 0$, so that only small n contribute: at small t for the heat kernel, and only at large resolution for the Poisson kernel $h(r) = P_\varepsilon(\theta - r) + P_\varepsilon(\theta + r)$. That kernel is not in $\mathcal{T}$, but for $\varepsilon > \frac{1}{2}$ its transform $g(x) = \pi^{-1} e^{-\varepsilon|x|} \cos \theta x$ makes the prime sum converge absolutely; for smaller ε the prime side is Selberg's smoothed sum (Section 3.5.1). The spectral side is cheap where the test function is concentrated at the low zeros: at large t, and at small ε on a window. In the proofs of the lower rungs in Section 5.1 the two forms meet at $t = 1/20000$, where the inequalities are checked in interval arithmetic. The Thorin condition requires positivity at every scale, and neither side is cheap at every scale. The Pólya route of Section 2.1 is, but only on a window of finite height, and that restriction is the barrier of Section 5.2.

5.4 Spacings, irregularity and the cell distance

On a window of finite height the errors of the Pólya route, measured by the cell distance $d_{\mathrm{cell}}(\varepsilon)$ of Section 3.2, are set by the spacings of the zeros, which follow the random matrix prediction already for the first 1700 zeros.

5.4.1 The cell distance to first order

We use the cells $C_k = (b_{k-1}, b_k]$ of Section 3.2.2, with $\Theta = 200$ and $K = N(\Theta) = 79$, and the first order formula for $\mu_\varepsilon(C_k) - 1$ derived there, whose remainder is $O(\varepsilon^3)$ (Remark 3.2.3); Exercise 5.10 gives its leading term. The formula needs only that the zeros below Θ be simple and on the line.

Under RH the formula becomes a sum over the ordinates. By Proposition 1.2.3, $\rho_\varepsilon(\theta) = \sum_j [P_\varepsilon(\theta - \gamma_j) + P_\varepsilon(\theta + \gamma_j)]$ over all positive ordinates, and the kernel $P_\varepsilon(\cdot - \gamma)$ has distribution function $F_\varepsilon(y - \gamma)$, where $F_\varepsilon(y) = \frac{1}{2} + \frac{1}{\pi} \arctan(y/\varepsilon) = \mathbf{1}_{y \geq 0} - \varepsilon/(\pi y) + O(\varepsilon^3/|y|^3)$ as $\varepsilon/|y| \to 0$. No ordinate lies on a cell boundary, so

$$\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi} \sum_j \Big[\frac{1}{b_k - \gamma_j} - \frac{1}{b_{k-1} - \gamma_j} + \frac{1}{b_k + \gamma_j} - \frac{1}{b_{k-1} + \gamma_j} \Big] + O(\varepsilon^3),$$

and by (1.2.1) with $s = i\theta$, $\Xi'/\Xi(\theta) = \sum_j [(\theta - \gamma_j)^{-1} + (\theta + \gamma_j)^{-1}]$ under RH, so the two expansions agree. Summing absolute values,

$$d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\, \varepsilon + O(\varepsilon^3),$$
$$\kappa_{\mathrm{cell}} = \frac{1}{\pi} \sum_k \Big| \sum_j \Big[\frac{1}{b_k - \gamma_j} - \frac{1}{b_{k-1} - \gamma_j} + \frac{1}{b_k + \gamma_j} - \frac{1}{b_{k-1} + \gamma_j} \Big] \Big|.$$

Truncated to the first 1700 ordinates, the zero sum gives 5.415, against $\kappa_{\mathrm{cell}} = 5.43665$ from the form in Ξ'/Ξ and the observed slope $d_{\mathrm{cell}}(0.005)/0.005 = 5.43664$. The difference of 0.4 per cent between 5.415 and 5.43665 is the contribution of the zeros above γ_{1700}. In the truncated formula the cell errors range from $-0.225\,\varepsilon$ to $+0.250\,\varepsilon$ (Figure 5.4), with the two largest positive values at the end cells: at the last because Θ is not a midpoint between ordinates, and at the first because below b_1 there is only γ_1 and then the reflections $-\gamma_j$, which do not balance the zeros above b_1.

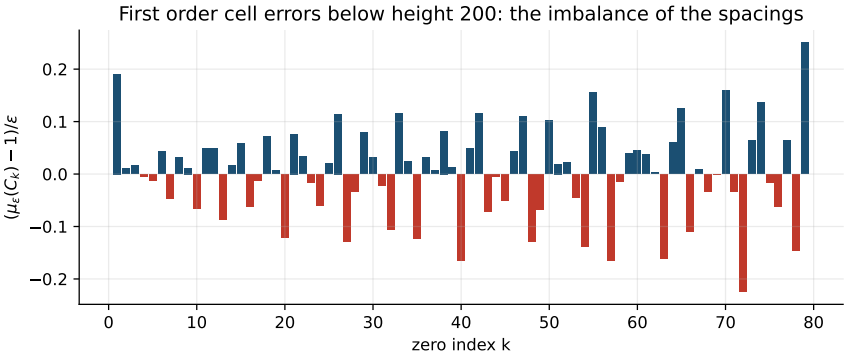


Figure 5.4: First order cell errors $(\mu_\varepsilon(C_k)-1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates.

The atoms $\theta = k \geq 1$ of the sinh benchmark of Section 1.3 and their reflections form the lattice $\mathbb{Z}$ without the point 0. For the full lattice every cell $(k-\frac{1}{2}, k+\frac{1}{2}]$ would have mass exactly one, by periodicity, so the deficit of each cell is the mass that the missing kernel at 0 would have put in it; the atom free interval $(0, \frac{1}{2}]$, which the benchmark counts as a cell of mass zero, receives the mass $\frac{1}{2}$ of the full lattice less that of the missing kernel:

$$\mu_\varepsilon\big((k-\tfrac{1}{2}, k+\tfrac{1}{2}]\big) = 1 - \frac{1}{\pi}\arctan\frac{\varepsilon}{\varepsilon^2+k^2-\frac{1}{4}}, \qquad \mu_\varepsilon\big((0,\tfrac{1}{2}]\big) = \frac{1}{\pi}\arctan(2\varepsilon).$$

The interior cells therefore do not balance: each loses $\varepsilon/(\pi(k^2-\frac{1}{4})) + O(\varepsilon^3)$, while $(0, \frac{1}{2}]$ gains $2\varepsilon/\pi + O(\varepsilon^3)$. As $\sum_{k\leq N}(k^2-\frac{1}{4})^{-1} = 2 - 1/(N+\frac{1}{2})$, the cell distance of N atoms is $(4 - 1/(N+\frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$, which is $1.26694\,\varepsilon$ for $N = 50$, the constant 1.267 of Section 1.3. With the convention used for the zeros, a first cell $(0, \frac{3}{2}]$, it would be $(\frac{4}{3} - 1/(N+\frac{1}{2}))/\pi = 0.418$. For the zeros no cell is balanced, and the errors in Figure 5.4 do not decay along the window.

The constant vanishes exactly when the sum of $1/(b-x)$ over the ordinates and their reflections vanishes at every cell boundary b, as it does when these points are symmetric about every boundary. The lattice $\gamma_j = (j-\frac{1}{2})h$, whose reflections complete $(\mathbb{Z}+\frac{1}{2})h$, has every cell error zero, whereas the lattice of the benchmark

does not. For the zeros every ordinate contributes to a cell error, with a weight that decays like $|C_k|/(\gamma_j - b_k)^2$. The two ordinates nearest an interior boundary are symmetric about it, so the asymmetry comes first from the next gaps on either side, and it is small when those gaps are equal.

5.4.2 Spacings of the zeros and the Thorin density

At height γ the mean spacing of the zeros is $\Delta = 2\pi/\log(\gamma/2\pi)$, as in Section 5.2.2. We unfold by $\tilde{\gamma} = \vartheta(\gamma)/\pi$. The 1699 normalised spacings $\tilde{\gamma}_{k+1} - \tilde{\gamma}_k$ of the first 1700 zeros, which lie below height 2198, have variance 0.146 and minimum 0.089. Figure 5.5 compares their histogram with the Wigner surmise for the Gaussian unitary ensemble, $p_{\mathrm{W}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, and with the Poisson law e^{-s} of independent points. The histogram follows the surmise already at this height, including the quadratic repulsion at small spacings that Montgomery's pair correlation conjecture predicts [203, pp. 183 and 184] and that Odlyzko observed at large height [219]. The variance is below the surmise value $3\pi/8 - 1 = 0.178$; the spacing distribution approaches the unitary one only slowly with height [203, p. 185], [35, Section 3].

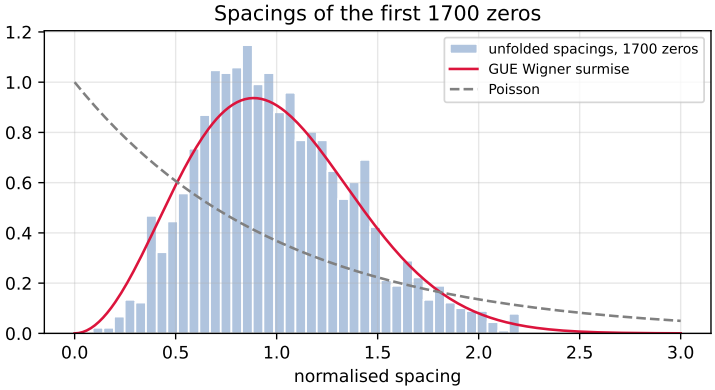


Figure 5.5: The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law.

Repulsion affects the Thorin density in two ways. First, it makes very small gaps rare but not absent. The smallest normalised spacing among the first 1700 zeros, 0.089, separates the ordinates 1977.17 and 1977.27, the pair that Section 5.2 moves off the line in Figure 5.3. By the Cauchy kernel criterion of Exercise 2.11, an isolated pair of ordinates at distance g in θ gives two peaks of ρ_ε only for $\varepsilon < \sqrt{3}g/2$. The resolution needed to separate every pair on a window is therefore set by the smallest gap, and repulsion controls it: if the N normalised spacings were independent, with density $p_{\mathrm{W}}(s) \propto s^2$ near zero, the smallest would be of order $N^{-1/3}$, which is 0.084 for $N = 1699$, close to the observed 0.089.

Second, repulsion sets the scale of the local visibility condition of Section 5.2.2: to first order in ε, an off line zero at distance d from the line makes ρ_ε negative at its ordinate if and only if $2/d^2$ exceeds $\sum_j \delta_j^{-2}$, where δ_j are the distances from its ordinate to the other ordinates. The sum is dominated by the nearest neighbours, and repulsion makes it typically of order Δ^{-2}, because $\mathbb{E}s^{-2}$ is finite for the surmise, where it equals $8/\pi$, and infinite for the Poisson law. With neighbours on a lattice of spacing Δ, the density at the ordinate stays positive at small resolution once d exceeds the threshold $(\sqrt{6}/\pi)\Delta \approx 0.78\,\Delta$ of that section. Since $d < \frac{1}{2}$, a zero hidden in this way can exist only where $\Delta < 0.64$, that is above height about 1.1×10^5. Hidden here means hidden at the ordinate: beside it the density can be negative even when the condition predicts a positive value there (Exercise 5.14).

The behaviour of the cell constant at large height is open (Section 12.2). In unfolded units the cell errors depend mainly on the nearby ordinates, so the sine kernel process of the unitary ensemble would predict their mean and fluctuations per cell; we have not computed that prediction. Computing the cell errors with Odlyzko's zeros near zero number 10^{12} [219], 10^{21} or 10^{22} [223, 224], at heights about 2.7×10^{11}, 1.4×10^{20} and 1.4×10^{21}, would compare the Thorin density with the spacing statistics of the zeros at large height. It needs the ordinates, which define the cells, and the hybrid evaluator of Section 2.2.2 for Ξ'/Ξ at the cell boundaries. Section 11.4.5 proposes the same comparison at large height for the positivity margins of the density, its values at the midpoints between consecutive ordinates.

5.5 Exercises

On Section 5.1.

5.1. Show that $c_0c_2 - c_1^2 = \frac{1}{2}\sum_{j,l}(a_j - a_l)^2 e^{-(a_j+a_l)t}$ and deduce that log convexity of W holds under RH.

5.2. Recompute the first row of Table 5.1 with the first 100 zeros and explain why no printed digit changes.

5.3. Show that for a real exponent a the summand of the Williamson measure U_k belonging to a is the inverse gamma density of ka/Γ_k with $\Gamma_k \sim \mathrm{Gamma}(k,1)$, and compute its variance for $k > 2$.

5.4. Let $\widetilde{W}(t) = e^{-t} + 2e^{-10t}\cos t$, the heat trace of the three exponents 1 and $10 \pm i$. Show that $\widetilde{W} > 0$ and $\widetilde{W}' < 0$ on $(0,\infty)$, using $\cos t > 0$ and $\sin t \geq 0$ for $t \leq 1$ and $2\sqrt{101}\,e^{-9} < 1$ for $t \geq 1$, so that the first two rungs of Theorem 5.1.1 hold. With $\theta = \arctan\frac{1}{10}$, show that $\frac{10}{101} < \theta < \frac{1}{10}$, hence $\frac{2\pi}{3} < 24\theta < \pi$, and deduce that $(-1)^{24}\widetilde{W}^{(24)}(0) = 1 + 2\cdot 101^{12}\cos 24\theta < 0$,

so that $\widetilde{W}$ is not completely monotone. Then check numerically that the first order k at which $(-1)^k\widetilde{W}^{(k)}(t) < 0$ for some $t > 0$ is 12, near $t = 2.87$, and compare with the first failures of Table 5.4.

5.5. Show that the term of order $1/u$ of ε_ρ in Proposition 5.1.10 is $\frac{2t}{3u}(\tilde\gamma - u)^3$. For $u = 2$, $t = 1$ and $k = 4$, show that no zero lies in the disc $|\tilde\gamma - u| \le u/2$, so that $c_4(1) = e^{f_0}R$ is carried entirely by the remainder, and compare the bound on R with the contribution of the first zero.

5.6. Show that $m(0) > -6$ and, comparing the series for ψ with an integral, that $m(60) > 1$. Deduce that for $t \le 10^{-6}$ the archimedean term of (1.2.5) is at least $1/(4\sqrt{\pi t}) - 210/\pi > 55$ and the prime term at most $(\pi t)^{-1/2}e^{-(\log 2)^2/4t} < 0.01$. With the argument of the proof of Theorem 5.1.1 for $t \ge \tau$, taken at height 10^4 and from $t = 10^{-6}$, conclude that $W > 0$ on $(0, \infty)$ follows from the verification of RH below height 10^4 by estimates that can all be checked by hand.

On Section 5.3.

5.7. Verify the arithmetic form of $W(t)$ at $t = 0.01$ with the prime powers up to 10^4, and explain why so few suffice by showing that only $n = 2$ and $n = 3$ affect the first fifteen digits.

5.8. Show that m increases on $[0, \infty)$ and vanishes at a single point r_*, and deduce that $e^{r_*^2 t}A(t)$ decreases in t, so that A changes sign at most once; with Table 5.6, conclude that $A < 0$ for $t \ge 0.005$. Then show that for $t \ge 0.05$ the number of digits lost in the table equals $\log_{10} 2 + (\gamma_1^2 + \frac14)t/\log 10$ up to an exponentially small error, and identify the source of each term.

5.9. Show that the regularised constants of the gamma, pole and prime parts of the variogram form sum to zero if and only if $\xi'(\frac12) = 0$.

On Section 5.4.

5.10. Let $\ell = \Xi'/\Xi$. Show that $\xi(\frac12 + \varepsilon + ib) = \Xi(b - i\varepsilon)$, and deduce from the Taylor series of $\log \Xi$ about b that $\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi}[\ell(b_k) - \ell(b_{k-1})] + \frac{\varepsilon^3}{6\pi}[\ell''(b_k) - \ell''(b_{k-1})] + O(\varepsilon^5)$. Under RH, check that the term in ε^3 agrees with the expansion of F_ε to third order.

5.11. Prove the arctangent formulas for the cell masses of the sinh benchmark, and deduce that its cell distance for N atoms is $(4 - 1/(N + \frac12))\,\varepsilon/\pi + O(\varepsilon^3)$, and $(\frac43 - 1/(N + \frac12))\,\varepsilon/\pi + O(\varepsilon^3)$ when the first cell is $(0, \frac32]$.

5.12. Show that the Wigner surmise $p_{\mathrm W}$ has mean 1 and variance $3\pi/8 - 1$, and that $\mathbb{E}s^{-2} = 8/\pi$ under it, while $\mathbb{E}s^{-2} = \infty$ for the Poisson law.

5.13. Show that $\sum_{m\neq 0}(m\Delta)^{-2} = \pi^2/(3\Delta^2)$, deduce the lattice threshold $(\sqrt{6}/\pi)\Delta$ of the visibility condition, and find the height above which the threshold is less than $\frac{1}{2}$.

5.14. In a local model, put an off line pair at distance $d = 0.3$ from the line at the ordinate γ and zeros on the line at $\gamma+0.2+m$ and $\gamma-4-m$ for $m \geq 0$, and ignore the reflected kernels. At $\varepsilon = 0.005$, show that $2/d^2 < \sum_j \delta_j^{-2}$, so that the visibility condition predicts a positive density at γ, and check that $\rho_\varepsilon(\gamma) > 0$ while $\rho_\varepsilon(\gamma - 0.079) < 0$.

Chapter 6

The reciprocal zeta function

The reciprocal zeta function $1/\zeta(\frac{1}{2}+\sqrt{s})$ is, up to the factor $1-4s$ of the pole and a constant, the reciprocal Xi function times the gamma factor, and unlike ξ it has a Dirichlet series and an Euler product, from which we read its Thorin type measures.

Sections 6.1 and 6.2 compute the Thorin measure of the gamma factor, the Thorin measures of the normalised reciprocal zeta function and of the reciprocal Xi function under RH, and, unconditionally, the formal Thorin measures of these two functions in the Vere-Jones form of Section 6.1.3. After the stable tilt $e^{-c\sqrt{s}}$, with $c=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, the reciprocal of the gamma factor is a generalised gamma convolution, and multiplying the normalised reciprocal zeta function by it gives the floor $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ of the reciprocal Xi function (Section 6.2.4), in which the zeros are the only possible source of negativity.

The infinite divisibility of the floor, as proved here, rests on the positivity of the canonical function of the normalised reciprocal zeta function, which is Theorem 7.1.1 of Chapter 7. It also follows from the infinite divisibility of the clock (Theorem 5.1.1); what the floor adds is a splitting of its canonical function into an arithmetic part and a completely monotone archimedean part. The floor is a generalised gamma convolution exactly when RH holds, and the derivatives of its canonical function alternate in sign up to order 10^{10} by the methods of this chapter and up to order 1.16×10^{25} by the estimates of Section 5.1.

6.1 Thorin type forms of the reciprocal zeta function

Under RH the Thorin measure of the normalised reciprocal zeta function is dM, where $M-1$ is the argument function $S(\sqrt{t})$. Its heat trace, the canonical function w of Theorem 6.1.11, is explicit in the primes, is an absolutely convergent series of Jacobi theta functions, and gives a criterion for RH (Theorem 6.1.15).

6.1.1 The question and the three functions

Write $\sqrt{s}$ for the principal square root on the cut plane $\mathbb{C} \setminus (-\infty, 0]$, so that $\operatorname{Re}\sqrt{s} > 0$ there, and evaluate ζ at $z = \frac{1}{2} + \sqrt{s}$. A zero $\rho = \frac{1}{2} + i\gamma$ then corresponds to $\sqrt{s} = i\gamma$, that is to $s = -\gamma^2$, a point of the cut, and an atom of a Thorin measure at t produces a singularity at $s = -t$, also on the cut. The question of this chapter and of Chapter 7 is whether the zeros, the primes and the gamma factor assemble into a Thorin measure, and if so which one. The two chapters have their own coordinates, which hold throughout both unless stated otherwise: s is the variable u of Sections 1.2 and 1.3, t is the variable z of their Thorin measures, x is the time of the heat trace $W(x)$, τ is the ordinate, and z is the argument of ζ and ξ.

The Möbius series and the Euler product converge absolutely to the right of the parabola $x = \frac{1}{4} - y^2$, on

$$H_{1/2} = \{s : \operatorname{Re}\sqrt{s} > \tfrac{1}{2}\} = \{x + iy : \ x > \tfrac{1}{4} - y^2\},$$

and there

$$\frac{1}{\zeta(\frac{1}{2} + \sqrt{s})} = \sum_{n \geq 1} \frac{\mu(n)}{\sqrt{n}}\, e^{-\sqrt{s}\log n} = \prod_p \Big(1 - p^{-1/2} e^{-\sqrt{s}\log p}\Big). \tag{6.1.1}$$

Every term is built from $e^{-\lambda\sqrt{s}}$, which is completely monotone in s: it is the Laplace transform of the positive stable law of index $\frac{1}{2}$,

$$e^{-\lambda\sqrt{s}} = \int_0^\infty e^{-sx}\, \frac{\lambda}{2\sqrt{\pi}}\, x^{-3/2} e^{-\lambda^2/4x}\, dx. \tag{6.1.2}$$

The Thorin measure, however, is read from the logarithm. Recall from (1.3.1) that a generalised gamma convolution (GGC) has Laplace transform

$$f(s) = \exp\Big(-as - \int_{(0,\infty)} \log\big(1 + \tfrac{s}{t}\big)\, U(dt)\Big), \tag{6.1.3}$$

with $a \geq 0$ and $U \geq 0$ the Thorin measure; equivalently $-f'/f(s) = a + \int U(dt)/(s+t)$ is a Stieltjes function [39]. We allow U to be signed, with $\int_{(0,1)} |\log t|\, |U|(dt) + \int_{[1,\infty)} t^{-1} |U|(dt) < \infty$ so that the integral converges, and then call (6.1.3) a Thorin type representation.

Three functions of s run through both chapters. The function $1/\zeta(\frac{1}{2} + \sqrt{s})$ vanishes at $s = \frac{1}{4}$, where ζ has its pole, and equals $1/\zeta(\frac{1}{2}) < 0$ at $s = 0$, so it is not the Laplace transform of a probability law. Both defects are removed by the normalised reciprocal zeta function

$$\varphi(s) = \frac{\zeta(\frac{1}{2})}{(1 - 4s)\, \zeta(\frac{1}{2} + \sqrt{s})}, \qquad \varphi(0) = 1. \tag{6.1.4}$$

The other two are the gamma factor $G(s) = \Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$ and the reciprocal Xi function $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, and $\varphi = RG$ (Proposition 6.1.9). In both chapters G is the gamma factor, not the function G of Section 1.2, whose reciprocal is R, except in Section 7.1.3, where G is a Laguerre Pólya function; and $\theta(\tau) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{i\tau}{2}) - \frac{\tau}{2}\log\pi$ is the Riemann Siegel theta function, written ϑ elsewhere in the book (Section 2.2), so that $\theta'(\tau) = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2}) - \frac{1}{2}\log\pi$ [81, Section 6.5]. Table 6.1 lists the Thorin measures of the three functions and of the building blocks used below. In it $N(T)$ counts the zeros with $0 < \gamma \le T$, $S(T) = \pi^{-1}\arg\zeta(\frac{1}{2} + iT)$ is the argument function, $c_n = 2n + \frac{1}{2}$, $c = -\theta'(0)$ is the constant of (6.1.6), and U_λ is the sawtooth measure of Lemma 6.1.7.

Function	Thorin measure	Sign
φ	$dM(t)$, $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ (RH)	signed
G	$-\dfrac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}dt = \dfrac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\ge0}\Big(\dfrac{c_n}{t + c_n^2} - \dfrac{1}{2n+2}\Big)\Big\}dt$	signed
R	$\sum_{\gamma>0}\delta_{\gamma^2}$ (RH)	positive
$2/(1 + e^{-\lambda\sqrt{s}})$	$-U_\lambda = \sum_k \delta_{((2k+1)\pi/\lambda)^2} - \dfrac{\lambda}{4\pi\sqrt{t}}dt$	signed
$(1 + \sqrt{s}/\kappa)^{-1}$	$\dfrac{\kappa}{2\pi\sqrt{t}\,(t + \kappa^2)}dt$	positive
$\operatorname{sech}(\lambda\sqrt{s})$	$\sum_k \delta_{((k+\frac{1}{2})\pi/\lambda)^2}$	positive
$e^{-c\sqrt{s}}/G$	$\dfrac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}}dt$	positive

Table 6.1: Thorin measures of φ, G and R and of the building blocks used in this chapter and in Chapter 7, with the convention $\log f(s) = -\int \log(1 + s/t)\,U(dt)$ after normalising $f(0) = 1$. Here $b = \frac{1}{2}(\gamma_E + \log\pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$. The rows for φ and R are Theorem 6.1.10, the row for G is Theorem 6.2.1, the fourth row is Lemma 6.1.7, the fifth and sixth appear in Table 1.8, and the last is Theorem 6.2.12.

6.1.2 Stieltjes inversion and the Euler factors

Lemma 6.1.1 recovers the Thorin measure U of a function F on the cut plane from the boundary values of $-F'/F$ on $(-\infty, 0]$ and computes most of the Thorin measures of this chapter and of Chapter 7.

Lemma 6.1.1. *Let f be analytic on $\mathbb{C} \setminus (-\infty, 0]$ with $f(\bar{s}) = \overline{f(s)}$, $f(s) \to 0$ as $|s| \to \infty$ uniformly in $\arg s$, $|f(s)| \le C|s|^{-\alpha}$ near 0 for some $\alpha < 1$, and such that f extends continuously from the upper half plane to the points of $(-\infty, 0)$ apart from finitely many points $-t_0$, at each of which $f(s) - r/(s + t_0)$ is bounded near $-t_0$ in $\mathbb{C} \setminus (-\infty, 0]$ for some real r; $f(-t + i0)$ denotes the values of this extension*

and $f(-t-i0)=\overline{f(-t+i0)}$. *Then*

$$f(s)=\int_{(0,\infty)}\frac{m(dt)}{s+t},\qquad m(dt)=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)\,dt,$$

where the integral is the limit of the integrals over $(0,T]$ *as* $T\to\infty$, *absolutely convergent when* $\int_{(1,\infty)}t^{-1}|m|(dt)<\infty$, *and where each of the points* $-t_0$, *which we call a simple pole of* f, *contributes the atom* $r\,\delta_{t_0}$ *to* m.

Proof. Fix $s\in\mathbb{C}\backslash(-\infty,0]$ and integrate $f(z)/(z-s)$ over a keyhole contour around $(-\infty,0]$, which by Cauchy's formula gives $2\pi i\,f(s)$; the large circle vanishes by the decay at infinity and the small circle by the bound at the origin. By the continuity of the extension, the integrals along the two sides of the keyhole converge to the integrals of the boundary values as the keyhole closes onto the cut, after subtracting $r/(z+t_0)$ near each pole $-t_0$, where the remainder is bounded. The two sides of the cut combine to $\frac{1}{2\pi i}\{f(-t-i0)-f(-t+i0)\}=-\frac{1}{\pi}\operatorname{Im} f(-t+i0)$. At a simple pole the contour follows small semicircles above and below the cut, which together contribute $r/(s+t_0)$. □

The lemma excludes functions with infinitely many poles on the cut, such as $-R'/R$ under RH, which has a pole at every $-\gamma^2$; these need a direct argument from Hadamard's product, as in the proof of Theorem 6.1.10.

For $f(s)=\lambda e^{-\lambda\sqrt{s}}/(2\sqrt{s})=-\frac{d}{ds}e^{-\lambda\sqrt{s}}$ one has $\sqrt{-t+i0}=i\sqrt{t}$, and the lemma gives the density $\frac{\lambda}{2\pi\sqrt{t}}\cos(\lambda\sqrt{t})$, so that

$$e^{-\lambda\sqrt{s}}=1-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\frac{\lambda\cos(\lambda\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

This is not the Thorin measure of $e^{-\lambda\sqrt{s}}$, which is the positive density $\lambda/(2\pi\sqrt{t})$ (Table 1.8), but the Thorin type density that an exponential contributes when it sits in an exponent, as it does in the logarithm of every Euler factor of (6.1.1) and in $\log\zeta(\frac12+\sqrt{s})$.

Proposition 6.1.2. *For* $0<q<1$ *and* $\lambda>0$ *the Euler factor* $E(s)=1-qe^{-\lambda\sqrt{s}}$ *is analytic and zero free on* $\mathbb{C}\setminus(-\infty,0]$, *and*

$$\log\frac{E(s)}{1-q}=-\int_0^\infty\log\bigl(1+\tfrac{s}{t}\bigr)\,u(t)\,dt,$$

$$u(t)=-\frac{q\lambda}{2\pi\sqrt{t}}\,\frac{\cos(\lambda\sqrt{t})-q}{1-2q\cos(\lambda\sqrt{t})+q^2}=-\frac{\lambda}{2\pi\sqrt{t}}\sum_{k\ge1}q^k\cos(k\lambda\sqrt{t}).$$

Proof. On $\mathbb{C}\setminus(-\infty,0]$ we have $|e^{\lambda\sqrt{s}}|>1>q$, so E has no zeros. Put $f=-E'/E=-\frac{q\lambda}{2\sqrt{s}}(e^{\lambda\sqrt{s}}-q)^{-1}$, which is $O(|s|^{-1/2})$ at 0 and decays at infinity.

At $\sqrt{s} = i\tau$, $\tau = \sqrt{t}$, we get $f = \frac{iq\lambda}{2\tau}(e^{i\lambda\tau} - q)^{-1}$ and $\operatorname{Im} f = \frac{q\lambda}{2\tau}\,\frac{\cos\lambda\tau - q}{1 - 2q\cos\lambda\tau + q^2}$. Lemma 6.1.1 gives u; integrating $-E'/E = \int u(t)\,dt/(s+t)$ from 0 to s gives the logarithm, and the series is the Poisson kernel expansion. □

With $q = p^{-1/2}$ and $\lambda = \log p$, summing u over p formally gives

$$u_\zeta(t) = -\frac{1}{2\pi\sqrt{t}} \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} \cos(\sqrt{t}\log n) \quad \text{“}=\text{”} \quad \frac{1}{2\pi\sqrt{t}} \operatorname{Re}\frac{\zeta'}{\zeta}\big(\tfrac12 + i\sqrt{t}\big), \qquad (6.1.5)$$

a divergent series. Since $\frac{d}{d\tau}\arg\zeta(\frac12 + i\tau) = \operatorname{Re}\frac{\zeta'}{\zeta}(\frac12 + i\tau)$, the right side is formally the density of $dS(\sqrt{t})$, but the series converges only in the sense of distributions. Theorem 6.1.10 makes this precise.

6.1.3 The Lai and Vere-Jones identity and sawtooth measures

We trade each exponential for a hyperbolic secant, whose Thorin measure is a lattice of unit atoms, at the price of a Möbius inversion.

Lai and Vere-Jones [162] pose the completeness of the dilates $\operatorname{sech}(\theta x)$, $\theta \ge 0$, as an open problem for the hyperbolic secant law. The Möbius inversion of the alternating exponential series of the secant expands e^{-x} in the dilates $\operatorname{sech}(mx)$. This is Proposition 6.1.3, which we call the Lai and Vere-Jones identity, and the expansions of $1/\zeta$ and $\log\zeta$ built from it form the Vere-Jones representation. Here χ_4 is the nontrivial character modulo 4.

Proposition 6.1.3. *For* $\operatorname{Re} x > 0$,

$$\tfrac12 \operatorname{sech} x = \sum_{m \text{ odd}} \chi_4(m)\, e^{-mx}, \qquad e^{-x} = \tfrac12 \sum_{m\ge 1} \mu(m)\chi_4(m)\, \operatorname{sech}(mx).$$

Proof. The first is the geometric series $e^{-x}/(1+e^{-2x})$. Since χ_4 is completely multiplicative, $\sum_m \mu(m)\chi_4(m) \sum_k \chi_4(k) e^{-mkx} = \sum_r e^{-rx} \sum_{mk=r} \mu(m)\chi_4(m)\chi_4(k) = e^{-x}$, the rearrangement being justified by $\sum_{m,k} e^{-mk\operatorname{Re} x} < \infty$ for $\operatorname{Re} x > 0$. □

Lemma 6.1.4. *For* $\lambda > 0$, $\operatorname{sech}(\lambda\sqrt{s}) = \prod_{k\ge 0}\big(1 + s/\tau_k\big)^{-1}$ *with* $\tau_k = \big((k+\frac12)\pi/\lambda\big)^2$, *written* $\tau_k(\lambda)$ *when the dependence on* λ *matters. It is the Laplace transform of* $T_\lambda = \sum_k \tau_k^{-1} e_k$ *with* e_k *independent standard exponentials, the exit time of* $\sqrt{2}B$ *from* $(-\lambda, \lambda)$, *and it is a GGC with atomic Thorin measure* $\sum_k \delta_{\tau_k}$; *see [32] and Table 1.8.*

Theorem 6.1.5. *For* $s \in H_{1/2}$,

$$\frac{1}{\zeta(\frac12 + \sqrt{s})} = 1 + \sum_{N\ge 2} A_N \operatorname{sech}(\sqrt{s}\log N) = 1 + \sum_{N\ge 2} A_N\, \mathbb{E}\, e^{-sT_{\log N}},$$

where $A_N = \frac12 \mu(m)\chi_4(m)\mu(n)\, n^{-1/2}$ *if* $N = n^m$ *with* $n \ge 2$ *squarefree and* m *odd, and* $A_N = 0$ *otherwise. In particular* $A_N = \mu(N)/(2\sqrt{N})$ *for squarefree* N.

Proof. Keep the term $n = 1$ of (6.1.1) and apply Proposition 6.1.3 with $x = \sqrt{s}\log n$ to each $n \geq 2$. The pair (n, m) with n squarefree and $n^m = N$ is unique, since all exponents of N must equal m. Absolute convergence follows from $\sum_n n^{-1/2}\sum_m n^{-m\,\mathrm{Re}\,\sqrt{s}} < \infty$ for $\mathrm{Re}\,\sqrt{s} > \frac{1}{2}$. □

Each $\mathrm{sech}(\sqrt{s}\log N)$ is a solved model, by which we mean a function whose Thorin measure is explicit. Its atoms $\tau_k(\log N)$ are units at the squares of the odd multiples of $\pi/(2\log N)$, a lattice of spacing $\pi/\log N$ in the variable $\sqrt{t}$; we call them pseudo zeros. The coefficients A_N carry the signs of the Möbius function, so positivity is now a question about the coefficients alone.

Theorem 6.1.5 is additive, but a mixture of GGCs is not in general a GGC, and the Thorin measure is defined through the logarithm. To reach the logarithm we replace sech by $\log(1 + e^{-x})$, whose coefficients have as Dirichlet series Dirichlet's eta function $\eta(1 + v) = (1 - 2^{-v})\zeta(1 + v)$, not the Dedekind eta function of Sections 8.5 and 11.3.

Proposition 6.1.6. *Define* d_n *by* $\sum_n d_n n^{-v} = 1/\eta(1 + v)$, *that is* $d_n = \sum_{2^j m=n} \mu(m)/m$. *Then* $|d_n| \leq 1$ *and for* $\mathrm{Re}\,x > 0$

$$e^{-x} = \sum_{m\geq 1} d_m \log\bigl(1 + e^{-mx}\bigr).$$

Proof. Expand $\log(1 + e^{-y}) = \sum_k (-1)^{k+1} e^{-ky}/k$. The coefficients $(-1)^{k+1}/k$ have Dirichlet series $\eta(1 + v)$, so the Dirichlet convolution with d is $\delta_{r,1}$. If $2^\ell \| n$, a squarefree $m = n/2^j$ requires $j \in \{\ell - 1, \ell\}$, so $d_n = \mu(n)/n$ for odd n and $d_n = \mu(k)(2^\ell - 2^{\ell-1})/n = \mu(k)/(2k)$ for $n = 2^\ell k$ with $\ell \geq 1$ and k odd. In particular $|d_n| \leq 1$ and $\sum_n |d_n|/n < \infty$. □

Lemma 6.1.7. *For* $\lambda > 0$ *and* $s \in \mathbb{C} \setminus (-\infty, 0]$,

$$\log\bigl(1 + e^{-\lambda\sqrt{s}}\bigr) = \log 2 - \int \log\bigl(1 + \tfrac{s}{t}\bigr)\, U_\lambda(dt), \qquad U_\lambda(dt) = \frac{\lambda}{4\pi\sqrt{t}}\, dt - \sum_{k\geq 0} \delta_{\sigma_k(\lambda)},$$

with $\sigma_k(\lambda) = \bigl((2k + 1)\pi/\lambda\bigr)^2$. *Equivalently* $U_\lambda(dt) = -d_t \Sigma_\lambda(\sqrt{t})$, *where* $\Sigma_\lambda(\tau) = \#\{k \geq 0 : (2k + 1)\pi/\lambda \leq \tau\} - \lambda\tau/2\pi$ *is a sawtooth with* $|\Sigma_\lambda| \leq \frac{1}{2}$.

Proof. Write $1 + e^{-y} = 2e^{-y/2}\cosh(y/2)$ with $y = \lambda\sqrt{s}$. Lemma 6.1.4 gives $\log\cosh(\lambda\sqrt{s}/2) = \sum_k \log(1 + s/\sigma_k)$, and $\int_0^\infty \log(1 + s/t)\, t^{-1/2} dt = 2\pi\sqrt{s}$ gives $\lambda\sqrt{s}/2 = \int \log(1 + s/t)\frac{\lambda}{4\pi\sqrt{t}} dt$. □

The reciprocal of each block has Thorin measure $-U_\lambda$: unit atoms at the pseudo zeros, from $\mathrm{sech}(y/2)$, minus the density $\lambda/(4\pi\sqrt{t})$ that counts them on average, from $e^{y/2}$. The function Σ_λ is a toy model of $S(\tau)$, with $1 + e^{-\lambda\sqrt{s}}$ playing the role of a zeta function all of whose zeros lie on the line.

THEOREM 6.1.8. *Put* $a_n = \Lambda(n)/(\sqrt{n}\log n)$ *and* $B_N = \sum_{n^m=N} d_m a_n$. *For* $s \in H_{1/2}$,

$$\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N \log\big(1+N^{-\sqrt{s}}\big)$$

and

$$\frac{d}{ds}\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N\Big[\sum_{k\ge 0}\frac{1}{s+\sigma_k(\log N)} - \frac{\log N}{4\sqrt{s}}\Big].$$

The first values are $B_p = p^{-1/2}$, $B_4 = \frac14 + \frac{1}{2\sqrt{2}}$, $B_8 = -\frac{1}{6\sqrt{2}}$, $B_9 = \frac16 + \frac{1}{2\sqrt{3}}$.

Proof. For $\operatorname{Re} z > 1$, $\log\zeta(z) = \sum_n a_n n^{-(z-1/2)}$. Apply Proposition 6.1.6 with $x = \sqrt{s}\log n$ and regroup by $N = n^m$; since $|d_m| \le 1$ the double series converges absolutely on $H_{1/2}$. Differentiate with Lemma 6.1.7. □

Formally the Thorin measure of $1/\zeta(\frac12+\sqrt{s})$ is $-\sum_N B_N U_{\log N} = \sum_N B_N\, d\Sigma_{\log N}(\sqrt{t})$, a superposition of sawtooth measures. The series does not converge as a series of measures, but its Laplace transforms do; Proposition 6.1.13 and Corollary 6.1.14 make this exact.

6.1.4 The normalised reciprocal and its Thorin measure

The factorisation $\varphi = RG$, with $\xi(z) = \frac12 z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ as in Section 1.2, separates the zeros from the archimedean part.

PROPOSITION 6.1.9. *The function* φ *is meromorphic on* $\mathbb{C}\setminus(-\infty,0]$ *and*

$$\varphi(s) = R(s)\,G(s), \qquad R(s) = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{s})}, \qquad G(s) = \frac{\Gamma(\frac14+\frac{\sqrt{s}}{2})\,\pi^{-\sqrt{s}/2}}{\Gamma(\frac14)}.$$

It is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ *if and only if the Riemann Hypothesis holds. On* $(0,\infty)$ *it is analytic and positive, it is continuous at* 0 *with* $\varphi(0) = 1$, *and* $-\varphi'/\varphi$ *is analytic in a neighbourhood of* $(0,\infty)$.

Proof. With $z = \frac12 + \sqrt{s}$ we have $z(z-1) = s - \frac14$, so $(1-4s)\zeta(z) = -8\,\xi(z)\pi^{z/2}/\Gamma(z/2)$; evaluating at $s = 0$ fixes the constant. The gamma factor G is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ since $\operatorname{Re}(\frac14+\frac{\sqrt{s}}{2}) > 0$, so φ has no zeros in $\mathbb{C}\setminus(-\infty,0]$, and its poles there are those of R, namely the points $\sqrt{s} = \rho - \frac12$ with $\operatorname{Re}\rho > \frac12$. By the functional equation there are none if and only if RH holds. On the real axis $\zeta(\frac12+x)$ is finite and nonzero for $x > 0$, $x \ne \frac12$, and $(1-4s)\zeta(\frac12+\sqrt{s}) \to -4$ as $s \to \frac14$. Since $\zeta(\frac12+x) < 0$ for $0 < x < \frac12$ and $\zeta(\frac12+x) > 0$ for $x > \frac12$, while $1-4s$ changes sign at $s = \frac14$ and $\zeta(\frac12) < 0$, the function φ is positive on $(0,\infty)$. □

THEOREM 6.1.10. *Assume RH and let $N(T)$ count the zeros $\frac12+i\gamma$ with $0<\gamma\le T$, with multiplicity, as do the sums over γ below. Then for $s\in\mathbb{C}\setminus(-\infty,0]$*

$$\log\varphi(s)=-\int_{(0,\infty)}\log\bigl(1+\tfrac{s}{t}\bigr)\,dM(t),\qquad M(t)=N(\sqrt t)-\frac{\theta(\sqrt t)}{\pi},$$

that is

$$dM(t)=\sum_{\gamma>0}\delta_{\gamma^2}-\frac{\theta'(\sqrt t)}{2\pi\sqrt t}\,dt,\qquad -\frac{\varphi'}{\varphi}(s)=\int\frac{dM(t)}{s+t}.$$

By the Riemann von Mangoldt formula $M(t)=1+S(\sqrt t)$ for $\sqrt t$ not an ordinate, with $S(0^+)=-1$.

Proof. Under RH, Hadamard's product gives $R(s)=\prod_{\gamma>0}(1+s/\gamma^2)^{-1}$, a GGC with Thorin measure $\sum\delta_{\gamma^2}$, and $-R'/R(s)=\sum_{\gamma>0}1/(s+\gamma^2)$. For G put $g=-G'/G=-\frac{1}{4\sqrt s}\{\psi(\frac14+\frac{\sqrt s}{2})-\log\pi\}$, analytic on $\mathbb{C}\setminus(-\infty,0]$, $O(|s|^{-1/2})$ at 0 and $O(|s|^{-1/2}\log|s|)$ at infinity. At $\sqrt s=i\tau$, $g=\frac{i}{4\tau}\{\psi(\frac14+\frac{i\tau}{2})-\log\pi\}$, so $\operatorname{Im} g=\theta'(\tau)/(2\tau)$, and Lemma 6.1.1 gives the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$. Add the two Stieltjes representations and integrate from 0 to s; Fubini applies because $|dM|$ integrates $\min(1,1/t)(1+|\log t|)$. Finally $N(T)=\theta(T)/\pi+1+S(T)$ [324, Section 9.3]. □

Under RH the Thorin measure of φ has the unit atoms at the squared ordinates, the Thorin measure of R, which is a GGC exactly when RH holds (Proposition 1.2.1), and the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$ of the gamma factor, positive for $t<\tau_0^2$ and negative beyond, where $\tau_0\approx6.28984$ is the point at which θ attains its minimum. Between consecutive ordinates, all of which exceed τ_0, the function M decreases; hence dM is not positive, and φ is not a GGC even under RH. Theorem 7.1.3 proves more, unconditionally.

Near the origin $M(t)\sim c\sqrt t/\pi$, with

$$c=-\theta'(0)=\frac{\log\pi-\psi(\frac14)}{2}=\frac{\zeta'(\frac12)}{\zeta(\frac12)}\approx2.68609,\tag{6.1.6}$$

the last equality from $\xi'(\frac12)=0$. The function M rises to $-\theta(\tau_0)/\pi\approx1.12394$ at $t=\tau_0^2$, falls to ≈0.55025 just before γ_1^2, jumps by one, and thereafter is a staircase with mean 1 whose continuous parts decrease (Figure 6.1; Table 7.7 gives these constants to more digits). As $s\to0$, $\varphi(s)=1-c\sqrt s+O(s)$.

6.1.5 Lévy density, theta functions and a criterion for RH

Under RH, integrating by parts in Theorem 6.1.10 and using $s/(s+t)=\int_0^\infty(1-e^{-sx})te^{-tx}dx$ turns the Thorin type representation of φ into a Lévy Khintchine

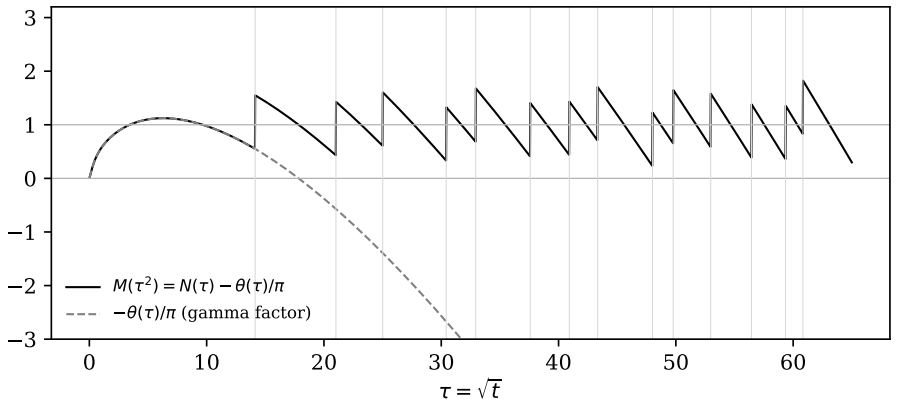


Figure 6.1: The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of φ. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone.

one,

$$\log \varphi(s) = -\int_0^\infty \bigl(1 - e^{-sx}\bigr)\,\frac{w(x)}{x}\,dx,$$
$$w(x) = \int e^{-tx}\,dM(t), \qquad \frac{w(x)}{x} = \int_0^\infty e^{-tx} M(t)\,dt.$$

Theorem 6.1.11. *Define, unconditionally,*

$$w(x) = e^{x/4} - P(x), \qquad P(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x}, \qquad x > 0.$$

Then $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)\,dx$ *for* $s > \frac{1}{4}$, *and for every* $x > 0$

$$\begin{aligned} w(x) &= W(x) - W_0(x), \qquad W(x) = \tfrac{1}{2} \sum_\rho e^{x(\rho - \frac{1}{2})^2}, \\ W_0(x) &= \frac{1}{\pi} \int_0^\infty e^{-x\tau^2} \theta'(\tau)\,d\tau, \end{aligned} \tag{6.1.7}$$

the sum running over all nontrivial zeros with multiplicity. Here W *is the heat trace of Section 1.2 in the variable* x*; under RH,* $W(x) = \sum_{\gamma > 0} e^{-x\gamma^2}$ *and* $w(x) = \int e^{-tx} dM(t)$. *The function* w *satisfies* $w(0^+) = 1$ *and* $w(x) \sim c/(2\sqrt{\pi x})$ *as* $x \to \infty$.

Proof. By logarithmic differentiation, $-\varphi'/\varphi = \frac{1}{s - 1/4} + \frac{1}{2\sqrt{s}} \frac{\zeta'}{\zeta}(\frac{1}{2} + \sqrt{s})$. The first term is the transform of $e^{x/4}$; for the second use $\zeta'/\zeta = -\sum \Lambda(n) n^{-z}$ and

$\frac{1}{2\sqrt{s}}e^{-a\sqrt{s}} = \int_0^\infty e^{-sx}(4\pi x)^{-1/2}e^{-a^2/4x}dx$, with Tonelli justified on $s > \frac{1}{4}$. Identity (6.1.7) is Weil's explicit formula for the even test function $h(r) = e^{-xr^2}$, whose Fourier partner is $g(u) = (4\pi x)^{-1/2}e^{-u^2/4x}$: the pole contributes $h(\pm\frac{i}{2}) = e^{x/4}$, the archimedean place contributes W_0, and the primes contribute P. It is (1.2.5) of Section 1.2 with $t = x$. Under RH, Theorem 6.1.10 gives the same Laplace transform for $\int e^{-tx}dM(t)$ and uniqueness identifies the two. The limit $w(0^+) = 1$ holds because $P(x) = O(x^{-1/2}e^{-(\log 2)^2/4x})$ as $x \to 0$. The behaviour at infinity needs no hypothesis: every term of W has modulus at most $e^{-x(\gamma^2-1/4)}$ with $\gamma > 14$, so W decays exponentially, while $W_0(x) \sim -c/(2\sqrt{\pi x})$ by Lemma 6.1.12 below. □

We call w the canonical function of φ: when φ is infinitely divisible, $w(x)/x$ is its Lévy density, and φ is infinitely divisible exactly when $w \geq 0$. The proof of Theorem 7.1.1 uses the prime side $e^{x/4} - P(x)$ for small x and the finitely many verified zeros for large x, and it rests on the following integral formula for W_0.

Lemma 6.1.12. *For $x > 0$,*

$$W_0(x) = \frac{F(x)}{4\sqrt{\pi x}}, \qquad F(x) = 2\theta'(0) + \int_0^\infty \frac{e^{-u/4}\big(1 - e^{-u^2/16x}\big)}{1 - e^{-u}}\,du.$$

The function F is strictly decreasing, $F(0^+) = +\infty$ and $F(\infty) = 2\theta'(0) = \psi(\frac{1}{4}) - \log\pi < 0$. Hence W_0 has exactly one zero x_*, with $W_0 > 0$ on $(0, x_*)$ and $W_0 < 0$ on (x_*, ∞).

Proof. Insert $\psi(z) = \int_0^\infty \big(e^{-u}/u - e^{-zu}/(1-e^{-u})\big)du$ with $z = \frac{1}{4} + \frac{i\tau}{2}$ into θ', and use $\int_0^\infty e^{-x\tau^2}\cos(\tau u/2)\,d\tau = \frac{1}{2}\sqrt{\pi/x}\,e^{-u^2/16x}$; the interchange is justified after writing the ψ integrand as $e^{-u}/u - e^{-u/4}/(1-e^{-u})$ plus $e^{-u/4}(1-\cos(\tau u/2))/(1-e^{-u})$, both absolutely integrable against $e^{-x\tau^2}$. The terms free of τ reassemble into $\psi(\frac{1}{4}) - \log\pi$. The integrand of F is positive and decreasing in x; as $x \to 0$ it increases to $e^{-u/4}/(1-e^{-u})$, which is not integrable at 0. □

For the sawtooth measure of Lemma 6.1.7 define its heat trace

$$h_\lambda(x) = -\int e^{-tx}\,U_\lambda(dt) = \sum_{k\geq 0} e^{-x\sigma_k(\lambda)} - \frac{\lambda}{4\sqrt{\pi x}} = \frac{\lambda}{2\sqrt{\pi x}}\sum_{j\geq 1}(-1)^j e^{-j^2\lambda^2/4x},$$

the last form by Poisson summation. The first form is a sum over the pseudo zeros minus their smooth count and the second a sum over the multiples $j\lambda$, so this Jacobi theta identity is the explicit formula of the toy zeta function $1 + e^{-\lambda\sqrt{s}}$.

Proposition 6.1.13. *For every $x > 0$, with absolutely convergent sums,*

$$w(x) = e^{x/4} + \sum_{N\geq 2} B_N\, h_{\log N}(x).$$

Proof. Substitute the theta form of h_λ and regroup the triple sum over (n, m, j) by $r = mj$. The coefficient of $e^{-r^2(\log n)^2/4x}$ is

$$-(4\pi x)^{-1/2} a_n \log n \sum_{mj=r} m\, d_m (-1)^{j+1},$$

and the inner sum is $\delta_{r,1}$ because its Dirichlet series is $\eta(v)^{-1}\cdot\eta(v)$. Since $a_n \log n = \Lambda(n)/\sqrt{n}$ the sum equals $-P(x)$. The bound $|m\, d_m| \le m$ against Gaussian decay gives absolute convergence. □

Proposition 6.1.13 is the Vere-Jones form of Theorem 6.1.11. Unconditionally, w is the Laplace transform, term by term, of the formal series $\delta_{-1/4} + \sum_N B_N\{\sum_k \delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}dt\}$ of sech atom lattices, with all the arithmetic in the weights B_N; under RH it is also the Laplace transform of dM. The atom at $-\frac{1}{4}$ comes from the pole of ζ and lies outside $(0,\infty)$, where Thorin measures live, so under RH the sawtooth series must cancel it. Two consequences follow. The first makes the sawtooth expansion of S rigorous after Gaussian averaging.

Corollary 6.1.14. *Assume RH. For every $x > 0$,*

$$\int_0^\infty S(\tau)\, 2\tau e^{-x\tau^2}\, d\tau = \frac{e^{x/4} - 1}{x} + \sum_{N\ge 2} B_N \int_0^\infty \Sigma_{\log N}(\tau)\, 2\tau e^{-x\tau^2}\, d\tau,$$

and each integral on the right equals $h_{\log N}(x)/x$.

Proof. By Theorem 6.1.11,

$$\int_0^\infty e^{-tx}(1 + S(\sqrt{t}))dt = w(x)/x,$$

and $\int_0^\infty e^{-tx}\Sigma_\lambda(\sqrt{t})\, dt = h_\lambda(x)/x$ by parts since $\Sigma_\lambda(0) = 0$. Subtract $\int e^{-tx}dt = 1/x$ and apply Proposition 6.1.13; substitute $t = \tau^2$. □

The second consequence is a criterion for RH.

Theorem 6.1.15. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) There is a real signed Borel measure m on $[0,\infty)$ with $\int |m|(dt)/(1+t) < \infty$ and $w(x) = \int e^{-tx} m(dt)$ for all $x > 0$.

In that case $m = dM$.

Proof. If RH holds take $m = dM$; the atoms satisfy $\sum(1+\gamma^2)^{-1} < \infty$ and the density is $O(t^{-1/2}(1+t)^{-1}\log(2+t))$ after division by $1+t$. Conversely, given m, put $J(s) = \int m(dt)/(s+t)$, analytic on $\mathbb{C} \setminus (-\infty, 0]$. For $s > \frac{1}{4}$ Fubini gives

$\int_0^\infty e^{-sx}w(x)dx = J(s)$, so $-\varphi'/\varphi = J$ on $(\frac{1}{4},\infty)$ and hence on all of $\mathbb{C}\setminus(-\infty,0]$ by analytic continuation. A zero or pole of φ in $\mathbb{C}\setminus(-\infty,0]$ would be a pole of $-\varphi'/\varphi$, so φ has none, and Proposition 6.1.9 gives RH. Uniqueness of the Stieltjes transform gives $m = dM$. □

6.1.6 Across the critical strip

Fix σ and evaluate ζ on the line $\operatorname{Re} z = \sigma$ through $z = \sigma + \sqrt{s}$. A zero $\rho = \beta + i\gamma$ then sits at $s = (\rho - \sigma)^2$: if $\beta > \sigma$ it gives a pole of $1/\zeta(\sigma + \sqrt{s})$ in $\mathbb{C}\setminus(-\infty,0]$, if $\beta = \sigma$ a point of the cut, and if $\beta < \sigma$ no singularity, only a Poisson kernel of width $\sigma - \beta$ in the Thorin density. On each line one can ask whether a normalised reciprocal of ζ is infinitely divisible, and whether the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ has the stronger GGC property.

To the right of $\operatorname{Re} z = 1$ the first question has a classical answer, in the variable τ rather than s. The Dirichlet series converges, $\zeta(\sigma+i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau \log N_\sigma}$ for the zeta distribution $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and the Lévy measure of $\log N_\sigma$ is the prime measure $\sum_n \Lambda(n)n^{-\sigma}(\log n)^{-1}\delta_{\log n}$. The Vere-Jones representation writes this measure as a superposition of the alternating lattices $\sum_j (-1)^{j+1}j^{-1}\delta_{jn\log p}$, the signed Lévy measures of fair coins, one for each prime power p^n, with weight the necklace polynomial $b_n(p^{-\sigma})$ (Theorem 7.1.8). At $\sigma = 0$ the weight $b_n(1)$ is 1 when n is a power of 2 and 0 otherwise, and the identity becomes Euler's binary product, with every block a solved model of unit multiplicity; for $\sigma > 0$ the weights are fractional and some are negative.

Inside the strip $|\zeta(\sigma+i\tau)|$ is unbounded, so the characteristic function picture breaks down; the Vere-Jones representation survives in the variable s. For $0 < \sigma < 1$ put $\varphi_\sigma(s) = \zeta(\sigma)/\{(1 - s/(1-\sigma)^2)\zeta(\sigma + \sqrt{s})\}$, which is φ at $\sigma = \frac{1}{2}$. For $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma$ is the Laplace transform of a function w_σ, given by the theta series of Proposition 6.1.13 (Section 6.1.5) with $e^{(1-\sigma)^2 x}$ in place of $e^{x/4}$ and the weights $b_n(p^{-\sigma})$ in place of $B_{p^n} = b_n(p^{-1/2})$ (Theorem 7.1.12). When $w_\sigma \geq 0$ this is a Lévy representation, $\log\varphi_\sigma(s) = -\int(1 - e^{-sx})w_\sigma(x)x^{-1}dx$. Theorem 7.1.1 proves $w_{1/2} = w > 0$, and numerically $w_\sigma > 0$ on nine lines from $\sigma = 0.05$ to $\sigma = 0.99$, for $10^{-4} \leq x \leq 10^3$ (Section 7.1); whether infinite divisibility holds on every line of the strip is open (Problem 7.8).

The Thorin property is where the zeros enter: see Proposition 7.1.13 and Theorem 7.1.14 of Section 7.1.2.3.

Chapters 9 to 11 return to the completed functions R_σ, written there $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, with the basepoint α in place of σ and the variable u of the book in place of s; they are not the normalised functions φ_σ of this section.

6.2 The gamma factor, the Meixner law and the sine squared lemma

On the critical line $|G|^2$ is proportional to the density of a Meixner hypergeometric law, whose characteristic function is a GGC transform in the square of its variable and whose orthogonal polynomials satisfy a local Riemann Hypothesis. The sine squared lemma turns the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. It shows that after a stable tilt the reciprocals of $\zeta(\beta+\sqrt{s})$, for $\beta>1$, and of the gamma factor are GGC transforms, and it shows how positivity is lost at the centre.

6.2.1 The gamma factor in Vere-Jones form

Recall from Section 6.1.1 the constants $c_n = 2n+\frac12$ for $n\ge 0$ and $b=\frac12(\gamma_E+\log\pi)\approx 0.860973$, where γ_E is Euler's constant.

Theorem 6.2.1. *For $s\in\mathbb{C}\setminus(-\infty,0]$,*

$$G(s)=e^{-b\sqrt{s}}\prod_{n\ge0}\frac{e^{\sqrt{s}/(2n+2)}}{1+\sqrt{s}/c_n},$$

and the Thorin measure of G is

$$U_G(dt)=\frac{1}{2\pi\sqrt{t}}\Big\{b+\sum_{n\ge0}\Big(\frac{c_n}{t+c_n^2}-\frac{1}{2n+2}\Big)\Big\}dt=-\frac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

Each factor $(1+\sqrt{s}/\kappa)^{-1}$ is the Laplace transform of a GGC, with positive Thorin density $q_\kappa(t)=\dfrac{\kappa}{2\pi\sqrt{t}\,(t+\kappa^2)}$, a Poisson kernel in $\sqrt{t}$; each factor $e^{a\sqrt{s}}$ with $a>0$ has the negative Thorin density $-a/(2\pi\sqrt{t})$.

Proof. Weierstrass gives $\log\Gamma(z_0+v)-\log\Gamma(z_0)=-\gamma_E v-\log(1+v/z_0)+\sum_{n\ge1}\{v/n-\log(1+v/(n+z_0))\}$. Take $z_0=\frac14$, $v=\sqrt{s}/2$, note $v/(n+\frac14)=\sqrt{s}/c_n$, and shift the index of v/n. For q_κ: $\frac{d}{ds}\log(1+\sqrt{s}/\kappa)=\frac{1}{2\sqrt{s}(\sqrt{s}+\kappa)}$, and Lemma 6.1.1 with $\sqrt{-t+i0}=i\sqrt{t}$ gives the density $\frac{\kappa}{2\pi\sqrt{t}(t+\kappa^2)}$. For $e^{a\sqrt{s}}$ use $\int_0^\infty\log(1+s/t)t^{-1/2}dt=2\pi\sqrt{s}$. To sum the Thorin densities of the factors, write the bracket with index n in U_G as $(\frac{1}{c_n}-\frac{1}{2n+2})-\frac{t}{c_n(t+c_n^2)}$: both parts are nonnegative, and over n the first sums to a constant and the second to $O(\log(2+t))$, so Tonelli's theorem applies to each part against $\log(1+s/t)\frac{dt}{2\pi\sqrt{t}}$ for $s>0$, and analytic continuation extends the sum to $\mathbb{C}\setminus(-\infty,0]$; the same split justifies the termwise integration against e^{-tx} in Proposition 6.2.2. The identification with θ' is the partial fraction series $\operatorname{Re}\psi(\frac14+\frac{i\tau}{2})=-\gamma_E+\sum_{n\ge0}\{\frac{1}{n+1}-\frac{2c_n}{c_n^2+\tau^2}\}$. □

Formally, the gamma factor is thus a renormalised infinite convolution of GGC laws. Write X_φ and X_R for random variables with Laplace transforms φ and R; both laws exist and are infinitely divisible, by Theorem 7.1.1 and by Theorem 5.1.1 [252, Theorem A]. Write Y_κ for the variable with $\mathbb{E}e^{-sY_\kappa} = \kappa/(\kappa+\sqrt{s})$, and L_a for the positive stable variable of index $\frac{1}{2}$ with $\mathbb{E}e^{-sL_a} = e^{-a\sqrt{s}}$. Theorem 6.2.1 and Proposition 6.1.9 then say formally

$$X_\varphi \overset{d}{=} X_R + L_b + \sum_{n\ge 0} \big(Y_{c_n} \ominus L_{1/(2n+2)}\big), \tag{6.2.1}$$

where $A \ominus B$ stands for a formal variable whose Laplace transform is the quotient of those of A and B. The sum of the Y_{c_n} diverges almost surely because $\sum 1/c_n = \infty$, and the stable deconvolutions are the compensators that make the product of the Laplace transforms converge.

Figure 6.2 shows the Thorin density of G against the first three kernels. It changes sign at τ_0, so the reciprocal $1/G$ has a Thorin density that is negative near the origin.

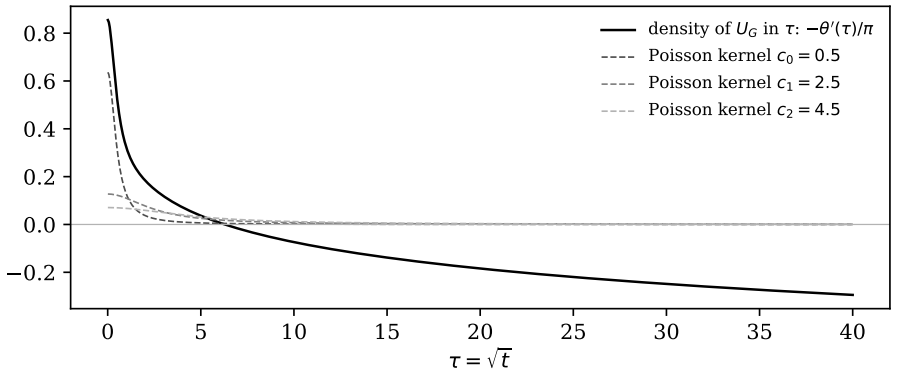


Figure 6.2: The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2+\tau^2))$ of Theorem 6.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels.

Proposition 6.2.2. *For $x > 0$,*

$$W_0(x) = -\int e^{-tx}\, U_G(dt) = -\frac{b}{2\sqrt{\pi x}} + \sum_{n\ge 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Proof. $\int_0^\infty e^{-tx} q_\kappa(t)\,dt = \frac{\kappa}{\pi}\int_0^\infty \frac{e^{-xv^2}}{v^2+\kappa^2}dv = \frac{1}{2}e^{\kappa^2 x}\operatorname{erfc}(\kappa\sqrt{x})$ and $\int_0^\infty e^{-tx}\frac{dt}{2\pi\sqrt{t}} = \frac{1}{2\sqrt{\pi x}}$. □

THEOREM 6.2.3. *For every $x > 0$, with absolutely convergent sums,*

$$W(x) = e^{x/4} + \sum_{N\geq 2} B_N\, h_{\log N}(x) - \frac{b}{2\sqrt{\pi x}} + \sum_{n\geq 0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2}e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Equivalently, $W(x)$ is the Laplace transform, taken term by term, of the formal series

$$U_R = \delta_{-1/4} + \sum_{N\geq 2} B_N\Big\{\sum_{k\geq 0}\delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}\,dt\Big\} - \frac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\geq 0}\Big(\frac{c_n}{t+c_n^2} - \frac{1}{2n+2}\Big)\Big\}dt,$$

the formal Thorin measure of $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$. Under RH, R is a GGC, and its Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2} = dN(\sqrt{t})$, where $N(T)$ counts the zeros with $0 < \gamma \leq T$, is a positive measure whose Laplace transform is also W.

Proof. $W = w + W_0$ by (6.1.7); substitute Proposition 6.1.13 for w and Proposition 6.2.2 for W_0. Term by term, $\delta_{-1/4}$ has Laplace transform $e^{x/4}$, the bracket with index N has transform $h_{\log N}(x)$ by the definition of h_λ, and the last density is $-U_G$, whose transform is $W_0(x)$ by Proposition 6.2.2. Under RH, Hadamard's product gives $R(s) = \prod_{\gamma>0}(1 + s/\gamma^2)^{-1}$, as in the proof of Theorem 6.1.10, and $W(x) = \sum_{\gamma>0} e^{-x\gamma^2}$ by Theorem 6.1.11. □

In U_R the gamma factor gives negative Poisson kernels of widths c_n in the variable $\sqrt{t}$, with poles at $\sqrt{t} = \pm ic_n$, each compensated by a positive flat density. RH is the statement that the atom at $-\frac{1}{4}$, the sech lattices of the primes and the gamma factor terms combine into unit atoms at the γ^2 and nothing else. They combine only through Laplace transforms: every partial sum of U_R carries the atom at $-\frac{1}{4}$ and the Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$ of R under RH does not, so, like the series after Proposition 6.1.13, U_R does not converge to that measure as a series of measures. By Theorem 6.1.15, R has a Thorin type representation only if RH holds, since adding U_G to its measure would give a measure m as in condition (ii) there.

On the cut the gamma factor has modulus and phase

$$G(-\tau^2 + i0) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})}\, e^{i\theta(\tau)}.$$

The phase is the Riemann Siegel theta function, and it is $-\pi$ times the distribution function of the Thorin measure: $U_G((0,\tau^2]) = -\theta(\tau)/\pi$, since $\operatorname{Im}\log(1 + s/t) = \pi$

for $t < \tau^2$ at $s = -\tau^2 + i0$. The modulus is the logarithmic potential of the same measure,

$$\log \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})} = -\int \log\Big|1 - \frac{\tau^2}{t}\Big| \, U_G(dt), \tag{6.2.2}$$

the integral converging absolutely.

The square of the modulus is proportional to the density of a Meixner hypergeometric law, with parameter $a = \frac{1}{4}$; Lai and Vere-Jones [162] study this family and its case $a = \frac{1}{2}$, the hyperbolic secant law. Since $\int |\Gamma(a+ix)|^2 e^{iux} dx = 2\pi\Gamma(2a)(2\cosh\frac{u}{2})^{-2a}$,

$$p(\tau) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2}{2^{3/2}\pi^{3/2}} \quad \text{is a probability density with} \quad \int p(\tau) e^{iv\tau} d\tau = \operatorname{sech}(v)^{1/2}. \tag{6.2.3}$$

In this section p denotes this Meixner density. Since $\Xi(\tau) = -\frac{1}{2}(\tau^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|Z(\tau)$, with Hardy's function $Z(\tau) = e^{i\theta(\tau)}\zeta(\frac{1}{2} + i\tau)$ of Section 2.2, the Meixner density is the archimedean part of Ξ^2 on the critical line:

$$\Xi(\tau)^2 = \tfrac{1}{4}\big(\tau^2 + \tfrac{1}{4}\big)^2 \, 2^{3/2}\pi \, p(\tau) \, Z(\tau)^2.$$

6.2.2 From Lévy to Thorin: the sine squared lemma

For a function of the form $f(0)/f(\sqrt{s})$ we call f its primal function. The sine squared lemma of [250, Lemma 1] converts the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. The Vere-Jones representation produces such Lévy measures, sums of atoms at the points $k \log N$, so the lemma applies to them.

Lemma 6.2.4 ([250, Lemma 1]). *Suppose that for some measure μ on $(0, \infty)$ and some α with $\int_0^\infty (1 \wedge x) e^{-\alpha x} \, \mu(dx) < \infty$,*

$$\log \frac{f(\alpha + s)}{f(\alpha)} - s\frac{f'(\alpha)}{f(\alpha)} = \int_0^\infty \big(e^{-sx} - 1 + sx\big) e^{-\alpha x} \, \frac{\mu(dx)}{x}, \qquad s \geq 0.$$

Then, for $s \geq 0$,

$$\log \frac{f(\alpha)}{f(\alpha + s)} + s\frac{f'(\alpha)}{f(\alpha)} = -\int_0^\infty \big(1 - e^{-\frac{1}{2}s^2 t}\big) \frac{\nu_\alpha(t)}{t} \, dt,$$

$$\nu_\alpha(t) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-tz} \Big\{ \int_0^\infty 2\sin^2\big(x\sqrt{z/2}\big) e^{-\alpha x} \, \mu(dx) \Big\} \frac{dz}{\sqrt{\pi z}}.$$

The map $\mu \mapsto \nu_\alpha$ is linear and positivity preserving.

We include the proof by subordination, because its two stages, a Lévy density in s followed by a Thorin density, reappear in Section 6.2.4.

LEMMA 6.2.5 (Lemma 6.2.4 in the scaling $s = u^2$). *Let $\tilde{\mu}$ be a signed measure on $(0,\infty)$ with $\int (1\wedge x)|\tilde{\mu}|(dx) < \infty$, and suppose*

$$\log \frac{f(u)}{f(0)} - u\frac{f'(0)}{f(0)} = \int_0^\infty \big(e^{-ux} - 1 + ux\big)\frac{\tilde{\mu}(dx)}{x}, \qquad u \geq 0.$$

Then for $s \in \mathbb{C}\setminus(-\infty,0]$

$$\log \frac{f(0)}{f(\sqrt{s})} + \sqrt{s}\,\frac{f'(0)}{f(0)} = -\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U(t)\,dt,$$
$$U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{t}}{2}\Big)\tilde{\mu}(dx).$$

If $\tilde{\mu} \geq 0$ then $U \geq 0$, and $s \mapsto f(0)e^{\sqrt{s}f'(0)/f(0)}/f(\sqrt{s})$ is the Laplace transform of a GGC. Since $\sqrt{s} = \int \log(1+s/t)\,\frac{dt}{2\pi\sqrt{t}}$, the Thorin measure of $f(0)/f(\sqrt{s})$ itself is $U(t)\,dt + \frac{f'(0)}{f(0)}\frac{dt}{2\pi\sqrt{t}}$.

Proof. Subordination by the first passage law (6.1.2) gives, for $x, u > 0$,

$$e^{-ux} - 1 + ux = \int_0^\infty \big(1-e^{-u^2y}\big)\big(1-e^{-x^2/4y}\big)\frac{x}{\sqrt{4\pi y^3}}\,dy,$$

because $\int_0^\infty (1-e^{-u^2y})\frac{x}{\sqrt{4\pi y^3}}dy = ux$. Integrating against $\tilde{\mu}(dx)/x$ gives a Lévy form in $s = u^2$ with Lévy density $k(y)/y$, where $k(y) = (4\pi y)^{-1/2}\int(1 - e^{-x^2/4y})\tilde{\mu}(dx)$ is the canonical function. Finally $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos(x\sqrt{t})\}\frac{dt}{\sqrt{\pi t}}$, so $k(y) = \int e^{-yt}U(t)dt$, and Frullani turns $\int(1-e^{-sy})k(y)\frac{dy}{y}$ into $\int \log(1+s/t)U(t)dt$. All kernels in these steps are nonnegative, so by Tonelli's theorem the steps hold with $|\tilde{\mu}|$ in place of $\tilde{\mu}$, and there every integral is finite, since $0 \leq (e^{-ux}-1+ux)/x \leq \min(\frac{u^2x}{2}, u)$ and $\int(1\wedge x)|\tilde{\mu}|(dx) < \infty$; this justifies each interchange for the signed $\tilde{\mu}$. The analytic continuation to $\mathbb{C}\setminus(-\infty,0]$ is immediate. □

Lemma 6.2.5 is (1.2.9) of Section 1.2 integrated against $\tilde{\mu}(dx)/x$; the letters s and u are exchanged between the two sections. We call $\tilde{\mu}$ the primal Lévy measure of $f(0)/f(\sqrt{s})$. Positivity comes from the kernel. The signed cosine density that Lemma 6.1.1 produces for a single exponential in an exponent (Section 6.1.2) is $-\cos = 2\sin^2 - 1$, and Lemma 6.2.5 separates it into the positive part $2\sin^2$ and a flat part -1, which is absorbed into the linear tilt $u\,f'(0)/f(0)$, a stable law of index $\frac{1}{2}$.

6.2.2.1 Zeta to the right of the line, the Vere-Jones blocks and the centre

PROPOSITION 6.2.6. *For $\beta > 1$ put $\tilde\mu_\beta = \sum_{n\ge2} \Lambda(n) n^{-\beta} \delta_{\log n}$. Then*

$$\frac{\zeta(\beta)}{\zeta(\beta+\sqrt{s})}\, e^{\sqrt{s}\,\zeta'(\beta)/\zeta(\beta)} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{t}\Big) U_\beta(t)\, dt\Big),$$

$$U_\beta(t) = \frac{1}{2\pi\sqrt{t}} \sum_{n\ge2} \frac{\Lambda(n)}{n^\beta}\, 2\sin^2\Big(\frac{\sqrt{t}\log n}{2}\Big) \ge 0,$$

so the left side is the Laplace transform of a GGC. Adding the tilt, the Thorin measure of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is $-\frac{1}{2\pi\sqrt{t}} \sum \Lambda(n) n^{-\beta} \cos(\sqrt{t}\log n)\, dt$, which is the Euler product density (6.1.5) *at β in place of $\frac12$.*

Proof. $\log\zeta(\beta+u) = \sum \Lambda(n) n^{-\beta} e^{-u\log n}/\log n$, so the hypothesis of Lemma 6.2.5 holds with $\tilde\mu_\beta$, which has finite mass for $\beta > 1$. □

The tilt is needed: without it the Thorin density of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is negative near $t = 0$. For ξ no tilt is needed, since $s \mapsto \xi(\alpha)/\xi(\alpha+\sqrt{s})$ is a GGC transform for every $\alpha \ge 1$ (Theorem 1.2.2); at $\alpha = 1$, Section 9.1 obtains its Thorin density from a Lévy form of $\log\xi(1+u)$ by the same lemma.

Apply the lemma to one block $f_\lambda(u) = 1 + e^{-\lambda u}$ of Theorem 6.1.8, shifted by $\alpha > 0$. Since $\log(1+e^{-\lambda(\alpha+u)}) = \sum_k (-1)^{k+1} q^k e^{-k\lambda u}/k$ with $q = e^{-\lambda\alpha}$, the primal Lévy measure is the alternating lattice $\tilde\mu_{\lambda,\alpha} = \lambda \sum_{k\ge1} (-1)^{k+1} q^k \delta_{k\lambda}$.

PROPOSITION 6.2.7. *For $\alpha > 0$ the sine squared transform of $\tilde\mu_{\lambda,\alpha}$ is positive:*

$$U_{\lambda,\alpha}(t) = \frac{\lambda}{2\pi\sqrt{t}}\, \frac{q(1-q)\big(1-\cos\lambda\sqrt{t}\big)}{(1+q)\big(1+2q\cos\lambda\sqrt{t}+q^2\big)} \ge 0,$$

so $\frac{1+e^{-\lambda\alpha}}{1+e^{-\lambda(\alpha+\sqrt{s})}}\, e^{-\sqrt{s}\lambda q/(1+q)}$ is the Laplace transform of a GGC. As $\alpha \downarrow 0$, $U_{\lambda,\alpha}(t)\, dt \to \sum_{k\ge0} \delta_{\sigma_k(\lambda)}$ vaguely, the tilt density tends to $-\frac{\lambda}{4\pi\sqrt{t}}$, and the Thorin measure of $2/(1+e^{-\lambda\sqrt{s}})$ is recovered as $\sum_k \delta_{\sigma_k(\lambda)} - \frac{\lambda}{4\pi\sqrt{t}} dt = -U_\lambda$, the sawtooth measure of Lemma 6.1.7.

Proof. $\sum_k (-1)^{k+1} q^k = \frac{q}{1+q}$ and $\sum_k (-1)^{k+1} q^k \cos k\omega = \frac{q\cos\omega+q^2}{1+2q\cos\omega+q^2}$; their difference is $\frac{q(1-q)(1-\cos\omega)}{(1+q)(1+2q\cos\omega+q^2)} \ge 0$. As $q \uparrow 1$ this is a Poisson kernel in $\omega = \lambda\sqrt{t}$ concentrating at $\omega = (2k+1)\pi$ with mass π per period, which becomes $\sum_k \delta_{\sigma_k(\lambda)}$ after the change of variable. The tilt is $f'(\alpha)/f(\alpha) = -\lambda q/(1+q) \to -\lambda/2$. □

The negative part of the sawtooth measure is entirely the tilt. Summing the blocks with the weights B_N gives $\log\zeta(\frac12+\alpha+\sqrt{s})$, and the sum of Thorin densities $\sum_N B_N U_{\log N,\alpha}$ converges absolutely for $\alpha > \frac12$, where it equals $U_{1/2+\alpha}$ of Proposition 6.2.6. Positivity is then visible termwise in the prime form. It is

not visible blockwise in the Vere-Jones form, because infinitely many weights are negative: $B_{p^3} = \frac{1}{3}(p^{-3/2} - p^{-1/2}) < 0$ for every prime p.

At $\beta = \frac{1}{2}$ the prime Lévy measure $\tilde{\mu}_{1/2} = \sum \Lambda(n) n^{-1/2} \delta_{\log n}$ has infinite mass, and both the image and the tilt in Lemma 6.2.5 diverge. The divergences cancel, and the heat trace makes the cancellation exact. Since $\int_0^\infty e^{-xt}\, 2\sin^2(\frac{\sqrt{t}L}{2}) \frac{dt}{2\pi\sqrt{t}} = \frac{1-e^{-L^2/4x}}{2\sqrt{\pi x}}$, the heat trace of the sine squared image is formally

$$\frac{1}{2\sqrt{\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(1 - e^{-(\log n)^2/4x}\big) = \frac{1}{2\sqrt{\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} - P(x).$$

Replace the divergent constant by its analytic continuation $\sum \Lambda(n) n^{-1/2} \to -\zeta'(\frac{1}{2})/\zeta(\frac{1}{2}) = -c$, and add the tilt $\frac{c}{2\sqrt{\pi x}}$ (the heat trace of $\frac{c}{2\pi\sqrt{t}} dt$) and the heat trace $e^{x/4}$ of the pole atom $\delta_{-1/4}$. The result is $e^{x/4} - P(x) = w(x)$, in agreement with Theorem 6.1.11. So at the centre Lemma 6.2.5 still produces the correct Thorin measure dM, but only after a regularisation that subtracts an infinite positive constant from a positive density, and that is where the sign of dM is lost.

6.2.2.2 The inverse direction: cosine Lévy forms

The Frullani identity behind Lemma 6.2.5 also runs in the opposite direction, from a Thorin measure in s to a symmetric Lévy measure in $u = \sqrt{s}$: for $\beta > 0$ and real u, $\int_0^\infty 2\sin^2(\frac{ux}{2}) e^{-\beta x} \frac{dx}{x} = \frac{1}{2}\log(1 + u^2/\beta^2)$, hence

$$\int \log\Big(1 + \frac{s}{t}\Big) U(dt) = \int_0^\infty 2\sin^2\Big(\frac{ux}{2}\Big) K_U(x)\, \frac{dx}{x}, \qquad K_U(x) = 2\int e^{-\sqrt{t}\,x}\, U(dt). \tag{6.2.4}$$

If $U \geq 0$ then $u \mapsto \exp(-\int \log(1 + u^2/t) U(dt))$ is the characteristic function of a symmetric infinitely divisible law with Lévy density $K_U(|x|)/(2|x|)$. Section 10.1 applies this dictionary to $\xi(\alpha)/\xi(\alpha + |u|)$, which is the characteristic function of a normal scale mixture at every basepoint α.

Proposition 6.2.8. *Under* (6.2.4)*:*

(i) the Meixner law (6.2.3) *has Lévy density* $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$ *on* $\mathbb{R}$*, and its characteristic function, as a function of* v^2*, is a GGC transform with Thorin measure* $\frac{1}{2}\sum_{k \geq 0} \delta_{((k+1/2)\pi)^2}$*, since* $\operatorname{sech}(v)^{1/2} = \prod_k (1 + v^2/((k + \frac{1}{2})\pi)^2)^{-1/2}$*;*

(ii) the sawtooth measure has $K_{U_\lambda}(x) = \dfrac{\lambda}{\pi x} - \dfrac{1}{\sinh(\pi x/\lambda)}$*;*

(iii) under RH, the Thorin measure $U = \sum_{\gamma > 0} \delta_{\gamma^2}$ *of* R *has* $K_U(x) = 2\sum_{\gamma > 0} e^{-\gamma x}$*, Cramér's function [65, Chapter I], and* $\xi(\frac{1}{2})/\xi(\frac{1}{2} + u)$ *is the characteristic function of a sum of independent Laplace variables of scales* $1/\gamma$*.*

PROPOSITION 6.2.9. *For every $x > 0$ the series $\sum_N B_N K_{U_{\log N}}(x)$ diverges.*

Proof. B_N vanishes unless N is a prime power. Grouping the powers of a fixed p, $\sum_k B_{p^k}/k = \sum_{j,m} d_m\, p^{-j/2}/(j^2 m) = \frac{12}{\pi^2} p^{-1/2} + O(p^{-1})$, using $\sum_m d_m/m = 1/\eta(2) = 12/\pi^2$. Since $K_{U_\lambda}(x) = \frac{\pi x}{6\lambda} + O((x/\lambda)^3)$, the group for p equals $\frac{2x}{\pi}\frac{p^{-1/2}}{\log p}(1 + o(1))$, and $\sum_p p^{-1/2}/\log p = \infty$. The grouping changes the order of summation, but in the order of increasing N the partial sums are dominated by the prime terms $B_p K_{U_{\log p}}(x) \sim \frac{\pi x}{6}\frac{p^{-1/2}}{\log p}$, so the series diverges in that order as well. □

The inverse direction fails to sum over primes for the reason that Lemma 6.2.5 fails at the centre: without Gaussian damping the weights $p^{-1/2}$ are not summable. The Poisson kernel $e^{-\sqrt{t}\,x}$ is the stable $\frac{1}{2}$ average of the heat kernel, $e^{-\sqrt{t}\,x} = \int_0^\infty e^{-ty}\frac{x}{2\sqrt{\pi}} y^{-3/2} e^{-x^2/4y} dy$, and the average extends over $y \gtrsim (\log N)^2$, where the block heat trace $h_{\log N}(y)$ has lost its Gaussian factor $e^{-(\log N)^2/4y}$ and is close to $-\log N/(4\sqrt{\pi y})$. Under the heat kernel alone each block heat trace is a theta function with Gaussian decay in $\log N$, and the series converges (Proposition 6.1.13, Theorem 6.2.3).

6.2.3 The Meixner law and its orthogonal polynomials

Let Y have the density p of (6.2.3), so that $\mathbb{E}e^{ivY} = \operatorname{sech}(v)^{1/2}$.

PROPOSITION 6.2.10. *(i) $Y \overset{d}{=} B(T)$, where B is a standard Brownian motion and T is independent with $\mathbb{E}e^{-\lambda T} = \operatorname{sech}(\sqrt{2\lambda})^{1/2}$, the variable $C_{1/2}$ of Biane, Pitman and Yor [32], so that Y is the value at time $\frac{1}{2}$ of the process $\widehat{C}_t = B(C_t)$ of Pitman and Yor [244, (1) and (4)]. The time $T = \sum_{k\ge 0} \frac{8\,\Gamma_k}{(2k+1)^2\pi^2}$, with Γ_k independent gamma variables of shape $\frac{1}{2}$ and unit scale, is a GGC with Thorin measure $\frac{1}{2}\sum_k \delta_{(k+1/2)^2\pi^2/2}$.*

(ii) Y is infinitely divisible with Lévy density $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$, one half of the Lévy density $1/(2|x|\sinh(\pi|x|/2))$ of the process $\widehat{C}$ [244, Table 5]; Y has a Meixner law [301, 101]. As $|\tau| \to \infty$, $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2} e^{-\pi|\tau|/2}$, matching the singularities of $\operatorname{sech}(v)^{1/2}$ at $v = \pm i\pi/2$.

(iii) p is not a Pólya frequency function of infinite order.

Proof. (i) $\mathbb{E}e^{ivB(T)} = \mathbb{E}e^{-v^2T/2} = \operatorname{sech}(v)^{1/2}$, and $\operatorname{sech}(\sqrt{2\lambda})^{1/2} = \prod_k (1 + 2\lambda/((k+\frac{1}{2})\pi)^2)^{-1/2}$. (ii) is Proposition 6.2.8(i); the tail follows from Stirling, $|\Gamma(\frac{1}{4} + ix)|^2 \sim 2\pi|x|^{-1/2}e^{-\pi|x|}$ with $x = \tau/2$. (iii) By Schoenberg's theorem [300, Theorem 1] the bilateral Laplace transform of a PF_∞ density is the reciprocal of an entire function of Laguerre Pólya class, so the reciprocal of its characteristic function is

entire. Here $1/\mathbb{E}e^{ivY} = \cosh(v)^{1/2}$, which has branch points at the simple zeros $v = i(k+\frac{1}{2})\pi$ of cosh and is not entire. □

The time T of (i), which is not the clock variable of Section 1.2, is twice the Meixner time of Table 1.8. The exponent $\frac{1}{2}$ in $\operatorname{sech}(v)^{1/2}$ is the Meixner parameter $2a = \frac{1}{2}$, forced by the factor $\Gamma(z/2)$ of ξ at $z = \frac{1}{2}$. For $2a$ a positive integer the Meixner law is PF_∞; the zeta function sits at a value where it is not, although its characteristic function is still a GGC transform in v^2.

Define the measure ν on $(0,\infty)$ by

$$\nu(dy) = \frac{e^{y/2}}{\sinh y}\,dy = 2\sum_{k\ge 0} e^{-c_k y}\,dy,$$

with $c_k = 2k + \frac{1}{2}$ as in Theorem 6.2.1.

PROPOSITION 6.2.11. *The function* $\Phi_\Gamma(\lambda) = \psi(\frac{1}{4}+\frac{\lambda}{2}) - \psi(\frac{1}{4}) = \int_0^\infty (1-e^{-\lambda y})\,\nu(dy)$ *is a Bernstein function. Its boundary values on the imaginary axis are*

$$\operatorname{Re}\Phi_\Gamma(i\tau) = 2\big(\theta'(\tau) - \theta'(0)\big) = \int_0^\infty (1-\cos\tau y)\,\nu(dy),$$

$$\operatorname{Im}\Phi_\Gamma(i\tau) = -\frac{d}{d\tau}\log p(\tau) = \int_0^\infty \sin(\tau y)\,\nu(dy).$$

Consequently $2\theta(\tau) + \tau\log\pi$ *and* $\log p(\tau)$ *are, up to constants, the imaginary and real parts of* $2\log\Gamma(\frac{1}{4}+\frac{i\tau}{2})$*, the boundary value of a function analytic in the right half plane. So* $2\theta(\tau)+\tau\log\pi$*, which is* 2π *times the smooth part* θ/π *of the zero counting function plus a linear term, is the harmonic conjugate of the Meixner log density. The smooth heat trace is the heat trace of* ν*:*

$$W_0(x) = \frac{\theta'(0)}{2\sqrt{\pi x}} + \frac{1}{4\sqrt{\pi x}}\int_0^\infty \big(1-e^{-y^2/4x}\big)\,\nu(dy),$$

which is Lemma 6.1.12 after the substitution $u = 2y$*; the function* F *of that lemma decreases because* ν *is positive.*

Proof. Substitute $u = 2y$ in $\psi(z) = \int_0^\infty (e^{-u}/u - e^{-zu}/(1-e^{-u}))du$ with $z = \frac{1}{4}+\frac{\lambda}{2}$; note $2e^{-y/2}/(1-e^{-2y}) = e^{y/2}/\sinh y$. Taking $\lambda = i\tau$ gives the boundary values, with $\theta' = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4}+\frac{i\tau}{2}) - \frac{1}{2}\log\pi$ and $\frac{d}{d\tau}\log|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = -\operatorname{Im}\psi(\frac{1}{4}+\frac{i\tau}{2})$. For W_0 integrate $\theta'(\tau) = \theta'(0) + \frac{1}{2}\int(1-\cos\tau y)\nu(dy)$ against $\frac{1}{\pi}e^{-x\tau^2}$. □

THEOREM 6.2.12. *Let* $c = -\theta'(0) = \frac{1}{2}(\log\pi - \psi(\frac{1}{4})) \approx 2.68609$ *be the constant of* (6.1.6)*. Then*

$$\frac{e^{-c\sqrt{s}}}{G(s)} = \frac{\Gamma(\frac{1}{4})\,\pi^{\sqrt{s}/2}\,e^{-c\sqrt{s}}}{\Gamma(\frac{1}{4}+\frac{\sqrt{s}}{2})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U_\Gamma(t)\,dt\Big),$$

with

$$U_\Gamma(t) = \frac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}} = \frac{1}{2\pi\sqrt{t}} \sum_{k\geq 0} \frac{t}{c_k\,(c_k^2 + t)} \geq 0.$$

So $e^{-c\sqrt{s}}/G(s)$ is the Laplace transform of a GGC, and its Thorin distribution function is $(\theta(\sqrt{t}) - \theta'(0)\sqrt{t})/\pi$, a convex increasing function of $\sqrt{t}$.

Proof. With $f(u) = \Gamma(\frac{1}{4} + \frac{u}{2})\pi^{-u/2}$ we have $f'(0)/f(0) = \frac{1}{2}\psi(\frac{1}{4}) - \frac{1}{2}\log\pi = \theta'(0) = -c$, and $\frac{d}{du}[\log f(u) - uf'(0)/f(0)] = \frac{1}{2}\Phi_\Gamma(u) = \frac{1}{2}\int(1 - e^{-uy})\nu(dy)$. Integrating from 0, $\log f(u)/f(0) - uf'(0)/f(0) = \int(e^{-uy} - 1 + uy)\frac{\tilde{\mu}(dy)}{y}$ with $\tilde{\mu} = \frac{1}{2}\nu \geq 0$. Lemma 6.2.5 gives $U_\Gamma(t) = \frac{1}{2\pi\sqrt{t}}\int 2\sin^2(\frac{y\sqrt{t}}{2})\frac{1}{2}\nu(dy) = \frac{1}{2\pi\sqrt{t}}\sum_k \int_0^\infty (1 - \cos y\sqrt{t})e^{-c_k y}dy$, which is the series stated, and Proposition 6.2.11 identifies it with θ'. □

Multiplying $1/G$ by $e^{-c\sqrt{s}}$ adds the Thorin density $\frac{c}{2\pi\sqrt{t}}$, and $U_\Gamma = -U_G + \frac{c}{2\pi\sqrt{t}}$, which is the identity $c - b = \sum_k(\frac{1}{c_k} - \frac{1}{2k+2})$, has no negative part. This is the positive counterpart of Theorem 6.2.1.

6.2.3.1 Meixner Pollaczek polynomials and the local Riemann Hypothesis

Let $\psi_{2m}(x) = H_{2m}(\sqrt{2\pi}\,x)e^{-\pi x^2}$ be the even Hermite functions, eigenfunctions of the Fourier transform $\hat{f}(y) = \int f(x)e^{-2\pi ixy}dx$ with eigenvalue $(-1)^m$. Define the Mellin transform $\mathcal{M}f(s) = \int_0^\infty x^{s-1}f(x)dx$ and $\Gamma_\mathbb{R}(s) = \pi^{-s/2}\Gamma(s/2)$. From here to the end of Section 6.2.3, s is the Mellin variable, the argument of ζ, and not the variable of the clock.

THEOREM 6.2.13. *For each $m \geq 0$, $Q_m(s) = \mathcal{M}\psi_{2m}(s)/\Gamma_\mathbb{R}(s)$ is a polynomial of degree m satisfying $Q_m(1-s) = (-1)^m Q_m(s)$, and*

$$Q_m(\tfrac{1}{2} + i\tau) = \kappa_m\, P_m^{(1/4)}\big(\tfrac{\tau}{2}; \tfrac{\pi}{2}\big),$$

where $P_m^{(\lambda)}(x;\phi)$ are the Meixner Pollaczek polynomials and $\kappa_m \neq 0$. These are the orthogonal polynomials of the Meixner law p. Consequently all zeros of Q_m lie on $\operatorname{Re} s = \frac{1}{2}$ and are simple: the Mellin transform of every even Hermite function satisfies the Riemann Hypothesis.

Proof. Expanding $H_{2m}(\sqrt{2\pi}\,x) = \sum_j h_j x^{2j}$ gives

$$\mathcal{M}\psi_{2m}(s) = \frac{1}{2}\Gamma_\mathbb{R}(s)\sum_j h_j \pi^{-j}(s/2)_j$$

with Pochhammer symbols, a polynomial of degree m times $\Gamma_\mathbb{R}(s)$. Tate's local functional equation [318, Theorem 2.4.1 and Section 2.5], $\mathcal{M}f(s)/\Gamma_\mathbb{R}(s) = \mathcal{M}\hat{f}(1-$

$s)/\Gamma_{\mathbb{R}}(1-s)$ for even f, applied to the eigenfunction ψ_{2m} gives $Q_m(1-s) = (-1)^m Q_m(s)$, so $i^{-m}Q_m(\frac{1}{2}+i\tau)$ is a real polynomial $R_m(\tau)$. By Mellin Plancherel,

$$\int_0^\infty \psi_{2m}\psi_{2n}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} Q_m\overline{Q_n}\,|\Gamma_{\mathbb{R}}(\tfrac{1}{2}+i\tau)|^2 d\tau = \sqrt{2}\,i^{m-n}\int_{\mathbb{R}} R_m(\tau)R_n(\tau)\,p(\tau)\,d\tau,$$

since $|\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)|^2 = \pi^{-1/2}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = 2^{3/2}\pi\,p(\tau)$. The left side vanishes for $m \neq n$, so (R_m) is the orthogonal polynomial sequence of p, which with $x = \tau/2$ is the Meixner Pollaczek family with $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$ [152, Section 9.7]. Orthogonal polynomials of a measure with infinite support have real simple zeros. □

The conclusion is the archimedean local Riemann Hypothesis of Bump, Choi, Kurlberg and Vaaler [50], seen through the Meixner law. We make no priority claim for this route; Meixner Pollaczek expansions of Ξ in the literature are reviewed in Section 6.2.3.2.

6.2.3.2 Zeta in Meixner coordinates

Kuznetsov [158, 159] expands Ξ and Dirichlet L functions in Meixner Pollaczek polynomials; Inoue [126] studies the symmetric expansion of the completed zeta function, the multiplicative structure of its coefficients and the zeros of the approximating polynomials; Romik [290] expands Ξ itself in the symmetric Meixner Pollaczek basis $P_n^{(3/4)}(x;\frac{\pi}{2})$, together with the Hermite and continuous Hahn bases, and derives coefficient asymptotics; and Candelpergher [52] gives related polynomial expansions whose polynomials have their roots on the line. The expansions below are included for comparison with the Thorin picture and are not claimed as new.

The map $\mathcal{U}f(\tau) = \mathcal{M}f(\frac{1}{2}+i\tau)/\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)$ is, by the Mellin Plancherel identity in the proof of Theorem 6.2.13, a unitary map from $L^2(0,\infty)$ onto $L^2(\sqrt{2}\,p(\tau)d\tau)$, carrying the even Hermite functions to the Meixner Pollaczek polynomials. Dilations become multipliers, $\mathcal{U}[f(n\,\cdot)] = n^{-1/2-i\tau}\mathcal{U}f$. The function $g(x) = (2\pi^2x^4 - 3\pi x^2)e^{-\pi x^2} = \frac{1}{2}(xD)(xD+1)e^{-\pi x^2}$, with $D = d/dx$, has $\mathcal{U}g = \frac{1}{4}s(s-1)$, a Meixner Pollaczek combination of degree two, and its dilation sum Ω is, up to the factor $\frac{1}{2}x^{-1/2}$, the Pólya kernel Φ of Section 1.2 in the variable $x = e^u$:

$$\Omega(x) = \sum_{n\geq 1} g(nx) = \tfrac{1}{2}x^{-1/2}\Phi(\log x), \qquad \mathcal{U}\Omega = \tfrac{1}{4}s(s-1)\zeta(s)\big|_{s=\frac{1}{2}+i\tau}.$$

So in Meixner coordinates $\frac{1}{4}s(s-1)\zeta(s)$ on the critical line is the image of a function in $L^2(0,\infty)$ and has a convergent Meixner Pollaczek expansion

$$\tfrac{1}{4}s(s-1)\zeta(s) = \sum_{m\geq 0} a_{2m}\,Q_m(s), \qquad a_{2m} = \frac{\langle \Omega, \psi_{2m}\rangle}{\langle \psi_{2m}, \psi_{2m}\rangle},$$

in $L^2(p)$. Each Q_m satisfies the local Riemann Hypothesis. We ask whether the property survives in the partial sums.

The coefficients decay faster than geometrically, but the polynomials grow, and the expansion converges slowly as $|\tau|$ grows: near the first zero $\frac{1}{2} + 14.1347i$ the partial sums are still far from zero at $N = 24$. The partial sums $\sum_{m \le N} a_{2m} Q_m(s)$ mix polynomials of both parities under $s \mapsto 1-s$, so they have no reason to vanish on the critical line, and for $N = 12$ and $N = 24$ they do not: all their zeros lie off the line. The asymmetric expansion therefore does not test whether the local Riemann Hypothesis survives the superposition; the symmetric expansion does.

The symmetric choice is the real even function $F(\tau) = (\tau^2 + \frac{1}{4})Z(\tau)$, proportional to $\Xi(\tau)/|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|$, expanded in the even Meixner Pollaczek polynomials $P_{2m}^{(1/4)}(\tau/2; \frac{\pi}{2})$, normalised as in [152, Section 9.7]. Its truncations, the orthogonal projections of F in $L^2(p)$ onto the even polynomials of degree at most $2M$, are real even polynomials. The coefficients alternate in sign from $m = 3$ on, and from $m = 4$ on their modulus decreases slowly, to about 0.01 at $m = 20$. The slow decay reflects the polynomial growth of F against the exponential decay of p: the expansion converges in $L^2(p)$, but its truncations approximate F poorly at the ordinates. The real zeros of the truncations do not track the ordinates: the zero nearest 14.13 is 14.50, 14.32, 14.16, 13.73, 13.54, 14.84 for the degrees 38, 40, 42, 50, 60, 80, and there are eight to ten real zeros in $[0, 40]$ against six ordinates. Neither expansion suggests a route from the local Riemann Hypothesis to the global one; such a route must use the arithmetic of the full dilation sum, not only the orthogonality of its archimedean pieces.

The factorisation of Ξ^2 in Section 6.2.1 induces a factorisation of the autocorrelation $\Phi * \Phi$ of the Pólya kernel, whose Fourier transform is Ξ^2.

PROPOSITION 6.2.14. *The integral* $\mathcal{C}(v) = \int p(\tau)\,|\zeta(\frac{1}{2} + i\tau)|^2 e^{iv\tau}\,d\tau$ *converges absolutely, and*

$$(\Phi * \Phi)(v) = 2^{-3/2}\Big(\frac{d^2}{dv^2} - \frac{1}{4}\Big)^2 \mathcal{C}(v).$$

Formally $\mathcal{C} = \operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$, *with*

$$\mathcal{A} = \sum_{m,n \ge 1} \frac{\delta_{\log(m/n)}}{\sqrt{mn}},$$

where $\operatorname{sech}(v)^{1/2}$ *is the characteristic function of the Meixner law and* $\mathcal{A}$, *whose Fourier transform is formally* $|\zeta(\frac{1}{2} + i\tau)|^2$, *is the autocorrelation of the measure* $\sum_n n^{-1/2}\delta_{\log n}$. *The measure* $\mathcal{A}$ *is positive and positive definite, and the series for* $\mathcal{C}$ *diverges: truncated at* $m, n \le X$ *it equals* $X \int \operatorname{sech}(v-r)^{1/2} e^{-|r|/2}\,dr + O(X^{1/2})$. *This growth comes from the pole of* ζ, *and the factor* $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ *does not remove it, since* $(\frac{d^2}{dv^2} - \frac{1}{4})e^{-|v|/2} = -\delta_0$.

Proof. Since Φ is even, $\int(\Phi * \Phi)(v)e^{i\tau v}dv = \Xi(\tau)^2 = \frac{1}{4}(\tau^2 + \frac{1}{4})^2\pi^{-1/2}|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2|\zeta(\frac{1}{2} + i\tau)|^2$, and $\pi^{-1/2}|\Gamma|^2 = 2^{3/2}\pi\, p$. Fourier inversion divides by 2π, the factor $(\tau^2 + \frac{1}{4})^2$ becomes $(\frac{d^2}{dv^2} - \frac{1}{4})^2$, and the inverse transform of the even function $p|\zeta|^2$ is $\mathcal{C}/(2\pi)$; the constant is $\frac{1}{4}\,2^{3/2}\pi/(2\pi) = 2^{-3/2}$. The integral $\mathcal{C}$ converges absolutely because $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2}e^{-\pi|\tau|/2}$ (Proposition 6.2.10) and $\zeta(\frac{1}{2} + i\tau)$ grows at most polynomially. Formally $|\zeta|^2 = |\sum n^{-1/2-i\tau}|^2 = \sum_{m,n}(mn)^{-1/2}e^{i\tau\log(m/n)}$, and $\int p(\tau)e^{i\tau(v+r)}d\tau = \operatorname{sech}(v + r)^{1/2}$, which gives $\mathcal{C} = \operatorname{sech}^{1/2} * \mathcal{A}$. For the truncation put $D_X(\tau) = \sum_{n\le X} n^{-1/2-i\tau}$; the truncated series is exactly $\int p(\tau)|D_X(\tau)|^2e^{iv\tau}d\tau$. For $|\tau| \le X$, $D_X(\tau) = \zeta(\frac{1}{2} + i\tau) + X^{1/2-i\tau}/(\frac{1}{2} - i\tau) + O(X^{-1/2})$ [324, Theorem 4.11], and for $|\tau| > X$ the bound $|D_X|^2 \le 4X$ meets the exponential decay of p. Since $\int e^{-|r|/2}e^{i\tau r}\,dr = (\frac{1}{4} + \tau^2)^{-1}$, the squared pole term contributes $X\int p(\tau)(\frac{1}{4} + \tau^2)^{-1}e^{iv\tau}d\tau = X\int \operatorname{sech}(v - r)^{1/2}e^{-|r|/2}\,dr$, up to an exponentially small error from $|\tau| > X$; the remaining terms contribute $O(X^{1/2})$ uniformly in v. Finally $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ turns $X\int \operatorname{sech}(v - r)^{1/2}e^{-|r|/2}\,dr$ into $-X(\frac{d^2}{dv^2} - \frac{1}{4})\operatorname{sech}(v)^{1/2}$, which is not zero. □

Both factors of $\mathcal{C}$ are positive, the Meixner factor being a GGC transform in v^2 and $\mathcal{A}$ a positive measure, but convolution with a positive measure does not preserve the GGC property, so the GGC structure of the Meixner factor does not pass to the product. The lattice sum also needs a counterterm, the pole term of the approximate functional equation, which comes from the same pole of ζ that produced the atom at $-\frac{1}{4}$ in Proposition 6.1.13. Problem 7.6 of Section 7.2 asks for the regularised form of $\operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$ and whether it has a Thorin or Lee and Yang structure in v.

6.2.4 An infinitely divisible floor for the reciprocal Xi function

Multiplying $\varphi = RG$ by the GGC transform $e^{-c\sqrt{s}}/G$ of Theorem 6.2.12 gives $e^{-c\sqrt{s}}R$, which we call the floor of the reciprocal Xi function.

Theorem 6.2.15. *Unconditionally, $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density*

$$\frac{W(x) + c/(2\sqrt{\pi x})}{x}, \qquad W(x) = \tfrac{1}{2}\sum_{\rho} e^{x(\rho-\frac{1}{2})^2},$$

and $W(x) + c/(2\sqrt{\pi x}) > 0$ for all $x > 0$. The following are equivalent: (i) RH; (ii) this law is a GGC; (iii) $W(x) + c/(2\sqrt{\pi x})$ is completely monotone; (iv) W is completely monotone. The proof uses $w > 0$, which is Theorem 7.1.1 of Section 7.1 and rests on the verification of RH to height 50, an explicit bound for $N(T)$ [109] and the enclosures E1 to E3.

Proof. $e^{-c\sqrt{s}}R = \varphi \cdot (e^{-c\sqrt{s}}/G)$ is the product of the transform of an infinitely divisible law (Theorem 7.1.1) and a GGC (Theorem 6.2.12), so it is infinitely divisible, and its Lévy density is the sum $\{w(x) + \int e^{-xt}U_\Gamma(t)dt\}/x$. By Proposition 6.2.2 and Theorem 6.2.12, $\int e^{-xt}U_\Gamma(t)dt = W_0(x) + \frac{c}{2\sqrt{\pi x}}$, and $w + W_0 = W$ by (6.1.7). Positivity holds because $w > 0$ and $U_\Gamma \geq 0$. For the equivalences: GGC means the Lévy density times x is completely monotone, so (ii)⇔(iii). Under RH $W = \sum e^{-x\gamma^2}$ is completely monotone, so (i)⇒(iv)⇒(iii). If (ii) holds, the transform is analytic on $\mathbb{C} \setminus (-\infty, 0]$ and agrees with $e^{-c\sqrt{s}}R$ on $(0, \infty)$, so by uniqueness of analytic continuation R has no poles in $\mathbb{C} \setminus (-\infty, 0]$; since R has no zeros and G is analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, $\varphi = RG$ is analytic and zero free there, and Proposition 6.1.9 gives RH. □

Infinite divisibility alone is not new: by Theorem 5.1.1 [252, Theorem A], $W > 0$ and $W' < 0$ on $(0, \infty)$, so R itself is infinitely divisible and self decomposable, and so is its product with the stable factor $e^{-c\sqrt{s}}$; Section 11.1 draws the consequences of self decomposability. What the floor adds is the splitting of its canonical function $k = W + c/(2\sqrt{\pi x})$ as $k = w + \int e^{-xt}U_\Gamma(t)\,dt$, an arithmetic part plus a completely monotone archimedean part contributed by the Meixner law (Theorem 6.2.12). In law the floor adds to X_R an independent first passage time L_c of $\sqrt{2}B$ to the level c. Everything that could fail therefore lies in w, and RH is the upgrade of k from positivity, the lowest rung of the ladder of Section 5.1, to complete monotonicity, its top.

6.2.4.1 The sine squared lemma at the floor

As in the proof of Lemma 6.2.5, from here to the end of the section x is the variable of the primal Lévy measure, and the time of the heat trace is written y.

The floor is $e^{-c\sqrt{s}}R(s) = f(0)/f(\sqrt{s})$ with primal function $f(u) = e^{cu}\xi(\frac{1}{2}+u)$, and $f'(0)/f(0) = c$ because $\xi'(\frac{1}{2}) = 0$. The tilt contributes the linear term cu to $\log f$, which cancels in $\log f(u)/f(0) - uf'(0)/f(0) = \log \xi(\frac{1}{2}+u)/\xi(\frac{1}{2})$, so the primal Lévy measure of the floor is that of ξ at the centre. Collecting the three factors of $\xi(\frac{1}{2}+u) = \frac{1}{2}(u^2 - \frac{1}{4})\,\pi^{-1/4} f_\Gamma(u)\,\zeta(\frac{1}{2}+u)$, where $f_\Gamma(u) = \pi^{-u/2}\Gamma(\frac{1}{4}+\frac{u}{2})$, gives formally

$$\tilde{\mu}_\xi(dx) = -2\cosh\left(\tfrac{x}{2}\right)dx + \sum_{k\geq 0} e^{-c_k x}\,dx + \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx). \tag{6.2.5}$$

The first term comes from $u^2 - \frac{1}{4} = (u+\frac{1}{2})(u-\frac{1}{2})$, each linear factor $1 + u/a$ contributing $-e^{-ax}dx$ by Frullani (with $a = -\frac{1}{2}$ by continuation); the second is $\frac{1}{2}\nu$ (Proposition 6.2.11 and the proof of Theorem 6.2.12); the third is the prime measure of Proposition 6.2.6 at $\beta = \frac{1}{2}$. By the explicit formula, as distributions on $x > 0$,

$$\tilde{\mu}_\xi(dx) = -\sum_\rho e^{(\rho-\frac{1}{2})x}\,dx, \qquad \text{which under RH is} \qquad -2\sum_{\gamma>0}\cos(\gamma x)\,dx. \tag{6.2.6}$$

PROPOSITION 6.2.16. *Apply the two stages of Lemma 6.2.5 to $\tilde{\mu}_\xi$, regularising $\int_0^\infty \cos(\gamma x)dx = 0$ and the growing term of (6.2.5) by analytic continuation.*

(i) Subordination stage: $(4\pi y)^{-1/2}\int_0^\infty (1-e^{-x^2/4y})\,\tilde{\mu}_\xi(dx) = W(y)$, the heat trace over the zeros. With the tilt, the canonical function of the floor is $k(y) = W(y) + c/(2\sqrt{\pi y})$, as in Theorem 6.2.15, and its positivity is proved.

(ii) Bernstein stage: $U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2(\frac{x\sqrt{t}}{2})\,\tilde{\mu}_\xi(dx)$ equals $\sum_{\gamma>0}\delta_{\gamma^2}$ under RH, and the Thorin measure of the floor is $U + \frac{c}{2\pi\sqrt{t}}dt$.

The two stages are linked by $W(y) = \int_0^\infty e^{-yt}U(t)\,dt$. The floor is a GGC exactly when $U + \frac{c}{2\pi\sqrt{t}}\,dt \geq 0$, that is, when k is completely monotone; by Theorem 6.2.15 this happens exactly under RH, and then $U = \sum_{\gamma>0}\delta_{\gamma^2} \geq 0$.

Proof. (i) For $\tilde{\mu} = -2\cos(\gamma x)dx$,

$$\int_0^\infty (1-e^{-x^2/4y})(-2\cos\gamma x)\,dx = 2\int_0^\infty e^{-x^2/4y}\cos\gamma x\,dx = \sqrt{4\pi y}\,e^{-\gamma^2 y}.$$

Without RH the same computation with $e^{(\rho-\frac{1}{2})x}$ gives $e^{y(\rho-\frac{1}{2})^2}$, so the result is W in general. (ii) $\int_0^\infty (1-\cos x\sqrt{t})(-2\cos\gamma x)\,dx = \pi\,\delta(\sqrt{t}-\gamma)$, and $\frac{1}{2\pi\sqrt{t}}\pi\,\delta(\sqrt{t}-\gamma) = \delta(t-\gamma^2)$. The link is the identity $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos x\sqrt{t}\}\frac{dt}{\sqrt{\pi t}}$ used in the proof of Lemma 6.2.5. □

The lemma preserves positivity, so if $\tilde{\mu}_\xi$ were a positive Lévy measure the floor would be a GGC and RH would follow. But $\tilde{\mu}_\xi$ is not positive. Its pole part $-2\cosh(x/2)\,dx$ is negative, and at large x it cancels the mean density $e^{x/2}dx$ of the prime atoms, leaving the oscillating sum over zeros (6.2.6). At the centre the lemma turns the question into one about a signed input whose sine squared image is positive if and only if RH holds.

6.2.4.2 Complete monotonicity to finite order

Positivity at stage (i) of Proposition 6.2.16 is Theorem 6.2.15; positivity at stage (ii) is complete monotonicity of $k(y) = W(y) + c/(2\sqrt{\pi y})$, that is $(-1)^n k^{(n)} \geq 0$ for every n.

THEOREM 6.2.17. *Let $k(y) = W(y) + c/(2\sqrt{\pi y})$. For every $n \leq 10^{10}$ and every $y > 0$,*

$$(-1)^n\,k^{(n)}(y) > 0.$$

Consequently, by Williamson's theorem, for each $1 \leq n \leq 10^{10}$ there is a positive measure Ω_n on $(0,\infty)$ with $k(y) = \int (1-ty)_+^{n-1}\,\Omega_n(dt)$. For $n \geq 1$ the proof uses the verification of RH to height $T_0 = 3\times 10^{12}$ [247], an explicit bound for $N(T)$ [109] and one interval enclosure (E4, Section 7.2.3), and no other information about zeros.

Proof. The case $n = 0$ is the positivity in Theorem 6.2.15. Fix $1 \le n \le 10^{10}$ and put $y_n = 0.05/n$.

Small y. Write $k = w + (W_0 + \frac{c}{2\sqrt{\pi y}})$. By Theorem 6.2.12 the bracket is $\int e^{-ty} U_\Gamma(t)\,dt$ with $U_\Gamma \ge 0$, so $(-1)^n$ times its nth derivative is $L_n(y) = \int t^n e^{-ty} U_\Gamma(t) dt$, decreasing in y. Since $\theta'(\tau) - \theta'(0) \ge \frac{2\tau^2}{\frac14+\tau^2} \ge 1$ for $\tau \ge \frac12$ (the $k = 0$ term of the series in Theorem 6.2.12),

$$L_n(y) = \frac{1}{\pi}\int_0^\infty \tau^{2n} e^{-y\tau^2}\big(\theta'(\tau) - \theta'(0)\big)d\tau \ge \frac{\Gamma(n+\frac12)}{2\pi}\, y^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)}.$$

On the other side $(-1)^n w^{(n)} = (-1)^n 4^{-n} e^{y/4} - (-1)^n P^{(n)}$. P is analytic in $\operatorname{Re} y > 0$, and on the circle $|z - y| = y/2$ one has $|z| \ge y/2$ and $\operatorname{Re}(1/z) \ge 2/(3y)$, so Cauchy's estimate gives

$$\begin{aligned} |P^{(n)}(y)| &\le n!\,(2/y)^n\,(2\pi y)^{-1/2} \sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y} \\ &\le C_0\, n!\,(2/y)^n\,(2\pi y)^{-1/2} e^{-(\log 2)^2/6y}, \end{aligned}$$

where $C_0 = 0.54993$ bounds $e^{(\log 2)^2/6y}\sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y}$ for every $y \le 0.05$, each term being increasing in y (enclosure E4). Divide by the leading term of the lower bound for L_n and use Gautschi's inequality $n!/\Gamma(n+\frac12) < \sqrt{n+1}$ [225, (5.6.4)]. For $y \le y_n$ we have $(\log 2)^2/6y \ge 1.6015\,n$, so the ratio is at most

$$r_n = C_0\sqrt{2\pi}\sqrt{n+1}\, e^{(\log 2 - 1.6015)n},$$

which decreases in n, since $r_{n+1}/r_n \le \sqrt{3/2}\, e^{\log 2 - 1.6015} < \frac12$, and satisfies $r_n \le r_1 < 0.786$ (enclosure E4). Hence, for $y \le y_n$,

$$(-1)^n k^{(n)}(y) \ge (1 - r_n)\frac{\Gamma(n+\frac12)}{2\pi} y_n^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)} - 4^{-n} e^{y/4}.$$

The first term is at least $0.214\,\Gamma(\frac32)\,20^{3/2}/(2\pi) > 2.6$, since $\Gamma(n+\frac12)(20n)^{n+\frac12}$ increases in n, and the two negative terms total less than 0.27. So $(-1)^n k^{(n)} > 0$ on $(0, y_n]$ for every $n \ge 1$.

Large y. Here $(-1)^n k^{(n)} = (-1)^n W^{(n)} + \frac{c\,\Gamma(n+\frac12)}{2\pi} y^{-n-\frac12}$. Zeros with $0 < \gamma \le T_0$ lie on the line and contribute $\gamma^{2n} e^{-y\gamma^2} > 0$. Any other zero has $\gamma > T_0$ and contributes at most $(\gamma+1)^{2n} e^{-y(\gamma^2 - \frac14)}$ in modulus. Let $y \ge y_n$. Since $n/y \le 20n^2 < T_0^2$, the function $(\gamma+1)^{2n} e^{-y\gamma^2}$ decreases for $\gamma \ge T_0$, so by $N(t+1) - N(t) \le 2\log(t+1)$, which follows from [109] as in the proof of Theorem 7.1.1, the zeros with $T_0 + j < \gamma \le T_0 + j + 1$ contribute at most $b_j = 2\log(T_0 + j + 1)\,(T_0 + j + 1)^{2n} e^{-y((T_0+j)^2 - \frac14)}$. For $n \le 10^{10}$ we have $b_{j+1}/b_j \le \exp\big(\frac{2n}{T_0} + \frac{1}{T_0 \log T_0} - \frac{2T_0+1}{20n}\big) < 10^{-13}$, so the tail is at most $2b_0 = 4\log(T_0+1)\,(T_0+1)^{2n} e^{-y(T_0^2 - \frac14)}$. Its ratio to

the positive term $\frac{c\,\Gamma(n+\frac{1}{2})}{2\pi} y^{-n-\frac{1}{2}}$ is a constant multiple of $y^{n+\frac{1}{2}} e^{-y(T_0^2-\frac{1}{4})}$, which decreases for $y \geq y_n$ because $T_0^2 - \frac{1}{4} > 20n(n+\frac{1}{2})$. At $y = y_n$, using $\Gamma(n+\frac{1}{2}) \geq \Gamma(\frac{3}{2})$ and $y_n < 1$, the logarithm of the ratio is at most

$$\log \frac{8\pi \log(T_0+1)}{c\,\Gamma(\frac{3}{2})} + 2n \log(T_0+1) - \frac{T_0^2 - \frac{1}{4}}{20n} < -4.4 \times 10^{13},$$

because the left side increases with n, and at $n = 10^{10}$ its first term is below 6 while $y_n(T_0^2 - \frac{1}{4}) \approx 4.5 \times 10^{13}$ exceeds $2n \log(T_0+1) < 5.8 \times 10^{11}$ by a wide margin. So the tail is smaller than the positive term.

Since $k(\infty) = 0$, Williamson's theorem [345] gives Ω_n. □

The large y branch covers $y > 0.05/n$, and there the bound of the proof lets the positive term $c\,\Gamma(n+\frac{1}{2}) y^{-n-\frac{1}{2}}/(2\pi)$ contributed by the tilt dominate the zeros above T_0 whenever $y_n(T_0^2 - \frac{1}{4}) > 2n \log(T_0+1) + 6$. This sufficient condition holds while $n < 8.8 \times 10^{10}$, and we state the safe round number 10^{10}.

Section 5.1.6 compares the reach of Theorem 6.2.17 with that of Theorems 5.1.6 and 5.1.13.

Chapter 7

Infinite divisibility and truncations of the reciprocal zeta function

Section 7.1 proves with no hypothesis the positivity of the canonical function w of Theorem 6.1.11, which the proof of Theorem 6.2.15 used, so that φ itself is infinitely divisible, and classifies φ in the Bondesson hierarchy: infinitely divisible, but not self decomposable, not in the Bondesson class and not a generalised gamma convolution.

Section 7.2 studies the truncations of this lattice. One sided truncations fail at zeros where $|S|$ is large, while symmetrised truncations keep their zeros in the strip on the line in every window computed. Conjecture S asserts this for every size; for small sizes it reduces to a finite computation, and for infinitely many sizes it would imply RH. The coordinates of Chapter 6, declared in Section 6.1.1, hold in this chapter too: in the variable s of the clock a zero $\frac{1}{2} + i\gamma$ sits at $s = -\gamma^2$ on the cut.

7.1 Infinite divisibility, necklaces and solved models

By Theorem 6.1.11, $-\varphi'/\varphi$ is the Laplace transform of the canonical function $w = e^{x/4} - P = W - W_0$.

7.1.1 Infinite divisibility and the class of φ

Whether φ is infinitely divisible is a question about the sign of w, since by Theorem 6.1.11 the candidate Lévy density of φ is $w(x)/x$. Recall that $w(0^+) = 1$ and $w(x) \sim c/(2\sqrt{\pi x})$ as $x \to \infty$, where $c = -\theta'(0) \approx 2.68609$ is the constant of (6.1.6). The proof uses the verification of RH to height 50 [11, 247], the explicit Riemann von Mangoldt bound of Hasanalizade, Shen and Wong [109], and three interval enclosures, $F(0.004) < -0.111$ for the function F of Lemma 6.1.12, $P(0.004) < 2 \times 10^{-13}$ and $E(0.004) < 0.0011$ for the sum E defined in the proof, computed by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E3).

THEOREM 7.1.1. *$w(x) > 0$ for every $x > 0$. Consequently, unconditionally, φ is the Laplace transform of an infinitely divisible probability law on $(0,\infty)$ with zero drift and Lévy density $w(x)/x$:*

$$\frac{\zeta(\frac12)}{(1-4s)\,\zeta(\frac12+\sqrt{s})} = \exp\Big(-\int_0^\infty \big(1-e^{-sx}\big)\frac{w(x)}{x}\,dx\Big), \qquad \operatorname{Re} s > 0.$$

Proof. Fix $x_1 = 0.004$.

Small x. Each term $x^{-1/2}e^{-(\log n)^2/4x}$ increases in x while $(\log n)^2/4x > \frac12$, which holds for $x \le x_1$ and $n \ge 2$. Hence $P(x) \le P(x_1) < 2\times 10^{-13}$ (enclosure E2: the term $n=2$ in interval arithmetic, and $n \ge 3$ through $(\log n)^2/4x_1 \ge \frac{\log 3}{4x_1}\log n$), and $w(x) = e^{x/4} - P(x) > 0$ on $(0, x_1]$.

Large x. By Lemma 6.1.12 and $F(x_1) < 0$, for $x \ge x_1$ we have $-W_0(x) \ge |F(x_1)|/(4\sqrt{\pi x})$. Here $F(x_1) < -0.111$ by enclosure E1. In W the zeros with $0 < \gamma \le 50$ lie on the line [11] and contribute $e^{-x\gamma^2} > 0$. Any other zero has $\gamma > 50$ and contributes at most $e^{-x(\gamma^2-1/4)}$ in modulus. The explicit Riemann von Mangoldt formula $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \le E(T)$ for $T \ge e$, with $E(T) = 2.5167 + \frac{1}{50T}$ for $T \le X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T} < 0.3\log T$ for $T > X_0$ [109, (1.7), (2.7) and Corollary 1.4], gives $N(t+1) - N(t) \le 2\log(t+1)$ for $t \ge 50$: the main term grows by at most $\frac{1}{2\pi}\log\frac{t+1}{2\pi}$, and while $t+1 \le X_0$ the two error terms add at most $5.0334 + \frac{1}{25t}$, together less than 5.37 at $t = 50$ against $2\log 51 = 7.86$, with a gap that widens as t grows, since its derivative exceeds $(2 - \frac{1}{2\pi})/(t+1)$; for $t+1 > X_0$ the error terms add at most $0.6\log(t+1)$, and $\frac{1}{2\pi} + 0.6 < 2$. So all such zeros together contribute at most

$$E(x) = \sum_{j\ge 0} 2\log(51+j)\, e^{-x((50+j)^2 - \frac14)}.$$

Enclosure E3 gives $E(x_1) < 0.0011$, against $|F(x_1)|/(4\sqrt{\pi x_1}) > 0.247$. Since $(50+j)^2 - \frac14 > 1/(2x_1)$ for every j, $\sqrt{x}\,E(x)$ decreases on $[x_1,\infty)$, so $E(x) < |F(x_1)|/(4\sqrt{\pi x}) \le -W_0(x)$ for all $x \ge x_1$, and $w = W - W_0 > 0$ on $[x_1,\infty)$.

Representation. By Theorem 6.1.11, $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)dx$ for $s > \frac14$. Since $w \ge 0$, Landau's theorem applies [340, Chapter II, Theorem 5b]: if the Laplace transform of a nonnegative function has abscissa of convergence σ_c, then the transform cannot be continued analytically to a neighbourhood of σ_c. Here $-\varphi'/\varphi$ is analytic near every point of $(0,\infty)$ by Proposition 6.1.9. Hence $\sigma_c \le 0$, the integral converges for $\operatorname{Re} s > 0$, and equals the continuation of $-\varphi'/\varphi$ there. Integrating over $[a,s]$ for real $0 < a < s$ with Tonelli, and letting $a \downarrow 0$ with $\varphi(a) \to 1$ and monotone convergence, gives the formula for real $s > 0$; then $\int \min(1,x)\, w(x)x^{-1}dx < \infty$ because the left side is finite, and analytic continuation gives the formula for $\operatorname{Re} s > 0$. □

Theorem 7.1.1 is the main unconditional positive result of Chapter 6 and of this chapter, and its proof has a wide margin (Figure 7.1). The only place where

w comes close to a sign change is at infinity, where it decays like $x^{-1/2}$, and there the sign is fixed by $\theta'(0) < 0$.

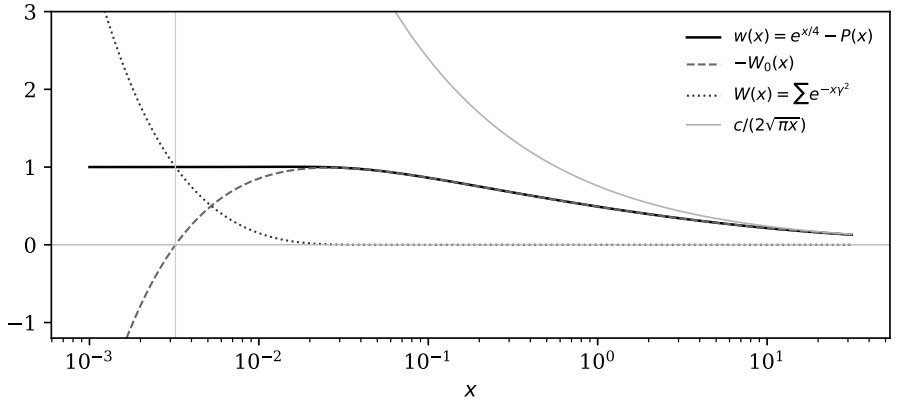


Figure 7.1: The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace W over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where W_0 changes sign. To the left, w differs from $e^{x/4}$ by less than 2×10^{-13}; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to W. The faint curve is the asymptote $c/(2\sqrt{\pi x})$.

In arithmetic terms, Theorem 7.1.1 is the Gaussian smoothed Chebyshev bound

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x} < \sqrt{4\pi x}\; e^{x/4} \qquad (x > 0).$$

The right side is the same smoothing applied to dy in place of $d\psi(y)$. To see what the primes alone give, write $\psi(y) = y + \Delta(y)$ and put $I(x) = \frac{1}{2} e^{x/4} \operatorname{erfc}(\frac{\sqrt{x}}{2})$.

Proposition 7.1.2. *For $x > 0$,*

$$w(x) = I(x) - Q_\Delta(x), \qquad Q_\Delta(x) = \frac{1}{\sqrt{4\pi x}} \int_{1^-}^{\infty} y^{-1/2} e^{-(\log y)^2/4x}\, d\Delta(y),$$

with $I(x) \sim (\pi x)^{-1/2}$ and $w(x) \sim \frac{c}{2}(\pi x)^{-1/2}$ as $x \to \infty$. Hence $Q_\Delta(x) \sim -(\frac{c}{2} - 1)(\pi x)^{-1/2}$ with $\frac{c}{2} - 1 \approx 0.343$. On the other hand, any bound $|\Delta(y)| \leq A\sqrt{y} \log^2 y$ for $y \geq 2$, including the one that RH implies, yields only $|Q_\Delta(x)| \leq A\,(x/2 + O(\sqrt{x}))$ after integration by parts.

Proof. With $d\Delta = d\psi - dy$, the main term is $(4\pi x)^{-1/2} \int_1^\infty y^{-1/2} e^{-(\log y)^2/4x} dy = (4\pi x)^{-1/2} \int_0^\infty e^{u/2 - u^2/4x} du = \frac{1}{2} e^{x/4}(1 + \operatorname{erf} \frac{\sqrt{x}}{2})$, and $e^{x/4}$ minus this is I. The asymptotics of w are Theorem 6.1.11. For the bound, integrate by parts and

use $\frac{d}{dy}\{y^{-1/2}e^{-(\log y)^2/4x}\} = -y^{-3/2}e^{-(\log y)^2/4x}(\frac{1}{2} + \frac{\log y}{2x})$ and $\int_0^\infty u^2 e^{-u^2/4x}(\frac{1}{2} + \frac{u}{2x})du = \frac{\sqrt{\pi}}{2}(2\sqrt{x})^3 \cdot \frac{1}{4} + 4x$; the boundary term at 1^- and the range $1 \le y < 2$ contribute $O(x^{-1/2})$. □

In this Gaussian weighting the error term Δ of the prime number theorem has a definite negative bias at large x, $Q_\Delta(x) \approx -0.343(\pi x)^{-1/2}$, which is $I + W_0$ up to an exponentially small term, and $w > 0$ is the statement that $Q_\Delta(x) < I(x)$ for all x. Pointwise bounds on Δ, which oscillates with both signs, cannot give this at large x: the margin that keeps w positive there is of order $x^{-1/2}$, while a pointwise bound on Δ controls Q_Δ only up to order x; the function $D(y) = -A\sqrt{y}\log^2 y$, in place of Δ, gives $Q_D(x) = -A\big(x/2 + 2\sqrt{x/\pi}\big)$ by the computation in the proof, so the bound of Proposition 7.1.2 cannot be improved using the size of Δ alone. Mean square information on $\Delta(y)/\sqrt{y}$ in the logarithmic scale, of the kind Cramér obtained under RH [66], controls Gaussian smoothed quantities much better, and we make no claim that such information is insufficient.

7.1.1.1 The place of φ in the Bondesson hierarchy

Recall from Section 1.3.2 the classes of laws on $[0,\infty)$ defined through the Lévy density $k(x)/x$: ID, SD (k nonincreasing), BO ($k(x)/x$ completely monotone) and GGC (k completely monotone). They are nested as GGC $\subset$ SD $\subset$ ID and GGC $\subset$ BO $\subset$ ID [39, 315].

THEOREM 7.1.3. *$\varphi \in$ ID but $\varphi \notin$ SD, $\varphi \notin$ BO, and φ is not a mixture of exponentials. In particular φ is not a generalised gamma convolution.*

Proof. ID is Theorem 7.1.1.

SD. If φ is not infinitely divisible there is nothing to prove, and if it is then $k = w$ by Theorem 6.1.11. Now $w = e^{x/4} - P$, and P together with every one of its derivatives vanishes at 0^+, so $w'(0^+) = \frac{1}{4} > 0$ and w strictly increases at the origin.

BO. Membership means that the Lévy density $w(x)/x$ equals $\int e^{-tx}m(dt)$ for some $m \ge 0$. Then $-\varphi'/\varphi(s) = \int_0^\infty xe^{-sx}\int e^{-tx}m(dt)\,dx = \int m(dt)/(s+t)^2$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, so φ has no zeros or poles there and RH follows from Proposition 6.1.9. Under RH, Theorem 6.1.11 gives $w(x) = \int e^{-tx}dM(t)$, with $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ as in Theorem 6.1.10, and since $M(0^+) = 0$ an integration by parts gives $w(x)/x = \int e^{-tx}M(t)\,dt$. Uniqueness of Laplace transforms gives $M\,dt = m \ge 0$, that is $S(\tau) \ge -1$ for almost all τ. But by Selberg's central limit theorem [303], $S(\tau) < -1$ for a positive proportion of $\tau \in [T, 2T]$ once T is large, so $M < 0$ on a set of positive measure. Numerically, for $527.71 \le \tau \le 527.90$, just below $\gamma_{289} = 527.9036\ldots$, ball arithmetic gives $N(\tau) = 288$ and $-1.143 < S(\tau) < -1.008$, so $M < 0$ there, and $S(\gamma_{289}^-) = -1.14548\ldots$ (`verification_reciprocal_zeta/certify_k9_k2.py`).

Mixtures of exponentials. The law is a mixture of exponentials exactly when it has a completely monotone density, that is when φ is a Stieltjes function, which requires $\operatorname{Im}\varphi(-t+i0)$ to have one sign. With $\zeta(\frac{1}{2}+i\tau) = Z(\tau)e^{-i\theta(\tau)}$ we get $\operatorname{Im}\varphi(-\tau^2+i0) = \zeta(\frac{1}{2})\sin\theta(\tau)/\{(1+4\tau^2)Z(\tau)\}$. On $(0,\gamma_1)$, Z has no zero while θ decreases through $-\pi$ at $\tau_\pi = 3.43622\ldots$, so $\sin\theta$ changes sign.

GGC. Since GGC is contained in SD and in BO, either clause excludes it. □

The theorem is unconditional.

The failures have two causes, and neither involves a zero off the line. Self decomposability and the GGC property fail because of the pole: the factor $1-4s$ that removes it puts $e^{x/4}$ into w, so the canonical function increases near the origin. The Bondesson property fails because the negative Thorin density of the gamma factor meets the fluctuations of the zeros about their smooth count, measured by S, which has large negative excursions: the first zero with $S(\gamma^-) < -1$ is $\gamma_{127} = 282.4651$, where $S(\gamma_{127}^-) = -1.0063$, and the most negative value among the first 460 zeros is at γ_{289} (the seven such zeros are listed after Theorem 7.2.3, and these values of S are enclosed in ball arithmetic by the program cited in the proof). This failure would occur even if RH were true, and since S is unbounded below, by Selberg and Tsang [303, 330], it recurs infinitely often. Both causes disappear for the completed function R, which is infinitely divisible and self decomposable with no hypothesis, by Theorem 5.1.1 [252, Theorem A]. What Theorem 7.1.1 adds there is the floor of Theorem 6.2.15: the canonical function of $e^{-c\sqrt{s}}R(s)$ splits into the arithmetic part w and a completely monotone archimedean part (Section 6.2).

7.1.1.2 The density of the law

Let X_φ be, as in Section 6.2, a random variable with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$. Differentiating the representation of Theorem 7.1.1 gives $-\varphi'(s) = \varphi(s)\int_0^\infty e^{-sx}w(x)\,dx$, which says that the measure $x\Pr(X_\varphi \in dx)$ is the convolution of the law of X_φ with $w(x)\,dx$. Since $\varphi(s)\to 0$ as $s\to\infty$, the law has no atom at 0, so it is absolutely continuous, and its density f satisfies Steutel's equation [315, Proposition III.4.16]

$$x\,f(x) = \int_0^x f(x-y)\,w(y)\,dy, \qquad x>0. \tag{7.1.1}$$

The right side is continuous in x by dominated convergence, since f is integrable and w is bounded and continuous on $[0,\infty)$, so f may be taken continuous on $(0,\infty)$. Near the origin, $w(y) = 1+O(y)$ gives $f(x) = \bar f(x)(1+O(x))$ with $\bar f(x) = x^{-1}\int_0^x f$, and $\bar f(x) > 0$ for every $x>0$, since otherwise $s\varphi(s)\to 0$, whereas $s\varphi(s)\to -\zeta(\frac{1}{2})/4 > 0$. As $\bar f' = (f-\bar f)/x$, this gives $(\log\bar f)' = O(1)$ near 0, so $\bar f$ and f have a common positive limit $f(0^+)$, and dominated convergence in $s\varphi(s) = \int_0^\infty e^{-y}f(y/s)\,dy$, f being bounded, gives $f(0^+) = \lim_{s\to\infty} s\varphi(s) = -\zeta(\frac{1}{2})/4 \approx 0.36509$.

At infinity the same equation gives $f(x) \sim w(x)/x \sim \frac{c}{2\sqrt{\pi}} x^{-3/2} \approx 0.75773\, x^{-3/2}$, in three steps. First, $xf(x) \le \sup w$. Second, split the integral in (7.1.1), written as $\int_0^x f(u)w(x-u)\,du$, at $u = x/2$: the part $u \le x/2$ is at most $\sup_{y \ge x/2} w(y) = O(x^{-1/2})$, and the part $u \ge x/2$ is at most $\sup_{[x/2,x]} f \cdot \int_0^{x/2} w = O(x^{-1})\, O(x^{1/2})$, so $f(x) = O(x^{-3/2})$. Third, with this bound the part $u \ge x/2$ is $O(x^{-1}) = o(w(x))$, while the part $u \le x/2$ is $w(x)(1 + o(1))$ by dominated convergence, since $w(x-u)/w(x) \to 1$ for fixed u and stays below 2 for $u \le x/2$ and large x. Hence $xf(x) = w(x)(1 + o(1))$. It follows that $\Pr(X_\varphi > x) \sim \frac{c}{\sqrt{\pi}} x^{-1/2}$ and $\mathbb{E}X_\varphi = \infty$, and since $\varphi(s) = 1 - c\sqrt{s} + O(s)$ gives $\varphi(s/n^2)^n \to e^{-c\sqrt{s}}$, the variable X_φ lies in the domain of attraction of the positive stable law of index $\frac{1}{2}$ that generates the building blocks (6.1.2).

THEOREM 7.1.4. *$f(0^+) = -\zeta(\frac{1}{2})/4$ and $f'(0^+) = f(0^+)\, w'(0^+) = -\zeta(\frac{1}{2})/16 \approx 0.09127 > 0$. In particular f is not monotone, so X_φ is not a mixture of uniform laws on intervals $[0, a]$, and f is not log convex and not completely monotone.*

Proof. The value of $f(0^+)$ was found after (7.1.1). Since P and all its derivatives vanish at 0^+, w is continuously differentiable on $[0, \infty)$ with $w(0^+) = 1$ and $w'(0^+) = \frac{1}{4}$. Write (7.1.1) as $xf(x) = \int_0^x f(u)\, w(x-u)\, du$. Since f is continuous on $(0, \infty)$ and bounded near 0, the right side is differentiable, with derivative $f(x)w(0^+) + \int_0^x f(u)w'(x-u)\,du$, so f is differentiable on $(0, \infty)$ and

$$f'(x) = \frac{1}{x} \int_0^x f(u)\, w'(x-u)\, du \longrightarrow f(0^+)\, w'(0^+) \qquad (x \downarrow 0),$$

because f and w' are continuous at 0. A nonincreasing density is required for a mixture of uniforms by Khintchine's theorem, and log convex or completely monotone densities are nonincreasing. □

PROPOSITION 7.1.5. *$w' > 0$ on $(0, 0.004]$ and $w' < 0$ on $[4.52, \infty)$. On $[0.004, 4.6]$ a grid evaluation of w' (computed, not proved) finds exactly one sign change, at $x_m = 0.015851\ldots$, where $w(x_m) = 1.003409\ldots$.*

Proof. On $(0, 0.004]$, $w' = \frac{1}{4}e^{x/4} - P'(x)$ and $0 \le P'(x) \le P'(0.004) < 2 \times 10^{-9}$, since each prime term of P' increases in x there. For $x \ge 0.004$ write $w' = W' - W_0'$. The zeros on the line contribute $-\sum \gamma^2 e^{-x\gamma^2} < 0$ to W'. Every zero below height 3×10^{12} lies on the line [247], so an off line zero contributes at most $(\gamma^2 + \frac{1}{4})e^{-x(\gamma^2 - 1/4)}$ with $\gamma > 3 \times 10^{12}$. With $N(t+1) - N(t) \le 2\log(t+1)$ from the proof of Theorem 7.1.1, these contributions together are less than $e^{-10^{22}} \gamma_1^2 e^{-x\gamma_1^2}$ for $x \ge 0.004$; hence $W' < 0$. By Proposition 6.2.11, with $\nu(dy) = 2\sum_{k \ge 0} e^{-(2k + \frac{1}{2})y}\, dy$

the measure defined there,

$$-W_0'(x) = \frac{D(x)}{4\sqrt{\pi}\,x^{3/2}}, \qquad D(x) = \theta'(0) + \frac{1}{2}\int_0^\infty g\Big(\frac{y^2}{4x}\Big)\nu(dy),$$
$$g(u) = 1 - (1-2u)e^{-u}.$$

Since $g(0) = 0$, $g'(0) = 3$, g is concave on $[0, \frac{5}{2}]$ and $g \le 1 + 2e^{-3/2} < 3u$ beyond, $g(u) \le 3u$ for all $u \ge 0$. With $\int y^2\nu(dy) = 4\sum_{k\ge 0}(2k+\frac{1}{2})^{-3} < 32.34$ this gives $D(x) \le \theta'(0) + 12.13/x < 0$ for $x \ge 4.52$, and then both parts of w' are negative. □

The computed statement comes from an evaluation of w' on $[0.004, 4.6]$ from the first 300 zeros and the integral for W_0', on a grid of spacing 10^{-4} up to 0.05 and 5×10^{-3} beyond. The prime and zero evaluations of w' agree to 10^{-11} at $x = 0.01$.

Numerically f is unimodal as well: it rises from 0.365089 to its maximum 0.366883 at the mode $x = 0.024046$ and then decreases to 0 like $0.75773\,x^{-3/2}$, so its mode lies to the right of the maximum of w. Talbot and de Hoog inversion agree to twelve digits at every point used. Unimodality of f remains open (Section 7.2.4). Since X_φ is infinitely divisible but not self decomposable, Yamazato's theorem that every self decomposable law is unimodal [347], [297, Theorem 53.1] does not apply. Section 11.1 applies the theorem to the law of X_R, which is self decomposable, and to the shifted clocks T_α of Theorem 1.2.2 with $\alpha \ge 1$ (Corollary 11.1.2).

7.1.1.3 Closure operations and the harmonic mean

GGC is closed under independent sums and weak limits, and, by a theorem of Bondesson [40], under independent products, so Theorem 7.1.3 gives at once the following.

Corollary 7.1.6. *X_φ is not a sum or a product of independent GGC variables, nor a weak limit of such.*

The reciprocal in φ suggests a fourth operation, the harmonic mean. Closure under it would be the statement

(H) if V_1 and V_2 are independent GGC variables, then so is $2V_1V_2/(V_1+V_2)$,

which would have widened Corollary 7.1.6; it is false.

Proposition 7.1.7. *(H) is false. If E_1 and E_2 are independent standard exponentials, then $H = 2E_1E_2/(E_1+E_2)$ has density*

$$f_H(h) = e^{-h}\int_0^\infty \Big(1 + \frac{h}{2u}\Big)e^{-u-h^2/4u}\,du, \qquad h > 0,$$

and H is not a GGC.

Proof. For $h > 0$, $2E_1E_2 > h(E_1 + E_2)$ holds if and only if $E_1, E_2 > \frac{h}{2}$ and $(2E_1 - h)(2E_2 - h) > h^2$: both factors cannot be negative, since $E_1, E_2 < \frac{h}{2}$ forces $|2E_1 - h|, |2E_2 - h| < h$. Substituting $E_1 = \frac{h}{2} + u$, $E_2 = \frac{h}{2} + v$ gives

$$\Pr(H > h) = e^{-h} \iint_{uv > h^2/4} e^{-u-v}\, du\, dv = e^{-h} \int_0^\infty e^{-u-h^2/4u}\, du,$$

so differentiating under the integral, which is justified by the bound $\frac{h}{2u} e^{-u-h^2/4u} \le u^{-1/2} e^{-u} \sup_{y>0} y e^{-y^2}$, locally uniform in h, gives f_H. In terms of the modified Bessel functions of the second kind the two integrals are $h\,\mathrm{K}_1(h)$ and $h\{\mathrm{K}_0(h) + \mathrm{K}_1(h)\}$, by $\int_0^\infty u^{-\nu-1} e^{-u-h^2/4u}\, du = 2(h/2)^{-\nu}\mathrm{K}_\nu(h)$ [225, (10.32.10)]. Splitting the integral at $u = h$ shows that f_H is bounded with $f_H(0^+) = 1$.

Suppose H were a GGC with drift a and Thorin measure U of total mass β. Since $f_H > 0$ on $(0, \infty)$ and a GGC satisfies $H \ge a$ almost surely, $a = 0$. Since f_H is bounded, $s\,\mathbb{E}e^{-sH} = \int_0^\infty e^{-y} f_H(y/s)\, dy \to f_H(0^+) = 1$, so $-\log \mathbb{E}e^{-sH} = \log s + o(1)$, while Fatou applied to $\int \log(1 + s/t)\, U(dt)$ gives $\liminf_{s\to\infty}(-\log \mathbb{E}e^{-sH})/\log s \ge \beta$. Hence $\beta \le 1$.

Now let V be any GGC with zero drift and $\beta \le 1$, and $\mathcal{L}(s) = \mathbb{E}e^{-sV}$. For $\operatorname{Im} s > 0$ each $1 + s/t$ lies in the upper half plane, so $\operatorname{Im} \log \mathcal{L}(s) = -\int \arg(1 + s/t)\, U(dt) \in [-\pi\beta, 0) \subseteq [-\pi, 0)$, and $\mathcal{L} > 0$ on $(0, \infty)$: $\mathcal{L}$ is a Stieltjes function [299, Corollary 7.4]. In its representation $\mathcal{L}(s) = b_0 + b_1/s + \int (s + r)^{-1} m(dr)$ [299, Definition 2.1] we have $b_0 = \lim_{s\to\infty} \mathcal{L}(s) = 0$ and $b_1 = 0$, since $\mathcal{L}(0^+) = 1$. So $\mathcal{L}(s) = \int_0^\infty e^{-sx} \big(\int e^{-rx} m(dr) \big) dx$ and the density of V is $\int e^{-rx} m(dr)$, completely monotone and in particular nonincreasing.

But f_H is not nonincreasing: $f_H(h) - 1 \sim h \log(1/h)$ as $h \downarrow 0$, and f_H attains 1.11125 at $h = 0.09765$ against $f_H(0^+) = 1$. So H is not a GGC. □

There is also a structural obstruction. The factorisation of φ into R and the gamma factor is a factorisation of Laplace transforms, that is a convolution of laws, while (H) acts on the variables themselves. The gamma factor is not a Laplace transform at all, since it tends to infinity with s, and (6.2.1) shows that it enters only through deconvolution by stable laws. The one place where (H) could have acted is the positive part: R is GGC if and only if RH holds, so an unconditional representation $X_R \overset{d}{=} 2V_1V_2/(V_1 + V_2)$ with V_1, V_2 independent GGC variables built from the primes and the gamma factor would have proved RH under (H). Proposition 7.1.7 removes the implication, and we know of no candidate representation.

7.1.2 The Vere-Jones weights and the Lévy representation across the strip

Throughout this subsection σ is the real part of the argument z of ζ.

7.1.2.1 Necklace polynomials and the sign rule

Let $M_k(q) = \frac{1}{k}\sum_{d|k}\mu(k/d)\,q^d$ be the necklace polynomial [202].

THEOREM 7.1.8. *$B_N = 0$ unless $N = p^n$ is a prime power, and then*

$$B_{p^n} = b_n\big(p^{-1/2}\big), \qquad b_n(q) = \sum_{i\ge 0,\ 2^i\mid n} M_{n/2^i}(q).$$

Equivalently, for $|y| < 1$ and $0 < q \le 1$,

$$\frac{1}{1-qy} = \prod_{n\ge 1}\big(1+y^n\big)^{b_n(q)}, \tag{7.1.2}$$

so that $\zeta(\frac12+u) = \prod_p\prod_n(1+p^{-nu})^{b_n(p^{-1/2})}$ for $\operatorname{Re} u > \frac12$. At $q = 1$, $b_n(1) = 1$ if n is a power of 2 and 0 otherwise, and (7.1.2) becomes Euler's binary identity $\frac{1}{1-y} = \prod_{j\ge0}(1+y^{2^j})$.

Proof. From the definition of B_N in Theorem 6.1.8, $B_{p^n} = \sum_{jm=n} d_m\,a_{p^j}$ with $a_{p^j} = q^j/j$. Proposition 6.1.6 gives $y^j = \sum_m d_m \log(1+y^{mj})$; multiplying by q^j/j and summing over j gives $\sum_n b_n(q)\log(1+y^n) = \sum_j q^j y^j/j = -\log(1-qy)$, which is (7.1.2). The cyclotomic identity $\frac{1}{1-qy} = \prod_k(1-y^k)^{-M_k(q)}$ and $\log(1+y^n) = \log(1-y^{2n}) - \log(1-y^n)$ give, comparing coefficients of $\log(1-y^n)$, the recursion $b_n = M_n + b_{n/2}$ for even n and $b_n = M_n$ for odd n, whose solution is the stated sum. At $q = 1$, $M_k(1) = \frac{1}{k}\sum_{d|k}\mu(k/d) = \delta_{k,1}$. □

The identity (7.1.2) is a 2 adic variant of the cyclotomic identity of Metropolis and Rota [202]; we have not found it stated in this form. Its exponents include $b_2 = \frac{q^2+q}{2}$, $b_3 = \frac{q^3-q}{3}$, $b_4 = \frac{q^4}{4} + \frac{q^2}{4} + \frac{q}{2}$, $b_6 = \frac{q^6+q^3-q^2-q}{6}$, which give B_4, B_8 and B_9 of Theorem 6.1.8 and, at $p = 2$, B_{16} and B_{64}.

PROPOSITION 7.1.9. *For every prime p, with $q = p^{-1/2}$: $b_{2^a}(q) > 0$ for all $a \ge 1$, and for odd $n > 1$, $b_n(q) \ne 0$ and the sign of $b_n(q)$ is $(-1)^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors of n.*

Proof. For $n = 2^a$, $b_n = q + \sum_{j=1}^{a}\frac{q^{2^j}-q^{2^{j-1}}}{2^j} = \frac{q}{2} + \frac{q^2}{4} + \cdots + \frac{q^{2^{a-1}}}{2^a} + \frac{q^{2^a}}{2^a} > 0$. For odd n, $b_n = M_n$. Put $\ell = \operatorname{rad}(n)$ and $\tilde q = q^{n/\ell} \le q \le 2^{-1/2}$; then $nM_n(q) = \sum_{k|\ell}\mu(\ell/k)\tilde q^k = (-1)^{\omega(n)}\big(\tilde q + \sum_{k|\ell,k>1}\pm\tilde q^k\big)$, and the divisors $k > 1$ of the odd squarefree ℓ are distinct odd integers at least 3, so $\sum_{k>1}\tilde q^k < \sum_{m\ \text{odd}\ge3}\tilde q^m = \tilde q\cdot\frac{\tilde q^2}{1-\tilde q^2} \le \tilde q$. □

For even n other than powers of 2 the sign depends on how the 2 adic parts combine. Among $2 \le n \le 60$ there are 42 negative exponents for each of $p = 2, 3, 5, 7$.

The weights B_N and d_m also govern the truncation of the sawtooth expansion of S. The sawtooth Σ_λ of Lemma 6.1.7 has the Fourier series $\Sigma_\lambda(\tau) = \sum_{j\geq 1}(-1)^j \frac{\sin(j\lambda\tau)}{\pi j}$, so the formal series $\sum_N B_N \Sigma_{\log N}(\tau)$ is a double series in (N, j) whose frequencies are $j \log N$.

Proposition 7.1.10. *Truncate the double series by frequency, keeping the terms with $j \log N \leq L$. Then*

$$\sum_{\substack{N,j \\ j\log N\leq L}} B_N\,(-1)^j \frac{\sin(j\tau\log N)}{\pi j} = -\frac{1}{\pi}\sum_{n\leq e^L} \frac{\Lambda(n)}{\sqrt{n}\log n}\sin(\tau\log n),$$

the classical truncated series for $S(\tau)$. Truncation by $N \leq N_0$ instead is not equivalent: it keeps the harmonics $j \log N > \log N_0$ of each retained sawtooth without the terms of larger N that cancel them.

Proof. With $N = n^m$, the frequency is $r \log n$ with $r = mj$, and the constraint $j \log N \leq L$ is $r \log n \leq L$, which does not separate m from j. The coefficient of $\sin(r\tau \log n)$ is $-\frac{a_n}{\pi}\sum_{mj=r} d_m \frac{(-1)^{j+1}}{j}$, and $\sum_{mj=r} d_m (-1)^{j+1} j^{-1}$ has Dirichlet series $\eta(1+v)^{-1}\eta(1+v) = 1$, so it is $\delta_{r,1}$. □

So the sawtooth expansion contains nothing that the classical series lacks once it is truncated consistently, and truncating it by the index N is worse. Numerically, on a uniform grid of step 10^{-3} in $\tau \in [10, 50]$ and with the Cesàro weight $1 - \log N/\log N_0$, the root mean square error against $S(\tau)$ is 0.228, 0.223, 0.218 for $N_0 = 10^3, 10^4, 10^5$, against 0.146, 0.128, 0.115 for the classical series with the same weight, the root mean square of S itself being 0.320 (`sawtooth_truncation.py`).

7.1.2.2 The cyclotomic form, the prime number theorem and a trace formula

The cyclotomic identity gives a second factorisation, into the blocks $1 - N^{-u}$:

$$\frac{1}{\zeta(\frac{1}{2}+u)} = \prod_p \prod_{k\geq 1} \bigl(1 - p^{-ku}\bigr)^{M_k(p^{-1/2})}, \qquad \operatorname{Re} u > \tfrac{1}{2}.$$

Each block $1 - e^{-\lambda u} = 2e^{-\lambda u/2}\sinh(\lambda u/2)$ is again a solved model, now with a sinh lattice: with $u = \sqrt{s}$, $\frac{\sinh(\lambda\sqrt{s}/2)}{\lambda\sqrt{s}/2} = \prod_{m\geq 1}\bigl(1 + \frac{s}{(2\pi m/\lambda)^2}\bigr)$, a reciprocal GGC with unit atoms at $(2\pi m/\lambda)^2$. The sinh blocks are the local zeta factors on the line $\sigma = 0$, and unlike the sech blocks they vanish at $u = 0$. Normalising the product at $u \to 0$ therefore requires two identities.

Proposition 7.1.11. *For $0 < q < 1$, summing over k in increasing order,*

$$\sum_{k\geq 1} M_k(q) = 0, \qquad \sum_{k\geq 1} M_k(q)\log k = \log(1-q).$$

Both follow from the prime number theorem in the forms $\sum_m \mu(m)/m = 0$ *and* $\sum_m \mu(m)\log m/m = -1$.

Proof. $\sum_{k\le K} M_k(q) = \sum_d \frac{q^d}{d}\sum_{m\le K/d}\frac{\mu(m)}{m}$. The inner sums are bounded by 1 in absolute value and tend to 0, so dominated convergence gives 0. Similarly $\sum_{k\le K} M_k(q)\log k = \sum_d \frac{q^d}{d}\sum_{m\le K/d}\frac{\mu(m)\log(dm)}{m} \to \sum_d \frac{q^d}{d}(-1) = \log(1-q)$, using $\sum \mu(m)/m = 0$ for the $\log d$ part. □

Heuristically, the two identities are what make the cyclotomic product of solved models converge at the edge $u \to 0$: $\prod_k (1-y^k)^{-M_k(q)}$ behaves near $y = 1$ like $(1-y)^{-\sum M_k}\prod_k k^{-M_k}$, and the finite limit $\frac{1}{1-q}$ requires both.

In heat trace form the Vere-Jones representation becomes a trace formula. Proposition 6.1.13 and Poisson summation give, for every $x > 0$,

$$\sum_{\gamma>0} e^{-x\gamma^2} - W_0(x) = e^{x/4} + \sum_p \sum_{n\ge1} b_n(p^{-1/2})\Big[\sum_{k\ge0} e^{-x\left(\frac{(2k+1)\pi}{n\log p}\right)^2} - \frac{n\log p}{4\sqrt{\pi x}}\Big], \quad (7.1.3)$$

where the left side is written under RH and is $W(x) - W_0(x)$ in general, and the double series converges absolutely once each bracket is written in its theta form. The left side is the heat trace of the zeros net of its smooth part; the right side is the pole plus the heat traces of the arithmetic progressions $\frac{(2k+1)\pi}{n\log p}$, the pseudo zeros of the blocks, each net of its own mean. So in heat trace the zeros of ζ are a signed superposition of arithmetic progressions with spacings $2\pi/(n\log p)$. Heuristically, constructive interference of the progressions places the mass at the ordinates γ, and the negative weights of Proposition 7.1.9 remove the mass from the other points of the progressions.

The trace formula does not give positivity of the Thorin measure of R. The obstruction is explicit: the weights $b_n(p^{-1/2})$ are not nonnegative integers, and the passage from the binary product at $q = 1$ to the critical line at $q = p^{-1/2}$ is a deformation of the exponents in which integrality and positivity are lost together. A proof of RH along these lines would have to show that the interference of the progressions in (7.1.3), with exactly these weights, produces a positive measure; Section 7.2.4 poses the deformation from $q = 1$ to $q = p^{-1/2}$ as a problem.

7.1.2.3 The Lévy and Thorin representations across the strip

For $\sigma > 1$ the Vere-Jones representation is a statement about the classical zeta distribution (Section 6.1.6): $\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau\log N_\sigma}$ where $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and $\log N_\sigma$ is compound Poisson with Lévy measure $\nu_\sigma = \sum_n \frac{\Lambda(n)}{n^\sigma \log n}\delta_{\log n}$. Theorem 7.1.8 decomposes this Lévy measure as

$$\nu_\sigma = \sum_p \sum_{n\ge1} b_n(p^{-\sigma})\,\nu^{\mathrm{alt}}_{n\log p}, \qquad \nu^{\mathrm{alt}}_\lambda = \sum_{k\ge1}\frac{(-1)^{k+1}}{k}\,\delta_{k\lambda},$$

where $\nu_\lambda^{\mathrm{alt}}$ is the formal Lévy measure of the fair coin λB, $B \in \{0,1\}$, whose characteristic function $\frac12(1+e^{-i\lambda\tau})$ is the sech block. For $\sigma > 1$ there is also a positive decomposition: by $\frac{1}{1-qy} = \prod_j (1+q^{2^j}y^{2^j})$, the exponent of p in N_σ has independent binary digits with $\Pr(\text{digit } j=1) = \frac{p^{-2^j\sigma}}{1+p^{-2^j\sigma}}$. The Vere-Jones form keeps the coins fair and moves $p^{-\sigma}$ into the multiplicities; the binary digit form keeps unit multiplicities and biases the coins.

Inside the critical strip neither form survives as a characteristic function, since $|\zeta(\sigma+i\tau)|$ is unbounded in τ for $0<\sigma<1$ and the Dirichlet series diverges (Section 6.1.6); the heat kernel supplies the replacement. For $0<\sigma<1$ put

$$\varphi_\sigma(s) = \frac{\zeta(\sigma)}{(1-s/(1-\sigma)^2)\,\zeta(\sigma+\sqrt{s})}, \qquad \varphi_{1/2} = \varphi.$$

THEOREM 7.1.12. *For $0<\sigma<1$ and $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\,dx$, where w_σ has the three equivalent forms*

$$\begin{aligned} w_\sigma(x) &= e^{(1-\sigma)^2 x} - \frac{1}{\sqrt{4\pi x}} \sum_{n\ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x} \\ &= e^{(1-\sigma)^2 x} + \sum_p \sum_{n\ge 1} b_n(p^{-\sigma})\, h_{n\log p}(x) \\ &= \tfrac12 e^{(1-\sigma)^2 x} \operatorname{erfc}\big((1-\sigma)\sqrt{x}\big) + \sum_{\gamma>0} \operatorname{Re}\Big[e^{a_\gamma^2 x} \operatorname{erfc}(a_\gamma\sqrt{x})\Big] \\ &\qquad + \frac{\gamma_E + \log\pi}{4\sqrt{\pi x}} + \sum_{k\ge 0}\Big[\tfrac12 e^{a_k^2 x}\operatorname{erfc}(a_k\sqrt{x}) - \frac{1}{4(k+1)\sqrt{\pi x}}\Big], \end{aligned}$$

with h_λ the Jacobi theta function of Proposition 6.1.13, $a_\gamma = \sigma - \frac12 - i\gamma$ (the last form written under RH; in general γ runs over $-i(\rho-\frac12)$), and $a_k = 2k+2+\sigma$. As $x\to\infty$, $w_\sigma(x) \sim \frac{\zeta'(\sigma)/\zeta(\sigma)}{2\sqrt{\pi x}}$, with $\zeta'(\sigma)/\zeta(\sigma) > 0$ on $(0,1)$. If $w_\sigma \ge 0$, then φ_σ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density $w_\sigma(x)/x$:

$$\log\varphi_\sigma(s) = -\int_0^\infty \big(1-e^{-sx}\big)\frac{w_\sigma(x)}{x}\,dx, \qquad \operatorname{Re} s > 0.$$

Proof. The first form is the argument of Theorem 6.1.11 with the pole now at $s=(1-\sigma)^2$. For the second, regroup $\sum_n b_n(q)\, n\log p \sum_j (-1)^j e^{-(jn\log p)^2/4x}$ by $r = jn$: with $nb_n = \sum_{ab=n} a\,d_a q^b$ the coefficient of $e^{-(r\log p)^2/4x}$ is $\sum_{b|r} q^b \sum_{ja=r/b} a\,d_a(-1)^j = -q^r$, since the inner sum is $-\delta_{r/b,1}$ (Proposition 6.1.13); this is the prime p part of the first form. For the third, insert the Hadamard product of ζ'/ζ into $-\varphi_\sigma'/\varphi_\sigma = \frac{1}{s-(1-\sigma)^2} + \frac{1}{2\sqrt{s}}\frac{\zeta'}{\zeta}(\sigma+\sqrt{s})$; the pole terms combine to $\frac{1}{2\sqrt{s}(\sqrt{s}+1-\sigma)}$, and each $\frac{1}{\sqrt{s}(\sqrt{s}+a)}$ is the transform of $e^{a^2x}\operatorname{erfc}(a\sqrt{x})$. At $\sigma=\frac12$ this reduces to $w = W - W_0$ with Proposition 6.2.2.

The asymptotic follows from the third form term by term. Uniformly for $|\arg a| \le \frac{3\pi}{4} - \epsilon$ and $|a|\sqrt{x} \ge 1$,

$$\tfrac{1}{2} e^{a^2 x} \operatorname{erfc}(a\sqrt{x}) = \frac{1}{2a\sqrt{\pi x}} + O\big(|a|^{-3} x^{-3/2}\big);$$

every a in the third form has $|\arg a| < \frac{\pi}{2} + \frac{1}{14}$, and $\sum |a|^{-3}$ converges. So $w_\sigma(x) = C_\sigma/(2\sqrt{\pi x}) + O(x^{-3/2})$ with

$$C_\sigma = \frac{1}{1-\sigma} + \sum_\rho \frac{1}{\sigma - \rho} + \frac{\gamma_E + \log \pi}{2} + \sum_{k \ge 0} \Big(\frac{1}{2k+2+\sigma} - \frac{1}{2k+2} \Big),$$

the sum over ρ taken in conjugate pairs, and $C_\sigma = \zeta'(\sigma)/\zeta(\sigma)$ by the Hadamard product of ξ. Infinite divisibility follows from $w_\sigma \ge 0$ by Landau's theorem as in Theorem 7.1.1, since φ_σ is analytic near $(0, \infty)$: $\zeta(\sigma + x) \ne 0$ for $x > 0$ and the pole at $x = 1 - \sigma$ is cancelled. □

Numerically w_σ is positive on every line we tried. We evaluated it on 300 points of $x \in [10^{-4}, 10^3]$, from the prime form for $x \le 0.02$ and the zero form beyond, for $\sigma = 0.05, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 0.9, 0.99$. In every case the minimum is attained at the largest x, where $2\sqrt{\pi x}\, w_\sigma(x)$ is close to $\zeta'(\sigma)/\zeta(\sigma)$ up to $\sigma = 0.75$ and falls short of it as $\sigma \to 1$, where the pole makes the approach slow and the asymptotic has not been reached at $x = 10^3$. Whether the Lévy density is positive on every vertical line of the strip is open (Problem 7.8).

For the Thorin representation two families have to be distinguished: φ_σ, which carries the gamma factor and the pole, and the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$. For $\sigma > \frac{1}{2}$ the second is, with no hypothesis, the Laplace transform of an infinitely divisible law obtained by tilting and subordinating the law of X_R (Theorem 9.2.5 of Section 9.2 and the proof of Proposition 10.1.1 of Section 10.1, where R_σ is written φ_α). The first never has the Thorin property.

PROPOSITION 7.1.13. *For every $\sigma \in (0,1)$, φ_σ is not the Laplace transform of a GGC, and if it is infinitely divisible it is not self decomposable.*

Proof. The argument is that of Theorem 7.1.3. By Theorem 7.1.12, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\, dx$ for $s > (1-\sigma)^2$, with

$$w_\sigma = e^{(1-\sigma)^2 x} - P_\sigma, \qquad P_\sigma(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x}.$$

Every derivative of P_σ vanishes at 0^+, so $w_\sigma(x) = 1 + (1-\sigma)^2 x + o(x)$ and w_σ strictly increases near the origin. If φ_σ were a GGC transform, then $-\varphi_\sigma'/\varphi_\sigma = a + \int U(dt)/(s+t)$ with $a \ge 0$ and $U \ge 0$ satisfying the Thorin integrability conditions, so that $k_U(x) = \int e^{-tx} U(dt)$ is integrable near 0 and, by Tonelli,

$\int U(dt)/(s+t) = \int_0^\infty e^{-sx} k_U(x)\,dx$ for $s > (1-\sigma)^2$. Letting $s \to \infty$ gives $a = 0$, and uniqueness of the Laplace transform gives $w_\sigma = k_U$, which is completely monotone and so nonincreasing, a contradiction. If φ_σ is infinitely divisible, the same uniqueness identifies its Lévy density as $w_\sigma(x)/x$, and self decomposability would require w_σ to be nonincreasing. □

As on the critical line, the obstruction is the factor $1 - s/(1-\sigma)^2$ that removes the pole, and it says nothing about the zeros. Beyond it, for $\frac{1}{2} < \sigma < 1$ with $\sup_\rho \operatorname{Re}\rho < \sigma$, the Thorin density $\frac{1}{2\pi\sqrt{t}} \operatorname{Re}\frac{\zeta'}{\zeta}(\sigma + i\sqrt{t})$ given by Lemma 6.1.1 also takes negative values on every line we examined (the minimum of $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\tau)$ over $\tau \in (0,60]$ is -1.049, -0.995, -0.867, -0.769, -0.721 at $\sigma = 0.55, 0.6, 0.75, 0.9, 0.99$), but the proposition does not need this.

The completed functions R_σ carry the zeros. Theorem 7.1.14 is Theorem 1.2.2 of Section 1.2 in the variable of this chapter.

THEOREM 7.1.14. *Let $\sigma \geq \frac{1}{2}$. Then $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is the Laplace transform of a GGC if and only if ζ has no zero ρ with $\operatorname{Re}\rho > \sigma$. In that case its Thorin measure is the Poisson smoothed zero density*

$$U_\sigma(dt) = \frac{1}{2\pi\sqrt{t}} \sum_\rho \frac{\sigma - \beta_\rho}{(\sigma-\beta_\rho)^2 + (\sqrt{t} - \gamma_\rho)^2}\,dt,$$

the sum running over all nontrivial zeros $\rho = \beta_\rho + i\gamma_\rho$, counted with multiplicity, of both signs of γ_ρ, and taken symmetrically; each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ contributes a unit atom at γ_ρ^2 instead of its kernel. Consequently R_σ is a GGC for every $\sigma \geq 1$, unconditionally, and

$$\Theta := \sup_\rho \operatorname{Re}\rho = \inf\{\sigma \geq \tfrac{1}{2} :\ R_\sigma \text{ is a GGC}\}.$$

RH is the statement $\Theta = \frac{1}{2}$, that is, that $R_{1/2} = R$ is a GGC.

Proof. A zero with $\operatorname{Re}\rho > \sigma$ gives $\sqrt{s} = \rho - \sigma$ on the principal branch and hence a pole of R_σ in $\mathbb{C} \setminus (-\infty, 0]$, so R_σ is not a GGC. Suppose there is none. Write the zeros with positive ordinate, counted with multiplicity, as $\rho = \beta + i\gamma$, put $a = \sigma - \beta \geq 0$, and let $\mathcal{P}_a(x) = a/\{\pi(a^2 + x^2)\}$, for $a > 0$, be the Cauchy density of scale a (the Poisson kernel P_a of Section 1.2). Hadamard's product [71, Chapter 12], with each ρ paired with $\bar\rho$, gives $\xi(z) = \xi(0)\prod_{\gamma>0}(1 - z/\rho)(1 - z/\bar\rho)$, absolutely convergent since $\sum_\rho |\rho|^{-2} < \infty$, and hence for $s > 0$ and $q = \sqrt{s}$

$$R_\sigma(s) = \prod_{\gamma>0} \frac{a^2 + \gamma^2}{(q+a)^2 + \gamma^2}.$$

For $a > 0$ let $U_{a,\gamma}(dt) = \{\mathcal{P}_a(\sqrt{t} - \gamma) + \mathcal{P}_a(\sqrt{t} + \gamma)\}\,dt/(2\sqrt{t})$, and let $U_{0,\gamma} = \delta_{\gamma^2}$. The substitution $t = \tau^2$, the identity $1/(s + \tau^2) = (\pi/q)\mathcal{P}_q(\tau)$ and $\mathcal{P}_a * \mathcal{P}_q = \mathcal{P}_{a+q}$

give

$$\int_0^\infty \frac{U_{a,\gamma}(dt)}{s+t} = \frac{q+a}{q\{(q+a)^2+\gamma^2\}} = \frac{d}{ds}\log\{(q+a)^2+\gamma^2\},$$

which also holds for $a = 0$. Integrating over $(0, s]$ with Tonelli, $\int \log(1 + s/t)\, U_{a,\gamma}(dt) = \log\{((q+a)^2+\gamma^2)/(a^2+\gamma^2)\}$, which is nonnegative and at most $(s + 2\sigma\sqrt{s})/\gamma^2$. Since $\sum_{\gamma>0}\gamma^{-2} < \infty$, Tonelli's theorem shows that $U_\sigma = \sum_{\gamma>0} U_{a,\gamma}$ is a positive measure with $\int \log(1 + s/t)\, U_\sigma(dt) = -\log R_\sigma(s) < \infty$, which gives the Thorin integrability conditions, and that $R_\sigma(s) = \exp\{-\int \log(1 + s/t)\, U_\sigma(dt)\}$ for $s > 0$. So R_σ is the Laplace transform of a GGC with Thorin measure U_σ, whose density is the stated sum, grouped by the pairs $\rho, \bar\rho$, and each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ gives the atom $\delta_{\gamma_\rho^2}$. For $\sigma \geq 1$ the hypothesis holds by the Euler product.

For $\sigma > \Theta$ the density also follows from Lemma 6.1.1: then $-R_\sigma'/R_\sigma(s) = \frac{1}{2\sqrt{s}}\frac{\xi'}{\xi}(\sigma + \sqrt{s})$ is $O(|s|^{-1/2})$ near 0, $O(|s|^{-1/2}\log|s|)$ at infinity uniformly on $\mathbb{C} \setminus (-\infty, 0]$ (by the partial fraction formula for ζ'/ζ [71, Chapter 15], since every zero lies at distance at least $\sigma - \Theta$ from the half plane $\operatorname{Re} z \geq \sigma$), and continuous up to the cut away from 0, and at $\sqrt{s} = i\tau$ the lemma gives the density $\frac{1}{2\pi\tau}\operatorname{Re}\frac{\xi'}{\xi}(\sigma + i\tau) = \frac{1}{2\pi\tau}\sum_\rho \operatorname{Re}\frac{1}{\sigma+i\tau-\rho}$, summed symmetrically. □

The quasi Riemann Hypothesis $\Theta < 1$, announced in October 2026 with $\Theta \leq \frac{7}{8}$ (Section 12.1.2), is the statement that R_σ is a GGC for some $\sigma < 1$; it is step 7 of the path to RH in Section 12.1 (Table 12.1), and short of it Section 11.4 shows with no hypothesis that for $\frac{1}{2} < \sigma < 1$, in the density $\rho_{\sigma-1/2}$ of (1.2.7), the Cauchy kernels of the zeros to the right of σ, if there are any, carry below height T a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$. Numerically $\min_{0<\tau\leq 70}\operatorname{Re}\frac{\xi'}{\xi}(\sigma + i\tau)$ is 4.6×10^{-5}, 0.0046, 0.0116, 0.0231 at $\sigma = 0.501, 0.6, 0.75, 1$, attained as $\tau \to 0$.

7.1.3 Pólya and Cardon theory for the solved models

Pólya's theory of Fourier transforms with only real zeros, and its extension by Cardon, is built from the objects that the Vere-Jones step produces: cosines, hyperbolic cosines and their products. The solved models are special cases of that theory; ζ is not, because its multiplicities are neither integers nor nonnegative. Throughout, Φ is the Pólya kernel and $\Xi(\tau) = \int \Phi(t)\cos(\tau t)\, dt$, as in (1.2.3) with the variable written t; Φ is even, analytic in the strip $|\operatorname{Im} t| < \pi/4$, and has a natural boundary on the lines $\operatorname{Im} t = \pm\pi/4$ [284, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The Pólya variable X of Section 1.2 has density $\Phi/\Xi(0)$ and moment generating function $\mathbb{E}e^{hX} = \xi(\frac{1}{2} + h)/\xi(\frac{1}{2})$, by (1.2.4). In this subsection G denotes a function of the Laguerre Pólya class, not the gamma factor.

7.1.3.1 Solved models and the Laguerre Pólya class

On the critical line the block $1+N^{-\sqrt{s}}$ becomes

$$1+N^{-i\tau}=2e^{-i\tau\ell/2}\cos\tfrac{\ell\tau}{2},\qquad \ell=\log N,$$

and $\cos(\ell\tau/2)=\mathbb{E}e^{i\tau(\ell/2)\varepsilon}$ with $\varepsilon=\pm1$ equally likely. A product of blocks with nonnegative integer multiplicities m_N is, up to the phase, the characteristic function of the Rademacher sum $\sum_N\sum_{j\le m_N}\frac{\log N}{2}\varepsilon_{N,j}$, and all its zeros are real, lying on the lattices $(2k+1)\pi/\log N$. Cardon's theorem on exponential sums [53] extends this: if $a_1,\dots,a_n$ and $b_1,\dots,b_n$ are positive and G is a real entire function of genus 0 or 1 with only real zeros, then

$$\sum_{\varepsilon\in\{\pm1\}^n}G\Big(i\sum_k a_k\varepsilon_k\Big)\exp\Big(iz\sum_k b_k\varepsilon_k\Big),$$

with the same signs in both sums, has only real zeros.

The function ζ is not of this form. Formally $\zeta(\frac12+i\tau)=\prod_N(1+N^{-i\tau})^{B_N}$, but the exponents B_N are not integers, and some are negative: by Theorem 7.1.8, at prime cubes $B_{p^3}=\frac13p^{-3/2}-\frac13p^{-1/2}<0$.

In the kernel variable the solved models act as multipliers. By Pólya's Hilfssatz II [273, Hilfssatz II], if $a>0$, b is real and G is a real entire function of genus 0 or 1 with only real zeros, at least one of them, then $e^{ib}G(z+ia)+e^{-ib}G(z-ia)$ has only real zeros. With $G=\Xi$ this says that replacing $\Phi(t)$ by $\Phi(t)\cosh(at)$, a solved model in the variable t, cannot create nonreal zeros: if Ξ has only real zeros, so does the transform of the new kernel. For $a\ge\frac12$ the conclusion needs no hypothesis: the transform of $\Phi(t)\cosh(at)$ is $\frac12(\xi(s+a)+\xi(s-a))$ with $s=\frac12+iz$, and Lagarias proved that for $a\ge\frac12$ its zeros all lie on the critical line and are simple [161]. Products of such factors converge to Gaussians, $\prod_{k\le n}\cosh(ct/\sqrt{n})\to e^{c^2t^2/2}$, and this is de Bruijn's route to the Gaussian multipliers $e^{\lambda t^2}$ [73]. In the variable t, multiplying $\Phi(t)$ by $e^{\lambda t^2/4}$ is the de Bruijn Newman flow at time λ (Section 8.4), the flow of Theorem 7.2.15, so the multiplier $e^{\lambda t^2}$ is the flow at time 4λ. The solved models are therefore the elementary steps of the de Bruijn Newman flow, and the bound $\Lambda\le0.2$ [247] says that the flow at time 0.2, applied to Φ or, in the normalisation of Rodgers and Tao, to Φ_{dBN} defined before Theorem 7.2.15, already produces a kernel whose transform has only real zeros.

Proposition 7.1.15. *Let G be a real even entire function of order less than 2 with $G(0)\neq0$. Then G is in the Laguerre Pólya class if and only if $s\mapsto G(0)/G(i\sqrt{s})$ is the Laplace transform of a GGC whose Thorin measure is $\sum_k m_k\delta_{\alpha_k^2}$ with integers $m_k\ge1$, where $\pm\alpha_k$ are the zeros of G with multiplicities m_k.*

Proof. By Hadamard, $G(z)=G(0)\prod_k(1-z^2/\alpha_k^2)^{m_k}$ with no exponential factor, since G is even and of order less than 2. All α_k are real if and only if

$G(0)/G(i\sqrt{s}) = \prod_k (1 + s/\alpha_k^2)^{-m_k}$ is a product of gamma transforms, which is the stated GGC. If some α_k^2 is not positive, the product has a zero or a pole off $(-\infty, 0]$, which a GGC transform cannot have. □

The Laguerre Pólya class consists of the locally uniform limits of real polynomials with only real zeros, and a real entire function of order less than 2 belongs to it exactly when its zeros are real. In the language of Proposition 7.1.15, Pólya's multiplier theorem [54, Proposition 1], for even G of order less than 2, reads: if f is even and integrable with $f(t) = O(e^{-|t|^b})$ for some $b > 2$, $\int f(t)e^{izt}dt$ has only real zeros, and $G(0)/G(i\sqrt{s})$ is an integer atom GGC, then $\int G(it)f(t)e^{izt}dt$ has only real zeros. RH is the statement that $R = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is an integer atom GGC. The Thorin measures in the Pólya theory are always integer valued.

7.1.3.2 Lee and Yang, and Newman

Pólya's Hilfssatz II has a physical reading, to which Kac drew attention in his comments on Pólya's 1926 paper [280, pp. 424 to 426]: it is the one spin case of the Lee and Yang circle theorem for the Ising ferromagnet [171]. Cardon's exponential sums make the correspondence explicit. For $\varepsilon \in \{\pm 1\}^n$ the sum

$$\mathcal{Z}_n(z) = \sum_{\varepsilon} G\Big(i \sum_k a_k \varepsilon_k\Big) \exp\Big(iz \sum_k b_k \varepsilon_k\Big)$$

is a partition function. The spins are weighted by $G(i \sum_k a_k \varepsilon_k)$, a function of a magnetisation, and they feel the purely imaginary external fields izb_k. For the Gaussian member of the Laguerre Pólya class, $G(x) = e^{-\beta x^2/2}$, which has genus 2 and lies outside Cardon's theorem, the weight is $\exp(\beta(\sum_k a_k \varepsilon_k)^2/2)$, the Curie Weiss ferromagnet with couplings $\beta a_j a_k \geq 0$. The zeros of $\mathcal{Z}_n$ in z are then real by the Lee and Yang theorem [171], since off the real axis the fields izb_k have real parts of one sign. A general even G in the class is a limit of products of such Gaussians and of factors $1 + M^2/\alpha^2$ in the magnetisation $M = \sum_k a_k \varepsilon_k$, and Cardon's theorem is the Lee and Yang theorem for this family of ferromagnetic interactions of the magnetisation.

Newman made the connection with the Riemann Hypothesis precise [210]. In a 1974 paper on generalised Ising systems [208] he had characterised the single spin laws with the Lee and Yang property: a symmetric single spin law ρ with $\int e^{bt^2}\rho(dt) < \infty$ for every b has the Lee and Yang property, that ferromagnetic systems of ρ spins have zeros only on the imaginary field axis, exactly when its moment generating function $\int e^{ht}\rho(dt)$ has only purely imaginary zeros, that is when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class. In Thorin language this reads as follows.

PROPOSITION 7.1.16. *Let ρ be a symmetric law with Gaussian moments of all orders. Then ρ has the Lee and Yang property if and only if $s \mapsto 1/\int e^{\sqrt{st}}\rho(dt)$ is*

the Laplace transform of a GGC with zero drift whose Thorin measure has integer atoms. The atoms are at the squared zeros α_k^2 of $\int e^{izt}\rho(dt)$, with the multiplicities of the zeros.

Proof. By Newman's theorem [208] the Lee and Yang property holds exactly when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class, and then, by Hadamard, $\int e^{ht}\rho(dt) = e^{bh^2}\prod_k(1+h^2/\alpha_k^2)^{m_k}$ with $b \geq 0$. In fact $b = 0$: for every $\epsilon > 0$, $e^{ht} \leq e^{\epsilon h^2 + t^2/4\epsilon}$ gives $\int e^{ht}\rho(dt) \leq C_\epsilon e^{\epsilon h^2}$ for real h, while the right side is at least e^{bh^2}. With $h = \sqrt{s}$ the reciprocal is $\prod_k(1+s/\alpha_k^2)^{-m_k}$, a GGC with zero drift and Thorin measure $\sum m_k \delta_{\alpha_k^2}$. Conversely, as in Proposition 7.1.15, a GGC transform with zero drift and integer atoms m_k at α_k^2 is $\prod_k(1+s/\alpha_k^2)^{-m_k}$, whose reciprocal at $s = -z^2$ is $\prod_k(1-z^2/\alpha_k^2)^{m_k}$, a function in the Laguerre Pólya class. □

Apply this to the Pólya variable X: since $\mathbb{E}e^{hX} = \xi(\frac{1}{2}+h)/\xi(\frac{1}{2})$, we have $1/\mathbb{E}e^{\sqrt{s}X} = R(s)$, the reciprocal Xi function of Section 6.1.1. Three statements therefore coincide: RH; the law of X is a Lee and Yang single spin law; R is a GGC with no drift whose Thorin measure has an atom at each γ^2, of mass the multiplicity of the zero. The de Bruijn Newman flow at time λ multiplies $\Phi(t)$ by $e^{\lambda t^2/4}$, which in Ising terms adds the ferromagnetic self interaction $e^{\lambda t^2/4}$ to the Pólya variable viewed as a spin, and Λ is the least λ for which the resulting spin is Lee and Yang. Newman's conjecture $\Lambda \geq 0$, proved by Rodgers and Tao [289], says that no antiferromagnetic self interaction can be tolerated.

7.1.3.3 A Cardon route

Cardon's Fourier theorem [54, Theorem 1] gives a constructive sufficient condition. If

$$\Phi(t)\,dt = \Xi(0)\,G(it)\,\Pr\Big(\sum_k a_k\varepsilon_k \in dt\Big), \tag{7.1.4}$$

where the ε_k are independent fair signs, the $a_k > 0$ are nonincreasing, and G is real, even, entire of order less than 2 with only real zeros, then Ξ has only real zeros. In the Ising reading, (7.1.4) says that the Pólya variable is the magnetisation of a weighted Ising paramagnet with weights a_k, tilted by a ferromagnetic interaction $G(iM)$ of its magnetisation $M = \sum_k a_k\varepsilon_k$. Since $|G(it)| \geq |G(0)|$ for even G in the class, the law of $\sum_k a_k\varepsilon_k$ must carry the doubly exponential decay of Φ, and among Rademacher laws the random harmonic series does.

Proposition 7.1.17. *For $\kappa > -1$ let $H_\kappa = \sum_{k\geq 1}\varepsilon_k/(2(k+\kappa))$, with characteristic function $\hat{f}_\kappa(v) = \prod_k \cos(v/(2(k+\kappa)))$ and density f_κ.*

(i) As $s \to \infty$, $\log \mathbb{E}e^{sH_\kappa} = \frac{s}{2}\log s + A(\kappa)s + o(s)$ with $A(\kappa) = A(0) - \frac{1}{2}(\psi(1+\kappa)+\gamma_E)$ (here ψ is the digamma function), $A(0) = -0.1485757\ldots$, so $\log\Pr(H_\kappa > u) = -\frac{1}{2}e^{2u-2A(\kappa)-1}(1+o(1))$ as $u \to \infty$. Since $\log\Phi(t) =$

$-\pi e^{2t} + \frac{9}{2}t + O(1)$ *as* $t \to +\infty$, *the double exponential terms agree exactly when* $A(\kappa) = -\frac{1}{2}(1 + \log 2\pi)$, *that is at* $\kappa^* = 6.61845\ldots$.

(ii) $\limsup_{|v|\to\infty} |v|^{-1} \log|\hat{f}_\kappa(v)| = -\pi/4$ *for every* κ. *Hence* f_κ *is analytic in the strip* $|\operatorname{Im} t| < \pi/4$, *which is exactly the strip of analyticity of* Φ.

Proof. (i) Put $\ell_\kappa(s) = \log \mathbb{E}e^{sH_\kappa} = \sum_{k\geq 1} \log\cosh \frac{s}{2(k+\kappa)}$. For $\kappa = 0$ put $\lambda = s/2$ and split the sum at $k = \lambda$, using $\log\cosh y = y - \log 2 + \log(1 + e^{-2y})$ for $k \leq \lambda$. Since $\sum_{k\leq\lambda} 1/k = \log\lambda + \gamma_E + O(1/\lambda)$, and since $k \mapsto \log(1 + e^{-2\lambda/k})$ and $k \mapsto \log\cosh(\lambda/k)$ are monotone, so that their sums differ from the corresponding integrals by $O(1)$,

$$\ell_0(s) = \lambda\log\lambda + (\gamma_E - \log 2)\lambda + \lambda\int_1^\infty \frac{\log(1+e^{-2y})}{y^2}\,dy + \lambda\int_0^1 \frac{\log\cosh y}{y^2}\,dy + O(1),$$

which is $\frac{s}{2}\log s + A(0)s + O(1)$ with $A(0) = \frac{1}{2}\big(\gamma_E - 2\log 2 + \int_1^\infty y^{-2}\log(1+e^{-2y})\,dy + \int_0^1 y^{-2}\log\cosh y\,dy\big)$. For general κ the mean value theorem gives $s^{-1}(\ell_\kappa - \ell_0)(s) = \sum_k \tanh(s\xi_k)\big(\frac{1}{2(k+\kappa)} - \frac{1}{2k}\big)$ with ξ_k between $\frac{1}{2(k+\kappa)}$ and $\frac{1}{2k}$, and since $\tanh(s\xi_k) \to 1$, dominated convergence gives the limit $\frac{1}{2}\sum_k(\frac{1}{k+\kappa} - \frac{1}{k}) = -\frac{1}{2}(\psi(1+\kappa) + \gamma_E)$. For the tail put $s_u = e^{2u - 2A(\kappa) - 1}$. Chebyshev's exponential inequality at $s = s_u$ gives $\log\Pr(H_\kappa > u) \leq \ell_\kappa(s_u) - s_u u = -\frac{1}{2}s_u(1 + o(1))$. For the lower bound tilt the law of H_κ by $e^{sx - \ell_\kappa(s)}$. By convexity of ℓ_κ the tilted mean is $\ell_\kappa'(s) = \frac{1}{2}\log s + A(\kappa) + \frac{1}{2} + o(1)$, and the tilted variance is $\ell_\kappa''(s) = \sum_k (2(k+\kappa))^{-2}\operatorname{sech}^2\frac{s}{2(k+\kappa)} = O(1/s)$, by comparison with an integral. Given $\delta > 0$, choose s with $\ell_\kappa'(s) = u + \delta/2$; by Chebyshev's inequality the tilted law gives probability at least $\frac{1}{2}$ to $(u, u+\delta]$ for large u, so $\log\Pr(H_\kappa > u) \geq \ell_\kappa(s) - s(u+\delta) - \log 2 = -\frac{1}{2}s_u e^{\delta}(1+\delta)(1+o(1))$. Letting $\delta \to 0$ gives the tail.

(ii) Since $\hat{f}_\kappa$ is even we take $v \to +\infty$. First, $\int_0^\infty \log|\cos y|\, y^{-2}dy = -\pi/2$: as $\log|\cos|$ is even and π periodic and $\sum_{n\in\mathbb{Z}}(y + n\pi)^{-2} = \csc^2 y$, the integral equals $\int_0^{\pi/2}\log(\cos y)\csc^2 y\,dy$, and an integration by parts against $-\cot y$ turns this into $-\int_0^{\pi/2}\cot y\tan y\,dy = -\pi/2$. For the upper bound fix $M, Y > 0$ and put $g_M = \max(\log|\cos|, -M)$, which is continuous, of bounded variation on $[0, Y]$ and $O(y^2)$ at 0. The points $y_k = v/(2(k+\kappa))$ have spacing at most $2y_k^2/v$, so $v^{-1}\sum_{y_k\leq Y} g_M(y_k) \to \frac{1}{2}\int_0^Y g_M(y)y^{-2}dy$ as a Riemann sum. Since $\log|\cos| \leq g_M \leq 0$, this gives $\limsup_{v\to\infty} v^{-1}\log|\hat{f}_\kappa(v)| \leq \frac{1}{2}\int_0^Y g_M(y)y^{-2}dy$, and letting $Y \to \infty$ and then $M \to \infty$ gives the bound $-\pi/4$. For the lower bound, the same Riemann sum argument, applied to the bounded continuous function $\Psi(y)/y$ with

$\Psi(y) = \int_0^y \log|\cos u|\,du$, gives

$$\int_0^V \log|\hat f_\kappa(v)|\,dv = \sum_k 2(k+\kappa)\Psi\Big(\frac{V}{2(k+\kappa)}\Big) = \frac{V^2}{2}\int_0^\infty \frac{\Psi(y)}{y^3}\,dy + o(V^2) = -\frac{\pi V^2}{8} + o(V^2),$$

since an integration by parts gives $\int_0^\infty \Psi(y)y^{-3}dy = \frac12\int_0^\infty \log|\cos y|\,y^{-2}dy = -\pi/4$. If $\log|\hat f_\kappa(v)| \le -(\pi/4+\epsilon)v$ for all large v, the left side would be at most $-(\pi/8+\epsilon/2)V^2 + O(1)$. Finally, $|\hat f_\kappa(v)| \le C_\epsilon e^{-(\pi/4-\epsilon)|v|}$ makes the Fourier integral of $\hat f_\kappa$ converge locally uniformly in $|\operatorname{Im} t| < \pi/4 - \epsilon$, so f_κ is analytic there. □

The coincidence of the strips in (ii) has a common source: both densities are Fourier transforms of functions that decay at the exponential rate $\pi/4$ on the real axis. For f_κ this is (ii). For Φ, $|\Xi(\tau)| \le e^{-\pi|\tau|/4+o(|\tau|)}$ by Stirling's formula, since $\zeta(\frac12+i\tau)$ grows at most polynomially, and the strip ends at $|\operatorname{Im} t| = \pi/4$ because Φ is a theta series in e^{2t}, which reaches the imaginary axis there.

With both leading orders matched, the question is the ratio

$$r(t) = \frac{\Phi(t)}{\Xi(0) f_{\kappa^*}(t)},$$

which must equal $G(it)$. We computed f_{κ^*} on the real axis by a saddle point contour integral of $\mathbb{E}e^{sH_{\kappa^*}}$, and on the imaginary axis by the Fourier integral $f_{\kappa^*}(ix) = \frac1\pi\int_0^\infty \hat f_{\kappa^*}(v)\cosh(vx)\,dv$. The two methods agree at the origin, $f_{\kappa^*}(0) = 2.104533$, to seven digits.

Proposition 7.1.18. *The ratio r satisfies the following necessary conditions for* (7.1.4) *(computed).*

(i) Growth. r is even, $r(0) = 0.853936$, and $\log r(t)$ becomes linear with slope approaching $\frac72$: the increments over steps of 0.1 fall from 0.3534 to 0.3514 between $t = 1.5$ and $t = 2.2$. The slope is explained exactly: Φ carries the prefactor $e^{9t/2}$, and the saddle point prefactor $\sqrt{s^*/\pi} \approx e^t$ of f_{κ^*} removes one unit. So G would have exponential type $\frac72$, compatible with order one.

(ii) Stieltjes signs. With $L(s) = \log r(\sqrt{s}) - \log r(0)$, a Laguerre Pólya G requires $L'(s) = \sum m_k/(s+\alpha_k^2)$, a Stieltjes function. On a grid of 81 points in $s \in [0,4]$ the difference quotients of L give $L' > 0$, $L'' < 0$ and $L''' > 0$ everywhere, and the first order quotients fall from 4.07 on the first grid interval to 0.94 on the last.

(iii) The imaginary axis. The continuation $G(x) = r(-ix)$, computed independently from $\Phi(ix)$ and $f_{\kappa^*}(ix)$, is real, agrees with r at second order

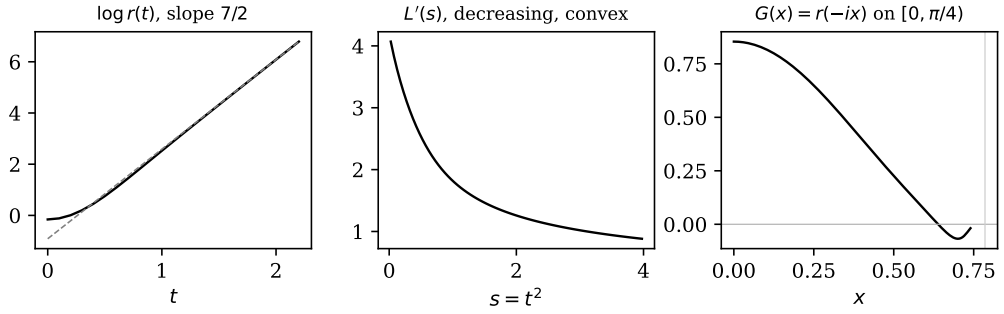


Figure 7.2: The Cardon ratio for the random harmonic law H_{κ^*}. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$.

> *(*$\log(G(0.1)/G(0)) = -0.0423$ *against* $\log(r(0.1)/r(0)) = 0.0417$*), decreases from* 0.8539*, has a simple zero at* $\alpha_1 = 0.6388$*, and satisfies the Laguerre inequality* $G'^2 - GG'' \geq 2.4 > 0$ *on* $[0, 0.735]$*.*

Everything visible inside the strip is consistent with G being a Laguerre Pólya function of exponential type $\frac{7}{2}$. The decisive question cannot be settled inside the strip.

Proposition 7.1.19. *If* (7.1.4) *holds with* G *entire, then*

$$f_{\kappa^*}(t) = \Phi(t)/(\Xi(0)G(it)),$$

so f_{κ^*} *continues meromorphically exactly as far as* Φ *does, and the two share their singular behaviour on the lines* $\operatorname{Im} t = \pm\pi/4$*, apart from the zeros of* G*. Since* Φ *has a natural boundary on these lines, coming from the Jacobi theta function at the edge of its modular domain,* f_{κ^*} *must have the same natural boundary, with singular behaviour proportional to that of* Φ *through a factor that continues to an entire function.*

The first sentence of Proposition 7.1.19 is immediate. The second uses the classical fact that the theta function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ has the line $\operatorname{Re} x = 0$ as a natural boundary [284, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], together with $\Phi(t) = 4x^{9/4}\omega''(x) + 6x^{5/4}\omega'(x)$ at $x = e^{2t}$.

So the random harmonic law fits the Pólya kernel inside the strip, and the question is the boundary, where the two functions have different origins: Φ is a modular theta series and f_{κ^*} a Rademacher product whose characteristic function has real zeros at $v = (2j+1)\pi(k+\kappa^*)$, increasingly dense. We see no reason for the Rademacher product to reproduce the modular singularities of Φ, and this

suggests that the right Ising system for the Pólya variable is not the harmonic paramagnet but one whose magnetisation law carries theta function structure.

7.1.3.4 The Pólya variable as the logarithm of a GGC

PROPOSITION 7.1.20. *For $0 < q < 1$ the triple product*

$$\sum_{n\in\mathbb{Z}} q^{n^2} e^{inv} = \prod_{m\geq 1} (1-q^{2m})\big(1 + 2q^{2m-1}\cos v + q^{4m-2}\big)$$

exhibits the discrete Gaussian law $\Pr(J=n) \propto q^{n^2}$ as the law of $\sum_m (B_m - B'_m)$ with independent Bernoulli variables of parameter $q^{2m-1}/(1+q^{2m-1})$. Each factor vanishes at $v = \pi \pm i(2m-1)\log(1/q)$, so the characteristic function has no real zeros and the moment generating function has no purely imaginary zeros: the triple product spins are not Lee and Yang spins, and neither is the discrete Gaussian.

Proof. $1 + 2q^{2m-1}\cos v + q^{4m-2} = (1+q^{2m-1}e^{iv})(1+q^{2m-1}e^{-iv})$, and each factor, normalised, is the characteristic function of a Bernoulli variable or its negative. The zeros are those of $1 + q^{2m-1}e^{\pm iv}$. □

The triple product places the theta function in the field variable. For Φ the theta function enters multiplicatively, through the modular variable e^{2t}; the identity of Biane, Pitman and Yor [32], going back to Williams [344] (Section 1.2), makes this exact.

THEOREM 7.1.21. *Let $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponentials e_k, e'_k, so that $\mathbb{E}e^{-sS_2} = \big(\sqrt{2s}/\sinh\sqrt{2s}\big)^2$ and S_2 is a GGC with Thorin measure $2\sum_k \delta_{\pi^2k^2/2}$. Then*

$$2\xi(z) = \mathbb{E}\Big[\Big(\frac{\pi S_2}{2}\Big)^{z/2}\Big].$$

Consequently, with $Y = \frac{1}{2}\log(\pi S_2/2)$, $2\Xi(\tau) = \mathbb{E}[e^{Y/2}e^{i\tau Y}]$, and the Pólya variable X is Y tilted by $e^{Y/2}$: $\Phi(t)\,dt = \frac{1}{2}e^{t/2}\Pr(Y \in dt)$.

Proof. This is the identity of [32, Section 2]; the density of S_2 is a theta series by the partial fraction expansion of $(x/\sinh x)^2$ and Poisson summation, and its Mellin transform is 2ξ. □

Section 9.1.1 tilts Y to every real basepoint σ: with no hypothesis, $\xi(\sigma+z)/\xi(\sigma)$ is the moment generating function of $\frac{1}{2}\log(\frac{\pi}{2}S_2)$ under the law of S_2 size biased by $S_2^{\sigma/2}$.

Theorem 7.1.21 shows where the modular boundary comes from. It is the quadratic lattice $\pi^2k^2/2$ of the Thorin atoms of S_2, through Jacobi's theta function and Poisson summation, that produces the modular natural boundary at $\operatorname{Im} t =$

$\pm\pi/4$: the density of S_2 is analytic in $\operatorname{Re} x > 0$ with natural boundary on $\operatorname{Re} x = 0$, and $x = \frac{2}{\pi}e^{2t}$. The Thorin structure thus enters twice: R is the reciprocal moment generating function of the Pólya variable in $\sqrt{s}$, and S_2 is a GGC in the usual sense. RH is the statement that every zero of $w \mapsto \mathbb{E}[(\pi S_2/2)^w]$ in the strip $0 < \operatorname{Re} w < \frac{1}{2}$ lies on the line $\operatorname{Re} w = \frac{1}{4}$.

This suggests a multiplicative Lee and Yang property. We say that a positive random variable V has *Mellin zeros on a line* if $\mathbb{E}V^w$, continued, vanishes only on one vertical line. The solved models of the Vere-Jones step have exact multiplicative analogues.

PROPOSITION 7.1.22. *Let $V = \sum_{k\le n} c_k e_k$ with independent standard exponentials e_k and distinct $c_k > 0$. Then $\mathbb{E}V^{u-1} = \Gamma(u)\sum_k A_k c_k^u$ with $A_k = c_k^{n-2}/\prod_{j\neq k}(c_k - c_j)$.*

(i) For $n = 2$ the zeros are $u = 2\pi i m/\log(c_1/c_2)$, $m \neq 0$: all on the line $\operatorname{Re} u = 0$. The two exponential sum is a multiplicative solved model.

(ii) For $n = 3$ with $c = (1, \frac{1}{4}, \frac{1}{9})$, a truncation of the quadratic lattice with single atoms, the zeros in $0 < \operatorname{Im} u < 30$ have real parts 0.488, −1.268, 0.300, 0.387, −1.318, 0.457, 0.136, −1.129, 0.500 (computed): they do not lie on a line.

Proof. The density of V is $\sum_k A_k e^{-x/c_k}$ by partial fractions of $\prod_k (1 + c_k s)^{-1}$, and its Mellin transform is the stated exponential polynomial times $\Gamma(u)$. For $n = 2$ it is proportional to $c_1^u - c_2^u$. For $n = 3$ the zeros are those of $1.5 - 2.4\cdot 4^{-u} + 0.9\cdot 9^{-u}$, which has none with $|\operatorname{Re} u| \ge 8$; the argument principle on $[-8, 8] \times [0.01, 30]$ counts nine, none lies in $0 < \operatorname{Im} u < 0.01$, and Newton's method locates them. □

So the Lee and Yang mechanism survives the passage from additive to multiplicative magnetisations for two components and fails at three.

7.1.4 Harmonic means and scale mixtures of the reciprocal Xi function

The reciprocal Xi function $1/G(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the Laplace transform of the clock T (Section 1.2), is a harmonic mean of transforms. For $x \neq 0$ put $L_x(s) = 1/\cosh(x\sqrt{s})$. From $\cosh z = \prod_{k\ge 1}\left(1 + z^2/(\pi^2(k - \frac{1}{2})^2)\right)$, L_x is a GGC transform with Thorin atoms $\pi^2(k - \frac{1}{2})^2/x^2$; probabilistically, if C is the exit time of Brownian motion from $(-1, 1)$, then $\mathbb{E}e^{-\theta C} = 1/\cosh\sqrt{2\theta}$ and $L_x(s) = \mathbb{E}\exp(-\frac{1}{2}x^2 sC)$.

PROPOSITION 7.1.23. $\mathbb{E}e^{-sT} = 1/\mathbb{E}[1/L_X(s)]$ *with X the Pólya variable: the reciprocal Xi transform is the harmonic mean, over the Pólya density, of Brownian exit time transforms. By Jensen's inequality, $\mathbb{E}e^{-sT} \le \mathbb{E}\exp(-\frac{1}{2}sX^2C)$, with C independent of X.*

Proof. By (1.2.4), $1/\mathbb{E}e^{-sT} = G(s) = \mathbb{E}\cosh(\sqrt{s}X) = \mathbb{E}[1/L_X(s)]$, and $y \mapsto 1/y$ is convex. □

The arithmetic mean on the right is the Laplace transform of the mixture $\frac{1}{2}X^2C$. A harmonic mean of transforms is not obviously a transform at all. If harmonic means of GGC transforms were GGC, Proposition 7.1.23 would make the clock a GGC, which is RH (Proposition 1.2.1). That fails: a harmonic mean of GGC transforms need not even be a transform.

Example 7.1.24. The harmonic mean of the GGC transforms $(1+s)^{-2}$ and $(1+2s)^{-2}$ is $1/(1+3s+\frac{5}{2}s^2)$, with poles at $-0.6 \pm 0.2i$. Its inverse transform $2e^{-0.6t}\sin(0.2t)$ is negative on $(5\pi, 10\pi)$, so the harmonic mean is not even the Laplace transform of a positive measure.

Example 7.1.25. Every scale mixture $\Gamma_2 Y$ of a Gamma(2) variable Γ_2 by an independent $Y \geq 0$ is infinitely divisible [156], and it is a GGC if Y is, since products of independent GGC variables are GGC [40]. Without a hypothesis on Y it need not be: let Y equal 1 or $\lambda > 0$, $\lambda \neq 1$, with probability $\frac{1}{2}$ each. The transform $\mathbb{E}(1+sY)^{-2}$ has numerator $(1+\lambda s)^2 + (1+s)^2$, which vanishes at non real s. A GGC transform has no zeros off $(-\infty, 0]$, so $\Gamma_2 Y$ is infinitely divisible but not a GGC.

A last negative result excludes Gamma laws as the outer layer of the clock itself, as Theorem 11.2.10 does for $|X|$, X^2 and S_2.

Theorem 7.1.26. *For no $a > 0$ and no $Y \geq 0$ independent of Γ_a does the clock T have the law of $\Gamma_a Y$.*

Proof. Suppose it did. Then $Y > 0$ almost surely, since $\mathbb{E}e^{-sT} = 1/G(s) \to 0$ as $s \to \infty$, and $\Gamma_a Y$ has density $t^{a-1}g(t)/\Gamma(a)$ with $g(t) = \mathbb{E}[Y^{-a}e^{-t/Y}]$ completely monotone and not identically zero, so $g \geq g(1) > 0$ on $(0,1]$ and $\mathbb{E}e^{-sT} \geq c_1 s^{-a}$ for large s. But $G(s) \geq \cosh(x_0\sqrt{s})\Pr(|X| \geq x_0)$ for every $x_0 > 0$, so $\mathbb{E}e^{-sT} \leq c_2 e^{-x_0\sqrt{s}}$ with $c_2 = 2/\Pr(|X| \geq x_0)$, and the two bounds contradict each other as $s \to \infty$. □

7.2 Truncations, finite conditions and the road to RH

Cutting the series for S_2 in Theorem 7.1.21 after n terms gives variables whose Mellin transforms are explicit, each a gamma factor times an exponential polynomial, with zeros that converge to the points $\rho/2$; a condition on these zeros, holding for infinitely many n, implies RH. The one sided condition fails because the truncated zeros approach the critical line from a side that changes at zeros where $|S(\gamma)|$ is large, with $S(t) = \pi^{-1}\arg\zeta(\frac{1}{2}+it)$ as in Appendix C. Symmetrising by the functional equation removes this mechanism.

7.2.1 Truncations of the Biane, Pitman and Yor lattice

In this subsection w is the complex variable of a Mellin transform and $u = w + 1$.

7.2.1.1 Truncated lattices and Conjecture T

Let e_k, e'_k be independent standard exponential variables, and put $\mathcal{S} = \sum_{k\geq 1} k^{-2}(e_k + e'_k) = \frac{\pi^2}{2} S_2$, with S_2 the variable of Theorem 7.1.21, and $\mathcal{S}_n = \sum_{k\leq n} k^{-2}(e_k + e'_k)$, a GGC with finitely many Thorin atoms. Put $F(w) = \mathbb{E}\mathcal{S}^{\overline{w}}$ and $F_n(w) = \mathbb{E}\mathcal{S}_n^w$ for $\operatorname{Re} w > -2$. By Theorem 7.1.21,

$$F(w) = 2\pi^w \, \xi(2w),$$

so the zeros of F are the points $w = \rho/2$, and RH says they lie on $\operatorname{Re} w = \frac{1}{4}$. Each F_n is explicit. Define α_k and β_k, which depend on n, by the partial fraction expansion

$$\prod_{j\leq n}\Big(1+\frac{\lambda}{j^2}\Big)^{-2} = \sum_{k\leq n} k^{-2}\Big\{\alpha_k\Big(1+\frac{\lambda}{k^2}\Big)^{-1} + \beta_k\Big(1+\frac{\lambda}{k^2}\Big)^{-2}\Big\},$$

so that $\beta_k = k^2 \prod_{j\leq n,\, j\neq k}(1-k^2/j^2)^{-2}$ and $\alpha_k = -2\beta_k \sum_{j\leq n,\, j\neq k} k^2/(j^2-k^2)$; for $n = 2$ they are $\beta = (\frac{16}{9}, \frac{4}{9})$ and $\alpha = (-\frac{32}{27}, \frac{32}{27})$. The left side is the Laplace transform of $\mathcal{S}_n$, whose density is therefore $\sum_k (\alpha_k + \beta_k k^2 x) e^{-k^2 x}$, and taking moments gives, with $u = w + 1$, $F_n(u-1) = \Gamma(u) Q_n(u)$, where $Q_n(u) = \sum_{k\leq n} k^{-2u}(\alpha_k + \beta_k u)$.

Theorem 7.2.1. *If for infinitely many n the function F_n has no zeros in the half plane $\operatorname{Re} w > \frac{1}{4}$, then the Riemann Hypothesis holds.*

Proof. For $\operatorname{Re} w > -2$ the variables $\mathcal{S}_n^w$ are dominated by an integrable variable, since $\mathcal{S}_n \geq e_1 + e'_1$ and $\mathcal{S}_n \leq \mathcal{S}$, so $F_n \to F$ locally uniformly on $\operatorname{Re} w > -2$. If F had a zero w_0 with $\operatorname{Re} w_0 > \frac{1}{4}$, Hurwitz's theorem would give zeros of F_n near w_0 for all large n. Hence every zero $\rho/2$ of F has $\operatorname{Re}\rho \leq \frac{1}{2}$, and the symmetry $\rho \mapsto 1-\rho$ gives $\operatorname{Re}\rho = \frac{1}{2}$. □

Conjecture T. For every $n \geq 1$, $F_n(w) = \mathbb{E}\big[\big(\sum_{k\leq n} k^{-2}(e_k + e'_k)\big)^w\big]$ has no zeros with $\operatorname{Re} w > \frac{1}{4}$.

Conjecture T is a stability statement, of Hermite Biehler or Lee and Yang type [171]. It is false (Corollary 7.2.4), and we state it because the mechanism of its failure also defeats, for large n, the Hermite Biehler inequality on which the bounds of Theorems 7.2.8 and 7.2.10 for the symmetrised truncations rest.

At small n the evidence supports it: Table 7.1 finds no zero of Q_n with $\operatorname{Re} u > \frac{5}{4}$ in any rectangle searched. The zeros are counted by the argument principle and located by Newton's method from a grid; the counts and the zeros found agree.

n	region, in $u = w + 1$	zeros with $\operatorname{Re} u > \frac{5}{4}$
$2, 3, 4, 5, 6, 8, 12, 16, 24, 32$	$\frac{5}{4} < \operatorname{Re} u < 6,\ 0.3 < \operatorname{Im} u < 40$	none
$8, 16, 32$	$\frac{5}{4} < \operatorname{Re} u < 6,\ 40 < \operatorname{Im} u < 100$	none
$48, 64$	$\frac{5}{4} < \operatorname{Re} u < 6,\ 0.3 < \operatorname{Im} u < 30$	none
$16, 32, 64$	$6 < \operatorname{Re} u < 40,\ 0.3 < \operatorname{Im} u < 40$	none

Table 7.1: Argument principle counts for Conjecture T: rectangles searched and zeros of Q_n found with $\operatorname{Re} u > \frac{5}{4}$.

For fixed n and large $|\operatorname{Im} u|$ the zeros move far to the left, because the extreme coefficients β_1 and β_n differ by a factor of order 16^n, up to a power of n. The intermediate heights are therefore where Conjecture T is at risk, and they are the heights that Table 7.1 searches; the failure found in Corollary 7.2.4 lies above them (Remark 7.2.5). An experiment shows that the absence of such zeros at small n is not a consequence of the GGC structure alone: replacing the weights k^{-2} by $k^{-2}(1 + \eta_k)$, with independent η_k uniform on $[-0.15, 0.15]$, one of two draws at $n = 16$ gives a zero with $\operatorname{Re} u > \frac{5}{4}$.

7.2.1.2 How the truncated zeros reach the line, and why Conjecture T fails

Proposition 7.2.2. *Let ρ be a simple zero of ξ and put*

$$\kappa(\rho) = \frac{\rho\,\xi(\rho - 2)}{2\pi\,\xi'(\rho)} = \frac{\rho\,\xi(3 - \rho)}{2\pi\,\xi'(\rho)}.$$

There is a constant C such that for all large n, F_n has exactly one zero $w_n(\rho)$ in the disc $|w - \frac{\rho}{2} - \frac{\kappa(\rho)}{n}| < C/n^2$; thus $w_n(\rho) = \frac{\rho}{2} + \frac{\kappa(\rho)}{n} + O(n^{-2})$.

Proof. Write $\mathcal{S} = \mathcal{S}_n + \mathcal{T}_n$ with $\mathcal{T}_n = \sum_{k>n} k^{-2}(e_k + e'_k)$, independent of $\mathcal{S}_n$, so that $\mathbb{E}\mathcal{T}_n = 2\sum_{k>n} k^{-2} = \frac{2}{n} + O(n^{-2})$ and $\mathbb{E}\mathcal{T}_n^2 = O(n^{-2})$. For $a > 0$, $b \geq 0$ and $0 < \operatorname{Re} w < 2$, Taylor's formula with integral remainder and $|(a + tb)^{w-2}| = (a + tb)^{\operatorname{Re} w - 2} \leq a^{\operatorname{Re} w - 2}$ give

$$\left|(a + b)^w - a^w - wb\,a^{w-1}\right| \leq \tfrac{1}{2}|w(w - 1)|\,b^2 a^{\operatorname{Re} w - 2},$$
$$\left|(a + b)^{w-1} - a^{w-1}\right| \leq |w - 1|\,b\,a^{\operatorname{Re} w - 2}.$$

Since $\mathcal{S}_n \geq e_1 + e'_1$, a $\Gamma(2)$ variable, $\mathbb{E}\mathcal{S}_n^{\operatorname{Re} w - 2} \leq \Gamma(\operatorname{Re} w)$. Taking expectations with $a = \mathcal{S}_n$ and $b = \mathcal{T}_n$, and using independence,

$$F(w) = F_n(w) + w\,\mathbb{E}\mathcal{T}_n\,F_n(w - 1) + O(n^{-2}), \qquad F_n(w - 1) = F(w - 1) + O(n^{-1}),$$

uniformly on compact subsets of $0 < \operatorname{Re} w < 2$. Hence $F_n(w) = F(w) - \frac{2}{n} wF(w - 1) + O(n^{-2})$ uniformly there. Put $w_0 = \rho/2$, so $0 < \operatorname{Re} w_0 < \frac{1}{2}$, and recall

$F(w_0) = 0$ and $F'(w_0) = 4\pi^{w_0}\xi'(\rho) \neq 0$. For $|w - w_0| = O(1/n)$, Taylor expansion of F and of $wF(w-1)$ about w_0 gives

$$F_n(w) = F'(w_0)\Big(w - w_0 - \frac{\kappa(\rho)}{n}\Big) + O(n^{-2}), \qquad \frac{\kappa(\rho)}{n} = \frac{2}{n}\,\frac{w_0 F(w_0 - 1)}{F'(w_0)},$$

and $F(w_0 - 1) = 2\pi^{w_0 - 1}\xi(\rho - 2)$ turns the second identity into the stated κ. On the circle $|w - w_0 - \kappa/n| = C/n^2$ the linear term has modulus $|F'(w_0)|C/n^2$, which exceeds the $O(n^{-2})$ error once C is larger than its constant divided by $|F'(w_0)|$. Rouché's theorem gives exactly one zero inside for all large n. The second form of κ follows from $\xi(s) = \xi(1 - s)$. □

If $\operatorname{Re}\kappa(\rho) > 0$ the zero $w_n(\rho)$ lies to the right of the critical line for all large n. This happens at zeros where $|S(\gamma)|$ is large, the first being γ_{213} (Theorem 7.2.3 and the computation after it), and it is why Conjecture T fails.

For each of the first 100 zeros $\kappa(\rho)$ has negative real part, starting with $\kappa(\rho_1) = -4.762 - 1.944i$, $\kappa(\rho_2) = -8.673 - 0.627i$, $\kappa(\rho_3) = -10.149 - 3.872i$ and $\kappa(\rho_4) = -15.165 + 3.705i$. So these truncated zeros approach the critical line from the left, at distance about $-\operatorname{Re}\kappa(\rho)/n$, which grows with the height. Figure 7.3 shows the approach for the first four zeros, computed by locating the zeros of F_n directly at each n rather than from the expansion, an independent check on Proposition 7.2.2 at low height.

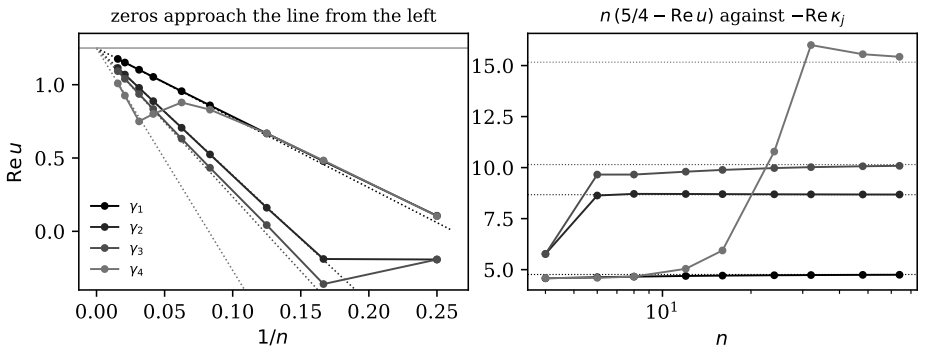


Figure 7.3: Left: real parts of the zeros of F_n near $\rho_j/2$, in the variable $u = w + 1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against n, converging to $-\operatorname{Re}\kappa(\rho_j)$.

THEOREM 7.2.3. *Let $\rho = \frac{1}{2} + i\gamma$ be a simple zero on the critical line, let θ be the Riemann Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ Hardy's function, and*

put $a(\gamma) = e^{i\theta(\gamma)}\zeta(\frac{5}{2} + i\gamma)$. *Then*

$$\operatorname{Re}\kappa(\rho) = \frac{(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma) - 4\gamma \operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)},$$
$$\operatorname{Im}\kappa(\rho) = \frac{4\gamma \operatorname{Im} a(\gamma) + (\gamma^2 - \frac{15}{4})\operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)}.$$

Proof. On the line $\xi(\frac{1}{2} + it) = \Xi(t)$ is real, so $\xi'(\rho) = -i\,\Xi'(\gamma)$, and $\Xi(t) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ gives $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|\overline{Z'(\gamma)}$ at a zero. Since $\xi(3-\rho) = \xi(\frac{5}{2} - i\gamma)$ and ξ is real on the real axis, $\xi(3-\rho) = \overline{\xi(\frac{5}{2} + i\gamma)}$, and $\Gamma(\frac{5}{4} + \frac{i\gamma}{2}) = (\frac{1}{4} + \frac{i\gamma}{2})\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$ with $\pi^{-i\gamma/2}\Gamma(\frac{1}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|e^{i\theta(\gamma)}$. Substituting, the moduli of Γ cancel, $(\frac{1}{2} + i\gamma)(\frac{1}{4} - \frac{i\gamma}{2}) = \frac{1}{2}(\gamma^2 + \frac{1}{4})$ cancels the remaining factor, and $\kappa = -i(\frac{5}{2} - i\gamma)(\frac{3}{2} - i\gamma)\overline{a(\gamma)}/(4\pi^2 Z'(\gamma))$, whose real and imaginary parts are as stated. □

For large γ the term $(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma)$ decides the sign of $\operatorname{Re}\kappa(\rho)$. Since $|\zeta(\frac{5}{2} + i\gamma) - 1| \le \zeta(\frac{5}{2}) - 1 < 0.35$, $a(\gamma)$ is a positive multiple of $e^{i\theta(\gamma)}$ up to a phase below 0.35 in modulus, and up to that phase the sign of $\operatorname{Re}\kappa(\rho)$ is the sign of $\sin\theta(\gamma)/Z'(\gamma)$. Let S_{mid} be the midpoint of the jump of S at the nth zero γ_n. Then $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\mathrm{mid}})$ and the sign of $Z'(\gamma_n)$ is $(-1)^{n+1}$, so the sign of $\sin\theta(\gamma_n)/Z'(\gamma_n)$ is that of $-\cos(\pi S_{\mathrm{mid}})$: negative when $|S_{\mathrm{mid}}| < \frac{1}{2}$ and positive when $\frac{1}{2} < |S_{\mathrm{mid}}| < \frac{3}{2}$. The phase of $\zeta(\frac{5}{2} + i\gamma)$ moves the threshold $\frac{1}{2}$ by at most about 0.11 either way. For the first 460 zeros the sign of $\operatorname{Re}\kappa$ agrees with the sign of $\operatorname{Im} a(\gamma)/Z'(\gamma)$ in every case, and $\operatorname{Re}\kappa > 0$ occurs three times, at

$$\gamma_{213} = 415.4552, \quad \gamma_{289} = 527.9036, \quad \gamma_{379} = 650.6687,$$

with $\kappa = 233.96 - 2298.91i$, $604.16 + 3074.08i$, $1164.60 - 4296.91i$.

At these three zeros the midpoints of the jumps are 0.598, −0.645 and 0.664. Seven of the first 460 zeros have $S(\gamma^-) < -1$: besides γ_{289} these are γ_{127}, γ_{196}, γ_{233}, γ_{368}, γ_{380} and γ_{401}, with midpoints −0.506, −0.505, −0.537, −0.556, −0.538 and −0.514, and $\operatorname{Re}\kappa$ is negative at every one of them. Three further zeros have midpoints above $\frac{1}{2}$ and $\operatorname{Re}\kappa < 0$: γ_{136}, γ_{256} and γ_{399}, at 0.507, 0.510 and 0.539. The largest $|S_{\mathrm{mid}}|$ with $\operatorname{Re}\kappa < 0$ is 0.556, at γ_{368}, and the smallest with $\operatorname{Re}\kappa > 0$ is 0.598, at γ_{213}, so the threshold on the midpoint lies between about 0.56 and 0.60, inside the range $\frac{1}{2} \pm 0.11$ allowed by the phase of $\zeta(\frac{5}{2} + i\gamma)$.

Corollary 7.2.4. *Conjecture T is false: for all sufficiently large n, F_n has a zero with $\operatorname{Re} w > \frac{1}{4}$ near $\rho_{213}/2$.*

Proof. The first 1.5×10^9 zeros, ρ_{213} among them, lie on the critical line and are simple [334], and ball arithmetic applied to the formula of Theorem 7.2.3 gives $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ (Section 7.2.3). By Proposition 7.2.2, for

all large n the function F_n has a zero w_n with $|w_n - \frac{\rho_{213}}{2} - \frac{\kappa(\rho_{213})}{n}| < C/n^2$, so $\operatorname{Re} w_n > \frac{1}{4} + \frac{233.95}{n} - \frac{C}{n^2} > \frac{1}{4}$ once $n > C/233.95$. □

REMARK 7.2.5. The constants in Proposition 7.2.2 depend on ρ, and the first order regime requires n large compared with $|\kappa(\rho)|$. For the zero ρ_{213} of Corollary 7.2.4, $|\kappa(\rho_{213})| = 2311$, so the phenomenon begins at n of order 10^3 or beyond, larger than every n computed in this section.

The failure is intrinsic to the method, not to the particular lattice. Any truncation whose first order error is a multiple of $F(w-1)$ produces the displacement $\kappa(\rho)/n$, and the phase of $\kappa(\rho)$ is governed by $S(\gamma)$. By Selberg's central limit theorem [303] S takes values beyond $\pm\frac{1}{2}$ for a positive proportion of heights, so one expects a positive proportion of zeros to be approached from the right. One sided Hurwitz arguments therefore cannot succeed for truncations of this kind; what is needed is a scheme in which symmetry keeps the zeros on the line.

7.2.1.3 Symmetrised truncations and Conjecture S

The functional equation supplies that symmetry. Put

$$K_n(w) = \pi^{-w} F_n(w) + \pi^{w-\frac{1}{2}} F_n(\tfrac{1}{2} - w), \qquad Z_n(y) = K_n(\tfrac{1}{4} + iy).$$

PROPOSITION 7.2.6. *$K_n(\frac{1}{2} - w) = K_n(w)$, $K_n(\bar{w}) = \overline{K_n(w)}$, so Z_n is real for real y and the zeros of K_n off the line $\operatorname{Re} w = \frac{1}{4}$ come in quadruples. As $n \to \infty$, $K_n \to 4\,\xi(2w)$ locally uniformly on $-\frac{7}{4} < \operatorname{Re} w < \frac{9}{4}$. Every simple zero $\frac{1}{4} + i\frac{\gamma}{2}$ of $\xi(2w)$ on the line is, for all large n, the limit of a zero of K_n lying exactly on the line.*

Proof. The symmetries are immediate from the definition and the reality of F_n on the real axis. The limit follows from $F_n \to F = 2\pi^w \xi(2w)$ and $\xi(1-2w) = \xi(2w)$; since $F_n \to F$ on $\operatorname{Re} w > -2$, the convergence holds on $-2 < \operatorname{Re} w < \frac{5}{2}$, which contains the strip of the statement. Near a simple zero y_0 of the real function $4\xi(\frac{1}{2} + 2iy)$, Z_n converges uniformly and so changes sign near y_0; by Hurwitz, K_n has exactly one zero near $\frac{1}{4} + iy_0$, which must be the real zero of Z_n, since an off line zero would bring its mirror image $\frac{1}{2} - \bar{w}$ with it. □

The Hurwitz argument now gives the following.

THEOREM 7.2.7. *If for infinitely many n all zeros of K_n in the strip $0 < \operatorname{Re} w < \frac{1}{2}$ lie on the line $\operatorname{Re} w = \frac{1}{4}$, then RH holds.*

Proof. A zero $\rho/2$ of $\xi(2w)$ off the line lies in the strip $0 < \operatorname{Re} w < \frac{1}{2}$, since $0 < \operatorname{Re} \rho < 1$, and by Hurwitz it is the limit of zeros of K_n in the strip, which would be off the line for large n. □

Conjecture S. For every $n \geq 1$, all zeros of K_n in $0 < \operatorname{Re} w < \frac{1}{2}$ lie on $\operatorname{Re} w = \frac{1}{4}$.

The symmetrised zeros move along the line. The first order argument of Proposition 7.2.2, applied to $K_n = 2\operatorname{Re}[\pi^{-w}F_n(w)]$ on the line, gives

$$w_n^{\mathrm{sym}}(\rho) = \frac{\rho}{2} + \frac{i\,\operatorname{Im}\kappa(\rho)}{n} + O(n^{-2}) :$$

the symmetrisation removes the real part of the displacement and keeps the imaginary part.

n	window in $\operatorname{Im} w$	zeros in $-4 < \operatorname{Re} w < 4.5$: total, on the line	zeros of ξ
1	$(0.5, 40)$, $(40, 70)$	23, 21; 27, 27	21; 27
2	$(0.5, 40)$, $(40, 70)$	23, 21; 27, 27	21; 27
3	$(0.5, 40)$, $(40, 70)$	23, 21; 27, 27	21; 27
4, 5, 6, 8, 16	$(0.5, 40)$, $(40, 70)$	21, 21; 27, 27	21; 27
32	$(0.5, 40)$	21, 21	21
8	$(70, 140)$, $(195, 215)$	78, 78; 27, 27	78; 26

Table 7.2: Zeros of the symmetrised truncations K_n in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of ξ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for K_8 exceeds that for ξ by one; see the text.

Table 7.2 counts the zeros of K_n in windows of height. For $n = 1$ the one off line quadruple lies at $\operatorname{Re} w = -2.001, 2.501$, $\operatorname{Im} w = \pm 2.471$, and for $n = 2$ at $\operatorname{Re} w = -2.949, 3.449$, $\operatorname{Im} w = \pm 3.261$; for $n = 3$ it is similar. These quadruples have $|\operatorname{Re} w - \frac{1}{4}| > 2$, so they lie outside the strip $0 < \operatorname{Re} w < \frac{1}{2}$ of Theorem 7.2.7 and do not bear on Conjecture S. For $n \geq 4$ every zero found lies on the line.

The window $195 < \operatorname{Im} w < 215$ matters because it contains $\gamma_{213}/2 = 207.73$, where the unsymmetrised truncations are forced off the line. There the count for K_8 exceeds that for ξ by one: the zeros of ξ with $390 < \gamma < 430$ are γ_{196} to γ_{221}, twenty six of them, the nearest excluded ordinates being $\gamma_{195} = 388.85$ and $\gamma_{222} = 430.33$. The difference is not a contradiction: at this height and $n = 8$ the first order displacement along the line, $\operatorname{Im}\kappa(\rho)/n$, is of order 300 for ρ_{213}, so K_8 is far from its limit and its zeros need not correspond one to one with those of ξ. What matters for Conjecture S is that every zero found in the window lies on the line.

The symmetrised truncations are the lattice analogue of the approximations $\zeta_N(s) = \sum_{m \leq N} m^{-s} + \chi(s) \sum_{m \leq N} m^{s-1}$ derived from the approximate functional equation, with χ the factor in $\zeta(s) = \chi(s)\zeta(1-s)$. Spira [309, Section 6] and Gonek and Montgomery [96] studied their zeros, and Gonek and Montgomery proved that for $N \leq T^{o(1)}$ almost all zeros of ζ_N with ordinates in $[0, T]$ lie on

the critical line and are simple. A proof of Conjecture S for infinitely many n would require controlling the zeros of Z_n uniformly in height, including the range of heights comparable with n, where K_n is neither close to $4\xi(2w)$ nor dominated by a single exponential term; this is the same difficulty as for ζ_N.

7.2.1.4 A single term bound for small n

For small n, Conjecture S reduces to a finite computation, because at large heights the gamma factor makes $\log|A_n|$ increase across the strip. Here $A_n(w) = \pi^{-w}F_n(w)$, so that $K_n(w) = A_n(w) + A_n^*(w)$ with $A_n^*(w) = A_n(\frac{1}{2} - w)$ and $|A_n^*(\sigma + iy)| = |A_n(\frac{1}{2} - \sigma + iy)|$. If $\sigma \mapsto \log|A_n(\sigma + iy)|$ is increasing on $(0, \frac{1}{2})$, then $|A_n| > |A_n^*|$ for $\frac{1}{4} < \sigma < \frac{1}{2}$, and K_n has no zeros there at height y; by symmetry none to the left of the line either. The derivative is $\operatorname{Re}\frac{A_n'}{A_n} = -\log\pi + \operatorname{Re}\psi(w+1) + \operatorname{Re}\frac{Q_n'}{Q_n}(w+1)$, and for large y the digamma term dominates.

THEOREM 7.2.8. *Let $Q_n(u) = \sum_k k^{-2u}(\alpha_k + \beta_k u)$ as above and, for $y > |\alpha_1|/|\beta_1|$, let $\delta_n(y) = \sum_{k\geq 2} k^{-2}\frac{|\beta_k|y+|\alpha_k|}{|\beta_1|y-|\alpha_1|}$ and*

$$\delta_n'(y) = \sum_{k\geq 2} k^{-2}\frac{2\log k\,(|\beta_k|y + |\alpha_k|) + |\beta_k|}{|\beta_1|y - |\alpha_1|} + \delta_n(y)\,\frac{|\beta_1|}{|\beta_1|y - |\alpha_1|},$$

which bounds a derivative in the proof and is not the derivative of δ_n. If $\delta_n(y) < 1$ and

$$\log\frac{y}{\pi} - \frac{1}{y} - \frac{|\beta_1|}{|\beta_1|y - |\alpha_1|} - \frac{\delta_n'(y)}{1 - \delta_n(y)} > 0,$$

then K_n has no zeros off the line in $0 < \operatorname{Re} w < \frac{1}{2}$ at any height $\geq y$. The condition holds for all $y \geq Y_n$, with $Y_1 = 4.81$, $Y_2 = 5.85$, $Y_3 = 7.11$, $Y_4 = 9.08$, $Y_5 = 12.8$, $Y_6 = 20.8$, $Y_8 = 104.1$ and $Y_{10} = 21232$.

Proof. $|\psi(z) - \log z| \leq 1/|z|$ for $\operatorname{Re} z \geq 1$, so $\operatorname{Re}\psi(w+1) \geq \log y - \frac{1}{y}$ at height y. Write $Q_n(u) = (\alpha_1 + \beta_1 u)(1 + \delta(u))$. On $1 \leq \operatorname{Re} u \leq \frac{3}{2}$, $|u| \geq y$, we have $|k^{-2u}| \leq k^{-2}$, so $|\delta(u)| \leq \delta_n(y)$, and differentiating the quotient $\delta(u)$ term by term gives $|\delta'(u)| \leq \delta_n'(y)$; both bounds decrease in y. Then $\operatorname{Re}\frac{Q_n'}{Q_n} \geq -\frac{|\beta_1|}{|\alpha_1+\beta_1 u|} - \frac{|\delta'|}{1-|\delta|}$, and the condition makes $\operatorname{Re}(A_n'/A_n) > 0$ on the strip at every height $\geq y$. The values of Y_n come from evaluating these bounds with the exact coefficients. □

COROLLARY 7.2.9. *Conjecture S holds for $n = 1, 2, 3, 4, 5, 6, 8$ above the heights Y_n of Theorem 7.2.8, and is verified numerically below them, by a floating point argument principle count and not by an enclosure.*

Proof. Above Y_n this is Theorem 7.2.8. Below it, the argument principle counts cover the strip $0 < \operatorname{Re} w < \frac{1}{2}$ for $0 < \operatorname{Im} w < 6, 7, 8$ when $n = 1, 2, 3$, and the windows of Table 7.2 cover $0 < \operatorname{Im} w < 70$ for $4 \leq n \leq 6$ and $0 < \operatorname{Im} w < 140$ for

$n = 8$, with $0 < \operatorname{Im} w < 0.6$ checked separately and $K_n > 0$ on the real segment; in each case the total count in the strip equals the number of sign changes of Z_n on the line. □

The single term bound of Theorem 7.2.8 deteriorates quickly: the lattice coefficients $|\beta_k/\beta_1|$ grow with n, $Y_{10} = 21232$, and for $n \geq 12$ the bound fails because $\delta_n > 1$.

7.2.1.5 Torus bounds and where they stop

Write $u = x + iy$ and $Q_n(u) = u\,P_n(u) + R_n(u)$ with the Dirichlet polynomials $P_n(u) = \sum_{k\leq n} \beta_k k^{-2u}$ and $R_n(u) = \sum_{k\leq n} \alpha_k k^{-2u}$. At height y the values of P_n depend on y only through the prime phases $(p^{-2iy})_{p\leq n}$, which lie on the torus $\mathbb{T}^{\pi(n)}$. Replacing them by independent phases $\omega = (\omega_p)_{p\leq n} \in \mathbb{T}^{\pi(n)}$ gives the torus function

$$P_n(x,\omega) = \sum_{k\leq n} \beta_k k^{-2x} \exp\Big(-i \sum_{p\leq n} v_p(k)\,\omega_p\Big),$$

and in the same way the torus function $P_n'(x,\omega)$ of $P_n'(u) = -\sum_k 2\log k\,\beta_k k^{-2u}$. Their extremes over the torus bound the values at every height. Put $r(x) = \sum_k |\alpha_k| k^{-2x}$ and $r'(x) = \sum_k 2\log k\,|\alpha_k| k^{-2x}$.

THEOREM 7.2.10. *Let* $m_n = \min |P_n(x,\omega)|$, $M_n = \max\{-\operatorname{Re}(P_n'/P_n)(x,\omega)\}$ *and* $\ell_n = \max |P_n'/P_n|(x,\omega)$, *over* $\omega \in \mathbb{T}^{\pi(n)}$ *and the band* $1.2 \leq x \leq 1.3$, *and let*

$$B_n(y) = \max_{0.3\leq\sigma\leq\frac12,\ \omega} \frac{\log\big(|P_n(\frac32-\sigma,\omega)| + r(\frac32-\sigma)/y\big) - \log\big(|P_n(1+\sigma,\omega)| - r(1+\sigma)/y\big)}{2\sigma - \frac12}.$$

If $m_n > r(1.2)/y$,

$$\log\frac{y}{\pi} - \frac{2}{y} - M_n - \frac{\ell_n r(1.2) + r'(1.2) + r(1.2)/y}{y\,(m_n - r(1.2)/y)} > 0 \quad \textit{and} \quad \log\frac{y}{\pi} - \frac{1}{y} > B_n(y),$$

then K_n *has no zeros off the line in* $0 < \operatorname{Re} w < \frac12$ *at height* y, *and the same holds at every larger height.*

Proof. For $\frac14 < \sigma \leq 0.3$ both σ and $\frac12 - \sigma$ lie in $[0.2, 0.3]$, and the first condition makes $\operatorname{Re}(A_n'/A_n) > 0$ there: $\operatorname{Re}\psi(w+1) \geq \log y - \frac1y$, $\operatorname{Re}\frac1u > 0$, and $\frac{Q_n'}{Q_n} = \frac1u + \frac{P_n' + R_n'/u - R_n/u^2}{P_n + R_n/u}$ differs from $\frac1u + \frac{P_n'}{P_n}$ by at most the stated correction. So $\log|A_n|$ increases on $[\frac12 - \sigma, \sigma]$ and $|A_n| > |A_n^*|$. For $0.3 \leq \sigma \leq \frac12$ compare directly: $\log|\Gamma(1+\sigma+iy)| - \log|\Gamma(\frac32 - \sigma + iy)| \geq (2\sigma - \frac12)(\log y - \frac1y)$ by integrating $\operatorname{Re}\psi$, $|1+\sigma+iy| > |\frac32 - \sigma + iy|$, and $|P_n + R_n/u|$ is bounded below on the right by $|P_n| - r/y$ and above on the left by $|P_n| + r/y$; the second condition then gives $|A_n(\sigma+iy)| > |A_n(\frac12 - \sigma + iy)|$. Both conditions improve as y increases. □

Table 7.3 lists bounds for the torus extremes (Remark 7.2.12) and the smallest integer heights at which the first condition holds with these bounds. For $n = 7$ and $9 \le n \le 15$ we use a single band variant: if $m_n > r(1)/y$ and the first condition of Theorem 7.2.10 holds with m_n, M_n, ℓ_n the extremes over the band $1 \le x \le \frac{3}{2}$ and with $r(1)$, $r'(1)$ in place of $r(1.2)$, $r'(1.2)$, then $\operatorname{Re}(A_n'/A_n) > 0$ on $0 < \operatorname{Re} w < \frac{1}{2}$ at height y, so $\log|A_n|$ increases across the strip and K_n has no zeros off the line there; the condition improves as y increases. The proof is the first part of the proof of Theorem 7.2.10 with the band widened, since r and r' decrease in x and $\operatorname{Re}\psi(w+1) \ge \log y - \frac{1}{y}$ on the whole strip.

n	16	17	18	19	20	21	22	23	24	32	36
$m_n \ge$	1.062	1.003	0.947	0.895	0.846	0.800	0.757	0.716	0.677	0.361	0.203
$M_n \le$	0.862	0.895	0.926	0.956	0.985	1.012	1.039	1.064	1.089	1.259	1.331
$\ell_n \le$	3.600	3.948	4.310	4.689	5.086	5.504	5.954	6.473	7.075	17.203	33.568
$r(1.2) \le$	7.522	7.776	8.009	8.223	8.420	8.600	8.834	9.062	9.280	10.675	11.313
$r'(1.2) \le$	4.648	5.019	5.361	5.675	5.961	6.222	6.650	7.078	7.489	10.181	11.577
threshold	30	34	37	41	46	50	56	62	69	213	558

Table 7.3: Bounds for the torus extremes over the band $1.2 \le x \le 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 7.2.10 holds with these values. The bounds are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 7.2.12); the programs are in `verification_reciprocal_zeta/torus/`.

n	7	9	10	11	12	13	14	15
$m_n \ge$	1.403	1.052	0.892	0.744	0.607	0.482	0.367	0.261
$M_n \le$	0.576	0.715	0.777	0.834	0.887	0.937	0.984	1.028
$\ell_n \le$	1.796	3.193	4.167	5.428	7.121	9.511	13.140	19.314
$r(1) \le$	4.848	5.888	6.291	6.629	6.971	7.445	7.886	8.294
$r'(1) \le$	2.014	3.096	3.520	3.869	4.282	5.032	5.746	6.418
threshold	14	24	31	41	57	84	131	229

Table 7.4: As Table 7.3, for the single band variant: bounds for the extremes over the band $1 \le x \le \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values.

Corollary 7.2.11. *Conjecture S holds for $n = 16, \dots, 24, 32, 36$ above the thresholds of Table 7.3 and for $n = 7$ and $9 \le n \le 15$ above those of Table 7.4, and for all these n it is verified numerically below the thresholds, by floating point argument principle counts. With Corollary 7.2.9 this covers every $n \le 24$ and $n = 32, 36$.*

Proof. Above the thresholds this is Theorem 7.2.10, with the bounds of Table 7.3 and the second condition of Remark 7.2.12, and its single band variant, with the

bounds of Table 7.4. The hypothesis $m_n > r/y$ and the first condition, here and in the variant, hold with the true values wherever they hold with these bounds, because both improve as m_n increases and as M_n, ℓ_n and the values of r and r' decrease. Below the thresholds the argument principle count in $0 < \operatorname{Re} w < \frac{1}{2}$ equals the number of sign changes of Z_n on the line, with the following heights and counts:

n	7	9	10	11	12	13	14	15	16	17
height	15	24	32	43	58	86	132	230	500	35
zeros	3	9	14	23	36	64	116	242	649	17
n	18	19	20	21	22	23	24	32	36	
height	250	43	47	51	57	62	110	215	570	
zeros	269	23	26	30	35	40	90	221	763	

Each height is at least the threshold for the same n, with equality for $n = 9$ and $n = 23$, so the counts and the theorem together cover all heights. For $n = 36$ the sampling step on the line is 0.02, cross checked by the argument principle. Every count equals the number of zeros of ξ with ordinate below twice the height, except for $n = 9$, where the ninth zero of K_9, at $\operatorname{Im} w = 23.82$, is the image of $\gamma_9/2 = 24.003$ displaced below height 24. □

Remark 7.2.12. The bounds of Tables 7.3 and 7.4 are proved by a branch and bound over the band and the half torus $0 \le \omega_2 \le \pi$, which suffices because $P_n(x, -\omega) = \overline{P_n(x, \omega)}$. On a box with centre (x_c, ω_c) and half widths h in x and h_p in ω_p, the kth term of P_n differs from its value at the centre by at most $\beta_k k^{-2x_c}\{(k^{2h} - 1) + k^{2h} \min(2, \sum_p v_p(k) h_p)\}$, and the resulting enclosures of P_n, P_n' and their partial derivatives bound $|P_n|$ from below and $-\operatorname{Re}(P_n'/P_n)$ and $|P_n'/P_n|$ from above on the box; for M_n and ℓ_n the proved bound for m_n serves as a lower bound for $|P_n|$. Since $2\sigma - \frac{1}{2} > 0$, the second condition of Theorem 7.2.10 at height y is equivalent to

$$\Big(|P_n(1+\sigma,\omega)| - \frac{r(1+\sigma)}{y}\Big) \exp\Big(\big(2\sigma - \tfrac{1}{2}\big)\Big(\log\frac{y}{\pi} - \frac{1}{y}\Big)\Big) > |P_n(\tfrac{3}{2} - \sigma, \omega)| + \frac{r(\frac{3}{2} - \sigma)}{y}$$

for $0.3 \le \sigma \le \frac{1}{2}$ and all ω, which also gives the positivity of $|P_n(1+\sigma,\omega)| - r(1+\sigma)/y$ that $B_n(y)$ needs. The same branch and bound, applied to the difference of the two sides, proves this at the heights 29, 33, 36, 40, 45, 49, 55, 61, 68, 211, 555 for $n = 16, \dots, 24, 32, 36$, each below the threshold, and so at every larger height. The searches run in double precision and only choose the boxes: every final box, about 1.1×10^8 in all, is proved again in ball arithmetic with Arb [130], with exact coefficients and on the exact box.

With the band $1.2 \le x \le 1.3$ the method stops between $n = 40$ and $n = 44$. The band minimum m_n falls to 0.20 at $n = 36$, 0.056 at $n = 40$ and 0 at $n = 44$, so at $n = 44$ the hypothesis $m_n > r(1.2)/y$ of Theorem 7.2.10 fails at every height. This reflects the functions and not only the bound: the torus function $P_n(x, \omega)$ then vanishes at a point of the band, and since by Kronecker's theorem the prime phases at height y come arbitrarily close to every point of the torus, A_n has zeros with real part near $[0.2, 0.3]$ at arbitrarily large heights, where $\log|A_n|$ cannot increase across the strip. On the line $x = \frac{5}{4}$ itself the minimum is still 0.29 at $n = 44$ and 0.19 at $n = 48$, so these torus zeros lie off the line, and a narrower band may postpone the stop. A torus zero with $x > \frac{5}{4}$ would end every comparison of this kind: it gives zeros of A_n with $\operatorname{Re} w > \frac{1}{4}$ at arbitrarily large heights, where $|A_n(w)| > |A_n^*(w)|$ fails. On which side of the line the torus zeros at $n = 44$ lie is not determined here.

The monotonicity used in Theorem 7.2.8 implies the Hermite Biehler inequality $|A_n(w)| > |A_n(\frac{1}{2} - \bar{w})|$ for $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, and the inequality holds far beyond the proven range. On a grid of $\sigma \in \{0.26, 0.3, 0.4, 0.49\}$ and $y \le 100$ it holds for $n = 8, 16, 24$ at every height above 3.6, and it fails only below 3.6, where the strip contains no zeros. For $n = 8, 16$ it also holds for $200 \le y \le 212$, around $\gamma_{213}/2$. It cannot hold for all large n: by Corollary 7.2.4, for large n the function A_n has a zero w with $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, where the inequality fails. Quantitatively, near a zero $\rho/2$ the first order margin is governed by the residue of $w\xi(2w-2)/\xi(2w)$, which is a positive multiple of $\kappa(\rho)$, so at zeros with $\operatorname{Re}\kappa(\rho) > 0$ the inequality reverses once n is large compared with $|\kappa(\rho)|$. A proof of Conjecture S for infinitely many n must establish real rootedness without a one sided inequality.

7.2.2 What is proved, and how far it reaches

No result of Chapter 6 or of this chapter proves RH. In Table 7.5 every entry in the middle column is a positivity statement about a transform, or about finitely many derivatives. Every entry on the left is a positivity statement about all derivatives, equivalently about a measure. The question is whether the middle column can be pushed to the left. So far it reaches Weil positivity on the Bernstein cone and $(-1)^n W^{(n)} > 0$ up to a finite order, and neither the de Bruijn Newman flow nor a GGC multiplier closes the gap. Chapters 9 to 11 measure the same gap from the right of the centre: $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a GGC for every $\sigma \ge 1$ with no hypothesis (Theorem 7.1.14; Proposition 9.1.2 computes its Thorin measure at $\sigma = 1$), and for $\frac{1}{2} < \sigma < 1$ zero density estimates show that in the density $\rho_{\sigma-1/2}$ of (1.2.7) the Cauchy kernels of the zeros to the right of σ carry, below height T, a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$ (Theorem 11.4.2).

Equivalent form of RH	Proved with no unproved hypothesis	Computation the middle column uses
$R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is a GGC, Thorin measure $\sum \delta_{\gamma^2}$ [249, Proposition 3.2]	$e^{-c\sqrt{s}}/G$ is a GGC (Theorem 6.2.12); $e^{\sqrt{s}\zeta'/\zeta(\beta)}\zeta(\beta)/\zeta(\beta + \sqrt{s})$ is a GGC for $\beta > 1$ (Proposition 6.2.6)	none
$W(x) = \frac{1}{2}\sum_\rho e^{x(\rho-\frac{1}{2})^2}$ is completely monotone	$W > 0$ (proof of Theorem 7.2.14) and $(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.13); $W + c/(2\sqrt{\pi x})$ is positive (Theorem 6.2.15) and n monotone for $n \leq 10^{10}$ (Theorem 6.2.17)	positivity: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}
$e^{-c\sqrt{s}}R$ is a GGC	it is infinitely divisible (Theorem 6.2.15)	RH to 50 and enclosures
w is the Laplace transform of a real measure on $[0, \infty)$ (Theorem 6.1.15)	$w > 0$ (Theorem 7.1.1)	RH to 50 and enclosures
the sine squared image of $\tilde{\mu}_\xi$ is positive, equivalently Weil's distribution is positive definite (Proposition 7.2.13)	its subordination stage is positive; Weil positivity holds on the Bernstein cone (Theorem 7.2.14), against every Gaussian (its proof) and against $r^{2n}e^{-xr^2}$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.13); the canonical function k of the floor is n monotone for every $n \leq 10^{10}$ (Theorem 6.2.17), which is strictly weaker than n monotonicity of W	Bernstein cone: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}

Table 7.5: Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none.

7.2.2.1 Weil's criterion in Thorin coordinates

Proposition 6.2.16 left open which signed primal measures have a positive sine squared image. For $\tilde{\mu}_\xi$ the answer is Weil's criterion.

PROPOSITION 7.2.13. *Let $D(x) = \sum_\rho e^{(\rho-\frac{1}{2})x}$ be Weil's distribution, whose action on a test function h is $\sum_\rho \hat{h}(\gamma_\rho)$ with $\gamma_\rho = -i(\rho - \frac{1}{2})$. Then R is the Laplace transform of a GGC, with Thorin measure the sine squared image of $\tilde{\mu}_\xi$, if and only if D is positive definite, that is, $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for every $h = g * \tilde{g}$ with g smooth and compactly supported. Both conditions are equivalent to RH.*

Proof. By Proposition 1.2.1, R is the Laplace transform of a GGC if and only if RH holds, and its Thorin measure is then $\sum_{\gamma>0} \delta_{\gamma^2}$, which is the sine squared image of $\tilde{\mu}_\xi$ by Proposition 6.2.16(ii). Weil's theorem [338, 38] states that RH holds if and only if $\sum_\rho \hat{h}(\gamma_\rho) \geq 0$ for all $h = g * \tilde{g}$ with g smooth and compactly supported, that is, if and only if D is positive definite. □

A formal computation shows how the two formulations correspond. For a positive measure U on $(0,\infty)$ with $\int U(dt)/(1+t) < \infty$, the primal measure $\tilde{\mu}_U(dx) = -2\big(\int \cos(\sqrt{t}\,x)\,U(dt)\big)dx$ on $x > 0$, understood as a distribution, has sine squared image U: since $\int_0^\infty (1-\cos x\sqrt{z})(-2\cos bx)\,dx = \pi\delta(\sqrt{z}-b)$, $\tilde{\mu}_U$ maps to $\int \delta(z-t)U(dt) = U$. The map of Lemma 6.2.5 is injective on the class of that lemma: if $\int_0^\infty \sin^2(x\sqrt{z}/2)\,\nu(dx) = 0$ for every $z > 0$, then $\psi(t) = \int(1-\cos xt)\,\nu(dx)$ vanishes for all real t, and $\psi(t) - \frac{1}{2}\int_{t-1}^{t+1}\psi(s)\,ds = -\int \cos(xt)\big(1 - \frac{\sin x}{x}\big)\nu(dx)$ shows that the finite measure $(1-\frac{\sin x}{x})\nu(dx)$ has vanishing cosine transform, so $\nu = 0$. Carried over to distributions, these two facts say that a primal measure has a positive sine squared image if and only if $-\tilde{\mu}$ is the restriction to $x > 0$ of an even positive definite distribution on $\mathbb{R}$, that is, by the Bochner Schwartz theorem, the Fourier transform of a positive measure. For ξ, $-\tilde{\mu}_\xi$ is the restriction to $x > 0$ of D, by (6.2.6); D is even because $\rho \mapsto 1-\rho$ permutes the zeros, and the explicit formula (6.2.5) is its prime side. The computation is formal, for two reasons: Lemma 6.2.5 is proved only for primal measures with $\int(1\wedge x)|\tilde{\mu}|(dx) < \infty$, and when RH fails D grows exponentially and is not tempered, so the Bochner Schwartz theorem applies only in Weil's own formulation, with compactly supported test functions. Riesz's criterion has a Thorin form of the same kind (Theorem 10.2.7 of Section 10.2).

The unconditional positivity results of Chapter 6, of this chapter and of Section 5.1 are partial positivity statements of Weil type. The gamma part is a positive definite archimedean piece (Theorem 6.2.12). The heat trace results are Weil positivity along the Gaussian family: with h Gaussian, $\sum_\rho \hat{h}(\gamma_\rho) = 2W(x)$, and $W(x) > 0$ for every $x > 0$ unconditionally (Theorem 5.1.1 [252, Theorem A]). Weil positivity against $r^{2n}e^{-xr^2}$ is $(-1)^n W^{(n)} \geq 0$, which Theorem 5.1.6 [249, Theorem 7.10] proves for every $n \leq 10^{17}$ and Theorem 5.1.13 for every $n \leq 1.16 \times 10^{25}$.

Theorem 6.2.17 proves the corresponding statement for the canonical function $k = W + c/(2\sqrt{\pi x})$ of the floor, for $n \le 10^{10}$, and not for W itself. The known unconditional cases of Weil positivity, due to Yoshida [349, Theorem 1] and to Bombieri [38], concern test functions of small support; the Gaussian family, in which Theorems 5.1.6, 5.1.13 and 6.2.17 apply, is the opposite regime.

7.2.2.2 The Bernstein cone and the resolution of finite order tests

Proposition 7.2.13 identifies the Thorin positivity of R with Weil positivity for all test functions $\hat{h} = |\hat{g}|^2$. A cone of test functions of another kind, completely monotone in r^2, can be settled unconditionally.

THEOREM 7.2.14. *Let* $\hat{h}(r) = \int_0^\infty e^{-yr^2} g(y)\,dy$ *with* $g \ge 0$, $g \not\equiv 0$ *and* $\int_0^1 g(y)\, y^{-1/2}\big(1 + \log(1/y)\big)\,dy + \int_1^\infty g(y)\,dy < \infty$, *so that* $\hat{h}$ *is a completely monotone function of* r^2. *Then*

$$\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)\, g(y)\,dy > 0,$$

where $\gamma_\rho = -i(\rho - \frac{1}{2})$ *and the sum is over all nontrivial zeros. In particular Weil's functional is positive for every test function of Thorin type* $\hat{h}(r) = \prod_k (1 + r^2/t_k)^{-1}$, *a nonempty product with* $t_k > 0$ *and* $\sum_k 1/t_k < \infty$, *for* $\hat{h}(r) = \operatorname{sech}(cr)$ *with* $c > 0$, *and for* $\hat{h}(r) = (1 + r^2/a)^{-m}$ *with* $a > 0$ *and* $m > \frac{1}{2}$. *The proof uses the verification of RH to height* 50 *and the interval enclosures of Theorem 7.1.1 together with the enclosures E5 and E6 of Section 7.2.3.*

Proof. For each $y > 0$, $\sum_\rho e^{-y\gamma_\rho^2} = \sum_\rho e^{y(\rho - \frac{1}{2})^2} = 2W(y)$, and $W(y) > 0$ for every $y > 0$. For $y \le 0.003$ this comes from the prime side: $W = w + W_0$ with $w > 0$ (Theorem 7.1.1) and $W_0 \ge 0$, because the function F of Lemma 6.1.12 satisfies $F(0.003) > 0.0329$ (enclosure E6) and is decreasing. For $y \ge 0.003$ it comes from the zeros below height 50, which lie on the line and contribute $e^{-y\gamma^2} > 0$: the first alone contributes more than $e^{-0.003 \cdot 14.14^2} > 0.548$ at $y = 0.003$, the off line tail above height 50 is below 0.0162 there (enclosure E5, with the bound $N(t+1) - N(t) \le 2\log(t+1)$ of the proof of Theorem 7.1.1), and the ratio of the tail to the first term decreases as y grows. The interchange of sum and integral is justified by Fubini's theorem: $\sum_\rho |e^{y(\rho - \frac{1}{2})^2}| \le e^{y/4} \sum_\rho e^{-y(\operatorname{Im}\rho)^2}$, which is $O(y^{-1/2}(1 + \log(1/y)))$ for $y \le 1$ by the Riemann von Mangoldt formula and bounded for $y \ge 1$, and the integrability condition makes this majorant integrable against g. The Thorin type functions are completely monotone in r^2 because $\prod(1 + s/t_k)^{-1}$ and $\operatorname{sech}(c\sqrt{s})$ are GGC transforms. The examples satisfy the integrability condition. A nonempty product $\prod_k (1 + s/t_k)^{-1}$ is the Laplace transform of $\sum_k e_k/t_k$, with e_k independent standard exponential variables, and the density g of this sum is at most t_1, the maximum of the density of e_1/t_1; the function $\operatorname{sech}(c\sqrt{s}) =$

$\prod_{k\ge1}\bigl(1+4c^2s/((2k-1)^2\pi^2)\bigr)^{-1}$ is such a product; and for $(1+r^2/a)^{-m}$ the function $g(y)=a^m y^{m-1}e^{-ay}/\Gamma(m)$ satisfies the condition because $m>\frac12$. □

The Bernstein cone consists of test functions that decrease in $|r|$ and so cannot concentrate near a given height. This is why positivity on it does not imply RH, and it matches the resolution analysis of Table 7.6: an off line zero at height γ and distance δ from the line is detected only by test functions able to localise at γ. Theorem 5.1.13 gives $(-1)^nW^{(n)}>0$ for $n\le1.16\times10^{25}$, which is Weil positivity for $r^{2n}e^{-yr^2}$ and, after integration against g where the integral converges, for the image of the Bernstein cone under multiplication by r^{2n}.

Complete monotonicity of W is Bernstein's condition, and its Post Widder form is

$$L_n[W](t)=\frac{(-1)^n}{n!}\Bigl(\frac{n}{t}\Bigr)^{n+1}W^{(n)}\Bigl(\frac{n}{t}\Bigr)\ge0\qquad\text{for all } n,t.$$

$L_n[W]$ is the Thorin measure $\sum\delta_{\gamma^2}$ seen through a kernel of relative width about $n^{-1/2}$, and an off line quadruple $\rho=\frac12\pm\delta\pm i\gamma$ contributes $2\,\mathrm{Re}\{(n/t)^{n+1}z^ne^{-zn/t}/n!\}$ with $z=\gamma^2-\delta^2-2i\delta\gamma$. In Table 7.6 the quadruple is inserted midway between two consecutive true ordinates among the first 300 zeros.

γ	$\delta=0.4$	$\delta=0.2$	$\delta=0.1$	$\delta=0.05$
27.7	724	1448	2896	5792
60.1	11585	23170	46340	92681
120.1	32768	65536	131072	262144

Table 7.6: Smallest order n^* at which an inserted off line quadruple at height γ and distance δ makes $\min_t L_n[W](t)$ negative.

In every row the detection order doubles when δ halves, within the resolution $\sqrt2$ of the scan, and across the rows it grows roughly like γ^2: the ratio $n^*\delta/\gamma^2$ lies between 0.38 and 1.3 in Table 7.6. So an off line zero at height γ first becomes visible at order about γ^2/δ, and a zero above the verified height T_0 of Theorem 6.2.17 at order about T_0^2/δ. Theorem 5.1.13 reaches orders of this size, every order up to $1.29(T_0-23)^2$ for $T_0=3\times10^{12}$ (Section 5.1.6), while the method of Theorem 6.2.17, whose Cauchy estimate loses a factor 2^n, certifies n monotonicity only up to orders proportional to T_0. Finite order tests therefore cannot prove RH: each order sees only a bounded height, and the height it sees tends to infinity only with the order.

7.2.2.3 Weak limits and the de Bruijn Newman flow

The class of GGC laws is closed under convolution and weak limits [39, Section 3.1]. So RH is equivalent to exhibiting R as a limit of Laplace transforms of GGC

laws, and positivity then passes to the limit at every order simultaneously. The de Bruijn Newman flow gives a family of candidates. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$, with Φ the Pólya kernel (1.2.3); Φ_{dBN} is the kernel of Section 8.4, in the rescaled variable of Rodgers and Tao [289]. Let $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with Λ defined in this normalisation, and put

$$R_t(s) = \frac{H_t(0)}{H_t(2i\sqrt{s})}, \qquad R_0 = R.$$

THEOREM 7.2.15. *Let Λ be the de Bruijn Newman constant. For every $t \geq \Lambda$, R_t is the Laplace transform of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the zeros of H_t. In particular R_t is a GGC for every $t \geq 0.2$, unconditionally. RH holds if and only if R_t is a GGC for arbitrarily small $t > 0$.*

Proof. For $t \geq \Lambda$ all zeros of H_t are real by definition of Λ, and H_t is even and entire of order one with the Hadamard product $H_t(z) = H_t(0)\prod_k(1 - z^2/z_k^2)$, so $R_t(s) = \prod_k(1 + 4s/z_k^2)^{-1}$, an infinite convolution of exponential laws. The bound $\Lambda \leq 0.2$ is [247], improving $\Lambda \leq 0.22$ [282]. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t \to R$ pointwise on $(0,\infty)$; if each R_t is a GGC transform, so is the limit, and Proposition 6.1.9 gives RH. Conversely RH means $\Lambda \leq 0$, so every R_t with $t \geq 0$ is a GGC. By Rodgers and Tao [289], $\Lambda \geq 0$, so RH is equivalent to $\Lambda = 0$. □

Conversely, for $t < \Lambda$, and so for every $t < 0$ [289], R_t is not the Laplace transform of a GGC (Corollary 4.2.5), and Proposition 11.5.12 turns this into a test for the properties of the Pólya kernel that could imply RH.

The zeros $z_k(t)$ move by the backward heat flow, and the Thorin measure of R_t can cease to be positive only by two real zeros colliding and leaving the real line, which is the collision picture of Rodgers and Tao; Section 8.4 computes the flow numerically. What Chapter 6 and this chapter add concerns $t = 0$: any failure there comes from zeros above the verified height. For $t \neq 0$ no prime side is known for the Thorin measure of R_t, and changing the test function does not give one. Replacing the Gaussian e^{-xr^2} of Theorem 6.1.11 by a t dependent test function gives a linear statistic $\sum_\rho h(\gamma_\rho)$ of the zeros of ξ, whereas the heat trace $\sum_k e^{-xz_k(t)^2/4}$ of the Thorin measure of R_t depends on those zeros nonlinearly: along the equation of motion of the zeros (Section 8.4.1) its derivative in t is a sum over pairs of zeros. No Euler product for H_t is known either.

7.2.2.4 No GGC multiplier can manufacture positivity

PROPOSITION 7.2.16. *Let V be the Laplace transform of any GGC. Then $V \cdot R$ is the Laplace transform of a GGC if and only if RH holds. For φ only one direction holds for every V: if $V \cdot \varphi$ is a GGC transform then RH holds. The converse depends on V: it fails for $V \equiv 1$ (Theorem 7.1.3) and holds for $V = e^{-c\sqrt{s}}/G$, for which $V\varphi$ is the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15.*

Proof. If RH holds, R and V are GGC transforms and so is their product. Conversely, if VR is a GGC transform, then $-(VR)'/(VR)$ and $-V'/V$ are Stieltjes functions, analytic on $\mathbb{C} \setminus (-\infty, 0]$, so $-R'/R$ is analytic there and R has no zeros or poles in $\mathbb{C} \setminus (-\infty, 0]$. By Proposition 6.1.9 this is RH. The same argument applies to $V \cdot \varphi$, since $\varphi = RG$ with G analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. For $V = e^{-c\sqrt{s}}/G$, a GGC transform by Theorem 6.2.12, $V\varphi = e^{-c\sqrt{s}}R$, which is a GGC under RH by Theorem 6.2.15. □

Multipliers can remove obstructions that come from normalisation, as the floor does, but not the obstruction that comes from the zeros. Thorin measures add under multiplication, so a multiplier can hide negative Thorin mass. What RH forbids, however, is Thorin mass off the half line, the complex atoms at $-(\rho - \frac{1}{2})^2$ contributed by zeros off the critical line, and no GGC factor can cancel a singularity of R in the cut plane. This is why the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15, which uses the best available multiplier $e^{-c\sqrt{s}}/G$, is infinitely divisible but can be shown to be a GGC only by proving RH. Section 11.1 finds the same for the general theory of infinitely divisible laws: each GGC criterion it examines is, for R, equivalent to RH (Section 11.1.5).

7.2.3 Numerical verification

The computations of Chapter 6 and of this chapter have three levels of certainty. The inputs to Theorems 7.1.1, 6.2.15, 6.2.17 and 7.2.14 are enclosed in outward rounded interval arithmetic by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E6, recorded in `verification_reciprocal_zeta/enclosures_report.md`; `code/README.md` lists the scripts). The bounds $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ of Corollary 7.2.4 are enclosed in ball arithmetic with Arb [130] by `verification_reciprocal_zeta/certify_k9_k2.py`, which also encloses the values of S quoted with Theorem 7.1.3. The torus bounds of Tables 7.3 and 7.4 and the second condition of Theorem 7.2.10 are proved by a branch and bound whose final boxes are enclosed in ball arithmetic with Arb (Remark 7.2.12); the programs are in `verification_reciprocal_zeta/torus`. The constants of Table 7.7 are computed in floating point with `mpmath` at 20 to 30 digits, and the argument principle counts of Section 7.2.1 in floating point with 40 or more digits. The torus minima quoted for $n \geq 40$ after Remark 7.2.12, and the threshold at $n = 40$ in Problem 7.2, are estimated by optimisation and are not certified.

The function w at $x = 10^{-3}, 10^{-2}, 0.1, 1, 10$ is 1.00025, 1.00249, 0.86251, 0.49140, 0.21524; its initial increase rules out self decomposability (Theorem 7.1.3). Talbot inversion of φ gives the density f of the law with Laplace transform φ: $f(0.05) = 0.36222$, $f(1) = 0.13508$, $f(16) = 0.0087321$, and $x^{3/2}f(x) = 0.5589, 0.6382, 0.6908, 0.7220, 0.7392$ at $x = 16, 32, 64, 128, 256$, approaching 0.75773.

Quantity	Value
$c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$	2.6860917096
τ_0, $\theta(\tau_0)$, $\max M = -\theta(\tau_0)/\pi$	6.2898360, −3.5309728, 1.1239436
$M(\gamma_1^{2-})$	0.5502528
τ_π with $\theta(\tau_\pi) = -\pi$	3.4362182
x_* with $W_0(x_*) = 0$	0.0032173906
$F(0.003)$, $F(0.0035)$, $F(0.004)$	0.0366850, −0.0442285, −0.1145624
$P(0.004)$	1.99×10^{-13}
$\min S$ over the first 300 ordinates	$S(\gamma_{289}^{-}) = -1.1454808$
$\max S$ over the first 300 ordinates	1.0975638, at $\gamma = 415.4552$

Table 7.7: Constants used in Sections 6.1.4 and 7.1.1.

Three steps remain open, in increasing order of difficulty. First, write the formal Thorin measure of R_t in Vere-Jones form, extending Theorem 6.2.3 with t dependent weights. Second, prove Conjecture S for infinitely many n (Problems 7.1 and 7.2). Third, find a family of positive admissible primal Lévy measures whose sine squared images converge to the image U of $\tilde{\mu}_\xi$ in Proposition 6.2.16.

7.2.4 Open problems

Problem 7.1. Conjecture S. Prove that the symmetrised truncations K_n have all their zeros in $0 < \operatorname{Re} w < \frac{1}{2}$ on the line $\operatorname{Re} w = \frac{1}{4}$ for infinitely many n; by Theorem 7.2.7 this implies RH. Corollaries 7.2.9 and 7.2.11 give its status for $n \le 24$ and $n = 32, 36$. Since Conjecture T and the Hermite Biehler inequality for A_n fail for large n, a proof must establish real rootedness without a one sided inequality, for instance by an interlacing argument in n, or by following the zeros of Z_n continuously in a parameter interpolating between n and $n+1$ and excluding collisions of zeros on the line.

Problem 7.2. Beyond the torus method. Corollary 7.2.11 rests on torus bounds proved by a branch and bound up to $n = 36$; at $n = 40$ the threshold, estimated by optimisation, is already near 4900, and with the band $1.2 \le x \le 1.3$ the method ends between $n = 40$ and $n = 44$, where the torus minimum of P_n in the band reaches zero. Extend Theorem 7.2.10 with narrower bands around $x = \frac{5}{4}$, and find the first n at which P_n has torus zeros with $x \ge \frac{5}{4}$.

Problem 7.3. From finite order to all orders. Theorem 6.2.17 proves that $k = W + c/(2\sqrt{\pi y})$ is n monotone for $n \le 10^{10}$, with the bound on n coming only from the height of verified zeros; the small y branch holds for every n, on $(0, 0.05/n]$. Can the two regimes be made to overlap for all n? The large y branch bounds the zeros above T_0 in absolute value, and the proof applies it where $yT_0^2 > 2n \log T_0$. No bound of this kind works where yT_0^2 is below about n: the peak of $\gamma^{2n} e^{-y\gamma^2}$, at

$\gamma^2 = n/y$, then lies above T_0, where the zeros are dense enough that their contributions, taken in absolute value, outweigh the positive term $c\,\Gamma(n+\frac{1}{2})y^{-n-\frac{1}{2}}/(2\pi)$ and the verified zeros together. So this asks for a prime side estimate for $(-1)^n k^{(n)}$ on a range $y \le y_n$ with y_n growing at least linearly in n; a range that decreases more slowly than $1/n$ raises the order reached but reaches only finitely many orders. In the proof of Theorem 6.2.17 the obstruction is the factor 2^n in the Cauchy estimate for $P^{(n)}$ on the circle of radius $y/2$. A circle of radius close to y, as in the proof of Theorem 5.1.6, loses only a factor of order n (Section 5.1.6), but that proof also stops at a finite order, and Proposition 7.1.2 suggests that at large y the obstruction cannot be removed without the explicit formula.

Problem 7.4. Beyond the Bernstein cone. Theorem 7.2.14 proves Weil positivity for test functions completely monotone in r^2, under an integrability condition. Find the largest cone of test functions, closed under the operations that preserve the Thorin structure, on which positivity can be proved unconditionally, and determine how far it can localise in height.

Problem 7.5. Deforming the binary product. By Theorem 7.1.8 each Euler factor at $\operatorname{Re} s = \sigma$ is $\prod_n (1+p^{-nu})^{b_n(p^{-\sigma})}$, with integer multiplicities 0 and 1 at $\sigma = 0$ and necklace multiplicities at $\sigma = \frac{1}{2}$. Follow the pseudo spectral trace formula (7.1.3) along the path $q = p^{-\sigma}$, $0 \le \sigma \le \frac{1}{2}$, and determine where and how the positivity of the interference pattern of the progressions can fail.

Problem 7.6. Regularising the autocorrelation. Proposition 6.2.14 factors $\Phi * \Phi$, formally, into the Meixner characteristic function and the positive lattice measure $\mathcal{A}$. The lattice series diverges: truncated at $m, n \le X$ it grows linearly in X, through the square of the pole term of the partial sums of ζ, and the factor $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ does not remove this growth. Find the regularised form of $\operatorname{sech}^{1/2} * \mathcal{A}$ and decide whether it has a Thorin or Lee and Yang structure in v.

Problem 7.7. Unimodality. Proposition 7.1.5 gives numerical evidence that w is unimodal. Prove it, and prove that f is unimodal, with mode near 0.0240, for instance from Steutel's equation (7.1.1) and the unimodality of w. Yamazato's theorem [347], [297, Theorem 53.1], that every self decomposable law is unimodal, shows with no hypothesis that X_R, the variable with Laplace transform R, is unimodal (Corollary 11.1.2 of Section 11.1), but it does not apply to X_φ, whose density is f, since X_φ is infinitely divisible and not self decomposable (Theorem 7.1.3).

Problem 7.8. Infinite divisibility across the strip. Is $w_\sigma > 0$ for every $\sigma \in (0,1)$ and every $x > 0$? For $\frac{1}{2} < \sigma < 1$ the method of Theorem 7.1.1 may apply: in the third form of Theorem 7.1.12 the terms of the zeros below the verified height are Poisson kernels of positive width, so the question reduces to the sign of the explicit pole and archimedean terms at large x, whose limit is governed by $\zeta'(\sigma)/\zeta(\sigma) > 0$. For the completed functions the answer is known: R_σ is infinitely divisible for every

$\sigma > \frac{1}{2}$ with no hypothesis (Theorem 9.2.5 and the proof of Proposition 10.1.1). Since φ_σ differs from R_σ by the gamma factor and elementary factors in $\sqrt{s}$, this does not settle the question for w_σ.

Problem 7.9. Meixner Pollaczek truncations. The symmetric expansion of $(\tau^2 + \frac{1}{4})Z(\tau)$ captures the first ordinate at degree 40 but converges slowly. Is there a weighting of its coefficients, or a Jensen type construction from them, whose polynomials are real rooted and whose zeros converge to the ordinates?

Problem 7.10. Nyman Beurling. Theorem 6.1.5 writes $1/\zeta(\frac{1}{2}+\sqrt{s})$ in the span of $\{\mathrm{sech}(\sqrt{s}\log N)\}$ with Möbius weights on $H_{1/2}$. Relate the completeness question of Lai and Vere-Jones for the dilates $\{\mathrm{sech}(\lambda x)\}$ to the Nyman Beurling criterion [218, 29], which Section 10.1.8 sets beside it, and decide whether the sech expansion admits an L^2 version on the boundary $\mathrm{Re}\,\sqrt{s} = \frac{1}{2}$.

Problem 7.11. Product representations of X_R. By the closure properties behind Corollary 7.1.6, an unconditional representation of X_R built from independent GGC variables by sums, products and weak limits would make R a GGC and prove RH (Proposition 1.2.1); for X_φ the corollary rules out such a representation. The harmonic mean is not available, because Proposition 7.1.7 shows it is not a closure operation for the class. The deconvolution (6.2.1) of Section 6.2.1 gives X_R, formally, as X_φ minus a stable law and a compensated series of the variables Y_{c_n}. Can the compensation be absorbed into a product?

Chapter 8

Probability, hitting times and other criteria

The clock $1/G$ is also the Laplace transform of a random variable, whose law the Pólya route of Chapter 2 computes from the same kernel as the Thorin density. This chapter treats the clock as a random variable, through its law, its Kent approximants and its Brownian boundaries, and reads two classical equivalents of RH that the Pólya density computes directly, Li's criterion and the de Bruijn Newman flow, in terms of the Thorin measure.

8.1 The clock as a random variable

The clock variable T of Section 1.2, with $\mathbb{E}e^{-uT} = 1/G(u)$, exists unconditionally and is self decomposable (Theorem 5.1.1). Under RH it is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (Proposition 1.2.1), and if the zeros are also simple, the theorems of Kent and Krein make it the hitting time of a level by a diffusion whose Dirichlet spectrum is $\{\gamma^2\}$ [145], [252, Section 16.3].

8.1.1 The law of the clock and its decay rate

The Laplace transform $\mathbb{E}e^{-uT} = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is available from the Pólya route at every complex u other than its poles, untilted for $|\operatorname{Im}\sqrt{u}| < 120$ and tilted above (Section 2.1.3). Talbot's contour inversion [316], as implemented in `mpmath` [205], then gives the density f of T and its survival function $\bar{F}(t) = P(T > t)$ with no reference to the zeros.

Under RH, with simple zeros, the density also has the spectral expansion [145]

$$f(t) = \sum_k R_k e^{-\gamma_k^2 t}, \qquad P(T>t) = \sum_k \frac{R_k}{\gamma_k^2} e^{-\gamma_k^2 t}, \qquad R_k = \frac{2i\gamma_k\,\xi(\frac{1}{2})}{\xi'(\frac{1}{2}+i\gamma_k)},$$

whose coefficients are the residues of the transform at its poles $u = -\gamma_k^2$. The first step of the proof of Theorem 4.3.22 derives it by a contour shift, with

$R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ in the role of c_n there, and the Pólya route supplies both the ordinates, as its Newton roots, and the values of ξ' at them. The residues alternate in sign (Exercise 8.1) and grow like $e^{\pi\gamma_k/4}$ up to powers of γ_k and the factor $1/|\zeta'(\rho_k)|$, because Stirling's formula puts the factor $e^{-\pi\gamma_k/4}$ into $|\Xi'(\gamma_k)| = |\xi'(\rho_k)|$; thus $R_1 = 1.0164 \times 10^4$ and $R_2 = -1.1775 \times 10^6$. The series converges for $t > 0$ but cancels heavily at small t, so near the origin Talbot's contour is the better tool (Exercise 8.2). Computed with 50 digits from the 79 zeros below 200, the spectral expansion agrees with the Talbot values to 24 digits for $0.010 \le t \le 0.12$. Beyond that range the Talbot values lose relative accuracy as $\bar F(t)$ falls, down to six digits at $t = 0.3$, and the spectral expansion is the accurate one (Figure 8.1).

The primes give a second expansion of f, with no hypothesis. Write $\xi = \eta\zeta$ with the gamma factor $\eta(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, and put $\phi_\eta(u) = \xi(\frac12)/\eta(\frac12 + \sqrt{u})$.

Proposition 8.1.1. *Let $c > \frac14$, let g_1 be the inverse Laplace transform of ϕ_η along $\operatorname{Re} u = c$, and let $q_\lambda(x) = \frac{\lambda}{2\sqrt{\pi}} x^{-3/2} e^{-\lambda^2/4x}$ for $x > 0$, both extended by 0 to $x \le 0$. Then for every $t > 0$*

$$f(t) = \sum_{n\ge1} \frac{\mu(n)}{\sqrt{n}}\, g_n(t), \qquad g_n = g_1 * q_{\log n}, \qquad |g_n(t)| \le \sup_{0<x\le t} |g_1(x)|\, \operatorname{erfc}\Big(\frac{\log n}{2\sqrt{t}}\Big),$$

so the series converges absolutely, faster than any power of n. Moreover $g_1(t) = 2\xi(\frac12)e^{t/4} + O(e^{\varepsilon t})$ as $t \to \infty$ for every $\varepsilon > 0$.

Proof. On $\operatorname{Re} u = c$ we have $\operatorname{Re}\sqrt{u} \ge \sqrt{c} > \frac12$, so the series (6.1.1) for $1/\zeta(\frac12 + \sqrt{u})$ converges absolutely there, with the sum of the absolute values of its terms at most $\zeta(\frac12 + \sqrt{c})$. By Stirling's formula $|\phi_\eta(u)|$ decays faster than any power of $|u|$ on vertical lines in $\operatorname{Re} u > 0$, uniformly on compact ranges of $\operatorname{Re} u$; hence $|g_1(x)| \le C_c e^{cx}$, the Laplace transform of g_1 is ϕ_η on $\operatorname{Re} u > c$, and f, whose transform $\phi_\eta(u)/\zeta(\frac12 + \sqrt{u})$ is integrable on the line, is its Bromwich integral along $\operatorname{Re} u = c$. Dominated convergence inverts the series term by term. The term of n is $\phi_\eta(u)e^{-\sqrt{u}\log n}$, and $e^{-\lambda\sqrt{u}}$ is the transform of q_λ by (6.1.2), which gives the convolution; the bound follows from $\int_0^t q_\lambda = \operatorname{erfc}(\lambda/2\sqrt{t})$. Finally ϕ_η is analytic off $(-\infty, 0]$ except for a simple pole at $u = \frac14$, where $\eta(\frac12 + \sqrt{u}) = \frac12(u - \frac14) + O((u-\frac14)^2)$, so the residue is $2\xi(\frac12)$; moving the contour to $\operatorname{Re} u = \varepsilon$ gives the asymptotic of g_1. □

So g_1 is not a density: it grows like $2\xi(\frac12)e^{t/4}$. The same contour shift gives $g_n(t) = 2\xi(\frac12)n^{-1/2}e^{t/4} + O_n(e^{\varepsilon t})$, so for fixed N the partial sum over $n \le N$ is $2\xi(\frac12)e^{t/4}\sum_{n\le N}\mu(n)/n + O_N(e^{\varepsilon t})$ and grows, while f decays like $e^{-\gamma_1^2 t}$. For large t the series therefore rests on the cancellation $\sum_n \mu(n)/n = 0$, the prime number theorem. For small t it converges very fast. Computed by Talbot inversion at 30

digits (`bn_tail.py`), the terms with $n \le 5$ ($n \le 7$ at $t = 0.04$) agree with the Talbot value of f to at least 20 digits at $t = 0.008, 0.012, 0.016, 0.02, 0.03, 0.04$. Relative to f, the term of the prime 2 is 1×10^{-26} at $t = 0.008$, 3×10^{-9} at $t = 0.02$ and 2.8 percent at $t = 0.04$. The two expansions are complementary. For relative accuracy 10^{-8} the spectral expansion needs 50 zeros at $t = 0.008$, 10 at $t = 0.02$ and 2 at $t = 0.06$ (among $1, 2, 5, 10, 20, 50, 120$ zeros), while the Möbius series needs only $n = 1$ up to $t = 0.02$, and $n \le 3$ at $t = 0.04$.

The law is unimodal with no hypothesis (Corollary 11.1.2) and sharply concentrated. Its mean and variance are $\mathbb{E}T = 0.0231050$ and $\operatorname{Var} T = 3.71726 \times 10^{-5}$ (Section 1.2) and its median is 0.0220. At the other end, with no hypothesis, $P(T \le y) \le \exp\{-(1 + o(1))(\log \frac{1}{y})^2/16y\}$ as $y \downarrow 0$, so the density vanishes to infinite order at the origin: Chernoff's bound $P(T \le y) \le e^{uy}/G(u)$, with $\log G(u) \sim \frac{\sqrt{u}}{4} \log u$ by Stirling's formula for ξ, is optimised at $\sqrt{u} \approx \log(1/y)/4y$.

PROPOSITION 8.1.2. *RH holds if and only if the density f of T is a Pólya frequency function of infinite order, that is* $\det[f(x_i - y_j)]_{i,j \le r} \ge 0$ *for every r and all $x_1 < \cdots < x_r$, $y_1 < \cdots < y_r$.*

Proof. Under RH, f is the density of $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 1.2.1), exponential densities are Pólya frequency functions, and the class is closed under convolution and weak limits. Conversely, by Schoenberg's theorem [300] the bilateral Laplace transform of an integrable Pólya frequency function of infinite order is $1/\Psi$ with $\Psi(u) = Ce^{-cu^2+bu} \prod_k (1 + \lambda_k u)e^{-\lambda_k u}$, $c \ge 0$, λ_k real. Here the transform is $1/G$, so $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ has only real zeros. Its zeros are $u = -a_k$ with $a_k = -(\rho - \frac{1}{2})^2$, which is real only if $\rho - \frac{1}{2}$ is real or purely imaginary, and ξ has no real zeros, so every ρ is on the critical line. □

This is the one sided form of the criterion of Proposition 16.5.1 for the two sided density. Under RH with simple zeros, assuming that the spectral expansion converges absolutely for $t > 0$ and may be differentiated termwise, the opposite signs of R_1 and R_2 make the upper tail of f log concave: $f(t) = R_1 e^{-\gamma_1^2 t}(1 - \varepsilon(t)) + O(e^{-\gamma_3^2 t})$ with $\varepsilon = |R_2/R_1| e^{-\Delta t}$ and $\Delta = \gamma_2^2 - \gamma_1^2$, so $(\log f)''(t) = -\Delta^2 \varepsilon/(1 - \varepsilon)^2 + O(e^{-(\gamma_3^2 - \gamma_1^2)t}) < 0$ for large t. On 91 logarithmically spaced points of $[0.006, 0.08]$, which carry all but 6×10^{-6} of the mass, a floating point Talbot computation at 25 digits (no interval arithmetic) gives $(\log f)'' < 0$ with maximum -0.026, the first two orders of hyperbolic complete monotonicity of f. These are necessary conditions for Proposition 8.1.2 tested in the bulk of the law, not a proof.

The hazard $h = f/\bar{F}$ tends to the principal eigenvalue γ_1^2. Under RH with simple zeros the leading term of the spectral expansion dominates, and $h(t) - \gamma_1^2 = O(e^{-(\gamma_2^2 - \gamma_1^2)t})$ (Exercise 8.3). This is the quasi stationary behaviour of a hitting time; Roberts [287, Section 2] obtains approximations of Brownian boundary hitting laws from approximations of the conditioned diffusion. The Talbot side shows the limit with no zero as input: $h(0.05) = 199.72$, $h(0.1) = 199.7904544$

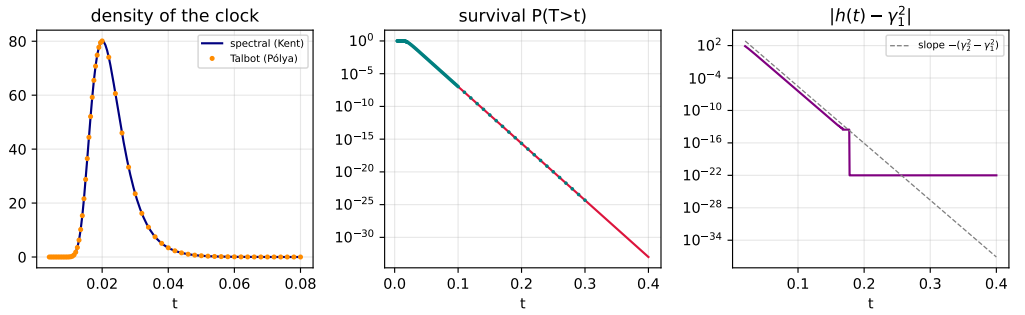


Figure 8.1: The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from γ_1^2, whose logarithm has slope $-(\gamma_2^2 - \gamma_1^2)$ (dashed). Beyond $t \approx 0.17$ the last curve reflects the double precision of the plotting script, not the hazard.

and $h(0.25) = 199.79045483$, the last equal to γ_1^2 to the eleven digits shown. The rate of approach is the spectral gap: a fit of $\log|h(t) - \gamma_1^2|$ on $[0.05, 0.12]$ has slope -242.12, against $-(\gamma_2^2 - \gamma_1^2) = -242.14$.

The clock also generates a natural exponential family in the sense of Barndorff-Nielsen. The cumulant generating function $K(\theta) = \log \mathbb{E}e^{\theta T} = \log\{\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{-\theta})\}$ is finite exactly for $\theta < \gamma_1^2$ and $K'(\theta) \to \infty$ as $\theta \uparrow \gamma_1^2$, so the family is regular and steep with natural parameter space $(-\infty, \gamma_1^2)$, whose boundary is the first zero; under RH the tilted laws are $\sum_\gamma e_\gamma/(\gamma^2 - \theta)$. The saddlepoint approximation of Barndorff-Nielsen and Cox [17], $\hat f(y) = \exp\{K(\hat\theta) - \hat\theta y\}/\sqrt{2\pi K''(\hat\theta)}$ with $K'(\hat\theta) = y$, needs ξ only on the real axis $s \ge \frac{1}{2}$ (for $y < \mathbb{E}T$) and on the segment of the critical line from $\frac{1}{2}$ to $\frac{1}{2} + i\gamma_1$ (for $y > \mathbb{E}T$), so it never passes the first zero. It is not normalised (it integrates to 0.98492) and agrees with the Talbot density to about 0.5 percent in the lower tail and within 2.5 percent on $[0.011, 0.035]$ (Table 8.1); in the upper tail the ratio tends to $e/\sqrt{2\pi} = 1.0844$, the error of the approximation for an exponential density, because the tail is $R_1 e^{-\gamma_1^2 y}$. As $y \downarrow 0$ the exponent $K(\hat\theta) - \hat\theta y$ is $-(1 + o(1))(\log\frac{1}{y})^2/16y$, the Chernoff exponent above, but slowly: at $y = 0.006$ the leading term gives -272 against -33.0 for the exact $\log f$, while the saddlepoint, which keeps all of ξ, is within 0.5 percent of f. The tail formula of Lugannani and Rice [190], built from the same $\hat\theta$, $K(\hat\theta)$ and $K''(\hat\theta)$, is within 1.4 percent of the Talbot values of $P(T \le y)$ and within 2.4 percent of those of $P(T > y)$ at the sampled y (`bn_tail.py`, part a).

y	$\hat{\theta}$	$f(y)$ (Talbot)	$\hat{f}(y)$	ratio
0.0060	-15909.8	5.165×10^{-15}	5.138×10^{-15}	0.9949
0.0120	-1674.1	1.7018	1.6953	0.9962
0.0201	-114.5	80.1084	78.5287	0.9803
0.0239	19.4	61.4522	59.9432	0.9754
0.0401	148.8	3.3491	3.4507	1.0303
0.0800	182.8	0.0012	0.0013	1.0762

Table 8.1: The saddlepoint density of the clock against the Talbot inversion of its Laplace transform, both from ξ alone.

terms	γ_1^2	γ_2^2	γ_3^2	γ_4^2
2	199.86679	427.28		
3	199.79031	442.129	618.93	
4	199.790455	441.9255	625.63	907.24
true	199.7904548	441.9261506	625.5429969	925.6730873

Table 8.2: Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise 10^{-16}, 10^{-30} and 10^{-45}: the error is model truncation, not noise.

8.1.2 The cost of the zeros in time and in samples

We fit sums of p exponentials to $P(T > t)$ by Prony's method, from 40 values at $t = 0.03, 0.032, \ldots, 0.108$ carrying relative noise from 10^{-16} down to 10^{-45} (Table 8.2). The error of the fitted atoms comes from the neglected terms, whose relative size on a window starting at t_0 is $e^{-(\gamma_{p+1}^2 - \gamma_1^2)t_0}$ up to a factor set by the residues. Resolving the kth atom therefore needs data accurate to about $(\gamma_k^2 - \gamma_1^2)t_0 / \log 10$ digits, a number that grows without bound in k.

Sampling meets the same cancellation in another currency. By the identity $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Section 1.2, with $S_2 = \sum_{k \geq 1} \frac{2}{\pi^2 k^2} \Gamma_k$ for independent Gamma$(2,1)$ variables Γ_k, the variable $Y = \frac{1}{2}\log(\pi S_2/2)$ satisfies

$$2\,\Xi(z) = \mathbb{E}\big[e^{Y/2}\cos(zY)\big].$$

$\Xi(z)$ is the mean of a random variable with finite variance, since $\mathbb{E}e^Y = 2\xi(1) = 1$, so a sample mean estimates it at every z. We draw $n = 10^6$ copies of S_2 and compare the estimates with $\Xi(z)$ in Table 8.3, a check of the identity on the critical line; Table 9.1 checks it at real s (Section 9.1.1).

The standard error settles at 3.5×10^{-4}, the standard deviation of $e^{Y/2}\cos(zY)/2$ over $\sqrt{n}$, while $|\Xi(z)|$ decays like $e^{-\pi z/4}$. The sample size needed to resolve the sign of Ξ therefore grows like $e^{\pi z/2}$. Even the first sign change, at $z = 14.13$, needs about 10^6 samples: between the first two zeros $|\Xi|$ is at most

z	Monte Carlo	standard error	$\Xi(z)$	n with standard error $\lvert\Xi(z)\rvert$
0	0.497061	5.4×10^{-5}	0.497121	1.2×10^{-2}
2	0.453136	7.3×10^{-5}	0.453100	2.6×10^{-2}
5	0.275888	2.4×10^{-4}	0.275550	0.77
8	0.103756	3.4×10^{-4}	0.103303	11
10	0.038401	3.5×10^{-4}	0.037968	86
20	0.000314	3.5×10^{-4}	-3.67×10^{-5}	9.3×10^{7}
30	0.000289	3.5×10^{-4}	-1.50×10^{-8}	5.5×10^{14}

Table 8.3: Monte Carlo estimates of $\Xi(z)$ from Williams' representation with 10^6 samples. The last column is the sample size at which the standard error equals $\lvert\Xi(z)\rvert$.

8.0×10^{-4} (at $z = 15.59$), and resolving it at two standard errors takes about 8×10^5 samples. Sampling is the right tool for the law of the Pólya variable itself, for its moments and for the Bayesian inverse problems of [252, Section 19.2], and the wrong tool for the zeros. Devroye's exact samplers for theta type laws [76] make sampling efficient, but the sample size needed for the zeros still grows like $e^{\pi z/2}$.

8.1.3 Kent approximants with a Weyl tail

In the rest of this section we write w for the variable u of the clock. Kent's theorem [145] identifies the first hitting time of a level by a diffusion started at a reflecting or entrance end of its interval with the GGC $\prod_n (1 + w/\lambda_n)^{-1}$, where the λ_n are the Dirichlet eigenvalues of the interval up to that level. Krein's inverse spectral theory [79] builds a string, and so a generalised diffusion, from any increasing sequence $\lambda_n > 0$ with $\sum 1/\lambda_n < \infty$. The GGCs that arise as such hitting times are therefore exactly those without drift whose Thorin measure consists of unit atoms at distinct points. By Proposition 1.2.1 the clock is one of them exactly when RH holds and the zeros are simple, and its diffusion then has Dirichlet spectrum $\{\gamma^2\}$.

The approximants of the clock that Kent's theorem suggests are the transforms $1/G_N(w)$ of the partial products $G_N(w) = \prod_{k \le N} (1 + w/\gamma_k^2)$, each the transform of the hitting time of a finite string with N point masses. They are admissible in the closure criterion of Corollary 4.1.2, and they are the approximating sequence, under RH, in the proof of [252, Proposition 18.10]. They converge slowly. With $\tau_1(N) = \sum_{k>N} \gamma_k^{-2}$, the relative error $G(w)/G_N(w) - 1$ of the clock transform lies, under RH, between $w\tau_1(N)$ and $e^{w\tau_1(N)} - 1$.

We keep the verified zeros as atoms and replace the others by their average density. By the Riemann von Mangoldt formula $N(t) = \vartheta(t)/\pi + 1 + S(t)$ of Section 2.2, the zeros above the cutoff Θ_N have density $\vartheta'(\gamma)/\pi \approx \frac{1}{2\pi} \log \frac{\gamma}{2\pi}$,

which is positive for $\gamma > 6.3$. So

$$G_N^{\mathrm{W}}(w) = \prod_{k \le N} \Big(1 + \frac{w}{\gamma_k^2}\Big) \exp\Big\{ \int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big) \frac{\vartheta'(\gamma)}{\pi}\, d\gamma \Big\},$$

with W for Weyl, has reciprocal equal to the Laplace transform of a GGC with N unit atoms and the continuous Thorin density $\vartheta'(\sqrt{z})/(2\pi\sqrt{z})$ on $z > \Theta_N^2$. For Θ_N we take the Gram point g_{N-1} of Section 2.2.1, where $\vartheta(\Theta_N)/\pi + 1 = N$. For the values of N used here it lies between γ_N and γ_{N+1}, so $S(\Theta_N) = 0$.

Table 8.4 compares the two approximants with the clock computed from ξ.

N	$w = 1$	$w = 100$	$w = 1000$	$w = 10^4$
1	1.8×10^{-2} / 1.3×10^{-4}	4.8 / 1.0×10^{-2}	$\gg 1$ / 3.3×10^{-2}	$\gg 1$ / 4.3×10^{-2}
20	7.2×10^{-3} / 4.3×10^{-7}	1.04 / 4.2×10^{-5}	$\gg 1$ / 3.7×10^{-4}	$\gg 1$ / 1.5×10^{-3}
79	3.6×10^{-3} / 1.5×10^{-7}	0.43 / 1.5×10^{-5}	34 / 1.5×10^{-4}	$\gg 1$ / 1.2×10^{-3}
300	1.6×10^{-3} / 7.4×10^{-9}	0.17 / 7.4×10^{-7}	3.95 / 7.4×10^{-6}	$\gg 1$ / 7.2×10^{-5}
1700	5.0×10^{-4} / 2.9×10^{-11}	0.051 / 2.9×10^{-9}	0.64 / 2.9×10^{-8}	142 / 2.9×10^{-7}

Table 8.4: Relative error of the clock transform: partial product over N zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above 10^3. The partial product values agree with [252, Section 18.2.1] where both are given.

At the four points $w = 1, 100, 1000, 10^4$ the table exhibits an explicit positive Thorin measure whose Thorin transform $\int \log(1 + w/z)\, U(dz)$ matches $\log G(w)$ to within 3×10^{-7}. This is a finite instance of the weak Thorin condition of [252, Section 18.2.3], which asks for such a measure, to within any $\eta > 0$, at every finite set of points, and which is equivalent to RH by the closure criterion of Corollary 4.1.2. A finite instance proves nothing: agreement for $w \le 10^4$ cannot detect off line zeros above the verified height, which change $\log G$ there by far less than the error. The error of the Weyl tail grows linearly in w, and to first order it is a weighted average of S beyond the cutoff,

$$\log G(w) - \log G_N^{\mathrm{W}}(w) \approx w\big(\tau_1(N) - \tau_1^{\mathrm{W}}(N)\big) = 2w \int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma^3}\, d\gamma,$$

where $\tau_1^{\mathrm{W}}(N) = \int_{\Theta_N}^{\infty} \gamma^{-2} \vartheta'(\gamma) \pi^{-1} d\gamma$ is the Weyl estimate of $\tau_1(N)$ and the identity is an integration by parts that uses $S(\Theta_N) = 0$.

8.1.4 Moment matching and the error at large w

The error of the Weyl tail comes from the first order term of $\log(1 + w/\gamma^2) = w/\gamma^2 - w^2/(2\gamma^4) + \cdots$. For w small compared with Θ_N^2 the error of $\log G_N^{\mathrm{W}}$ is approximately w times $\tau_1(N) - \tau_1^{\mathrm{W}}(N)$, and the second order term is $-w^2/2$ times the corresponding difference for $\tau_2(N) = \sum_{k>N} \gamma_k^{-4}$. Both true tail sums

are available without any zero beyond γ_N, because the power sums $p_m = \sum_k a_k^{-m}$, with $a_k = -(\rho_k - \frac12)^2$ as in Section 1.2 and equal to $\sum_{\gamma>0} \gamma^{-2m}$ under RH, are Taylor coefficients of ξ at the centre: if $\xi(\frac12 + s)/\xi(\frac12) = 1 + c_2 s^2 + c_4 s^4 + \cdots$, then

$$p_1 = \sum_k a_k^{-1} = c_2 = 0.023104993115418970 7889,$$
$$p_2 = \sum_k a_k^{-2} = c_2^2 - 2c_4 = 0.000037172599285269 6862,$$

and $\tau_m(N) = p_m - \sum_{k\le N} \gamma_k^{-2m}$. Here $p_1 = \mathbb{E}T$ and $p_2 = \operatorname{Var} T$ (Section 1.2). We therefore replace the Weyl density $\vartheta'(\gamma)/\pi$ on $[\Theta_N, \infty)$ by $\lambda\,\vartheta'(\gamma)/\pi$ on $[\Theta'_N, \infty)$, and choose the two parameters λ and Θ'_N so that the tail reproduces $\tau_1(N)$ and $\tau_2(N)$ exactly. The tail density remains positive, so the approximant is still a GGC. The adjustments are small: for $N = 79$ the cutoff moves from 199.649 to 199.675 and $\lambda = 1.000059$.

N	tail	$w=1$	$w=10^2$	$w=10^4$	$w=10^5$	$w=10^6$	$w=10^7$
79	Weyl	1.5×10^{-7}	1.5×10^{-5}	1.2×10^{-3}	4.3×10^{-3}	5.8×10^{-3}	6.0×10^{-3}
	matched	2.6×10^{-17}	2.6×10^{-11}	1.8×10^{-5}	3.1×10^{-3}	6.1×10^{-2}	0.47
300	Weyl	7.4×10^{-9}	7.4×10^{-7}	7.2×10^{-5}	5.5×10^{-4}	1.7×10^{-3}	2.1×10^{-3}
	matched	2.3×10^{-20}	2.3×10^{-14}	2.2×10^{-8}	1.5×10^{-5}	1.8×10^{-3}	2.8×10^{-2}
1700	Weyl	2.9×10^{-11}	2.9×10^{-9}	2.9×10^{-7}	2.8×10^{-6}	2.4×10^{-5}	9.3×10^{-5}
	matched	3.5×10^{-23}	1.1×10^{-17}	7.0×10^{-13}	3.6×10^{-10}	2.5×10^{-7}	5.0×10^{-5}

Table 8.5: Relative error $|G_{\mathrm{a}}/G - 1|$ of the approximant G_{a} of G with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta_N^2 = 3.99\times10^4$, 2.95×10^5 and 4.83×10^6 for $N = 79$, 300 and 1700.

Table 8.5 shows two regimes. For w below Θ_N^2 moment matching gains ten to twelve digits at $w = 1$ and about two as w approaches Θ_N^2. Well beyond Θ_N^2 the gain disappears, and at $w = 10^7$ with $N = 79$ the matched tail is worse than the plain Weyl tail. For $w \gg \Theta_N^2$ the matched tail differs from the Weyl tail mainly by $(\lambda-1)\int_{\Theta'_N}^{\infty} \log(1 + w/\gamma^2)\,\vartheta'(\gamma)\pi^{-1} d\gamma$, which grows like $(\lambda-1)\frac{\sqrt{w}}{2}\log\frac{\sqrt{w}}{2\pi}$, while the error of the Weyl tail tends to a constant.

Under RH, since $S(\Theta_N) = 0$ and no ordinate lies in $(\gamma_N, \Theta_N]$, the true tail sum minus its Weyl estimate is a Stieltjes integral against S, and an integration by parts gives

$$\log G_N^{\mathrm{W}}(w) - \log G(w) = -\int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big)\, dS(\gamma) = -\int_{\Theta_N}^{\infty} \frac{2w\,S(\gamma)}{\gamma(\gamma^2 + w)}\, d\gamma$$
$$\longrightarrow\ -2\int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma}\, d\gamma \qquad (w \to \infty).$$

The limit is $-\ell_N$, with $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as in Exercise 8.5. It is finite because $\int_0^t S(u)\,du = O(\log t)$ (Littlewood; Section 2.2.1), and in general it is not zero. The measured errors of the Weyl tail level off at about 6×10^{-3} for $N = 79$ and 2×10^{-3} for $N = 300$.

The constant is not an obstruction of principle. It depends only on $\gamma_1, \dots, \gamma_N$ and on ξ along the real axis, and Stirling's formula gives ℓ_N in closed form (Exercise 8.5). It is also a floor only for the Weyl density. Adding to ϑ'/π a continuous ω with compact support in (Θ_N, ∞) and $\int \omega = 0$ changes the limit by $-2\int \omega(\gamma)\log\gamma\,d\gamma$, and two wide bumps of opposite sign placed far apart give this any value while the density stays positive, so a suitable positive tail removes the constant.

8.2 Brownian boundaries and Williams' structure

Under RH with simple zeros, Kent's theorem makes the clock variable T the hitting time of a level by a diffusion, but it describes that diffusion only through its spectrum, the set $\{\gamma^2\}$ that is to be found (Section 8.1.3). We look instead for a representation by Brownian motion itself. Unconditionally, T is not a Brownian first passage time to an independent random level for any constant drift (Section 8.2.1). Across a curved boundary it is one: such a boundary exists by a theorem of Ekström and Janson, and we compute it (Section 8.2.2). Williams' representation of the Pólya kernel through Bessel hitting times explains the tilted contour of Section 2.1.3, but it transfers no positivity to the clock (Section 8.2.3).

8.2.1 The clock is not a first passage time to a random level

The first passage time of Brownian motion with drift $c \geq 0$ to a level $a > 0$ has the inverse Gaussian law, with transform $\exp\{a(c - \sqrt{c^2 + 2u})\}$. Mixing over an independent random level with law F gives the transforms $\int \exp\{a(c - \sqrt{c^2 + 2u})\}F(da)$, a class of first passage laws.

Proposition 8.2.1. *For no drift $c \geq 0$ and no probability law F on $(0, \infty)$ is the clock variable T equal in law to the first passage time of Brownian motion with drift c to an independent level $A \sim F$.*

Proof. On the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, $q \in \mathbb{R}$, we have $c^2 + 2u = (c + iq)^2$, so $\operatorname{Re}\sqrt{c^2 + 2u} = c$ and every mixture transform has modulus at most one there. Suppose that T had such a law, so that the clock transform $1/G(u) = \mathbb{E}e^{-uT}$ and the mixture transform agree for $u \geq 0$. The mixture transform is analytic and bounded by one on the region $D = \{u : \operatorname{Re}\sqrt{c^2 + 2u} > c\}$, and continuous up to its boundary, which is the parabola. The region D is the image of the half plane $\operatorname{Re} v > c$ under $v \mapsto (v^2 - c^2)/2$, so it is connected, and it contains the right half plane. By the identity theorem $1/G$ coincides with the mixture transform

on D, so it has no pole in D and is bounded by one there, and by continuity $|1/G| \le 1$ on the parabola away from the zeros of $\xi(\frac12+\sqrt{u})$. For the clock, however, $\sqrt{u} = (c+iq)/\sqrt{2} + O(1/q)$ as $q \to \infty$ (for $c = 0$ the parabola is the negative axis, approached from above, and $\sqrt{u} = i|q|/\sqrt{2}$), so $|\xi(\frac12+\sqrt{u})|$ decays like $\exp(-\pi|q|/(4\sqrt{2}))$ times a power, by Stirling's formula and the polynomial growth of ζ in vertical strips. Hence $|\xi(\frac12)/\xi(\frac12+\sqrt{u})|$ grows exponentially along the parabola, and the transforms cannot agree. □

The proof covers the nonincreasing lines $a - ct$, $c \ge 0$; an increasing line, with $c < 0$, gives a passage time that is infinite with probability $1 - \mathbb{E}e^{2cA} > 0$, which the clock is not. Expanding $G(u) = 1 + m_1 u + m_2 u^2 + \cdots$ with $m_1 = \mathbb{E}T$ and $m_1^2 - 2m_2 = \operatorname{Var} T$ gives $|G(u)|^2 = 1 - q^2(m_1 - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola, so the clock transform already exceeds one at small q whenever

$$c < c_* = \sqrt{\mathbb{E}T/\operatorname{Var} T} = \sqrt{0.0231050/3.71726 \times 10^{-5}} = 24.93.$$

8.2.2 A curved Brownian boundary for the clock

Finding a boundary whose first passage time has a prescribed law is the inverse first passage problem [56, Theorem 1]. By the theorems of Ekström and Janson [82, Theorems 2.3, 8.2 and 9.2], stated in full in Section 9.2.4, the problem has a solution for the clock, which we call the exact boundary. Since the density of the clock is continuous and positive on $(0,\infty)$ (Exercise 8.10), the same theorems make the exact boundary continuous on $(0,\infty)$ and the unique continuous solution b with $\liminf_{t\downarrow 0} b(t) \ge 0$ of the Volterra integral equation (8.2.1) of the first passage density. We call the solution of the discretised Volterra equation the Volterra boundary.

8.2.2.1 The Roberts Shortland tangent boundary

Roberts and Shortland [288] approximate the hazard of a Brownian first passage across a smooth boundary b by the hazard of the tangent line at the current time, for which the Bachelier Lévy formula is exact:

$$h_{\tan}(t) = \frac{\alpha(2\pi t^3)^{-1/2}\exp\{-(\alpha+\beta t)^2/2t\}}{\mathcal{N}\big((\alpha+\beta t)/\sqrt{t}\big) - e^{-2\alpha\beta}\mathcal{N}\big((\beta t-\alpha)/\sqrt{t}\big)},$$
$$\beta = b'(t),\ \alpha = b(t) - tb'(t),$$

where $\mathcal{N}$ is the standard normal distribution function; we write $\overline{\mathcal{N}} = 1 - \mathcal{N}$ for its tail. We use the formula in reverse. Setting $h_{\tan}(t) = h(t)$, the computed hazard of the clock, defines an ordinary differential equation for b, which we integrate by the fourth order Runge Kutta method with step 10^{-4} from $t_0 = 0.011$. The starting value $b(t_0) = 0.4023$ is the constant level with the crossing mass of the clock, $2\overline{\mathcal{N}}(b(t_0)/\sqrt{t_0}) = P(T \le t_0) = 1.25 \times 10^{-4}$.

The boundary falls steeply at first, with slope -87.5 at t_0, and then becomes asymptotically linear with slope tending to $-\sqrt{2}\,\gamma_1 = -19.99$, since an inverse Gaussian hazard tends to $\beta^2/2$.

8.2.2.2 The boundary from the Volterra equation

The Volterra equation of the first passage density determines the exact boundary (Section 8.2.2 and [82, Theorems 9.1 and 9.2]), and solving its discretisation for the clock needs three zones: small times, the bulk and the right tail.

For a continuous boundary b with $b(0+) > 0$, the first passage density f of Brownian motion started at 0 satisfies the Volterra equation of the first kind [240]

$$\begin{aligned} P\big(B_t \ge b(t)\big) &= \int_0^t f(s)\, P\big(B_t \ge b(t) \mid B_s = b(s)\big)\, ds, \\ \text{that is} \quad \overline{\mathcal{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) &= \int_0^t f(s)\, \overline{\mathcal{N}}\Big(\frac{b(t)-b(s)}{\sqrt{t-s}}\Big) ds, \end{aligned} \tag{8.2.1}$$

with $\overline{\mathcal{N}}$ the standard normal tail. In the inverse problem we know f, the density of the clock computed in Section 8.1, and solve for b one time step at a time, since once b is known at the earlier grid points the discretised equation at time t is a single scalar equation for $b(t)$. We discretise with 2900 steps on $[0.009, 0.3]$, midpoint quadrature in s, and a scalar root search at each step.

Before $t_0 = 0.009$ the clock has crossed with probability only $P(T \le t_0) = 1.42 \times 10^{-7}$. Its density is super flat: it vanishes to all orders at $t = 0$, like $\exp(-(1+o(1))\log^2(1/t)/(16t))$, as the growth $\log G(u) \sim \frac{1}{4}\sqrt{u}\log u$ at infinity dictates; de Bruijn's exponential Tauberian theorem [33, Theorem 4.12.9] gives this rate for $P(T \le t)$, and the density has the same rate because the clock is unimodal (Corollary 11.1.2). So the exact boundary is unbounded as $t \downarrow 0$: a path above $b(t)$ at time t has crossed by then, so $\overline{\mathcal{N}}(b(t)/\sqrt{t}) \le P(T \le t)$, which forces $b(t) \ge (1+o(1))\log(1/t)/(2\sqrt{2})$. In this first zone we replace the exact boundary by a constant level, fixed by the reflection principle, $2\overline{\mathcal{N}}(b/\sqrt{t_0}) = P(T \le t_0)$, which gives $b(t_0) = 0.4992$. The Volterra boundary then falls from 0.4992 to 0.3908 at $t = 0.011$. The tangent boundary starts there at 0.4023, the constant level with the same crossing mass by time 0.011; a boundary that decreases on $[0, 0.011]$ with that crossing mass must end below this level, so the gap is no defect of the tangent approximation.

In the bulk, between $t = 0.011$ and $t = 0.05$, the Volterra and tangent boundaries agree closely: 0.0721 against 0.0710 at $t = 0.02$, and -0.5894 against -0.5954 at $t = 0.05$. Here there is no cancellation: both sides of (8.2.1) are of order one.

In the right tail the equation cancels. Solved in the form (8.2.1), the computation fails at $t = 0.0937$, where the clock has survival probability 3.8×10^{-7}: both sides of the equation are within that amount of $P(T \le t) \approx 1$. In general the absolute error of the quadrature must stay below $P(T > t)$, which is 5×10^{-25}

at $t = 0.3$, beyond double precision. The cure is to subtract the equation from $P(T \le t) = \int_0^t f$, which gives the complementary form

$$\mathcal{N}\Big(\frac{b(t)}{\sqrt{t}}\Big) - P(T > t) = \int_0^t f(s)\,\mathcal{N}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big)\,ds,$$

in which every term is small and computed without cancellation, provided the survival function itself is known to relative accuracy. The spectral expansion of Section 8.1 supplies it to full working precision, even where $P(T > t)$ is of order 10^{-25}. With the switch made where $P(T > t)$ falls below one half, the solution runs smoothly to $t = 0.3$.

In this zone the boundary is nearly linear with slope $\beta = -\sqrt{2}\gamma_1 = -19.99$. A linear boundary does not suffice. Its crossing time has survival function of order $t^{-3/2}e^{-\beta^2 t/2}$, while that of the clock is asymptotic to a multiple of $e^{-\gamma_1^2 t}$. The power $t^{-3/2}$ is removed by a logarithmic correction: for $b(t) = \beta t + a + \kappa \log t$ the Gaussian factor $e^{-b(t)^2/2t}$ acquires $t^{-\kappa\beta}$, and $\kappa\beta = -\frac{3}{2}$ cancels the power. So the right tail boundary is

$$b(t) \approx \beta t + a + \frac{3}{2|\beta|}\log t, \qquad \beta = -\sqrt{2}\,\gamma_1, \qquad \frac{3}{2|\beta|} = 0.0750.$$

Fitting a on $[0.15, 0.3]$ gives $a = 0.578$ with largest residual 0.0095. The slope of the Volterra boundary over $[0.2, 0.3]$ is -19.77. It lies between the slope -19.99 without the logarithm and the slope $-19.99 + 0.075/0.25 = -19.69$ of the asymptote at $t = 0.25$.

We simulate $n = 2 \times 10^5$ Brownian paths against both boundaries with step 2×10^{-5} and a Brownian bridge crossing correction; sampling noise alone gives a Kolmogorov distance of about $0.87/\sqrt{n} \approx 0.002$. For the Volterra boundary the Kolmogorov distance to the law of the clock is 0.0020, at the level of the Monte Carlo noise, and the mean crossing time is 0.023085 against $\mathbb{E}T = 0.0231050$. For the tangent boundary the distance is 0.0059 and the mean 0.023021 (Figure 8.2). The right tail itself cannot be checked by plain simulation, since only about one path in ten million survives past $t = 0.1$.

Wald's identities constrain the exact boundary as a whole. Since $\mathbb{E}T$ is finite, the first passage time H across it satisfies $\mathbb{E}B_H = 0$ and $\mathbb{E}B_H^2 = \mathbb{E}H$, and since $B_H = b(H)$ and H has the law of the clock, $\int_0^\infty b(t) f(t)\,dt = 0$ and $\int_0^\infty b(t)^2 f(t)\,dt = \mathbb{E}T$.

8.2.3 Williams' Gamma(2) structure: the tilt, a benchmark and its limits

Williams [344] and Biane, Pitman and Yor [32] showed that the variable $S_2 = \sum_{n\ge1} \frac{2}{\pi^2 n^2}\Gamma_n$ of Section 8.1.2 is a sum of two independent hitting times of level one by the three dimensional Bessel process started at 0, and that $2\xi(s) =$

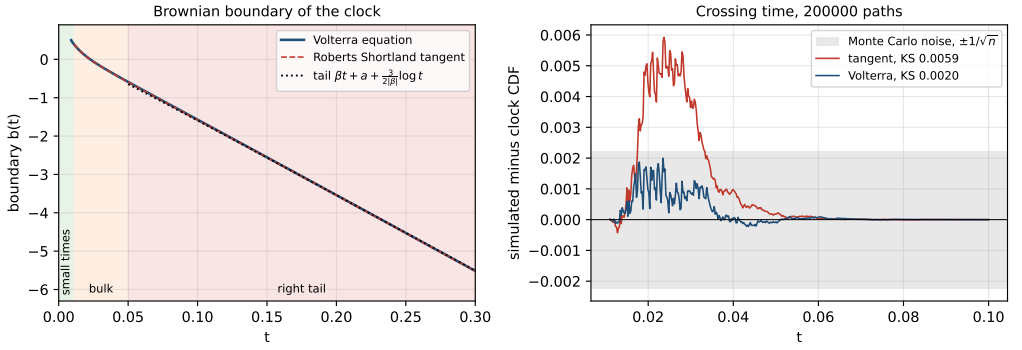


Figure 8.2: Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function.

$\mathbb{E}[(\pi S_2/2)^{s/2}]$, so that the Pólya variable is $Y = \frac{1}{2}\log(\pi S_2/2)$ tilted by $e^{Y/2} = (\pi S_2/2)^{1/4}$ (Section 1.2; [252, Section 16.2.1]). The Thorin measure of S_2 has atoms of mass two at the points $\pi^2 n^2/2$, the Gamma(2) shape of a sum of two copies.

First, the tilt. Write y for the variable of the Pólya kernel, so that the value y of Y corresponds to the value e^{2y} of $\pi S_2/2$. Moving the line of integration from the real axis to the line $\operatorname{Im} y = \eta$, as in Proposition 2.1.1, rotates e^{2y} by the angle 2η into the complex plane. The density of S_2, a theta series in its argument, continues analytically to the right half plane, and the boundary of that half plane, the natural boundary of the theta function [284, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], is reached exactly at $\eta = \pi/4$. So the admissible tilts of Proposition 2.1.1 are the rotations of Williams' variable within the half plane of analyticity of its density, and the saving $e^{-\eta\theta}$ is the factor gained by that rotation.

Second, a benchmark. Since $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$, the Thorin measure of $\pi^2 S_2/2$ is $2\sum_n \delta_{n^2}$, and in the square root variable its Cauchy flow $P_\varepsilon * 2\sum_n \delta_n$ is explicit [252, Section 16.4]. On this measure the distances of Section 3.2 separate the effects of the method from those of the arithmetic, and they need no separate computation: $\pi^2 S_2/2$ is the sum of two independent copies of the sinh clock of Section 1.3.4, so its distances follow from the closed forms of that section (Section 12.2.3).

Third, the limits. Williams' atoms are the Dirichlet eigenvalues $\pi^2 n^2/2$ of an interval, and they live on the Laplace side of the kernel. The zeros of ξ live on the Mellin side of the same variable, and they reappear as unit atoms only on the Laplace side of the reciprocal. The Gamma(2) structure therefore gives positivity

and an exact sampler [76] on the kernel side, but it transfers no positivity to the Thorin measure of the clock. In particular it does not give a positive Gamma(2) scale mixture representation of the Pólya density, a route that [252, Section 12.6] excludes.

8.3 Keiper Li coefficients from the Pólya density

The Pólya density that computes the law of the clock also computes the coefficients of Li's criterion [176], which states that RH holds if and only if

$$\lambda_n = \sum_{\rho} \Big[1 - \Big(1 - \frac{1}{\rho} \Big)^n \Big] \geq 0 \qquad \text{for every } n \geq 1,$$

the sum over the zeros taken in conjugate pairs. Keiper [143] introduced the coefficients λ_n and computed them before Li proved the criterion, and Bombieri and Lagarias [37] derived an arithmetic expression for them from the explicit formula and related the criterion to Weil's.

8.3.1 Taylor coefficients of $\log \xi$ at $s = 1$

Li [176] expresses the coefficients as derivatives at $s = 1$,

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \Big[s^{n-1} \log \xi(s) \Big]_{s=1}.$$

With $\log \xi(1+w) = \sum_{k \geq 0} a_k w^k$ and $(1+w)^{n-1} = \sum_j \binom{n-1}{j} w^j$, comparing the coefficients of w^n gives

$$\lambda_n = n \sum_{j=0}^{n-1} \binom{n-1}{j} a_{n-j}, \tag{8.3.1}$$

so the computation reduces to the Taylor coefficients of $\log \xi$ at $s = 1$. The zeros nearest to $s = 1$ are $\frac{1}{2} \pm 14.1347i$, so $\log \xi(1+w)$ is analytic in the disc $|w| < R$, where $R = |\rho_1 - 1| = 14.1436$.

The sum (8.3.1) is exact, but its terms can be larger than λ_n. The Hadamard product $\xi(1+w) = \xi(1) \prod_\rho (1 - w/(\rho - 1))$ gives $m a_m = -\sum_\rho (\rho - 1)^{-m}$ for $m \geq 2$, and for large m the two zeros nearest to $s = 1$ dominate this sum, so that $|m a_m|$ is at most about $2R^{-m}$. Since $n\binom{n-1}{m-1} = m\binom{n}{m}$, the terms of (8.3.1) are $\binom{n}{m} m a_m$ with $m = n - j$, and for large n their absolute sum is at most about $2(1 + 1/R)^n$. Cancellation therefore costs at most about $\log_{10} \big(2(1 + 1/R)^n / \lambda_n \big)$ digits. This grows like $n \log_{10}(1 + 1/R) \approx 0.03n$ and is about one digit at $n = 100$, where $2(1 + 1/R)^{100} \approx 1.9 \times 10^3$ and $\lambda_{100} = 118.6$.

We compute the a_k by the trapezoid rule for the Cauchy integral on the circle $|w| = 6$,

$$a_k \approx \frac{1}{N r^k} \sum_{j=0}^{N-1} \log \xi\big(1 + r e^{2\pi i j/N}\big) e^{-2\pi i j k/N}, \qquad r = 6,\ N = 512,$$

which converges geometrically. The rule returns $a_k + a_{k+N}r^N + a_{k+2N}r^{2N} + \cdots$, so aliasing changes a_k by a relative amount of about $(r/R)^N \approx 10^{-191}$. Rounding errors in the sum are amplified in a_k by about $(R/r)^k$, which is 10^{37} at $k = 100$ and is absorbed by the working precision.

Each value $\xi(1+w) = \xi(\frac{1}{2} + (\frac{1}{2} + w))$ comes from the Pólya route at 600 bits; the imaginary part of $\frac{1}{2}+w$ never exceeds 6, so the Pólya sum loses almost nothing to cancellation. The principal branch of the logarithm is the analytic one, because $|\arg \xi(1+w)| < 1$ on the circle: in the Hadamard product (1.2.1) with $s = \frac{1}{2} + w$, so that $|s| \le 6.5$, the factor $1 + s^2/\gamma^2$ of the zeros $\frac{1}{2} \pm i\gamma$ has argument at most $\arcsin(6.5^2/\gamma^2) \le 1.01 \cdot 6.5^2/\gamma^2$, and since $\sum_{\gamma>0} \gamma^{-2} = 0.0231$ these arguments add up to less than 0.99.

The first coefficient has the closed form $\lambda_1 = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, where γ_E is Euler's constant. The Pólya computation gives

$$\lambda_1 = 0.023095708966121033814311025,$$

in agreement with the closed form to all twenty five digits shown. The same number $\lambda_1 = \sum_\rho \rho^{-1} = (\xi'/\xi)(1)$ reappears at the basepoint one (Section 9.1.2): the shifted clock T_1 there, with Laplace transform $\xi(1)/\xi(1+\sqrt{u}) = 1 - \lambda_1\sqrt{u} + O(u)$ near $u = 0$, has a tail of order $x^{-1/2}$, and λ_1/π is its Thorin density at the origin in the chart $\theta = \sqrt{z}$. Table 8.6 lists further values with Keiper's asymptotic $\lambda_n \approx \frac{n}{2}(\log n - 1 - \log 2\pi + \gamma_E)$ [143].

n	λ_n (Pólya)	Keiper asymptotic
1	0.023095708966121033814	−1.13
2	0.092345735228046670386	−1.57
3	0.207638920554324803 79	−1.74
5	0.575542714461177452 43	−1.63
10	2.279339363193157743 7	0.21
20	8.769276872093215129 6	7.35
50	43.531096488374019299	41.28
100	118.60377537679132993	117.23

Table 8.6: Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large n.

8.3.2 Li's criterion as positivity of the Thorin functional

Li's criterion is a statement about the Thorin functional U of Section 5.3, which has a unit atom at $-(\rho - \frac{1}{2})^2$ for each pair $\{\rho, 1-\rho\}$ of zeros, on the line or off it. For a zero ρ put $q = 1 - 1/\rho$. Then $1 - 1/(1-\rho) = 1/q$, so the pair contributes $2 - q^n - q^{-n}$ to λ_n; and if $a = -(\rho - \frac{1}{2})^2$, with the square root chosen so that $\rho - \frac{1}{2} = i\sqrt{a}$, then $q = (2\sqrt{a} + i)/(2\sqrt{a} - i)$. This proves the following.

Proposition 8.3.1. *Without any hypothesis on the zeros,*

$$\lambda_n = \sum_{\{\rho,1-\rho\}} \varphi_n\Big(-\big(\rho-\tfrac12\big)^2\Big) = \langle U, \varphi_n\rangle,$$

where

$$\varphi_n(z) = 2 - \Big(\frac{2\sqrt{z}+i}{2\sqrt{z}-i}\Big)^n - \Big(\frac{2\sqrt{z}-i}{2\sqrt{z}+i}\Big)^n$$

is a rational function of z, and $\varphi_n(z) = 4\sin^2\big(n\arctan\frac{1}{2\sqrt{z}}\big) \ge 0$ for $z > 0$.

The two fractions are exchanged when $\sqrt{z}$ changes sign, so φ_n is a rational function of z, with a pole of order n at $z = -\frac14$. At infinity $\varphi_n(z) \sim n^2/z$, so the sum converges absolutely.

For $z > 0$ the two fractions are conjugate numbers of modulus one. On the critical line, for $\rho = \frac12 + i\gamma$ with $\gamma > 0$, the number $1 - 1/\rho = (\rho-1)/\rho$ has modulus one and argument $\phi_\gamma = 2\arctan\frac{1}{2\gamma}$, and the conjugate zero gives $-\phi_\gamma$. Adding the contributions of ρ and $\bar\rho$,

$$\lambda_n = \sum_{\gamma>0} 2\big(1-\cos(n\phi_\gamma)\big) = \sum_{\gamma>0} 4\sin^2\Big(n\arctan\frac{1}{2\gamma}\Big) \qquad \text{under RH.}$$

Li's criterion thus says that RH holds if and only if $\langle U, \varphi_n\rangle \ge 0$ for every n, that is, the Thorin functional is nonnegative on the cone spanned by the φ_n.

The test function φ_n is about n^2/z for $\sqrt{z} \gg n$ and oscillates between 0 and 4 for $\sqrt{z} \lesssim n/\pi$, so λ_n weighs the zeros below height of order n at full size and the higher zeros by about n^2/γ^2 (Figure 8.3). Every computed λ_n is positive, as it must be: every zero below height 3×10^{12} lies on the line [247] and contributes a nonnegative term, while a zero at height $\gamma \gg n$ contributes only about n^2/γ^2 in absolute value, so the zeros above that height cannot change the sign of λ_n for $n \le 100$.

An off line zero makes $|1-1/\rho| \ne 1$. For $\rho = \frac12 - d + i\gamma$ with $d > 0$ one has $|1-1/\rho|^2 = 1 + 2d/((\frac12-d)^2+\gamma^2)$, so for large n the term $(1-1/\rho)^n$ grows like e^{nd/γ^2}, and this is how a failure of RH would eventually make λ_n negative (Exercise 8.14). Since the exponent is nd/γ^2, the index at which this happens grows with the height of the zero, so finitely many coefficients, like the finite tests of Section 5.2 (Table 5.5), cannot exclude an off line zero at great height.

Summing $4\sin^2(n\arctan\frac{1}{2\gamma})$ over the first 1700 zeros, computed with `mpmath` [205] as in Section 3.2, and adding the tail above γ_{1700} with the smooth zero density $\log(\gamma/2\pi)/2\pi$ gives the Pólya values with relative residuals near 5×10^{-6} for $n \le 20$ and 10^{-5} at $n = 100$. The absolute residual grows like n^2 and comes from the cutoff: at γ_{1700} the zero count exceeds its smooth part by about one half, so the smooth tail counts about half a term, $\frac12\varphi_n(\gamma_{1700}^2) \approx n^2/(2\gamma_{1700}^2)$, too many.

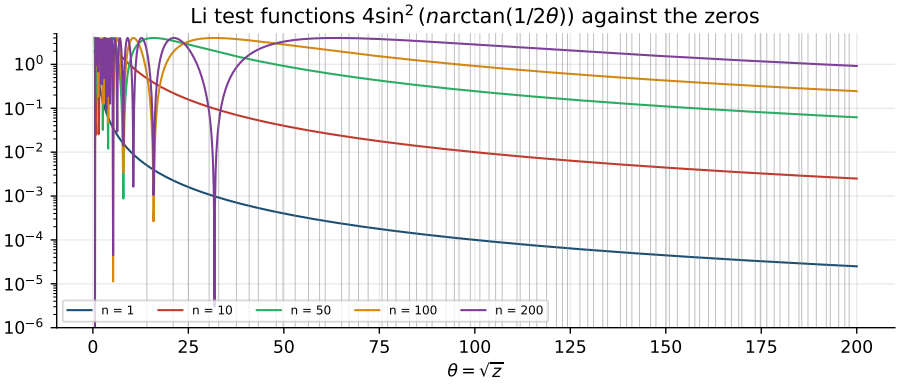


Figure 8.3: The Li test functions $4\sin^2(n \arctan \frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger n probes higher zeros.

As a diagnostic the coefficients are poorly localised, so λ_n behaves like the heat trace and Bernstein diagnostics of Table 1.7 and unlike the Thorin density, which is local in the ordinate.

Weil's criterion has the same form as Li's, with the functions of the class $\mathcal{T}$ of Section 5.3 that are nonnegative on $(0, \infty)$ in place of the φ_n (Section 3.5.2), but the two cones are different. The pole of φ_n lies at $-\frac{1}{4}$, the point at which (5.3.1) evaluates its test functions, inside the region where $\mathcal{T}$ requires holomorphy, and φ_n decays only like $1/z$. So φ_n is not in $\mathcal{T}$, $\langle U, \varphi_n \rangle$ is defined through the zeros, and the prime side of λ_n is the regularised arithmetic formula of Bombieri and Lagarias [37].

8.4 The de Bruijn Newman flow from the Pólya density

The de Bruijn Newman flow reads RH from a deformation of the Pólya kernel. Multiplying the kernel by a Gaussian factor turns Ξ into a family of entire functions H_t that evolve by the backward heat equation. We follow two pairs of zeros backward in time to their collisions and compare the collision times with the isolated pair estimate and with the lower bound of Csordas, Smith and Varga.

8.4.1 The flow, the motion of its zeros and its Thorin form

In the normalisation of Rodgers and Tao [289], which Theorem 7.2.15 also uses,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du,$$
$$\Phi_{\mathrm{dBN}}(u) = \sum_{n\geq 1}\left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right)\exp(-\pi n^2 e^{4u}),$$

so that $H_0(z) = \frac{1}{8}\Xi(z/2)$. If $v = 2u$ denotes the variable of (1.2.3), then $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(v)$ and

$$8H_t(2\theta) = \int_{\mathbb{R}} \Phi(v)\, e^{tv^2/4}\cos(\theta v)\,dv,$$

so the flow multiplies the kernel of (1.2.3) by $e^{tv^2/4}$, and a zero z of H_t corresponds to the ordinate $\theta = z/2$.

De Bruijn showed that H_t has only real zeros for $t \geq \frac{1}{2}$ and that the property persists forward in time [73]. Newman showed that there is a constant $\Lambda > -\infty$, at most $\frac{1}{2}$ by de Bruijn's theorem, such that H_t has only real zeros exactly for $t \geq \Lambda$, and conjectured $\Lambda \geq 0$ [210], which Rodgers and Tao proved [289]. Ki, Kim and Lee sharpened de Bruijn's bound to $\Lambda < \frac{1}{2}$ [150], the Polymath project proved $\Lambda \leq 0.22$ [282], and Platt and Trudgian improved this to $\Lambda \leq 0.2$ [247]. De Bruijn's bound comes from a theorem about strips: if the zeros of H_0 lie in $|\operatorname{Im} z| \leq \Delta$, then H_t has only real zeros for $t \geq \Delta^2/2$ [73, Theorem 13], and the critical strip gives $\Delta = 1$. A zero free half plane $\operatorname{Re} s > \alpha$ gives $\Delta = 2\alpha - 1$, hence $\Lambda \leq 2(\alpha - \frac{1}{2})^2$, which improves 0.2 only if $\alpha < \frac{1}{2} + \sqrt{0.1} \approx 0.816$. RH is the inequality $\Lambda \leq 0$, so by the theorem of Rodgers and Tao it is equivalent to $\Lambda = 0$.

While the zeros are simple they move according to

$$\frac{dz_k}{dt} = 2\sum_{j\neq k}\frac{1}{z_k - z_j},$$

the sum running over all other zeros, positive and negative, with z_j and $-z_j$ taken together. Differentiating $H_t(z_k(t)) = 0$ in t and using the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ gives $dz_k/dt = \partial_z^2 H_t/\partial_z H_t$ at z_k, and the Hadamard product of the even function H_t turns this into the display (Exercise 8.15). The equation is the gradient flow of unit charges on the line with logarithmic interaction, so forward in time the zeros repel each other, spread and become more regular. Backward in time neighbouring zeros approach each other, and a close pair collides and then leaves the real axis as a conjugate pair. A pair at distance D that is isolated from the other zeros obeys $dD/dt \approx 4/D$, so $D^2 \approx D_0^2 + 8t$ with $D_0 = D(0)$, and the pair collides at $t \approx -D_0^2/8$. RH is the statement that no two zeros collide at a positive time.

Let $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$, where u is the variable of the clock (written s in Chapters 6 and 7), not the variable of integration in H_t, so that R_0 is the clock

$1/G$. By Theorem 7.2.15, for every $t \geq \Lambda$, and so for every $t \geq 0.2$, R_t is the Laplace transform of a GGC whose Thorin atoms move with the zeros of H_t. Put $\xi_t(s) = 8H_t\big(-2i(s-\tfrac{1}{2})\big)$, so that $\xi_t(\tfrac{1}{2}+i\theta) = 8H_t(2\theta)$ and $\xi_0 = \xi$, and

$$\rho_{\varepsilon,t}(\theta) = \pi^{-1}\,\mathrm{Re}(\xi_t'/\xi_t)\big(\tfrac{1}{2}+\varepsilon+i\theta\big).$$

Each zero z of H_t gives the zero $\frac{1}{2}+iz/2$ of ξ_t, and for every t the function $\rho_{\varepsilon,t}$ is the sum of the Poisson kernels $P_{\frac{1}{2}+\varepsilon-\beta}(\theta-\gamma)$ over the zeros $\beta+i\gamma$ of ξ_t. For $t \geq \Lambda$ every β equals $\frac{1}{2}$ and $\rho_{\varepsilon,t}$ is the Thorin density of R_t at resolution ε. As t decreases through a collision, $\rho_{\varepsilon,t}$ shows the off line signature of Section 3.3: two Poisson peaks merge, and after the collision the pair contributes the kernels $P_{\varepsilon+d(t)} + P_{\varepsilon-d(t)}$ at its ordinate, where $d(t)$, half the imaginary part of the pair in z, grows from zero; the second kernel is negative once $d(t) > \varepsilon$. The flow thus produces off line configurations of known geometry, on which the diagnostics of Chapters 3 to 5 can be calibrated; Section 12.2.3 lists this calibration among the open computations.

8.4.2 Two collisions computed from the Pólya kernel

The Pólya route evaluates H_t at any t by multiplying the node values $\Phi(v_j)$ by $e^{tv_j^2/4}$ and summing. The factor is entire, so the error analysis of Section 2.1 carries over. Among the 79 zeros below height 200 the closest pair is $\gamma_{71} = 184.8745$, $\gamma_{72} = 185.5988$, at distance 0.7243 in θ, that is $D_0 = 1.4486$ in z. We follow the pair by the sign changes of $H_t(2\theta)$ on a grid and refine each zero by Newton's method (Table 8.7).

t	zeros in $\theta = z/2$	sign changes
0.2	184.676, 185.590	2
0.1	184.771, 185.598	2
0.0	184.874, 185.599	2
−0.05	184.931, 185.595	2
−0.1	184.992, 185.587	2
−0.15	185.059, 185.573	2
−0.2	185.136, 185.550	2
−0.3	none	0

Table 8.7: The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; t is the de Bruijn Newman time.

Bisection on t for the disappearance of the two sign changes gives the collision time

$$t_* = -0.2880,$$

with a grid error of at most 6.1×10^{-5}. Near a collision the term $4/D$ dominates the motion of the pair, so $D^2 \approx 8(t - t_*)$ and the gap in θ is about $\sqrt{2(t - t_*)}$; on a grid of step h the sign changes therefore vanish at a time at most $h^2/2$ above t_*. The isolated pair estimate is $-D_0^2/8 = -0.262$: the neighbouring zeros, which the estimate ignores, delay the collision. Since nonreal zeros persist backward in time, H_t has nonreal zeros for every t below the collision time, so $\Lambda \geq t_* \approx -0.288$.

Lehmer's pair near height 7005, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ (Section 2.2.3.1), is much closer: $D_0 = 2 \times 0.0377 = 0.0754$, and its isolated pair estimate is $-D_0^2/8 = -7.1 \times 10^{-4}$. The tilted Pólya route evaluates H_t at that height at the same cost as Ξ. With the parameters of Section 2.2.3.1 we track the two sign changes of the pair on a grid of step 0.0025 in θ over $[7005.03, 7005.13]$. Both sign changes are present at $t = 0$ and absent at $t = -0.004$, and twelve bisection steps on this interval give -7.10×10^{-4}, which by the argument above lies above the collision time by at most $0.0025^2/2 \approx 3 \times 10^{-6}$. Newton's method for the double zero, $H_t(2\theta) = \partial_\theta H_t(2\theta) = 0$ in the unknowns θ and t, started from this value, gives

$$t_* = -7.11 \times 10^{-4}$$

at $\theta = 7005.0817$. Here the isolated pair estimate is accurate: for a pair this close the neighbours hardly matter. The bound $\Lambda \geq t_*$ is much sharper than the one from the pair below height 200.

Pólya's function $K_{i\theta/2}(2\pi)$ behaves differently under the same flow. Down to $t = -60$ none of its real zeros below height 90 collide, and as t decreases from 0 to -0.5 they rise almost as a rigid block, at speeds within 2% of the mean speed $\pi/8$ that the two functions share, while the zeros of Ξ rise at speeds between 0.004 and 0.735; yet by Newman's theorem the de Bruijn Newman constant of Pólya's function is finite (Section 8.7.6).

8.4.3 The Hankel test along the flow

The Stieltjes moment test of Section 5.1 (Table 5.2) needs only the Taylor coefficients of Ξ at the centre, and along the flow these are moments of the reweighted kernel [261]. Putting $\theta = i\sqrt{u}$ in the formula for $8H_t(2\theta)$ of Section 8.4.1 gives

$$G_t(u) = \frac{H_t(2i\sqrt{u})}{H_t(0)} = \sum_{k \geq 0} \frac{M_k(t)\, u^k}{(2k)!\, M_0(t)}, \qquad M_k(t) = \int_{\mathbb{R}} v^{2k} e^{tv^2/4} \Phi(v)\, dv,$$

and G_0 is the function G of Section 5.1. Write $\pm\theta_j(t)$ for the zeros of $\theta \mapsto H_t(2\theta)$, so that $\theta_j(0) = \gamma_j$ while the zeros are on the line. Since H_t is even and entire of order one, G_t has order $\frac{1}{2}$ and $G_t(u) = \prod_j (1 + u/a_j(t))$ with $a_j(t) = \theta_j(t)^2$, so the Newton identities turn the $M_k(t)$ into the moments $m_n(t) = \sum_j a_j(t)^{-n-1}$, exactly as at $t = 0$. If every $\theta_j(t)$ is real, the $m_n(t)$ are the moments of the positive measure $\nu_t = \sum_j a_j(t)^{-1} \delta_{1/a_j(t)}$, and every Hankel matrix $\mathcal{H}_N(t) = [m_{i+j}(t)]_{i,j<N}$

is positive definite. The converse is proved as at $t = 0$: if every $\mathcal{H}_N(t)$ is positive definite, every $a_j(t)$ is real, and it is positive because H_t has no zeros on the imaginary axis: $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, the kernel (1.2.3) being strictly positive. So at every time t all the matrices $\mathcal{H}_N(t)$ are positive definite exactly when H_t has only real zeros. The Cholesky factor gives the Jacobi matrix $J_N(t)$, whose eigenvalues return the lowest zeros of H_t as J_N returns the γ_j at $t = 0$ (Table 8.8).

t	$\theta_1(t)$	$\theta_2(t)$	$\theta_3(t)$	$\theta_5(t)$	$\theta_{10}(t)$
0.2	14.07681	20.94669	24.95221	32.88801	49.72988
0	14.13473	21.02204	25.01086	32.93506	49.77383
−0.4	14.25083	21.17365	25.12774	33.02595	49.85213
−1.0	14.42563	21.40341	25.30189	33.15277	49.93350

Table 8.8: Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \dots, m_{59}(t)$; at $t = 0$ they are γ_1, γ_2, γ_3, γ_5 and γ_{10}. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \dots, \theta_{12}(t)$ to within 3.1×10^{-7} of the zeros that Newton's method finds on H_t, and $\theta_{13}(t)$ only to within 5.2×10^{-4}.

Going back from $t = 0$, the matrix $\mathcal{H}_N(t)$ stays positive definite until a time t_N^* at which it ceases to be. Since $\mathcal{H}_N(t)$ is the leading block of $\mathcal{H}_{N+1}(t)$, $t_N^* \le t_{N+1}^*$. Under RH every t_N^* is negative, because $\mathcal{H}_N(0)$ is positive definite, and $t_N^* \to 0$: for each $t < 0$ the function H_t has nonreal zeros by the theorem of Rodgers and Tao, so some $\mathcal{H}_N(t)$ is not positive definite, and then $t_{N'}^* \ge t$ for every $N' \ge N$. With the trapezoid rule of step $1/600$ at $100 + 16N$ digits, the grid $t = 0, -0.05, \dots, -4$ and bisection to 10^{-7} give no failure of order 10 down to $t = -4$, and

$$t_{20}^* = -1.8991865, \qquad t_{30}^* = -1.2853067,$$
$$t_{40}^* = -1.2760788, \qquad t_{50}^* = -1.0325048;$$

for $N = 30$ a second discretisation (step $1/550$, 650 digits) gives a bracket that overlaps the first in $[-1.28530669, -1.28530664]$.

Each failure registers a collision. We follow the real zeros of H_t that start at $2\gamma_j$, $j \le 64$, back to $t = -2.05$ by the equation of motion of Section 8.4.1, on the trapezoid sum for H_t, and locate each collision as a double zero, $H_t(z) = \partial_z H_t(z) = 0$, by Newton's method in the two unknowns z and t; the backward heat equation, which holds exactly for the trapezoid sum, gives the derivatives in t. Thirteen of the pairs below index 59 collide in this time range, among them those that start at (γ_k, γ_{k+1}) for $k = 9$, 13, 19, 24, 27 and 34, at

$$t = -1.87791,\ -1.28527,\ -1.27554,\ -1.14387,\ -0.88108,\ -0.38456$$

and the heights $\theta = 49.664$, 60.491, 77.027, 88.466, 95.540 and 111.638; the other seven start at $k = 40$, 43, 45, 48, 51, 53 and 57. To see which collision a failure

registers, we continue each double zero into the complex plane, starting from $z_c \pm i\sqrt{2(t_c - t)}$ near the collision, and remove the contribution $2\,\mathrm{Re}\,\theta^{-2n-2}$ of the conjugate pair $\theta = z/2$, $\bar\theta$ from $m_n(t)$. At $t_N^* - \delta$, with $\delta = 0.01$ (for $N = 40$, $\delta = 0.0046$, half the time to the collision of $(\gamma_{13}, \gamma_{14})$), the only one of the thirteen pairs whose removal makes $\mathcal{H}_N(t)$ positive definite again is the one that starts at (γ_9, γ_{10}) for $N = 20$, at $(\gamma_{13}, \gamma_{14})$ for $N = 30$, at $(\gamma_{19}, \gamma_{20})$ for $N = 40$ and at $(\gamma_{27}, \gamma_{28})$ for $N = 50$. The same holds at $t_N^* - 10^{-5}$, except that there removing $(\gamma_{34}, \gamma_{35})$ also restores the order 50 matrix. The failures thus come 0.021, 3.7×10^{-5}, 5.3×10^{-4} and 0.151 after the collisions they register.

A Hankel test of finite order is thus a window on the low part of the spectrum. The order 30 matrix is positive definite at $t = -1.276, -1.278, \ldots, -1.284$ and -1.2852, after the collision of $(\gamma_{19}, \gamma_{20})$ at -1.27554 and before that of $(\gamma_{13}, \gamma_{14})$ at -1.28527: it passes the pair that is too high and too light for its order and registers the lower one. A conjugate pair at height θ enters $m_n(t)$ with weight of order θ^{-2n-2}, against $\theta_1(t)^{-2n-2}$ for the lowest zero, as at $t = 0$, and increasing the order widens the window.

The whole curve $N \mapsto t_N^*$ up to $N = 50$ follows from one elimination per time. Let $n_{\mathrm{fail}}(t)$ be the index of the first nonpositive pivot of Gaussian elimination on $\mathcal{H}_{50}(t)$, that is, the least $N \le 50$ for which $\mathcal{H}_N(t)$ is not positive definite, so that $t_N^* = \sup\{t < 0 : n_{\mathrm{fail}}(t) \le N\}$. On the grid of step 0.01 it is a staircase [261] (Table 8.9). To see what each step registers, the table gives, at one time on each step, the least order at which each complex pair breaks the test alone, that is, with the contributions of all the other complex pairs removed from the moments. A pair alone breaks the test at a definite order, a higher pair at a higher order, and n_{fail} is close to the least of these orders. The order at which a pair is seen falls as the pair moves away from the axis: $(\gamma_{27}, \gamma_{28})$, which collides at -0.881, breaks order 50 at $t = -1.09$ and order 39 at -1.29, and $(\gamma_{34}, \gamma_{35})$, which collides earlier, at -0.385, but lies higher, breaks the test only from $t = -1.28$ on, at order 50 and then 47. Removing the pair with the least order moves n_{fail} up to about the next one: at $t = -1.30$ from 27 to 35, when $(\gamma_{13}, \gamma_{14})$ is removed, at -1.28 from 39 to 41, when $(\gamma_{19}, \gamma_{20})$ is removed, and at -1.09 from 50 to no failure, when $(\gamma_{27}, \gamma_{28})$ is removed. With all thirteen pairs removed, no order up to 50 fails at any of these times.

A rule suggested in [261] is that the collision of (γ_k, γ_{k+1}) is registered from order about $2k + 2$ on, so that $t_N^* \approx \max\{t_{\mathrm{coll}}(k) : 2k + 2 \le N\}$, where $t_{\mathrm{coll}}(k)$ is the collision time of the pair. It holds for $N = 30$ and 40 ($k = 13$ and 19), roughly for $N = 20$ ($k = 9$, delay 0.021), and fails at $N = 50$: it gives the collision of $(\gamma_{24}, \gamma_{25})$ at -1.144, while the order 50 matrix registers $(\gamma_{27}, \gamma_{28})$, for which $2k + 2 = 56$, at -1.0325, before $(\gamma_{24}, \gamma_{25})$ has collided. Table 8.9 shows why: a pair beyond the window of the rule enters it after a delay, and t_N^* is the latest of the collision times of the pairs inside the window and of the times at which pairs above it have moved far enough from the axis.

			order broken by (γ_k, γ_{k+1}) alone				
t	$n_{\mathrm{fail}}(t)$	at $t =$	$k = 13$	19	24	27	34
$[-1.03, 0]$	none						
$[-1.14, -1.04]$	50	-1.09	real	real	real	50	> 50
$[-1.18, -1.15]$	46	-1.16	real	real	44	45	> 50
$[-1.27, -1.19]$	45	-1.23	real	real	44	45	> 50
-1.28	39	-1.28	real	38	40	41	50
-1.29	28	-1.29	28	35	37	39	47
-1.30	27	-1.30	27	35	37	39	47

Table 8.9: The first failing order $n_{\mathrm{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01 and, at one time on each step, the least order at which the pair that starts at (γ_k, γ_{k+1}) breaks the test alone (real: the pair has not yet collided; > 50: no order up to 50 fails). The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone.

The curve rises to 0 through the collision times of ever closer pairs. Following every real zero that starts below index 215 in the same way (200 digits, step 1/200), 33 of the first 120 gaps close on $[-2.05, 0]$, 28 of them after $t = -1.6$, and 50 of the first 200 gaps close after $t = -1.6$. The latest collisions among the first 200 gaps are those of $(\gamma_{186}, \gamma_{187})$, at distance 0.498, at $t = -0.1310$, of $(\gamma_{159}, \gamma_{160})$ at -0.1695 and of $(\gamma_{135}, \gamma_{136})$ at -0.2015, and below height 230 those of $(\gamma_{91}, \gamma_{92})$ at -0.2821 and of the closest pair below height 200, $(\gamma_{71}, \gamma_{72})$, at -0.2880, the value of Section 8.4.2. A collision close to $t = 0$ needs a pair as close as Lehmer's (Section 8.4.2), far beyond the reach of a Hankel test at the centre.

Further back in time the low pairs leave the axis one after another, and Theorem 5.1.7 turns the test into a count. Its proof applies at every time to the zeros of H_t, none of which is purely imaginary: the numbers $\iota_N(t)$ and $\iota'_N(t)$ of negative eigenvalues of $\mathcal{H}_N(t)$ and of $\mathcal{H}'_N(t) = [m_{i+j+1}(t)]_{i,j<N}$ do not decrease with N, never exceed the number of distinct conjugate pairs $\theta, \bar\theta$ of zeros with $\mathrm{Re}\,\theta > 0$, and tend to it, so that each negative eigenvalue certifies a pair. The signs of the pivots of one Gaussian elimination give $\iota_1(t), \dots, \iota_{30}(t)$, and $n'_{\mathrm{fail}}(t)$ is to $\mathcal{H}'_N(t)$ what $n_{\mathrm{fail}}(t)$ is to $\mathcal{H}_N(t)$. We follow the real zeros below $\theta = 88$, those that start at $\gamma_1, \dots, \gamma_{24}$, back to $t = -20$ and locate each collision as a double zero as above. Below $\theta = 88$ the collisions found above are those of (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$, and after them come those of $(\gamma_{22}, \gamma_{23})$, $(\gamma_{15}, \gamma_{16})$, (γ_4, γ_5), (γ_7, γ_8), $(\gamma_{17}, \gamma_{18})$ and (γ_2, γ_3), at

$$t = -2.32414,\ -2.75904,\ -3.95479,\ -4.03382,\ -7.31401,\ -12.49451$$

and the heights $\theta = 84.875$, 67.406, 33.099, 43.403, 72.746 and 27.321. Each pair is followed into the complex plane as above (Table 8.10). Removing the contribution

of one pair from the moments shows what the first failure registers: at $t = -2$ and -3 the pair that starts at (γ_9, γ_{10}), whose removal moves n_{fail} from 19 to 21 and from 18 to 21, and from $t = -4$ to -12 the pair that starts at (γ_4, γ_5), whose removal raises n_{fail} by two or three, from 11 to 14 at $t = -4$; at these times the removal of any other pair that starts below $\theta = 88$ leaves n_{fail} unchanged. Both pairs are registered soon after their collisions, at the orders 19 and 11, close to $2k + 2 = 20$ and 10. At $t = -2$ the three pairs that start at (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$ account for every failure up to order 30: with them removed none remains, and alone they break the test at the orders 19, 21 and 30. From $t = -3$ on, pairs that start above $\theta = 88$ enter as well, since with all the pairs that start below $\theta = 88$ removed the test still fails, from order 24 at $t = -3$, 21 at -4, 15 at -8 and 11 at -15 and -20. At $t = -15$ and -20 the lowest pair is no longer the first to be seen: no single removal raises n_{fail}, and the pair that starts at (γ_4, γ_5) alone breaks the test as early as the lowest pair alone or earlier, at order 4 against 5 at $t = -20$. The two discretisations of the moments, step 1/300 at 600 digits and step 1/450 at 900 digits, give the same counts and the same removal tests, and at the orders 10 and 30 the eigenvalues of the diagonally scaled matrices give the same counts as the pivots.

t	pair	θ	real	collided	n_{fail}	n'_{fail}	$(\iota_{10}, \iota'_{10})$	$(\iota_{30}, \iota'_{30})$
-2	$(9, 10)$	$49.715 \pm 0.249i$	8	3	19	19	$(0, 0)$	$(2, 2)$
-3	$(9, 10)$	$50.139 \pm 0.792i$	8	5	18	17	$(0, 0)$	$(3, 3)$
-4	$(4, 5)$	$33.115 \pm 0.151i$	3	6	11	10	$(0, 1)$	$(5, 5)$
-5	$(4, 5)$	$33.478 \pm 0.745i$	3	7	9	9	$(1, 1)$	$(6, 6)$
-6	$(4, 5)$	$33.841 \pm 1.071i$	3	7	9	8	$(1, 1)$	$(6, 6)$
-8	$(4, 5)$	$34.572 \pm 1.592i$	3	8	8	8	$(1, 1)$	$(6, 6)$
-10	$(4, 5)$	$35.306 \pm 2.052i$	3	8	8	8	$(1, 1)$	$(6, 7)$
-12	$(4, 5)$	$36.043 \pm 2.492i$	3	8	8	8	$(1, 1)$	$(7, 7)$
-15	$(2, 3)$	$28.202 \pm 1.166i$	1	9	6	5	$(2, 3)$	$(8, 8)$
-20	$(2, 3)$	$29.967 \pm 2.178i$	1	9	5	5	$(3, 3)$	$(8, 8)$

Table 8.10: The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k + 1)$ by the zeros γ_k, γ_{k+1} it starts from, and its position θ; the number of real zeros below it; the number of collisions so far among $\gamma_1, \ldots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\text{fail}}(t)$ and $n'_{\text{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30.

The programs are `jacobi_dbn.py` (the moments and the collisions below index 59), `dbn_collisions.py` (the census of the first 200 gaps) and `dbn_hankel_long.py` (the collisions and counts down to $t = -20$).

8.4.4 Biane's theorem and Laguerre's quantity

The collision times of Section 8.4.3 can be predicted from H_0 alone, before the flow is followed, through a quantity that free probability singles out. The material comes from Section 8 of the first author's note [261], except Corollaries 8.4.12 and 8.4.14 and the halved constant of Remark 8.4.10, which are not in the note; the numbers are recomputed here from H_t itself.

Let μ be a probability measure on $\mathbb{R}$, $G_\mu(z) = \int \mu(dx)/(z-x)$ its Cauchy transform, and σ_t the semicircle law of variance $t > 0$, with density $(2\pi t)^{-1}(4t - x^2)^{1/2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$. The free convolution $\mu \boxplus \sigma_t$ is the law of $A + S$ for free random variables A and S with laws μ and σ_t. Put

$$F_\mu(u) = \int \frac{\mu(dx)}{(u-x)^2} \in (0,\infty], \qquad v_t(u) = \inf\Big\{v \geq 0 : \int \frac{\mu(dx)}{(u-x)^2 + v^2} \leq \frac{1}{t}\Big\},$$
$$\psi_t(u) = u + t\int \frac{(u-x)\,\mu(dx)}{(u-x)^2 + v_t(u)^2}.$$

Theorem 8.4.1 (Biane [30]). *Let μ be a probability measure on $\mathbb{R}$ and $t > 0$.*

(a) ψ_t is a homeomorphism of $\mathbb{R}$, and $\mu \boxplus \sigma_t$ has a continuous density p_t with $p_t(\psi_t(u)) = v_t(u)/(\pi t)$ for every real u. For $\operatorname{Im} z > 0$, $G_{\mu\boxplus\sigma_t}(z) = G_\mu(\omega_t(z))$, where $\omega_t(z) = z - tG_{\mu\boxplus\sigma_t}(z)$, and ω_t is the inverse of the map $w \mapsto w + tG_\mu(w)$ on the region $\{u + iv : v > v_t(u)\}$, which that map sends onto the upper half plane.

(b) Let $U_t = \{u : F_\mu(u) > 1/t\}$. The interior of the support of $\mu \boxplus \sigma_t$ is $\psi_t(U_t)$, the support is its closure, the density is real analytic on $\psi_t(U_t)$, and the number of components of U_t does not increase with t.

(c) Let (a,b) be a bounded component of $\mathbb{R}\setminus\operatorname{supp}\mu$ with $\mu(\{a\}) > 0$ and $\mu(\{b\}) > 0$. Then F_μ is strictly convex on (a,b) and tends to $+\infty$ at both ends; let u_0 be its minimiser and

$$t_0 = \frac{1}{F_\mu(u_0)} = \Big(\min_{a<u<b} \int \frac{\mu(dx)}{(u-x)^2}\Big)^{-1}. \tag{8.4.1}$$

For $t < t_0$ the set $\{u \in (a,b) : F_\mu(u) \leq 1/t\}$ is an interval $[a(t), b(t)]$ with $a(t) < b(t)$, and ψ_t maps $(a(t), b(t))$ onto a component of the complement of the support of $\mu \boxplus \sigma_t$. As t increases to t_0, $a(t)$ increases and $b(t)$ decreases to u_0, and at t_0 the gap closes.

Proof. Part (a) is Lemma 4, Proposition 2 and Corollaries 2 and 3 of [30], where σ_t has R transform tz as here; part (b) is Proposition 3, its proof, and Corollary 4 there. For (c), μ does not charge (a,b), so $F_\mu''(u) = 6\int (u-x)^{-4}\mu(dx) > 0$ there, and the atoms make F_μ tend to $+\infty$ at a and b. The sublevel sets of a strictly convex function are intervals; $\{F_\mu \leq 1/t\} \cap (a,b)$ contains u_0 in its interior when

$1/t > F_\mu(u_0)$, shrinks as t grows, and reduces to u_0 at $t = t_0$. Its interior is a component of $\mathbb{R} \setminus \overline{U_t}$, because the points of (a, b) just outside it lie in U_t, and ψ_t, a homeomorphism, maps it onto a component of the complement of $\overline{\psi_t(U_t)}$, which is the support by (b). This is the discussion following Corollary 5 of [30]. □

The time (8.4.1) is Biane's threshold. For a real entire function f write

$$\mathcal{L}_f(x) = -\frac{d^2}{dx^2}\log|f(x)| = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2},$$

Laguerre's quantity. If f has genus at most one and zeros z_j, then $\mathcal{L}_f(x) = \sum_j (x - z_j)^{-2}$. For the normalised counting measure $\mu_p = n^{-1}\sum_j \delta_{z_j}$ of the zeros of a polynomial p of degree n with only real simple zeros, $F_{\mu_p} = \mathcal{L}_p/n$, and by Theorem 8.4.1(c) the gap (z_k, z_{k+1}) survives in $\mu_p \boxplus \sigma_t$ until $t_0 = n/\min_{(z_k, z_{k+1})} \mathcal{L}_p$.

For Ξ, whose zeros are $\pm\gamma_j$, and for $H_0(z) = \frac{1}{8}\Xi(z/2)$,

$$\mathcal{L}_{H_0}(z) = \tfrac{1}{4}\mathcal{L}_\Xi(z/2), \qquad \mathcal{L}_\Xi(x) = \sum_j \Big[\frac{1}{(x-\gamma_j)^2} + \frac{1}{(x+\gamma_j)^2}\Big] = \sum_j \frac{2(x^2+\gamma_j^2)}{(x^2-\gamma_j^2)^2}.$$

Both are computed from Ξ and its first two derivatives, without the zeros. Since $\Xi(x) = \Xi(0)G(-x^2)$, with $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.1,

$$\mathcal{L}_\Xi(x) = 2\frac{G'}{G}(-x^2) - 4x^2\Big(\frac{G'}{G}\Big)'(-x^2) = 2\frac{d}{dx}\Big(x\,\frac{G'}{G}(-x^2)\Big),$$
$$\frac{G'}{G}(u) = \sum_j \frac{1}{u + \gamma_j^2},$$

and G'/G is the Stieltjes transform of the Thorin measure $\sum_j \delta_{\gamma_j^2}$ of the clock $1/G$ (under RH; the identities hold in general with γ_j^2 replaced by $a_j = -(\rho_j - \frac{1}{2})^2$).

Lemma 8.4.2. *Let p be a real polynomial of degree $n \geq 2$ with only real simple zeros and critical points $c_1 < \cdots < c_{n-1}$, and let s be real with $s\,\mathcal{L}_p(c_i) < 1$ for every i. Then $p + sp''$ has only real simple zeros, one in each of the intervals $(-\infty, c_1)$, (c_1, c_2), …, (c_{n-1}, ∞).*

Proof. At a critical point $p'(c_i) = 0$, so $p''(c_i)/p(c_i) = (\log|p|)''(c_i) = -\mathcal{L}_p(c_i)$, and $(p + sp'')(c_i) = p(c_i)\big(1 - s\mathcal{L}_p(c_i)\big)$ has the sign of $p(c_i)$. There is exactly one critical point between consecutive zeros of p, so the signs of $p(c_1), \ldots, p(c_{n-1})$ alternate. The polynomial $p + sp''$ has degree n and the sign of p at $\pm\infty$, which is opposite to the sign of $p(c_1)$ at $-\infty$ and to that of $p(c_{n-1})$ at $+\infty$, since p has one zero to the left of c_1 and one to the right of c_{n-1}. So $p + sp''$ changes sign in each of the n intervals and has n real zeros there. □

Since $e^{sD^2} = \lim_N (1 + \frac{s}{N} D^2)^N$, the lemma suggests, without proving it, that the gap (z_k, z_{k+1}) of H_t stays open while $|t| < 1/\mathcal{L}_{H_0}(c_k)$, where c_k is the critical point of H_0 in the gap. With $z_k = 2\gamma_k$ and u_k the minimiser of $\mathcal{L}_{H_0}$ over the gap, this gives two predictions of the collision time from H_0 alone, the critical point form and Biane's form,

$$\tau_c = -\frac{1}{\mathcal{L}_{H_0}(c_k)} = -\frac{4}{\mathcal{L}_{\Xi}(c_k/2)},$$
$$\tau_B = -\frac{1}{m(0)}, \qquad m(0) = \min_{(z_k, z_{k+1})} \mathcal{L}_{H_0} = \mathcal{L}_{H_0}(u_k).$$

For an isolated pair, $\mathcal{L} = 8/D_0^2$ at the midpoint of the gap $D_0 = z_{k+1} - z_k$, and both equal the isolated pair estimate $-D_0^2/8$ of Section 8.4.1, which is the exact collision time of $z^2 - a^2$ under e^{sD^2}, since $e^{sD^2}(z^2 - a^2) = z^2 - a^2 + 2s$. Section 8.4.5 turns both predictions into theorems for functions with only real zeros.

Remark 8.4.3. Biane's theorem and the collisions concern opposite directions of the heat flow. For polynomials of degree n the finite free convolution of Marcus, Spielman and Srivastava [199] is $p \boxplus_n q = \hat{q}(D)p$ when $q = \hat{q}(D)x^n$, so $e^{-aD^2}p = p \boxplus_n e^{-aD^2}x^n$. The polynomial $e^{-aD^2}x^n$ is a rescaled Hermite polynomial [199] and has zeros with mean square $2a(n-1)$, and for large n their distribution is close to the semicircle law σ_{2an}; in the limit finite free convolution becomes free convolution [199]. So $\mu \mapsto \mu \boxplus \sigma_t$ is the limit of the forward flow e^{-aD^2} with $a \approx t/(2n)$, that is of H_t with t increasing: the zeros repel, and a gap between two groups of zeros fills in as the groups spread, which is what Theorem 8.4.1(c) describes. Zeros collide only in the other direction, under e^{sD^2} with $s = -t > 0$. What carries over is the functional. Biane's threshold for μ_p is $n/\min \mathcal{L}_p$, and Theorem 8.4.8 below shows that under e^{sD^2} a gap of a function with only real zeros cannot close before $s = 1/\min \mathcal{L}$ while all the zeros stay real. For the pair $\pm a$ the two times are a^2 and $a^2/2$, and the factor 2 is the n of the normalisation of μ_p.

Table 8.11 compares the two predictions with the collision times of the census. Every collision of the census, all 58 among the first 200 gaps on $[-2.05, 0]$, comes later in the backward flow than both predictions and the isolated pair estimate: $t_{\text{coll}}/\tau_B \geq 1.083$, $t_{\text{coll}}/\tau_c \geq 1.104$ and $t_{\text{coll}}/(-D_0^2/8) \geq 1.056$, all three for $(186, 187)$, which collides at $t = -0.1310$. Over the 28 pairs of the table, t_{coll}/τ_B lies in $[1.114, 1.719]$, t_{coll}/τ_c in $[1.181, 1.951]$ and $t_{\text{coll}}/(-D_0^2/8)$ in $[1.077, 1.518]$, and $|\tau_B|$ falls short of $|t_{\text{coll}}|$ by 25.5% on average. The ratio t_{coll}/τ_B is close to 1 when the two zeros dominate their gap, and it is very nearly an increasing function of the share $6Q(0)/m(0)$ of the other zeros in Biane's functional (Section 8.4.5): ordered by this share, the ratios in the table rise with only small inversions, while the gap alone orders them much less well (compare $(114, 115)$ and $(13, 14)$ in the table). Real zeros outside the gap pull the two zeros apart while they fall together.

Over all 58 collisions t_{coll}/τ_B lies in $[1.083, 2.329]$, and the largest ratios, 2.042 to 2.329, belong to the five collisions with $6Q(0)/m(0) > 0.9$, all at $t < -1.89$.

pair	$\gamma_{k+1}-\gamma_k$	t_{coll}	$-D_0^2/8$	τ_c	τ_B	t_{coll}/τ_B	$6Q/m$	frozen
$(91,92)$	0.7158	−0.2821	−0.2562	−0.2369	−0.2455	1.149	0.250	−0.2826
$(71,72)$	0.7243	−0.2880	−0.2623	−0.2305	−0.2517	1.144	0.242	−0.2883
$(97,98)$	0.7370	−0.3040	−0.2716	−0.2454	−0.2587	1.175	0.286	−0.3047
$(63,64)$	0.8175	−0.3686	−0.3341	−0.3122	−0.3199	1.152	0.256	−0.3694
$(34,35)$	0.8451	−0.3846	−0.3571	−0.3115	−0.3452	1.114	0.200	−0.3852
$(53,54)$	0.8717	−0.4231	−0.3800	−0.3361	−0.3624	1.167	0.277	−0.4244
$(114,115)$	0.9307	−0.5380	−0.4331	−0.3933	−0.3997	1.346	0.462	−0.5364
$(93,94)$	0.9763	−0.5533	−0.4766	−0.4179	−0.4471	1.237	0.371	−0.5589
$(106,107)$	0.9650	−0.5691	−0.4656	−0.3620	−0.4306	1.321	0.451	−0.5724
$(75,76)$	1.0531	−0.6572	−0.5545	−0.4537	−0.5166	1.272	0.410	−0.6648
$(111,112)$	1.0745	−0.7102	−0.5772	−0.4715	−0.5310	1.338	0.481	−0.7240
$(117,118)$	1.0571	−0.7653	−0.5588	−0.4318	−0.5029	1.522	0.599	−0.7676
$(78,79)$	1.1388	−0.8235	−0.6485	−0.5695	−0.5930	1.389	0.513	−0.8318
$(27,28)$	1.2193	−0.8811	−0.7433	−0.6313	−0.6937	1.270	0.400	−0.8861
$(45,46)$	1.2588	−0.9600	−0.7923	−0.6731	−0.7325	1.311	0.452	−0.9751
$(82,83)$	1.2050	−0.9734	−0.7260	−0.6170	−0.6533	1.490	0.601	−0.9985
$(80,81)$	1.2288	−1.0044	−0.7550	−0.5459	−0.6808	1.475	0.589	−1.0283
$(87,88)$	1.1991	−1.0343	−0.7190	−0.5805	−0.6390	1.619	0.667	−1.0540
$(100,101)$	1.2456	−1.0435	−0.7757	−0.5739	−0.6963	1.499	0.614	−1.0797
$(104,105)$	1.2476	−1.0485	−0.7783	−0.6591	−0.6983	1.502	0.617	−1.0858
$(57,58)$	1.2524	−1.1004	−0.7842	−0.6484	−0.7031	1.565	0.621	−1.0984
$(24,25)$	1.3838	−1.1439	−0.9575	−0.7980	−0.8898	1.286	0.424	−1.1578
$(40,41)$	1.3100	−1.1519	−0.8580	−0.7345	−0.7771	1.482	0.566	−1.1458
$(65,66)$	1.3427	−1.2447	−0.9014	−0.6381	−0.8066	1.543	0.631	−1.2737
$(19,20)$	1.4401	−1.2755	−1.0370	−0.7719	−0.9573	1.332	0.461	−1.2837
$(13,14)$	1.4847	−1.2853	−1.1022	−0.9064	−1.0334	1.244	0.374	−1.2947
$(51,52)$	1.4218	−1.3190	−1.0107	−0.7548	−0.9126	1.445	0.582	−1.3684
$(48,49)$	1.3875	−1.4611	−0.9626	−0.7573	−0.8497	1.719	0.703	−1.4679

Table 8.11: The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time t_{coll} from the census of Section 8.4.3 (`dbn_collisions.py`), against predictions computed from H_0: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1}-\gamma_k)^2/2$, the critical point and Biane forms τ_c and τ_B, and the frozen predictor of Section 8.4.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of H_t.

8.4.5 Collision brackets from Laguerre's quantity

Write $s = -t$ for the backward time, so that $f_s = H_{-s}$ solves $\partial_s f_s = f_s''$; for a real polynomial p, $f_s = e^{sD^2}p = \sum_{k\ge 0} s^k p^{(2k)}/k!$ solves the same equation. In this subsection f_s, $s \ge 0$, is a family of real entire functions of genus at most one, smooth in (z,s), with $\partial_s f_s = f_s''$, and $(z_k(s), z_{k+1}(s))$ is a gap between two adjacent simple real zeros, which move continuously until they meet, if they do.

On the gap put $q = f_s'/f_s$ and $\mathcal{L} = \mathcal{L}_{f_s} = -q'$. For a point x of the gap write $x_j = 1/(x - z_j)$, where z_j runs over the zeros of f_s, and $S_k = \sum_j x_j^k$ for $k \geq 2$, a real and absolutely convergent sum; then

$$\mathcal{L}(x) = S_2, \qquad \mathcal{L}'(x) = -2S_3, \qquad \mathcal{L}''(x) = 6S_4.$$

When all zeros are real, q falls from $+\infty$ to $-\infty$ across the gap and vanishes at a single critical point $c(s)$; we write $\ell(s) = \mathcal{L}(c(s))$. In all cases $\mathcal{L} \to +\infty$ at both ends of the gap, and we write $m(s)$ for its minimum over the gap, attained at a minimiser $u(s)$, where $\mathcal{L}'(u) = 0$, that is $S_3 = 0$.

LEMMA 8.4.4. *On the gap $\partial_s \mathcal{L} = \mathcal{L}'' + 2q\mathcal{L}' - 2\mathcal{L}^2$. If all zeros of f_s are real and simple, then $c(s)$ is differentiable with $\dot{c} = -\mathcal{L}'(c)/\mathcal{L}(c)$, and*

$$\dot{\ell} = 6S_4 - 2S_2^2 - \frac{4S_3^2}{S_2} \qquad \text{at } x = c(s).$$

Proof. With $w = \log f_s$ the heat equation gives $\partial_s w = f_s''/f_s = w'' + w'^2$, hence $\partial_s q = q'' + 2qq'$ and, with $\mathcal{L} = -q'$, $\partial_s \mathcal{L} = -q''' - 2q'^2 - 2qq'' = \mathcal{L}'' - 2\mathcal{L}^2 + 2q\mathcal{L}'$. Since $\partial_z q = -\mathcal{L} < 0$ at c, the implicit function theorem applied to $q(c(s), s) = 0$ gives $\dot{c} = -\partial_s q/\partial_z q = (q'' + 2qq')/\mathcal{L} = -\mathcal{L}'/\mathcal{L}$ at c, where $q = 0$. Hence $\dot{\ell} = \mathcal{L}'\dot{c} + \partial_s \mathcal{L} = \mathcal{L}'' - 2\mathcal{L}^2 - \mathcal{L}'^2/\mathcal{L}$ at c, and $\mathcal{L} = S_2$, $\mathcal{L}' = -2S_3$, $\mathcal{L}'' = 6S_4$. □

LEMMA 8.4.5. *For every real square summable sequence (x_j) with power sums $S_k = \sum_j x_j^k$,*

$$E = 3S_2^3 - 6S_2S_4 + 4S_3^2 \geq 0,$$

and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.

Proof. By homogeneity take $S_2 = 1$. Let $S_k^+ = \sum_{x_j > 0} x_j^k$ and $S_k^- = \sum_{x_j < 0} |x_j|^k$, and $\alpha = (S_2^+)^{1/2}$, $\beta = (S_2^-)^{1/2}$, so that $\alpha^2 + \beta^2 = 1$, $S_3 = S_3^+ - S_3^-$ and $S_4 = S_4^+ + S_4^-$. Every positive term is at most α, so $S_4^+ \leq \alpha S_3^+$ and $S_3^+ \leq \alpha S_2^+ = \alpha^3$; likewise $S_4^- \leq \beta S_3^-$ and $S_3^- \leq \beta^3$. Therefore $E \geq g(S_3^+, S_3^-)$ with

$$g(p, n) = 3 - 6(\alpha p + \beta n) + 4(p - n)^2 \qquad \text{on the box } [0, \alpha^3] \times [0, \beta^3].$$

The function g is convex, its Hessian being $\begin{pmatrix} 8 & -8 \\ -8 & 8 \end{pmatrix}$, and its gradient $\big(-6\alpha + 8(p - n), -6\beta - 8(p - n)\big)$, whose components add up to $-6(\alpha + \beta) < 0$, never vanishes; so its minimum over the box lies on the boundary. On the edge $p = 0$, $g = 3 - 6\beta n + 4n^2 \geq 3 - \frac{9}{4}\beta^2 \geq \frac{3}{4}$, and the edge $n = 0$ is symmetric. At the corner (α^3, β^3), using $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$ and $1 - 4\alpha^2\beta^2 = (\alpha^2 - \beta^2)^2$,

$$g = 4(\alpha^3 - \beta^3)^2 - 3(\alpha^2 - \beta^2)^2 = (\alpha - \beta)^2\big[4(\alpha^2 + \alpha\beta + \beta^2)^2 - 3(\alpha + \beta)^2\big],$$

and with $\sigma = \alpha+\beta$, $\alpha\beta = (\sigma^2-1)/2$, the bracket is $(\sigma^2+1)^2-3\sigma^2 = \sigma^4-\sigma^2+1 > 0$. On the edge $p = \alpha^3$, g is a convex quadratic in n with vertex at $n = \alpha^3 + \frac{3}{4}\beta$. If the vertex lies at or beyond β^3, the minimum over the edge is at the corner. Otherwise $\beta^3 - \frac{3}{4}\beta > \alpha^3 \geq 0$ forces $\beta^2 > \frac{3}{4}$ and $\alpha < \frac{1}{2}$, and the vertex value is

$$\tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3\beta \geq h(\alpha) = \tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3;$$

$h'(\alpha) = \alpha(\frac{9}{2} - 18\alpha - 24\alpha^2)$, so h increases up to $\alpha \approx 0.198$ and decreases after it, and on $[0, \frac{1}{2}]$ it is at least $\min(h(0), h(\frac{1}{2})) = \min(\frac{3}{4}, \frac{3}{16}) > 0$. The edge $n = \beta^3$ is symmetric. Hence $E \geq 0$. Equality forces the corner with $\alpha = \beta$, and $S_4^+ = \alpha S_3^+$ with $S_3^+ = \alpha^3$, so every positive term equals α and their squares add up to α^2: there is exactly one positive term, α, and likewise exactly one negative term, $-\alpha$. Conversely $E = 0$ for $\{a, -a\}$. □

THEOREM 8.4.6 (critical point bound). *If all zeros of f_σ are real and simple for $0 \leq \sigma < s$, then $\dot\ell \leq \ell^2$ on $[0, s)$ and*

$$\ell(\sigma) \leq \frac{\ell(0)}{1 - \sigma\ell(0)} \qquad \big(0 \leq \sigma < \min(s, 1/\ell(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \geq 1/\mathcal{L}_p(c_k)$. The bound is attained by $p(z) = z^2 - a^2$.

Proof. By Lemma 8.4.4, $\dot\ell \leq \ell^2$ reads $6S_4 - 2S_2^2 - 4S_3^2/S_2 \leq S_2^2$, which is $E \geq 0$ divided by $S_2 > 0$, Lemma 8.4.5. Then $\frac{d}{d\sigma}(1/\ell) = -\dot\ell/\ell^2 \geq -1$, so $1/\ell(\sigma) \geq 1/\ell(0) - \sigma$, which is the bound. For the polynomial, all zeros are real and simple on $[0, s_1)$, and $\ell \geq (c - z_k)^{-2} + (z_{k+1} - c)^{-2} \geq 8/g^2$, where g is the length of the gap. If $s_1 < 1/\ell(0)$, then ℓ stays bounded on $[0, s_1)$, so g stays bounded below and the two zeros cannot meet at s_1. For $z^2 - a^2$, $e^{sD^2}p = z^2 - (a^2 - 2s)$, $c = 0$ and $\ell(s) = 2/(a^2 - 2s) = \ell(0)/(1 - s\ell(0))$, with the collision at $s = a^2/2 = 1/\ell(0)$. □

The theorem applies to every real rooted polynomial, in particular to every truncation $\prod_{j\leq J}(1 - z^2/z_j^2)$ of H_0. For H_t itself the hypothesis fails at every $t < 0$: by the theorem of Rodgers and Tao [289], H_t has nonreal zeros for every $t < 0$ (Section 8.4.1). The minimum of $\mathcal{L}$ over the gap gives the sharper Biane form; at the minimiser $S_3 = 0$, while $q(u)$ need not vanish.

LEMMA 8.4.7. *For every real square summable sequence with $S_3 = 0$, $2S_4 \leq S_2^2$, and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.*

Proof. In the notation of the proof of Lemma 8.4.5, without the normalisation, $S_3 = 0$ means $S_3^+ = S_3^- = \mu$, and for a nonzero sequence both α and β are then positive. Now $\mu \leq \min(\alpha^3, \beta^3)$ and $S_4 \leq \alpha S_3^+ + \beta S_3^- = (\alpha + \beta)\mu$. Let

$\lambda = \min(\alpha, \beta)$ and $\max(\alpha, \beta) = r\lambda$ with $r \geq 1$. Then $S_4 \leq (1+r)\lambda^4$ and $S_2^2 = (1+r^2)^2\lambda^4$, and $2(1+r) \leq (1+r^2)^2$ for $r \geq 1$, with equality only at $r = 1$, since both sides equal 4 there and the right side has derivative $4r(1+r^2) \geq 8 > 2$. Equality in $2S_4 \leq S_2^2$ forces $r = 1$, $\mu = \alpha^3$ and $S_4^+ = \alpha S_3^+$, hence one positive term α and one negative term $-\alpha$. □

THEOREM 8.4.8 (Biane form). *If all zeros of f_σ are real and simple for $0 \leq \sigma < s$, then the upper right Dini derivative of m satisfies $D^+m \leq m^2$ on $[0, s)$, and*

$$m(\sigma) \leq \frac{m(0)}{1 - \sigma m(0)} \qquad \big(0 \leq \sigma < \min(s, 1/m(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \geq 1/\min_{(z_k, z_{k+1})} \mathcal{L}_p$. This bound on s_1 contains that of Theorem 8.4.6, since $m(0) \leq \ell(0)$, and it is attained by $z^2 - a^2$.

Proof. Since $\mathcal{L} \to +\infty$ at both ends of the gap, the minimisers are interior and m is continuous. If u is a minimiser at time σ, then $m(\sigma + h) \leq \mathcal{L}(u, \sigma + h)$ for small $h > 0$, so by Lemma 8.4.4, with $\mathcal{L}'(u) = 0$,

$$D^+m(\sigma) \leq \partial_\sigma \mathcal{L}(u, \sigma) = \mathcal{L}''(u) - 2\mathcal{L}(u)^2 = 6S_4 - 2S_2^2 \leq S_2^2 = m(\sigma)^2,$$

the last step by Lemma 8.4.7, as $S_3 = 0$ at u. The lower right Dini derivative of the continuous function $1/m(\sigma) + \sigma$ is therefore nonnegative, so the function does not decrease, and $1/m(\sigma) \geq 1/m(0) - \sigma$. A collision forces $m \geq 8/g^2 \to \infty$, as in the proof of Theorem 8.4.6. For $z^2 - a^2$ the minimiser is the critical point 0, so $m = \ell$. □

Both theorems were also tested numerically. The program `wigbiane_poly.py` draws 2400 polynomials of degree 3 to 12 with only real zeros, 600 in each of four families (zeros uniform on an interval; Gaussian; uniform with one added zero at 2 to 30% of the mean spacing from another, a Lehmer type pair; and a perturbed lattice), follows the zeros of $e^{sD^2}p$ until at most one real zero is left, and locates each collision as a double zero by Newton's method in (z, s) at 40 digits. For the 2398 polynomials it completed (two failed in the location of a collision), the first collision satisfied $s_1\ell(0) \geq 1.0000136$ and $s_1 m(0) \geq 1.0000055$, both for a Lehmer type pair, with medians 1.061 and 1.039.

After the first collision Theorem 8.4.8 no longer applies, yet each of the 4574 later collisions of two zeros adjacent at $s = 0$ satisfied $s_c m(0) \geq 1.0000326$, although the quantity δ of Corollary 8.4.12 below was positive at some step before 806 of them.

For H_t the hypothesis of Theorem 8.4.8 fails, but the argument survives with a correction. At the minimiser u, replace every nonreal pair $a \pm ib$ by the double real point a, its shadow.

LEMMA 8.4.9. *(i) For every real square summable sequence,* $S_4 \le \frac{1}{2}S_2^2 + S_2^{1/2}|S_3|$.

(ii) If $w = 1/(d - ib)$ *with* $d \neq 0$ *and* b *real, then* $\bigl|2\operatorname{Re} w^k - 2d^{-k}\bigr| \le k(k+1)\,b^2/|d|^{k+2}$ *for every* $k \ge 1$.

Proof. (i) Replacing the sequence by its negative if necessary, let $S_3 = \varepsilon \ge 0$, so $S_3^+ = S_3^- + \varepsilon$ in the notation of the proof of Lemma 8.4.5. Then $S_4 \le \alpha S_3^+ + \beta S_3^- = (\alpha+\beta)S_3^- + \alpha\varepsilon$, where $S_3^- \le \beta^3$ and $S_3^- = S_3^+ - \varepsilon \le \alpha^3$. As in the proof of Lemma 8.4.7, $(\alpha+\beta)\min(\alpha,\beta)^3 \le \frac{1}{2}(\alpha^2+\beta^2)^2 = \frac{1}{2}S_2^2$, and $\alpha\varepsilon \le S_2^{1/2}|S_3|$. (ii) With $\epsilon = b/d$ and $\theta = \arctan\epsilon$, $w^k = d^{-k}(1-i\epsilon)^{-k}$ and $\operatorname{Re}(1-i\epsilon)^{-k} = A\cos k\theta$ with $A = (1+\epsilon^2)^{-k/2}$. Now $1 - A\cos k\theta = (1-A) + A(1-\cos k\theta)$, where $0 \le 1-A \le \frac{k}{2}\epsilon^2$ and $0 \le 1-\cos k\theta \le \frac{1}{2}k^2\theta^2 \le \frac{1}{2}k^2\epsilon^2$, so $|\operatorname{Re}(1-i\epsilon)^{-k} - 1| \le \frac{1}{2}(k+k^2)\epsilon^2$; multiply by $2|d|^{-k}$. □

REMARK 8.4.10. The coefficient of the second term in (i) can be halved, and $\frac{1}{2}$ is then best possible, since a single nonzero term gives $S_4 - \frac{1}{2}S_2^2 = \frac{1}{2}S_2^{1/2}|S_3|$. Normalise $S_2 = 1$ and $S_3 = \varepsilon \ge 0$ and put $\mu = S_3^-$, so that $S_4 \le \alpha(\mu+\varepsilon) + \beta\mu$ with $0 \le \mu \le \min(\beta^3, \alpha^3 - \varepsilon)$; the bound increases with μ. For $\varepsilon \le \alpha^3 - \beta^3$ it is at most $(\alpha+\beta)\beta^3 + \alpha\varepsilon$, which is linear in ε, and for $\varepsilon \ge \max(0, \alpha^3 - \beta^3)$ it is at most $\alpha^4 + \alpha^3\beta - \beta\varepsilon$, which decreases. So the bound minus $\frac{1}{2}(1+\varepsilon)$ is largest at $\varepsilon = 0$ or, when $\alpha > \beta$, at $\varepsilon = \alpha^3 - \beta^3$. At $\varepsilon = 0$ the bound is $(\alpha+\beta)\min(\alpha,\beta)^3 \le \frac{1}{2}$, as in the proof of Lemma 8.4.7. At $\varepsilon = \alpha^3 - \beta^3$ it is $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$, and $\alpha^4 + \beta^4 - \frac{1}{2} \le \frac{1}{2}(\alpha^3 - \beta^3)$ reads $(\alpha^2 - \beta^2)^2 \le \alpha^3 - \beta^3$, that is $(\alpha-\beta)(\alpha+\beta)^2 \le 1 + \alpha\beta$; with $\nu = 2\alpha\beta \in [0,1]$ this is $(1-\nu)^{1/2}(1+\nu) \le 1 + \frac{\nu}{2}$, which after squaring becomes $-\nu^2 - \nu^3 \le \frac{1}{4}\nu^2$. Hence

$$S_4 \le \tfrac{1}{2}S_2^2 + \tfrac{1}{2}S_2^{1/2}|S_3|,$$

and the coefficient 72 in Theorem 8.4.11 below may be replaced by 36; we keep the form of [261], which the computations below use. The bound (ii) is sharp as $b/d \to 0$, since $\operatorname{Re}(1-i\epsilon)^{-k} = 1 - \frac{1}{2}k(k+1)\epsilon^2 + O(\epsilon^4)$.

THEOREM 8.4.11 (local bound). *Let* u *be a minimiser of* $\mathcal{L}$ *over the gap at time* s, *with* $m = \mathcal{L}(u) > 0$, *and let* $B_p = \sum b^2|u-a|^{-p}$, *summed over the conjugate pairs* $a \pm ib$, $b > 0$, *of nonreal zeros of* f_s, *each pair once, be finite for* $p = 4, 5, 6$. *Then*

$$\mathcal{L}''(u) - 3m^2 \le \eta\, m^2, \qquad \eta = \frac{36\, mB_4 + 108\, B_4^2 + 72\,(m + 6B_4)^{1/2}B_5 + 120\, B_6}{m^2},$$

and $D^+ m \le (1+\eta)m^2$. *Hence, if the two zeros bounding the gap meet at* s_c, *and* $m > 0$ *and* $\eta \le \bar\eta$ *on* $[0, s_c)$, *then* $s_c \ge 1/\bigl((1+\bar\eta)m(0)\bigr)$.

Proof. Let S_k' be the power sums of the real sequence in which every nonreal pair $a \pm ib$ is replaced by its shadow, two terms $1/(u-a)$, and let $D_k = S_k - S_k'$.

By Lemma 8.4.9(ii) with $d = u - a$, $|D_2| \le 6B_4$, $|D_3| \le 12B_5$ and $|D_4| \le 20B_6$. Since $S_3 = 0$ at u, $S_3' = -D_3$, and Lemma 8.4.9(i) for the shadow sequence gives $S_4' \le \frac{1}{2}S_2'^2 + S_2'^{1/2}|D_3|$. With $S_4 = S_4' + D_4$, $0 \le S_2' = m - D_2 \le m + 6B_4$ and $S_2'^2 - m^2 = -2mD_2 + D_2^2$,

$$\begin{aligned} 6S_4 - 3m^2 &\le 3(S_2'^2 - m^2) + 6S_2'^{1/2}|D_3| + 6|D_4| \\ &\le 6m|D_2| + 3D_2^2 + 6S_2'^{1/2}|D_3| + 6|D_4| \le \eta m^2, \end{aligned}$$

and $\mathcal{L}''(u) = 6S_4$. As in the proof of Theorem 8.4.8, $D^+m \le \mathcal{L}''(u) - 2m^2 \le (1+\eta)m^2$, so the function $1/m(\sigma) + (1+\bar\eta)\sigma$ does not decrease on $[0, s_c)$. At the collision $m \to \infty$: the share of the two meeting zeros tends to $+\infty$, while that of the other zeros, which stay away from the point where the two meet, remains bounded. No nonreal zero tends to that point: if z_c is a zero of multiplicity k of f_{s_c}, then $f_{s_c-\sigma} = e^{-\sigma D^2} f_{s_c}$, so $\sigma^{-k/2} f_{s_c-\sigma}(z_c + \sqrt{\sigma}\, w)$ tends to a nonzero multiple of $H_k(w/\sqrt{2})$ as $\sigma \downarrow 0$, uniformly on compact sets, and by Hurwitz's theorem the k zeros of $f_{s_c-\sigma}$ near z_c lie close to $\sqrt{2\sigma}$ times the zeros of H_k, which are real and simple; so they are real, and real zeros only add to m. □

Corollary 8.4.12. *In the setting of Theorem 8.4.11 put $\delta = \mathcal{L}''(u)/m^2 - 3$, computed from f_s and its first four derivatives at u. Then $D^+m \le (1+\delta)m^2$ and $\delta \le \eta$, and $\delta \le 0$ when all zeros of f_s are real. Hence, if the two zeros bounding the gap meet at s_c, and $m > 0$ and $\delta \le \bar\delta$ on $[0, s_c)$ with $\bar\delta > -1$, then $s_c \ge 1/\big((1+\bar\delta)m(0)\big)$; if $\delta \le 0$ throughout, the collision cannot come before Biane's time $1/m(0)$, exactly as for functions with only real zeros.*

Proof. The first display of the proof of Theorem 8.4.8 gives $D^+m \le \mathcal{L}''(u) - 2m^2 = (1+\delta)m^2$, Theorem 8.4.11 gives $\delta \le \eta$, and Lemma 8.4.7 gives $\delta \le 0$ for real zeros. The rest is the comparison argument of the proof of Theorem 8.4.11. □

Write $m = P + Q$, where $P = (u - z_k)^{-2} + (u - z_{k+1})^{-2}$ is the share of the two zeros bounding the gap and Q the share of all the others, and let $W = \sum |u - z|^{-4}$ over the nonreal zeros z of f_s.

Theorem 8.4.13 (a deadline). *While the two zeros bounding the gap are real, the right derivative $m'(s^+)$ exists and is the least value of $\mathcal{L}''(u) - 2m^2$ over the minimisers u, and at every minimiser*

$$\mathcal{L}''(u) - 2m^2 \ge m^2 - 6mQ + 3Q^2 - 6W,$$

with Q and W taken at u. Suppose that $Q \le \rho$ and $W \le \bar W$ at every minimiser for as long as the two zeros bounding the gap are real, and that $m(0) > r_1$, where $r_{1,2} = 3\rho \pm (9\rho^2 + 6\bar W)^{1/2}$. Then the gap closes, one of the two zeros bounding it meeting another zero, at a time

$$s_c \le s^* = \frac{1}{r_1 - r_2} \log \frac{m(0) - r_2}{m(0) - r_1}.$$

If all zeros are real, $\bar{W} = 0$ *and, for* $\rho > 0$, $s^* = (6\rho)^{-1} \log\big(m(0)/(m(0) - 6\rho)\big) = 1/m(0) + 3\rho/m(0)^2 + 12\rho^2/m(0)^3 + \cdots$.

Proof. Near a given time the minimisers lie in a fixed compact subinterval of the gap, and by Danskin's theorem the right derivative of the minimum exists and equals the least value of $\partial_s \mathcal{L}$ over the minimisers, which is $\mathcal{L}''(u) - 2m^2 = 6S_4 - 2m^2$ by Lemma 8.4.4. In S_4 the two zeros bounding the gap contribute $x_k^4 + x_{k+1}^4 \geq \frac{1}{2}(x_k^2 + x_{k+1}^2)^2 = \frac{1}{2}P^2$, the other real zeros contribute nonnegative terms, and each nonreal zero z contributes $\mathrm{Re}(u - z)^{-4} \geq -|u - z|^{-4}$. So $\mathcal{L}''(u) - 2m^2 \geq 3P^2 - 6W - 2m^2$, and $P = m - Q$ gives the first inequality. While $m > 0$, $Q \leq \rho$ and $W \leq \bar{W}$, it gives $m'(s^+) \geq m^2 - 6\rho m - 6\bar{W} = (m - r_1)(m - r_2)$, so m stays above the solution y of $y' = (y - r_1)(y - r_2)$, $y(0) = m(0) > r_1 \geq 0$, which increases and tends to $+\infty$ as $s \to s^*$. If the gap were still open on $[0, s^*]$, its two zeros real, simple and apart, then m, being at most $\mathcal{L}$ at the midpoint of the gap, would be bounded on $[0, s^*]$, a contradiction. The first collision of either zero need not be with the other: one of them may first meet the zero beyond it, and the theorem then bounds the time of that collision. For a pair that is known to collide, as in the census, s^* bounds the collision time of the pair. □

Corollary 8.4.14. *Put* $\kappa = \mathcal{L}''(u) - 3P^2$, *computed from* f_s *at* u *and the two zeros bounding the gap, so that*

$$\kappa = 3\big(x_k^2 - x_{k+1}^2\big)^2 + 6 \sum_{j \neq k, k+1} x_j^4 \geq -6W,$$

with $\kappa \geq 0$ *when all zeros are real. Theorem 8.4.13 holds with* $6\bar{W}$ *replaced by* $\bar{\kappa} = \max\big(0, \sup(-\kappa)\big)$, *the supremum over the minimisers at the times before the collision, and this form is never weaker.*

Proof. The identity is $6(x_k^4 + x_{k+1}^4) - 3(x_k^2 + x_{k+1}^2)^2 = 3(x_k^2 - x_{k+1}^2)^2$, and the inequality is that of the proof of Theorem 8.4.13. That proof gives $m'(s^+) = \kappa + 3P^2 - 2m^2 \geq m^2 - 6mQ + 3Q^2 - \bar{\kappa}$, and $\bar{\kappa} \leq 6\bar{W}$. □

The two inequalities behind Theorems 8.4.11 and 8.4.13 were tested on random configurations as well. For 29620 configurations of 3 to 11 real zeros and 1 to 7 conjugate pairs, at the critical point u of $\mathcal{L}$ on a random gap, with $\mathcal{L}(u) > 0$, the quantity $6S_4 - 3S_2^2$ never exceeded $0.016\,\eta S_2^2$, and $6S_4 \geq 3P^2 - 6W$ held up to rounding (`wigbiane_ineq.py`). On the polynomials of `wigbiane_poly.py`, with ρ the largest value of Q at the minimisers before the collision, the deadline s^* of Theorem 8.4.13 is defined for 2308 of the 2398 first collisions, with $\bar{W} = 0$, and for 3322 of the 4574 later ones (3345 in the form of Corollary 8.4.14), and no collision came after it: the largest ratio s_c/s^* is $1 - 1.5 \times 10^{-11}$ for the first collisions and $1 - 3.1 \times 10^{-10}$ for the later ones.

Corollaries 8.4.12 and 8.4.14 bracket every collision time by quantities along the flow that need only H_t near the gap, while Theorems 8.4.11 and 8.4.13 need the nonreal zeros. The program `wigbiane_flow.py` (`code/README.md`) computes both kinds for each of the 58 collisions of the census of Section 8.4.3. It evaluates H_t and its first four z derivatives in ball arithmetic at 800 bits, refines the double zero $H_t = \partial_z H_t = 0$ by Newton's method in (z, t), and on a grid of about 140 times s in $[0, |t_{\mathrm{coll}}|)$ it follows the two zeros by Newton's method and computes the minimiser u, the critical point c and the quantities ℓ, m, Q, δ and κ, the last three in ball arithmetic, since near the collision m and P are large and nearly equal. For η and W it also follows, by Newton's method in the complex plane, the conjugate pairs born in the census collisions before the time considered (60 pairs, two of them from collisions among the zeros 203 to 206), with their mirror images $-a \pm ib$, and adds a model of the zeros above the census, $z = 2\gamma_j$ for $207 \le j \le 2000$, in which every such zero is half of a pair $a \pm ib$ with $b^2 = 4|t|$. Newton's method reproduces the census collision times to 5×10^{-12}.

First, $\delta < 0$ at every point of the grid on all 58 trajectories; the largest value is -3.1×10^{-8}, for $(186, 187)$. So the hypothesis of Corollary 8.4.12 with $\bar{\delta} = 0$ holds on the grid in spite of the nonreal zeros, and its conclusion, the lower bound $1/m(0)$, holds for all 58 collisions of the census, as the direct comparison of Section 8.4.4 shows ($t_{\mathrm{coll}}/\tau_B \ge 1.083$; Table 8.11). Likewise $\ell(s) \le \ell(0)/(1 - s\ell(0))$ for $s < 1/\ell(0)$ and $m(s) \le m(0)/(1 - sm(0))$ for $s < 1/m(0)$ at every point of the grid.

Second, $\kappa \ge 0$ at every point of the grid except on the trajectories of $(108, 109)$ and $(153, 154)$, where $\bar{\kappa} = 0.029$ and 0.071. The upper bound s^* of Corollary 8.4.14, with ρ the largest value of Q on the grid, is defined, $m(0) > r_1$, for 55 of the 58 collisions; it is not for $(153, 154)$, $(178, 179)$ and $(181, 182)$, for which $6\rho/m(0)$ is 0.986, 1.068 and 1.103. Every collision time of the census lies in its bracket $[1/m(0), s^*]$, with $s^* = \infty$ for those three, and Table 8.13 shows the seven pairs of the note [261]. The upper bound is tight when the pair dominates its gap: $s^*/|t_{\mathrm{coll}}|$ ranges from 1.0010, for $(186, 187)$ with $6\rho/m(0) = 0.151$, to 2.021, for $(48, 49)$ with $6\rho/m(0) = 0.965$.

Third, the correction η of Theorem 8.4.11, computed from the tracked pairs and the model above the census, is at most 3.25×10^{-5} for the seven pairs of Table 8.13, but it reaches 0.53 for $(153, 154)$ and exceeds 10^{-2} for four more collisions, all at $t < -1.89$. With $\bar{\eta}$ in place of $\bar{\delta}$ the lower bound still holds for all 58, the closest case being again $(186, 187)$, whose collision time exceeds the bound by the factor 1.083. For $(34, 35)$, $(71, 72)$, $(91, 92)$ and five other late collisions more than half of B_4 at the worst point comes from the model, so $\bar{\eta}$ depends on it, while δ and κ depend on no model. The model is not pessimistic in every respect: the tracked pairs leave the axis faster than an isolated pair would, with b^2 up to 3.39 times $2(t_c - t)$.

Freezing Q at its value at $s = 0$, that is, taking $\rho = Q(0)$ and $\bar{\kappa} = 0$ in

$s = -t$	0	0.2	0.4	0.6	0.8	1.0	1.2
$(13,14)$: $\ell(s)$	1.1033	1.2462	1.4539	1.7833	2.3846	3.8293	12.051
$\ell(0)/(1-s\ell(0))$	1.1033	1.4156	1.9748	3.2638	9.3992		
$m(s)$	0.9676	1.1099	1.3170	1.6457	2.2461	3.6898	11.911
$m(0)/(1-sm(0))$	0.9676	1.1998	1.5787	2.3071	4.2837	29.903	
$(19,20)$: $\ell(s)$	1.2955	1.4520	1.6768	2.0293	2.6688	4.2153	13.848
$\ell(0)/(1-s\ell(0))$	1.2955	1.7485	2.6889	5.8171			
$m(s)$	1.0446	1.1904	1.4029	1.7413	2.3644	3.8914	13.500
$m(0)/(1-sm(0))$	1.0446	1.3205	1.7944	2.7990	6.3583		

Table 8.12: Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13,14)$ and $(19,20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 8.4.6 and 8.4.8; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of H_t; `wigbiane_flow.py`.

pair	$m(0)$	$Q(0)$	ρ	$10^8\bar{\delta}$	$\bar{\eta}$	pairs	$1/m(0)$	s^*	$\lvert t_{\rm coll}\rvert$
$(13,14)$	0.9676	0.0604	0.0610	-9.0	2.8×10^{-7}	49	1.0334	1.2982	1.2853
$(19,20)$	1.0446	0.0803	0.0879	-12.8	2.4×10^{-6}	48	0.9573	1.3325	1.2755
$(24,25)$	1.1238	0.0794	0.0794	-10.2	3.2×10^{-5}	44	0.8898	1.1578	1.1439
$(27,28)$	1.4415	0.0962	0.1012	-10.2	4.0×10^{-6}	36	0.6937	0.9006	0.8811
$(34,35)$	2.8970	0.0968	0.0969	-4.3	1.7×10^{-9}	11	0.3452	0.3853	0.3846
$(71,72)$	3.9727	0.1605	0.1678	-5.5	2.0×10^{-9}	4	0.2517	0.2903	0.2880
$(91,92)$	4.0736	0.1701	0.1757	-5.7	3.1×10^{-9}	3	0.2455	0.2841	0.2821

Table 8.13: Two sided brackets for the collision time of seven pairs, in the units of H_t: the lower bound $1/m(0)$ of Corollary 8.4.12, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound s^* of Corollary 8.4.14, with ρ the largest value of Q and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 8.4.11 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. `wigbiane_flow.py`.

Corollary 8.4.14, gives a predictor of the collision time from H_0 alone,

$$t_{\rm coll} \approx -\frac{1}{6Q(0)} \log \frac{m(0)}{m(0) - 6Q(0)},$$

the last column of Table 8.11. It is not a bound, since Q grows along the flow ($\rho > Q(0)$ for 41 of the 58 collisions) and the predictor falls on both sides of $t_{\rm coll}$, but it is accurate when the pair dominates its gap. Over the 28 collisions of Table 8.11 its relative error is at most 0.32% for the six with $6Q(0)/m(0) < 0.3$ and 3.75% in all, for $(51,52)$; over the 52 collisions of the census with $6Q(0)/m(0) < 0.8$ it is at most 5.75%, for $(130,131)$. It fails when the other zeros carry most of $m(0)$: the six collisions with $6Q(0)/m(0) \geq 0.8$, all at $t < -1.87$, have errors from

10.4%, for $(68, 69)$, to 46.6%, for $(188, 189)$. To first order in $Q(0)$ the predictor is $1/m(0) + 3Q(0)/m(0)^2$ in absolute value, and if the two zeros sit symmetrically about u, so that $P(0) = 8/D_0^2$, it is

$$\frac{D_0^2}{8} + \frac{D_0^4 Q(0)}{32} + O\big(Q(0)^2\big).$$

The bound Λ_k of Csordas, Smith and Varga [70] in Section 8.4.6 has the expansion $-\Delta_k^2/8 - \Delta_k^4 g_k/64 - \cdots$ with $\Delta_k = D_0$: the same two terms with g_k in place of $2Q(0)$, and indeed $g_k \approx 2Q(0)$, since g_k adds up, over the other zeros, the inverse squares of their distances to each of the two zeros of the pair. For $(71, 72)$, $g_{71} = 0.33734$ and $2Q(0) = 0.3210$, and the frozen predictor gives -0.2883, against $t_{\mathrm{coll}} = -0.2880$ and $\Lambda_{71} = -0.3048$.

Both brackets rest on hypotheses along the flow, on δ or η for the lower bound and on Q and κ or W for the upper bound, which the computation checks on the grid of times but which cannot yet be guaranteed in advance.

Problem 8.1. Find a bound for how far the nonreal zeros of H_t, $t < 0$, travel, and for how close to a gap between real zeros they come. Such control would make the brackets unconditional statements about H_t, computed from H_0 alone.

Biane's work meets ξ in two more places. With Pitman and Yor [32], following Williams [344], he wrote $2\xi(s)$ as the Mellin transform of a sum of two Bessel hitting times, the identity behind the Pólya variable of Section 1.2. And in [31] he revisits the paper of Pólya [273], treated in full in Section 8.7.

8.4.6 Lehmer pairs and lower bounds for Λ

Every collision time is a lower bound for Λ, because nonreal zeros persist backward in time, and Lehmer pairs, pairs with unusually small D_0, collide at times close to zero. Csordas, Smith and Varga [70] turned this into a theorem that uses only the zeros of H_0. Write $x_j = 2\gamma_j$ for the positive zeros of H_0, $x_{-j} = -x_j$, $\Delta_k = x_{k+1} - x_k$ and

$$g_k = \sum_{j \neq k, k+1}{}' \Big[\frac{1}{(x_k - x_j)^2} + \frac{1}{(x_{k+1} - x_j)^2} \Big],$$

where the prime restricts the sum to nonzero integers j, of either sign. The pair (x_k, x_{k+1}) is called a Lehmer pair if $\Delta_k^2 g_k < \frac{4}{5}$, and then [70, Theorem 1]

$$\Lambda \geq \Lambda_k = \frac{\big(1 - \frac{5}{4}\Delta_k^2 g_k\big)^{4/5} - 1}{8 g_k} = -\frac{\Delta_k^2}{8} - \frac{\Delta_k^4 g_k}{64} - \cdots .$$

The leading term is the isolated pair collision time $-D_0^2/8$ with $D_0 = \Delta_k$. The binomial series of $(1-x)^{4/5}$ has only negative coefficients after the constant term, so every further term is negative and decreases in g_k, and so does Λ_k.

For the closest pair below height 200, $\Delta_{71} = 1.44863$ and, from the first 1700 zeros, $g_{71} = 0.33734$, so $\Delta^2 g = 0.708 < \frac{4}{5}$: the pair is a Lehmer pair in this sense, and it gives $\Lambda_{71} = -0.3048$. The omitted zeros above height 2197 increase g by less than 10^{-3}, which lowers Λ_{71} by less than 3×10^{-4}, so with the tail the bound is -0.305. The theorem is proved, but this evaluation of it is a floating point computation with zeros computed by `mpmath` [205]. The bound lies below the computed collision time -0.288, as expected. The theorem itself does not order the bound and the collision time: its proof, by contradiction, assumes that H_t has only real zeros for $\Lambda_k \le t \le 0$, which fails for every $t < 0$ because $\Lambda \ge 0$.

Table 8.14 lists the published bounds. The last four lower bounds before 2020, among them that of Csordas, Odlyzko, Smith and Varga [69], came from this theorem applied to Lehmer pairs of increasingly high quality, found at heights far greater than 7005. Rodgers and Tao proved $\Lambda \ge 0$ by a different argument, showing that $\Lambda < 0$ would force the zeros into a local equilibrium incompatible with Montgomery's pair correlation estimates [203].

bound	authors	year
$\Lambda > -\infty$	Newman	1976
$\Lambda > -50$	Csordas, Norfolk, Varga	1988
$\Lambda > -5$	te Riele	1991
$\Lambda > -0.385$	Norfolk, Ruttan, Varga	1992
$\Lambda > -0.0991$	Csordas, Ruttan, Varga	1991
$\Lambda > -4.379 \times 10^{-6}$	Csordas, Smith, Varga	1994
$\Lambda > -5.895 \times 10^{-9}$	Csordas, Odlyzko, Smith, Varga	1993
$\Lambda \ge -2.63 \times 10^{-9}$	Odlyzko	2000
$\Lambda > -1.15 \times 10^{-11}$	Saouter, Gourdon, Demichel	2011
$\Lambda \ge 0$	Rodgers, Tao	2020
$\Lambda \le \frac{1}{2}$	de Bruijn	1950
$\Lambda < \frac{1}{2}$	Ki, Kim, Lee	2009
$\Lambda \le 0.22$	Polymath	2019
$\Lambda \le 0.2$	Platt, Trudgian	2021

Table 8.14: Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [289, Table 1], upper bounds from [73], [150], [282] and [247]. Odlyzko's bound is the one computed in [222, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication.

8.4.7 The shielding deficit

Let $a < b$ be adjacent simple real zeros of H_t with gap $g = b - a$, and put

$$S_{a,b} = \sum_{\rho \ne a,b} \frac{1}{(a-\rho)(b-\rho)}, \qquad \mathrm{Def} = 2 - g^2 S_{a,b},$$

the sum running over all the other zeros ρ (conjugate pairs, if any, give a real sum). Subtracting the two equations of motion of Section 8.4.1 gives $\dot g = 4/g - 2gS_{a,b}$, that is

$$\frac{d(g^2)}{dt} = 4\,\mathrm{Def}. \tag{8.4.2}$$

For an isolated pair $S_{a,b} = 0$ and $\mathrm{Def} = 2$, which is the law $g^2 = g(0)^2 + 8t$ of Section 8.4.1. For real zeros $S_{a,b} > 0$, and $2xy \le x^2 + y^2$ gives $S_{a,b} \le g_k/2$ with the g_k of Csordas, Smith and Varga (computed here from the zeros of H_t) in Section 8.4.6, so $\mathrm{Def} \ge 2 - \Delta_k^2 g_k/2$. A regular lattice of gap g shields exactly: each side contributes $\sum_{m\ge1} 1/(m(m+1)) = 1$, the factor 2 being what the other zeros must supply. On each side number the exterior gaps outward from the pair, write them $g(1+\varepsilon_m)$ with $\varepsilon_m > -1$, and put $D_0 = 0$, $D_m = \sum_{j\le m}\varepsilon_j$. The mth exterior zero on that side is at distance $g(m + D_m)$ from the nearer end of the pair.

THEOREM 8.4.15 (Newton). *On one side, for every $N \ge 1$,*

$$1 - \sum_{m=1}^{N} \frac{1}{(m+D_m)(m+1+D_m)} = \sum_{m=1}^{N} \frac{\varepsilon_m}{(m+D_{m-1})(m+D_m)} + \frac{1}{N+1+D_N}.$$

If all zeros are real and the exterior zeros tend to infinity on both sides, then $\mathrm{Def} = \sum_{\text{sides}} \sum_{m\ge1} \varepsilon_m / \big((m+D_{m-1})(m+D_m)\big)$.

Proof. Put $y_m = m + D_{m-1}$ and $z_m = m + D_m$, so $z_m - y_m = \varepsilon_m$, $y_1 = 1$ and $y_{m+1} = z_m + 1$. The mth exterior zero contributes $1/\big(z_m(z_m+1)\big)$ to $g^2 S_{a,b}$, because $a-\rho = -g(z_m+1)$ and $b-\rho = -gz_m$. Now $\frac{1}{z_m(z_m+1)} + \frac{z_m - y_m}{y_m z_m} = \frac{1}{y_m} - \frac{1}{z_m+1}$, and summing over $m \le N$ gives $1/y_1 - 1/(z_N+1) = 1 - 1/(N+1+D_N)$. The last term tends to zero when the zeros tend to infinity; add the two sides. □

COROLLARY 8.4.16. *(a) If all $\varepsilon_m = 0$ then* $\mathrm{Def} = 0$ *and g is constant; the lattice of zeros of $e^{\kappa^2 t}\sin\kappa z$ is an example. (b) Keeping R exterior zeros on each side of a regular lattice leaves* $\mathrm{Def} = 2/(R+1)$*, the size of the omitted tail. (c) A single enlarged exterior gap $g(1+\delta)$ at position j contributes exactly $\delta/(j(j+\delta))$. (d) With* Def *as above and $d_{\nu,j}$ the exterior gaps on the sides $\nu = L, R$,*

$$[\mathrm{Def}]_+ \le g^{-1} \sum_{j\ge1} j^{-2}\big[(d_{L,j} - g)_+ + (d_{R,j} - g)_+\big],$$

and by (c) the constant j^{-2} is optimal to first order. (e) If the configuration has a smallest gap $g > 0$ (a finite configuration or a lattice) and g is that gap, then every $\varepsilon_m \ge 0$, so $\mathrm{Def} \ge 0$ *and the smallest gap does not shrink forward in time, with equality only for the lattice.*

Proof. (a) to (c) are read off from the theorem; in (b), $2(1-R/(R+1)) = 2/(R+1)$. For (d) let $P_m = \sum_{i\le m}(\varepsilon_i)_+$. Then $z_m \le m + P_m$ and $1/(z(z+1))$ decreases, so by the theorem applied to the sequence $(\varepsilon_i)_+$, one side contributes at most $1 - \sum_m \frac{1}{(m+P_m)(m+P_m+1)} = \sum_m \frac{(\varepsilon_m)_+}{(m+P_{m-1})(m+P_m)} \le \sum_m (\varepsilon_m)_+/m^2$. (e) is the theorem with $\varepsilon_m \ge 0$ and (8.4.2). □

Part (e) says nothing about H_t: its mean gap at height T is of order $1/\log T$, so there is no smallest gap. Being minimal among the neighbours is not enough either: take $g = 1$ and exterior zeros at distances 1, 1.01, 1.02 from the pair on both sides, so that the nearest exterior gaps equal g and the next ones are 0.01. Then $g^2 S_{a,b} \ge 1 + \frac{20000}{20301} + \frac{20000}{20604} = \frac{340039}{115039} > 2$, so Def < 0 and this gap shrinks forward. Finally, (8.4.2) integrates to an identity, $4\int \mathrm{Def}\, dt = \Delta(g^2)$, so a bound on the shielding budget that already contains the moving gap is circular; an estimate has to bound Def from the exterior zeros alone.

8.4.8 The Thorin atoms of the tilted clock

Multiplying the kernel by e^{tu^2} tilts the clock. For $t \ge \Lambda$ the function R_t of Section 8.4.1 is the Laplace transform of a GGC T_t, the tilted clock, with the Thorin atoms $a_j(t) = z_j(t)^2/4 = \theta_j(t)^2$ (Theorem 7.2.15), and $T_0 = T$ is the clock of Section 1.2, which exists unconditionally (Theorem 5.1.1).

Proposition 8.4.17. *For $t > 0$, with Z a standard normal variable,*

$$H_t(z) = \mathbb{E}\, H_0\big(z + i\sqrt{2t}\, Z\big), \qquad z \in \mathbb{C},$$
$$\frac{1}{R_t(\sigma^2)} = \frac{\mathbb{E}\, \xi\big(\frac12 + \sigma + \sqrt{t/2}\, Z\big)}{\mathbb{E}\, \xi\big(\frac12 + \sqrt{t/2}\, Z\big)}, \qquad \sigma \ge 0.$$

Proof. With $c = \sqrt{2t}$ we have

$$\mathbb{E}\cos\big((z + icZ)u\big) = \cos(zu)\,\mathbb{E}\cosh(cuZ) = \cos(zu)\, e^{tu^2},$$

because the part odd in Z has mean zero. Integrating against Φ_{dBN} gives the first identity, Fubini applying because $\Phi_{\mathrm{dBN}}(u)$ decays like $\exp(-\pi e^{4u})$; the same computation with $e^{tu^2}\Phi_{\mathrm{dBN}}$ in place of Φ_{dBN} gives $H_{t+s}(z) = \mathbb{E}\, H_t\big(z + i\sqrt{2s}\, Z\big)$ for real t and $s > 0$. For the second identity, $8H_0(\pm 2i\sigma) = \xi(\frac12 + \sigma)$ and Z is symmetric. We checked the first identity at $t = 0.3$ and $z = -4i$ to 40 digits (`xg_tilt.py`, `code/README.md`). □

So for $t > 0$ the tilt is Gaussian convolution, along the real axis of σ, of the even positive function $\sigma \mapsto \xi(\frac12 + \sigma)$, with variance $t/2$. Theorem 19.4.1(b) is the smoothing along the real axis of z, which lowers t.

The Newton identities of Section 8.4.3 give the mean and variance of T_t from the moments $M_k(t)$ of the reweighted kernel,

$$\mathbb{E}T_t = \sum_j \frac{1}{a_j(t)} = \frac{M_1(t)}{2M_0(t)}, \qquad \operatorname{Var} T_t = \sum_j \frac{1}{a_j(t)^2} = \frac{M_1(t)^2}{4M_0(t)^2} - \frac{M_2(t)}{12M_0(t)},$$

and in the variable a_j the motion of the zeros in Section 8.4.1 reads

$$\frac{da_j}{dt} = \frac{1}{2} + 2\sum_{k\neq j} \frac{a_j}{a_j - a_k}, \tag{8.4.3}$$

the term $\frac{1}{2}$ coming from the mirror zero $-z_j$ and the others from the pairs $\pm z_k$. The table gives the first four atoms at $t = 0$ and $t = 0.2$, with $\mathbb{E}T_t$ and $\operatorname{Var} T_t$, the difference quotient $\Delta a_j/\Delta t = (a_j(0.2) - a_j(0))/0.2$ and the rate $\dot a_j(0)$ from (8.4.3).

	$t = 0$	$t = 0.2$	$\Delta a_j/\Delta t$	$\dot a_j(0)$
a_1	199.7905	198.1565	-8.170	-8.193
a_2	441.9262	438.7638	-15.812	-15.872
a_3	625.5430	622.6126	-14.652	-14.652
a_4	925.6731	919.9043	-28.844	-29.069
$\mathbb{E}T_t$	0.0231049931	0.0232009711		
$\operatorname{Var} T_t$	3.71726×10^{-5}	3.77203×10^{-5}		

At $t = 0$ the a_j are γ_j^2, the Thorin atoms of T if RH holds (Proposition 1.2.1), and $\mathbb{E}T_0$ and $\operatorname{Var} T_0$ are the constants of Section 1.2. The rate is the sum (8.4.3) over the first 8000 zeros completed by the tail $-2a_j(\log(\gamma_{8000}/2\pi) + 1)/(2\pi\gamma_{8000})$; a central difference of the zeros of H_t at $t = \pm 0.01$ and ± 0.005, with Richardson extrapolation, reproduces it to 2×10^{-5}, the order of the error of the tail.

For large t the tilt concentrates the kernel. Under the probability density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $(0,\infty)$, with expectation $\mathbb{E}_t$, we have $1/R_t(s) = \mathbb{E}_t \cosh(2\sqrt{s}\,u)$ and $H_t(z)/H_t(0) = \mathbb{E}_t \cos(zu)$.

Proposition 8.4.18. *As $t \to \infty$ let u_t be the mode of $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$, so that $e^{4u_t} \sim tu_t/(2\pi)$. Then*

$$R_t\big(s/u_t^2\big) \to \frac{1}{\cosh(2\sqrt{s})}, \qquad \frac{T_t}{u_t^2} \Rightarrow 2C, \qquad u_t\, z_j(t) \to \pi\big(j - \tfrac{1}{2}\big),$$

where C is the exit time of Brownian motion from $(-1,1)$, whose Laplace transform is $1/\cosh\sqrt{2\theta}$.

Proof. Put $g(u) = tu^2 + \log\Phi_{\mathrm{dBN}}(u) = tu^2 + 9u + \log 2\pi^2 - \pi e^{4u} + o(1)$. The stationary point satisfies $e^{4u_t} = (2tu_t + 9)/(4\pi) + o(1)$ and $g''(u_t) = -8tu_t + O(t)$,

so under $\mathbb{E}_t$ the variable u has mean $u_t + o(1)$ and standard deviation of order $(tu_t)^{-1/2} \to 0$, with Gaussian tails in $(u-u_t)\sqrt{tu_t}$. Hence $u/u_t \to 1$ in probability with $\mathbb{E}_t e^{cu/u_t}$ bounded for each c, and $\mathbb{E}_t \cosh(2\sqrt{s}\,u/u_t) \to \cosh(2\sqrt{s})$. The Laplace transform $1/\cosh(2\sqrt{s})$ is that of $2C$, and the zeros follow from Hurwitz's theorem applied to $z \mapsto H_t(z/u_t)/H_t(0) \to \cos z$, uniformly on compacts. □

The convergence is fast: at $t = 300$, 1000 and 3000 the mode is $u_t = 0.9602$, 1.3418, 1.6707, and $u_t z_1(t) = 1.57334$, 1.57112, 1.57086 against $\pi/2 = 1.57080$. The path of clocks T_t thus runs from the Brownian exit time $2C$, after the scaling by u_t^2, at $t = \infty$ to T at $t = 0$, and consists of GGCs for $t \geq \Lambda$ (Theorem 7.2.15). The limit shape is a fixed shape of the flow: the zeros of $e^{\kappa^2 t}\cos\kappa z$ are the equally spaced points $\pi(j - \frac{1}{2})/\kappa$ at all times [213].

Gaussian smoothing of a reciprocal GGC transform in the variable $\sigma = \sqrt{s}$ is again one, and not only for the tilt.

THEOREM 8.4.19 (Gaussian smoothing of reciprocal GGC transforms). *Let $a_j > 0$ with $\sum_j a_j^{-1} < \infty$, and let $M(\sigma) = \prod_j (1 + \sigma^2/a_j)$, the reciprocal of the Laplace transform, at $s = \sigma^2$, of the GGC with the discrete Thorin measure $\sum_j \delta_{a_j}$ and no drift; M is an even entire function of genus zero in σ^2. For $c > 0$ put*

$$M_c(\sigma) = \frac{\mathbb{E}\, M(\sigma + cZ)}{\mathbb{E}\, M(cZ)}.$$

Then $M_c(\sigma) = \prod_j (1 + \sigma^2/a_j(c))$ with real $a_j(c) > 0$, so that $1/M_c(\sqrt{s})$ is the Laplace transform of a GGC S_λ with the Thorin measure $\sum_j \delta_{a_j(c)}$, where $\lambda = 2c^2$. The atoms obey (8.4.3) *with t replaced by λ while they are simple, and*

$$\frac{d}{d\lambda}\mathbb{E}S_\lambda = (\mathbb{E}S_\lambda)^2 - \frac{3}{2}\operatorname{Var} S_\lambda.$$

Proof. Write $\lambda = 2c^2$, $D = d/d\sigma$ and $u(\lambda, \sigma) = \mathbb{E}M(\sigma + cZ) = e^{\lambda D^2/4} M(\sigma)$, so that $\partial_\lambda u = \frac{1}{4}\partial_\sigma^2 u$, the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ in the variable $\sigma = iz/2$. The function $f(\tau) = M(i\tau) = \prod_j (1 - \tau^2/a_j)$ is an even real entire function of genus at most one with only real zeros, and $u(\lambda, i\tau) = \mathbb{E}f(\tau + icZ) = e^{-\lambda D_\tau^2/4} f(\tau)$ with $D_\tau = d/d\tau$. By Pólya's theorem that e^{-aD^2}, $a > 0$, preserves the class of real entire functions of genus at most one with only real zeros [275] (for H_t it is Theorem 19.4.1(c)), $u(\lambda, i\tau)$ has only real zeros in τ. It is even and $u(\lambda, 0) = \mathbb{E}M(cZ) \geq 1$, so $M_c = u/u(\lambda, 0)$ has the product form with real $a_j(c) > 0$. For a single atom, $\mathbb{E}\big(1 + (\sigma + cZ)^2/a\big) = (1 + c^2/a)\big(1 + \sigma^2/(a + c^2)\big)$: the atom moves from a to $a + \lambda/2$, the drift $\frac{1}{2}$ of (8.4.3). In general the atoms obey (8.4.3) by the computation of Section 8.4.1, which uses only the heat equation, the product form above and the simplicity of the zeros.

For the mean write $u = c_0 + c_2\sigma^2 + c_4\sigma^4 + \cdots$, so that $\dot{c}_0 = \frac{1}{2}c_2$ and $\dot{c}_2 = 3c_4$. Since $u/c_0 = \prod_j (1 + \sigma^2/a_j(\lambda)) = 1 + m\sigma^2 + \frac{1}{2}(m^2 - p)\sigma^4 + \cdots$ with $m = \sum_j a_j^{-1}$

and $p = \sum_j a_j^{-2}$, we have $c_2 = mc_0$ and $c_4 = \frac{1}{2}(m^2 - p)c_0$, and

$$\dot m = \frac{\dot c_2}{c_0} - m\,\frac{\dot c_0}{c_0} = \frac{3}{2}(m^2 - p) - \frac{1}{2}m^2 = m^2 - \frac{3}{2}p.$$

Finally $\mathbb{E}S_\lambda = m$ and $\operatorname{Var} S_\lambda = p$. □

Take $M(\sigma) = 1/R_t(\sigma^2)$. The proof of Proposition 8.4.17 gives

$$M_c(\sigma) = \frac{1}{R_{t+2c^2}(\sigma^2)},$$

so for $t \geq \Lambda$ the identity becomes

$$\frac{d}{dt}\mathbb{E}T_t = (\mathbb{E}T_t)^2 - \frac{3}{2}\operatorname{Var} T_t.$$

For the tilt it also follows without the zeros from $dM_k/dt = M_{k+1}/4$ and the formulas for $\mathbb{E}T_t$ and $\operatorname{Var} T_t$ above. At $t = 0$ both sides equal 4.7808×10^{-4} and at $t = 0.2$ they equal 4.8170×10^{-4}.

The clock $T_{0.2}$ is an unconditional GGC (Theorem 7.2.15). By Proposition 8.4.17 its Laplace transform is the reciprocal of the Gaussian smoothing of $\xi(\frac{1}{2} + \sigma)$ with variance 0.1 in σ, normalised at $\sigma = 0$. It is close to T: the means differ by 9.6×10^{-5} on 0.0231, and $R_{0.2}(s)/R_0(s) - 1$ is -9.6×10^{-5} at $s = 1$, -9.3×10^{-4} at $s = 10$ and -7.6×10^{-3} at $s = 100$. In this sense T is a GGC up to a smoothing of variance 0.1, and RH is the statement that the variance can be taken to be zero.

8.4.9 High order Laguerre expressions along the flow

This subsection and Section 8.4.7 report results of T. Newton [213], rewritten in the notation of the book; the Lambert form of Proposition 8.4.21 is the authors' own. Put $\nu_t(u) = \frac{1}{2}e^{tu^2}\Phi_{\mathrm{dBN}}(u)$ on $\mathbb{R}$, so that $H_t(z) = \int_{\mathbb{R}} \nu_t(u)e^{izu}\,du$, and define the Laguerre expressions $L_{n,t}(x)$ by

$$H_t(x + iy)H_t(x - iy) = \sum_{n\geq 0} L_{n,t}(x)\, y^{2n}.$$

Then $L_{1,t} = H_t'^2 - H_tH_t'' = H_t^2\mathcal{L}_{H_t}$ with Laguerre's quantity of Section 8.4.4. With $w = u + v$ and $b = u - v$ the odd powers of b cancel under $(u, v) \mapsto (v, u)$, and

$$L_{n,t}(x) = \frac{1}{(2n)!}\iint b^{2n}\,\nu_t(u)\nu_t(v)\cos(xw)\,du\,dv. \tag{8.4.4}$$

Since H_t is even, $H_t(iy) = H_t(0)/R_t(y^2/4)$, so $\sum_n L_{n,t}(0)y^{2n} = H_t(0)^2R_t(y^2/4)^{-2}$: the $L_{n,t}(0)$ are the Taylor coefficients of the square of the reciprocal tilted clock. By (8.4.4), $L_{n,t}(0) > 0$. Let $\mathbb{P}_{n,t}$ be the probability law on $\mathbb{R}^2$ with density

proportional to $b^{2n}\nu_t(u)\nu_t(v)$, with expectation $\mathbb{E}_{n,t}$. Then $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, and with $k_n = (n+1)(2n+1)$ and $k_n(2n)!/(2n+2)! = \frac{1}{2}$, the central ratio is

$$c_n(t) = \frac{k_n L_{n+1,t}(0)}{L_{n,t}(0)} = \tfrac{1}{2}\,\mathbb{E}_{n,t}[b^2].$$

THEOREM 8.4.20 (Newton). $\lim_{n\to\infty} \sup_{0\le t\le 1/2} \mathbb{E}_{n,t}[w^2] = 0$. *Consequently, for every* $X < \infty$ *there is* N_X *with* $L_{n,t}(x) > 0$ *for all* $n \ge N_X$, $|x| \le X$ *and* $0 \le t \le \frac{1}{2}$.

Proof. Let $V = -\log \Phi_{\mathrm{dBN}}$, an even smooth function with $V''(u) \sim 16\pi e^{4u}$ as $u \to \infty$ (the first theta summand dominates). So V'' is continuous, bounded below by $-K$ and tends to infinity; Newman's monotonicity of V'' [211] is not needed. Given $b = d$, the density of w is proportional to $e^{-Q(w)}$ with $Q(w) = V(\frac{w+d}{2}) + V(\frac{w-d}{2}) - \frac{t}{2}(w^2+d^2)$ and $Q'' = \frac{1}{4}[V''(\frac{w+d}{2}) + V''(\frac{w-d}{2})] - t$. One of the two arguments has modulus at least $|d|/2$, so for $|d| \ge D_0$ and $t \le \frac{1}{2}$ we get $Q'' \ge \kappa(D_0) := \frac{1}{4}[\inf_{|x|\ge D_0/2} V'' - K] - \frac{1}{2}$, which tends to infinity with D_0. Since V is even, Q is even, $Q'(0) = 0$ and $wQ'(w) \ge \kappa w^2$; integration by parts gives $\mathbb{E}[wQ'(w) \mid b = d] = 1$, so $\mathbb{E}[w^2 \mid b = d] \le \kappa^{-1}$. For $|d| \le D_0$ the conditional second moment is at most a constant $B(D_0)$ uniform in t, because $1 \le e^{t(u^2+v^2)} \le e^{(u^2+v^2)/2}$ and both integrals are continuous in d on a compact set. Finally $\nu_0 \le \nu_t \le \nu_{1/2}$, and comparing the weight b^{2n} on $\{|b| \le D_0\}$ with that on $\{D_0+1 \le |b| \le D_0+2\}$ gives $\mathbb{P}_{n,t}(|b| \le D_0) \le C(D_0)\big(D_0/(D_0+1)\big)^{2n}$ uniformly in t. Hence $\mathbb{E}_{n,t}[w^2] \le B(D_0)C(D_0)(\frac{D_0}{D_0+1})^{2n} + \kappa(D_0)^{-1}$; let $n \to \infty$ and then $D_0 \to \infty$. For the second claim, $L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw) \ge 1 - \frac{x^2}{2}\mathbb{E}_{n,t}[w^2]$. □

Newton states that $c_n(t) = (\frac{1}{8} + o(1))(\log n)^2$ uniformly in $t \in [0, \frac{1}{2}]$. The Laplace calculation behind it gives the better centring $\frac{1}{8}\mathcal{W}(n/\pi)^2$, with $\mathcal{W}$ the Lambert function of Section 11.5.3.

PROPOSITION 8.4.21. *Uniformly in* $t \in [0, \frac{1}{2}]$, $c_n(t) = \frac{1}{8}\,\mathcal{W}(n/\pi)^2 + o(\log n)$.

Sketch. For large $|b|$ only the first theta summand matters, and $\log \nu_t(u) + \log \nu_t(v) = -2\pi e^{2b}\cosh 2w + O(b^2)$. Integrating out w (a Gaussian of width of order e^{-b}) leaves a marginal $-2\pi e^{2b} + O(b^2)$, uniformly in t. The law of b has log density $2n\log b - 2\pi e^{2b} + O(b^2)$, maximal where $2b\,e^{2b} = n/\pi$ up to an $O(b^2/n)$ shift, that is at $b_n = \frac{1}{2}\mathcal{W}(n/\pi) + o(1)$, with width $O(\sqrt{b_n/n})$. So $\mathbb{E}_{n,t}[b^2] = b_n^2 + o(\log n)$. □

Since $\mathcal{W}(y) = \log y - \log\log y + o(1)$, the proposition is a statement about the second order term, and the relative error of $\frac{1}{8}(\log n)^2$ is of order $\log\log n/\log n$. The numerical values below, computed from the moment convolution $L_{n,t}(0) = \frac{1}{4}\sum_j M_j M_{n-j}/((2j)!(2n-2j)!)$ with $M_k = \int u^{2k} e^{tu^2}\Phi_{\mathrm{dBN}}(u)\,du$ (`code/newton_central_ratio.py`), suggest that the error is $o(1)$.

n	$c_n(\frac12)$	$c_n(0)$	$\frac18\mathcal{W}(n/\pi)^2$
10^2	0.82596	0.82413	0.80112
10^3	2.32317	2.32216	2.31510
10^4	4.86185	4.86152	4.85998
10^5	8.52286	8.52278	8.52250
10^6	13.35216	13.35215	13.35210

The difference $c_n(\frac12) - \frac18\mathcal{W}(n/\pi)^2$ is 0.025, 0.0081, 0.0019, 0.00035 and 0.00006, and $c_n(0)$ differs from $c_n(\frac12)$ by 3×10^{-4} at $n=10^4$. At $n=10^5$, $\frac18(\log n)^2 = 16.6$ against $c_n = 8.52$, so an effective version of Theorem 8.4.22 must use the Lambert centring.

Theorem 8.4.22 (Newton). *For every $X<\infty$, uniformly in $0\le t\le\frac12$ and $|x|\le X$,*

$$\frac{k_n L_{n+1,t}(x)}{L_{n,t}(x)} = c_n(t)\,(1+o(1)) = \tfrac18(\log n)^2(1+o(1)).$$

Proof. With $r_n(t,x) = L_{n,t}(x)/L_{n,t}(0) = \mathbb{E}_{n,t}\cos(xw)$, Theorem 8.4.20 gives $r_n\to1$ uniformly, and $k_nL_{n+1,t}(x)/L_{n,t}(x) = c_n(t)\,r_{n+1}/r_n$; the second equality is Proposition 8.4.21 with $\mathcal{W}(n/\pi)\sim\log n$, used as stated. □

The result has the shape of the theorem of Griffin, Ono, Rolen and Zagier [99] (Problem 16.4): all large orders at each fixed window, for the Laguerre expressions along the whole flow. The index N_X is not effective, and effectivity in X is the open problem here: it would turn the positivity of all $L_{n,t}$ on $|x|\le X$ into a finite computation in $n<N_X$.

8.4.10 Forward persistence of real zeros

The motion $\dot z_j = 2\sum_{k\ne j}1/(z_j-z_k)$ of the zeros of H_t (Section 8.4.1) says nothing at a collision, where the sum diverges. The lemma below and the barrier argument after it come from the authors' discussion with a collaborator.

Lemma 8.4.23. *Let c be a real zero of multiplicity $m\ge1$ of H_{t^*}. For $t<t^*$ close to t^*, the m zeros of H_t near c are $c+i\sqrt{2(t^*-t)}\,x_k+o(\sqrt{t^*-t})$, $k=1,\dots,m$, with x_k the roots of He_m, so at most one of them is real, and for $t>t^*$ close to t^* all m of them are real and simple. Consequently real zeros of H_t never collide as t increases; a real zero stays real, and a simple real zero is simple except at times when zeros arriving from the complex plane meet it; and nonreal zeros of H_τ cannot accumulate at a real point as τ decreases to a limit τ_1.*

Proof. Put $t=t^*+s$ and expand $H_t(c+w)=\sum a_{jk}w^js^k$ (a convergent series, since $H_t(z)$ is an entire function of $(z,t)\in\mathbb{C}^2$, by the decay of Φ), and give w weight 1 and s weight 2. The lowest weighted degree part P satisfies $\partial_sP=-\partial_w^2P$, because $\partial_tH_t=-\partial_z^2H_t$ and the two sides have the same weighted degree after the

shift, so $P(w,s) = e^{-s\partial_w^2}P(w,0)$. The part of P with s^0 is $a\,w^d$, and $a \neq 0$, since otherwise $P = 0$; as $H_{t^*}(c+w) = a_{m0}w^m + \cdots$ with $a_{m0} \neq 0$, we get $d = m$ and $P = a\,h_m$ with $h_m(w,s) = e^{-s\partial_w^2}w^m$. For $s = \mp\sigma$ the substitution $w = \sqrt{2\sigma}\,x$ turns h_m into $(2\sigma)^{m/2}e^{\pm\partial_x^2/2}x^m$. For $s > 0$ this is $(2s)^{m/2}\mathrm{He}_m(x)$, with $\mathrm{He}_m = e^{-\partial_x^2/2}x^m$ the Hermite polynomial, whose m roots are real and simple. For $s < 0$ put $w = i\sqrt{2|s|}\,x$ instead: $h_m(w,s) = (i\sqrt{2|s|})^m\mathrm{He}_m(x)$, so the zeros of h_m are $i\sqrt{2|s|}$ times the roots of He_m, simple and purely imaginary, with $w = 0$ among them exactly for odd m. Now write $w = \sqrt{|s|}\,\omega$ (respectively $\sqrt{s}\,\omega$): every monomial $w^j s^k$ becomes $(\operatorname{sgn} s)^k|s|^{(j+2k)/2}\omega^j$, so $|s|^{-m/2}H_t(c+\sqrt{|s|}\,\omega) \to a\,h_m(\omega,\pm1)$ uniformly on compact sets as $s \to 0$, and the limit has m simple zeros. By Rouché's theorem, in small discs around the zeros of the limit the function has exactly one zero each; by Hurwitz's theorem applied to $H_t(c+w)$ on a fixed circle $|w| = r$, there are exactly m zeros in $|w| < r$, so these are all of them. For $s < 0$ the discs are disjoint, and all lie off the real axis except the one at $\omega = 0$ (odd m), which contains exactly one zero; for $s > 0$ the zeros are real because they are simple, near a real simple zero of the limit, and the zero set is symmetric under conjugation. The consequences follow: two real zeros meeting at c at time t^* would give two real zeros near c for $t < t^*$; a real zero stays real by the symmetry (for a multiple one, by the case $t > t^*$), since a simple zero near a real simple zero is unique; and nonreal zeros of H_{τ_k}, $\tau_k \downarrow \tau_1$, with a limit point $c \in \mathbb{R}$ would give nonreal zeros near c just after τ_1, against the case $s > 0$. □

For a height X let $0 = t_0 < t_1 < \cdots < t_M = 0.2$ and let $Q_i = [u_i, v_i] \times [-1,1]$ be rectangles in the z plane with $u_i < X < v_i$. The hypothesis $B(X)$ is:

(B0) every zero of H_0 with $|\operatorname{Re} z| \le u_0$ is real and simple (Platt and Trudgian [247] verified this to 6×10^{12} in z);

(B1) for each i, H_{t_i} has no zero on ∂Q_i, and the number of zeros of H_{t_i} in Q_i (counted with multiplicity by the argument principle) is at most the number of sign changes of H_{t_i} observed on $[u_i, v_i]$;

(B2) for each i and every $t \in [t_i, t_{i+1}]$, $|H_t - H_{t_i}| < |H_{t_i}|$ on ∂Q_i.

Proposition 8.4.24. *Assume $B(X)$. Then for every $\lambda \in [0, 0.2]$ every zero of H_λ with $|\operatorname{Re} z| \le X$ is real and simple.*

Proof. Fix i. By (B2) and Rouché's theorem, H_t has no zero on ∂Q_i and the same number N_i of zeros in Q_i for all $t \in [t_i, t_{i+1}]$. At t_i a sign change is a real zero of odd multiplicity, so by (B1) the N_i zeros are real and simple. Let A be the set of $t \in [t_i, t_{i+1}]$ at which the N_i zeros in Q_i are real and simple. It contains t_i, and it is open in $[t_i, t_{i+1}]$, since a simple real zero persists as a simple zero, real by symmetry. Let τ be the supremum of the t with $[t_i, t] \subset A$ and take $t'_k \uparrow \tau$ in A: the zeros of H_τ in Q_i are limits of real zeros (Hurwitz's theorem, with no zero on the boundary), hence real, and two of them cannot coincide, since the N_i

distinct real zeros of $H_{t_k'}$ near the common point would contradict Lemma 8.4.23; so there are N_i distinct real zeros, all simple, and $\tau \in A$. As A is open, $\tau = t_{i+1}$, so $A = [t_i, t_{i+1}]$, and every zero of H_t in Q_i is real and simple for $t \in [t_i, t_{i+1}]$. The same holds for the mirror image $-Q_i$.

Now let $\lambda \in [0, 0.2]$ and suppose H_λ has a nonreal zero z_0 with $|\operatorname{Re} z_0| \le X$. By de Bruijn's strip theorem [73, Theorem 13], the zeros of H_t satisfy $|\operatorname{Im} z| \le \sqrt{1-2t}$ for $0 \le t \le \frac{1}{2}$ (and $|\operatorname{Im} z| < 1$ at $t = 0$), so no zero meets the lines $\operatorname{Im} z = \pm 1$ and every zero of H_τ with $|\operatorname{Re} z| \le X$ lies in the compact set $K = \{|\operatorname{Re} z| \le X,\ |\operatorname{Im} z| \le 1\}$. Let F be the set of $\tau \in [0, \lambda]$ at which H_τ has a nonreal zero in K; then $\lambda \in F$. For $\tau \in [t_i, t_{i+1}]$ the first part shows that such a zero has $|\operatorname{Re} z| < u_i < X$ (otherwise it would lie in Q_i or $-Q_i$), and $|\operatorname{Im} z| < 1$, so it is an interior point of K and F is open in $[0, \lambda]$ by Hurwitz's theorem. If $\tau_k \downarrow \tau$ with $\tau_k \in F$, a limit point of the nonreal zeros is a zero of H_τ in K, and it is nonreal by the case $s > 0$ of Lemma 8.4.23. Hence $\tau_0 = \inf\{\tau : [\tau, \lambda] \subset F\}$ lies in F, openness forces $\tau_0 = 0$, and $0 \in F$ gives a nonreal zero of H_0 with $|\operatorname{Re} z| \le X$; by (B0) it has $|\operatorname{Re} z| > u_0$, so it lies in $\pm Q_0$, against the first part. Hence every zero of H_λ with $|\operatorname{Re} z| \le X$ is real. The constants in $B(X)$ do not change if X is replaced by $X' \in (X, \min_i v_i)$, so the same holds for $|\operatorname{Re} z| \le X'$. For simplicity at $\lambda = 0$ use (B0) and the first part. For $\lambda > 0$, a real zero c of multiplicity $m \ge 2$ with $|c| \le X$ would by Lemma 8.4.23 have at least $m - 1 \ge 1$ nonreal zeros of H_t near c for $t < \lambda$ close to λ, with $|\operatorname{Re} z| \le X'$, which is impossible. □

The hypothesis $B(X)$ has not been verified at any height X, and no computation is claimed. The verification intended is the count barrier of the collaborator: evaluate H_{t_i} on the boundary of Q_i and on its real diameter at the grid times, and check (B1) and (B2) there. The status of the problem is recorded in Remark 14.15.1.

Two results of Newton [213] bear on the verification of $B(X)$. The first is the local form of (B2). Let $t_* \ge 0$ and let $a < b$ be adjacent real simple zeros of H_{t_*} with $g_* = b - a$; let $0 < r < g_*/2$ and let Ω be a conjugation invariant Jordan domain containing the closed discs D_a, D_b of radius r about a, b and the segment between them, with no other zero of H_{t_*} in $\overline{\Omega}$. Let $\Gamma = \partial D_a \cup \partial D_b \cup \partial\Omega$, $Y = \max_\Gamma |\operatorname{Im} z|$, $m = \min_\Gamma |H_{t_*}|$ and $\psi_Y(\sigma) = \int_0^\infty \Phi_{\mathrm{dBN}}(u) e^{t_* u^2 + Yu}(1 - e^{-\sigma u^2})\, du$.

Proposition 8.4.25 (Newton). *If $\psi_Y(\sigma) < m$, then for $0 \le \tau \le \sigma$ the function $H_{t_* - \tau}$ has exactly one zero in each of D_a, D_b, both real and simple, and no other zero in Ω; so the gap satisfies $g \ge g_* - 2r$ and, by* (8.4.2), *$4 \int \mathrm{Def} \le g_*^2 - (g_* - 2r)^2$ over the interval.*

Proof. Since $|\cos(zu)| \le e^{|\operatorname{Im} z| u}$, on Γ we have $|H_{t_* - \tau} - H_{t_*}| \le \psi_Y(\tau) \le \psi_Y(\sigma) < m \le |H_{t_*}|$. By Rouché's theorem $H_{t_* - \tau}$ has as many zeros in D_a, D_b and Ω as H_{t_*}, namely 1, 1 and 2. A single zero in a disc centred on the real axis is real by conjugation symmetry and is simple, and Ω has no further zero. □

This is the mechanism of (B2) in backward time, with a rectangle replaced by a pair of discs. The second result shows what it cannot do. The decay of H_t along the real axis is that of Ξ: for $0<\delta<\pi/8$, Φ_{dBN} is analytic in $|\operatorname{Im} u|\le\delta$, where $|\exp(-\pi n^2 e^{4(u+i\eta)})|=\exp(-\pi n^2 e^{4u}\cos 4\eta)$ and, Φ_{dBN} being even, $\Phi_{\mathrm{dBN}}(u+i\eta)=\Phi_{\mathrm{dBN}}(-u-i\eta)$ decays as fast as $u\to-\infty$, and shifting the contour $u\mapsto u+i\delta$ in $H_t(z)=\int_{\mathbb{R}}\nu_t(u)e^{izu}\,du$ gives, for $\operatorname{Re} z=x>0$ and $|\operatorname{Im} z|\le 1$,

$$|H_t(z)|\le C(\delta,t)\,e^{-\delta x},\qquad C(\delta,t)=\tfrac12\int_{\mathbb{R}}|\Phi_{\mathrm{dBN}}(u+i\delta)|\,e^{t(u^2-\delta^2)+|u|}\,du<\infty.$$

On the other hand $\psi_Y(\sigma)\ge\psi_0(\sigma)$ for every $Y\ge0$, and $\psi_0(\sigma)>0$ does not depend on the height.

Proposition 8.4.26 (Newton, with the mechanism corrected). *Fix $t_*\ge0$, $\sigma>0$ and $0<\delta<\pi/8$. For every real zero $a\ge1$ of H_{t_*} and every contour Γ containing a point $a+i\eta$ with $0<\eta\le1$, we have $\min_\Gamma|H_{t_*}|\le C(\delta,t_*)e^{-\delta a}<\psi_0(\sigma)\le\psi_Y(\sigma)$ as soon as $a>\delta^{-1}\log\bigl(C(\delta,t_*)/\psi_0(\sigma)\bigr)$. So no criterion of the form $\psi_Y(\sigma)<m$ certifies a fixed step $\sigma>0$ at every height, and the step it can certify at height a is $O(e^{-\delta a})$.*

Proof. $|H_{t_*}(a+i\eta)|\le C(\delta,t_*)e^{-\delta a}$ by the display, and $\psi_Y\ge\psi_0>0$. For $\sigma\le1$, $1-e^{-x}\ge x/(1+x)$ gives $\psi_0(\sigma)\ge\sigma\int_0^\infty\Phi_{\mathrm{dBN}}(u)e^{t_*u^2}u^2(1+u^2)^{-1}\,du$, so a certified step is $O(e^{-\delta a})$. □

Newton proved this for pairs with gaps tending to 0, but the obstruction is the decay $e^{-\pi x/8}$ of H_t and not the gaps: it holds at every zero. It concerns bounds through the absolute value of the integrand $\Phi_{\mathrm{dBN}}e^{tu^2}(1-e^{-\sigma u^2})\cos(zu)$; bounds on $H_{t_*-\tau}-H_{t_*}$ that use H itself are not excluded. Jensen's inequality for the convex map $x\mapsto e^{-\sigma x}$ gives $\psi_Y(\sigma)\le A(1-e^{-\sigma M/A})\le 2A\sigma M/(2A+\sigma M)$, with A and M the mass and first moment of the image of $\Phi_{\mathrm{dBN}}(u)e^{t_*u^2+Yu}\,du$ under $x=u^2$ (the second step is $2\operatorname{artanh}(x/2)\ge x$); it improves the linear bound σM only when σM is comparable to A, and it bounds backward steps only: the forward profile $\int(e^{\tau x}-1)\,\mu(dx)$ of (B2) is convex in x, and Jensen then gives a lower bound.

The practical consequence concerns (B2). On ∂Q_i the minimum of $|H_{t_i}|$ is at most $C(\delta,t_i)e^{-\delta v_i}$, with $v_i>X$, while an absolute value bound for $|H_t-H_{t_i}|$ is at least $\psi_0(t-t_i)$, of order $t-t_i$ by $1-e^{-x}\ge x/(1+x)$. A step certified that way must therefore be $O(e^{-\delta X})$ for every $\delta<\pi/8$, about 10^{-3400} at $X=2\times10^4$, and (B2) cannot be checked by it at an appreciable height. The perturbation has to be bounded through H: by Taylor's formula in t with $\partial_t^k H_t=(-1)^k H_t^{(2k)}$ and Cauchy estimates from $|H_t|$ on a larger contour at the same height, in ball arithmetic, or by a Riemann–Siegel type approximation with relative error, as in the Polymath project [282].

8.5 Beyond zeta: the Dedekind eta kernel, $L(s,\chi_{12})$ and Ramanujan's Δ

The Pólya route uses little that is special to ζ. The zero density identity of Section 3.4 needs a completed function that is positive on the real axis and zero free on $\operatorname{Re} s = 1$, and the sine squared decomposition (3.1.1) needs an Euler product. The completed L functions of the real character χ_{12} modulo 12 and of Ramanujan's cusp form Δ have these properties, and their kernels, built from the Dedekind eta function, are positive by the product formula. From the eta kernel alone, the zero density identity and a count of sign changes place every zero of $L(s,\chi_{12})$ below height 100 on the critical line, which is the generalised Riemann hypothesis (GRH) for χ_{12} on that window. The first author's note on the Thorin measures of the Dedekind eta function asks what the eta function says about GRH for χ_{12} [255, Theorem 12]. Section 11.3 returns to $L(s,\chi_{12})$ at the basepoint one.

8.5.1 The eta kernel and GRH for $L(s,\chi_{12})$ below height 100

The character χ_{12} is the even primitive character modulo 12: $\chi_{12}(n) = 1$ for $n \equiv \pm 1$, -1 for $n \equiv \pm 5 \pmod{12}$, and 0 otherwise. Euler's pentagonal number theorem gives

$$\eta(ix) = e^{-\pi x/12}\prod_{n\ge 1}\bigl(1-e^{-2\pi n x}\bigr) = \sum_{n\ge 1}\chi_{12}(n)\,e^{-\pi n^2 x/12}, \qquad x>0,$$

so the theta series of χ_{12} is the Dedekind eta function on the imaginary axis. The completed L function

$$\xi(s,\chi_{12}) = \Bigl(\frac{12}{\pi}\Bigr)^{s/2}\Gamma\Bigl(\frac{s}{2}\Bigr)L(s,\chi_{12}) = \int_0^\infty \eta(ix)\,x^{s/2}\frac{dx}{x}$$

satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ because $\eta(i/x) = \sqrt{x}\,\eta(ix)$, the case $z = ix$ of $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$. With $x = e^{2u}$,

$$\xi\Bigl(\frac12 + s,\chi_{12}\Bigr) = \int_{\mathbb{R}} K(u)\,e^{su}\,du, \qquad K(u) = 2e^{u/2}\eta\bigl(ie^{2u}\bigr),$$

and the modular transformation makes K even. The product formula makes $\eta(ix) > 0$, so K is positive and $K/\xi(\frac12,\chi_{12})$ is a probability density, the analogue of the Pólya density: if Y has this density, then $\mathbb{E}e^{sY} = \xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ for every complex s. Since $\eta(ie^{2u})$ is analytic for $|\operatorname{Im} 2u| < \pi/2$, K is analytic in the same strip $|\operatorname{Im} u| < \pi/4$ as Φ, and like Φ it decays doubly exponentially, so the trapezoid analysis of Section 2.1 applies unchanged.

The construction extends to every real primitive character χ modulo m, whose root number is always one. For even χ the kernel is $2e^{u/2}\sum_{n\ge1}\chi(n)e^{-\pi n^2 e^{2u}/m}$; for odd χ, whose completed function is $(m/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$, it is

$2e^{3u/2}\sum_{n\ge1} n\chi(n)e^{-\pi n^2e^{2u}/m}$. Both kernels are even and analytic in $|\operatorname{Im} u|<\pi/4$, but their positivity has to be checked case by case.

The two polynomial evaluator of Section 2.1, run with the eta kernel, agrees with `mpmath`'s Dirichlet L function [205] to sixteen digits. Sign changes of $\xi(\frac12+i\theta,\chi_{12})$ on a grid of step 0.02, refined by Newton's method, give 67 zeros below height 100, the first of them at

$$3.8046276,\ 6.6922233,\ 8.8905930,\ 11.1883927,\ 12.9661788,\ 15.1814809,\ \dots.$$

The Thorin density at resolution ε, $\rho_\varepsilon(\theta)=\pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta,\chi_{12})$ as in (1.2.7), is again a sum of Poisson kernels over the zeros, by the Hadamard product (Proposition 1.2.3).

The zero density identity, Proposition 3.4.2, holds for $\xi(s,\chi_{12})$ with the same proof, since $\xi(s,\chi_{12})$ is entire, satisfies $\xi(s,\chi_{12})=\xi(1-s,\chi_{12})$, is positive on the real axis because K is, and has no zeros with $\operatorname{Re} s\ge1$. Its hypotheses are also the same. In it $N_{\chi_{12}}(T)$ counts the zeros with $0<\operatorname{Im}\rho\le T$, and $\nu(\sigma,T)$ those among them with $\operatorname{Re}\rho>\sigma$; $M_\varepsilon(T)=\int_0^T\rho_\varepsilon(\theta)\,d\theta$ is the Thorin mass up to height T, and $\tau_\varepsilon(T)$ is the change of argument of $\xi(s,\chi_{12})$ from $\frac12+\varepsilon+iT$ to $\frac12+iT$, divided by π. Table 8.15 evaluates the identity at $T=100$.

ε	$M_\varepsilon(100)$	$\tau_\varepsilon(100)$	$M_\varepsilon+\tau_\varepsilon=N-2\nu$
0.5	67.604974186	−0.604974186	67.00000000000003
0.25	67.476432836	−0.476432836	67.00000000010355
0.1	67.308171973	−0.308171973	67.00000000014663
0.05	67.185437421	−0.185437421	67.00000000009639
0.02	67.080023774	−0.080023774	67.00000000005207

Table 8.15: The zero density identity for $L(s,\chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100)-2\nu(\frac12+\varepsilon,100)$; the row $\varepsilon=\frac12$ gives $N_{\chi_{12}}(100)$.

The row $\varepsilon=\frac12$ gives $N_{\chi_{12}}(100)=67$, and with the 67 sign changes every zero below height 100 is simple and lies on the critical line. The other rows give $\nu(\frac12+\varepsilon,100)=0$ without using the sign changes. This is GRH for χ_{12} below height 100, computed, not certified: the argument is followed on a grid as in Section 3.4, and its increments are summed in double precision. The deviations from 67, at most 1.5×10^{-10}, are far below the one half that fixes the integer. GRH for small moduli has long been verified to far greater heights by other methods, by Rumely [293] and later by Platt [245].

8.5.2 Ramanujan's Δ and the cost of the three evaluators

The same construction works for Ramanujan's cusp form $\Delta(z)=\eta(z)^{24}=\sum_{n\ge1}\tau(n)q^n$, $q=e^{2\pi iz}$, of weight 12 and level 1. Its L function, nor-

malised so that the critical line is $\operatorname{Re} s = \frac{1}{2}$, is $L(s,\Delta) = \sum_{n\geq 1} \tau(n) n^{-s-11/2}$, and the completed function $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ satisfies $\xi(s,\Delta) = \xi(1-s,\Delta)$. The Mellin transform of $\Delta(iy)$ gives the Pólya type representation

$$\xi\Big(\frac{1}{2}+s,\Delta\Big) = \int_{\mathbb{R}} K_\Delta(u)\, e^{su}\, du,$$
$$K_\Delta(u) = e^{6u}\,\Delta(ie^u) = e^{6u}e^{-2\pi e^u}\prod_{n\geq 1}\big(1-e^{-2\pi n e^u}\big)^{24}.$$

The kernel is even because $\Delta(i/y) = y^{12}\Delta(iy)$, positive by the product formula, and analytic in the strip $|\operatorname{Im} u| < \pi/2$.

We computed $\tau(n)$ for $n \leq 3000$, the range needed for the prime weights of Section 8.5.3, and evaluated the kernel by its q expansion. At 60 digits the trapezoid evaluator finds the first zeros on the critical line at

$$9.222379,\ 13.907550,\ 17.442777,\ 19.656513,\ 22.336104,\ \ldots,$$

in agreement with the known values [188, modular form 1.12.a.a], and 20 sign changes below height 50, the number of zeros there.

The cost of the three evaluators, for Δ, ζ and $L(s,\chi_{12})$ at 60 digits (settings, digits lost per unit height and time per point), is compared in `code/README.md`.

8.5.3 The sine squared lemma without a pole

The eta based L functions are entire, so two consequences of the pole of ζ disappear: the negative pole piece of the sine squared decomposition, and the factor that Chapters 6 and 7 need to remove the pole, which by Proposition 7.1.13 alone prevents the normalised reciprocal zeta function from being a GGC. A floor in the sense of Theorem 6.2.15 is still needed for χ_{12}, since its tilt, the factor $e^{-c\sqrt{u}}$ in the variable u of the clock (written $e^{-c\sqrt{s}}$ in Section 6.2), removes the negative part of the Thorin density of the gamma factor, and the gamma factor $(12/\pi)^{s/2}\Gamma(s/2)$ of $\xi(s,\chi_{12})$ is that of ξ times $12^{s/2}$: Theorem 6.2.12 holds for it with $c = 2.6861$ replaced by $c - \frac{1}{2}\log 12 = 1.4436$. The positivity of such a floor would need an analogue of Theorem 7.1.1.

For $f = \chi_{12}$ or Δ write $\xi(s,f) = L_\infty(s,f)L(s,f)$, with L_∞ the gamma factor, and $-(L'/L)(s,f) = \sum_{n\geq 2}\Lambda_f(n)n^{-s}$ for $\operatorname{Re} s > 1$. At a basepoint $\alpha > 1$ the sine

squared lemma gives, as (3.1.1) does for ζ,

$$v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,f) = \underbrace{\frac{\xi'}{\xi}(\alpha,f)}_{\text{drift}} + \underbrace{\operatorname{Re}\frac{L'_\infty}{L_\infty}(\alpha+i\theta,f) - \frac{L'_\infty}{L_\infty}(\alpha,f)}_{\text{archimedean } \geq 0} + \underbrace{\sum_{n\geq 2}\Lambda_f(n)\,n^{-\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)}_{\text{primes}},$$

where $v_\alpha(\theta)$ is π times the Thorin density at resolution $\alpha-\frac12$, and the archimedean piece is nonnegative because the digamma function $\psi=\Gamma'/\Gamma$ satisfies $\operatorname{Re}\psi(x+iy)\geq\psi(x)$ for $x>0$. For ζ, whose gamma factor $\pi^{-s/2}\Gamma(1+\frac{s}{2})$ absorbs the factor s of ξ, the remaining factor $s-1$ adds the pole piece $\operatorname{Re}\frac{1}{s-1}-\frac{1}{\alpha-1}$, which is $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ at $s=\alpha+i\theta$, and the identity becomes (3.1.1).

The prime weights are $\Lambda(n)=\log p$ at $n=p^k$ for ζ, all positive; $\Lambda_{\chi_{12}}(p^k)=\chi_{12}(p)^k\log p$ for χ_{12}, of both signs and zero at $p=2$ and 3; and $\Lambda_\Delta(p^k)=(\alpha_p^k+\bar\alpha_p^k)\log p$ for Δ, where the Satake parameters α_p and $\bar\alpha_p$ are the roots of $X^2-\tau(p)p^{-11/2}X+1$. Deligne's theorem $|\tau(p)|\leq 2p^{11/2}$ [75] makes them complex conjugates of modulus one, so the weights of Δ lie in $[-2\log p,2\log p]$ and take both signs.

We computed the three decompositions at $\alpha=2$ for $0\leq\theta\leq 40$, with prime powers up to 3000, and compared their sum with the left side computed from the kernel. They agree to 5×10^{-4} for ζ and 9×10^{-6} for Δ, the difference being the truncation of the prime sum. For χ_{12} the largest difference, 2.5×10^{-5} at $\theta=40$, is an error of the kernel evaluator, whose truncation at $|u|\leq 2.5$ is too short at that height. Of the 466 prime power weights up to 3000, none is negative for ζ, 222 are negative for χ_{12} (whose 18 weights at powers of 2 and 3 vanish) and 230 for Δ.

On $0\leq\theta\leq 40$ the prime piece of $L(s,\chi_{12})$ ranges over $[-0.218,0.053]$, and that of $L(s,\Delta)$ over $[-0.261,0.222]$ (Figure 8.4, left). Their left sides are positive, with minima at $\theta=0$ equal to the drifts, 0.466 and 0.167, so the positivity comes from the drift and the archimedean piece, not from the primes.

For each of the three functions its Riemann hypothesis, RH for ζ and GRH for the other two, is the statement that v_α stays positive for every $\alpha>\frac12$; this is Lagarias's criterion [160], whose proof from the Hadamard product applies unchanged to $\xi(s,\chi_{12})$ and $\xi(s,\Delta)$. Below $\alpha=1$ the prime piece no longer converges absolutely. For $L(s,\chi_{12})$, Section 11.3.2 shows that the extension of the decomposition down to the centre is GRH for χ_{12}. At resolution 0.1 the Thorin densities of $L(s,\chi_{12})$ and $L(s,\Delta)$ have computed minima 0.011 and 0.004 below height 40 (Figure 8.4, right), positive as GRH predicts.

The distances of Section 3.2, with the zeros of $L(s,\chi_{12})$ or $L(s,\Delta)$ as the reference atoms, Theorem 4.3.5 and the flow of Section 8.4 carry over unchanged,

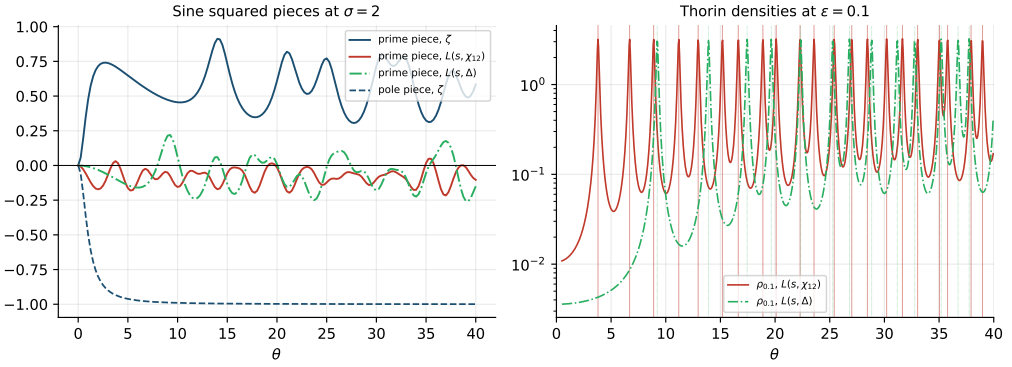


Figure 8.4: Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written σ in the plot, for ζ, $L(s,\chi_{12})$ and $L(s,\Delta)$, and the pole piece of ζ. For ζ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s,\chi_{12})$ and $L(s,\Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines).

the last two because they use only the Hadamard product, positivity on the real line and the decay of the kernel; we have not computed the first two for these functions, and Section 8.5.5 computes the flow for Δ. Theorem 4.3.5 also needs the bound $|\arg z| < \pi/4$ of Lemma 4.3.4 for the zeros z of $\xi(\frac{1}{2}+iz, f)$ with $\operatorname{Re} z > 0$, and it holds here too: the zeros lie in the open critical strip, so $|\operatorname{Im} z| < \frac{1}{2}$, while the lowest ordinates, 3.80 and 9.22, exceed $\frac{1}{2}$. The arithmetic side, however, offers no termwise positivity, even far to the right of the critical line, so the domination of the pole by positive primes that Sections 3.1 and 3.5 formulate for ζ has no direct analogue. Far enough to the right the absolute values do suffice: the drift and the archimedean piece outweigh the prime piece bounded by the absolute values of its weights for $\alpha > 1.8349$ for $L(s,\chi_{12})$ and for $\alpha > 2.2823$ for the Dirichlet beta function $L(s,\chi_4)$, and by Kronecker's theorem no better bound from the sizes of the weights holds at all large heights (Propositions 8.5.25 and 8.5.26). So neither eta based function is theoretically easier than ζ, although $L(s,\chi_{12})$ is the cheapest of the three per zero.

8.5.4 Maxima of exponentials

The product formula has a probabilistic reading, which the first author's note [256] develops. The factor $1-e^{-2\pi ny}$ is the distribution function of an exponential variable with rate $2\pi n$, so

$$F(y) = \prod_{n\geq 1} \left(1-e^{-2\pi ny}\right)^{24} = e^{2\pi y}\Delta(iy), \qquad F_1(y) = \prod_{n\geq 1} \left(1-e^{-2\pi ny}\right) = e^{\pi y/12}\eta(iy)$$

are the distribution functions of the maxima M^* and M_1 of independent exponential variables.

8.5.4.1 The max representation

LEMMA 8.5.1. *Let $E_{n,j}$, $n \ge 1$, $1 \le j \le 24$, be independent, with $E_{n,j}$ exponential of rate $2\pi n$, and put $M^* = \sup_{n,j} E_{n,j}$ and $M^{(j)} = \sup_n E_{n,j}$. Almost surely these suprema are attained and finite, $\Pr(M^* \le y) = F(y)$ for $y > 0$, and each $M^{(j)}$ has the distribution function F_1. So $M^* = \max_{j\le 24} M^{(j)}$ is the maximum of 24 independent copies of M_1.*

Proof. For $y > 0$, $\sum_{n,j} \Pr(E_{n,j} > y) = 24 \sum_n e^{-2\pi n y} < \infty$, so by the Borel Cantelli lemma only finitely many $E_{n,j}$ exceed y, almost surely. Taking y rational and below $E_{1,1}$, the supremum is the largest of finitely many values. By independence $\Pr(M^* \le y) = \lim_N \prod_{n\le N}(1 - e^{-2\pi n y})^{24} = F(y)$, and in the same way each $M^{(j)}$ has the distribution function F_1; the $M^{(j)}$ are independent because they are built from disjoint sets of the $E_{n,j}$. □

The modular relations $\Delta(i/y) = y^{12}\Delta(iy)$ and $\eta(i/y) = \sqrt{y}\,\eta(iy)$ become self dualities of the two laws,

$$F(y) = y^{-12} e^{2\pi y - 2\pi/y} F(1/y), \qquad F_1(y) = y^{-1/2} e^{\pi y/12 - \pi/(12y)} F_1(1/y), \quad (8.5.1)$$

so the lower tails are thin: $\Pr(M^* \le y) \sim y^{-12} e^{-2\pi/y}$ and $\Pr(M_1 \le y) \sim y^{-1/2} e^{-\pi/(12y)}$ as $y \downarrow 0$. Since $F(1/y) \le 1$, the first relation also gives $F(y) \le y^{-12} e^{2\pi y - 2\pi/y}$ for every $y > 0$, a bound that increases with y, because its logarithmic derivative is $(2\pi y^2 - 12y + 2\pi)/y^2$ and $144 < 16\pi^2$.

The density of M^* is Ramanujan's logarithmic derivative. Expanding $1/(1 - e^{-2\pi n y})$ gives $(\log F)'(y) = 24 \sum_n 2\pi n\, e^{-2\pi n y}/(1 - e^{-2\pi n y}) = 48\pi \sum_{m\ge 1} \sigma(m) e^{-2\pi m y}$, with $\sigma(m)$ the sum of the divisors of m, so that, with Eisenstein's series $E_2(z) = 1 - 24 \sum_{m\ge 1} \sigma(m) e^{2\pi i m z}$,

$$f^*(y) = F(y)\,(\log F)'(y) = 2\pi F(y)\big(1 - E_2(iy)\big). \quad (8.5.2)$$

Taking the logarithmic derivative of (8.5.1) gives the functional equation of the density,

$$f^*(1/y) = y^{12} e^{2\pi/y - 2\pi y}\Big(\big(2\pi y^2 - 12y + 2\pi\big)F(y) - y^2 f^*(y)\Big),$$

which in terms of E_2 is its quasi modularity, $E_2(i/y) = -y^2 E_2(iy) + 6y/\pi$ (`maxexp_checks.py`).

THEOREM 8.5.2. *For every complex s,*

$$\xi(s,\Delta) = (2\pi)^{-s-\frac{11}{2}}\, \mathbb{E}\,\Gamma\big(s + \tfrac{11}{2}, 2\pi M^*\big), \qquad L(s,\Delta) = \mathbb{E}\,Q\big(s + \tfrac{11}{2}, 2\pi M^*\big), \quad (8.5.3)$$

with $\Gamma(a,x)=\int_x^\infty e^{-v}v^{a-1}dv$ *and* $Q(a,x)=\Gamma(a,x)/\Gamma(a)$, *and both expectations converge absolutely. In particular* $L(s,\Delta)=0$ *at the points* $s=-\frac{11}{2}-k$, $k=0,1,2,\dots$, *where* $Q(-k,x)=0$.

Proof. Put $a=s+\frac{11}{2}$, so that $\xi(s,\Delta)=\int_0^\infty \Delta(iy)y^{a-1}dy$. Inserting $\Delta(iy)=e^{-2\pi y}\Pr(M^*\le y)=e^{-2\pi y}\mathbb{E}\,\mathbf{1}\{M^*\le y\}$ and integrating in y first gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{a-1}dy=(2\pi)^{-a}\mathbb{E}\,\Gamma(a,2\pi M^*)$. Fubini's theorem applies: with $\sigma=\operatorname{Re}a$, Tonelli's theorem gives $\mathbb{E}\int_{M^*}^\infty e^{-2\pi y}y^{\sigma-1}dy=\int_0^\infty \Delta(iy)y^{\sigma-1}dy$, which is finite for every real σ, since $\Delta(iy)$ decays exponentially as $y\to\infty$ and like $y^{-12}e^{-2\pi/y}$ as $y\downarrow 0$. The same computation gives $\mathbb{E}|\Gamma(a,2\pi M^*)|\le \mathbb{E}\,\Gamma(\sigma,2\pi M^*)<\infty$. Dividing by $(2\pi)^{-a}\Gamma(a)$, with $1/\Gamma$ entire, gives the second identity, in which $Q(a,x)$, for fixed $x>0$, is the entire function $\Gamma(a,x)/\Gamma(a)$ of a. At $a=-k$ it vanishes, since $\Gamma(-k,x)$ is finite and $1/\Gamma(-k)=0$. □

Corollary 8.5.3 (Gamma mixture and mixed Poisson forms). *Let* G_a *be a gamma variable of shape* $a>0$ *and scale one, and* N *a Poisson process of unit rate, both independent of* M^*. *For real* $s>-\frac{11}{2}$, *with* $a=s+\frac{11}{2}$,

$$L(s,\Delta)=\Pr\big(G_a>2\pi M^*\big)=\mathbb{E}F\Big(\frac{G_a}{2\pi}\Big)=\mathbb{E}\prod_{n\ge1}\big(1-e^{-nG_a}\big)^{24},$$

and for every integer $k\ge1$

$$L\big(k-\tfrac{11}{2},\Delta\big)=\Pr\big(N(2\pi M^*)\le k-1\big)=\mathbb{E}\Big[e^{-2\pi M^*}\sum_{j<k}\frac{(2\pi M^*)^j}{j!}\Big].$$

Proof. For real $a>0$, $Q(a,x)=\Pr(G_a>x)$, which gives the first form; conditioning on G_a instead gives $\Pr(M^*<G_a/2\pi)=\mathbb{E}F(G_a/2\pi)$, since the law of M^* has no atoms. For an integer k the kth point of N has the law of G_k, so $\Pr(G_k>x)=\Pr(N(x)\le k-1)$. □

So $L(-\frac{9}{2},\Delta)=\mathbb{E}e^{-2\pi M^*}=0.037441281268515541 74$ is a value of the Laplace transform of M^*, and the central value $L(\frac{1}{2},\Delta)=0.792122838646030569 36$ is the probability that a Poisson clock of rate 2π rings at most five times before the time M^*. Write $P_k=\Pr(N(2\pi M^*)\le k-1)=L(k-\frac{11}{2},\Delta)$. The functional equation $\xi(s,\Delta)=\xi(1-s,\Delta)$ at $s=k-\frac{11}{2}$ becomes the family of identities

$$(2\pi)^{-k}(k-1)!\,P_k=(2\pi)^{k-12}(11-k)!\,P_{12-k},\qquad 1\le k\le 11,\tag{8.5.4}$$

both sides being $\xi(k-\frac{11}{2},\Delta)$.

Remark 8.5.4. The identities (8.5.4) follow from the self duality (8.5.1), since $(2\pi)^{-k}(k-1)!\,P_k=\int_0^\infty e^{-2\pi y}F(y)y^{k-1}dy$ and the substitution $y\mapsto 1/y$ turns this integral into the one with $12-k$ in place of k. They are five relations, since k and $12-k$ give the same one and $k=6$ gives none, and they do not

imply the self duality. Let g be smooth and not identically zero, with compact support in $(0,1)$, such that $g(\log y)$ is orthogonal on $(1,e)$ to the five functions $e^{-2\pi y}F(y)(y^{k-1}-y^{11-k})$, $k\le 5$. For small ε, $\tilde F(y)=F(y)\big(1+\varepsilon g(\log y)\big)$ is again a distribution function, because F' has a positive lower bound and F is below one on $[1,e]$, and it satisfies (8.5.4), because the perturbation changes the difference of the two sides of the kth identity by ε times the integral of $g(\log y)$ against the kth function. It is not self dual: on $(1,e)$ self duality would give $\tilde F(y)=y^{-12}e^{2\pi y-2\pi/y}\tilde F(1/y)=F(y)$, since $g(\log(1/y))=0$. With g even and supported in $(-1,1)$ instead, the same construction gives a self dual distribution function other than F, so self duality does not determine the law of M^* either. What does determine it is self duality together with the form of F as a power series in $e^{-2\pi y}$. If $F(y)=\sum_{n\ge0}c_ne^{-2\pi ny}$ for $y>0$, with $c_0=1$, and F satisfies (8.5.1), then $f(z)=\sum_{n\ge0}c_ne^{2\pi i(n+1)z}$ is holomorphic on the upper half plane, satisfies $f(z+1)=f(z)$, and satisfies $f(-1/z)=z^{12}f(z)$ on the imaginary axis, hence everywhere by analytic continuation. So f is a cusp form of weight 12 for $\mathrm{SL}_2(\mathbb{Z})$, a multiple of Δ because these cusp forms make up a space of dimension one [128, Chapter 14], and $c_0=1$ gives $f=\Delta$. This argument is the simplest case of Hecke's converse theorem [113]. The distribution function of a maximum of independent exponential variables whose rates are integer multiples of 2π is a power series of this kind, so M^* is the only such maximum with the self duality (8.5.1).

Cutting the integral at $y=1$ and folding $(0,1)$ onto $(1,\infty)$ by modularity gives in the same way the symmetric form

$$\xi(s,\Delta)=A(s)+A(1-s),\qquad A(s)=\int_1^\infty \Delta(iy)\,y^{s+\frac92}\,dy, \tag{8.5.5}$$

$$A(s)=(2\pi)^{-s-\frac{11}{2}}\,\mathbb{E}\,\Gamma\big(s+\tfrac{11}{2},2\pi\max(M^*,1)\big)=\sum_{n\ge1}\tau(n)\,\frac{\Gamma(s+\frac{11}{2},2\pi n)}{(2\pi n)^{s+11/2}},$$

where the last form, Hecke's, comes from expanding $\Delta(iy)=\sum_n\tau(n)e^{-2\pi ny}$ instead. Hecke's form is the analogue for Δ of Riemann's second proof of the functional equation [81, Chapter 1], which cuts the Mellin integral of the theta function $\omega(x)=\sum_{n\ge1}e^{-\pi n^2x}$ of Section 1.2 at $x=1$ in the same way.

Proposition 8.5.5 (Riemann). *For every complex $s\ne0,1$,*

$$\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\zeta(s)=-\frac{1}{s}-\frac{1}{1-s}+\sum_{n\ge1}\Big[\frac{\Gamma(\frac{s}{2},\pi n^2)}{(\pi n^2)^{s/2}}+\frac{\Gamma(\frac{1-s}{2},\pi n^2)}{(\pi n^2)^{(1-s)/2}}\Big].$$

Proof. This is Riemann's second proof of the functional equation, cited above: split the Mellin integral of ω at $x=1$, apply the theta transformation on $(0,1)$, and integrate term by term on $(1,\infty)$. □

The three formulas are incomplete gamma continuations of Mellin integrals, but they randomise different things. In Riemann's and Hecke's the threshold is the fixed point $x = 1$ or $y = 1$, and the rates πn^2 or $2\pi n$ carry weights: one each for ζ, at the price of the pole terms, and the signed coefficients $\tau(n)$ for Δ. In (8.5.3) there is a single rate 2π, coming from the factor $q = e^{-2\pi y}$ of Δ, and the threshold M^* is random with a positive law; the symmetric form (8.5.5) is Riemann's formula with the sum over n replaced by an expectation over $\max(M^*, 1)$, and with no pole terms.

At $s = \frac{1}{2}$, $\frac{1}{2} + 5i$ and 3 the max representation, the symmetric form and Hecke's formula agree to 25 digits. On the critical line the symmetric form gives $\xi(\frac{1}{2} + i\theta, \Delta) = 2\,\mathrm{Re}\,A(\frac{1}{2} + i\theta)$. It also suggests a Hermite Biehler test, the route that Lagarias studied for ξ [161]: if $|A(s)| > |A(1-s)|$ held throughout $\mathrm{Re}\,s > \frac{1}{2}$, there could be no zeros there. The test fails. On $\mathrm{Re}\,s = 0.6$ the two halves have equal modulus at $\theta = 10.26$, 14.67, 18.04, 20.36 and 22.99, and the dominant half alternates between these ordinates, which separate the zeros 9.22, 13.91, 17.44, 19.66 and 22.34 (`maxlaw.py`, `maxexp_hb.py`). So the cut at $y = 1$ gives no Hermite Biehler function.

The cut can be moved. For every $c > 0$ the same folding gives $\xi(s, \Delta) = A_c(s) + A_{1/c}(1-s)$ with $A_c(s) = \int_c^\infty \Delta(iy) y^{s+9/2} dy$, and c may be chosen separately on each horizontal ray. The freedom is in any case illusory. As $c \to 0$, $A_{1/c}(1-s) \to 0$ and $A_c(s) \to \xi(s, \Delta)$ uniformly on compact sets, so domination for small c only says that $\xi(s, \Delta)$ does not vanish there, and the certificate turns into an evaluation of ξ; and at a zero $\frac{1}{2} + i\gamma$ the two halves have equal modulus for every c, since their sum vanishes. A Hermite Biehler function that could prove GRH for Δ would have to come from the structure of the law of M^*, not from tuning a cut. The representation is exact and has no signs, but like the kernel it restates the problem without locating the zeros.

8.5.4.2 Other theta products

The construction needs a modular form whose expansion on the imaginary axis is a power of q times a product of factors $1 - q^a$, $a > 0$, with nonnegative exponents, so that the product is a distribution function; eta products with nonnegative exponents qualify.

For χ_{12} the Mellin integral of Section 8.5.1, with $\eta(ix) = e^{-\pi x/12} F_1(x)$, gives in the same way

$$L(s, \chi_{12}) = \mathbb{E}\, Q\Big(\frac{s}{2}, \frac{\pi}{12} M_1\Big), \qquad \xi(s, \chi_{12}) = A_{12}(s) + A_{12}(1-s),$$

with $A_{12}(s) = \int_1^\infty \eta(ix) x^{s/2-1} dx = (\frac{\pi}{12})^{-s/2} \mathbb{E}\,\Gamma(\frac{s}{2}, \frac{\pi}{12}\max(M_1, 1))$; the second exponent comes from $\eta(i/x) = \sqrt{x}\,\eta(ix)$. At $s = 1$, where $Q(\frac{1}{2}, x) = \mathrm{erfc}\,\sqrt{x}$, the class number formula reads $\mathbb{E}\,\mathrm{erfc}\,\sqrt{\pi M_1/12} = L(1, \chi_{12}) = \log(2+\sqrt{3})/\sqrt{3}$; at $s = 2$, where $Q(1, x) = e^{-x}$, $L(2, \chi_{12}) = \mathbb{E} e^{-\pi M_1/12} = \pi^2/(6\sqrt{3})$ is a value

of the Laplace transform of M_1, which Section 8.5.4.4 computes in closed form; and at the centre $L(\frac{1}{2},\chi_{12}) = \Pr(12G_{1/4} > \pi M_1) = 0.498557002457815436157$. At $s = 0.6 + 5i$ the max representation, both forms of the symmetric form and `mpmath`'s Dirichlet L function agree to 25 digits, as do the three values above with their closed forms (`maxexp_checks.py`, `maxlaw.py`).

For the elliptic curve of conductor 11 the newform is $f(z) = \eta(z)^2\eta(11z)^2 = q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2 = \sum_{n\ge1} a_n q^n$ [188, modular form 11.2.a.a], so $f(iy) = e^{-2\pi y}F_{11}(y)$, where F_{11} is the distribution function of the maximum M_{11} of independent exponential variables, two of each rate $2\pi n$ and two of each rate $22\pi n$. The Fricke relation $f(i/(11y)) = 11y^2 f(iy)$ makes $f(iy)$ decay faster than any power at both ends, and the proof of Theorem 8.5.2 gives, in the arithmetic normalisation $L(s,E) = \sum_n a_n n^{-s}$ with centre $s = 1$,

$$L(s,E) = \mathbb{E}\,Q(s, 2\pi M_{11}) \quad \text{for every complex } s, \qquad L(1,E) = \mathbb{E}e^{-2\pi M_{11}}.$$

The max representation gives $L(1,E) = 0.2538418608559106843378$, equal to all digits to the rapidly convergent series $2\sum_n a_n n^{-1}e^{-2\pi n/\sqrt{11}}$ of root number one.

Jacobi's triple product gives $\vartheta_4(x) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-\pi n^2 x} = \prod_{m\ge1}(1-e^{-2\pi m x})(1-e^{-\pi(2m-1)x})^2$, a distribution function with no prefactor: that of the maximum M_4 of independent exponential variables, one of each rate $2\pi m$ and two of each rate $\pi(2m-1)$. The partial sums of the alternating series lie between -1 and 1, so the series can be integrated term by term against $\theta e^{-\theta y}$, and the partial fraction expansion of $z/\sinh z$ gives

$$\mathbb{E}e^{-\theta M_4} = \sum_{n\in\mathbb{Z}}(-1)^n\frac{\theta}{\theta+\pi n^2} = \frac{\sqrt{\pi\theta}}{\sinh\sqrt{\pi\theta}} = \prod_{n\ge1}\Big(1+\frac{\theta}{\pi n^2}\Big)^{-1},$$

so $M_4 \overset{d}{=} \sum_{n\ge1} E_n/(\pi n^2)$ with independent standard exponential variables E_n: a GGC with Thorin measure $\sum_n \delta_{\pi n^2}$. Since $\Pr(K \le k) = \sum_n (-1)^n e^{-2n^2k^2} = \vartheta_4(2k^2/\pi)$, $\pi M_4 = 2K^2$ has the Kolmogorov law of Table 1.8. With no prefactor the Mellin transform gives moments instead of incomplete gamma functions: $\mathbb{E}M_4^{s/2} = \frac{s}{2}\int_0^\infty y^{s/2-1}(1-\vartheta_4(y))\,dy$ for $\operatorname{Re} s > 0$, and integrating $1-\vartheta_4(y) = 2\sum_{n\ge1}(-1)^{n+1}e^{-\pi n^2 y}$ term by term, for $\operatorname{Re} s > 1$ and then by analytic continuation, gives

$$(1-2^{1-s})\,\zeta(s) = \frac{\mathbb{E}\big[(2K^2)^{s/2}\big]}{2\,\Gamma(1+\frac{s}{2})}$$

for every complex s, both sides being entire, since $\vartheta_4(y) = y^{-1/2}\sum_n e^{-\pi(n+\frac12)^2/y}$ makes every moment of M_4 finite. This is a case of the dictionary of Biane, Pitman and Yor [32]. At $s = 1$ it gives $\mathbb{E}K = \sqrt{\pi/2}\,\log 2 = 0.868731160636159141830$, which quadrature of $1-\vartheta_4$ reproduces to 2×10^{-23}; at $s = 3$ and $\frac12 + 5i$ the two sides agree to 25 digits.

For ζ itself the construction stops. The theta function $\vartheta_3(x) = 1 + 2\omega(x) = \prod_{m\ge1}(1-q^{2m})(1+q^{2m-1})^2$, $q = e^{-\pi x}$, has factors larger than one and is not a distribution function, and the transformation $\vartheta_4(1/x) = \sqrt{x}\,\vartheta_2(x)$ leads to $\vartheta_2(x) = \sum_n e^{-\pi(n+\frac12)^2 x}$, which is not one either. So the max structure reaches ζ only through the factor $1-2^{1-s}$, in moment form. It does not reach it through $\zeta_{\mathbb{Q}(\sqrt{3})}(s) = \zeta(s)L(s,\chi_{12})$: the eta law represents the factor $L(s,\chi_{12})$ only, and no product of this kind can represent a Dedekind zeta function, since an eta product with nonnegative exponents decays faster than any power of y at both ends of the imaginary axis, so its Mellin transform is entire, as in the proof of Theorem 8.5.2, while $\zeta_{\mathbb{Q}(\sqrt{3})}$ has a pole at $s=1$.

8.5.4.3 Gamma shapes and infinite divisibility at real points

The Pólya kernel of ζ and the kernel K_Δ of Δ differ in their gamma shapes. On the logarithmic scale the function $x^a e^{-x}$ of $x = ce^{ku}$ is proportional to the density of $(\log G_a - \log c)/k$, a log gamma density of shape a, and its Fourier transform $\int_{\mathbb{R}} x^a e^{-x} e^{i\theta u}du = k^{-1}c^{-i\theta/k}\Gamma(a+i\theta/k)$ has no zeros. With $x = 2\pi e^u$, $K_\Delta(u) = e^{6u}\Delta(ie^u) = (2\pi)^{-6}x^6e^{-x}F(e^u)$ is a single gamma kernel of shape 6 times the distribution function of $\log M^*$, and the gamma kernel's own transform is $(2\pi)^{-6-i\theta}\Gamma(6+i\theta)$, so every zero of $\xi(\frac12+i\theta,\Delta)$ is produced by the multiplier $F(e^u)$. The Pólya kernel (1.2.3) is, with $x_n = \pi n^2 e^{2u}$ and the gamma densities $g_k(x) = x^{k-1}e^{-x}/\Gamma(k)$,

$$\Phi(u) = 2e^{u/2}\sum_{n\ge1}\big(2x_n^2 - 3x_n\big)e^{-x_n} = 2e^{u/2}\sum_{n\ge1}\big(4g_3(x_n) - 3g_2(x_n)\big),$$

a signed mixture of Gamma(2) and Gamma(3) densities over the scales πn^2. As a function of $x = e^{2u}$ the sum is, up to the factor $x/2$, the density $2\sum_n(2\pi^2n^4x - 3\pi n^2)e^{-\pi n^2 x}$ of $\pi S_2/2$ in the representation $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Williams and of Biane, Pitman and Yor (Section 8.2.3). On the logarithmic scale, with the weight $e^{u/2} = (x_n/\pi n^2)^{1/4}$, the two terms have shapes $\frac94$ and $\frac54$, while the theta function itself contributes $x_n^{1/4}e^{-x_n}$, of shape $\frac14$.

Remark 8.5.6. Let $D = -x\,d/dx$. Integrating by parts, $\int_0^\infty (Df)(x)x^{s/2-1}dx = \frac{s}{2}\int_0^\infty f(x)x^{s/2-1}dx$ whenever the boundary terms vanish, so the operator $2D(D-\frac12)$ corresponds to the factor $\frac12 s(s-1)$ of $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$. It annihilates $-\frac12$ and $\frac12 x^{-1/2}$, the two terms by which $\omega(x)$ and $x^{-1/2}\omega(1/x)$ differ, and it maps e^{-x} to $(2x^2-3x)e^{-x}$. Hence $[2D(D-\frac12)\omega](x)$, which is also $2D(D-\frac12)$ applied to $x^{-1/2}\omega(1/x)$, decays faster than any power at both ends,

$$\xi(s) = \int_0^\infty \big[2D(D-\tfrac12)\omega\big](x)\,x^{s/2-1}dx \quad \text{for every } s,$$
$$\Phi(u) = 2e^{u/2}\big[2D(D-\tfrac12)\omega\big](e^{2u}),$$

the first for $\operatorname{Re} s > 1$ by the integration by parts and then for every s by analytic continuation, and the second by $x = e^{2u}$. So the gamma shapes of the terms $e^{-\pi n^2 x}$ of ω, one in x and $\frac{1}{4}$ on the logarithmic scale, are raised by one and by two, to the Gamma(2) and Gamma(3) of the Pólya kernel, by the operator that removes the poles of ζ. For Δ, whose L function is entire, no such operator is needed and the kernel keeps the single shape 6; the cancellation between the Gamma(2) and Gamma(3) terms of Φ is replaced by the positive multiplier $F(e^u)$.

At real points the Gamma(2) shape enters through Kristiansen's theorem [156]: every scale mixture of Gamma(2) laws is infinitely divisible. Since $G_a \overset{d}{=} G_2 B$ for $0 < a < 2$, with B a beta variable of parameters a and $2 - a$ independent of G_2, the scale mixtures of Gamma(a) laws with $a \leq 2$ are scale mixtures of Gamma(2) laws.

Proposition 8.5.7. *For real $a > 0$ let $W_a = G_a/(2\pi M^*)$, with G_a independent of M^*. Then W_a is a scale mixture of Gamma(a) laws, the mixing law being that of $1/(2\pi M^*)$, and $L(a - \frac{11}{2}, \Delta) = \Pr(W_a > 1)$. For $0 < a \leq 2$, that is $-\frac{11}{2} < s \leq -\frac{7}{2}$, W_a is infinitely divisible. In the same way $L(s, \chi_{12}) = \Pr\left(12G_{s/2}/(\pi M_1) > 1\right)$ for real $s > 0$, where $12G_{s/2}/(\pi M_1)$ is infinitely divisible for $s \leq 4$, and is a mixture of exponential laws for $s \leq 2$, in particular at the centre $s = \frac{1}{2}$.*

Proof. The identities are Corollary 8.5.3 and its analogue for M_1. For $a \leq 2$, $W_a = G_2 \cdot B/(2\pi M^*)$ is a scale mixture of Gamma(2) laws, infinitely divisible by Kristiansen's theorem. For $s \leq 2$, $G_{s/2} \overset{d}{=} G_1 B'$ with B' a beta variable of parameters $\frac{s}{2}$ and $1 - \frac{s}{2}$ (or $B' = 1$ at $s = 2$), so $12G_{s/2}/(\pi M_1)$ is a scale mixture of exponential laws, which is infinitely divisible by the theorem of Goldie and Steutel [95, 312]. □

The proposition says nothing about zeros: the representation $\Pr(W_a > 1)$ holds for real a only, since on the critical line the shape $6 + i\theta$ is not that of a probability law; for Δ the centre $a = 6$ lies outside Kristiansen's range, while for χ_{12} the centre shape $\frac{1}{4}$ lies inside it. The two Gamma(2) structures are also different in kind: the one in the Pólya kernel is a kernel shape forced by the poles of ζ, and the one in Proposition 8.5.7 is a mixing law.

8.5.4.4 The law of M_1

Put $b_n = \pi(n^2 - 1)/12$; the pentagonal number theorem of Section 8.5.1 reads $F_1(y) = \sum_{n \geq 1} \chi_{12}(n) e^{-b_n y}$, so the rates in this expansion are $b_{6k \pm 1} = \pi k(3k \pm 1)$, 2π times the generalised pentagonal numbers $k(3k \pm 1)/2$.

Proposition 8.5.8. *For $\theta > 0$ let $w = \sqrt{1 - 12\theta/\pi}$, positive or purely imaginary.*

Then

$$\mathbb{E}e^{-\theta M_1} = \frac{\pi(1-w^2)\sin(\pi w/3)}{2\sqrt{3}\,w\cos(\pi w/2)} = \prod_{k\ge 1} \frac{1+\theta/b_{6k}}{(1+\theta/b_{6k-1})(1+\theta/b_{6k+1})}.$$

Proof. Since $\sum_n \theta/(\theta+b_n) < \infty$, $\theta e^{-\theta y}F_1(y)$ can be integrated term by term, and $\theta + b_n = \pi(n^2-w^2)/12$ gives $\mathbb{E}e^{-\theta M_1} = \sum_{n\ge1}\chi_{12}(n)\theta/(\theta+b_n) = (1-w^2)\sum_{n\ge1}\chi_{12}(n)/(n^2-w^2)$. The function $g(w) = \pi\sin(\pi w/3)/(2\sqrt{3}\,w\cos(\pi w/2))$ is even and meromorphic, its only poles are simple and at the odd integers prime to 3, and its residue at $w=n$ is $-\chi_{12}(n)/(2n)$, as for the series, so their difference is entire. On the circles $|w| = 2k$, $|\cos(\pi w/2)|$ is bounded below, so $g = O(1/k)$ there, and the series is $O(k^{-1}\log k)$; by the maximum modulus principle the difference vanishes. For the product, insert $\sin(\pi w/3) = \frac{\pi w}{3}\prod_{m\ge1}(1-\frac{w^2}{9m^2})$ and $\cos(\pi w/2) = \prod_{n\ \mathrm{odd}}(1-\frac{w^2}{n^2})$. The factors with $n = 3m$, m odd, cancel, $1-w^2$ cancels the factor $n=1$, and what remains is a constant times the factors $1-w^2/n^2$ with $n = 6k$ above and $n = 6k\pm1$ below. The value 1 at $\theta = 0$ fixes the constant, and $(1-w^2/n^2)/(1-1/n^2) = 1+\theta/b_n$. □

At $\theta = \pi/12$, where $w = 0$, the closed form gives $\mathbb{E}e^{-\pi M_1/12} = \pi^2/(6\sqrt{3})$, which is $L(2,\chi_{12})$ by Section 8.5.4.2.

Corollary 8.5.9 (A Rényi type identity). *$M_1 \overset{d}{=} \sum_{k\ge1} Y_k$ with independent Y_k, where Y_k is exponential of rate b_{6k-1} with probability p_k and of rate b_{6k+1} with probability $1-p_k$, and*

$$p_k = \Big(1 - \frac{b_{6k-1}}{b_{6k}}\Big)\frac{b_{6k+1}}{b_{6k+1}-b_{6k-1}} = \frac{(12k-1)(3k+1)}{2(36k^2-1)},$$

so $p_1 = \frac{22}{35}$, $p_2 = \frac{161}{286}$, and $p_k \downarrow \frac12$.

Proof. For $\alpha<\gamma<\beta$ the partial fractions $\frac{1+\theta/\gamma}{(1+\theta/\alpha)(1+\theta/\beta)} = p\frac{\alpha}{\alpha+\theta} + (1-p)\frac{\beta}{\beta+\theta}$ hold with $p = (1-\alpha/\gamma)\beta/(\beta-\alpha)$ and $1-p = \alpha(\beta-\gamma)/(\gamma(\beta-\alpha))$, both in $(0,1)$; take $\alpha = b_{6k-1} = \pi k(3k-1)$, $\gamma = b_{6k} = \pi(36k^2-1)/12$ and $\beta = b_{6k+1} = \pi k(3k+1)$ in each factor of Proposition 8.5.8, and use the continuity theorem for Laplace transforms. □

By Frullani's integral, as in the first reading of Section 1.3.1, the product gives

$$-\log\mathbb{E}e^{-\theta M_1} = \int_0^\infty \big(1-e^{-\theta t}\big)\frac{k_1(t)}{t}\,dt, \quad k_1(t) = \sum_{k\ge1}\big(e^{-b_{6k-1}t} + e^{-b_{6k+1}t} - e^{-b_{6k}t}\big),$$

so M_1 is infinitely divisible, with no drift and Lévy density $k_1(t)/t$; this also follows from Corollary 8.5.9, since mixtures of exponential laws are infinitely divisible [95, 312]. The Lévy density is completely monotone, since $(e^{-b_{6k-1}t} - e^{-b_{6k}t})/t =$

$\int_{b_{6k-1}}^{b_{6k}} e^{-zt}dz$, so M_1 lies in the Bondesson class BO of Section 1.3.2. It is not a GGC: its Thorin measure would have to be $U_1 = \sum_k(\delta_{b_{6k-1}}+\delta_{b_{6k+1}}-\delta_{b_{6k}})$, negative at the points b_{6k} where χ_{12} vanishes between $6k-1$ and $6k+1$. Concretely, the transform vanishes at $\theta = -b_{6k}$, where $w = 6k$, so $-(\log \mathbb{E}e^{-\theta M_1})'$ has residue -1 there, and if M_1 were a GGC the Stieltjes inversion (1.3.2) of this function would give $U(\{b_{6k}\}) = -1$. The poles of the transform are at $\theta = -b_n$, $n \equiv \pm 1 \pmod 6$, $n \geq 5$, and its zeros at $-b_{6k}$: each zero lies between its own two poles $-b_{6k\pm1}$, but zeros and poles do not alternate, since $-b_7$ and $-b_{11}$, for instance, are consecutive poles. The next lemma reads Lemma 6.2.5 and the cosine dictionary (6.2.4) for a signed Thorin measure.

LEMMA 8.5.10. *Let $X \geq 0$ have Laplace exponent $-\log \mathbb{E}e^{-\theta X} = \int_{(0,\infty)} \log(1+\theta/z)\, U(dz)$ for a signed measure U with $\int \log(1+1/z)\,|U|(dz) < \infty$, and put $N(z) = U((0,z])$, with $N(z)\log(1+u^2/z^2) \to 0$ as $z \to \infty$ for real u. Then $-\log \mathbb{E}e^{-\theta X} = \int_0^\infty (1-e^{-\theta t})\nu(t)\,dt$ for $\theta \geq 0$, with the signed function $\nu(t) = t^{-1}\int e^{-zt}U(dz) = \int_0^\infty e^{-zt}N(z)\,dz$, and for real u*

$$\begin{aligned} -\log\left|\mathbb{E}e^{iuX}\right|^2 &= \int_0^\infty 4\sin^2\Big(\frac{ut}{2}\Big)\nu(t)\,dt \\ &= \int \log\Big(1+\frac{u^2}{z^2}\Big)U(dz) = \int_0^\infty N(z)\,\frac{2u^2}{z(z^2+u^2)}\,dz. \end{aligned} \tag{8.5.6}$$

The law of X is a GGC if and only if $U \geq 0$, and it lies in BO, that is, it is infinitely divisible with a completely monotone Lévy density, if and only if $N \geq 0$.

Proof. Frullani's integral $\log(1+\theta/z) = \int_0^\infty (1-e^{-\theta t})e^{-zt}t^{-1}dt$ gives the Lévy form, the exchange of integrals being justified by $\log(1+\theta/z) \leq C_\theta \log(1+1/z)$, and integrating e^{-zt} against dN by parts gives the second expression of ν. The first equality of (8.5.6) is twice the real part of the Lévy form at $\theta = -iu$, since $1-\cos ut = 2\sin^2(ut/2)$; the second is the Frullani identity $\int_0^\infty 4\sin^2(ut/2)e^{-zt}t^{-1}dt = \log(1+u^2/z^2)$; the third is an integration by parts, with $\frac{d}{dz}\log(1+u^2/z^2) = -2u^2/(z(z^2+u^2))$, whose boundary terms vanish by the assumptions. The signed function ν is determined by the law of X, since $t\nu(t)$ is the inverse Laplace transform of the derivative of $-\log \mathbb{E}e^{-\theta X}$, and so is the signed measure U, by the Stieltjes inversion of $\int U(dz)/(\theta+z)$. If $U \geq 0$ the law is a GGC, and if it is a GGC, its Laplace exponent grows more slowly than θ, so it has no drift, and its Thorin measure is U. If $N \geq 0$, ν is completely monotone and the law is in BO; if the law is in BO, its Lévy density is $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, it equals ν, and the uniqueness of Laplace transforms gives $\mu(dz) = N(z)\,dz$, so $N \geq 0$, N being right continuous. □

For M_1 the Thorin measure is $U_1 = \sum_{n\geq 2} a(n)\delta_{b_n}$ with weights of sine squared form,

$$a(n) = \sin^2\frac{\pi n}{2} + \frac{4}{3}\sin^2\frac{\pi n}{3} - 1 = \mathbf{1}\{2 \nmid n\} - \mathbf{1}\{3 \mid n\} = \frac{1}{6} - \frac{1}{2}\cos \pi n - \frac{2}{3}\cos\frac{2\pi n}{3},$$

equal to 1 for $n \equiv \pm 1 \pmod 6$, to -1 for $6 \mid n$, and to 0 otherwise.

PROPOSITION 8.5.11. *The canonical function $k_1(t) = t\nu_1(t)$ of M_1 has the theta form*

$$k_1(t) = \tfrac12 e^{\pi t/12}\Big(1 + \frac{\vartheta_0 - 3\vartheta_{1/2} - 4\vartheta_{1/3}}{\sqrt{3t}}\Big) - 1, \qquad \vartheta_\alpha = \sum_{m\in\mathbb{Z}} e^{-12\pi(m+\alpha)^2/t},$$

so $\nu_1(t) = (2\sqrt3)^{-1}t^{-3/2} - (2t)^{-1} + O(t^{-1/2})$ as $t \downarrow 0$ and $\nu_1(t) \sim e^{-2\pi t}/t$ as $t \to \infty$. Its Bondesson function $N_1(z) = U_1((0,z])$ is nonnegative, and $N_1(z) \sim \frac13\sqrt{3z/\pi}$ as $z \to \infty$.

Proof. Since a is even, $a(0) = -1$ and $a(1) = 1$,

$$k_1(t) = \sum_{n\ge2} a(n)e^{-b_n t} = \tfrac12 e^{\pi t/12}\Big(1 + \sum_{n\in\mathbb{Z}} a(n)e^{-\pi n^2 t/12}\Big) - 1.$$

Poisson summation, $\sum_{n\in\mathbb{Z}} \cos(2\pi n\alpha)e^{-\pi n^2 v} = v^{-1/2}\sum_{m\in\mathbb{Z}} e^{-\pi(m+\alpha)^2/v}$ with $v = t/12$, applied to the three terms of the cosine form of $a(n)$, gives the theta form. As $t \downarrow 0$, $\vartheta_0 = 1 + O(e^{-12\pi/t})$, while $\vartheta_{1/2}$ and $\vartheta_{1/3}$ are $O(e^{-4\pi/(3t)})$, so $k_1(t) = (2\sqrt{3t})^{-1} - \frac12 + O(\sqrt t)$; as $t \to \infty$ the first term $e^{-b_5 t} = e^{-2\pi t}$ of the series dominates. Since $b_{6k-1} < b_{6k} < b_{6k+1}$, every negative atom of U_1 is preceded by its own positive atom, so $N_1 \ge 0$. Up to z there are about $X/3$ integers $n \equiv \pm1 \pmod 6$ and $X/6$ multiples of 6 with $b_n \le z$, where $X = \sqrt{12z/\pi + 1}$, so $N_1(z) = X/6 + O(1) \sim \frac13\sqrt{3z/\pi}$. □

The Lévy density has the index $\frac12$ of a Brownian first passage time, matching $-\log \mathbb{E}e^{-\theta M_1} \sim \sqrt{\pi\theta/3}$, which the lower tail $F_1(y) \sim y^{-1/2}e^{-\pi/(12y)}$ gives; this index comes from modularity. By (8.5.6), the modulus of the characteristic function is

$$\big|\mathbb{E}e^{iuM_1}\big|^{-2} = \prod_{k\ge1} \frac{(1+u^2/b_{6k-1}^2)(1+u^2/b_{6k+1}^2)}{1+u^2/b_{6k}^2} = \exp\int_0^\infty N_1(z)\,\frac{2u^2\,dz}{z(z^2+u^2)},$$

the exponential of a positive kernel integrated against a nonnegative function. Figure 8.5 (left) shows N_1 against its asymptote.

The sine squared form of $a(n)$ splits U_1 into two positive measures: $\sin^2(\pi n/2)$ selects the odd n, and $1 - \frac43\sin^2(\pi n/3)$ the multiples of 3.

THEOREM 8.5.12. *With independent standard exponential variables E_n and E'_m, let $S = \sum_{n\ odd,\, n\ge3} E_n/b_n$ and $Z = \sum_{m\ge1} E'_m/b_{3m}$, with Z independent of M_1. Then S and Z are GGCs,*

$$\mathbb{E}e^{-\theta S} = \frac{\pi}{4}\,\frac{1-w^2}{\cos(\pi w/2)}, \qquad \mathbb{E}e^{-\theta Z} = \frac{\sqrt3\, w/2}{\sin(\pi w/3)},$$

and $M_1 + Z \stackrel{d}{=} S$. *Equivalently* $U_1 = U_S - U_Z$, *with* $U_S = \sum_{n \text{ odd}, n\geq 3} \delta_{b_n}$ *and* $U_Z = \sum_{m\geq 1} \delta_{b_{3m}}$, *the atoms at* b_{3m} *with* m *odd cancelling.*

Proof. With $1 + \theta/b_n = (n^2 - w^2)/(n^2 - 1)$, the product $\cos(\pi w/2) = \prod_{n \text{ odd}}(1 - w^2/n^2)$ gives $\prod_{n \text{ odd}, n\geq 3}(1 + \theta/b_n) = \frac{4}{\pi}\cos(\pi w/2)/(1 - w^2)$, the constant $\frac{\pi}{4} = \prod_{n \text{ odd}, n\geq 3}(1 - n^{-2})$ being the limit at $w = 1$; and $\sin(\pi w/3)/(\pi w/3) = \prod_m(1 - w^2/9m^2)$ gives $\prod_m(1 + \theta/b_{3m}) = \frac{2}{\sqrt{3}}\sin(\pi w/3)/w$. The quotient of the two transforms is the closed form of Proposition 8.5.8, so $\mathbb{E}e^{-\theta(M_1+Z)} = \mathbb{E}e^{-\theta S}$. □

The rates are shifted eigenvalues of Brownian hitting times [32]. With $\lambda_j = \pi^2(2j - 1)^2/8$, the rates of the exit time C of Brownian motion from $(-1, 1)$, whose transform is $1/\cosh\sqrt{2\theta} = \prod_j(1 + \theta/\lambda_j)^{-1}$, $b_{2j-1} = \frac{2}{3\pi}(\lambda_j - \lambda_1)$, so $S \stackrel{d}{=} \frac{3\pi}{2}\sum_{j\geq 2} E_j/(\lambda_j - \lambda_1)$: the variable $C = \sum_j E_j/\lambda_j$ with its ground mode $j = 1$ removed and the others shifted by λ_1. With $\mu_m = \pi^2 m^2/2$, the rates of the first passage T of the three dimensional Bessel process from 0 to 1, whose transform is $\sqrt{2\theta}/\sinh\sqrt{2\theta}$, $b_{3m} = \frac{3}{2\pi}(\mu_m - \frac{\pi^2}{18})$, so $Z \stackrel{d}{=} \frac{2\pi}{3}T^{(\pi^2/18)}$, where $T^{(c)}$ has the law of T tilted by e^{cT}, a proper law since $\frac{\pi^2}{18} < \mu_1$. By Kent's theorem with Krein's inverse spectral theory (Section 8.1.3), S and Z, GGCs without drift whose Thorin measures are unit atoms at distinct points, are hitting times of levels by generalised diffusions started at a reflecting or entrance end, so

$$M_1 + \big(\text{a tilted Bessel first passage time}\big) \stackrel{d}{=} \big(\text{the reduced Brownian exit time } S\big),$$

and M_1 itself is not a hitting time from such an end, since those are GGCs. Every first hitting time of a one dimensional diffusion lies in BO [146], as M_1 does, and from an interior point the transform is a quotient of the kind discussed in Section 11.1.4; whether M_1 is the hitting time of some level by some diffusion started at an interior point we do not know.

8.5.4.5 The law of M^*

The law of M^* has no such product, but its transform $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*} = \theta\int_0^\infty e^{-\theta y}F(y)\,dy$ is meromorphic: since $F(y) = \sum_{n\geq 1}\tau(n)e^{-2\pi(n-1)y}$, for any $c > 0$

$$\psi^*(\theta) = \theta\int_0^c e^{-\theta y}F(y)\,dy + \theta\sum_{n\geq 1}\tau(n)\frac{e^{-(\theta+2\pi(n-1))c}}{\theta + 2\pi(n-1)},$$

with at most simple poles at $\theta = -2\pi m$, $m \geq 1$, of residue $-2\pi m\,\tau(m + 1)$. Just to the right of a pole ψ^* has the sign of the residue and just to the left the opposite sign, so it has a real zero between $-2\pi(m+1)$ and $-2\pi m$ whenever $\tau(m+1)\tau(m+2) > 0$. The first such pair is $\tau(6) = -6048$, $\tau(7) = -16744$, and `maxlaw.py`, with $c = 0.3$ at 40 digits, finds the zero at $\theta = -34.4840085134049704852 = -2\pi \times 5.48830041253$. It is the only sign change of ψ^* on $(-15\pi, 0)$, and the

argument principle on the rectangle $|\operatorname{Re}\theta| \le 15\pi$, $|\operatorname{Im}\theta| \le 60$, which contains the seven poles $-2\pi,\dots,-14\pi$, gives winding number -6, so it is the only zero there. As for M_1, the zero shows that M^* is not a GGC.

The script `maxexp_psizeros.py` certifies the zeros in ball arithmetic, with $c = \frac{3}{100}$, the series to $n = 1500$, its tail bounded by Deligne's bound $|\tau(n)| \le d(n)n^{11/2} \le 2n^6$, and the integral bounded by $F(y) \le y^{-12}e^{2\pi y - 2\pi/y}$, which is increasing.

PROPOSITION 8.5.13. *1. For each of the* 25 *values* $m \le 60$ *with* $\tau(m+1)\tau(m+2) > 0$, ψ^* *changes sign in an interval of length* 2×10^{-20} *inside the gap* $(-2\pi(m+1), -2\pi m)$*; the zeros begin* $\theta/2\pi = -5.48830041$, -8.66373325, -11.4203774, -13.2533839, -14.7973058, *and the last is* -60.7670708.

2. ψ^* *has exactly one zero in the disc of radius* 10^{-7} *about each of the ten points*

$$\begin{aligned}
&-155.5940822775511402804 \pm 2.7631188861212609656\,i,\\
&-215.6211277570694988944 \pm 2.9647292148515237348\,i,\\
&-307.4008549057072052743 \pm 3.2283287024583906394\,i,\\
&-322.6135369899940182007 \pm 3.0449254921259967464\,i,\\
&-347.8816009468099010890 \pm 2.7171007701068929577\,i,
\end{aligned}$$

and it lies within 10^{-19} *of that point.*

Proof. For (1), the values of ψ^* at the two ends of each interval are balls of opposite signs. For (2), let z_0 be the approximation to the zero computed by Newton's method at 230 bits; the point shown is z_0 rounded to nineteen decimals, at distance at most 6.1×10^{-20} from it. With $A \ge |\psi^*(z_0)|$, $B \le |\psi^{*\prime}(z_0)|$ and $C \ge |\psi^{*\prime\prime}|$ on the square of half side r about z_0, each computed as a ball, $A + 2Cr^2 < Br/2$ holds for $r = 10^{-6}$ and for $r = 10^{-29}$. On the circle $|\theta - z_0| = r$, inside the square, $|\psi^*(\theta) - \psi^{*\prime}(z_0)(\theta - z_0)| \le A + Cr^2/2 < Br$, so by Rouché's theorem ψ^* has as many zeros in the disc $|\theta - z_0| < r$ as $\psi^{*\prime}(z_0)(\theta - z_0)$, namely one. The disc of radius 10^{-7} about the point shown lies in the disc of radius 10^{-6} about z_0, and the disc of radius 10^{-29} about z_0 lies within 10^{-19} of the point shown. The zeros come in conjugate pairs because ψ^* is real on the real axis. □

The argument principle on the rectangles $|\operatorname{Re}\theta| \le 2\pi(k+\frac12)$, with $|\operatorname{Im}\theta| \le 30$ for $k = 15$, 25, 40 and 60 and $|\operatorname{Im}\theta| \le 100$ for $k = 60$, and on a grid of cells, 2π wide and at most 5 high, covering $-2\pi \times 60.5 \le \operatorname{Re}\theta \le 2\pi \times 5.5$, $\frac12 \le \operatorname{Im}\theta \le 30$, finds exactly these zeros: the winding number on the largest rectangle, which contains the 60 poles $-2\pi,\dots,-120\pi$, is -26, so it contains 34 zeros, the 24 real zeros with $m \le 59$ and the ten nonreal zeros of the proposition. This count, unlike the zeros themselves, is computed and not certified: the change of argument is followed along the contours with steps halved until it is below 0.3, in ball arithmetic whose radii stay below a relative 10^{-46}. Every real zero lies in a gap where the residues

force it, every gap with $\tau(m+1)\tau(m+2) < 0$ is free of real zeros, and the five nonreal pairs have their real parts in such gaps, those with $m = 24$, 34, 48, 51 and 55 ($\operatorname{Re}\theta/2\pi = -24.764$, -34.317, -48.924, -51.346, -55.367), at heights between 2.72 and 3.23 (Figure 8.5, right). In a gap where the residues have opposite signs, ψ^* has the same sign at both ends, so its real zeros there come in pairs; the computation finds none, and the five nonreal pairs sit in such gaps, as if a pair of real zeros had collided and left the axis.

COROLLARY 8.5.14. *M^* is not in the Bondesson class BO; in particular it is not a GGC.*

Proof. If M^* were in BO, its Lévy density would be $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, so $-(\log\psi^*)'(\theta) = d + \int \mu(dz)(\theta+z)^{-2}$, with $d \geq 0$ the drift, for $\theta > 0$. The right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $\int \mu(dz)(1+z)^{-2} < \infty$ and $|\theta + z|$ is bounded below by a multiple of $1 + z$ for θ in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. Then the meromorphic function $\psi^{*\prime}/\psi^*$ would have no poles in $\mathbb{C} \setminus (-\infty, 0]$, and ψ^* no zeros there, against Proposition 8.5.13(2). □

Nor does M^* have a signed Thorin measure in the sense of Lemma 8.5.10: for a signed measure U on $(0,\infty)$, $-(\log\psi^*)'(\theta) = \int U(dz)(\theta+z)^{-1}$ would again be holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. Unit atoms at the points $2\pi m$ with $\tau(m+1) \neq 0$, less unit atoms at the real zeros, account for the poles and the real zeros of ψ^*, but no measure on the positive axis accounts for the nonreal zeros. Whether M^* is infinitely divisible remains open. If it is, with Lévy measure ν^*, then $-(\log\psi^*)'(\theta) = d + \int t\, e^{-\theta t}\nu^*(dt)$ is completely monotone on $(0,\infty)$: the cumulants $\kappa_j(\theta) = (-1)^j(\log\psi^*)^{(j)}(\theta)$ of M^* tilted by $e^{-\theta M^*}$ are all positive. At $\theta = 0.5$, 2, 8 and 30 the first eight are positive, as the first author's note found. As $\theta \to \infty$ the Laplace method, at the saddle point $y_0 = \sqrt{2\pi/\theta}$ of $e^{-\theta y - 2\pi/y}$ and with the lower tail $F(y) \sim y^{-12}e^{-2\pi/y}$, gives

$$\psi^*(\theta) \sim 2^{-1/2}(2\pi)^{-21/4}\,\theta^{25/4}\,e^{-2\sqrt{2\pi\theta}},$$

so $-\log\psi^*(\theta) = 2\sqrt{2\pi\theta} - \frac{25}{4}\log\theta + \frac{21}{4}\log 2\pi + \frac{1}{2}\log 2 + o(1)$. If M^* is infinitely divisible, it has no drift, and Karamata's Tauberian theorem turns the leading term into $\nu^*((t,\infty)) \sim 2\sqrt{2}\,t^{-1/2}$ as $t \downarrow 0$.

The first author conjectures that every maximum of independent exponential variables is infinitely divisible [256]; M_1, M_4, M^* and M_{11} are of this kind. For finitely many rates $\lambda_1, \ldots, \lambda_n$ the transform is rational: with $\lambda_S = \sum_{i\in S}\lambda_i$,

$$\mathbb{E}e^{-\theta M} = \theta\int_0^\infty e^{-\theta y}\prod_{i\leq n}(1 - e^{-\lambda_i y})\,dy = \sum_{S\neq\emptyset}(-1)^{|S|+1}\frac{\lambda_S}{\theta+\lambda_S},$$

with simple poles at the values $-\lambda_S$ whose grouped coefficients do not vanish, generically $2^n - 1$ of them, and, since $\mathbb{E}e^{-\theta M} \sim n!\,\lambda_1\cdots\lambda_n\theta^{-n}$, with n fewer

zeros. Writing the poles as $-p$ and the zeros as $-z$, Frullani's integral gives $-\log \mathbb{E}e^{-\theta M} = \int_0^\infty (1-e^{-\theta t})k(t)t^{-1}dt$ with $k(t) = \sum_p e^{-pt} - \sum_z e^{-zt}$, as long as every zero has $\operatorname{Re} z > 0$, and by the uniqueness of the Lévy measure the law is infinitely divisible if and only if $k \geq 0$. In particular infinite divisibility forces every zero to have $\operatorname{Re} z$ at least the smallest pole $p_{\min}$, since otherwise k would change sign at large t. For two rates $a < b$ the transform is $ab(a+b+2\theta)/((a+\theta)(b+\theta)(a+b+\theta))$ and the canonical function is $e^{-at}+e^{-bt}+e^{-(a+b)t}-e^{-(a+b)t/2}$, positive by the inequality of arithmetic and geometric means; the signed Thorin measure $\delta_a + \delta_b + \delta_{a+b} - \delta_{(a+b)/2}$ has a nonnegative distribution function, so this maximum is in BO and is not a GGC, and $e^{at}k(t) = 1 - x + x^2 + x^{2b/(b-a)} > \frac{3}{4}$ with $x = e^{-(b-a)t/2}$, a bound approached as $b/a \downarrow 1$. In `maxexp_checks.py`, k is evaluated at 181 points of $p_{\min}t \in [10^{-3}, 40]$ for 300 families of random rational rates in $[0.1, 5]$ for each $n = 2, 3, 4, 5$ and 40 families with $n = 6$. In every family k is positive at every point, $e^{p_{\min}t}k(t) \geq 0.53$, and every zero has $\operatorname{Re} z > p_{\min}$. Nonreal zeros are common: they occur in 140, 222, 266 and 40 of the families with 3, 4, 5 and 6 rates, up to 2, 8, 22 and 52 of them. So nonreal zeros, which exclude BO, do not seem to exclude infinite divisibility. The truncations of M_1, with the rates $2\pi n$, $n \leq N$, have only real zeros for $N = 4, \ldots, 10$, and those of M^*, with each rate $2\pi n$, $n \leq N$, taken 24 times, have 24 real zeros for $N = 2$ and 72 zeros, 52 of them nonreal, for $N = 3$; in each case k is positive on the grid. None of this is a proof.

Question 8.5.15. Is every maximum of independent exponential variables infinitely divisible? In particular, is M^*?

For the clock the relaxation from GGC to BO gains nothing. If its Lévy density $W(t)/t$ were completely monotone, equal to $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, then the formula $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ of Section 1.2 would give $(G'/G)(u) = \int \mu(dz)(u+z)^{-2}$, holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. But G'/G has a pole at every zero $-a_k = (\rho_k - \frac{1}{2})^2$ of G, and this point lies in $(-\infty, 0]$ only when ρ_k is on the critical line. So the clock is in BO if and only if RH holds (Table 1.6). The signed Thorin measures of the max laws come from zeros of their Laplace transforms, and the transform $1/G$ of the clock has none. The two laws of this section show the two ways out of the Thorin class: M_1 leaves it through real zeros, each preceded by its own pole, and stays in BO, while M^* leaves BO too, through zeros off the real axis, the way a zero of ξ off the critical line would take the clock out of both classes.

8.5.5 The de Bruijn Newman flow for Δ

The flow of Section 8.4 needs only an even kernel that decays faster than any Gaussian, so it applies to K_Δ:

$$H_t^\Delta(\theta) = \int_0^\infty e^{tu^2} K_\Delta(u)\cos(\theta u)\,du, \qquad 2H_0^\Delta(\theta) = \xi\big(\tfrac{1}{2} + i\theta, \Delta\big).$$

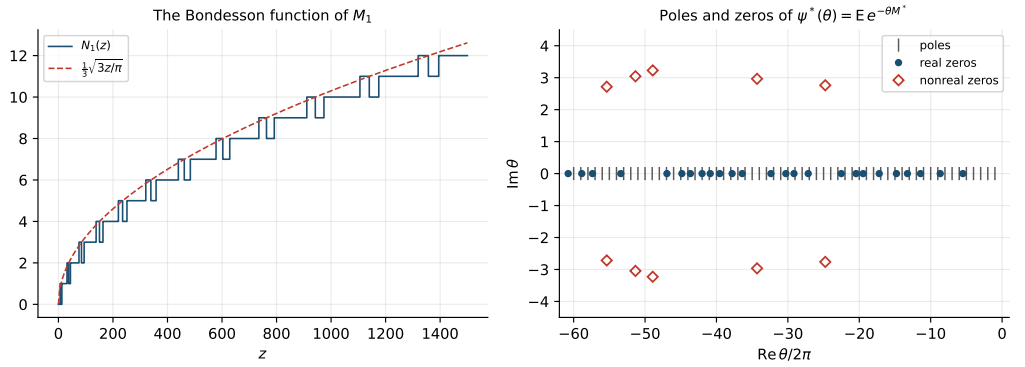


Figure 8.5: Left: the Bondesson function N_1 of M_1, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \le 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi \times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\,\mathrm{Re}\,\theta| \le 2\pi \times 60.5$ and $|\,\mathrm{Im}\,\theta| \le 100$. Drawn by `maxexp_figs.py` from `maxexp_psizeros.json`.

In the variable θ the times of Section 8.4, whose variable is $z = 2\theta$, are divided by four, and the isolated pair estimate for two zeros at distance D is $-D^2/8$. The zeros of H_0^Δ lie in the strip $|\,\mathrm{Im}\,\theta| < \frac{1}{2}$, so by de Bruijn's theorem [73] every zero of H_t^Δ is real for $t \ge \frac{1}{8}$. Dobner's theorem for the extended Selberg class [77] gives the other side: Δ has a de Bruijn Newman constant Λ_Δ, and H_t^Δ has nonreal zeros for every $t < 0$. So $0 \le \Lambda_\Delta \le \frac{1}{8}$, and GRH for Δ is the statement $\Lambda_\Delta = 0$.

On the critical line the kernel is a probability density up to normalisation, as the eta kernel is in Section 8.5.1: if V has the density $K_\Delta(v)/\xi(\frac{1}{2},\Delta)$ on $\mathbb{R}$, then $\mathbb{E}e^{i\theta V} = \xi(\frac{1}{2}+i\theta,\Delta)/\xi(\frac{1}{2},\Delta)$, and GRH for Δ is the statement that this characteristic function has only real zeros. In the language of Section 8.5.4, e^V has the density proportional to $y^5\Delta(iy) = y^5 e^{-2\pi y}F(y)$, so V is the logarithm of $G_6/2\pi$ conditioned on $G_6/2\pi \ge M^*$, with G_6 a gamma variable of shape 6 independent of M^*; its law is symmetric by the self duality (8.5.1), with no symmetrisation. The variable V is not infinitely divisible, nor even quasi infinitely divisible, because its characteristic function vanishes at $\theta = 9.2223793999$; this is the argument of Proposition 11.1.3(1) for the Pólya variable. Since $1-\cos x = 2\sin^2(x/2)$ and K_Δ is even,

$$\xi\big(\tfrac{1}{2},\Delta\big) - \xi\big(\tfrac{1}{2}+i\theta,\Delta\big) = 4\int_0^\infty K_\Delta(v)\sin^2\Big(\frac{\theta v}{2}\Big)dv,$$

so the zeros on the critical line are the solutions of $\mathbb{E}[2\sin^2(\theta V/2)] = 1$. At the level of logarithms, the factorisation $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ splits $\log|\xi(\frac{1}{2}+i\theta,\Delta)/\xi(\frac{1}{2},\Delta)|^2$ into $\log|\Gamma(6+i\theta)/\Gamma(6)|^2$ and $\log|L(\frac{1}{2}+i\theta,\Delta)/L(\frac{1}{2},\Delta)|^2$.

The first has the positive sine squared form given by Binet's formula,

$$\log\Big|\frac{\Gamma(6+i\theta)}{\Gamma(6)}\Big|^2 = -\int_0^\infty 4\sin^2\Big(\frac{\theta u}{2}\Big)\frac{e^{-6u}}{u(1-e^{-u})}\,du,$$

the Lévy Khintchine form of the infinitely divisible variable $\log G_6$, and the second carries every zero. To the right of the critical strip the second has a sine squared form too: for $\sigma > 1$, expanding $\log L(s,f) = \sum_{n\ge 2}\Lambda_f(n)n^{-s}/\log n$ gives

$$\log\Big|\frac{L(\sigma+i\theta,f)}{L(\sigma,f)}\Big|^2 = -\sum_{n\ge 2}\frac{4\Lambda_f(n)}{n^\sigma\log n}\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

for $f = \chi_{12}$ and Δ, and for ζ with Λ in place of Λ_f. Since $\int_\sigma^\infty n^{-\alpha}d\alpha = n^{-\sigma}/\log n$, the right side is minus twice the integral of the prime piece of the decomposition of Section 8.5.3 over the basepoints $\alpha > \sigma$. For ζ the weights are positive, which is the infinite divisibility of log of the zeta distribution (Section 6.1.6). For Δ they are $8\cos(k\vartheta_p)/(kp^{k\sigma})$ at $n = p^k$, where $\alpha_p = e^{i\vartheta_p}$, $0 \le \vartheta_p \le \pi$, and they take both signs. The Sato Tate law for Δ, proved by Barnet-Lamb, Geraghty, Harris and Taylor [22], makes the angles ϑ_p equidistributed for the measure $\frac{2}{\pi}\sin^2\vartheta\,d\vartheta$ on $[0,\pi]$, under which $\cos k\vartheta < 0$ with probability $\frac{1}{2}$ for $k \ne 2$ and $\frac{1}{2}+\frac{1}{\pi}$ for $k = 2$. So even where the Euler product converges absolutely the sine squared form of $\log|L(\sigma+i\theta,\Delta)|$ has a signed measure, and on the critical line the series diverges: positivity lives in the kernel and in the gamma factor, not in the arithmetic factor.

The kernel factors as $K_\Delta(u) = e^{-2\pi\cosh u}H(u)$, with $H(u) = \sqrt{F(e^u)F(e^{-u})}$ because $K_\Delta(u)^2 = K_\Delta(u)K_\Delta(-u)$, and $\log H(u) = -12\sum_m \frac{\sigma(m)}{m}(e^{-2\pi m e^u} + e^{-2\pi m e^{-u}})$. The expansion comes from $-\log F(y) = 24\sum_{m\ge1}\frac{\sigma(m)}{m}e^{-2\pi m y}$, so $-\log F$ is completely monotone, the Laplace transform of the positive measure with mass $24\sigma(m)/m$ at each point $2\pi m$; for η the masses are $\sigma(m)/m$, and every maximum of independent exponential variables has this property, since $-\log(1-e^{-\lambda y}) = \sum_k e^{-k\lambda y}/k$. The transform $K_{i\theta}(2\pi)$ of Pólya's factor has only real zeros [273], the first at $\theta = 9.768770083509977867$ (`maxlaw.py`), and the multiplier H moves it to 9.2223793999. Pólya's theorem on universal factors [73, p. 197], which covers the Gaussian multiplier e^{tu^2} of the flow for $t > 0$, does not apply to H, which is bounded by one and supplies half of the doubly exponential decay of K_Δ. The positivity of the measure of $-\log F$ is automatic and does not by itself decide where the zeros are; whether some multiplier of the same form moves zeros off the line is the following question, and any answer must account for Δ, χ_{12} and the conductor 11 curve, whose kernels have this form.

QUESTION 8.5.16. For which $a > 0$, $c_m \ge 0$ and $\beta_m > 0$ does the Fourier transform of $e^{-a\cosh u}\exp\big(-\sum_m c_m(e^{-\beta_m e^u} + e^{-\beta_m e^{-u}})\big)$ have only real zeros?

The kernel K_Δ is strictly log concave, as the first author's note [260] shows. For $c > 0$, $\log(1-e^{-ce^u})$ is the logarithm of the distribution function of $\log E$, with

E exponential of rate c, and it is strictly concave in u: with $x = ce^u$ its second derivative is $x(e^x - 1 - xe^x)/(e^x - 1)^2 < 0$. So $\log F(e^u)$, a sum of such terms, is strictly concave (in the language of Section 8.5.4, $\log M^*$ has a log concave distribution function), and since $\log K_\Delta(u) = 6u - 2\pi e^u + \log F(e^u)$ and K_Δ is even,

$$-\big(\log K_\Delta\big)''(u) > 2\pi e^{|u|}, \qquad u \in \mathbb{R},$$

where the ratio of the two sides tends to 1 as $|u| \to \infty$. The multiplier H is log concave as well, since $\log H(u) = \frac{1}{2}\big(\log F(e^u) + \log F(e^{-u})\big)$. In the same way $\log K(u) = \log 2 + \frac{u}{2} - \frac{\pi}{12}e^{2u} + \log F_1(e^{2u})$ gives $-(\log K)''(u) > \frac{\pi}{3}e^{2|u|}$ for the kernel of Section 8.5.1. The Pólya kernel Φ is log concave too [58], though it is not a product of this kind. Log concavity alone does not put the zeros of a transform on a line, or in any strip. The function $(1 - |u|)_+^2$ is log concave, and its transform $4(\theta - \sin\theta)/\theta^3$ vanishes at the nonzero roots of $\sin\theta = \theta$, which are nonreal, the first at $7.4977 + 2.7687i$, with imaginary parts that grow like $\log|\theta|$. Its convolution with $e^{-\cosh u}$ is even, positive, entire and log concave [298], and decays doubly exponentially, like K_Δ; its transform $8(\theta - \sin\theta)K_{i\theta}(1)/\theta^3$ has the same nonreal zeros.

In the language of Section 8.5.4 the flow is a family of Gaussian tilts: for every real t, $e^{tv^2}K_\Delta(v)$ is integrable, because K_Δ decays doubly exponentially, so after normalisation it is the density of the law of V tilted by e^{tV^2}, and $H_t^\Delta(\theta)/H_t^\Delta(0)$ is its characteristic function. The question $\Lambda_\Delta = 0$ asks whether the smallest tilt whose characteristic function has only real zeros is $t = 0$.

With `dbn_delta.py` (Gauss Legendre quadrature at 40 digits) the 21 zeros below height 52, from 9.2224 to 51.1566, stay real for $t = 0.05, 0.1, \ldots, 0.5$ and all move down. Backward in time each pair of neighbours is followed through the critical point of H_t^Δ between them until the critical value changes sign, and each such event is tested by direct sign counts: a collision at (t_*, θ_*) leaves two sign changes on $[\theta_* - 1.2, \theta_* + 1.2]$ at $t_* + 0.005$ and none at $t_* - 0.005$, and the colliding pair is identified by the number of sign changes below $\theta_* - 1.2$, since real zeros keep their order. Table 8.16 lists the seven collisions. The numbers of sign changes on $(0.5, 52]$ at eleven values of t from 0 to -1.5 fall by two at each collision and once by one, between $t = -0.2$ and -0.3, where the highest zero, followed back from 51.1566, passes 52 at $t \approx -0.29$.

The script `maxexp_flow.py` recomputes the table independently, at 70 digits and with a different tracking, and finds the same seven collisions, with the same t_* and θ_* to all digits of the table, and the highest zero passing 52 between $t = -0.29$ and -0.30. Figure 8.6 shows the trajectories.

The low zeros of Δ are robust: below height 33, the window of the first author's note, the first pair to leave the line, $(25.27, 26.80)$, does so only at $t = -0.407$. Collisions do not come in order of height, however. The pairs near 37.6 and 49.2 collide at -0.159 and -0.126, and under GRH for Δ Dobner's theorem forces collisions in every interval $(t, 0)$, higher up; so no window of heights decides whether

pair at $t = 0$	$-D^2/8$	t_*	θ_*
48.3591, 49.2761	−0.1051	−0.12622	49.18
36.7415, 37.7539	−0.1281	−0.15864	37.60
45.0401, 46.1973	−0.1674	−0.24305	46.19
25.2746, 26.8044	−0.2925	−0.40713	27.26
40.2190, 41.7305	−0.2856	−0.48271	42.49
31.1782, 32.7749	−0.3187	−0.48637	33.30
17.4428, 19.6565	−0.6126	−0.98116	21.09

Table 8.16: The collisions of zeros of H_t^{Δ} below height 52, in the order in which they occur as t decreases: the pair at $t = 0$, the isolated pair estimate with D the gap of the pair, and the time t_* and height θ_* of the collision, computed by `dbn_delta.py`.

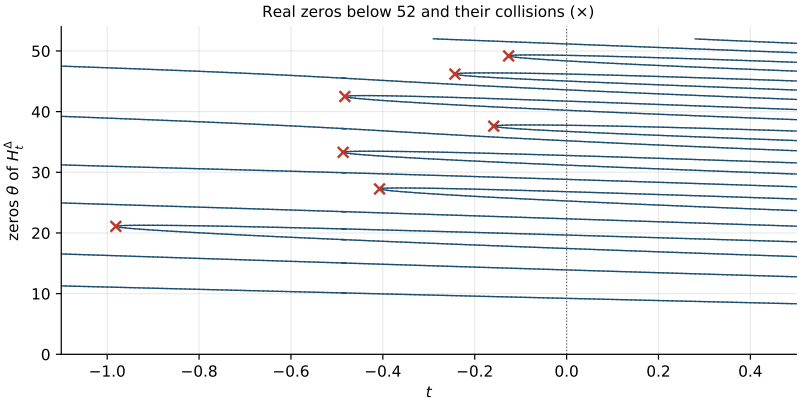


Figure 8.6: The real zeros of H_t^{Δ} below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 8.16; all zeros move down as t grows. Drawn by `maxexp_figs.py` from `maxexp_flow.json`.

$\Lambda_{\Delta} = 0$ (Dobner's theorem already gives $\Lambda_{\Delta} \ge 0$), and what a computation can show is how fast the collision times approach 0 with height. The neighbours delay every collision beyond the isolated pair estimate, by factors from 1.20 to 1.69. In the variable θ the closest pair of ζ below height 200 collides at -0.072, and Lehmer's pair near height 7005 at -1.8×10^{-4} (Section 8.4.2).

8.5.6 A Pólya kernel for imaginary quadratic fields

This subsection follows the last part of the first author's note on the squarefree series [259].

Let $d < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{d})$, χ_d the Kronecker symbol of d, a real primitive odd character modulo $|d|$, h the class number of K

and w the number of its roots of unity. Then $\zeta_K(s) = \zeta(s)L(s,\chi_d) = \sum_{n\ge1} a_n n^{-s}$, where $a_n = \sum_{m|n} \chi_d(m) \ge 0$ is the number of ideals of norm n, and we put

$$\Lambda_K(s) = \Big(\frac{\sqrt{|d|}}{2\pi}\Big)^s \Gamma(s)\,\zeta_K(s), \qquad \xi_K(s) = 2s(s-1)\Lambda_K(s),$$

the function ξ_K of Section 10.3.3 with $r_1 = 0$ and $r_2 = 1$. The duplication formula gives $\xi_K(s) = \frac{2}{\sqrt{|d|}}\,\xi(s)\,\xi(s,\chi_d)$, with $\xi(s,\chi_d) = (|d|/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_d)$ the completed function of Section 8.5.1, so $\xi_K(s) = \xi_K(1-s)$ because the root number of χ_d is one. Hence $\Lambda_K(s) = \Lambda_K(1-s)$, and Λ_K is meromorphic with simple poles at $s = 1$ and $s = 0$ only, with residues h/w and $-h/w$, by the class number formula $L(1,\chi_d) = 2\pi h/(w\sqrt{|d|})$ and the symmetry. Let

$$\vartheta_K(x) = \sum_{n\ge1} a_n\, e^{-2\pi n x/\sqrt{|d|}}, \qquad \Psi(y) = y(y-2)\,e^{-y}, \qquad H_K(x) = \big(x^2\vartheta_K'(x)\big)',$$

and $\Phi_K(u) = 2e^{u/2}H_K(e^u)$.

THEOREM 8.5.17. (i) $\vartheta_K(1/x) = x\,\vartheta_K(x) + \frac{h}{w}\,(x-1)$ *for* $x > 0$.

(ii) $H_K(x) = \sum_{n\ge1} a_n \Psi(2\pi n x/\sqrt{|d|})$ *and* $H_K(1/x) = x\,H_K(x)$. *The kernel* Φ_K *is even, analytic in the strip* $|\operatorname{Im} u| < \frac{\pi}{2}$, *and decays doubly exponentially.*

(iii) $\xi_K(\frac12 + s) = \int_{\mathbb R} \Phi_K(u)\,e^{su}\,du$ *for every* $s \in \mathbb C$; *in particular* $\xi_K(\frac12 + i\theta) = \int_{\mathbb R} \Phi_K(u)\cos(\theta u)\,du$.

(iv) $\Phi_K > 0$ *on* $\mathbb R$ *if and only if* $H_K > 0$ *on* $[1, \sqrt{|d|}/\pi]$. *In particular* $\Phi_K > 0$ *when* $|d| < \pi^2$, *that is for* $d = -3, -4, -7, -8$.

(v) *If* $\Phi_K > 0$, *then* $\zeta_K(\frac12) < 0$ *and* $L(\frac12,\chi_d) > 0$. *In particular* $L(\frac12,\chi_d) > 0$ *for* $d = -3, -4, -7, -8$.

Proof. (i) For $c > 1$, Mellin inversion of $\int_0^\infty e^{-2\pi n x/\sqrt{|d|}} x^{s-1}dx = \Gamma(s)\big(\sqrt{|d|}/2\pi n\big)^s$ gives $\vartheta_K(x) = \frac{1}{2\pi i}\int_{(c)} \Lambda_K(s) x^{-s} ds$. Move the line of integration to $\operatorname{Re} s = 1 - c$, which the exponential decay of Γ and the polynomial growth of ζ_K on vertical strips allow; the residues at $s = 1$ and $s = 0$ are $\frac{h}{w}x^{-1}$ and $-\frac{h}{w}$. On the new line substitute $s \mapsto 1 - s$ and use $\Lambda_K(1-s) = \Lambda_K(s)$: the integral becomes $x^{-1}\vartheta_K(1/x)$. So $\vartheta_K(x) = \frac{h}{w}(x^{-1} - 1) + x^{-1}\vartheta_K(1/x)$, which is (i).

(ii) Termwise, $\big(x^2 \frac{d}{dx} e^{-\kappa x}\big)' = (\kappa^2 x^2 - 2\kappa x)e^{-\kappa x} = \Psi(\kappa x)$. Differentiating (i) gives $\vartheta_K'(1/x) = -x^2\big(\vartheta_K(x) + x\vartheta_K'(x) + \frac{h}{w}\big)$, so $g(x) = x^2\vartheta_K'(x)$ satisfies $g(1/x) = -\big(\vartheta_K(x) + x\vartheta_K'(x) + \frac{h}{w}\big)$, and differentiating again gives $g'(1/x) = x^2\big(2\vartheta_K'(x) + x\vartheta_K''(x)\big) = x\,g'(x)$, that is $H_K(1/x) = xH_K(x)$. Then $\Phi_K(-u) = 2e^{-u/2}H_K(e^{-u}) = 2e^{u/2}H_K(e^u) = \Phi_K(u)$. Since $a_n \le n$, the series for $H_K(e^u)$ converges locally uniformly where $\operatorname{Re} e^u > 0$, that is for $|\operatorname{Im} u| < \frac{\pi}{2}$; and $\Phi_K(u) = O\big(e^{5u/2}\exp(-2\pi e^u/\sqrt{|d|})\big)$ as $u \to \infty$, and likewise as $u \to -\infty$ by evenness.

(iii) By (i), $x\vartheta_K(x) \to \frac{h}{w}$ and $x^2\vartheta_K'(x) \to -\frac{h}{w}$ as $x \to 0$, while ϑ_K and ϑ_K' decay exponentially at infinity. For $\operatorname{Re} z > 1$ two integrations by parts, whose boundary

terms vanish, give $\int_0^\infty H_K(x)x^{z-1}dx = z(z-1)\int_0^\infty \vartheta_K(x)x^{z-1}dx = z(z-1)\Lambda_K(z)$. With $x=e^u$ and $z=\frac12+s$, $\int_{\mathbb R}\Phi_K(u)e^{su}du = 2\int_0^\infty H_K(x)x^{z-1}dx = \xi_K(z)$. Both sides are entire in s, the left by (ii), so the identity holds for every s.

(iv) By (ii), $\Phi_K>0$ exactly when $H_K>0$ on $(0,\infty)$, and by the symmetry exactly when $H_K>0$ on $[1,\infty)$. For $x\ge\sqrt{|d|}/\pi$ every argument $2\pi nx/\sqrt{|d|}$ is at least $2n\ge2$, where $\Psi\ge0$, and the term $n=4$ is positive, since $a_4 = 1+\chi_d(2)+\chi_d(2)^2\ge1$; so $H_K>0$ there. If $|d|<\pi^2$ the interval $[1,\sqrt{|d|}/\pi]$ is empty.

(v) By (iii), $\xi_K(\frac12)=\int_{\mathbb R}\Phi_K>0$, while

$$\xi_K\big(\tfrac12\big) = -\tfrac12\Lambda_K\big(\tfrac12\big) = -\tfrac12\big(\sqrt{|d|}/2\pi\big)^{1/2}\sqrt{\pi}\,\zeta_K\big(\tfrac12\big).$$

So $\zeta_K(\frac12)<0$, and $L(\frac12,\chi_d)=\zeta_K(\frac12)/\zeta(\frac12)>0$ because $\zeta(\frac12)<0$. □

When $\Phi_K>0$, the normalised kernel $\Phi_K/\xi_K(\frac12)$ is the density of a symmetric variable X_K with $\mathbb Ee^{sX_K}=\xi_K(\frac12+s)/\xi_K(\frac12)$, the Pólya variable of K, and its tilts $V_{K,\alpha}$, with $\mathbb Ee^{sV_{K,\alpha}}=\xi_K(\alpha+s)/\xi_K(\alpha)$, have the means $\frac{\xi_K'}{\xi_K}(\alpha)$. The zero part of the constant C_K of Theorem 10.3.7 is then $\mathbb EV_{K,1}-2\,\mathbb EV_{K,2}$, as over $\mathbb Q$ in Theorem 10.3.5; by (iv) this applies to the Gaussian field $d=-4$ of Section 10.3.3. If moreover the Riemann hypothesis holds for ζ_K, then $1/\mathbb E\cosh(\sqrt{u}\,X_K)=\xi_K(\frac12)/\xi_K(\frac12+\sqrt{u})$ is the Laplace transform of a GGC, the clock of K (Proposition 10.3.6).

The kernel Φ_K is tied to the kernel of χ_d in Section 8.5.1,

$$K_{\chi_d}(u) = 2e^{3u/2}\sum_{n\ge1} n\chi_d(n)\,e^{-\pi n^2e^{2u}/|d|}, \qquad \xi\big(\tfrac12+s,\chi_d\big) = \int_{\mathbb R} K_{\chi_d}(u)\,e^{su}\,du.$$

Since the two sided Laplace transform turns convolution into multiplication, the identity $\xi_K=\frac{2}{\sqrt{|d|}}\xi\cdot\xi(\cdot,\chi_d)$ and (iii) give $\Phi_K=\frac{2}{\sqrt{|d|}}\,\Phi * K_{\chi_d}$, with Φ the Pólya kernel (1.2.3). As $\Phi>0$, the kernel Φ_K is positive whenever K_{χ_d} is, and if Φ_K takes a negative value, so does K_{χ_d}. For $d=-4$ Jacobi's identity $\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}=q^{1/8}\prod_{n\ge1}(1-q^n)^3$ gives $K_{\chi_{-4}}(u)=2e^{3u/2}\eta(ie^{2u})^3>0$, positive by the product formula as the kernel of χ_{12} is.

The script `sqfree_quadratic.py` checks the theorem at 30 digits. For $d=-4$, where $h/w=\frac14$, the trapezoid rule with step $1/200$ gives $\int\Phi_K(u)\cos(\theta u)\,du = 0.48753311500024448536$ at $\theta=0$ and 0.25376667901239784985 at $\theta=2.5$, equal to $\xi_K(\frac12+i\theta)$ in every digit shown. The values $\zeta_K(\frac12)=-0.70223711$, -0.97506623, -1.6744215 and -1.6070054 for $d=-3,-4,-7,-8$ are negative, as (v) requires.

The sign of H_K on $[1,\sqrt{|d|}/\pi]$ depends strongly on the first few weights: Ψ is negative on $(0,2)$ and positive beyond, and the weights $a_2=1+\chi_d(2)$ and $a_3=1+\chi_d(3)$ vanish when 2 and 3 are inert in K. The next theorem makes this precise.

THEOREM 8.5.18. *Let $d<0$ be a fundamental discriminant with $\chi_d(2)=\chi_d(3)=-1$, that is $d\equiv 5 \pmod{24}$. Then $H_K(x)<0$ at $x=1.42\sqrt{|d|}/2\pi$, so Φ_K takes negative values, and so does the kernel K_{χ_d}. There are infinitely many such d, for instance $d=-p$ for every prime $p\equiv 19 \pmod{24}$.*

Proof. For a fundamental discriminant $d<0$, $\chi_d(2)=-1$ exactly when $d\equiv 5 \pmod 8$, and $\chi_d(3)=-1$ exactly when $d\equiv 2 \pmod 3$; together they give $d\equiv 5 \pmod{24}$. The function $n\mapsto a_n$ is multiplicative, with $a_{p^k}=\frac12(1+(-1)^k)$ when $\chi_d(p)=-1$ and $a_{p^k}\le k+1$ for every p. Hence $a_1=1$, $a_2=a_3=0$ and $a_n\le a_n^*$, where $a_n^*=0$ unless 2 and 3 divide n to even powers, and otherwise a_n^* is the number of divisors of n with its factors 2 and 3 removed. At $x=1.42\sqrt{|d|}/2\pi$ the nth argument of Ψ in Theorem 8.5.17(ii) is $1.42n$, which exceeds 2 for $n\ge 2$, where $\Psi>0$. So

$$\begin{aligned} H_K(x) &\le \Psi(1.42)+\sum_{n\ge 4} a_n^*\,\Psi(1.42n) \\ &= -0.19907566+0.13917904=-0.05989662<0, \end{aligned}$$

the sum taken to $n=2000$ and the rest bounded, with $a_n^*\le 2\sqrt{n}$, by 10^{-1225} (`sqfree_quadratic.py`). Then $\Phi_K(\log x)<0$, and K_{χ_d} takes negative values by the convolution identity above. If $p\equiv 19 \pmod{24}$ is prime, then $p\equiv 3 \pmod 4$, so $-p$ is a fundamental discriminant, and $-p\equiv 5 \pmod{24}$; by Dirichlet's theorem there are infinitely many such primes. □

Table 8.17 shows the discriminants discussed in the first author's note. The four of class number one with 2 and 3 inert, $d=-19,-43,-67,-163$, take the least value of H_K on $[1,\sqrt{|d|}/\pi]$ at the centre $x=1$, the two composite ones with $d\equiv 5 \pmod{24}$ inside the interval, and most of the positive kernels at its right end $\sqrt{|d|}/\pi$, where the term $n=1$ vanishes. All twenty values of $\zeta_K(\frac12)$ computed are negative, so $L(\frac12,\chi_d)>0$ for all of them whether or not the kernel is positive: Theorem 8.5.17(v) is a sufficient condition only.

The script also scans all 6079 fundamental discriminants $-20000\le d<0$. It evaluates H_K in double precision on 3000 geometrically spaced points of $[1,\sqrt{|d|}/\pi]$, refines the least value by golden section search, and would recompute at 30 digits any least value within 10^{-8} of zero; there is none. The kernel is positive for 2738 of the discriminants. Table 8.18 sorts them by the splitting of 2 and 3: the proportion of positive kernels falls as either prime goes from split to ramified to inert. By blocks of 2000 in $|d|$ the proportion falls from 337/611 for $|d|\le 2000$ to between 0.41 and 0.45 in each of the five blocks above 10000.

QUESTION 8.5.19. Does the proportion of fundamental discriminants $-X\le d<0$ with $\Phi_K>0$ tend to a limit as $X\to\infty$, and is the limit positive? Is Φ_K positive for infinitely many d, for instance among the d in which 2 and 3 split, where the proportion up to 20000 is 676/767?

d	$\chi_d(2)$	$\chi_d(3)$	$\Phi_K>0$	least H_K on $[1,\sqrt{\lvert d\rvert}/\pi]$	at $x=$	$\zeta_K(\frac12)$
-11	-1	1	yes	0.14236	1.0557	-1.4480539
-15	1	0	yes	0.40606	1.2328	-2.6973176
-20	0	1	yes	0.287033	1.4235	-2.4529153
-23	1	1	yes	0.4634	1.5266	-3.5856997
-24	0	0	yes	0.225215	1.2409	-2.2955534
-31	1	-1	yes	0.349008	1.7723	-3.2253855
-39	1	0	yes	0.409694	1.9878	-4.0547932
-47	1	1	yes	0.46368	2.1822	-4.8243519
-71	1	1	yes	0.470667	2.6821	-5.9925928
-191	1	1	yes	0.470667	4.3991	-8.1483722
-19	-1	-1	no	-0.0596605	1	-1.1747378
-43	-1	-1	no	-0.214707	1	-0.76635571
-67	-1	-1	no	-0.251274	1	-0.54460945
-163	-1	-1	no	-0.315818	1	-0.10008066
-427	-1	-1	no	-0.181949	3.1887	-0.20993694
-1003	-1	-1	no	-0.228309	3.9479	-0.64248774

Table 8.17: The Pólya kernel Φ_K of $K=\mathbb{Q}(\sqrt{d})$ for the discriminants named in the first author's note: the characters at 2 and 3, the sign of the kernel, the least value of H_K on $[1,\sqrt{\lvert d\rvert}/\pi]$, which decides it by Theorem 8.5.17(iv), with the point where it is attained, and $\zeta_K(\frac12)=\zeta(\frac12)L(\frac12,\chi_d)$. Computed by `sqfree_quadratic.py`.

	$\chi_d(3)=1$	$\chi_d(3)=0$	$\chi_d(3)=-1$
$\chi_d(2)=1$	676/767	334/507	412/757
$\chi_d(2)=0$	489/753	178/514	131/757
$\chi_d(2)=-1$	397/761	121/500	0/763

Table 8.18: Positive Pólya kernels Φ_K among the 6079 fundamental discriminants $-20000\le d<0$, by the splitting of 2 and 3 in $K=\mathbb{Q}(\sqrt{d})$ ($\chi_d(p)=1$ split, 0 ramified, -1 inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 8.5.18. Computed by `sqfree_quadratic.py`.

Question 8.5.20. For real quadratic fields and for fields of higher degree the gamma factor $\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}$ replaces the exponentials in ϑ_K by other kernels, for $\Gamma_{\mathbb{R}}(s)^2$ a Bessel function K_0. Is there a criterion for the positivity of the Pólya kernel of ξ_K as simple as Theorem 8.5.17(iv)?

8.5.7 The Dirichlet beta function and the eta cube kernel

The L function of the odd character χ_4 modulo 4,

$$L(s,\chi_4)=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)^s}=\prod_p\bigl(1-\chi_4(p)p^{-s}\bigr)^{-1},\qquad \operatorname{Re}s>1,$$

is the Dirichlet beta function, with $L(1,\chi_4)=\frac{\pi}{4}$ and $L(2,\chi_4)$ Catalan's constant. It is the character χ_{-4} of Section 8.5.6, whose kernel is the cube of the Dedekind eta function and positive by the product formula. This subsection carries the Thorin framework over to it, following the first author's note [253]. The comparison across vertical lines of the note, which applies to ξ and $L(s,\chi_{12})$ as well, is in Section 9.1.6.

8.5.7.1 The kernel, the exit time and the zeros

Jacobi's identity $\eta(\tau)^3=q^{1/8}\prod_{m\ge1}(1-q^m)^3=\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}$, with $q=e^{2\pi i\tau}$ (Section 8.5.6), gives at $\tau=ix$ the theta series of χ_4,

$$\eta(ix)^3=\sum_{n\ge1}\chi_4(n)\,n\,e^{-\pi n^2x/4}=e^{-\pi x/4}\prod_{m\ge1}\bigl(1-e^{-2\pi mx}\bigr)^3,\qquad x>0,$$

and the transformation $\eta(i/x)=\sqrt{x}\,\eta(ix)$ gives $\eta(i/x)^3=x^{3/2}\eta(ix)^3$. In the normalisation of Section 8.5.1 for odd characters the completed function is

$$\xi(s,\chi_4)=\Bigl(\frac{4}{\pi}\Bigr)^{(s+1)/2}\Gamma\Bigl(\frac{s+1}{2}\Bigr)L(s,\chi_4)=\int_0^\infty\eta(ix)^3\,x^{(s+1)/2}\,\frac{dx}{x},$$

the integral by termwise integration for $\operatorname{Re}s>1$, since $\int_0^\infty e^{-\pi n^2x/4}x^{(s+1)/2-1}dx=\Gamma(\frac{s+1}{2})(4/\pi n^2)^{(s+1)/2}$, and for every s by analytic continuation, since $\eta(ix)^3$ decays like $e^{-\pi x/4}$ as $x\to\infty$ and, by the transformation, like $x^{-3/2}e^{-\pi/(4x)}$ as $x\to0$. The transformation gives $\xi(s,\chi_4)=\xi(1-s,\chi_4)$, the root number of χ_4 being one, and $\xi(s,\chi_4)$ is entire of order one, with no factor $s(s-1)$ to remove. It is $2/\sqrt{\pi}$ times the function Λ of [253], and $\xi(1,\chi_4)=\frac{4}{\pi}L(1,\chi_4)=1$.

Proposition 8.5.21. *The kernel $K_4(u)=2e^{3u/2}\eta(ie^{2u})^3$ satisfies $\xi(\frac12+s,\chi_4)=\int_{\mathbb R}K_4(u)\,e^{su}\,du$ for every complex s. It is even, positive and analytic in the strip $|\operatorname{Im}u|<\pi/4$, it has the symmetric product form*

$$K_4(u)=2\exp\Bigl(-\frac{\pi}{4}\cosh2u\Bigr)\prod_{m\ge1}\bigl(1-e^{-2\pi me^{2u}}\bigr)^{3/2}\bigl(1-e^{-2\pi me^{-2u}}\bigr)^{3/2},$$

so that $K_4(u)=2\exp\bigl(\frac32|u|-\frac{\pi}{4}e^{2|u|}\bigr)\bigl(1+O(e^{-2\pi e^{2|u|}})\bigr)$, and $K_4(0)=2\eta(i)^3=\Gamma(\frac14)^3/(4\pi^{9/4})=0.906767655167731$.

Proof. With $x = e^{2u}$, $x^{(s+1)/2}dx/x = 2e^{(s+1)u}du$, so the Mellin form with $\frac12 + s$ in place of s is the stated integral. The product formula makes every factor of $K_4(u) = 2e^{3u/2}e^{-\frac{\pi}{4}e^{2u}}\prod_m(1-e^{-2\pi m e^{2u}})^3$ positive. The transformation gives $K_4(-u) = 2e^{-3u/2}\eta(ie^{-2u})^3 = 2e^{-3u/2}e^{3u}\eta(ie^{2u})^3 = K_4(u)$. Hence $K_4(u) = \sqrt{K_4(u)K_4(-u)}$, and in the product $K_4(u)K_4(-u)$ the factors $e^{\pm 3u/2}$ cancel while $e^{-\frac{\pi}{4}(e^{2u}+e^{-2u})} = e^{-\frac{\pi}{2}\cosh 2u}$, which gives the symmetric form. For $u \to +\infty$ the product for $K_4(u)$ itself gives the asymptotic, and evenness gives it for $u \to -\infty$. The function $\eta(ie^{2u})$ is analytic where $\operatorname{Re} e^{2u} > 0$, that is for $|\operatorname{Im} u| < \pi/4$. Finally $\eta(i) = \Gamma(\frac14)/(2\pi^{3/4})$. □

The evaluator of Section 2.1 applies unchanged to K_4. The script `dbeta_thorin.py` computes $\xi(\frac12 + s, \chi_4)$ from K_4 by the trapezoid rule with step $1/64$ on $|u| \le 3$ and agrees with the Dirichlet L function of Arb [130] in all fourteen digits printed at $s = 0$, $\frac12$, $0.1 + 5i$ and $0.1 + 20i$; for instance $\xi(\frac12, \chi_4) = 0.98071361405771$, $\xi(0.6 + 5i, \chi_4) = 0.080871220951695 + 0.011736751015145i$ and $\xi(0.6 + 20i, \chi_4) = (-2.3183027026127 - 0.20613003173368i) \times 10^{-6}$. Also $L(\frac12, \chi_4) = 0.667691457189609$.

Remark 8.5.22 (The exit time). By Theorem 4.3.20, $\frac{4}{\pi}\eta(4it/\pi)^3$ is the density of $X_C = \sum_k e_k/(2k-1)^2$, which is $\frac{\pi^2}{8}$ times the exit time τ of Brownian motion from $(-1,1)$ (Section 4.2.2). With $x = 4t/\pi$, $\eta(ix)^3$ is therefore the density on $(0,\infty)$ of $Z = \frac{4}{\pi}X_C = \frac{\pi}{2}\tau$ [260, Section 3], and

$$\xi(s,\chi_4) = \mathbb{E}\big[Z^{(s-1)/2}\big], \qquad \mathbb{E}e^{-uZ} = \operatorname{sech}\sqrt{\pi u} = \prod_{k\ge1}\Big(1 + \frac{4u}{\pi(2k-1)^2}\Big)^{-1}.$$

So Z is a GGC with Thorin measure $\sum_k \delta_{\pi(2k-1)^2/4}$; its total mass $\xi(1,\chi_4) = 1$ is the value $L(1,\chi_4) = \frac{\pi}{4}$, and its mean $\xi(3,\chi_4) = \frac{4}{\pi}\sum_k(2k-1)^{-2} = \frac{\pi}{2}$ is the value $L(3,\chi_4) = \pi^3/32$. The transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$ says that $1/Z$ has the density $x^{-2}\eta(i/x)^3 = x^{-1/2}\eta(ix)^3$, that is $\mathbb{E}g(1/Z) = \mathbb{E}[Z^{-1/2}g(Z)]$ for $g \ge 0$, and with $g(z) = z^{(1-s)/2}$ this is the functional equation. The primal variable Y_4, with density $K_4/\xi(\frac12,\chi_4)$ and $\mathbb{E}e^{sY_4} = \xi(\frac12+s,\chi_4)/\xi(\frac12,\chi_4)$, is $\frac12\log Z$ under the size biasing by $Z^{-1/4}$, since $\mathbb{E}[Z^{-1/4}e^{(s/2)\log Z}]/\mathbb{E}[Z^{-1/4}] = \mathbb{E}[Z^{s/2-1/4}]/\mathbb{E}[Z^{-1/4}]$. So the analogue of the Pólya variable is half the logarithm of a Brownian exit time biased by $Z^{-1/4}$, as for ζ it is half the logarithm of the variable $\pi S_2/2$ of Williams and of Biane, Pitman and Yor biased by $(\pi S_2/2)^{1/4}$ (Section 8.2.3). In the notation of Section 11.5.5.4, $Z = 1/\mathcal{B}_{\chi_{-4}}$ (Proposition 11.5.19).

The basepoint one has an explicit drift. By the functional equation $b_1 = \frac{\xi'}{\xi}(1,\chi_4) = -\frac{\xi'}{\xi}(0,\chi_4)$. With $L(s,\chi_4) = 4^{-s}\big(\zeta(s,\frac14) - \zeta(s,\frac34)\big)$, $\zeta(0,a) = \frac12 - a$ and Lerch's formula $\partial_s\zeta(0,a) = \log\Gamma(a) - \frac12\log 2\pi$ [225, §25.11], one has $L(0,\chi_4) = \frac12$ and $L'(0,\chi_4) = \log\big(\Gamma(\frac14)/2\Gamma(\frac34)\big)$, and with $\psi(\frac12) = -\gamma_E - 2\log 2$ and $\Gamma(\frac14)\Gamma(\frac34) =$

$\pi\sqrt{2}$,

$$b_1 = 3\log 2 + \frac{5}{2}\log\pi + \frac{\gamma_E}{2} - 4\log\Gamma\Big(\frac{1}{4}\Big) = 0.0777839899618, \tag{8.5.7}$$

and so, since $b_1 = \frac{1}{2}\log\frac{4}{\pi} - \frac{\gamma_E}{2} + \frac{4}{\pi}L'(1,\chi_4)$, $L'(1,\chi_4) = \frac{\pi}{4}\big(\gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac{1}{4})\big) = 0.192901316797$. At the centre $\xi'(\frac{1}{2},\chi_4) = 0$, and the variance of the primal variable is $V_4 = \operatorname{Var} Y_4 = (\log\xi)''(\frac{1}{2},\chi_4) = 0.156029649890$.

Since $K_4 > 0$, $\xi(\frac{1}{2}+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}du > 0$ for every real s, so $\xi(s,\chi_4)$ has no real zeros. The script `dbeta_zeros.py` computes the zeros of $L(s,\chi_4)$ below height 100 in ball arithmetic with Arb. The function $\xi(\frac{1}{2}+i\theta,\chi_4)$ is real; its sign is certified on a grid of step 1/32, and each sign change is refined by bisection to 10^{-12}. The argument principle along $2 \to 2+100i \to \frac{1}{2}+100i$, on pieces halved until the enclosure of $\xi(s,\chi_4)$ over each piece lies in a disc excluding zero (1633 pieces), gives $N(100) = 50$ for the number of zeros with $0 < \gamma \le 100$, counted with multiplicity, the remaining sides of the rectangle $[-1,2]\times[0,100]$ following by the functional equation and $\xi(\bar{s},\chi_4) = \overline{\xi(s,\chi_4)}$. This equals the number of sign changes, so all fifty zeros are simple and on the critical line. The zero counting function satisfies

$$N(T) = \frac{T}{2\pi}\log\frac{2T}{\pi e} + \frac{1}{8} + \frac{1}{\pi}\arg L\big(\tfrac{1}{2}+iT,\chi_4\big) + \frac{1}{48\pi T} + O(T^{-3}), \tag{8.5.8}$$

with the argument continued from $L(2,\chi_4) > 0$ along $2 \to 2+iT \to \frac{1}{2}+iT$, for T not an ordinate. Indeed, by the argument principle on the rectangle $[-1,2]\times[0,T]$, the functional equation, the reflection $\xi(\bar{s},\chi_4) = \overline{\xi(s,\chi_4)}$ and the positivity of $\xi(s,\chi_4)$ on the real axis, $\pi N(T)$ is the change of $\arg\xi(s,\chi_4)$ along $2 \to 2+iT \to \frac{1}{2}+iT$, and the expansion of the gamma factor by Stirling's series [225, §5.11] in the proof of Proposition 8.5.30 below, at $\alpha = \frac{1}{2}$, gives (8.5.8); compare [71, Chapter 16], where the main term is derived with an error $O(\log T)$. Under GRH for χ_4, $V_4 = \sum_\gamma \gamma^{-2}$ over all the ordinates of both signs; twice the sum over the fifty computed ones is 0.1395621, and the main term $\frac{1}{\pi}(\log\frac{2T}{\pi}+1)/T$ of the rest, from $dN(t) \approx \frac{1}{2\pi}\log\frac{2t}{\pi}\,dt$, is 0.0164 at $T = 100$, together 0.1560.

8.5.7.2 The Lévy form and the sine squared decomposition

The Lévy form of Proposition 11.3.2 for χ_{12} has an analogue for χ_4 at every basepoint $\alpha \ge 1$.

Proposition 8.5.23. *For $\alpha \ge 1$,*

$$\log\frac{\xi(\alpha+s,\chi_4)}{\xi(\alpha,\chi_4)} = b_\alpha s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_\alpha(dx), \qquad b_\alpha = \frac{\xi'}{\xi}(\alpha,\chi_4),$$

with the signed Lévy measure

$$\Pi_\alpha(dx) = \frac{e^{-\alpha x}}{2x\sinh x}\,dx + \sum_{n\ge 2}\frac{\chi_4(n)\Lambda(n)}{n^\alpha\log n}\,\delta_{\log n}(dx),$$

whose canonical measure $x\Pi_\alpha(dx)$ has the positive density $e^{-\alpha x}/(2\sinh x)$ and the atoms $\chi_4(n)\Lambda(n)n^{-\alpha}$ at $\log n$. For $\alpha > 1$ the form holds for $\operatorname{Re} s > 1-\alpha$, with absolute convergence. At $\alpha = 1$ it holds for $\operatorname{Re} s > 0$, the prime sum converging conditionally, as a limit over $n \le X$.

Proof. The factor $(4/\pi)^{(s+1)/2}$ is linear in log and is absorbed by the drift. Binet's formula [225, §5.9], in the integrated form

$$\log\frac{\Gamma(z+v)}{\Gamma(z)} - v\psi(z) = \int_0^\infty (e^{-vx} - 1 + vx)\frac{e^{-zx}}{x(1-e^{-x})}\,dx$$

for $\operatorname{Re} z > 0$ and $\operatorname{Re}(z+v) > 0$, which one checks by differentiating twice in v, gives at $z = \frac{\alpha+1}{2}$, $v = \frac{s}{2}$ and after $x \mapsto 2x$ the density $\frac{e^{-(\alpha+1)x}}{x(1-e^{-2x})} = \frac{e^{-\alpha x}}{2x\sinh x}$, for $\operatorname{Re} s > -1-\alpha$. For $\operatorname{Re}(\alpha+s) > 1$ the Euler product gives $\log L(\alpha+s,\chi_4) = \sum_n \chi_4(n)\Lambda(n)n^{-\alpha-s}/\log n$, and subtracting the value and the derivative at $s=0$ gives the atoms; for $\alpha > 1$ the sum converges absolutely for $\operatorname{Re} s > 1-\alpha$, because $|e^{-sx}-1+sx| \le e^{-x\operatorname{Re} s}+1+|s|x$ and $\sum_n \Lambda(n)n^{-\alpha}\big(n^{-\operatorname{Re} s}+1+|s|\log n\big)/\log n < \infty$ there. At $\alpha = 1$ the linear coefficient is $-\sum_n \chi_4(n)\Lambda(n)/n = \frac{L'}{L}(1,\chi_4)$ and the constant term is $\sum_n \chi_4(n)\Lambda(n)/(n\log n) = \log L(1,\chi_4)$; both series converge by the prime number theorem for arithmetic progressions [71, Chapter 20], as in the proof of Proposition 11.3.2. □

The archimedean density is that of ξ shifted by one basepoint: at the basepoint $\alpha+1$ the gamma density $e^{-(\alpha+2)x}/(1-e^{-2x})$ of $x\Pi_{\alpha+1}$ is the gamma density of ξ at the basepoint α (Section 9.1.4), because the gamma factor $\Gamma(\frac{s+1}{2})$ at $\alpha+1$ and the gamma factor $\Gamma(1+\frac{s}{2})$ of ξ, which absorbs the factor s, at α have the same argument $\frac{\alpha+2+s}{2}$. There is no pole term, and the atoms carry the sign of $\chi_4(n)$. As for χ_{12} in Section 11.3.2, the tilted variable $V_\alpha^{(4)}$ with moment generating function $\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4)$ is quasi infinitely divisible for $\alpha > 1$, the Euler product writing its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions, and it is not infinitely divisible: by the uniqueness of the Lévy measure of a quasi infinitely divisible law [181], the atom $-3^{-\alpha}$ of Π_α at $\log 3$, reflected to $-\log 3$, would have to be positive. At $\alpha = 1$, $V_1^{(4)}$ is not quasi infinitely divisible, since the atoms of each sign have infinite mass, $\sum_{p\equiv 1}1/p$ and $\sum_{p\equiv 3\,(4)}1/p$ both diverging, and cancel only conditionally; the argument of Proposition 11.1.3(3) applies.

The sine squared lemma (1.2.9) turns the Lévy form into the Thorin density, the bracket being $\operatorname{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$ [249, Theorem 8.2]. With $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,\chi_4)$ and $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$ as in (9.1.7),

$$v_\alpha(\theta) = b_\alpha + G_\alpha(\theta) + \varpi_\alpha(\theta), \qquad \alpha \ge 1, \tag{8.5.9}$$

where the gamma piece and the prime piece are

$$G_\alpha(\theta) = \int_0^\infty \frac{1-\cos\theta x}{2\sinh x}\, e^{-\alpha x}dx = \frac{1}{2}\Big[\operatorname{Re}\psi\Big(\frac{\alpha+1+i\theta}{2}\Big) - \psi\Big(\frac{\alpha+1}{2}\Big)\Big]$$
$$= \sum_{k\ge 0} q_{\alpha+1+2k}(\theta) \ge 0,$$
$$\varpi_\alpha(\theta) = \sum_{n\ge 2} \frac{\chi_4(n)\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big),$$

the series for G_α from $1/(2\sinh x) = \sum_{k\ge 0} e^{-(2k+1)x}$, and ϖ_α converging conditionally at $\alpha = 1$. So $G_\alpha = \sum_{k\ge 1} q_{(\alpha-1)+2k}$ is the gamma piece of ξ at the basepoint $\alpha - 1$ in (9.1.7), without its pole term $-q_{\alpha-2}$. For $\alpha \ge 1$ the Thorin representation is

$$\log\frac{\xi(\alpha+\sqrt{u},\chi_4)}{\xi(\alpha,\chi_4)} = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\, v_\alpha(\theta)\, d\theta, \qquad u \ge 0, \tag{8.5.10}$$

by Proposition 8.5.24 below.

The first zero of $\xi(\frac{1}{2}+i\theta,\chi_4)$ is $\gamma_1 = 6.0209489$.

Figure 8.7 shows the density and the decomposition. On the left, v_α at $\alpha = 1$, 0.75 and 0.55. Its minimum is b_α, at the origin (0.0777840, 0.0389784 and 0.0078013), as Theorem 9.1.18 proves for every $\alpha \in (\frac{1}{2}, 4.70]$ on the verified range and for every $\alpha \ge 1$ at every height. Near each ordinate v_α rises to about $1/(\alpha-\frac{1}{2})$, the height of the Cauchy kernel of a zero on the line, plus the contribution of the other zeros. As $\alpha \downarrow \frac{1}{2}$ the margin at the origin shrinks linearly, $b_\alpha \approx (\alpha-\frac{1}{2})V_4$. On the right, at $\alpha = \frac{3}{2}$, the prime piece ϖ_α is negative on 71% of the grid of step 0.01 on $[0,30]$, with maximum 0.33562 at $\theta = 6.0$, next to γ_1; the minimum of v_α on the grid is $b_{3/2} = 0.15421515$, at the origin. For ξ the same figure would show a positive prime piece and the negative pole piece (Figure 8.4), while for χ_4 the negative input is spread over the primes $p \equiv 3 \pmod 4$.

Proposition 8.5.24. *For every $\alpha \ge 1$, $u \mapsto \xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$ is the Laplace transform of a GGC with Thorin density $v_\alpha(\theta)/\pi > 0$ in the chart $\theta = \sqrt{z}$, and* (8.5.10) *holds. For $\alpha > \frac{1}{2}$ it is the Laplace transform of a GGC if and only if no zero of $L(s,\chi_4)$ has real part greater than α. GRH for χ_4 holds if and only if $v_\alpha \ge 0$ for every $\alpha > \frac{1}{2}$, and the Thorin measures at resolution $\varepsilon = \alpha - \frac{1}{2}$ converge vaguely, as $\varepsilon \downarrow 0$, to the sum of the point masses δ_{γ^2} over the zeros on the critical line, which is the Thorin measure of $\xi(\frac{1}{2},\chi_4)/\xi(\frac{1}{2}+\sqrt{u},\chi_4)$ under GRH.*

Proof. The proofs of Theorem 1.2.2, of Lagarias's criterion [160] and of the vague limit stated after Remark 1.2.4 use only the Hadamard product of an entire function of order one whose zeros are symmetric under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1-\rho$, with $\sum_\rho |\rho|^{-2} < \infty$; they apply to $\xi(s,\chi_4)$ [71, Chapter 12], as Section 8.5.3 and Proposition 11.3.1 apply them to $\xi(s,\chi_{12})$. Positivity at $\alpha \ge 1$ needs every zero to have

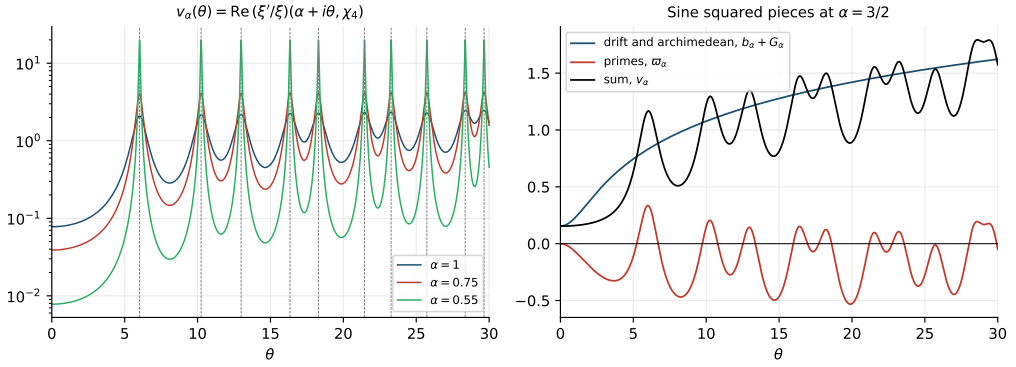


Figure 8.7: Left: $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta, \chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1$, 0.75, 0.55 on a logarithmic scale, with the ordinates of the zeros of $L(s,\chi_4)$ (dashed). Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece ϖ_α, of both signs, and their sum v_α. Drawn by `dbeta_figs.py`.

real part less than one, which is the absence of zeros of $L(s,\chi_4)$ on $\mathrm{Re}\,s = 1$ [71, Chapter 14] with the Euler product and the functional equation. □

At the basepoint one the clock $T_1^{(4)}$, with Laplace transform $1/\xi(1+\sqrt{u},\chi_4)$, therefore has the expansion $1 - b_1\sqrt{u} + O(u)$ at the origin, and as for ξ in (9.1.3) the Tauberian theorem [86, XIII.5] gives the tail $P(T_1^{(4)} > x) \sim (b_1/\sqrt{\pi})\,x^{-1/2} = 0.0438849\,x^{-1/2}$, with b_1 from (8.5.7).

Positivity at $\alpha \geq 1$ uses the zeros, not the Lévy measure. Bounding the prime weights by their absolute values gives positivity only far to the right, and no such bound does better.

Proposition 8.5.25. *For $\alpha > 1$ and real θ,*

$$v_\alpha(\theta) \geq A_4^\infty(\alpha,\theta) - S_4(\alpha),$$
$$A_4^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi\Big(\frac{\alpha+1+i\theta}{2}\Big),$$
$$S_4(\alpha) = \sum_{n\ \mathrm{odd}} \frac{\Lambda(n)}{n^\alpha} = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log 2}{2^\alpha - 1},$$

and $A_4^\infty(\alpha,\theta)$ increases in $|\theta|$. The bound is positive at every θ exactly when $\alpha > \alpha_c(\chi_4) = 2.2823106368$, the root of $A_4^\infty(\alpha,0) = S_4(\alpha)$. For χ_{12} the same bound, with $A_{12}^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{12}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi(\frac{\alpha+i\theta}{2})$ and $S_{12}(\alpha) = S_4(\alpha) - \log 3/(3^\alpha - 1)$, is positive at every θ exactly when $\alpha > \alpha_c(\chi_{12}) = 1.8348743295$.

Proof. For $\mathrm{Re}\,s > 1$,

$$\frac{\xi'}{\xi}(s,\chi_4) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\psi\Big(\frac{s+1}{2}\Big) + \frac{L'}{L}(s,\chi_4),$$

and $|\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)| \le \sum_n |\chi_4(n)|\Lambda(n)n^{-\alpha} = S_4(\alpha)$. The series $\operatorname{Re}\psi(x+iy)-\psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$ shows that A_4^∞ increases in $|\theta|$. Since $A_4^\infty(\alpha,0)$ increases in α and $S_4(\alpha)$ decreases, the root is unique, and `dbeta_thorin.py` locates it by bisection with certified signs. For χ_{12} the primes 2 and 3 drop out of the sum. □

Proposition 8.5.26. *For $\alpha>1$ and every $T>0$,*

$$\inf_{\theta\ge T}\Big(v_\alpha(\theta)-A_4^\infty(\alpha,\theta)\Big) = \inf_{\theta\ge T}\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -S_4(\alpha),$$

so no lower bound for $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)$ better than $-S_4(\alpha)$ holds at all large heights. The bound of Proposition 8.5.25 certifies $v_\alpha(\theta)>0$ exactly for $|\theta|>t_0(\alpha)$, where $A_4^\infty(\alpha,t_0)=S_4(\alpha)$, for $1<\alpha\le\alpha_c$ (so that $t_0(\alpha_c)=0$), and at every real θ for $\alpha>\alpha_c$, and

$$\log t_0(\alpha) = \frac{2}{\alpha-1} - 2\gamma_E - \log\frac{8}{\pi} + O(\alpha-1), \qquad \alpha\downarrow 1,$$

with $-2\gamma_E-\log\frac{8}{\pi} = -2.0891$.

Proof. From the Euler product, $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -\sum_p \log p\,\operatorname{Re}\frac{w_p}{1-w_p}$ with $w_p=\chi_4(p)p^{-\alpha-i\theta}$, and on the circle $|w|=r<1$ the maximum of $\operatorname{Re}\frac{w}{1-w}=\operatorname{Re}\frac{1}{1-w}-1$ is $\frac{r}{1-r}$, at $w=r$; this gives the lower bound $-S_4(\alpha)$. Since the numbers $\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem [107, Chapter 23] gives, for every P and $\eta>0$, a height $\theta\ge T$ at which $|\chi_4(p)p^{-i\theta}-1|<\eta$ for all odd $p\le P$, and the primes beyond P contribute at most $\sum_{p>P}\log p\,p^{-\alpha}/(1-p^{-\alpha})$, which is small for P large. For the asymptotic, $S_4(\alpha)=\frac{1}{\alpha-1}-\gamma_E-\log 2+O(\alpha-1)$ and $A_4^\infty(\alpha,t)=\frac12\log\frac{2t}{\pi}+O(t^{-2})$ as $t\to\infty$, uniformly for $1\le\alpha\le2$, from $\psi(z)=\log z-\frac{1}{2z}+O(|z|^{-2})$; solving $A_4^\infty(\alpha,t_0)=S_4(\alpha)$ gives the formula, since $t_0(\alpha)\to\infty$. □

α	$S_4(\alpha)$	$t_0(\alpha)$	$\log t_0(\alpha)-\frac{2}{\alpha-1}$
2	0.33891193	2.36723	−1.1383
1.75	0.57950941	4.71677	−1.1155
1.5	1.1261406	14.8728	−1.3005
1.25	2.9637961	589.44	−1.6208
1.1	8.8348984	7.41336×10^{7}	−1.8786
1.05	18.784554	3.25219×10^{16}	−1.9793

Table 8.19: The height $t_0(\alpha)$ above which the bound of Proposition 8.5.25 certifies $v_\alpha>0$, for basepoints below $\alpha_c(\chi_4)=2.2823$; the last column tends to $-2\gamma_E-\log\frac{8}{\pi}=-2.0891$ (Proposition 8.5.26). Computed by `dbeta_thorin.py`.

So for $1<\alpha<\alpha_c$ the absolute values of the weights reduce the positivity of v_α to a computation on $[0,t_0(\alpha)]$, but t_0 grows like $e^{2/(\alpha-1)}$ (Table 8.19). At $\alpha=1$ the prime piece converges only conditionally, by the prime number theorem for progressions, whose analytic input is the absence of zeros of $L(s,\chi_4)$ on $\operatorname{Re} s=1$; the positivity of v_1, which Proposition 8.5.24 draws from that input, cannot come from the sizes of the weights.

8.5.7.3 The Gaussian field

The pairing of the primes $p\equiv 1$ and $p\equiv 3 \pmod 4$ is their splitting in $K=\mathbb{Q}(i)$. Accordingly $\zeta_K(s)=\zeta(s)L(s,\chi_4)$, and in the normalisation of Section 10.3.3,

$$\xi_K(s)=2s(s-1)\pi^{-s}\Gamma(s)\zeta_K(s)=\xi(s)\,\xi(s,\chi_4),$$

by the duplication formula $\Gamma(s)=2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$. Write v_α^K and v_α^ξ for the brackets $\operatorname{Re}(\xi_K'/\xi_K)(\alpha+i\theta)$ and $\operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$, and v_α as before for χ_4.

Proposition 8.5.27. *On every vertical line, $v_\alpha=v_\alpha^K-v_\alpha^\xi$. For $\alpha>1$ the canonical measure of ξ_K at the basepoint α is*

$$x\Pi_\alpha^K(dx)=\Big(\frac{e^{-(\alpha+1)x}}{1-e^{-x}}-e^{-(\alpha-1)x}\Big)dx+\sum_{n\ge 2}\frac{(1+\chi_4(n))\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

and the canonical measure of Proposition 8.5.23 is $x\Pi_\alpha=x\Pi_\alpha^K-x\Pi_\alpha^\xi$, with $x\Pi_\alpha^\xi$ that of Section 9.1.4: the pole densities $-e^{-(\alpha-1)x}$ cancel, the gamma densities differ by $e^{-\alpha x}/(2\sinh x)$, and the atoms by $\chi_4(n)\Lambda(n)n^{-\alpha}$.

Proof. Take logarithmic derivatives in $\xi_K=\xi\cdot\xi(\cdot,\chi_4)$. For the measures write $\xi_K(s)=(s-1)\cdot 2\pi^{-s}\Gamma(s+1)\zeta_K(s)$: Binet's formula for $\Gamma(s+1)$ at α gives the density $e^{-(\alpha+1)x}/(x(1-e^{-x}))$, the factor $s-1$ gives $-e^{-(\alpha-1)x}/x$ as for ξ, and $-\frac{\zeta_K'}{\zeta_K}=-\frac{\zeta'}{\zeta}-\frac{L'}{L}$ has the coefficients $(1+\chi_4(n))\Lambda(n)\ge 0$. Finally $\frac{e^{-(\alpha+1)x}}{1-e^{-x}}-\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}=e^{-\alpha x}\Big(\frac{1}{e^x-1}-\frac{1}{e^{2x}-1}\Big)=\frac{e^{-\alpha x}e^x}{e^{2x}-1}=\frac{e^{-\alpha x}}{2\sinh x}$. □

Corollary 8.5.28. (i) *For every $\alpha\ge 1$ there are independent GGC variables H_α^ξ and $H_\alpha^{(4)}$, with Laplace transforms $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ and $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, whose sum $H_\alpha^K=H_\alpha^\xi+H_\alpha^{(4)}$ is the GGC with Laplace transform $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ and Thorin density v_α^K/π.*

(ii) *GRH for χ_4 holds if and only if $v_\alpha^K\ge v_\alpha^\xi$ on every line $\operatorname{Re} s=\alpha>\frac12$: the Thorin density of ξ_K dominates that of ξ at every basepoint.*

(iii) *Under RH, GRH for χ_4 holds if and only if, for every $\alpha>\frac12$, $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ is the Laplace transform of $H_\alpha^\xi+H$ for some GGC H independent of H_α^ξ.*

Proof. (i) By Theorem 1.2.2 and Proposition 8.5.24 both factors are GGC transforms for $\alpha \geq 1$, and transforms of independent sums multiply. (ii) By Proposition 8.5.27, $v_\alpha^K \geq v_\alpha^\xi$ is $v_\alpha \geq 0$, and Proposition 8.5.24 applies. (iii) Under RH, H_α^ξ is a GGC for every $\alpha > \frac{1}{2}$ (Theorem 1.2.2). If GRH holds for χ_4, take $H = H_\alpha^{(4)}$. Conversely, the transform of H is then $\xi(\alpha, \chi_4)/\xi(\alpha + \sqrt{u}, \chi_4)$, and if it is a GGC transform for every $\alpha > \frac{1}{2}$, Proposition 8.5.24 gives GRH. □

Neither v_α^K nor v_α^ξ is positive term by term: each has the pole piece $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ of (3.1.1). The factorisation is the right language for the pairing of the primes, but it does not by itself lower α_c: by Proposition 8.5.26 any argument that uses only the sizes of the weights reaches no further. Table 8.20 shows the three brackets at $\alpha = \frac{3}{2}$.

θ	0	2	5	10	20
$v_{3/2}^K(\theta)$	0.200351	0.225243	0.685189	1.32766	1.49863
$v_{3/2}^\xi(\theta)$	0.0461359	0.0470396	0.0526739	0.102889	0.606114
$v_{3/2}(\theta)$	0.154215	0.178204	0.632515	1.22477	0.892517

Table 8.20: The brackets of $\xi_{\mathbb{Q}(i)}$, ξ and $\xi(\cdot, \chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 8.5.27). Computed by `dbeta_thorin.py`.

The same holds for every real primitive character χ_D, with D a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$: ξ_K is a constant multiple of $\xi \cdot \xi(\cdot, \chi_D)$, the constant being $2/\sqrt{|D|}$ for $D < 0$ (Section 8.5.6) and 2 for $D > 0$, since then $\xi_K(s) = s(s-1)D^{s/2}\Gamma_{\mathbb{R}}(s)^2\zeta_K(s)$ and $\xi(s, \chi_D) = D^{s/2}\Gamma_{\mathbb{R}}(s)L(s, \chi_D)$. So $v_\alpha^{(\chi_D)} = v_\alpha^K - v_\alpha^\xi$; the GGC property at every $\alpha \geq 1$, the equivalence of GRH for χ_D with the domination of the Thorin density of ξ by that of ξ_K, and the localisation of Corollary 8.5.31 hold for $\xi(s, \chi_D)$ with the same proofs. For χ_{12}, $K = \mathbb{Q}(\sqrt{3})$, and the numbers are: $\alpha_c(\chi_{12}) = 1.8348743295$ (Proposition 8.5.25); $V_{12} = (\log \xi)''(\frac{1}{2}, \chi_{12}) = 0.332923608212$; and at $\alpha = 1$, 0.75 and 0.6 the drift b_α is 0.1650833, 0.0830566 and 0.0332812, the minimum of v_α on a grid of step 0.01 on $[0, 60]$, attained at the origin (`dbeta_minimum.py`; Theorem 9.1.18). The script `dbeta_zeros.py` certifies the first five ordinates of $L(s, \chi_{12})$, 3.8046276331, 6.6922233205, 8.8905929587, 11.1883927451 and 12.9661788080, all simple and on the critical line, in agreement with Section 8.5.1; their largest gap, the first, 2.8876, lies below $2/\sqrt{V_{12}} = 3.4662$, and below the value 3.3189 at $a = \frac{1}{2}$ of the threshold of Remark 9.1.20 for χ_{12}.

Question 8.5.29. Is there a construction of $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ from the Gaussian primes themselves, two of norm p for each $p \equiv 1$ and one of norm p^2 for each $p \equiv 3 \pmod 4$, that persists below the basepoint one? By Proposition 8.5.26 such a construction must see the phases $p^{-i\theta}$, not only the sizes of the weights; below

$\alpha = 1$ its existence is equivalent to GRH for χ_4 under RH, by Corollary 8.5.28(iii). In the language of Section 8.5.7.4, the decomposition is the additivity $\arg \xi_K = \arg \xi + \arg \xi(\cdot, \chi_4)$ on vertical lines, and the domination of Corollary 8.5.28(ii) says that $\arg \xi_K - \arg \xi$ is nondecreasing on every line $\operatorname{Re} s = \alpha > \frac{1}{2}$.

8.5.7.4 The Thorin measure as an argument

PROPOSITION 8.5.30. *Let $\alpha > \frac{1}{2}$ be such that no zero of $\xi(s,\chi_4)$ lies on the line $\operatorname{Re} s = \alpha$; this holds for every $\alpha \geq 1$. Then for $\theta > 0$*

$$\frac{1}{\pi}\int_0^\theta v_\alpha(t)\,dt = \frac{1}{\pi}\arg \xi(\alpha + i\theta, \chi_4),$$

with the argument continuous along the segment from α to $\alpha + i\theta$ and zero at α; for $\alpha \geq 1$ the left side is the mass of $(0,\theta^2]$ under the Thorin measure of Proposition 8.5.24. As $\theta \to \infty$, with $a = \frac{\alpha+1}{2}$,

$$\frac{1}{\pi}\arg \xi(\alpha+i\theta,\chi_4) = \frac{\theta}{2\pi}\log\frac{2\theta}{\pi e} + \frac{\alpha}{4} + \frac{1}{\pi}\arg L(\alpha+i\theta,\chi_4) - \frac{a(a-1)+\frac{1}{6}}{\pi\theta} + O(\theta^{-3}),$$

with $\arg L$ continuous along the same segment. As $\alpha \downarrow \frac{1}{2}$, for θ not an ordinate, the left side tends to the number of zeros on the critical line with $0 < \gamma < \theta$, which is $N(\theta)$ under GRH; at $\alpha = \frac{1}{2}$ the expansion is the counting formula (8.5.8).

Proof. Since $\frac{d}{dt}\log \xi(\alpha+it,\chi_4) = i\frac{\xi'}{\xi}(\alpha+it,\chi_4)$, the derivative of $\arg \xi(\alpha+it,\chi_4) = \operatorname{Im}\log \xi(\alpha+it,\chi_4)$ is $v_\alpha(t)$, and $\xi(\alpha,\chi_4) > 0$. The expansion is $\frac{\theta}{2}\log\frac{4}{\pi}$ plus Stirling's series $\log\Gamma(z) = (z-\frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi + \frac{1}{12z} + O(|z|^{-3})$ [225, §5.11] at $z = a + \frac{i\theta}{2}$: with $y = \frac{\theta}{2}$,

$$\operatorname{Im}\log\Gamma(a+iy) = y\log y - y + \frac{\pi}{2}\Big(a-\frac{1}{2}\Big) - \Big(\frac{a(a-1)}{2} + \frac{1}{12}\Big)\frac{1}{y} + O(y^{-3}),$$

from $\arg z = \frac{\pi}{2} - \frac{a}{y} + O(y^{-3})$ and $\log|z| = \log y + \frac{a^2}{2y^2} + O(y^{-4})$; and $\frac{\theta}{2}\log\frac{4}{\pi} + \frac{\theta}{2}\log\frac{\theta}{2} - \frac{\theta}{2} = \frac{\theta}{2}\log\frac{2\theta}{\pi e}$, $\frac{1}{2}(a-\frac{1}{2}) = \frac{\alpha}{4}$. For the limit, write v_α by the Hadamard product, as in Proposition 1.2.3, as the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$ of $(\alpha-\beta)/((\alpha-\beta)^2 + (t-\gamma)^2)$; each term integrates over $[0,\theta]$ to $\arctan\frac{\theta-\gamma}{\alpha-\beta} + \arctan\frac{\gamma}{\alpha-\beta}$, which for a zero on the line tends to π if $0 < \gamma < \theta$ and to 0 if $\gamma > \theta$ or $\gamma < 0$, while the two terms of an off line pair $\frac{1}{2} \pm \delta + i\gamma$ have the limits $\mp\big(\arctan\frac{\theta-\gamma}{\delta} + \arctan\frac{\gamma}{\delta}\big)$, which cancel. The terms are dominated by π and by $C\theta/\gamma^2$ for $|\gamma| > 2\theta + 1$, uniformly for $\frac{1}{2} < \alpha \leq 1$, so dominated convergence applies. At $\alpha = \frac{1}{2}$, $a = \frac{3}{4}$ and $a(a-1) + \frac{1}{6} = -\frac{1}{48}$. □

The Thorin measure at resolution $\varepsilon = \alpha - \frac{1}{2}$ is therefore the zero counting function smoothed by the Cauchy kernel of width ε, the analogue for χ_4 of the

cumulative Thorin mass of ξ (Section 3.4.1). At $\alpha = 0.75$ the difference between the left side and the first three terms of the expansion is $-0.0182/\theta$ at $\theta = 50$, 100 and 200, against $-(a(a-1)+\frac{1}{6})/\pi = -11/(192\pi) = -0.018237$ at $a = \frac{7}{8}$ (`dbeta_thorin.py`).

COROLLARY 8.5.31. *GRH for χ_4 holds if and only if, for every $\alpha > \frac{1}{2}$, $\theta \mapsto \arg \xi(\alpha + i\theta, \chi_4)$ is nondecreasing. With no hypothesis, the heights $\theta > 0$ at which it decreases lie in the union of the windows $|\theta - \gamma| < \sqrt{\delta^2 - (\alpha - \frac{1}{2})^2}$, each of length less than 1, over the zeros $\frac{1}{2} + \delta + i\gamma$ with $\delta > \alpha - \frac{1}{2}$, and there are none at heights below $2.5 \times 10^7 - \frac{1}{2}$.*

Proof. The derivative is $v_\alpha(\theta)$, and Proposition 8.5.24 gives the equivalence. By the Hadamard product, v_α is a sum of positive Cauchy kernels, one for each zero on the line and two for each off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta \le \alpha - \frac{1}{2}$, plus, for each pair with $\delta > \alpha - \frac{1}{2}$, the difference $\pi\big(P_{\varepsilon+\delta} - P_{\delta-\varepsilon}\big)(\theta \mp \gamma)$ with $\varepsilon = \alpha - \frac{1}{2}$, which is negative exactly where $|\theta \mp \gamma| < \sqrt{\delta^2 - \varepsilon^2}$ (proof of Proposition 3.3.1(ii)); for $\theta > 0$ and $\gamma > 0$ the reflected term at $\theta + \gamma$ is positive, since $\theta + \gamma > \gamma > \delta$. A window has length $2\sqrt{\delta^2 - \varepsilon^2} < 2\delta < 1$. Platt's verification of GRH for the modulus 4 to height $10^8/4$ [245] puts every zero off the line above 2.5×10^7, and so every window above $2.5 \times 10^7 - \frac{1}{2}$. □

A window holds a decrease only if the negative term of its pair outweighs all the positive terms there, and the decrease is bounded: the negative part of the term $\pi(P_{\varepsilon+\delta} - P_{\delta-\varepsilon})(\theta - \gamma)$ of a pair has integral $4 \arctan \sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - \pi < \pi$, by Proposition 3.3.1(iii), so over any interval the argument falls by less than π times the number of pairs whose windows meet it, whatever the other zeros, and the fall caused by a pair tends to zero as $(\alpha - \frac{1}{2})/\delta \to 0$. This answers the quantitative half of a question of the first author's note. No lower bound for the positive terms over a window, such as a mean value estimate for the contribution of the zeros on the line, can exclude this for every pair: at the centre $\theta = \gamma$ of its window the pair contributes $\frac{1}{\varepsilon+\delta} - \frac{1}{\delta-\varepsilon} = -\frac{2\varepsilon}{\delta^2-\varepsilon^2}$, which is unbounded below as $\delta \downarrow \varepsilon$, that is as the zero approaches the line $\operatorname{Re} s = \alpha$ from the right, while by Lemma 11.4.9 a zero far enough to the right of the line cannot produce a decrease against a given positive floor. Outside the strips $|\theta - \gamma| < (1 + \sqrt{2})\delta$ around the ordinates of the zeros off the line, Theorem 9.1.15 gives the argument a rate of increase that grows like $\log \theta$ (Corollary 9.1.16). Figure 8.8 shows the smoothing.

8.5.7.5 The centre

For ζ the positive structure at the centre is the Thorin measure of the clock, not a structure of the Pólya variable itself (Proposition 11.1.3(1)). The same holds for χ_4. The characteristic function $\xi(\frac{1}{2} + i\theta, \chi_4)/\xi(\frac{1}{2}, \chi_4)$ of Y_4 is real and changes sign at $\gamma_1 = 6.0209489$ and at each of the other ordinates below height 100; Hardy's

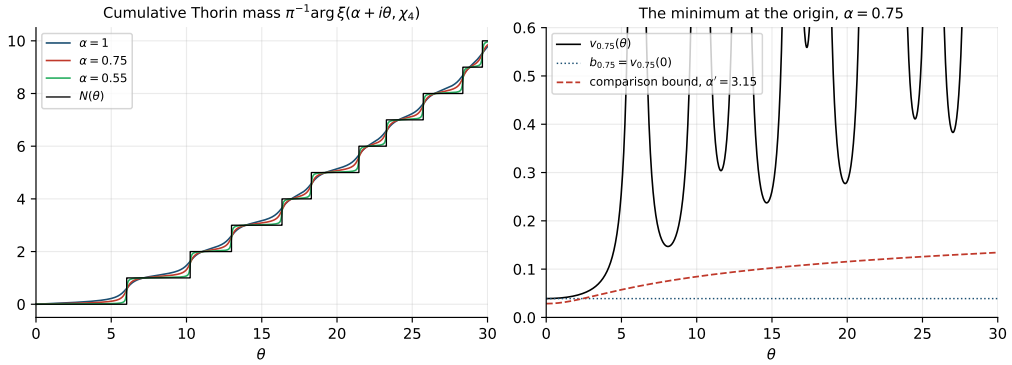


Figure 8.8: Left: the cumulative Thorin mass $\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4)=\frac{1}{\pi}\int_0^\theta v_\alpha$ for $\alpha=1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s,\chi_4)$. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75}=0.0389784$ at the origin, and the lower bound of Corollary 9.1.16 with $\alpha'=3.15$, which exceeds $b_{0.75}$ for $\theta>2.434$ (`dbeta_minimum.py`). Drawn by `dbeta_figs.py`.

argument for ζ [324, Chapter 10], applied to the theta series $\eta(ix)^3$ of χ_4, gives infinitely many such sign changes. So Y_4 is not quasi infinitely divisible, and in particular not infinitely divisible: a quasi infinitely divisible characteristic function never vanishes. Nor is $\xi(\frac12+w,\chi_4)/\xi(\frac12,\chi_4)$ a positive mixture $\int M(d\lambda)/(w+\lambda)^2$ of Gamma(2) transforms, for a more elementary reason: such a mixture decreases on $w>0$, while $\xi(\frac12+w,\chi_4)=\int_{\mathbb{R}}K_4(u)\cosh(wu)\,du$ increases there. A one sided object remains, the half kernel $K_4(u)\mathbf{1}\{u>0\}$, and it is not a positive gamma mixture either.

Proposition 8.5.32. *For no $a>0$ is $u^{1-a}K_4(u)$ completely monotone on $u>0$; in particular the half kernel is not a positive mixture of Gamma(a) densities $u^{a-1}e^{-\lambda u}$, for any shape a. The theta function $\eta(ix)^3$ is not completely monotone on $x>0$.*

Proof. If $g(u)=\int_{[0,\infty)}e^{-\lambda u}M(d\lambda)$ with $M\ge0$ and $M\ne0$, choose L with $M([0,L])>0$; then $g(u)\ge M([0,L])e^{-Lu}$, so g decays at most exponentially, while $u^{1-a}K_4(u)$ decays like $\exp(-\frac{\pi}{4}e^{2u})$ (Proposition 8.5.21). The Jacobi series $\eta(ix)^3=\sum_{n\ge0}(-1)^n(2n+1)e^{-\pi(2n+1)^2x/4}$ is the Laplace transform of the signed measure with the weights $(-1)^n(2n+1)$ at the points $\pi(2n+1)^2/4$, which by the uniqueness of the Laplace transform is its only representing measure, and it is not positive, so Bernstein's theorem excludes complete monotonicity. □

The decay argument applies to the kernel of every completed L function of degree one, all of which decay like $\exp(-ce^{2|u|})$; for ξ it shows that no positive

mixture of Gamma(a) densities represents the Pólya kernel on the half line, for any shape a, as [249, Section 13.6] observes (see also [252, Section 12.6]).

The positive structure is in the reciprocal of the product. With $q = e^{-2\pi x}$,

$$\frac{e^{-\pi x/4}}{\eta(ix)^3} = \prod_{m\ge 1} \left(1-q^m\right)^{-3} = \exp\Big(3\sum_{N\ge 1} \sigma_{-1}(N)\, e^{-2\pi N x}\Big), \qquad \sigma_{-1}(N) = \sum_{d\mid N} \frac{1}{d}, \tag{8.5.11}$$

from $-\log(1-q^m) = \sum_k q^{mk}/k$; so the logarithm of the left side is completely monotone, and hence so is the left side. In the language of Section 8.5.4, $\prod_m (1 - e^{-2\pi m x})^3$ is the distribution function of the maximum M_3 of independent exponential variables, three of each rate $2\pi m$. The representation of Section 8.5.4 also carries over: since $\eta(ix)^3 = e^{-\pi x/4}\Pr(M_3 \le x)$ and $x^{c-1}e^{-\pi x/4}(\pi/4)^c/\Gamma(c)$ is the density of $\frac{4}{\pi}G_c$, with G_c a gamma variable of shape c independent of M_3, the Mellin form gives $\xi(s,\chi_4) = (\frac{4}{\pi})^c\Gamma(c)\Pr(\frac{4}{\pi}G_c \ge M_3)$ with $c = \frac{s+1}{2}$, that is

$$L(s,\chi_4) = \Pr\big(4G_{(s+1)/2} \ge \pi M_3\big), \qquad s > -1,$$

which at $s = 1$ is $\mathbb{E}e^{-\pi M_3/4} = \frac{\pi}{4}\int_0^\infty \eta(ix)^3 dx = \frac{\pi}{4}$.

The half kernel has the Laplace transform

$$\int_0^\infty e^{-wu} K_4(u)\, du = A_4\big(\tfrac{1}{2} - w\big), \tag{8.5.12}$$

where

$$A_4(s) = \int_1^\infty \eta(ix)^3 x^{(s+1)/2}\frac{dx}{x} = \sum_{n\ge 1} \chi_4(n)\, n\, \frac{\Gamma(\frac{s+1}{2}, \frac{\pi n^2}{4})}{(\pi n^2/4)^{(s+1)/2}}$$

is an incomplete Mellin transform of η^3, and the transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$, applied to the integral over $(0,1)$, gives $\xi(s,\chi_4) = A_4(s) + A_4(1-s)$ for every s. This is the analogue of Riemann's formula (Proposition 8.5.5) and of the symmetric form (8.5.5) for Δ, with no pole terms, since $\eta(ix)^3$ has no constant term.

Question 8.5.33. Can the complete monotonicity of (8.5.11), and of its analogue $\prod_m (1-q^m)^{-1}$ for the kernel of χ_{12}, be turned into a positive representation on the reciprocal side, in the spirit of Thorin's construction for the reciprocal of a GGC transform?

8.6 The inverse problem: recovering the atoms by nonnegative deconvolution

The inverse problem is to recover the zeros of ζ from its Thorin density alone. Under RH the density is the Poisson smoothing of a positive measure with an atom at each zero, so the recovery is a deconvolution under a positivity constraint, and

RH on a window becomes a feasibility question. The monograph [252, Section 19.3] poses it as a Bayesian problem with a positivity prior and a posterior predictive check, and notes that the experiment had not been run because ρ_ε was available only from the primes, at resolutions above one half. The Pólya route of Section 2.1 supplies ρ_ε at small resolution to about 60 digits below height 200 (Section 2.1.1); the experiment uses it in double precision.

8.6.1 Recovering the zeros by nonnegative least squares

Write $P_\varepsilon(x) = \varepsilon/(\pi(\varepsilon^2+x^2))$ and $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$. Under RH, $\rho_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ on $\theta > 0$ (Proposition 1.2.3); since P_ε has Fourier transform $e^{-\varepsilon|t|}$, recovering $U^{\pm}_{\mathrm{RH}}$ is severely ill posed. An off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta < \varepsilon$ contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at γ, and $P_{\varepsilon-\delta}$ is sharper than the Poisson smoothing at level ε of any positive measure, since $P_{\varepsilon-\delta} = P_\varepsilon * \lambda$ would give λ the exponentially growing Fourier transform $e^{\delta|t|}$. Positivity of the recovered measure is therefore the test.

Fix a data window $[\theta_a, \theta_b]$ with sample points $\theta_1 < \cdots < \theta_n$, and candidate atom locations $x_1 < \cdots < x_m$ on a grid over a candidate window $[x_a, x_b]$ that contains $[\theta_a, \theta_b]$. With $A_{ij} = P_\varepsilon(\theta_i - x_j) + P_\varepsilon(\theta_i + x_j)$ and data $y_i = \rho_\varepsilon(\theta_i) - b_i$, where b_i is the contribution of the zeros outside the candidate window, the problem is

$$\min_{w \geq 0} \ \|Aw - y\|_2.$$

Under a Gaussian likelihood with a flat prior on the positive orthant its solution is the maximum a posteriori estimate, and the minimal residual is the discrepancy statistic of the posterior predictive check: up to the error of the grid, it vanishes when the data are the Poisson smoothing of a positive measure. The active set algorithm of Lawson and Hanson [169] solves the problem in finitely many steps.

We took $\varepsilon = 0.1$, the Pólya computed $\rho_{0.1}$ on $[15, 55]$ at 1601 points, and candidates on $[5, 65)$ at spacing 0.02. The background b_i is the Poisson smoothing of the zeros outside $[5, 65]$ among the first 1700 ordinates, the list of Section 3.2 without its smooth tail, which would add about 3×10^{-5} to each sample. The nonnegative solution concentrates on clusters, runs of adjacent grid points with weights above 10^{-3}, one at each zero in the data window (Table 8.21 and Figure 8.9, left): every ordinate is recovered to within 10^{-5}, with mass one to within half a per cent. The remaining mass, 0.071 of the total 10.094, lies in small clusters at the edges of the data window (0.025), which absorb the tails of the zeros at 14.13 and 56.45 just outside it, and in weights below 10^{-3} (0.046). The residual norm is 0.106. It is set by the grid: the true atoms lie between grid points, and a positive combination of two neighbouring Cauchy kernels reproduces an off grid kernel only to second order in the grid spacing.

k	γ_k	c_k	$10^6(c_k-\gamma_k)$	mass
2	21.022040	21.022034	−5.5	1.0013
3	25.010858	25.010859	1.6	1.0027
4	30.424876	30.424867	−8.8	1.0020
5	32.935062	32.935070	8.7	1.0020
6	37.586178	37.586171	−7.4	1.0023
7	40.918719	40.918723	3.8	1.0008
8	43.327073	43.327068	−5.5	1.0025
9	48.005151	48.005145	−6.1	1.0024
10	49.773832	49.773840	7.9	1.0028
11	52.970321	52.970330	8.8	1.0042

Table 8.21: The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero γ_k in the data window, the centre of mass c_k of its cluster, the error of the centre and the total mass of the cluster.

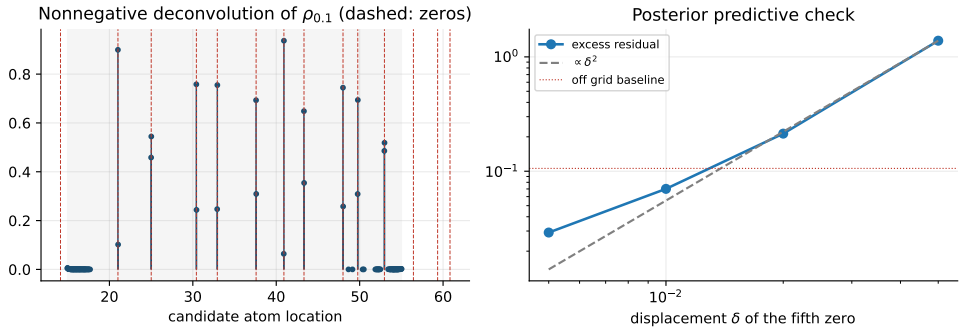


Figure 8.9: Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. The zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement δ of the fifth zero, the δ^2 law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$.

8.6.2 A displaced zero and the threshold of detection

To calibrate the check we displaced the fifth zero, $\gamma_5 = 32.935$, in the sense of Proposition 3.3.1(i): we added to the data the change per displaced zero, $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$, centred at γ_5, and refitted. Table 8.22 gives the residuals.

For $\delta \geq 0.02$ the excess residual grows roughly like δ^2, as the signature does (Proposition 3.3.1(i) and [252, Proposition 19.7]): it rises from 0.21 at $\delta = 0.02$ to 1.39 at $\delta = 0.05$, a factor 6.5 for a factor 2.5 in δ, against $2.5^2 = 6.25$. At the smallest displacements it grows more slowly, by a factor 2.4 from $\delta = 0.005$ to 0.01, because there it is dominated by the cross term between the signature and

δ	residual $r = \|Aw - y\|$	excess over baseline	total mass
0	0.1058	0	10.0942
0.005	0.1098	0.029	10.0947
0.01	0.1269	0.070	10.0963
0.02	0.2380	0.213	10.1031
0.05	1.3891	1.385	10.1566

Table 8.22: Nonnegative deconvolution of $\rho_{0.1}$ with the signature g of a displaced fifth zero added to the data. Here r is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$.

the off grid misfit at γ_5, which has the same shape. The off grid baseline therefore limits detection: the excess reaches it between $\delta = 0.01$ and 0.02 (Figure 8.9, right). This is the square root law that (3.3.1) states for the matched distance (Section 3.3.2), whatever limits the residual.

For $\delta \geq \varepsilon$ the check is unreliable. At $\delta = \varepsilon$ the zero $\frac{1}{2} + \delta + i\gamma_5$ lies on the line of evaluation, where its kernel is a point mass that the samples miss, and the residual, 0.099, falls below the baseline. For $\delta > \varepsilon$ the atom at γ_5 leaves the data and is replaced by the signed term $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ of total mass zero, whose negative part, of mass $\varepsilon/(\pi\delta) + O(\varepsilon^3/\delta^3)$ by Proposition 3.3.1(iii), fades as δ grows. The unit of mass that leaves the data is what the zero density identity of Proposition 3.4.2 counts, and zeros farther from the line than ε are left to that identity.

As [252, Section 19.3.7] emphasises, this is a model check at finite height and not a route to a proof. What the Pólya route adds is the small resolution, where the signature $(\delta/\varepsilon)^2$ of Proposition 3.3.1(i) is largest for a given δ. The recovery and the calibration are computed, at one resolution and on one window.

The direction of this section is the hard one: Poisson smoothing, which moves the basepoint to the right, preserves positivity, while deconvolution, which moves it back towards the critical line, does not (Section 11.4.7).

8.7 Pólya's Bessel function: hitting times, a clock and the flow

Pólya replaced the kernel Φ of (1.2.3) by its asymptote Φ^* and proved that the transform Ξ^* has only real zeros [273] (Section 13.4.1). The building block of his proof is the Macdonald function

$$P(\theta) = K_{i\theta/2}(2\pi) = \int_{\mathbb{R}} e^{-2\pi \cosh 2u}\, e^{i\theta u}\, du, \tag{8.7.1}$$

which has only real zeros and whose zero counting function has the main term of the Riemann von Mangoldt formula. Biane [31] showed that Ξ and P are

Mellin transforms of two Brownian hitting times, and that the Thorin measures of these hitting times are spectral measures of the radial Brownian motions on the unitary group $SU(2)$ and on the hyperbolic space $SL_2(\mathbb{C})/SU(2)$. This section follows the first author's note on Biane's paper [254]. For P every statement of the Thorin programme is a theorem with an explicit witness: the zeros are the Dirichlet spectrum of a Liouville operator, the reciprocal $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ is a GGC whose Thorin atoms are the squared zeros, its clock is the hitting time of an explicit diffusion, and its Poisson trace is completely monotone because it is the trace of a semigroup. The model also marks where the analogy stops: by a theorem of Newman, Pólya's function has a finite de Bruijn Newman constant, although its zeros show no sign of colliding in the range we compute.

8.7.1 Two hitting times and Biane's dictionary

As $u \to \pm\infty$ only the term $n = 1$ of (1.2.3) matters, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2}\exp(-\pi e^{2|u|})$. Pólya replaced Φ by the even function with the same asymptotics on both sides,

$$\Phi^*(u) = 8\pi^2 \cosh\tfrac{9u}{2}\, e^{-2\pi\cosh 2u},$$
$$\Xi^*(\theta) = \int_{\mathbb{R}} \Phi^*(u) e^{i\theta u}\, du = 4\pi^2\Big[P\big(\theta - \tfrac{9i}{2}\big) + P\big(\theta + \tfrac{9i}{2}\big)\Big],$$

since the factor $e^{\pm 9u/2}$ shifts the argument of the transform by $\mp\frac{9i}{2}$. This is the form $G(z+ia)+G(z-ia)$ of Pólya's Hilfssatz II (Section 7.1.3) with $G = P$ and $a = \frac{9}{2}$: Pólya proved that P, an entire function of order one, has only real zeros, and the Hilfssatz gives the same for Ξ^*. In the index of the Bessel function the shift is $\frac{9}{4}$, since $P(\theta \mp \frac{9i}{2}) = K_{i\theta/2\pm 9/4}(2\pi)$, which is the form of Ξ^* in Section 13.4.1. Figure 8.10 compares the three kernels. As a pointwise approximation Φ^* is poor near the origin, $\Phi^*(0) = 8\pi^2 e^{-2\pi} = 0.147447$ against $\Phi(0) = 0.893394$, and the kernel $e^{-2\pi\cosh 2u}$ of P is narrower still; what the three share is the doubly exponential tail, which governs the zero counting functions. Table 8.23 lists the zeros below height 60 and the counts up to height 1000.

The value 2π of the argument is forced. Pólya's asymptotics for the zeros of $y \mapsto K_{iy}(2a)$, which Biane derives by stationary phase [31, Section 3.5], give $\frac{T}{\pi}\log\frac{T}{a} - \frac{T}{\pi} + O(1)$ zeros with $0 < y \le T$; for P the zeros are at $\theta = 2y$ with $a = \pi$, and the count in $(0,T]$ becomes $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1) = N_0(T) + O(1)$. The constant $a = \pi$ comes from the factor $e^{-\pi e^{2u}}$ of the term $n = 1$.

Let $H_a^{(c)}$ be the first passage time of $B_t + ct$ to the level $a > 0$, for a standard Brownian motion B and a drift $c \ge 0$, as in Appendix C. Its density is $a(2\pi\tau^3)^{-1/2}e^{-(a-c\tau)^2/2\tau}$, the inverse Gaussian law of Table 1.8, and its Laplace transform is $e^{a(c-\sqrt{c^2+2u})}$.

Proposition 8.7.1. *Let $H = H_a^{(c)}$ with $a = c = \sqrt{2\pi}$.*

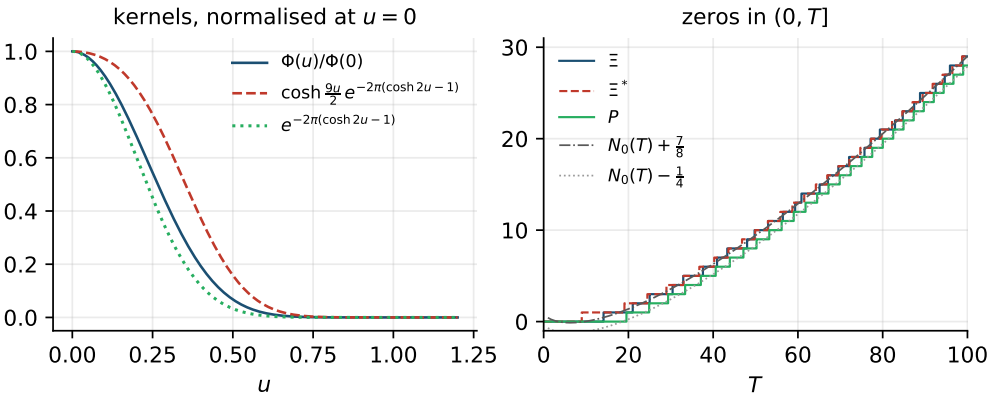


Figure 8.10: Left: the kernel Φ of (1.2.3), Pólya's asymptote Φ^* and the kernel $e^{-2\pi\cosh 2u}$ of P, each divided by its value at $u = 0$. Right: the zero counting functions of Ξ, Ξ^* and P on $(0, 100]$, against $N_0(T) + \frac{7}{8}$ and $N_0(T) - \frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 8.7.8, with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Drawn by `biane_figs.py` from `biane_polya.json`.

n	Ξ	Ξ^*	P
1	14.135	8.993	19.538
2	21.022	19.065	24.897
3	25.011	24.475	29.370
4	30.425	28.975	33.383
5	32.935	33.007	37.099
6	37.586	36.737	40.600
7	40.919	40.249	43.936
8	43.327	43.595	47.140
9	48.005	46.807	50.234
10	49.774	49.908	53.236
11	52.970	52.916	56.158
12	56.446	55.843	59.009
13	59.347	58.700	

T	Ξ	Ξ^*	P	$N_0(T)$
60	13	13	12	11.998
100	29	29	28	28.127
240	102	102	101	100.946
1000	649	649	647	647.741

Table 8.23: Left: the positive zeros below 60 of Ξ, of Pólya's Ξ^* and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0, T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed by `biane_polya.py`.

(i) H *has the density* $\tau^{-3/2}e^{2\pi}e^{-\pi(\tau+1/\tau)}$ *on* $(0,\infty)$*, and* $1/H$ *has the law of* H *size biased by* H*.*

(ii) $\mathbb{E}H^s = 2e^{2\pi}K_{s-\frac12}(2\pi)$ *for every* $s\in\mathbb{C}$*, so that* $\mathbb{E}H^s = \mathbb{E}H^{1-s}$*.*

(iii) $P(\theta) = \frac12 e^{-2\pi}\,\mathbb{E}H^{\frac12+\frac{i\theta}{2}}$ *for every* $\theta\in\mathbb{C}$*.*

(iv) $U = \frac12\log H$ *has the density* $2e^{2\pi}e^{-u}e^{-2\pi\cosh 2u}$*, and the law of* U *size biased by* e^U *has the density* $e^{-2\pi\cosh 2u}/K_0(2\pi)$*, whose characteristic function is* $P(\theta)/P(0)$*.*

Proof. (i) With $a=c=\sqrt{2\pi}$, $(a-c\tau)^2/2\tau = \pi(\tau+1/\tau)-2\pi$ and $a/\sqrt{2\pi}=1$. The density of $1/H$ at σ is σ^{-2} times that of H at $1/\sigma$, which is $\sigma^{-1/2}e^{2\pi}e^{-\pi(\sigma+1/\sigma)}$, that is σ times the density of H at σ; in particular $\mathbb{E}H=1$. (ii) The substitution $\tau=e^w$ and $K_\nu(x)=\int_0^\infty e^{-x\cosh w}\cosh(\nu w)\,dw$ [225, (10.32.9)] give $\int_0^\infty \tau^{\nu-1}e^{-\pi(\tau+1/\tau)}\,d\tau = 2K_\nu(2\pi)$, here with $\nu=s-\frac12$; the integral converges for every complex ν, and $K_\nu=K_{-\nu}$ gives the symmetry. (iii) is (ii) at $s=\frac12+\frac{i\theta}{2}$. (iv) The substitution $\tau=e^{2u}$ turns the density of (i) into $2e^{2\pi}e^{-3u}e^{-2\pi\cosh 2u}e^{2u}$; multiplying by e^u and dividing by $\mathbb{E}H^{1/2}=2e^{2\pi}K_0(2\pi)$ gives the last density, and its characteristic function is $\int e^{-2\pi\cosh 2u}e^{i\theta u}du/K_0(2\pi) = P(\theta)/P(0)$ by (8.7.1). □

The parallel with Ξ is exact. By the identity (9.1.1) of Biane, Pitman and Yor [32], with $V=\frac{\pi}{2}S_2$, $2\xi(s)=\mathbb{E}V^{s/2}$, so $\Xi(\theta)=\frac12\mathbb{E}V^{\frac14+\frac{i\theta}{2}}$, and the kernel is $\Phi(u)=\frac12 e^{u/2}f_Y(u)$ with $Y=\frac12\log V$: the Pólya law of (1.2.4) is the law of Y size biased by $e^{Y/2}$, as the law of (iv) is that of U size biased by e^U. In both cases the evenness of the kernel is a reflection symmetry of the hitting time: $\tau\mapsto 1/\tau$ for H, which is (i), and the functional equation $\mathbb{E}V^{s/2}=\mathbb{E}V^{(1-s)/2}$ for V, which comes from Jacobi's theta transformation.

Both hitting times are GGCs, with the Thorin measures of Table 1.8. For $H_a^{(c)}$ the Thorin measure is $a\,\{\pi\sqrt{2z-c^2}\}^{-1}\mathbf{1}_{z>c^2/2}\,dz$, and the Lévy density, t^{-1} times its Laplace transform, is $a\,e^{-c^2t/2}(2\pi t^3)^{-1/2}$. For the Bessel hitting time, $\mathbb{E}e^{-uJ_a}=a\sqrt{2u}/\sinh(a\sqrt{2u})$ with $J_a\overset{d}{=}a^2J$, so $J_a+J_a'=a^2S_2$ has Thorin measure $2\sum_{n\ge1}\delta_{\pi^2n^2/2a^2}$ and Lévy density $2t^{-1}\sum_n e^{-\pi^2n^2t/2a^2}$. The Thorin measure can also be read from the continuation of $\Psi(u)=-\log\mathbb{E}e^{-uX}$ across the cut: integrating the Stieltjes inversion (1.3.2),

$$U\big((0,z)\big)=\frac{1}{\pi}\operatorname{Im}\Psi(-z+i0)\qquad(z>0\text{ not an atom}),$$

because each factor $\log(1+u/w)$ with $w<z$ contributes $i\pi$ at $u=-z+i0$. For $H_a^{(c)}$, $\Psi(-z+i0)=a\big(i\sqrt{2z-c^2}-c\big)$ for $z>\frac{c^2}{2}$, so $U((0,z))=\frac{a}{\pi}\sqrt{2z-c^2}$; for J_a+J_a', $\Psi(-\frac{\lambda^2}{2}+i0)=2\log\frac{\sin\lambda a}{\lambda a}$, which gains $2\pi i$ each time λa passes a multiple of π. Biane's dictionary identifies these Thorin measures with spectral

measures. In his normalisation [31, (2.1) to (2.3)] the generator $L_1 = \frac{1}{2}\frac{d^2}{dx^2}$ on $[0,\pi]$ with Dirichlet conditions has the eigenfunctions $\sin(kx)$, $k = 1, 2, \ldots$, with eigenvalues $-\frac{k^2}{2}$, and the spectral measure $\sigma_1 = \frac{1}{\pi}\sum_{n\geq 1}\delta_{n^2/2}$; the generator $L_2 = \frac{1}{2}\frac{d^2}{dx^2} + \frac{d}{dx}$ on $[0,\infty)$ with a Dirichlet condition at 0, selfadjoint in $L^2(e^{2x}dx)$, has the generalised eigenfunctions $e^{-x}\sin(kx)$, $k > 0$, with eigenvalues $-\frac{k^2+1}{2}$, and the spectral measure $\sigma_2(d\lambda) = d\lambda/(\pi\sqrt{2\lambda - 1})$ on $\lambda > \frac{1}{2}$.

THEOREM 8.7.2 (Biane's dictionary, with corrected constants). *(i) The Thorin measure of $J_\pi + J'_\pi = \pi^2 S_2$ is $2\pi\sigma_1$.*

(ii) The Thorin measure of $H_1^{(1)}$ is σ_2.

(iii) The radial generator $\frac{1}{2}\frac{d^2}{dx^2} + \cot x\frac{d}{dx}$ of Brownian motion on the unit sphere $S^3 \cong SU(2)$ is the Doob transform of L_1 by its ground state $\sin x$, $\sin^{-1}(L_1 + \frac{1}{2})\sin$, and the radial generator $\frac{1}{2}\frac{d^2}{dx^2} + \coth x\frac{d}{dx}$ of Brownian motion on $\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$ is the Doob transform of L_2 by the harmonic function $1 - e^{-2x}$.

Proof. (i) With $a = \pi$ the atoms $\pi^2 n^2/2a^2$ are at $n^2/2$, with mass $2 = 2\pi\cdot\frac{1}{\pi}$. (ii) With $a = c = 1$ the density $a\{\pi\sqrt{2z - c^2}\}^{-1}$ on $z > \frac{c^2}{2}$ is that of σ_2. (iii) For f smooth, $\frac{1}{2}(f\sin)'' = \sin\cdot(\frac{1}{2}f'' + \cot x\, f' - \frac{1}{2}f)$, so $\sin^{-1}(L_1 + \frac{1}{2})(f\sin) = \frac{1}{2}f'' + \cot x\, f'$; and $L_2 h = 0$ for $h = 1 - e^{-2x}$, while $h^{-1}L_2(hf) = \frac{1}{2}f'' + (1 + h'/h)f'$ with $1 + h'/h = 1 + \frac{2e^{-2x}}{1-e^{-2x}} = \coth x$. The eigenfunctions are checked the same way: $L_2(e^{-x}s) = e^{-x}(\frac{1}{2}s'' - \frac{1}{2}s)$. □

For Pólya's function itself the parameters are $a = c = \sqrt{2\pi}$ rather than $a = c = 1$; Brownian scaling gives $H_a^{(c)} \overset{d}{=} \frac{a}{c}H^{(\sqrt{ac})}_{\sqrt{ac}}$, so the level and the drift can always be made equal, which is the reciprocal symmetry of Proposition 8.7.1(i), but their common value $\sqrt{ac}$ is $\sqrt{2\pi}$ for P and 1 in the dictionary. Part (iii) corrects the direction of the Doob transforms in the note [254], which has L_1 and L_2 as the transforms of the radial generators; Biane states them in the direction of (iii) [31, Section 2.4].

REMARK 8.7.3 (Biane's constants). Biane's $K_\mu(x) = \int_0^\infty t^{\mu-1}e^{-\frac{x}{2}(t+t^{-1})}dt$ is twice the standard Macdonald function, and his kernel (3.1) is the Φ of (1.2.3); with that kernel $\Xi(z) = 2\int_0^\infty \Phi(u)\cos(zu)\,du$ holds for the standard $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, so the factor $\frac{1}{2}$ is missing from his displayed definition of ξ, and his Mellin formula $\mathbb{E}W_a^s = 2(2a^2/\pi)^s\xi(2s)$ also uses the standard ξ (both sides are $\frac{2}{3}$ at $s = 1$, $a = 1$). In his Section 4.3 the Lévy measure of $W_a = J_a + J'_a$ should be $2t^{-1}\sum_n e^{-\pi^2 n^2 t/2a^2}dt$, with the factor t^{-1} and the $\frac{1}{2}$ in the exponent, and that of the drifted first passage time should carry the level x as a factor. His Thorin measure (4.1) should have its atoms at $\pi^2 n^2/2a^2$, not at

n^2/a^2; the density (4.2) should be $x\,dc/(\pi\sqrt{2c-a^2})$, which for $x=1$ is his density divided by $\sqrt{2}$, and the identity offered for it should read $(\pi t)^{-1/2}e^{-a^2t/2} = \int_{a^2/2}^{\infty} e^{-ct}\,dc/(\pi\sqrt{c-a^2/2})$, with t in place of t^3. The measure (2.3) lives on $(\frac{1}{2},\infty)$, not on $[1,\infty)$. In Propositions 2.3 and 2.4 the drift and the harmonic function are oriented for the cone opposite to the one stated: with the cone $\{y_1 \geq \cdots \geq y_n\}$ the drift is $\rho' = (n-1, n-3, \ldots, 1-n)$ and the positive harmonic function is $\prod_{i<j}(1-e^{-2(y_i-y_j)}) = e^{-\langle\rho',y\rangle}\prod_{i<j} 2\sinh(y_i-y_j)$; the Vandermonde function $\prod_{i>j}(x_i-x_j)$ of his Proposition 2.1 is reversed in the same way, and is positive on the cone as $\prod_{i<j}(x_i-x_j)$. The eigenvalue of Proposition 2.6 is discussed in Remark 8.7.5. With these corrections the dictionary of his Section 4.5 is the exact statement of Theorem 8.7.2.

8.7.2 The Wick rotation and the curvature family

Biane closes his paper by observing that ξ and Pólya's function are related, through $SU(2)$ and $SL_2(\mathbb{C})/SU(2)$, "in this non obvious way" [31, Section 4.5]. The relation is a Wick rotation, the substitution of iy for a real variable y in a function analytic in y. Wick's device [339] rotates the time axis, $t = -i\tau$, which turns the unitary group e^{-itH} of quantum mechanics into the heat semigroup $e^{-\tau H}$. The rotation keeps the spectrum of H and exchanges oscillation and positivity: probability laws, Laplace transforms and completely monotone functions live on the Euclidean side, Fourier transforms and their zeros on the real time side. A Thorin measure is spectral data that can be read on either side. Through Cartan's duality between a symmetric space G/K of noncompact type and its compact dual U/K the substitution $y \mapsto iy$ reverses the sign of the curvature. Biane's paper contains the rotation in four places: in space (the sign of the curvature), in the Laplace variable, in time (heat kernels) and in the Mellin index.

Let $H_n = \{y \in \mathbb{R}^n : \sum_i y_i = 0\}$ and $\mathcal{C} = \{y \in H_n : y_1 \geq \cdots \geq y_n\}$. Biane describes three matrix valued Brownian motions of type A [31, Section 2]. For traceless Hermitian Brownian motion the ordered eigenvalues perform Brownian motion in $\mathcal{C}$ conditioned in Doob's sense never to reach the boundary: the semigroup of Brownian motion killed at $\partial\mathcal{C}$ is the alternating sum $p_t^0(x,y) = \sum_{w\in S_n}\varepsilon(w)p_t(x,wy)$ (the reflection principle), the Vandermonde product $\prod_{i<j}(y_i-y_j)$ is its positive harmonic function vanishing on $\partial\mathcal{C}$, and the Doob transform is Dyson's Brownian motion [80]. For Brownian motion g_t on $SL_n(\mathbb{C})$, the logarithms of the singular values, the radial part of Brownian motion on $SL_n(\mathbb{C})/SU(n)$, perform Brownian motion in $\mathcal{C}$ with generator $\frac{1}{2}\Delta + \langle\nabla\log\tilde{h},\nabla\rangle$, $\tilde{h}(y) = \prod_{i<j}\sinh(y_i-y_j)$ [132], which is Brownian motion with drift $\rho' = (n-1, n-3, \ldots, 1-n)$, killed at $\partial\mathcal{C}$, transformed by the harmonic function of Remark 8.7.3. For Brownian motion on $SU(n)$ the eigenvalues $e^{i\theta_j}$, with $\sum_j\theta_j = 0$, live in the alcove $\theta_1 \geq \cdots \geq \theta_n$, $\theta_1 - \theta_n \leq 2\pi$, a fundamental domain of the affine Weyl group $S_n \ltimes ((2\pi\mathbb{Z})^n \cap H_n)$; the killed semigroup is the alternating sum

over that group, and the eigenvalue process is the Doob transform by the Dirichlet ground state $\prod_{j>k}(e^{i\theta_j}-e^{i\theta_k})$, which is a constant multiple of $\prod_{j<k}\sin\frac{\theta_j-\theta_k}{2}$ because $\sum_j\theta_j=0$.

The three Doob functions are one function at three values of a curvature parameter. For real κ put $s_\kappa(z)=\sin(\sqrt{\kappa}z)/\sqrt{\kappa}$, so that $s_{-1}=\sinh$, $s_0(z)=z$ and $s_1=\sin$, and

$$h_\kappa(y)=\prod_{i<j}s_\kappa(y_i-y_j),\qquad |\rho|^2=\sum_{j=1}^{n}(n+1-2j)^2=\frac{n(n^2-1)}{3}.$$

PROPOSITION 8.7.4. *For every real κ and every $n\geq 2$, $\Delta h_\kappa=-\kappa|\rho|^2h_\kappa$ on H_n.*

Proof. For $\kappa=-1$, write $\prod_{i<j}2\sinh(y_i-y_j)=\prod_{i<j}e^{-(y_i+y_j)}(e^{2y_i}-e^{2y_j})$. The Vandermonde determinant gives $\prod_{i<j}(e^{2y_i}-e^{2y_j})=\det(e^{2(n-j)y_i})$, and on H_n the factor $\prod_{i<j}e^{-(y_i+y_j)}=e^{-(n-1)\sum_i y_i}$ is one, so the exponents $2(n-j)$ may be replaced by $2(n-j)-(n-1)=n+1-2j$: $\prod_{i<j}2\sinh(y_i-y_j)=\sum_{w\in S_n}\varepsilon(w)e^{\langle w\rho',y\rangle}$, the Weyl denominator formula. Each $w\rho'$ lies in H_n and has $|w\rho'|^2=|\rho|^2$, so each exponential is an eigenfunction of the Laplacian of H_n with eigenvalue $|\rho|^2$, and $\Delta h_{-1}=|\rho|^2h_{-1}$. This identity between entire functions of $y\in H_n\otimes\mathbb{C}$ continues analytically. For $\kappa<0$, $h_\kappa(y)$ is a constant times $h_{-1}(\sqrt{-\kappa}\,y)$, and the scaling multiplies the Laplacian by $-\kappa$; for $\kappa>0$, $s_\kappa(z)=\sinh(i\sqrt{\kappa}z)/(i\sqrt{\kappa})$, so $h_\kappa(y)$ is a constant times $h_{-1}(i\sqrt{\kappa}\,y)$, and the substitution multiplies the Laplacian by $(i\sqrt{\kappa})^2=-\kappa$. At $\kappa=0$, h_0 is the Vandermonde product, the limit of h_κ as $\kappa\to 0$, since $s_\kappa(z)=z-\kappa z^3/6+O(\kappa^2)$, and it is harmonic. □

So Biane's three semigroups are one formula,

$$q_t^{(\kappa)}(x,y)=\frac{h_\kappa(y)}{h_\kappa(x)}\,e^{\kappa|\rho|^2t/2}\,p_t^0(x,y),\qquad \kappa\in\{-1,0,1\},$$

with generator $\frac12\Delta+\langle\nabla\log h_\kappa,\nabla\rangle$, where p^0 is the killed kernel of the chamber $\mathcal{C}$ for $\kappa\leq 0$ and of the alcove $\{y_1\geq\cdots\geq y_n,\ y_1-y_n\leq\pi\}$ for $\kappa=1$; for $\kappa=1$ the coordinates are the half angles $y=\theta/2$, in which h_1 is the positive Dirichlet ground state of the alcove. The rotation $y\mapsto iy$ sends h_κ to a constant times $h_{-\kappa}$, since $s_\kappa(iz)=is_{-\kappa}(z)$: it exchanges the noncompact and the compact Doob functions and reverses the sign of the ground energy, $\frac12\Delta h_\kappa=-\frac{\kappa}{2}|\rho|^2h_\kappa$, which is $+\frac12|\rho|^2$ for $SL_n(\mathbb{C})/SU(n)$ and $-\frac12|\rho|^2=\frac{n-n^3}{6}$ for $SU(n)$ in the half angles.

REMARK 8.7.5 (The eigenvalue of Biane's Proposition 2.6). Biane endows the Lie algebra of $SU(n)$ with the form $\langle A,B\rangle=-\operatorname{Tr}(AB)$, in which the angles θ_j perform standard Brownian motion in H_n, and states that the ground state $\prod_{j<k}\sin\frac{\theta_j-\theta_k}{2}$ has the eigenvalue $\frac{n-n^3}{6}$ for $\frac12\Delta$ [31, Proposition 2.6]. By Proposition 8.7.4 in the half angles, the eigenvalue for $\frac12\Delta_\theta$ is $\frac14\cdot\frac{n-n^3}{6}=\frac{n-n^3}{24}$; more generally, for

the form $-c\,\mathrm{Tr}(AB)$ it is $\frac{n-n^3}{24c}$. Biane's value holds for $c=\frac14$, that is when the half angles are standard Brownian motion. His rank one discussion uses a third normalisation, $c=\frac12$, in which $SU(2)$ is the unit sphere S^3, the angle $x\in[0,\pi]$ has the generator $\frac12\frac{d^2}{dx^2}+\cot x\frac{d}{dx}$ of Theorem 8.7.2(iii), and the ground energy is $\frac{2-8}{12}=-\frac12$. `biane_polya.py` confirms $\frac12\Delta_\theta h/h=-\frac14,-1,-\frac52$ for $n=2,3,4$.

In rank one the curvature family is the radial part of Brownian motion on the three dimensional space form of curvature κ, with generator $\frac12\frac{d^2}{dx^2}+\sqrt{\kappa}\cot(\sqrt{\kappa}x)\frac{d}{dx}$: the sphere of radius $\kappa^{-1/2}$ for $\kappa>0$, the Bessel process of dimension three for $\kappa=0$, and the hyperbolic space of curvature κ for $\kappa<0$. Let $J_a^{(\kappa)}$ be its hitting time of the radius a from the centre, with $a<\pi/\sqrt{\kappa}$ when $\kappa>0$; then $J_a^{(0)}=J_a$, whose Laplace transform is classical [57]. For $\kappa<0$ the generator is that of the Bessel process of dimension three with drift $\sqrt{-\kappa}$ [295, (2.5)], the norm of a three dimensional Brownian motion with drift of that length.

Proposition 8.7.6. *For $u\ge0$,*

$$\mathbb{E}e^{-uJ_a^{(\kappa)}}=\frac{\sqrt{2u-\kappa}\;s_\kappa(a)}{\sinh\left(a\sqrt{2u-\kappa}\right)},$$

the Laplace transform of a GGC with Thorin measure $\sum_{n\ge1}\delta_{\pi^2n^2/2a^2-\kappa/2}$. Equivalently, the law of $J_a^{(\kappa)}$ is the exponential tilt $e^{\kappa t/2}\,\Pr(J_a\in dt)/\mathbb{E}e^{\kappa J_a/2}$ of the law of J_a.

Proof. The generator is the Doob transform of $\frac12\frac{d^2}{dx^2}$ by $s=s_\kappa$, since $s''=-\kappa s$ gives $s^{-1}(\frac12\frac{d^2}{dx^2}+\frac{\kappa}{2})(sf)=\frac12f''+\frac{s'}{s}f'$ and $s'/s=\sqrt{\kappa}\cot(\sqrt{\kappa}x)$. Hence $f(x)=\sinh(x\sqrt{2u-\kappa})/s_\kappa(x)$ satisfies $\frac12f''+\frac{s'}{s}f'=s^{-1}(\frac12g''+\frac{\kappa}{2}g)=uf$ with $g=\sinh(x\sqrt{2u-\kappa})$, and f is, up to a constant factor, the positive increasing solution that stays bounded at 0, with $f(0+)=\sqrt{2u-\kappa}$. Hence $\mathbb{E}_xe^{-uJ_a^{(\kappa)}}=f(x)/f(a)$ for $0<x<a$, and letting $x\downarrow0$ gives $\mathbb{E}e^{-uJ_a^{(\kappa)}}=f(0+)/f(a)$, which is the formula; for $\kappa>2u$ the square roots are imaginary and the formula is read through $\sinh(ix)/i=\sin x$. As a function of u the denominator divided by $\sqrt{2u-\kappa}$ is entire of order $\frac12$, with zeros where $a\sqrt{2u-\kappa}=i\pi n$, that is at $u=-(\pi^2n^2/2a^2-\kappa/2)$, all negative under the hypothesis on a; Hadamard's product in genus zero gives the Thorin measure. The tilt is the formula at $u-\frac{\kappa}{2}$ for $\kappa=0$, divided by its value at $u=0$, since $\sqrt{-\kappa}/\sinh(a\sqrt{-\kappa})=1/s_\kappa(a)$. □

Three consequences clarify the dictionary. First, Riemann's clock J_a+J_a' is the flat member $\kappa=0$, and its link with $SU(2)$ in Theorem 8.7.2 is spectral rather than geometric: the atoms $\pi^2n^2/2a^2$ are the Dirichlet spectrum of $\frac12\frac{d^2}{dx^2}$ on $[0,a]$, which for $a=\pi$ is the spectrum of the radial generator of $SU(2)$ shifted by its ground energy $\frac12$. Second, the hyperbolic member $\kappa=-1$ has the discrete atoms $\pi^2n^2/2a^2+\frac12$; Pólya's continuous measure comes instead from a first passage over

a level on the half line, a Dirichlet problem on $[0,\infty)$ with continuous spectrum. Third, the tilt destroys the reciprocal symmetry of $J_a + J_a'$ that encodes the functional equation. With $W = J_1^{(\kappa)} + J_1'^{(\kappa)}$, the function $(\frac{\pi}{2})^s \mathbb{E}W^s$ equals $2\xi(2s)$ for $\kappa = 0$ and is symmetric under $s \mapsto \frac{1}{2} - s$: it takes the value 0.9963110325 at both $s = 0.1$ and $s = 0.4$. For $\kappa = 1$ the two values are 1.002917197 and 1.023703077. A curvature interpolation between the two clocks must therefore be sought in a class that keeps the symmetry (Question 8.7.20).

Replacing u by a negative value in $\mathbb{E}e^{-uX}$ continues the transform across the origin, and this is the rotation of the Laplace variable used in the inversion formula above. For the curvature family the reciprocal of the transform becomes, for $u < \frac{\kappa}{2}$, the damped sine $\sin(a\sqrt{\kappa - 2u})/(\sqrt{\kappa - 2u}\, s_\kappa(a))$, whose zeros sit at minus the Thorin atoms and slide with κ; for the drifted clock the reciprocal $e^{a(\sqrt{c^2+2u}-c)}$ becomes $e^{-ac}e^{ia\sqrt{2|u|-c^2}}$ beyond the branch point $u = -\frac{c^2}{2}$, which has constant modulus and no zeros.

The rotation of time relates the heat kernels of the two dual spaces. The radial heat kernels of $\frac{1}{2}\Delta$, relative to the Riemannian volume, are

$$p_t^{\mathbb{H}^3}(r) = \frac{e^{-t/2}}{(2\pi t)^{3/2}} \frac{r}{\sinh r} e^{-r^2/2t},$$
$$p_t^{S^3}(\omega) = \frac{e^{t/2}}{(2\pi t)^{3/2}} \sum_{k\in\mathbb{Z}} \frac{\omega + 2\pi k}{\sin \omega} e^{-(\omega+2\pi k)^2/2t},$$

and the term $k = 0$ of the second is, up to the constant factor $(-1)^{-3/2}$, the first after $r \mapsto i\omega$, $t \mapsto -t$, a genuine Wick rotation of time; the sum over k is the method of images over the affine Weyl group, which makes the kernel periodic.

The fourth rotation acts on the Mellin index. The substitution $s = \frac{1}{2} + i\theta$ turns real Mellin moments into Fourier transforms of a tilted law, and Biane's contour shift $w \mapsto w + \frac{i\pi}{2}$ [31, Section 3.5], with $\cosh(w + \frac{i\pi}{2}) = i \sinh w$, turns the damped integrand $e^{zw - 2a\cosh w}$ into an oscillating one on which stationary phase locates the zeros. For P this is the map $\mu \mapsto i\mu$ on the index of K_μ, and Section 8.7.5 shows that the index is the eigenvalue of a boost, which the rotation turns into an angular momentum. Atoms of a Thorin measure on the Euclidean side become poles after the rotation, and poles require discrete spectrum, which for these radial Brownian motions, with no potential, means a compact space. That is why ξ, whose time side Thorin measure is atomic, is attached to $SU(2)$, while P, whose time side measure has a density, is attached to $\mathbb{H}^3$.

8.7.3 The zeros of P as a Dirichlet spectrum

Put $\psi_\mu(x) = K_\mu(e^x)$. Since $\frac{d}{dx} = r\frac{d}{dr}$ for $r = e^x$, the modified Bessel equation $r^2K'' + rK' - (r^2 + \mu^2)K = 0$ becomes

$$-\psi_\mu'' + e^{2x}\psi_\mu = -\mu^2\psi_\mu,$$

a Schrödinger equation with the Liouville potential e^{2x}, in which the index enters only through the energy $-\mu^2$. Let $x_0 = \log 2\pi$, and let $A_P = -\frac{d^2}{dx^2} + e^{2x}$ act in $L^2([x_0,\infty))$ with a Dirichlet condition at x_0. Biane recalls this spectral interpretation of the zeros of the Macdonald function as well known (Titchmarsh [323]) and notes that Pólya does not mention it [31, Section 3.3]. We write $\nu_1 < \nu_2 < \cdots$ for the positive zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta = \pm 2\nu_n$.

Proposition 8.7.7. *(i) A_P is selfadjoint, has compact resolvent, and $A_P \geq 4\pi^2$.*

(ii) A number λ is an eigenvalue of A_P if and only if $\lambda = \nu^2$ with $\nu > 0$ and $K_{i\nu}(2\pi) = 0$, and the eigenfunction is then $K_{i\nu}(e^x)$. In particular every zero of P is real, of absolute value at least 4π, and the zeros of P are the numbers $\pm 2\nu_n$ with ν_n^2 running over the spectrum of A_P.

(iii) Every eigenvalue of A_P is simple, and every zero of $\mu \mapsto K_\mu(2\pi)$ and of P is simple. More precisely, for $\nu > 0$ and every $y \in \mathbb{R}$,

$$\psi'_{i\nu}(y)\,\partial_\nu\psi_{i\nu}(y) - \psi_{i\nu}(y)\,\partial_\nu\psi'_{i\nu}(y) = -2\nu\int_y^\infty \psi_{i\nu}(x)^2\,dx < 0. \tag{8.7.2}$$

Proof. (i) The endpoint x_0 is regular, and the potential is bounded below, so the equation is in the limit point case at $+\infty$ and the Dirichlet realisation is selfadjoint; since $e^{2x} \to \infty$, the resolvent is compact. For f in the form domain, $\langle A_P f, f\rangle = \int_{x_0}^\infty (|f'|^2 + e^{2x}|f|^2)\,dx \geq e^{2x_0}\|f\|^2 = 4\pi^2\|f\|^2$. (ii) For every complex μ, $K_\mu(r) = \sqrt{\pi/2r}\,e^{-r}(1 + O(1/r))$ as $r \to \infty$ [225, (10.40.2)], so ψ_μ is a nonzero solution in L^2 near $+\infty$, and in the limit point case it is the only one up to a constant factor. Hence $-\mu^2$ is an eigenvalue exactly when $\psi_\mu(x_0) = K_\mu(2\pi) = 0$. By (i) such an eigenvalue is real and at least $4\pi^2$, so $\mu = \pm i\nu$ with $\nu \geq 2\pi$; since $P(\theta) = K_{i\theta/2}(2\pi)$, the zeros of P are $\theta = \pm 2\nu$. (iii) Two eigenfunctions with the same eigenvalue both vanish at x_0, so they are proportional. For (8.7.2), put $\phi = \partial_\nu\psi_{i\nu}$; differentiating $\psi''_{i\nu} = (e^{2x} - \nu^2)\psi_{i\nu}$ in ν gives $\phi'' = (e^{2x} - \nu^2)\phi - 2\nu\psi_{i\nu}$, so that $(\psi'_{i\nu}\phi - \psi_{i\nu}\phi')' = 2\nu\psi_{i\nu}^2$. Both functions and their derivatives decay like $\exp(-e^x)$, and integrating from y to ∞ gives the identity. The function $\psi_{i\nu}$ is real valued and not identically zero, so the right side is negative. At $y = x_0$ and $\nu = \nu_n$ the second term on the left vanishes, so $\partial_\nu K_{i\nu}(2\pi) \neq 0$ at ν_n, which is the simplicity of the zero. □

The proposition is Pólya's theorem that P has only real zeros, obtained from selfadjointness, and A_P is an explicit Hilbert Pólya operator for P. Biane also writes the equation in Dirac form [31, Section 3.4]. With $D = \frac{d}{dx}$ and $\mathcal{A} = D + \frac{1}{2} + e^x$, whose formal adjoint is $\mathcal{A}^\dagger = -D + \frac{1}{2} + e^x$, the system $\mathcal{A}f = \gamma g$, $\mathcal{A}^\dagger g = \gamma f$ gives $\mathcal{A}^\dagger\mathcal{A}f = \gamma^2 f$, and a direct computation gives $\mathcal{A}^\dagger\mathcal{A} = A_P + \frac{1}{4}$. With the Dirichlet condition on f the Dirac eigenvalues are therefore $\gamma = \pm(\nu_n^2 + \frac{1}{4})^{1/2}$,

not $\pm\nu_n$; the shift $\frac14$, which Biane writes as $\gamma^2-\frac14$, is missing from the note [254]. In the variable $u=e^x$ the system reads $(u\frac{d}{du}+\frac12+u)f=\gamma g$, $(-u\frac{d}{du}+\frac12+u)g=\gamma f$, and since $u\frac{d}{du}+\frac12$ is i times the symmetrised dilation operator $\frac12(up+pu)$, $p=-i\frac{d}{du}$, it is the Hamiltonian xp of Berry and Keating [28] perturbed by the potential u, as Biane remarks.

The zeros of P obey an exact quantisation rule. Let I_μ be the modified Bessel function of the first kind, and for real ν put

$$\vartheta_P(\nu)=\operatorname{Im}\log\Gamma(1+i\nu)-\nu\log\pi-\arg{}_0F_1(;1+i\nu;\pi^2), \tag{8.7.3}$$

with the branches continuous in ν and equal to zero at $\nu=0$, where ${}_0F_1(;1;\pi^2)=I_0(2\pi)>0$. Write $N_P(T)$ for the number of zeros of P in $(0,T]$ and

$$x(T)=\frac{1}{\pi}\vartheta_P\big(\tfrac{T}{2}\big),\qquad N_0(T)=\frac{T}{2\pi}\log\frac{T}{2\pi e}.$$

Proposition 8.7.8. *(i) For $\nu>0$, $I_{i\nu}(2\pi)\neq0$ and*

$$K_{i\nu}(2\pi)=\frac{\pi\,|I_{i\nu}(2\pi)|\,\sin\vartheta_P(\nu)}{\sinh\pi\nu}.$$

(ii) $\vartheta_P(\nu)>0$ for $\nu>0$, $\vartheta_P(\nu_n)=n\pi$ for every n, and the number of zeros of $\nu\mapsto K_{i\nu}(2\pi)$ in $(0,\nu)$ is $\lceil\vartheta_P(\nu)/\pi\rceil-1$. Equivalently, $N_P(T)=\lfloor x(T)\rfloor$ for every $T>0$ that is not a zero of P.

(iii) As $\nu\to\infty$,

$$\vartheta_P(\nu)=\nu\log\frac{\nu}{\pi}-\nu+\frac{\pi}{4}+\frac{\pi^2-\frac{1}{12}}{\nu}+O(\nu^{-3}),$$

so that $x(T)=N_0(T)+\frac14+\big(2\pi-\frac{1}{6\pi}\big)\frac{1}{T}+O(T^{-3})$ as $T\to\infty$.

Proof. (i) For real $r>0$, $I_{-i\nu}(r)=\overline{I_{i\nu}(r)}$, and $K_\mu=\pi(I_{-\mu}-I_\mu)/(2\sin\mu\pi)$ gives $K_{i\nu}(r)=-\pi\operatorname{Im}I_{i\nu}(r)/\sinh\pi\nu$. The series $I_{i\nu}(r)=(r/2)^{i\nu}\,{}_0F_1(;1+i\nu;r^2/4)/\Gamma(1+i\nu)$ shows that, for $r>0$,

$$\vartheta(\nu,r)=\operatorname{Im}\log\Gamma(1+i\nu)-\nu\log\frac{r}{2}-\arg{}_0F_1\big(;1+i\nu;\tfrac{r^2}{4}\big)$$

satisfies $I_{i\nu}(r)=|I_{i\nu}(r)|e^{-i\vartheta(\nu,r)}$, and $\vartheta_P(\nu)=\vartheta(\nu,2\pi)$; this gives the formula. If $I_{i\nu}(r)$ vanished, so would $I_{-i\nu}(r)$, against the Wronskian $I_\mu I'_{-\mu}-I'_\mu I_{-\mu}=-2\sin(\mu\pi)/(\pi r)$ [225, (10.28.1)], which is not zero for $\mu=i\nu$. So ${}_0F_1$ does not vanish on $[0,\infty)\times(0,\infty)$, and ϑ is defined there as a continuous function with $\vartheta(0,r)=0$. (ii) The same Wronskian, written for $I=I_{i\nu}$ and $\bar I=I_{-i\nu}$, gives $\operatorname{Im}(I'/I)=\sinh(\pi\nu)/(\pi r|I|^2)$, hence $\partial_r\vartheta(\nu,r)=-\sinh(\pi\nu)/(\pi r|I_{i\nu}(r)|^2)<0$. As $r\to\infty$, $I_{i\nu}(r)\sim e^r/\sqrt{2\pi r}$ and $K_{i\nu}(r)\sim\sqrt{\pi/2r}\,e^{-r}$ uniformly for ν in compact sets, so $\sin\vartheta(\nu,r)=\sinh(\pi\nu)K_{i\nu}(r)/(\pi|I_{i\nu}(r)|)\to0$ and $\cos\vartheta(\nu,r)>0$ for

large r; by continuity in ν from $\nu = 0$, $\vartheta(\nu, r) \to 0$. So for fixed $\nu > 0$ the function $r \mapsto \vartheta(\nu, r)$ decreases from $\vartheta_P(\nu)$ at $r = 2\pi$ to 0 at $r = \infty$; in particular $\vartheta_P(\nu) > 0$, and $r \mapsto K_{i\nu}(r)$ has exactly $\lceil \vartheta_P(\nu)/\pi \rceil - 1$ zeros in $(2\pi, \infty)$, one at each crossing of a multiple of π. On the other hand, by (8.7.2) at a zero y of $\psi_{i\nu}$, $\psi'_{i\nu}(y)\,\partial_\nu \psi_{i\nu}(y) < 0$, so every zero of $x \mapsto K_{i\nu}(e^x)$ moves to the right as ν increases; no zero comes in from $+\infty$, where $K_{i\nu}(r) > 0$ uniformly for ν in compact sets, and zeros cannot appear in pairs, since they are simple. So the number of zeros in (x_0, ∞) is zero for small ν and increases by one exactly when ν passes some ν_n, where a zero enters through x_0: it equals the number of $\nu_n < \nu$. Comparing the two counts gives the statement, and $\vartheta_P(\nu_n) \in \pi\mathbb{Z}$ with the count passing from $n-1$ to n gives $\vartheta_P(\nu_n) = n\pi$. (iii) Stirling's series gives $\operatorname{Im} \log \Gamma(1 + i\nu) = \nu \log \nu - \nu + \frac{\pi}{4} - \frac{1}{12\nu} + O(\nu^{-3})$, and with $b = 1 + i\nu$, $\log {}_0F_1(;b;z) = z/b - z^2/(2b^2(b+1)) + O(b^{-3})$, so $\arg {}_0F_1(;1+i\nu;\pi^2) = \operatorname{Im} \frac{\pi^2}{1+i\nu} + O(\nu^{-3}) = -\frac{\pi^2}{\nu} + O(\nu^{-3})$. With $\nu = T/2$, $\frac{1}{\pi}(\nu \log \frac{\nu}{\pi} - \nu) = N_0(T)$. □

Writing $s_P(T) = \{x(T)\} - \frac{1}{2}$ for the sawtooth,

$$N_P(T) = N_0(T) - \frac{1}{4} - s_P(T) + \Big(2\pi - \frac{1}{6\pi}\Big)\frac{1}{T} + O(T^{-3}),$$

against the Riemann von Mangoldt formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$ of Section 2.2, which with $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3})$ reads

$$N(T) = N_0(T) + \frac{7}{8} + S(T) + \frac{1}{48\pi T} + O(T^{-3}).$$

The two counting functions share the main term N_0, but their mean constant terms differ by $\frac{7}{8} + \frac{1}{4} = \frac{9}{8}$, and the fluctuation of N_P is the bounded sawtooth of a quantisation rule where that of N is $S(T)$. These two facts govern the comparison of the two Poisson traces in Section 10.1.5. Table 8.23 shows the effect: $N(1000) - N_P(1000) = 2$, with $N_0(1000) = 647.741$. The note [254] states that the semiclassical count of A_P is exactly the main term of the Riemann von Mangoldt formula. The count is

$$N_{\mathrm{WKB}}(\nu) = \frac{1}{\pi}\int_{x_0}^{\log\nu} \sqrt{\nu^2 - e^{2x}}\,dx = \frac{1}{\pi}\Big(\nu \operatorname{arccosh}\frac{\nu}{2\pi} - \sqrt{\nu^2 - 4\pi^2}\Big),$$

and its expansion gives $N_{\mathrm{WKB}}(T/2) = N_0(T) + \frac{2\pi}{T} + O(T^{-3})$: it agrees with the main term up to $2\pi/T$, not exactly. Its phase $\varphi(\nu) = \pi N_{\mathrm{WKB}}(\nu) + \frac{\pi}{4}$ is the phase of Debye's expansion of $K_{i\nu}(2\pi)$ for large ν, the term $\frac{\pi}{4}$ being the Maslov correction at the turning point, and by (iii) $\vartheta_P(\nu) - \varphi(\nu) = -\frac{1}{12\nu} + O(\nu^{-3})$.

`biane_polya.py` computes the 1516 zeros $\nu_n \le 1000$ from (8.7.3) with mpmath [205].

8.7.4 The index side: a GGC with the squared zeros as Thorin atoms, and its clock

Biane's Thorin measures in Theorem 8.7.2 live on the time side: they describe the hitting time itself, in the energy variable of its Laplace transform. The Thorin programme of this book works on the index side, with the reciprocal $1/\mathbb{E}\cosh(\sqrt{u}\,X)$ of the moment generating function of the Pólya law, (1.2.4), whose Thorin atoms would be the squared ordinates. The same hitting time carries both measures, through the Mellin transforms of Proposition 8.7.1 and (9.1.1).

Theorem 8.7.9. *For $u \geq 0$,*

$$\frac{K_0(2\pi)}{K_{\sqrt{u}}(2\pi)} = \frac{\mathbb{E}H^{1/2}}{\mathbb{E}H^{1/2+\sqrt{u}}} = \prod_{n\geq 1}\Big(1+\frac{u}{\nu_n^2}\Big)^{-1},$$

with $H = H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ as in Proposition 8.7.1. This is the Laplace transform of the GGC $T_P = \sum_{n\geq 1} e_n/\nu_n^2$, with independent standard exponentials e_n, no drift and Thorin measure $\sum_n \delta_{\nu_n^2}$.

Proof. $K_\mu(2\pi) = \int_0^\infty e^{-2\pi\cosh t}\cosh(\mu t)\,dt$ is entire and even in μ, so $w \mapsto K_{\sqrt{w}}(2\pi)$ is entire. Since $|\cosh(\mu t)| \leq \cosh(t\,\mathrm{Re}\,\mu)$, $|K_\mu(x)| \leq K_{\mathrm{Re}\,\mu}(x)$, and for $\nu > 0$, $K_\nu(x) = \frac{1}{2}\int_0^\infty s^{\nu-1}e^{-\frac{x}{2}(s+1/s)}ds \leq \frac{1}{2}\Gamma(\nu)(2/x)^\nu$. Hence $\log|K_{\sqrt{w}}(2\pi)| = O(|w|^{1/2}\log|w|)$, the function has order at most $\frac{1}{2}$ in w, and Hadamard's theorem gives a product of genus zero over its zeros. By Proposition 8.7.7 these are the points $w = -\nu_n^2$, each simple. So $K_{\sqrt{u}}(2\pi) = K_0(2\pi)\prod_n(1+u/\nu_n^2)$, and Proposition 8.7.1(ii) with $s = \frac{1}{2}+\sqrt{u}$ gives the middle expression. Each factor $(1+u/\nu_n^2)^{-1}$ is the Laplace transform of e_n/ν_n^2, and $\sum_n \nu_n^{-2} < \infty$. □

The theorem is an instance of two results of the book. In the language of Section 13.1.3, the law U of Proposition 8.7.1(iv), with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, has the moment generating function $L_U(s) = K_{s/2}(2\pi)/K_0(2\pi)$, which by Proposition 8.7.7 vanishes only at $s = \pm 2i\nu_n$; so U is ferromagnetic, with $c_k = 2\nu_k$ and $b = 0$, and Theorem 13.1.4 gives the clock $T_U = \sum_k e_k/(4\nu_k^2) = T_P/4$, the same variable in the normalisation of the ordinate θ. And Proposition 7.1.15, applied to the even function P of order one, which is in the Laguerre Pólya class by Proposition 8.7.7, gives the GGC $P(0)/P(i\sqrt{s}) = K_0(2\pi)/K_{\sqrt{s}/2}(2\pi)$ with atoms at the squared zeros $4\nu_n^2$. The mean and variance of T_P follow from $K_\mu(x) = k_0(x) + \frac{\mu^2}{2}k_2(x) + \frac{\mu^4}{24}k_4(x) + \cdots$, with $k_{2j}(x) = \int_0^\infty t^{2j}e^{-x\cosh t}dt$:

$$\mathbb{E}T_P = \sum_n \frac{1}{\nu_n^2} = \frac{k_2(2\pi)}{2k_0(2\pi)} = 0.0741239453481,$$

$$\mathrm{Var}\,T_P = \sum_n \frac{1}{\nu_n^4} = 2.25103144196\times 10^{-4},$$

where the sums over the 1516 zeros $\nu_n \le 1000$, with the tail described next, agree with these values.

Table 8.24 compares the closed forms with the products over the zeros, for P, for Pólya's Ξ^* and for Ξ. The tail beyond the last zero V is $\int_V^\infty \log(1 + u/\nu^2)\, dN(\nu) - \frac{1}{2}\log(1+u/V^2)$, with dN the smooth density of zeros, $\pi^{-1}\vartheta'_P$ from Proposition 8.7.8(iii) for P and $\frac{1}{2\pi}\log\frac{T}{2\pi}$ for Ξ^* and Ξ; the subtracted half term accounts for the last zero sitting at the lower end of the integral. The note [254] truncated at the zeros below a fixed height and continued with the plain density from the last zero; its agreement to three or four digits, which the last column reproduces, is limited by that tail, not by the identities, which hold to the accuracy of the full computation. For Ξ the agreement is a check, not a test, since RH is verified far beyond this range.

function	u	closed form	product	difference	truncated
P	0.5	0.0370339026997	$\nu_n \le 1000$	-4.6×10^{-11}	1.1×10^{-5}
	2	0.147801988307		-1.8×10^{-10}	4.5×10^{-5}
	5	0.367871890035		-4.6×10^{-10}	1.1×10^{-4}
	100	6.62276702182		-9.1×10^{-9}	2.3×10^{-3}
Ξ^*	10	0.3044209914	$\gamma_n^* \le 1000$	1.6×10^{-8}	8.8×10^{-5}
	60	1.661131318		9.4×10^{-8}	5.3×10^{-4}
	200	4.771881099		3.1×10^{-7}	1.8×10^{-3}
Ξ	10	0.2292377633	1700 zeros	1.6×10^{-7}	3.4×10^{-5}
	60	1.328022599		9.8×10^{-7}	2.0×10^{-4}
	200	4.11056511		3.3×10^{-6}	6.8×10^{-4}

Table 8.24: Index side products. The closed forms are $\log(K_{\sqrt{u}}(2\pi)/K_0(2\pi))$, $\log(\Xi^*(i\sqrt{u})/\Xi^*(0))$ and $\log(\xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2}))$; the next column says which zeros enter the product $\sum\log(1+u/z^2)$, whose difference from the closed form, after the tail of the text is added, is the fifth column. The last column is the difference for the truncation of the note: the 137 zeros $\nu_n < 150$, the 102 zeros of Ξ^* below 240 and 300 zeros of Ξ, each with the plain density from the last zero. Computed by `biane_polya.py`.

Pólya's Ξ^* belongs to the same class. The law with density proportional to Φ^* is ferromagnetic (Section 13.4.1), so by Theorem 13.1.4, or by Proposition 7.1.15, $\Xi^*(0)/\Xi^*(i\sqrt{u}) = \prod_n (1 + u/\gamma_n^{*2})^{-1}$ is the Laplace transform of a GGC whose Thorin measure $\sum_n \delta_{\gamma_n^{*2}}$ sits at the squares of the positive zeros γ_n^* of Ξ^*; the measure is positive. In the language of Pólya's Hilfssatz II this is the closure of the class under the cosine tilt: the kernel of $G(\theta + ia) + G(\theta - ia)$ is $2\cosh(au)$ times that of G, and Pólya's passage from P to Ξ^* is this closure with $a = \frac{9}{2}$ in the variable θ (Section 7.1.3); the note gives the shift as $\frac{9}{4}$, which is its value in the index of the Bessel function.

The random variable T_P can be realised pathwise. The index of the

Bessel equation enters as a centrifugal term $\mu^2/2r^2$, so a Feynman Kac weight $\exp(-\frac{u}{2}\int dt/R_t^2)$ along a radial path, the clock of the Hartman Watson law and of Brownian winding, shifts the index from 0 to $\sqrt{u}$.

THEOREM 8.7.10. *Let R be the diffusion on $(0,\infty)$ with generator*

$$\mathcal{G} = \frac{1}{2}\frac{d^2}{dr^2} + \Big(\frac{1}{2r} - \frac{K_1(r)}{K_0(r)}\Big)\frac{d}{dr},$$

the radial part of planar Brownian motion killed at rate $\frac{1}{2}$ and Doob transformed by K_0. For $r > b > 0$ let T_b be the hitting time of b and $\tau_r = \frac{1}{2}\int_0^{T_b} dt/R_t^2$ under $\Pr_r$. Then $T_b < \infty$ almost surely and, for $u \geq 0$,

$$\mathbb{E}e^{-u\tau_r} = \frac{K_{\sqrt{u}}(r)}{K_0(r)} \cdot \frac{K_0(b)}{K_{\sqrt{u}}(b)}.$$

As $r \to \infty$, τ_r converges in law to τ_∞ with $\mathbb{E}e^{-u\tau_\infty} = K_0(b)/K_{\sqrt{u}}(b)$. For $b = 2\pi$, $\tau_\infty \stackrel{d}{=} T_P$.

Proof. Let $\mathcal{A} = \frac{1}{2}(\frac{d^2}{dr^2} + \frac{1}{r}\frac{d}{dr} - 1)$, the generator of the radial part of planar Brownian motion killed at rate $\frac{1}{2}$. The Bessel equation gives $\mathcal{A}K_\mu = \frac{\mu^2}{2r^2}K_\mu$, in particular $\mathcal{A}K_0 = 0$, and since $K_0' = -K_1$, $\mathcal{G}f = K_0^{-1}\mathcal{A}(K_0 f)$. Hence $v = K_{\sqrt{u}}/K_0$ satisfies $\mathcal{G}v = \frac{u}{2r^2}v$. For real order $K_{\sqrt{u}} \geq K_0$, so $v \geq 1$, and $v(r) \to 1$ as $r \to \infty$ because $K_\mu(r) = \sqrt{\pi/2r}\, e^{-r}(1 + O(1/r))$ uniformly for bounded μ; so v is bounded on $[b,\infty)$. The scale density of $\mathcal{G}$ is $s'(r) = 1/(rK_0(r)^2)$, which is not integrable at ∞, so from $r > b$ the diffusion reaches b before any level $M > r$ with probability $(s(M) - s(r))/(s(M) - s(b)) \to 1$, and $T_b < \infty$ almost surely; the drift is $-1 + \frac{1}{8r^2} + O(r^{-3})$ at infinity. By Itô's formula $v(R_{t\wedge T_b})\exp(-\frac{u}{2}\int_0^{t\wedge T_b} ds/R_s^2)$ is a bounded martingale, and optional stopping gives $v(b)\mathbb{E}_r e^{-u\tau_r} = v(r)$. The limit follows from $v(r) \to 1$ and the continuity theorem for Laplace transforms, and the last statement from Theorem 8.7.9. □

In the logarithmic coordinate $X = \log R$, run on the clock $\tau = \frac{1}{2}\int dt/R_t^2$, Itô's formula gives

$$dX_\tau = -2e^{X_\tau}\frac{K_1(e^{X_\tau})}{K_0(e^{X_\tau})}\,d\tau + \sqrt{2}\,dW_\tau,$$

a diffusion with generator $\frac{d^2}{dx^2} + 2\frac{\psi'}{\psi}\frac{d}{dx} = -\psi^{-1}A_P\,\psi$, where $\psi(x) = K_0(e^x)$ is the ground solution of $-\psi'' + e^{2x}\psi = 0$. Its drift is about $-2e^x$, so $+\infty$ is an entrance boundary, and τ_∞ is its hitting time of $x_0 = \log 2\pi$ from $+\infty$. This is the Kent form of Section 8.1.3: by Kent's theorem [145] the hitting time from an entrance boundary is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues λ_n of the generator, here those of A_P, which are the ν_n^2. For Pólya's function Theorem 8.7.10 therefore solves the inverse problem that Sections 9.2.1 and 8.1.3 pose for ξ: it exhibits

the diffusion, here with the Liouville potential, whose hitting time has its Thorin atoms at the squared zeros, without Krein's inverse spectral theory. For Ξ one wants a half line operator with Dirichlet spectrum $\{\gamma^2/4\}$, the counterpart of $\{\nu_n^2\} = \{\theta_n^2/4\}$ over the zeros $\theta_n = 2\nu_n$ of P; the potential e^{2x} on $[\log 2\pi, \infty)$ has the right Weyl law (Proposition 8.7.8), so it is the natural reference to perturb, and the perturbation must account for the difference between Φ and Pólya's asymptote seen in Figure 8.10 (Question 8.7.17).

COROLLARY 8.7.11. *For real λ, with τ_∞ the clock of Theorem 8.7.10 for $b = 2\pi$ and a standard Brownian motion B independent of it,*

$$\frac{K_0(2\pi)}{K_\lambda(2\pi)} = \mathbb{E}e^{i\lambda B_{2\tau_\infty}}.$$

Proof. $\mathbb{E}e^{i\lambda B_{2\tau}} = \mathbb{E}e^{-\lambda^2\tau}$ for an independent random time τ; apply Theorem 8.7.10 with $u = \lambda^2$. □

For Ξ the corresponding identity $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\lambda) = \mathbb{E}e^{i\lambda B_{2T}}$ holds with no hypothesis, with the clock T of Theorem 5.1.1, and the variable B_{2T} is the variable $Y_{1/2}$ of Section 10.1.4; what RH adds is that T is the GGC $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 1.2.1). The note [254] states the identity for Ξ under RH and says that for Riemann's function the random time exists if and only if RH holds; the time exists unconditionally, and only its GGC form is equivalent to RH.

REMARK 8.7.12 (The Hartman Watson law). Theorem 8.7.10 is the K counterpart of a formula of Yor [348] for the Hartman Watson law [108]: for planar Brownian motion started at distance r from the origin and conditioned on $|B_t| = \rho$, $\mathbb{E}[\exp(-\frac{u}{2}\int_0^t ds/|B_s|^2)] = I_{\sqrt{u}}(r\rho/t)/I_0(r\rho/t)$. There the endpoint is fixed and the function is I; here the path is killed at rate $\frac{1}{2}$, transformed by K_0 and stopped at a level, and the function is K. Pitman and Yor develop the general theory of such Bessel laws [243].

`biane_clock.py` simulates the clock. With $R_0 = 2\pi e^{2.5} = 76.544867$, so that X starts at $\log 2\pi + 2.5$ as in the note, the exact law of τ_{R_0} has, from the expansion of $\log v$ in u, mean 0.06763390577 and standard deviation 0.01499747823, against 0.07412394535 and 0.01500343775 for τ_∞. The script runs 100000 paths of the Euler scheme for X with step $\Delta = 4\times10^{-6}$, the step of the note, and records two hitting times on the same paths: the plain one, the first step that ends below $\log 2\pi$, and one with the Brownian bridge correction, under which a step from x to x', both above the level $\ell = \log 2\pi$, crosses it with probability $\exp(-(x-\ell)(x'-\ell)/\Delta)$, the variance of the step being 2Δ. The plain scheme overshoots the barrier and is late by about $0.5826\sqrt{2\Delta}/|\beta|$, with $\beta = -13.5317$ the drift at the barrier and $0.5826 = -\zeta(\frac{1}{2})/\sqrt{2\pi}$ the constant of Broadie, Glasserman and Kou [49], that is by 1.218×10^{-4}. Table 8.25 shows that the plain mean is 2.9 standard errors late, by 1.39×10^{-4}, and that the corrected one agrees with the exact mean. The

exact Laplace transform is computed with Arb and the exact densities by the fixed Talbot contour [316, 1]. Figure 8.11 shows both comparisons. The note's own run, 40000 plain paths with mean 0.067695 ± 0.000075, is consistent with the exact mean and with the predicted lateness.

	mean	standard deviation	z of the mean	quantiles 0.01, 0.5, 0.99
exact	0.0676339	0.0149975		
plain Euler	0.0677726	0.0150584	2.91	0.04190, 0.06559, 0.11391
bridge corrected	0.0676538	0.0150221	0.42	0.04185, 0.06548, 0.11364
u	25	100	200	400
exact $\mathbb{E}e^{-u\tau_{R_0}}$	0.196345	2.54238×10^{-3}	1.65093×10^{-5}	2.91684×10^{-9}
z, bridge corrected	-0.25	0.36	0.72	0.12
z, plain Euler	-2.81	-1.84	-0.76	-0.48

Table 8.25: The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step 4×10^{-6}, with and without the bridge correction, and the Laplace transform, with the z scores (Monte Carlo $-$ exact)/standard error. The standard error of the mean is 4.75×10^{-5}; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed by `biane_clock.py`.

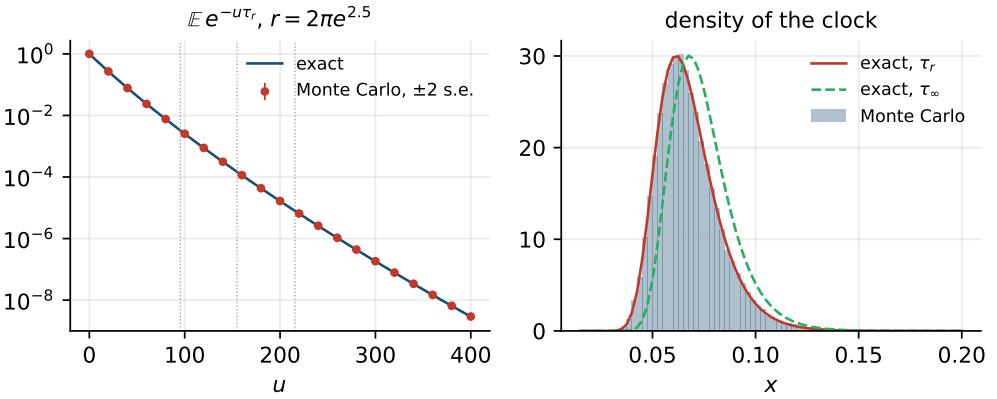


Figure 8.11: The clock from $R_0 = 2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2 = 95.4289$, $\nu_2^2 = 154.965$, $\nu_3^2 = 215.648$ of τ_∞. Right: the histogram of the corrected hitting times against the exact densities of τ_{R_0} and of τ_∞, by Talbot inversion. Drawn by `biane_figs.py` from `biane_clock.json`.

The cosine dictionary (6.2.4) turns each Thorin measure of this section into a Lévy measure of the reciprocal transform, in the variable conjugate to the square root of the energy.

Proposition 8.7.13. *For real λ, $a>0$, $c>0$ and $\ell>0$,*

$$2\log\frac{\sinh\lambda a}{\lambda a}=\int_0^\infty(1-\cos\lambda v)\,\frac{4\,dv}{v\,(e^{\pi v/a}-1)},$$

$$\ell\big(\sqrt{c^2+\lambda^2}-c\big)=\int_0^\infty(1-\cos\lambda v)\,\frac{2\ell c}{\pi}K_1(cv)\,\frac{dv}{v},$$

$$\log\frac{K_\lambda(2\pi)}{K_0(2\pi)}=\int_0^\infty 4\sin^2\Big(\frac{\lambda v}{2}\Big)\,\Theta_P(v)\,\frac{dv}{v},\qquad \Theta_P(v)=\sum_{n\ge1}e^{-\nu_n v}=\operatorname{tr}e^{-v\sqrt{A_P}}.$$

The left sides are $-\log\mathbb{E}e^{-\lambda^2W/2}$ for $W=J_a+J_a'$ and $W=H_\ell^{(c)}$, and $-\log\mathbb{E}e^{-\lambda^2T_P}$.

Proof. By Frullani's integral, $\log(1+\lambda^2/\beta^2)=\int_0^\infty(1-\cos\lambda v)\,2e^{-\beta v}dv/v$. For $W=J_a+J_a'$ the Thorin measure $2\sum_n\delta_{\pi^2n^2/2a^2}$ of Section 8.7.1 gives, at $u=\lambda^2/2$, the rates $\beta=\pi n/a$ with weight 2, and $\sum_n e^{-\pi nv/a}=(e^{\pi v/a}-1)^{-1}$. For $H_\ell^{(c)}$ the density $\ell\{\pi\sqrt{2z-c^2}\}^{-1}$ on $z>\frac{c^2}{2}$ becomes, under $\beta=\sqrt{2z}$, the measure $\ell\beta\,d\beta/(\pi\sqrt{\beta^2-c^2})$ on (c,∞), and $\int_c^\infty e^{-\beta v}\beta(\beta^2-c^2)^{-1/2}d\beta=cK_1(cv)$. For T_P take Theorem 8.7.9 at $u=\lambda^2$, with rates ν_n, and use $1-\cos=2\sin^2(\cdot/2)$. □

The Bessel clock has the Planck measure, a geometric superposition of the frequencies $\pi n/a$, and the drifted clock has the Macdonald measure, absolutely continuous because its spectrum is. The dictionary acts on Thorin measures, not on Biane's Lévy measures in the time variable, which are heat traces. For Pólya's function the third identity is the exact analogue of the Lévy Khintchine form of $\xi(\frac12)/\xi(\frac12+u)$ in Section 10.1.4, with the Poisson trace $K_{1/2}$ replaced by the wave trace Θ_P of the selfadjoint operator A_P, which is completely monotone with no hypothesis. In the variable θ, where the zeros of P are $2\nu_n$, the trace is $K^P(y)=\Theta_P(2y)=\sum_n e^{-2\nu_n y}$, and $\log(K_{\lambda/2}(2\pi)/K_0(2\pi))=\int_0^\infty 4\sin^2(\lambda y/2)K^P(y)\,dy/y$ has the form of the formula for ξ with K^P in place of $K_{1/2}$; Section 10.1.5 compares the two traces.

8.7.5 Rindler modes and a Klein Gordon criterion

Undoing the Wick rotation of the angle in the Euclidean plane of Theorem 8.7.10 gives a Lorentzian picture in which Pólya's function is a standard object of quantum field theory. Put

$$F(x,w)=\exp(-e^x\cosh w),\qquad\text{so that}\qquad K_\mu(e^x)=\frac12\int_{\mathbb{R}}F(x,w)\,e^{\mu w}\,dw.$$

Since $\partial_x^2F=(e^{2x}\cosh^2w-e^x\cosh w)F$ and $\partial_w^2F=(e^{2x}\sinh^2w-e^x\cosh w)F$,

$$\Big(\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial w^2}-e^{2x}\Big)F=0.$$

In the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ of the Rindler wedge $X_1 > |X_0|$, in which $\rho = e^x$ is the proper distance from the edge of the wedge and w the rapidity, $\partial_{X_1}^2 - \partial_{X_0}^2 = e^{-2x}(\partial_x^2 - \partial_w^2)$, so the equation is $(\partial_{X_1}^2 - \partial_{X_0}^2 - 1)F = 0$ with $F = e^{-X_1}$: the Klein Gordon equation of unit mass in $1+1$ dimensional Minkowski space, solved by a plane wave. The index μ is the eigenvalue of the boost generator ∂_w, and $K_{i\omega}(\rho)$ is the Rindler mode of boost frequency ω [88]. The zeros of $P(\theta) = K_{i\theta/2}(2\pi)$ are therefore, with $\theta = 2\omega$, the boost frequencies of the normal modes of a unit mass scalar field in the Rindler wedge with a Dirichlet mirror on the hyperbola $\rho = 2\pi$, the configuration of 't Hooft's brick wall model of a black hole horizon [116]. Biane's Dirac system of Section 8.7.3 is the corresponding Dirac field, close to the Rindler space models of Sierra for the Hamiltonian xp [305].

The Wick rotation $w = i\phi$ turns the wedge into the Euclidean plane in polar coordinates (ρ, ϕ), the boost into a rotation, the Klein Gordon operator into $\Delta - 1$, and the plane wave e^{-X_1} into $e^{-\rho\cos\phi}$, which is harmonic for $\frac{1}{2}(\Delta - 1)$, the generator of planar Brownian motion killed at rate $\frac{1}{2}$ of Theorem 8.7.10. The boost frequency becomes the angular momentum μ, and the radial operator $\frac{1}{2}(\partial_\rho^2 + \rho^{-1}\partial_\rho - \mu^2/\rho^2 - 1)$ carries the index as the centrifugal term that the Feynman Kac weight of the clock produces.

THEOREM 8.7.14 (A Klein Gordon criterion). *Let g be even, positive and continuous with $\int g(w)e^{c|w|}dw < \infty$ for every c, and $G(t) = \int g(w)e^{itw}dw$. Suppose that there are $x_0 \in \mathbb{R}$, a real continuous function V on $[x_0, \infty)$ bounded below, and $F \in C^2([x_0, \infty) \times \mathbb{R})$ such that*

(a) $(\partial_x^2 - \partial_w^2 - V(x))F = 0$ and $F(x_0, \cdot) = g$;

(b) F and its partial derivatives of order at most two are $O(e^{-c|w|})$ for every $c > 0$, uniformly for x in compact subsets of $[x_0, \infty)$, and $\Psi_\mu(x) = \int F(x, w)e^{\mu w}dw$ lies in $L^2((x_0, \infty))$ for every $\mu \in \mathbb{C}$;

(c) Ψ_μ is not identically zero for any $\mu \in \mathbb{C}$.

Then every zero of G is real, and its square is an eigenvalue of $-\frac{d^2}{dx^2} + V$ on $[x_0, \infty)$ with a Dirichlet condition at x_0.

Proof. By (b), Ψ_μ is twice differentiable under the integral sign, and two integrations by parts in w, whose boundary terms vanish, give $\Psi_\mu'' = \int(\partial_w^2 F + VF)e^{\mu w}dw = (\mu^2 + V)\Psi_\mu$. Since V is bounded below, the operator $-\frac{d^2}{dx^2} + V$ is in the limit point case at ∞, and its Dirichlet realisation at x_0 is selfadjoint. If $G(t) = 0$, then $\Psi_{it}(x_0) = G(t) = 0$ by (a), and by (b) and (c) Ψ_{it} is an eigenfunction with eigenvalue $-(it)^2 = t^2$, which is therefore real. So t is real or purely imaginary, and $t = -i\mu$ with real μ is excluded because $G(-i\mu) = \int g(w)e^{\mu w}dw > 0$. □

For Pólya's function the hypotheses hold in the rapidity variable $w = 2u$: with $g(w) = e^{-2\pi\cosh w}$, $G(t) = 2K_{it}(2\pi) = 2P(2t)$, $V(x) = e^{2x}$, $x_0 = \log 2\pi$, $F = \exp(-e^x \cosh w)$ and $\Psi_\mu(x) = 2K_\mu(e^x)$, and the theorem gives Proposition 8.7.7(ii) again. `biane_polya.py` checks the equation for F. The criterion says that the reality of the zeros of an even Fourier transform follows from embedding its kernel as the boundary value of a field that satisfies a Klein Gordon equation with a real, time independent potential, which decays fast enough in the rapidity.

For Ξ a Klein Gordon extension of Φ would prove RH (Question 8.7.18). The scaling matters. For P the eigenvalues of A_P are $\nu_n^2 = \theta_n^2/4$, and the function whose zeros they are, as a function of the spectral parameter μ at the boundary x_0, is the Jost function $\mu \mapsto \Psi_\mu(x_0) = 2K_\mu(2\pi) = 2P(-2i\mu)$. A potential asymptotic to e^{2x} at $+\infty$ has the Weyl law of Proposition 8.7.8, which matches the zeros of Ξ only in the same scaling, $g(w) = \Phi(w/2)$ and $G(t) = 2\Xi(2t)$, and then the Jost function is $\mu \mapsto 2\Xi(-2i\mu) = 2\xi(\frac{1}{2} + 2\mu)$. The note [254] proposes $\mu \mapsto \xi(\frac{1}{2} + \mu)$, which corresponds to the eigenvalues γ^2, whose density is half that of the Weyl law of e^{2x} (a potential growing like e^{4x} would match them). Inverse spectral theory reconstructs an operator from its eigenvalues and norming constants, a potential in the forms of Gelfand and Levitan or of Marchenko and a string in Krein's form, and under RH and the simplicity of the zeros Krein's theory realises the eigenvalues $\gamma^2/4$ (Section 8.1.3); the content of the question is that the operator be of the form $-\frac{d^2}{dx^2} + V$ and that V admit a Klein Gordon solution whose boundary value is $\Phi(w/2)$, which is a statement about the whole function ξ and not only about its zeros.

8.7.6 The de Bruijn Newman flow of P against that of Ξ

For real t put

$$P_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4} e^{-2\pi\cosh 2u}\, e^{i\theta u}\, du,$$

$$\Xi_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4}\, \Phi(u)\cos(\theta u)\, du = 8H_t(2\theta),$$

so that $P_0 = P$, $\Xi_0 = \Xi$, and t is the time of the de Bruijn Newman flow in the normalisation of Section 8.4.1. Decreasing t is the direction in which real zeros can be lost: RH is the statement that Ξ_t keeps only real zeros down to $t = 0$, and by the theorem of Rodgers and Tao [289] it has nonreal zeros for every $t < 0$.

Proposition 8.7.15. *There is a constant Λ_P with $-\infty < \Lambda_P \le 0$ such that P_t has only real zeros if and only if $t \ge \Lambda_P$.*

Proof. Let S be the set of t for which P_t has only real zeros. By Proposition 8.7.7, $0 \in S$. If $t \in S$ and $t' > t$, then $t' \in S$ by de Bruijn's theorem on the Gaussian factor $e^{\lambda u^2}$, $\lambda > 0$ [73, Theorem 13], which applies to the kernel $e^{tu^2/4} e^{-2\pi\cosh 2u}$

because it is $O(e^{-|u|^b})$ for every $b > 2$. If $t_k \in S$ decrease to t, then $P_{t_k} \to P_t$ locally uniformly on $\mathbb{C}$ by dominated convergence, with $|e^{t_k u^2/4} e^{i\theta u}| \le e^{t_1 u^2/4} e^{|\operatorname{Im}\theta||u|}$, and $P_t(0) > 0$, so by Hurwitz's theorem every zero of P_t is a limit of real zeros and $t \in S$. Hence $S = [\Lambda_P, \infty)$ with $\Lambda_P \in [-\infty, 0]$. Suppose $\Lambda_P = -\infty$. Then, for the even probability measure ρ with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, the Fourier transform of $e^{\lambda u^2}\rho(du)$ has only real zeros for every real λ, and by Newman's theorem [210], [212, Theorem 10], ρ is either $\frac{1}{2}(\delta_{u_0} + \delta_{-u_0})$ or has a density $Ku^{2m}\exp(-\alpha u^4 - \beta u^2)\prod_j (1 + u^2/a_j^2)e^{-u^2/a_j^2}$, with $K > 0$, $m \ge 0$ an integer, $a_j > 0$, $\sum_j a_j^{-4} < \infty$, and $\alpha > 0$, or $\alpha = 0$ and $\beta + \sum_j a_j^{-2} > 0$. The inequality $(1 + y)e^{-y} \ge e^{-y^2/2}$ for $y \ge 0$ bounds such a density below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$, while $e^{-2\pi\cosh 2u}$ decays faster than e^{-Cu^4} for every C. This contradiction gives $\Lambda_P > -\infty$. □

The proof is Newman's proof that $\Lambda > -\infty$ [210], Theorem 9 of [212], which uses only the fact that the kernel decays faster than e^{-Cu^4} for every C; it applies verbatim to Φ^*, and to every even, positive kernel with that decay whose Fourier transform has only real zeros. The note [254] conjectures $\Lambda_P = -\infty$, on the evidence of the computation below; Proposition 8.7.15 shows that the conjecture is false.

`biane_flow.py` computes both transforms by the trapezoid rule with step $1/250$ on $[0, 2]$ at 90 digits. The kernels are analytic in $|\operatorname{Im} u| < \pi/4$ (the kernel of P_t is entire) and below 10^{-68} beyond $u = 2$, and at $t = 0$ the sums reproduce Arb's $K_{i\theta/2}(2\pi)$ and mpmath's Ξ at test points up to $\theta = 90$ with relative errors below 1.4×10^{-45}. The real zeros in $(0, 90)$ are located by sign changes on a grid of step 0.05 and refined, for $t = 0, -0.5, \ldots, -20$ and $t = -21, \ldots, -60$. Figure 8.12 shows the result. The real zeros of Ξ_t in the window fall from 25 at $t = 0$ to 2 at $t = -40$, and they are lost in pairs, neighbouring zeros colliding and leaving the axis: the first collisions in the window, as t decreases, are those of the census of Section 8.4.3, beginning with $(\gamma_{24}, \gamma_{25})$ at $t = -1.144$ and height 88.47 near the top of the window. For P_t nothing of the kind occurs down to $t = -60$: from each time of the grid to the next every real zero increases, and the number in the window falls only as zeros leave it through $\theta = 90$.

Sign changes count only real zeros, so the script also counts all zeros in the box $0.5 \le \operatorname{Re}\theta \le 60.3$, $|\operatorname{Im}\theta| \le 3$ by the argument principle, on a contour refined until the argument changes by less than 0.5 between neighbours. For P_t the box holds 12, 11, 8 and 3 zeros at $t = 0$, -10, -30 and -60, all of them real. For Ξ_t it holds 13, 12 and 12 zeros at $t = 0$, -2 and -4, of which 13, 10 and 8 are real. Three of the collisions below height 60, those listed in the note [254], are located

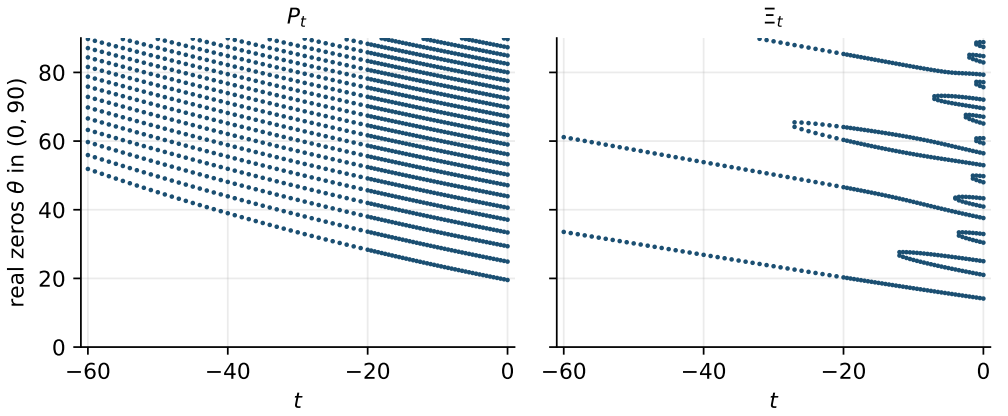


Figure 8.12: The real zeros in $(0, 90)$ of P_t (left) and of Ξ_t (right) for $-60 \le t \le 0$, on a grid (step 0.5 above $t = -20$, step 1 below). The zeros of P_t drift upward and leave the window one at a time; those of Ξ_t collide in pairs. Drawn by `biane_figs.py` from `biane_flow.json`.

by Newton's method for the double zero $\Xi_t = \partial_\theta \Xi_t = 0$ in (θ, t):

$$
\begin{aligned}
(\gamma_9, \gamma_{10}) &: \; t_c = -1.877909073, \; \theta_c = 49.66423465; \\
(\gamma_4, \gamma_5) &: \; t_c = -3.954792524, \; \theta_c = 33.09873381; \\
(\gamma_2, \gamma_3) &: \; t_c = -12.49451313, \; \theta_c = 27.3212277.
\end{aligned}
$$

The first agrees with the census of Section 8.4.3 (-1.87791 at height 49.664); the other two lie below its range $[-2.05, 0]$. The note presents the three as the first collisions below height 60, but the computation, on its grid of step 0.5 in t, shows a fourth before $t = -12.5$: the zeros γ_7 and γ_8 are at 43.263 and 43.522 at $t = -4$ and have left the axis at $t = -4.5$, and the isolated pair estimate of Section 8.4.1 puts their collision at about $t = -4.03$. In the variable θ a simple zero moves by

$$
\frac{d\theta_k}{dt} = \frac{\partial_\theta^2 F_t(\theta_k)}{4\, \partial_\theta F_t(\theta_k)} = \frac{1}{2} \sum_{j \neq k} \frac{1}{\theta_k - \theta_j},
$$

for $F_t = P_t$ and for $F_t = \Xi_t$, which is the equation of motion of Section 8.4.1 with $z = 2\theta$. As t decreases every two zeros attract each other; a pair collides when its mutual attraction overcomes the difference of the pulls of all the other zeros on its two members.

The mean motion can be predicted. For $\theta > 4\pi$ the integral $P(\theta)$ is dominated by the conjugate saddle points $u_\pm = \pm \frac{x}{2} + \frac{i\pi}{4}$ of $i\theta u - 2\pi \cosh 2u$, with $x = \operatorname{arccosh}(\theta/4\pi)$, where the exponent is $-\frac{\pi\theta}{4} \pm i(\frac{1}{2}\theta x - 2\pi \sinh x)$. So $P(\theta)$ is $e^{-\pi\theta/4}$ times a slowly varying amplitude times a cosine of phase $\varphi(\theta) = \frac{1}{2}\theta x - 2\pi \sinh x$

plus a constant; since $\varphi'(\theta) = \frac{x}{2}$, the zeros have density $\frac{x}{2\pi} = \frac{1}{2\pi}\log\frac{\theta}{2\pi} + O(\theta^{-2})$. To first order in t the factor $e^{tu^2/4}$ adds to the phase the imaginary part $\frac{\pi t}{16}x$ of $\frac{t}{4}u_+^2$, so the number of zeros below a fixed height changes at the rate $\frac{x}{16}$ per unit of t, and each zero moves with

$$\frac{d\theta}{dt} \approx -\frac{x/16}{x/2\pi} = -\frac{\pi}{8},$$

whatever its height. The equation of motion gives the same value: with the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$ on both half lines, $\frac{1}{2}\sum_{j\neq k}(\theta_k - \theta_j)^{-1}$ becomes $\frac{1}{4\pi}$ times the principal value integral of $\log|s|/(\theta_k - s)$, which is $-\frac{\pi^2}{2}$ for $\theta_k > 0$. Over $-0.5 \le t \le 0$ the 24 real zeros of P in the window rise at mean speeds that decrease with the height from 0.3990 (the first zero) to 0.3949 (the last), towards $\pi/8 = 0.3927$, while the 25 zeros of Ξ rise at speeds between 0.0040, for γ_{25}, held back by γ_{24}, which is closing on it, and 0.7350, for γ_{19}, which is closing on γ_{20}, with mean 0.3647 (`biane_figs.py`). The zeros of P move almost as a rigid block; those of Ξ fluctuate about the same mean motion, and the fast and slow members of the close pairs are the ones that collide.

In the language of Lee and Yang [171] (Section 7.1.3 and Proposition 7.1.16), the single spin law with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$ has the Lee and Yang property, since its moment generating function $K_{h/2}(2\pi)/K_0(2\pi)$ vanishes only at imaginary h; it is the ferromagnetic law U of Section 8.7.4. Newman's theorem says that this property survives every Gaussian factor e^{-bu^2}, $b > 0$, only for two point laws and for the densities of the proof of Proposition 8.7.15, which include the φ^4 laws $e^{-au^4-bu^2}$ that Simon and Griffiths realised as limits of Ising ferromagnets [306]. The law with density proportional to $e^{-2\pi\cosh 2u-bu^2}$ has the property exactly when its Fourier transform, a multiple of P_{-4b}, has only real zeros, that is when $b \le -\Lambda_P/4$. So the Simon Griffiths type construction for $e^{-2\pi\cosh 2u-bu^2}$ at every real b that the note proposes cannot exist.

8.7.7 Pólya's function and Ξ side by side

Table 8.26 follows the comparison in the note [254], corrected in three rows: the index side clock of Ξ exists with no hypothesis, the de Bruijn Newman constant of P is finite, and the computation of the flow describes a window of heights and times, not the whole flow. The Wick rotation of Section 8.7.2 relates the two entries of each of the second, third and fourth rows.

The statements about Ξ in the right column and their relatives, Proposition 1.2.1, Corollary 10.1.4, Proposition 7.1.15 and the GGC form of the clock T discussed after Corollary 8.7.11, are proved by Hadamard factorisation, Bernstein's theorem and integral identities. No property of the primes enters their proofs, and none could, since each is an equivalence between two statements about the zeros of one entire function: they reorganise RH, as Section 12.2 says of the

	Pólya's $P(\theta) = K_{i\theta/2}(2\pi)$	Riemann's $\Xi(\theta)$
kernel	$e^{-2\pi\cosh 2u}$, (8.7.1)	$\Phi(u)$, (1.2.3)
hitting time	$H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$, Brownian motion with drift (Proposition 8.7.1)	$\frac{\pi}{2}S_2$, three dimensional Bessel process, (9.1.1)
symmetric space	$\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$	$SU(2) \cong S^3$
Thorin measure of the hitting time (energy)	continuous: σ_2 for $H^{(1)}_1$	discrete: $2\pi\sigma_1$ for $\pi^2 S_2$ (Theorem 8.7.2)
index side Thorin measure (angular momentum)	$\sum_n \delta_{\nu_n^2}$, a theorem (Theorem 8.7.9)	$\sum_\gamma \delta_{\gamma^2}$ if and only if RH (Proposition 1.2.1)
index side clock	τ_∞, a hitting time of an explicit diffusion (Theorem 8.7.10)	a random time T (Theorem 5.1.1); no diffusion is known (Question 8.7.17)
trace of the sine squared form	$\Theta_P = \operatorname{tr} e^{-v\sqrt{A_P}}$, completely monotone (Proposition 8.7.13)	$K_{1/2}$, completely monotone if and only if RH (Corollary 10.1.4)
operator for the zeros	$-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ (Proposition 8.7.7)	not known
Klein Gordon extension	$\exp(-e^x \cosh w)$ (Theorem 8.7.14)	not known (Question 8.7.18)
de Bruijn Newman constant	$-\infty < \Lambda_P \leq 0$ (Proposition 8.7.15)	$\Lambda \geq 0$ [289]; RH if and only if $\Lambda = 0$
real zeros in $(0, 90)$ for $-60 \leq t \leq 0$	no collision; mean speeds 0.395 to 0.399 on $[-0.5, 0]$	collisions in pairs from $t = -1.144$; mean speeds 0.004 to 0.735 on $[-0.5, 0]$

Table 8.26: Pólya's function and Ξ. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open.

programme as a whole, and do not approach it. The proofs of the entries on the left that concern the zeros all go through the selfadjoint operator A_P, which is exactly what the right column lacks.

The note [254] proves part (a) of the following proposition; part (b) is the strengthening that Newman's theorem gives.

PROPOSITION 8.7.16. *Let $\mathcal{P}$ be a property of even, positive, integrable kernels such that*

(i) if $\mathcal{P}(g)$ holds, then $\int g(u)e^{i\theta u}du$ has only real zeros, and

(ii) there is $\varepsilon > 0$ such that $\mathcal{P}(g)$ implies $\mathcal{P}(e^{-\varepsilon u^2/4}g)$.

(a) If (ii) holds for $g = \Phi$, then $\mathcal{P}(\Phi)$ is false.

(b) If (ii) holds for all g with the same ε, then $\mathcal{P}(g)$ is false for every kernel g that is $O(e^{-|u|^b})$ for some $b > 2$ and satisfies $g(u)e^{Cu^4} \to 0$ as $|u| \to \infty$ for every C; in particular $\mathcal{P}$ fails for Φ, for Φ^* and for $e^{-2\pi\cosh 2u}$.

Proof. (a) If $\mathcal{P}(\Phi)$ held, then by (ii) and (i) the transform of $e^{-\varepsilon u^2/4}\Phi$, which is $\Xi_{-\varepsilon}$ of Section 8.7.6, would have only real zeros, so $\Lambda \le -\varepsilon < 0$, against the theorem of Rodgers and Tao [289]. (b) If $\mathcal{P}(g)$ held, then by induction $\mathcal{P}(e^{-k\varepsilon u^2/4}g)$ would hold for $k = 0, 1, 2, \dots$, so the transform of $e^{-k\varepsilon u^2/4}g$ would have only real zeros for each k. This kernel is again $O(e^{-|u|^b})$, so by de Bruijn's theorem [73, Theorem 13] the transform of $e^{tu^2/4}g$ would have only real zeros for every $t \ge -k\varepsilon$, and hence for every real t. By Newman's theorem, as in the proof of Proposition 8.7.15, $g/\int g$ would be a density of the form given there, which is bounded below by $Ku^{2m}\exp(-C'u^4 - \beta u^2)$ for some C', against the decay of g. □

Part (b), which uses only de Bruijn's theorem and Newman's theorem of 1976, shows that the obstruction has nothing to do with the position of Φ on the boundary of the class of kernels with real zeros: a property that survives a fixed amount of damping for all kernels fails for P as well, although the zeros of P are real for a structural reason, the selfadjointness of A_P. So the reading in the note, that Pólya's kernel tolerates every Gaussian damping, that P has slack and Ξ none, is to be replaced by a quantitative one: P has the finite slack $|\Lambda_P|$, whose size is not known, and Ξ has none if RH holds. The note lists among the excluded arguments a ferromagnetic or Lee and Yang construction and a Simon Griffiths representation. The Lee and Yang property of a single spin law is preserved by the ferromagnetic factor e^{bu^2}, $b > 0$, which runs the flow forward, but by (b) it is not preserved by a fixed damping except on Newman's class. A Lee and Yang representation of the law with density proportional to Φ, which is equivalent to RH (Proposition 7.1.16), is therefore not excluded by the proposition, and neither is a Simon Griffiths type representation of that law.

Three questions raised by the note are not equivalences and can be studied without assuming what is to be proved. The first two ask for an object, a diffusion or a potential, not for the zeros, and part (b) does not apply to them, because the existence of such an object is not preserved by a fixed Gaussian damping of every kernel: for P it is lost at every $t < \Lambda_P$, where P_t has nonreal zeros. The third is a question about an explicit kernel with no arithmetic in it, and it measures the slack of Pólya's function. A fourth concerns the interpolation between the two clocks of Section 8.7.2.

Question 8.7.17. Is there a diffusion on a half line, with an entrance boundary, whose hitting time of a level from the entrance boundary has the Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + 2\sqrt{u})$, the transform of the clock T of Theorem 5.1.1 scaled by 4, as τ_∞ has the transform $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ in Theorem 8.7.10? By Kent's theorem [145] such a hitting time is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues $\lambda_n > 0$ of

the generator, so a positive answer implies RH; under RH and the simplicity of the zeros, Krein's inverse spectral theory gives a generalised diffusion with the eigenvalues $\gamma^2/4$ (Section 8.1.3). The question asks for one constructed without RH, for example as a perturbation of the diffusion X of Section 8.7.4, whose potential e^{2x} has the right Weyl law.

Question 8.7.18. Does $g(w) = \Phi(w/2)$ admit a Klein Gordon extension in the sense of Theorem 8.7.14, with a real potential V asymptotic to e^{2x} at $+\infty$? By the theorem a positive answer implies RH. Under RH the question asks whether $\mu \mapsto 2\xi(\frac{1}{2} + 2\mu)$ is, up to normalisation, the Jost function of such a potential whose Klein Gordon solution has the boundary value g. A first numerical step is to reconstruct a potential from the first few thousand ordinates and a choice of norming constants by the method of Gelfand and Levitan or of Marchenko, and to test whether it is stable under truncation and close to e^{2x}.

Question 8.7.19. What is the de Bruijn Newman constant Λ_P of Pólya's function, and that of Ξ^*, which is also finite and at most 0 by the proof of Proposition 8.7.15? At what height do the first nonreal zeros of P_t appear as t decreases through Λ_P, and do they appear, as for Ξ_t, through the collision of two real zeros?

Question 8.7.20. Is there a one parameter family of laws joining the drifted first passage time and the sum of two Bessel hitting times, each with a reciprocal symmetry that encodes a functional equation, as $\tau \mapsto 1/\tau$ does for H in Proposition 8.7.1(i) and Jacobi's transformation does for S_2, along which the index side Thorin measure can be followed? The curvature family of Proposition 8.7.6 is an Esscher tilt and breaks the symmetry. Positivity of the index side Thorin measure at the Bessel end is RH.

8.8 Exercises

On Section 8.1.

8.1. Assuming that the zeros are on the critical line and simple, show that the residues of the clock transform at $u = -\gamma_k^2$ are $R_k = 2i\gamma_k \xi(\frac{1}{2})/\xi'(\frac{1}{2} + i\gamma_k)$ and that they alternate in sign. (They are the coefficients c_n of Theorem 4.3.22.)

8.2. Explain why the spectral expansion of the density diverges at $t = 0$ while the Talbot inversion does not, and compare the abscissa of convergence in Theorem 4.3.22.

8.3. Show that the hazard of a finite sum of exponentials with rates $\lambda_1 < \lambda_2 < \cdots$ tends to λ_1 and that $h(t) - \lambda_1$ decays like $e^{-(\lambda_2 - \lambda_1)t}$.

8.4. Sample Williams' variable with 10^5 draws and estimate $\Xi(5)$; compare the standard error with Table 8.3.

8.5. Assume RH, let Θ_N be the cutoff of Section 8.1.3, and assume that $S(\Theta_N) = 0$, as for the values of N used there. Show that $\log G(w) - \log G_N^{\mathrm{W}}(w)$ tends to $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as $w \to \infty$. Then, using Stirling's formula for $\log\xi(\frac12 + x)$ as $x \to +\infty$ and $\int_0^\infty \log(1 + x^2/t^2)\log\frac{t}{2\pi}\,dt = \pi x \log\frac{x}{2\pi} - \pi x$, show that

$$\ell_N = 2\sum_{k\le N}\log\gamma_k - \log\xi(\tfrac12) + \tfrac14\log\tfrac{\pi}{2} - \frac{1}{\pi}\int_0^{\Theta_N}\log t\,\log\frac{t}{2\pi}\,dt + \frac{2}{\pi}\int_{\Theta_N}^{\infty}\Big(\vartheta'(t) - \frac12\log\frac{t}{2\pi}\Big)\log t\,dt,$$

the terms in $\log x$ cancelling because $\vartheta(\Theta_N) = (N-1)\pi$. So ℓ_N is determined by $\gamma_1, \dots, \gamma_N$ and $\xi(\frac12)$.

On Section 8.2.

8.6. Show that on the parabola $u = \frac12(-q^2 + 2icq)$ one has $\operatorname{Re}\sqrt{c^2 + 2u} = c$, and hence that every mixture of inverse Gaussian laws with drift c has transform bounded by one there.

8.7. Derive the small q expansion $|G(u)|^2 = 1 - q^2(\mathbb{E}T - c^2\operatorname{Var}T) + O(q^3)$ on the parabola.

8.8. Show that the hazard of the first passage time of Brownian motion across the line $\alpha + \beta t$, with $\alpha > 0$ and $\beta < 0$, which has the inverse Gaussian law with drift $|\beta|$, tends to $\beta^2/2$ with correction $3/(2t)$.

8.9. Implement the tangent boundary ODE with a different starting time and check that the asymptotic slope is unchanged.

8.10. Show that the density f of the clock is continuous and positive on $(0,\infty)$. (The Laplace transform of T equals $1/G$ on $\operatorname{Re} u \ge 0$, by analytic continuation from $(0,\infty)$ and continuity, so the characteristic function is $1/G(-iy)$; Stirling's formula and the bound $|\zeta(s)| \ge 1/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$, applied on the rays $\frac12 + re^{\pm i\pi/4}$, show that it decays faster than any power of $|y|$. Next, $\log G(u) = o(u)$ as $u \to \infty$, so $P(T < t) > 0$ for every $t > 0$; $1/G$ has poles, so the law of T has unbounded support; and the law is unimodal by Corollary 11.1.2, so f itself, which is the derivative of the distribution function at every point, is nondecreasing up to a mode and nonincreasing after it.)

On Section 8.3.

8.11. Derive formula (8.3.1) from the derivative representation of λ_n.

8.12. Show that $\lambda_1 = \sum_\rho \rho^{-1} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, and that under RH $\lambda_1 = \sum_{\gamma>0}(\gamma^2+\frac14)^{-1}$.

8.13. Show that under RH $\lambda_n = \sum_{\gamma>0} 4\sin^2(n\arctan\frac{1}{2\gamma})$, and that the terms are of order n^2/γ^2 for $\gamma \gg n$.

8.14. For a hypothetical zero $\rho = \frac12 + d + i\gamma$ with $d>0$, show that $|1-1/(1-\bar\rho)|>1$, and estimate the index n at which its contribution would make λ_n negative.

On Section 8.4.

8.15. Show that H_t satisfies $\partial_t H_t = -\partial_z^2 H_t$ and derive the equation of motion of a simple zero.

8.16. For an isolated pair at distance D_0, solve $dD/dt = 4/D$ and find the collision time.

8.17. Modify the Pólya evaluator to compute H_t and reproduce Table 8.7.

8.18. Explain why the collision time of a pair of zeros gives numerical evidence for, but not a proof of, a lower bound on Λ.

On Section 8.5.7.

8.19. Show that $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$ is even, using $\eta(i/x) = \sqrt{x}\,\eta(ix)$, and deduce the functional equation $\xi(s,\chi_4) = \xi(1-s,\chi_4)$ from $\xi(\frac12+s,\chi_4) = \int_{\mathbb R} K_4(u)e^{su}\,du$.

8.20. Show that for real $\sigma>0$ the series $\sum_{n\ge0}(-1)^n(2n+1)^{-\sigma}$ lies between $1-3^{-\sigma}$ and 1, and deduce that $L(s,\chi_4)$ has no zeros on $(0,1)$ without using the kernel.

8.21. For $0<\varepsilon<\delta$ show that $\pi(P_{\varepsilon+\delta}+P_{\varepsilon-\delta})$ integrates over the interval $|x|<\sqrt{\delta^2-\varepsilon^2}$, where it is negative, to $\pi - 4\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)}$, which lies in $(-\pi,0)$ and tends to 0 as $\varepsilon/\delta\to0$; this is the bound on the fall of the argument after Corollary 8.5.31.

8.22. Show that $g(c,u) = 2c(3u^2-c^2)/(c^2+u^2)^3$, the second derivative of $c/(c^2+u^2)$ in u, has $\partial_c g(c,u) = 6(u^4-6c^2u^2+c^4)/(c^2+u^2)^4$, which is positive exactly when $|c|<(\sqrt2-1)|u|$ or $|c|>(\sqrt2+1)|u|$, and deduce that $g(a+\delta,u)+g(a-\delta,u)>0$ for $a,\delta>0$ with $a+\delta<(\sqrt2-1)|u|$, the step for the zeros off the line in the proof of Theorem 9.1.18(ii).

On Section 8.7.

8.23. Show that $1/H$ has the law of H size biased by H, for $H = H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ with the density of Proposition 8.7.1(i), and deduce $\mathbb{E}H^s = \mathbb{E}H^{1-s}$ without computing the Mellin transform.

8.24. Show that $F(x,w) = \exp(-e^x \cosh w)$ satisfies $(\partial_x^2 - \partial_w^2 - e^{2x})F = 0$, and that in the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ one has $\partial_x^2 - \partial_w^2 = e^{2x}(\partial_{X_1}^2 - \partial_{X_0}^2)$, so that $F = e^{-X_1}$ solves the Klein Gordon equation of unit mass.

8.25. Show that $(1+y)e^{-y} \geq e^{-y^2/2}$ for $y \geq 0$, and deduce that each density of Newman's theorem in the proof of Proposition 8.7.15 is bounded below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$.

8.26. Show that the principal value integral of $\log|s|/(1-s)$ over $\mathbb{R}$, taken with symmetric limits at $\pm\infty$, is $-\frac{\pi^2}{2}$, and deduce the mean speed $-\frac{\pi}{8}$ of the zeros under the flow in Section 8.7.6 from the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$.

8.27. For $0 < y < \pi$ show, as claimed in Subsection 10.1.5, that $\gamma^n e^{-\gamma y}$, for $\gamma \leq n/\pi$, is at most $\exp(n(\log\frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its maximum over $\gamma > 0$, and evaluate the exponent for $n = 10^6$ at $y = 1$ and at $y = 3$.

Chapter 9

The basepoint one

We now work at the resolution $\varepsilon = \frac{1}{2}$ of the Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2, the basepoint one. The shifted clocks T_α, with $\mathbb{E}e^{-uT_\alpha} = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, carry the Thorin programme. At a basepoint α to the right of every zero, T_α is a GGC (Theorem 1.2.2 and Proposition 1.2.3), and from $\alpha = 1$ on its Thorin density is positive with no hypothesis, because no zero lies on the line $\operatorname{Re} s = 1$ or to its right.

We work the basepoint one out completely, as a test bed for the tools of the book. We ask three questions of T_1: what its Thorin measure and its law are, whether its Thorin density comes from the sine squared lemma, and whether it is a hitting time to which Kent's theorem applies.

For the Thorin measure and the Lévy form of $\log \xi(1+s)$ alike, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH.

9.1 The Thorin measure and the Lévy form at the basepoint one

The scripts of the chapter are in the folders `basepoint_one/xi` and `basepoint_one/process_map` of the archive, except those of Section 9.1.6, which are in `code/`.

9.1.1 The Mellin side: Bessel hitting times and size biasing

Let J be the first time a three dimensional Bessel process started at 0 reaches the level one. Its Laplace transform is

$$\mathbb{E}e^{-uJ} = \frac{\sqrt{2u}}{\sinh\sqrt{2u}} = \prod_{n\geq 1}\Big(1 + \frac{2u}{\pi^2 n^2}\Big)^{-1}, \qquad \mathbb{E}J = \frac{1}{3},$$

so $J \stackrel{d}{=} \sum_n \frac{2}{\pi^2 n^2} e_n$ with independent standard exponentials e_n. This is a GGC with unit Thorin atoms at the points $\pi^2 n^2/2$, which are the Dirichlet eigenvalues of the Bessel process on $[0,1]$, so J is the simplest instance of Kent's theorem (Section 8.1.3): up to a change of scale the Bessel process is the string diffusion with that spectrum [145]. With J' an independent copy of J, the sum $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2} \Gamma_n$, with independent Gamma$(2,1)$ variables Γ_n, is Williams' Gamma(2) structure [344] of Section 8.2.3, and Biane, Pitman and Yor [32] proved the identity recalled in Section 1.2 and Theorem 7.1.21,

$$2\xi(s) = \mathbb{E}\Big[\Big(\frac{\pi}{2} S_2\Big)^{s/2}\Big], \qquad s \in \mathbb{C}. \tag{9.1.1}$$

Table 9.1 checks (9.1.1) at real s. Table 8.3 checks the same identity on the critical line.

s	$\mathbb{E}(\pi S_2/2)^{s/2}$, Monte Carlo	standard error	$2\xi(s)$
0.5	0.99415	0.00017	0.99424
1	0.99981	0.00034	1
2	1.04677	0.00074	1.04720
3	1.14713	0.00127	1.14788

Table 9.1: The identity (9.1.1) of Biane, Pitman and Yor, checked with 4×10^5 draws of S_2 by `comp.py`. The sample mean of S_2 is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$.

Put $Y = \frac{1}{2}\log(\frac{\pi}{2} S_2)$, as in Section 1.2. By (9.1.1), $2\xi(\alpha + w) = \mathbb{E}[e^{\alpha Y} e^{wY}]$, so for every real α

$$\frac{\xi(\alpha + w)}{\xi(\alpha)} = \mathbb{E} e^{w V_\alpha}, \qquad P(V_\alpha \in dy) = \frac{e^{\alpha y} P(Y \in dy)}{\mathbb{E} e^{\alpha Y}}.$$

Since the Pólya variable X is Y tilted by $e^{Y/2}$, the variable V_α has the density $e^{(\alpha - \frac{1}{2})y} p(y)\, \xi(\frac{1}{2})/\xi(\alpha)$, the Esscher tilt of the Pólya density p of (1.2.4); so $V_{1/2} = X$, and we write $V = V_1$ for the tilt at the basepoint one. Equivalently, V_α is $\frac{1}{2}\log(\frac{\pi}{2} S_2)$ under the law of S_2 size biased by $S_2^{\alpha/2}$: by $S_2^{1/4}$ at the centre, by $S_2^{1/2}$ at the basepoint one, where the weight e^Y needs no normalisation because $\mathbb{E} e^Y = 2\xi(1) = 1$, and by $S_2^{(1-\varepsilon)/2}$ at $1-\varepsilon$. Every basepoint thus gives a probability law with no hypothesis on this side, and nothing distinguishes $\alpha = 1$ except the weight. The law of $-V_\alpha$, with characteristic function $\xi(\alpha - it)/\xi(\alpha)$, is Nakamura's complete Riemann zeta distribution [206]. He proved that RH holds if and only if, for every $\frac{1}{2} < \alpha < 1$, this characteristic function has a Lévy Khintchine form with a signed measure, with no integrability condition on the measure. The theory of infinite divisibility does distinguish it: V_α is quasi infinitely divisible, though not infinitely divisible, for every $\alpha > 1$, and it is not quasi infinitely divisible at $\alpha = 1$ or at the centre (Proposition 11.1.3; for $\alpha = 1$ see also Section 9.1.4).

REMARK 9.1.1 (Ostrovsky's Barnes beta laws). The companion $C_2 = \sum_{k\ge1} \frac{2}{\pi^2(k-\frac12)^2}(e_k + e'_k)$ of S_2, with independent standard exponentials, has $\mathbb{E}e^{-uC_2} = \operatorname{sech}^2\sqrt{2u}$ [32], and its Mellin transform is again ξ:

$$\mathbb{E}[C_2^q] = \frac{2^{2q+2}-1}{q+1}\Big(\frac{2}{\pi}\Big)^{q+1}\xi(2q+2),$$

which we checked at $q = -0.3$, $\frac12$, 1, 2 to 10^{-12} by quadrature of the Laplace transform. The Lévy density $\frac{1}{4t}\operatorname{sech}^2\frac{t}{4}$ of Proposition 9.1.9 is

$$\frac{1}{4t}\operatorname{sech}^2\frac{t}{4} = \frac{1}{4t}\,\mathbb{E}e^{-t^2C_2/32},$$

a mixture of Gaussians in t whose mixing law has Mellin transform $\xi(2q+2)$. The positivity comes from the gamma side, and the zeros of ξ enter only through the Mellin transform of C_2. More generally the Barnes beta laws $\beta_{M,N}(a,b)$ of [191] have Lévy density $e^{-b_0t}\prod_{j=1}^N(1-e^{-b_jt})/\big(t\prod_{i=1}^M(1-e^{-a_it})\big)$; for $M=N=2$, $a=(1,1)$ and $b=(b_0,\frac12,\frac12)$ it is $\frac{e^{-(b_0-\frac12)t}}{4t}\operatorname{sech}^2\frac{t}{4}$, the density above times $e^{-(b_0-\frac12)t}$, and it is again not that of a GGC, since t times it is the Laplace transform of the signed measure $\sum_{j\ge1}(-1)^{j+1}j\,\delta_{b_0-\frac12+\frac{j}{2}}$.

9.1.2 The Thorin measure and the law of T_1

For a real basepoint $\alpha \ge \frac12$ we write $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, so that $\varphi_{1/2} = 1/G$ is the clock, and T_α for a random variable with $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha(u)$ for $u \ge 0$, whenever one exists. Theorem 1.2.2 provides it, as a GGC, for $\alpha \ge 1$, and Theorem 9.2.5 in Section 9.2 provides it for every $\alpha > \frac12$. As in Section 1.2, $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$.

PROPOSITION 9.1.2. *The function $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$ is the Laplace transform of a GGC T_1 with no drift, $\log\varphi_1(u) = -\int_0^\infty \log(1+u/z)\,U_1(dz)$, and in the variable $\theta = \sqrt{z}$ its Thorin measure has density*

$$\rho_{1/2}(\theta) = \frac{1}{\pi}v_1(\theta) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(1+i\theta) = \frac{1}{\pi}\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2}, \tag{9.1.2}$$

the sum running over all nontrivial zeros $\rho = \beta + i\gamma$ in conjugate pairs. Every term is positive, so U_1 is a positive measure with no hypothesis on the zeros.

Proof. This is Theorem 1.2.2 at $\alpha = 1$. First, $-\frac{d}{du}\log\varphi_1(u) = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(1+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$, because the zeros of $\xi(1+w)$ lie at $w = \rho-1$ with $\operatorname{Re} w = \beta - 1 < 0$ and so are never reached by the principal branch $w = \sqrt{u}$; Stieltjes inversion (1.3.2) of its jump across the cut, where $\sqrt{u} = \pm i\theta$, gives the density (1.2.6) in z, which is (9.1.2) in θ because $dz = 2\theta\,d\theta$. Second, by (1.2.2) with conjugate pairs grouped, $\operatorname{Re}1/(1+i\theta-\rho) = (1-\beta)/((1-\beta)^2+(\theta-\gamma)^2) =$

$\pi P_{1-\beta}(\theta-\gamma)$, as in the proof of Proposition 1.2.3, and every term is positive because $\beta<1$ for all zeros, which is the nonvanishing of ζ on the line $\operatorname{Re} s=1$. Third, there is no drift because $\log\xi(1+\sqrt{u})$ grows like $\frac12\sqrt{u}\log\sqrt{u}$, more slowly than u. □

Under RH every width equals $\frac12$, and (1.2.8) at $\varepsilon=\frac12$ reads

$$\rho_{1/2}(\theta)=\sum_{\gamma>0}\Big(P_{1/2}(\theta-\gamma)+P_{1/2}(\theta+\gamma)\Big),$$

the RH measure blurred by the Cauchy kernel of width one half. The value at the origin is explicit, $\rho_{1/2}(0)=\frac1\pi\sum_\rho\frac1\rho=\frac1\pi\big(1+\frac{\gamma_E}{2}-\frac12\log4\pi\big)=\frac{0.0230957089661}{\pi}=0.0073515925$, where γ_E is Euler's constant, and it is the minimum of $\rho_{1/2}$, attained only at the origin (Proposition 9.1.11). The cumulative mass $M_{1/2}(T)=\int_0^T\rho_{1/2}(\theta)\,d\theta$ is π^{-1} times the argument of ξ along the line $\operatorname{Re} s=1$, and since no zero lies on or to the right of that line, Proposition 3.4.2 gives the exact unconditional count $N(T)=M_{1/2}(T)+\tau_{1/2}(T)$, in which $\tau_{1/2}(T)$, the change of argument of ξ from $1+iT$ to $\frac12+iT$ divided by π, is $O(\log T)$, like the argument $\arg\zeta(\frac12+iT)$ of Section 2.2.1 [324]. Numerically $\int_0^{100}\rho_{1/2}=29.111$ against $N(100)=29$, so that $\tau_{1/2}(100)=-0.111$. Table 9.2 lists values of the density, and Figure 9.1 shows it with its mass.

θ	$\rho_{1/2}(\theta)$ from ξ'/ξ	from the arithmetic form	from 1700 zeros, $\beta=\frac12$
0	0.0073515925	0.0073515925	0.0071935
5	0.0084053053	0.0084053053	0.0082472
14.134725	0.6473699273	0.6473699273	0.6472119
20	0.1418590769	0.1418590769	0.1417010
30	0.4070956035	0.4070956035	0.4069375
100	0.2308349819	0.2308349819	0.2306766

Table 9.2: The Thorin density $\rho_{1/2}=v_1/\pi$ of T_1 at the basepoint one, computed from ξ'/ξ, from the arithmetic form $v_1(\theta)=\frac{1}{1+\theta^2}-\frac12\log\pi+\frac12\operatorname{Re}\psi(\frac{1+i\theta}{2})+\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with ζ'/ζ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta=\frac12$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant 1.6×10^{-4}.

Because $\xi'(1)\neq0$, the Laplace transform of T_1 is not differentiable at the origin:

$$\varphi_1(u)=1-\frac{\xi'}{\xi}(1)\sqrt{u}+O(u)=1-\Big(\sum_\rho\frac1\rho\Big)\sqrt{u}+O(u),\qquad u\downarrow0.$$

By the Tauberian theorem for Laplace transforms [86, XIII.5], T_1 has infinite mean

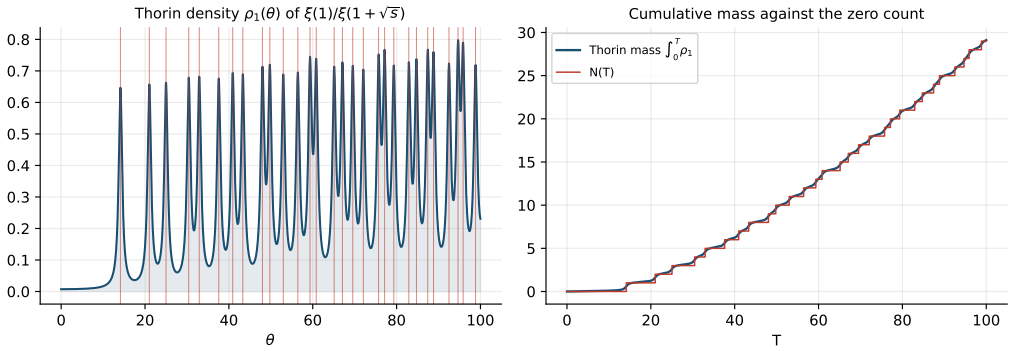


Figure 9.1: Left: the Thorin density $\rho_{1/2}(\theta) = v_1(\theta)/\pi$ of T_1 on $[0, 100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T \rho_{1/2}$ against the zero count $N(T)$. The plot writes ρ_1 for $\rho_{1/2}$ and s for u.

and

$$P(T_1 > x) \sim \frac{\sum_\rho \rho^{-1}}{\Gamma(\frac{1}{2})}\, x^{-1/2} = 0.01303\, x^{-1/2}, \qquad x \to \infty. \tag{9.1.3}$$

Equivalently, near the origin the Thorin density in z behaves like $\rho_{1/2}(0)/(2\sqrt{z})$, which makes $\int U_1(dz)/z$ diverge. At the centre, by contrast, $\xi'(\frac{1}{2}) = 0$, the clock has the finite mean $\mathbb{E}T = \sum_k a_k^{-1} = 0.0231050$, which is $\sum_{\gamma>0} \gamma^{-2}$ under RH; at the basepoint one the relevant constant is $\sum_\rho \rho^{-1} = 0.0230957$, and it multiplies $\sqrt{u}$.

9.1.3 The sine squared lemma at the basepoint one

At a basepoint $\alpha > 1$ the sine squared lemma (1.2.9), applied to the canonical measure of $\log \xi$, writes v_α as the sum of a drift, an archimedean piece, a pole piece and a prime piece, (3.1.1). The pole piece is the image under the lemma of the measure $e^{-(\alpha-1)x}dx$ and it enters with a minus sign; the prime piece is positive. As $\alpha \downarrow 1$ both diverge, because $\sum \Lambda(n)/n$ diverges (Section 3.1.2). So the lemma cannot be applied to the two pieces separately at the basepoint one. It can be applied to their difference. Throughout Section 9.1, $\psi(y) = \sum_{n \le y} \Lambda(n)$ is Chebyshev's function, whose argument is always the real variable $y \ge 1$ or the prime cutoff X, while at every other argument, as in $\psi(\frac{3}{2})$ or $\psi(\frac{1+i\theta}{2})$, $\psi = \Gamma'/\Gamma$ is the digamma function; from here on X always denotes the cutoff, not the Pólya variable of Section 9.1.1.

Proposition 9.1.3. *Let* $\nu_1 = \sum_n \Lambda(n) n^{-1} \delta_{\log n} - \mathbf{1}_{(0,\infty)}(x)\, dx$, *the prime measure*

compensated by Lebesgue measure. For every $\theta \neq 0$,

$$v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\theta}{2}\Big) + \gamma_E + \lim_{X\to\infty}\int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)\nu_1(dx), \tag{9.1.4}$$

that is, the limit of $\sum_{n\le X}\Lambda(n)n^{-1}2\sin^2(\frac{1}{2}\theta\log n) - \big(\log X - \sin(\theta\log X)/\theta\big)$. The limit exists by the prime number theorem, and at $\theta = 0$ the formula reduces to $v_1(0) = \sum_\rho \rho^{-1}$.

Proof. On the line $\operatorname{Re} s = 1$ the pole term $1/(s-1) = 1/(i\theta)$ is purely imaginary, so $v_1(\theta) = \operatorname{Re}\frac{1}{1+i\theta} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$. For $\operatorname{Re} s \ge 1$ with $s \neq 1$, partial summation gives

$$-\sum_{n\le X}\frac{\Lambda(n)}{n^s} + \int_1^X y^{-s}dy = -\frac{\psi(X)-X}{X^s} - s\int_1^X (\psi(y)-y)\,y^{-s-1}dy - 1,$$

and the prime number theorem with the de la Vallée Poussin error term, $\psi(y)-y = O(y\exp(-c\sqrt{\log y}))$, makes the boundary term vanish and the integral converge absolutely as $X\to\infty$, uniformly for $\operatorname{Re} s \ge 1$. For $\operatorname{Re} s > 1$ the limit is $\zeta'(s)/\zeta(s) + 1/(s-1)$, and by continuity the same holds on $\operatorname{Re} s = 1$, $s \neq 1$. Taking real parts at $s = 1+i\theta$, where $\operatorname{Re}\int_1^X y^{-1-i\theta}dy = \sin(\theta\log X)/\theta$, gives $\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta) = \lim_X\big[-\sum_{n\le X}\Lambda(n)n^{-1}\cos(\theta\log n) + \sin(\theta\log X)/\theta\big]$. Writing $-\cos = 2\sin^2 - 1$ and using $\sum_{n\le X}\Lambda(n)/n = \log X - \gamma_E + o(1)$, again a consequence of the prime number theorem, gives (9.1.4), since $\int_0^{\log X} 2\sin^2(\theta x/2)dx = \log X - \sin(\theta\log X)/\theta$. At $\theta = 0$ the bracket vanishes and $1 - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{1}{2}) + \gamma_E = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$. □

So the Thorin density comes from the sine squared lemma, but its positivity does not. Since ν_1 is signed, its sine squared image is not termwise positive. Positivity comes from the zero side, Proposition 9.1.2, through $\beta < 1$, and written arithmetically it is the inequality

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(1+it) < \frac{1}{1+t^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+it}{2}\Big),$$

whose right side grows like $\frac{1}{2}\log(t/2\pi)$: the inequality of Hadamard and de la Vallée Poussin that underlies the proof of the prime number theorem [324].

Proposition 9.1.4 (de la Vallée Poussin in Thorin form). *For $\alpha > 1$ let $c_\alpha = -\frac{\zeta'}{\zeta}(\alpha)$ and let $Z(\alpha,y) = \sum_{n\ge 2}\Lambda(n)n^{-\alpha}(1-\cos(y\log n))$ be the prime part of the Thorin density $g(\alpha,y) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+iy)$, so that $Z(\alpha,y) = c_\alpha + g(\alpha,y) - \operatorname{Re}\mathcal{A}(\alpha+iy)$ with $\mathcal{A}$ as in Theorem 16.12.1(ii).*

(i) $0 \le Z(\alpha,y) \le 2c_\alpha$ and $4Z(\alpha,y) + Z(\alpha,2y) \le 8c_\alpha$.

(ii) If $\beta_0 + i\gamma_0$ is a zero, then $\dfrac{4}{\alpha-\beta_0} \le 3c_\alpha + 4\operatorname{Re}\mathcal{A}(\alpha+i\gamma_0) + \operatorname{Re}\mathcal{A}(\alpha+2i\gamma_0)$. *With $\alpha = 1 + \frac{1}{4\log\gamma_0}$ this gives $1-\beta_0 \ge c/\log\gamma_0$ for $\gamma_0 \ge \gamma^*$, with absolute constants $c, \gamma^* > 0$.*

Proof. (i) Termwise, $0 \le 1-\cos \le 2$, and $3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \ge 0$ gives $3c_\alpha + 4(c_\alpha - Z(\alpha,y)) + (c_\alpha - Z(\alpha,2y)) \ge 0$. (ii) For $\alpha > 1$ every term $(\alpha-\beta)/|\alpha+iy-\rho|^2$ of g is positive, so $g(\alpha,\gamma_0) \ge 1/(\alpha-\beta_0)$ and $g(\alpha,2\gamma_0) \ge 0$; insert these into (i) through $Z = c_\alpha + g - \operatorname{Re}\mathcal{A}$. Then $c_\alpha = \frac{1}{\alpha-1} + O(1)$ and $\operatorname{Re}\mathcal{A}(\alpha+i\tau) = \frac{1}{2}\log\tau + O(1)$ for $1 < \alpha \le 2$, $\tau \ge 2$, and with $x = (1-\beta_0)\log\gamma_0$ the inequality reads $4/(x+\frac{1}{4}) \le \frac{29}{2} + o(1)$, so $x \ge \frac{8}{29} - \frac{1}{4} - o(1) = \frac{3}{116} - o(1)$. □

The sine squared lemma is the lower bound $Z \ge 0$ at one frequency. A zero close to $\operatorname{Re} s = 1$ makes $Z(\alpha,\gamma_0)$ large, and what excludes it is the upper bound in (i), which needs the two frequencies y and $2y$. At $\alpha = \frac{3}{2}$, in floating point over $0 < y \le 100$, $\max_y Z/(2c_\alpha) = 0.716$ and $\max_y\{4Z(\alpha,y)+Z(\alpha,2y)\}/(8c_\alpha) = 0.813$.

The same lemma measures how far it alone reaches. For $\alpha > 1$ let $D(\alpha) = \frac{\xi'}{\xi}(\alpha) - \frac{1}{\alpha-1} = \frac{1}{2}\psi(1+\frac{\alpha}{2}) - \frac{1}{2}\log\pi + \frac{\zeta'}{\zeta}(\alpha)$, the logarithmic derivative of ξ without the pole.

Proposition 9.1.5. *The function D is increasing and its root is $\alpha_0 = 5.4962215670\ldots$. For every $\alpha \ge \alpha_0$, $u \mapsto \xi(\alpha)/\xi(\alpha+\sqrt{u})$ is the Laplace transform of a GGC, and the proof uses only the sine squared lemma* (1.2.9)*, with no information on the zeros.*

Proof. With $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ and $a = \alpha - 1$, write $\xi(\alpha)/\xi(\alpha+s) = \frac{a}{a+s}/G(s)$, where $G(s) = \pi^{-s/2}\Gamma(1+\frac{\alpha+s}{2})\zeta(\alpha+s)/(\Gamma(1+\frac{\alpha}{2})\zeta(\alpha))$. Differentiating $\log\Gamma$ twice, and expanding $\log\zeta$ by the Euler product, gives $\log G(s) = D(\alpha)s + \int_0^\infty (e^{-sx} - 1 + sx)\, e^{-\alpha x}\big(\frac{dx}{e^{2x}-1} + \sum_n \Lambda(n)\delta_{\log n}\big)\frac{1}{x}$, a positive measure. Put $s = \sqrt{u}$. By (1.2.9), the compensated integral makes $\exp(-\cdot)$ the transform of a GGC with positive Thorin density. The pole factor is $\exp(-\log(1+\sqrt{u}/a))$, and $\log(1+\sqrt{u}/a) = \frac{a}{\pi}\int_0^\infty \log(1+u/\theta^2)\frac{d\theta}{a^2+\theta^2}$ (both sides vanish at $u=0$ and have the derivative $1/(2\sqrt{u}\,(a+\sqrt{u}))$), so it is a GGC factor as well. Finally $e^{-D\sqrt{u}} = \exp(-\frac{D}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta)$ is a GGC transform exactly when $D \ge 0$. The function D increases because ψ increases and $\frac{\zeta'}{\zeta}(\alpha) = -\sum_n \Lambda(n)n^{-\alpha}$ increases, and $D(\alpha) \to \infty$, so $D \ge 0$ exactly for $\alpha \ge \alpha_0$. □

This is weaker than Theorem 1.2.2, which gives the Thorin condition for every $\alpha \ge 1$, and its interest is only the measure it provides. Below α_0 the factor $e^{|D|\sqrt{u}}$ is not a Laplace transform, so the factorwise use of the lemma stops there. Keeping the pole and Gamma factors together reaches lower in floating point, as follows, and the condition below that holds only because the three parts combine.

Remark 9.1.6. In the variable $y = \sqrt{z}$ the Thorin density of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is $\pi^{-1}g(\alpha,y)$ with $g = P + D + A + Z$, where $P = \frac{\alpha-1}{(\alpha-1)^2+y^2}$ comes from the

pole, $A = \frac{1}{2}\big(\operatorname{Re}\psi(1+\frac{\alpha+iy}{2}) - \psi(1+\frac{\alpha}{2})\big) \ge 0$ from the Gamma factor, and $Z = \sum_{n\ge 2}\Lambda(n)n^{-\alpha}(1-\cos(y\log n)) \ge 0$ from the primes. Dropping Z leaves $m(\alpha) = \min_y\{P+A\} + D(\alpha)$, which is -0.0133 at $\alpha = 3.5$, 0.0638 at 4, 0.1650 at 5 and 0.2012 at 5.5, and changes sign at $\alpha_1 = 3.5747764\ldots$, with the minimum in y at $y = 3.4506$. Where $m(\alpha) \ge 0$ the Thorin condition holds with only $\Lambda(n) \ge 0$ used, but this is computed in floating point (20 digits) at the sampled points and not proved on $[\alpha_1, \alpha_0)$, where m need not be monotone; a sweep in ball arithmetic is open. No bound on the primes uniform in y goes below α_1, since $\inf_y Z(\alpha, y) = 0$ for every $\alpha > 1$: the numbers $\log p$ are independent over $\mathbb{Q}$, so by Kronecker's theorem for any finite set of primes there are arbitrarily large y at which every $\cos(y\log p^k)$ is close to 1, and the tail of Z is uniformly small. The slice version of this obstruction is in Section 12.2.6. For every $\alpha > 1$ a march with step $0.9\,g/L$ and a Lipschitz bound $L(\alpha)$ for g on $[0, Y(\alpha)]$, where $A(\alpha, Y) + D(\alpha) = 0$, checks $g > 0$ from values of ζ on $\operatorname{Re} s = \alpha$ alone; this is not new, since Theorem 1.2.2 gives $\alpha \ge 1$ from the Hadamard product, and it measures the price of avoiding the zeros: $Y(\alpha)$ grows like $e^{2/(\alpha-1)}$, and $\alpha = 1.3$ took 21 931 evaluations (451 s, floating point, no interval arithmetic).

For $F = \mathbb{Q}(\sqrt{-3})$, with $\xi_F(s) = s(s-1)(\sqrt{3}/2\pi)^s\Gamma(s)\zeta_F(s)$ and $\zeta_F = \zeta\,L(\cdot, \chi_{-3})$, the same splitting has the drift $D_F(\alpha) = \psi(1+\alpha) + \log\frac{\sqrt{3}}{2\pi} + \frac{\zeta_F'}{\zeta_F}(\alpha)$, negative on $(1, \alpha_0^F)$ with root $\alpha_0^F = 3.3098807\ldots$, and the merged threshold, in floating point at sampled y as for ζ, is $\alpha_1^F = 2.38426$. The zeros give the Thorin property of $\xi_F(\alpha)/\xi_F(\alpha+\sqrt{u})$ for every $\alpha \ge 1$ (Theorem 21.5.8(a)), and the half plane $\operatorname{Re} s > \frac{7}{8}$ that [226] claims for every Dirichlet L function, which has not been refereed (Section 12.1.2), would give it for every $\alpha > \frac{7}{8}$.

The lemma also acts on the heat trace. Write $p_1 = \int_0^\infty W(t)\,dt = \frac{1}{2}(\log\xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and $F(v) = \xi(\frac{1}{2}+\sqrt{v})/\xi(\frac{1}{2})$. Theorem 5.1.1 gives $\log F(v) = \int_0^\infty (1-e^{-vt})W(t)\frac{dt}{t}$ with $W > 0$, hence $p_1 v - \log F(v) = \int_0^\infty (e^{-vt} - 1 + vt)W(t)\frac{dt}{t}$.

Proposition 9.1.7. *Unconditionally, for $s \ge 0$,*

$$\Psi(s) := \frac{\xi(\frac{1}{2}+s^{1/4})}{\xi(\frac{1}{2})}\,e^{-p_1\sqrt{s}} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{s}{z}\Big)U_*(dz)\Big),$$

where

$$U_*(dz) = \frac{dz}{\pi\sqrt{z}}\int_0^\infty \sin^2\Big(\frac{t\sqrt{z}}{2}\Big)W(t)\,dt \ge 0,$$

so Ψ is the Laplace transform of a GGC with a Thorin measure that is absolutely continuous. Under RH, $\Psi(s) = \prod_{\gamma>0}(1+\sqrt{s}/\gamma^2)\,e^{-\sqrt{s}/\gamma^2}$.

Proof. Apply (1.2.9) with $x = t$ and $s = v$ inside the last integral above, and Tonelli's theorem (the integrand is positive): $p_1 v - \log F(v) = \int_0^\infty \log(1+v^2/z)\,U_*(dz)$. Put $v = \sqrt{s}$. Under RH $F(v) = \prod_{\gamma>0}(1+v/\gamma^2)$. □

REMARK 9.1.8 (The square root is forced). For every real α, $q \mapsto \xi(\alpha)/\xi(\alpha+q)$ is not $\int_{[0,\infty)} e^{-qt}\mu(dt)$ for a positive measure $\mu \neq 0$ on any half plane. For real $q \to \infty$, $\xi(\alpha+q)$ grows like $\exp(\frac{q}{2}\log q)$, so the function decays faster than e^{-qT} for every T, while $\int e^{-qt}\mu(dt) \geq \mu([0,T])e^{-qT}$; hence $\mu([0,T]) = 0$ for all T. (Landau's theorem [340, Chapter II, Theorem 5b] gives it too: the function has no real singularity, since ξ has no real zeros, so the abscissa would be $-\infty$ and the function entire, but it has poles.) No law on $(0,1]$ has Mellin transform $\xi(\alpha)/\xi(\alpha+q)$, and the clock starts at the variable $u = q^2$.

The construction of Proposition 9.1.7 is explicit for a law of Barnes type. Let $Y_{\mathfrak{b}} \geq 0$ be infinitely divisible with Lévy density $\nu(t) = \operatorname{sech}^2(t/4)/(4t)$, so that $\mathbb{E}Y_{\mathfrak{b}} = \frac{1}{4}\int_0^\infty \operatorname{sech}^2(t/4)\,dt = 1$, and let $\mathfrak{b} = e^{-Y_{\mathfrak{b}}}$; written as $e^{-t/2}(1-e^{-t/2})^2/(t(1-e^{-t})^2)$, ν is the Lévy density of a Barnes beta law of Ostrovsky [191].

PROPOSITION 9.1.9. *(i) For* $\operatorname{Re} q > -\frac{1}{2}$,

$$\mathbb{E}\mathfrak{b}^q = \frac{\mathsf{G}(q+1)^2\mathsf{G}(\frac{1}{2})\mathsf{G}(\frac{3}{2})}{\mathsf{G}(q+\frac{1}{2})\mathsf{G}(q+\frac{3}{2})} = \frac{2^{2q^2}(2\pi)^{-q}\mathsf{G}(1+q)^4}{\mathsf{G}(1+2q)},$$

with G *the Barnes function;* $\mathbb{E}\mathfrak{b} = 2/\pi$, *and* $\mathbb{E}\mathfrak{b}^q \sim Cq^{-1/4}$. *Since* $t\nu(t) = \sum_{k\geq 1}(-1)^{k+1}k\,e^{-kt/2}$, $Y_{\mathfrak{b}}$ *is not a GGC.*

(ii) For $s \geq 0$, $\mathbb{E}\mathfrak{b}^{\sqrt{s}} = e^{-\sqrt{s}}/\Psi_{\mathfrak{b}}(s)$, *where* $\Psi_{\mathfrak{b}}(s) = \exp(-\int_0^\infty \log(1+\frac{s}{z})\,\omega_{\mathfrak{b}}(z)\,dz)$ *is the Laplace transform of a GGC with Thorin density*

$$\omega_{\mathfrak{b}}(z) = \frac{1}{2\pi\sqrt{z}} - \frac{1}{\sinh(2\pi\sqrt{z})} > 0.$$

(iii) For $-\frac{1}{2} < \operatorname{Re} w < \frac{1}{2}$, $\int_0^\infty z^{w-1}\omega_{\mathfrak{b}}(z)\,dz = -4(2\pi)^{-2w}(1-2^{-2w})\Gamma(2w)\zeta(2w)$, *whose zeros in that strip are the points* $\rho/2$ *and* $\pi ik/\log 2$, $k \neq 0$.

Proof. (i) By $\log\Gamma(z) = \int_0^\infty[(z-1)e^{-t} - (e^{-t}-e^{-zt})/(1-e^{-t})]\,dt/t$, both $\log\mathbb{E}\mathfrak{b}^{q+1} - \log\mathbb{E}\mathfrak{b}^q = -\int_0^\infty e^{-qt}(1-e^{-t})\nu(t)\,dt$ and $\log\frac{\Gamma(q+1)^2}{\Gamma(q+\frac{1}{2})\Gamma(q+\frac{3}{2})}$ equal $-\int_0^\infty e^{-qt}\frac{e^{-t/2}(1-e^{-t/2})^2}{t(1-e^{-t})}\,dt$. The first expression for $\mathbb{E}\mathfrak{b}^q$ satisfies the same recursion, by $\mathsf{G}(z+1) = \Gamma(z)\mathsf{G}(z)$, and both sides equal 1 at $q = 0$, so their quotient is 1 periodic on $q > -\frac{1}{2}$. Both sides are $q^{-1/4}(C+o(1))$ as $q \to \infty$: the left because $\nu(t) = \frac{1}{4t} + O(1)$ near 0 and decays exponentially, the right by Stirling's formula for G. So the quotient has a limit, is constant, and equals 1; it extends to $\operatorname{Re} q > -\frac{1}{2}$ by analyticity. The second expression follows from the multiplication formula for G at $z = q+\frac{1}{2}$. (ii) The sine squared lemma (1.2.9) with $L(dx) = \nu(x)dx$ gives the Thorin density $\frac{1}{\pi\sqrt{z}}\int_0^\infty \sin^2(\frac{t\sqrt{z}}{2})\frac{\operatorname{sech}^2(t/4)}{4}\,dt$, and $\int_0^\infty \cos(vt)\operatorname{sech}^2(t/4)\,dt = 8\pi v/\sinh(2\pi v)$, from $\int_{\mathbb{R}} e^{i\omega x}\operatorname{sech}^2 x\,dx = \pi\omega/\sinh(\pi\omega/2)$, turns it into $\omega_{\mathfrak{b}}$; positivity is $\sinh x > x$.

(iii) With $x = 2\pi\sqrt{z}$ the integral is $2(2\pi)^{-2w}\int_0^\infty x^{2w-1}(\frac{1}{x} - \frac{1}{\sinh x})\,dx$, and $\int_0^\infty x^{s-1}(\frac{1}{\sinh x} - \frac{1}{x})\,dx = 2(1-2^{-s})\Gamma(s)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, the continuation of $\int_0^\infty x^{s-1}\,dx/\sinh x = 2(1-2^{-s})\Gamma(s)\zeta(s)$ past $s = 1$. In the strip $\Gamma(2w)$ has no zeros, its pole at $w = 0$ cancels against $1 - 2^{-2w}$, and $\zeta(2w)$ vanishes only at $2w = \rho$. □

Numerically (25 digits), $\log \mathbb{E}\mathfrak{b} = -0.4515827052894548647$ from the Lévy integral and from both closed forms, and (ii) holds at $s = 10$ and 100 to 10^{-14}. The GGC $\Psi_{\mathfrak{b}}$ is a ratio of Barnes values and contains no ζ; ζ enters only through the Mellin transform of its Thorin density, in Riemann's kernel $\frac{1}{x} - \frac{1}{\sinh x}$, which places no constraint on the zeros. The clock is not of this type: here $t\nu(t)$ is bounded near 0, while $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ (Section 5.1).

The lemma runs outward only: from a positive Lévy density in the variable v it gives a Thorin measure in $\sqrt{\ }$ of that variable, here $v = \sqrt{s}$ and the argument $s^{1/4}$ of ξ. At the argument $\sqrt{s}$ the Lévy measure of $\log \xi(\frac{1}{2} + v)$ is the signed Weil distribution, so the lemma does not apply, and the Thorin form there is equivalent to RH (Theorem 1.2.2). The continuation of Ψ to $\mathbb{C}\setminus(-\infty, 0]$ through the integral is zero free, which says $\xi(\frac{1}{2} + w) \neq 0$ for $|\arg w| < \frac{\pi}{4}$, and this is empty, since every zero has $w = \rho - \frac{1}{2}$ with $|\arg w - \frac{\pi}{2}| \leq \arctan(1/2\gamma_1)$.

9.1.4 The Lévy form of $\xi(1+s)$

Proposition 9.1.10 (Lévy form). *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s)}{\xi(1)} = b_1 s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_1(dx), \tag{9.1.5}$$

with drift $b_1 = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{3}{2}) + \gamma_E = \sum_\rho \rho^{-1} = 0.0230957089661$ *and the signed Lévy measure*

$$\Pi_1(dx) = \Big(\frac{e^{-3x}}{1-e^{-2x}} - 1\Big)\frac{dx}{x} + \sum_{n\geq 2} \frac{\Lambda(n)}{n\log n}\,\delta_{\log n}(dx),$$

the integral against the last two pieces being taken as $\lim_{X\to\infty}\int_0^{\log X}$.

Proof. Write $\xi(z) = (z-1)\pi^{-z/2}\Gamma(1+\frac{z}{2})\zeta(z)$ and put $z = 1+s$, so that $\xi(1+s) = \pi^{-(1+s)/2}\Gamma(\frac{3}{2}+\frac{s}{2})\,s\zeta(1+s)$ and $\xi(1) = \frac{1}{2}$. Three classical formulas give the three pieces. The factor $\pi^{-s/2}$ contributes $-\frac{s}{2}\log\pi$. Binet's formula [225, §5.9], $\log\Gamma(a+w) - \log\Gamma(a) = w\psi(a) + \int_0^\infty (e^{-wt} - 1 + wt)\frac{e^{-at}}{t(1-e^{-t})}dt$, with $a = \frac{3}{2}$, $w = \frac{s}{2}$ and $t = 2x$, contributes $\frac{s}{2}\psi(\frac{3}{2})$ and the density $e^{-3x}/(x(1-e^{-2x}))$. For the last factor, the Euler product gives $\log\zeta(1+s) = \sum_n \frac{\Lambda(n)}{n\log n}e^{-s\log n}$ for $\operatorname{Re} s > 0$, and Frullani's integral gives $\log s = \int_0^\infty (e^{-x} - e^{-sx})\frac{dx}{x}$. Their sum is $\log(s\zeta(1+s))$, which is analytic at $s = 0$ with value 0 and derivative γ_E, and subtracting the

linear term puts it in the compensated form $\gamma_E s + \int (e^{-sx} - 1 + sx)\, x^{-1}\nu_1(dx)$, where $x^{-1}\nu_1(dx) = \sum \frac{\Lambda(n)}{n\log n}\delta_{\log n} - \frac{dx}{x}$. The compensated integral converges as $X \to \infty$ because the prime atoms and the measure dx/x cancel at first order, by the prime number theorem in the form $\sum_{n\le X} \Lambda(n)/n = \log X - \gamma_E + o(1)$, and at zeroth order, by the logarithm of Mertens' product theorem [107, Theorem 429], $\sum_{p^k \le X} 1/(kp^k) = \log\log X + \gamma_E + o(1)$, which does not need the prime number theorem (the logarithm of the product also contains the terms with $p \le X < p^k$, whose sum is $O(X^{-1/2})$). The drift is $\frac{d}{ds}\log \xi(1+s)|_{s=0} = \frac{\xi'}{\xi}(1)$. □

The variable $V = V_1$ of Section 9.1.1 has $\mathbb{E}e^{sV} = \xi(1+s)/\xi(1)$, so (9.1.5) has the form of a Lévy Khintchine formula for the moment generating function of V, with Π_1, reflected to the negative half line by $x \mapsto -x$, in the place of the Lévy measure. That measure is signed: beyond $x = 0.28120$ its absolutely continuous part is negative and the only positive mass is the prime atoms. So the uniqueness theorem for Lévy triplets does not apply to Π_1 directly. Nor is Π_1 the signed Lévy measure of a quasi infinitely divisible law in the sense of Lindner, Pan and Sato [181], a law whose characteristic function is the quotient of two infinitely divisible ones: that would require the positive and negative parts of Π_1 to be Lévy measures, of finite mass away from the origin, while on $(1,\infty)$ the positive part is the prime atoms and the negative part exceeds $0.94\,dx/x$, both of infinite mass, and they cancel only conditionally, which is why (9.1.5) holds only as a compensated limit. Proposition 11.1.3(3) proves that no other signed measure represents the characteristic function of V, so V is not quasi infinitely divisible, and hence not infinitely divisible. The left panel of Figure 9.2 shows the canonical measure $x\Pi_1(dx)$.

In the variable $\theta = \sqrt{z}$ the sine squared lemma (1.2.9) reads

$$e^{-sx} + sx - 1 = \frac{2x}{\pi}\int_0^\infty \log\Big(1 + \frac{s^2}{\theta^2}\Big)\sin^2\Big(\frac{x\theta}{2}\Big)d\theta, \qquad \operatorname{Re} s > 0,\ x > 0,$$

and the drift converts through $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta = \pi s$. Applying both to (9.1.5) and substituting $s = \sqrt{u}$ gives $\log\varphi_1(u) = -\int_0^\infty \log(1+u/\theta^2)\rho_{1/2}(\theta)\,d\theta$ with

$$v_1(\theta) = b_1 + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)\, x\Pi_1(dx) = b_1 + A(\theta) + Q(\theta), \tag{9.1.6}$$

where the drift b_1 is the stable part, $A(\theta) = \frac{1}{2}\big[\operatorname{Re}\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})\big]$ is the gamma piece and $Q(\theta) = \lim_{X\to\infty}\int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ is the compensated prime piece. This is Proposition 9.1.3 again: the two constant terms $-\frac{1}{2}\log\pi + \gamma_E$ and $\frac{1}{2}\psi(\frac{3}{2})$ combine into b_1, and $\frac{1}{2}\operatorname{Re}\psi(\frac{3+i\theta}{2}) = \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \frac{1}{1+\theta^2}$ accounts for the term $1/(1+\theta^2)$.

The conversion has three readings, each visible in the right panel of Figure 9.2. First, the drift $b_1 = \sum_\rho \rho^{-1}$ becomes the constant Thorin density b_1/π, which is

the Thorin density of the positive stable one half factor $e^{-b_1\sqrt{u}}$. So φ_1 is $e^{-b_1\sqrt{u}}$ times the transform of a GGC with Thorin density $(A+Q)/\pi$ (Proposition 9.1.11). The stable factor plays the role that the stable tilt plays for the floor of Section 6.2.4. At $u = (2k-1)^2$ the factorisation splits each coefficient $1/\zeta(2k)$ of Riesz's function (10.1.5), up to an explicit factor, into a stable value and a GGC value (Section 10.2.3). Second, the gamma factor contributes $A(\theta) \geq 0$, smooth and growing like $\frac{1}{2}\log(\theta/2)$: the average density of zeros. Third, the primes and the pole contribute $Q(\theta)$, which oscillates with both signs, between -1.47 and 1.04 on $[0, 100]$ and carries the peaks at the ordinates.

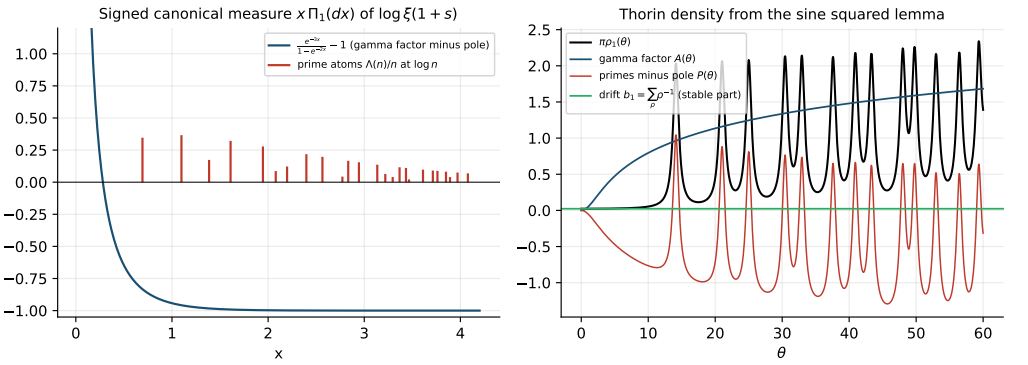


Figure 9.2: Left: the signed canonical measure $x\Pi_1(dx)$ of $\log\xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}} - 1\big)dx$, which is negative beyond $x = 0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n \leq 60$. Right: the bracket $v_1(\theta) = \pi\rho_{1/2}(\theta)$ on $[0, 60]$ assembled by the sine squared lemma from the drift b_1, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes ρ_1 for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$. Drawn by `figs3.py`.

The first reading needs $A + Q \geq 0$, that is, $v_1 \geq b_1$, at every height.

PROPOSITION 9.1.11. *With no hypothesis on the zeros, $v_1(\theta) > v_1(0) = b_1$ for every real $\theta \neq 0$; the proof uses a computation of the zeros below height* 2000 *in ball arithmetic and, above that height, explicit bounds for the number of zeros up to a given height [109, 11]. Consequently $T_1 \stackrel{d}{=} S + T_1^\circ$, where S and T_1° are independent, S is positive stable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and T_1° is a GGC with no drift whose Thorin density in the chart $\theta = \sqrt{z}$ is $(v_1 - b_1)/\pi = (A+Q)/\pi$.*

Proof. By (9.1.2), $v_1(\theta) = \pi\sum_\rho P_{1-\beta}(\theta - \gamma)$ is a sum of positive terms, with $\pi P_a(x) = a/(a^2 + x^2)$, and v_1 is even, because ξ is real on the real axis. We treat three ranges of $\theta \geq 0$.

On $[0, 13.5]$ we use convexity. Every zero has $|\gamma| \geq \gamma_1 = 14.134725$: the certified zeros of the next step, computed in order from the first, show that none

has $0 < |\gamma| < \gamma_1$, and none is real, because $\zeta(\sigma) < 0$ for $0 < \sigma < 1$. Since $\pi P_a''(x) = 2a(3x^2 - a^2)/(a^2 + x^2)^3$, the kernel P_a is strictly convex where $|x| > a/\sqrt{3}$, and for $|\theta| \le 13.5$ every zero has $|\theta - \gamma| \ge \gamma_1 - 13.5 > 0.63 > 1/\sqrt{3} > (1-\beta)/\sqrt{3}$. So v_1 is strictly convex on $[-13.5, 13.5]$, and, being even, it has there its unique minimum at $\theta = 0$ and increases on $[0, 13.5]$. By (1.2.2), differentiated twice,

$$v_1''(0) = -\Big(\frac{\xi'}{\xi}\Big)''(1) = \sum_\rho \frac{2(1-\beta)\big(3\gamma^2 - (1-\beta)^2\big)}{\big((1-\beta)^2 + \gamma^2\big)^3} > 0.$$

On $[13.5, 2000]$ we use the nearest zero. The script `v1min.py` of the folder `riesz_program/code` computes the first 1800 zeros in ball arithmetic with Arb [130], which isolates them in order by Gram's law and Rosser's rule (Section 2.2.1), each applied only in the range where it is known to hold [47, Section 3], so that none is missed. They reach height 2304.3, all are simple and on the critical line, and consecutive ordinates differ by at most $\gamma_2 - \gamma_1 < 6.8874$. Every $\theta \in [13.5, 2000]$ therefore lies within 3.4437 of an ordinate, half the largest gap, and a zero on the critical line at distance x contributes $\pi P_{1/2}(x) = \frac12/(\frac14 + x^2)$, which exceeds b_1 for $x < 4.6259$. All other terms being positive,

$$v_1(\theta) \ge \frac{1/2}{1/4 + 3.4437^2} > 0.041, \qquad 13.5 \le \theta \le 2000.$$

For $\theta \ge 2000$ we count zeros. Write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). Hasanalizade, Shen and Wong [109, (1.7) and Corollary 1.4] give $|S(T)| \le 2.5167$ for $0 \le T \le X_0 = 30\,610\,046\,000$, from Platt's database of the zeros, and $|S(T)| \le 0.1095 \log T + 0.2042 \log\log T + 3.0305$ for every $T \ge e$, and the smooth part $\vartheta(T)/\pi + 1$ of $N(T)$ differs from $\frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac78$ by at most $\frac{1}{50T}$ for $T \ge \frac57$ [109, (2.7)]. So

$$\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78\Big| \le E(T)$$
$$= \begin{cases} 2.5167 + \frac{1}{50T}, & e \le T \le X_0, \\ 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}, & T > X_0, \end{cases}$$

and $E(T) < 0.3 \log T$ for $T > X_0$, since this holds at X_0 and $0.3 - 0.1095 - 0.2042/\log X_0 > 0$. The bound $0.112\log T + 0.278\log\log T + 2.510 + 0.2/T$ of Trudgian [329], which is often quoted, has an incomplete proof [23, 109]. Put $H = 15$. The window $(\theta - H, \theta + H]$ holds $N(\theta + H) - N(\theta - H)$ zeros, counted with multiplicity, and since the main term has derivative $\frac{1}{2\pi}\log\frac{T}{2\pi}$, their number is at least

$$n(\theta) = \frac{H}{\pi}\log\frac{\theta - H}{2\pi} - E(\theta - H) - E(\theta + H).$$

While $\theta + H \le X_0$, n increases, because E decreases there, and $n(2000) > 22.44$; for $\theta + H > X_0$, both values of E are below $0.3\log(\theta + H)$, and $n(\theta) > \frac{H}{\pi}\log\frac{\theta - H}{2\pi} -$

$0.6\log(\theta+H) > 90.$ So for every $\theta \ge 2000$ the window holds at least 23 zeros. Since $\xi(1-\bar{s}) = \overline{\xi(s)}$, the map $\rho \mapsto 1-\bar{\rho}$ sends each zero to a zero of the same multiplicity; it keeps the ordinate and exchanges β and $1-\beta$, so it permutes the zeros of the window, and the sum of $1-\beta$ over them is half their number. Each of them has $0 < 1-\beta < 1$ and $|\theta-\gamma| \le H$, so its term $(1-\beta)/((1-\beta)^2+(\theta-\gamma)^2)$ in $v_1(\theta)$ exceeds $(1-\beta)/(1+H^2)$, and all other terms are positive. Hence

$$v_1(\theta) > \frac{23}{2(1+H^2)} = \frac{23}{452} > 0.05, \qquad \theta \ge 2000,$$

wherever the zeros lie. Backlund's classical constants 0.137, 0.443 and 4.35, valid for $T \ge 200$ [11], give in the same way the increasing bound $E_B(T) = 0.137\log T + 0.443\log\log T + 4.35 + 0.2/T$ in place of E and at least $\frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 2E_B(\theta+H)$ zeros in the window, a number that increases with θ and exceeds 14.89 at $\theta = 2000$; the window then holds at least 15 zeros, and the bound becomes $15/452 > 0.033$.

It remains to split φ_1. Since $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta = \sqrt{u}$, Proposition 9.1.2 gives

$$\varphi_1(u) = e^{-b_1\sqrt{u}} \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big(v_1(\theta)-b_1\big)\,d\theta\Big\}, \qquad u \ge 0,$$

where the integral is finite because $0 \le v_1 - b_1 \le v_1$ and $\int_0^\infty \log(1+u/\theta^2)\,v_1(\theta)\,d\theta = -\pi\log\varphi_1(u)$ is finite by Theorem 1.2.2 at $\alpha = 1$. The image U_1° of $\pi^{-1}(v_1-b_1)\,d\theta$ under $\theta \mapsto \theta^2 = z$ is a positive measure with $\int \log(1+u/z)\,U_1^\circ(dz) < \infty$, which is equivalent to the integrability condition in (1.3.1), so the second factor is the Laplace transform of a GGC T_1° with no drift. Hence $T_1 \stackrel{d}{=} S + T_1^\circ$, and $v_1 - b_1 = A + Q$ by (9.1.6). □

The stable summand carries the infinite mean and the whole tail: $P(S>t) \sim b_1/\sqrt{\pi t}$ is the tail (9.1.3) of T_1, while T_1° has finite mean and infinite variance. The Laplace exponent of T_1° is $\log(\xi(1+\sqrt{u})/\xi(1)) - b_1\sqrt{u} = \frac{1}{2}(\log\xi)''(1)\,u + O(u^{3/2})$ as $u \downarrow 0$, so that

$$\mathbb{E}T_1^\circ = \frac{1}{2}(\log\xi)''(1) = -\frac{1}{2}\sum_\rho \rho^{-2} = 0.0230772,$$

and its Thorin density $(v_1-b_1)/\pi$ vanishes at the origin to exactly second order, since v_1 is even and $v_1''(0) > 0$; for instance $v_1(1) - b_1 = 1.12\times10^{-4}$ and $v_1(5) - b_1 = 3.31\times10^{-3}$. Near $z = 0$ the Thorin measure U_1° therefore has a density of exact order $\sqrt{z}$, and the variance $\int z^{-2}\,U_1^\circ(dz)$ is infinite.

For every $\alpha \ge 1$ the same construction gives the Lévy form of $\log(\xi(\alpha+s)/\xi(\alpha))$

with drift $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and

$$x\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)dx + \sum_n \frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)x\Pi_\alpha(dx),$$

which for $\alpha > 1$ is (3.1.1). For $\alpha > 1$ every piece converges on its own, and the pole contribution is the finite negative term $-\theta^2/((\alpha-1)((\alpha-1)^2+\theta^2))$. Expanding $1/(e^{2x}-1) = \sum_{k\ge1} e^{-2kx}$ in the archimedean piece of (3.1.1), and writing $q_c(\theta) = \int_0^\infty (1-\cos\theta x)e^{-cx}\,dx = \theta^2/(c(c^2+\theta^2))$, gives

$$v_\alpha(\theta) - b_\alpha = \sum_{k\ge1} q_{\alpha+2k}(\theta) - q_{\alpha-1}(\theta) + \sum_{n\ge2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big), \qquad \alpha > 1. \tag{9.1.7}$$

Only the pole term is negative. Near the origin $q_c(\theta) \sim \theta^2/c^3$, and the pole term outweighs the gamma terms when $\alpha < 7.789$, where $\frac18\zeta(3,\frac{\alpha}{2}+1) = (\alpha-1)^{-3}$; below that basepoint the arithmetic form needs the primes to keep $v_\alpha - b_\alpha$ positive near the origin.

Proposition 9.1.12. *With no hypothesis on the zeros, $v_\alpha(\theta) > v_\alpha(0) = b_\alpha$ for every $\alpha \ge 1$ and every real $\theta \ne 0$; the proof uses the zeros below height* 2000 *of Proposition 9.1.11, the explicit bounds of [109] for the number of zeros above that height, and a computation in ball arithmetic. Consequently, for every $\alpha \ge 1$, $T_\alpha \overset{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, which is the law of the first passage time of Brownian motion to the level $b_\alpha/\sqrt{2}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\mathbb{E}T_\alpha^\circ = \frac12(\log\xi)''(\alpha)$.*

Proof. At $\alpha = 1$ this is Proposition 9.1.11, so let $\alpha > 1$. Since v_α is even, it suffices to take $\theta > 0$. A zero $\rho = \beta + i\gamma$ contributes $a/(a^2+(\theta-\gamma)^2)$ to $v_\alpha(\theta)$, with $a = \alpha - \beta \in (\alpha-1,\alpha)$, and we put $\theta_0(\alpha) = \gamma_1 - \alpha/\sqrt{3}$. Four steps cover all $\alpha > 1$.

Near the origin, for $\alpha \le 9$, we use convexity. If $0 < \theta \le \theta_0(\alpha)$, every zero has $|\theta-\gamma| \ge \gamma_1 - \theta \ge \alpha/\sqrt{3} > a/\sqrt{3}$, so, as in the proof of Proposition 9.1.11, v_α is strictly convex on $[-\theta_0(\alpha),\theta_0(\alpha)]$, and $v_\alpha(\theta) > v_\alpha(0)$ there. Since $\theta_0(9) = 8.94 > 0$, it remains to treat $\theta > \theta_0(\alpha)$ for $\alpha \le 9$, and every $\theta > 0$ for $\alpha > 9$.

For $1 < \alpha \le 3$ we use the zeros. Here $\theta_0(\alpha) \ge \theta_0(3) = 12.40$ and $\gamma_1 - 12.40 < 3.4437$, so by the computation in the proof of Proposition 9.1.11 every $\theta \in [\theta_0(\alpha), 2000]$ lies within 3.4437 of the ordinate of a zero on the critical line, whose term exceeds $r(\alpha) = (\alpha-\frac12)/((\alpha-\frac12)^2 + 3.4437^2)$. For $\theta \ge 2000$ the window $(\theta-15,\theta+15]$ holds at least 20 zeros, the sum of a over them is $\alpha - \frac12$ times their number, and each term exceeds $a/(\alpha^2+225)$, so that $v_\alpha(\theta) > w(\alpha) = 20(\alpha-\frac12)/(\alpha^2+225)$. The functions r and w increase on $[1,3]$, and so does b_α,

whose derivative $\sum_\rho(\gamma^2-a^2)/(a^2+\gamma^2)^2$ is positive for $\alpha<\gamma_1$. It therefore suffices that $r(\alpha_k)>b_{\alpha_{k+1}}$ and $w(\alpha_k)>b_{\alpha_{k+1}}$ at the points $\alpha_k=1+k/256$, and in ball arithmetic the smallest margins are 0.0180 and 0.0209.

For $3\le\alpha\le7$ we use the primes. Adding and subtracting $q_\alpha(\theta)$ in (9.1.7), dropping the prime terms and using $q_c(\theta)<1/c$ gives

$$\begin{aligned} v_\alpha(\theta)-b_\alpha > L(\alpha,\theta) &= \sum_{k\ge0} q_{\alpha+2k}(\theta) - \frac{1}{\alpha} - \frac{1}{\alpha-1} \\ &= \frac12\Big[\operatorname{Re}\psi\Big(\frac{\alpha+i\theta}{2}\Big)-\psi\Big(\frac{\alpha}{2}\Big)\Big] - \frac{1}{\alpha} - \frac{1}{\alpha-1}, \end{aligned}$$

which increases in θ. Since $|\psi'(z)|\le\psi'(\operatorname{Re}z)$ and $\psi'(x)\le1/x+1/x^2$ for $x>0$, $|\partial L/\partial\alpha|\le M(\alpha)=1/\alpha+3/\alpha^2+1/(\alpha-1)^2$, which decreases in α. It therefore suffices that $L(\alpha_k,\theta_0(\alpha_{k+1}))>M(\alpha_k)/256$ at the points $\alpha_k=3+k/256$ of $[3,7]$; the smallest margin is 0.0228, at $\alpha=7$, and the same check succeeds up to $\alpha=7.707$.

For $\alpha\ge7$ the gamma terms suffice. Since $q_c(\theta)$ decreases in c, $\sum_{k\ge1}q_{\alpha+2k}(\theta)\ge\frac12\int_{\alpha+2}^\infty q_c(\theta)\,dc=\frac14\log(1+\theta^2/(\alpha+2)^2)$, so with $A=\alpha+2$, $B=\alpha-1$ and $s=\theta^2$,

$$v_\alpha(\theta)-b_\alpha\ge\Phi_\alpha(s)=\frac14\log\Big(1+\frac{s}{A^2}\Big)-\frac{s}{B(B^2+s)}.$$

Here $\Phi_\alpha(0)=0$, and $\Phi'_\alpha(s)=1/(4(A^2+s))-B/(B^2+s)^2$ is nonnegative exactly when $(B^2+s)^2\ge4B(A^2+s)$, the two sides differing by a function that increases in s when $B\ge2$. For $\alpha\ge9$ they are in this order already at $s=0$, because $B^3-4A^2=(\alpha-1)^3-4(\alpha+2)^2$ is 28 at $\alpha=9$ and increases beyond, so $\Phi_\alpha(s)>0$ for every $s>0$, and no zero enters. For $7\le\alpha\le9$ and $\theta>\theta_0(\alpha)\ge\theta_0(9)$, we have $s>s_9=\theta_0(9)^2=79.90$, $6\le B\le8$ and $A\le11$, so $(B^2+s_9)^2\ge(36+s_9)^2>32(121+s_9)\ge4B(A^2+s_9)$; hence Φ_α increases on $[s_9,\infty)$, where $\Phi_\alpha(s_9)\ge\frac14\log(1+s_9/121)-s_9/(6(36+s_9))>0.0118$.

The checks of the last three steps are carried out in ball arithmetic with Arb [130] (`repro/valpha_min.py`). The factorisation follows from the inequality as in the proof of Proposition 9.1.11, Theorem 1.2.2 giving $\int_0^\infty\log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta<\infty$, and the mean is the coefficient of u in $\log(\xi(\alpha+\sqrt{u})/\xi(\alpha))-b_\alpha\sqrt{u}=\frac12(\log\xi)''(\alpha)\,u+O(u^{3/2})$. □

As at the basepoint one, $v_\alpha-b_\alpha$ starts as $c(\alpha)\theta^2$, with $c(\alpha)=-\frac12(\xi'/\xi)''(\alpha)>0$, so at every basepoint above one the Thorin density of T_α° vanishes to second order at the origin, T_α° has finite mean and infinite variance, and S_α carries the whole tail $P(T_\alpha>t)\sim b_\alpha/\sqrt{\pi t}$.

For $\alpha<1$ the Euler product no longer converges near $\operatorname{Re}s=\alpha$, and the prime atoms $\Lambda(n)n^{-\alpha}$ can be compensated only if the primes are well enough distributed. So the route from the Lévy form through the sine squared lemma computes the Thorin measure unconditionally on $\alpha\ge1$.

9.1.5 Below the basepoint one

The construction of Section 9.1.4 can be written down for any basepoint $\alpha = 1-\varepsilon$ with $0 < \varepsilon < \frac{1}{2}$, and the question is whether it converges. Formally nothing changes except the pole term, which now grows: with $b_\alpha = \frac{\xi'}{\xi}(\alpha)$,

$$x\Pi_\alpha(dx) = \frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx \;+\; \Big(\sum_n \frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx) - e^{\varepsilon x}\,dx\Big),$$

and the converted Thorin density would be π^{-1} times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty}\Big[\sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big) - \int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)e^{\varepsilon x}dx\Big], \quad (9.1.8)$$

with $A_\alpha(\theta) = \frac{1}{2}\big[\operatorname{Re}\psi(\frac{\alpha+2+i\theta}{2}) - \psi(\frac{\alpha+2}{2})\big]$. The prime piece and the pole piece now each grow like $X^\varepsilon/\varepsilon$, and whether their difference converges is a question about how well $\psi(y)$ is approximated by y.

Proposition 9.1.13. *Let $\Theta = \sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros and let $\frac{1}{2} < \alpha < 1$.*

1. *If $\Theta < \alpha$, the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx} - 1 + sx)\Pi_\alpha(dx)$ holds for $\operatorname{Re} s > 0$, the compensated limit converging, and the sine squared lemma converts it into (9.1.8), which then equals $v_\alpha(\theta)$ and is positive. The error of the compensated sum with prime cutoff X is $O(X^{\Theta-\alpha+\delta})$ for every $\delta > 0$.*

2. *Conversely, if the Lévy form holds for $\operatorname{Re} s > 0$, then $\xi(\alpha+s)$ has no zeros with $\operatorname{Re} s > 0$, so $\Theta \le \alpha$.*

Proof. If $\Theta < \alpha$, the explicit formula gives $\psi(y) - y = O(y^{\Theta+\delta})$ for every $\delta > 0$ [324]. Then the Stieltjes integral $\int_1^\infty y^{-w}\,d(\psi(y)-y)$, which equals $-\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ for $\operatorname{Re} w > 1$, converges after one integration by parts for $\operatorname{Re} w > \Theta$ and provides the analytic continuation there, with tail beyond X of order $X^{\Theta-\operatorname{Re} w+\delta}$. Taking $w = \alpha + s$ with $\operatorname{Re} s \ge 0$ and integrating twice in s from $s = 0$ gives the compensated Lévy form, the gamma and polynomial factors being handled exactly as in Proposition 9.1.10. For fixed θ the sine squared lemma may be applied under the limit, because the compensated measure integrated against $2\sin^2(\theta x/2)$ is a combination of the same convergent Stieltjes integrals at $w = \alpha$ and $w = \alpha \pm i\theta$. Positivity follows from the Poisson form of Proposition 1.2.3, as in Proposition 9.1.2, with widths $\alpha - \beta > 0$. Conversely, suppose the Lévy form holds for $\operatorname{Re} s > 0$. Write ν_α for the bracket in $x\Pi_\alpha$, so that the compensated part of the right side is the limit of $\int_0^{\log X}(e^{-sx} - 1 + sx)\,x^{-1}\nu_\alpha(dx)$; its part over $[0,1]$ is an entire function of s. For real $s > 1-\alpha$ the integral $\int_1^\infty e^{-sx}x^{-1}\nu_\alpha(dx)$ converges absolutely, so the convergence of the compensated limit at two such values of s gives that of $\int_1^{\log X}\nu_\alpha(dx)$ and of $\int_1^{\log X} x^{-1}\nu_\alpha(dx)$. The function $y \mapsto \int_1^y \nu_\alpha(dx)$ is

therefore bounded on $[1,\infty)$, and one integration by parts against it shows that $\int_1^{\log X} e^{-sx}x^{-1}\nu_\alpha(dx)$ converges uniformly on compact subsets of $\operatorname{Re} s > 0$. So the compensated limit converges locally uniformly, the right side of the Lévy form is analytic in $\operatorname{Re} s > 0$, hence so is $\log \xi(\alpha+s)$, and $\xi(\alpha+s)$ cannot vanish there. □

So the extension to $\xi(1-\varepsilon+s)$ exists exactly when ζ has no zeros in the half plane $\operatorname{Re} s > 1-\varepsilon$, up to the boundary case $\Theta = 1-\varepsilon$. At the basepoint one the convergence of the compensated sum was the prime number theorem; at the basepoint $1-\varepsilon$ it is the estimate $\psi(y) - y = O(y^{1-\varepsilon+\delta})$, which is the zero free half plane $\operatorname{Re} s > 1-\varepsilon$; and as $\varepsilon \uparrow \frac{1}{2}$ it is $\psi(y) - y = O(y^{1/2+\delta})$, which is RH. The Lévy form route is an arithmetic restatement of step 7 of the path in Table 12.1, not a way around it.

`below.py` evaluates (9.1.8) with primes up to 10^7 at $\theta = \gamma_1$ and compares it with $v_\alpha(\gamma_1)$ computed directly. On the range of the computation the primes behave as RH predicts, $\psi(y) - y = O(y^{1/2}\log^2 y)$, so the errors should decay like $X^{1/2-\alpha}$, and they do (Figure 9.3, left): the decay is fast at $\alpha = 0.9$, slow at $\alpha = 0.6$ and absent at the centre. At $\alpha = \frac{1}{2}$ and $\theta = \gamma_1$ the compensated sum grows like $\log X$ (Figure 9.3, right), which is the atom δ_{γ_1} of the RH measure forming as the cutoff increases; away from the zeros it oscillates without converging. The exponent in $X^{\Theta-\alpha}$ thus measures the largest real part of a zero, but only up to the computable range of X, so it can never certify Θ.

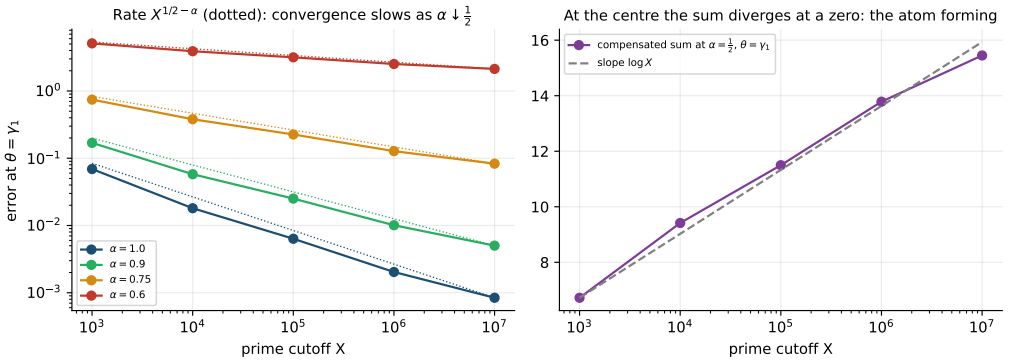


Figure 9.3: Left: absolute error of the compensated formula (9.1.8) at $\theta = \gamma_1$ against the prime cutoff X for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming. Drawn by `figs4.py`.

Writing the tail of the compensated sum beyond X as a Stieltjes integral against $d(\psi(y) - y)$ and inserting the explicit formula $\psi(y) = y - \sum_\rho y^\rho/\rho -$

$\log 2\pi - \frac{1}{2}\log(1-y^{-2})$ [71, Chapter 17] gives, for $\Theta < \alpha$ and X not a prime power,

$$\begin{aligned}&(\text{formula (9.1.8) at cutoff } X) - v_\alpha(\theta)\\ &\quad= \sum_\rho \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)}\Big] + O(X^{-1-\alpha}), \quad (9.1.9)\end{aligned}$$

the sum taken over conjugate pairs. Every term has size $X^{\beta-\alpha}$, which makes the rate $X^{\Theta-\alpha}$ visible term by term, and at $\theta = \gamma$ the third term has denominator $\beta-\alpha$, so under RH it approaches $X^{1/2-\alpha}/(1-2\alpha)$ and at the centre it becomes the $\log X$ growth of Figure 9.3. Table 9.3, computed by `errformula.py`, compares the observed error with (9.1.9) truncated to the 1700 zeros below height 2198. The explicit formula accounts for the error to four or five decimal places for $\alpha \geq 0.75$ and to three at $\alpha = 0.6$, and the hybrid of primes up to X plus the first 1700 zeros reduces the error of the primes alone by factors between about 700 and 11000.

α	X	observed error	predicted by (9.1.9)	error after correction
1	10^5	-6.352×10^{-3}	-6.349×10^{-3}	-2.9×10^{-6}
1	10^7	-8.419×10^{-4}	-8.431×10^{-4}	1.2×10^{-6}
0.9	10^7	-5.014×10^{-3}	-5.020×10^{-3}	6.1×10^{-6}
0.75	10^7	-8.297×10^{-2}	-8.304×10^{-2}	6.8×10^{-5}
0.6	10^5	-3.1668	-3.1665	-2.9×10^{-4}
0.6	10^7	-2.1285	-2.1292	7.7×10^{-4}

Table 9.3: The error of the compensated sum (9.1.8) at $\theta = \gamma_1$ with primes up to X, its prediction from the explicit formula (9.1.9) over the first 1700 zeros, and what remains after subtracting the prediction.

This also sharpens the logic of Proposition 9.1.13. A zero with $\beta > \alpha$ would contribute a term of size $X^{\beta-\alpha}$ that grows with the cutoff, so the compensated sum at the basepoint α would diverge, and the size and phase of that growth would locate the zero. The hybrid formula is a zero detector with the same structure as Turing's method (Section 2.2.1): primes up to X and zeros up to height T must balance to within the tail of both.

If instead of compensating one continues the pole contribution analytically, its value at $\alpha = 1-\varepsilon$ is $\operatorname{Re}\frac{1}{-\varepsilon+i\theta} + \frac{1}{\varepsilon} = \frac{\theta^2}{\varepsilon(\varepsilon^2+\theta^2)}$, which is positive, since the pole now lies to the right of the basepoint. The single negative term of the decomposition (3.1.1) above the basepoint one has become positive. Below the line $\operatorname{Re} s = 1$, therefore, the negativity that RH must control is carried by the primes, not by the pole.

9.1.6 The minimum at the origin below the basepoint one

Propositions 9.1.11 and 9.1.12 put the minimum of v_α at the origin for every $\alpha \geq 1$, with no hypothesis on the zeros. Below the basepoint one the margin at the origin

vanishes linearly, $b_\alpha \approx (\alpha - \frac{1}{2})(\log\xi)''(\frac{1}{2})$, and Table 9.6 found the minimum at the origin on a window by computation. This subsection proves it on the range of heights where the zeros are known to lie on the critical line, up to $\frac{1+\sqrt{2}}{2}$ below its top, for every basepoint $\alpha \in (\frac{1}{2}, 7.35]$, and shows that RH is equivalent to it at every height for every $\alpha \in (\frac{1}{2}, 1)$. The tool is a comparison across vertical lines, from the first author's note on the Dirichlet beta function [253]. We state it for $f = \xi$, for $f = \xi(\cdot, \chi_4)$ of Section 8.5.7 and for $f = \xi(\cdot, \chi_{12})$ of Section 8.5.1, and write $v_\alpha(\theta) = \mathrm{Re}(f'/f)(\alpha + i\theta)$, $b_\alpha = v_\alpha(0) = (f'/f)(\alpha)$ and $V_f = (\log f)''(\frac{1}{2})$.

Each f is entire of order one, real on the real axis, with $f(s) = f(1-s)$ and no real zeros. Its zeros therefore come in groups: a zero $\frac{1}{2} + i\gamma$ on the critical line, or a pair $\frac{1}{2} \pm \delta + i\gamma$ with $0 < \delta < \frac{1}{2}$. At resolution $\varepsilon = \alpha - \frac{1}{2}$ and offset $u = \theta - \gamma$, a zero on the line contributes $\varepsilon/(\varepsilon^2 + u^2)$ to $v_\alpha(\theta)$, and a pair contributes

$$k_\delta(\varepsilon, u) = \frac{\varepsilon - \delta}{(\varepsilon - \delta)^2 + u^2} + \frac{\varepsilon + \delta}{(\varepsilon + \delta)^2 + u^2} = \frac{2\varepsilon(w + c)}{(w + c)^2 + \kappa^2}, \tag{9.1.10}$$

where $w = \varepsilon^2$, $c = u^2 - \delta^2$ and $\kappa = 2\delta|u|$, since the numerator of the sum is $2\varepsilon(\varepsilon^2 + u^2 - \delta^2)$ and its denominator is $(\varepsilon^2 + \delta^2 + u^2)^2 - 4\varepsilon^2\delta^2 = (w+c)^2 + \kappa^2$. The sum over the groups converges absolutely, since $\sum_\rho |\rho|^{-2} < \infty$. So $v_\alpha(\theta)/(\alpha - \frac{1}{2})$ is the sum over the groups of the ratios $1/(w + u^2)$ for the zeros on the line and

$$r_\delta(w, u) = \frac{k_\delta(\varepsilon, u)}{\varepsilon} = 2\,\mathrm{Re}\,\frac{1}{w + c + i\kappa}$$

for the pairs, as functions of $w = (\alpha - \frac{1}{2})^2$. The ratio of a zero on the line decreases in w; that of a pair need not.

Lemma 9.1.14 (Comparison). *Let $0 < \delta < \frac{1}{2}$ and let u be real. As a function of $w > 0$, the ratio $r_\delta(w, u)$*

(a) is nonincreasing on $(0, \infty)$ if $|u| \geq (1 + \sqrt{2})\delta$;

(b) is nonincreasing on $[2\delta^2, \infty)$ for every u, in particular on $w \geq \frac{1}{2}$, that is for $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, whatever δ;

(c) if $0 < |u| < (1 + \sqrt{2})\delta$, increases exactly on the interval $\max\big(0, 2\delta^2 - (|u| + \delta)^2\big) < w < 2\delta^2 - (|u| - \delta)^2$, which begins at 0 if and only if $|u| \geq (\sqrt{2} - 1)\delta$; at $u = 0$ it equals $2/(w - \delta^2)$, which has a pole at $w = \delta^2$.

In particular the constant $1 + \sqrt{2}$ in (a) cannot be lowered.

Proof. For $u \neq 0$, $\partial_w r_\delta = -2\,\mathrm{Re}(w+c+i\kappa)^{-2} = -2\big((w+c)^2 - \kappa^2\big)/\big((w+c)^2 + \kappa^2\big)^2$, which is positive exactly when $-\kappa < w + c < \kappa$, that is when $2\delta^2 - (|u| + \delta)^2 < w < 2\delta^2 - (|u| - \delta)^2$, since $-\kappa - c = 2\delta^2 - (|u| + \delta)^2$ and $\kappa - c = 2\delta^2 - (|u| - \delta)^2$. The right end is at most $2\delta^2$, which gives (b) for $u \neq 0$; it is positive exactly when

$\big||u|-\delta\big| < \sqrt{2}\delta$, that is when $|u| < (1+\sqrt{2})\delta$, which gives (a) and the first part of (c); and the left end is at most 0 exactly when $|u|+\delta \ge \sqrt{2}\delta$. At $u=0$, $c=-\delta^2$ and $\kappa = 0$, and $2/(w-\delta^2)$ decreases on $w > \delta^2$, which completes (b). Finally $2\delta^2 < \frac{1}{2}$ because $\delta < \frac{1}{2}$. □

THEOREM 9.1.15 (Comparison across vertical lines). *Let $\frac{1}{2} < \alpha < \alpha'$ and let θ be real. Suppose that every zero $\frac{1}{2}+\delta+i\gamma$ of f with $\delta > (\alpha-\frac{1}{2})/\sqrt{2}$ satisfies $|\theta-\gamma| \ge (1+\sqrt{2})\delta$. Then*

$$\frac{v_\alpha(\theta)}{\alpha-\frac{1}{2}} \;\ge\; \frac{v_{\alpha'}(\theta)}{\alpha'-\frac{1}{2}}.$$

The hypothesis holds at every θ under RH for ξ, or GRH for χ_4 or χ_{12}; at every θ when $\alpha \ge \frac{1}{2}+\frac{1}{\sqrt{2}}$, with no hypothesis; and at every θ with $|\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, where H_0 is a height below which every zero of f is known to lie on the critical line: $H_0 = 3\times 10^{12}$ for ξ [247], and $H_0 = 10^8/q$ for $\xi(\cdot,\chi_q)$, that is 2.5×10^7 for χ_4 and $10^8/12 > 8.33\times 10^6$ for χ_{12} [245].

Proof. A zero on the line has the ratio $1/(w+u^2)$, which decreases in w. A pair with $\delta \le (\alpha-\frac{1}{2})/\sqrt{2}$ has $2\delta^2 \le (\alpha-\frac{1}{2})^2$, so its ratio is nonincreasing on $[(\alpha-\frac{1}{2})^2, (\alpha'-\frac{1}{2})^2]$ by Lemma 9.1.14(b), and a pair with $|\theta-\gamma| \ge (1+\sqrt{2})\delta$ by Lemma 9.1.14(a). Summing over the groups, which converges absolutely, gives the inequality. Under RH there are no pairs. If $\alpha \ge \frac{1}{2}+\frac{1}{\sqrt{2}}$, every pair has $\delta < \frac{1}{2} \le (\alpha-\frac{1}{2})/\sqrt{2}$. If $|\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, every zero off the line has $|\gamma| > H_0$, so that $|\theta-\gamma| > \frac{1+\sqrt{2}}{2} > (1+\sqrt{2})\delta$. □

COROLLARY 9.1.16 (Explicit lower bound). *For $\sigma > 1$ and real θ, $v_\sigma(\theta) \ge A_f^\infty(\sigma,\theta) - S_f(\sigma)$, where*

$$\begin{aligned}
A_\xi^\infty(\sigma,\theta) &= -\tfrac{1}{2}\log\pi + \tfrac{1}{2}\operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),\\
A_{\chi_4}^\infty(\sigma,\theta) &= \tfrac{1}{2}\log\tfrac{4}{\pi} + \tfrac{1}{2}\operatorname{Re}\psi\big(\tfrac{\sigma+1+i\theta}{2}\big),\\
A_{\chi_{12}}^\infty(\sigma,\theta) &= \tfrac{1}{2}\log\tfrac{12}{\pi} + \tfrac{1}{2}\operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),\\
S_f(\sigma) &= \sum_{(n,q)=1} \frac{\Lambda(n)}{n^\sigma},
\end{aligned}$$

with $q = 1$, 4 and 12 for the three functions, so that $S_\xi = -\zeta'/\zeta$, $S_{\chi_4}(\sigma) = S_\xi(\sigma) - \log 2/(2^\sigma-1)$ and $S_{\chi_{12}}(\sigma) = S_{\chi_4}(\sigma) - \log 3/(3^\sigma-1)$; and $A_f^\infty(\sigma,\theta)$ increases in $|\theta|$. Hence, under the hypothesis of Theorem 9.1.15 with $\alpha' > 1$,

$$\begin{aligned}
v_\alpha(\theta) \;&\ge\; \frac{\alpha-\frac{1}{2}}{\alpha'-\frac{1}{2}}\Big(A_f^\infty(\alpha',\theta) - S_f(\alpha')\Big)\\
&=\; \frac{\alpha-\frac{1}{2}}{2\alpha'-1}\log|\theta| + O(1), \qquad |\theta|\to\infty.
\end{aligned}$$

Proof. For $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s),$$

where $\operatorname{Re}(\frac{1}{s} + \frac{1}{s-1}) \geq 0$ and $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\theta) = -\sum_n \Lambda(n) n^{-\sigma}\cos(\theta\log n) \geq -S_\xi(\sigma)$; for the two L functions the bound is that of Proposition 8.5.25. The monotonicity in $|\theta|$ follows from the series $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\geq 0} y^2/\big((x+k)((x+k)^2+y^2)\big)$, and $\operatorname{Re}\psi(x+iy) = \log|y| + O(1)$ for fixed $x > 0$. Then apply Theorem 9.1.15 between α and α'. □

Remark 9.1.17 (A bound of order $\log\theta$ outside a sparse set). For fixed $\alpha > \frac{1}{2}$ the hypothesis of Theorem 9.1.15 fails only at heights within $(1+\sqrt{2})\delta < \frac{1+\sqrt{2}}{2}$ of the ordinate of a zero $\frac{1}{2} + \delta + i\gamma$ with $\delta > (\alpha - \frac{1}{2})/\sqrt{2}$. For $f = \xi$ the number of such zeros with $0 < \gamma \leq T$ is $N(\sigma_0, T)$ with $\sigma_0 = \frac{1}{2} + (\alpha - \frac{1}{2})/\sqrt{2} > \frac{1}{2}$, which is $O(T^{3(1-\sigma_0)/(2-\sigma_0)}\log^5 T) = o(T)$ by Ingham's estimate [124] (Section 11.4.1). So Corollary 9.1.16 with $\alpha' = \alpha + 1$ gives $v_\alpha(\theta)/\pi \geq c\log\theta$, for every $c < (\alpha - \frac{1}{2})/(\pi(2\alpha+1))$ and all large θ, outside a set of heights of density zero: this is the lower bound of the first of the four questions of Section 11.4.7, for the density v_α/π itself. More directly, by the proof of Corollary 11.4.7, $v_\alpha(\theta) \leq 0$ only within $\sqrt{\delta^2 - (\alpha - \frac{1}{2})^2} < \frac{1}{2}$ of the ordinate of a zero $\frac{1}{2} + \delta + i\gamma$ with $\frac{1}{2} + \delta > \alpha$, so the set of $\theta \in [0, T]$ with $v_\alpha(\theta) \leq 0$ has measure at most $N(\alpha, T + \frac{1}{2}) \leq T^{e(\alpha)+o(1)}$, with $e(\alpha) < 1$ of Theorem 11.4.2. The exceptional set, however, contains the window of every zero to the right of the line $\operatorname{Re} s = \alpha$, since such a zero has $\delta > \alpha - \frac{1}{2} > (\alpha - \frac{1}{2})/\sqrt{2}$. So the bound gives no information where Lemma 11.4.9 would need it, and it does not by itself confine a failure of positivity to the thin strips of that question.

Theorem 9.1.18 (Minimum at the origin). *Let V_f, γ_1, H_0 and $\alpha_{\max}$ be as in Table 9.4.*

(i) For every $\alpha > \frac{1}{2}$, $0 < b_\alpha \leq (\alpha - \frac{1}{2})V_f$; the ratio $b_\alpha/(\alpha - \frac{1}{2})$ decreases in α and tends to V_f as $\alpha \downarrow \frac{1}{2}$.

(ii) For $\frac{1}{2} < \alpha \leq \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for $0 < |\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$.

(iii) For $\frac{1}{2} + \frac{1}{\sqrt{2}} \leq \alpha \leq \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for every $\theta \neq 0$.

(iv) For $f = \xi(\cdot, \chi_4)$ and $f = \xi(\cdot, \chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$ and every $\theta \neq 0$.

(v) For $f = \xi(\cdot, \chi_4)$ and $f = \xi(\cdot, \chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \geq 1$ and every $\theta \neq 0$.

(vi) RH, respectively GRH for χ_4 or for χ_{12}, holds if and only if $v_\alpha(\theta) > b_\alpha$ for every $\alpha \in (\frac12, 1)$ and every $\theta \neq 0$.

Proof. (i) The function $f(\frac12 + s) = \int_{\mathbb{R}} K(u) e^{su}\, du$ is the two sided Laplace transform of a positive even kernel K: the Pólya kernel Φ for ξ (Section 1.2), the kernel K_4 of Proposition 8.5.21, and the eta kernel of Section 8.5.1. So $\log f(\frac12 + s)$ is, up to a constant, the cumulant generating function of a nondegenerate law; it is even and strictly convex in real s, and $b_\alpha = (\log f)'(\alpha) = \int_{1/2}^{\alpha} (\log f)''(r)\, dr > 0$. At $\theta = 0$ the offset of a group with ordinate γ has $|u| = |\gamma| \geq \gamma_1 > \frac{1+\sqrt2}{2} > (1+\sqrt2)\delta$, so by Lemma 9.1.14(a) the ratio of every group is nonincreasing in w, strictly for the zeros on the line, and $b_\alpha/(\alpha - \frac12)$, their sum, decreases in α. As $w \downarrow 0$ the ratios increase to γ^{-2} for a zero on the line and to $2\,\mathrm{Re}(|\gamma| + i\delta)^{-2}$ for a pair, and by monotone convergence the sum tends to the sum of these limits, which is $-\sum_\rho (\frac12 - \rho)^{-2} = (\log f)''(\frac12) = V_f$.

(ii) Put $\alpha' = \alpha_{\max}$ and let t_1 be its value at $\alpha_{\max}$ in Table 9.4, where $A_f^\infty(\alpha', t_1) - S_f(\alpha') > (\alpha' - \frac12) V_f$ is certified in ball arithmetic; since A_f^∞ increases in $|\theta|$, the inequality holds for $|\theta| \geq t_1$. For $t_1 \leq |\theta| \leq H_0 - \frac{1+\sqrt2}{2}$, Corollary 9.1.16 and (i) give $v_\alpha(\theta) > (\alpha - \frac12) V_f \geq b_\alpha$ (for $\alpha = \alpha'$ the comparison is an identity). For $|\theta| \leq t_1$ we show that v_α is strictly convex on $[-t_1, t_1]$; as it is even, it then exceeds $v_\alpha(0) = b_\alpha$ at $0 < |\theta| \leq t_1$. The term $\varphi(\theta - \gamma)$, $\varphi(x) = a/(a^2 + x^2)$ with $a = \alpha - \frac12$, of a zero on the line is strictly convex where $|\theta - \gamma| > a/\sqrt3$, since $\varphi''(x) = 2a(3x^2 - a^2)/(a^2 + x^2)^3$. Below H_0 all zeros lie on the line and have $|\gamma| \geq \gamma_1$, and for $|\theta| \leq t_1$, $|\theta - \gamma| \geq \gamma_1 - t_1 > (\alpha' - \frac12)/\sqrt3 \geq a/\sqrt3$ by the convexity margin of the table. Every zero above H_0 has $|\theta - \gamma| > H_0 - t_1 > 10^6$. For a pair there, the second derivative in θ of $k_\delta(a, \theta - \gamma)$ is $g(a + \delta, u) + g(a - \delta, u)$ with $g(c, u) = 2c(3u^2 - c^2)/(c^2 + u^2)^3$, the second derivative of $c/(c^2 + u^2)$ in u; since $\partial_c g(c, u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2 + u^2)^4 > 0$ for $|c| < (\sqrt2 - 1)|u|$, and $|a \pm \delta| < \alpha_{\max} < (\sqrt2 - 1)10^6$, the odd function $g(\cdot, u)$ increases on an interval containing $\pm(a \pm \delta)$, so that $g(a + \delta, u) > g(\delta - a, u) = -g(a - \delta, u)$. The sum of the second derivatives converges uniformly on $[-t_1, t_1]$, so $v_\alpha'' > 0$ there.

(iii) For $\alpha \geq \frac12 + \frac{1}{\sqrt2}$ the hypothesis of Theorem 9.1.15 holds at every θ, so the bound of (ii) holds for every $|\theta| \geq t_1$, and the convexity argument for $|\theta| \leq t_1$ is unchanged.

(iv) It remains to treat $\alpha > \alpha_{\max}$, and there we use the primes. For χ_4, $v_\alpha(\theta) - b_\alpha = G_\alpha(\theta) + \varpi_\alpha(\theta)$ by (8.5.9), with $G_\alpha(\theta) = \frac12[\mathrm{Re}\,\psi(\frac{\alpha+1+i\theta}{2}) - \psi(\frac{\alpha+1}{2})]$; for χ_{12} the same holds with $G_\alpha(\theta) = \frac12[\mathrm{Re}\,\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$ and the weights $\chi_{12}(n)\Lambda(n)$ (Section 8.5.3). Dropping the positive weights and bounding $1 - \cos x$ by 2 and

by $x^2/2$ gives $\varpi_\alpha(\theta) \geq -R_\alpha(\theta)$ with $R_\alpha(\theta) \leq \min\big(2S^-(\alpha), \frac{1}{2}\theta^2 M_2(\alpha)\big)$, where

$$S^-(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)}{n^\alpha} = \frac{1}{2}\Big(S_f(\alpha) + \frac{L'}{L}(\alpha)\Big),$$
$$M_2(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)\log^2 n}{n^\alpha} = \frac{1}{2}\Big(S_f''(\alpha) + \Big(\frac{L'}{L}\Big)''(\alpha)\Big).$$

The function $G_\alpha(\theta)$ is a sum of terms $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$, so it increases in $|\theta|$ and decreases in α, and $G_\alpha(\theta)/\theta^2$ decreases in $|\theta|$; and S^- and M_2 decrease in α. So on a cell $s_0 \leq \alpha \leq s_1$ it suffices that $G_{s_1}(T) > \max\big(2S^-(s_0), \frac{1}{2}T^2 M_2(s_0)\big)$ for one $T > 0$: then $G_\alpha(\theta) \geq G_{s_1}(T) > 2S^-(\alpha)$ for $|\theta| \geq T$, and $G_\alpha(\theta) \geq \theta^2 G_{s_1}(T)/T^2 > \frac{1}{2}\theta^2 M_2(\alpha)$ for $0 < |\theta| \leq T$. The script `dbeta_minimum.py` checks this in ball arithmetic on cells of width $\frac{1}{32}$ covering $[\sigma_2, 20]$, with $T \in \{0.25, 0.5, \ldots, 4\}$, where $\sigma_2 = 3.71875$ for χ_4 and 2.8125 for χ_{12}; the cells just below σ_2 fail. For $\alpha \geq 20$ take $T = 2$: since q_c decreases in c, $G_\alpha(2) \geq \frac{1}{2}\int_{\alpha+1}^\infty q_c(2)\,dc = \frac{1}{4}\log(1 + 4/(\alpha+1)^2) \geq 1/((\alpha+1)^2+4)$, and since every n with $\chi(n) = -1$ is at least 3, $S^-(\alpha)$ and $M_2(\alpha)$ are at most $3^{-(\alpha-20)}$ times their values at 20. With $K = \max(2S^-(20), 2M_2(20))$ it therefore suffices that $K((\alpha+1)^2+4)3^{-(\alpha-20)} < 1$ for $\alpha \geq 20$; the left side decreases in α, and at $\alpha = 20$ it is 3.4×10^{-7} for χ_4 and 3.9×10^{-11} for χ_{12}. As $\alpha_{\max} = 4.70 > 3.71875$ for χ_4 and $3.10 > 2.8125$ for χ_{12}, (iii) and this argument cover every $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$.

(v) By (ii) and (iv) it remains to treat $1 \leq \alpha < \frac{1}{2} + \frac{1}{\sqrt{2}}$ and $|\theta| > H_0 - \frac{1+\sqrt{2}}{2}$, and by symmetry $\theta > 0$. Let n be the number of zeros with $|\gamma - \theta| \leq h$, counted with multiplicity, where $h = 15$ for χ_4 and $h = 10$ for χ_{12}. Every zero has $0 < \beta < 1 \leq \alpha$, so every term of $v_\alpha(\theta)$ is positive, and a zero in the window contributes $a/(a^2+(\theta-\gamma)^2) \geq a/(\alpha^2+h^2)$, with $a = \alpha - \beta \in (0, \alpha)$; the window contains both members of each pair $\frac{1}{2} \pm \delta + i\gamma$, whose values of a add up to $2(\alpha - \frac{1}{2})$, so the values of a over the window add up to $n(\alpha - \frac{1}{2})$. Hence $v_\alpha(\theta) > n(\alpha - \frac{1}{2})/(\alpha^2+h^2)$, which is at least $(\alpha - \frac{1}{2})V_f \geq b_\alpha$ as soon as $n \geq V_f(\alpha^2+h^2)$. Since $\alpha^2 < (\frac{1}{2} + \frac{1}{\sqrt{2}})^2 = 1.4571\ldots$, it suffices that $n > 35.335$ for χ_4 and $n > 33.778$ for χ_{12}. By the explicit counting formula of Bennett, Martin, O'Bryant and Rechnitzer [23, Theorem 1.1], for a primitive character χ of conductor $q > 1$, $T \geq \frac{5}{7}$ and $\ell = \log(q(T+2)/(2\pi)) > 1.567$, the number $N(T, \chi)$ of zeros with $|\gamma| \leq T$ satisfies

$$\Big|N(T,\chi) - \frac{T}{\pi}\log\frac{qT}{2\pi e} + \frac{\chi(-1)}{4}\Big| \leq E(T) = 0.22737\,\ell + 2\log(1+\ell) - 0.5.$$

As f has no real zeros and its zeros are symmetric about the real axis, $N(\theta + h, \chi) - N(\theta - h, \chi)$ is at most twice n, whatever the convention for zeros at the ends of the window, and since E increases in T,

$$n \;\geq\; \frac{1}{2\pi}\int_{\theta-h}^{\theta+h} \log\frac{qt}{2\pi}\,dt - E(\theta+h) \;\geq\; \frac{h}{\pi}\log\frac{q(\theta-h)}{2\pi} - E(\theta+h).$$

The right side increases in θ, its derivative being at least $\big(h/\pi - 0.22737 - 2/(1+\ell)\big)/(\theta+h+2)$ with $\ell = \ell(\theta+h) > 16$, which is positive. At $\theta = H_0 - \frac{1+\sqrt{2}}{2}$, where $qH_0 = 10^8$ for both characters, $\log(q(\theta-h)/(2\pi)) = 16.58280\ldots$ and $E(\theta+h) = 9.00428\ldots$, so that $n \geq 79.177 - 9.005 > 70$ for χ_4 and $n \geq 52.785 - 9.005 > 43$ for χ_{12}, well above the 36 and 34 zeros needed. The script `dbeta_window.py` checks these inequalities and the hypotheses of the formula in ball arithmetic.

(vi) Under RH, or GRH, the hypothesis of Theorem 9.1.15 holds at every θ, and the proof of (ii) applies with $H_0 = \infty$ for every $\alpha < 1 < \alpha_{\max}$. Conversely, the condition and (i) give $v_\alpha \geq b_\alpha > 0$ on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$, and $v_\alpha > 0$ for $\alpha \geq 1$ (Theorem 1.2.2, Proposition 8.5.24, and Proposition 11.3.1 with Section 8.5.3); so $\operatorname{Re}(f'/f)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$, which is RH, or GRH, by Lagarias's criterion [160]. □

	ξ	$\xi(\cdot,\chi_4)$	$\xi(\cdot,\chi_{12})$
$V_f = (\log f)''(\frac{1}{2})$	0.04620998623	0.1560296499	0.3329236082
first ordinate γ_1	14.1347251	6.0209489	3.8046276
verified height H_0	3×10^{12}	2.5×10^7	$10^8/12$
comparison line α' of [253]	4.75	3.15	2.30
t_1 at α'	9.236	2.439	1.685
convexity margin $\gamma_1 - t_1 - (\alpha' - \frac{1}{2})/\sqrt{3}$	2.4450	2.0520	1.0804
α' minimising t_1, and t_1 there	4.770, 9.2353	3.320, 2.4176	2.395, 1.6717
$\alpha_{\max}$	7.35	4.70	3.10
t_1 at $\alpha_{\max}$	10.124	3.574	2.245
convexity margin at $\alpha_{\max}$	0.0559	0.0221	0.0585
cells of the proof of (iv) pass from $\sigma_2 =$		3.71875	2.8125

Table 9.4: Constants of Theorem 9.1.18. For a comparison line $\alpha' > 1$, t_1 is the root of $A_f^\infty(\alpha', t) = S_f(\alpha') + (\alpha' - \frac{1}{2})V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of t_1 is located on a grid of step 0.005; $\alpha_{\max}$ is the largest α' in steps of 0.05 keeping the margin above 0.02. For ξ the range $\alpha > \alpha_{\max}$ is covered by Proposition 9.1.12. Computed by `dbeta_minimum.py`; the ordinates of $L(s,\chi_4)$ and $L(s,\chi_{12})$ are certified by `dbeta_zeros.py`.

For the two L functions, part (v) gives the factorisation of Proposition 9.1.12 at every basepoint $\alpha \geq 1$: for the clock T_α with Laplace transform $f(\alpha)/f(\alpha+\sqrt{u})$, $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\frac{1}{2}(\log f)''(\alpha)$. The proof is that of Proposition 9.1.12: the Thorin representation $\log(f(\alpha+\sqrt{u})/f(\alpha)) = \frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\, v_\alpha(\theta)\, d\theta$ holds at every $\alpha \geq 1$ (Proposition 8.5.24 and its proof, which applies to $\xi(\cdot,\chi_{12})$), and $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\, d\theta = \sqrt{u}$.

Part (vi) sharpens Lagarias's criterion, which asks only for $v_\alpha > 0$: RH is equivalent to the Thorin density on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$ being smallest at the origin and only there, and by (ii) this holds with no hypothesis for $|\theta| \le 3 \times 10^{12} - 1.21$. In the language of Proposition 9.1.12, b_α/π is the density of the positive stable part of the clock, so (vi) says that RH holds exactly when, for every $\alpha > \frac{1}{2}$, T_α splits as $S_\alpha + T_\alpha^\circ$ with T_α° a GGC with no drift; under RH every T_α is a GGC (Theorem 1.2.2), and the proof of the factorisation is that of Proposition 9.1.12. Part (i) also shows that the stable part is present at every basepoint: in the chart $z = \theta^2$ the density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ behaves like $b_\alpha/(2\pi\sqrt{z})$ as $z \downarrow 0$, with $b_\alpha > 0$ for every $\alpha > \frac{1}{2}$. For ξ, part (iii) gives a second proof of Proposition 9.1.12 for $\frac{1}{2} + \frac{1}{\sqrt{2}} \le \alpha \le 7.35$ that uses of the zeros only the first ordinate and the verification height, not the zeros below height 2000; for $1 \le \alpha < \frac{1}{2} + \frac{1}{\sqrt{2}}$ the comparison reaches only the verified height, and there and for $\alpha > 7.35$ Proposition 9.1.12 is still needed. For the two L functions no zero beyond the first is used: the zeros enter only through the verification heights of Platt and the counting formula of Bennett, Martin, O'Bryant and Rechnitzer.

Remark 9.1.19 (The margin at the origin). For ξ, parts (i) and (ii) explain the margin of Section 3.1. For every $\varepsilon \in (0, 6.85]$ the minimum of $\rho_\varepsilon = v_{1/2+\varepsilon}/\pi$ over $0 \le \theta \le 3 \times 10^{12} - 1.21$ is $\rho_\varepsilon(0) = b_{1/2+\varepsilon}/\pi$, attained only at the origin, and $\rho_\varepsilon(0) \le \varepsilon V_\xi/\pi$, with $V_\xi/\pi = 2\pi^{-1}\sum_k a_k^{-1} = 0.0147091$, the constant of the computed minimum $0.0147\,\varepsilon$ (Section 3.1 and Table 9.6). By (i), $\rho_\varepsilon(0)/\varepsilon$ increases to V_ξ/π as $\varepsilon \downarrow 0$.

Remark 9.1.20 (The nearest zero under GRH). Under RH, or GRH, there is a shorter route to the minimum at the origin on a finite range of heights, which the first author's note gives first for χ_4 [253]. Let $0 < a \le \frac{1}{2}$ and assume that every zero of f lies on the critical line. Then $v_{1/2+a}(\theta) = \sum_\gamma a/(a^2 + (\theta-\gamma)^2)$ over all ordinates, and $b_{1/2+a} = \sum_\gamma a/(a^2+\gamma^2) \le aV_f$, since $V_f = \sum_\gamma \gamma^{-2}$. If every gap between consecutive positive ordinates up to γ_N is less than $2\sqrt{1/V_f - a^2}$, then $v_{1/2+a}(\theta) > b_{1/2+a}$ for $0 < |\theta| \le \gamma_N$. Indeed, on $|\theta| < \gamma_1 - a/\sqrt{3}$ the sum is even and strictly convex; within $a/\sqrt{3}$ of an ordinate the nearest term alone is at least $3/(4a) > aV_f$, as $a^2 \le \frac{1}{4} < 3/(4V_f)$ for each of the three functions; and otherwise θ lies between two consecutive ordinates at distance at most half their gap g from one of them, whose term is at least $a/(a^2 + g^2/4)$, which exceeds aV_f exactly when $g < 2\sqrt{1/V_f - a^2}$. For χ_4 the threshold is 4.963 at $a = \frac{1}{2}$ and tends to $2/\sqrt{V_{\chi_4}} = 5.063$ as $a \downarrow 0$, and the largest gap below height 100 is the first, 4.2228 (Section 8.5.7.1), so under GRH the conclusion holds for $|\theta| \le 98.75$. Under GRH the gaps tend to zero, since the argument of $L(\frac{1}{2} + iT, \chi_4)$ is $O(\log T/\log\log T)$ by Littlewood's argument for ζ [324, Theorem 14.13], so the gap condition can fail only on a bounded range of heights. Theorem 9.1.18 needs no gaps: with no hypothesis its part (ii) gives the conclusion for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, and under RH, or GRH, its part (vi) gives it at every height.

Remark 9.1.21 (Complete monotonicity in the resolution). Under RH, or GRH, the ratio of Theorem 9.1.15 is $v_{1/2+\sqrt{w}}(\theta)/\sqrt{w} = \sum_\gamma 1/(w+(\theta-\gamma)^2)$, the sum over all ordinates, a Stieltjes function of $w > 0$ and in particular completely monotone. With no hypothesis the ratio is then a Stieltjes function at no height, and it fails to be completely monotone near every zero off the line. The ratio of a pair is $r_\delta(w,u) = 2\int_0^\infty e^{-wx}e^{-cx}\cos(\kappa x)\,dx$ for $w > -c$, so the representing function of the whole ratio is $\sum e^{-(\theta-\gamma')^2x}$, over the zeros γ' on the line, plus $2e^{-cx}\cos\kappa x$ for each pair. If one pair has $\kappa \neq 0$ and $c = u^2 - \delta^2$ smaller than $(\theta-\gamma')^2$ for every other zero and than the value of c of every other pair, its term dominates the representing function as $x \to \infty$ and changes sign, so by Bernstein's theorem the ratio is not completely monotone at that height; at $u = 0$ the pair contributes the pole $2/(w-\delta^2)$. So, in answer to a question of the first author's note [253], no constant plays for complete monotonicity the role that $1+\sqrt{2}$ plays in Lemma 9.1.14: whether the ratio is completely monotone at a height near a zero off the line depends on the neighbouring zeros, not only on the distance to that zero in units of δ.

9.1.7 The heat trace and the Chebyshev error

The compensated sums of Sections 9.1.3 and 9.1.5 converge only conditionally. At $\alpha = 1$ an explicit bound for Chebyshev's error $\psi(y) - y$ bounds the error of a truncation at X only by an amount that grows linearly with the height θ. Averaging over the height against a Gaussian changes this. For $\alpha \geq 1$ the heat trace of T_α is the Laplace transform $k_\alpha(t) = \pi^{-1}\int_0^\infty e^{-t\theta^2}v_\alpha(\theta)\,d\theta$ of its Thorin measure in the variable $z = \theta^2$. The average turns the kernel $2\sin^2(x\theta/2)$ of the sine squared lemma (1.2.9) into a kernel that is positive, increasing and bounded, and one integration by parts then turns an explicit bound for $\psi(y) - y$ into an error bound that holds for all t at once.

Lemma 9.1.22. *For $x \geq 0$ and $t > 0$,*

$$\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}\,2\sin^2\Big(\frac{x\theta}{2}\Big)\,d\theta = \frac{g_t(x)}{2\sqrt{\pi t}}, \qquad g_t(x) = 1 - e^{-x^2/4t}.$$

Proof. Write $2\sin^2(x\theta/2) = 1-\cos x\theta$ and use $\int_0^\infty e^{-t\theta^2}\cos x\theta\,d\theta = \frac{1}{2}\sqrt{\pi/t}\,e^{-x^2/4t}$. □

The kernel g_t increases from 0 to 1 on $[0,\infty)$, it satisfies $g_t(x) \leq \min(1, x^2/4t)$, and for fixed x it decreases in t.

Proposition 9.1.23 (Heat kernel form). *For $\alpha \geq 1$ and $t > 0$,*

$$2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \bar{A}_\alpha(t) + \int_{[1,\infty)} g_t(\log y)\,y^{-\alpha}\,d\big(\psi(y) - y\big), \tag{9.1.11}$$

where the gamma piece

$$\bar{A}_\alpha(t) = \int_0^\infty g_t(x)\,\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx = \sum_{k\ge1}\Big(\frac{1}{c_k} - \sqrt{\pi t}\,e^{c_k^2 t}\operatorname{erfc}\big(c_k\sqrt{t}\big)\Big), \qquad c_k = \alpha+2k,$$

is positive and the Stieltjes integral converges absolutely after one integration by parts. For $\alpha > 1$ that integral is $\sum_n \Lambda(n)n^{-\alpha}g_t(\log n) - L_\alpha(t)$, with the Lebesgue piece

$$L_\alpha(t) = \int_0^\infty g_t(x)\,e^{-(\alpha-1)x}\,dx = \frac{1}{\alpha-1} - \sqrt{\pi t}\,e^{(\alpha-1)^2 t}\operatorname{erfc}\big((\alpha-1)\sqrt{t}\big).$$

Proof. Let $\alpha > 1$ and write $v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2(x\theta/2)\,x\Pi_\alpha(dx)$, as at the end of Section 9.1.4. Near the origin $|x\Pi_\alpha|$ has density about $1/(2x)$, and away from it the gamma and pole densities and the prime atoms have finite mass, so $\int_0^\infty \min(1,x^2)\,|x\Pi_\alpha|(dx) < \infty$. By Lemma 9.1.22 the integral of $e^{-t\theta^2}2\sin^2(x\theta/2)$ against $d\theta\,|x\Pi_\alpha|(dx)$ is $\frac{\sqrt{\pi}}{2\sqrt{t}}\int g_t\,d|x\Pi_\alpha| < \infty$, so the order of integration may be exchanged, and the lemma gives $2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \int_0^\infty g_t(x)\,x\Pi_\alpha(dx)$. The gamma density gives $\bar{A}_\alpha$, whose series follows from $e^{-(\alpha+2)x}/(1-e^{-2x}) = \sum_{k\ge1}e^{-c_k x}$ and $\int_0^\infty g_t(x)e^{-cx}\,dx = \frac{1}{c} - \sqrt{\pi t}\,e^{c^2t}\operatorname{erfc}(c\sqrt{t})$ for $c>0$, the case $c=\alpha-1$ being L_α; the prime atoms and the pole density give the Stieltjes integral, written in the variable $y = e^x$.

At $\alpha = 1$ the same computation applies to $v_1 = b_1 + A + Q$ of (9.1.6), in which $Q(\theta) = \lim_{X\to\infty} Q_X(\theta)$ with $Q_X(\theta) = \int_{[1,X]} 2\sin^2(\frac12\theta\log y)\,y^{-1}d(\psi(y)-y)$. Integration by parts gives

$$Q_X(\theta) = w(X)\big(\psi(X)-X\big) - \int_1^X \big(\psi(y)-y\big)\,w'(y)\,dy, \qquad w(y) = \frac{2\sin^2(\frac12\theta\log y)}{y},$$

and since $|w'(y)| \le (2+|\theta|)\,y^{-2}$, while $\int_1^\infty |\psi(y)-y|\,y^{-2}dy < \infty$ by the prime number theorem with the de la Vallée Poussin error term, $|Q_X(\theta)| \le C(1+|\theta|)$ for all X. By dominated convergence the limit in X may therefore be taken outside the integral over θ, and for each X Lemma 9.1.22, applied to the finitely many atoms and to Lebesgue measure on $[0,\log X]$, gives $\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}Q_X(\theta)\,d\theta = (2\sqrt{\pi t})^{-1}\int_{[1,X]} g_t(\log y)\,y^{-1}d(\psi(y)-y)$; the gamma piece A gives $\bar{A}_1$ as before. For every $\alpha \ge 1$ the same integration by parts, now with $|\frac{d}{dy}(g_t(\log y)\,y^{-\alpha})| \le (g_t'(\log y)+\alpha)\,y^{-\alpha-1}$ and g_t' bounded, shows that the Stieltjes integral converges absolutely after one integration by parts, the boundary terms vanishing because $g_t(0) = 0$ and $y^{-\alpha}(\psi(y)-y) \to 0$. □

For $\alpha > 1$ every prime power enters (9.1.11) with the positive weight $\Lambda(n)n^{-\alpha}g_t(\log n)$, and the other terms are explicit. Keeping the primes up to X

and dropping the rest therefore gives a lower bound with no error term,

$$2\sqrt{\pi t}\,k_\alpha(t) \ge b_\alpha + \bar{A}_\alpha(t) + \sum_{n\le X} \frac{\Lambda(n)}{n^\alpha}\, g_t(\log n) - L_\alpha(t), \qquad \alpha > 1, \tag{9.1.12}$$

which uses only $\Lambda(n) \ge 0$, the positivity of the coefficients of $-\zeta'/\zeta$. By Table 9.5, the right side is positive for every t once $\alpha \ge 3.191$ with no prime atoms, the primes then entering only through the drift b_α, and once $\alpha \ge 1.533$, 1.367 and 1.284 with the primes up to 10^3, 10^5 and 10^7; the thresholds, to four decimals, are 3.1900, 1.5320, 1.3667 and 1.2840. The same argument pointwise in θ, applied to the sine squared form of v_α with no prime atoms, needs $\alpha \ge 3.575$ (the threshold is 3.5748). At α equal to the four decimal value less 5×10^{-5} the right side is negative at one t, or one θ. From the smaller of the four decimal value plus 5×10^{-5} and the bound in the table up to $\alpha = 10$, cells cover every α and every $t > 0$, or $\theta \ge 0$, and the right side is positive on each cell: $\bar{A}_\alpha(t)$, the prime sum and $L_\alpha(t)$ are monotone in α and in t, so each is bounded at a corner of the cell or, near the minimum over t, by a mean value form in t, and b_α by its mean value form in α. No finite X reaches $\alpha = 1$: for fixed t, $L_\alpha(t)$ increases to $\int_0^\infty g_t(x)\,dx = \infty$ as $\alpha \downarrow 1$, while the other terms of (9.1.12) stay bounded.

At the basepoint one the primes and the pole must be kept together. Splitting the Stieltjes integral at X, and writing $\int_0^{\log X} g_t(x)\,dx = \log X - \sqrt{\pi t}\,\mathrm{erf}\big(\log X/(2\sqrt{t})\big)$,

$$2\sqrt{\pi t}\,k_1(t) = b_1 + \bar{A}_1(t) + \sum_{n\le X} \frac{\Lambda(n)}{n}\, g_t(\log n) - \Big(\log X - \sqrt{\pi t}\,\mathrm{erf}\,\frac{\log X}{2\sqrt{t}}\Big) + R_X(t), \tag{9.1.13}$$

with $R_X(t) = \int_{(X,\infty)} g_t(\log y)\, y^{-1} d(\psi(y) - y)$. Suppose that $|\psi(y) - y| \le \omega(\log y)\, y$ for $y \ge X$. Integrating by parts, with $|\frac{d}{dy}(g_t(\log y)/y)| \le (g_t'(\log y) + g_t(\log y))\, y^{-2}$, $g_t \le 1$ and $\int_{\log X}^\infty g_t' = 1 - g_t(\log X)$,

$$\begin{aligned} |R_X(t)| &\le g_t(\log X)\,\frac{|\psi(X) - X|}{X} + \big(1 - g_t(\log X)\big) \sup_{x \ge \log X} \omega(x) \\ &\quad + \int_{\log X}^\infty \omega(x)\,dx \le \eta_X \end{aligned} \tag{9.1.14}$$

for every $t > 0$, where $\eta_X = \sup_{x\ge\log X} \omega(x) + \int_{\log X}^\infty \omega(x)\,dx$, the last step because $|\psi(X) - X| \le \omega(\log X) X$. Explicit forms of the prime number theorem supply ω: $|\psi(y) - y| \le 0.94\sqrt{y}$ for $11 < y \le 10^{19}$ [51]; $|\psi(y) - y| < 1.93378\times10^{-8}\, y$ for $y \ge 10^{19}$ and $|\psi(y) - y| < 6.95\times10^{-9}\, y$ for $y \ge e^{46}$ [48]; and $|\psi(y) - y| \le 8.86\, y(\log y)^{1.514} e^{-0.8288\sqrt{\log y}}$ for $\log y \ge 3000$ [131]. The constants alone would not do, since ω must be integrable, and the decaying bound of Johnston and Yang closes the integral. With $X = 10^7$ they give $\eta_X < 9.2\times10^{-4}$.

PROPOSITION 9.1.24. *For every $t > 0$, $2\sqrt{\pi t}\, k_1(t) > 0.021$. The proof uses the primes up to 10^7, the bounds for $\psi(y) - y$ quoted above and a computation in ball arithmetic.*

Proof. By (9.1.13) and (9.1.14) it suffices that the right side of (9.1.13), with $R_X(t)$ replaced by $-\eta_X$, exceed 0.021 for every t. The three terms of it that depend on t, the gamma piece, the prime sum and the bracket, all decrease in t, because g_t does, so on a cell $[t_0, t_1]$ the value obtained by evaluating the first two at t_1 and the bracket at t_0 is a lower bound. The script `heat_primes.py` evaluates it in ball arithmetic with Arb [130], on 65137 cells that cover $[10^{-10}, 10^8]$ and are refined until each bound exceeds 0.021. It bounds $\bar{A}_1$ from below by the terms of its series with $k \le 1000$ and $c_k\sqrt{t} \le 25$, all terms being positive, and the prime sum from below by grouping the prime powers above 10^4 into blocks of width 5×10^{-4} in $\log n$, each weighted by g_t at its left end. For $t \le 10^{-10}$ the same monotonicity, with the bracket at most $\log X$, gives the lower bound 2.86, and for $t \ge 10^8$ dropping $\bar{A}_1$ and the prime sum gives 0.0221. □

From the zero side, Proposition 9.1.11 gives $v_1 \ge b_1$, hence $2\sqrt{\pi t}\, k_1(t) > b_1$ for every t, where $b_1 = 0.0230957$. So Proposition 9.1.24 is weaker than what the zeros give. Its interest is its input, the primes up to 10^7 and published bounds for $\psi(y) - y$, with no zero computed here. It does not avoid the zeros altogether, since those bounds are themselves proved from them: from the verification of RH to a finite height, now 3×10^{12} [247], and from explicit zero free regions and zero density estimates.

input from the primes	positivity	basepoints
$\Lambda(n) \ge 0$	$v_\alpha(\theta) > 0$ for every θ	$\alpha \ge 3.575$
$\Lambda(n) \ge 0$	$k_\alpha(t) > 0$ for every t	$\alpha \ge 3.191$
primes up to 10^3	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.533$
primes up to 10^5	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.367$
primes up to 10^7	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.284$
primes up to 10^7 and bounds for $\psi(y) - y$	$2\sqrt{\pi t}\, k_1(t) > 0.021$ for every t	$\alpha = 1$

Table 9.5: What the primes show about the Thorin density and the heat trace of T_α, with no information on the zeros beyond what the published bounds for $\psi(y) - y$ contain. The first five rows are proved in ball arithmetic by `heat_thresholds.py` on cells covering every α from the bound shown to 10 and every time, or height; for $\alpha \ge 10$ positivity is immediate: $b_\alpha = \mathbb{E}V_\alpha$ increases with α, its derivative being $\operatorname{Var} V_\alpha$, $b_{10} = 0.391$, and the pole term is at most $\frac{1}{\alpha-1} \le \frac{1}{9}$ in both forms. The last row is Proposition 9.1.24, also in ball arithmetic. Below $\alpha = 1$ this route needs an explicit quasi Riemann hypothesis as input.

Below the basepoint one this method stops, and the reason is Proposition 9.1.13 again. For $\frac{1}{2} < \alpha < 1$ the integration by parts bounds the tail by a boundary term

plus $\int_{\log X}^{\infty} \omega(x)\, e^{(1-\alpha)x}(g_t'(x) + \alpha g_t(x))\, dx$, and since $g_t(x) \to 1$ as $x \to \infty$ for every t, this is finite only if $\int^{\infty} \omega(x) e^{(1-\alpha)x}\, dx < \infty$, that is, only if ω decays exponentially. Every explicit unconditional bound decays more slowly, like $e^{-c\sqrt{x}}$ from the zero free region of de la Vallée Poussin or like $e^{-cx^{3/5}(\log x)^{-1/5}}$ from that of Vinogradov and Korobov, and so do the published bounds above beyond their finite ranges. The half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 (Section 12.1.2) would give $\psi(y) - y = O(y^{7/8+\delta})$ for every $\delta > 0$ [324], enough in principle for every $\alpha > \frac{7}{8}$; its constant has not been made explicit, and an explicit one would let this method pass a little below the basepoint one. A bound with $\int^{\infty} \omega(x) e^{(1-\alpha)x}\, dx < \infty$ is itself a quasi Riemann hypothesis: it makes $\int_1^{\infty} y^{-w}\, d(\psi(y) - y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ converge for $\operatorname{Re} w \geq \alpha$ after one integration by parts, so it forces $\Theta \leq \alpha$, and it follows from $\Theta < \alpha$. Along this route the step below the basepoint one therefore needs a quasi Riemann hypothesis as input, and reaching every $\alpha > \frac{1}{2}$ needs RH itself. The obstruction belongs to the route. Whether $k_\alpha > 0$ just below $\alpha = 1$ needs any hypothesis is not settled by it, since $k_\alpha > 0$ is weaker than $v_\alpha \geq 0$ and only zeros with real part above α contribute negatively to v_α.

9.2 Hitting times at the basepoints

The third question, whether T_1 is a hitting time, is probabilistic.

We keep the notation of Section 9.1: $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of T_α whenever that variable exists, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha + i\theta)$, and $\rho_{1/2} = v_1/\pi$ is the Thorin density of T_1 in the chart $\theta = \sqrt{z}$. In addition B is a standard Brownian motion, $H_a^{(c)}$ is the first passage time of $B_t + ct$ to the level $a \geq 0$, and $H_a = H_a^{(0)}$; this H has nothing to do with the de Bruijn Newman flow H_t of Section 8.4.

9.2.1 Kent's theorem and Brownian first passage to a random level

The first candidate is the hitting time of a level by a diffusion. By Kent's theorem [145] and Krein's inverse spectral theory [79], recalled in Section 8.1.3, the first hitting times of levels by diffusions started at a reflecting or entrance end are exactly the GGCs without drift whose Thorin measures consist of unit atoms at distinct points, the Dirichlet eigenvalues of the interval up to the level. The Thorin measure of φ_1 is absolutely continuous, so T_1 is not such a hitting time. It is the clock at the centre that has the discrete form: under RH its Thorin measure has an atom at each γ^2, of mass the multiplicity of the zero, so that it is such a hitting time exactly when the zeros are also simple. Brownian motion offers a second candidate, a first passage to a random level, and Proposition 8.2.1 excludes it for the clock.

Proposition 9.2.1. *For no drift $c \geq 0$ and no random level $L \geq 0$ independent of the Brownian motion is T_1 equal in law to the first passage time $H_L^{(c)}$. (For $c < 0$ the first passage time is infinite with positive probability, so equality is impossible.)*

Proof. Conditionally on L, the Laplace transform of $H_L^{(c)}$ is $\exp\{-L(\sqrt{c^2+2u}-c)\}$, so $\mathbb{E}e^{-uH_L^{(c)}} = \mathbb{E}\exp\{-L(\sqrt{c^2+2u}-c)\}$ extends analytically, and bounded by one, to the region $R_c = \{u : \operatorname{Re}\sqrt{c^2+2u} > c\}$, which contains the right half plane and lies inside $\mathbb{C}\setminus(-\infty,0]$. The function φ_1 is analytic there as well, by the proof of Proposition 9.1.2, so if the two agreed on $\operatorname{Re} u > 0$ they would agree on R_c, and $|\varphi_1| \leq 1$ on R_c and on its boundary, the parabola $u_q = \frac{1}{2}(-q^2+2icq)$, $q \in \mathbb{R}$, on which $\sqrt{c^2+2u_q} = c+iq$, as in the proof of Proposition 8.2.1. On that parabola $\operatorname{Re}\sqrt{u_q}$ stays bounded while $\operatorname{Im}\sqrt{u_q} \sim q/\sqrt{2}$, so by Stirling's formula $|\xi(1+\sqrt{u_q})|$ decays exponentially and $|\varphi_1(u_q)| \to \infty$. For $c = 0$ the parabola is the cut itself and $|\varphi_1(-t^2)| = |\xi(1)/\xi(1+it)|$, which equals 13.0, 1.2×10^4, 2.0×10^{10} and 1.5×10^{23} at $t = 10, 20, 40, 80$, as computed by `compute.py`. □

A subordinator with a drift as well as a stable one half part escapes this obstruction: on the cut, where $\sqrt{u} = it$, its exponent $u+\sqrt{u} = -t^2+it$ has negative real part, so mixtures of its transforms can grow.

9.2.2 Subordination of the centre clock

Let T be the clock variable of Section 1.2, with $\mathbb{E}e^{-uT} = \varphi_{1/2}(u) = 1/G(u)$, a positive random variable that exists with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000); it is the clock of [249], and we call it the centre clock.

Theorem 9.2.2. *Let $\widetilde{T}$ have the tilted law $P(\widetilde{T} \in dt) = e^{-t/4}P(T \in dt)/\mathbb{E}e^{-T/4}$, and let $(\sigma_\ell)_{\ell\geq0}$ be an independent subordinator with Laplace exponent $u+\sqrt{u}$. Then*

$$T_1 \overset{d}{=} \sigma_{\widetilde{T}} \overset{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}},$$

where H_a is the first passage time of a standard Brownian motion to the level a, independent of $\widetilde{T}$.

Proof. For $\operatorname{Re}\sqrt{u} \geq 0$ the number $\frac{1}{2}+\sqrt{u}$ has positive real part and $(\frac{1}{2}+\sqrt{u})^2 = \frac{1}{4}+u+\sqrt{u}$, so $\xi(1+\sqrt{u}) = \xi\big(\frac{1}{2}+\sqrt{\frac{1}{4}+u+\sqrt{u}}\big)$ with the principal root. Hence

$$\begin{aligned}\varphi_1(u) &= \frac{\xi(1)}{\xi(1+\sqrt{u})} = \frac{\varphi_{1/2}(\frac{1}{4}+u+\sqrt{u})}{\varphi_{1/2}(\frac{1}{4})}\\ &= \frac{\mathbb{E}\big[e^{-T/4}e^{-(u+\sqrt{u})T}\big]}{\mathbb{E}e^{-T/4}} = \mathbb{E}e^{-(u+\sqrt{u})\widetilde{T}} = \mathbb{E}e^{-u\sigma_{\widetilde{T}}}.\end{aligned}$$

Finally $\sigma_\ell \overset{d}{=} \ell + H_{\ell/\sqrt{2}}$, because $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$. □

The subordinator σ has a classical description: a subordinator with exponent $\kappa_1 u + \kappa_2\sqrt{2u}$ is the inverse local time at zero of a Brownian motion that is sticky at zero, the drift κ_1 measuring the time spent at zero per unit of local time [127, Section 5.7, p. 187]; here $\kappa_1 = 1$ and $\kappa_2 = 1/\sqrt{2}$. So T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$.

Corollary 9.2.3. *Assume RH and that the zeros are simple. Then $\widetilde{T}$ is Kent's hitting time for a Krein string whose Dirichlet spectrum is $\{\gamma^2 + \frac14\} = \{|\rho|^2\}$, and T_1 is the first hitting time of the level y_* by the process $D(\ell_t)$, where D is that string diffusion started at its left end, y_* is the level whose hitting time by D is $\widetilde{T}$, and ℓ_t is the local time at zero of an independent sticky Brownian motion B. Equivalently, T_1 is the first hitting time of the line $\{y = y_*\}$ by the two dimensional Markov process $(D(\ell_t), B_t)$.*

Proof. Under RH, $\mathbb{E}e^{-u\widetilde{T}} = \varphi_{1/2}(\frac14+u)/\varphi_{1/2}(\frac14) = \prod_{\gamma>0}(\gamma^2+\frac14)/(\gamma^2+\frac14+u)$, the product running over the ordinates $\gamma > 0$ repeated according to multiplicity. Since the zeros are simple, this is a GGC without drift with unit atoms at the distinct points $|\rho|^2$, so Kent's theorem and Krein's inverse theory provide the string. The first time $D(\ell_t)$ reaches y_* is the first t with $\ell_t \ge \widetilde{T}$, which is $\sigma_{\widetilde{T}}$. □

The tail (9.1.3) follows from the theorem: $P(H_a > x) \sim a\sqrt{2/(\pi x)}$, so $P(T_1 > x) \sim \mathbb{E}[\widetilde{T}]/\sqrt{\pi x}$, and $\mathbb{E}\widetilde{T} = -\frac{d}{du}\log\varphi_{1/2}(\frac14+u)\big|_{u=0} = \frac{\xi'}{\xi}(1) = \sum_\rho \rho^{-1}$.

9.2.3 The density of T_1 and harmonic measure

Conditioning on $\widetilde{T}$ in Theorem 9.2.2 and using the density of H_a gives the density of T_1 as an expectation over the tilted centre clock,

$$f_{T_1}(x) = \mathbb{E}\Big[\frac{\widetilde{T}}{2\sqrt{\pi}}\,(x-\widetilde{T})^{-3/2}\exp\Big(-\frac{\widetilde{T}^2}{4(x-\widetilde{T})}\Big)\,\mathbf{1}\{\widetilde{T}<x\}\Big], \tag{9.2.1}$$

since H_a has density $a(2\pi h^3)^{-1/2}e^{-a^2/2h}$ and $a = \widetilde{T}/\sqrt{2}$. Figure 9.4 compares (9.2.1), averaged over 4×10^5 draws of $\widetilde{T}$, against numerical inversion of φ_1 by Talbot's method [316] and shows the whole density. At the points computed the two agree to about 0.1%, except near the mode, at $x = 0.024$, where the Monte Carlo average is within 1%. For large x the density behaves like $\frac{1}{2\sqrt{\pi}}\sum_\rho \rho^{-1}\,x^{-3/2} = 0.00652\,x^{-3/2}$, the derivative of (9.1.3). Near the origin the density is super flat, inherited from the centre clock through $\widetilde{T}$.

The Cauchy kernels of Proposition 9.1.2 and the stable one half subordinator of Theorem 9.2.2 are both laws of planar Brownian motion at the moment it first reaches a line. Let $B = (B^1, B^2)$ be a planar Brownian motion started at a zero $\rho = \beta + i\gamma$, identified with the point (β,γ), and let H be the first time B^1 reaches

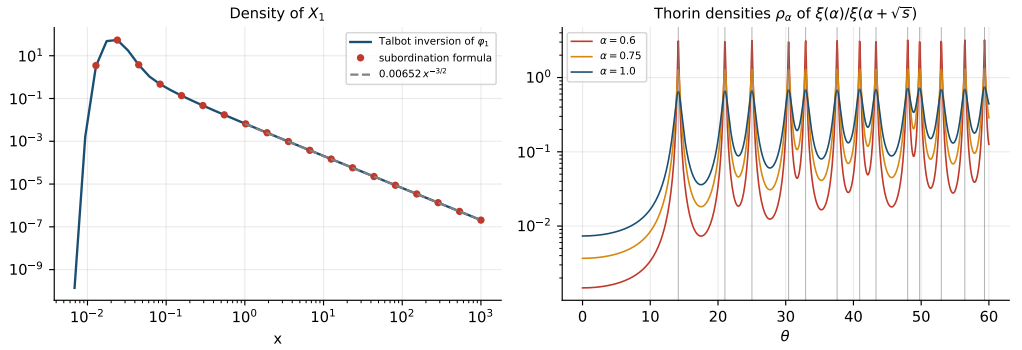


Figure 9.4: Left: the density of T_1 by Talbot inversion of φ_1 and by the subordination formula (9.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities v_α/π at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as α decreases, and for $\alpha < 1$ they are Thorin densities of T_α only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write X_1 for T_1, ρ_α for v_α/π and s for u. Drawn by `figs2.py`.

the line $\operatorname{Re} s = 1$, a distance $1 - \beta$ away, that is, the first passage time $H_{1-\beta}$ of the one dimensional Brownian motion $B^1 - \beta$. Then B^2_H has the Cauchy law with centre γ and width $1 - \beta$, which is the harmonic measure of the half plane, and H has Laplace transform $e^{-(1-\beta)\sqrt{2u}}$, the stable one half law. So, exactly,

$$\rho_{1/2}(\theta)\,d\theta = \sum_\rho P\big(B^2_H \in d\theta \mid B_0 = \rho\big).$$

The times H have the same stable one half law, up to scale, as the stable part of the subordinator σ in Theorem 9.2.2; the correspondence is of type rather than of scale, since in the theorem the stable part runs per unit of local time $\widetilde{T}$ rather than per unit of distance.

9.2.4 The exact Brownian boundary and Wald's identity

The inverse first passage question asks for a boundary b such that the first time $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ at which a standard Brownian motion started at zero crosses b has the law of T_1. A deterministic curved boundary always exists. Anulova proved this for two sided symmetric boundaries [4], and Chen, Cheng, Chadam and Saunders for one sided boundaries when the survival function is continuous and positive [56, Theorem 1]. By the theorems of Ekström and Janson [82, Theorems 2.3, 8.2 and 9.2], every law on $(0, \infty)$ is the law of H_b for a lower semicontinuous boundary b, which is continuous on every interval where the density of the law is bounded below by a positive constant; when that holds on every compact subinterval of $(0, \infty)$, b is the unique continuous solution of

the Volterra equation (8.2.1) with $\liminf_{t\downarrow 0} b(t) \geq 0$. The density (9.2.1) of T_1 is continuous and positive on $(0,\infty)$, so the boundary of T_1 is continuous there, and it can be computed from the law of T_1 alone through (8.2.1), with f the density of T_1, used in the complementary form of Section 8.2.2.2, with $P(T_1 > t)$ in place of $P(T > t)$, once the survival probability falls below one half; `rs1.py` solves it. We computed the distribution function of T_1 by Talbot inversion of $\varphi_1(u)/u$ for $t < 0.03$, where T_1 is super flat, and by the subordination representation of Theorem 9.2.2 beyond, in the form

$$P(T_1 > t) = \mathbb{E}\Big[\mathbf{1}\{\widetilde{T} \geq t\} + \mathbf{1}\{\widetilde{T} < t\}\Big(2\mathcal{N}\Big(\frac{\widetilde{T}}{\sqrt{2(t-\widetilde{T})}}\Big) - 1\Big)\Big],$$

with $\mathcal{N}$ the standard normal distribution function, as in Section 8.2.2.2. This form is smooth and gives the survival function to relative accuracy even at $t = 100$, where $\sqrt{t}\,P(T_1 > t) = 0.013037$ against the limit 0.013030. As at the centre, the boundary is computed over the three zones of Section 8.2.2.2.

In the small time zone the law of T_1 near the origin is that of the tilted centre clock, and the boundary starts from the reflection level 0.505 at $t = 0.009$. In the bulk, from $t = 0.011$ to $t = 0.05$, the boundary falls steeply, to 0.363 at $t = 0.012$ and 0.111 at $t = 0.02$. Here the tangent approximation, taken in its density form, equates the first passage density of the tangent line with the density of T_1, where the boundary of Roberts and Shortland [288] in Section 8.2.2.1 equates hazards. Its solution, the boundary b with $(b - tb')(2\pi t^3)^{-1/2}e^{-b^2/2t} = f(t)$ computed by `rs1c.py`, agrees with the exact boundary to 10^{-4} at $t = 0.012$ and to 1.5×10^{-3} at $t = 0.02$.

The right tail is where the basepoint one differs completely from the centre. At the centre the survival function is exponential and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1$ (Section 8.2.2.2). At the basepoint one the survival function is the stable one half tail $0.01303\,t^{-1/2}$, and a Brownian motion survives that long only below a boundary that stops moving. The exact boundary flattens at the level $b_\infty = -0.19277$: it stays within 10^{-3} of this level from $t = 0.056$ on and within 10^{-4} from $t = 1.2$ on, up to the end of the computation at $t = 100$; that $b_\infty = \lim_{t\to\infty} b(t)$ is observed, not proved. The tangent approximation fails here: it overshoots downward, to -0.332 at $t = 0.05$ against the exact -0.188, and has no solution beyond $t = 0.091$. It treats the crossing density at time t as if the path had just started, so a flat boundary at level a would have to produce the density $a\,(2\pi)^{-1/2}t^{-3/2}$, forcing $a = \sqrt{2\pi} \times 0.00652 = 0.0163 > 0$; but by the time the tail is reached the tangent boundary is already negative, and a negative linear boundary can only produce an exponential tail.

The positive number 0.0163 does appear in the exact picture, in a different role. Optional stopping for the martingale $B_{t\wedge H_b}$ gives $\mathbb{E}[b(H_b)\mathbf{1}\{H_b \leq t\}] = -\mathbb{E}[B_t\mathbf{1}\{H_b > t\}]$, and once the boundary is flat, a survivor at depth y below it survives a further time r with probability about $y\sqrt{2/(\pi r)}$. The next lemma

makes the matching of these two facts exact. For a continuous monotone boundary that is positive at 0 the limit is Theorem 2 of Novikov [216]; the lemma assumes boundedness in place of monotonicity.

LEMMA 9.2.4. *Let b be continuous on $(0,\infty)$ and bounded on $[1,\infty)$, and let $H_b = \inf\{t>0 : B_t \ge b(t)\}$ be positive almost surely. Then $b(H_b) = B_{H_b}$ is integrable and $\sqrt{t}\,P(H_b>t) \to \sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$ as $t\to\infty$.*

Proof. Since b is bounded above on $[1,\infty)$ and $\limsup_t B_t = \infty$, H_b is finite, and $B_{H_b} = b(H_b)$ by continuity. Choose $u_0 > 0$ with $P(H_b > u_0) > 0$; on that event $b > B$ on $(0,u_0]$, so b, which is also continuous on $[u_0,1]$ and bounded on $[1,\infty)$, is bounded below on $(0,\infty)$, say by $-K$. Optional stopping gives $A(s) = \mathbb{E}[b(H_b)\mathbf{1}\{H_b \le s\}] = -\mathbb{E}[B_s\mathbf{1}\{H_b > s\}]$. Fix $s \ge 1$, let c_1 and c_2 be the infimum and the supremum of b on $[s,\infty)$, and let $g(y,\tau) = (2\mathcal{N}(y/\sqrt{\tau}) - 1)^+$, the probability that a Brownian motion started at zero stays below the level y up to time τ. By the Markov property at time s, on $\{H_b > s\}$ the conditional probability of $\{H_b > t\}$ given the path up to time s lies between $g(c_1 - B_s, t-s)$ and $g(c_2 - B_s, t-s)$, and $c_2 - B_s > 0$ there. Since $\sqrt{t}\,g(y,t-s) \to \sqrt{2/\pi}\,y^+$ as $t\to\infty$, and $\sqrt{t}\,g(y,t-s) \le 2y^+$ for $t \ge 2s$, dominated convergence and $y^+ \ge y$ give

$$\liminf_{t\to\infty} \sqrt{t}\,P(H_b>t) \ge \sqrt{\tfrac{2}{\pi}}\big(c_1 P(H_b>s) + A(s)\big),$$
$$\limsup_{t\to\infty} \sqrt{t}\,P(H_b>t) \le \sqrt{\tfrac{2}{\pi}}\big(c_2 P(H_b>s) + A(s)\big).$$

The upper bound at $s=1$ and the lower bound at every s bound $A(s)$ uniformly, so, since $b(H_b) \ge -K$, monotone convergence makes $b(H_b)$ integrable with $A(s) \to \mathbb{E}\,b(H_b)$. As $s\to\infty$, c_1 and c_2 stay bounded and $P(H_b > s) \to 0$, so both bounds tend to $\sqrt{2/\pi}\,\mathbb{E}\,b(H_b)$. □

The exact boundary of T_1 is continuous; its boundedness on $[1,\infty)$, which the lemma needs, is observed, not proved. Granting it, the lemma and the tail (9.1.3) give

$$\mathbb{E}\,b(T_1) = \lim_{t\to\infty} \mathbb{E}\big[(b_\infty - B_t)\mathbf{1}\{H_b > t\}\big] = \sqrt{\frac{\pi}{2}} \times \frac{\sum_\rho \rho^{-1}}{\sqrt{\pi}} = \frac{1}{\sqrt{2}} \sum_\rho \frac{1}{\rho} = 0.0163311,$$

the first equality by optional stopping, since $P(H_b > t) \to 0$. So, if the boundary is bounded on $[1,\infty)$, Wald's identity $\mathbb{E}B_{H_b} = 0$, which holds for every crossing time of finite mean, fails here, and the defect is exactly $\sum_\rho \rho^{-1}/\sqrt{2}$. Numerically, integrating the computed boundary against the law of T_1 gives $\mathbb{E}\,b(T_1) = 0.0163370$. The defect is also the level of the stable summand of T_1: since $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$, the variable S of Proposition 9.1.11 has the law of the first passage time $H_{b_1/\sqrt{2}}$ (Section 10.2.4).

We simulated 1.5×10^5 Brownian paths against the exact boundary on $[0, 1]$, with a Brownian bridge crossing correction, using `rs1d.py`. The Kolmogorov distance between the simulated and the exact survival functions is 0.0019, at the level of the Monte Carlo noise. Figure 9.5, drawn by `figs5.py`, shows the boundary in its three zones and the survival check.

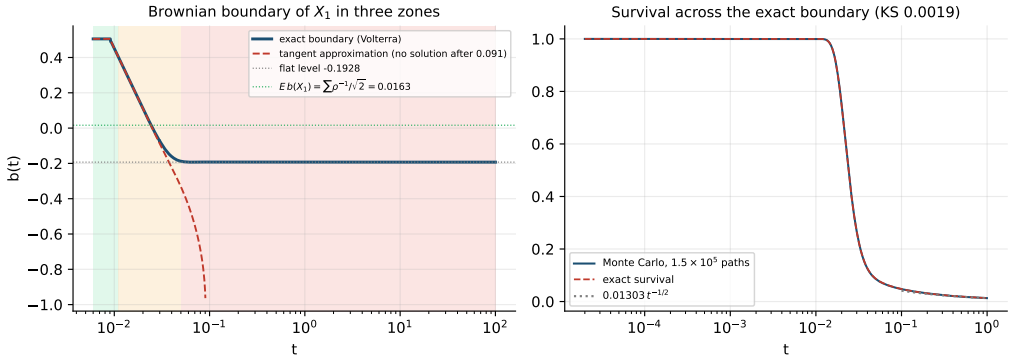


Figure 9.5: Left: the exact Brownian boundary of T_1, written X_1 in the plot, from the Volterra equation over the three zones of Section 8.2.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level -0.19277, and the value $\mathbb{E}\, b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function of T_1 and the tail $0.01303\, t^{-1/2}$; the Kolmogorov distance is 0.0019.

9.2.5 Every basepoint above the centre

Nothing in Theorem 9.2.2 is special to the basepoint one. Since $(\alpha - \frac{1}{2} + \sqrt{u})^2 = (\alpha - \frac{1}{2})^2 + u + 2(\alpha - \frac{1}{2})\sqrt{u}$, the same computation applies at every basepoint to the right of the centre.

Theorem 9.2.5. *For every $\alpha > \frac{1}{2}$, with no hypothesis on the zeros (the existence of the centre clock, Theorem 5.1.1, uses the verification of RH below height 1000), $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha}$, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and $\sigma^{(\alpha)}$ is an independent subordinator with exponent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, that is $T_\alpha \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$. The law has infinite mean and $P(T_\alpha > x) \sim \frac{\xi'}{\xi}(\alpha)\,(\pi x)^{-1/2}$. The variable T_α is a GGC if and only if no zero has $\beta > \alpha$, in particular for every $\alpha \geq 1$, and when every zero has $\beta < \alpha$ its Thorin density in the chart $\theta = \sqrt{z}$ is $\frac{1}{\pi}\sum_\rho (\alpha - \beta)/((\alpha - \beta)^2 + (\theta - \gamma)^2)$.*

Proof. The subordination is the computation of Theorem 9.2.2 with $\frac{1}{4}$ replaced by

$(\alpha-\frac12)^2$ and $\sqrt{u}$ by $2(\alpha-\frac12)\sqrt{u}$, and H_a has exponent $a\sqrt{2u}$ with $a=\sqrt{2}(\alpha-\frac12)\ell$. The tail follows as in Section 9.1.2 from $\varphi_\alpha(u)=1-\frac{\xi'}{\xi}(\alpha)\sqrt{u}+O(u)$. The Thorin density and the GGC criterion are the threshold theorem, Theorem 1.2.2: when every zero has $\beta<\alpha$ each Cauchy term is positive, and a zero with $\beta>\alpha$ puts a pole of φ_α at the point $(\rho-\alpha)^2$ of $\mathbb{C}\setminus(-\infty,0]$ (its principal root is $\rho-\alpha$, since $\mathrm{Re}(\rho-\alpha)>0$), where the Laplace transform of a GGC is holomorphic, whatever the sign of the density near its ordinate (Remark 1.2.4). In the boundary case, when some zero has $\beta=\alpha$ and none has $\beta>\alpha$, every zero has $\beta<\alpha'$ for $\alpha'>\alpha$, so $T_{\alpha'}$ is a GGC; as $\alpha'\downarrow\alpha$, $\varphi_{\alpha'}\to\varphi_\alpha$ pointwise on $[0,\infty)$, so $T_{\alpha'}\to T_\alpha$ in law, and T_α is a GGC by the closure theorem of Section 4.1.1. There the Cauchy kernel of a zero $\alpha+i\gamma$, $\gamma>0$, of multiplicity m becomes an atom: with its conjugate it contributes the factor $(1+u/\gamma^2)^{-m}$ to φ_α, a Thorin atom of mass m at $z=\gamma^2$. □

Corollary 9.2.6. *Let $\alpha>\frac12$ and $a=\alpha-\frac12$. Then T_α is infinitely divisible, with no drift and Lévy measure $W_\alpha(x)\,dx/x$, where*

$$W_\alpha(x)=\frac{ax}{\sqrt{\pi}}\int_0^x (x-s)^{-3/2}e^{-a^2xs/(x-s)}\,W(s)\,ds$$
$$=\mathbb{E}\Big[(1+4xV)^{-1/2}\,W\Big(\frac{x}{1+4xV}\Big)\Big],$$

with $V=a^2/(2Z^2)$ and Z standard normal. Hence $W_\alpha>0$ and $W_\alpha'<0$ on $(0,\infty)$, so that T_α is self decomposable; and on every interval $(0,T]$ on which W is completely monotone, so is W_α.

Proof. By Theorem 9.2.5, T_α is the subordinator $\sigma_s^{(\alpha)}=s+H_{\sqrt{2}as}$ evaluated at the independent time $\widetilde{T}_\alpha$, which is infinitely divisible with no drift and Lévy measure $e^{-a^2s}W(s)\,ds/s$, and $H_{\sqrt{2}as}$ has the density $(as/\sqrt{\pi})y^{-3/2}e^{-a^2s^2/y}$. By Bochner's subordination [297, Theorem 30.1], T_α is infinitely divisible with no drift and Lévy measure $\int_0^\infty P(\sigma_s^{(\alpha)}\in dx)\,e^{-a^2s}W(s)\,ds/s$, which, since $a^2s^2/(x-s)+a^2s=a^2xs/(x-s)$, is $W_\alpha(x)\,dx/x$ with the first form of W_α. The substitution $s=x/(1+4xv)$ turns the integral into an average against the density $\frac{a}{2\sqrt{\pi}}v^{-3/2}e^{-a^2/4v}$ of V, which is the second form. There the derivative in x of $(1+4xv)^{-1/2}W(x/(1+4xv))$ is $-2v(1+4xv)^{-3/2}W+(1+4xv)^{-5/2}W'$, negative because $W>0$ and $W'<0$ (Theorem 5.1.1), and differentiation under the average is legitimate, because $W^{(n)}(s)=O(s^{-n-1/2}\log(1/s))$ as $s\to0$ makes every derivative in x of the integrand $O(\log(1+V))$ on compact subsets of $(0,\infty)$, and $\mathbb{E}\log(1+V)<\infty$. Finally, $x\mapsto x/(1+4xv)$ has a completely monotone derivative and maps $(0,T]$ into itself, and $(1+4xv)^{-1/2}$ is completely monotone, so Faà di Bruno's formula and Leibniz's rule carry complete monotonicity of W on $(0,T]$ to every term of the average. □

With Theorem 5.1.9, W_α is therefore completely monotone on $(0,1.29]$ for every $\alpha>\frac12$. When no zero has real part above α, in particular for every $\alpha\ge1$, T_α is a

GGC and W_α is the Laplace transform k_α of its Thorin measure (Section 11.1.6); when some zero has real part above α, T_α is not a GGC, so W_α is not completely monotone on $(0,\infty)$, although it is on $(0, 1.29]$.

Table 9.6, computed by `improve.py`, and the right panel of Figure 9.4 give the constants and the densities for three basepoints. As $\alpha \downarrow \frac{1}{2}$ the tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$ tends to zero, because $\xi'(\frac{1}{2}) = 0$, and the law acquires the finite mean $\sum_k a_k^{-1}$ of the centre clock; at the same time the subordinator tends to pure drift and the Cauchy widths to $\frac{1}{2} - \beta$, which is where RH is decided.

α	$\frac{\xi'}{\xi}(\alpha)$	tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$	$\min_{[0,60]} \rho_{\alpha-1/2}$
0.6	0.00462092	0.002607	0.00147
0.75	0.01155134	0.006517	0.00368
1	0.02309571	0.013030	0.00735

Table 9.6: Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of T_α, and the minimum on $[0, 60]$, attained at the origin, of the density $\rho_{\alpha-1/2} = v_\alpha/\pi$, the Thorin density of T_α when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. At $\alpha = 1$ the minimum at the origin holds at every height (Proposition 9.1.11), as it does at every $\alpha \geq 1$ (Proposition 9.1.12); for $\frac{1}{2} < \alpha < 1$ the minimum at the origin holds on the verified window $|\theta| \leq 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)), and at every height for every such α it is equivalent to RH (part (vi) of that theorem).

9.2.6 Boundaries across the basepoints

We solved the Volterra equation (8.2.1) [240] at $\alpha = 0.6$ and 0.75, with the law of T_α from Theorem 9.2.5. As at the basepoint one, the distribution function comes from Talbot inversion of $\varphi_\alpha(u)/u$ for $t < 0.03$ and from the subordination identity beyond, here with 3×10^5 draws of $\widetilde{T}_\alpha$, made under RH as a sum of exponential variables over the first 200 zeros and a gamma variable with the mean and variance of the remaining terms. The bottom left panel of Figure 11.2 in Section 11.1 draws the four boundaries together.

In the small time zone and the bulk the boundaries at the different basepoints are close: they start near the reflection level 0.50, and at $t = 0.02$ they are at 0.072 for the centre and at 0.080, 0.092 and 0.111 for $\alpha = 0.6$, 0.75 and 1. They differ in the right tail. At the centre the survival function is exponential with rate γ_1^2, and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1 = -19.99$ and the logarithmic correction $\frac{3}{2\sqrt{2}\gamma_1} \log t$; the slope of the computed boundary over $[0.2, 0.3]$ is -19.77 (Section 8.2.2.2). Above the centre the exact boundaries flatten: each stays within 0.011 of its level at $t = 100$, the end of the computation, from $t = 0.1$ on, and within 10^{-3} from $t = 1$ on. As $\alpha \downarrow \frac{1}{2}$ the flat level moves down, towards the linear boundary of the centre.

The Wald type identity of Section 9.2.4 extends to every basepoint above the centre. The density of T_α is again continuous and positive on $(0,\infty)$, so its boundary is continuous [82, Theorem 8.2], and as computed it is bounded up to $t = 100$. Granting boundedness at infinity, which here too is observed, not proved, Lemma 9.2.4 and the stable tail $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ of Theorem 9.2.5 give $\mathbb{E}\, b(T_\alpha) = b_\alpha/\sqrt{2}$, so Wald's identity $\mathbb{E}B_{H_b} = 0$ fails by exactly $b_\alpha/\sqrt{2}$. Table 9.7 compares the identity and the tail constant with the computed values; all six pairs agree to within 6×10^{-4} in relative terms. The reason given in Section 9.2.4 for the failure of the tangent approximation in the right tail holds at every basepoint above the centre.

α	flat level	$\mathbb{E}\, b(T_\alpha)$	$b_\alpha/\sqrt{2}$	$\sqrt{100}\, P(T_\alpha > 100)$	$b_\alpha/\sqrt{\pi}$
0.6	−0.33384	0.0032687	0.0032675	0.0026083	0.0026071
0.75	−0.25537	0.0081710	0.0081680	0.0065203	0.0065171
1	−0.19277	0.016337	0.016331	0.013037	0.013030

Table 9.7: Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\, b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of T_α, against the Wald type identity, and the computed $\sqrt{100}\, P(T_\alpha > 100)$ against the tail constant of Theorem 9.2.5. Computed by `bnd_alpha.py` for $\alpha = 0.6$ and 0.75 and by `rs1.py` for $\alpha = 1$.

Chapter 10

The critical strip and Riesz's criterion

Chapter 9 read the reciprocal Xi function at a basepoint α as a Laplace transform on the half line, $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ for $u \geq 0$. At the basepoint one its Thorin measure is the Poisson smoothing of the zeros and its Lévy form carries the primes, and by Theorem 9.2.5 it is, for every $\alpha > \frac{1}{2}$ and with no hypothesis, the Laplace transform of a positive random variable T_α obtained by subordinating the centre clock. The same function has two further readings, as the characteristic function of a normal scale mixture and through the completeness of the translates of the tilted Pólya kernels. Both run across the whole critical strip, not only to the right of the zeros, and so show how much of what holds with no hypothesis at the basepoint one survives inside the strip. Section 10.1 develops the two readings in turn and ends with Riesz's criterion, which reads the zero free half planes through growth rather than completeness. Section 10.2 writes that criterion in the language of the clocks, where it becomes the statement that every T_α with $\alpha > \frac{1}{2}$ is a GGC, and shows why none of the resulting representations of its coefficients supplies the growth bound that the criterion needs. Section 10.3 turns to the squarefree integers, whose Dirichlet series $\zeta(s)/\zeta(2s)$ carries the same reciprocal $1/\zeta(2s)$: the constant of $\sum_{n \leq x} \mu(n)^2/n$ is a combination of two tilted Pólya means, and smoothing the count by a gamma law gives criteria for RH of the same kind as Riesz's.

For the scale mixtures and the completeness statements, as for the Thorin measure and the Lévy form in Chapter 9, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH. Chapters 9 to 11 prove no new zero free region.

10.1 Normal scale mixtures and completeness across the strip

The zero free half planes of Chapter 9 have two readings, the second being completeness (Section 10.1.6). The first evaluates the Laplace transform φ_α at u^2. Since $\varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$, the reciprocal of ξ along the real axis is a charac-

teristic function, that of a normal scale mixture, at every basepoint and with no hypothesis (Section 10.1.1); what the zeros decide is the class of the law.

We keep the notation of Chapter 9: T is the centre clock and $T_{1/2} = T$, $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac12)^2 T}$, H_a is the first passage time of a standard Brownian motion to the level a, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$. When every zero lies to the left of the line $\mathrm{Re}\, s = \alpha$, the Thorin density of T_α in the chart $\theta = \sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (1.2.7) (Theorem 1.2.2); when a zero lies to the right of that line, T_α is not a GGC and v_α/π is not a Thorin density for it (Proposition 11.4.3). Throughout, Z is a standard normal variable independent of everything else. The scripts are in the folders `basepoint_one/scale_mixture` and `basepoint_one/wiener` of the archive.

10.1.1 The reciprocal on the real axis is a characteristic function

If $X \geq 0$ is independent of Z, then $\mathbb{E}e^{iu\sqrt{2X}Z} = \mathbb{E}e^{-u^2 X}$ for real u. The centre clock T of [249] is infinitely divisible and self decomposable with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000). For $\alpha > \frac12$, Theorem 9.2.5 writes $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha} \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac12)\widetilde{T}_\alpha}$, where $\sigma^{(\alpha)}$ is an independent subordinator with Laplace exponent $u + 2(\alpha - \frac12)\sqrt{u}$. With $X = T_\alpha$ the evaluation gives the following.

Proposition 10.1.1. *For every $\alpha \geq \frac12$ the function $u \mapsto \varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$ is the characteristic function of $Y_\alpha = \sqrt{2T_\alpha}\, Z$. The law of Y_α is symmetric, unimodal and infinitely divisible with no hypothesis. At the centre it has variance $2\mathbb{E}T = 2\sum_k a_k^{-1} = 0.046210$. For $\alpha > \frac12$ its characteristic function has a corner at the origin, $\xi(\alpha)/\xi(\alpha + |u|) = 1 - b_\alpha|u| + O(u^2)$, so Y_α lies in the domain of attraction of the Cauchy law, with $P(|Y_\alpha| > y) \sim 2b_\alpha/(\pi y)$ as $y \to \infty$.*

Proof. The first statement is the evaluation with $X = T_\alpha$. Symmetry and unimodality hold for every normal scale mixture. The tilt $\widetilde{T}_\alpha$ of the infinitely divisible law of T is infinitely divisible, with Lévy measure $e^{-(\alpha-\frac12)^2 t} W(t)\, dt/t$, and a subordinator run for an independent infinitely divisible time is infinitely divisible, because a Bernstein function composed with a Bernstein function is again one [299, Corollary 3.8]. So T_α is infinitely divisible, and Y_α, which is a Brownian motion run for the independent time $2T_\alpha$, is infinitely divisible for the same reason. At the centre $\xi'(\frac12) = 0$, so $\xi(\frac12)/\xi(\frac12 + |u|) = 1 - \mathbb{E}T\, u^2 + O(u^4)$, with $\mathbb{E}T = \sum_k a_k^{-1}$ by (1.2.1). For $\alpha > \frac12$ the expansion follows from $\xi(\alpha + |u|) = \xi(\alpha)(1 + b_\alpha|u| + O(u^2))$. The tail comes from that of T_α: by Theorem 9.2.5, $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ as $t \to \infty$, so that, since $P(T_\alpha > t) \leq 1$, also $P(T_\alpha > t) \leq Kt^{-1/2}$ for all $t > 0$ and some constant K. Conditioning on Z, $yP(|Y_\alpha| > y) = \mathbb{E}\big[yP(T_\alpha > y^2/(2Z^2) \mid Z)\big]$, and the integrand is at most $\sqrt{2}K|Z|$ and tends to $\sqrt{2/\pi}\, b_\alpha|Z|$ as $y \to \infty$; by dominated convergence $yP(|Y_\alpha| > y) \to \sqrt{2/\pi}\, b_\alpha \mathbb{E}|Z| = 2b_\alpha/\pi$. □

At the centre $Y_{1/2} = Z\sqrt{2T}$ is the second member of the van Dantzig pair of Section 1.2; it has the same variance as the Pólya variable X, the first member, as the relation $\mathbb{E}e^{-\theta^2 T} = 1/\mathbb{E}e^{\theta X}$ requires.

With no hypothesis the law of $Y_{1/2}$ is also self decomposable. For $0 < c < 1$ the self decomposability of T (Theorem 5.1.1) gives $T \stackrel{d}{=} cT' + R_c$ with $T' \stackrel{d}{=} T$ and $R_c \geq 0$ independent, and then $Y_{1/2} \stackrel{d}{=} \sqrt{c}\,Y' + Z'\sqrt{2R_c}$, where $Y' = Z\sqrt{2T'}$ has the law of $Y_{1/2}$ and Z' is standard normal and independent of (Z, T', R_c).

Under RH each factor $(1 + u^2/\gamma^2)^{-1}$ of $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ is a characteristic function, that of L/γ for a standard Laplace variable L, and a zero off the line would break this. The quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, with $\gamma > 0$ and $0 < \delta < \frac{1}{2}$, contributes to $\xi(\frac{1}{2})/\xi(\frac{1}{2}+|u|)$, by (1.2.1), the factor $g(u) = |1+u^2/(\gamma-i\delta)^2|^{-2}$, which is positive and integrable with $g(0) = 1$, and

$$\frac{1}{2\pi}\int_{\mathbb{R}} g(u)e^{-iux}\,du = \frac{\gamma^2+\delta^2}{4\gamma\delta}\,e^{-\gamma|x|}\big(\gamma\sin\delta|x| + \delta\cos\delta|x|\big).$$

For $x > 0$ the derivative of the right side is $-\frac{(\gamma^2+\delta^2)^2}{4\gamma\delta}e^{-\gamma x}\sin\delta x$, so its least value, attained at $x = \pi/\delta$, is $-e^{-\pi\gamma/\delta}$ times its value $(\gamma^2 + \delta^2)/(4\gamma)$ at $x = 0$. Since g is integrable, it could only be the characteristic function of the law with this density, which is negative there. So g is not a characteristic function, and $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ would still be one, as $W > 0$ ensures, but no longer factor by factor. The defect is tiny. Every ordinate is at least γ_1 and $\delta < \frac{1}{2}$, so the least value exceeds -9.5×10^{-39}, the value of the formula at $\gamma = \gamma_1$ and $\delta = \frac{1}{2}$; and a zero off the line would lie above T_0, where $e^{-\pi\gamma/\delta} < e^{-2\pi T_0}$.

Away from the centre neither V_α nor Y_α belongs to a pair. Take $\alpha > \frac{1}{2}$; for V_α the case $\alpha < \frac{1}{2}$ follows, since $V_\alpha \stackrel{d}{=} -V_{1-\alpha}$. The tilted Pólya variable V_α of Section 1.2 has characteristic function $\varphi(\theta) = \xi(\alpha + i\theta)/\xi(\alpha)$, and $1/\varphi(i\theta) = \xi(\alpha)/\xi(\alpha - \theta)$ exceeds 1 for $0 < \theta < 2\alpha - 1$, because $|\alpha - \theta - \frac{1}{2}| < \alpha - \frac{1}{2}$ there and $\xi(\frac{1}{2} + s) = \xi(\frac{1}{2})\mathbb{E}\cosh(sX)$ increases with $|s|$; at $\alpha = 0.75$ and $\theta = 0.25$ it is 1.00145. So it is not a characteristic function. By its Cauchy tail Y_α has no exponential moments, so it cannot be a first member. As a second member its partner φ would satisfy $\varphi(i\theta) = \xi(\alpha+|\theta|)/\xi(\alpha)$, and a moment generating function finite on the whole line is analytic, while $\xi(\alpha + |\theta|)/\xi(\alpha) = 1 + b_\alpha|\theta| + O(\theta^2)$ has a corner at $\theta = 0$, since $b_\alpha > 0$ (Proposition 10.1.1).

10.1.2 From the Lévy form in $|u|$ to the Thorin form in u^2

For $\alpha \geq 1$ the function $\log(\xi(\alpha + s)/\xi(\alpha))$ has, for $\operatorname{Re} s > 0$, the Lévy form (9.1.5) with b_α and Π_α in place of b_1 and Π_1, where $x\Pi_\alpha(dx) = \big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\big)dx + \sum_n \Lambda(n)n^{-\alpha}\delta_{\log n}(dx)$ is the signed measure that Section 9.1.4 gives for every

basepoint $\alpha \geq 1$. Putting $s = |u|$ gives

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = e^{-b_\alpha |u|} \exp\Big\{ -\int_0^\infty \big(e^{-|u|x} - 1 + |u|x\big)\Pi_\alpha(dx)\Big\}, \tag{10.1.1}$$

a representation in $|u|$. The first factor is the characteristic function of a Cauchy law of scale b_α. The second is not in Lévy Khintchine form, because $e^{-|u|x}$ is a function of $|u|$ rather than of u. The sine squared lemma (1.2.9) converts it. In the chart $\theta = \sqrt{z}$ and at $s = |u|$, as in the derivation of (9.1.6), it writes $e^{-|u|x} + |u|x - 1$ as $\frac{2x}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\sin^2(x\theta/2)\,d\theta$ for $x > 0$, and the drift converts through $|u| = \frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\,d\theta$. Both identities follow from $\int_0^\infty \log(1+u^2/\theta^2)\cos(x\theta)\,d\theta = \pi(1-e^{-|u|x})/x$.

By Section 1.3.2, a symmetric law is an extended generalised gamma convolution (EGGC) if and only if it is the law of $Z\sqrt{2T'}$ with T' a GGC independent of Z.

THEOREM 10.1.2. *Let $\alpha \geq 1$, or more generally let $\alpha > \frac{1}{2}$ be such that $\operatorname{Re}\rho < \alpha$ for every zero ρ. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big) v_\alpha(\theta)\,d\theta\Big\}, \tag{10.1.2}$$

where

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big) x\Pi_\alpha(dx) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta) > 0,$$

the first expression being available for $\alpha \geq 1$, and as the compensated limit (9.1.8) *(in which $\varepsilon = 1-\alpha$) when $\sup_\rho \operatorname{Re}\rho < \alpha$, so Y_α is a symmetric EGGC with Thorin density v_α/π in the chart θ. With no hypothesis on the zeros, for $\frac{1}{2} < \alpha < 1$ the law of Y_α is an EGGC if and only if ζ has no zeros with real part greater than α, and at the centre if and only if RH holds. Under RH,*

$$\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+|u|)} = \prod_{\gamma>0}\Big(1+\frac{u^2}{\gamma^2}\Big)^{-1}, \qquad Y_{1/2} \stackrel{d}{=} \sum_k \frac{L_k}{\gamma_k},$$

with independent standard Laplace variables L_k, of density $\frac{1}{2}e^{-|y|}$.

Proof. For $\alpha \geq 1$, insert the two identities of the lemma into (10.1.1) and exchange the order of integration, which is justified for $\alpha > 1$ by absolute convergence and at $\alpha = 1$ by the compensation argument of Proposition 9.1.10. This gives the first expression for v_α, and the second is its identification, (9.1.6) at $\alpha = 1$, which is Proposition 9.1.3 of Section 9.1.3, and (3.1.1) for $\alpha > 1$; for $\frac{1}{2} < \alpha < 1$ with $\sup_\rho \operatorname{Re}\rho < \alpha$ the first expression holds in the compensated form (9.1.8) of Proposition 9.1.13, in which $\varepsilon = 1-\alpha$. In every case (10.1.2) is the Thorin representation of T_α given by Theorem 1.2.2, $\varphi_\alpha(u) = \exp\{-\int \log(1+$

$u/z)\,U_\alpha(dz)\}$ with U_α as in (1.2.6), evaluated at u^2 in the chart $z=\theta^2$, and $v_\alpha>0$ because every Cauchy kernel of Proposition 1.2.3 then has positive width $\alpha-\beta$. Since T_α is a GGC, $Y_\alpha=Z\sqrt{2T_\alpha}$ is an EGGC. Conversely, if $Y_\alpha\overset{d}{=}Z\sqrt{2T'}$ with T' a GGC, then $\mathbb{E}e^{-u^2T'}=\mathbb{E}e^{-u^2T_\alpha}$ for all real u, so $T'\overset{d}{=}T_\alpha$. Hence Y_α is an EGGC exactly when T_α is a GGC, which by Theorem 1.2.2 happens exactly when $\alpha\ge\sup_\rho\operatorname{Re}\rho$. For $\frac12<\alpha<1$ this is the absence of zeros with real part greater than α, and at the centre, by the symmetry $\rho\mapsto1-\rho$, it is RH (Proposition 1.2.1). Under RH the Hadamard product (1.2.1) at $s=|u|$ is the product displayed, and each factor $(1+u^2/\gamma^2)^{-1}$ is the characteristic function of L/γ, as in Proposition 6.2.8(iii). □

The hypothesis that every zero lies to the left of the line $\operatorname{Re}s=\alpha$ cannot be dropped from (10.1.2). By Proposition 1.2.3, v_α/π is, off the ordinates of the zeros on the line $\operatorname{Re}s=\alpha$ and whatever the zeros do, the sum over the zeros with $\gamma>0$ of $P_{\alpha-\beta}(\theta-\gamma)+P_{\alpha-\beta}(\theta+\gamma)$, with $P_{-c}=-P_c$, and by the semigroup property of the Cauchy kernels $\int_0^\infty\frac{2s}{s^2+\theta^2}[P_c(\theta-\gamma)+P_c(\theta+\gamma)]\,d\theta=\frac{2(s+|c|)\operatorname{sgn}c}{(s+|c|)^2+\gamma^2}$ for $s>0$ and real $c\ne0$. Integrating in s from 0 to $|u|$ shows that a zero $\rho=\beta+i\gamma$ with $\gamma>0$ contributes to the right side of (10.1.2) the factor $((\alpha-\beta)^2+\gamma^2)/((|u|+\alpha-\beta)^2+\gamma^2)$ when $\beta<\alpha$, which is its factor in $\xi(\alpha)/\xi(\alpha+|u|)$, but the factor $((|u|+\beta-\alpha)^2+\gamma^2)/((\beta-\alpha)^2+\gamma^2)$ when $\beta>\alpha$. If no zero lies on the line $\operatorname{Re}s=\alpha$, the integral with the signed density therefore converges and equals $\xi(\alpha)/\xi(\alpha+|u|)$ multiplied by

$$\prod_{\beta>\alpha,\ \gamma>0}\frac{\big((|u|+\beta-\alpha)^2+\gamma^2\big)\big((|u|-\beta+\alpha)^2+\gamma^2\big)}{\big((\beta-\alpha)^2+\gamma^2\big)^2},$$

which exceeds one for $u\ne0$ as soon as there is such a zero, since each factor equals $1+u^2(u^2+2\gamma^2-2(\beta-\alpha)^2)/((\beta-\alpha)^2+\gamma^2)^2$ and $0<\beta-\alpha<\frac12<\gamma$. The series of Proposition 1.2.3 may be integrated term by term, because the absolute value of the term of a zero integrates to $\log(((|\alpha-\beta|+|u|)^2+\gamma^2)/((\alpha-\beta)^2+\gamma^2))=O(\gamma^{-2})$. A zero on the line $\operatorname{Re}s=\alpha$ contributes instead an atom of mass one at its ordinate, as every zero does at the centre under RH.

The drift b_α becomes, through $|u|=\frac1\pi\int_0^\infty\log(1+u^2/\theta^2)\,d\theta$, the constant Thorin density b_α/π, which is also the value $v_\alpha(0)/\pi$. At the basepoint one this constant is the minimum of v_1/π, attained only at $\theta=0$ (Proposition 9.1.11), and `comp.py` confirms it on $[0,200]$. So Y_1 splits as an independent sum of a Cauchy variable of scale $b_1=\sum_\rho\rho^{-1}=0.0230957$ and a symmetric EGGC with Thorin density $(v_1-b_1)/\pi$ (Figure 10.1, left). This is the image at u^2 of the factorisation $T_1\overset{d}{=}S+T_1^\circ$ of Proposition 9.1.11 into a positive stable variable of index one half and a GGC with Thorin density $(A+Q)/\pi$: since $\mathbb{E}e^{-u^2S}=e^{-b_1|u|}$, the Cauchy summand is $Z\sqrt{2S}$, and the EGGC summand is $\sqrt{2T_1^\circ}$ times an independent standard normal variable, with characteristic function $e^{b_1|u|}\xi(1)/\xi(1+$

$|u|) = \exp\{-\frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)(v_1-b_1)(\theta)\,d\theta\}$. Table 10.1 checks (10.1.2); at $\alpha = 0.75$ the formula presupposes that no zero has real part 0.75 or more.

u	Thorin form, $\alpha=1$	$\xi(1)/\xi(1+\lvert u\rvert)$	Thorin form, $\alpha=0.75$	$\xi(0.75)/\xi(0.75+\lvert u\rvert)$
0.5	0.982809	0.982838	0.988506	0.988520
1	0.954904	0.954930	0.965969	0.965980
2	0.871156	0.871171	0.891317	0.891317
5	0.507984	0.507962	0.536812	0.536769
20	0.0003491	0.0003487	0.0004116	0.0004110

Table 10.1: The Thorin form (10.1.2) against the reciprocal Xi function on the real axis, computed by `comp.py`.

10.1.3 Extended generalised gamma convolutions in the critical strip

Thorin introduced generalised gamma convolutions [319, 320] and extended them to laws on the whole line [321]. Bondesson [39, Section 7.1] calls the resulting class the extended generalised gamma convolutions, and Section 1.3.2 recalls it. A symmetric law without Gaussian part is an EGGC exactly when its characteristic function has the form

$$\exp\Big\{-\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)m(d\theta)\Big\}$$

with a positive measure m on $(0,\infty)$ that satisfies the integrability conditions of a Thorin measure in the chart $z=\theta^2$; under $z=\theta^2$ it is the Thorin measure of the mixing GGC T' in the representation $Z\sqrt{2T'}$ recalled before Theorem 10.1.2. Whether T_α is a GGC is decided by the location of the zeros and not by the sign of v_α on the single line $\operatorname{Re} s=\alpha$: for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

Theorem 10.1.3. *Let $\frac12<\alpha<1$ and write $\varepsilon=\alpha-\frac12$.*

1. *With no hypothesis, $\xi(\alpha)/\xi(\alpha+|u|)$ is the characteristic function of the infinitely divisible, symmetric, unimodal law Y_α. If every zero has real part less than α, then* (10.1.2) *holds with the positive density $v_\alpha(\theta)/\pi = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(\theta-\gamma)^2}$.*
2. *Y_α is an EGGC if and only if no zero has $\beta>\alpha$, that is, if and only if ζ has no zeros in the half plane $\operatorname{Re} s>\alpha$.*
3. *Let $T_0\ge 2$ be a height up to which all zeros are known to lie on the critical line. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \prod_{0<\gamma\le T_0}\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}\times \mathcal{E}_{\alpha,T_0}(u),$$

where the finite product is the characteristic function of an EGGC whose Thorin density $\sum_{0<\gamma\le T_0}[P_\varepsilon(\theta-\gamma)+P_\varepsilon(\theta+\gamma)]$ *is positive, and the remainder satisfies* $|\log \mathcal{E}_{\alpha,T_0}(u)| \le C(|u|+u^2)\frac{\log T_0}{T_0}$ *for an absolute constant* C.

Proof. (1) is Proposition 10.1.1 and Theorem 10.1.2, the sum being that of Proposition 1.2.3. (2) is part of Theorem 10.1.2: Y_α is an EGGC exactly when T_α is a GGC, and a zero with $\beta > \alpha$ puts a pole of φ_α at the point $(\rho-\alpha)^2$ of $\mathbb{C}\setminus(-\infty,0]$, where the Laplace transform of a GGC is holomorphic (Theorem 1.2.2). (3) Group the zeros into those on the line below T_0 and the rest. Each pair $\frac12 \pm i\gamma$ contributes the factor $\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}$, whose Thorin density is the pair of Cauchy kernels of width ε at $\pm\gamma$, by the computation after Theorem 10.1.2, and which is therefore an EGGC characteristic function. Every other zero has $|\gamma| > T_0$, and a pair $\rho, \bar\rho$ with $\gamma > T_0$ contributes $-\log(1+x)$ to $\log \mathcal{E}_{\alpha,T_0}(u)$, with $x = (2(\alpha-\beta)|u|+u^2)/((\alpha-\beta)^2+\gamma^2)$. Since $|\alpha-\beta| < 1$, we have $x \ge -(\alpha-\beta)^2/((\alpha-\beta)^2+\gamma^2) > -\frac12$ and $|x| \le 2(|u|+u^2)/\gamma^2$, so the contribution is at most $4(|u|+u^2)/\gamma^2$ in absolute value. Summing over the zeros with $\gamma > T_0$ gives at most $4(|u|+u^2)\sum_{\gamma>T_0}\gamma^{-2}$, and by partial summation this sum is at most $2\int_{T_0}^\infty N(t)t^{-3}\,dt$. For $t \ge 2$ we have $N(t) \le \frac{t}{2\pi}\log t$: no zero has $0<\gamma<\gamma_1$, and for $t \ge \gamma_1$ the explicit bound of [109], recalled in the proof of Proposition 9.1.11, gives $N(t) \le \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac78 + E(t)$, where $\frac78 + E(t) < \frac{t}{2\pi}\log(2\pi e)$: up to X_0 the left side is at most 3.40 and the right side is at least 6.38, its value at $t=\gamma_1$, and beyond X_0 the left side is less than $\frac78 + 0.3\log t$ and the right side exceeds $0.45\,t$. So $\sum_{\gamma>T_0}\gamma^{-2} \le \frac{1}{\pi}\int_{T_0}^\infty t^{-2}\log t\,dt = \frac{1+\log T_0}{\pi T_0} \le \frac{\log T_0}{T_0}$, the last step because $T_0 \ge 2$, and the bound holds with $C=4$. □

With the verification height $T_0 = 3\times10^{12}$ of Platt and Trudgian [247], which also enters Corollary 4.3.7, the remainder is of relative size at most 8×10^{-11} for $|u| \le 1$, so Y_α is, to that accuracy in characteristic function, an EGGC for every $\alpha > \frac12$, whether or not RH holds; Theorem 11.4.6 gives the same statement for the Laplace transform of T_α with explicit constants. Grouping each zero above T_0 with its reflection in the critical line makes the contribution of every off line pair nonnegative below $T_0-\frac12$, so $v_\alpha > 0$ on $[0, T_0-\frac12]$ for every $\alpha > \frac12$ (Corollary 11.4.7). A zero above T_0 with $\beta > \alpha$ would stop T_α from being a GGC, and Y_α from being an EGGC, whether or not it made v_α negative near its ordinate (Exercise 1.7), and no finite computation excludes such a zero (Section 3.1.1). What can be said with no hypothesis is in Theorem 11.4.2.

Table 10.2 checks the product over the computed zeros against $\xi(\alpha)/\xi(\alpha+|u|)$ at $\alpha = 0.6$ and 0.75. On $[0,200]$ the Thorin density is positive at both basepoints, and in both cases its minimum is at $\theta = 0$ and equals the Cauchy floor b_α/π: 0.0014709 at $\alpha = 0.6$ and 0.0036769 at $\alpha = 0.75$ (Figure 10.1, left).

α	u	Thorin integral (10.1.2)	product over zeros	$\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$
0.6	1	0.9726796	0.9726812	0.9726812
0.6	5	0.5549761	0.5549190	0.5549193
0.6	20	0.0004547	0.0004539	0.0004539
0.75	1	0.9659691	0.9659798	0.9659798
0.75	5	0.5368120	0.5367687	0.5367691
0.75	20	0.0004116	0.0004110	0.0004110

Table 10.2: The EGGC representation in the strip, computed by `strip.py`: the Thorin integral over $\theta \le 200$ plus the Weyl tail, and the product of the factors of Theorem 10.1.3(3) over the 1700 computed zeros plus the Weyl tail.

10.1.4 The density, the Lévy measure and the Poisson trace

Because $\xi(\alpha)/\xi(\alpha+|u|)$ is integrable, Y_α has the density

$$f_{Y_\alpha}(y) = \frac{1}{\pi}\int_0^\infty \cos(uy)\,\frac{\xi(\alpha)}{\xi(\alpha+u)}\,du,$$

which `dens2.py` computes. At the origin $f_{Y_\alpha}(0) = 1.8994,\ 1.8680,\ 1.8225,\ 1.7507$ for $\alpha = \frac12,\ 0.6,\ 0.75,\ 1$. The corner of the characteristic function at the origin gives the tail $f_{Y_\alpha}(y) \sim b_\alpha/(\pi y^2)$ for $\alpha > \frac12$ (Figure 10.1, middle). At the centre there is no corner, and the density decays like $e^{-\gamma_1|y|}$, the tail of L_1/γ_1 under RH; unconditionally, the characteristic function continues to the meromorphic function $\xi(\frac12)/\xi(\frac12+u)$ of u, whose poles nearest the real axis are $\pm i\gamma_1$.

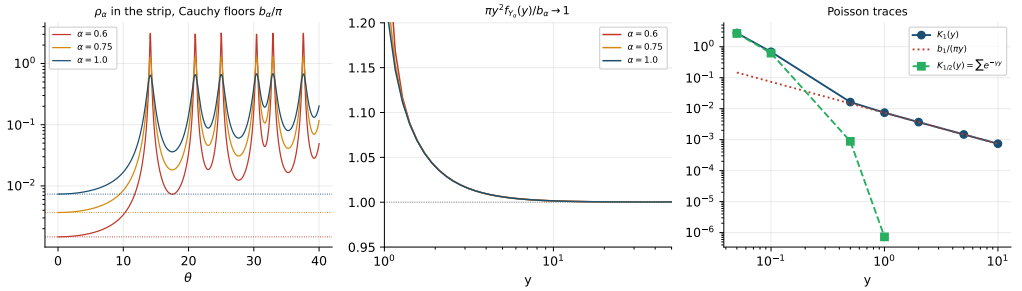


Figure 10.1: Left: the Thorin densities v_α/π at $\alpha = 0.6,\ 0.75,\ 1$, written ρ_α in the plot, and their Cauchy floors b_α/π (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces K_1 and $K_{1/2}$ against $b_1/(\pi y)$.

Frullani's integral gives $\log(1+u^2/\theta^2) = \int_{\mathbb{R}}(1-\cos uy)\,e^{-\theta|y|}\,\frac{dy}{|y|}$, which is the sine squared lemma read in the other direction, since $1-\cos uy = 2\sin^2(uy/2)$, and which is the cosine dictionary (6.2.4) of Section 6.2. Inserting it in (10.1.2)

puts the characteristic function of Y_α in Lévy Khintchine form in u:

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{ -\int_{\mathbb{R}} 2\sin^2\Big(\frac{uy}{2}\Big)\,\frac{K_\alpha(|y|)}{|y|}\,dy\Big\},$$
$$K_\alpha(y) = \frac{1}{\pi}\int_0^\infty e^{-\theta y} v_\alpha(\theta)\,d\theta.$$

So the Lévy density of Y_α is $K_\alpha(|y|)/|y|$, as Section 1.3.2 gives for a symmetric EGGC; in the normalisation of (6.2.4), $K_\alpha = \frac{1}{2}K_{U_\alpha}$. We call K_α the Poisson trace, because at the centre, where under RH the Thorin measure in the chart θ is $\sum_{\gamma>0}\delta_\gamma$, it is $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$, half of Cramér's function [65, Chapter I] of Proposition 6.2.8(iii): the sum over the zeros of the Poisson kernel in place of the heat kernel of the heat trace $W(t) = \sum_k e^{-a_k t}$, which is $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH. Since $Y_{1/2} = B_{2T}$ for an independent Brownian motion B, and T has Lévy density $W(t)/t$ (Theorem 5.1.1), the Lévy density of $Y_{1/2}$ is $\int_0^\infty (4\pi t)^{-1/2} e^{-y^2/4t}\, W(t)\, t^{-1} dt$, so

$$K_{1/2}(y) = \int_0^\infty \frac{y}{\sqrt{4\pi t^3}}\, e^{-y^2/4t}\, W(t)\,dt = \mathbb{E}\, W(H_{y/\sqrt{2}}), \qquad y>0.$$

The Poisson trace is the heat trace averaged over the first passage time of Brownian motion to the level $y/\sqrt{2}$, that is, subordinated by the stable one half first passage kernel, whose Laplace transform $e^{-y\sqrt{a}}$ is the stable one half average of the heat kernel recalled after Proposition 6.2.9. This defines $K_{1/2}$ unconditionally, as $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ with the principal root, and makes it computable from the explicit formula (1.2.5) for W.

Corollary 10.1.4. *RH holds if and only if the Poisson trace $K_{1/2}$ is completely monotone, and if and only if the heat trace W is completely monotone. For $\alpha \geq 1$ the Poisson trace K_α is completely monotone with no hypothesis, and $K_\alpha(y) \sim b_\alpha/(\pi y)$ as $y\to\infty$, so that the Lévy density of Y_α is asymptotic to $b_\alpha/(\pi y^2)$, the Lévy density of the Cauchy component.*

Proof. For $\alpha \geq 1$, K_α is the Laplace transform of the positive density v_α/π, hence completely monotone by Bernstein's theorem, and since v_α is continuous with $v_\alpha(0) = b_\alpha$, $yK_\alpha(y) \to b_\alpha/\pi$ as $y\to\infty$. At the centre, W is completely monotone exactly when T is a GGC (Section 1.2), that is exactly when RH holds (Proposition 1.2.1). Under RH, $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$ is completely monotone. Conversely, suppose that $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure ν. For $w>0$ the unconditional formula $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ gives

$$\int_0^\infty e^{-wy}\, y\, K_{1/2}(y)\,dy = \sum_k \frac{1}{(w+\sqrt{a_k})^2} = \int_{[0,\infty)} \frac{\nu(d\theta)}{(w+\theta)^2}.$$

The first equality integrates term by term, which is legitimate because $\sqrt{a_k} = \gamma_k - i(\beta_k - \frac{1}{2})$ has real part γ_k when $\rho_k = \beta_k + i\gamma_k$ is taken with $\gamma_k > 0$, so the absolute values of the terms integrate to $\sum_k (w+\gamma_k)^{-2} < \infty$. The second is Tonelli's theorem, and it shows that the right side is finite for $w > 0$. The middle expression is meromorphic in $w \in \mathbb{C}$, with a double pole at each point $-\sqrt{a_k}$, while the right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $|w+\theta|$ is bounded below by a positive multiple of $1+\theta$ for w in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. The two sides agree for $w > 0$, hence on $\mathbb{C} \setminus (-\infty, 0]$ with the poles removed, so no pole $-\sqrt{a_k}$ lies in $\mathbb{C} \setminus (-\infty, 0]$. Since $\operatorname{Re}\sqrt{a_k} > 0$, every $\sqrt{a_k}$ is therefore real and positive, that is, every $a_k = -(\rho_k - \frac{1}{2})^2$ is positive, which is RH. □

The traces K_1 and $K_{1/2}$ are drawn in Figure 10.1, right.

10.1.5 The complete monotonicity conditions of the Poisson trace

By Corollary 10.1.4, RH is the complete monotonicity of the Poisson trace $K_{1/2}$. This subsection follows the first author's note on Biane's paper [254]. It compares $K_{1/2}$ with the corresponding trace of Pólya's function $P(\theta) = K_{i\theta/2}(2\pi)$ of Section 8.7, which is completely monotone with no hypothesis.

Write the zeros as $\rho_k = \beta_k + i\gamma_k$ with $\gamma_k > 0$, so that $\sqrt{a_k} = \gamma_k + i\delta_k$ with $\delta_k = \frac{1}{2} - \beta_k \in (-\frac{1}{2}, \frac{1}{2})$, and put $\varphi_k = \arctan(\delta_k/\gamma_k)$. The series $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ converges absolutely, with all its derivatives, for $y > 0$, and the zeros ρ_k and $1-\bar{\rho}_k$ have conjugate values of $\sqrt{a_k}$. Hence

$$(-1)^n K_{1/2}^{(n)}(y) = \sum_k \operatorname{Re}\left((\sqrt{a_k})^n e^{-y\sqrt{a_k}}\right) = \sum_k |\sqrt{a_k}|^n \, e^{-\gamma_k y} \cos(n\varphi_k - \delta_k y), \tag{10.1.3}$$

with each zero counted once, with its multiplicity; a zero on the line contributes $\gamma_k^n e^{-\gamma_k y} > 0$. So $K_{1/2}$ is completely monotone, and by Corollary 10.1.4 RH holds, exactly when every one of the quantities (10.1.3) is nonnegative, for $n = 0, 1, 2, \ldots$ and $y > 0$.

Proposition 10.1.5. *The term of a zero with $\delta_k \neq 0$ in* (10.1.3) *is positive whenever*

$$n \arctan \frac{|\delta_k|}{\gamma_k} + |\delta_k| \, y < \frac{\pi}{2}.$$

If $\delta_k > 0$ and $\delta_k y < \frac{\pi}{2}$, the first order n at which the term is negative is the least integer with $n\varphi_k - \delta_k y > \frac{\pi}{2}$, that is the least integer above $(\frac{\pi}{2} + \delta_k y)/\varphi_k$ when this ratio is not an integer. The partner $1 - \bar{\rho}_k$ has the same term.

Proof. $|n\varphi_k - \delta_k y| \le n|\varphi_k| + |\delta_k| y$, and the cosine is positive on $(-\frac{\pi}{2}, \frac{\pi}{2})$. For $\delta_k > 0$ the argument $n\varphi_k - \delta_k y$ increases with n from $-\delta_k y \in (-\frac{\pi}{2}, 0)$, and the cosine is first negative when it passes $\frac{\pi}{2}$. The partner has $-\delta_k$ and $-\varphi_k$, so its argument is the negative of this one. □

`biane_traces.py` finds the first negative order by evaluating the sign factor; Table 10.3 compares it with the formula, and the note's values are the same.

γ	δ	y	first negative order	$(\frac{\pi}{2}+\delta y)/\varphi$
50	0.25	1	365	364.16
50	0.25	0.3	330	329.16
200	0.1	1	3342	3341.59
14.13	0.5	0.5	52	51.48

Table 10.3: The first order at which the term of a hypothetical zero $\frac{1}{2}-\delta+i\gamma$ in (10.1.3) is negative, against Proposition 10.1.5. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed by `biane_traces.py`.

THEOREM 10.1.6. *Suppose that every zero with $0<\gamma_k\le H$ lies on the critical line and that every zero has $|\delta_k|\le\Delta$, with $0<\Delta\le\frac{1}{2}$. Then for $0<y<\pi/(2\Delta)$ and every integer n with $0\le n<\pi H/(2\Delta)$, every term of* (10.1.3) *is positive, and $(-1)^nK^{(n)}_{1/2}(y)>0$. Since $0<\beta_k<1$, $\Delta=\frac{1}{2}$ is admissible with no hypothesis, and with $H=T_0=3\times10^{12}$ [247] this covers $0<y<\pi$ and every $n<\pi T_0=9.42\times10^{12}$.*

Proof. A zero with $\gamma_k\le H$ has $\delta_k=0$, and its term $\gamma_k^n e^{-\gamma_k y}$ is positive. A zero with $\gamma_k>H$ has φ_k of the sign of δ_k, $n|\varphi_k|\le n|\delta_k|/\gamma_k\le n\Delta/H<\frac{\pi}{2}$ and $|\delta_k|y\le\Delta y<\frac{\pi}{2}$. So $n\varphi_k-\delta_k y$, a difference of two numbers of the same sign and of absolute value less than $\frac{\pi}{2}$, lies in $(-\frac{\pi}{2},\frac{\pi}{2})$, and the cosine in (10.1.3) is positive. The series converges absolutely, so its sum is positive. □

Every condition that could fail has $y\ge\pi$ or $n\ge\pi H$, which is above 9.4×10^{12}; there, complete monotonicity is a characterisation of RH, not a test that a computation could run to a failure.

REMARK 10.1.7. The rectangle uses that φ_k and δ_k have the same sign, which the sum bound of Proposition 10.1.5 discards; the observation is from the first author's note on Ostrovsky's laws and the quasi Riemann hypothesis. Under $(\mathrm{Q}_{7/8})$ of Section 16.12, which rests on the unrefereed half plane of [226], $\Delta=\frac{3}{8}$ and the theorem holds for $0<y<\frac{4\pi}{3}=4.1888$ and $n<\frac{4\pi}{3}T_0=1.26\times10^{13}$. Liu's claimed half plane $\operatorname{Re}s>0.874957$ [192], also unrefereed, would give $y<4.1893$. The half plane at $\frac{7}{8}$ changes the constants by the factor $\frac{4}{3}$ and nothing else.

The weight of the zero γ_k in the condition (n,y) is about $\gamma_k^n e^{-\gamma_k y}$, whose logarithm $n\log\gamma-\gamma y$ is largest at

$$\gamma_*=\frac{n}{y},$$

with a width of order $\sqrt{n}/y$. The condition (n, y) is therefore a smoothed probe of the zeros near the height n/y. Among the first 300 ordinates the dominant one is, in every case tested, the ordinate nearest to n/y, for instance 98.831 for $(n, y) = (100, 1)$ (`biane_traces.py`). Figure 10.2 shows two weight profiles and two sign factors. By Proposition 10.1.5 a zero at height γ can make its term negative only when $n \gtrsim \frac{\pi}{2}\gamma/|\delta| \geq \pi\gamma$, that is when $\gamma \lesssim n/\pi$, below the peak at n/y when $y < \pi$; and since $\gamma^n e^{-\gamma y}$ increases up to n/y, its value at $\gamma \leq n/\pi$ is at most $\exp(n(\log \frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its value at the peak, which is exponentially small in n. Turning this comparison into a proof would need a bound for the number of zeros off the line near each height.

For the whole sum the first failure lies far above the termwise order. We replaced two adjacent ordinates among the first 500 by a pair $\frac{1}{2} \pm \delta + i\bar{\gamma}$ at their midpoint, and located by bisection over integers the order n^* at which the minimum of $(-1)^n K_{1/2}^{(n)}(y)$ over $y \in [0.85, 1.15]\, n/\bar{\gamma}$ first becomes negative; the bisection assumes that the sign is monotone in n, and at sampled lower orders no negative value occurs for $y \leq 30$ or $y \leq 5n/\bar{\gamma}$. For $\bar{\gamma} = 54.7$, 125.9, 272.5 and $\delta = 0.1$, 0.375 the ratio $n^*\delta/\bar{\gamma}^2$ lies between 0.63 and 1.19, the law $n^* \approx \gamma^2/\delta$ of the heat trace (Table 7.6), against the termwise order $\pi\bar{\gamma}/(2\delta)$ near $y = 0$.

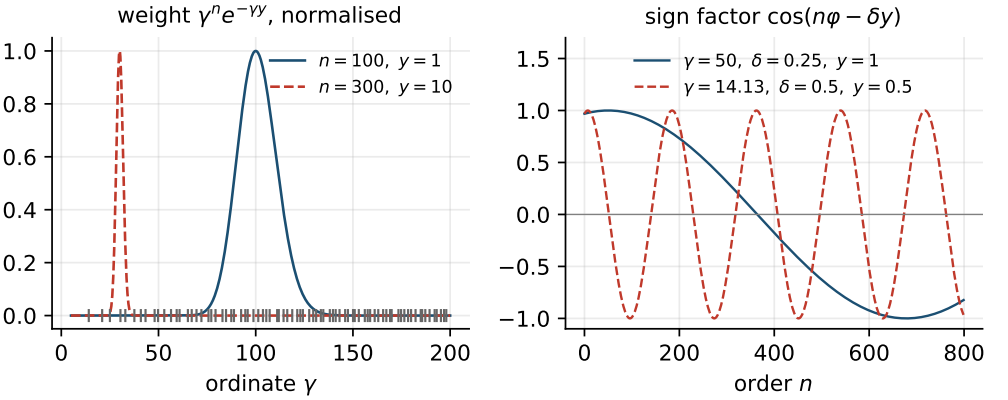


Figure 10.2: Left: the weight $\gamma^n e^{-\gamma y}$ of the condition (n, y) in (10.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52, at which it is first negative. Drawn by `biane_figs.py`.

Proposition 10.1.8. *The following are equivalent:*

(i) RH;

(ii) $K_{1/2}$ is completely monotone on $(0, \infty)$;

(iii) for every $x > 0$ the function $y \mapsto K_{1/2}(x+iy)$ is positive definite on $\mathbb{R}$;

(iv) for some $x > 0$ the function $y \mapsto K_{1/2}(x+iy)$ is bounded on $[0,\infty)$.

Under these conditions $|K_{1/2}(x+iy)| \le K_{1/2}(x)$.

Proof. The series $K_{1/2}(z) = \sum_k e^{-z\sqrt{a_k}}$ converges absolutely and locally uniformly for $\operatorname{Re} z > 0$, since $|e^{-z\sqrt{a_k}}| = e^{-\gamma_k \operatorname{Re} z + \delta_k \operatorname{Im} z} \le e^{-\gamma_k \operatorname{Re} z + |\operatorname{Im} z|/2}$, and defines an analytic function there. (i) implies (ii) by Corollary 10.1.4. If (ii) holds, Bernstein's theorem gives $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure ν, the integral converges for $\operatorname{Re} z > 0$ and agrees there with $K_{1/2}(z)$ by analytic continuation, and $K_{1/2}(x+iy) = \int e^{-iy\theta}e^{-x\theta}\nu(d\theta)$ is the Fourier transform of the finite positive measure $e^{-x\theta}\nu(d\theta)$, which gives (iii) and the bound. A positive definite function f satisfies $|f(y)| \le f(0)$, so (iii) implies (iv). Suppose (iv) holds at x. For $\operatorname{Re} w > \frac12$ the bound above allows term by term integration, and

$$\int_0^\infty e^{-wy} K_{1/2}(x+iy)\,dy = \sum_k \frac{e^{-x\sqrt{a_k}}}{w + i\sqrt{a_k}}.$$

The right side is a meromorphic function of $w \in \mathbb{C}$, since $\sum_k e^{-x\gamma_k} < \infty$, with a pole at each point $-i\sqrt{a_k} = \delta_k - i\gamma_k$, of residue $e^{-x\sqrt{a_k}}$ times the multiplicity, which is not zero. Since $K_{1/2}(x+iy)$ is bounded, the left side is analytic in $\operatorname{Re} w > 0$, so no pole has $\delta_k > 0$. The zeros ρ_k and $1-\bar\rho_k$ have opposite δ_k, so every δ_k is 0, which is RH. □

Written out, (iii) says that for every $x > 0$, all real $y_1, \dots, y_m$ and all complex $c_1, \dots, c_m$,

$$\sum_{j,l} c_j \bar c_l\, K_{1/2}\big(x + i(y_j - y_l)\big) = \sum_k e^{-x\gamma_k} \Big| \sum_j c_j e^{-i\gamma_k y_j} \Big|^2 \ge 0,$$

the equality holding under RH.

One would like to turn each condition into a statement about primes with the explicit formula of Section 1.2, the display before (1.2.5). The test function of the condition (n, y) is $h(r) = |r|^n e^{-y|r|}$, with

$$g(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r) e^{-irx}\,dr = \frac{n!}{\pi} \operatorname{Re} \frac{1}{(y+ix)^{n+1}},$$

which decays only like a power of $|x|$ and has one sign for large $|x|$. The prime side $-2\sum_{m\ge2} \Lambda(m) m^{-1/2} g(\log m)$ then diverges, since $\sum_p p^{-1/2}(\log p)^{-j} = \infty$ for every j; the formula holds for g decaying faster than $e^{-|x|/2}$, which goes with h analytic in a strip $|\operatorname{Im} r| < \frac12 + \epsilon$, and this h is not even analytic at $r = 0$. Smoothing in the variable of the zeros removes the divergence without changing

the problem: the Gaussian weight e^{-r^2t} in place of $e^{-y|r|}$ gives the heat trace $W(t)$, whose explicit formula (1.2.5) converges absolutely, and whose complete monotonicity is again equivalent to RH (Corollary 10.1.4). The note [254] identifies this smoothing with the de Bruijn Newman flow, but the Gaussian weight acts on the test function, while the flow multiplies the kernel Φ by $e^{tu^2/4}$ and moves the zeros.

By Proposition 8.7.13,

$$\log\frac{K_{\lambda/2}(2\pi)}{K_0(2\pi)} = \int_0^\infty 4\sin^2\Big(\frac{\lambda y}{2}\Big)K^P(y)\,\frac{dy}{y},$$
$$K^P(y) = \sum_n e^{-2\nu_n y} = \operatorname{tr} e^{-2y\sqrt{A_P}},$$

the sum over the zeros $\theta_n = 2\nu_n$ of P, so K^P is the counterpart of $K_{1/2}$ in the formula for $\xi(\frac12)/\xi(\frac12+u)$, and it is completely monotone because A_P is selfadjoint and positive.

Proposition 10.1.9. *Let* $M(y) = \frac{1}{2\pi y}\big(\log\frac{1}{2\pi y} - \gamma_E\big)$ *and* $c = 2\pi - \frac{1}{6\pi} - \frac{1}{48\pi} = 6.2235022035$. *As* $y \to 0+$:

(i) $K_{1/2}(y) = M(y) + \frac78 + O(y\log\frac1y)$, *and under RH* $K_{1/2}(y) = M(y) + \frac78 + \frac{1}{48\pi}y\log\frac1y + o(y\log\frac1y)$;

(ii) $K^P(y) = M(y) - \frac14 + \big(2\pi - \frac{1}{6\pi}\big)y\log\frac1y + O(y)$;

(iii) $D(y) = K_{1/2}(y) - K^P(y) = \frac98 + O(y\log\frac1y)$, *and under RH* $D(y) = \frac98 - c\,y\log\frac1y + o(y\log\frac1y)$.

Proof. (i) The real part of the term $e^{-y\sqrt{a_k}}$ is $e^{-\gamma_k y}\cos(\delta_k y) = e^{-\gamma_k y} + O(y^2e^{-\gamma_k y})$, and the imaginary parts cancel in pairs, so $K_{1/2}(y) = \sum_k e^{-\gamma_k y} + O(y^2\sum_k e^{-\gamma_k y})$, and the sum will turn out to be $O(y^{-1}\log\frac1y)$. By Abel summation $\sum_k e^{-\gamma_k y} = y\int_0^\infty e^{-Ty}N(T)\,dT$, with $N(T) = \vartheta(T)/\pi + 1 + S(T)$ [324, Section 9.3]. The Riemann Siegel function satisfies $\vartheta(T)/\pi = N_0(T) - \frac18 + \frac{1}{48\pi T} + O(T^{-3})$ with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$, so $N = N_0 + \frac78 + S + E$ with $E(T) = \frac{1}{48\pi T} + O(T^{-3})$ for $T \ge 1$ and E bounded on $(0,1]$. Now $\int_0^\infty Te^{-Ty}\log T\,dT = y^{-2}(1 - \gamma_E - \log y)$ gives $y\int_0^\infty e^{-Ty}N_0(T)\,dT = M(y)$; the constant gives $\frac78$; $y\int_1^\infty e^{-Ty}\frac{dT}{48\pi T} = \frac{1}{48\pi}y\log\frac1y + O(y)$, and the rest of E gives $O(y)$. With $S_1(T) = \int_0^T S(t)\,dt$, an integration by parts gives $y\int_0^\infty e^{-Ty}S(T)\,dT = y^2\int_0^\infty e^{-Ty}S_1(T)\,dT$, which is $O(y\log\frac1y)$ since $S_1(T) = O(\log T)$ [324, Theorem 9.9(A)]. Under RH there are no terms $O(y^2e^{-\gamma_k y})$, and $S_1(T) = O(\log T/(\log\log T)^2)$ [324, Theorem 14.13] makes the last integral $o(y\log\frac1y)$. (ii) By Proposition 8.7.8, the number of zeros $\theta_n \le T$ is $\lfloor x(T)\rfloor$, with $x(T) = N_0(T) + \frac14 + (2\pi - \frac{1}{6\pi})T^{-1} + O(T^{-3})$ for $T \ge 1$. Write $\lfloor x\rfloor = x - \frac12 + (\frac12 - \{x\})$. The terms N_0, $\frac14 - \frac12$ and $(2\pi - \frac{1}{6\pi})T^{-1}$ give $M(y)$, $-\frac14$

and $(2\pi - \frac{1}{6\pi}) y \log \frac{1}{y} + O(y)$ as in (i). The sawtooth $\frac{1}{2} - \{x(T)\}$ has integral over each interval on which x increases by one equal to $O(T^{-1}(\log T)^{-3})$, because such an interval has length about $2\pi/\log T$ and $x'(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} + O(T^{-2})$ varies over it by $O(T^{-1}(\log T)^{-2})$ relative to its size, so its integral from 0 to T is bounded, and another integration by parts makes its contribution $O(y)$. (iii) is (i) minus (ii). □

`biane_traces.py` tests the proposition. It takes the first 30000 ordinates from Arb, up to 25755.348, cuts at $G = 25754.905$, midway between the last two, and adds the tail $\int_G^\infty e^{-Ty}\vartheta'(T)\,dT/\pi - e^{-Gy}S(G)$, with $S(G) = -0.569$ exact; the neglected term $y\int_G^\infty e^{-Ty}S(T)\,dT$ equals $y^2\int_G^\infty e^{-Ty}\int_G^T S(t)\,dt\,dT$, and Trudgian's bound $|\int_{t_1}^{t_2} S(t)\,dt| \le 2.067 + 0.059\log t_2$ for $t_2 > t_1 > 168\pi$ [327], with Jensen's inequality for the logarithm, bounds it by $ye^{-Gy}(0.059\log(G + 1/y) + 2.067)$; the script records twice this bound, 4.1×10^{-5} and 4.3×10^{-5} at $y = 10^{-4}$ and 10^{-5}, so the entries of Table 10.4 at $y = 10^{-4}$ and 10^{-5} may be off by up to two units in the fifth decimal. The zeros of P come from the phase condition of Proposition 8.7.8, solved by Newton's method for all 30132 zeros to just above the cut, with the tail beyond G added. The factor e^{-Gy} that weights the tails is 0.076 at $y = 10^{-4}$ and 0.77 at $y = 10^{-5}$, so the last row of Table 10.4 rests mostly on the smooth tails. The constant $\frac{9}{8}$ appears clearly, and D approaches it along $\frac{9}{8} - cy\log\frac{1}{y}$.

y	$K_{1/2}(y)$	$K^P(y)$	$K_{1/2} - M$	$K^P - M$	$D(y)$	$\frac{9}{8} - cy\log\frac{1}{y}$
0.05	2.686015	2.230883	0.837782	0.382650	0.455132	0.192803
0.02	12.772471	12.033884	0.860279	0.121691	0.738587	0.638070
0.01	35.723862	34.836495	0.867696	−0.019670	0.887366	0.838397
0.005	92.647265	91.663043	0.871374	−0.112848	0.984222	0.960130
0.003	180.932794	179.901902	0.872836	−0.158057	1.030892	1.016540
0.001	715.903738	714.816981	0.874286	−0.212471	1.086757	1.082010
10^{-4}	10815.847446	10814.727704	0.874931	−0.244811	1.119742	1.119268
10^{-5}	144797.380092	144796.255761	0.874994	−0.249338	1.124332	1.124283

Table 10.4: The Poisson trace $K_{1/2}$ of the zeros of ζ, Pólya's trace K^P, their distances from the common main term M, which tend to $\frac{7}{8}$ and $-\frac{1}{4}$, and their difference D, which tends to $\frac{9}{8}$, against the prediction of Proposition 10.1.9 under RH. Computed by `biane_traces.py`.

The note [254] states that $K_{1/2}(y) = M(y) + O(\log\frac{1}{y})$, which is true but loses the constant, and it reports $D(y)/\log\frac{1}{y} = 0.195,\ 0.194,\ 0.193,\ 0.193$ at $y = 0.003,\ 0.005,\ 0.01,\ 0.02$, from which it concludes that D grows like $0.19\log\frac{1}{y}$. The accurate values of $D/\log\frac{1}{y}$ in this range, 0.177, 0.186, 0.193, 0.189, are also close to 0.19, but that is a coincidence of the range: by Proposition 10.1.9, D tends to $\frac{9}{8}$, and $D/\log\frac{1}{y}$ is 0.098 at $y = 10^{-5}$.

K^P is completely monotone for a structural reason, as the trace of the semigroup $e^{-2y\sqrt{A_P}}$. By Corollary 10.1.4, RH says that $K_{1/2} = K^P + D$ is completely monotone as well, where D is bounded, tends to $\frac{9}{8}$ while both traces grow like $\frac{1}{2\pi y}\log\frac{1}{y}$, and carries the whole difference between Φ and Pólya's kernel, the arithmetic included. Complete monotonicity constrains the derivatives of every order, and no bound on the size of D can decide it.

10.1.6 Wiener's theorem and the basepoint kernels

The zero free half planes have a second reading, as completeness statements, through Wiener's closure theorem [341], for which Kac [134] and Korevaar [154] gave short alternative treatments: the translates $f(\cdot - x)$, $x \in \mathbb{R}$, of a function $f \in L^1(\mathbb{R})$ span a dense subspace of $L^1(\mathbb{R})$ if and only if its Fourier transform $\hat{f}(\theta) = \int f(u)e^{i\theta u}du$ has no real zeros.

Let Φ be Pólya's kernel (1.2.3), so that $\xi(\frac{1}{2}+s) = \int_{\mathbb{R}} \Phi(u)e^{su}du$ for every complex s by (1.2.4), and for $\alpha \geq \frac{1}{2}$ put

$$\Phi_\alpha(u) = e^{(\alpha-\frac{1}{2})u}\,\Phi(u), \qquad \hat{\Phi}_\alpha(\theta) = \int_{\mathbb{R}} \Phi_\alpha(u)e^{i\theta u}du = \xi(\alpha + i\theta). \tag{10.1.4}$$

Up to the factor $\xi(\alpha)$, Φ_α is the density of the tilted Pólya variable V_α of Section 9.1.1, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and it is integrable for every α because Φ decays doubly exponentially. The real zeros of $\hat{\Phi}_\alpha$ are exactly the ordinates of the zeros of ζ on the line $\operatorname{Re} s = \alpha$, so Wiener's theorem gives the following.

Proposition 10.1.10. *1. The translates of Φ_α are dense in $L^1(\mathbb{R})$ if and only if ζ has no zeros on the line $\operatorname{Re} s = \alpha$.*

2. At the centre they are not dense. For every finite combination $h = \sum_k c_k\Phi(\cdot - x_k)$ and every $g \in L^1(\mathbb{R})$,

$$\|g - h\|_1 \;\geq\; \sup_\gamma |\hat{g}(\gamma)|,$$

the supremum over the ordinates of the zeros on the critical line.

3. At the basepoint one they are dense, with no hypothesis, because $\xi(1+i\theta) \neq 0$.

4. For $\alpha_0 \in [\frac{1}{2}, 1)$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha_0$ if and only if the translates of Φ_α are dense in $L^1(\mathbb{R})$ for every $\alpha > \alpha_0$. In particular RH holds if and only if they are dense for every $\alpha > \frac{1}{2}$.

5. If the translates of Φ_α are dense, approximation near the zeros is expensive: if $h = \sum_k c_k\Phi_\alpha(\cdot - x_k)$ and $\|g-h\|_1 \leq \delta$, then $\sum_k |c_k| \;\geq\; (|\hat{g}(\gamma)| - \delta)/|\xi(\alpha + i\gamma)|$ for every ordinate γ.

Proof. (1), (3) and (4) are Wiener's theorem applied to $\hat{\Phi}_\alpha(\theta) = \xi(\alpha + i\theta)$, together with the nonvanishing of ζ on the line $\operatorname{Re} s = 1$. For (2) and (5), $\hat{h}(\theta) = \hat{\Phi}_\alpha(\theta) \sum_k c_k e^{i\theta x_k}$, so at the centre $\hat{h}(\gamma) = 0$ at every ordinate of a zero on the line, while $|\hat{g}(\theta) - \hat{h}(\theta)| \le \|g - h\|_1$ for every θ; for (5), $|\hat{h}(\gamma)| \le |\xi(\alpha + i\gamma)| \sum_k |c_k|$. $\square$

Wiener's proof of the prime number theorem runs through statement (3), and the proposition shows the whole family interpolating between that theorem at $\alpha = 1$ and RH at $\alpha = \frac{1}{2}$. It is a restatement rather than a new criterion. By statement (4), completeness at every $\alpha' > \alpha$ is the condition $\alpha \ge \sup_\rho \operatorname{Re} \rho$ under which T_α is a GGC (Theorem 1.2.2) and Y_α an EGGC (Theorem 10.1.2), and, by Lagarias's argument [160], it is also the positivity of $v_{\alpha'}$ for every $\alpha' > \alpha$ (Remark 1.2.4). What the completeness form adds is an explicit, computable obstruction at the centre, statement (2), and an explicit cost of approximation above it, statement (5): near the centre $|\xi(\alpha + i\gamma)| \approx (\alpha - \frac{1}{2})|\xi'(\frac{1}{2} + i\gamma)|$ at the ordinate γ of a simple zero, so the coefficients needed to reproduce $\hat{g}$ at an ordinate grow like $1/(\alpha - \frac{1}{2})$.

We tested the dichotomy with `wiener.py`. We approximated the Gaussian $g(u) = (2\pi\kappa^2)^{-1/2} e^{-u^2/2\kappa^2}$ with $\kappa = 0.15$, for which $\hat{g}(\gamma_1) = e^{-\kappa^2\gamma_1^2/2} = 0.10565$, by combinations of N equally spaced translates of Φ_α on $[-1.5, 1.5]$, minimising the L^1 error on $[-3, 3]$ exactly by linear programming. Table 10.5 shows the results. At the centre every approximant has $\hat{h}(\gamma_1) = 0$ up to rounding, the computed values staying below 1.2×10^{-8} while the coefficient sums grow to 4×10^9, and the best error stalls at 0.1345, above the barrier 0.1056 of statement (2), however many translates are used. Above the centre the error falls below the value at which the centre stalls and keeps decreasing, to below the barrier at $\alpha = 0.75$ and $\alpha = 1$. The remaining error is limited by conditioning rather than by completeness. With 80 translates the coefficient sums reach 3×10^7 to 2×10^8 above the centre, as statement (5) and the rapid decay of $\hat{\Phi}_\alpha$ require, and double precision linear programming cannot go much further. The closer α is to $\frac{1}{2}$, the larger the coefficients and the slower the approach, which is the completeness counterpart of the Cauchy kernels sharpening onto the zeros.

N	$\alpha = \frac{1}{2}$	$\alpha = 0.6$	$\alpha = 0.75$	$\alpha = 1$
10	0.4705	0.4698	0.4664	0.4540
20	0.1345	0.1221	0.1060	0.0852
40	0.1345	0.1152	0.0907	0.0614
80	0.1365	0.1139	0.0878	0.0570
$\lvert\hat{h}(\gamma_1)\rvert$ at $N = 80$	0	0.0167	0.0377	0.0635

Table 10.5: Best L^1 error in approximating a Gaussian of width 0.15 by N translates of Φ_α, computed by `wiener.py`. At the centre the error cannot fall below $|\hat{g}(\gamma_1)| = 0.1056$, and $\hat{h}(\gamma_1)$ vanishes exactly; the computed value is 1.2×10^{-8}.

10.1.7 Salem's criterion and its Pólya analogue

Wiener's theorem has a dual form: for $k \in L^1(\mathbb{R})$, the convolution equation $\int k(x-y)f(y)\,dy = 0$ for all x has a bounded solution $f \neq 0$ if and only if $\hat{k}$ has a real zero. Salem [294] used it to give one of the classical equivalents of RH: RH holds if and only if, for every $\frac{1}{2} < \sigma < 1$, the equation

$$\int_{\mathbb{R}} \frac{e^{-\sigma y} f(y)}{e^{e^{x-y}} + 1}\, dy = 0 \qquad (x \in \mathbb{R})$$

has no bounded solution other than $f = 0$. Multiplied by $e^{\sigma x}$, it is the convolution equation with the kernel $k_\sigma(u) = e^{\sigma u}/(e^{e^u}+1)$, the Fermi Dirac function $1/(e^w+1)$ in the logarithmic variable $w = e^u$, and that substitution gives its Fourier transform,

$$\hat{k}_\sigma(\theta) = \int_{\mathbb{R}} \frac{e^{\sigma u} e^{i\theta u}}{e^{e^u}+1}\, du = \int_0^\infty \frac{w^{\sigma+i\theta-1}}{e^w+1}\, dw = \Gamma(\sigma+i\theta)\,\eta(\sigma+i\theta),$$
$$\eta(s) = (1-2^{1-s})\zeta(s),$$

where η is Dirichlet's alternating zeta function, not the Dedekind eta function of Section 11.3. So for $\frac{1}{2} < \sigma < 1$ the real zeros of $\hat{k}_\sigma$ are exactly the ordinates of the zeros of ζ on the line $\operatorname{Re} s = \sigma$, and Salem's criterion is Proposition 10.1.10(4) with the Fermi Dirac kernel in place of the tilted Pólya kernel (10.1.4).

Corollary 10.1.11. *RH holds if and only if, for every $\frac{1}{2} < \alpha < 1$, the equation $\int_{\mathbb{R}} \Phi_\alpha(x-y)f(y)\,dy = 0$ for all x has no bounded solution $f \neq 0$. At the centre the equation has the explicit bounded solutions $f(y) = \cos(\gamma y)$, one for each ordinate γ of a zero on the critical line, because $\Xi(\gamma) = 0$ there; at the basepoint one it has none.*

Proof. By the dual form of Wiener's theorem with $k = \Phi_\alpha$, a bounded solution $f \neq 0$ exists exactly when $\hat{\Phi}_\alpha(\theta) = \xi(\alpha+i\theta)$ has a real zero, that is, when ζ has a zero on the line $\operatorname{Re} s = \alpha$. That no such zero exists for any $\alpha \in (\frac{1}{2}, 1)$ means that no zero has $\frac{1}{2} < \operatorname{Re}\rho < 1$, which, since no zero lies on the line $\operatorname{Re} s = 1$, is RH by the symmetry $\rho \mapsto 1-\rho$. At the centre Φ is even, so $\int_{\mathbb{R}} \Phi(x-y)\cos(\gamma y)\,dy = \Xi(\gamma)\cos(\gamma x)$, which vanishes when $\frac{1}{2}+i\gamma$ is a zero; at the basepoint one $\xi(1+i\theta) \neq 0$ for every real θ. □

The factor $1-2^{1-s}$ in η vanishes at $s = 1 + 2\pi i k/\log 2$ with $k \neq 0$, so Salem's kernel has real Fourier zeros on the line $\sigma = 1$, at $\theta = 9.0647, 18.1294, 27.1942, \ldots$, and its translates are not complete there, whereas the Pólya family is complete at $\alpha = 1$ because $\xi(1+i\theta) \neq 0$. The left panel of Figure 10.3, computed by `salem_riesz.py`, shows $|\eta(\sigma+i\theta)|$. At $\sigma = \frac{1}{2}$ it dips to zero at every ordinate, and at $\sigma = 0.6$ and 0.75 the dips are shallow. The depth of the dips plays for Salem's kernel the role that $|\xi(\alpha+i\gamma)|$ plays in the cost of approximation of Proposition 10.1.10(5), and it is of order $\sigma - \frac{1}{2}$ near each ordinate.

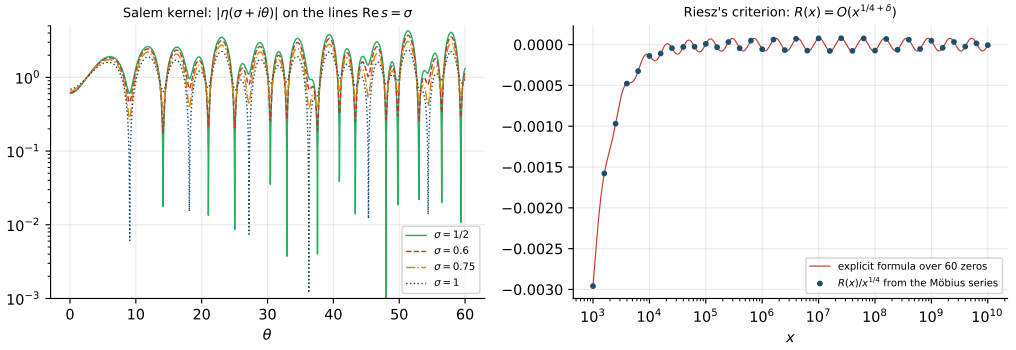


Figure 10.3: Left: $|\eta(\sigma + i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma + i\theta)$, at $\sigma = \frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (10.1.7) over 60 zeros (`figs_wiener.py`).

10.1.8 Riesz's criterion and multiplicative completeness

Riesz [285] found the first criterion of this type, in terms of growth rather than completeness. With the Möbius function $\mu(n)$, which is unrelated to the Thorin measure μ_ε of Section 1.2, put

$$R(x) = -\sum_{k\ge1} \frac{(-x)^k}{(k-1)!\,\zeta(2k)} = x\sum_{n\ge1} \frac{\mu(n)}{n^2}\, e^{-x/n^2}, \tag{10.1.5}$$

the second form following from $1/\zeta(2k) = \sum_n \mu(n) n^{-2k}$. Riesz proved that RH holds if and only if $R(x) = O(x^{1/4+\delta})$ for every $\delta > 0$. The Mellin transform explains why: for $\frac{1}{2} < \mathrm{Re}\, s < 1$,

$$\int_0^\infty R(x)\, x^{-s-1} dx = \frac{\Gamma(1-s)}{\zeta(2s)}, \tag{10.1.6}$$

whose poles in the strip $0 < \mathrm{Re}\, s < \frac{1}{2}$ sit at $s = \rho/2$, and shifting the contour gives, when the zeros are simple, the explicit formula

$$R(x) = \sum_\rho \frac{\Gamma(1-\rho/2)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k\ge1} \frac{\Gamma(1+k)}{2\zeta'(-2k)}\, x^{-k}, \tag{10.1.7}$$

in which the sum over the zeros is the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence of heights $T_\nu \to \infty$. The same argument gives the basepoint version: for $\frac{1}{2} \le \alpha < 1$, ζ has no zeros in $\mathrm{Re}\, s > \alpha$ if and only if $R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. Proposition 10.2.1 carries out the proof in detail, for a twisted form of R: the zero free half plane bounds $1/\zeta(\sigma + it)$ by a power of $|t|$ on $\sigma \ge \alpha + 2\delta$,

by the Borel Carathéodory lemma as in [324, Section 14.2] with the line $\operatorname{Re} s = \alpha$ in place of the critical line.

In the language of this book, $R(x)/x = \sum_n \mu(n) n^{-2} e^{-x/n^2}$ is the Laplace transform of the signed Möbius measure $\sum_n \mu(n) n^{-2} \delta_{1/n^2}$. It is a heat trace on the arithmetic side, with the Möbius function in place of the zeros, and it is the same Möbius series that underlies the representation of the reciprocal zeta function in Section 6.1 (Theorem 6.1.5). The coefficients $1/\zeta(2k)$ of the power series of R are, up to elementary factors, values of the Laplace transforms of the centre clock T and of T_1 (Section 10.2.2).

We computed $R(x)$ with `salem_riesz.py` from the Möbius series with $n \le N = 10^7$, approximating

$$R(x) \approx x\Big[\sum_{n \le N} \mu(n) n^{-2}\big(e^{-x/n^2} - 1\big) + \frac{6}{\pi^2}\Big]$$

to avoid the slowly converging tail, for $10 \le x \le 10^{12}$. Beyond $x \approx 10^{10}$ the truncation at $N = 10^7$ limits the accuracy: the omitted terms are about $-x^2 \sum_{n>N} \mu(n) n^{-4} \approx 9 \times 10^{-26} x^2$, that is 3×10^{-8} in $R(x)/x^{1/4}$ at $x = 10^{10}$ and 9×10^{-5} at $x = 10^{12}$. For $10^4 \le x \le 10^{10}$ the ratio $R(x)/x^{1/4}$ oscillates with amplitude about 1.1×10^{-4}, and for $10^5 \le x \le 10^{10}$ with amplitude 7.8×10^{-5}, which is the amplitude $|\Gamma(1-\rho_1/2)/\zeta'(\rho_1)|$ of the term of the first zero. The larger swing near $x = 10^4$ comes from the term $\Gamma(2) x^{-1}/(2\zeta'(-2)) = -16.42\, x^{-1}$ of the first trivial zero, which contributes -1.6×10^{-4} to the ratio at $x = 10^4$. The explicit formula (10.1.7) with the first 60 zeros and the first three trivial terms reproduces the computed values to 3.1×10^{-8} on $[10^3, 10^{10}]$, against a signal of size up to 3×10^{-3} (`riesz_explicit.py`, output `riesz_explicit.json`; Figure 10.3, right). The largest difference, at $x = 10^{10}$, is the truncation error just described, and on $[10^3, 10^9]$ the difference is at most 6.2×10^{-10}. Section 10.2.1 twists the Möbius series by $(x/n^2)^{ih}$ so that it responds to the zeros near height $2h$.

Wiener's theorem concerns translates. Its multiplicative counterpart concerns dilates, and there completeness is equivalent to RH itself. Nyman and Beurling [218, 29] proved that RH holds if and only if the constant function 1 lies in the $L^2(0,1)$ closure of the span of the functions $r_\lambda(x) = \{\lambda/x\} - \lambda\{1/x\}$, $0 < \lambda \le 1$, where $\{\cdot\}$ is the fractional part, and Báez-Duarte [12] showed that $\lambda = 1/n$ suffices. Under $x = e^{-u}$ this is a closure problem for translates in a weighted L^2 space on the half line, and the Mellin transform of r_λ is $-\zeta(s)(\lambda^s - \lambda)/s$, so the zeros of ζ in the half plane $\operatorname{Re} s > \frac{1}{2}$ are exactly what obstructs the closure. The same framework governs the dilates $\operatorname{sech}(nx)$ of the Lai and Vere-Jones identity (Proposition 6.1.3), which lies behind the expansion of Theorem 6.1.5: the Mellin transform of sech is $2\Gamma(s) L(s, \chi_4)$, and completeness of systems of dilates is controlled by where the associated Dirichlet series vanishes, in the sense of Beurling's periodic dilation problem and the work of Hedenmalm, Lindqvist and Seip [114]. How the two questions are related is the Nyman Beurling problem of Section 7.2.4. Each of these

completeness questions is by itself equivalent to RH or to the generalised Riemann hypothesis. The additive family of Proposition 10.1.10 is the probabilistic one, and it reaches RH only as $\alpha \downarrow \frac{1}{2}$, in the same way as every other construction of Chapters 9 to 11.

10.2 Riesz's criterion and the Thorin path

Section 10.1.8 introduced Riesz's criterion [285] for the Riesz function (10.1.5) and its basepoint form, of which RH is the case $\alpha = \frac{1}{2}$. Hardy and Littlewood [106] gave a companion criterion with $\zeta(2k+1)$ in place of $\zeta(2k)$ and the exponent $-\frac{1}{4}$, and the constructions of this section apply to it in the same way.

We develop the criterion in two directions. The first is computational (Section 10.2.1). The second reads the coefficients $1/\zeta(2k)$ through the clocks of Chapter 9. None of this supplies the growth bound (Section 10.2.6).

We keep the notation of Chapter 9 and Section 10.1 for $T = T_{1/2}$, T_α, φ_α, H_a, b_α and v_α; in particular $b_1 = \sum_\rho \rho^{-1} = 0.0230957$. We write $\widetilde{T} = \widetilde{T}_1$ for the centre clock tilted by $e^{-T/4}$, as in Theorem 9.2.2, $\psi(y) = \sum_{n \le y} \Lambda(n)$ for Chebyshev's function, and

$$\kappa_k = \frac{2k(2k-1)(k-1)!}{\pi^k}, \qquad c_k = \frac{1}{\kappa_k\, \zeta(2k)},$$

so that $\xi(2k) = \frac{1}{2}\kappa_k \zeta(2k)$ and $c_k = \xi(1)/\xi(2k)$. The scripts named below are in `riesz_program/code` of the archive, their outputs in `riesz_program/data`.

10.2.1 Which zeros the Riesz function sees

On the range computed in Section 10.1.8 the Riesz function behaves exactly as RH predicts, and its fluctuations are accounted for by the known zeros, in fact by very few of them. In the explicit formula (10.1.7) a zero ρ enters $R(x)$ through the term $r_\rho x^{\rho/2}$ with the weight $r_\rho = \Gamma(1-\rho/2)/(2\zeta'(\rho))$. By Stirling's formula $|\Gamma(\frac{3}{4} - \frac{i}{2}\gamma)|$ decays like $e^{-\pi\gamma/4}$. The weight is 3.9×10^{-5} for the first zero, whose term and that of its conjugate give the amplitude 7.8×10^{-5} found in Section 10.1.8, 2.1×10^{-17} for the tenth, 1.5×10^{-49} for the fiftieth and 2.2×10^{-81} for the hundredth (`sens.py`, output `sens.json`; Figure 10.4, left). An off line zero at height 1000 would enter with a weight near 10^{-340}, and no feasible range of x would reveal it. Riesz's criterion is equivalent to RH, but as a computational test it examines only the bottom of the spectrum.

The remedy is to change the kernel. For real h put

$$R_h(x) = x \sum_{n \ge 1} \frac{\mu(n)}{n^2} \Big(\frac{x}{n^2}\Big)^{ih} e^{-x/n^2}, \qquad x > 0,$$

so that $R_0 = R$.

PROPOSITION 10.2.1. *For every real h and $\frac{1}{2} < \operatorname{Re} s < 1$,*

$$\int_0^\infty R_h(x)\, x^{-s-1} dx = \frac{\Gamma(1-s+ih)}{\zeta(2s)}.$$

For $\frac{1}{2} \le \alpha < 1$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha$ if and only if $R_h(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. In particular RH holds if and only if $R_h(x) = O(x^{1/4+\delta})$ for every $\delta > 0$.

Proof. Term by term, $\int_0^\infty x^{1+ih} n^{-2-2ih} e^{-x/n^2} x^{-s-1} dx = n^{-2s}\Gamma(1-s+ih)$, and summing against $\mu(n)$ gives $\Gamma(1-s+ih)\sum_n \mu(n) n^{-2s}$; the interchange is justified because, with $\sigma = \operatorname{Re} s$, the integrals of the absolute values sum to $\Gamma(1-\sigma)\zeta(2\sigma) < \infty$. Suppose that ζ has no zeros in $\operatorname{Re} s > \alpha$. Then $1/\zeta(2s)$ is analytic in $\operatorname{Re} s > \frac{\alpha}{2}$. By the argument of [324, Section 14.2] with the line of real part α in place of the critical line, as in Section 10.1.8, $|1/\zeta(2s)| \le |2 \operatorname{Im} s|^{C_\delta}$ on $\operatorname{Re} s \ge \frac{\alpha}{2} + \delta$, $|\operatorname{Im} s| \ge 5$, while $|\Gamma(1-s+ih)|$ decays like $e^{-\pi|\operatorname{Im} s - h|/2}$ in vertical strips. So the Mellin inversion integral $R_h(x) = \frac{1}{2\pi i}\int_{(c)} \Gamma(1-s+ih)\zeta(2s)^{-1} x^s\, ds$, with $\frac{1}{2} < c < 1$, can be moved to the line $\operatorname{Re} s = \frac{\alpha}{2} + \delta$, which gives $R_h(x) = O(x^{\alpha/2+\delta})$. Conversely, if $R_h(x) = O(x^{\alpha/2+\delta})$ as $x \to \infty$, then, since also $|R_h(x)| \le \zeta(2)\, x$, the Mellin integral converges absolutely for $\frac{\alpha}{2} + \delta < \operatorname{Re} s < 1$ and continues $\Gamma(1-s+ih)/\zeta(2s)$ analytically there. As Γ has no zeros, $\zeta(2s)$ has no zeros with $\alpha + 2\delta < \operatorname{Re} 2s < 2$; since $\delta > 0$ is arbitrary and ζ has no zeros with $\operatorname{Re} s \ge 1$, it has none with $\operatorname{Re} s > \alpha$. □

Shifting the contour further gives, when the zeros are simple, the twisted form of the explicit formula (10.1.7), with the sum over the zeros again the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence $T_\nu \to \infty$:

$$R_h(x) = \sum_\rho \frac{\Gamma(1-\rho/2+ih)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k \ge 1} \frac{\Gamma(1+k+ih)}{2\zeta'(-2k)}\, x^{-k}. \tag{10.2.1}$$

The heights T_ν are those of [324, Theorem 9.7], on which $1/\zeta(\sigma + iT_\nu) = O(T_\nu^A)$ for $-1 \le \sigma \le 2$; with the functional equation for $\operatorname{Re} 2s \le -1$ this makes the integrals over the horizontal segments $\operatorname{Im} 2s = \pm T_\nu$ tend to 0. On the line $\operatorname{Re} s = -q - \frac{1}{2}$ we write $1/\zeta(2s) = \pi^{1/2-2s}\Gamma(s)/(\Gamma(\frac{1}{2}-s)\zeta(1-2s))$; since $|\Gamma(1-s+ih)\Gamma(s)| \le e^{\pi|h|/2}\pi/\cosh(\pi \operatorname{Im} s)$, $|\Gamma(\frac{1}{2}-s)| \ge q!\,(\pi t/\sinh \pi t)^{1/2}$ with $t = \operatorname{Im} s$, and $|\zeta(1-2s)| \ge 2 - \zeta(4)$, the integral over that line is at most $Ce^{\pi|h|/2}\pi^{2q} x^{-q-1/2}/q!$, which tends to 0 as $q \to \infty$.

What the twist changes is the weight. Since $|\Gamma(\frac{3}{4} + i(h - \gamma/2))|$ is largest when $\gamma \approx 2h$ and decays like $e^{-\pi|h-\gamma/2|/2}$ away from it, R_h sees the zeros in a band of width about ten around height $2h$. At $h = 50$ the largest weight among the first 300 zeros is 0.124, at $\gamma = 101.3$, where the untwisted weight is 2.9×10^{-35} (Figure 10.4, left). With `twisted.py` we computed $R_{50}(x)$ from the Möbius series

with μ sieved to 10^7 at forty values of x in $[10^4, 10^{10}]$, and compared it with (10.2.1) over the first 300 zeros and three trivial terms. The two agree to 1.2×10^{-8} in $|R_h(x)|/x^{1/4}$, against a signal of size up to 0.27, and the zeros with $|\gamma - 100| < 15$ carry 99.9997% of the modulus of the explicit formula, in the median over the forty values of x (Figure 10.4, right).

An off line zero with real part β near height $2h$ would contribute a term of size $x^{\beta/2}$, growing relative to the background $x^{1/4}$ by the factor $x^{(\beta-1/2)/2}$: a factor 3.2 between $x = 10^{10}$ and $x = 10^{20}$ when $\beta = 0.6$.

Remark 10.2.2. Twisted Riesz functions are not competitive with direct zero finding as a verification method. Their value is independence: they are computed from $\mu(n)$ only, with no evaluation of ζ, so agreement between R_h and the explicit formula over computed zeros is a check of the zeros that shares no code or method with their computation.

The cost of $R(x)$ or $R_h(x)$ is dominated by sieving $\mu(n)$ to $N = K\sqrt{x}$. Terms with $n < \sqrt{x}/12$ are below e^{-144} and can be omitted, and the terms with $n > N$ are summed through $\sum_n \mu(n)n^{-2} = 6/\pi^2$, or $1/\zeta(2+2ih)$ for R_h: the script computes $R(x)/x$ as $\sum_{\sqrt{x}/12 \le n \le N} \mu(n)n^{-2}e^{-x/n^2} + \big(6/\pi^2 - \sum_{n \le N} \mu(n)n^{-2}\big)$, and the neglected remainder $\sum_{n>N} \mu(n)n^{-2}(e^{-x/n^2} - 1) \approx -x \sum_{n>N} \mu(n)n^{-4}$ is of order $xN^{-7/2}$ under square root cancellation in the Möbius tail, that is, of order $K^{-7/2}$ in $R(x)/x^{1/4}$. That cancellation is proved in the range used: the Mertens function $\sum_{n \le t} \mu(n)$ is less than $0.571\sqrt{t}$ in absolute value for $33 \le t \le 10^{16}$ [117, Section 6.1], and partial summation then bounds the remainder in $R(x)/x^{1/4}$ by 1.1×10^{-8} for $K = 200$ and $1 \le x \le 10^{12}$. The script `riesz_large.py` implements this with a segmented sieve and extended precision. With $K = 200$ it reproduces the explicit formula over 60 zeros and three trivial terms to about 2×10^{-9} in $R(x)/x^{1/4}$ at $x = 10^8$, 10^{10}, 10^{11} and 10^{12} (`check_large.py`). Its twisted option agrees as well: at $x = 10^{10}$ with $h = 50$ it gives $R_{50}(x) = 37.441416 - 26.042495\,i$, within 3.1×10^{-9} of (10.2.1) over 300 zeros in $|R_h(x)|/x^{1/4}$ (`check_large_twisted.py`).

10.2.2 The Riesz coefficients through the centre clock

The Riesz coefficients are values of two Laplace transforms: that of the centre clock T of [249] and that of the basepoint one clock T_1 (Proposition 9.1.2).

Proposition 10.2.3. *For every integer $k \ge 1$, with no hypothesis,*

$$\frac{1}{\zeta(2k)} = \frac{\kappa_k}{2\xi(\frac{1}{2})}\,\mathbb{E}\,e^{-(2k-\frac{1}{2})^2 T} = \kappa_k\,\mathbb{E}\,e^{-(2k-1)^2 T_1}.$$

Proof. From the definition of ξ, $\xi(2k) = k(2k-1)\pi^{-k}\Gamma(k)\zeta(2k) = \frac{1}{2}\kappa_k\zeta(2k)$. Since $2k - \frac{1}{2} > 0$, the Laplace transform of T at $(2k-\frac{1}{2})^2$ is $\varphi_{1/2}((2k-\frac{1}{2})^2) = \xi(\frac{1}{2})/\xi(2k)$; similarly $\mathbb{E}e^{-(2k-1)^2 T_1} = \xi(1)/\xi(2k)$, with $\xi(1) = \frac{1}{2}$. □

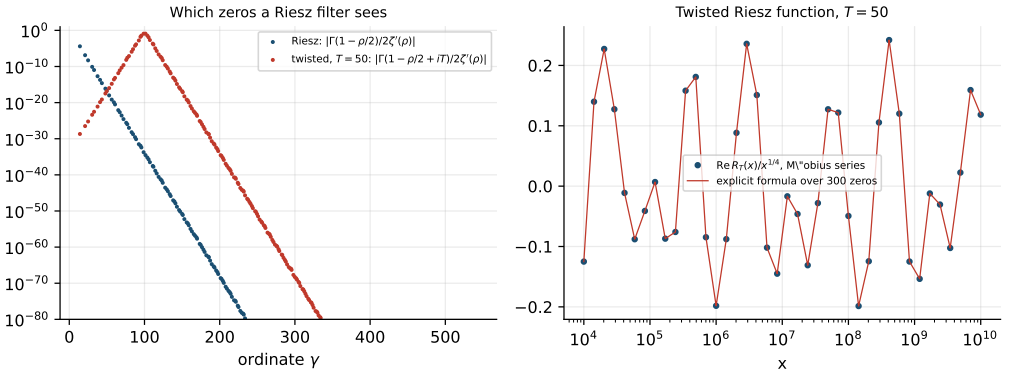


Figure 10.4: Left: the weights $|\Gamma(1-\rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h=50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (10.2.1) over 300 zeros. The plot writes T for h. Data from `twisted.py`; drawn by `figs_strategy.py`.

Proposition 10.2.4. *For every $x>0$, with no hypothesis,*

$$R(x) = -\frac{1}{\xi(\frac12)}\,\mathbb{E}\Big[\sum_{k\ge1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-\frac12)^2T}\Big].$$

Proof. Since $\kappa_k/(2(k-1)!) = k(2k-1)\pi^{-k}$, Proposition 10.2.3 turns the term $-(-x)^k/((k-1)!\,\zeta(2k))$ of Riesz's series into $-\xi(\frac12)^{-1}k(2k-1)(-x/\pi)^k\,\mathbb{E}e^{-(2k-\frac12)^2T}$. The exchange of sum and expectation is justified by Fubini's theorem, because

$$\sum_{k\ge1} k(2k-1)\Big(\frac{x}{\pi}\Big)^k \mathbb{E}\,e^{-(2k-\frac12)^2T} = \xi(\tfrac12)\sum_{k\ge1}\frac{x^k}{(k-1)!\,\zeta(2k)} < \infty.$$

□

The terms of the series averaged in Proposition 10.2.4 decay like e^{-4k^2T} rather than like $1/(k-1)!$. Table 10.6, computed by `riesz_thorin.py` (output `riesz_thorin.json`), checks both propositions with 4×10^5 draws of T, generated under RH as $\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ: the first 200 zeros are used exactly, and the remaining terms are replaced by a gamma variable with the same mean and variance, computed from the exact power sums $\sum\gamma^{-2}$ and $\sum\gamma^{-4}$. The coefficients agree within about one standard error, and so do the values of R from the Möbius series, the power series and the clock average.

The clock average is also far better conditioned than the power series. Riesz's power series is numerically hopeless for large x: its largest term is about $e^x\sqrt{x/2\pi}$,

k	$1/\zeta(2k)$	via the clock	standard error
1	0.6079271	0.6079115	0.0000131
2	0.9239384	0.9238153	0.0001046
3	0.9829526	0.9826581	0.0002603
4	0.9959392	0.9954609	0.0004615

x	$R(x)$, Möbius	clock average	standard error
0.5	0.1252199	0.1252287	0.0000074
1	0.0439818	0.0440007	0.0000157
2	−0.2677934	−0.2677918	0.0000026
5	−0.7705597	−0.7706050	0.0000397

Table 10.6: Left: the Riesz coefficients from the clock, Proposition 10.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 10.2.4.

that is $10^{4344.5}$ at $x = 10^4$, while $R(10^4)$ is of order 10^{-3}. In the clock average the largest term of the inner series, $\max_k k(2k-1)(x/\pi)^k e^{-(2k-\frac{1}{2})^2 T}$, is about 10^{81} at $x = 10^4$ for the typical value $T = 0.023$, and its logarithm grows like $(\log x)^2/(16T)$ instead of like x (Figure 10.5, right).

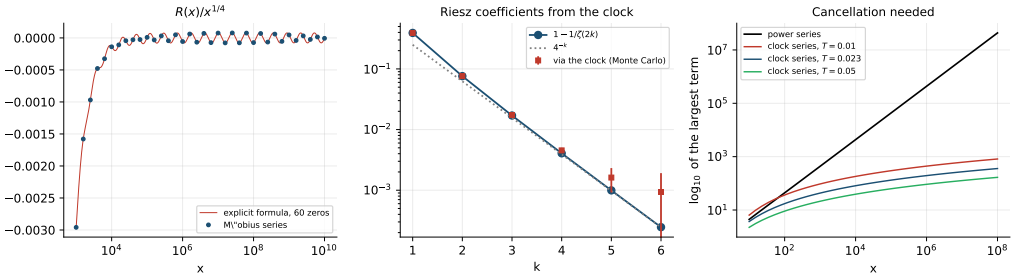


Figure 10.5: Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 10.3. Middle: $1 - 1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 10.2.3), against 4^{-k}. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 10.2.4 for $T = 0.01$, 0.023 and 0.05.

10.2.3 The sine squared lemma and a GGC form of Riesz's criterion

The computation of Proposition 10.2.3 works at every basepoint between the centre and one. For $\alpha > \frac{1}{2}$ the clock T_α exists with no hypothesis (Theorem 9.2.5), and it is a GGC exactly when ζ has no zeros with $\operatorname{Re} s > \alpha$ (Theorem 1.2.2).

PROPOSITION 10.2.5 (Riesz coefficients as GGC values). *For every $\alpha \in [\frac12, 1]$ and every integer $k \geq 1$,*

$$\frac{1}{\zeta(2k)} = \frac{\kappa_k}{2\xi(\alpha)}\, \mathbb{E}\, e^{-(2k-\alpha)^2 T_\alpha}.$$

At $\alpha = 1$, with no hypothesis, T_1 is a GGC with Thorin density $\rho_{1/2} = v_1/\pi$ in the chart $\theta = \sqrt{z}$, and

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big) \rho_{1/2}(\theta)\, d\theta \Big\}.$$

Proof. Since $2k-\alpha > 0$, the Laplace transform of T_α at $(2k-\alpha)^2$ is $\xi(\alpha)/\xi(2k)$, and $\xi(2k) = \frac12 \kappa_k \zeta(2k)$. At $\alpha = 1$, $\xi(1) = \frac12$, and the Thorin representation $\log \varphi_1(u) = -\int_0^\infty \log(1 + u/\theta^2) \rho_{1/2}(\theta)\, d\theta$ of Proposition 9.1.2, taken at $u = (2k-1)^2$, gives the second formula. □

REMARK 10.2.6. All the Riesz coefficients are samples of one GGC Laplace transform: $c_k = \varphi_1((2k-1)^2)$, where $\varphi_1(u) = \mathbb{E}e^{-uT_1}$. As a Laplace transform φ_1 is completely monotone, and this already makes $\log \varphi_1$ convex in u, by Hölder's inequality; so the slopes $(\log c_{k+1} - \log c_k)/8k$ increase with k. As the transform of a GGC, φ_1 is also hyperbolically completely monotone [39, Theorem 6.1.1]: for each $u > 0$ the function $v \mapsto \varphi_1(uv)\varphi_1(u/v)$ is a completely monotone function of $v + 1/v$. In particular it decreases as v moves away from 1, and differentiating $\log \varphi_1(uv) + \log \varphi_1(u/v)$ in $\log v$ shows that $t \mapsto \log \varphi_1(e^t)$ is concave. At the sample points this gives inequalities in the other direction: the slopes $(\log c_{k+1} - \log c_k)/\log \frac{2k+1}{2k-1}$ decrease with k, and $c_j c_l \leq c_k^2$ whenever $(2j-1)(2l-1) = (2k-1)^2$, for instance $c_1 c_5 \leq c_2^2$.

The Thorin density v_1/π is built from the primes by the sine squared lemma (1.2.9). Proposition 9.1.10 writes $\log(\xi(1+s)/\xi(1))$ in the Lévy form (9.1.5), with the drift b_1 and a signed Lévy measure Π_1 made of the gamma density, the pole of ζ and the prime atoms, and the lemma converts it term by term into the Thorin form $v_1 = b_1 + A + Q$ of (9.1.6), with the gamma piece A and the compensated prime piece Q. The Lévy form in $s = 2k-1$ is

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ -b_1(2k-1) - \int_0^\infty \big(e^{-(2k-1)x} - 1 + (2k-1)x\big)\, \Pi_1(dx) \Big\},$$

and in the Thorin form in $u = (2k-1)^2$ the drift splits off a positive stable one half factor:

$$\frac{1}{\zeta(2k)} = \kappa_k\, e^{-b_1(2k-1)}\, \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big) \big(A(\theta) + Q(\theta)\big)\, d\theta \Big\}. \tag{10.2.2}$$

This is the factorisation $T_1 \overset{d}{=} S + T_1^\circ$ of Proposition 9.1.11 evaluated at $u = (2k-1)^2$: the middle factor is $\mathbb{E}e^{-(2k-1)^2 S}$ for the positive stable variable S of

index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and the last factor is $\mathbb{E}e^{-(2k-1)^2 T_1^\circ}$ for the independent GGC T_1° with Thorin density $(A+Q)/\pi = (v_1 - b_1)/\pi$.

Inserting Proposition 10.2.5 at $\alpha = 1$ into the power series gives the Riesz function as a GGC mixture,

$$R(x) = -2\,\mathbb{E}\Big[\sum_{k\geq 1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-1)^2 T_1}\Big], \tag{10.2.3}$$

where the exchange of sum and expectation is justified as in Proposition 10.2.4, since $\sum_k 2k(2k-1)(x/\pi)^k c_k = \sum_k x^k/((k-1)!\,\zeta(2k)) < \infty$. The variable T_1 has two realisations. It is the tilted centre clock run through the inverse local time of a sticky Brownian motion (Theorem 9.2.2 and the paragraph after it), and, like every GGC without drift [129], it is the Wiener gamma integral (11.1.4) against the inverse of the Poisson smoothed zero count $M_{1/2}(\theta) = \pi^{-1}\int_0^\theta v_1$ (Section 11.1.6).

THEOREM 10.2.7. *For $\frac{1}{2} \leq \alpha < 1$ the following are equivalent:*

1. *$R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$;*
2. *ζ has no zeros with $\operatorname{Re} s > \alpha$;*
3. *$T_{\alpha'}$ is a GGC for every $\alpha' > \alpha$;*
4. *for every $\alpha' > \alpha$, $v_{\alpha'}(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha' + i\theta) \geq 0$ at every real θ for which $\alpha' + i\theta$ is not a zero of ξ;*
5. *for every $\alpha' > \alpha$ and every real θ the compensated sine squared sum*

$$\lim_{L\to\infty}\Big[\sum_{\log n \leq L} \frac{\Lambda(n)}{n^{\alpha'}}\, 2\sin^2\Big(\frac{\theta\log n}{2}\Big) - \int_0^L 2\sin^2\Big(\frac{\theta x}{2}\Big)e^{(1-\alpha')x}\,dx\Big]$$

converges.

In particular RH holds if and only if T_α is a GGC for every $\alpha > \frac{1}{2}$.

Proof. Write $\Theta = \sup\operatorname{Re}\rho$ and $D(y) = \psi(y) - y$. Conditions (1) and (2) are equivalent by the basepoint version of Riesz's criterion (Section 10.1.8, and Proposition 10.2.1 with $h = 0$). Conditions (2) and (3) are equivalent by Theorem 1.2.2: $\varphi_{\alpha'}$ is the Laplace transform of a GGC exactly when $\alpha' \geq \Theta$, and this holds for every $\alpha' > \alpha$ exactly when $\Theta \leq \alpha$, which is (2).

(2) implies (4), because if $\Theta \leq \alpha < \alpha'$ every Cauchy kernel in $v_{\alpha'}(\theta) = \sum_\rho(\alpha' - \beta)/((\alpha'-\beta)^2 + (\theta-\gamma)^2)$ has positive width (Proposition 1.2.3). Conversely, let $\rho_0 = \beta_0 + i\gamma_0$ be a zero of multiplicity m with $\beta_0 > \alpha$. The zeros of ξ are isolated, so only finitely many lie on the segment $\{\sigma + i\gamma_0 : 0 \leq \sigma \leq 1\}$. Let n_0 be the number of those other than ρ_0, counted with multiplicity, and choose $\eta \in (0, \beta_0 - \alpha)$ with

$|\beta-\beta_0|\ge 2\eta$ for each of them. For $\alpha'\in(\beta_0-\eta,\beta_0)$ the point $\alpha'+i\gamma_0$ is not a zero, and Proposition 1.2.3 gives

$$v_{\alpha'}(\gamma_0)=\frac{m}{\alpha'-\beta_0}+\sum_{\rho\neq\rho_0,\ \gamma=\gamma_0}\frac{1}{\alpha'-\beta}+\sum_{\gamma\neq\gamma_0}\frac{\alpha'-\beta}{(\alpha'-\beta)^2+(\gamma-\gamma_0)^2}.$$

The middle sum is at most n_0/η in absolute value. Since α' and β lie in $(0,1)$, each term of the last sum is at most $(\gamma-\gamma_0)^{-2}$ in absolute value, and $\sum_{\gamma\neq\gamma_0}(\gamma-\gamma_0)^{-2}<\infty$: finitely many zeros have $0<|\gamma-\gamma_0|<1$, and for $|\gamma-\gamma_0|\ge 1$ we have $(\gamma-\gamma_0)^{-2}\le 5/(4+(\gamma-\gamma_0)^2)\le 5(2-\beta)/((2-\beta)^2+(\gamma-\gamma_0)^2)$, a series bounded by $5v_2(\gamma_0)<\infty$. Hence $v_{\alpha'}(\gamma_0)\le m/(\alpha'-\beta_0)+n_0/\eta+\sum_{\gamma\neq\gamma_0}(\gamma-\gamma_0)^{-2}<0$ for α' close to β_0, and (4) fails at (α',γ_0).

(2) implies (5). This is part 1 of Proposition 9.1.13 at the basepoint α', since $\Theta\le\alpha<\alpha'$, and the argument is short. Choose σ with $\Theta<\sigma<\alpha'$, so that $D(y)=O(y^\sigma)$ [324]. With $f(y)=y^{-\alpha'}\,2\sin^2(\frac12\theta\log y)$ the bracket in (5) is

$$\int_{(1,e^L]} f\,dD=f(e^L)D(e^L)-\int_1^{e^L}D(y)f'(y)\,dy,$$

since $f(1)=0$. The boundary term tends to 0, and the integral converges absolutely because $|f'(y)|\le(2\alpha'+|\theta|)y^{-\alpha'-1}$.

(5) implies (2). Suppose (5), and let ρ_0 be a zero of multiplicity m with $\operatorname{Re}\rho_0>\alpha$. Fix $\alpha'\in(\alpha,\operatorname{Re}\rho_0)$ and $\theta\notin\mathcal{E}=\{\theta:\rho_0+i\theta$ or $\rho_0-i\theta$ is a zero$\}$, a countable set that contains 0, and let $M(t)=\int_{(0,t]}(1-\cos\theta s)\,dD(e^s)$, a function of locally bounded variation. The bracket in (5) is $J(L)=\int_{(0,L]}e^{-\alpha' t}dM(t)$; by hypothesis it converges as $L\to\infty$, so J is bounded. For $\operatorname{Re}w>\alpha'$, integration by parts gives

$$\int_{(0,L]}e^{-wt}\,dM(t)=e^{-(w-\alpha')L}J(L)+(w-\alpha')\int_0^L e^{-(w-\alpha')t}J(t)\,dt,$$

which converges as $L\to\infty$, locally uniformly in w, to an analytic function $F_\theta(w)$ on $\operatorname{Re}w>\alpha'$. For $\operatorname{Re}w>1$ the integral converges absolutely, and writing $1-\cos\theta t=1-\frac12e^{i\theta t}-\frac12e^{-i\theta t}$ gives

$$F_\theta(w)=g(w)-\tfrac12 g(w-i\theta)-\tfrac12 g(w+i\theta),$$
$$g(w)=\int_{(1,\infty)}y^{-w}\,dD(y)=-\frac{\zeta'}{\zeta}(w)-\frac{1}{w-1}.$$

The right side, call it $\widetilde{F}_\theta(w)$, is meromorphic in the plane with a discrete set $\mathcal{P}$ of poles. On the connected open set $\Omega=\{\operatorname{Re}w>\alpha'\}\setminus\mathcal{P}$ both $\widetilde{F}_\theta$ and F_θ are analytic, and they agree on $\operatorname{Re}w>1$, so $\widetilde{F}_\theta=F_\theta$ on Ω by the identity theorem. The function g is analytic at 1, and in $\operatorname{Re}w>0$ its only poles are the zeros of ζ,

each simple with residue minus its multiplicity. Since $\theta \notin \mathcal{E}$, $g(w \pm i\theta)$ is analytic at ρ_0, so $\widetilde{F_\theta}$ has a pole at ρ_0 with residue $-m \neq 0$, and $|F_\theta(w)| \to \infty$ as $w \to \rho_0$ in Ω. This contradicts the continuity of F_θ at ρ_0, where $\operatorname{Re}\rho_0 > \alpha'$. Hence no zero has real part greater than α, which is (2). □

10.2.4 The coefficients as killing probabilities

For a positive random variable X and an independent exponential variable E_λ of rate λ, $\mathbb{E}e^{-\lambda X} = P(X < E_\lambda)$, the probability that X occurs before a killing at rate λ. Propositions 10.2.3 and 10.2.5 therefore say that every Riesz coefficient is a killing probability, in two ways and with no hypothesis:

$$\frac{1}{\zeta(2k)} = \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big) = \frac{\kappa_k}{2\xi(\frac{1}{2})} \, P\big(T < E_{(2k-\frac{1}{2})^2}\big). \tag{10.2.4}$$

By Theorem 9.2.2, $T_1 \stackrel{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$, and by the paragraph after it T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$. So $c_k = 1/(\kappa_k \zeta(2k))$ is the probability that this hitting happens before an independent killing at rate $(2k-1)^2$.

The factorisation (10.2.2) has the same reading. Since $\mathbb{E}e^{-\lambda H_a} = e^{-a\sqrt{2\lambda}}$, the stable factor is

$$e^{-b_1(2k-1)} = P\big(H_{b_1/\sqrt{2}} < E_{(2k-1)^2}\big),$$

the probability that Brownian motion reaches the level $b_1/\sqrt{2} = 0.016331$ before it is killed. So $S \stackrel{d}{=} H_{b_1/\sqrt{2}}$ and $T_1 \stackrel{d}{=} H_{b_1/\sqrt{2}} + T_1^\circ$ with T_1° an independent GGC. Thus $c_k = P(H_{b_1/\sqrt{2}} < E_{(2k-1)^2})\, P(T_1^\circ < E'_{(2k-1)^2})$, with independent exponential clocks E and E' of the same rate. If the exact Brownian boundary b of T_1 is bounded at infinity, as Section 9.2.4 observes but does not prove, Lemma 9.2.4 and the tail of T_1 make the level $b_1/\sqrt{2} = \sum_\rho \rho^{-1}/\sqrt{2}$ its Wald defect $\mathbb{E}\, b(T_1)$.

Under RH, if the zeros are also simple, more is true. The centre clock is then a GGC with unit atoms at the distinct points γ^2 (Proposition 1.2.1), hence, by Kent's theorem and Krein's inverse spectral theory (Section 8.1.3), the hitting time of a level by the diffusion of a Krein string with Dirichlet spectrum $\{\gamma^2\}$ (Table 4.1). The second form of (10.2.4) then says that $1/\zeta(2k)$ is proportional to the probability that this Riemann string diffusion reaches its level before it is killed at rate $(2k-\frac{1}{2})^2$. At the basepoint one, Corollary 9.2.3 gives the first form the same reading under the same hypotheses.

The Riesz function becomes an alternating sum of killing probabilities,

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!}\, \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big),$$

and the criterion asks that these alternating sums grow no faster than $x^{1/4+\delta}$. The killing rates $(2k-\alpha)^2$ that the criterion needs, with k in the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$, are

complex, and at a complex rate the Laplace transform is no longer a probability; this is the hitting time form of the obstruction of Section 10.2.6.

We checked these identities with `killing.py` (output `killing.json`), using 10^6 draws of $T_1 = \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ generated under RH as in Section 9.2.3. Table 10.7 compares the killing probabilities with c_k, and the Brownian hitting probability with $e^{-b_1(2k-1)}$; all agree within two standard errors. With the averaged killing probabilities for $k \le 30$ from a second run of 10^6 draws (`killing2.py`), the alternating sum gives $R(0.5) = 0.125242$, $R(1) = 0.043990$, $R(2) = -0.267895$ and $R(3) = -0.526639$, against the exact 0.125220, 0.043982, -0.267793 and -0.526356.

k	$P(T_1 < E)$, drawn	$\mathbb{E}e^{-(2k-1)^2 T_1}$	$c_k = \frac{1}{\kappa_k \zeta(2k)}$	$P(H_{b_1/\sqrt{2}} < E)$	$e^{-b_1(2k-1)}$
1	0.95481 ± 0.00021	0.954928 ± 0.000112	0.954930	0.97699 ± 0.00015	0.97717
2	0.75965 ± 0.00043	0.759835 ± 0.000161	0.759909	0.93352 ± 0.00025	0.93306
3	0.50738 ± 0.00050	0.507865 ± 0.000156	0.507962	0.89098 ± 0.00031	0.89094
4	0.28834 ± 0.00045	0.288646 ± 0.000122	0.288731	0.85059 ± 0.00036	0.85072
5	0.14179 ± 0.00035	0.141473 ± 0.000080	0.141535	0.81195 ± 0.00039	0.81232
6	0.06045 ± 0.00024	0.060638 ± 0.000045	0.060679	0.77555 ± 0.00042	0.77565

Table 10.7: Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; E is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of 10^6 draws.

10.2.5 Half Cauchy variables, the sech clock and a duality

Bourgade, Fujita and Yor [43] derived Euler's formulae for $\zeta(2n)$ from products of Cauchy variables, and the construction gives a representation of $\zeta(2n)$ dual to that of $1/\zeta(2n)$. Let C and C' be independent standard Cauchy variables, so that $|C|$ and $|C'|$ are half Cauchy. By Section 4.2.2, $\mathbb{E}|C|^s = \sec(\pi s/2)$ for $|\operatorname{Re} s| < 1$, so $\sec^2(\pi s/2) = \mathbb{E}|CC'|^s$ is the moment generating function of $\log|CC'|$ on $|s| < 1$. Differentiating the series $\tan z = \sum_{n\ge1} 2^{2n}(2^{2n}-1)|B_{2n}|z^{2n-1}/(2n)!$ and inserting Euler's formula $\zeta(2n) = (2\pi)^{2n}|B_{2n}|/(2(2n)!)$ gives

$$\sec^2\frac{\pi s}{2} = \sum_{n\ge1} \frac{8(2n-1)(1-2^{-2n})\,\zeta(2n)}{\pi^2}\, s^{2n-2},$$

and comparing coefficients with the moment generating function,

$$\zeta(2n) = \frac{\pi^2}{8(2n-1)(1-2^{-2n})}\,\frac{\mathbb{E}\big[(\log|CC'|)^{2n-2}\big]}{(2n-2)!}, \qquad n \ge 1. \tag{10.2.5}$$

Let τ be the exit time of a standard Brownian motion from $(-1,1)$, so that $\mathbb{E}e^{-\lambda\tau} = 1/\cosh\sqrt{2\lambda}$ (Section 4.2.2), and let B be a Brownian motion independent of τ. Then $\mathbb{E}e^{i\theta B_\tau} = \mathbb{E}e^{-\theta^2\tau/2} = \operatorname{sech}\theta$, and since $\log|C|$ has characteristic

function $\operatorname{sech}(\pi t/2)$ (Section 4.2.2),

$$\log|C| \stackrel{d}{=} \frac{\pi}{2} B_\tau .$$

We call τ the sech clock. By the product $\cosh\sqrt{2\lambda} = \prod_{k\geq 1}\left(1+8\lambda/((2k-1)^2\pi^2)\right)$ it is a GGC with unit Thorin atoms at $(2k-1)^2\pi^2/8$, the GGC $\sum_k e_k/(2k-1)^2$ of Section 4.2.2 multiplied by $8/\pi^2$, and it is Kent's hitting time for the interval: the hitting time of 1 by reflected Brownian motion started at 0, whose eigenvalues on $[0,1]$, with reflection at 0 and killing at 1, are these atoms. So its spectrum is proportional to the very points $(2k-1)^2$ at which the Riesz coefficients sample the Laplace transform of T_1. For two independent copies, $\log|CC'| \stackrel{d}{=} \frac{\pi}{2}B_{\tau+\tau'}$. Combining (10.2.5) with Proposition 10.2.5 gives a duality between the two clocks,

$$\frac{\mathbb{E}\left[(\log|CC'|)^{2k-2}\right]}{(2k-2)!}\,\mathbb{E}\,e^{-(2k-1)^2T_1} = \frac{8(2k-1)(1-2^{-2k})}{\pi^2\kappa_k}, \tag{10.2.6}$$

in which a moment of Brownian motion stopped at a sum of two independent sech clocks, times a killing probability of the basepoint one clock, is an elementary constant.

We checked the moment formula (10.2.5) exactly through the Taylor coefficients of $\sec^2$, where it reproduces $\zeta(2)$, $\zeta(4)$, $\zeta(6)$ and $\zeta(8)$ to ten digits. The duality (10.2.6) holds with the killing probabilities simulated in the second run of Section 10.2.4: the products are 0.95508, 1.87524, 2.06191 and 1.63889 for $k=1,\dots,4$, against 0.95493, 1.87500, 2.06167 and 1.63851.

The duality is structural rather than quantitative: $\zeta(2k)$ as a moment lives at the integers, while Riesz's criterion needs $1/\zeta(2s)$ inside the strip. The half Cauchy law also enters the book through Bosch's theorem [41] (Section 4.2), whose dictionary, Table 4.1, pairs the odd integers $2k-1$ with the ordinates γ.

10.2.6 Why the representation cannot supply the growth bound

The representations of Sections 10.2.2 to 10.2.5 are exact, and all of them except the Kent string readings of Section 10.2.4 hold with no hypothesis. They do not supply the bound $R(x) = O(x^{1/4+\delta})$, for four reasons.

1. The integer values carry almost no information. The growth of R is fixed by the Mellin transform $\Gamma(1-s)/\zeta(2s)$, that is, by the analytic interpolation of $k \mapsto 1/\zeta(2k)$ to complex k with $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$. At the integers $1/\zeta(2k) = 1 - 4^{-k} + O(9^{-k})$, and these values, which Proposition 10.2.3 represents, determine the interpolation only through its analytic continuation.

2. The representation interpolates only on one side of the line $\operatorname{Re} 2k = \alpha$, and as an expectation only on a cone. For complex k the expectation $\mathbb{E}e^{-(2k-\alpha)^2T_\alpha}$ is finite exactly when $\operatorname{Re}(2k-\alpha)^2 \geq 0$, because T_α has a tail of order $t^{-1/2}$

(Theorem 9.2.5) and no exponential moments; there it equals $\varphi_\alpha((2k-\alpha)^2) = \xi(\alpha)/\xi(\alpha+\sqrt{(2k-\alpha)^2})$, and the principal square root equals $2k-\alpha$ only when $\operatorname{Re}(2k-\alpha)>0$. So the representation through T_α returns $\xi(\alpha)/\xi(2k)$ as an expectation on the cone $|\operatorname{Im} 2k| \le \operatorname{Re} 2k - \alpha$, and on the rest of the half plane $\operatorname{Re} 2k > \alpha$ only through the analytic continuation of φ_α, which is analytic there exactly when no zero lies to the right of α. It is a GGC representation with no hypothesis only for $\alpha \ge 1$, that is, on the half plane $\operatorname{Re} 2k > 1$ of absolute convergence, where the criterion carries no information. To reach $\frac12 < \operatorname{Re} 2k < 1$ one needs a basepoint $\alpha < 1$, and T_α is a GGC there exactly when ζ has no zeros in $\operatorname{Re} s > \alpha$.

3. The centre clock's Laplace integral converges only up to its first singularity, which under RH is at $-\gamma_1^2$. In the k plane, with $u = (2k-\frac12)^2$, the region $\operatorname{Re} u > -\gamma_1^2$ is a neighbourhood of the segment $|\operatorname{Im} 2k| < \gamma_1$ of the critical line $\operatorname{Re} 2k = \frac12$, not the strip; beyond it one needs analytic continuation, which is again a statement about the zeros.

4. Positivity does not survive. Each coefficient becomes a positive average, but the series is alternating, and the bound requires, at $x = 10^4$ and $T = 0.023$, the cancellation of terms as large as 10^{81} after averaging. No monotonicity or positivity argument controls that cancellation; it is carried by the oscillating terms $x^{\rho/2}$ of the explicit formula.

What would help is what the basepoint version of the criterion (Proposition 10.2.1 with $h=0$) requires: an analytic, and ideally positive, representation of $1/\zeta(2s)$ on whole vertical lines of the strip $\frac14 < \operatorname{Re} s < \frac12$. A representation of the Thorin kind that is positive there is a GGC at a basepoint below one, that is, a zero free half plane. The Thorin representation of the Riesz coefficients is therefore an exact reformulation, with better cancellation and a probabilistic reading, that reaches the criterion only through the same step that every other construction of Chapters 9 to 11 requires. At a basepoint $\frac12 < \alpha < 1$, zero density estimates bound the number of Mellin poles to the right of the line $\operatorname{Re} s = \alpha/2$ up to any given height, as they bound the negative part of v_α/π (Theorem 11.4.2), but a single such pole breaks the bound $R(x) = O(x^{\alpha/2+\delta})$ for small δ.

10.3 The squarefree series and the clock

Riesz's criterion (Sections 10.1.8 and 10.2) reads RH from a function whose Mellin transform is $\Gamma(1-s)/\zeta(2s)$. The squarefree integers carry the same reciprocal $1/\zeta(2s)$. Their Dirichlet series is

$$D(s) = \sum_{n\ge1} \frac{\mu(n)^2}{n^s} = \prod_p \left(1+p^{-s}\right) = \frac{\zeta(s)}{\zeta(2s)}, \qquad \operatorname{Re} s > 1, \tag{10.3.1}$$

whose singularities in the half plane $\operatorname{Re} s > 0$ are the pole at $s = 1$ and the poles at the points $\rho/2$ at which $\zeta(\rho/2) \neq 0$ (all of them under RH), and the number $Q(x)$ of squarefree integers up to x oscillates about $6x/\pi^2$ on the scale $x^{1/4}$ that these poles set when they lie on the line $\operatorname{Re} s = \frac{1}{4}$, that is under RH. This section follows the first author's note [259], which starts from a Math StackExchange thread [201] on the partial sums of $\mu(n)^2 n^{-\alpha}$ over $\mathbb{Q}$ and over number fields. Write $\Delta(x) = Q(x) - 6x/\pi^2$ for the error. The bound $\Delta(x) = O(x^{1/4+\varepsilon})$ is open even under RH, but it is equivalent to RH once the count is smoothed by an exponential or a Gamma(2) law, and in mean square; the order $x^{1/4}$ is attained with no hypothesis; and the bound for the count smoothed by a Gamma(n) law loses a factor $n^{1/8}$, which still gives $\Delta(x) = O(x^{2/5+\varepsilon})$ under RH (Sections 10.3.5 and 10.3.6). The Pólya kernels of the Dedekind zeta functions of imaginary quadratic fields, whose positivity can fail, are in Section 8.5.6 of Chapter 8.

We keep the notation of Section 1.2 for the Pólya variable X, with $\mathbb{E}e^{sX} = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, the clock T, $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2})$, the numbers $a_k = -(\rho_k - \frac{1}{2})^2$, the heat trace $W(t) = \sum_k e^{-a_k t}$ and the measure U_{RH}, and the notation of Section 9.1.1 for the tilted variables V_α, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and $b_\alpha = \frac{\xi'}{\xi}(\alpha) = \mathbb{E}V_\alpha$. Throughout, μ is the Möbius function, $M(t) = \sum_{d \le t} \mu(d)$ is the Mertens function, $\{y\}$ is the fractional part of y, and

$$Q(x) = \sum_{n \le x} \mu(n)^2, \qquad \Delta(x) = Q(x) - \frac{6}{\pi^2}\, x.$$

10.3.1 The thread: Perron's formula and an elementary method

The question asks for the behaviour as $x \to \infty$ of

$$S_\alpha(x) = \sum_{N(\mathfrak{a}) \le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})^\alpha},$$

the sum running over the integral ideals $\mathfrak{a}$ of a number field K, with μ the Möbius function of ideals and N the norm, and, in a comment, for its expansion at $\alpha = 1$. Over $\mathbb{Q}$, $S_\alpha(x) = \sum_{n \le x} \mu(n)^2 n^{-\alpha}$, and another comment points to Abel summation from the classical count

$$Q(x) = \frac{6}{\pi^2}\, x + O(\sqrt{x}). \tag{10.3.2}$$

The accepted answer applies Perron's formula over K, and the other answer carries out the elementary method over $\mathbb{Q}$. Neither evaluates the constant at $\alpha = 1$.

Since $\mu(\mathfrak{p}^k)^2$ is 1 for $k \le 1$ and 0 for $k \ge 2$, the Euler factor of $\sum_{\mathfrak{a}} \mu(\mathfrak{a})^2 N(\mathfrak{a})^{-s}$ at a prime ideal $\mathfrak{p}$ is $1 + N(\mathfrak{p})^{-s} = (1 - N(\mathfrak{p})^{-2s})/(1 - N(\mathfrak{p})^{-s})$, so the series is $\zeta_K(s)/\zeta_K(2s)$, with ζ_K the Dedekind zeta function of K, and

$$S_\alpha(x) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \frac{\zeta_K(\alpha+s)}{\zeta_K(2\alpha+2s)}\, x^s\, \frac{ds}{s}, \qquad \sigma > \max\{1-\alpha, 0\}. \tag{10.3.3}$$

The integrand of (10.3.3) has a simple pole at $s = 1-\alpha$, a pole at $s = 0$, and poles at the points $s = \rho/2 - \alpha$ for the zeros ρ of ζ_K, trivial and nontrivial, at which $\zeta_K(\rho/2) \neq 0$. Let $c_K = \lim_{s\to 1}(s-1)\zeta_K(s) = 2^{r_1}(2\pi)^{r_2}hR/(w\sqrt{|d_K|})$ be the residue of the class number formula.

The case $\alpha \neq 1$. The residue at $s = 1-\alpha$ is $c_K x^{1-\alpha}/((1-\alpha)\zeta_K(2))$, as the accepted answer states, and the residue at $s = 0$, which it lists among the poles but does not compute, is $\zeta_K(\alpha)/\zeta_K(2\alpha)$, the value of the continued series. Over $\mathbb{Q}$, for $\alpha > \frac12$ and $\alpha \neq 1$, the two residues are the main terms of

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \frac{\zeta(\alpha)}{\zeta(2\alpha)} + O\big(x^{\frac12-\alpha}\big). \tag{10.3.4}$$

For a proof, Abel summation and (10.3.2) give, for $\alpha > \frac12$, $\alpha \neq 1$,

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \Big(\alpha\int_1^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt - \frac{6}{\pi^2}\,\frac{\alpha}{1-\alpha}\Big) + \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{10.3.5}$$

The last two terms are $O(x^{\frac12-\alpha})$, and the bracket is $D(\alpha)$, because $D(s) = s\int_1^\infty Q(t)t^{-s-1}dt = \frac{6}{\pi^2}\frac{s}{s-1} + s\int_1^\infty \Delta(t)t^{-s-1}dt$ for $\operatorname{Re} s > 1$ and the right side is analytic in $\operatorname{Re} s > \frac12$ apart from $s = 1$. For $\frac12 < \alpha < 1$ the constant is negative (Corollary 10.3.4). Write $\mathcal{E}_\alpha(x)$ for the left side of (10.3.4) minus its two main terms, so that by (10.3.5)

$$\mathcal{E}_\alpha(x) = \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{10.3.6}$$

The case $\alpha = 1$. The poles at $1-\alpha$ and at 0 merge into a double pole at $s = 0$ of $\zeta_K(1+s)\zeta_K(2+2s)^{-1}e^{s\log x}/s$. The accepted answer computes the coefficient of $\log x$ and says that the constant term is much more complicated, at least when $K \neq \mathbb{Q}$. It is explicit. Define the Euler Kronecker constant γ_K of K by $\zeta_K(s) = c_K/(s-1) + c_K\gamma_K + O(s-1)$ [120], so that $\gamma_{\mathbb{Q}} = \gamma_E$. With $1/\zeta_K(2+2s) = \zeta_K(2)^{-1}\big(1 - 2s\,\zeta_K'(2)/\zeta_K(2) + O(s^2)\big)$ the residue at $s = 0$ is

$$\frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big), \qquad C_K = \gamma_K - 2\,\frac{\zeta_K'(2)}{\zeta_K(2)}. \tag{10.3.7}$$

The elementary method. The identity $\mu(n)^2 = \sum_{d^2\mid n}\mu(d)$, the coefficientwise form of $D(s) = \zeta(s)\cdot\zeta(2s)^{-1}$, gives

$$Q(x) = \sum_{d\le\sqrt{x}} \mu(d)\Big\lfloor \frac{x}{d^2}\Big\rfloor = x\sum_{d\le\sqrt{x}}\frac{\mu(d)}{d^2} - \sum_{d\le\sqrt{x}}\mu(d)\Big\{\frac{x}{d^2}\Big\}, \tag{10.3.8}$$

and since $\sum_{d>\sqrt{x}} d^{-2} = O(x^{-1/2})$ and the last sum has at most $\sqrt{x}$ terms, each in $[0,1)$, this is (10.3.2). The second answer then sets

$$B(x) = \sum_{n\le x} \Big(\mu(n)^2 - \frac{6}{\pi^2}\Big) = O(\sqrt{x}),$$
$$D_0(s) = \sum_{n\ge 1} \frac{\mu(n)^2 - 6/\pi^2}{n^s} = D(s) - \frac{6}{\pi^2}\,\zeta(s).$$

By Abel summation, $D_0(s) = \lim_{x\to\infty}\big(B(x)x^{-s} + s\int_1^x B(t)t^{-s-1}dt\big)$ converges for $\operatorname{Re} s > \frac12$, so $\sum_{n\le x}\mu(n)^2/n = \frac{6}{\pi^2}(\log x + \gamma_E) + D_0(1) + o(1)$, and there the answer stops. Since $D_0(s) = \zeta(s)\big(\zeta(2s)^{-1} - \zeta(2)^{-1}\big)$ and ζ has a simple pole with residue 1 at $s=1$,

$$D_0(1) = \lim_{s\to 1}\zeta(s)\Big(\frac{1}{\zeta(2s)} - \frac{1}{\zeta(2)}\Big) = -\frac{2\zeta'(2)}{\zeta(2)^2} = 0.69298946940360442669,$$

in agreement with (10.3.7) at $K=\mathbb{Q}$, where $c_{\mathbb{Q}} = 1$ and $\gamma_{\mathbb{Q}} = \gamma_E$.

Proposition 10.3.1. (i) *Let K be a number field of degree $n \ge 2$. As $x\to\infty$,*

$$\sum_{N(\mathfrak{a})\le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} = \frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big) + O\big(x^{-1/n}\log x\big).$$

(ii) *Over $\mathbb{Q}$,*

$$\sum_{n\le x} \frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\big(\log x + C\big) + O\big(x^{-1/2}\big),$$
$$C = \gamma_E - 2\,\frac{\zeta'(2)}{\zeta(2)} = 24\log A - \gamma_E - 2\log 2\pi, \tag{10.3.9}$$

where A is the Glaisher Kinkelin constant; $C = 1.71713765109059847340624$1 and $\frac{6}{\pi^2}C = 1.0438945157119382974$. Moreover $\mathcal{E}_1(x) = \sum_{n\le x}\mu(n)^2/n - \frac{6}{\pi^2}(\log x + C)$ is given by (10.3.6) *with $\alpha = 1$, and*

$$\int_1^\infty \frac{\Delta(t)}{t^2}\,dt = \frac{6}{\pi^2}\,(C-1) = 0.43596741385791166874.$$

Proof. (i) The number of ideals of norm at most y is $c_K y + O(y^{1-1/n})$ [168, Chapter VI], so by Abel summation $L(y) = \sum_{N(\mathfrak{c})\le y} N(\mathfrak{c})^{-1} = c_K(\log y + \gamma_K) + O(y^{-1/n})$ for $y \ge 1$, the constant being the one in the Laurent expansion of ζ_K at $s=1$.

Since $\mu(\mathfrak{a})^2 = \sum_{\mathfrak{b}^2\mid\mathfrak{a}} \mu(\mathfrak{b})$, as one checks on prime powers,

$$\begin{aligned}\sum_{N(\mathfrak{a})\le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} &= \sum_{N(\mathfrak{b})\le\sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\, L\Big(\frac{x}{N(\mathfrak{b})^2}\Big)\\ &= c_K \sum_{N(\mathfrak{b})\le\sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\big(\log x + \gamma_K - 2\log N(\mathfrak{b})\big)\\ &\quad + O\Big(x^{-1/n} \sum_{N(\mathfrak{b})\le\sqrt{x}} N(\mathfrak{b})^{-2+2/n}\Big).\end{aligned}$$

The last sum is $O(\log x)$, since $-2+2/n \le -1$. Completing the sums over $\mathfrak{b}$ costs $O(x^{-1/2}\log x)$, which is $O(x^{-1/n}\log x)$ as $n \ge 2$, since $\sum_{N(\mathfrak{b})>Y} N(\mathfrak{b})^{-2}\log N(\mathfrak{b}) = O(Y^{-1}\log Y)$, and the completed sums are $\sum_{\mathfrak{b}} \mu(\mathfrak{b})N(\mathfrak{b})^{-2} = 1/\zeta_K(2)$ and $\sum_{\mathfrak{b}} \mu(\mathfrak{b})\log N(\mathfrak{b})\, N(\mathfrak{b})^{-2} = \zeta_K'(2)/\zeta_K(2)^2$, which give (10.3.7).

(ii) By Abel summation, $\sum_{n\le x}\mu(n)^2/n = Q(x)/x + \int_1^x Q(t)t^{-2}dt$, and inserting $Q(t) = 6t/\pi^2 + \Delta(t)$ gives

$$\sum_{n\le x}\frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\log x + \frac{6}{\pi^2} + \int_1^\infty \frac{\Delta(t)}{t^2}\,dt + \frac{\Delta(x)}{x} - \int_x^\infty \frac{\Delta(t)}{t^2}\,dt,$$

whose last two terms are $O(x^{-1/2})$ by (10.3.2). So the formula holds with some constant, and the constant is identified through D: for $\operatorname{Re} s > 1$, $\int_1^\infty \Delta(t)t^{-s-1}dt = D(s)/s - \frac{6}{\pi^2}\frac{1}{s-1}$, and $D(s) = \frac{6}{\pi^2}\zeta(s) + D_0(s) = \frac{6}{\pi^2}\big(\frac{1}{s-1} + \gamma_E - 2\frac{\zeta'(2)}{\zeta(2)}\big) + O(s-1)$, so that $\int_1^\infty \Delta(t)t^{-2}dt = \frac{6}{\pi^2}(C-1)$ by continuity at $s=1$. This gives (10.3.9) and the formula for $\mathcal{E}_1$, and the classical evaluation $\zeta'(2)/\zeta(2) = \gamma_E + \log 2\pi - 12\log A$ gives the second form of C. □

The sieve `sqfree_sieve.py` gives $Q(2\times10^8) = 121585426$. Table 10.8 shows that $\mathcal{E}_1(x)$ is of order $x^{-3/4}$ and $\Delta(x)$ of order $x^{1/4}$, the sizes that the poles at $\rho/2-1$ and $\rho/2$ predict under RH (Section 10.3.4).

10.3.2 The Pólya form of the series and the constant

Lemma 10.3.2. *For all s,*

$$D(s) = \Gamma_{\mathrm{sq}}(s)\,\frac{\xi(s)}{\xi(2s)}, \qquad \Gamma_{\mathrm{sq}}(s) = \frac{2s-1}{s-1}\,2^s\pi^{-(s+1)/2}\,\Gamma\Big(\frac{s+1}{2}\Big).$$

Proof. By the definition of ξ, $\zeta(s)/\zeta(2s) = \frac{\xi(s)}{\xi(2s)}\cdot\frac{2(2s-1)}{s-1}\,\pi^{-s/2}\,\frac{\Gamma(s)}{\Gamma(s/2)}$, and the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ gives the factor Γ_{sq}. □

The script `sqfree_constants.py` checks the lemma at $s = 3$, $0.3+2.1i$, $\frac{3}{4}$ and -1.5, with relative errors below 5×10^{-41}.

x	$Q(x)$	$\Delta(x)/x^{1/4}$	$\mathcal{E}_1(x)$	$\mathcal{E}_1(x)\,x^{3/4}$	$\mathcal{E}_{3/4}(x)\,x^{1/4}$
10^3	608	0.0130	-9.130×10^{-4}	-0.162	-0.0291
10^4	6083	0.3729	3.031×10^{-4}	0.303	0.0296
10^5	60794	0.0725	-2.308×10^{-6}	-0.0130	-0.00031
10^6	607926	-0.0348	-1.665×10^{-6}	-0.0526	-0.00177
10^7	6079291	0.3553	1.729×10^{-6}	0.308	0.00561
10^8	60792694	-0.1619	-1.737×10^{-7}	-0.174	-0.00174

Table 10.8: The sharp counts: the number $Q(x)$ of squarefree integers up to x, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (10.3.9), which confirms the constant C, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (10.3.4) at $\alpha=\frac34$ multiplied by $x^{1/4}$. Computed by `sqfree_sieve.py`; the sums of reciprocals are checked in extended precision.

PROPOSITION 10.3.3. *For every s with $\xi(2s)\neq0$,*

$$\sum_{n\geq1}\frac{\mu(n)^2}{n^s}=\Gamma_{\mathrm{sq}}(s)\,\frac{\mathbb{E}e^{(s-\frac12)X}}{\mathbb{E}e^{(2s-\frac12)X}},$$

the left side read as the continuation $D(s)$ where the series diverges.

Proof. Lemma 10.3.2 and (1.2.4), which gives $\xi(s)=\xi(\frac12)\mathbb{E}e^{(s-\frac12)X}$ and $\xi(2s)=\xi(\frac12)\mathbb{E}e^{(2s-\frac12)X}$. □

COROLLARY 10.3.4. *For real $\alpha\notin\{1,\frac12,-1,-3,-5,\dots\}$ the sign of $\zeta(\alpha)/\zeta(2\alpha)$ is the sign of $\frac{2\alpha-1}{\alpha-1}\Gamma(\frac{\alpha+1}{2})$. It is positive for $\alpha>1$ and for $-1<\alpha<\frac12$, negative for $\frac12<\alpha<1$, and equal to $(-1)^k$ on the interval $(-2k-1,-2k+1)$ for $k\geq1$.*

Proof. At real α the ratio of moment generating functions in Proposition 10.3.3 is positive, and so is $2^\alpha\pi^{-(\alpha+1)/2}$. The rational factor is positive except on $(\frac12,1)$, and $\Gamma(\frac{\alpha+1}{2})$ has the sign $(-1)^k$ when $\frac{\alpha+1}{2}\in(-k,-k+1)$, that is when $\alpha\in(-2k-1,-2k+1)$. □

At the even negative integers both $\zeta(\alpha)$ and $\zeta(2\alpha)$ vanish and the quotient keeps its sign through them. At $\alpha=\frac34$, inside the interval where the series diverges, $\Gamma_{\mathrm{sq}}(\frac34)=-1.34612167295$, $\mathbb{E}e^{X/4}=1.00144503253351$ and $\mathbb{E}e^{X}=1.02335500958202$, the last two by the trapezoid rule on the Pólya kernel (1.2.3); the ratio 0.978590052481 multiplies $\Gamma_{\mathrm{sq}}(\frac34)$ to $-1.31730127858=\zeta(\frac34)/\zeta(\frac32)$.

THEOREM 10.3.5. *The constant of* (10.3.9) *is*

$$C=2+\log2-\tfrac12\log\pi-\tfrac12\gamma_E+b_1-2b_2=2+\log2-\tfrac12\log\pi-\tfrac12\gamma_E+\mathbb{E}V_1-2\,\mathbb{E}V_2.$$

With no hypothesis on the zeros,

$$\begin{aligned} b_1 - 2b_2 = \frac{G'}{G}\Big(\frac14\Big) - 6\,\frac{G'}{G}\Big(\frac94\Big) &= \sum_k \Big(\frac{1}{\frac14 + a_k} - \frac{6}{\frac94 + a_k}\Big) \\ &= \int_0^\infty \big(e^{-t/4} - 6e^{-9t/4}\big)\, W(t)\, dt, \end{aligned} \tag{10.3.10}$$

the sum running over the pairs $\{\rho, 1-\rho\}$ *and converging absolutely. Under RH,* $a_k = \gamma_k^2$ *and* $b_1 - 2b_2 = \int \big((z+\frac14)^{-1} - 6(z+\frac94)^{-1}\big)\, U_{\mathrm{RH}}(dz)$*, a combination of two values of the Stieltjes transform of the Thorin measure of the clock.*

Proof. By the proof of Proposition 10.3.1, $(s-1)D(s) = \frac{6}{\pi^2}\big(1 + C(s-1) + O((s-1)^2)\big)$, so C is the derivative of $\log((s-1)D(s))$ at $s=1$. By Proposition 10.3.3,

$$\begin{aligned} \log\big((s-1)D(s)\big) = \log(2s-1) + s\log 2 - \tfrac{s+1}{2}\log\pi + \log\Gamma\big(\tfrac{s+1}{2}\big) \\ + \log \mathbb{E}e^{(s-\frac12)X} - \log \mathbb{E}e^{(2s-\frac12)X}. \end{aligned}$$

At $s=1$ the derivatives of the first four terms are 2, $\log 2$, $-\frac12\log\pi$ and $\frac12\psi(1) = -\frac12\gamma_E$, with ψ the digamma function here. The derivative of $\log \mathbb{E}e^{(s-\frac12)X}$ at $s=1$ is $\mathbb{E}Xe^{X/2}/\mathbb{E}e^{X/2}$, the mean of X tilted by $e^{X/2}$, which is $\mathbb{E}V_1 = b_1$ (Section 9.1.1); likewise the last term gives $-2\mathbb{E}V_2 = -2b_2$. For the zero form, differentiating $\log G(w^2) = \log\xi(\frac12 + w) - \log\xi(\frac12)$ gives $\frac{\xi'}{\xi}(\frac12+w) = 2w\frac{G'}{G}(w^2)$, so $b_1 = \frac{G'}{G}(\frac14)$ and $b_2 = 3\frac{G'}{G}(\frac94)$. Then $\frac{G'}{G}(u) = \sum_k (u+a_k)^{-1}$ by (1.2.1), and this is $\int_0^\infty e^{-ut}W(t)\,dt$ for $u \ge 0$, by termwise integration, since $\operatorname{Re} a_k \ge \gamma_1^2 - \frac14 > 0$ and $\sum_k |a_k|^{-1} < \infty$. Under RH, $\frac{G'}{G}(u) = \int (u+z)^{-1} U_{\mathrm{RH}}(dz)$ by Proposition 1.2.1. □

The elementary part is $2 + \log 2 - \frac12\log\pi - \frac12\gamma_E = 1.832174405184478792$, and $b_1 = 0.023095708966121033814$, which is $1 + \frac12\gamma_E - \frac12\log 4\pi = \sum_\rho \rho^{-1}$, and $b_2 = 0.069066231530000676225$, by the closed forms of ξ'/ξ and by numerical differentiation, which agree to every digit printed (`sqfree_constants.py`). So $b_1 - 2b_2 = -0.11503675409388031864$, and the elementary part plus the zero part reproduces C to all 25 digits.

The variance identity of the first author's note, $\sum_k a_k^{-1} = \frac12 \operatorname{Var} X$, with $a_k = \gamma_k^2$ under RH, is the formula $\mathbb{E}T = \sum_k a_k^{-1} = \xi''(\frac12)/(2\xi(\frac12))$ of Section 1.2, and the alternation of the even cumulants of X, which the note derives under RH, holds with no hypothesis (Section 13.1.2).

10.3.3 Number fields

Let K be a number field with r_1 real and r_2 pairs of complex embeddings and discriminant d_K, and put $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(\frac{s}{2})$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_K(s) = s(s-1)\,|d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s).$$

Then ξ_K is entire of order one, $\xi_K(s) = \xi_K(1-s)$ by Hecke's functional equation [168, Chapter XIII], ξ_K is real on the real axis, and its zeros are the nontrivial zeros of ζ_K, all in the strip $0 < \operatorname{Re} s < 1$; for $K = \mathbb{Q}$, $\xi_{\mathbb{Q}} = 2\xi$. For $K = \mathbb{Q}(i)$, where $\zeta_K(s) = \zeta(s)L(s,\chi_{-4})$, the duplication formula gives $\xi_K(s) = \xi(s)\,\xi(s,\chi_{-4})$ exactly, with $\xi(s,\chi_{-4}) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_{-4})$ the completed function of the odd character modulo 4 in the normalisation of Section 8.5.1.

PROPOSITION 10.3.6. *Suppose $\zeta_K(\frac12) \neq 0$ and let $G_K(u) = \xi_K(\frac12 + \sqrt{u})/\xi_K(\frac12)$. Then $1/G_K$ is the Laplace transform of a GGC if and only if every zero of ζ_K in the critical strip lies on the line $\operatorname{Re} s = \frac12$, the Riemann hypothesis for ζ_K. Its drift is then 0 and its Thorin measure is $\sum_{\gamma>0}\delta_{\gamma^2}$, over the ordinates of the zeros counted with multiplicity. A real zero $\beta \in (\frac12, 1)$ of ζ_K would be a pole of $1/G_K$ at the point $(\beta - \frac12)^2$ of the positive axis.*

Proof. Since $\xi_K(\frac12 + w)$ is even and entire of order one, G_K is entire of order $\frac12$, and Hadamard's theorem gives $G_K(u) = \prod_k (1 + u/a_{K,k})$ with $a_{K,k} = -(\rho_k - \frac12)^2$, one factor for each pair $\{\rho, 1-\rho\}$ of zeros of ξ_K, and $\sum_k |a_{K,k}|^{-1} < \infty$. If every zero lies on the line, every $a_{K,k} = \gamma_k^2$ is positive and $1/G_K$ is a convergent product of gamma Laplace transforms, a GGC with the stated Thorin measure and no drift. Conversely, the Laplace transform of a GGC is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, so G_K has no zero there, and its zeros $(\rho - \frac12)^2$ lie in $(-\infty, 0]$; since $\rho \neq \frac12$, this means $\rho - \frac12 \in i\mathbb{R} \setminus \{0\}$, that is $\operatorname{Re}\rho = \frac12$. A real zero β gives the zero $(\beta - \frac12)^2 > 0$ of G_K. □

The hypothesis $\zeta_K(\frac12) \neq 0$ fixes the normalisation. Armitage found number fields whose Dedekind zeta function vanishes at the centre [8]; there the zero of $\xi_K(\frac12 + w)$ at $w = 0$ has even order $2m$, and the proposition applies to $u^{-m}\xi_K(\frac12 + \sqrt{u})$ normalised at $u = 0$.

THEOREM 10.3.7. *Let*

$$A_K = 2 + \tfrac12 \log|d_K| + r_1\big(\log 2 - \tfrac12 \log\pi - \tfrac12\gamma_E\big) + r_2\big(2 - \gamma_E - \log 2\pi\big).$$

Then the constant of (10.3.7) *is*

$$C_K = A_K + \frac{\xi_K'}{\xi_K}(1) - 2\,\frac{\xi_K'}{\xi_K}(2) = A_K + \sum_k \Big(\frac{1}{\frac14 + a_{K,k}} - \frac{6}{\frac94 + a_{K,k}}\Big),$$

the sum running over the pairs $\{\rho, 1-\rho\}$ of zeros of ξ_K, with multiplicity, a zero of order $2m$ at $s = \frac12$ counting as m pairs with $a_{K,k} = 0$. Under the Riemann hypothesis for ζ_K, and $\zeta_K(\frac12) \neq 0$, the zero part is $\frac{G_K'}{G_K}(\frac14) - 6\frac{G_K'}{G_K}(\frac94)$, the same combination of values of the Stieltjes transform of the Thorin measure of Proposition 10.3.6 as over $\mathbb{Q}$.

Proof. From the definition of ξ_K,

$$(s-1)\frac{\zeta_K(s)}{\zeta_K(2s)} = \frac{\xi_K(s)}{\xi_K(2s)} \cdot 2(2s-1)\,|d_K|^{s/2}\,\frac{\Gamma_{\mathbb{R}}(2s)^{r_1}\Gamma_{\mathbb{C}}(2s)^{r_2}}{\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}},$$

and C_K is the logarithmic derivative of the left side at $s=1$, by the Laurent expansions behind (10.3.7). With $\Gamma'_{\mathbb{R}}/\Gamma_{\mathbb{R}}(s) = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2})$, $\Gamma'_{\mathbb{C}}/\Gamma_{\mathbb{C}}(s) = -\log 2\pi + \psi(s)$, $\psi(1) = -\gamma_E$, $\psi(\frac{1}{2}) = -\gamma_E - 2\log 2$ and $\psi(2) = 1-\gamma_E$, the logarithmic derivative of the elementary factor at $s=1$ is A_K, and that of $\xi_K(s)/\xi_K(2s)$ is $\frac{\xi'_K}{\xi_K}(1) - 2\frac{\xi'_K}{\xi_K}(2)$. The Hadamard product of $\xi_K(\frac{1}{2}+w)$, which is even in w, gives $\frac{\xi'_K}{\xi_K}(\frac{1}{2}+w) = \sum_k 2w/(w^2+a_{K,k})$, and the values at $w=\frac{1}{2}$ and $w=\frac{3}{2}$ give the zero form. □

For the Gaussian field $\mathbb{Q}(i)$, with $r_2=1$ and $d_K=-4$, `sqfree_constants.py` gives $c_K/\zeta_K(2) = 3/(2\pi L(2,\chi_{-4})) = 0.521269392989129$, with $L(2,\chi_{-4})$ Catalan's constant, and

$$\gamma_K = \gamma_E + \frac{L'}{L}(1,\chi_{-4}) = 0.822825249678847033,$$
$$\frac{\zeta'_K}{\zeta_K}(2) = -0.48089570872664776864,$$

and $C_K = 1.784616667132142570284$, where $\frac{L'}{L}(1,\chi_{-4}) = \gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac{1}{4}) = 0.24560958477731417239$. In Theorem 10.3.7, $A_K = 4 - \gamma_E - \log\pi = 2.2780544492490669653$ and the zero part is the sum of the zero part $b_1 - 2b_2 = -0.11503675409388$ of ζ and the zero part $\frac{\xi'_\chi}{\xi_\chi}(1) - 2\frac{\xi'_\chi}{\xi_\chi}(2) = -0.378401028023044$ of $\xi_\chi = \xi(\cdot,\chi_{-4})$: the zeros of ζ and of $L(s,\chi_{-4})$ simply add their Thorin atoms, and the two sides of the theorem differ by 4.6×10^{-41}. The 289 zeros of $L(s,\chi_{-4})$ below height 400, located in Arb, give -0.365385920572 for the L part, and their smooth tail -0.013015062 brings the sum within 4.6×10^{-8} of the closed form.

10.3.4 The Thorin atoms in the squarefree count

Put $s = \frac{1}{4} + \frac{1}{2}\sqrt{u}$, so that $2s = \frac{1}{2} + \sqrt{u}$. Lemma 10.3.2 and the clock give, for $u \geq 0$ and with no hypothesis,

$$D\Big(\frac{1}{4} + \frac{\sqrt{u}}{2}\Big) = \Gamma_{\mathrm{sq}}\Big(\frac{1}{4} + \frac{\sqrt{u}}{2}\Big)\,\frac{\xi(\frac{1}{4}+\frac{1}{2}\sqrt{u})}{\xi(\frac{1}{2})}\,\mathbb{E}e^{-uT},$$

since $1/G$ is the Laplace transform of the clock T (Section 1.2). The poles of D in the strip $0 < \operatorname{Re} s < \frac{1}{2}$ are therefore among the points where G vanishes, $u = -a_k$, that is $s = \rho_k/2$, and they are all of these points when $\zeta(\rho_k/2) \neq 0$, as under RH; under RH they are the atoms γ_k^2 of U_{RH}, placed at $u = -\gamma_k^2$, that is at $s = \frac{1}{4} \pm \frac{i}{2}\gamma_k$.

At a simple zero ρ, $\zeta(2s) = 2\zeta'(\rho)(s-\rho/2) + O((s-\rho/2)^2)$, so the residue of $D(s)x^s/s$ at $s=\rho/2$ is $R_\rho x^{\rho/2}$, and formally

$$\Delta(x) \approx 1 + \sum_{\rho} R_\rho\, x^{\rho/2} + \sum_{k=1,3,5,\dots} r_k\, x^{-k}, \qquad R_\rho = \frac{\zeta(\rho/2)}{\rho\,\zeta'(\rho)}, \qquad r_k = \frac{\zeta(-k)}{-2k\,\zeta'(-2k)}, \tag{10.3.11}$$

where $1 = D(0)$ is the residue at $s=0$ and $r_k x^{-k}$ is the residue at the trivial pole $s=-k$. For the first zero $R_{\rho_1} = 0.028803 - 0.0961835i$ and $|R_{\rho_1}| = 0.100404$, with $\zeta(\rho_1/2) = 1.03142 + 0.452556i$, and for the next four zeros $|R_\rho| = 0.069179$, 0.0338528, 0.0260772 and 0.0464698; $r_1 = -1.3684327776202059$, $r_3 = 0.23541450677954054$ and $r_5 = -0.020962860062791976$ (`sqfree_constants.py`, `sqfree_atoms.py`). Under RH every term of the sum over ρ has size $x^{1/4}$, which is why $\Delta(x) = O(x^{1/4+\varepsilon})$ is the natural conjecture and why $\Delta(x)/x^{1/4}$ stays of order one in Table 10.8; Corollary 10.3.11 shows that the exponent $\frac14$ cannot be lowered. The series over ρ is not expected to converge absolutely, and the bound $O(x^{1/4+\varepsilon})$ is open even under RH: the known conditional exponents are larger than $\frac14$, among them $\frac{7}{22}$ of Baker and Pintz [13], since improved, and the obstacle is the control of $1/\zeta$ near the critical line.

Figure 10.6 compares $\Delta(x)$ on $[100, 3000]$ with the right side of (10.3.11) truncated after the first K zeros, both divided by $x^{1/4}$ and sampled on the grid $x = 100 + j/20$. The Pearson correlation of the two curves is 0.828, 0.866, 0.915, 0.927 and 0.945 for $K = 10$, 20, 60, 100 and 200, and the root mean square of their difference falls from 0.149 to 0.0868, against 0.304 for $\Delta(x)/x^{1/4}$ itself; what remains includes the unit jumps of Q, which no finite sum of atoms reproduces.

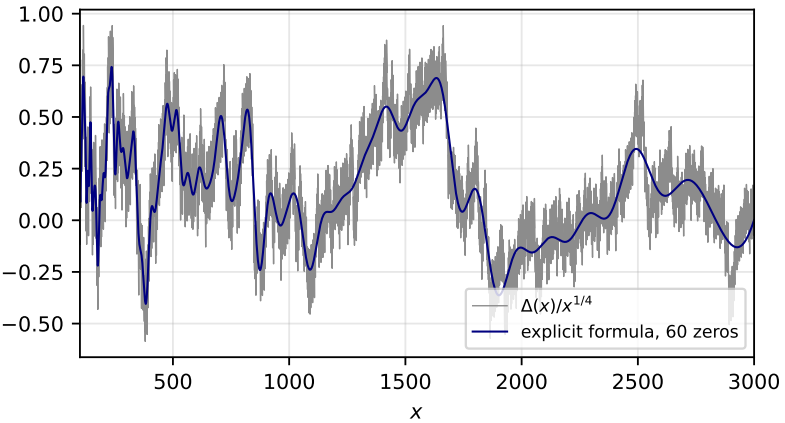


Figure 10.6: The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (10.3.11) with the first 60 zeros and the trivial terms at $s=-1$, -3 and -5, divided by $x^{1/4}$ (dark). Drawn by `sqfree_atoms.py`.

For the sum $\sum_{n\le x}\mu(n)^2/n$ of Proposition 10.3.1 the same atoms appear at $s=\rho/2-1$, with residues $\zeta(\rho/2)x^{\rho/2-1}/((\rho-2)\zeta'(\rho))$ of size $x^{-3/4}$ under RH (of modulus $0.09991\,x^{-3/4}$ for the first zero), inside the error $O(x^{-1/2})$. Likewise the error $\mathcal{E}_\alpha$ of (10.3.4), for $\frac12<\alpha<1$, has atoms of size $x^{1/4-\alpha}$. By (10.3.6), a pointwise bound of that order in either case would follow from $\Delta(x)=O(x^{1/4+\varepsilon})$, and it is open for the same reason.

10.3.5 Smoothed counts

Smoothing the count removes the obstacle. For $x>0$ let

$$Q_1(x)=\sum_{n\ge1}\mu(n)^2e^{-n/x}=\sum_{d\ge1}\frac{\mu(d)}{e^{d^2/x}-1},\qquad Q_2(x)=\sum_{n\ge1}\mu(n)^2\Big(1+\frac{2n}{x}\Big)e^{-2n/x},$$

the second form of Q_1 by $\mu(n)^2=\sum_{d^2\mid n}\mu(d)$. If $G_{1,x}$ is exponential and $G_{2,x}$ has the Gamma(2) law, both with mean x, then $P(G_{1,x}\ge n)=e^{-n/x}$ and $P(G_{2,x}\ge n)=(1+2n/x)e^{-2n/x}$, so $Q_1(x)=\mathbb{E}Q(G_{1,x})$ and $Q_2(x)=\mathbb{E}Q(G_{2,x})$. Since $\int_0^\infty e^{-n/x}x^{-s-1}dx=\Gamma(s)n^{-s}$ and $\int_0^\infty(1+2n/x)e^{-2n/x}x^{-s-1}dx=(1+s)\Gamma(s)(2n)^{-s}$, the Mellin transforms are

$$\begin{aligned}\int_0^\infty Q_1(x)\,x^{-s-1}dx&=\Gamma(s)D(s),\\ \int_0^\infty Q_2(x)\,x^{-s-1}dx&=(1+s)\Gamma(s)\,2^{-s}D(s),\qquad \operatorname{Re}s>1.\end{aligned}\tag{10.3.12}$$

The kernels $\Gamma(s)$ and $(1+s)\Gamma(s)2^{-s}$ have no zeros in $\operatorname{Re}s>0$ and decay exponentially on vertical lines.

Lemma 10.3.8. *If $D(s)=\zeta(s)/\zeta(2s)$ is analytic in the half plane $\operatorname{Re}s>\frac14$ apart from $s=1$, then RH holds.*

Proof. Let ρ be a zero with $\beta=\operatorname{Re}\rho>\frac12$. Then $\operatorname{Re}(\rho/2)>\frac14$, so the pole of $1/\zeta(2s)$ at $s=\rho/2$ must be cancelled, that is $\zeta(\rho/2)=0$. Since $\frac14<\operatorname{Re}(\rho/2)<\frac12$, the functional equation and the reflection $s\mapsto\bar s$ make $\rho'=1-\bar\rho/2$ a zero, with $\operatorname{Re}\rho'=1-\beta/2>\frac12$ and $\operatorname{Im}\rho'=\operatorname{Im}\rho/2$. Iterating gives zeros ρ_j whose real parts tend to $\frac23$, the fixed point of $\beta\mapsto1-\beta/2$, and whose imaginary parts are $\operatorname{Im}\rho/2^j$. Since $\zeta<0$ on $(0,1)$, $\operatorname{Im}\rho\ne0$, so the ρ_j are distinct and accumulate at $\frac23$, where ζ is analytic and not identically zero, a contradiction. □

Theorem 10.3.9. (i) *RH holds if and only if $Q_1(x)=\frac{6}{\pi^2}x+O(x^{1/4+\varepsilon})$ for every $\varepsilon>0$, and if and only if $Q_2(x)=\frac{6}{\pi^2}x+O(x^{1/4+\varepsilon})$ for every $\varepsilon>0$.*

(ii) *Under RH, if the zeros are simple, then*

$$Q_1(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho A_\rho\, x^{\rho/2} + \frac{c_1 \log x + c_2}{x} + O(x^{-2}),$$
$$Q_2(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho B_\rho\, x^{\rho/2} + \frac{2c_1}{x} + O(x^{-2}),$$
$$A_\rho = \frac{\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)}, \qquad B_\rho = \Big(1+\frac{\rho}{2}\Big)\frac{2^{-\rho/2}\,\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)},$$

the sums over ρ taken over $|\operatorname{Im}\rho| < 2T_j$ along a sequence $T_j \to \infty$, with

$$c_1 = -\frac{\zeta(-1)}{2\zeta'(-2)} = -1.36843277762021,$$
$$c_2 = c_1\Big(1 - \gamma_E + \frac{\zeta'(-1)}{\zeta(-1)} - \frac{\zeta''(-2)}{\zeta'(-2)}\Big) = -0.339381226129599.$$

Proof. (i) Suppose RH. By Section 14.2 of [324], $1/\zeta(\sigma+it) = O(|t|^\varepsilon)$ uniformly for $\sigma \ge \frac12 + \varepsilon$, and $\zeta(\sigma+it)$ grows at most polynomially in $|t|$ on vertical strips, while $\Gamma(s)$ decays exponentially. Move the line of integration in $Q_1(x) = \frac{1}{2\pi i}\int_{(2)} \Gamma(s)D(s)x^s\,ds$ to $\operatorname{Re} s = \frac14 + \varepsilon$. The only pole crossed is $s=1$, with residue $\Gamma(1)/\zeta(2)\cdot x = 6x/\pi^2$; the point $s=\frac12$ is regular because $\zeta(2s)$ has a pole there. On the new line $|\Gamma(s)D(s)|$ is integrable and the integral is $O(x^{1/4+\varepsilon})$. The same argument with the kernel $(1+s)\Gamma(s)2^{-s}$ treats Q_2.

Conversely, suppose $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$. For $\operatorname{Re} s > 1$, by (10.3.12),

$$\Gamma(s)D(s) - \frac{6}{\pi^2}\,\frac{1}{s-1} = \int_1^\infty \Big(Q_1(x) - \frac{6}{\pi^2}x\Big)x^{-s-1}dx + \int_0^1 Q_1(x)\,x^{-s-1}dx.$$

The first integral converges and is analytic in $\operatorname{Re} s > \frac14$, by the hypothesis, and the second is entire, since $Q_1(x) = O(e^{-1/x})$ as $x \to 0$. As Γ has no zeros, D is analytic in $\operatorname{Re} s > \frac14$ apart from $s=1$, and Lemma 10.3.8 gives RH. The same argument applies to Q_2.

(ii) By Theorem 9.7 of [324] there are heights $T_j \in [j, j+1]$ at which $1/\zeta(\sigma \pm 2iT_j) = T_j^{O(1)}$ uniformly for $-1 \le \sigma \le 2$, and on the left of the line $\operatorname{Re} s = -\frac12$ the functional equation bounds $1/\zeta(2s)$ polynomially. Move the line of integration to $\operatorname{Re} s = -\frac52$ along the rectangles of height T_j; the exponential decay of Γ makes the horizontal sides vanish as $j \to \infty$, and the integral on the new line is $O(x^{-5/2})$. The poles crossed are $s=1$; the points $\rho/2$, with the residues $A_\rho x^{\rho/2}$ and $B_\rho x^{\rho/2}$; $s=0$, where $\Gamma(s)D(s)$ and $(1+s)\Gamma(s)2^{-s}D(s)$ have residue $D(0) = \zeta(0)/\zeta(0) = 1$; $s=-1$; and $s=-2$, a simple pole of both, which gives a multiple of x^{-2}. At $s=-1$, Γ has a simple pole and $1/\zeta(2s)$ another, from the trivial zero -2 of ζ, so

$\Gamma(s)D(s) = c_1(s+1)^{-2} + c_2(s+1)^{-1} + O(1)$, with c_1 and c_2 from the expansions $\Gamma(s) = -(s+1)^{-1}(1 + (1-\gamma_E)(s+1) + \cdots)$, $\zeta(s) = \zeta(-1)(1 + \frac{\zeta'(-1)}{\zeta(-1)}(s+1) + \cdots)$ and $\zeta(2s) = 2\zeta'(-2)(s+1)(1 + \frac{\zeta''(-2)}{\zeta'(-2)}(s+1) + \cdots)$, and the residue of $\Gamma(s)D(s)x^s$ is $x^{-1}(c_1 \log x + c_2)$. The factor $1+s$ makes the pole of $(1+s)\Gamma(s)2^{-s}D(s)$ at $s=-1$ simple, with residue $2c_1$ times x^{-1}. □

At the even negative integers $\Gamma(s)D(s)$ has simple poles, since $\zeta(s)$ vanishes there together with $\zeta(2s)$, and at the odd ones double poles: `sqfree_smooth.py` computes the Laurent coefficients at $s = -1, \dots, -6$ by the trapezoid rule on circles of radius 0.3 and reproduces the closed forms of c_1 and c_2 within 5×10^{-46}. For Q_2 the double pole at $s=-1$ becomes a simple one, with residue $2c_1$.

The coefficients A_ρ are tiny, $|A_{\rho_1}| = 1.648 \times 10^{-5}$ and 6.29×10^{-82} at the hundredth zero, because $|\Gamma(\rho/2)|$ decays like $e^{-\pi\gamma/4}$. `sqfree_smooth.py` checks part (ii) for Q_1: the smoothed error is the sum of the atom terms and the trivial terms to the working precision. Table 10.10 in Section 10.3.7 does the same for Q_2.

The Perron kernel x^s/s decays only like $1/|t|$, so the atom series for $Q(x)$ has no exponential damping, and bounding its partial sums would need control of $\sum_{|\gamma|\le T} |\zeta(\rho/2)/(\rho\zeta'(\rho))|$, which is out of reach.

10.3.6 Mean square, and how much smoothing the sharp count needs

Two further results bracket the sharp problem. The first settles it on average; it is presumably folklore. For $\operatorname{Re} s > 1$ let

$$F(s) = \int_1^\infty \Delta(x)\, x^{-s-1} dx = \frac{D(s)}{s} - \frac{6}{\pi^2}\,\frac{1}{s-1},$$

the second form because $\int_1^\infty Q(x)x^{-s-1}dx = D(s)/s$.

Theorem 10.3.10. *RH holds if and only if $\int_X^{2X} \Delta(x)^2\,dx = O(X^{3/2+\varepsilon})$ for every $\varepsilon > 0$, that is, if and only if the root mean square of Δ on every range $[X, 2X]$ is $O(X^{1/4+\varepsilon})$.*

Proof. Suppose RH. Then F continues analytically to $\operatorname{Re} s > \frac14$: the poles of D at the points $\rho/2$ lie on the line $\operatorname{Re} s = \frac14$, $s = \frac12$ is regular, and the pole at $s=1$ cancels. By Section 14.2 of [324], under RH $\zeta(\sigma+it) = O(|t|^{\max(0,\frac12-\sigma)+\varepsilon})$ and $1/\zeta(2\sigma+2it) = O(|t|^\varepsilon)$ uniformly for $\sigma \ge \frac14+\varepsilon$ and $|t| \ge 1$, so $F(\sigma+it) = O(|t|^{-3/4+2\varepsilon})$ uniformly there. For $w > 2$ let $I(w) = \int_1^\infty \Delta(x)^2 x^{-w-1}dx$, the Laplace transform in $v = \log x$ of the nonnegative function $\Delta(e^v)^2$. Plancherel's theorem for the function $v \mapsto \Delta(e^v)e^{-wv/2}$ on $(0,\infty)$, whose Fourier transform is $t \mapsto F(\frac{w}{2}+it)$, and the symmetry $F(\bar s) = \overline{F(s)}$ give

$$I(w) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\Big(\frac{w}{2}+it\Big)\,F\Big(\frac{w}{2}-it\Big)\,dt, \qquad w > 2.$$

The right side is analytic in w on $\operatorname{Re} w > \frac12$, since the integrand is analytic there and $O(|t|^{-3/2+4\varepsilon})$ locally uniformly. So I, the Laplace transform of a nonnegative function, continues analytically to a neighbourhood of every point of $(\frac12,\infty)$, and Landau's theorem [340, Chapter II, Theorem 5b] puts its abscissa of convergence at or below $\frac12$. Hence $I(w) < \infty$ for every $w > \frac12$, and $\int_X^{2X} \Delta^2\,dx \le (2X)^{w+1} I(w) = O(X^{3/2+\varepsilon})$ with $w = \frac12 + \varepsilon$.

Conversely, by the Cauchy Schwarz inequality the hypothesis gives $\int_X^{2X} |\Delta(x)| x^{-\sigma-1} dx = O(X^{1/4+\varepsilon-\sigma})$, so, summing over $X = 2^j$, the integral defining F converges absolutely for $\operatorname{Re} s > \frac14$. Then $D(s) = s\big(F(s) + \frac{6}{\pi^2}\frac{1}{s-1}\big)$ is analytic in $\operatorname{Re} s > \frac14$ apart from $s = 1$, and Lemma 10.3.8 gives RH. □

Table 10.9 gives the root mean square of Δ on dyadic ranges, computed exactly from the sieve: on $[n, n+1)$ the error decreases linearly from $\Delta_n = Q(n) - 6n/\pi^2$ to $\Delta_n - 6/\pi^2$, so $\int_n^{n+1} \Delta^2 = \Delta_n^2 - \frac{6}{\pi^2}\Delta_n + \frac{12}{\pi^4}$, and the extremes over real x are taken at integers or as left limits there. Over the eighteen ranges $[X, 2X]$ with $X = 10^4 \cdot 2^j$, $j = 0, \dots, 13$, and $X = 10^4, \dots, 10^8$, the root mean square divided by $X^{1/4}$ lies between 0.2337 and 0.3606 and the largest $|\Delta(x)|/X^{1/4}$ between 0.6851 and 1.0361; on $[10^3, 2\times10^8]$ the largest value of $\Delta(x)/x^{1/4}$ is 0.9807, near $x = 47523$, and the least -0.8685, near $x = 80157$.

X	$\big(\frac1X\int_X^{2X}\Delta^2\big)^{1/2}/X^{1/4}$	$\max_{[X,2X]}\lvert\Delta(x)\rvert/X^{1/4}$	near $x =$
10^4	0.2701	0.8979	18471
10^5	0.2494	0.6851	128599
10^6	0.2779	0.8382	1618611
10^7	0.3541	0.9931	17199381
10^8	0.2812	0.8636	144151023

Table 10.9: Root mean square and largest absolute value of the squarefree error Δ on the ranges $[X, 2X]$, divided by $X^{1/4}$, over real x; the largest value on $[10^7, 2\times10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed by `sqfree_sieve.py`.

Corollary 10.3.11. (i) *Under RH, for every $\varepsilon > 0$ the set of $x \in [X, 2X]$ with $|\Delta(x)| > X^{1/4+\varepsilon}$ has measure $O(X^{1-\varepsilon})$.*

(ii) *With no hypothesis,*

$$\limsup_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \ge |R_{\rho_1}|, \qquad \liminf_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \le -|R_{\rho_1}|,$$

where $R_{\rho_1} = \zeta(\rho_1/2)/(\rho_1\zeta'(\rho_1))$ and $|R_{\rho_1}| = 0.100404$. The same holds for $Q_2(x) - \frac{6}{\pi^2}x - 1$ with $|B_{\rho_1}| = 9.944\times10^{-5}$ in place of $|R_{\rho_1}|$.

Proof. (i) is Chebyshev's inequality applied to Theorem 10.3.10. For (ii) suppose that $\Delta(x) \le c\,x^{1/4}$ for $x \ge x_0$, with $0 < c < |R_{\rho_1}|$. The function $L(s) =$

$\int_{x_0}^{\infty}(cx^{1/4}-\Delta(x))x^{-s-1}dx$ is the Laplace transform in $\log x$ of a nonnegative function, and for $\operatorname{Re} s>1$

$$L(s)=\frac{c\,x_0^{1/4-s}}{s-\frac14}-F(s)+E(s),$$

with E entire. The right side has no singularity on the real half line $(\frac14,\infty)$: a real pole of D there would be a real zero of ζ in $(\frac12,\infty)$, of which there are none, $s=\frac12$ is regular, and the pole at $s=1$ cancels. By Landau's theorem the integral $L(s)$ converges for $\operatorname{Re} s>\frac14$, so F is analytic there (which already gives RH, by Lemma 10.3.8), and $|L(\sigma+it)|\le L(\sigma)$ for $\sigma>\frac14$. As $\sigma\downarrow\frac14$, $L(\sigma)=c/(\sigma-\frac14)+O(1)$, since F is analytic at $s=\frac14$, where $\zeta(2s)=\zeta(\frac12)\neq0$. At $s=\sigma+\frac{i}{2}\gamma_1$ the first and last terms of the display are $O(1)$, while $F(s)=R_{\rho_1}/(\sigma-\frac14)+O(1)$, because $D(s)/s$ has a simple pole at $\rho_1/2$ with residue R_{ρ_1} (the zero ρ_1 is simple and $\zeta(\rho_1/2)\neq0$). Hence $|R_{\rho_1}|\le c$, a contradiction. The lower bound is the same argument applied to $c\,x^{1/4}+\Delta(x)$, and the smoothed case is the same argument with the kernel $(1+s)\Gamma(s)2^{-s}$, whose residue at $\rho_1/2$ is B_{ρ_1}. □

The second result measures the cost of sharpening. Let $G_{n,x}$ have the Gamma(n) law with mean x, so that $G_{n,x}$ concentrates at x in a window of width $x/\sqrt{n}$; $Q_1(x)=\mathbb{E}Q(G_{1,x})$ and $Q_2(x)=\mathbb{E}Q(G_{2,x})$.

Proposition 10.3.12. *Under RH, uniformly in $n\ge1$,*

$$\mathbb{E}Q(G_{n,x})=\frac{6}{\pi^2}\,x+O_\varepsilon\big(x^{1/4+\varepsilon}\,n^{1/8+\varepsilon}\big).$$

Proof. With rate $\lambda=n/x$, $P(G_{n,x}\ge m)=\Gamma(n,\lambda m)/\Gamma(n)$, whose Mellin transform in λm is $\Gamma(s+n)/(s\Gamma(n))$, so $\mathbb{E}Q(G_{n,x})=\frac{1}{2\pi i}\int_{(2)}D(s)\frac{\Gamma(s+n)}{s\Gamma(n)}\big(\frac{x}{n}\big)^s ds$. Move the line to $\operatorname{Re} s=c=\frac14+\varepsilon$, collecting $6x/\pi^2$ at $s=1$. On the new line $|\Gamma(n+c+it)|/\Gamma(n)=O\big(n^c\exp(-\kappa\min(t^2/n,|t|))\big)$ for an absolute $\kappa>0$, since $\Gamma(n+c)/\Gamma(n)=O(n^c)$ and $|\Gamma(n+c+it)/\Gamma(n+c)|^2=\prod_{k\ge0}\big(1+t^2/(n+c+k)^2\big)^{-1}$; the factors with $k<n$ give $e^{-\kappa t^2/n}$ when $|t|\le n$, and those with $k\le|t|$ give $e^{-\kappa|t|}$ when $|t|>n$. Under RH, $|D(c+it)|=O(|t|^{1/4+\varepsilon})$ for $|t|\ge1$ (Section 14.2 of [324] and the functional equation). The remaining integral is therefore $O\big(x^c\int(1+|t|)^{-3/4+\varepsilon}e^{-\kappa\min(t^2/n,|t|)}dt\big)=O(x^cn^{1/8+\varepsilon})$, the integral being dominated by $|t|\lesssim\sqrt{n}$. □

Balancing the window against the loss gives a bound for the sharp count.

Corollary 10.3.13. *Under RH, $\Delta(x)=O(x^{2/5+\varepsilon})$ for every $\varepsilon>0$.*

Proof. Since Q is nondecreasing and counts integers, $|Q(y)-Q(x)|\le|y-x|+1$, so $|\mathbb{E}Q(G_{n,x})-Q(x)|\le\mathbb{E}|G_{n,x}-x|+1\le x/\sqrt{n}+1$. With Proposition 10.3.12,

$\Delta(x) = O(x^{1/4+\varepsilon}n^{1/8+\varepsilon} + x/\sqrt{n})$, and $n = \lceil x^{6/5} \rceil$ balances the two terms at $x^{2/5+O(\varepsilon)}$. $\square$

The exponent $\frac{2}{5}$ is weaker than the known conditional exponents. If the loss $n^{1/8}$ were replaced by n^{δ}, the same balance would give $\Delta(x) = O(x^{1-3/(8\delta+4)+\varepsilon})$, which tends to $x^{1/4+\varepsilon}$ as $\delta \to 0$; conversely $\Delta(x) = O(x^{1/4+\varepsilon})$ gives $|\mathbb{E}Q(G_{n,x}) - \frac{6}{\pi^2}x| \le \mathbb{E}|\Delta(G_{n,x})| = O(x^{1/4+\varepsilon})$ uniformly in n, by Jensen's inequality. So the sharp conjecture amounts to removing the loss, which needs cancellation in $\int_{|t|\le T} D(\frac{1}{4}+it)x^{it}\,dt/t$ and not only a bound on the size of the integrand.

10.3.7 Gamma(2) smoothing and the Mertens function

The elementary method rests on the terms $\mu(d)(\lfloor x/d^2 \rfloor - x/d^2) = -\mu(d)\{x/d^2\}$ of (10.3.8). Following the idea of Steutel and Thiemann of splitting a random variable into its integer and fractional parts [314], we randomise x by a gamma law. Theorem 10.3.18 shows that here shape two is the largest for which the average is taken under a positive law.

Let Y have the Gamma(ν) law with rate λ, let $a > 0$ and $y = \lambda a$. Then $\mathbb{E}\lfloor Y/a \rfloor = \sum_{m\ge1} P(Y \ge ma) = \sum_{m\ge1} \Gamma(\nu, my)/\Gamma(\nu)$ and $\mathbb{E}Y/a = \nu/y$, so the expected fractional part is

$$\mathbb{E}\{Y/a\} = g_\nu(y) = \frac{\nu}{y} - \sum_{m\ge1} \frac{\Gamma(\nu, my)}{\Gamma(\nu)}.$$

For $\nu = 1$, Steutel and Thiemann observed that $\lfloor Y/a \rfloor$ is geometric and $\{Y/a\}$ an independent truncated exponential variable, with $\mathbb{E}\lfloor Y/a \rfloor = 1/(e^y - 1)$ and $g_1(y) = 1/y - 1/(e^y - 1)$.

Lemma 10.3.14. *For $\nu = 2$,*

$$\mathbb{E}\lfloor Y/a \rfloor = \frac{1}{e^y - 1} + \frac{y}{4\sinh^2(y/2)}, \qquad g_2(y) = \frac{2}{y} - \frac{1}{e^y - 1} - \frac{y}{4\sinh^2(y/2)}.$$

For $0 < |y| < 2\pi$, with the Bernoulli numbers B_{2k},

$$g_2(y) = \frac{1}{2} + \sum_{k\ge2} \frac{(2k-2)B_{2k}}{(2k)!}\, y^{2k-1} = \frac{1}{2} - \frac{y^3}{360} + \frac{y^5}{7560} - \cdots,$$

$g_2(y) = 2/y + O(ye^{-y})$ as $y \to \infty$, and for every $\nu > 0$

$$\int_0^\infty g_\nu(y)\, y^{s-1}\, dy = -\frac{\Gamma(s+\nu)\,\zeta(s)}{s\,\Gamma(\nu)}, \qquad 0 < \operatorname{Re} s < 1,$$

which is $-(1+s)\Gamma(s)\zeta(s)$ for $\nu = 2$.

Proof. For $\nu = 2$, $P(Y \geq ma) = (1+my)e^{-my}$, and $\sum_m e^{-my} = 1/(e^y-1)$, $\sum_m mye^{-my} = y/(4\sinh^2(y/2))$. For the expansion use $1/(e^y-1) = 1/y - \frac12 + \sum_{k\geq 1} B_{2k} y^{2k-1}/(2k)!$ and $y/(4\sinh^2(y/2)) = -y\frac{d}{dy}(e^y-1)^{-1}$. For the Mellin transform write $g_\nu(y) = \mathbb{E}\{Y/y\}$ with Y of law Gamma(ν,1), substitute $u = Y/y$, and use the classical formula $\int_0^\infty \{u\}u^{-s-1}du = -\zeta(s)/s$ for $0 < \operatorname{Re} s < 1$ [324, Section 2.1]: the transform is $\mathbb{E}Y^s \cdot (-\zeta(s)/s)$, with $\mathbb{E}Y^s = \Gamma(s+\nu)/\Gamma(\nu)$. □

The terms of the elementary method now read $\mathbb{E}\big[\mu(d)\big(\lfloor G_{2,x}/d^2\rfloor - G_{2,x}/d^2\big)\big] = -\mu(d)g_2(2d^2/x)$ for $G_{2,x}$ of the Gamma(2) law with mean x, and since $0 \leq g_2(y) \leq \min(1, 2/y)$ the sum over all d converges absolutely, with no truncation at $d^2 \leq x$:

$$Q_2(x) = \sum_{n\geq 1} \mu(n)^2 \Big(1 + \frac{2n}{x}\Big) e^{-2n/x} = \frac{6}{\pi^2}\, x - \sum_{d\geq 1} \mu(d)\, g_2\Big(\frac{2d^2}{x}\Big). \tag{10.3.13}$$

Theorem 10.3.15. *Let*

$$f_2(y) = -2g_2'(y) = \frac{4}{y^2} - \frac{y\cosh(y/2)}{2\sinh^3(y/2)}, \qquad y > 0.$$

Then $f_2 > 0$, $f_2(y) = y^2/60 + O(y^4)$ as $y \to 0$ and $f_2(y) \sim 4/y^2$ as $y \to \infty$, and f_2 is the density of a probability law $\mathcal{V}_2$ with $P(\mathcal{V}_2 \geq y) = 2g_2(y)$ and

$$\mathbb{E}\mathcal{V}_2^s = -2\,\Gamma(s+2)\,\zeta(s), \qquad -3 < \operatorname{Re} s < 1.$$

For every $x > 0$,

$$Q_2(x) = \frac{6}{\pi^2}\, x - \frac12\, \mathbb{E}M\Big(\sqrt{x\mathcal{V}_2/2}\Big).$$

Moreover f_2 is twice the difference between the continuous and the integer mixtures of the Gamma(2) densities $\gamma_{2,\lambda}(y) = \lambda^2 y e^{-\lambda y}$ over their rate:

$$f_2(y) = 2\Big(\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda - \sum_{m\geq 1} \gamma_{2,m}(y)\Big) = -2\int_0^\infty \Big(\{\lambda\} - \frac12\Big)\, \partial_\lambda \gamma_{2,\lambda}(y)\, d\lambda.$$

Proof. With $z = y/2$, $f_2(y) > 0$ is $(\sinh z/z)^3 > \cosh z$, Lazarević's inequality [170]. The expansion at 0 follows from Lemma 10.3.14. The mass is $2(g_2(0+) - g_2(\infty)) = 1$ and the tail is $2g_2(y)$. Integrating by parts in the Mellin formula of Lemma 10.3.14 gives $\mathbb{E}\mathcal{V}_2^s = 2s\int_0^\infty y^{s-1} g_2(y)\, dy = -2s(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$, and the behaviour at 0 and at infinity extends the integral and the identity to $-3 < \operatorname{Re} s < 1$. For the representation, $P(\sqrt{x\mathcal{V}_2/2} \geq d) = P(\mathcal{V}_2 \geq 2d^2/x) = 2g_2(2d^2/x)$, so $\frac12 \mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = \sum_d \mu(d) g_2(2d^2/x)$, the interchange being justified by absolute convergence, and (10.3.13) gives the formula. Finally $\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda = 2/y^2$ and $\sum_m \gamma_{2,m}(y) = y\sum_m m^2 e^{-my} = y\cosh(y/2)/(4\sinh^3(y/2))$, and the last form is the Euler Maclaurin formula $\int_0^\infty h - \sum_{m\geq 1} h(m) = -\int_0^\infty (\{\lambda\} - \frac12) h'(\lambda)\, d\lambda$ for $h = \gamma_{2,\cdot}(y)$, which vanishes at $\lambda = 0$. □

The exponential case is the same with

$$f_1(y) = -2g_1'(y) = \frac{2}{y^2} - \frac{1}{2\sinh^2(y/2)} = 2\Big(\int_0^\infty \lambda e^{-\lambda y}\,d\lambda - \sum_{m\ge 1} m e^{-my}\Big) > 0,$$

positive because $\sinh z > z$, with $f_1(0+) = \frac16$, $\mathbb{E}\mathcal{V}_1^s = -2\Gamma(s+1)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, and

$$Q_1(x) = \frac{6}{\pi^2}\,x - \frac12\,\mathbb{E}M\Big(\sqrt{x\mathcal{V}_1}\Big).$$

Corollary 10.3.16. (i) *With no hypothesis, $Q_\nu(x) = \frac{6}{\pi^2}x + o(x^{1/2})$ for $\nu = 1, 2$.* (ii) *Under RH, $|Q_2(x) - \frac{6}{\pi^2}x| \le \frac12\mathbb{E}|M(\sqrt{x\mathcal{V}_2/2})| = O(x^{1/4+\varepsilon})$, and likewise for Q_1; this is a second proof of the forward half of Theorem 10.3.9(i).*

Proof. (i) $|M(t)| \le t$ and $M(t) = o(t)$, and $\mathbb{E}\mathcal{V}_\nu^{1/2} < \infty$, so dominated convergence gives $\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = o(x^{1/2})$, and likewise for $\mathcal{V}_1$. (ii) RH gives $M(t) = O(t^{1/2+\varepsilon})$ [324, Section 14.25], and $\mathbb{E}\mathcal{V}_\nu^{1/4+\varepsilon} < \infty$. □

Table 10.10 checks (10.3.13) and Theorem 10.3.9(ii) for Q_2. It computes $Q_2(x) = \sum_d \mu(d)\,\mathbb{E}\lfloor G_{2,x}/d^2\rfloor = \sum_d \mu(d)\big(1/(e^t-1) + t/(4\sinh^2(t/2))\big)$, $t = 2d^2/x$, whose terms with $t > 140$ are below 10^{-55}, and this agrees with the direct sum over n, in double precision, within 10^{-10} up to $x = 10^6$. The sum (10.3.13) truncated at $d \le 2\times10^5$, as in the first author's note, loses 2.2×10^{-4} at $x = 10^6$, because $g_2(2d^2/x)$ is close to x/d^2 for large d and the tail $x\sum_{d>2\times10^5}\mu(d)/d^2$ is left out.

x	$Q_2(x) - \frac{6}{\pi^2}x$	$d \le 2\cdot10^5$	$\frac{Q_2(x) - \frac{6}{\pi^2}x - 1}{x^{1/4}}$	100 atoms	trivial terms	residual
10^3	0.997161220898	0.997161002208	-5.048×10^{-4}	-1.0571×10^{-4}	-2.73307×10^{-3}	-1.0×10^{-23}
10^4	1.00094524529	1.00094305839	9.452×10^{-5}	1.21889×10^{-3}	-2.73648×10^{-4}	-1.3×10^{-30}
10^5	0.996620193442	0.99659832449	-1.9006×10^{-4}	-3.35244×10^{-3}	-2.73683×10^{-5}	-1.5×10^{-37}
10^6	1.00610290943	1.00588421991	1.9299×10^{-4}	6.10565×10^{-3}	-2.73686×10^{-6}	-1.8×10^{-41}
10^7	0.992114718377		-1.4022×10^{-4}	-7.88501×10^{-3}	-2.73687×10^{-7}	-3.9×10^{-42}
10^8	1.00440085379		4.4009×10^{-5}	4.40088×10^{-3}	-2.73687×10^{-8}	-2.2×10^{-38}

Table 10.10: The Gamma(2) smoothed count: $Q_2(x) - 6x/\pi^2$, the sum (10.3.13) truncated at $d \le 2\times10^5$, the normalised error, the contribution $2\operatorname{Re}\sum_{k\le100} B_{\rho_k}x^{\rho_k/2}$ of the first hundred atoms, the residues at $s = -1,\dots,-6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s = 0$. Computed by `sqfree_smooth.py` at 45 digits.

10.3.8 Gamma shapes: two is the threshold

For every real shape $\nu > 0$, $f_\nu = -2g_\nu'$ is a signed density of total mass $2g_\nu(0+) = 1$, and the argument of Theorem 10.3.15 gives, for Y of the Gamma(ν) law with

rate λ,

$$\mathbb{E}Q(Y) = \frac{6}{\pi^2}\,\mathbb{E}Y - \frac{1}{2}\int_0^\infty M\Big(\sqrt{y/\lambda}\Big)\,f_\nu(y)\,dy,$$
$$\int_0^\infty y^s f_\nu(y)\,dy = -\frac{2\,\Gamma(s+\nu)\,\zeta(s)}{\Gamma(\nu)}, \qquad -\nu < \operatorname{Re} s < 1.$$

Write $\gamma_a(t) = t^{a-1}e^{-t}/\Gamma(a)$ for the Gamma(a) density, and let

$$\mathcal{R}_\nu(y) = y\sum_{m\ge1}\gamma_{\nu+1}(my) = \frac{y^{\nu+1}}{\Gamma(\nu+1)}\sum_{m\ge1} m^\nu e^{-my},$$
$$q_\nu(y) = \frac{\Gamma(\nu+1)\,\big(1-\mathcal{R}_\nu(y)\big)}{y^{\nu+1}},$$

so that $\mathcal{R}_\nu(y)$ is the right Riemann sum with step y of the Gamma($\nu+1$) density, whose integral is 1.

Lemma 10.3.17. *For $\nu > 0$ and $y > 0$,*

$$f_\nu(y) = \frac{2\nu\big(1-\mathcal{R}_\nu(y)\big)}{y^2} = \frac{2y^{\nu-1}}{\Gamma(\nu)}\,q_\nu(y),$$

so $f_\nu \ge 0$ exactly when $\mathcal{R}_\nu \le 1$. For $0 < y < 2\pi$,

$$q_\nu(y) = -\sum_{k\ge0}\zeta(-\nu-k)\,\frac{(-y)^k}{k!},$$

and $\mathcal{R}_1(y) = (y/2)^2/\sinh^2(y/2)$, $\mathcal{R}_2(y) = (y/2)^3\cosh(y/2)/\sinh^3(y/2)$.

Proof. With Y of the Gamma(ν,1) law and $c = 1/y$, $g_\nu(y) = \mathbb{E}\{cY\} = c\nu - \sum_{m\ge1} P(Y \ge m/c)$, whose derivative in c is $\nu - \sum_m (m/c^2)p_\nu(m/c)$, with $p_\nu = \gamma_\nu$ the density of Y. Since $tp_\nu(t) = \nu\gamma_{\nu+1}(t)$, the derivative is $\nu(1-\mathcal{R}_\nu(1/c))$, and returning to y gives $g'_\nu(y) = -\nu(1-\mathcal{R}_\nu(y))/y^2$. The series is the expansion $\sum_m m^\nu e^{-my} = \Gamma(\nu+1)y^{-\nu-1} + \sum_{k\ge0}\zeta(-\nu-k)(-y)^k/k!$ of the polylogarithm, valid for $0 < y < 2\pi$. □

So $\mathcal{R}_1 \le 1$ is $\sinh z \ge z$ and $\mathcal{R}_2 \le 1$ is Lazarević's inequality. Near the origin f_ν has the sign of $q_\nu(0+) = -\zeta(-\nu)$, and at an even integer $\nu = 2m$, where $\zeta(-2m) = 0$, the sign of $q_{2m}(y) = \zeta(-2m-1)y + O(y^2)$. Since ζ is positive on $(-4,-2)$, $(-8,-6)$, … and $\zeta(-2m-1) = -B_{2m+2}/(2m+2)$, f_ν is negative near the origin exactly when $\nu \in (4j+2, 4j+4]$ for some $j \ge 0$.

Theorem 10.3.18. $\{\nu > 0 :\ f_\nu \ge 0 \text{ on } (0,\infty)\} = (0,2]$. *Thus two is the largest shape for which the gamma smoothed squarefree error is a positive average of the Mertens function, and the obstruction is the trivial zero of ζ at -2.*

Proof. Let $\nu > 2$ and suppose $f_\nu \geq 0$. Since $f_\nu(y) \sim 2\nu/y^2$ at infinity, $\int_1^\infty y^s f_\nu(y)\,dy$ converges and is analytic for $\operatorname{Re} s < 1$. The function $m(s) = -2\Gamma(s+\nu)\zeta(s)/\Gamma(\nu)$ is analytic in a neighbourhood of $(-\nu, 1)$, so $\int_0^1 y^s f_\nu(y)\,dy = m(s) - \int_1^\infty y^s f_\nu(y)\,dy$, a Laplace transform in $-\log y$ of a nonnegative function, continues analytically along the real interval $(-\nu, 1)$. By Landau's theorem [340, Chapter II, Theorem 5b] it converges on all of $(-\nu, 1)$, and there $m(s) = \int_0^\infty y^s f_\nu(y)\,dy > 0$. But $\Gamma(s+\nu) > 0$ on $(-\nu, 1)$, while $\zeta(s) > 0$ on $(-4, -2)$, which meets $(-\nu, 1)$ because $\nu > 2$, so $m(s) < 0$ there, a contradiction.

Let $0 < \nu \leq 2$. The cases $\nu = 1$ and $\nu = 2$ are the inequalities above. Otherwise write $\nu = k + \mu$ with $k \in \{0, 1\}$ and $0 < \mu < 1$. Bernstein's formula $t^\mu = \frac{\mu}{\Gamma(1-\mu)} \int_0^\infty (1 - e^{-st}) s^{-\mu-1} ds$, applied to $t = mh$ in $\sum_m (mh)^{k+\mu} e^{-mh}$ and to t in $\int_0^\infty t^{k+\mu} e^{-t} dt$, together with $h \sum_m (mh)^k e^{-mhc} = k!\,\mathcal{R}_k(hc)/c^{k+1}$ and $\int_0^\infty t^k e^{-ct} dt = k!/c^{k+1}$ for $c = 1 + s$, gives

$$\Gamma(\nu+1)\big(\mathcal{R}_\nu(h) - 1\big) = \frac{\mu\, h^{k+1}}{\Gamma(1-\mu)} \int_0^\infty \big(q_k(h(1+s)) - q_k(h)\big)\, s^{-\mu-1}\, ds, \qquad h > 0.$$

So $\mathcal{R}_\nu \leq 1$ follows if q_k is nonincreasing. For $k = 0$, $q_0(y) = 1/y - 1/(e^y - 1) = g_1(y)$, which decreases because $f_1 > 0$. For $k = 1$, $q_1(y) = 1/y^2 - 1/(4\sinh^2(y/2)) = f_1(y)/2$, and with $z = y/2$

$$q_1'(y) = \frac{1}{4}\Big(\frac{\cosh z}{\sinh^3 z} - \frac{1}{z^3}\Big) < 0$$

by Lazarević's inequality. □

The script `sqfree_gammanu.py` evaluates $1 - \mathcal{R}_\nu$ for twenty shapes at 40 digits, by the series of Lemma 10.3.17 for $y \leq 2.5$ and by the direct sum beyond, on 400 geometrically spaced points in $[10^{-4}, 10^{0.4}]$ and 799 equally spaced points in $(2.5, 60]$. For the ten shapes $\nu = 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 1.9, 1.99, 2$ no grid point gives a negative value; on the range $[0.02, 15]$ the least values, from 2.68×10^{-3} at $\nu = 0.25$ to 6.90×10^{-10} at $\nu = 2$, are taken at its left end. Table 10.11 shows the shapes above 2: the negative set starts at the origin for the shapes in $(2, 4]$ and in $(6, 8]$, as the sign rule predicts.

10.3.9 The Euler product of the squarefree law

For $\sigma > 1$ let N_σ have the law $P(N_\sigma = n) = \mu(n)^2 n^{-\sigma} \zeta(2\sigma)/\zeta(\sigma)$ on the squarefree integers, the squarefree analogue of the zeta distribution of Section 6.1.6. By the Euler product in (10.3.1), $\log N_\sigma = \sum_p B_p \log p$ with independent Bernoulli variables B_p of success probability $p^{-\sigma}/(1 + p^{-\sigma})$, and

$$\mathbb{E} e^{-it \log N_\sigma} = \prod_p \frac{1 + p^{-\sigma-it}}{1 + p^{-\sigma}}, \qquad \log\big(1 + p^{-s}\big) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k}\, p^{-ks}.$$

ν	$-\zeta(-\nu)$	least value of $1-\mathcal{R}_\nu$	at $y=$	grid points with $1-\mathcal{R}_\nu<0$
2.01	-3.012×10^{-4}	-7.310×10^{-10}	0.0273	from 10^{-4} to 0.0362
2.1	-2.7295×10^{-3}	-4.941×10^{-6}	0.265	from 10^{-4} to 0.347
2.5	-8.517×10^{-3}	-1.0763×10^{-3}	1.187	from 10^{-4} to 1.512
3	-8.333×10^{-3}	-7.2304×10^{-3}	2.144	from 10^{-4} to 2.716
3.5	-4.441×10^{-3}	-0.019006	2.9745	from 10^{-4} to 3.723
4	0	-0.035249	3.726	from 10^{-4} to 4.659
5	3.968×10^{-3}	-0.076713	5.084	from 1.716 to 6.386
6	0	-0.12519	6.323	from 3.076 to 7.969
7	-4.167×10^{-3}	-0.17715	7.4915	from 10^{-4} to 1.234, from 4.227 to 9.409
8	0	-0.23064	8.614	from 10^{-4} to 2.269, from 5.307 to 10.85

Table 10.11: Gamma(ν) smoothing above the threshold of Theorem 10.3.18: the limit $-\zeta(-\nu)$ of q_ν at 0, the least value of $1-\mathcal{R}_\nu$, which has the sign of f_ν, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1-\mathcal{R}_\nu<0$. Computed by `sqfree_gammanu.py`.

So the would be Lévy measure of $\log N_\sigma$ is the signed measure

$$\sum_p \sum_{k\geq 1} \frac{(-1)^{k+1}}{k}\, p^{-k\sigma}\, \delta_{k\log p}$$

of finite total variation, a sum over the primes of alternating lattices like the fair coin lattices of Section 6.1.6, here with the geometric weights $p^{-k\sigma}$ of a biased coin: the law is quasi infinitely divisible, and by the uniqueness of the signed Lévy measure it is not infinitely divisible [181]. Its negative masses, at even k, come from the factor $1/\zeta(2s)$, whose weights $-2/k$ at even k outweigh the weights $1/k$ of $\zeta(s)$.

10.3.10 Questions

Question 10.3.19. Prove $\Delta(x)=O(x^{1/4+\varepsilon})$ under RH. Equivalent forms above: remove the loss $n^{1/8}$ in Proposition 10.3.12; or show that the exceptional set of Corollary 10.3.11(i), with $X^{1/4+\varepsilon}$ replaced by a constant multiple, is empty for large X. By that corollary, the latter would follow from a bound $O(X^{1/4+\varepsilon})$ for the oscillation of Δ on the subintervals of $[X,2X]$ of length $X^{1-\varepsilon/2}$.

Question 10.3.20. Sharpen the constant $|R_{\rho_1}|=0.1004$ of Corollary 10.3.11(ii) with a Fejér kernel over many zeros, as Ingham did for $M(x)$ [125], and compare it with the observed $\max|\Delta(x)|/x^{1/4}$, about 1 up to 2×10^8 (Table 10.9). Is $\limsup|\Delta(x)|/x^{1/4}=\infty$, as the linear independence of the ordinates would suggest?

Chapter 11

Infinite divisibility, the eta analogue and almost generalised gamma convolutions

The five sections of this chapter read the laws of Chapters 9 and 10, the shifted clocks T_α, the normal scale mixtures Y_α and the tilted Pólya variables V_α, in other settings; the question throughout is what can be said below the basepoint one without a zero free half plane, which none of the five sections supplies.

11.1 Infinite divisibility and a map of processes

The general theory of infinitely divisible laws is that of two monographs: Sato's book on Lévy processes and infinitely divisible distributions [297] and Bondesson's book on generalised gamma convolutions [39].

None of the results of this section supplies positivity of the Thorin measure uniformly in the height (Section 11.1.5). The laws are also the marginals of processes. Self decomposability makes the centre clock the stationary law of an Ornstein Uhlenbeck process, and the Thorin measure at the basepoint one makes T_1 the integral of a gamma process against the inverse of the Poisson smoothed zero count; Section 11.1.6 derives both. Section 11.1.7 adds Brownian motion run on the clocks and places these processes and those of Chapter 9 on one map, sorted by what each needs: nothing, a zero free half plane, or RH.

We use the notation of Chapter 9. The centre clock T has Laplace transform $\varphi_{1/2}(u) = 1/G(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$, and for $\alpha > \frac{1}{2}$ the shifted clock T_α has Laplace transform $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ and the representation $T_\alpha \stackrel{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$ of Theorem 9.2.5, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and H_a is the first passage time of an independent standard Brownian motion to the level a. For $\alpha \geq 1$, T_α is a GGC whose Thorin density in the chart $\theta = \sqrt{z}$ is v_α/π, with $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ (Theorem 1.2.2 and (1.2.7)); at the basepoint one this is the density $\rho_{1/2} = v_1/\pi$ of (9.1.2). We write $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, so that $b_1 = \sum_\rho \rho^{-1} = 0.0230957089661$, and V_α for the tilted Pólya variable of Section 9.1.1, the Esscher tilt of the Pólya density p of (1.2.4) with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha + s)/\xi(\alpha)$, so

that $V_{1/2}$ is the Pólya variable X and V_1 is the variable V of Section 9.1.4.

11.1.1 Self decomposability: the driving process and unimodality

The Laplace exponent of the centre clock is $\log G(u) = \int_0^\infty (1 - e^{-ut})\, W(t)\, dt/t$, where $W(t) = \sum_k e^{-a_k t}$, with $a_k = -(\rho_k - \frac{1}{2})^2$, is the heat trace of the zeros, equal to $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH and to the arithmetic expression (1.2.5) in any case (Section 1.2). Since $W > 0$ and $W' < 0$ on $(0, \infty)$ with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), T is infinitely divisible with Lévy measure $W(t)\, dt/t$, and it is self decomposable. A law on $[0, \infty)$ is self decomposable if and only if it is the law of $\int_0^\infty e^{-t}\, dZ_t$ for a subordinator Z with $\mathbb{E}\log(1 + Z_1) < \infty$, its background driving Lévy process [346, 133], [297, Theorem 17.5]. The Lévy measure of a self decomposable law on the half line has the form $k(x)\, dx/x$ with k nonincreasing [297, Corollary 15.11], and the Lévy measure of Z then gives mass $k(x)$ to (x, ∞), so that it has the density $-k'(x)$ when k is differentiable. For the centre clock $k = W$, and self decomposability, which is the unconditional statement that W is nonincreasing, gives the following.

Proposition 11.1.1. *With no hypothesis, $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$ for a subordinator Z with Lévy density $h(x) = -W'(x)$, which is $\sum_{\gamma>0} \gamma^2 e^{-\gamma^2 x}$ under RH and is given for every $x > 0$ by the derivative of the explicit formula* (1.2.5) *for W. The condition $\mathbb{E}\log(1 + Z_1) < \infty$ holds because h decays exponentially. Moreover RH holds if and only if h is completely monotone.*

Proof. The representation is the theorem of Wolfe and of Jurek and Vervaat applied to the self decomposable law of T, and the Lévy density of Z can be checked directly. If $\Psi_Z(u) = -\log \mathbb{E}e^{-uZ_1}$, the Laplace exponent of $\int_0^\infty e^{-t}\, dZ_t$ is $\int_0^\infty \Psi_Z(ue^{-t})\, dt = \int_0^u \Psi_Z(w)\, dw/w$, so that $\Psi_Z(u) = u(\log G)'(u)$. By Section 1.2, $(\log G)'(u) = \int_0^\infty e^{-ut} W(t)\, dt$, and since $W(t) = \int_t^\infty h(x)\, dx$ with $h > 0$, Fubini's theorem gives

$$u(\log G)'(u) = \int_0^\infty h(x) \int_0^x ue^{-ut}\, dt\, dx = \int_0^\infty \big(1 - e^{-ux}\big) h(x)\, dx.$$

Here $\int_1^\infty h = W(1)$ and $\int_0^1 xh(x)\, dx = \int_0^1 (W(y) - W(1))\, dy$, which is finite because $W(y)\, dy/y$ is a Lévy measure, so h is the Lévy density of a subordinator without drift. It is not integrable at the origin, since $\int_t^1 h = W(t) - W(1)$ and $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1), so Z makes infinitely many small jumps. As $x \to \infty$, $h(x) \sim \gamma_1^2 e^{-\gamma_1^2 x}$, which gives $\mathbb{E}\log(1 + Z_1) < \infty$ [297, Theorem 25.3]. For the last statement, h is completely monotone exactly when W is, since $W(x) = \int_x^\infty h$, and complete monotonicity of the heat trace is equivalent to RH (Section 1.2 and Table 1.6). □

t	$W(t)$, zeros	$W(t)$, explicit formula
0.005	0.537191964091	0.537191964091
0.01	0.149698011254	0.149698011254
0.02	0.0185412917313	0.0185412917313
0.05	$4.587835048574 \times 10^{-5}$	$4.587835048574 \times 10^{-5}$
0.1	$2.104799795822 \times 10^{-9}$	$2.104799798080 \times 10^{-9}$
t	$-W'(t)$, zeros	$-W'(t)$, explicit formula
0.005	165.1022811964	165.1022811964
0.01	33.72924560562	33.72924560562
0.02	3.741068142123	3.741068142123
0.05	$9.166117863113 \times 10^{-3}$	$9.166117863113 \times 10^{-3}$
0.1	$4.205189085540 \times 10^{-7}$	$4.205189085735 \times 10^{-7}$

Table 11.1: The heat trace W and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (1.2.5) and its derivative with the primes below 2×10^4. Computed by `weil.py` in the folder `sato_bondesson`.

Table 11.1 checks the explicit formula and its derivative against the zero sums.

At $t = 0.1$ the terms of the explicit formula cancel to about nine digits (Table 5.6), and `weil.py` takes the weights $\log p$ in double precision; the cancellation magnifies their rounding errors into the differences of 2×10^{-18} in W and 2×10^{-17} in h that the rows at $t = 0.1$ show. With the weights in multiple precision both explicit columns agree with the zero sums at $t = 0.1$ to every digit shown, as in Table 1.4, so the cancellation costs working precision and nothing more.

Yamazato's theorem states that every self decomposable law on the line is unimodal [347], [297, Theorem 53.1]. The centre clock and every shifted clock T_α, $\alpha > \frac{1}{2}$, are self decomposable with no hypothesis (Theorem 5.1.1 and Corollary 9.2.6), so we obtain the following.

Corollary 11.1.2. *The centre clock is unimodal with no hypothesis, and so is T_α for every $\alpha > \frac{1}{2}$.*

The computed densities (Figure 11.1) have modes near 0.020 for the centre clock and 0.024 for T_1, the latter located only to the spacing of the geometric grid of the Talbot inversion behind the figure, whose points next to 0.024 are 0.0175 and 0.0328. Yamazato's theorem does not reach the reciprocal zeta law of Section 7.1, which is infinitely divisible but not self decomposable (Theorem 7.1.3), and the unimodality of its density remains open (Section 7.2.4). The density of the centre clock is also C^∞, by Orey's condition [297, Proposition 28.3], since $\int_0^r tW(t)\,dt \geq cr^{3/2}$ by the expansion of W at 0 (Section 5.1).

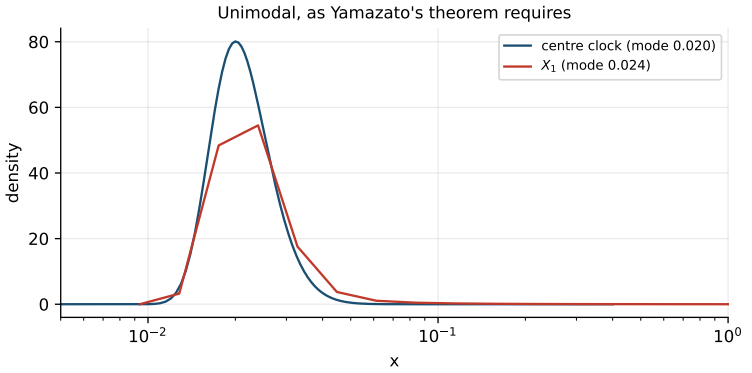


Figure 11.1: The densities of the centre clock T and of the shifted clock T_1, written X_1 in the plot, on a logarithmic axis. The density of T is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 8.1; that of T_1 on the 40 point geometric grid of the Talbot inversion, which is why its curve is piecewise linear. Assembled by `comp.py` and drawn by `figs.py`, both in the folder `sato_bondesson`.

11.1.2 Moments, the tail and the two ends of the Thorin measure

For a GGC with Thorin measure U the Lévy density is $x^{-1}\int e^{-xz}\,U(dz)$ (Section 1.3.1), so the Lévy measure ν_{T_1} of T_1 and its tail are, in the chart $\theta = \sqrt{z}$,

$$\nu_{T_1}(dx) = \frac{dx}{x}\int_0^\infty e^{-x\theta^2}\rho_{1/2}(\theta)\,d\theta, \qquad \nu_{T_1}((x,\infty)) = \int_0^\infty \rho_{1/2}(\theta)\,E_1(x\theta^2)\,d\theta, \tag{11.1.1}$$

where $E_1(y) = \int_y^\infty e^{-w}\,dw/w$ is the exponential integral. As $x \to \infty$ only θ near zero contributes, and $\int_0^\infty E_1(w^2)\,dw = \sqrt{\pi}$ gives $\nu_{T_1}((x,\infty)) \sim \rho_{1/2}(0)\sqrt{\pi/x} = b_1(\pi x)^{-1/2}$. The tail is regularly varying of index $-\frac{1}{2}$, so the Lévy measure is subexponential. By the theorem of Embrechts, Goldie and Veraverbeke [84], an infinitely divisible law on $[0,\infty)$ is subexponential if and only if its Lévy measure, restricted to $(1,\infty)$ and normalised, is subexponential, and if and only if the tail of the law is asymptotic to the tail of the Lévy measure. Hence

$$P(T_1 > x) \sim \nu_{T_1}((x,\infty)) \sim \frac{b_1}{\sqrt{\pi x}} = 0.01303\,x^{-1/2}, \qquad x \to \infty. \tag{11.1.2}$$

Theorem 9.2.5 gives the infinite mean and the tail of T_α for every $\alpha > \frac{1}{2}$ by the Tauberian route, and since T_α is infinitely divisible for every $\alpha > \frac{1}{2}$ (Section 11.1.4), the theorem of Embrechts, Goldie and Veraverbeke, read from the tail of the law to that of the Lévy measure, shows that the Lévy tail of T_α is asymptotic to $b_\alpha(\pi x)^{-1/2}$ as well, wherever the zeros lie.

Table 11.2 compares the survival function of T_1 from Section 9.2.4 with the Lévy tail computed from (11.1.1) and the Thorin density on $[0, 100]$. In the bulk of the law the two differ widely: at $x = 0.02$, just below the mode, the survival probability is 0.747 while the Lévy tail is 0.101.

x	$P(T_1 > x)$	$\nu_{T_1}((x,\infty))$	ratio	$b_1(\pi x)^{-1/2}$
0.1	4.7639×10^{-2}	4.1558×10^{-2}	1.1463	4.1206×10^{-2}
0.5	1.8907×10^{-2}	1.8458×10^{-2}	1.0243	1.8428×10^{-2}
1	1.3199×10^{-2}	1.3041×10^{-2}	1.0121	1.3030×10^{-2}
10	4.1272×10^{-3}	4.1209×10^{-3}	1.0015	4.1206×10^{-3}
100	1.3037×10^{-3}	1.3030×10^{-3}	1.0005	1.3030×10^{-3}

Table 11.2: The subexponential principle for T_1: the tail of the law is asymptotic to the tail of its Lévy measure. The ratio is that of the second column to the third, and the last column is the common asymptote of (11.1.2). Computed by `comp.py` in the folder `sato_bondesson`.

Bondesson's results on the two ends of the Thorin measure explain the shape of T_1: for a GGC the behaviour of the Thorin measure near zero governs the right tail, and its total mass governs the density at the origin [39]. At the basepoint one the Thorin measure U_1 of T_1 (Proposition 9.1.2) has $U_1((0,z]) \sim (b_1/\pi)\sqrt{z}$ as $z \downarrow 0$, which gives the expansion $1 - \varphi_1(u) \sim b_1\sqrt{u}$ of Section 9.1. At the other end the total mass is infinite, since $U_1((0, Y^2]) = M_{1/2}(Y)$ differs from the number $N(Y)$ of zeros with ordinates in $(0, Y]$ by the term $\tau_{1/2}(Y) = O(\log Y)$ of Section 9.1.2, and this is why T_1, like the centre clock (Section 8.2.2.2), is super flat at the origin: its density vanishes faster than any power. Indeed, for every $z_0 > 0$ the Thorin form gives $\varphi_1(u) \le (1 + u/z_0)^{-U_1((0,z_0])}$, and $U_1((0,z_0]) \to \infty$ as $z_0 \to \infty$, so $\varphi_1(u)$ decays faster than any power of u. Markov's inequality $P(T_1 \le x) \le e\,\varphi_1(1/x)$ carries this to the distribution function, and since the density f_{T_1} is nondecreasing up to the mode (Corollary 11.1.2), the bound $x f_{T_1}(x) \le P(T_1 \le 2x)$ for small x carries it to the density.

11.1.3 Uniqueness and quasi infinite divisibility

Lindner, Pan and Sato [181] call a law quasi infinitely divisible if its characteristic function is the quotient of two infinitely divisible characteristic functions, equivalently if it has a Lévy Khintchine form with a signed measure $\nu^+ - \nu^-$ whose parts ν^+ and ν^- are both Lévy measures, so in particular of finite mass away from the origin; the Gaussian coefficient is still nonnegative [181, Lemma 2.7], and the triplet is again unique. Their theory places the tilted Pólya variables V_α precisely. The signed measures in question are those of the Lévy forms of Section 9.1.4: for $\alpha \ge 1$,

$$\log \frac{\xi(\alpha + s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_\alpha(dx),$$

which at $\alpha = 1$ is (9.1.5), a compensated limit, and which for $\alpha > 1$ converges absolutely when $\operatorname{Re} s > 1 - \alpha$. For $\alpha > 1$, read at $s = it$, it is a Lévy Khintchine form for the characteristic function of V_α, with the measure Π_α carried to the negative half line by the reflection $x \mapsto -x$. We write $x_* = 0.28120$ for the root of $e^{-3x} = 1 - e^{-2x}$.

PROPOSITION 11.1.3. *1. The Pólya variable $X = V_{1/2}$ is not quasi infinitely divisible, since its characteristic function $\Xi(t)/\Xi(0)$ vanishes at the ordinates.*

2. For every $\alpha > 1$, V_α is quasi infinitely divisible and not infinitely divisible. Its signed Lévy measure is the reflection to the negative half line of

$$\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)\frac{dx}{x} + \sum_n \frac{\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

whose atoms have total mass $\log\zeta(\alpha)$ and whose absolutely continuous part is positive on $(0, x_*)$ and negative beyond. The negative part of Π_α is therefore carried by (x_*, ∞), and its mass on $(1,\infty)$ is $E_1(\alpha-1) - \sum_{k\ge 0} E_1(\alpha+2+2k)$: it is dominated by the pole term $e^{-(\alpha-1)x}dx/x$, of mass $E_1(\alpha-1)$ on $(1,\infty)$, but is not equal to it.

3. At $\alpha = 1$, V_1 is not quasi infinitely divisible. The candidate measure Π_1 of Proposition 9.1.10 has the pole term $-dx/x$ and the prime atoms $\Lambda(n)/(n\log n)$ at $\log n$, each of infinite mass on $(1,\infty)$; since the gamma density $e^{-3x}(1-e^{-2x})^{-1}/x$ is less than $0.06/x$ there, the positive and the negative part of Π_1 both have infinite mass on $(1,\infty)$, and its Lévy form converges only as a compensated limit, which exists by the prime number theorem.

Items (1) and (2) are due to Nakamura [206]: (2), with the same measure, is his Theorem 1.4, and (1) is the remark after it. The argument of (1) applied to the moment generating function shows that no V_α, $\alpha \in \mathbb{R}$, is infinitely divisible, because $\xi(\alpha + w)/\xi(\alpha)$ is entire with zeros [297, Theorem 25.17].

Proof. (1) A quasi infinitely divisible characteristic function is the exponential of the finite exponent of its Lévy Khintchine form, so it never vanishes, while $\mathbb{E}e^{itX} = \Xi(t)/\Xi(0)$ by (1.2.4), which vanishes at $t = 14.134725$ and at every other ordinate of a zero on the critical line.

(2) Let $\alpha > 1$. Every piece of the Lévy form then converges absolutely, and each piece is a Lévy measure: the positive density $e^{-(\alpha+2)x}(1-e^{-2x})^{-1}dx/x$, the pole term and the atoms. Removing the factor $s(s-1)$ from $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ gives, for $\operatorname{Re} s > 1-\alpha$,

$$\frac{\xi(\alpha+s)}{\xi(\alpha)}\cdot\frac{\alpha}{\alpha+s}\cdot\frac{\alpha-1}{\alpha-1+s} = \pi^{-s/2}\,\frac{\Gamma(\frac{\alpha+s}{2})}{\Gamma(\frac{\alpha}{2})}\,\frac{\zeta(\alpha+s)}{\zeta(\alpha)}.$$

Let e_1 and e_2 be standard exponential variables, $\Gamma_{\alpha/2}$ a gamma variable of shape $\frac{\alpha}{2}$ and unit scale, and $\mathcal{Z}_\alpha$ a variable with the zeta law $P(\mathcal{Z}_\alpha = n) = n^{-\alpha}/\zeta(\alpha)$, all independent of each other and of V_α. Since $\mathbb{E}e^{-se_1/\alpha} = \alpha/(\alpha+s)$, $\mathbb{E}\Gamma_{\alpha/2}^{s/2} = \Gamma(\frac{\alpha+s}{2})/\Gamma(\frac{\alpha}{2})$ and $\mathbb{E}\mathcal{Z}_\alpha^{-s} = \zeta(\alpha+s)/\zeta(\alpha)$, the identity says that

$$V_\alpha - \frac{e_1}{\alpha} - \frac{e_2}{\alpha-1} \stackrel{d}{=} \frac{1}{2}\log\Gamma_{\alpha/2} - \frac{1}{2}\log\pi - \log\mathcal{Z}_\alpha.$$

The right side is infinitely divisible: $\frac{1}{2}\log\Gamma_{\alpha/2}$ has, by Binet's formula [225, §5.9] as in the proof of Proposition 9.1.10, the Lévy measure $e^{-\alpha x}(1-e^{-2x})^{-1}dx/x$ reflected to the negative half line, and by the Euler product $-\log\mathcal{Z}_\alpha$ is compound Poisson with Lévy measure $\sum_n \Lambda(n)(n^\alpha \log n)^{-1}\delta_{-\log n}$, of total mass $\log\zeta(\alpha)$. The variable $-e_1/\alpha - e_2/(\alpha-1)$ is infinitely divisible as well, with Lévy measure $(e^{-\alpha x} + e^{-(\alpha-1)x})\,dx/x$ reflected, so the law of V_α convolved with an infinitely divisible law is infinitely divisible, and V_α is quasi infinitely divisible. Its signed Lévy measure is the difference of the two Lévy measures, which is Π_α reflected, since $e^{-\alpha x}(1-e^{-2x})^{-1} - e^{-\alpha x} = e^{-(\alpha+2)x}(1-e^{-2x})^{-1}$. The absolutely continuous part of Π_α has the density $e^{-(\alpha-1)x}\big(e^{-3x}(1-e^{-2x})^{-1} - 1\big)x^{-1}$, whose sign is that of $e^{-3x} - (1-e^{-2x})$ whatever α is: positive on $(0, x_*)$ and negative beyond, where it is at least $-e^{-(\alpha-1)x}/x$, whose integral over $(1,\infty)$ is $E_1(\alpha-1)$; expanding $(1-e^{-2x})^{-1}$ in powers of e^{-2x} and integrating termwise over $(1,\infty)$ gives the mass of the negative part there. If V_α were infinitely divisible, uniqueness of the triplet [297, Theorem 8.1], applied to the identity in law above, would make its Lévy measure equal to Π_α reflected, which is negative on the reflection of $(x_*, \log 2)$, an interval that carries no atom. So V_α is not infinitely divisible.

(3) Suppose that V_1 is quasi infinitely divisible, so that $\mathbb{E}e^{itV_1} = \exp\Psi_{V_1}(t)$ with

$$\Psi_{V_1}(t) = ict - \frac{A}{2}t^2 + \int_{\mathbb{R}} \big(e^{ity} - 1 - ity\mathbf{1}\{|y|\le 1\}\big)\,\mu(dy),$$

where A and c are real and μ is a signed measure on $\mathbb{R}\setminus\{0\}$ with $\int \min(1,y^2)\,|\mu|(dy) < \infty$. As in (1), the characteristic function never vanishes, so $\xi(1+it) \neq 0$ for real t; and $\xi(1+s) \neq 0$ for $\operatorname{Re} s > 0$ by the Euler product. So $\ell(s) = \log(\xi(1+s)/\xi(1))$, with $\ell(0) = 0$, is a continuous logarithm on the closed half plane $\operatorname{Re} s \ge 0$, holomorphic inside, and $\Psi_{V_1}(t) = \ell(it)$, since both are continuous logarithms of $\xi(1+it)/\xi(1)$ that vanish at $t=0$. For $r>0$ put $\Delta_r F(t) = F(t+r) + F(t-r) - 2F(t)$. Then

$$\Delta_r\Psi_{V_1}(t) = \int_{\mathbb{R}} e^{ity}\,\lambda_r(dy), \qquad \lambda_r(dy) = -Ar^2\delta_0(dy) + (2\cos ry - 2)\,\mu(dy),$$

and λ_r is a finite signed measure because $|2\cos ry - 2| \le \min(4, r^2y^2)$. On the other side, let $\check{\Pi}_1$ be the reflection of Π_1 to the negative half line. For $\sigma > 0$ the variable $V_{1+\sigma}$ has the characteristic function $\exp(\ell(\sigma+it) - \ell(\sigma))$, and by

(2) its signed Lévy measure is the reflection of $\Pi_{1+\sigma}(dx) = e^{-\sigma x}\Pi_1(dx)$, that is $e^{\sigma y}\check{\Pi}_1(dy)$. The second difference removes the drift and the compensator, so

$$\Delta_r \ell(\sigma + i\,\cdot\,)(t) = \int_{\mathbb{R}} e^{ity}\, m_\sigma(dy), \qquad m_\sigma(dy) = e^{\sigma y}(2\cos ry - 2)\,\check{\Pi}_1(dy),$$

where m_σ is a finite signed measure, the atoms contributing at most $4\log\zeta(1+\sigma)$ to its total variation. As $\sigma \downarrow 0$, m_σ tends to $m_0 = (2\cos ry - 2)\,\check{\Pi}_1$ in the space of tempered distributions, by dominated convergence, because $|m_0|$ is locally finite and $|m_0|((-R,0)) = O(R)$ as $R \to \infty$: the factor $2\cos ry - 2$ tames $\check{\Pi}_1$ near the origin, and $\Lambda(n) \le \log n$ bounds the atom at $-\log n$ by $1/n$. So the Fourier transforms converge as tempered distributions. On the other hand $\Delta_r \ell(\sigma + it) \to \Delta_r \Psi_{V_1}(t)$ uniformly for t in compact sets, because ℓ is uniformly continuous on compact subsets of the closed half plane. Hence the Fourier transforms of m_0 and λ_r agree on smooth test functions of compact support, which are dense in the Schwartz space, so they agree as tempered distributions, and since the Fourier transform is injective there, $m_0 = \lambda_r$ as measures. The mass at the origin gives $A = 0$. On a compact set $K \subset \mathbb{R} \setminus \{0\}$, $2\cos ry - 2 < 0$ once $r < 2\pi/\max_K |y|$, so $\mu = \check{\Pi}_1$ on K, and hence on $\mathbb{R} \setminus \{0\}$. But the atom of Π_1 at $\log p$ has mass $1/p$, so $|\mu|((-\infty,-1)) \ge \sum_{p\ge 3} 1/p = \infty$, and $|\mu|$ is not a Lévy measure. This contradiction proves (3). □

The proof of (3) uses about the primes only the Euler product, and not the prime number theorem: the nonvanishing of ζ on the line $\operatorname{Re} s = 1$, the first step towards that theorem, enters only through the hypothesis, which forces it.

Section 9.1.4 reads the conclusion of (3) from the Lévy form itself, and the proof of (3) supplies the uniqueness on which that reading rests. The same argument, with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, shows that the tilted eta variable of Section 11.3.2 is neither quasi infinitely divisible nor infinitely divisible, as that section argues.

11.1.4 Bondesson's classes and Thorin Bernstein functions

The class of HCM densities is closed under products and quotients of independent variables and under powers X^q with $|q| \ge 1$ [39, Section 5.1], and together with the characterisation of GGCs by hyperbolic complete monotonicity [39, Theorem 6.1.1] (Section 1.3.2) these are the tools behind the Bosch threshold of Theorem 4.3.5.

Bondesson treats mixtures of exponential laws and their generalised convolutions. This is the natural class for Kent's hitting times started from an interior point, whose Laplace transform is a ratio of two products with interlacing zeros. The law of T_1 is not a mixture of exponentials, since a mixture of exponentials has a completely monotone density while that of T_1 vanishes at the origin and has an interior mode (Figure 11.1). This is consistent with the conclusion of Section 9.2

that T_1 is a subordinated rather than a direct Kent hitting time (Section 9.2.1 and Theorem 9.2.2).

In the language of Schilling, Song and Vondraček [299, Definition 9.10], a law on $[0,\infty)$ is a GGC exactly when its Laplace exponent is a Thorin Bernstein function, equivalently when the derivative of the exponent is a Stieltjes function, the function f_H of Section 1.3.1. The subordination identity of Theorem 9.2.5 reads

$$\begin{aligned} -\log\varphi_\alpha(u) &= \log G\big((\alpha-\tfrac12)^2+f_\alpha(u)\big)-\log G\big((\alpha-\tfrac12)^2\big),\\ f_\alpha(u) &= u+2(\alpha-\tfrac12)\sqrt{u}, \end{aligned} \tag{11.1.3}$$

with f_α, the Laplace exponent of the subordinator $\sigma^{(\alpha)}$ of that theorem, a complete Bernstein function. The derivative $\frac{d}{du}[-\log\varphi_\alpha(u)] = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(\alpha+\sqrt{u})$ is a Stieltjes function exactly when no zero lies to the right of α, that is when $\alpha \ge \sup_\rho \operatorname{Re}\rho$, which is the criterion of Theorem 1.2.2: a zero with $\operatorname{Re}\rho > \alpha$ makes $\xi(\alpha+\sqrt{u})$ vanish at the point $u=(\rho-\alpha)^2$ of the cut plane $\mathbb{C}\setminus(-\infty,0]$, where a Stieltjes function is holomorphic.

Proposition 11.1.4. *For every $\alpha \ge 1$ the composition $\log G((\alpha-\frac12)^2+f_\alpha) - \log G((\alpha-\frac12)^2)$ is a Thorin Bernstein function with no hypothesis, while $\log G$ itself is a Thorin Bernstein function if and only if RH holds. For $\frac12 < \alpha < 1$ the composition is a Thorin Bernstein function if and only if ζ has no zeros with real part greater than α.*

Proof. By (11.1.3) the composition is the Laplace exponent of T_α, so it is a Thorin Bernstein function exactly when T_α is a GGC, which by Theorem 1.2.2 happens exactly when no zero has real part greater than α. For $\alpha \ge 1$ there is none, since ζ has no zeros on or to the right of the line $\operatorname{Re} s = 1$. The exponent $\log G$ is a Thorin Bernstein function exactly when T is a GGC, which is RH by Proposition 1.2.1. □

For every $\alpha > \frac12$ the composition is at least a Bernstein function: $u \mapsto \log G((\alpha-\frac12)^2+u) - \log G((\alpha-\frac12)^2)$ is the Laplace exponent of the tilted clock $\widetilde{T}_\alpha$, and a Bernstein function composed with a Bernstein function is again one [299, Corollary 3.8]. So T_α is infinitely divisible for every $\alpha > \frac12$ with no hypothesis, and the location of the zeros decides only whether it is a GGC. The proposition says that composing with the complete Bernstein function f_α creates the Thorin property from a function only known to be Bernstein.

General composition theorems do not settle this. For a Bernstein function κ and $a \ge 0$ let $\kappa_a(u) = \kappa((\sqrt{u}+a)^2) - \kappa(a^2)$, so that the composition in (11.1.3) is κ_a for $\kappa = \log G$ and $a = \alpha - \frac12$. If κ is a Thorin Bernstein function then so is every κ_a, because the shift turns a drift cu into $c(u+2a\sqrt{u})$ and each factor $(1+u/t)^{-1}$ of the Thorin form into $(a^2+t)/((\sqrt{u}+a)^2+t)$, the Laplace transform of a GGC whose Thorin density in θ is the pair of Cauchy kernels of width a at $\pm\sqrt{t}$, as in Section 9.2.2. Applied to κ_a, with $\kappa_{a+a'} = (\kappa_a)_{a'}$, this shows that

the shifts a for which κ_a is a Thorin Bernstein function form a half line, and for $\kappa = \log G$ Proposition 11.1.4 identifies it as $[a_*, \infty)$ with $a_* = \sup_\rho \operatorname{Re}\rho - \frac{1}{2}$; RH is the statement $a_* = 0$. For a fixed shift the implication cannot be reversed. The function $\kappa(u) = 2\log(1+u) + \log(1 + (120u + 16u^2)/289)$ is a Bernstein function, since κ' is the Laplace transform of $2e^{-x} + 2e^{-15x/4}\cos 2x \geq 0$, and it is not a Thorin Bernstein function, since κ' has poles at $-\frac{15}{4} \pm 2i$, where a Stieltjes function is holomorphic; yet

$$\kappa_{1/2}(u) = \log\left(1 + \frac{u}{4}\right) + 2\log\frac{4((\sqrt{u}+\frac{1}{2})^2+1)}{5} + \log\frac{(\sqrt{u}+1)^2+4}{5}$$

is a Thorin Bernstein function, the exponent of an exponential law plus the exponents of GGCs whose Thorin densities are pairs of Cauchy kernels, of width $\frac{1}{2}$ at ± 1 and of width 1 at ± 2; the function $u \mapsto \kappa(u/4a^2)$ gives the same example at any shift $a > 0$.

The Dirichlet mean representation of James, Roynette and Yor [129] writes a GGC whose Thorin measure has finite total mass c as the product $\Gamma_c \mathcal{M}$ of a gamma variable Γ_c of shape c and an independent Dirichlet mean $\mathcal{M}$. It applies to the truncated lattice variables $\mathcal{S}_n$ of Section 7.2.1, whose Thorin measures have total mass $2n$, and to the finite sums of exponential variables of Proposition 7.1.22, but not to T_α or, under RH, to T, whose Thorin measures have infinite mass.

11.1.5 What these results cannot give

Every GGC criterion in these books is, for the reciprocal Xi function at the centre, equivalent to RH: the HCM property of the Laplace transform and the Stieltjes property of the derivative of the exponent (Theorem 4.2.4), the Pick function property (Corollary 4.3.6), and the complete monotonicity of the Lévy density of the driving process (Proposition 11.1.1). The consequences obtained in Sections 11.1.1 to 11.1.4 are mostly unconditional. Table 11.3 collects them with their sources.

They are unconditional for one of two reasons. Some use only the infinite divisibility and self decomposability of the centre clock, that is the inequalities $W > 0$ and $W' < 0$, which hold wherever the zeros above height 1000 lie (Theorem 5.1.1). The others use the region $\operatorname{Re} s \geq 1$, through positivity of the Thorin measure at a basepoint $\alpha \geq 1$, which rests on the nonvanishing of ζ on the line $\operatorname{Re} s = 1$ (Theorem 1.2.2), or through the Euler product, as in Proposition 11.1.3. So none of the results supplies positivity of the Thorin measure uniformly in the height at a basepoint $\alpha < 1$, in the sense of Theorem 1.2.2 that T_α is a GGC, which is the zero free half plane $\operatorname{Re} s > \alpha$ of step 7 of the path in Table 12.1, and still less at the centre, where it is RH itself. That positivity, uniform in the height, is the one ingredient still missing on the path to RH; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would supply it for every $\alpha > \frac{7}{8}$, by arithmetic from outside the theory (Section 12.1.2).

result	source	consequence here	status
background driving Lévy process	Wolfe; Jurek and Vervaat; Sato, Theorem 17.5	centre clock driven by a subordinator with Lévy density $-W'$; RH if and only if it is completely monotone	proved
Yamazato's theorem	Yamazato; Sato, Theorem 53.1	centre clock and T_α, $\alpha > \frac{1}{2}$, unimodal	proved
Urbanik's classes L_n	Urbanik; Sato 1980; Berg and Forst	centre clock in L_2, not in L_3; its driving process and the driving process of that self decomposable, with unimodal marginals	proved
moment theorem	Sato, Theorem 25.3	$\mathbb{E}T_\alpha = \infty$ for $\alpha \geq 1$	proved
subexponential tails	Embrechts, Goldie, Veraverbeke	$P(T_\alpha > x) \sim b_\alpha(\pi x)^{-1/2}$; ratio 1.0005 at $x = 100$	proved, checked
uniqueness, quasi infinite divisibility	Sato, Theorem 8.1; Lindner, Pan, Sato	V_α quasi infinitely divisible for $\alpha > 1$, not at $\alpha = 1$; Pólya variable not quasi infinitely divisible	proved
HCM characterisation and closure	Bondesson	Bosch threshold for powers	proved
Thorin measure at the two ends	Bondesson	$x^{-1/2}$ tail and super flat origin	proved
Thorin Bernstein composition	Schilling, Song, Vondraček	smoothing by f_α creates the Thorin property for $\alpha \geq 1$	proved
Dirichlet means	James, Roynette, Yor	applies to truncations only	remark

Table 11.3: What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function. Theorem numbers, here and in the text, refer to the first edition of Sato's book [297].

11.1.6 Wiener gamma integrals and the Riemann Ornstein Uhlenbeck process

A GGC is a superposition of gamma laws, and a gamma process realises the superposition pathwise. Let $(\Gamma_t)_{t\geq 0}$ be a standard gamma process, the subordinator with $\mathbb{E}e^{-u\Gamma_t} = (1+u)^{-t}$. For a measurable $f \geq 0$ the Wiener integral $\int_0^\infty f(t)\, d\Gamma_t$ has Laplace exponent $\int_0^\infty \log(1 + uf(t))\, dt$. Let U be the Thorin measure of a GGC without drift, $M(\theta) = U((0, \theta^2])$ its cumulative mass in the chart $\theta = \sqrt{z}$, and $M^{-1}(t) = \inf\{\theta : M(\theta) > t\}$ the right continuous inverse. The image of Lebesgue measure on $(0, M(\infty))$ under M^{-1} is dM, so the substitution $t = M(\theta)$

turns $\int_0^\infty \log(1+u/M^{-1}(t)^2)\,dt$ into $\int \log(1+u/z)\,U(dz)$, and the GGC is the Wiener gamma integral

$$\int_0^\infty \frac{d\Gamma_t}{M^{-1}(t)^2}, \tag{11.1.4}$$

which is the Wiener gamma representation of the class [129]. At the centre, under RH, M is the zero counting function N, $M^{-1}(t)=\gamma_k$ for $k-1\le t<k$, the ordinates being counted in increasing order and with multiplicity, and (11.1.4) is $\sum_k(\Gamma_k-\Gamma_{k-1})/\gamma_k^2$, the series $\sum_{\gamma>0} e_\gamma/\gamma^2$ of Proposition 1.2.1. At a basepoint α the count is replaced by the Poisson smoothed count $M_{\alpha-1/2}(\theta)=\pi^{-1}\int_0^\theta v_\alpha$ of Section 3.4, which is π^{-1} times the argument of ξ along the line $\operatorname{Re} s=\alpha$. For $\alpha\ge 1$, $v_\alpha>0$ and T_α is a GGC, so with no hypothesis T_α is (11.1.4) with $M=M_{\alpha-1/2}$. For $\frac12<\alpha<1$ the representation holds exactly when T_α is a GGC, that is when no zero has real part greater than α (Theorem 9.2.5), and then M is $M_{\alpha-1/2}$ plus a unit jump at the ordinate of each zero on the line $\operatorname{Re} s=\alpha$ (Theorem 7.1.14). It needs $M_{\alpha-1/2}$ to be nondecreasing, that is $v_\alpha\ge 0$ on the line $\operatorname{Re} s=\alpha$, but for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

Self decomposability gives a second process. If Z is a subordinator with $\mathbb{E}\log(1+Z_1)<\infty$, the Ornstein Uhlenbeck type process $dF_t=-F_t\,dt+dZ_t$, that is $F_t=e^{-t}F_0+\int_0^t e^{-(t-r)}\,dZ_r$, has a unique stationary law, the law of $\int_0^\infty e^{-t}\,dZ_t$ [297, Theorem 17.5 and Corollary 17.9]. By Proposition 11.1.1 the driving process of the centre clock is the subordinator with Lévy density $h=-W'$, which the explicit formula computes from the primes. So there is, with no hypothesis, an Ornstein Uhlenbeck process F whose stationary law is the centre clock, which we call the Riemann Ornstein Uhlenbeck process. Under RH the Laplace exponent of Z is $\Psi_Z(u)=u(\log G)'(u)=\sum_{\gamma>0} u/(\gamma^2+u)$, so the driver is a superposition of independent compound Poisson processes, one for each zero counted with multiplicity, each with rate one and exponential jumps of mean γ^{-2}.

The simulation by `ou.py` is exact for the 1700 computed zeros. The zeros above the 1700th enter as a drift of the driver, which adds to the stationary law the mean $\mathbb{E}T-\sum_{k\le 1700}\gamma_k^{-2}$ of their components. The time average is 0.023094 against $\mathbb{E}T=\sum_k a_k^{-1}=0.023105$ (Section 9.1.2). The empirical Laplace transform of the stationary law is 0.79520 ± 0.00096, 0.32828 ± 0.00180 and 0.016404 ± 0.00030 at $u=10$, 50 and 200, with standard errors from 100 batch means, against $\varphi_{1/2}(u)=\xi(\frac12)/\xi(\frac12+\sqrt{u})=0.79514$, 0.32827 and 0.016399. The stationary process at time $\tau\ge0$ is $e^{-\tau}$ times its value at time 0 plus a variable independent of that value, and the centre clock has finite variance, so the autocorrelation at lag τ is $e^{-\tau}$ with no hypothesis. The estimates at lags 0.5, 1 and 2 are 0.602, 0.352 and 0.119 against 0.607, 0.368 and 0.135; the deficits are within 1.6 standard errors of the estimates, which are about 0.008, 0.010 and 0.013 by batch estimates over 20 blocks of the same run. The top right panel of Figure 11.2 shows ten units of time of the path.

For $\alpha \geq 1$ the shifted clock T_α is a GGC, hence self decomposable, and has its own driver: by (11.1.1), with v_α/π in place of $\rho_{1/2}$, its Lévy measure is $k_\alpha(x)\,dx/x$ with $k_\alpha(x) = \pi^{-1}\int_0^\infty e^{-x\theta^2} v_\alpha(\theta)\,d\theta$, and its driver has the completely monotone Lévy density $-k'_\alpha(x) = \pi^{-1}\int_0^\infty \theta^2 e^{-x\theta^2} v_\alpha(\theta)\,d\theta$. For $\frac{1}{2} < \alpha < 1$ the process exists as well, because T_α is self decomposable with no hypothesis (Corollary 9.2.6); its driver has the Lévy density $-W'_\alpha$, which is completely monotone exactly when T_α is a GGC.

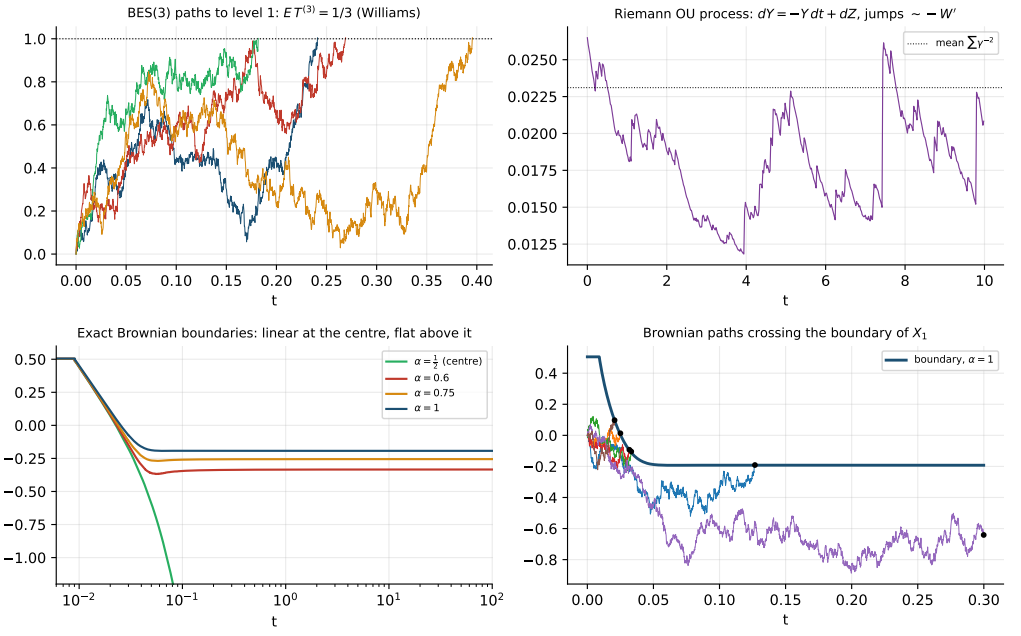


Figure 11.2: Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable S_2 of Section 9.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process F, written Y in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum\gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of T_α at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 9.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of T_1, written X_1 in the plot (Section 9.2.4). Drawn by `figs.py` in the folder `basepoint_one/process_map`.

11.1.7 Subordinated Brownian motion and the map of processes

Running an independent standard Brownian motion B on a clock gives a symmetric law with

$$\mathbb{E}e^{iuB_{T_\alpha}} = \mathbb{E}e^{-u^2T_\alpha/2} = \frac{\xi(\alpha)}{\xi(\alpha + |u|/\sqrt{2})}, \qquad u \in \mathbb{R},$$

so B_{T_α} has the law of $Y_\alpha/\sqrt{2}$, the normal scale mixture of Proposition 10.1.1, and what Section 10.1 proves for Y_α carries over with a factor $\sqrt{2}$.

Table 11.4 arranges the processes of Chapter 9 and of this section by basepoint; we write $a = \alpha - \frac{1}{2}$. The Mellin side is the Bessel picture of Williams [344] and of Biane, Pitman and Yor [32] (Sections 1.2 and 9.1.1). The clock side starts from the centre clock, which under RH with simple zeros is Kent's hitting time for a string with spectrum $\{\gamma^2\}$, and passes through the tilt e^{-a^2T} and the sticky Brownian subordination of Theorem 9.2.5 to T_α. From T_α it continues to the Wiener gamma integral of Section 11.1.6, to Brownian motion run on the clock, and to the Brownian boundaries of Section 9.2.6.

Three conclusions come out of the map. First, the Mellin side and the clock side differ in kind. The Bessel picture and the size biased variables V_α built from it are unconditional at every basepoint, and on that side the zeros enter only as zeros, of the characteristic function $\Xi(t)/\Xi(0)$ of the Pólya variable and of $\xi(\alpha+s)$ in the Lévy forms of Section 11.1.3; the positivity statements equivalent to RH, the GGC property of the centre clock and the EGGC property of $Y_{1/2}$, live on the clock side, in the reciprocal.

Second, on the clock side the Ornstein Uhlenbeck process at the centre and, at every basepoint above it, the tilted and subordinated clock, Brownian motion run on the clock and the Brownian boundary exist with no hypothesis. The Kent strings need RH at every basepoint, and simple zeros as well, because a Kent hitting time is a GGC whose Thorin measure consists of unit atoms at distinct points (Section 9.2.1). The shifted clock T_α is never a Kent hitting time for $\alpha > \frac{1}{2}$: when it is a GGC, each pair of zeros $\rho, \bar\rho$ with $\beta < \alpha$, such as those on the critical line, contributes to its Thorin measure the pair of Cauchy kernels of width $\alpha - \beta$ at $\pm\gamma$ (Section 11.1.4), so that measure is not a sum of atoms.

Third, the Bessel process is the template. Its hitting time of the level one is Kent's hitting time for a string with spectrum $\{\pi^2n^2/2\}$, and under RH with simple zeros the centre clock is Kent's hitting time for a string with spectrum $\{\gamma^2\}$.

11.2 Free probability and the Upsilon transform

Wigner's semicircle law [343] is the limit law of the free central limit theorem [336, 214]. This section, which follows the first author's note [261], applies the Bercovici Pata bijection [24] and the Upsilon transform [21, 19] to the centre

	$\alpha = \frac{1}{2}$	$\frac{1}{2} < \alpha < 1$	$\alpha = 1$	$\alpha > 1$
Mellin side	Pólya variable, $S_2^{1/4}$ bias	$S_2^{\alpha/2}$ bias	$S_2^{1/2}$ bias	$S_2^{\alpha/2}$ bias
Lévy form of $\log \xi(\alpha + s)$	none: the characteristic function vanishes at the ordinates	exists if $\sup \operatorname{Re} \rho < \alpha$, only if $\sup \operatorname{Re} \rho \le \alpha$	compensated limit, by the prime number theorem	absolutely convergent; quasi infinitely divisible
clock law	T, infinitely divisible and self decomposable	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$	$\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$
GGC	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes
Thorin measure	$\sum_{\gamma>0} \delta_{\gamma^2}$ (RH)	Cauchy widths $\alpha - \beta$	Cauchy widths $1 - \beta$	Cauchy widths $\alpha - \beta$
Kent string	spectrum γ^2 (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)	spectrum $\lvert\rho\rvert^2$ on the sticky clock (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)
tail of the clock	exponential, rate γ_1^2	$b_\alpha(\pi x)^{-1/2}$	$0.01303\, x^{-1/2}$	$b_\alpha(\pi x)^{-1/2}$
Brownian boundary	asymptotically linear, slope $-\sqrt{2}\gamma_1$	flat: -0.3338 ($\alpha = 0.6$), -0.2554 ($\alpha = 0.75$)	flat at -0.19277	flat
Wald defect $\mathbb{E}\, b(T_\alpha)$	0	$b_\alpha/\sqrt{2}$	$\sum_\rho \rho^{-1}/\sqrt{2}$	$b_\alpha/\sqrt{2}$
B_{T_α}	finite variance	Cauchy domain	Cauchy domain, tail $0.0104/y$	Cauchy domain
Ornstein Uhlenbeck driver	jump density $-W'$	jump density $-W'_\alpha$	jump density $-k'_1$	jump density $-k'_\alpha$
symmetric EGGC Y_α	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes

Table 11.4: The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of S_2 that produces V_α (Section 9.1.1); $\widetilde{T}_\alpha$, H and the sticky clock are those of Theorem 9.2.5 and Corollary 9.2.3, with $\widetilde{T} = \widetilde{T}_1$; and the boundaries b and their Wald defects are those of Sections 9.2.4 and 9.2.6. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 9.2.4). All other entries hold as stated.

clock T of Section 11.1. With no hypothesis, T has an Upsilon image $\Upsilon(T)$, a GGC whose Thorin measure has the distribution function $u \mapsto W(1/u)$, and a free counterpart $\Lambda_{\mathrm{BP}}(T)$, a freely self decomposable law on $[0,\infty)$; and RH holds exactly when T is itself the Upsilon image of an infinitely divisible law (Theorem 11.2.3).

11.2.1 The Bercovici Pata bijection and the Upsilon transform

As in Section 8.4.4, $G_\mu(z) = \int \mu(dx)/(z-x)$ is the Cauchy transform of a probability measure μ on $\mathbb{R}$ and σ_t the semicircle law of variance t; the subscript distinguishes G_μ from the function G of the clock. The Cauchy transform maps the upper half plane $\mathbb{C}^+$ into the lower half plane $\mathbb{C}^-$. The function $1/G_\mu$ has a right inverse $(1/G_\mu)^{-1}$ on a region $\Gamma_{\alpha,M} = \{z : |\operatorname{Re} z| < \alpha \operatorname{Im} z,\ \operatorname{Im} z > M\}$, and the Voiculescu transform $\varphi_\mu(z) = (1/G_\mu)^{-1}(z) - z$ linearises free additive convolution: $\varphi_{\mu\boxplus\nu} = \varphi_\mu + \varphi_\nu$ on a common region [336, 25]. Here $\mu \boxplus \nu$ is the law of $X+Y$ for free self adjoint variables X and Y with laws μ and ν [214]. The free cumulant transform $\mathcal{C}_\mu(z) = z\varphi_\mu(1/z)$, defined for $z \in \mathbb{C}^-$ with $1/z \in \Gamma_{\alpha,M}$, is additive under $\boxplus$, and μ is freely infinitely divisible if, for every n, it is the n fold free convolution of a law with itself. A freely infinitely divisible law has a Voiculescu transform that extends analytically to all of $\mathbb{C}^+$ [25], so its free cumulant transform is analytic on all of $\mathbb{C}^-$.

A law μ on $\mathbb{R}$ is infinitely divisible exactly when its cumulant function $C_\mu(u) = \log \int e^{iux}\mu(dx)$ has the Lévy Khintchine form

$$C_\mu(u) = i\eta u - \tfrac{1}{2}Au^2 + \int_{\mathbb{R}} \left(e^{iut} - 1 - iut\,\mathbf{1}_{[-1,1]}(t)\right)\Pi(dt), \qquad u \in \mathbb{R}, \quad (11.2.1)$$

with $\eta \in \mathbb{R}$, $A \geq 0$ and a Lévy measure Π, that is $\Pi(\{0\}) = 0$ and $\int \min(1,t^2)\,\Pi(dt) < \infty$; the triplet (η, A, Π) is unique [297, Theorem 8.1]. A law is freely infinitely divisible exactly when

$$\mathcal{C}_\mu(z) = \eta z + Az^2 + \int_{\mathbb{R}} \Big(\frac{1}{1-tz} - 1 - tz\,\mathbf{1}_{[-1,1]}(t)\Big)\,\Pi(dt), \qquad z \in \mathbb{C}^-, \quad (11.2.2)$$

with a triplet of the same kind, again unique [25, 24], [6, Proposition 6].

Definition 11.2.1. The Bercovici Pata bijection Λ_{BP} maps the infinitely divisible law with triplet (η, A, Π) in (11.2.1) to the freely infinitely divisible law with the same triplet in (11.2.2) [24], [6, Definition 7].

The map is a homomorphism, $\Lambda_{\mathrm{BP}}(\mu * \nu) = \Lambda_{\mathrm{BP}}(\mu) \boxplus \Lambda_{\mathrm{BP}}(\nu)$, and a homeomorphism for weak convergence [24], [6, Section 3.2]. When Π has moments of all orders, comparing the expansions of e^{iut} and $1/(1-tz)$ in (11.2.1) and (11.2.2) shows that the free cumulants of $\Lambda_{\mathrm{BP}}(\mu)$ are the classical cumulants of μ: both are $A\mathbf{1}_{\{n=2\}} + \int t^n\,\Pi(dt)$ for $n \geq 2$. The normal law $N(0,t)$, with triplet $(0,t,0)$, goes to the semicircle law σ_t [24], [6, Section 3.2 and Remark 8].

Barndorff-Nielsen and Thorbjørnsen [20] proved that Λ_{BP} maps the self decomposable laws onto the freely self decomposable laws, those μ that for every $c \in (0,1)$ satisfy $\mu = D_c\mu \boxplus \mu_c$ for some law μ_c, $D_c\mu$ being the law of cX when X has law μ; and Hasebe and Thorbjørnsen [110] proved that freely self decomposable laws are unimodal, the free analogue of Yamazato's theorem (Corollary 11.1.2). By [6, Theorem 9] a freely infinitely divisible law μ is the image of an infinitely divisible law on $[0,\infty)$ if and only if $\mu^{\boxplus t}$ is carried by $[0,\infty)$ for every $t > 0$; these are the free regular laws. If μ is infinitely divisible on $[0,\infty)$, with drift $d \geq 0$ and Lévy measure Π on $(0,\infty)$, $\int \min(1,x)\,\Pi(dx) < \infty$, then (11.2.2) reduces to [6, (7)]

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(z) = dz + \int_{(0,\infty)} \Big(\frac{1}{1-zx} - 1\Big)\,\Pi(dx) = dz + \int_{(0,\infty)} \frac{zx}{1-zx}\,\Pi(dx). \quad (11.2.3)$$

The right side is analytic on $\mathbb{C} \setminus (0,\infty)$, and we use it to extend $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}$ to the negative half line.

The Upsilon transform [21, 19] maps an infinitely divisible law μ with Lévy process (X_t), $X_1 \sim \mu$, to the law $\Upsilon(\mu)$ of $\int_0^1 \log(1/t)\,dX_t$. It is infinitely divisible, with

$$C_{\Upsilon(\mu)}(u) = \int_0^1 C_\mu\big(u\log(1/t)\big)\,dt = \int_0^\infty C_\mu(ux)\,e^{-x}\,dx. \quad (11.2.4)$$

Barndorff-Nielsen, Maejima and Sato [19] proved that Υ maps the infinitely divisible laws onto the Goldie Steutel Bondesson class (the infinitely divisible laws whose Lévy measure has a completely monotone density in $|x|$ on each half line), and the self decomposable laws onto the Thorin class. On $[0,\infty)$ these are the class BO and the GGCs of Section 1.3.2. Barndorff-Nielsen and Thorbjørnsen [21] found the link with the free side:

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(i\zeta) = C_{\Upsilon(\mu)}(\zeta), \qquad \zeta < 0. \quad (11.2.5)$$

11.2.2 The Upsilon transform on the half line

Let μ be an infinitely divisible law on $[0,\infty)$ with drift $d \geq 0$ and Lévy measure Π on $(0,\infty)$,

$$\Psi_\mu(s) = ds + \int_{(0,\infty)} \big(1 - e^{-sx}\big)\,\Pi(dx), \qquad \int \min(1,x)\,\Pi(dx) < \infty.$$

If μ is self decomposable, $\Pi(dx) = k(x)\,dx/x$ with k nonincreasing, right continuous and $k(\infty) = 0$, the canonical function of μ; if μ is a GGC with Thorin measure U, then $k(x) = \int e^{-tx}\,U(dt)$ (Section 1.3.1). The cumulants of μ, when finite, are $\kappa_1 = d + \int x\,\Pi(dx)$ and $\kappa_n = \int x^n\,\Pi(dx)$ for $n \geq 2$.

Proposition 11.2.2. *Let μ be as above.*

(a) $\Upsilon(\mu)$ *is the infinitely divisible law on* $[0,\infty)$ *with drift* d*, Laplace exponent*

$$\Psi_{\Upsilon(\mu)}(s) = \int_0^\infty \Psi_\mu(sx)\, e^{-x}\, dx = ds + \int_{(0,\infty)} \frac{sy}{1+sy}\, \Pi(dy) \tag{11.2.6}$$

and completely monotone Lévy density $\ell(z) = \int y^{-1} e^{-z/y}\, \Pi(dy)$*. Every infinitely divisible law on* $[0,\infty)$ *with a completely monotone Lévy density, that is every law in BO, is* $\Upsilon(\mu)$ *for exactly one* μ*. Moreover* $\mathcal{C}_{\Lambda_{\mathrm{BP}(\mu)}}(-s) = -\Psi_{\Upsilon(\mu)}(s)$ *for* $s > 0$*, and* $\kappa_n(\Upsilon(\mu)) = n!\, \kappa_n(\mu)$ *whenever* $\int x^n\, \Pi(dx) < \infty$*.*

(b) $\Upsilon(\mu)$ *is a GGC if and only if* μ *is self decomposable. In that case* $\Upsilon(\mu)$ *has drift* d *and the Thorin measure* $\widetilde{U}$ *with* $\widetilde{U}((0,u]) = k(1/u)$*, the image of the measure* $-dk$ *under* $v \mapsto 1/v$*, and* Υ *is a bijection from the self decomposable laws on* $[0,\infty)$ *onto the GGCs, with inverse given by* $k(v) = \widetilde{U}((0,1/v])$*.*

(c) If μ *is a GGC with Thorin measure* U*, then* $\widetilde{U}$ *is the mixture over* $U(dt)$ *of the laws of* t/E*, with* E *standard exponential:*

$$\widetilde{U}(du) = u^{-2}(-k')(1/u)\, du, \qquad \Psi_{\Upsilon(\mu)}(s) = ds + \int e^{t/s} E_1(t/s)\, U(dt), \tag{11.2.7}$$

and $\int u^p\, \widetilde{U}(du) = \Gamma(1-p) \int t^p\, U(dt)$ *for* $p < 1$*.*

Proof. (a) The first equality in (11.2.6) is (11.2.4) for Laplace transforms: for a subordinator, $\mathbb{E} \exp(-s \int_0^1 f(t)\, dX_t) = \exp(-\int_0^1 \Psi_\mu(sf(t))\, dt)$ for every bounded measurable $f \geq 0$, by approximation with step functions and independence of increments, and the case $f = \log(1/t)$ follows by monotone convergence. Since $\int_0^\infty (1 - e^{-sxy}) e^{-x} dx = sy/(1+sy)$, Fubini's theorem gives the second equality, and since

$$\frac{sy}{1+sy} = \int_0^\infty \left(1 - e^{-sz}\right) y^{-1} e^{-z/y}\, dz,$$

it gives $\Psi_{\Upsilon(\mu)}(s) = ds + \int_0^\infty (1 - e^{-sz})\, \ell(z)\, dz$. The function ℓ is a mixture of exponentials, hence completely monotone, and it is a Lévy density: $\int_0^\infty \min(1,z)\, y^{-1} e^{-z/y} dz = y(1 - e^{-1/y})$, which lies between $(1-e^{-1}) \min(1,y)$ and $\min(1,y)$, so that $\int \min(1,z)\, \ell(z)\, dz$ is comparable with $\int \min(1,y)\, \Pi(dy) < \infty$. Conversely, let λ be infinitely divisible on $[0,\infty)$ with drift d and completely monotone Lévy density $\ell(z) = \int_{[0,\infty)} e^{-zr}\, \sigma(dr)$ (Bernstein's theorem). Integrability of ℓ on $(1,\infty)$ forces $\sigma(\{0\}) = 0$. Let Π be the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$. Then $\int y^{-1} e^{-z/y}\, \Pi(dy) = \int e^{-zr}\, \sigma(dr) = \ell(z)$, the comparison above shows that Π is a Lévy measure with $\int \min(1,y)\, \Pi(dy) < \infty$, and $\lambda = \Upsilon(\mu)$ for the law μ with drift d and Lévy measure Π. Uniqueness: ℓ determines σ by the uniqueness of Laplace transforms, hence Π, and the drift is read off from Ψ at infinity. By (11.2.3), $\mathcal{C}_{\Lambda_{\mathrm{BP}(\mu)}}(-s) = -ds - \int sx(1+sx)^{-1}\, \Pi(dx) = -\Psi_{\Upsilon(\mu)}(s)$. Finally $\int z^n \ell(z)\, dz = \int \Pi(dy) \int_0^\infty z^n y^{-1} e^{-z/y} dz = n! \int y^n\, \Pi(dy)$, which gives

$\kappa_n(\Upsilon(\mu)) = n!\,\kappa_n(\mu)$ for $n \geq 2$; for $n = 1$ the drift, which is the same for both laws, is added on both sides.

(b) Let μ be self decomposable with canonical function k. With $r = 1/y$,

$$z\,\ell(z) = \int_0^\infty \frac{z}{y^2}\, e^{-z/y}\, k(y)\, dy = \int_0^\infty z\, e^{-zr}\, k(1/r)\, dr = \int_{(0,\infty)} e^{-zr}\, d_r\big[k(1/r)\big],$$

by an integration by parts whose boundary terms vanish: $k(1/r) \to k(\infty) = 0$ as $r \to 0$, and $e^{-zr}k(1/r) \to 0$ as $r \to \infty$ because $yk(y) \leq \int_0^y k \to 0$ as $y \to 0$, k being nonincreasing and integrable near 0. The function $r \mapsto k(1/r)$ is nondecreasing, so $\widetilde{U} = d_r[k(1/r)]$ is a positive measure with $\widetilde{U}((0,u]) = k(1/u)$, and $z\ell(z) = \int e^{-zr}\,\widetilde{U}(dr)$ says that $\Upsilon(\mu)$ is a GGC with Thorin measure $\widetilde{U}$ (Section 1.3.1). Conversely, let $\Upsilon(\mu)$ be a GGC with drift d and Thorin measure $\widetilde{U}$, and put $k(v) = \widetilde{U}((0,1/v])$, a nonincreasing function with $k(\infty) = 0$ (its values at its jumps do not matter). By Fubini's theorem $\int_0^1 k(y)\,dy = \int \min(1,1/r)\,\widetilde{U}(dr)$ and $\int_1^\infty k(y)\,dy/y = \int_{(0,1)} \log(1/r)\,\widetilde{U}(dr)$, both finite by the integrability conditions of (1.3.1) and because $\widetilde{U}$ is finite on compact subsets of $(0,\infty)$. So $k(x)\,dx/x$ is a Lévy measure, the self decomposable law μ' with drift d and canonical function k exists, and by the first part $\Upsilon(\mu')$ has drift d and Thorin measure $\widetilde{U}$. Hence $\Upsilon(\mu') = \Upsilon(\mu)$, and $\mu = \mu'$ by the uniqueness in (a). The same construction applied to an arbitrary GGC shows that Υ maps the self decomposable laws onto the GGCs, and (a) gives injectivity.

(c) Now $k(x) = \int e^{-tx}\,U(dt)$, so $\widetilde{U}((0,u]) = k(1/u) = \int e^{-t/u}\,U(dt) = \int \Pr(t/E \leq u)\,U(dt)$, and differentiating in u, $\widetilde{U}(du) = u^{-2}\int te^{-t/u}\,U(dt)\,du = u^{-2}(-k')(1/u)\,du$. Since $\Psi_\mu(s) = ds + \int \log(1+s/t)\,U(dt)$ and, by an integration by parts, $\int_0^\infty \log(1+sx/t)\,e^{-x}dx = \int_0^\infty e^{-x}(x+t/s)^{-1}dx = e^{t/s}E_1(t/s)$, the formula for $\Psi_{\Upsilon(\mu)}$ follows from (11.2.6) by Fubini's theorem. Finally $\int u^p\,\widetilde{U}(du) = \int \mathbb{E}[(t/E)^p]\,U(dt)$ and $\mathbb{E}E^{-p} = \Gamma(1-p)$ for $p < 1$. □

Part (b) is the half line reading of $\Upsilon(\mathrm{SD}) = $ Thorin class [19], which the note [261] pointed out through (11.2.7). For the standard exponential law, $k(x) = e^{-x}$ and $\Upsilon(\mathrm{Exp}(1))$ has Lévy density $\int_0^\infty y^{-2}e^{-z/y-y}dy = 2z^{-1/2}K_1(2\sqrt{z})$.

11.2.3 The Upsilon image and the free counterpart of the clock

We apply Proposition 11.2.2 to the centre clock T, whose Laplace transform is $1/G(u)$ with $G(u) = \xi(\frac{1}{2}+\sqrt{u})/\xi(\frac{1}{2}) = \prod_k(1+u/a_k)$ (1.2.1). By Section 11.1.1, T is infinitely divisible with no drift and Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$, and with no hypothesis self decomposable with canonical function W (Theorem 5.1.1); $h = -W'$ is the Lévy density of its driving subordinator (Proposition 11.1.1). The Stieltjes moments of Section 5.1 are $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_k a_k^{-n-1}$, and since $-\log G(u) = \sum_{n\geq 1}(-1)^n m_{n-1}u^n/n$ near $u = 0$, the cumulants of T are $\kappa_n(T) = (n-1)!\,m_{n-1}$.

THEOREM 11.2.3. *(a) With no hypothesis, $\Upsilon(T)$ is a GGC with no drift and Thorin measure $\widetilde{U}(du) = u^{-2}h(1/u)\,du$, so that $\widetilde{U}((0,u]) = W(1/u) > 0$ for every $u > 0$. Its Laplace exponent is*

$$\Psi_{\Upsilon(T)}(s) = \int_0^\infty \log G(sx)\, e^{-x}\, dx = \sum_k e^{a_k/s} E_1(a_k/s), \qquad s > 0, \quad (11.2.8)$$

with the principal branch of E_1; its Lévy density is the completely monotone function $\ell(z) = z^{-1}\int_0^\infty W(z/w)\, e^{-w}\, dw$; its cumulants are $\kappa_n(\Upsilon(T)) = n!\,(n-1)!\,m_{n-1}$; and $\int u^{-n}\,\widetilde{U}(du) = n!\,m_{n-1}$ for $n \geq 1$. Moreover $\mathbb{E}e^{\lambda\Upsilon(T)} = \infty$ for every $\lambda > 0$, while $\mathbb{E}e^{\lambda T} < \infty$ exactly when $\lambda < \gamma_1^2$.

(b) With no hypothesis, $\Lambda_{\mathrm{BP}}(T)$ is free regular, freely self decomposable and unimodal. Its free cumulant transform is

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(z) = \int_0^\infty \frac{z\,W(x)}{1 - zx}\, dx, \qquad z \in \mathbb{C}^-, \quad (11.2.9)$$

where the right side continues it analytically to $\mathbb{C}\backslash(0,\infty)$, with $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ for $s > 0$; its free cumulants are $(n-1)!\,m_{n-1}$, the cumulants of T. Its Cauchy transform satisfies

$$\Psi_{\Upsilon(T)}\big(-G_{\Lambda_{\mathrm{BP}}(T)}(z)\big) = 1 - z\,G_{\Lambda_{\mathrm{BP}}(T)}(z), \qquad z \in \mathbb{C}^+, \quad (11.2.10)$$

where $\Psi_{\Upsilon(T)}$ is continued analytically to $\mathbb{C} \setminus (-\infty, 0)$ by the right side of (11.2.6), and its support lies in $[c,\infty)$, where $c = \max_{s>0}(\Psi_{\Upsilon(T)}(s) - 1)/s > 0$.

(c) RH holds if and only if T is the Upsilon image of an infinitely divisible law on $[0,\infty)$. In that case $T = \Upsilon(\mu_0)$, where μ_0 is the self decomposable law with no drift and canonical function $N(v^{-1/2})$, that is with Lévy measure $N(x^{-1/2})\,dx/x$; μ_0 is neither a GGC nor in BO. With no hypothesis, T is not the Upsilon image of a law in BO.

Proof. (a) Proposition 11.2.2(b) applies with $k = W$, which is continuously differentiable with $W' = -h$, so $\widetilde{U}((0,u]) = W(1/u)$ and $\widetilde{U}(du) = u^{-2}h(1/u)\,du$; the drift is zero because T has none. The first form of $\Psi_{\Upsilon(T)}$ is (11.2.6) with $\Psi_T = \log G$. For the second, $\operatorname{Re} a_k > 0$ and hence $\operatorname{Re}(1/a_k) > 0$ for every k, so $w = sx/a_k$ has $\operatorname{Re} w \geq 0$, and $\log(1+w) = \int_0^1 w\,d\tau/(1+\tau w)$ with $|1+\tau w| \geq 1$ gives $|\log(1+w)| \leq |w|$. Since $\sum_k |a_k|^{-1} < \infty$, $\sum_k \int_0^\infty |\log(1 + sx/a_k)|\, e^{-x} dx \leq s\sum_k |a_k|^{-1} < \infty$, the series $\log G(sx) = \sum_k \log(1 + sx/a_k)$ can be integrated term by term, and each term gives $e^{a_k/s}E_1(a_k/s)$ by the integration by parts in the proof of Proposition 11.2.2(c), valid for $\operatorname{Re} a_k > 0$ because $e^z E_1(z) = \int_0^\infty e^{-w}(w+z)^{-1}dw$ for $\operatorname{Re} z > 0$. The Lévy density of Proposition 11.2.2(a) is $\ell(z) = \int_0^\infty y^{-2}e^{-z/y}W(y)\,dy$, and the substitution $y = z/w$

gives the stated form. The cumulants follow from $\kappa_n(\Upsilon(T)) = n!\,\kappa_n(T)$, and $\int u^{-n}\widetilde{U}(du) = \int_0^\infty v^n h(v)\,dv = n\int_0^\infty v^{n-1}W(v)\,dv = n!\,m_{n-1}$, by the substitution $v = 1/u$ and an integration by parts whose boundary terms vanish because $W(v) = O(v^{-1/2}\log(1/v))$ at the origin and decays exponentially at infinity. By Sato's moment theorem [297, Theorem 25.17], an infinitely divisible law has $\mathbb{E}e^{\lambda X} < \infty$ exactly when its Lévy measure integrates $e^{\lambda x}$ over $(1,\infty)$. For $\Upsilon(T)$, $z\ell(z) = \int e^{-zr}\widetilde{U}(dr) \geq e^{-z\varepsilon}W(1/\varepsilon)$ for every $\varepsilon > 0$, and with $\varepsilon < \lambda$ the integral $\int_1^\infty e^{\lambda z}\ell(z)\,dz$ diverges. For T, every zero other than the first has $\operatorname{Re} a_k \geq \gamma_2^2 - \frac{1}{4} > \gamma_1^2$, so $W(t) = e^{-\gamma_1^2 t}(1+o(1))$ as $t\to\infty$, and $\int_1^\infty e^{\lambda t}W(t)\,dt/t$ converges exactly when $\lambda < \gamma_1^2$.

(b) T is infinitely divisible on $[0,\infty)$ with no drift and Lévy measure $\Pi(dx) = W(x)\,dx/x$, $\int \min(1,x)\,\Pi(dx) < \infty$, so $\Lambda_{\mathrm{BP}}(T)$ is free regular by [6, Theorem 9], freely self decomposable by [20], since T is self decomposable, and unimodal by [110]. Formula (11.2.9) is (11.2.3) with this Π, and $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ is Proposition 11.2.2(a). The right sides of (11.2.9) and (11.2.6) give $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(w) = -\Psi_{\Upsilon(T)}(-w)$ on all of $\mathbb{C}\setminus(0,\infty)$. Since Π has moments of all orders, the free cumulants are the classical cumulants of T (Section 11.2.1). Write G_Λ for $G_{\Lambda_{\mathrm{BP}}(T)}$. On a region where $(1/G_\Lambda)^{-1}(1/G_\Lambda(z)) = z$, the definition of φ gives $\varphi(1/G_\Lambda(z)) = z - 1/G_\Lambda(z)$, that is $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(G_\Lambda(z)) = zG_\Lambda(z) - 1$; both sides are analytic on $\mathbb{C}^+$, which G_Λ maps into $\mathbb{C}^-$, so the identity holds on $\mathbb{C}^+$, and it is (11.2.10). For the support, put $x(s) = (\Psi_{\Upsilon(T)}(s)-1)/s$ for $s>0$. Then $s^2x'(s) = f(s) = s\Psi'_{\Upsilon(T)}(s) - \Psi_{\Upsilon(T)}(s) + 1$, with $f(0+) = 1$, $f'(s) = s\Psi''_{\Upsilon(T)}(s) < 0$, and $f(s)\to-\infty$ as $s\to\infty$, since $s\Psi' - \Psi = -\int(1 - e^{-sz} - sze^{-sz})\ell(z)\,dz$ and ℓ is not integrable at the origin. So x increases from $-\infty$ to its maximum $c = x(s^*) = \Psi'_{\Upsilon(T)}(s^*)$ on $(0,s^*]$ and decreases to 0 on $[s^*,\infty)$, because $\Psi_{\Upsilon(T)}(s)/s\to 0$; in particular $c>0$. For $x<0$, which lies outside the support, $G_\Lambda(x) < 0$ and (11.2.10) extends by continuity, so $S = -G_\Lambda(x) > 0$ solves $x(S) = x$, whose only positive solution is $s(x)$, the inverse of $x(\cdot)$ on $(0,s^*)$. This inverse is analytic on a neighbourhood V of $(-\infty,c)$, a union of discs centred on that half line that do not meet $[c,\infty)$, and $-s$ agrees with G_Λ on $(-\infty,0)$, hence on the connected set $V\setminus[0,c)$. So G_Λ extends analytically across $(-\infty,c)$ with real values there, and by Stieltjes inversion $\Lambda_{\mathrm{BP}}(T)$ gives no mass to $(-\infty,c)$.

(c) By Proposition 11.2.2(a), the Upsilon images of the infinitely divisible laws on $[0,\infty)$ are the laws of the class BO, and the clock is in BO if and only if RH holds (Section 8.5.4 and Table 1.6). Under RH, $W(z)/z = \sum_{\gamma>0}\int_{\gamma^2}^\infty e^{-zr}\,dr = \int_0^\infty e^{-zr}N(\sqrt{r})\,dr$, so in the proof of Proposition 11.2.2(a) $\sigma(dr) = N(\sqrt{r})\,dr$, and the image of $r^{-1}\sigma(dr)$ under $r\mapsto 1/r$ is $N(x^{-1/2})\,dx/x$. Its canonical function $N(v^{-1/2})$ is nonincreasing, so μ_0 is self decomposable, and by Proposition 11.2.2(b) the Thorin measure of $\Upsilon(\mu_0)$ has the distribution function $u\mapsto N(\sqrt{u})$, that is, it is $\sum_{\gamma>0}\delta_{\gamma^2}$, as Proposition 1.2.1 requires. The canonical function and the Lévy density of μ_0 vanish on (γ_1^{-2},∞) without vanishing identically, so neither is

completely monotone: μ_0 is not a GGC and not in BO. If $T = \Upsilon(\mu)$ with μ in BO, then T is in BO, RH holds, and $\mu = \mu_0$ by the uniqueness in Proposition 11.2.2(a), which is impossible. □

The preimage μ_0 of Theorem 11.2.3(c) has a reading in terms of the Dickman law D, the law known from smooth numbers with $\mathbb{E}e^{-sD} = \exp(-\operatorname{Ein}(s))$ and $\operatorname{Ein}(s) = \int_0^s (1-e^{-v})\,dv/v$. Integrating $\operatorname{Ein}(ax) = \int_0^1 (1-e^{-axv})\,dv/v$ against e^{-x} gives $\int_0^\infty \operatorname{Ein}(ax)e^{-x}dx = \log(1+a)$, so $\Upsilon(D/\gamma^2)$ is the exponential law of rate γ^2, and since $\operatorname{Ein}(s/\gamma^2) = \int_0^{\gamma^{-2}} (1-e^{-sy})\,dy/y$, under RH

$$\mu_0 = \text{law of } \sum_{\gamma>0} D_\gamma/\gamma^2, \qquad D_\gamma \text{ independent Dickman variables,}$$

with Lévy measure $\sum_\gamma \mathbf{1}_{(0,\gamma^{-2})}(y)\,dy/y = N(y^{-1/2})\,dy/y$ and all jumps at most γ_1^{-2}. With no hypothesis $\psi_0(s) = \sum_k \operatorname{Ein}(s/a_k)$ is entire and $\int_0^\infty \psi_0(sx)e^{-x}dx = \log G(s)$ for $s > 0$: termwise integration is justified because $\operatorname{Re}(sx/a_k) > 0$ and $|\operatorname{Ein}(z)| \le |z|$ for $\operatorname{Re} z \ge 0$, since $|1-e^{-w}| \le |w|$ there. Under RH the density of μ_0, by Fourier inversion because the Laplace transform grows along the negative axis, has mass 0.99993 on $[0.008, 0.045]$ and mean 0.023104, against $p_1 = 0.0231050$.

Proposition 11.2.4. *Let $\mathcal{C}_\xi(z) = \log\{\xi(\frac12)/\xi(\frac12 + \sqrt{-z})\} = \sum_{n\ge1} m_{n-1}z^n/n$ for $|z| < \gamma_1^2$. RH holds if and only if there is a freely infinitely divisible probability measure ν_ξ whose free cumulant transform is $\mathcal{C}_\xi$. Under RH, $\nu_\xi = \Lambda_{\mathrm{BP}}(\mu_0)$, its free cumulants are m_{n-1}/n, its free Lévy measure is $N(y^{-1/2})\,dy/y$, its support is compact and it is freely self decomposable.*

Proof. Under RH, (11.2.5) gives $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu_0)}(i\zeta) = C_{\Upsilon(\mu_0)}(\zeta) = C_T(\zeta) = \mathcal{C}_\xi(i\zeta)$, and Λ_{BP} preserves Lévy measures and maps self decomposable laws to freely self decomposable ones. Conversely let ν_ξ exist. Since $\mathcal{C}_\xi$ is analytic at 0, ν_ξ has compact support and its free cumulants are m_{n-1}/n. For $n \ge 3$ they are the moments $\int t^n \rho(dt)$ of its free Lévy measure ρ, and $|m_{n-1}|/n \le \gamma_1^{-2(n-1)} \sum_k |a_k|^{-1}$ because $|a_k| = \gamma^2 + \delta^2 \ge \gamma_1^2$, so $t^2\rho$ lives on $[-\gamma_1^{-2}, \gamma_1^{-2}]$. Then $\sum_{n\ge3} m_{n-1}z^n/n = -\sum_k \{\log(1 - z/a_k) + z/a_k + z^2/2a_k^2\} = \int \frac{z^3t^3}{1-zt}\,\rho(dt)$ is analytic off a subset of the real axis, while the left side has a logarithmic singularity at every a_k. So every a_k is real, and RH holds since ξ has no real zeros. □

The Cauchy transform of ν_ξ solves $\mathcal{C}_\xi(G_{\nu_\xi}(z))/G_{\nu_\xi}(z) + 1/G_{\nu_\xi}(z) = z$, and so the density follows from ξ alone. Solved at 30 digits along $z = x + i\eta$ with η down to 10^{-11}, it is supported on an interval close to $[0.0166, 0.0348]$ with steep edges and its mode is at 0.0193, where it equals 98.6. The trapezoid rule on a grid of spacing 0.00025 gives mass 0.99992, mean 0.0231057 and variance 1.8592×10^{-5}, against $p_1 = 0.0231050$ and $m_1/2 = 1.8586 \times 10^{-5}$. This is consistent with RH and is no evidence beyond the zeros already verified, since the low cumulants are fixed by the first zeros and the existence in Proposition 11.2.4 concerns all of them

at once. The Upsilon preimage is due to Barndorff-Nielsen, Maejima and Sato [19] and the link between the two sides to Barndorff-Nielsen and Thorbjørnsen [21].

Under RH, $\widetilde{U}$ spreads the atom at γ^2 into the law of γ^2/E (Proposition 11.2.2(c)), and in the chart $\theta = \sqrt{u}$

$$\widetilde{U}((0,\theta^2]) = W(\theta^{-2}) = \sum_{\gamma>0} e^{-\gamma^2/\theta^2} = \mathbb{E}\, N(\theta\sqrt{E}),$$

with E standard exponential, a smoothing of the zero count. With no hypothesis, the asymptotic $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1) and the Riemann von Mangoldt formula give $W(\theta^{-2})/N(\theta) \to \sqrt{\pi}/2 = 0.8862$ as $\theta \to \infty$. Under RH the expectation gives the next term: $\mathbb{E}N_0(\theta\sqrt{E})$, with $N_0(\theta) = (\theta/2\pi)\log(\theta/2\pi e) + \frac{7}{8}$, equals $(\theta/4\sqrt{\pi})\big(\log(\theta/2\pi e) + \frac{1}{2}\psi(\frac{3}{2})\big) + \frac{7}{8}$, where $\psi(\frac{3}{2}) = 2-\gamma_E-2\log 2 = 0.0365$; so the ratio of $W(\theta^{-2})$ to the smooth count $N_0(\theta)$ exceeds $\sqrt{\pi}/2$ by a relative amount of about $\psi(\frac{3}{2})/(2\log(\theta/2\pi e))$.

The free counterpart behaves differently at the left end. The clock and its Upsilon image are infinitely divisible with no drift and infinite Lévy measure, so their supports are $[0,\infty)$ [297, Section 24], and their densities are super flat at the origin; the free law has a hard edge at $c > 0$. Since $\Psi''_{\Upsilon(T)}(s^*) \neq 0$, the solution of $x(s) = x$ leaves the real axis as a square root at $x = c$, so the density $\operatorname{Im} s/\pi$ of $\Lambda_{\mathrm{BP}}(T)$ vanishes like $(x-c)^{1/2}$ there. At the other end, as $z \to \infty$ only the first zero matters in $\ell(z) = z^{-1}\int_0^\infty W(z/w)e^{-w}dw$, the saddle point of $e^{-\gamma_1^2 z/w - w}$ lies at $w = \gamma_1\sqrt{z}$, and

$$\ell(z) \sim 2\gamma_1 z^{-1/2} K_1(2\gamma_1\sqrt{z}) \sim \sqrt{\pi}\,\gamma_1^{1/2} z^{-3/4} e^{-2\gamma_1\sqrt{z}};$$

under RH, $\ell(z) = 2z^{-1/2}\sum_{\gamma>0}\gamma K_1(2\gamma\sqrt{z})$ exactly. The computed density of $\Upsilon(T)$ is 2.552 times ℓ at $x = 0.2$ and 1.828 times ℓ at $x = 0.4$, so the Lévy density takes over the tail only slowly.

Table 11.5 checks (11.2.8). The integral is computed from ξ alone, by quadrature at 30 digits, and the sums use one of two tails: the tail of order one, $s(\log(\gamma_N/2\pi)+1)/(2\pi\gamma_N)$, which is $s\int_{\gamma_N}^\infty \theta^{-2}dN_0(\theta)$ and the choice of [261], and the full smooth tail $\int_{\gamma_N}^\infty \phi(\theta^2/s)\,dN_0(\theta) + \phi(\gamma_N^2/s)(N_0(\gamma_N) - N)$, where $\phi(z) = e^z E_1(z)$ and N is the number of zeros used. The note's Table 1, the integral and the sums over 300 zeros without and with the tail of order one at $s = 1$, 10 and 100, is reproduced to every digit shown (0.0230681050, 0.0214685871 and 0.0230714840 at $s = 1$). With 1700 zeros and the full tail the difference is $-6.29 \times 10^{-11}s$ at every s up to 10^3, that is, it comes from the smooth tail of $\sum\gamma^{-2}$ beyond $\gamma_{1700} = 2197.26$, where $\phi(\gamma^2/s) \approx s/\gamma^2$.

Table 11.6 and the right panel of Figure 11.3 compare the three laws with the Lévy measure $W(t)\,dt/t$. The density of T is the fixed Talbot inversion of $1/G$ at 25 digits on the grid 0.002(0.0005)0.15. That of $\Upsilon(T)$ is the fixed Talbot inversion of $e^{-\Psi_{\Upsilon(T)}}$, with $\Psi_{\Upsilon(T)}$ from the 1700 zeros and the tail $u\tau_1 - u^2\tau_2 + 2u^3\tau_3$, τ_j

		sum over the zeros minus the integral				
s	$\int_0^\infty \log G(sx)\, e^{-x}\, dx$	300 zeros	300, tail 1	300, full tail	1700, tail 1	1700, full tail
1	0.0230681049610841	-0.0016	3.38×10^{-6}	-1.32×10^{-9}	1.2×10^{-7}	-6.29×10^{-11}
10	0.227587470315411	-0.016	3.39×10^{-5}	-1.32×10^{-8}	1.2×10^{-6}	-6.29×10^{-10}
100	2.07560251202393	-0.16	3.54×10^{-4}	-1.32×10^{-7}	1.23×10^{-5}	-6.29×10^{-9}
10^3	15.2403221009594	-1.6	4.96×10^{-3}	-1.31×10^{-6}	1.51×10^{-4}	-6.29×10^{-8}
10^4	87.4852790298693	-15.8	0.187	-1.24×10^{-5}	4.28×10^{-3}	-6.27×10^{-7}

Table 11.5: The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (11.2.8) from ξ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed by `wigfree_upsilon.py`.

the smooth tail of $\sum \gamma^{-2j}$ with its boundary term; it runs in double precision from $x = 0.0125$ on and at 30 digits below, where double precision loses the cancellation. That of $\Lambda_{\mathrm{BP}}(T)$ is $\operatorname{Im} s/\pi$, with s the solution of $\Psi_{\Upsilon(T)}(s) = 1 + xs$ in the upper half plane given by (11.2.10), found by Newton's method on 500 points of $[c, 0.15]$, spaced quadratically near c, with residuals below 4.6×10^{-15}. Here $c = 0.01529224618$ and $s^* = 387.9402$.

	T	$\Upsilon(T)$	$\Lambda_{\mathrm{BP}}(T)$
total mass	1.00000000	1.00000053	1.00000021
mean	0.023104993	0.023104991	0.023104991
variance	3.71726×10^{-5}	7.43455×10^{-5}	3.71726×10^{-5}
third central moment	2.8835×10^{-7}	1.7301×10^{-6}	2.8835×10^{-7}
mode	0.0200	0.0185	0.017711
density at the mode	80.1042	72.6771	91.3878
left end of the support	0	0	0.01529224618

Table 11.6: The clock T, its Upsilon image and its free counterpart, from the densities of Figure 11.3. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of T and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about 7×10^{-5}. Computed by `wigfree_upsilon.py`.

11.2.4 The semicircle as a local model for the Jacobi parameters

Section 5.1 builds from the Stieltjes moments $m_0, \dots, m_{2N-1}$ the Jacobi matrix J_N of the measure $\nu = \sum_j a_j^{-1} \delta_{1/a_j}$, with diagonal entries $\alpha_1, \dots, \alpha_N$ and positive off diagonal entries $\beta_1, \dots, \beta_{N-1}$, the recurrence coefficients of the orthogonal polynomials of ν, computed from the Taylor coefficients of Ξ at the centre alone.

For constant parameters $\alpha_n = \alpha$ and $\beta_n = \beta$ the Jacobi matrix is $\alpha + \beta(S + S^*)$

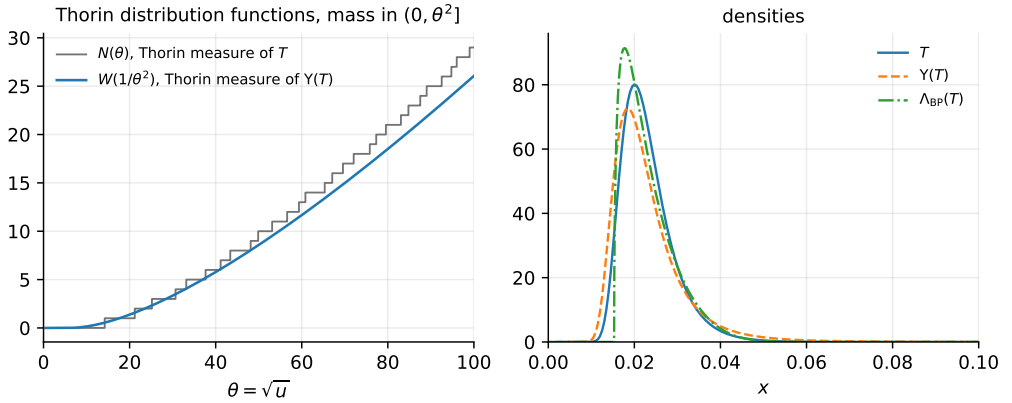


Figure 11.3: Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of T in the same chart. Right: the densities of T, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed by `wigfree_upsilon.py` and drawn by `wigfree_plots.py`.

on $\ell^2(\{0, 1, 2, \dots\})$, with S the unilateral shift, and the spectral measure of its first basis vector is the semicircle law on $[\alpha - 2\beta, \alpha + 2\beta]$. The $N \times N$ truncations have the eigenvalues $\alpha + 2\beta\cos(\pi k/(N+1))$, $1 \le k \le N$, whose distribution tends to the arcsine law with density $\big(\pi\sqrt{4\beta^2 - (x-\alpha)^2}\big)^{-1}$ on the same interval, the density of states. If the parameters vary slowly, the eigenvalues are counted by integrating local arcsine laws over the rows. For recurrence coefficients that vary on the scale of the matrix size, $\alpha_{n,N} \to a(n/N)$ and $\beta_{n,N} \to b(n/N)$, this is the theorem of Kuijlaars and Van Assche [157]: the zero distribution of the orthogonal polynomials tends to $\int_0^1 \omega_{[a(t)-2b(t),\,a(t)+2b(t)]}\,dt$, with $\omega_{[A,B]}$ the arcsine law on $[A, B]$. The parameters of ν decrease to zero with no such scaling, and the count needed below is that of the largest eigenvalues, an edge statement; for ν the local model is a heuristic. Since ν has its atoms in $(0, \gamma_1^{-2}]$ and accumulates only at the origin, the local interval $[\alpha_n - 2\beta_n, \alpha_n + 2\beta_n]$ should reach down to zero, which suggests $\alpha_n \approx 2\beta_n$. Write $L(n) = \alpha_n + 2\beta_n$ for its right end, taken to decrease in n, and $n(L)$ for the inverse function.

The eigenvalues of J_N above a level $x > 0$ approximate the atoms $1/\gamma_j^2$ above x, of which there are $N(x^{-1/2})$. In the local model row n contributes the mass $\frac{2}{\pi}\arccos\sqrt{x/L(n)}$ of the arcsine law on $[0, L(n)]$ above x, and nothing when $x \ge L(n)$, so that

$$N\big(x^{-1/2}\big) = \int_0^\infty \frac{2}{\pi}\arccos\sqrt{x/L(n)}\,{}_+\,dn = \frac{1}{\pi}\int_x^\infty n(L)\,\frac{\sqrt{x}\,dL}{L\sqrt{L-x}}, \qquad (11.2.11)$$

the second form by an integration by parts in L. With $u = 1/L$ and $w = 1/x$ it

becomes Abel's equation

$$\int_0^w \frac{f(u)\,du}{\sqrt{w-u}} = \pi N(\sqrt{w}), \qquad f(u) = u^{-1/2}\,n(1/u). \tag{11.2.12}$$

PROPOSITION 11.2.5. *Let N be nondecreasing and right continuous on $[0,\infty)$, with $N = 0$ on $[0,\theta_0)$ for some $\theta_0 > 0$. Then*

$$n(L) = \int_{(0,L^{-1/2})} \frac{dN(T)}{\sqrt{1-LT^2}} \tag{11.2.13}$$

gives a locally integrable solution f of (11.2.12), with $N(\sqrt{w})$ replaced by its left limit at the jumps of N, and it is the only locally integrable solution. If $dN(T) = c\,dT$ on $(0,\infty)$, then $n(L) = \frac{\pi}{2}cL^{-1/2}$, so that $L_n = (\pi c/2n)^2$, which is $\gamma_{2n/\pi}^{-2}$ for the ordinates $\gamma_k = k/c$ of that count.

Proof. For $0 < a < b$ the substitution $u = 1/L$ gives

$$\begin{aligned}\int_a^b \frac{dL}{L\sqrt{(L-a)(b-L)}} &= \int_{1/b}^{1/a} \frac{du}{\sqrt{(1-au)(bu-1)}} \\ &= \frac{1}{\sqrt{ab}} \int_{1/b}^{1/a} \frac{du}{\sqrt{(1/a-u)(u-1/b)}} = \frac{\pi}{\sqrt{ab}}.\end{aligned}$$

Insert (11.2.13) in the right side of (11.2.11), which is equivalent to (11.2.12), and exchange the integrals, all terms being positive. A point T of the support of dN contributes only when $T < x^{-1/2}$, through $L \in (x, T^{-2})$, and with $b = T^{-2}$ and $\sqrt{1-LT^2} = \sqrt{(b-L)/b}$ its contribution is

$$\frac{\sqrt{x}}{\pi} \int_x^b \frac{\sqrt{b}\,dL}{L\sqrt{(L-x)(b-L)}} = 1.$$

So the right side is N of the open interval $(0, x^{-1/2})$, the left limit of N at $x^{-1/2}$. The function f is locally integrable: n vanishes for $L > \theta_0^{-2}$, and for $0 < a < b$, by Fubini's theorem, $\int_a^b n(L)\,dL \le \int_{(0,a^{-1/2})} dN(T) \int_0^{T^{-2}} dL/\sqrt{1-LT^2} = 2\int_{(0,a^{-1/2})} T^{-2}\,dN(T) < \infty$. If f_1 and f_2 are two locally integrable solutions, $g = f_1 - f_2$ satisfies $\int_0^w g(u)(w-u)^{-1/2}du = 0$ for every w; integrating against $(y-w)^{-1/2}$ over $w \in (0,y)$ and using $\int_u^y dw/\sqrt{(y-w)(w-u)} = \pi$ gives $\pi\int_0^y g = 0$ for every y, so $g = 0$ almost everywhere. Finally $\int_0^{L^{-1/2}} c\,dT/\sqrt{1-LT^2} = cL^{-1/2}\arcsin 1$. □

The local model therefore predicts $\alpha_n \approx 2\beta_n \approx \frac{1}{2}L_n$, with $n(L_n) = n$ [261]. For the zeros of ξ we use the smooth density $dN_0(T) = (2\pi)^{-1}\log(T/2\pi)\,dT$ on $T > 2\pi$, for which the substitution $T = L^{-1/2}\sin\varphi$ gives

$$n(L) = \frac{1}{2\pi\sqrt{L}} \int_{\varphi_0}^{\pi/2} \log\frac{\sin\varphi}{2\pi\sqrt{L}}\,d\varphi, \qquad \sin\varphi_0 = 2\pi\sqrt{L}.$$

As $L \to 0$, $\int_0^{\pi/2} \log\sin\varphi\, d\varphi = -\frac{\pi}{2}\log 2$ gives $n(L) = (4\sqrt{L})^{-1}\log(1/(4\pi\sqrt{L})) + O(1)$, while the rule of Proposition 11.2.5 for a linear count, applied with the ordinates of the smooth count, $N_0(\gamma_r) = r$ with $N_0(T) = (T/2\pi)\log(T/2\pi e) + \frac{7}{8}$, gives $\frac{1}{2}L_n = \frac{1}{2}\gamma_{2n/\pi}^{-2}$ and corresponds to $n = (4\sqrt{L})^{-1}\log(1/(2\pi e\sqrt{L}))$ to the same order. The two rules differ by a factor that tends to one only logarithmically; this is why the cruder rule $\alpha_n \approx \frac{1}{2}\gamma_{2n/\pi}^{-2}$, which treats the count as linear, is off by a factor near 1.3.

Table 11.7 compares the computed parameters with the two rules. The program `wigfree_jacobi.py` computes $M_0, \dots, M_{100}$ at 1000 digits by the trapezoid rule of step $1/600$ on $[0, 3.6]$, the moments $m_0, \dots, m_{99}$, and the Cholesky factorisation of $\mathcal{H}_{50}$. The matrix $L^{-1}[m_{i+j+1}]L^{-\top}$ is tridiagonal to 1.3×10^{-836}, and its entries change by at most 2.3×10^{-429} relatively when the factorisation runs at 600 digits. The first parameters are irregular, $\alpha_1, \dots, \alpha_5 = (1.61, 3.50, 1.80, 1.22, 1.07) \times 10^{-3}$ and $\beta_1, \dots, \beta_5 = (1.91, 1.15, 0.697, 0.554, 0.498) \times 10^{-3}$, and the model is meant for larger n. Over $17 \le n \le 49$ the ratio $\alpha_n/\frac{1}{2}L_n$ has mean 1.033 and standard deviation 0.060 (range 0.940 to 1.178), and $2\beta_n/\frac{1}{2}L_n$ has mean 1.012 and standard deviation 0.060 (range 0.893 to 1.134); the columns α_n, $2\beta_n$, $\frac{1}{2}L_n$ and these two ratios reproduce Table 6 of [261] to the digits shown there. Over the same range $\alpha_n/(2\beta_n)$ has mean 1.021 and standard deviation 0.047, and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac{1}{4}$ in the local model, has mean 0.2502 and standard deviation 0.013 (mean 0.256 and standard deviation 0.059 over $1 \le n \le 49$). The cruder rule falls short: α_n and $2\beta_n$ exceed it by the factors 1.286 and 1.26 on average, with standard deviations 0.081 and 0.078.

Two cautions apply. The rule $n(L_n) = n$ and the midpoint rule $n(L_n') = n - \frac{1}{2}$ differ by a shift of n of order one, which the asymptotics cannot see, and they move the mean ratios by two per cent, to 1.011 and 0.991, within the scatter; the data do not decide between them. And the scatter itself, about six per cent, is not explained by this computation: the note attributes it to the discreteness of the zeros, which the smooth density ignores, and that remains to be tested (Problem 12.3). So the local semicircle model predicts the size of the Jacobi parameters to a few per cent over $17 \le n \le 49$ with no fitted constant, but the agreement is not a proof of an asymptotic law.

11.2.5 Gamma(2), Laguerre polynomials and finite free convolution

Williams' variable [344] $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma(2) variables Γ_n, the sum of two independent copies of the hitting time $J = \sum_n \frac{2}{\pi^2 n^2} e_n$ of Section 9.1.1, satisfies $2\xi(s) = \mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$ (9.1.1) [32], and the RH content of this identity lies on its Mellin side (Theorem 7.1.21 and Section 8.2.3). The cumulants of S_2 are $\kappa_m = 2(m-1)!\,(2/\pi^2)^m\zeta(2m)$.

	$10^5\times$			ratio to $\frac12 L_n$		$10^5\times$	ratio to $\frac12 L'_n$		$10^5\times$	
n	α_n	$2\beta_n$	$\frac12 L_n$	α_n	$2\beta_n$	$\frac12 L'_n$	α_n	$2\beta_n$	$\frac12\gamma^{-2}_{2n/\pi}$	$\frac{\beta_n^2}{\alpha_n\alpha_{n+1}}$
5	107.3	99.54	84.84	1.27	1.17	94.11	1.14	1.06	61.8	0.2665
9	56.96	51.12	45.73	1.25	1.12	48.70	1.17	1.05	34.5	0.2548
13	30.20	32.52	30.07	1.00	1.08	31.48	0.959	1.03	23.2	0.2611
17	21.71	21.51	21.83	0.995	0.986	22.64	0.959	0.950	17.0	0.2362
21	18.50	16.55	16.82	1.10	0.983	17.34	1.07	0.954	13.3	0.2479
25	15.90	14.58	13.50	1.18	1.08	13.85	1.15	1.05	10.7	0.2588
29	11.25	11.45	11.16	1.01	1.03	11.41	0.986	1.00	8.93	0.2550
33	9.530	9.336	9.424	1.01	0.991	9.615	0.991	0.971	7.59	0.2647
37	8.704	8.163	8.101	1.07	1.01	8.249	1.06	0.990	6.56	0.2463
41	6.998	7.053	7.062	0.991	0.999	7.179	0.975	0.982	5.74	0.2618
45	6.498	6.150	6.228	1.04	0.987	6.323	1.03	0.973	5.08	0.2481
49	5.957	6.172	5.547	1.07	1.11	5.625	1.06	1.10	4.54	0.2738

Table 11.7: The Jacobi parameters of ν against the local semicircle model: α_n and $2\beta_n$; half the right end $\frac12 L_n$ of the local interval from (11.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L'_n) = n - \frac12$; the cruder rule $\frac12\gamma^{-2}_{2n/\pi}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac14$ in the local model. Computed by `wigfree_jacobi.py`.

Proposition 11.2.6. *For every complex s,*

$$\mathbb{E}\Big[\Big(\frac{\pi}{2}J\Big)^{s/2}\Big] = \frac{2(1-2^{1-s})\,\xi(s)}{s-1} = s\,\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)(1-2^{1-s})\,\zeta(s), \tag{11.2.14}$$

the value at $s = 1$ being $2\xi(1)\log 2 = \log 2$. Equivalently $J \overset{d}{=} V^{-2}S_2$, with V uniform on $[1,2]$ and independent of S_2. The function (11.2.14) *vanishes at the zeros of ξ and at the points $1 + 2\pi i k/\log 2$, $k \neq 0$, of the line* $\operatorname{Re} s = 1$, *and, unlike the zero set of ξ, its zero set is not symmetric under $s \mapsto 1 - s$.*

Proof. This is the multiplicative relation of Biane, Pitman and Yor between J, their Σ_1, and S_2, their Σ_2 [32, Table 1 and (43) to (45)]; we give the computation. For $\operatorname{Re} p < 0$, $\int_0^\infty \lambda^{-p-1}e^{-\lambda x}d\lambda = \Gamma(-p)x^p$ and $\mathbb{E}e^{-\lambda J} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$ give, with $r = \sqrt{2\lambda}$,

$$\mathbb{E}J^p = \frac{2^{p+1}}{\Gamma(-p)}\int_0^\infty \frac{r^{-2p}}{\sinh r}\,dr = \frac{2^{p+2}(1-2^{2p-1})\,\Gamma(1-2p)\,\zeta(1-2p)}{\Gamma(-p)},$$

by $1/\sinh r = 2\sum_{k\geq 0} e^{-(2k+1)r}$, which gives $\int_0^\infty r^{z-1}/\sinh r\,dr = 2(1-2^{-z})\Gamma(z)\zeta(z)$ for $\operatorname{Re} z > 1$. Put $p = s/2$ with $\operatorname{Re} s < 0$. The functional equation $\zeta(1-s) = \pi^{1/2-s}\Gamma(\frac{s}{2})\zeta(s)/\Gamma(\frac{1-s}{2})$, the duplication formula $\Gamma(1-s) = 2^{-s}\pi^{-1/2}\Gamma(\frac{1-s}{2})\Gamma(1-\frac{s}{2})$ and $\Gamma(1-\frac{s}{2})/\Gamma(-\frac{s}{2}) = -\frac{s}{2}$ turn this into $\mathbb{E}J^{s/2} = -2^{1-s/2}s(1-2^{s-1})\pi^{-s}\Gamma(\frac{s}{2})\zeta(s)$, and multiplying by $(\pi/2)^{s/2}$ gives

(11.2.14). Both sides are entire, since J has moments of all real orders and the zero of $1-2^{1-s}$ at $s=1$ cancels the pole of ζ, so the identity holds for every s. Since $\mathbb{E}V^{-s}=\int_1^2 v^{-s}dv=(1-2^{1-s})/(s-1)$, the right side is $\mathbb{E}V^{-s}\,\mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$. The factor $1-2^{1-s}$ vanishes exactly at $1+2\pi ik/\log 2$; on the line $\operatorname{Re}s=0$, the image of $\operatorname{Re}s=1$ under $s\mapsto 1-s$, it equals -1 at the corresponding points $2\pi ik/\log 2$ and ξ has no zeros, so the zero set is not symmetric. □

The elementary factor $(1-2^{1-s})/(s-1)$ adds zeros on the line $\operatorname{Re}s=1$, the first at height $2\pi/\log 2=9.0647$, the zeros of the alternating zeta function that Section 10.1.7 meets.

Proposition 11.2.7. *Let $z_1,z_2,\dots$ be independent standard complex Gaussian vectors in $\mathbb{C}^2$, with $\mathbb{E}z_nz_n^*=I_2$, let $c_n=2/(\pi^2n^2)$ and $W=\sum_n c_nz_nz_n^*$. Then $\operatorname{tr}W\overset{d}{=}S_2$, so that $2\xi(s)=\mathbb{E}(\frac{\pi}{2}\operatorname{tr}W)^{s/2}$, and*

$$\mathbb{E}\det(x-W)=x^2-2m_1x+m_1^2-m_2,\qquad m_k=\sum_n c_n^k,\quad m_1=\tfrac{1}{3},\ m_2=\tfrac{2}{45},$$

with the real roots $m_1\pm\sqrt{m_2}=0.122515$ and 0.544152. This is the finite free convolution $\boxplus_2$ of the polynomials $\mathbb{E}\det(x-c_nz_nz_n^*)=x(x-2c_n)$, and the mixed characteristic polynomial of Marcus, Spielman and Srivastava [198] of the matrices c_nI_2.

Proof. $|z_n|^2$ is the sum of two independent standard exponential variables, so $\operatorname{tr}W=\sum_n c_n|z_n|^2\overset{d}{=}S_2$. For a 2×2 matrix, $\det(x-W)=x^2-x\operatorname{tr}W+\det W$ and $\det W=\frac{1}{2}((\operatorname{tr}W)^2-\operatorname{tr}W^2)$. Here $\mathbb{E}\operatorname{tr}W=2m_1$ and $\mathbb{E}(\operatorname{tr}W)^2=\operatorname{Var}S_2+(\mathbb{E}S_2)^2=2m_2+4m_1^2$, while $\mathbb{E}\operatorname{tr}W^2=\sum_{n,k}c_nc_k\mathbb{E}|z_n^*z_k|^2=6m_2+2(m_1^2-m_2)$, since $\mathbb{E}|z_n|^4=6$ and $\mathbb{E}|z_n^*z_k|^2=\operatorname{tr}(I_2I_2)=2$ for $n\neq k$. Hence $\mathbb{E}\det W=m_1^2-m_2$. The terms $c_nz_nz_n^*$ are independent and unitarily invariant in law, with $\mathbb{E}\det(x-c_nz_nz_n^*)=x^2-2c_nx$, so by [199, Theorem 1.2] and bilinearity the expected characteristic polynomial of a finite partial sum is the $\boxplus_2$ convolution of these polynomials; with the coefficient formula of $\boxplus_2$, adding $x(x-2c)$ to a polynomial x^2-Ax+B gives $x^2-(A+2c)x+B+Ac$, so the constant term is $2\sum_{n<k}c_nc_k=m_1^2-m_2$ in the limit. □

The reading of the exponent two of $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$ as the size of the matrix is special to the sinh family of [32].

In the normalisation of [199] write a polynomial of degree d as $p(x)=\sum_k(-1)^k\binom{d}{k}e_kx^{d-k}$; the multiplicative convolution $\boxtimes_d$ multiplies the normalised coefficients, $e_k(p\boxtimes_d q)=e_k(p)e_k(q)$ [199, Definition 1.4], and by Szegő's theorem it preserves the property of having only nonnegative roots [199, Theorem 1.6].

Lemma 11.2.8. *Let $a>0$ and $(a)_k=a(a+1)\cdots(a+k-1)$.*

(i) *The reversed Laguerre polynomial* $q_a(x) = x^d L_d^{(a-1)}(1/x)$ *has* d *positive simple roots and normalised coefficients* $e_k = e_0/(a)_k$, $e_0 = \Gamma(d+a)/(d!\,\Gamma(a))$. *Hence* $p \boxtimes_d (q_a/e_0)$ *has the normalised coefficients* $e_k(p)/(a)_k$, *and it has only real roots whenever the roots of* p *are real and of one sign.*

(ii) $\{1/(a)_k\}_{k\ge 0}$ *is a multiplier sequence: the function*

$$\sum_k \frac{x^k}{k!\,(a)_k} = \Gamma(a) x^{(1-a)/2} I_{a-1}(2\sqrt{x})$$

is entire of order $\frac{1}{2}$ *with only real negative zeros, at* $-j_{a-1,m}^2/4$.

(iii) *A density* f *on* $(0,\infty)$ *is a scale mixture of Gamma*(a) *densities if and only if* $x^{1-a} f(x)$ *is completely monotone; for* $a = 2$, *if and only if* $f(x)/x$ *is.*

Proof. (i) By [225, (18.5.12)],

$$x^d L_d^{(a-1)}(1/x) = \sum_k (-1)^k \binom{d+a-1}{d-k} \frac{x^{d-k}}{k!},$$

and $\binom{d+a-1}{d-k}/(k!\binom{d}{k}) = \Gamma(d+a)/(d!\,\Gamma(k+a))$, which is $e_0/(a)_k$. The zeros of $L_d^{(a-1)}$ are positive and simple, those of an orthogonal polynomial for the weight $x^{a-1}e^{-x}$ on $(0,\infty)$ [225, Section 18.2(vi)], and the roots of q_a are their reciprocals. The statement on $\boxtimes_d$ is Szegő's theorem for roots of one sign; for nonpositive roots apply it to $(-1)^d p(-x)$, whose normalised coefficients are $(-1)^k e_k(p)$. (ii) The series is that of the modified Bessel function [225, (10.25.2)]. For $x = -y^2/4 < 0$ it equals $\Gamma(a)(y/2)^{1-a} J_{a-1}(y)$, and for $a-1 > -1$ the zeros of $J_{a-1}(y)/y^{a-1}$ are real [225, Section 10.21(i)]. A real entire function of order less than one with only real zeros is in the Laguerre Pólya class, and by the theorem of Pólya and Schur [281] a sequence whose exponential generating function is in that class with coefficients of one sign is a multiplier sequence. (iii) $f(x) = \int_0^\infty \theta^{-a} x^{a-1} e^{-x/\theta}\, \varpi(d\theta)/\Gamma(a)$ says that $x^{1-a} f(x)$ is the Laplace transform of a positive measure in the variable $1/\theta$; conversely Bernstein's theorem writes a completely monotone $x^{1-a} f(x)$ in this form. □

Let X be the Pólya variable, with density $p = \Phi/\Xi(0)$ for the kernel Φ of (1.2.3) and moment generating function $\mathbb{E}e^{sX} = G(s^2)$ (1.2.4), and write $G(u) = \sum_k b_k u^k$, so that $b_k = \mathbb{E}X^{2k}/(2k)! = M_k/((2k)!\,M_0)$ with $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$, as in Sections 5.1 and 13.3.1.

Proposition 11.2.9. *For* $a > 0$ *let* $G_a(u) = \sum_k b_k u^k/(a)_k$. *If RH holds, then for every* $a > 0$ *the function* G_a *is in the Laguerre Pólya class with only real negative zeros, and all its Jensen polynomials* $\sum_{i=0}^d \binom{d}{i}(n+i)!\, b_{n+i} x^i/(a)_{n+i}$, $d, n \ge 0$, *have only real negative zeros. Multiplying the coefficients by* $(a)_k$ *instead does not preserve real zeros: it turns* $(1+u)^2 = 1+2u+u^2$ *into* $1+2au+a(a+1)u^2$, *whose discriminant is* $-4a < 0$.

Proof. Under RH, $G(u) = \prod_j (1+u/\gamma_j^2)$ with $\sum_j \gamma_j^{-2} < \infty$, so G is in the Laguerre Pólya class, and by Pólya's theorem its Jensen polynomials $J_{d,n}(x) = \sum_i \binom{d}{i}(n+i)!\, b_{n+i} x^i$ of Section 13.3.1 have only real zeros. Since $(a)_{n+i} = (a)_n (a+n)_i$, the Jensen polynomial of G_a of degree d and shift n is $1/(a)_n$ times the image of $J_{d,n}$ under the multiplier sequence $\{1/(a+n)_i\}_i$ of Lemma 11.2.8(ii), so it has only real zeros, negative because its coefficients are positive. For $n = 0$, $\binom{d}{i}\, i!\, d^{-i} \to 1$ gives $\sum_i \binom{d}{i}\, i!\, b_i x^i/((a)_i d^i) \to G_a(x)$ uniformly on compact sets as $d \to \infty$, so by Hurwitz's theorem G_a has only real negative zeros; since $(a)_k \geq a(k-1)!$, G_a has order at most $\frac{1}{3}$, and it is in the Laguerre Pólya class. The example is a direct computation. □

In terms of Lemma 11.2.8(i), dividing the normalised coefficients by $(a)_k$ is the multiplicative convolution with q_a/e_0, so peeling a Gamma(a) layer, the operation that would pass from the moments of a Gamma(a) scale mixture to those of its mixing law, keeps zeros real, while the inverse operation can destroy them. The note calls the peeling a free Poisson operation. The example does not show that no argument could pass from real zeros of some G_a to real zeros of G, as the note puts it.

Lemma 11.2.8(iii) makes the Gamma(2) mixture route of [252, Section 12.6], which Section 8.2.3 recalls as excluded, a statement about complete monotonicity.

THEOREM 11.2.10. *For no $a > 0$ is $|X|$, X^2 or S_2 a scale mixture of Gamma(a) laws.*

Proof. By Lemma 11.2.8(iii) a mixture would make $g(x) = x^{1-a} f(x)$ completely monotone, with f the density. A completely monotone $g \not\equiv 0$ is positive and log convex, so $g(x) \geq g(1) \exp\big((x-1) g'(1)/g(1)\big)$ for $x \geq 1$: it decays at most exponentially. For $v \geq 0$, dropping the negative terms of (1.2.3) and using $e^{2v} \geq 1$ gives

$$\Phi(v) \leq 4\pi^2 e^{9v/2} \exp(-\pi e^{2v}) \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)},$$

a constant multiple of $e^{9v/2} \exp(-\pi e^{2v})$. The densities of $|X|$ and X^2 are $2\Phi(v)/\Xi(0)$ and $\Phi(\sqrt{w})/(\Xi(0)\sqrt{w})$, so $v^{1-a}\Phi(v)$ and $w^{1/2-a}\Phi(\sqrt{w})$ would have to be completely monotone, and both decay faster than any exponential. For S_2 the obstruction is at the origin. By Chebyshev's inequality in exponential form, for $\lambda > 0$, $\Pr(S_2 \leq x) \leq e^{\lambda x} \mathbb{E} e^{-\lambda S_2} = e^{\lambda x} \big(\sqrt{2\lambda}/\sinh\sqrt{2\lambda}\big)^2$, and $\lambda = 2/x^2$ gives

$$\Pr(S_2 \leq x) \leq \frac{16\, e^{-2/x}}{x^2 (1 - e^{-4/x})^2} \leq 17\, x^{-2} e^{-2/x}, \qquad 0 < x \leq 1.$$

A mixture would give, g being nonincreasing and $g(1) > 0$, $f(x) \geq g(1) x^{a-1}$ and $\Pr(S_2 \leq x) \geq g(1) x^a/a$ for $0 < x \leq 1$, which is impossible as $x \downarrow 0$. □

The Hankel test sees the failure at low order. If $Y = \Theta\,\Gamma$ with Γ a Gamma(a) variable independent of $\Theta \geq 0$, then $\mathbb{E}Y^k = (a)_k \mathbb{E}\Theta^k$, so $\mu_k = \mathbb{E}Y^k/(a)_k$ must be a Stieltjes moment sequence, and both $[\mu_{i+j}]_{i,j<N}$ and $[\mu_{i+j+1}]_{i,j<N}$ must be positive semidefinite for every N. Table 11.8 gives the least order N at which each fails to be positive definite, by Cholesky factorisation at 400 digits, with the same orders at 250. The moments of X^2 are M_k/M_0; those of $|X|$ come from a tanh sinh rule, since odd powers of $|v|$ are not smooth at the origin; those of S_2 follow from its cumulants. The table of [261] is the Hamburger row, the first matrix alone.

a	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	4	6	8	12	20
$\lvert X\rvert$, Stieltjes	2	2	2	2	2	3	3	3	4	4	5
$\lvert X\rvert$, Hamburger	2	2	2	3	3	3	3	4	4	5	6
X^2, Stieltjes	2	3	3	4	4	4	5	5	6	7	9
X^2, Hamburger	2	3	4	4	4	5	5	6	6	7	9
S_2, both	2	2	2	2	2	2	2	3	4	6	12

Table 11.8: The first order N at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma(a) scale mixture, for $Y = |X|$, X^2 and S_2: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed by `wigfree_gamma2.py`.

The theorem also settles which tool applies. Kristiansen proved Steutel's conjecture that scale mixtures of Gamma(2) laws are infinitely divisible [156]; since $x^{\alpha-2}e^{-x}$ is completely monotone for $\alpha \leq 2$, Lemma 11.2.8(iii) makes Gamma(α) itself a scale mixture of Gamma(2) laws, and the theorem covers scale mixtures of Gamma(α) laws for every $\alpha \leq 2$. By Theorem 11.2.10 it cannot be applied to $|X|$, X^2 or S_2. The Gamma(2) structure that is true is Williams' convolution, a GGC and hence infinitely divisible directly.

11.2.6 The heat flow and finite free convolution

Remark 8.4.3 identifies the forward heat flow on polynomials with finite free convolution by a Hermite polynomial and, as the degree grows, with free convolution by a semicircle law. The time is the variance of the semicircle law. For a real polynomial p of degree $d \geq 2$ write $p_t = e^{-(t/2)D^2}p$, a finite sum, and

$$q_t(x) = e^{-(t/2)D^2}x^d = t^{d/2}H_d\big(x/\sqrt{t}\big), \qquad t > 0,$$

with $q_0 = x^d$ and $H_d = e^{-D^2/2}x^d$ the Hermite polynomial of Remark 8.4.3, whose zeros are real. The time t is twice the time of Section 8.4.1, where $\partial_t H_t = -\partial_z^2 H_t$.

Proposition 11.2.11. *Let p be a real polynomial of degree $d \geq 2$ and $s, t \geq 0$.*

(i) $p_t = p \boxplus_d q_t$.

(ii) $q_s \boxplus_d q_t = q_{s+t}$.

(iii) *If p has only real roots, then so has p_t, and the largest root of p_t is at most the largest root of p plus $\sqrt{t}\,h_d$, where h_d is the largest zero of H_d.*

(iv) *While the roots $z_1(t), \dots, z_d(t)$ of p_t are simple, $\dot z_k = \sum_{j\neq k}(z_k - z_j)^{-1}$.*

Proof. (i) In operator form, $p \boxplus_d q = \hat p(D)\hat q(D)x^d$ when $p = \hat p(D)x^d$ and $q = \hat q(D)x^d$ [199, Definition 1.1 and (2)]. For $q = q_t$ the operator $\hat q(D)$ acts on polynomials of degree at most d as $e^{-(t/2)D^2}$, and operators in D commute, so $p \boxplus_d q_t = e^{-(t/2)D^2}\hat p(D)x^d = p_t$. (ii) is (i) for $p = q_s$, since $e^{-(t/2)D^2}e^{-(s/2)D^2} = e^{-((s+t)/2)D^2}$; it is the identity $a^d H_d(x/a) \boxplus_d b^d H_d(x/b) = c^d H_d(x/c)$, $c^2 = a^2 + b^2$, of [199, Section 1.2]. (iii) The roots of q_t are $\sqrt{t}$ times those of H_d, and by Walsh's theorem [199, Theorem 1.3] the convolution $p \boxplus_d q$ of two polynomials of degree d with only real roots has only real roots, the largest at most the sum of the largest roots of p and q. (iv) Differentiate $p_t(z_k(t)) = 0$ in t and use $\partial_t p_t = -\frac12 p_t''$: $\dot z_k = \frac12 p_t''(z_k)/p_t'(z_k)$. Writing $p_t = (x - z_k)r$ gives $p_t'(z_k) = r(z_k)$ and $p_t''(z_k) = 2r'(z_k)$, so that $\dot z_k = r'(z_k)/r(z_k) = \sum_{j\neq k}(z_k - z_j)^{-1}$. □

For a conjugate pair $x \pm iy$, $y > 0$, the partner contributes $-1/(2y)$ to $\dot y$, and each real root r contributes $\operatorname{Im}(x + iy - r)^{-1} = -y/((x-r)^2 + y^2) < 0$: the pair is drawn to the axis by its partner and pulled further by every real root. On its own, $(x-1)^2 + 0.64$ becomes $(x-1)^2 + 0.64 - t$, which reaches the axis at $t = 0.64$. The example of the note adds four real roots,

$$p(x) = (x+3)(x+1)\big(x - \tfrac12\big)(x-2)\big((x-1)^2 + 0.64\big).$$

The discriminant of p_t in x is a polynomial of degree 15 in t whose only root in $(0, 2)$ is $t^* = 0.3203291143987806$. Before t^* the roots are distinct, so the pair stays off the axis, and since $p_t = e^{-((t-t^*)/2)D^2}p_{t^*}$ for $t > t^*$, part (iii) keeps all roots real afterwards: the pair meets the axis at t^*, at the point 1.25743, in about half the time it needs on its own. The program `wigfree_heat.py` computes the discriminant exactly.

The Hermite polynomial is the finite free counterpart of the semicircle law: H_d is the expected characteristic polynomial of a symmetric Gaussian matrix [199, Section 1.2], and the roots of $H_d(\sqrt{d}\,x)$, the roots of $q_{1/d}$, are close to the quantiles x_k of σ_1 given by $\sigma_1([-2, x_k]) = (k - \frac12)/d$. The largest deviation from the quantiles is at the edge, and its products with $d^{2/3}$, from 0.3906 to 0.4962 for $d = 20$ to 320, grow slowly, as they should at the edge scale $d^{-2/3}$ of the extreme zeros. In the limit $d \to \infty$ finite free convolution becomes free convolution [7], [199, Section 1.2]: if the root distributions of polynomials p_d with only real roots tend to a law μ in moments, those of $p_d \boxplus_d q_{t/d}$ tend to $\mu \boxplus \sigma_t$.

For H_t the identification is a heuristic. The motion of the zeros is the same, up to the factor two in the time (Section 8.4.1), but H_t has infinitely many zeros,

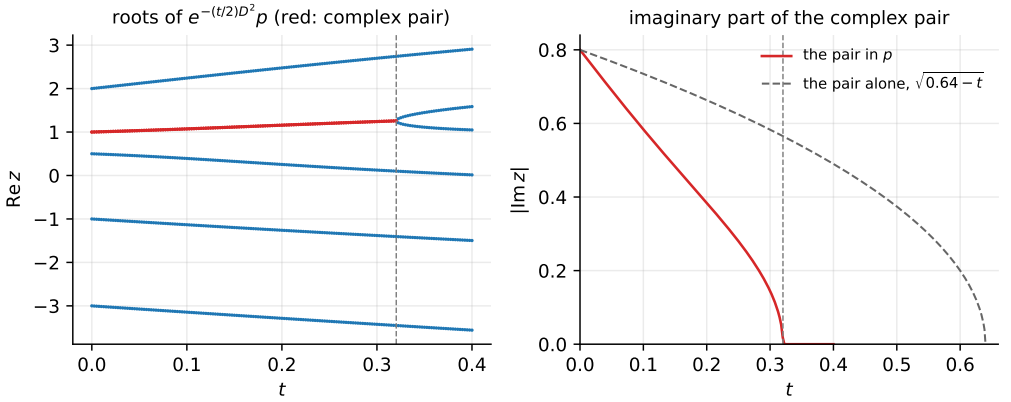


Figure 11.4: The flow $e^{-(t/2)D^2}p$ for p with the roots -3, -1, $\frac{1}{2}$, 2 and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed by `wigfree_heat.py` and drawn by `wigfree_plots.py`.

with a density that grows logarithmically, while the limit theorem concerns the distribution of all d roots of a polynomial after scaling; for H_t it would have to be applied to windows of zeros, with the zeros outside a window acting as an external field, and no theorem of this kind is known. De Bruijn's theorem that real zeros of H_t stay real forward in time [73] is the counterpart of (iii), proved directly. In Thorin terms the forward flow is the path of clocks R_t of Section 8.4.1, in the time of H_t (Theorem 7.2.15).

The Jensen polynomials of Section 13.3.1 meet the Hermite polynomials directly. Griffin, Ono, Rolen and Zagier proved that for each degree d the Jensen polynomials $J_{d,n}$, after an affine change of variable and a normalisation depending on n, tend to the Hermite polynomial of degree d as $n \to \infty$, and this is how they obtain real zeros for all large shifts [99]. In the language of this subsection the Jensen polynomials of large shift are asymptotically finite free Gaussian; Problem 12.1 asks for an exact finite free structure at every shift.

These limits see the zeros only through their distribution. Moving $o(d)$ of the d roots changes no limit law, so a single nonreal pair is invisible to free convolution limits, and these limits cannot decide RH by themselves [261].

11.3 The Dedekind eta analogue at the basepoint one

The L function of the real character χ_{12} modulo 12, to which Section 8.5 carried the pipeline of the book, has no pole. Its completed function is entire and is a

Mellin transform of the Dedekind eta function, whose product formula makes the kernel positive. We repeat for $L(s,\chi_{12})$ the calculations that Chapter 9 made for ζ at the basepoint one.

The scripts are in the folder `basepoint_one/eta` of the archive.

11.3.1 The completed L function and its Thorin measure

We use the completed function $\xi(s,\chi_{12}) = (12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12})$, its kernel K and the variable Y with density $K/\xi(\frac12,\chi_{12})$ of Section 8.5.1; the function satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ and is entire. We computed its zeros on the critical line by sign changes of $\xi(\frac12+it,\chi_{12})e^{\pi t/4}$ with `zeros12.py`, finding 297 below height 340; Section 8.5.1 lists the first of them. In this section $b_\alpha = (\xi'/\xi)(\alpha,\chi_{12})$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_{12})$, and φ_α, T_1, $\widetilde{T}$, Π, A, A_α, Q and $a_k = -(\rho_k-\frac12)^2$ are formed with $\xi(s,\chi_{12})$ and its zeros $\rho=\beta+i\gamma$ in place of ξ. So is the tilted eta variable V_α, the Esscher tilt of Y by $e^{(\alpha-\frac12)Y}$, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s,\chi_{12})/\xi(\alpha,\chi_{12})$. The constants we need, computed by `base.py`, are

$$L(1,\chi_{12}) = \frac{\log(2+\sqrt3)}{\sqrt3} = 0.760346, \qquad L'(1,\chi_{12}) = 0.362495,$$

$$b_1 = \frac{\xi'}{\xi}(1,\chi_{12}) = \frac12\log\frac{12}{\pi} + \frac12\psi\Big(\frac12\Big) + \frac{L'}{L}(1,\chi_{12}) = \sum_\rho \frac1\rho = 0.1650833123,$$

where ψ is the digamma function. The Taylor coefficients of $\xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ give the power sums $p_1 = \sum_k a_k^{-1} = 0.1664618$ and $p_2 = \sum_k a_k^{-2} = 0.0055985$, which are $\sum_{\gamma>0}\gamma^{-2}$ and $\sum_{\gamma>0}\gamma^{-4}$ under GRH; p_1 is the mean of the centre clock.

Proposition 11.3.1. *The function $\varphi_1(u) = \xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$ is the Laplace transform of a GGC with no drift, with Thorin density in $\theta=\sqrt{z}$*

$$\frac1\pi v_1(\theta) = \frac1\pi\,\mathrm{Re}\,\frac{\xi'}{\xi}(1+i\theta,\chi_{12}) = \frac1\pi\sum_\rho \frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2} > 0,$$

the sum over the nontrivial zeros $\rho=\beta+i\gamma$ of $L(s,\chi_{12})$ in conjugate pairs. Positivity is unconditional because $L(1+it,\chi_{12})\neq0$.

Proof. The proof is that of Proposition 9.1.2, with the Hadamard product of the entire function $\xi(s,\chi_{12})$ in place of that of ξ. □

The minimum on $[0,60]$ is at the origin, $v_1(0)/\pi = b_1/\pi = 0.0525477$, and the cumulative mass tracks the zero count: $\int_0^{60} v_1/\pi = 35.81$ against $N_{\chi_{12}}(60) = 36$, where $N_{\chi_{12}}(T)$ counts the zeros with $0<\gamma\le T$ as in Section 8.5.1. Table 11.9, computed by `base.py`, compares direct values with the Poisson sum over the 297 computed zeros; the nearly constant difference of 1.11×10^{-3} is the contribution of the zeros above height 340. Figure 11.5 compares the density with that of ζ: on

this window the zeros of $L(s,\chi_{12})$ are more than twice as dense, and its density is correspondingly higher. As for ζ, $\varphi_1(u) = 1 - b_1\sqrt{u} + O(u)$, so T_1 has infinite mean and $P(T_1 > x) \sim b_1(\pi x)^{-1/2} = 0.09314\, x^{-1/2}$.

θ	$v_1(\theta)/\pi$ from $\xi'/\xi(\cdot,\chi_{12})$	from 297 zeros, $\beta = \frac{1}{2}$
0	0.0525477	0.0514359
3.8046276	0.6889076	0.6877958
5	0.1852757	0.1841637
10	0.2694467	0.2683341
30	0.7445558	0.7434362
60	0.7298854	0.7287413

Table 11.9: The Thorin density v_1/π of $\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$, computed from $\xi'/\xi(\cdot,\chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s,\chi_{12})$ below height 340 with $\beta = \frac{1}{2}$.

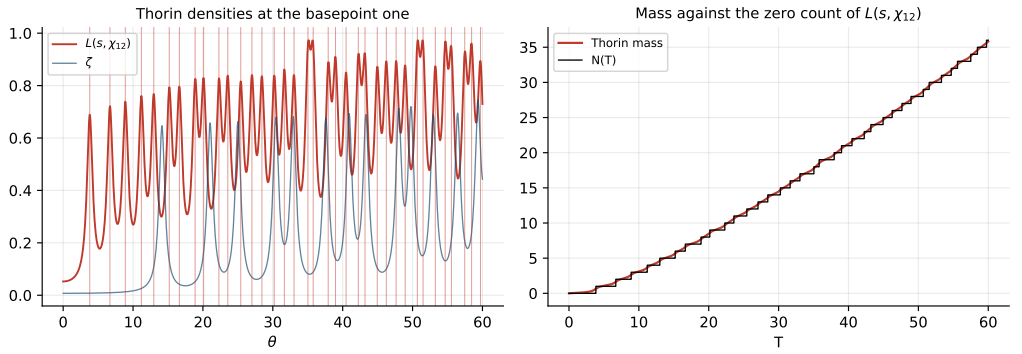


Figure 11.5: Left: the Thorin densities at the basepoint one for $L(s,\chi_{12})$ and for ζ on $[0,60]$, with the ordinates of the zeros of $L(s,\chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s,\chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

11.3.2 The Lévy form without a pole, at and below the basepoint one

Proposition 11.3.2. *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s,\chi_{12})}{\xi(1,\chi_{12})} = b_1 s + \int_0^\infty (e^{-sx} - 1 + sx)\,\Pi(dx),$$

where, with $\Lambda_{\chi_{12}}(n) = \chi_{12}(n)\Lambda(n)$,

$$\Pi(dx) = \frac{e^{-x}}{x(1-e^{-2x})}\,dx + \sum_{n\geq 2} \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,\delta_{\log n}(dx),$$

the prime sum converging conditionally, as a limit over $n \leq X$.

Proof. Binet's formula [225, §5.9] for $\log\Gamma(\frac{1}{2}+\frac{s}{2})$, in the integrated form of the proof of Proposition 9.1.10 with $a = \frac{1}{2}$, gives the gamma density; the factor $(12/\pi)^{s/2}$ and the linear terms combine into b_1; and for $\operatorname{Re} s > 0$ the Euler product gives

$$\log L(1+s,\chi_{12}) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,e^{-s\log n}.$$

The linear coefficient is $-\sum_n \Lambda_{\chi_{12}}(n)/n = \frac{L'}{L}(1,\chi_{12})$, which converges by the prime number theorem for arithmetic progressions [71, Chapter 20], in the form $\psi(y,\chi_{12}) = O(y\,e^{-c\sqrt{\log y}})$, where $\psi(y,\chi_{12}) = \sum_{n\leq y}\Lambda_{\chi_{12}}(n)$ is the Chebyshev function of χ_{12}. □

The form differs from that of ζ in two ways. There is no pole, so no compensating measure: the prime atoms converge on their own at the basepoint one. And the measure is signed for a different reason. For ζ the gamma density minus the pole term dx/x is negative beyond $x = 0.281$ and the prime atoms are positive; here the gamma density $e^{-x}/(x(1-e^{-2x}))$ is positive everywhere and the atoms carry the sign of $\chi_{12}(n)$ (Figure 11.6, left). Table 11.10, computed by `arith.py`, checks the form.

The tilted eta variable V_1, with moment generating function $\xi(1+s,\chi_{12})/\xi(1,\chi_{12})$, is not quasi infinitely divisible. Its positive and its negative atoms each have infinite total mass, since $\sum_{\chi_{12}(p)=1} 1/p$ and $\sum_{\chi_{12}(p)=-1} 1/p$ both diverge by Dirichlet's theorem, and they cancel only conditionally, so the argument of Proposition 11.1.3(3) applies with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, as the discussion after its proof explains. At every basepoint $\alpha > 1$, by contrast, V_α is quasi infinitely divisible: there the atoms of each sign have finite total mass, and the Euler product writes its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions.

The sine squared lemma in the form of Section 9.1.4, with $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta = \pi s$, converts the Lévy form, with $s = \sqrt{u}$, into

$$v_1(\theta) = b_1 + A(\theta) + Q(\theta),$$

with

$$A(\theta) = \tfrac{1}{2}\big[\operatorname{Re}\psi(\tfrac{1+i\theta}{2}) - \psi(\tfrac{1}{2})\big], \qquad Q(\theta) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big).$$

s	X	$b_1 s$	gamma piece	primes	total	exact
0.5	10^5	0.08254	0.121808	−0.082565	0.121785	0.122329
0.5	10^7	0.08254	0.121808	−0.081931	0.122419	0.122329
1	10^7	0.16508	0.409141	−0.254214	0.320011	0.320099
2	10^7	0.33017	1.270363	−0.689198	0.911332	0.911001

Table 11.10: The Lévy form of Proposition 11.3.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to X.

The drift becomes the stable one half factor $e^{-b_1\sqrt{u}}$, the gamma factor the smooth positive growth A, and the primes the oscillating Q, now without any divergent partner. With primes to 10^7 the formula gives 0.84633 at $\theta = 10$ against the exact 0.84649.11.6 (right) shows the three pieces: the prime piece ranges over both signs, and the positivity of the sum comes from the drift and the gamma factor, as Section 8.5.3 found at the basepoint two.

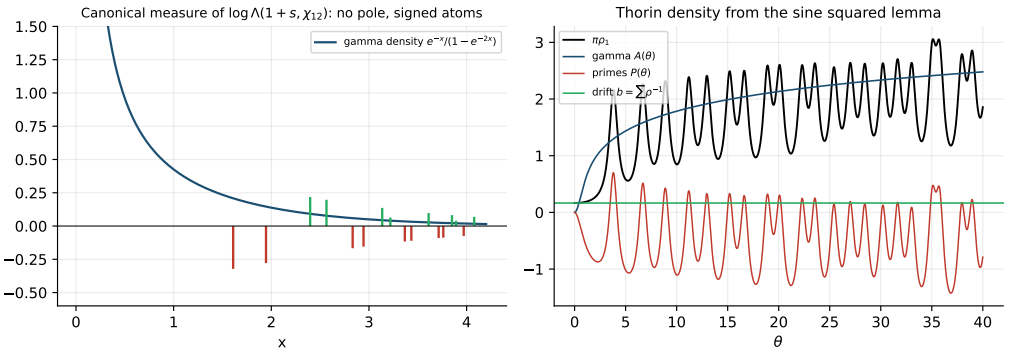


Figure 11.6: Left: the canonical measure $x\Pi(dx)$ of $\log\xi(1+s,\chi_{12})$, written with $\Lambda(1+s,\chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1-e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0,40]$, written $\pi\rho_1$ in the plot, as the sum of the drift b_1, written b, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

Below the basepoint one, at $\alpha = 1-\varepsilon$, the converted Thorin density is π^{-1} times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty} \sum_{n\le X} \frac{\Lambda_{\chi_{12}}(n)}{n^\alpha}\, 2\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

with $A_\alpha(\theta) = \frac{1}{2}[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$. Exactly as for ζ in Proposition 9.1.13, the limit exists and gives the positive Thorin density v_α/π when $L(s,\chi_{12})$ has no zeros with $\operatorname{Re} s > \alpha'$ for some $\alpha' < \alpha$, and conversely its existence for $\operatorname{Re} s > 0$ in the

Lévy form forces the absence of zeros with $\operatorname{Re} s > \alpha$. The error at cutoff X is given by the same explicit formula,

$$\sum_{\rho} \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)} \Big] + O(X^{-\alpha}\log X),$$

because $\psi(y,\chi_{12}) = -\sum_\rho y^\rho/\rho + O(\log y)$ has no main term [71, Chapter 19].

For ζ the prime piece and the pole piece each grow like $X^{1-\alpha}/(1-\alpha)$ and must cancel; for $L(s,\chi_{12})$ the prime piece stays bounded. Table 11.11, computed by `arith.py`, shows the errors, which decay like $X^{1/2-\alpha}$, and what remains after subtracting the explicit formula over the 297 computed zeros. The correction removes most of the error at $\theta = \gamma_1$; at $\theta = 10$, between zeros, it helps less, because 297 zeros are fewer than the 1700 used for ζ and the missing zeros above height 340 dominate what remains. At the centre and the first ordinate the sum grows like $\log X$: the atom δ_{γ_1} forming (Figure 11.7).

α	θ	X	prime piece	error	error after zero correction
1	γ_1	10^5	0.6964	-5.55×10^{-3}	4.2×10^{-6}
1	γ_1	10^7	0.7016	-4.15×10^{-4}	2.5×10^{-5}
0.75	γ_1	10^5	2.0933	-2.13×10^{-1}	1.4×10^{-4}
0.75	γ_1	10^7	2.2471	-5.92×10^{-2}	1.4×10^{-3}
0.6	γ_1	10^5	4.8292	-3.10	1.1×10^{-3}
0.6	γ_1	10^7	6.0670	-1.86	1.6×10^{-2}
0.6	10	10^7	-2.4525	-5.68×10^{-2}	2.6×10^{-2}

Table 11.11: The sine squared formula below the basepoint one for $L(s,\chi_{12})$, with primes up to X: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$.

11.3.3 The hitting time and its boundary

We computed the Brownian boundary $b(t)$ whose crossing time has the law of T_1 from the Volterra equation (8.2.1), with the distribution function from Talbot inversion for $t < 0.12$ and, under GRH, from the subordination representation of Theorem 9.2.2 beyond; its time scale is about seven times that of ζ, the ratio of the means of the two centre clocks. In the right tail the exact boundary flattens at -0.2747, the Brownian form of the stable one half tail.

11.3.4 The two examples compared

Table 11.12 sets the two basepoint one examples side by side. Structurally the eta case is the cleaner of the two: below the basepoint one its prime sums stay bounded, so the computation is far better conditioned. Nothing essential becomes

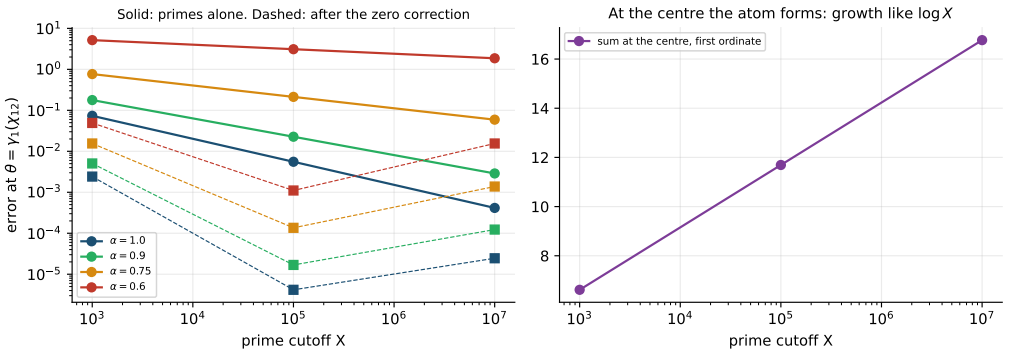


Figure 11.7: Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff X, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming. Drawn by `figs.py` of `basepoint_one/eta`.

easier, however. The negativity that ζ concentrates in the pole piece is carried here by the signs of $\chi_{12}(p)$; and, under GRH for χ_{12}, the probabilistic picture is the same, with every constant scaled by the denser zeros.

The comparison confirms the two conclusions that fix the shape of the path to RH. First, at the basepoint one the constructions for the reciprocal Xi function are, with two exceptions, unconditional and explicit. The exceptions are the Krein string representation of Corollary 9.2.3, which assumes RH and simple zeros, and the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$: Lemma 9.2.4 gives it for a boundary bounded at infinity, and the boundedness of the computed Brownian boundary is observed, not proved. Second, below the basepoint one each positive statement for ζ holds, up to the boundary case, exactly when no zero lies to the right of the line $\operatorname{Re} s = \alpha$ (Propositions 9.1.13 and 10.1.10, Theorems 9.2.5 and 10.1.3). The eta example shows that the pole changes how the prime sums converge, not where they stop.

11.4 Almost generalised gamma convolutions

Below the basepoint one, as Section 11.3.4 concluded, every positive object built so far from the shifted clock T_α, whose Laplace transform is $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, is equivalent, up to the boundary case, to the absence of zeros in the half plane $\operatorname{Re} s > \alpha$. These equivalences are restatements rather than reductions. Since no zero free half plane was known before 2026, and none is known to the left of the line $\operatorname{Re} s = \frac{7}{8}$ (Section 12.1.2), the question that remains is quantitative: how negative can the Thorin density be below the basepoint one? Zero density estimates answer it with no hypothesis: with those of Ingham, of Guth and Maynard and of Huxley,

	ζ	$L(s,\chi_{12})$
kernel	theta series	$\eta(ix)$, a positive product
pole	at $s=1$	none
first zero	14.134725	3.804628
drift $b_1=\sum_\rho \rho^{-1}$	0.0230957	0.1650833
tail constant $b_1/\sqrt{\pi}$	0.01303	0.09314
mean of the centre clock $\sum_k a_k^{-1}$	0.0231050	0.1664618
Lévy measure, continuous part	gamma minus pole, negative beyond 0.281	gamma only, positive
Lévy measure, prime atoms	all positive	signs of $\chi_{12}(n)$
compensation at $\alpha=1$	primes against dx/x (prime number theorem)	none needed
prime piece at $\alpha=0.6$, $X=10^7$, $\theta=\gamma_1$	about 1537, cancelled by the pole	6.07
flat level of the exact boundary	-0.19277	-0.27471
$\mathbb{E}\,b(T_1)=b_1/\sqrt{2}$	0.016331	0.116732
tangent approximation fails at	$t=0.091$	$t=1.83$

Table 11.12: The two basepoint one examples side by side: the reciprocal Xi function of Chapter 9 and the reciprocal completed L function of χ_{12} of this section.

every T_α with $\alpha>\frac12$ is a GGC up to a defect carried by the zeros to the right of α, whose Cauchy kernels carry a vanishing fraction of the mass of the others, with a power saving; we call this property almost GGC.

With ξ again in place of $\xi(\cdot,\chi_{12})$, we keep the notation of Sections 9.1 and 11.1: $v_\alpha(\theta)=\mathrm{Re}(\xi'/\xi)(\alpha+i\theta)$, and the Thorin density at the basepoint α in the chart $\theta=\sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (1.2.7) at the resolution $\alpha-\frac12$; in this section the basepoint, not the resolution, is the index. $P_w(x)=w/(\pi(w^2+x^2))$ is the Poisson kernel of Section 1.2, $P_{-w}=-P_w$, and P_0 is the unit point mass, as in Proposition 3.3.1. A zero is $\rho=\beta+i\gamma$; $N(T)$ counts the zeros with $0<\gamma\le T$, and $N(\sigma,T)$ those among them with $\beta\ge\sigma$, so that $N(\sigma,T)\ge\nu(\sigma,T)$ in the notation of Proposition 3.4.2.

11.4.1 Mass bookkeeping and the almost GGC theorem

The Poisson form of Proposition 1.2.3 splits according to the side of the line $\operatorname{Re} s = \alpha$ on which each zero lies, $v_\alpha = v_\alpha^+ - v_\alpha^-$, with

$$\begin{aligned} \frac{v_\alpha^+(\theta)}{\pi} &= \sum_{\beta\le\alpha,\ \gamma>0} \big[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\big], \\ \frac{v_\alpha^-(\theta)}{\pi} &= \sum_{\beta>\alpha,\ \gamma>0} \big[P_{\beta-\alpha}(\theta-\gamma) + P_{\beta-\alpha}(\theta+\gamma)\big]. \end{aligned} \tag{11.4.1}$$

Both parts are positive, and a zero on the line $\operatorname{Re} s = \alpha$, if there is one, contributes a unit atom at its ordinate to the positive part. Each pair of kernels has mass one on $(0,\infty)$. A zero with $\beta > \alpha$ has the partner $1-\bar\rho = 1-\beta+i\gamma$, whose real part $1-\beta < \frac12 < \alpha$ puts it in the positive part. Write $m_\alpha^\pm(T) = \pi^{-1}\int_0^T v_\alpha^\pm(\theta)\,d\theta$ for the positive and negative masses on $[0,T]$.

Lemma 11.4.1 (Mass bookkeeping). *Let $\alpha > \frac12$, and suppose that $N(\alpha,t) \le t^{e+o(1)}$ as $t\to\infty$ for some $e<1$. Then*

$$m_\alpha^-(T) \le N(\alpha,2T) + O\big(T^{e-1+o(1)}\big), \qquad m_\alpha^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} + O(T).$$

Proof. A zero with $\beta>\alpha$ and $\gamma \le 2T$ contributes at most its total mass one. For a zero with $\gamma > 2T$ put $w = \beta-\alpha$, so that $0<w<\frac12$. Since $\int_0^T P_w(\theta+\gamma)\,d\theta = \int_{-T}^0 P_w(\theta-\gamma)\,d\theta$, its pair of kernels has mass on $[0,T]$ at most

$$\int_{-\infty}^{T} P_w(\theta-\gamma)\,d\theta \le \frac{w}{\pi(\gamma-T)} \le \frac{1}{\pi\gamma},$$

and $\sum_{\beta>\alpha,\ \gamma>2T}\gamma^{-1} = \int_{2T}^\infty t^{-1}dN(\alpha,t) \le \int_{2T}^\infty N(\alpha,t)t^{-2}dt \ll T^{e-1+o(1)}$. The positive part contains the kernels of all zeros with $\beta\le\alpha$, among them every zero on the critical line and every partner $1-\bar\rho$. Its mass on $[0,T]$ is therefore $N(T)$ up to the $N(\alpha,T) = O(T^{e+o(1)})$ zeros to the right of α, whose kernels it lacks, and up to the tails of the kernels of the zeros near height T, which change it by $O(\log^2 T)$. Since $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, the second claim follows. □

Theorem 11.4.2 (Almost GGC). *For every $\frac12 < \alpha < 1$, with no hypothesis,*

$$\frac{m_\alpha^-(T)}{m_\alpha^+(T)} \le T^{e(\alpha)-1+o(1)}, \qquad e(\alpha) = \begin{cases} \dfrac{3(1-\alpha)}{2-\alpha}, & \frac12 < \alpha < \frac{7}{10}, \\[2mm] \dfrac{15(1-\alpha)}{3+5\alpha}, & \frac{7}{10} \le \alpha < 1, \end{cases}$$

and $e(\alpha)<1$ throughout, so the negative part v_α^-/π of (11.4.1), the Cauchy kernels of the zeros to the right of α, carries on $[0,T]$ a vanishing fraction of the mass of

the positive part, with a power saving. For $\alpha \geq 1$ no zero lies to the right of α, and $m_\alpha^- = 0$. Under the density hypothesis $e(\alpha) = 2(1-\alpha)$. RH holds if and only if v_α^- vanishes identically for every $\alpha > \frac{1}{2}$.

Proof. Combine Lemma 11.4.1 with three zero density estimates: Ingham's $N(\sigma,T) \ll T^{3(1-\sigma)/(2-\sigma)}\log^5 T$ [124]; Guth and Maynard's $N(\sigma,T) \leq T^{15(1-\sigma)/(3+5\sigma)+o(1)}$ for $\frac{7}{10} \leq \sigma \leq \frac{4}{5}$ [104, Theorem 1.2]; and Huxley's $N(\sigma,T) \leq T^{3(1-\sigma)/(3\sigma-1)+o(1)}$ for $\sigma \geq \frac{3}{4}$ [119], whose exponent is at most $15(1-\sigma)/(3+5\sigma)$ when $\sigma \geq \frac{4}{5}$. Since $3(1-\alpha) < 2-\alpha$ exactly when $\alpha > \frac{1}{2}$, and $15(1-\alpha) < 3+5\alpha$ exactly when $\alpha > \frac{3}{5}$, the exponent is below one for every $\frac{1}{2} < \alpha < 1$. Finally, v_α^- vanishes identically exactly when no zero has $\beta > \alpha$, which is the GGC criterion of Theorem 1.2.2. □

Even the classical estimate of Bohr and Landau [36], $N(\sigma,T) = O(T)$ for fixed $\sigma > \frac{1}{2}$, gives a vanishing fraction $O(1/\log T)$; the zero density estimates improve this to a power of T. First, the theorem bounds the decomposition mass $m_\alpha^-(T)$, which overcounts the negative part $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ of the density itself, because the positive kernels of nearby zeros cancel much of each negative kernel. The partner of a zero already cancels part of it: with $\varepsilon = \alpha - \frac{1}{2}$ and $\delta = \beta - \frac{1}{2}$, twice the mass in Proposition 3.3.1(iii) gives the negative part of $P_{\alpha+\beta-1} - P_{\beta-\alpha} = P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ the mass $\frac{4}{\pi}\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - 1 = \frac{2}{\pi}\arcsin\frac{\varepsilon}{\delta}$, less than the mass one that the decomposition counts, and Section 11.4.4 shows the effect of the other zeros. Second, on the verified range the negative mass is essentially zero. All zeros with $0 < \gamma \leq T_0$ lie on the critical line [247], so for $T \leq T_0/2$ the only negative contributions come from zeros above T_0, and by the proof of Lemma 11.4.1 their total mass on $[0,T]$ is $O(T_0^{e(\alpha)-1+o(1)})$.

The left panel of Figure 11.9 draws the exponents. The saving $1 - e(\alpha)$ is small near the centre, where Ingham's exponent gives $(2\alpha-1)/(2-\alpha)$. Sharper exponents are known near $\alpha = 1$: the density hypothesis itself holds for $\alpha > \frac{25}{32}$ [45, p. 133], and Huxley's exponent [119] is 0.176 at $\alpha = 0.9$; those of the theorem suffice for what follows.

11.4.2 The canonical factorisation and explicit unconditional bounds

Proposition 11.4.3 (Canonical factorisation). *For every $\alpha > \frac{1}{2}$ and $u \geq 0$, $\varphi_\alpha(u) = \varphi_\alpha^+(u)\,D_\alpha(u)$, where*

$$\varphi_\alpha^+(u) = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big)\frac{v_\alpha^+(\theta)}{\pi}\,d\theta\Big\},$$

$$D_\alpha(u) = \prod_{\beta>\alpha,\ \gamma>0} \frac{(\beta-\alpha)^2+\gamma^2}{(\sqrt{u}-\beta+\alpha)^2+\gamma^2};$$

here φ_α^+ is the Laplace transform of a GGC T_α^+ with Thorin density v_α^+/π in the chart $\theta = \sqrt{z}$, and the product defining the defect D_α runs over the zeros to the right of the line $\operatorname{Re} s = \alpha$ and converges absolutely. The defect is identically one if and only if ζ has no zeros with $\operatorname{Re} s > \alpha$, and then $T_\alpha \stackrel{d}{=} T_\alpha^+$. Otherwise $D_\alpha(u) \to 0$ as $u \to \infty$, and T_α^+ is not the sum of T_α and an independent nonnegative variable.

Proof. Grouping the Hadamard product of ξ in conjugate pairs, as in the proof of Proposition 1.2.3, gives for $u \geq 0$

$$\varphi_\alpha(u) = \prod_{\gamma>0} \frac{(\alpha-\beta)^2+\gamma^2}{(\alpha-\beta+\sqrt{u})^2+\gamma^2},$$

over the zeros with $\gamma > 0$, counted with multiplicity; each factor is $1 + O((u + \sqrt{u})\gamma^{-2})$ and $\sum \gamma^{-2} < \infty$, so the product converges absolutely. For $\beta < \alpha$, with $w = \alpha - \beta > 0$, the factor is $(w^2+\gamma^2)/((w+\sqrt{u})^2+\gamma^2) = \exp\{-\int_0^\infty \log(1 + u/\theta^2)[P_w(\theta-\gamma) + P_w(\theta+\gamma)]\, d\theta\}$, the Laplace transform of a GGC whose Thorin density is the pair of Cauchy kernels of width w at $\pm\gamma$ (the computation after Theorem 10.1.2); for $\beta = \alpha$ it is $\gamma^2/(\gamma^2+u)$, an exponential law, whose Thorin measure is the atom at γ. The partial products over the zeros with $\beta \leq \alpha$ are therefore Laplace transforms of GGCs, they converge for every $u \geq 0$ to a limit continuous at the origin, and by the closure theorem of Section 4.1.1 the limit is the Laplace transform of a GGC, with Thorin density v_α^+/π. The remaining factors form D_α. If a zero lies to the right of α, put $w = \beta - \alpha \in (0, \frac{1}{2})$; its factor tends to zero as $u \to \infty$, and every factor of D_α is at most one for $u \geq 1$, because $(\sqrt{u} - w)^2 \geq w^2$ when $\sqrt{u} \geq 2w$. So $D_\alpha(u) \to 0$. If T_α^+ were $T_\alpha + S$ with $S \geq 0$ independent of T_α, then $\mathbb{E}e^{-uS} = \varphi_\alpha^+(u)/\varphi_\alpha(u) = 1/D_\alpha(u)$ would be unbounded, which is impossible. □

The proposition also says what the signed density v_α/π is. When every zero lies to the left of the line $\operatorname{Re} s = \alpha$ it is the Thorin density of T_α (Theorem 1.2.2); a zero on the line adds to the Thorin measure an atom at its ordinate, the term P_0 of (11.4.1), which the function v_α does not show. When a zero lies to the right of the line, T_α is not a GGC and v_α/π is not a signed Thorin density for it in any sense: the negative kernel $-P_w$ of such a zero integrates to the factor $((w+\sqrt{u})^2 + \gamma^2)/(w^2+\gamma^2)$, while the zero contributes $(w^2+\gamma^2)/((\sqrt{u}-w)^2+\gamma^2)$ to φ_α. The bookkeeping of Theorem 11.4.2 applies to the other reformulations in Chapters 9 to 11. In the normal scale mixture of Theorem 10.1.3 it bounds the negative part of the candidate EGGC density v_α/π; in Wiener's theorem (Proposition 10.1.10) it bounds the proportion of frequencies at which the Fourier transform of the tilted Pólya kernel can vanish; and in Riesz's criterion (Section 10.1.8) it bounds the number of Mellin poles to the right of $\alpha/2$.

The pole factors can be written down. Let U_α° denote the signed measure on $\theta > 0$ with density v_α/π and a unit atom at γ for every zero $\beta + i\gamma$, $\gamma > 0$, with $\beta = \alpha$ (the term P_0 of (11.4.1)), counted with multiplicity.

THEOREM 11.4.4 (Signed Thorin representation). *For every $\alpha \ge \frac12$ and $u \ge 0$,*

$$\varphi_\alpha(u) = \exp\Big\{ -\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big) U_\alpha^\circ(d\theta)\Big\} \prod_{\beta>\alpha,\ \gamma>0} \Big|1-\frac{u}{(\rho-\alpha)^2}\Big|^{-2},$$

the integral and the product converging absolutely. For a zero to the right of α, with $p=(\rho-\alpha)^2 = -c+id$, $c=\gamma^2-(\beta-\alpha)^2>0$ and $d=2\gamma(\beta-\alpha)>0$, the factor is

$$\Big|1-\frac{u}{p}\Big|^{-2} = \frac{|p|^2}{(u+c)^2+d^2} = |p|^2\int_0^\infty e^{-ut}\,\frac{e^{-ct}\sin dt}{d}\,dt,$$

the Laplace transform of a damped sinusoid, which changes sign, so the factor is not completely monotone.

Proof. Write $f_w(u) = (w^2+\gamma^2)/((w+\sqrt{u})^2+\gamma^2)$. By the Hadamard product, as in the proof of Proposition 11.4.3, $\varphi_\alpha(u) = \prod_{\gamma>0} f_{\alpha-\beta}(u)$ over all zeros with $\gamma>0$, absolutely convergent. For $w\ge 0$ that proof gives $f_w(u) = \exp\{-\int_0^\infty \log(1+u/\theta^2)[P_w(\theta-\gamma)+P_w(\theta+\gamma)]\,d\theta\}$, with the atom at γ for $w=0$. For $w=-\omega<0$ the kernels carry the sign $P_{-\omega}=-P_\omega$, so the same integral equals f_ω^{-1}, and a direct computation gives $f_{-\omega} = f_\omega^{-1}\,|p|^2/|p-u|^2$ with $p=(\omega+i\gamma)^2$, since $|p-u|^2 = ((\sqrt{u}-\omega)^2+\gamma^2)((\sqrt{u}+\omega)^2+\gamma^2)$. Multiplying over the zeros gives the formula, once the integrals are summed under the integral sign: the integral of each pair of kernels against $\log(1+u/\theta^2)$ is $\pm\log f_{|w|}$, which is $O_\alpha((u+\sqrt{u})\gamma^{-2})$, and $\sum\gamma^{-2}<\infty$. Finally $|p-u|^2=(u+c)^2+d^2$, and $c>0$ because $\beta-\alpha<\frac12<\gamma_1$. □

The theorem needs no hypothesis on the zeros: the atoms carry the zeros on the line $\operatorname{Re} s=\alpha$, which is why the case $\alpha=\frac12$ is included. It is the identity behind Proposition 11.4.3, which combines the pole factors with the negative kernels they cancel into D_α. At $\alpha=\frac12$ every term is explicit.

COROLLARY 11.4.5 (The critical line). *Unconditionally, $U_{1/2}^\circ = \sum_{\beta=1/2,\ \gamma>0}\delta_\gamma \ge 0$ and*

$$\varphi_{1/2}(u) = \prod_{\beta=1/2,\ \gamma>0} \frac{\gamma^2}{\gamma^2+u} \prod_{\beta>1/2,\ \gamma>0} \Big|1-\frac{u}{(\rho-\frac12)^2}\Big|^{-2}.$$

Proof. A zero ρ with $\beta=\frac12+\delta$, $\delta>0$, and its partner $1-\bar\rho$ have the same ordinate, and their kernels $P_{-\delta}+P_\delta$ cancel, so $v_{1/2}=0$ away from the atoms (the term by term form in the proof of Proposition 1.2.3, which holds at $\alpha=\frac12$ away from the ordinates). The atoms come from the zeros with $\beta=\frac12$, each with the factor $f_0=\gamma^2/(\gamma^2+u)$; apply Theorem 11.4.4. □

So on the critical line positivity of $U_{1/2}^\circ$ costs nothing, and RH is the statement that the product of damped sinusoids is empty. For $\frac12<\alpha<\sup_\rho\operatorname{Re}\rho$ the negative kernels of Proposition 11.4.3 are present, and whether they can be

outweighed on the single line $\operatorname{Re} s = \alpha$ is open, except near a zero, where they cannot (Remark 1.2.4).

THEOREM 11.4.6 (Explicit bounds from verification). *Let $T_0 = 3 \times 10^{12}$, below which all zeros lie on the critical line [247], and let $\Sigma_0 = \sum_{\gamma > T_0} \gamma^{-2}$, the sum over all zeros above T_0, counted with multiplicity. By the Riemann von Mangoldt formula with the explicit error term of Hasanalizade, Shen and Wong [109], recalled in the proof of Proposition 9.1.11, Σ_0 differs from $\frac{\log(T_0/2\pi)+1}{2\pi T_0} = 1.4797 \times 10^{-12}$ by less than 2×10^{-24}, so $\Sigma_0 < 1.48 \times 10^{-12}$. Then for every $\alpha > \frac{1}{2}$ and every $u \geq 0$:*

1. *$|\log D_\alpha(u)| \leq (u + \sqrt{u})\,\Sigma_0$, so that*

$$e^{-1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u) \leq \mathbb{E}e^{-uT_\alpha} \leq e^{1.5\times 10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u);$$

2. *the negative part of the density satisfies $v_\alpha^-(\theta)/\pi \leq 2\Sigma_0/\pi < 9.43 \times 10^{-13}$ for $0 \leq \theta \leq T_0/2$;*

3. *consequently $v_\alpha(\theta) > 0$ for all $0 \leq \theta \leq T_0/2$ whenever $\alpha - \frac{1}{2} > 6 \times 10^{-10}$.*

Proof. Every zero to the right of α has $\gamma > T_0$ and $\beta > \frac{1}{2}$. The zeros above T_0 off the critical line come in pairs $\beta + i\gamma$, $1 - \beta + i\gamma$ of equal multiplicity, by $\xi(s) = \xi(1-s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$, and at most one member of each pair lies to the right of α, so $\sum_{\beta>\alpha,\ \gamma>0} \gamma^{-2} \leq \Sigma_0/2$. (1) For a zero to the right of α put $w = \beta - \alpha \in (0, \frac{1}{2})$ and $x = (u - 2w\sqrt{u})/(w^2+\gamma^2)$, so that its factor in D_α is $1/(1+x)$. Since $u - 2w\sqrt{u} \geq -w^2$, we have $x \geq -w^2/\gamma^2 > -\frac{1}{2}$, and $|x| \leq (u + \sqrt{u})/\gamma^2$; for $x \geq -\frac{1}{2}$, $|\log(1+x)| \leq 2|x|$. Hence the logarithm of its factor is at most $2(u + \sqrt{u})/\gamma^2$ in absolute value, and summing gives $|\log D_\alpha(u)| \leq (u + \sqrt{u})\Sigma_0$; the inequalities for $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha^+(u) D_\alpha(u)$ follow. (2) For $\theta \leq T_0/2$ and $\gamma > T_0$, $P_w(\theta - \gamma) + P_w(\theta + \gamma) \leq \frac{2w}{\pi(\gamma-\theta)^2} \leq \frac{4}{\pi\gamma^2}$, and summing over the zeros to the right of α gives at most $\frac{4}{\pi} \cdot \frac{\Sigma_0}{2}$. (3) Write $a = \alpha - \frac{1}{2}$. If $a \geq \frac{1}{2}$, then $\alpha \geq 1$, no zero lies to the right of α, and $v_\alpha = v_\alpha^+ > 0$; so let $6 \times 10^{-10} < a < \frac{1}{2}$. The zeros below T_0 lie on the line and contribute to v_α^+/π the kernels $P_a(\theta - \gamma)$, and all other kernels of v_α^+ are positive, so $v_\alpha^+(\theta)/\pi \geq \frac{a}{\pi(a^2+\gamma_1^2)}$ at every θ within γ_1 of an ordinate below T_0. Every $\theta \in [0, T_0/2]$ is such a point. On $[0, \gamma_1]$ the ordinate γ_1 serves. On $[\gamma_1, 2000]$ one of the first 1800 ordinates serves: they are the certified zeros of the proof of Proposition 9.1.11, they reach height 2304.3, and consecutive ones differ by less than $6.8874 < 2\gamma_1$. For $2000 \leq \theta \leq T_0/2$, the count of that proof, with $H = 14$ in place of 15, shows that the window $(\theta - 14, \theta + 14]$ holds at least $n(\theta) = \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - E(\theta-14) - E(\theta+14)$ zeros; while $\theta + 14 \leq X_0$, n increases and $n(2000) > 20$, and for $\theta + 14 > X_0$, $n(\theta) > \frac{14}{\pi}\log\frac{\theta-14}{2\pi} - 0.6\log(\theta+14) > 80$, so the window holds a zero, which lies within $14 < \gamma_1$ of θ and, since $\theta + 14 < T_0$, on the line. The script `explicit_bounds.py`, with output `explicit_bounds.json`,

recomputes the 1800 zeros and checks these inequalities in ball arithmetic. Finally, since $a < \frac{1}{2}$,

$$\frac{a}{\pi(a^2+\gamma_1^2)} > \frac{a}{\pi(\frac{1}{4}+\gamma_1^2)} > \frac{6\times 10^{-10}}{\pi(\frac{1}{4}+\gamma_1^2)} > 9.5\times 10^{-13},$$

which is more than the bound $2\Sigma_0/\pi < 9.43\times 10^{-13}$ of (2) on v_α^-/π, so $v_\alpha = v_\alpha^+ - v_\alpha^- > 0$ on $[0, T_0/2]$. □

Theorem 11.4.6 restates the verification of RH to height 3×10^{12} in Thorin form. Part (3) of Theorem 10.1.3 is the same statement for the characteristic function of the normal scale mixture, with a remainder of relative size $\log T_0/T_0$, and the paragraph after its proof bounds the negative part of v_α/π for $\theta \le 10^9$ in the same way as part (2).

Part (3) of Theorem 11.4.6 treats every zero above T_0 as if its kernel could be negative wherever the density is evaluated. As the first author's note on the Dirichlet beta function observes [253], grouping each such zero with its reflection in the critical line does better, because the sum of the two kernels of an off line pair is negative only near its ordinate.

Corollary 11.4.7 (Positivity on the verified window). *Let* $T_0 = 3\times 10^{12}$ *and* $V_\xi = (\log\xi)''(\frac{1}{2}) = 0.04620998623$.

1. *For every* $\alpha > \frac{1}{2}$, $v_\alpha(\theta) > 0$ *for* $|\theta| \le T_0 - \frac{1}{2}$.

2. *For every* $\alpha \in (\frac{1}{2}, 7.35]$, $v_\alpha(\theta) > v_\alpha(0)$ *for* $0 < |\theta| \le T_0 - \frac{1+\sqrt{2}}{2}$, *and* $0 < v_\alpha(0) \le (\alpha - \frac{1}{2})V_\xi$.

Proof. (1) Put $\varepsilon = \alpha - \frac{1}{2}$. By the Hadamard product, as in Proposition 1.2.3, $v_\alpha(\theta)$ is the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$, counted with multiplicity, of $(\alpha-\beta)/((\alpha-\beta)^2+(\theta-\gamma)^2)$. A zero on the critical line contributes $\pi P_\varepsilon(\theta-\gamma) > 0$. The zeros off the line come in pairs $\frac{1}{2}\pm\delta+i\gamma$ with $0<\delta<\frac{1}{2}$ and equal multiplicities, and a pair contributes $\pi(P_{\varepsilon+\delta}+P_{\varepsilon-\delta})(\theta-\gamma)$, which is nonnegative if $\delta \le \varepsilon$ and, if $\delta > \varepsilon$, negative exactly where $|\theta-\gamma| < \sqrt{\delta^2-\varepsilon^2}$, by the proof of Proposition 3.3.1(ii); and $\sqrt{\delta^2-\varepsilon^2} < \delta < \frac{1}{2}$. Every zero off the line has $|\gamma| > T_0$ [247], so for $|\theta| \le T_0 - \frac{1}{2}$ every pair has $|\theta-\gamma| > \frac{1}{2}$ and contributes a nonnegative term, while the zero $\frac{1}{2}+i\gamma_1$ contributes a positive one.

(2) This is Theorem 9.1.18(i) and (ii) for ξ, with $\alpha_{\max} = 7.35$ and $H_0 = T_0$ (Table 9.4). □

Remark 11.4.8. The corollary does not touch the defect of part (1) of Theorem 11.4.6: a zero above T_0 to the right of the line $\operatorname{Re} s = \alpha$ stops T_α from being a GGC whatever v_α does on the window (Proposition 11.4.3), and near its own ordinate it can make v_α negative (Section 11.4.3).

The proof of part (1) gives, with no hypothesis, $|\log D_\alpha(u)| \le 2(u + \sqrt{u})\sum_{\beta>\alpha,\ \gamma>0}\gamma^{-2}$. With the verification height and the trivial count the sum is at most $\Sigma_0/2$, with $\Sigma_0 < 1.48\times10^{-12}$. The zero density estimates of Section 11.4.1 [124, 104, 119] improve this to $\sum_{\beta>\alpha}\gamma^{-2} \le \int_{T_0}^{\infty} t^{-2}dN(\alpha,t) \ll T_0^{e(\alpha)-2+o(1)}$, smaller by a further power of T_0 since $e(\alpha) < 1$.

11.4.3 Where negativity can come from: a cancellation lemma

The counting bound treats every zero to the right of α as if its kernel were entirely negative. Pointwise, a negative kernel is cancelled wherever the rest of the density exceeds it, and the kernel of width w has height at most $1/(\pi w)$.

Lemma 11.4.9 (Cancellation). *Let $\rho = \beta + i\gamma$ be a zero with $\beta > \alpha$, and let $m > 0$ be a lower bound on an interval I for the rest of the density, $v_\alpha/\pi + P_{\beta-\alpha}(\cdot - \gamma)$. If $\beta - \alpha \ge 1/(\pi m)$, the kernel of ρ cannot make v_α negative on I.*

Proof. On I, $v_\alpha/\pi \ge m - \max_x P_{\beta-\alpha}(x) = m - 1/(\pi(\beta-\alpha)) \ge 0$. □

The lemma has a heuristic consequence. At height γ the density v_α/π averages $\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, the density of the ordinates, and when $\alpha - \frac12$ is large compared with the mean spacing $2\pi/\log\gamma$ it stays close to that average. Taking this average for m in the lemma, a zero to the right of α can then produce negativity only if

$$\beta - \alpha < \frac{2\pi}{\pi\log(\gamma/2\pi)} = \frac{2}{\log(\gamma/2\pi)}.$$

Heuristically, then, positivity of the Thorin density at the basepoint α can be broken at height γ only by zeros in the thin strip $\alpha < \beta < \alpha + 2/\log(\gamma/2\pi)$. The local visibility condition of Section 5.2.2, which concerns small resolutions, has the same scale $1/\log\gamma$. The heuristic proves nothing, because a rigorous lower bound for the background near an arbitrary height would itself control ζ'/ζ in the strip. It suggests combining zero density estimates for thin strips with lower bounds for $\operatorname{Re}\xi'/\xi$ on average.

We tested the lemma at the heights 300 and 10^5 with `high.py`. The script computes the background $v_{0.6}/\pi$ from ξ'/ξ on a window of width 8 and adds the kernels of a single hypothetical pair of zeros at $\beta + i\gamma_0$ and $1 - \beta + i\gamma_0$, with γ_0 the centre of the window (Figure 11.8, middle and right). At height 300 the background has mean 0.51 and minimum 0.069; at height 10^5 it has mean 1.59, close to the density of ordinates $\frac{1}{2\pi}\log\frac{10^5}{2\pi} = 1.54$, and minimum 0.34. Table 11.13 shows the result. At both heights zeros with $\beta - \alpha \le 0.2$ produce negativity and zeros with $\beta - \alpha \ge 0.3$ are absorbed. At height 10^5 the heuristic threshold $2/\log(10^5/2\pi) = 0.21$ is close to the observed one, while the rigorous guarantee of Lemma 11.4.9 with the window minimum, $1/(\pi m) = 0.93$, is conservative. For a fixed width the negative dip is deeper at the greater height, because the kernel is

the same while the local minimum of the background near the hypothetical zero differs.

	height 300		height 10^5	
β	min of $v_{0.6}/\pi$	negative mass	min of $v_{0.6}/\pi$	negative mass
0.62	-13.54	0.521	-13.97	0.567
0.65	-4.16	0.299	-4.59	0.365
0.70	-1.24	0.121	-1.62	0.193
0.75	-0.41	0.041	-0.72	0.097
0.80	-0.10	0.008	-0.32	0.041
0.90	0.072	0	0.023	0
0.95	0.075	0	0.100	0

Table 11.13: A single hypothetical pair of zeros at real parts β and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from ξ'/ξ on a window of width 8 at the heights 300 and 10^5: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed by `high.py`.

11.4.4 Hypothetical off line zeros in the computed density

No off line zero is known, so the negative part cannot be observed. To see the mechanism, `almost.py` inserts fifteen hypothetical quadruples of zeros $0.75 \pm i\gamma_j$, $0.25 \pm i\gamma_j$, at the heights $\gamma_j = 20\, j^{5/3}$ below 2000, into the Poisson form of Proposition 1.2.3 together with the first 1700 ordinates, taken on the line; the zeros above height 2198 are omitted. The count of the hypothetical zeros grows like $T^{0.6}$, as a zero density estimate with $e = 0.6$ would allow. Table 11.14 compares the negative mass $\int_0^T \max(-v_\alpha/\pi, 0)\, d\theta$ of the resulting density with its positive mass and with the counting bound $N_{\text{off}}(2T)$ of Lemma 11.4.1, the number of hypothetical zeros with ordinate at most $2T$.

α	T	negative mass	positive mass	fraction	counting bound $N_{\text{off}}(2T)$
0.6	250	0.842	108.8	7.7×10^{-3}	6
0.6	1000	1.655	650.5	2.5×10^{-3}	15
0.6	2000	2.278	1519.2	1.5×10^{-3}	15
0.7	250	2.130	110.1	1.9×10^{-2}	6
0.7	1000	4.815	653.6	7.4×10^{-3}	15
0.7	2000	6.938	1523.8	4.6×10^{-3}	15
0.8	any	0		0	0

Table 11.14: Negative and positive mass of the density v_α/π on $[0, T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed by `almost.py`.

Three features are visible. The negative mass is well below the counting bound, because each negative kernel is partly cancelled by its partner and by the positive

kernels of the neighbouring zeros on the line: at $\alpha = 0.6$ the negative kernel has width 0.15 and height $1/(0.15\pi) = 2.1$, and most of it is cancelled, while at $\alpha = 0.7$ the width is 0.05 and the kernel, of height 6.4, appears as a sharp negative spike that takes the density to -5.39 at the height 292.40 (Figure 11.9; the minimum is computed by `almost_dip.py`). The fraction decreases with T at the rate the counting predicts. And at $\alpha = 0.8$, to the right of the hypothetical zeros, the density is positive everywhere, as Theorem 1.2.2 requires: a zero with real part β contributes a positive kernel at every basepoint $\alpha > \beta$.

A zero is most visible from basepoints just to its left. The left panel of Figure 11.8, computed by `more2.py`, shows the negative mass on $[0, 2000]$ as a function of α for the same fifteen heights and three real parts $\beta = 0.6$, 0.75 and 0.9. As α increases towards β the negative kernels narrow, cancellation fails, and the negative mass climbs towards the number of hypothetical zeros, reaching 7.55 at $\alpha = 0.85$ for $\beta = 0.9$. At $\alpha = \beta$ the zeros lie on the line of evaluation and the negative mass drops to zero. This is the Thorin counterpart of the fact that the translates of the tilted Pólya kernel fail to be complete exactly on the line through the zero (Proposition 10.1.10).

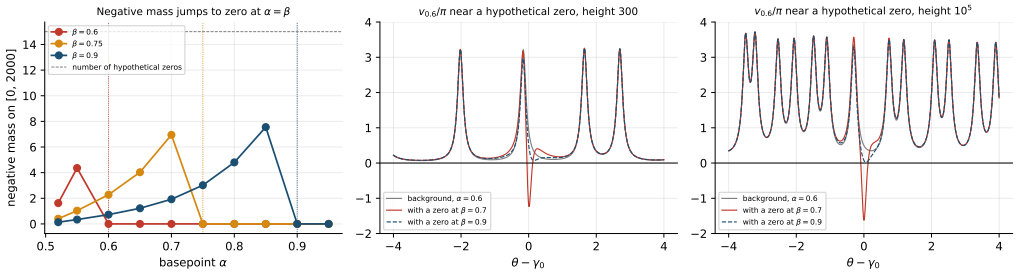


Figure 11.8: Left: negative mass of the density on $[0, 2000]$ against the basepoint α for fifteen hypothetical zeros at real part β (`more2.py`). Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and 10^5, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window (`high.py`). Drawn by `figs_almost.py`.

11.4.5 Positivity margins and the circular unitary ensemble

By Lemma 11.4.9, a zero to the right of α at height γ can make the density negative only if its distance to the line $\operatorname{Re} s = \alpha$ is below $1/(\pi m)$, where m is the local minimum of the rest of the density. The distribution of the local minima, the positivity margins, therefore determines how much room RH has at each height. The minima occur between consecutive ordinates. After unfolding the ordinates to unit mean spacing, $x_n = \vartheta(\gamma_n)/\pi + 1$ with ϑ the Riemann Siegel theta function,

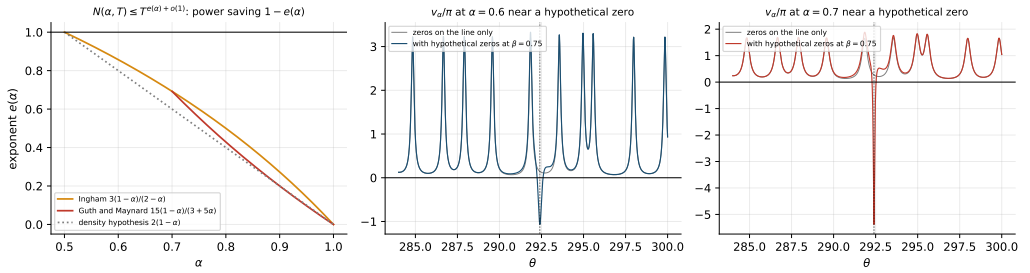


Figure 11.9: Left: the zero density exponents of Theorem 11.4.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [119]. Middle and right: the density v_α/π at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in `almost.py` with step 0.005. Drawn by `figs_almost.py`.

as in Section 5.4.2, the density at the resolution of κ mean spacings is

$$\bar{\rho}_\kappa(x) = \sum_n P_\kappa(x - x_n) = \sum_n \frac{\kappa}{\pi(\kappa^2 + (x - x_n)^2)},$$

whose mean is one.

The script `sens.py` evaluates $\bar{\rho}_\kappa$ at the 1600 midpoints between consecutive unfolded ordinates below height 2198, leaving out 50 mean spacings at each end, and the same statistic at about 3600 midpoints between the eigenvalues of twelve Haar distributed unitary matrices of size 400, unfolded and extended periodically. The circular unitary ensemble (CUE) of these matrices models the local statistics of the zeros. Table 11.15 shows the result. The medians agree to 1% at every resolution. In the lower tail the zeros have slightly larger margins than the CUE, a finite height effect in the expected direction, since the low zeros are more rigid than the limit of large matrices.

κ	zeta, 0.1%	CUE, 0.1%	zeta, 1%	CUE, 1%	zeta, median	CUE, median
0.1	0.1075	0.0919	0.1177	0.1095	0.3349	0.3343
0.25	0.2617	0.2178	0.2849	0.2659	0.7116	0.7092
0.5	0.4803	0.4047	0.5150	0.4645	0.9739	0.9796
1	0.7477	0.6552	0.7730	0.7313	1.0264	1.0240

Table 11.15: Quantiles of the normalised Thorin density $\bar{\rho}_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed by `sens.py`.

The next step is to repeat the comparison with Odlyzko's tables of zeros near the 10^{12}th zero [219] and near the 10^{21}st and 10^{22}nd zeros [223, 224], where the

finite height corrections are smaller, and to fit the lower tail of the margin distribution. The CUE lower tail is governed by large gaps, whose probability decays like a Gaussian in the gap length, so the smallest margin among the first N midpoints should decrease only like a power of $\sqrt{\log N}$. Converted through the cancellation lemma, this predicts the width, as a function of the height, of the strip to the right of each basepoint in which an off line zero could hide without being cancelled. That is the quantitative input that question (1) of Section 11.4.7 needs.

11.4.6 Moments of zeta, the Barnes function and zero density

The strength of Theorem 11.4.2 is set by the zero density exponents, which rest on mean value estimates on the critical line, Ingham's on the fourth moment of ζ. Keating and Snaith [141] modelled $\zeta(\frac{1}{2}+it)$ by the characteristic polynomial $Z_N(\theta)=\det(I-Ae^{-i\theta})$ of a Haar distributed unitary matrix A of size N, a matrix of the circular unitary ensemble of Section 11.4.5, with N in the role of $\log(T/2\pi)$. By analogy with the exact formula

$$\mathbb{E}|Z_N(\theta)|^{2k}=\prod_{j=1}^{N}\frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2}\sim\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\,N^{k^2},\qquad N\to\infty,$$

they conjectured that the moments of ζ grow in the same way, with $\log(T/2\pi)$ in place of N and an arithmetic factor a_k in front (Section 11.4.6.1). Here G is the Barnes G function, $\mathsf{G}(1+z)=\Gamma(z)\mathsf{G}(z)$ with $\mathsf{G}(1)=1$, written in sans serif to keep G for the clock of Section 1.2. Nikeghbali and Yor [215] gave the random matrix factor a probabilistic meaning in GGC language: the Mellin transform of $|Z_N|$ and the Barnes function are expressed through products and sums of independent gamma, beta and log gamma variables, and the reciprocal of G has a Lévy Khintchine type representation. The source of these representations is the decomposition of Bourgade, Hughes, Nikeghbali and Yor [44],

$$Z_N\overset{d}{=}\prod_{j=1}^{N}\Big(1+e^{i\omega_j}\sqrt{B_{1,j-1}}\Big),$$

with independent angles ω_j uniform on $[0,2\pi)$ and independent beta variables $B_{1,j-1}$ with parameters 1 and $j-1$, where $B_{1,0}=1$.

11.4.6.1 The Keating and Snaith moments

Write $I_k(T)=\frac{1}{T}\int_0^T|\zeta(\frac{1}{2}+it)|^{2k}dt$. The conjecture of Keating and Snaith [141], in the form of Conrey, Farmer, Keating, Rubinstein and Snaith [63], is

$$I_k(T)\sim a_k\,\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\Big(\log\frac{T}{2\pi}\Big)^{k^2},\qquad a_k=\prod_p\Big(1-\frac{1}{p}\Big)^{k^2}\sum_{m\ge0}\Big(\frac{\Gamma(m+k)}{m!\,\Gamma(k)}\Big)^2p^{-m},$$

the product of an arithmetic factor from the primes and the random matrix factor. It is a theorem for $k = 1$ and $k = 2$ and open for $k \geq 3$. For $k = 1$ Hardy and Littlewood [106] proved the leading term, and Ingham [122] the mean square with its secondary term, $I_1(T) = \log \frac{T}{2\pi} + 2\gamma_E - 1 + o(1)$; for $k = 2$ Ingham [122] proved the asymptotic with the leading coefficient $\frac{1}{2\pi^2}$. The local factor of a_k is $(1 - \frac{1}{p})^{k^2} {}_2F_1(k, k; 1; \frac{1}{p})$. Since ${}_2F_1(1, 1; 1; x) = (1 - x)^{-1}$ and ${}_2F_1(2, 2; 1; x) = (1 + x)(1 - x)^{-3}$, it equals 1 for $k = 1$ and $1 - p^{-2}$ for $k = 2$, so that $a_1 = 1$ and $a_2 = \prod_p (1 - p^{-2}) = 6/\pi^2$ exactly. The Euler product over the primes below 2×10^5 gives $a_3 = 0.0493218$ (`repro/ks_ak.py`); since the local factor for $k = 3$ is $1 - 9p^{-2} + O(p^{-3})$, the omitted primes lower it by a relative $9 \sum_{p > 2\times 10^5} p^{-2} \approx 4 \times 10^{-6}$, to about 0.0493217. With $\mathsf{G}(3)^2/\mathsf{G}(5) = \frac{1}{12}$ and $\mathsf{G}(4)^2/\mathsf{G}(7) = \frac{1}{8640}$ the leading coefficients are 1, $0.0506606 = \frac{1}{2\pi^2}$, which is Ingham's constant, and 5.709×10^{-6}, the value $42a_3/9!$ of [63].

Table 11.16, computed by `ks.py`, compares the moments with the leading terms. For $k = 1$ the full asymptotic is already accurate at $T = 1000$. For $k = 2$ and $k = 3$ the leading term is a small part of the moment at these heights, because the lower order terms of the polynomial in $\log T$ predicted in [63] dominate.

T	$I_1(T)$	$\log \frac{T}{2\pi} + 2\gamma_E - 1$	$I_2(T)$	leading, $k = 2$	$I_3(T)$	leading, $k = 3$
125	3.062	3.145	28.0	4.05	396	0.11
250	3.869	3.838	53.1	9.33	1136	0.71
500	4.550	4.531	84.8	18.6	2632	3.36
1000	5.211	5.224	127.2	33.5	5309	12.6

Table 11.16: Moments of ζ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k = 1$ and the leading Keating and Snaith terms for $k = 2$ and 3. Computed by `ks.py`.

11.4.6.2 The moments as exponential moments of the Thorin density

The densities v_σ are the derivatives of $\log|\xi|$ across the basepoints: $\frac{\partial}{\partial\sigma} \log|\xi(\sigma + it)| = \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it) = v_\sigma(t)$ wherever $\xi(\sigma + it) \neq 0$. Integrating from the critical line to the basepoint one, for every t such that ξ has no zero on the segment from $\frac{1}{2} + it$ to $1 + it$, which excludes only countably many t,

$$\log|\xi(\tfrac{1}{2} + it)| = \log|\xi(1 + it)| - \int_{1/2}^{1} v_a(t)\, da. \tag{11.4.2}$$

The moments concern ζ, not ξ. With the gamma factor $h(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, so that $\xi = h\zeta$, the identity becomes

$$|\zeta(\tfrac{1}{2} + it)|^{2k} = |\zeta(1 + it)|^{2k} \Big|\frac{h(1 + it)}{h(\frac{1}{2} + it)}\Big|^{2k} \exp\Big\{ - 2k \int_{1/2}^{1} v_a(t)\, da\Big\},$$

and Stirling's formula gives $|h(1+it)/h(\frac{1}{2}+it)| \sim (t/2\pi)^{1/4}$ as $t \to \infty$. So the Keating and Snaith moments are exponential moments, over t uniform on $[0,T]$, of the density $v_a(t)$ integrated across the basepoints from $\frac{1}{2}$ to 1, with the weight $|\zeta(1+it)|^{2k}|h(1+it)/h(\frac{1}{2}+it)|^{2k}$, which involves no zero of ζ. In the same terms, RH, which is the positivity of v_a for every $a > \frac{1}{2}$ (Lagarias's criterion, Remark 1.2.4), is the statement that $|\xi(\sigma+it)|$ increases with $\sigma \geq \frac{1}{2}$ for every t, a reformulation due to Sondow and Dumitrescu [308].

In the language of (11.4.2) the conjecture of Keating and Snaith is a large deviation statement for the integrated Thorin density. Selberg's central limit theorem [303] says that for t uniform on $[T,2T]$ the variable $\log|\zeta(\frac{1}{2}+it)|$ is approximately Gaussian with variance $\frac{1}{2}\log\log T$. Gaussian exponential moments would give $(\log T)^{k^2}$, and the constant $a_k\mathsf{G}(1+k)^2/\mathsf{G}(1+2k)$ is the correction to them. On $[500,1000]$ the sample variance of $\log|\zeta(\frac{1}{2}+it)|$ is 1.53, against $\frac{1}{2}\log\log 750 = 0.95$. The difference is a constant term, not slow convergence: the random matrix model predicts the variance $\frac{1}{2}\log\log\frac{t}{2\pi} + c + o(1)$ with $c = \frac{1}{2}\big(\gamma_E + 1 - \sum_p \sum_{m\geq 2}(\frac{1}{m} - \frac{1}{m^2})p^{-m}\big) = 0.70$ [141], a conjecture, and this gives 1.48 on $[500,1000]$, close to the sample value.

11.4.6.3 What the moments give

Ingham's exponent $3(1-\sigma)/(2-\sigma)$, which gives Theorem 11.4.2 its saving $(2\alpha-1)/(2-\alpha)$ for $\frac{1}{2} < \alpha < \frac{7}{10}$, rests on the fourth moment, $k=2$, proved by Ingham [122]. Bounds for all moments, that is the Lindelöf hypothesis in the form $\int_0^T |\zeta(\frac{1}{2}+it)|^{2k}dt \ll T^{1+\varepsilon}$ for every k, imply the density hypothesis $N(\sigma,T) \ll T^{2(1-\sigma)+\varepsilon}$, which would give the saving $2\alpha-1$, larger than that of Theorem 11.4.2 for $\frac{1}{2} < \alpha < \frac{9}{10}$. The Keating and Snaith asymptotics would give this and much more.

The GGC structure of Nikeghbali and Yor belongs to the random matrix model, not to ζ: it explains the conjecture without proving it, and only the second and fourth moments of ζ are known asymptotically. And moments, even the Lindelöf hypothesis, bound the number of zeros off the line but do not exclude them.

11.4.7 From almost to exactly

Theorem 11.4.2 cannot be upgraded to exact positivity by improving the exponents, or by the moment bounds of Section 11.4.6. Even the density hypothesis, or the Lindelöf hypothesis which implies it, leaves room for a sparse set of zeros off the line, and a single off line zero $\beta+i\gamma$ stops T_α from being a GGC for every α in $[\frac{1}{2},\beta)$ (Theorem 1.2.2) and makes v_α negative near γ for every α close enough to β (Section 11.4.3). The step from "almost" to "exactly" is a step from counting zeros to excluding them.

Four facts about the programme say that this step needs input from outside it. The smoothing runs the wrong way: when no zero lies to the right of α the density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the density at α by the kernel of width $\alpha' - \alpha$, and positivity propagates to the right; RH asks to propagate it to the left, which is deconvolution, and deconvolution does not preserve positivity. Arithmetic domination requires a zero free region: below the basepoint one the compensated Lévy form, which the sine squared lemma converts into the density, holds if $\psi(x) - x = O(x^{\alpha'})$ for some $\alpha' < \alpha$, and only if $\psi(x) - x = O(x^{\alpha+\delta})$ for every $\delta > 0$, where ψ is Chebyshev's function (Proposition 9.1.13). Either condition is a zero free half plane, none was known for any $\alpha < 1$ before the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026, which came from outside the programme (Section 12.1.2), and removing the pole does not help: for $L(s, \chi_{12})$ the prime sums need no compensation, and the boundary is the same (Section 11.3.2). There is no margin: along the de Bruijn Newman flow the reciprocal is a GGC for $t \geq 0.2$ [247] (Theorem 7.2.15 and Section 8.4), RH is the statement at $t = 0$, and $\Lambda \geq 0$ [289], so a proof must establish a closed condition at the boundary itself. And stronger positivity conditions can be false: Conrey and Li showed that a positivity condition in de Branges' approach to RH fails [62], so any positivity statement stronger than RH should be tested numerically before it is pursued (Section 12.2.4).

In the Stieltjes form (iv) of Theorem 4.2.4 the difference between counting zeros and excluding them is explicit. Split $G'/G = S_{\mathrm{on}} + R_{\mathrm{off}}$, where $S_{\mathrm{on}}(u) = \sum (u + \gamma^2)^{-1}$ runs over the zeros $\frac{1}{2} + i\gamma$ on the line with $\gamma > 0$ and is a Stieltjes function with measure U_{line}, and $R_{\mathrm{off}}(u) = \sum (u - (\rho - \frac{1}{2})^2)^{-1}$ runs over the zeros $\rho = \beta + i\gamma$ off the line with $\gamma > 0$. By Theorem 4.3.9, all the poles of R_{off} lie in a parabolic sliver $\mathcal{E}_{T_0}$ beside the negative axis, which begins at $-T_0^2 + T_0 + \frac{1}{4}$, and in the upper half plane outside it $\operatorname{Im} R_{\mathrm{off}} \leq 0$ and $\operatorname{Im}(G'/G) < 0$. The size of R_{off} away from the sliver needs no count of zeros: if $|u| \leq T_0^2/2$, a disc that misses $\mathcal{E}_{T_0}$, then every term has $|u - (\rho - \frac{1}{2})^2| \geq |\rho - \frac{1}{2}|^2 - |u| > \gamma^2/2$, since $\gamma > T_0$, so $|R_{\mathrm{off}}(u)| \leq 2\Sigma_0 < 3 \times 10^{-12}$ by Theorem 11.4.6.

The number of terms of R_{off}, on the other hand, is not known to be small. Zero density estimates bound the number of those with $|\beta - \frac{1}{2}| \geq \sigma - \frac{1}{2}$ and $\gamma \leq T$ by $T^{e(\sigma)+o(1)}$ with $e(\sigma) < 1$, for each fixed $\sigma > \frac{1}{2}$ (Theorem 11.4.2), which is $o(T)$; but zeros off the line may approach the line as γ grows, and for all of them together only a proportion is known. At most $(1 - \kappa + o(1))N(T)$ of the zeros up to height T lie off the line, with $\kappa > \frac{5}{12}$ in the last published record [283] and $\kappa > 0.6725$, counting only simple zeros on the line, in two preprints of 2026 [2, 164]; that is not $o(T)$, nor even $o(N(T))$. No count, however strong, touches the Thorin condition, which requires $R_{\mathrm{off}} \equiv 0$, because a single pair puts a pole of G'/G off the cut. Section 11.5.5 gives a test for arguments that claim to exclude such poles: since the de Bruijn Newman flow has nonreal zeros at every $t < 0$ [289], no property that the kernel $e^{tv^2/4}\Phi(v)$ keeps for some $t < 0$ can exclude them, so a proof must

use one that fails for every $t < 0$, and the Euler product and the explicit formula are the natural candidates.

The forms that the new input could take are those of Section 12.1. The first new theorem any of them would give is the GGC property of T_α at a single basepoint $\alpha < \frac{7}{8}$, that is $v_\alpha^- \equiv 0$, a zero free half plane beyond the one announced in 2026 (Section 12.1.2); positivity of v_α on the single line $\operatorname{Re} s = \alpha$ is not known to suffice (Remark 1.2.4).

Sections 11.4.2 and 11.4.3 make the remaining problem concrete. The growth of the defect is not part of it. For $u \geq 1$ every factor of D_α is at most one, and a factor with $\gamma \leq \sqrt{u}$ is at least $\gamma^2/(u+\gamma^2) \geq \gamma^2/(2u) \geq 1/u$, since $\gamma \geq \gamma_1 > \sqrt{2}$. Bounding the factors with $\gamma > \sqrt{u}$ as in the proof of Theorem 11.4.6 therefore gives, for $u \geq 1$,

$$\begin{aligned} |\log D_\alpha(u)| &\leq N(\alpha, \sqrt{u}) \log u + 2(u+\sqrt{u}) \sum_{\beta > \alpha,\ \gamma > \sqrt{u}} \gamma^{-2} \\ &\leq u^{e(\alpha)/2 + o(1)}, \qquad u \to \infty, \end{aligned}$$

by the exponents of Theorem 11.4.2. So bounds on the growth of the defect follow from zero density estimates alone, and what remains open is whether it is identically one. Four questions would each be a genuine step.

1. Cancellation on average: show that for fixed $\alpha > \frac{1}{2}$ the background density v_α/π is bounded below by $c \log \gamma$ outside a set of heights of density zero. With Lemma 11.4.9 this would confine any failure of positivity to zeros in the strips $\alpha < \beta < \alpha + O(1/\log \gamma)$. Remark 9.1.17 proves the lower bound for v_α/π itself, but its exceptional set contains a neighbourhood of the ordinate of every zero to the right of α, which is where Lemma 11.4.9 would need it.

2. Thin strip density: bound the number of zeros with $\alpha < \beta < \alpha + 2/\log \gamma$ and $\gamma \leq T$ by $o(T^{1-\delta})$ for some $\delta > 0$, uniformly in α. Combined with (1) this would give a pointwise almost positivity theorem.

3. Explicit defect: bound $\log D_\alpha(u)$ explicitly, as described at the end of Section 11.4.2.

4. Positivity at one basepoint: prove $D_\alpha \equiv 1$, that is $v_\alpha^- \equiv 0$, for a single $\alpha < \frac{7}{8}$, which is a zero free half plane beyond the one announced in 2026 (Section 12.1.2).

What the Thorin framework contributes is a precise target, the GGC property of one explicit clock T_α at one basepoint below $\frac{7}{8}$, together with computational tools that test candidate arguments against it: certified densities on windows, positivity margins calibrated against random matrix theory, and the explicit defect. Chapter 12 takes stock of the whole path, beginning with Section 12.1.

11.5 Pakes, the Lambert law and the Thorin condition

Anthony Pakes has worked on the objects that surround the Thorin condition: size biasing and the classical moment problem, the Laplace transforms of positive infinitely divisible laws, the Lambert function, and ladders of self decomposable laws. This section follows the first author's account of that work and of what it gives for the clock [262], verified, corrected where needed, and written in the notation of the book. In Pakes's language the question of Section 1.3 is whether $\log G$ is a Thorin Bernstein function, that is, a Bernstein function whose Lévy measure has the form $k(t)\,dt/t$ with k completely monotone [299, Chapter 8], or equivalently the Laplace exponent of a GGC. By Theorem 4.2.4 this holds if and only if $(\log G)' = G'/G$ is a Stieltjes function, and if and only if RH holds.

What the section adds is a dictionary between Pakes's language and that of the book, a few unconditional facts about the clock, the solved Lambert problem as a template whose single failing step for the clock is RH itself, a list of routes that are closed, and one property, the GGC property of the variable of Biane, Pitman and Yor, that passes the filter of Section 11.5.5 wherever it has been tested, but fails for most of the real Dirichlet L functions tested there, so that it does not detect the Euler product.

11.5.1 Two strands of Pakes's work

The first strand is the classical moment problem. For a random variable $X \geq 0$ with $0 < \mathbb{E}X < \infty$, the size biased variable $\widehat{X}$, length biased in Pakes's word, has the law $x\,P(X \in dx)/\mathbb{E}X$. Pakes and Khattree [237] characterise laws by in law equations linking $\widehat{X}$ and X, such as $X \stackrel{d}{=} \mathsf{U}\widehat{X}$ with U uniform on $(0,1)$ and independent of $\widehat{X}$, which holds exactly for the exponential laws. Pakes [228] studies size biasing for the laws with the moments of the log normal law, and in [229] he proves converse forms of the Carleman and Krein criteria and introduces size biasing and the stationary excess operation of order r; Berg [26] showed, correcting a remark in [229], that a law can be determinate in the sense of Stieltjes while its size biased and stationary excess versions of order one are not. Pakes, Hung and Wu [236] give criteria for the determination of a law by its moments, and Pakes [230] describes the structure of the Stieltjes classes of moment equivalent laws.

Size biasing meets infinite divisibility in Steutel's theorem: X is infinitely divisible if and only if $\widehat{X} \stackrel{d}{=} X + A$ for some $A \geq 0$ independent of X, and the law of A is then unique [313], [297, Theorem 51.1], [315]; in this form the theorem is [9, Theorem 11.1], and Pakes takes it for positive laws from Pakes, Sapatinas and Fosam [238]. In Laplace transforms, with $\Psi(u) = -\log \mathbb{E}e^{-uX}$, it reads $\mathbb{E}e^{-u\widehat{X}} = \mathbb{E}e^{-uX}\,\Psi'(u)/\Psi'(0)$, so that A has the Laplace transform $\Psi'(u)/\Psi'(0)$. In [232] Pakes assumes the decomposition and asks when A or $\widehat{X}$ is again infinitely

divisible; he shows that size biasing up to the fourth order preserves infinite divisibility for the Lambert law and for a compound Poisson law, and conjectures that it does so at every order.

For the laws of this book the moment problem is determinate. The moment generating function $\xi(\frac{1}{2}+z)/\xi(\frac{1}{2})$ of the Pólya density is entire (1.2.4). The clock has $\mathbb{E}e^{sT} < \infty$ exactly for $s < \gamma_1^2 = 199.7904548\ldots$: every zero has $|\rho - \frac{1}{2}| \geq |\operatorname{Im}\rho| \geq \gamma_1$, so $1/G$ is analytic in the disc $|u| < \gamma_1^2$ and has a pole at $-\gamma_1^2$, and by Landau's theorem [340, Chapter II, Theorem 5b] the abscissa of convergence is $-\gamma_1^2$. Under RH the measure $\nu = \sum_k a_k^{-1}\delta_{1/a_k}$ of the Stieltjes moments m_n is supported in $[0, \gamma_1^{-2}]$, so its moment problem is a determinate Hausdorff problem (Section 5.1), while the Thorin measure U_{RH} itself has unbounded support and infinite mass.

The second strand concerns the Laplace transforms of positive infinitely divisible laws, and the Lambert function $\mathcal{W}$ (Section 11.5.3). Pakes [231] and, independently, Kalugin and Jeffrey [135] proved that $\mathcal{W}$ is a Bernstein function, from a formula for the derivatives of $\mathcal{W}$ that involves a sequence of polynomials [235, (8)]. So $e^{-\mathcal{W}(u)} = \mathcal{W}(u)/u$ is the Laplace transform of an infinitely divisible law, the Lambert law. In the same paper, by Pakes's own account [235, Section 2], a complex variable argument shows that $\mathcal{W}'$ is a Stieltjes function, so that $\mathcal{W}$ is a Thorin Bernstein function and the Lambert law is a GGC. Kalugin, Jeffrey, Corless and Borwein [137] obtained Stieltjes representations independently and showed that $\mathcal{W}$ is a complete Bernstein function; the Thorin representation of $\mathcal{W}$ is the theorem proved below. Pakes gave two further proofs that $\mathcal{W}$ is Thorin Bernstein: from Nuttall's integral for the coefficients $n^n/n!$ of $\mathcal{W}'$, which yields an explicit quantile function of the Thorin measure [233], and from continuous state branching processes, one of whose martingale limits has the cumulant function $\mathcal{W}$ [234], which also gives the self decomposability of the Lambert law.

In [235] Pakes identifies the background driving Lévy process of the Lambert law: it is compound Poisson with unit rate and jump density $\int ze^{-xz}\,U_{\mathcal{W}}(dz)$, where $U_{\mathcal{W}}$ is the Thorin measure of $\mathcal{W}$. With $T_{\mathcal{W}}$ a variable with the Lambert law and J a jump of the driving process, this gives the in law relations $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}}+J)$ and $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, with U uniform on $(0,1)$, $\mathcal{S}$ the stationary excess operator and the factors on the right independent; and if both relations hold for a variable X and some $J \geq 0$ of finite mean $\mu > 0$, then $X \stackrel{d}{=} \mu T_{\mathcal{W}}$ [235, Theorem 1] (Remark 11.5.5). He embeds the Lambert law in a family of GGCs whose Laplace transforms solve Lambert's trinomial equation [235, Theorem 6], which gives one more proof that $\mathcal{W}$ is Thorin Bernstein [235, Corollary 2], and he introduces BDLP ladders: from an infinitely divisible $X_0 \geq 0$ with $\mathbb{E}(\log(1+X_0))^n < \infty$ for every n, the law of X_{n+1} is the self decomposable law driven by the Lévy process of X_n [235, Section 5]. In his indexing X_1 is self decomposable; this is a unit shift from Urbanik's classes L_n used in Section 11.5.4, in which L_0 is the class of self decomposable laws [333, 296, 27]. Along a ladder he finds the size biased

decomposition $\widehat{X}_{n+1} \stackrel{d}{=} X_{n+1} + A_n$, with A_n of density proportional to the tail of the Lévy measure of X_n [235, (53)], and this is the form in which Section 11.5.2 applies Steutel's theorem to the clock.

11.5.2 The Thorin question in Pakes's form

The Thorin question asks whether $\log G$ is a Thorin Bernstein function, that is, whether $(\log G)' = G'/G$ is a Stieltjes function (Theorem 4.2.4); then the Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, with no drift (Proposition 1.2.1), and Stieltjes inversion (1.3.2) recovers it from the boundary values of G'/G on the cut, or, in integrated form, from those of $\log G$ (Section 1.3.1). With no hypothesis, $\log G$ is a Bernstein function with Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$ is the heat trace (Theorem 5.1.1); under RH, $W(t) = \int e^{-tz}\, U_{\mathrm{RH}}(dz)$, the counterpart for the clock of the formula that gives the canonical function of the Lambert law as the Laplace transform of the density of its Thorin measure (Corollary 11.5.4; [235, (12)]).

The Bernstein property is weaker than the Thorin Bernstein property, but it is not blind to the zeros off the line. For $\log G$ it says that $W(t) \geq 0$ for every $t > 0$, and that would exclude, for instance, a zero $\rho = \frac{1}{2} + \delta + i\gamma$ off the line below all the other zeros: if $\gamma^2 - \delta^2$ were smaller than $\operatorname{Re} a_k$ for every other a_k, which is γ_k^2 for a zero on the line, the four zeros ρ, $1-\rho$, $\bar\rho$, $1-\bar\rho$ would contribute $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, which dominates as $t \to \infty$ and changes sign. The property holds with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), so it cannot decide RH.

Size biasing gives the Thorin question a form with one variable. The size biased clock $\widehat{T}$ has the law $t\,P(T \in dt)/\mathbb{E}T$, and $\mathbb{E}T = m_0 = \sum_k a_k^{-1} = 0.0231049931\ldots$, the first Stieltjes moment of Section 5.1.

Proposition 11.5.1. *With no hypothesis, $\widehat{T} \stackrel{d}{=} T + A$, with A independent of T and of density $W(t)/m_0$ on $(0,\infty)$. RH holds if and only if A is a mixture of exponential laws, and then $A \stackrel{d}{=} E/V$, with E standard exponential and independent of V, and V of law $m_0^{-1}\sum_{\gamma>0}\gamma^{-2}\delta_{\gamma^2}$, the ordinates counted with multiplicity.*

Proof. The Laplace transform of $\widehat{T}$ is

$$\frac{\mathbb{E}[Te^{-uT}]}{m_0} = -\frac{1}{m_0}\,\frac{d}{du}\,\frac{1}{G(u)} = \frac{1}{m_0}\,\frac{1}{G(u)}\,\frac{G'}{G}(u).$$

By Section 1.2, $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ for $u > 0$ with no hypothesis, and $W > 0$ (Theorem 5.1.1); letting $u \downarrow 0$, monotone convergence gives $\int_0^\infty W(t)\,dt = (G'/G)(0) = G'(0) = \sum_k a_k^{-1} = m_0$. So W/m_0 is a probability density with Laplace transform $(G'/G)(u)/m_0$, the Laplace transform of $\widehat{T}$ is the product of those of T and of a variable A with this density, and by the uniqueness of Laplace

transforms $\widehat{T} \stackrel{d}{=} T + A$ with A independent of T. A variable $A > 0$ is a mixture of exponential laws, $A \stackrel{d}{=} E/V'$ with $V' > 0$ independent of E, exactly when it has a completely monotone density: the density of E/V' is $\mathbb{E}[V'e^{-V't}]$, and conversely, by Bernstein's theorem, a completely monotone probability density is $\int_{[0,\infty)} e^{-vt}\,\sigma(dv)$ with $\int v^{-1}\sigma(dv) = 1$, which forces $\sigma(\{0\}) = 0$, and this is the density of E/V' with $P(V' \in dv) = v^{-1}\sigma(dv)$. The density W/m_0 is completely monotone exactly when RH holds (Section 1.2 and Table 1.6), and then $W(t) = \sum_{\gamma>0} e^{-\gamma^2 t} = m_0\mathbb{E}[Ve^{-Vt}]$ with V as stated. □

This is Steutel's decomposition of Section 11.5.1 with $\Psi = \log G$, and it is the identity that Pakes derives along BDLP ladders [235, (53)], here for the clock and the subordinator of Proposition 11.1.1 that drives it, whose Lévy density is $h = -W'$. Pakes's variable A_n has the density $x \mapsto \int_x^\infty \omega_n(y)\,dy$, the tail of the Lévy density ω_n of the driving law, normalised by the mean $\int_0^\infty x\,\omega_n(x)\,dx$, and for the clock the tail is $\int_t^\infty h(x)\,dx = W(t)$ and the mean is $\int_0^\infty xh(x)\,dx = \int_0^\infty W(t)\,dt = m_0$. The variable A has mean $\mathbb{E}A = m_1/m_0 = \operatorname{Var}T/\mathbb{E}T = 1.60885567 \times 10^{-3}$, from the Taylor coefficients of ξ at $\frac{1}{2}$, with no zero entering. Under RH, A is infinitely divisible, as a mixture of exponential laws [95, 312], and so is $\widehat{T} = T + A$; so under RH both parts of the question that Pakes asks in [232] have a positive answer for the clock. The weights $P(V = \gamma^2) = \gamma^{-2}/m_0$ of the first six ordinates are 0.21663, 0.09794, 0.06919, 0.04676, 0.03990 and 0.03064: the first six carry 0.50105, about a half. The first 1700 ordinates carry 0.978508 (`pakes_sizebias.py`).

Pakes's first strand, the moment problem, enters through the Taylor coefficients at the origin. For $|u| < \gamma_1^2$ and for $|u| < 1/e$ respectively,

$$(\log G)'(u) = \sum_{n\ge 0} m_n(-u)^n, \qquad \mathcal{W}'(u) = \sum_{n\ge 1} \frac{n^n}{n!}(-u)^{n-1},$$

with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; the second series is the derivative of the Lagrange series of $\mathcal{W}$ [64, (3.1)]. Under RH, $m_{n-1} = \int z^{-n}\,U_{\mathrm{RH}}(dz)$ for $n \ge 1$, just as $n^n/n! = \int z^{-n}\,U_{\mathcal{W}}(dz)$ (Corollary 11.5.6): both coefficient sequences are then moment sequences of positive measures. For the clock the positive definiteness of every $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is equivalent to RH (Section 5.1); the number of negative eigenvalues of $\mathcal{H}_N$ does not decrease with N and tends to the number of distinct zeros ρ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$ (Theorem 5.1.7).

Li's criterion is a positivity condition on the same zeros, read on the Thorin functional through the test functions φ_n of Section 8.3.2. It sees a zero off the line more slowly: the term of $\rho = \frac{1}{2} - d + i\gamma$, with $0 < d < \frac{1}{2}$, grows like e^{nd/γ^2}, so it can dominate only from n of order γ^2/d, which exceeds $2\gamma^2$, while in the data of Section 8.4.3 the Hankel test often registers a pair at an order close to

$2k+2$, set by its index k, at a cost in precision that grows linearly with the order (Section 5.1).

11.5.3 The Lambert law, a Thorin problem solved

Let $\mathcal{W}$ be the principal branch of Lambert's function, the solution of $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ that is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real and increasing on $(-1/e, \infty)$, with $\mathcal{W}(0) = 0$, $\mathcal{W}(-1/e) = -1$ and $|\operatorname{Im} \mathcal{W}| < \pi$. Its Lagrange series $\mathcal{W}(u) = \sum_{n\geq 1}(-n)^{n-1}u^n/n!$ has radius $1/e$, and $\mathcal{W}(u) = \log u - \log\log u + o(1)$ as $u \to \infty$ [64, (3.1), (4.19)]. We write $T_{\mathcal{W}}$ for a variable with the Lambert law. The proof that it is a GGC is a template that fails for the clock at exactly one step.

Proposition 11.5.2 ([64, Section 4, (4.4)], [136, (10), (11)]). *For $0 < b < \pi$ put $r(b) = b\, e^{-b\cot b}/\sin b$. Then r increases from $1/e$ to ∞ on $(0, \pi)$, and the boundary values of $\mathcal{W}$ on the cut, from the upper half plane, are $\mathcal{W}(-r(b)+i0) = -b\cot b + ib$.*

Proof. Put $w = \alpha + ib$ with $0 < b < \pi$ in $we^w = -x$, $x > 0$. The imaginary part gives $e^\alpha(\alpha \sin b + b\cos b) = 0$, so $\alpha = -b\cot b$, and then the real part gives $e^\alpha(\alpha\cos b - b\sin b) = -e^{-b\cot b}\, b/\sin b$, so that $x = r(b)$. Next, $(\log r)'(b) = 1/b - 2\cot b + b/\sin^2 b$ and $b\sin^2 b\,(\log r)'(b) = (b - \sin b\cos b)^2 + \sin^4 b > 0$, while $r(0+) = 1/e$ and $r(\pi-) = \infty$; so for each $x > 1/e$ there is exactly one point w of the curve $\{-b\cot b + ib : 0 < b < \pi\}$ with $we^w = -x$, namely the point with $r(b) = x$. This curve bounds the range of the principal branch in the upper half plane [64, Section 4], and $\mathcal{W}$ is continuous on the cut from above, with values on it; so $\mathcal{W}(-r(b) + i0) = -b\cot b + ib$. □

Kalugin, Jeffrey and Corless obtain the next theorem from the Stieltjes representation of $\mathcal{W}'$ [136, Theorem 3.3] and the characterisation of Thorin Bernstein functions in [299, Chapter 8].

Theorem 11.5.3 (Kalugin, Jeffrey and Corless [136, Theorem 6.2], [138]). *Let $U_{\mathcal{W}}$ be the image of the uniform law db/π on $(0, \pi)$ under r, a probability measure on $(1/e, \infty)$. Then for $u \in \mathbb{C} \setminus (-\infty, -1/e]$*

$$\mathcal{W}(u) = \int \log\Big(1 + \frac{u}{z}\Big)\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \log\Big(1 + \frac{u}{r(b)}\Big)\, db.$$

Hence $\mathcal{W}$ is a Thorin Bernstein function, and $T_{\mathcal{W}}$ is a GGC with no drift and Thorin measure $U_{\mathcal{W}}$, of total mass one, whose distribution function is $U_{\mathcal{W}}((1/e, y]) = \pi^{-1}\operatorname{Im}\mathcal{W}(-y + i0)$ for $y > 1/e$.

Proof. Let $\Psi(u)$ be the integral. Since $U_{\mathcal{W}}$ is carried by $(1/e, \infty)$, Ψ is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real on $(-1/e, \infty)$, and $\Psi(\bar u) = \overline{\Psi(u)}$; so is $\mathcal{W}$. Let $x > 1/e$, and let $b(x) \in (0, \pi)$ be the point with $r(b(x)) = x$. As $\eta \downarrow 0$, $1 + (-x + i\eta)/z$ tends to a negative number from the upper half plane when $z < x$ and to a positive number when $z > x$, and $U_{\mathcal{W}}$ has no atoms, so by dominated convergence $\operatorname{Im}\Psi(-x + i0) =$

$\pi\, U_{\mathcal{W}}((1/e, x)) = b(x)$; by Proposition 11.5.2 also $\operatorname{Im} \mathcal{W}(-x + i0) = b(x)$, which is the formula for the distribution function. Near $1/e$, $r(b) - 1/e \sim b^2/(2e)$, so $U_{\mathcal{W}}$ has a density asymptotic to $(e/2)^{1/2}\pi^{-1}(z - 1/e)^{-1/2}$ there, and a bounded continuous density on $[1/e + c, \infty)$ for every $c > 0$. Hence Ψ extends continuously to the closed upper half plane, and so does $\mathcal{W}$, with $\mathcal{W}(-1/e) = -1$. The difference $D = \mathcal{W} - \Psi$ is therefore continuous on the closed upper half plane and real on the real axis, where on $(-\infty, -1/e)$ the imaginary parts of the two boundary values agree. The Schwarz reflection principle continues D across $(-\infty, -1/e)$, the continuation coincides with D in the lower half plane because $D(\bar{u}) = \overline{D(u)}$, and the singularity at $-1/e$, where D is bounded, is removable. So D is entire, and $D(0) = 0$.

It remains to bound D, and one sided bounds on real parts suffice. First, $\operatorname{Re} \Psi(u) = \int \log|1 + u/z|\, U_{\mathcal{W}}(dz) \le \log(1 + e|u|)$, since $z > 1/e$. Next let $|u| = R \ge e$ and $w = \mathcal{W}(u)$, so that $|w|e^{\operatorname{Re} w} = R$ and $|\operatorname{Im} w| < \pi$. If $|w| > \log R + \pi$, then $|\operatorname{Re} w| > |w| - \pi > \log R$. This is impossible when $\operatorname{Re} w \ge 0$, since then $\operatorname{Re} w = \log R - \log|w| < \log R$; and when $\operatorname{Re} w < 0$ it gives $\log|w| - \log R > |w| - \pi$, that is $|w| - \log|w| < \pi - \log R \le \pi - 1$, which fails for $|w| > 1 + \pi$, since $x - \log x$ increases for $x > 1$ and exceeds $\pi - 1$ at $x = 1 + \pi$. Hence $|w| \le \log R + \pi$ and $\operatorname{Re} \mathcal{W}(u) = \log R - \log|w| \ge \log R - \log(\log R + \pi)$. By continuity both bounds hold on the whole circle, its point on the cut included, so

$$\max_{|u|=R} \operatorname{Re}\big(-D(u)\big) \le \log(1 + eR) - \log R + \log(\log R + \pi), \qquad R \ge e.$$

The Borel Carathéodory lemma for the entire function $-D$, which vanishes at the origin, gives $\max_{|u|=R/2}|D(u)| \le 2 \max_{|u|=R} \operatorname{Re}(-D(u))$, which is $O(\log\log R)$ by this bound. By Cauchy's estimates every Taylor coefficient of D of positive order vanishes, so $D \equiv D(0) = 0$. □

Corollary 11.5.4 (first part from [136, Theorem 6.5] and [138]). *The canonical function of $T_{\mathcal{W}}$ is*

$$k_{\mathcal{W}}(x) = \int e^{-xz}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi e^{-x\, r(b)}\, db,$$

so that $\mathcal{W}(u) = \int_0^\infty (1 - e^{-ux})\, k_{\mathcal{W}}(x)\, dx/x$ for $u \ge 0$. The symmetric law of $Z\sqrt{2T_{\mathcal{W}}}$, with Z standard normal and independent of $T_{\mathcal{W}}$, has characteristic function $e^{-\mathcal{W}(w^2)}$, is an EGGC (Section 1.3.2), and has Lévy density $k^{\mathrm{s}}_{\mathcal{W}}(|x|)/|x|$ with

$$k^{\mathrm{s}}_{\mathcal{W}}(x) = \int e^{-x\sqrt{z}}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi e^{-x\sqrt{r(b)}}\, db,$$

that is, $\mathcal{W}(w^2) = \int_0^\infty 4\sin^2(wx/2)\, k^{\mathrm{s}}_{\mathcal{W}}(x)\, dx/x$ for real w.

Proof. Insert Frullani's integral $\log(1 + u/z) = \int_0^\infty (1 - e^{-ux})e^{-zx}x^{-1}dx$ in Theorem 11.5.3 and exchange the integrals by Tonelli's theorem. For the symmetric

law, $\mathbb{E}e^{iwZ\sqrt{2T_{\mathcal{W}}}} = \mathbb{E}e^{-w^2 T_{\mathcal{W}}}$, and the cosine dictionary (6.2.4) with $U = U_{\mathcal{W}}$, for which $K_U = 2k^{\mathrm{s}}_{\mathcal{W}}$, gives the Lévy density $K_U(|x|)/(2|x|)$. □

Remark 11.5.5 (Self decomposability, after Pakes). Since $k_{\mathcal{W}}$ is nonincreasing, $T_{\mathcal{W}}$ is self decomposable, and as in Proposition 11.1.1 its driving subordinator Z has the Lévy density $-k'_{\mathcal{W}}(x) = \int z e^{-xz}\, U_{\mathcal{W}}(dz)$, of total mass $k_{\mathcal{W}}(0+) = 1$ and with mean $\int_0^\infty k_{\mathcal{W}}(x)\, dx = \int z^{-1}\, U_{\mathcal{W}}(dz) = 1$ (Corollary 11.5.6). So the driving process is compound Poisson with unit rate and jumps J of mean one; its Laplace exponent is $u\mathcal{W}'(u) = \mathcal{W}(u)/(1+\mathcal{W}(u))$, and $\mathbb{E}e^{-uJ} = 1/(1+\mathcal{W}(u))$ [235, (5), (6), (18)]. Stopping $T_{\mathcal{W}} = \int_0^\infty e^{-t}\, d\mathsf{Z}_t$ at the first jump of Z, at an exponential time τ, gives the perpetuity $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$, with $\mathsf{U} = e^{-\tau}$ uniform on $(0,1)$ and the three variables independent. The stationary excess $\mathcal{S}J$ has the density $P(J > x)$, since $\mathbb{E}J = 1$, and the Laplace transform $(1 - \mathbb{E}e^{-uJ})/u = \mathcal{W}'(u)$; since $\mathbb{E}e^{-u\mathsf{U}Y} = u^{-1}\int_0^u \mathbb{E}e^{-sY} ds$ for $Y \geq 0$ independent of U, also $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, and $\mathcal{S}J \stackrel{d}{=} J + T_{\mathcal{W}}$.

Corollary 11.5.6 ([136, Corollary 3.4], [137]; Nuttall's integral [217, 46]). *For $n \geq 0$,*

$$\int z^{-n}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \Big(\frac{\sin b\; e^{b\cot b}}{b}\Big)^n db = \frac{n^n}{n!} \qquad (0^0 = 1),$$

so $(n^n/n!)_{n\geq 0}$ is the moment sequence of the image of $U_{\mathcal{W}}$ under $z \mapsto 1/z$, a probability measure on $(0, e)$. Nuttall conjectured, and Bouwkamp proved, the identity with $\nu^\nu/\Gamma(\nu+1)$ for every real $\nu \geq 0$.

Proof. Differentiating Theorem 11.5.3 gives

$$\mathcal{W}'(u) = \int \frac{U_{\mathcal{W}}(dz)}{u+z} = \sum_{n\geq 1}(-u)^{n-1}\int z^{-n}\, U_{\mathcal{W}}(dz), \qquad |u| < 1/e,$$

the series converging uniformly in z since $z > 1/e$. Compare with the derivative of the Lagrange series $\mathcal{W}(u) = \sum_{n\geq 1}(-n)^{n-1}u^n/n!$, whose coefficients are $(-n)^{n-1}/(n-1)! = (-1)^{n-1}n^n/n!$. For $n = 0$ the integral is the total mass, one. □

The program `pakes_lambert.py` checks the section.

What transfers to the clock. The proof of Theorem 11.5.3 has four steps: the boundary values on the cut, the positivity of the boundary measure, the identification by reflection and a growth bound, and, for the symmetric law, the cosine dictionary (6.2.4). For the clock the first step is the argument count. Continued along the upper side of the negative axis, past the zeros of G on it by small half circles in the upper half plane, $\operatorname{Im}\log G(-x + i0) = \pi N_{\mathrm{line}}(\sqrt{x})$, where $N_{\mathrm{line}}(t)$

counts with multiplicity the zeros on the critical line with ordinate in $(0,t)$, because the four zeros ρ, $1-\rho$, $\bar\rho$, $1-\bar\rho$ of a zero off the line contribute to G the factor $(1+u/a)(1+u/\bar a)=|1+u/a|^2>0$ at real u; so $N_{\mathrm{line}}(\sqrt{x})$, the distribution function of U_{line} (Proposition 1.2.3), is read from the boundary values of $\log G$ with no hypothesis, and it is $N(\sqrt{x})$ under RH. The second step is then automatic, since a counting function is nondecreasing. The fourth step gives the Poisson trace $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$, which is $\sum_{\gamma>0}e^{-\gamma y}$ under RH, as the Lévy side of $Y_{1/2}=Z\sqrt{2T}$ (Section 10.1.4), while the heat trace W is the canonical function of T itself (Section 1.3.1); the two traces are linked with no hypothesis by subordination. The third step is the obstruction: $\mathcal{W}$ is analytic off $(-\infty,-1/e]$ by construction, whereas $\log G$ is analytic off $(-\infty,0]$ if and only if RH holds (Proposition 1.2.1 and the argument after it).

11.5.4 Multiply self decomposable laws and branching

The centre clock is self decomposable with no hypothesis (Theorem 5.1.1), so it is the integral $\int_0^\infty e^{-t}\,dZ_t$ of its driving subordinator Z (Proposition 11.1.1). The law of Z_1 may be self decomposable in its turn, with a driving process of its own, and so on. Read upwards, from the last driving process to the clock, such a chain is Pakes's BDLP ladder [235, Section 5], and how far it goes places a law in one of Urbanik's classes L_n. This subsection shows that for the centre clock the chain has exactly three steps, with no hypothesis on the zeros (Theorem 11.5.8), and that the clock is not the martingale limit of a supercritical continuous state branching process, the construction behind one of Pakes's proofs for the Lambert law (Theorem 11.5.11). Neither says anything about RH.

11.5.4.1 Urbanik's classes and the ladder functions

A law μ on $\mathbb{R}$ is self decomposable if for every $c\in(0,1)$ there is a law μ_c, its c component, with $\widehat\mu(z)=\widehat\mu(cz)\,\widehat\mu_c(z)$. Urbanik [333] let L_0 be the class of self decomposable laws, L_n for $n\ge1$ the class of laws in L_{n-1} all of whose components lie in L_{n-1}, and $L_\infty=\bigcap_n L_n$. Sato proved that L_∞ is the smallest class of laws that contains the stable laws and is closed under convolution and weak limits [296, Theorem 3.5]. In the language of Section 11.1.1, a law lies in L_n, $n\ge1$, exactly when it is self decomposable and the law at time one of its driving process lies in L_{n-1} [121].

For laws on $[0,\infty)$ Berg and Forst put the classes in analytic form. Let f be the Laplace exponent of an infinitely divisible law μ on $[0,\infty)$, so that $\int e^{-ux}\,\mu(dx)=e^{-f(u)}$, and let $(Sf)(u)=uf'(u)$. Then $\mu\in L_n$ if and only if $Sf, S^2f,\dots,S^{n+1}f$ are Bernstein functions [27, Theorem 1.3].

For the centre clock $f=\log G$, and S acts on the canonical function W through

the ladder functions

$$W^{[0]} = W, \qquad W^{[j+1]}(t) = -t\,\frac{d}{dt}W^{[j]}(t), \qquad \text{so that} \quad W^{[j]} = \Big(-t\frac{d}{dt}\Big)^j W.$$

For one exponential, $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)\,e^{-at}$, where $\beta_0 = 1$ and $\beta_{j+1}(u) = u\beta_j(u) - u\beta_j'(u)$:

$$\beta_1(u) = u, \quad \beta_2(u) = u^2 - u, \quad \beta_3(u) = u^3 - 3u^2 + u, \quad \beta_4(u) = u^4 - 6u^3 + 7u^2 - u.$$

In general $\beta_j(u) = \sum_{i=1}^{j}(-1)^{j-i}S(j,i)\,u^i = (-1)^j B_j(-u)$, with $S(j,i)$ the Stirling numbers of the second kind and B_j the Touchard polynomials, and since $\beta_j(u)e^{-u} = (-u\frac{d}{du})^j e^{-u}$ and $-u\frac{d}{du}$ multiplies Mellin transforms by s,

$$\int_0^\infty \beta_j(u)\,e^{-u}u^{s-1}\,du = s^j\,\Gamma(s) \qquad (s > 0). \tag{11.5.1}$$

The positive roots of β_3 are $(3 \mp \sqrt{5})/2 = 0.381966$ and 2.618034, and those of β_4 are 0.165757, 1.343380 and 4.490864, so that $\beta_4 < 0$ on $(0, 0.165757) \cup (1.343380, 4.490864)$. Since $W(t) = \sum_k e^{-a_k t}$ with $\operatorname{Re} a_k \geq \gamma_k^2 - \frac{1}{4} > 0$ for every zero (Section 11.1.1),

$$W^{[j]}(t) = \sum_k \beta_j(a_k t)\,e^{-a_k t}, \tag{11.5.2}$$

the series converging absolutely and locally uniformly on $(0,\infty)$.

Lemma 11.5.7. *For every $j \geq 0$ and $u \geq 0$,*

$$\Big(u\frac{d}{du}\Big)^j \log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\,W^{[j]}(t)\,\frac{dt}{t}. \tag{11.5.3}$$

For $n \geq 0$ the centre clock lies in L_n if and only if $W^{[n+1]} \geq 0$ on $(0,\infty)$, and then $W^{[1]}, \ldots, W^{[n]}$ are nonnegative too. In that case the driving processes $Z^{[1]} = Z$, $Z^{[2]}$, …, $Z^{[n+1]}$ exist, each the driving process of the law at time one of the one before; $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G(u)$ and Lévy density $W^{[j]}(t)/t$; and for $1 \leq j \leq n$ the law of $Z_s^{[j]}$ is self decomposable for every $s > 0$.

Proof. By (11.5.2), $W^{[j]}(t) = O(t^j e^{-\gamma_1^2 t})$ as $t \to \infty$, and by the explicit formula (11.5.4) below, $W^{[j]}(t) = O(t^{-1/2}\log(1/t))$ as $t \downarrow 0$. The case $j = 0$ of (11.5.3) is the Lévy Khintchine form of $\log G$ (Section 11.1.1). If it holds for j, then, since $ue^{-ut} = \frac{d}{dt}(1 - e^{-ut})$,

$$\begin{aligned} u\frac{d}{du}\int_0^\infty \big(1 - e^{-ut}\big)W^{[j]}(t)\,\frac{dt}{t} &= \int_0^\infty ue^{-ut}\,W^{[j]}(t)\,dt \\ &= \int_0^\infty \big(1 - e^{-ut}\big)\,W^{[j+1]}(t)\,\frac{dt}{t}, \end{aligned}$$

the boundary terms $(1-e^{-ut})W^{[j]}(t)$ vanishing at both ends by the two bounds. A function of the form (11.5.3) is a Bernstein function exactly when $W^{[j]} \geq 0$: if $W^{[j]} \geq 0$ it is a Lévy Khintchine form, and conversely, if it is a Bernstein function, its derivative $\int_0^\infty e^{-ut} W^{[j]}(t)\,dt$ is completely monotone, hence the Laplace transform of a nonnegative measure, which by the uniqueness of Laplace transforms is $W^{[j]}(t)\,dt$. So by the theorem of Berg and Forst the clock lies in L_n exactly when $W^{[1]}, \ldots, W^{[n+1]}$ are nonnegative. If $W^{[n+1]} = -t\,(W^{[n]})' \geq 0$, then $W^{[n]}$ is nonincreasing, and since it tends to 0 at infinity it is nonnegative; descending in the same way, so are $W^{[n-1]}, \ldots, W^{[1]}$. For the driving processes, the Laplace exponent of $\int_0^\infty e^{-t}\,dY_t$ is $\int_0^u \Psi_Y(w)\,dw/w$ when the subordinator Y has Laplace exponent Ψ_Y (proof of Proposition 11.1.1), so a self decomposable law with Laplace exponent f has a driving process with Laplace exponent $uf'(u)$. Starting from $f = \log G$ and using (11.5.3), $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G$ and Lévy density $W^{[j]}(t)/t$ for as long as the laws at time one are self decomposable, and the law of $Z_s^{[j]}$, with Lévy density $sW^{[j]}(t)/t$, is self decomposable exactly when $W^{[j]}$ is nonincreasing [297, Corollary 15.11], that is, when $W^{[j+1]} \geq 0$. □

11.5.4.2 The ladder index of the centre clock

THEOREM 11.5.8. *Unconditionally, that is, using only the verification of RH below height* 2197.3 *[247], $W^{[3]}(t) > 0$ for every $t > 0$, and $W^{[4]}(e^{-4.15}) < 0$. Consequently W, $W^{[1]}$ and $W^{[2]}$ are positive on $(0,\infty)$; the centre clock lies in L_2 and not in L_3; its driving process $Z = Z^{[1]}$ and the driving process $Z^{[2]}$ of Z are self decomposable at every time, and the driving process $Z^{[3]}$ of $Z^{[2]}$ is self decomposable at no time.*

Proof. By Lemma 11.5.7 it suffices to prove the two inequalities; the lower ladder functions are then positive, not only nonnegative, since $W^{[j]}$ strictly decreases to 0 when $W^{[j+1]} > 0$.

Large t. Let $\tau_1 = (3+\sqrt{5})/(2\gamma_1^2) = 0.0131039$, so that $\gamma_1^2\tau_1 = 2.618034$ is the largest root of β_3; beyond it β_3 is positive and increasing. The zeros with ordinates below $\Gamma = 2197.3$ lie on the line [247]; there are $N(\Gamma) = 1700$ of them, the last at 2197.2587, and `bdlp_ladder.py` encloses each one in ball arithmetic. For $t \geq \tau_1$ each of them contributes $\beta_3(\gamma^2 t)e^{-\gamma^2 t} \geq 0$ to $W^{[3]}(t)$, and the second contributes at least $\beta_3(\gamma_2^2\tau_1)e^{-\gamma_2^2 t} > 99.38\,e^{-\gamma_2^2 t}$, since $\gamma_2^2\tau_1 = 5.7910$. A zero with ordinate $\gamma > \Gamma$, wherever it lies, has $\operatorname{Re} a \geq \gamma^2 - \frac{1}{4}$ and $|a| \leq \gamma^2 + \frac{1}{4}$, so its term has modulus at most $f(\gamma) = B_3((\gamma^2+\frac{1}{4})t)\,e^{-(\gamma^2-\frac{1}{4})t}$, where the Touchard polynomial B_j has the moduli of the coefficients of β_j. Since $vB_j'(v) \leq jB_j(v)$, the logarithm of $B_j(v)e^{-v}$ decreases at a rate of at least $\frac{1}{2}$ for $v \geq 2j$, so for $(\Gamma^2+\frac{1}{4})t \geq 6$ the function f decreases and $f(\gamma) \leq f(\Gamma)e^{-(\gamma^2-\Gamma^2)t/2}$. Let

$$N^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + 0.1038\log T + 0.2573\log\log T + 9.3675,$$

an upper bound for $N(T)$ for $T \ge e$ [109, Corollary 1.2], whose derivative is at most $\frac{1}{2\pi}\log\frac{T}{2\pi} + \frac{0.4}{T}$ for $T \ge \Gamma$. Partial summation against $N \le N^+$, with $\gamma^2 - \Gamma^2 \ge 2\Gamma(\gamma - \Gamma)$ and $\log\frac{\gamma}{2\pi} \le \log\frac{\Gamma}{2\pi} + \frac{\gamma-\Gamma}{\Gamma}$, bounds the sum of these moduli by

$$\varepsilon(t) = f(\Gamma)\Big(N^+(\Gamma) - 1700 + \frac{L}{\Gamma t} + \frac{1}{2\pi\Gamma^3 t^2}\Big), \qquad L = \frac{1}{2\pi}\log\frac{\Gamma}{2\pi} + \frac{0.4}{\Gamma}.$$

The logarithmic derivative of $e^{\gamma_2^2 t}\varepsilon(t)$ is at most $3/t - (\Gamma^2 - \frac14 - \gamma_2^2) < 0$ for $t \ge \tau_1$, and $e^{\gamma_2^2\tau_1}\varepsilon(\tau_1) < 10^{-27458}$. So $W^{[3]}(t) \ge e^{-\gamma_2^2 t}\big(99.38 - e^{\gamma_2^2 t}\varepsilon(t)\big) > 0$ for $t \ge \tau_1$.

Small t. Applying $(-t\frac{d}{dt})^j$ term by term to (1.2.5), which the Gaussian factors in r and in $\log n$ allow, gives with no hypothesis

$$W^{[j]}(t) = \beta_j\big(-\tfrac{t}{4}\big)e^{t/4} + A_j(t) - P_j(t), \tag{11.5.4}$$

where

$$A_j(t) = \frac{1}{2\pi}\int_0^\infty \beta_j(r^2 t)\, e^{-r^2 t}\, m(r)\, dr,$$
$$P_j(t) = \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, e^{-v_n} q_j(v_n), \qquad v_n = \frac{(\log n)^2}{4t},$$

and $q_0 = 1$, $q_{j+1}(v) = (\frac12 - v)q_j(v) + vq_j'(v)$, because $-t\frac{d}{dt}$ maps $t^{-1/2}$ to $\frac12 t^{-1/2}$ and v_n to v_n; thus

$$q_3(v) = -v^3 + \tfrac92 v^2 - \tfrac{13}{4}v + \tfrac18, \qquad q_4(v) = v^4 - 8v^3 + \tfrac{29}{2}v^2 - 5v + \tfrac{1}{16}.$$

In the notation of the proof of Theorem 5.1.1, $\frac{1}{2\pi}\int_0^\infty e^{-r^2 t}\log\frac{r}{2\pi}\,dr = s_0(t)(\ell+\alpha_0)$, and since $-t\frac{d}{dt}$ multiplies s_0 by $\frac12$ and maps ℓ to $\frac12$, replacing $m(r)$ by $\log\frac{r}{2\pi}$ in A_j gives

$$\bar A_j(t) = 2^{-j} s_0(t)\,(\ell + \alpha_0 + j),$$

which also follows from (11.5.1) and its derivative in s at $s = \frac12$. For the difference we split the integral at $r = 5$. For $r \le 5$ we use $|\beta_j(u)|e^{-u} \le u$, valid for $j = 2, 3, 4$, and the bound $\int_0^5 r^2|m(r) - \log\frac{r}{2\pi}|\,dr < 0.404$ stated after the proof of Theorem 5.1.1; for $r \ge 5$ we use the bound $r^2|m(r) - \log\frac{r}{2\pi}| < 0.05$ stated there, and substitute $u = r^2 t$. This gives

$$|A_j(t) - \bar A_j(t)| \le E_j(t) = \frac{1}{2\pi}\Big(0.404\, t + 0.025\sqrt{t}\, J_j\Big),$$

where $J_j = \int_0^\infty |\beta_j(u)|\, e^{-u} u^{-3/2}\, du$ satisfies $J_3 < 1.3313$ and $J_4 < 1.8295$. Let $\tau_2 = 1/75$. For $t \le \tau_2$ the main term $\bar A_3(t)$ decreases in t, because $-t\frac{d}{dt}\bar A_3 = \bar A_4$ and $\ell + \alpha_0 + 4 > 0$; the modulus $|\beta_3(-\frac{t}{4})|e^{t/4}$ of the polar term and $E_3(t)$ increase; and so does the modulus of each term of $P_3(t)$, because $v_n \ge v_2 > 9$,

the polynomials q_3 and q_4 have no root in $[9,\infty)$, so that $q_3<0<q_4$ there, and $-t\frac{d}{dt}$ maps the nth term of P_3 to the nth term of P_4. At $t=\tau_2$ the four quantities are $\bar{A}_3(\tau_2)=0.357155$, 0.003378, $E_3(\tau_2)<0.001469$, and at most 0.057882 for the sum of the moduli of the terms of P_3, in which the terms with $n>200$ are bounded through $e^{-(\log n)^2/8t}\le n^{-\sigma}$, $\sigma=(\log 200)/8t$, as in the proof of Theorem 5.1.1. Hence $W^{[3]}(t)>0.2944$ for $t\le\tau_2$, and since $\tau_2>\tau_1$, $W^{[3]}>0$ on $(0,\infty)$.

Rung four. Let $t_0=e^{-4.15}$. Then $\gamma_1^2t_0=3.1496$ lies in the interval $(1.343380,4.490864)$ on which $\beta_4<0$. The first five zeros contribute -0.97604, 0.62189, 0.22811, 0.01295 and 0.00216 to $W^{[4]}(t_0)$, and all zeros together, with the tail bound above, give $W^{[4]}(t_0)=-0.11089729616255\pm10^{-23}$ in ball arithmetic. The explicit formula confirms the sign without the zeros: at t_0 the polar term is 0.004066, $\bar{A}_4(t_0)=0.228563$, $P_4(t_0)=0.344026$ and $E_4(t_0)<0.001928$, so that $W^{[4]}(t_0)<0.004066+0.228563+0.001928-0.344026<-0.109$. □

The ladder breaks where the first zero dominates. At t_0 its term, $\beta_4(3.1496)e^{-3.1496}=-0.97604$, is only partly cancelled by the others, which leave $W^{[4]}(t_0)=-0.11090$: the second zero, for which $\gamma_2^2t_0$ already exceeds the largest root 4.490864 of β_4, contributes 0.62189, and the zeros above it are too far out. On the arithmetic side the same failure is the prime 2: its term in $P_4(t_0)$, 0.343448, outweighs the archimedean main term $\bar{A}_4(t_0)=0.228563$. On $-\log t\in[0,10]$ a scan in double precision on a grid of step 0.001, with the sign at the ends of every interval certified from the zeros, finds no negative value of W, $W^{[1]}$, $W^{[2]}$ or $W^{[3]}$, and finds $W^{[4]}<0$ exactly for $-\log t\in(3.946,4.335)$, $W^{[5]}<0$ on $(3.443,4.126)$ and $W^{[6]}<0$ on $(3.146,3.773)\cup(4.403,5.051)$. Towards the origin the ladder functions approach fixed fractions of W: by (11.5.4), $W^{[j]}(t)=\bar{A}_j(t)+O(1)$ as $t\downarrow0$, so that $W^{[2]}/W\to\frac14$ and $W^{[3]}/W\to\frac18$. The two ratios fall from 5.73 and 5.95 at $-\log t=4.30$ to 0.614 and 0.404 at 8 and to 0.470 and 0.291 at 10, decreasing throughout $[4.30,10]$. The least values of $W^{[2]}$ and $W^{[3]}$ on $4.30\le-\log t\le8$ are 0.396547, at 4.30, and 0.410976, at 4.335.

Corollary 11.5.9. *With no hypothesis, for every $s>0$ the laws of Z_s and of $Z_s^{[2]}$ are self decomposable, hence unimodal, and the law of Z_s lies in L_1.*

Proof. The first statement is Theorem 11.5.8 with Lemma 11.5.7, and Yamazato's theorem [347], [297, Theorem 53.1] gives unimodality. The Laplace exponent of Z_s is $s\,u(\log G)'(u)$, so by (11.5.3) its images under S and S^2 correspond to $sW^{[2]}$ and $sW^{[3]}$, which are positive, and the theorem of Berg and Forst places Z_s in L_1. □

Remark 11.5.10. The perpetuity $T_{\mathcal{W}}\overset{d}{=}\mathsf{U}(T_{\mathcal{W}}+J)$ of Remark 11.5.5 comes from a driving process that is compound Poisson, and the argument there applies to any such process: if the driving process Y of a self decomposable law is compound Poisson with rate c and jumps distributed as J, stopping $X=\int_0^\infty e^{-t}\,dY_t$ at the

first jump gives $X \stackrel{d}{=} \mathsf{U}^{1/c}(X+J)$, with U uniform on $(0,1)$ and the three variables independent. The centre clock satisfies no relation of this form, for any $c>0$ and any $J \geq 0$. Such a J would have $\mathbb{E}\log(1+J) < \infty$, since $\log(1+\mathsf{U}^{1/c}J) \geq \log(1+J) + c^{-1}\log \mathsf{U}$ and $\mathbb{E}\log(1+\mathsf{U}^{1/c}J) \leq \mathbb{E}\log(1+T) \leq \mathbb{E}T = m_0$, and iterating the relation would write T as $\int_0^\infty e^{-t}\,dY_t$ with Y compound Poisson; but the driving process of a self decomposable law is unique in law, its Laplace exponent being $u(\log G)'(u)$, and that of the clock has the infinite Lévy measure $-W'(x)\,dx$ (Proposition 11.1.1).

The ladder of this subsection is not the derivative ladder of Theorems 5.1.1, 5.1.6, 5.1.9 and 5.1.13 and Table 1.6, which climbs the signs $c_k = (-1)^k W^{(k)} > 0$. Its rungs are alternating combinations of the same c_k, with Stirling numbers as coefficients:

$$W^{[j]}(t) = \sum_{i=1}^{j} (-1)^{j-i} S(j,i)\, t^i c_i(t),$$

$$W^{[3]} = t^3 c_3 - 3t^2 c_2 + t c_1, \qquad W^{[4]} = t^4 c_4 - 6t^3 c_3 + 7t^2 c_2 - t c_1.$$

Of the BDLP ladder RH gives only the first rung, since a GGC is self decomposable and a gamma law, a GGC, lies outside L_1. In the other direction, membership of the clock in L_∞ would imply RH, by [27, Section 4], but no finite rung does: for every n the canonical function $(\log \frac{1}{t})_+^n$, which vanishes on $[1,\infty)$ and so is not completely monotone, defines a law in L_n that is not a GGC, its ladder functions being $n!/(n-j)!\,(\log \frac{1}{t})_+^{n-j}$ for $j \leq n$ and the measure $n!\,\delta_1(dt)$ for $j = n+1$.

11.5.4.3 The ladder along the de Bruijn Newman flow

Theorem 7.2.15 places the centre clock in a family of laws. For every $t \geq \Lambda$, and so for every $t \geq 0.2$ with no hypothesis, the function $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ of Section 8.4.1 is the Laplace transform of a GGC whose Thorin measure has the atoms $a_k(t) = (z_k(t)/2)^2$, where $0 < z_1(t) < z_2(t) < \cdots$ are the zeros of H_t, and $R_0 = 1/G$. Since t now denotes the time of the flow, we write x for the variable of the canonical function $W(x;t) = \sum_k e^{-a_k(t)x}$ of R_t, whose ladder functions are

$$W^{[j]}(x;t) = \Big(-x\frac{\partial}{\partial x}\Big)^j W(x;t) = \sum_k \beta_j\big(a_k(t)x\big)\, e^{-a_k(t)x}.$$

Lemma 11.5.7 holds for these laws with the same proof, the bound for small x now coming from the number of zeros of H_t, so the law with Laplace transform R_t lies in L_n exactly when $W^{[n+1]}(\cdot\,;t) \geq 0$.

The program `heat_ladder.py` computes H_t by Gauss Legendre quadrature in ball arithmetic, its zeros on $(0,600]$ by a sign scan of step 0.2 and bisection, and the ladder functions $W^{[j]}(x;t)$, $j \leq 5$, from these zeros on the fixed window

$0 \le -\log x \le 8$, on a grid of step 0.001, for nine values of t (Table 11.17). Nothing needs computing for $-\log x \le 0$: the largest roots of $\beta_2, \dots, \beta_5$ are 1, 2.618034, 4.490864 and 6.510134, and $a_1(t)x \ge a_1(t) \ge 12.47$ in every row. The computation is numerical for $t > 0$. The quadrature is checked against a rule with 1024 nodes, with relative differences below 10^{-46} at $z = 100$, 300 and 590, but its error is not bounded, and the zeros above $z = 600$ are left out; an estimate from the density $\frac{1}{4\pi}\log\frac{z}{4\pi}$ of the zeros of H_0 puts their share of $W^{[3]}$ at $-\log x = 8$ below 2×10^{-9}, and their share of $W^{[5]}$ below 10^{-5}. At $t = 0$ the computed zeros are the numbers $2\gamma_k$ to double precision, and there are $N(300) = 138$ of them, the number that Arb's count of the zeros of ζ gives; H_0 agrees with $\Xi(z/2)/8$, computed from ζ, to 10^{-46}.

t	$z_1(t)$	a_2/a_1	$W^{[2]}/W$	$W^{[3]}/W$	$W^{[3]} < 0$	$W^{[4]} < 0$	n
0	28.2695	2.2119	0.6144	0.4038	none	$(3.946, 4.335)$	2
0.2	28.1536	2.2142	0.6131	0.4029	none	$(3.935, 4.333)$	2
1	27.6920	2.2239	0.6078	0.3993	none	$(3.893, 4.321)$	2
4	25.9874	2.2678	0.5887	0.3864	none	$(3.737, 4.263)$	2
10	22.7127	2.3915	0.5545	0.3630	none	$(3.426, 4.090)$	2
20	17.7111	2.7188	0.5069	0.3301	none	$(2.884, 3.675)$	2
30	13.3727	3.2530	0.4687	0.3030	$(2.983, 3.343)$	$(2.303, 3.153)$	1
40	9.7966	4.0960	0.4374	0.2807	$(2.236, 3.014)$	$(1.676, 2.593)$	1
50	7.0629	5.3383	0.4115	0.2620	$(1.563, 2.539)$	$(1.022, 2.035)$	1

Table 11.17: The ladder functions of R_t along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of H_t, the ratio a_2/a_1 of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest n with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: R_t lies in no class above L_n, and in L_n if the signs persist for $x < e^{-8}$. On the window W, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every t. Numerical for $t > 0$; the row $t = 0$ is Theorem 11.5.8. Computed by `heat_ladder.py`.

Along the flow the zeros move towards the origin and apart, as the table shows, and the number of zeros in $(0, 600]$ rises from 138 to 149. The set on which $W^{[4]} < 0$ follows the first atom to smaller values of $-\log x$. It lies where $a_1(t)x$ is in the interval $(1.343380, 4.490864)$ on which $\beta_4 < 0$, less the part on which the other atoms compensate, and at $t = 50$, where it covers most of that interval, the first atom alone almost decides the sign. Up to $t = 20$ the third ladder function stays nonnegative on the window, and R_t keeps the index two of the clock, as far as the window shows. Between $t = 27.8$ and $t = 27.9$ it turns negative. The program finds $W^{[3]} \ge 0$ on the window at $t = 21, \dots, 27$ and $t = 27.1, \dots, 27.8$, which gives $W^{[2]} \ge 0$ and $W^{[1]} \ge 0$ there as well, by the descent in the proof of Lemma 11.5.7 applied on $[e^{-8}, \infty)$; at $t = 27.9$ it finds $W^{[3]} < 0$ for $3.230 < -\log x < 3.307$, with least value -0.0015, and at $t = 30$, 40 and 50 on the intervals of the table.

So, by this computation, for these t the law with Laplace transform R_t, a GGC with no hypothesis, is not in L_2, and its index on the window drops to one. The mechanism is that of Theorem 11.5.8, one rung lower: the term of the first atom in $W^{[3]}$ is negative where $a_1(t)x$ lies in $(0.381966, 2.618034)$, and once the first two atoms are far enough apart the second no longer compensates it; at the threshold a_2/a_1 is between 3.113 and 3.119, against 2.212 for the clock. At $t = 50$ the ratio $W^{[2]}/W$, which for $t \le 40$ decreases across the whole window, has an interior minimum, 0.276 at $-\log x = 2.682$: the same mechanism has begun to act on the second rung, which a single atom already violates.

Below the window the ratios $W^{[2]}/W$ and $W^{[3]}/W$ tend to $\frac{1}{4}$ and $\frac{1}{8}$, at $t = 0$ by (11.5.4) and for every $t \ge 0.2$ by the count of the zeros of H_t. For fixed $t > 0$ the number $N_t(T)$ of zeros of H_t in $(0, T]$ is $\Psi(T) + \frac{t}{16}\log T + O(1)$, with $\Psi(T) = \frac{T}{4\pi}\log\frac{T}{4\pi} - \frac{T}{4\pi}$: Ki, Kim and Lee proved a formula of this kind for every $t > 0$ [150, Theorem 1.4], as recalled in [289, Section 3], and the Polymath project proved it in this form, with effective constants, for $0 < t \le \frac{1}{2}$ [282, Theorem 1.5]. An error $O(\log T)$ would do. Partial summation against $d\Psi$, with the substitution $u = z^2x/4$ and (11.5.1) at $s = \frac{1}{2}$, gives

$$W^{[j]}(x;t) = \frac{\log(1/x)}{2^{j+3}\sqrt{\pi x}}\big(1 + o(1)\big) \qquad (x \downarrow 0),$$

which at $t = 0$ is the leading part of $\bar{A}_j$. The approach is slow, as the table shows, so for $t > 0$ the signs below $x = e^{-8}$ are plausible but not proved. Either way the index is set by the positions and the spacing of the first zeros: it changes along the flow where every law is a GGC with no hypothesis, and it carries no information about RH.

11.5.4.4 No branching process realisation

One of Pakes's proofs that $\mathcal{W}$ is a Thorin Bernstein function obtains the Lambert law as the limit of a branching process [234] (Section 11.5.1). Let Y be a spectrally positive Lévy process with Laplace exponent $\Psi_Y(u) = -\log \mathbb{E}e^{-uY_1}$, with the sign of Proposition 11.1.1, and with finite Malthusian parameter $\Psi_Y'(0) \in (0, \infty)$. Lamperti's time change turns Y into a continuous state branching process $(\mathsf{B}_t)_{t\ge 0}$; started at $\mathsf{B}_0 = z > 0$ it has $\mathbb{E}\mathsf{B}_t = ze^{\Psi_Y'(0)t}$, and the nonnegative martingale $e^{-\Psi_Y'(0)t}\mathsf{B}_t$ converges almost surely to a limit M. By the branching property $\mathbb{E}e^{-u\mathsf{M}} = e^{-z\omega(u)}$ for a function ω that does not depend on z. Conditioning on B_s gives $\omega(u) = v_s\big(\omega(ue^{-\Psi_Y'(0)s})\big)$, where $e^{-zv_s(\theta)} = \mathbb{E}e^{-\theta\mathsf{B}_s}$ and $\partial_s v_s = \Psi_Y(v_s)$, and differentiating at $s = 0$ gives

$$\Psi_Y'(0)\, u\, \omega'(u) = \Psi_Y\big(\omega(u)\big), \tag{11.5.5}$$

which is [235, (68)]. For the Lambert law, Y is the compound Poisson process with unit rate and standard exponential jumps, $\Psi_Y(u) = u/(1+u)$ and $\Psi_Y'(0) = 1$, and

$\omega = \mathcal{W}$ solves (11.5.5), since differentiating $\mathcal{W}e^{\mathcal{W}} = u$ gives $u\mathcal{W}' = \mathcal{W}/(1+\mathcal{W})$ [235, Section 7].

For the centre clock, or λT with $\lambda > 0$, to be such a limit we need $z\omega(u) = \log G(\lambda u)$. Then (11.5.5) determines Ψ_Y on the range $[0,\infty)$ of ω:

$$\Psi_Y(w) = \frac{\Psi_Y'(0)}{z}\, g(zw), \qquad g = \Psi_Z \circ H = \frac{H}{H'},$$

where H is the inverse function of $\log G$ and $\Psi_Z(u) = u(\log G)'(u)$ is the Laplace exponent of the driving process of Proposition 11.1.1. The function g is analytic at 0, with $g(w) = \sum_{k\ge1} g_k w^k$ and $g_1 = 1$. For every spectrally positive Y the function $-\Psi_Y''(u) = \sigma^2 + \int_0^\infty x^2 e^{-ux}\,\Pi(dx)$, with σ^2 the Gaussian coefficient and Π the Lévy measure of Y, is completely monotone. Since $\Psi_Y^{(k)}(0) = \Psi_Y'(0)\, z^{k-1} k!\, g_k$, this forces

$$(-1)^{k-1} g_k \ge 0 \qquad (k \ge 2). \tag{11.5.6}$$

The coefficients g_k need no hypothesis. The Taylor series of ξ at $\frac12$, computed in ball arithmetic, gives that of $\log G(u) = \log\xi(\frac12 + \sqrt{u}) - \log\xi(\frac12) = \sum_{n\ge1}(-1)^{n+1} m_{n-1} u^n/n$, with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; reverting the series gives H, and composing gives $g = \Psi_Z \circ H$. The program `branch_coeffs.py` does this at 3200 bits for $k \le 60$, checks it against the second route $g = H/H'$, which agrees for $k \le 59$, and checks $m_1, \dots, m_{11}$ against the sums over the first 1700 zeros. Every sign for $k \le 60$ is certified.

THEOREM 11.5.11. *With no hypothesis, $(-1)^{15} g_{16} < 0$: in ball arithmetic $g_{16} = 1.49524474945112 \times 10^{-15} \pm 2 \times 10^{-30}$. Consequently no positive multiple of the centre clock is the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter, from any initial state.*

Proof. Such a limit would satisfy (11.5.6), which the sign of g_{16} violates. □

The signs keep failing for a structural reason, Pringsheim's theorem, the power series form of Landau's theorem [340, Chapter II, Theorem 5b]: if $(-1)^{k-1} g_k \ge 0$ for all large k and the series $\sum g_k w^k$ has a finite radius of convergence R, then its sum is singular at $w = -R$. But g is analytic on a neighbourhood of the negative axis. With no hypothesis, $(\log G)'(u) = \sum_k 1/(u + a_k) > 0$ for $-\gamma_1^2 < u \le 0$: a zero on the line contributes $1/(u+\gamma^2) > 0$, and the conjugate atoms a, $\bar a$ of an off line zero together contribute $2\,\mathrm{Re}(u+a)/|u+a|^2$, with $\mathrm{Re}(u+a) \ge \gamma^2 - \frac14 - \gamma_1^2 > 0$. So $\log G$ increases from $-\infty$ to 0 on $(-\gamma_1^2, 0]$, its inverse H is analytic on a neighbourhood of $(-\infty, 0]$, and so is $g = H/H'$. Hence (11.5.6) fails for infinitely many k unless g is entire.

Numerically the radius is finite, $R = 7.1565$, and the singularities that fix it lie off the axis. The function Ξ has a critical point $t^* = 15.5857085898$ between γ_1 and γ_2, certified by bisection on Ξ', and at $u^* = -t^{*2} = -242.914312$ the function $G(u) = \Xi(\sqrt{-u})/\Xi(0)$ has $G'(u^*) = 0$ and $G(u^*) = -0.00161241048$. Continued

from 0 around the zero $-\gamma_1^2$ through the upper half plane, $\log G$ reaches u^* with the value

$$\lambda^* = \log|G(u^*)| + i\pi = -6.43002502663 + i\pi,$$

of modulus $|\lambda^*| = 7.156453$ and argument $\arg\lambda^* = 2.687121$, and since $(\log G)'(u^*) = 0$, the inverse H has a square root branch point at λ^*, and by symmetry at $\bar\lambda^*$. Continuing H numerically along the segment from 0 to λ^* in 3000 steps keeps $|G| = e^{\operatorname{Re}\lambda} \geq 0.0016$ on the path, away from the zeros of G, and brings the preimage to within 2.86 of u^*, at a distance proportional to the square root of the distance from λ^* over the last steps, as at a simple critical point; so λ^* and $\bar\lambda^*$ are singularities of the principal branch of g (a numerical continuation, not a certificate). The root test agrees: $|g_k|^{1/k}$ is 0.1269, 0.1274, 0.1296, 0.1288 and 0.1317 at $k = 20$, 30, 40, 50 and 60, approaching $1/|\lambda^*| = 0.13973$ slowly, as the factor $k^{-3/2}$ of a square root singularity predicts. The conjugate pair makes $(-1)^{k-1}g_k$ oscillate in k with period $2\pi/(\pi - \arg\lambda^*) = 13.8253$.

The next critical points of Ξ do not matter. The second, $t^{**} = 22.0979772804$ between γ_2 and γ_3, has $G(-t^{**2}) = 1.56051 \times 10^{-5} > 0$ and $\log G(-t^{**2}) = -11.0679$, and the third, 26.2722473569 between γ_3 and γ_4, has $G = -1.34670 \times 10^{-6}$. So the branching obstruction, like the end of the ladder, comes from the bottom of the spectrum: the singularity that forces the changes of sign is created by the critical point of Ξ between γ_1 and γ_2.

11.5.5 The criticality filter

Along the de Bruijn Newman flow the function R_t of Theorem 7.2.15 is the Laplace transform of a GGC exactly when $t \geq \Lambda$ (Corollary 4.2.5), and $\Lambda \geq 0$ by the theorem of Rodgers and Tao [289]. So the Pólya kernel sits at the edge of a family of kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, as close to it as one wishes, whose transforms have nonreal zeros and for which the Thorin condition fails. This subsection turns that fact into a test for the properties of the kernel that might imply RH. It applies the test to two representations of the Pólya law as a mixture, and to the property that the first author's account singles out, the GGC property of the variable of Biane, Pitman and Yor, which passes the test at every $t < 0$ tested but does not detect the Euler product. Every inequality below that is called certified is checked in ball arithmetic [130] by `pakes_filter_cert.py`.

11.5.5.1 The filter

For an even, positive, integrable function g on $\mathbb{R}$ put

$$\widehat{g}(\theta) = \int_{\mathbb{R}} g(v)\cos(\theta v)\,dv, \qquad R_g(s) = \frac{\widehat{g}(0)}{\widehat{g}(i\sqrt{s})} = \frac{\int_{\mathbb{R}} g(v)\,dv}{\int_{\mathbb{R}} g(v)\cosh(\sqrt{s}\,v)\,dv}, \quad (11.5.7)$$

for the $s \geq 0$ at which the last integral converges. For the damped Pólya kernel $\Phi_t(v) = e^{tv^2/4}\Phi(v)$, with Φ as in (1.2.3), the transform $\widehat{\Phi_t}$ is the function Ξ_t

of Section 8.7.6, and R_{Φ_t} is the function R_t of Theorem 7.2.15, written φ_t in Corollary 4.2.5; at $t = 0$ it is the clock $1/G$.

PROPOSITION 11.5.12. *Let $\mathcal{P}$ be a property of even, positive, integrable functions on $\mathbb{R}$ that Φ_t has for some $t < 0$. Then $\mathcal{P}$ implies neither that $\widehat{g}$ has only real zeros nor that R_g is the Laplace transform of a GGC, for the functions g with $\mathcal{P}$. At every $t \le 0$ the kernel Φ_t*

(i) is positive and even, and strictly decreasing on $(0, \infty)$;

(ii) is log concave;

(iii) satisfies $\log \Phi_t(v) = -\pi e^{2|v|} + O(v^2)$ as $|v| \to \infty$;

(iv) extends analytically to the strip $|\operatorname{Im} v| < \pi/4$.

So no combination of (i) to (iv) implies that $\widehat{g}$ has only real zeros, or that R_g is the Laplace transform of a GGC.

Proof. Let $t < 0$ be such that Φ_t has $\mathcal{P}$. The zeros of H_t, and so those of $\Xi_t(\theta) = 8H_t(2\theta)$, are all real if and only if $t \ge \Lambda$ (de Bruijn [73], Newman [210]), and $\Lambda \ge 0$ [289], so Ξ_t has a zero $z \notin \mathbb{R}$. It is not on the imaginary axis either, since $\Xi_t(iy) = \int \Phi_t(v) \cosh(yv)\, dv > 0$ for real y. Hence $s_0 = -z^2$ has $\operatorname{Im} s_0 = -2 \operatorname{Re} z \operatorname{Im} z \ne 0$. As Ξ_t is even and entire, $s \mapsto \Xi_t(i\sqrt{s})$ is entire, and it vanishes at s_0, so R_t is meromorphic on $\mathbb{C}$ with a pole at $s_0 \notin (-\infty, 0]$. By (F7) of Section 4.2.1 an HCM function extends analytically to $\mathbb{C} \setminus (-\infty, 0]$, and the extension would agree with R_t off its poles; so R_t is not HCM, and by (F6) it is not the Laplace transform of a GGC.

For (i), Φ is positive and even, and $\Phi_t'(v) = e^{tv^2/4}\big(\Phi'(v) + \frac{t}{2} v\Phi(v)\big) < 0$ for $v > 0$, since $\Phi' < 0$ on $(0, \infty)$ [68, Theorem A] and $tv\Phi(v) \le 0$. For (ii), $(\log \Phi_t)'' = (\log \Phi)'' + \frac{t}{2} \le (\log \Phi)'' \le 0$, since Φ is log concave [58]. Properties (iii) and (iv) hold for Φ by (1.2.3), and the factor $e^{tv^2/4}$ is entire and adds $tv^2/4$ to the logarithm. □

A property that is to imply RH through the kernel alone must fail for Φ_t at every $t < 0$; we then say that it *passes the filter*. Passing is necessary and far from sufficient: it says only that the theorem of Rodgers and Tao does not rule the property out. For the properties (i), (ii) and (iv) the conclusion needs nothing so deep: the convolution of $(1 - |u|)_+^2$ with $e^{-\cosh u}$ in Section 8.5.5 is even, positive, entire, log concave and doubly exponentially decaying, and its transform has nonreal zeros. What the filter adds is a counterexample that has every property of the list and differs from Φ only by the factor $e^{tv^2/4}$, with t as close to 0 as one wishes.

The filter is the counterpart, for properties of the kernel, of the perturbation test of Section 12.2.6. There the function F of Theorem 5.2.1 keeps the zeros

of ξ below a height R, the order, the type and the functional equation, and has four zeros off the line above R; here Φ_t keeps every property of the kernel that a Gaussian factor $e^{tv^2/4}$ with $t<0$ preserves, and has nonreal zeros by the theorem of Rodgers and Tao, which does not say where. For $t>0$ the zeros of H_t obey a formula of Riemann and von Mangoldt type, $N_t(T)=\Psi(T)+\frac{t}{16}\log T+O(1)$ (Section 11.5.4.3); for $t<0$, the case the filter needs, we know of no count in the literature: Rodgers and Tao prove one for $\Lambda<t\le 0$, with the error $O(\log^2 T)$ [289, Section 3], as a step in showing that this range of t is empty. A proof of RH must therefore use, at some step, a property that Φ_t lacks for every $t<0$. The Euler product and the explicit formula are the natural candidates, as they are for F in Section 12.2.6, but this is a reading and not a theorem: no Euler product for H_t with $t\neq 0$ is known (see the discussion after Theorem 7.2.15), and none is known not to exist. The Lambert proof of Section 11.5.3 is untouched by the filter, since it concerns a single function, whose analyticity off the cut is forced by the equation $\mathcal{W}e^{\mathcal{W}}=u$, and no family of comparison functions sharing its properties is in sight.

11.5.5.2 Mixtures of arcsine laws

For $t\le 0$ let X_t be the variable with density $p_t=\Phi_t/\Xi_t(0)$, so that X_0 is the Pólya variable X. Since p_t is even and decreasing on $(0,\infty)$ by Proposition 11.5.12(i), Khinchine's theorem writes $X_t=UQ_t$ with U uniform on $[-1,1]$ and $Q_t\ge 0$ independent of U, of density $-2rp_t'(r)$; at $t=0$ this is the mixture of uniform laws of Section 13.3.2.

Proposition 11.5.13. *For every $t\le 0$ the function*

$$h_t(\rho)=-2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}},\qquad \rho>0,$$

is a positive probability density, and

$$p_t(v)=\frac{1}{\pi}\int_{|v|}^\infty \frac{h_t(\rho)\,d\rho}{\sqrt{\rho^2-v^2}},\qquad v\neq 0.$$

Equivalently, $X_t\overset{d}{=}\varrho_t\cos\Theta$ with Θ uniform on $[0,2\pi)$ and ϱ_t of density h_t independent of Θ: X_t is a mixture of arcsine laws, and it is the first coordinate of the rotation invariant vector $\varrho_t(\cos\Theta,\sin\Theta)$ in $\mathbb{R}^2$, as well as of a rotation invariant vector in $\mathbb{R}^3$.

Proof. By Archimedes' theorem the coordinates of a uniform point of the unit sphere in $\mathbb{R}^3$ are uniform on $[-1,1]$. Writing the point as $(B\cos\Theta,B\sin\Theta,V)$, with V uniform on $[-1,1]$, $B=\sqrt{1-V^2}$ and Θ uniform on $[0,2\pi)$ independent of V, the first coordinate gives $U\overset{d}{=}B\cos\Theta$, where B has density $b/\sqrt{1-b^2}$ on

$(0,1)$. Hence $X_t = UQ_t \overset{d}{=} (BQ_t)\cos\Theta$, and Q_t times the uniform point is a rotation invariant vector in $\mathbb{R}^3$ with first coordinate X_t. The density of $\varrho_t = BQ_t$ is

$$\int_\rho^\infty \big(-2rp_t'(r)\big)\,\frac{\rho/r}{\sqrt{1-\rho^2/r^2}}\,\frac{dr}{r} = -2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}} = h_t(\rho),$$

positive since $p_t' < 0$ on $(0,\infty)$. Given $\varrho_t = \rho$, the variable $\rho\cos\Theta$ has the arcsine density $1/(\pi\sqrt{\rho^2-v^2})$ on $(-\rho,\rho)$, which gives the formula for p_t. □

At $t = 0$ the proposition is the arcsine route of [252, Chapter 12] recalled in Section 5.2, which writes X as $2\sqrt{I}$ times the cosine of an independent angle; since $\mathbb{E}|\cos\Theta|^s \neq 0$ for $s > -1$, the Mellin transform of $|X|$ determines the law of the radius, so $\varrho_0 \overset{d}{=} 2\sqrt{I}$, and h_0 is the density of the variable I of that route after the change of variable. Both representations, the uniform mixture and the arcsine mixture, hold at every $t \le 0$, so by Proposition 11.5.12 neither can decide RH, or the Thorin condition, by itself: whatever is to be proved about Q_0 or ϱ_0 must use something that Q_t or ϱ_t lacks for $t < 0$. The scale mixtures of Gamma laws, the other mixture route of [252, Section 12.6], are excluded already at $t = 0$, for every shape (Theorem 11.2.10).

11.5.5.3 The variable of Biane, Pitman and Yor under damping

Let $\mathcal{B} = \frac{\pi}{2}S_2$, where $S_2 = J + J'$ is the sum of two independent hitting times of 1 by a three dimensional Bessel process started at 0, as in (9.1.1) and Theorem 7.1.21. Then

$$\mathbb{E}e^{-\lambda\mathcal{B}} = \Big(\frac{\sqrt{\pi\lambda}}{\sinh\sqrt{\pi\lambda}}\Big)^2 = \prod_{m\ge1}\Big(1+\frac{\lambda}{\pi m^2}\Big)^{-2}, \qquad \mathbb{E}\mathcal{B}^{s/2} = 2\xi(s), \tag{11.5.8}$$

so $\mathcal{B}$ is a GGC with Thorin measure $2\sum_m \delta_{\pi m^2}$, and its density is

$$f(b) = \sum_{m\ge1}\big(4\pi^2m^4b - 6\pi m^2\big)e^{-\pi m^2 b} = b^{-5/4}\,\Phi\big(\tfrac12\log b\big), \qquad b > 0.$$

More generally, for an even positive kernel g with $\int g(v)e^{|v|/2}\,dv < \infty$ let $\mathcal{B}_g$ have the density proportional to $b^{-5/4}g(\frac12\log b)$; the Pólya variable of g, of density $g/\widehat{g}(0)$, is $\frac12\log\mathcal{B}_g$ tilted by $\mathcal{B}_g^{1/4}$, and $\mathcal{B}_\Phi = \mathcal{B}$ (Theorem 7.1.21). For the damped kernels write $\mathcal{B}_t = \mathcal{B}_{\Phi_t}$: its density is proportional to $e^{t(\log b)^2/16}f(b)$, finite for every real t because $f(b)$ decays like $e^{-\pi b}$ as $b\to\infty$ and, up to a power of b, like $e^{-\pi/b}$ as $b\downarrow 0$, and $\mathbb{E}\mathcal{B}_t^{s/2} = \Xi_t(i(\frac12 - s))/\Xi_t(\frac{i}{2})$. Damping the kernel thus tilts $\mathcal{B}$ by a lognormal factor, and "$\mathcal{B}_g$ is a GGC" is a property of the kernel g that Φ has. The filter asks whether Φ_t lacks it for every $t < 0$.

Two tests are available for a positive variable $\mathcal{V}$ with $\ell(\lambda) = \mathbb{E}e^{-\lambda\mathcal{V}}$. Let $\kappa_k(\lambda) = (-1)^k\frac{d^k}{d\lambda^k}\log\ell(\lambda)$, $k \ge 1$, be the cumulants of the law $e^{-\lambda b}\,P(\mathcal{V} \in$

$db)/\ell(\lambda)$. The variable is infinitely divisible if and only if $-\log \ell$ is a Bernstein function, that is, if and only if $\kappa_k(\lambda) \geq 0$ for all $k \geq 1$ and $\lambda > 0$ [299]. If it is a GGC, with $-\log \ell(\lambda) = d\lambda + \int \log(1 + \lambda/z)\, U(dz)$, then for $k \geq 0$

$$c_k(\lambda) = \frac{\kappa_{k+1}(\lambda)}{k!} = d\,\delta_{k0} + \int \frac{U(dz)}{(\lambda + z)^{k+1}} = \int x^k\, \sigma_\lambda(dx), \tag{11.5.9}$$

where σ_λ is $d\,\delta_0$ plus the image of $(\lambda + z)^{-1}U(dz)$ under $z \mapsto (\lambda + z)^{-1}$. Hence the Hankel matrices $\mathcal{K}_n(\lambda) = [c_{i+j}(\lambda)]_{i,j=0}^{n-1}$ and $\mathcal{K}'_n(\lambda) = [c_{i+j+1}(\lambda)]_{i,j=0}^{n-1}$ are positive semidefinite for every n and $\lambda \geq 0$, since $\sum_{i,j} y_i y_j c_{i+j+r}(\lambda) = \int x^r \big(\sum_i y_i x^i\big)^2 \sigma_\lambda(dx)$ for $r = 0, 1$, and a negative eigenvalue of one of them proves that $\mathcal{V}$ is not a GGC. The smallest case is already useful.

LEMMA 11.5.14. *If $\mathcal{V}$ is a GGC with a finite fourth moment, its cumulants $\kappa_k = \kappa_k(0)$ satisfy $2\kappa_2\kappa_4 \geq 3\kappa_3^2$, with equality for the gamma laws.*

Proof. With $c_k = \kappa_{k+1}/k!$ the inequality reads $c_1 c_3 \geq c_2^2$, which is $\det \mathcal{K}'_2(0) \geq 0$. For a gamma law U is a single atom $\alpha\delta_z$ and $c_k = \alpha z^{-k-1}$, so equality holds. □

The program `pakes_filter.py` computes the moments

$$\int_{\mathbb{R}} e^{2kv}\, e^{-\lambda e^{2v}}\, e^{tv^2/4}\, e^{-v/2}\, \Phi(v)\, dv,$$

which are those of $\mathcal{B}_t$ under the law of the cumulants $\kappa_k(\lambda)$ up to normalisation, by the trapezoid rule on $[-5, 5]$. The program forms $\kappa_1, \ldots, \kappa_{26}$, at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200, and the matrices $\mathcal{K}_n(\lambda)$ and $\mathcal{K}'_n(\lambda)$ of orders $n \leq 12$. Dividing c_{i+j+r} by c^{i+j+r}, with $c = c_1/c_0$, is a congruence that keeps the inertia, and the eigenvalues quoted below are those of the matrices so scaled. The program `pakes_filter_cert.py` certifies the failures: it encloses the moments in balls (Arb, 256 bits, each tail bounded by 2^{-3000}), computes the cumulants and the scaled matrix M in ball arithmetic, and encloses $y^{\mathsf{T}} M y$ for a rational vector y close to the eigenvector of the least eigenvalue; an enclosure inside $(-\infty, 0)$ proves that M is not positive semidefinite. In every certificate below the cumulants used are certified positive and $y^{\mathsf{T}} M y$ agrees with the computed least eigenvalue to all the digits quoted.

Infinite divisibility shows nothing. In all 301 cases computed, the tilts $\mathcal{B}_t$ for the twenty values of t of the next paragraph, the reciprocals $1/\mathcal{B}_t$ for $t = -1$, 0, $\frac{1}{5}$, and the ten functions other than ξ of Table 11.19 below, on both sides, each at the seven values of λ, none of $\kappa_1, \ldots, \kappa_{26}$ is negative.

The GGC test separates $t = 0$ from every other value. It was run for $t = -20$, -10, -5, -2, -1, $-\frac{1}{2}$, $-\frac{1}{5}$, $-\frac{1}{10}$, $-\frac{1}{20}$, $-\frac{1}{50}$, $-\frac{1}{100}$, 0, $\frac{1}{100}$, $\frac{1}{50}$, $\frac{1}{20}$, $\frac{1}{10}$, $\frac{1}{5}$, $\frac{1}{2}$, 1 and 5. At $t = 0$ no matrix of order at most 12 has a negative eigenvalue, as the theorem requires, though the matrices are nearly singular. For every $t \neq 0$ some $\mathcal{K}'_n(\lambda)$ with $n \leq 12$ has a negative eigenvalue at each $\lambda \leq 50$. Table 11.18 shows eleven of the twenty values; the other nine give the same picture.

t	$\lambda=0$	$\frac{1}{2}$	2	10	20	50	eigenvalue at $\lambda=2$
-20	8	9	8	8	7	9	-1.582×10^{-7}
-5	8	8	7	7	8	10	-8.095×10^{-8}
-1	8	8	7	7	8	11	-1.533×10^{-8}
$-\frac{1}{2}$	8	8	7	7	9	11	-4.725×10^{-9}
$-\frac{1}{10}$	8	8	8	8	9	12	-1.800×10^{-9}
$-\frac{1}{100}$	9	9	8	8	10	12	-4.119×10^{-11}
0	none	none	none	none	none	none	1.247×10^{-13}
$\frac{1}{100}$	4	4	5	6	7	10	-2.357×10^{-5}
$\frac{1}{5}$	4	4	4	5	7	9	-1.101×10^{-4}
1	3	3	4	5	6	8	-2.102×10^{-3}
5	3	3	3	5	6	8	-2.141×10^{-3}

Table 11.18: The GGC test for the tilted variables $\mathcal{B}_t$ of Section 11.5.5.3: the least order $n\le12$ at which $\mathcal{K}_n'(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}_n'(2)$ at that order; for $t=0$, where no order fails, the last column gives $\mathcal{K}_7'(2)$. At $\lambda=200$ no matrix of order at most 12 fails, for any t. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}_9'(0)$ at $t=-\frac{1}{100}$, -2.899×10^{-12}. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

Proposition 11.5.15. *(i) $\mathcal{B}_t$ is not a GGC for $t=-20$, -5, -1, $-\frac{1}{2}$, $-\frac{1}{10}$, $-\frac{1}{100}$, $\frac{1}{100}$, $\frac{1}{5}$, 1 and 5.*

(ii) There is $t_0>0$ such that $\mathcal{B}_t$ is not a GGC for $0<|t|<t_0$.

Proof. (i) For each of these t the matrix $\mathcal{K}_n'(2)$ at the order n of Table 11.18 is certified not to be positive semidefinite.

(ii) Put $\ell_t(\lambda)=\mathbb{E}e^{-\lambda\mathcal{B}_t}$ and split $\int_0^\infty e^{-\lambda b}e^{t(\log b)^2/16}f(b)\,db$, the numerator of ℓ_t, at $b=1$. The part on $(0,1]$ is entire in $(t,\lambda)\in\mathbb{C}^2$, since $|e^{t(\log b)^2/16}|=e^{O((\log b)^2)}$ is negligible against $f(b)=O(b^{-7/2}e^{-\pi/b})$ as $b\downarrow0$, and the denominator, the value at $\lambda=0$, is analytic in t and positive for real t. On $[1,\infty)$ write f as the sum of its terms. Fix $M\ge1$ and x with $\pi M^2<x<\pi(M+1)^2$. The terms with $m>M$ give a function analytic in $t\in\mathbb{C}$ and $\operatorname{Re}\lambda>-\pi(M+1)^2$. For $m\le M$ turn the path $[1,\infty)$ into the ray $b=1+re^{-3\pi i/4}$, $r\ge0$, which avoids $(-\infty,0]$ and along which $(\log b)^2$ grows like $(\log r)^2$. Writing $\lambda+\pi m^2=u+iv$, the integrand decays along every ray $b=1+re^{i\alpha}$ with $-\frac{3\pi}{4}\le\alpha\le0$ when $v>u>0$, so there the rotated integral equals the original one; and it converges, and is analytic in (t,λ), on the half plane $v>u$, which contains $\lambda=-x$. So ℓ_t continues analytically, jointly in (t,λ) near $(0,-x)$, from the upper half plane across the negative axis at $-x$.

Suppose that $\mathcal{B}_t$ is a GGC, with Thorin measure U_t. Then $-\log\ell_t(\lambda)=d_t\lambda+\int\log(1+\lambda/z)\,U_t(dz)$ on $\mathbb{C}\setminus(-\infty,0]$, and as $\varepsilon\downarrow0$ its imaginary part at $\lambda=-y+i\varepsilon$ tends to $\pi\big(U_t((0,y))+\frac{1}{2}U_t(\{y\})\big)$. For t near 0, ℓ_t has no zero near

$-x$, since $\ell_0(-x) = \pi x/\sin^2\sqrt{\pi x} > 0$ by (11.5.8). So the limit is a real analytic function of y near x, U_t has no atom there, and $U_t((0,y)) = -\frac{1}{\pi}\arg \ell_t(-y+i0)$, with the argument continued from 0 on $(0,\infty)$ through the upper half plane, is nondecreasing in y; at $t=0$ it equals $2M$. Write $\ell_t = \ell_0 + t\big(\dot\ell - \dot\ell(0)\,\ell_0\big) + O(t^2)$, with $\dot\ell(\lambda) = \frac{1}{16}\int_0^\infty e^{-\lambda b}(\log b)^2 f(b)\,db$ continued in the same way; by joint analyticity the remainder is $O(t^2)$ together with its derivative in λ, uniformly near $-x$. Since ℓ_0 and $\dot\ell(0)$ are real there,

$$U_t((0,y)) = 2M - \frac{t}{\pi}\mathcal{J}(y) + O(t^2), \qquad \mathcal{J}(y) = \frac{\operatorname{Im}\dot\ell(-y+i0)}{\ell_0(-y)},$$

with the same expansion for the derivatives in y. In $\dot\ell(-y+i0)$ the part on $(0,1]$ and the terms with $m > M$ are real, and for $m \le M$ the integral over the rotated ray differs by a real number from $\int_0^\infty e^{-\sigma b}(\log b)^2(4\pi^2m^4b - 6\pi m^2)\,db$ with $\sigma = -s_m + i0$, $s_m = y - \pi m^2$, which is a combination of $\partial_a^2\big[\Gamma(a)\sigma^{-a}\big]$ at $a=2$ and $a=1$, with $\log\sigma = \log s_m + i\pi$. This gives

$$\operatorname{Im}\dot\ell(-y+i0) = \frac{\pi}{8}\sum_{m\le M}\Big(\frac{4\pi^2m^4(\log s_m - 1 + \gamma_E)}{s_m^2} + \frac{6\pi m^2(\log s_m + \gamma_E)}{s_m}\Big),$$

where γ_E is Euler's constant. At $y = 1.015\pi$, where $M=1$, the certificate program gives $\mathcal{J}'(y) = 8.52630079173 \pm 3.4\times10^{-12}$, and at $y = 2.47\pi$ it gives $\mathcal{J}'(y) = -0.0671813667518 \pm 10^{-14}$. So for small $t>0$ the derivative of $U_t((0,y))$ at $y = 1.015\pi$ is $-\frac{t}{\pi}\mathcal{J}'(1.015\pi) + O(t^2) < 0$, and for small $t<0$ the derivative at $y = 2.47\pi$ is negative; both contradict monotonicity. □

The program `pakes_characters.py` scans $\mathcal{J}'$ at 399 points in each of the first four gaps $(\pi M^2, \pi(M+1)^2)$: in each gap $\mathcal{J}'$ is positive on a left part and negative on a right part, for y/π in $[1.8925, 3.9925]$, $[5.5125, 8.9875]$, $[11.1, 15.9825]$ and $[18.6775, 24.9775]$, so every one of these gaps excludes both signs of t on its own.

Remark 11.5.16. The variable $\mathcal{B}$ is a GGC, but its density f is not HCM. Indeed f is analytic in $\operatorname{Re} b > 0$ with the natural boundary $\operatorname{Re} b = 0$ (the discussion after Theorem 7.1.21), while an HCM function extends analytically to $\mathbb{C}\setminus(-\infty,0]$ by (F7).

The GGC property of $\mathcal{B}_g$ passes the filter wherever it has been tested, and it is the only property in this subsection that does; whether $\mathcal{B}_t$ fails to be a GGC for every $t<0$ is open (Problem 11.2). It is not a property of the zeros alone: for $t \ge 0.2$ the zeros of Ξ_t are all real [247] and R_t is the Laplace transform of a GGC (Theorem 7.2.15), yet $\mathcal{B}_t$ is not a GGC at $t = \frac{1}{5}$, 1 and 5.

11.5.5.4 Theta kernels, characters and the function of Davenport and Heilbronn

The variable $\mathcal{B}_g$ can be formed for other Dirichlet series with a functional equation. Let $q \ge 3$, $p \in \{0,1\}$, and let $a = (a_n)_{n\ge1}$ be a real sequence of period q with

$a_1 = 1$ and $a_{q-n} = (-1)^p a_n$ whose theta series

$$\vartheta_a(x) = \sum_{n\ge1} n^p a_n e^{-\pi n^2 x/q}, \qquad x > 0,$$

satisfies $\vartheta_a(1/x) = x^{p+1/2}\vartheta_a(x)$. Then

$$\xi(s,a) = \int_0^\infty \vartheta_a(x)\, x^{(s+p)/2}\, \frac{dx}{x} = \Big(\frac{q}{\pi}\Big)^{(s+p)/2}\Gamma\Big(\frac{s+p}{2}\Big)\sum_{n\ge1}\frac{a_n}{n^s}$$

is entire with $\xi(s,a) = \xi(1-s,a)$, and $K_a(v) = 2e^{(p+1/2)v}\vartheta_a(e^{2v})$ is an even kernel with $\xi(\frac12 + s, a) = \int_{\mathbb{R}} K_a(v)e^{sv}\,dv$. The real primitive characters χ_d, d a fundamental discriminant, are of this kind, with $q = |d|$ and $p = 1$ for $d < 0$; for χ_{-4}, written χ_4 in Section 8.5.7, $\vartheta_a(x) = \eta(ix)^3$ and K_a is the kernel K_4 of Proposition 8.5.21. When $\vartheta_a > 0$, the variable $\mathcal{B}_a = \mathcal{B}_{K_a}$ has the density $b^{p/2-1}\vartheta_a(b)/\xi(0,a)$ and $\mathbb{E}\mathcal{B}_a^{s/2} = \xi(s,a)/\xi(0,a)$, and by the functional equation its reciprocal has the density $x^{(p-1)/2}\vartheta_a(x)/\xi(0,a)$. We test both. For χ_{-4} the reciprocal $1/\mathcal{B}_a$ is a GGC: it is $\frac{\pi}{2}$ times the exit time of Remark 8.5.22. Whether $\mathcal{B}_{\chi_{-4}}$ itself is a GGC is open, and HCM cannot decide it: the density of $1/\mathcal{B}_{\chi_{-4}}$ is not HCM (Theorem 4.3.20), so by (F2) neither is that of $\mathcal{B}_{\chi_{-4}}$.

Lemma 11.5.17. *Let $S_a = \sum_{n\ge2} n^p|a_n|e^{-\pi(n^2-1)/q}$. If $S_a < 1$, then $\vartheta_a > 0$ on $(0,\infty)$.*

Proof. For $x \ge 1$, $\vartheta_a(x) = e^{-\pi x/q}\big(1 + \sum_{n\ge2} n^p a_n e^{-\pi(n^2-1)x/q}\big) \ge e^{-\pi x/q}(1 - S_a) > 0$, and for $0 < x < 1$, $\vartheta_a(x) = x^{-p-1/2}\vartheta_a(1/x) > 0$. □

The certificate program encloses S_a in ball arithmetic: it is at most 0.659 for the nine characters and the function of Davenport and Heilbronn below, so these kernels are positive. Positivity can fail: for $d = -19$ and $d = -40$ the program `pakes_filter.py` finds $e^{\pi/q}\vartheta_a(1) = -0.567$ and -0.421. For $d = -19 \equiv 5 \pmod{24}$ this is a case of Theorem 8.5.18, and kernels of this kind go back to Pólya's examples recalled in [32, Section 6.1].

The function of Davenport and Heilbronn [72] is $f_{\mathrm{DH}}(s) = \sum_n a_n n^{-s}$ with a of period 5 and $(a_1,\dots,a_5) = (1, \varkappa, -\varkappa, -1, 0)$, where

$$\varkappa = \frac{\sqrt{10 - 2\sqrt5} - 2}{\sqrt5 - 1} = 0.2840790438404122960283.$$

With χ the character modulo 5 with $\chi(2) = i$, $f_{\mathrm{DH}} = \frac{1-i\varkappa}{2}L(s,\chi) + \frac{1+i\varkappa}{2}L(s,\bar\chi)$, and $\varkappa$ is chosen so that $(1+i\varkappa)/(1-i\varkappa)$ is the root number

$$\varepsilon(\chi) = 0.85065080835203993218 + 0.52573111211913360603\,i;$$

then $p = 1$ and $\vartheta_a(1/x) = x^{3/2}\vartheta_a(x)$. The coefficients are not multiplicative, since $a_6 = 1$ while $a_2 a_3 = -\varkappa^2$, so f_{DH} has no Euler product, and it has infinitely many zeros in $\operatorname{Re} s > 1$ [324, Chapter 10]. Zeros off the critical line inside the strip have been computed by Spira [310] and by Balanzario and Sánchez-Ortiz [15]; the program confirms two of them,

$$0.808517182456637 + 85.6993484853776\,i,$$
$$0.650830080609737 + 114.163342730757\,i,$$

where $|f_{\mathrm{DH}}| < 10^{-25}$. Modulo 5 there is no other function of the class built from χ and $\bar\chi$: a combination $\alpha L(s,\chi) + \beta L(s,\bar\chi)$ with $a_1 = 1$ whose completed function is w times its reflection must have $w = \pm 1$ and $\beta = w\,\varepsilon(\chi)\alpha$, and the choice $w = -1$ gives the coefficients $(1, -1/\varkappa, 1/\varkappa, -1, 0)$, whose theta series satisfies $\vartheta(1/x) = -x^{3/2}\vartheta(x)$, so that it vanishes at $x = 1$ and takes both signs.

Table 11.19 gives the test for ξ, for nine real characters and for f_{DH}. For the characters and for f_{DH} the moments are computed as for $\mathcal{B}_t$, from the kernels K_a, and at $\lambda = 0$ also from the values of L functions alone: $\mathbb{E}\mathcal{B}_a^k = \xi(2k,a)/\xi(0,a)$ and $\mathbb{E}\mathcal{B}_a^{-k} = \xi(2k+1,a)/\xi(0,a)$, with $\sum_n a_n n^{-s} = q^{-s}\sum_{r=1}^{q} a_r \zeta(s, r/q)$ through the Hurwitz zeta function. The two routes agree, and the failures marked in the table are certified, by the first route for $\lambda > 0$ and by the second at $\lambda = 0$.

function	S_a	$\mathcal{K}_7'(0)$	$\mathcal{K}_7'(2)$	$\mathcal{B}_a$	$1/\mathcal{B}_a$
ξ		6.2×10^{-16}	1.2×10^{-13}	GGC	fails at $\mathcal{K}_3'(0)$
χ_{-3}	0.086	2.4×10^{-8}	4.9×10^{-8}	passes	GGC
χ_{-4}	0.006	1.2×10^{-8}	4.6×10^{-8}	passes	GGC
χ_5	0.158	7.8×10^{-8}	7.1×10^{-8}	passes	fails at $\mathcal{K}_{10}'(0)$
χ_8	0.043	4.8×10^{-8}	3.8×10^{-8}	passes	fails at $\mathcal{K}_7'(2)$
χ_{12}	0.002	2.1×10^{-8}	4.2×10^{-8}	passes	fails at $\mathcal{K}_9'(10)$
χ_{24}	0.045	7.5×10^{-9}	2.0×10^{-8}	fails at $\mathcal{K}_{11}'(2)$	fails at $\mathcal{K}_6'(0)$
χ_{-7}	0.608	-0.149	-0.221	fails at $\mathcal{K}_3'(0)$	fails, zero
χ_{-8}	0.130	-0.057	-0.477	fails at $\mathcal{K}_2'(0)$	ID, fails, zero
χ_{13}	0.659	-0.0023	-0.256	fails at $\mathcal{K}_3'(0)$	fails at $\mathcal{K}_3'(0)$
f_{DH}	0.092	-1.2×10^{-5}	-1.4×10^{-4}	fails at $\mathcal{K}_6'(0)$	fails, zero

Table 11.19: The GGC test for the variables $\mathcal{B}_a$ of Section 11.5.5.4 and their reciprocals: the sum S_a of Lemma 11.5.17, the least eigenvalues of the scaled matrices $\mathcal{K}_7'(0)$ and $\mathcal{K}_7'(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}_n'(\lambda)$ with $n \le 12$ has a negative eigenvalue at $\lambda = 0$, $\frac12$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}_n'(\lambda)$": the least failing order, certified at that λ. "Zero": the Laplace transform vanishes on $(-\infty, 0)$ (Proposition 11.5.19). GGC for ξ by (11.5.8), for $1/\mathcal{B}_a$ with χ_{-3} and χ_{-4} by Proposition 11.5.19. All the functions but f_{DH} have an Euler product. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

Proposition 11.5.18. *Neither $\mathcal{B}_{\chi_{-8}}$ nor $1/\mathcal{B}_{\chi_{-8}}$ is a GGC.*

Proof. From the values of $L(s,\chi_{-8})$ at $s=0,2,4,6,8$ and at $s=3,5,7,9$, the certificate program gives

$$3\kappa_3^2-2\kappa_2\kappa_4=7.55465911093\pm4\times10^{-12}\qquad\text{and}\qquad 78.8854168657\pm2\times10^{-11}$$

for $\mathcal{B}_{\chi_{-8}}$, where $(\kappa_2,\kappa_3,\kappa_4)=(3.0767,15.997,123.54)$, and for its reciprocal. Both are positive, against Lemma 11.5.14. □

The same four numbers settle $1/\mathcal{B}_{\chi_{-7}}$ (27.8926758865) and $1/\mathcal{B}_a$ for f_{DH} (0.738104955084). So $L(s,\chi_{-8})$, which has an Euler product and whose zeros lie on the critical line to height $10^8/8$ at least [245], fails the test already at order two, on both sides.

Proposition 11.5.19. *Let $p=1$ and $\vartheta_a>0$, and put $w=\sqrt{\pi q\lambda}$ and $\Sigma_a(w)=\sum_{1\le n<q/2}a_n\sinh\frac{(q-2n)w}{q}$. Then*

$$\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx=\frac{\sqrt{q}\,\Sigma_a(w)}{\sinh w},\qquad \mathbb{E}e^{-\lambda/\mathcal{B}_a}=\frac{\sqrt{q}\,\Sigma_a(w)}{\xi(0,a)\sinh w},\tag{11.5.10}$$

and $\xi(0,a)=q^{-1/2}\sum_{1\le n<q/2}(q-2n)a_n$. In particular:

(i) for χ_{-4}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\operatorname{sech}\sqrt{\pi\lambda}$, and $1/\mathcal{B}_a$, which is $\frac{\pi}{2}$ times the exit time of Brownian motion from $(-1,1)$, is a GGC;

(ii) for χ_{-3}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=3\sinh(w/3)/\sinh w=\prod_{3\nmid m}\bigl(1+3\lambda/\pi m^2\bigr)^{-1}$, so $1/\mathcal{B}_a$ is a GGC with Thorin measure $\sum_{3\nmid m}\delta_{\pi m^2/3}$, and it is $\frac{3\pi}{2}$ times the hitting time of 1 by a three dimensional Bessel process started at $\frac13$;

(iii) for χ_{-8}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\cosh(w/4)/\cosh(w/2)=\prod_{k\ge1}(1+\lambda/4\alpha_k)/(1+\lambda/\alpha_k)$ with $\alpha_k=\frac{\pi}{8}(2k-1)^2$, so $1/\mathcal{B}_a$ is infinitely divisible, with Lévy density $\sum_k(e^{-\alpha_k x}-e^{-4\alpha_k x})/x>0$, and it is π times the hitting time of 1 by a reflected Brownian motion started at $\frac12$; it is not a GGC, because its Laplace transform vanishes at $\lambda=-\pi/2$;

(iv) if $a=\chi_d$ with $d<0$ and $\chi_d(2)=1$, that is $d\equiv1\pmod8$, the Laplace transform of $1/\mathcal{B}_a$ vanishes at a point of $(-4\pi/q,-\pi/q)$, so $1/\mathcal{B}_a$ is not a GGC; for χ_{-7} the point is $\lambda=-1.033610573$. The same holds for f_{DH}, at $\lambda=-2.046575671$.

Proof. By the functional equation $\vartheta_a(x)=x^{-3/2}\sum_n na_ne^{-\pi n^2/qx}$, and

$$\int_0^\infty e^{-\lambda x-c/x}x^{-3/2}\,dx=\sqrt{\pi/c}\,e^{-2\sqrt{c\lambda}}$$

with $c=\pi n^2/q$ gives $\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx=\sqrt{q}\sum_{n\ge1}a_ne^{-2nw/q}$ for $\lambda>0$. Summing over a period and pairing n with $q-n$, where $a_{q-n}=-a_n$, gives the first formula

of (11.5.10), which is [32, (96)] in our normalisation; at $\lambda = 0$ it gives $\int_0^\infty \vartheta_a = \xi(1,a) = \xi(0,a)$, and since $1/\mathcal{B}_a$ has the density $\vartheta_a/\xi(0,a)$ the second formula follows. (i) and (ii) follow from the product formulas for cosh and sinh, and [32, (92)] identifies the transforms in (ii) and (iii) with those of the hitting times. In (iii) Frullani's integral $\log(1+\lambda/c) = \int_0^\infty (1-e^{-\lambda x})e^{-cx}\,dx/x$ gives the Lévy density. A GGC transform ℓ cannot vanish on $(-\infty,0)$ where it extends meromorphically: for $0<\varepsilon\le 1$, $|\ell(-x+i\varepsilon)| \ge e^{dx-C}$ with $C = \int \log(1+(x+1)/z)\,U(dz) < \infty$, because $|1+(-x+i\varepsilon)/z| \le 1+(x+1)/z$. In (iii) the zero is that of $\cosh(w/4)$ at $w = 2\pi i$. In (iv) put $w = iy$; then $\Sigma_a(iy)/\sinh(iy) = \nu(y)/\sin y$ with $\nu(y) = \sum_{1\le n<q/2} a_n \sin\frac{(q-2n)y}{q}$, and since $a_{q-n} = -a_n$, $\nu(\pi) = \frac12\sum_{n=1}^{q-1} a_n \sin\frac{2\pi n}{q}$ and $\nu(2\pi) = -\frac12\sum_{n=1}^{q-1} a_n \sin\frac{4\pi n}{q}$. For an odd real primitive character the Gauss sum is $i\sqrt{q}$, so $\nu(\pi) = \frac{\sqrt{q}}{2}$ and $\nu(2\pi) = -\chi_d(2)\frac{\sqrt{q}}{2}$. If $\chi_d(2)=1$, ν vanishes at some $y_0 \in (\pi, 2\pi)$, where $\sin y_0 \ne 0$, and the transform vanishes at $\lambda = -y_0^2/\pi q$. For f_{DH}, $\nu(\pi) = \frac{\sqrt5}{2}$ and $\nu(2\pi) = -0.317610026505$. □

A survey extends the table. For each of the 26 fundamental discriminants d with $3 \le |d| \le 40$, the program `pakes_filter.py` tests the positivity of ϑ_{χ_d} and, where it holds, runs the test at 80 digits to order 10 at $\lambda = 0$, $\frac12$, 2, 10 and 20, on both sides. Of the 24 positive kernels, $\mathcal{B}_{\chi_d}$ passes only for $d = -4$, -3, 5, 8, 12 and 24 (the last fails at order 11, beyond the survey), and $1/\mathcal{B}_{\chi_d}$ only for $d = -4$ and -3. The 18 failures of $\mathcal{B}_{\chi_d}$ occur at order 2 or 3 for $d = -39$, -35, -31, -24, -23, -20, -15, -11, -8, -7, 13, 21, 29 and 37, and at orders 6 to 8 for $d = 17$, 28, 33 and 40; the failures of $1/\mathcal{B}_{\chi_d}$ for $d = -7$, -15, -23, -31 and -39 are cases of Proposition 11.5.19(iv).

Combinations of L functions do not change the picture. Modulo 7 the odd characters are χ_{-7} and a complex pair χ, $\bar\chi$, with $\chi(3) = e^{\pi i/3}$, and for real r

$$f_r = \big(1 - 2r\cos\tfrac{\phi}{2}\big)L(s,\chi_{-7}) + re^{-i\phi/2}L(s,\chi) + re^{i\phi/2}L(s,\bar\chi), \qquad e^{i\phi} = \varepsilon(\chi),$$

has real coefficients, $a_1 = 1$ and $\vartheta(1/x) = x^{3/2}\vartheta(x)$. For 17 values of r between -3 and 3 the kernel is positive for $-3 \le r \le 2$ and not for $r = 3$, and for each of the 16 positive kernels both $\mathcal{B}$ and $1/\mathcal{B}$ fail at $\lambda = 0$, at order 2 or 3. None of the characters whose variable passes has a complex character of the same conductor and parity to combine with, so the smallest family through a passing character would need a larger modulus.

The pattern is not that of the Euler product. All nine L functions of Table 11.19 have one, and GRH has been verified for them to height at least $10^8/q$ [245]; yet $\mathcal{B}_{\chi_d}$ fails for χ_{24}, χ_{-7}, χ_{-8} and χ_{13}, and for 18 of the 24 positive kernels of the survey. It fails for f_{DH}, which has zeros off the line, but it fails in the same way for χ_{-7} and χ_{13}, whose zeros are on the line as far as they have been computed. At $\lambda = 0$ the test of order n reads only the moments $\xi(2k,a)/\xi(0,a)$ with $1 \le k \le 2n$, whose numerators are values of Dirichlet series where they converge

absolutely, and for $\lambda > 0$ it reads Laplace transforms of the theta series: it sees the shape of the theta transform and not the arithmetic of the coefficients. Where the GGC property is proved, it comes from a product formula for that transform, $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$ for ξ and the products of Proposition 11.5.19(i) and (ii) for the reciprocals of χ_{-4} and χ_{-3}. The computations find no function with $\mathcal{B}_a$ a GGC and a zero off the line.

11.5.6 What survives, and open problems

Apart from the Lambert template of Section 11.5.3, the results of this section, and those of the other chapters that it uses, fall into three classes. The first consists of equivalences with RH: the Thorin Bernstein property of $\log G$ (Theorem 4.2.4), the mixture of exponential laws in the size biased clock (Proposition 11.5.1), the vanishing of the Hankel inertia indices (Theorem 5.1.7), and the GGC property of R_t at $t = 0$ (Theorem 7.2.15 and Proposition 1.2.1). The second consists of facts that hold with no hypothesis: the clock is infinitely divisible and self decomposable (Theorem 5.1.1), it lies in L_2 and not in L_3 (Theorem 11.5.8), and its driving subordinator has self decomposable, hence unimodal, laws (Corollary 11.5.9); the points of the upper half plane where $\mathrm{Im}(G'/G)$ could fail to be negative lie in a thin parabola beyond about $-T_0^2$ (Theorem 4.3.9); the clock is not the martingale limit of a supercritical continuous state branching process (Theorem 11.5.11); R_t is a GGC for $t \geq 0.2$ [247] and for no $t < 0$ (Corollary 4.2.5); and $\mathcal{B}_t$ is not a GGC for t near 0, $t \neq 0$ (Proposition 11.5.15). Being unconditional does not by itself prevent a statement from implying RH, since a proof of RH would be unconditional too. What prevents it is that none of these facts distinguishes ξ from functions with zeros off the line: the statements about the flow at $t \neq 0$ are removed by the filter of Section 11.5.5, infinite divisibility and self decomposability hold as well for the comparison function F of Section 12.2.6, the ladder index and the branching obstruction are set by the first zeros (Section 11.5.4), and the localisation uses only the zeros below the verification height. The third class consists of the routes that are closed. The Hankel route is closed for practical purposes, on the heuristic of Table 7.6. Properties of the kernel and its representations as mixtures are removed by the filter (Propositions 11.5.12 and 11.5.13), and the GGC property of the variable of Biane, Pitman and Yor does not detect the Euler product (Section 11.5.5.4).

11.5.6.1 Open problems

Problem 11.1. Find a natural class of functions $\xi(s, a)$ with positive kernels, containing ξ, in which the GGC property of $\mathcal{B}_a$ implies that every zero of $\xi(s, a)$ lies on the critical line, or a function in such a class whose variable $\mathcal{B}_a$ is a GGC and which has a zero off the line. Since $\mathcal{B}$ is a GGC, the first answer for a class containing ξ proves RH. No candidate for the second is known: the function of

Davenport and Heilbronn, the family f_r modulo 7 at the 16 tested values of r with a positive kernel and the tilts $\mathcal{B}_t$ at the eleven tested values of $t < 0$ all fail the test. The question is related to Question 8.7.20 and to the structural reason for R to be a GGC that the discussion of Table 7.6 asks for.

PROBLEM 11.2. Is $\mathcal{B}_t$ a GGC for no $t \neq 0$? For $t < 0$ this would be a property of the kernel that passes the filter at every t. Proposition 11.5.15 proves it for $0 < |t| < t_0$, through the first order change of the Thorin measure in the gaps between its atoms, and at the ten values of part (i).

PROBLEM 11.3. Which of the variables $\mathcal{B}_{\chi_d}$ for $d = -4$, -3, 5, 8 and 12, which pass the test to order 12, are GGCs? (For $d = 24$ the test fails at order 11.) On the reciprocal side, $1/\mathcal{B}_{\chi_d}$ is a GGC for $d = -3$ and -4 (Proposition 11.5.19); for $d = -4$ it is the exit time variable of Remark 8.5.22, whose density is not HCM (Theorem 4.3.20), so the question for $\mathcal{B}_{\chi_{-4}}$ cannot be settled through HCM. For even characters the reciprocal transform is the series $\frac{2}{\xi(0,a)} \sum_n a_n K_0\big(2n\sqrt{\pi\lambda/q}\big)$ of Bessel functions, and no product formula for it is known to us.

PROBLEM 11.4. With no hypothesis, the Li coefficients of Section 8.3.2 are

$$\lambda_n = \int_0^\infty k_n(t) W(t)\, dt = m_0\, \mathbb{E} k_n(A),$$

where W is the heat trace, A is the variable of Proposition 11.5.1, and k_n is the function with $\int_0^\infty e^{-zt} k_n(t)\, dt = \varphi_n(z)$. Since φ_n is a polynomial of degree n in $(z+\frac{1}{4})^{-1}$ with no constant term, $k_n(t) = e^{-t/4} p_n(t)$ with p_n a polynomial of degree $n-1$; for instance $k_1(t) = e^{-t/4}$ and $k_2(t) = (4-t)e^{-t/4}$. Which unconditional properties of W, or of A, force $\mathbb{E} k_n(A) \geq 0$ for every n, which is RH by Li's criterion [176, 37]? Positivity of W (Theorem 5.1.1) gives $n = 1$; complete monotonicity of W, which is RH, gives every n, since $\varphi_n \geq 0$ on $(0, \infty)$.

PROBLEM 11.5. Describe the ladder index of R_t along the de Bruijn Newman flow. The computation of Section 11.5.4.3 finds the index two of the clock up to $t = 27.8$ and one from $t = 27.9$ on, on its window. Is the index nonincreasing in t, and is it one for all large t? The answer carries no information about RH, but it would show how the spacing of the first zeros sets the index.

Chapter 12

Conclusions and open problems

The results of Chapters 1 to 11 can be placed on a single path, which runs along the basepoint from the region where positivity of the Thorin measure is a theorem to the centre, where it is RH.

Chapter 13 follows as a complement on the side of statistical mechanics, where RH is the statement that the Pólya law is ferromagnetic.

12.1 The Thorin approach in summary, and a path to RH

The path is a programme with its gap identified, not a proof.

12.1.1 The path along the basepoint

Figure 12.1 and Table 12.1 set out the path.

Section 11.3.4 draws the two conclusions that fix the shape of the path. At the basepoint one the constructions attached to the reciprocal Xi function are unconditional and explicit (Sections 9.1, 9.2 and 10.1, Table 11.3), with two exceptions: the Krein string representation of Corollary 9.2.3 assumes RH and simple zeros, and Lemma 9.2.4 gives the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$ only for a boundary bounded at infinity, while the boundedness of the computed boundary is observed, not proved. Below the basepoint one each construction holds, up to a boundary case, exactly when no zero lies to the right of α (Theorems 9.2.5 and 10.1.3, Propositions 9.1.13 and 10.1.10), and the theory of Sato and Bondesson reaches nothing there (Section 11.1.5).

12.1.2 The open steps

By Theorem 1.2.2, that is [252, Theorem 3.7 and Remark 3.10], step 7 is exactly a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < 1$, a quasi Riemann hypothesis. Until 2026 no such half plane was known. In October 2026 OpenAI released a proof, produced with one of its AI models, that $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{7}{8}$, and the same for every Dirichlet L function [226], together with a formal proof of both statements in the

step	statement	status
1	The clock is infinitely divisible and self decomposable; the heat trace is positive and decreasing	proved (Theorem 5.1.1, which is Theorem A of [252]), with the verification of RH below height 1000 as input
2	The heat trace is log convex, its leading Hankel blocks of size up to 3×10^7 are positive definite, derivative signs hold to order 1.16×10^{25}	proved, with verification height as input
3	The reciprocal at basepoint $\alpha \geq 1$ is a GGC with Poisson Thorin density, and at $\alpha = 1$ its Lévy form, hitting time, scale mixture and completeness are explicit; for every $\alpha > \frac{1}{2}$ the shifted clock T_α exists and is an almost GGC	proved (Theorems 1.2.2, 9.2.5 and 11.4.2; Chapters 9 to 11), with the verification of RH below height 1000 as input to Theorem 9.2.5
4	The normalised reciprocal zeta function is infinitely divisible; the canonical function of the floor of the reciprocal Xi function is an arithmetic part plus a completely monotone archimedean part, and RH is the complete monotonicity of the sum	proved (Chapters 6 and 7)
5	The Thorin density is computable at every resolution, with an explicit bound for each source of error, and is positive at every sample point of every window computed; the runs do not yet add these bounds to the ball radii (Section 12.2.2)	computed (Chapters 2 to 5)
6	Zero density $\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every computed ε, down to $\varepsilon = 0.005$	computed (Section 3.4)
7	For some $\alpha < 1$ the shifted clock T_α is a GGC, equivalently, the half plane $\operatorname{Re} s > \alpha$ is zero free (Theorem 1.2.2); then for every $\alpha' > \alpha$ the measure $U_{\alpha'}$ is the Thorin measure of $T_{\alpha'}$ and is positive at all heights	announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean, not yet refereed (Section 12.1.2); open for every $\alpha < \frac{7}{8}$
8	Positivity of U_α for every $\alpha > \frac{1}{2}$ (Lagarias), equivalently the Thorin condition	open; equivalent to RH

Table 12.1: The path, step by step.

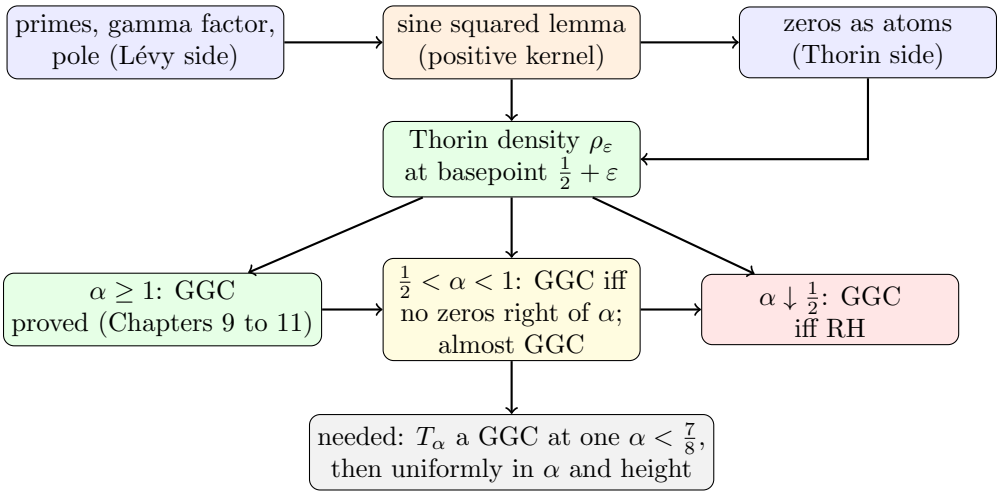


Figure 12.1: The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. The programme must carry the GGC property across the strip.

Lean proof assistant and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [227]. The argument proves a power saving for smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$, and it uses nothing from the Thorin picture. It has not yet been refereed. We rebuilt the formal proof from the public repository (commit `adc7f12`, Lean 4.34.1, Mathlib `d13f23b`): the Lean kernel accepts both proofs, which use only the axioms `propext`, `Classical.choice` and `Quot.sound`, and the comparator of the Lean Focused Research Organization, run without its sandbox, confirms that the theorems proved are the published challenge statements. If it stands, step 7 holds with $\alpha = \frac{7}{8}$, and the constructions of this book follow. By Theorem 1.2.2 the shifted clock T_α is a GGC for every $\alpha \geq \frac{7}{8}$, with Thorin density (1.2.6) and $v_\alpha > 0$ for $\alpha > \frac{7}{8}$, while at $\alpha = \frac{7}{8}$ each zero $\frac{7}{8} + i\gamma$ with $\gamma > 0$, which the announcement does not exclude, would add an atom at γ^2 (Theorem 7.1.14); for $\alpha > \frac{7}{8}$ the clock T_α is a Brownian first passage time plus an independent GGC with no drift, because the first two steps of the proof of Proposition 9.1.12 go through for $\frac{7}{8} < \alpha < 1$, where every zero has $0 < \alpha - \beta < \alpha$, with smallest margins 0.0137 and 0.0157 in ball arithmetic (`valpha_min.py`); by Proposition 9.1.13 the compensated Lévy form converges for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ at prime cutoff X for every $\delta > 0$, and the same holds for $L(s, \chi_{12})$ (Section 11.3.2); by Theorem 10.2.7 Riesz's function satisfies $R(x) = O(x^{7/16+\delta})$ for every $\delta > 0$; and by de Bruijn's theorem $\Lambda \leq \frac{9}{32}$ (Section 8.4), weaker than the bound 0.2, while for the cosine tilts of Section 13.4.2 his theorem for cosine factors gives $\Omega \leq \frac{3}{8}$, against $\frac{1}{2}$ with no hypothesis. Below $\frac{7}{8}$ nothing changes. Positivity of the Thorin density on the single line $\operatorname{Re} s = \alpha$ is a weaker statement

than a zero free half plane (Remark 1.2.4). So the first theorem the Thorin path needs beyond the announcement, a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < \frac{7}{8}$, is an open problem in its own right; the Lévy form, the scale mixture, completeness and Riesz's criterion (Sections 9.1, 10.1 and 10.2) restate it and do not get round it. The Thorin formulation offers two specific tools for it. The sine squared form, compensated as in Proposition 9.1.13, writes the required positivity as domination of the pole piece by the prime and archimedean pieces at basepoint α, a statement about a single explicit function of θ for each α; and the Poisson form shows that a zero to the right of α appears as a Cauchy kernel of negative width, whose signature Section 3.3 calibrates exactly.

Step 8 carries the GGC property from any fixed α to every $\alpha > \frac{1}{2}$. Weak limits cannot do this on their own (Section 4.1), because the limit of the Cauchy flow is positive whether or not RH holds. Three mechanisms could do it, each consistent with everything computed in this book:

1. a positive approximating family whose positivity is proved independently, such as zeta functions of Beurling generalised prime systems that provably have only real ordinates, finite Krein strings, or the partial products over verified zeros completed by the Weyl tail; once the first N zeros are verified, RH is equivalent to the GGC property of the tail factor left when their factors are removed from the clock [252, Corollary 18.11];

2. an operator whose spectrum is the set of ordinates, where positivity would follow from self adjointness, the Hilbert Pólya idea in Thorin form (Section 8.1.3);

3. a direct inequality for the variogram pairing that controls the pole by the primes and the gamma factor at every scale, the Weil form of step 8.

12.1.3 What computation contributes

Computation cannot supply the uniformity in the height that steps 7 and 8 require: neither finitely many heat trace tests (Theorem 5.2.1) nor positivity at a fixed resolution (Section 5.2.2) can exclude an off line zero above the verified height. What it can do is developed in Section 12.2.1. It calibrates the size of the positivity margins, which tells a proof how much room it has: the Thorin density at resolution ε has minimum $0.0147\,\varepsilon$ on the window below height 200 (Section 3.1), a margin that vanishes linearly and must be controlled uniformly. That this margin is the value at the origin is a theorem on the whole verified window, and at every height for every $\varepsilon < \frac{1}{2}$ it is equivalent to RH (Theorem 9.1.18).

Section 3.5 gives the variogram form of the domination inequality, on which the linear programming formulation of [252, Sections 18.2.8 and 18.2.9] works, the most concrete target for step 8. Both are posed at the centre, where the variogram form is Weil's criterion and the linear program asks whether the clock $1/G$ is a

GGC, so each is equivalent to RH. Section 4.1 and [252, Section 18.2.3] give the closure criterion in its weakest form, the target for approximating families in step 8. Section 11.4 closes with four questions between almost and exact positivity (Section 11.4.7), the last of which is step 7 below $\frac{7}{8}$, the GGC property at a single basepoint $\alpha < \frac{7}{8}$.

12.2 Open problems and lessons for the Thorin programme

The programme of computations below is aimed at the gap that Section 12.1 located, positivity uniform in the height, and the lessons that follow it are drawn for anyone working on the Thorin route to RH, including the first author's own programme [252]. The open problems on the reciprocal zeta function are stated in Section 7.2.4, those on the basepoint one and the critical strip in Section 11.4.7, and those of Pakes's framework in Section 11.5.6.1.

12.2.1 What computation can and cannot do on the path

Computation cannot supply uniformity in height (Section 12.1.3); what it can do falls under four aims. It can extend and certify: prove positivity of the Thorin density on explicit windows, at every resolution, with interval arithmetic. It can test intermediate conjectures: the four questions of Section 11.4.7 all have numerically testable consequences. It can calibrate: measure how much room positivity has, and compare with random matrix predictions, so that a proof knows which inequalities are sharp. And it can cross check: compute the same quantities from the primes, through the criteria of Riesz and Salem and the compensated Lévy form, independently of evaluations of ζ.

Five work packages follow: twisted Riesz functions as an arithmetic filter bank, the cross check (Section 10.2.1); certified Thorin densities at height (Section 12.2.2); the positivity margins against the circular unitary ensemble (Section 11.4.5); explicit tables of the almost GGC defect (Section 11.4.2); and testing stronger conjectures before pursuing them (Section 12.2.4).

12.2.2 Certified Thorin densities at height

At resolution $\varepsilon = \alpha - \frac{1}{2}$ the Thorin density $\rho_\varepsilon(\theta) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\theta)$ is a smooth function of θ. Its positivity on a window of ordinates is what the absence of zeros with $\beta > \alpha$ predicts there; since an off line zero need not make the density negative (Section 5.2.2), a certificate for the window should also count such zeros. Two computations do this. The first evaluates ρ_ε on a grid with rigorous error bounds, using the ball arithmetic Riemann Siegel and Euler Maclaurin evaluators of Arb [130], exposed in python flint, at moderate height and the Pólya route where it is faster, and bounds the variation between grid points by the derivative. The second counts the zeros to the right of the line $\operatorname{Re} s = \alpha$ through the zero

density identity $M_\varepsilon(T)+\tau_\varepsilon(T) = N(T)-2\nu(\frac{1}{2}+\varepsilon, T)$ of Proposition 3.4.2. It is the count of Section 3.4, which at $\varepsilon = \frac{1}{2}$ is Backlund's method. Section 3.4.3 followed the argument in steps of $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) without a bound on its change between grid points, so its counts are computed, not certified; with such a bound the count would certify a whole window. The deliverables are certified positivity of ρ_ε for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ on windows of length 10^4 at heights 10^6, 10^8 and 10^{10}, together with the minimum density on each window, which feeds the margin statistics of Section 11.4.5. Below the verified height 3×10^{12} the positivity itself needs no new computation: an off line pair makes the density negative only within $\frac{1}{2}$ of its ordinate, so $\rho_\varepsilon > 0$ on $[0, 3 \times 10^{12} - \frac{1}{2}]$ for every $\varepsilon > 0$ (Corollary 11.4.7); there the certificates test the evaluators against the verification and measure the margins.

Below height 200 the certificate is within reach now. The ball arithmetic of Section 2.1 certifies the rounding only. Adding the discretisation bound of Section 2.1.1 and of Proposition 2.1.1, bounds for the truncations of the theta series and of the u axis, a bound on the change of the argument between grid points, and an explicit bound on the remainder of Proposition 3.2.2, which is $O(\varepsilon^3)$ (Remark 3.2.3), turns positivity of ρ_ε on $0 < z \leq 200^2$ at each computed ε, the integer identity of Proposition 3.4.2 and the HCM derivative signs of Section 4.1.2 into theorems on that window.

Above that window the evaluator changes. The tilted contour gives 25 digits at height 5000 and resolves the Lehmer pair near height 7005 (Sections 2.1.3 and 2.2.3.1), but its number of nodes grows linearly with the height. Chirp transform batching of the Pólya route, and the hybrid of Section 2.2.2, a Riemann Siegel or Odlyzko Schönhage evaluator at $\frac{1}{2} + \varepsilon + i\theta$ with Odlyzko's zeros as the RH reference, would give the positivity margin, the matched total variation and the zero density scan near the 10^{12}th [219], 10^{21}st and 10^{22}nd zeros [223, 224]. The same zeros would measure the cell constant of Section 5.4 at large height, whose behaviour is open (Section 5.4.2), and allow a comparison of the cell errors with the prediction of the sine kernel process of the unitary ensemble.

12.2.3 Calibration against known configurations

The diagnostics of Chapters 3 to 5 are calibrated when they are run on configurations whose zeros are known exactly, and three such computations remain. Section 8.6 ran the posterior predictive check of [252, Section 19.3.7] at $\varepsilon = 0.1$ on one window and calibrated it with a displaced zero. Three extensions are open: smaller resolutions, where the signature of a displaced zero is largest and the Pólya route supplies ρ_ε to about 60 digits below height 200 (Section 2.1.1), whereas the deconvolution used double precision; continuous atom positions, which would remove the baseline of the grid; and larger heights. The square root law (3.3.1) of Section 3.3.2, which follows from Proposition 3.3.1, sets what the check can detect.

The de Bruijn Newman flow of Section 8.4 manufactures off line pairs in a

controlled way. Running these Thorin diagnostics along the flow through the collision of the Lehmer pair near 7005 would calibrate the zero density scan, the matched total variation and the negative mass against configurations that are known exactly.

Generalised prime systems of Beurling whose zeta functions provably have only real ordinates give positive Thorin measures by construction. Measuring how fast their clock transforms approach $1/G$, in the distances of Section 3.2 and along a sequence with an accumulation point as the closure criterion of Corollary 4.1.2 [252, Propositions 18.10 and 18.14] allows, is the computational side of that criterion; the Kent approximants of Section 8.1.3 are the first instance.

Two computations of Chapter 8 remain at their edges. The Brownian boundary of Section 8.2 reproduces the law of the clock in the bulk, to 0.0020 in Kolmogorov distance for the boundary of the Volterra equation and to 0.0059 for the tangent approximation of Roberts and Shortland (Section 8.2.2.2), but the zone of large times t is checked by no simulation, since only about one path in ten million survives past $t = 0.1$. Importance sampling with an exponentially tilted drift, or multilevel splitting, would check that zone and the logarithmic correction of the asymptote of the boundary directly. The Cauchy integral method of Section 8.3 extends to n in the thousands, at a cost of about $0.03n$ digits lost to cancellation. Comparing the Pólya values of the Keiper Li coefficients with the sums $\sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$ over the zeros to large n would test how the Li functional sees the zeros near height n.

12.2.4 Testing stronger conjectures before pursuing them

Conrey and Li showed that a positivity condition in de Branges' approach to RH is false even though RH may be true [62]. Any intermediate statement proposed on the Thorin path should therefore be tested numerically first. The candidates are a lower bound $\rho_{\alpha-1/2}(\theta) \geq c(\alpha) \log \theta$ outside a sparse set, the first of the four questions of Section 11.4.7; monotonicity or convexity of the margin in α; positivity of the Thorin density for α slightly below one, uniformly on large windows; hyperbolic complete monotonicity of the Laplace transform φ_α of T_α below one; and variogram inequalities at single scales. The first two are partly settled: the lower bound holds for $\rho_{\alpha-1/2}$ itself outside a set of heights of density zero, although not near the zeros where Lemma 11.4.9 would need it (Remark 9.1.17), and the margin divided by $\alpha - \frac{1}{2}$ decreases in α, with the ratio $\rho_{\alpha-1/2}(\theta)/(\alpha - \frac{1}{2})$ nonincreasing in α at every height outside the windows of the zeros off the line (Theorems 9.1.15 and 9.1.18). Each can be evaluated on the computed windows and at the heights of Odlyzko's tables, and any statement that fails there should be discarded.

The variogram inequalities can be tested in finite dimension. The reformulation of Section 3.5 poses finite dimensional feasibility problems: is the regularised pole mass dominated by the prime and archimedean mass on the cone of variograms of

positive definite functions of a given dimension? Solving them with explicit dual certificates at increasing dimension would measure how the margin behaves.

The cross checks test the zeros themselves. Agreement of the twisted Riesz functions (Proposition 10.2.1) with the explicit formula over the computed zeros is a check that shares no code or method with the computation of the zeros (Remark 10.2.2).

12.2.5 Open problems from free probability

The note [261] closes with eight problems, and five of them are settled or taken up in the book. They are settled by Theorem 5.1.1 through Theorem 11.2.3, by Propositions 11.2.7 and 11.2.9 with Theorem 11.2.10, or taken up in Sections 8.4.3 and 8.4.5. is the following.

Problem 12.1. Jensen polynomials as finite free products. The multiplicative convolution $\boxtimes_d$ multiplies normalised coefficients and keeps roots real and of one sign (Lemma 11.2.8(i)), and the normalised coefficients of the reversed Jensen polynomial $x^d J_{d,n}(1/x)$ are, up to the signs $(-1)^i$, the numbers $(n+i)!\, b_{n+i}$, $0 \le i \le d$. A factorisation $(n+i)!\, b_{n+i} = \alpha_i \beta_i$ into the normalised coefficients of two polynomials with only real roots of one sign would therefore give real zeros of $J_{d,n}$. Find such factorisations for every shift n at a fixed degree d, or show that none exists with factors of a given kind. The obvious split, a Gaussian multiplier sequence q^{i^2}, $0 < q \le 1$, times a residual close to the coefficients of $(1+x)^d$, fails, because a root of multiplicity d splits into nonreal roots under a generic perturbation; and by Proposition 11.2.9 the factor $1/(a)_i$ can be multiplied in but not divided out. A factorisation has to put real structure into both factors [261].

Problem 12.2. Laguerre and Meixner Pollaczek expansions. Theorem 11.2.10 excludes scale mixtures of gamma laws for $|X|$ and X^2, and Proposition 11.2.9 shows that removing a gamma layer keeps zeros real while adding one does not. Find a variable in which the Pólya density, or Ξ itself, has an expansion in Laguerre or Meixner Pollaczek polynomials with positive coefficients and truncations with only real zeros [261]. The symmetric Meixner Pollaczek expansions of Section 6.2.3.2 converge slowly, and a real rooted construction from their coefficients is a problem of Section 7.2.4.

Problem 12.3. Jacobi parameters. Under RH, where ν is a positive measure and all its Jacobi parameters exist, prove that $\alpha_n \sim 2\beta_n \sim \frac{1}{2} L_n$, with L_n given by (11.2.13) for the smooth count: a spectral form of the Riemann von Mangoldt formula. Table 11.7 supports it to a few per cent for $n \le 49$, but the local model of Section 11.2.4 is a heuristic, outside the scaling of [157]. Decide between the rules $n(L_n) = n$ and $n(L_n') = n - \frac{1}{2}$, which the data do not separate, and explain the scatter of the ratios, about six per cent, through the individual gaps between the zeros, as [261] proposes.

Problem 12.4. The order at which a collision is seen. Section 8.4.3 computes the failure times t_N^* of the Hankel test along the flow to $N = 50$ and explains their steps by collisions; the rule of [261] that the collision of (γ_k, γ_{k+1}) is registered from order $2k + 2$ on fails at $N = 50$ (Table 8.9). Find the rule that gives the order at which a pair is registered, as a function of the index k of the pair and of its distance from the axis, and follow the staircase of t_N^* to orders of a few hundred, towards the limit 0 that it has under RH.

Problem 12.5. Unconditional collision brackets. Section 8.4.5 brackets the collision times by quantities along the flow (Corollaries 8.4.12 and 8.4.14) and through the nonreal zeros (Theorems 8.4.11 and 8.4.13), and checks the hypotheses on a grid of times for the 58 collisions of the census. Find the a priori control of the nonreal zeros of H_t, $t < 0$, asked for in Problem 8.1, which would make the brackets statements about H_t computed from H_0 alone.

Problem 12.6. Moving the ladder without new zeros. Theorems 5.1.9 and 5.1.13 prove the derivative signs $(-1)^k W^{(k)} > 0$ on ranges that grow with the height to which RH is verified (Section 5.1.6), and complete monotonicity of W on $(0, \infty)$ is RH. Find a mechanism that extends these ranges with no new information on the zeros, for instance a functional equation for the law of T under Υ, as [261] suggests. Theorem 11.2.3(c) restricts the form such an equation can take: T is not the Upsilon image of a law in BO, and under RH it is the image of the law μ_0, which is neither a GGC nor in BO.

12.2.6 Lessons from the computations

Each lesson is a statement that a computation in the book established or refuted.

Equivalent statements are not equally computable. The Thorin condition has at least eight equivalent faces: positivity of the Thorin measure, complete monotonicity of the heat trace, Stieltjes form of G'/G, Hankel positivity at one time, HCM of the clock, Lagarias positivity, Weil positivity and Li positivity. Mathematically they are the same statement. Computationally they differ by many orders of magnitude in conditioning. The heat trace at moderate time loses about $87t$ digits to cancellation (Section 5.3); the normalised minors of the even Hankel block at $t = 0.05$ fall to 10^{-44} by size five (Section 5.1); the Keiper Li coefficients weight a zero at height γ by n^2/γ^2 until n reaches γ (Section 8.3); the Thorin density at resolution ε is local in the ordinate and sees each zero at full size $1/\varepsilon$ (Section 3.2). A computational attack should choose its face by conditioning, and the local face is the Thorin density.

The primes compute the wrong end: the sine squared lemma is the right way to understand the structure and the wrong way to compute near the line, where k digits at resolution ε need primes up to about $10^{2k/\varepsilon}$ even when no zero lies to the right of the line (Section 3.5.1). Any argument that proposes to establish positivity near the line by bounding prime sums must overcome that exponential,

and Section 7.1 shows the same obstruction in another form (Proposition 7.1.2).

The heat trace meets the same obstruction in its own terms. By Proposition 5.1.11, RH is the positivity of the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros at every height u and every $t>0$, and (5.1.3) writes the symmetric smoothing $\mathcal{G}(u,t)+\mathcal{G}(-u,t)$ as a polar term plus an archimedean term $A(u,t)$ minus a prime sum $P(u,t)$. Bounding $|\cos(u\log n)|$ by one bounds $P(u,t)$ by $S(t)/\sqrt{2\pi t}$, where $S(t)=\sum_{n\geq 2}\Lambda(n)n^{-1/2}e^{-(\log n)^2/8t}$ is, up to a factor, the prime term of (1.2.5) at time $2t$, and so proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t)>0$ for t below the time $t_0(u)$ at which the bound meets the archimedean and polar terms, which agrees to five digits with the root of $2S(t)=\log(u/2\pi)$ (computed by `phasefree.py`): t_0 is 0.45 at height 10^3, 1.72 at the verified height 3×10^{12} and 3.64 at 10^{50}, to the digits shown (proved in ball arithmetic by `phasefree_arb.py`), and it grows only doubly logarithmically in u, since $S(t)$ is asymptotic to $\sqrt{8\pi t}\,e^{t/2}$. These resolutions are too coarse to see the quadruples of Table 5.4: at their heights t_0 lies between 0.311 and 0.511, while within 8 of each moved pair they make $\mathcal{G}$ negative only for $t>1.18$ (both proved in ball arithmetic). Nor can any bound on P that holds uniformly in the height do better, since by Dirichlet's theorem on simultaneous approximation the phases $u\log n$ of any finite set of prime powers come arbitrarily close to multiples of 2π at arbitrarily large u, where $P(u,t)$ therefore comes arbitrarily close to $S(t)/\sqrt{2\pi t}$. Nor does positivity at a single resolution give a zero free strip: at fixed t the bound $S(t)/\sqrt{2\pi t}$ on P proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t)>0$ at every height u with $t_0(u)>t$, that is above about $2\pi e^{2S(t)}$, which is 8.1×10^{212} at $t=6$, and $\mathcal{G}(u,6)>0$ itself for every $u\geq 10^{213}$ (proved in ball arithmetic, with the bound on $|\mathcal{G}(-u,6)|$ above), whatever the zeros are, so a strip needs t to grow with the height. A proof along the heat trace must use the phases, uniformly in the height.

Limits do not create positivity: positivity must be proved at every resolution, not in the limit. A proposal that reaches the Thorin condition as a limit must say where the positivity of its approximants comes from.

The perturbation theorem gives a test that a proposed proof of RH must pass. Replace ξ by the function F of Theorem 5.2.1, with R above every height that the proof uses explicitly. A step that holds for F as well as for ξ cannot be the step that excludes zeros off the line, because F has four. The steps that survive include the Riemann von Mangoldt formula with an explicit error, which F satisfies with the error enlarged by one on $[A,B)$, the only heights where the two counts differ; the verified zeros below R; the small time expansion of the heat trace through the constant term, since $W_F-W=\left(\frac{1}{2}(B-A)^2+\frac{1}{8}\right)t+O(t^2)$; infinite divisibility, self decomposability and log convexity; the Hankel blocks of size N when $R\geq T_N$ and the derivative signs up to order 10^{17} when $R\geq 10^{12}$; and the order, type and functional equation. What separates ξ from F is arithmetic, the Euler product and the explicit formula with the prime weights $\Lambda(n)$, together with local counts of the zeros on the line in windows that contain the height g of the quadruple (Section 5.2.1). A proof through the derivative signs can do without them only

below the order at which the signs of F first fail, which is about $t^* R^2$, with t^* between 1.18 and 4.03 in the cases computed below height 2000 and, by the model of Section 5.2.1, of order one or more at larger heights, far beyond 10^{17}. The function F exists unconditionally when $R+50 \le 3\times 10^{12}$; for larger R the test still applies, because a correct proof of RH makes the hypothesis of Theorem 5.2.1(b) true for every R. The criticality filter of Section 11.5.5 is the companion test for properties of the Pólya kernel: a step that holds for $e^{tv^2/4}\Phi(v)$ with some $t < 0$, whose transform has nonreal zeros by the theorem of Rodgers and Tao, cannot be the step that excludes zeros off the line.

Chapter 13

Ferromagnetic random variables

A symmetric random variable is ferromagnetic if its moment generating function vanishes only on the imaginary axis. By (1.2.4) the moment generating function of the Pólya variable X is $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, so RH is the statement that one explicit law, the Pólya law with density $p = \Phi/\Xi(0)$, is ferromagnetic.

With no hypothesis, the Pólya law lies within total variation distance 1.3×10^{-4} of a law that is ferromagnetic by an explicit construction, and within 2×10^{-6} of one whose positivity, and hence ferromagnetism, rests on a computer assisted proof in ball arithmetic (Section 13.4).

The scripts are in the folder `ferro` (`code/README.md`). We use the units of (1.2.3) throughout, in which the zeros of the characteristic function of X are the ordinates γ and $\operatorname{Var} X = 2\mathbb{E}T = 0.0462099862$. The normalisation of Rodgers and Tao in Section 8.4 doubles the variable of the kernel, and statements quoted from the literature have been converted. As in Section 1.2, Z is a standard normal variable, the e_k are standard exponential variables, the L_k are standard Laplace variables, ε and the ε_k are fair random signs, U is uniform on $[-1,1]$, and all of them are independent of each other and of everything else.

13.1 Ferromagnetic laws and their clocks

13.1.1 Definition and background

Definition 13.1.1. A real random variable X is *ferromagnetic* if it is symmetric, $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$, and its moment generating function $L_X(z) = \mathbb{E}e^{zX}$, which is then entire, vanishes only on the imaginary axis. Equivalently, every zero of the characteristic function $\varphi_X(\theta) = \mathbb{E}e^{i\theta X} = L_X(i\theta)$ is real.

These are Newman's laws of type $\mathcal{L}$ [209].

The name comes from statistical mechanics. Lee and Yang proved that the partition function of an Ising ferromagnet, with spins ±1 and a common complex field, vanishes only when the field is purely imaginary [171], and Newman showed that a symmetric single spin law with Gaussian moments of all orders has the same

property in every ferromagnetic system exactly when its own moment generating function has only imaginary zeros [208]; Lieb and Sokal gave another proof and extended the theorem to spins with several components [179]. Section 7.1.3 recalls these results, and Proposition 7.1.16 restates Newman's theorem in the language of Thorin measures. Two smaller classes of single spin laws carry the correlation inequalities of the Ising model. The Griffiths Simon class consists of the weak limits of suitably scaled sums of ferromagnetically coupled Ising spins; it contains the laws with density proportional to $e^{-ax^4-bx^2}$, $a > 0$, and its laws inherit the Lee and Yang property and the inequalities of Griffiths, Kelly and Sherman and of Griffiths, Hurst and Sherman [306, 100]. The class of Ellis, Monroe and Newman consists of the laws with density $e^{-V(x)}$, V even and V' convex on $[0,\infty)$, and their weak limits; the Griffiths, Hurst and Sherman inequality holds for ferromagnets with such single spin laws [83].

13.1.2 The Hadamard form and the cumulants

Theorem 13.1.2 (Newman [209]). *Let X be symmetric with $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$. Then X is ferromagnetic if and only if*

$$L_X(s) = e^{bs^2} \prod_k \Big(1 + \frac{s^2}{c_k^2}\Big), \qquad b \ge 0, \quad c_k > 0, \quad \sum_k c_k^{-2} < \infty, \tag{13.1.1}$$

with each c_k repeated according to the multiplicity of the zero ic_k. If $\mathbb{E}e^{bX^2} < \infty$ for every $b > 0$, then $b = 0$.

Proof. Since $|L_X(z)| \le \mathbb{E}e^{|\operatorname{Re} z||X|} \le Ce^{|z|^2/4b_0}$, the function φ_X is entire of order at most 2; it is even and real on $\mathbb{R}$, and $\varphi_X(0) = 1$. Suppose that its zeros are real, at $\pm c_k$ with $c_k > 0$. Hadamard's theorem, with the primary factors of genus 2 paired over $\pm c_k$, gives $\varphi_X(\theta) = e^{-b'\theta^2} \prod_k (1 - \theta^2/c_k^2) e^{\theta^2/c_k^2}$ with b' real and $\sum_k c_k^{-4} < \infty$, so that, for real s,

$$\frac{\log L_X(s)}{s^2} = b' + \sum_k \frac{1}{c_k^2} \frac{\log(1+x_k) - x_k}{x_k}, \qquad x_k = \frac{s^2}{c_k^2}.$$

The function $(\log(1+x)-x)/x$ decreases from 0 to -1 on $(0,\infty)$, so by monotone convergence the sum tends to $-\sum_k c_k^{-2}$ as $s \to \infty$. If this sum diverged, $L_X(s)$ would tend to 0, which is impossible because $L_X(s) = \mathbb{E}\cosh(sX) \ge 1$. Hence $\sum_k c_k^{-2} < \infty$, the factors e^{θ^2/c_k^2} can be collected into the exponential, and (13.1.1) holds with $b = b' - \sum_k c_k^{-2}$. Then $\log L_X(s) = bs^2 + o(s^2)$, since $\log(1+x)/x$ is bounded by 1 and tends to 0, and $L_X \ge 1$ forces $b \ge 0$. The converse is clear, since the right side of (13.1.1) vanishes only at $\pm ic_k$. If $\mathbb{E}e^{bX^2} < \infty$ for every b, then $L_X(s) \le C_\epsilon e^{\epsilon s^2}$ for every $\epsilon > 0$, so $b = 0$, as in the proof of Proposition 7.1.16. □

Corollary 13.1.3 (Newman [209]). *If X is ferromagnetic, with* (13.1.1)*, then* $\operatorname{Var} X = 2b + 2\sum_k c_k^{-2}$*, and for* $m \ge 2$

$$\frac{\kappa_{2m}(X)}{(2m)!} = \frac{(-1)^{m+1}}{m} \sum_k c_k^{-2m}. \tag{13.1.2}$$

So the even cumulants alternate in sign: $\kappa_4 \le 0$, $\kappa_6 \ge 0$, *and so on. Moreover* $\mathbb{E}e^{sX} \le e^{s^2 \operatorname{Var} X/2}$ *for real* s, *and* $\mathbb{E}X^4 \le 3(\operatorname{Var} X)^2$.

Proof. The first two statements are the expansion of $\log L_X(s) = bs^2 + \sum_k \log(1 + s^2/c_k^2)$. The bound on $\mathbb{E}e^{sX}$ follows from $\log(1+x) \le x$, and the bound on $\mathbb{E}X^4 = \kappa_4 + 3\kappa_2^2$ from $\kappa_4 \le 0$. □

For the Pólya law the alternation holds with no hypothesis: by the formula for $\log G$ in Section 5.1, $(-1)^{n-1}\kappa_{2n}(X) = \frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt > 0$ for every n (Section 1.3), and under RH this is (13.1.2) with $c_k = \gamma_k$ and $b = 0$.

13.1.3 The clock of a ferromagnetic law

Theorem 13.1.4. *Let X be ferromagnetic, with* (13.1.1)*, and put*

$$T_X = b + \sum_k \frac{e_k}{c_k^2}, \qquad Y_X = Z\sqrt{2T_X},$$

the series converging almost surely and in mean.

(i) $\mathbb{E}e^{-uT_X} = 1/L_X(\sqrt{u})$ *for* $u \ge 0$, *and T_X is a GGC with drift b and Thorin measure* $\sum_k \delta_{c_k^2}$, *one unit atom for each pair of zeros* $\pm ic_k$.

(ii) $\mathbb{E}e^{i\theta Y_X} = 1/L_X(\theta)$ *for real* θ, *so that* $(\varphi_X, \varphi_{Y_X})$ *is a van Dantzig pair, and* $Y_X \overset{d}{=} \sqrt{2b}\,Z + \sum_k L_k/c_k$.

(iii) The moment generating function of Y_X is $1/\varphi_X(t)$ *for* $|t| < c_1$ *and infinite for* $|t| \ge c_1$, *unless X is Gaussian.*

(iv) Unless X is degenerate, Y_X has a Pólya frequency density of infinite order.

(v) $\kappa_m(T_X) = (-1)^{m+1} m!\,\kappa_{2m}(X)/(2m)!$ *for* $m \ge 1$; *in particular* $\mathbb{E}T_X = \frac{1}{2}\operatorname{Var} X$ *and* $\operatorname{Var} T_X = -\kappa_4(X)/12$.

Proof. (i) $\mathbb{E}e^{-ue_k/c_k^2} = (1 + u/c_k^2)^{-1}$; multiply over k and compare with (13.1.1) at $s = \sqrt{u}$. (ii) Conditionally on T_X, Y_X is centred normal with variance $2T_X$, so $\mathbb{E}e^{i\theta Y_X} = \mathbb{E}e^{-\theta^2 T_X} = 1/L_X(\theta)$, which is also the characteristic function of $\sqrt{2b}\,Z + \sum_k L_k/c_k$, since $\mathbb{E}e^{i\theta L} = (1+\theta^2)^{-1}$. (iii) $\mathbb{E}e^{tY_X} = \mathbb{E}e^{t^2 T_X} = e^{bt^2}\prod_k (1 - t^2/c_k^2)^{-1}$ while $t^2 < c_1^2$, and $\mathbb{E}e^{t^2 e_1/c_1^2} = \infty$ for $t^2 \ge c_1^2$. (iv) The Laplace density $\frac{1}{2}e^{-|x|}$ and the normal densities are Pólya frequency functions of infinite

order, and the class is closed under convolution and under the limits involved here. Equivalently, the bilateral Laplace transform of Y_X is the reciprocal of $\varphi_X(s) = e^{-bs^2}\prod_k(1 - s^2/c_k^2)$, which has Schoenberg's form [300]. (v) For u near 0, $\log \mathbb{E}e^{-uT_X} = -\log L_X(\sqrt{u}) = -\sum_m \kappa_{2m}(X)u^m/(2m)!$, and comparing coefficients with $\sum_m \kappa_m(T_X)(-u)^m/m!$ gives the formula. □

Part (i) is Proposition 7.1.16 with the Gaussian factor allowed. For the Pólya law under RH, $b = 0$, $c_k = \gamma_k$ and T_X is the clock T of Proposition 1.2.1, and $(\varphi_X, \varphi_{Y_X})$ is the van Dantzig pair of Section 1.2, with $Y_X = Z\sqrt{2T}$ the variable $Y_{1/2}$ of Proposition 10.1.1. Without RH the clock T still exists (Theorem 5.1.1), and so does the pair; what fails without RH is the product form of the clock. Konstantopoulos, Patie and Sarkar study van Dantzig pairs of this kind through the Laguerre Pólya class and Pólya frequency functions, and connect them with the Lee and Yang property [153]. The correspondence runs one way only: every ferromagnetic law gives a Pólya frequency law, but $\varphi(s) = 1 - s^2$, the reciprocal of the bilateral Laplace transform of the Laplace law, is not a characteristic function.

Part (v) holds for the Pólya law with no hypothesis, with the clock T of Theorem 5.1.1 in place of T_X: the identity $\log \mathbb{E}e^{-uT} = -\log G(u) = -\log \mathbb{E}e^{\sqrt{u}X}$ is all the proof uses. With the cumulants of Table 13.3 it gives $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = -\kappa_4/12 = 3.71725993 \times 10^{-5}$, the constants of Section 1.2.

13.1.4 Examples

Table 13.1 lists the basic ferromagnetic laws with their clocks. The variables are those of Biane, Pitman and Yor [32]: C is the exit time of Brownian motion from $(-1, 1)$, with $\mathbb{E}e^{-uC} = 1/\cosh\sqrt{2u}$, and J is the hitting time of 1 by a Bessel process of dimension 3 started at 0, with $\mathbb{E}e^{-uJ} = \sqrt{2u}/\sinh\sqrt{2u}$, as in Section 9.1.1. The first two rows are the classical van Dantzig pairs of Section 1.2.

law of X	$L_X(s)$	c_k	clock T_X	Y_X
fair sign ε	$\cosh s$	$(k - \frac{1}{2})\pi$	$C/2$	hyperbolic secant
uniform on $[-1, 1]$	$\sinh s/s$	$k\pi$	$J/2$	logistic, scale $1/\pi$
$c_\nu(1 - x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$	$\Gamma(\nu + 1)(2/s)^\nu I_\nu(s)$	$j_{\nu,k}$	$\frac{1}{2}\tau_\nu$	Bessel type
$N(0, \sigma^2)$	$e^{\sigma^2 s^2/2}$	none	$\sigma^2/2$	$N(0, \sigma^2)$
$e^{-2\pi\cosh 2x}/K_0(2\pi)$	$K_{s/2}(2\pi)/K_0(2\pi)$	$2\nu_k$	$T_P/4$	$Z\sqrt{T_P/2}$
Pólya, under RH	$\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$	γ_k	T	$Z\sqrt{2T}$

Table 13.1: Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function J_ν, and τ_ν is the hitting time of 1 by a Bessel process of dimension $2\nu + 2$ started at 0. In the fifth row ν_k are the zeros of $\nu \mapsto K_{i\nu}(2\pi)$ and T_P is the clock of Theorem 8.7.9.

The third row rests on Kent's theorem for Bessel processes [144]: for the Bessel process of dimension $2\nu + 2$ started at 0, the hitting time of 1 satisfies

$\mathbb{E}e^{-u\tau_\nu} = (\sqrt{2u})^\nu/(2^\nu\Gamma(\nu+1)I_\nu(\sqrt{2u}))$, so that $\mathbb{E}e^{-u\tau_\nu/2} = 1/L_X(\sqrt{u})$ and $T_X = \frac{1}{2}\tau_\nu = \sum_k e_k/j_{\nu,k}^2$. The zeros $j_{\nu,k}$ are real for $\nu > -1$, so every symmetric beta law $c_\nu(1-x^2)^{\nu-1/2}$ with $\nu > -\frac{1}{2}$ is ferromagnetic. The family contains the arcsine law ($\nu = 0$), the uniform law ($\nu = \frac{1}{2}$) and the semicircle law ($\nu = 1$), and it tends to the law of ε as $\nu \downarrow -\frac{1}{2}$, where $\tau_{-1/2} = C$.

13.1.5 Closure

Products of functions of the form (13.1.1) have the same form, so sums of independent ferromagnetic variables are ferromagnetic. The class is closed under convergence in law, and this is what the Gaussian factor in Definition 13.1.1 is for: the laws of $n^{-1/2}\sum_{k\le n}\varepsilon_k$ have Gaussian moments of all orders, and their limit $N(0,1)$ does not.

Theorem 13.1.5. *Let X_n be ferromagnetic and suppose that $X_n \to X$ in law. Then X is ferromagnetic,* $\operatorname{Var} X_n \to \operatorname{Var} X$, *and $L_{X_n} \to L_X$ locally uniformly on $\mathbb{C}$.*

Proof. Let $\sigma_n^2 = \operatorname{Var} X_n$. By Corollary 13.1.3, $\mathbb{E}X_n^4 \le 3\sigma_n^4$, so the Paley Zygmund inequality applied to X_n^2 gives $P(X_n^2 > \sigma_n^2/4) \ge (\frac{3}{4})^2\sigma_n^4/\mathbb{E}X_n^4 \ge \frac{3}{16}$. A sequence that converges in law is tight, so the σ_n are bounded, say by σ. By Corollary 13.1.3 again, $|L_{X_n}(z)| \le L_{X_n}(\operatorname{Re} z) \le e^{\sigma^2(\operatorname{Re} z)^2/2}$, so the L_{X_n} form a locally bounded, hence normal, family of entire functions, and for each real s the variables e^{sX_n} are uniformly integrable, since their second moments are bounded. Hence $\mathbb{E}e^{sX_n} \to \mathbb{E}e^{sX}$ for real s, which gives $\mathbb{E}e^{sX} \le e^{\sigma^2 s^2/2}$ and, averaging over $s = \sqrt{2b}Z$, $\mathbb{E}e^{bX^2} < \infty$ for $b < 1/2\sigma^2$. So L_X is entire, every locally uniform limit point of the L_{X_n} agrees with it on $\mathbb{R}$, and by Vitali's theorem $L_{X_n} \to L_X$ locally uniformly. Each L_{X_n} has no zero in the half plane $\operatorname{Re} z > 0$, and $L_X(0) = 1$, so by Hurwitz's theorem L_X has no zero there either; by symmetry it has none in $\operatorname{Re} z < 0$. The fourth moments are bounded, so the X_n^2 are uniformly integrable and the variances converge. □

Proposition 13.1.6. *Let X be ferromagnetic with law ρ, and let $\mathbb{E}e^{b_0X^2} < \infty$.*

(i) For every $a > 0$ the law $\cosh(ax)\,\rho(dx)/\mathbb{E}\cosh(aX)$ is ferromagnetic.

(ii) For every $0 < \lambda < b_0$ the law $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$ is ferromagnetic.

Proof. (i) The moment generating function of the tilted law is proportional to $L_X(z+a) + L_X(z-a)$. For w with $\operatorname{Re} w > 0$ and $a > 0$ we have $|w+a| > |w-a|$. Applied to $w = z \pm ic_k$ this gives $|c_k^2 + (z+a)^2| > |c_k^2 + (z-a)^2|$ when $\operatorname{Re} z > 0$, and $|e^{b(z+a)^2}| = e^{4ab\operatorname{Re} z}|e^{b(z-a)^2}|$. So for $\operatorname{Re} z > 0$ every factor of (13.1.1) has larger modulus at $z+a$ than at $z-a$, and $L_X(z+a) + L_X(z-a) \ne 0$ unless L_X is constant, in which case the sum is 2. The sum is even in z, so it has no zeros with $\operatorname{Re} z < 0$ either, and the tilted law satisfies the moment

condition because $\cosh(ax) \le e^{a|x|}$. (ii) By (i), applied n times, the law with density proportional to $\cosh(x\sqrt{2\lambda/n})^n$ with respect to ρ is ferromagnetic. Since $\cosh y \le e^{y^2/2}$, these densities are dominated by $e^{\lambda x^2}$, which is ρ integrable, and they converge to $e^{\lambda x^2}$, so the laws converge in total variation to $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$. Theorem 13.1.5 applies. □

Part (i) is the one spin case of the Lee and Yang theorem, and for the Pólya law, whose transform has genus 1, it is Pólya's Hilfssatz II of Section 7.1.3. For the Pólya law, part (ii) is the forward half of de Bruijn's theorem [73]: multiplying p by $e^{\lambda x^2/4}$ is the de Bruijn Newman flow at time λ (Section 8.4).

Corollary 13.1.7. *RH holds if and only if the Pólya law is a limit in law of ferromagnetic laws.*

Proof. If RH holds, the Pólya law is ferromagnetic by (1.2.4), and it is its own limit. Conversely, a limit in law of ferromagnetic laws is ferromagnetic by Theorem 13.1.5, and then every zero of Ξ is real. □

Let δ be the infimum of the total variation distances $\sup_A |P(X \in A) - P(Y \in A)|$ from the Pólya variable X to ferromagnetic variables Y; when Y has a density f the distance is $\mathrm{TV}(f,p) = \frac{1}{2}\int |f-p|$. If RH holds, $\delta = 0$; conversely, ferromagnetic laws that converge to the Pólya law in total variation converge in law, so $\delta = 0$ implies RH.

13.2 Laws on an interval

The Pólya law is symmetric and unimodal, and every such law is a mixture of uniform laws. This section asks which mixtures are ferromagnetic, for laws on an interval. Mixing towards the centre can destroy the property, as an atom at the origin shows; a concave shoulder guarantees it, and a convex one rules it out unless the density is linear on its support, a scaled triangle; and mixing towards the edges is always safe. The Pólya law falls between the two shoulder cases, which is one way to see why its zeros are hard.

13.2.1 Mixtures of uniform laws

By Khinchine's theorem [86], a random variable X is symmetric and unimodal about 0 if and only if $X \stackrel{d}{=} UR$ with $R \ge 0$ independent of U. Then $L_X(s) = \mathbb{E}[\sinh(sR)/(sR)]$, and if X has a density p that is absolutely continuous on $(0,\infty)$ and tends to 0 at infinity, R has the density $-2rp'(r)$. Williamson's theorem refines the representation [345]: a density on $(0,\infty)$ is k monotone exactly when it is a scale mixture of $k(1-x)_+^{k-1}$, so the symmetric k monotone densities are the scale mixtures of

$$f_k(x) = \tfrac{k}{2}(1-|x|)_+^{k-1},$$

of which f_1 is the uniform density and f_2 the triangle. The uniform law itself is ferromagnetic, with $L_U(s) = \sinh s/s$, and Viète's product

$$\frac{\sinh s}{s} = \prod_{k\ge 1} \cosh \frac{s}{2^k}, \qquad U \stackrel{d}{=} \sum_{k\ge 1} 2^{-k}\varepsilon_k,$$

shows why: the uniform variable is a sum of independent fair signs, and the property is that of a single spin. So is every sum $\sum_k c_k U_k$ of independent scaled uniform variables. Mixtures behave differently.

THEOREM 13.2.1. *For every $0 < q < 1$ the law $(1-q)\,U + q\,\delta_0$ is symmetric and unimodal but not ferromagnetic. Consequently, for every $0 < q < 1$ and every small enough $a > 0$, the mixture of the uniform laws on $[-1,1]$ and on $[-a,a]$ with weights $1-q$ and q is not ferromagnetic.*

Proof. The moment generating function $L(s) = (1-q)\sinh s/s + q$ vanishes exactly where $\sinh s/s = -c$, with $c = q/(1-q) > 0$. There is no real zero, since $\sinh s/s \ge 1$ on $\mathbb{R}$, and only finitely many imaginary ones, since $s = iy$ needs $\sin y/y = -c$ and $|\sin y/y| \le 1/|y|$. But L has infinitely many zeros. It is entire of order 1, so with finitely many zeros it would be $e^{\alpha s}P(s)$ with a polynomial P; evenness forces $\alpha = 0$, and L is not a polynomial, since it grows like e^s/s on the positive axis. So all but finitely many zeros lie off the imaginary axis. The second statement follows from Theorem 13.1.5, since the mixtures converge in law to $(1-q)U + q\delta_0$ as $a \downarrow 0$. □

Both components of the mixture are ferromagnetic, so the class is not closed under mixing, even of unimodal laws with a common mode. For $c > c_*$, where $c_* = -\min_y \sin y/y = 0.2172336\ldots$, the minimum being at the first positive root $y = 4.4934095\ldots$ of $\tan y = y$, there are no imaginary zeros at all.

PROPOSITION 13.2.2. *For $0 \le q \le 1$ the law $q\,\delta_0 + \frac{1-q}{2}(\delta_{-1} + \delta_1)$ is ferromagnetic if and only if $q \le \frac{1}{2}$ or $q = 1$.*

Proof. For $q = 1$ the law is δ_0, and $L = 1$ has no zeros. For $q < 1$ the zeros of $L(s) = (1-q)\cosh s + q$ solve $\cosh s = -c$ with $c = q/(1-q)$. If $c \le 1$, write $-c = \cos\theta_0$ with $\theta_0 \in [\frac{\pi}{2}, \pi]$; the solutions $s = \pm i\theta_0 + 2\pi i k$ are imaginary. If $c > 1$ the solutions are $s = \pm \operatorname{arccosh} c + i\pi(2k+1)$, which are not; for $q = 0.6$ the first is $0.9624237 + i\pi$. □

13.2.2 The shoulder theorem

Let p be a symmetric density on $[-1,1]$ that is absolutely continuous and non-increasing on $[0,1]$, with $p(1) = 0$, and put $g = -p' \ge 0$ on $(0,1)$. Integrating

$\varphi_X(z) = 2\int_0^1 p(y)\cos(zy)\,dy$ by parts turns the cosine transform of the density into a sine transform of its slope:

$$z\,\varphi_X(z) = 2\int_0^1 g(y)\sin(zy)\,dy. \tag{13.2.1}$$

Pólya proved that if f is positive, integrable and nondecreasing on $(0,1)$, then $\int_0^1 f(t)\cos(zt)\,dt$ and $\int_0^1 f(t)\sin(zt)\,dt$ have only real zeros [268].

THEOREM 13.2.3. *Let p be a symmetric density on $[-1,1]$, absolutely continuous and nonincreasing on $[0,1]$, with $p(1)=0$, and let $[-c,c]$, $0<c\le 1$, be its support.*

(i) If p is concave on $[0,1]$, then X is ferromagnetic.

(ii) If p is convex on $[0,1]$ and not linear on $[0,c]$, then X is not ferromagnetic. Moreover $\varphi_X \ge 0$ on $\mathbb{R}$, and its real zeros, if any, are double and lie in a lattice $(2\pi/s_0)\mathbb{Z}$ with $0<s_0<c$.

Proof. (i) If p is concave, then $c=1$ and g is nondecreasing on $(0,1)$. If $g>0$ on $(0,1)$, Pólya's theorem in its sine form and (13.2.1) show that φ_X has only real zeros. Otherwise p is constant near 0; then $(p+\epsilon(1-|x|))/(1+\epsilon)$ is concave with a positive slope, so it is ferromagnetic, and Theorem 13.1.5 applies as $\epsilon \downarrow 0$.

(ii) If p is convex, g is nonincreasing on $(0,c)$, so $g(y) = g(c-) + \mu((y,c))$ with a positive measure μ on $(0,c)$, which is not zero because p is not linear. Since $\int_0^s \sin(zy)\,dy = 2\sin^2(zs/2)/z$, Fubini's theorem in (13.2.1) gives, for real $z\neq 0$,

$$\frac{z^2}{4}\,\varphi_X(z) = g(c-)\sin^2\frac{zc}{2} + \int_{(0,c)} \sin^2\frac{zs}{2}\,\mu(ds) \;\ge 0.$$

A real zero z_0 needs $z_0 s \in 2\pi\mathbb{Z}$ for every s in the support of μ, so with any such $s_0 < c$ the real zeros lie in $(2\pi/s_0)\mathbb{Z}$. At a real zero the right side has vanishing first derivative and second derivative $\frac12 g(c-)c^2 + \frac12\int s^2\mu(ds) > 0$, because every cosine $\cos(z_0 s)$ that enters equals 1; so the zero is double. Suppose now that every zero of φ_X were real. Since φ_X is even and entire of exponential type, Hadamard's theorem would give $\varphi_X(z) = \prod_{m\in M}(1 - z^2/z_m^2)^2$ for a set M of positive integers, with $z_m = 2\pi m/s_0$, and hence

$$\varphi_X(iy) \le \prod_{m\ge 1}\Big(1 + \frac{y^2 s_0^2}{4\pi^2 m^2}\Big)^2 = \Big(\frac{\sinh(ys_0/2)}{ys_0/2}\Big)^2 \le e^{s_0|y|}.$$

But $p>0$ on $[0,c)$, so for $0<\epsilon<c-s_0$ and $y>0$, $\varphi_X(iy) = L_X(y) \ge \int_{c-\epsilon}^c e^{yx}p(x)\,dx \ge C_\epsilon e^{(c-\epsilon)y}$ with $C_\epsilon > 0$, which contradicts the bound as $y\to\infty$. □

COROLLARY 13.2.4. *(i) The law with density proportional to $1-|x|^b$ on $[-1,1]$, $b>0$, is ferromagnetic if and only if $b\ge 1$. In Khinchine's form, UR with R distributed as $\mathrm{Beta}(a,1)$ is ferromagnetic if and only if $a\ge 2$.*

(ii) The symmetric k monotone kernel f_k is ferromagnetic if and only if $k \le 2$.

(iii) The laws with density proportional to $(1-x^2)^m$, $0 < m \le 1$, which include the semicircle law, are ferromagnetic.

(iv) The product U_1U_2 of two independent uniform variables, with density $\frac{1}{2}\log(1/|x|)$, is not ferromagnetic.

Proof. (i) $1-x^b$ is concave on $[0,1]$ for $b \ge 1$ and convex and not linear for $b < 1$; the mixing variable has density $-2rp'(r) \propto r^b$, that of Beta$(b+1,1)$. This settles $a = b+1 > 1$. For $0 < a \le 1$ the density of UR is unbounded at 0 and the theorem does not apply, but Proposition 13.2.6 below excludes these laws: with $\mu_j = a/(a+2j)$, its case $j=1$ requires $a \ge \sqrt{10}-2 > 1$. (ii) f_1 is uniform and f_2 the triangle; for $k \ge 3$, $(1-x)^{k-1}$ is convex and not linear. (iii) $\frac{d^2}{dx^2}(1-x^2)^m = -2m(1-x^2)^{m-2}(1-(2m-1)x^2) \le 0$ on $[0,1]$ when $m \le 1$. (iv) The density is unbounded at the origin, so Theorem 13.2.3 does not apply, but directly $\varphi(z) = \int_0^1 \sin(zu)/(zu)\,du = \mathrm{Si}(z)/z$, and $\mathrm{Si}(z) > 0$ for $z > 0$. So φ has no real zero, while it has infinitely many zeros, by the argument in the proof of Theorem 13.2.1. □

13.2.3 Mixing towards the edges

Theorem 13.2.5. *Every symmetric density on $[-1,1]$ that is nondecreasing in $|x|$ is ferromagnetic. Equivalently, every mixture of the uniform laws on the shells $\{a \le |x| \le 1\}$, $0 \le a < 1$, is ferromagnetic.*

Proof. Here $\varphi_X(z) = 2\int_0^1 p(t)\cos(zt)\,dt$ with p nondecreasing on $(0,1)$. If $p > 0$ on $(0,1)$, Pólya's theorem in its cosine form [268] applies; otherwise add ϵ times the uniform density, normalise, and let $\epsilon \downarrow 0$ (Theorem 13.1.5). A density nondecreasing in $|x|$ is a mixture of the shell densities, and conversely. □

13.2.4 A moment test

Ferromagnetic laws satisfy inequalities between consecutive moments. If X is ferromagnetic, $L_X(\sqrt{x}) = \sum_j b_j x^j$, with $b_j = \mathbb{E}X^{2j}/(2j)!$, equals $e^{bx}\prod_k(1 + x/c_k^2)$, an entire function of x in the Laguerre Pólya class with only negative zeros. Its coefficients satisfy the Turán inequalities $(j!\,b_j)^2 \ge (j-1)!\,b_{j-1}(j+1)!\,b_{j+1}$, the limit as $n, K \to \infty$ of Newton's inequalities for the polynomials $(1 + bx/n)^n \prod_{k\le K}(1 + x/c_k^2)$, whose zeros are real; that is

$$b_j^2 \ge \Big(1 + \frac{1}{j}\Big) b_{j-1}b_{j+1}, \qquad j \ge 1. \tag{13.2.2}$$

For a mixture $X = UR$ the moments factor as $\mathbb{E}X^{2j} = \mu_j/(2j+1)$ with $\mu_j = \mathbb{E}R^{2j}$.

PROPOSITION 13.2.6. *If UR is ferromagnetic and R is not 0 almost surely, then for every $j \geq 1$*

$$\frac{2j+1}{2j+3} \leq \frac{\mu_j^2}{\mu_{j-1}\mu_{j+1}} \leq 1.$$

Proof. The moments μ_j are finite, since $\mathbb{E}(UR)^{2j} = \mu_j/(2j+1)$ and L_{UR} is entire, and positive, since $P(R > 0) > 0$. Substitute $b_j = \mu_j/(2j+1)!$ in (13.2.2): the factor is $(1 + \frac{1}{j})\frac{(2j+1)!^2}{(2j-1)!(2j+3)!} = \frac{2j+1}{2j+3}$. The upper bound is the Cauchy Schwarz inequality and holds for every R. □

The test is necessary and weak. For R distributed as $\mathrm{Beta}(a, 1)$, $\mu_j = a/(a + 2j)$; at $j = 1$ the condition is $a^2 + 4a - 6 \geq 0$, that is $a \geq \sqrt{10} - 2 = 1.1622777$, and for $j \geq 2$ it holds for every $a > 0$. Corollary 13.2.4(i) gives the true threshold $a \geq 2$.

13.2.5 SUMMARY

Table 13.2 collects the answers. The Pólya law is unbounded, so none of them applies to it.

law on $[-1, 1]$	ferromagnetic
uniform, and sums $\sum_k c_k U_k$	yes
symmetric beta $c_\nu(1 - x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$	yes
concave shoulder	yes
nondecreasing in $\lvert x\rvert$ (shell mixtures)	yes
convex shoulder, not linear on the support	no
$1 - \lvert x\rvert^b$	if and only if $b \geq 1$
k monotone kernel f_k	if and only if $k \leq 2$
uniform with an atom at 0, weight in $(0, 1)$	no
product of two uniform variables	no
spin one, $q\delta_0 + \frac{1-q}{2}(\delta_{-1} + \delta_1)$	if and only if $q \leq \frac{1}{2}$ or $q = 1$

Table 13.2: Which laws on $[-1, 1]$ are ferromagnetic. Every entry is proved in this section or in Section 13.1.

13.3 THE PÓLYA LAW

The Pólya variable X has the density $p = \Phi/\Xi(0)$, with $\Xi(0) = \xi(\frac{1}{2}) = 0.4971207782$, and $L_X(s) = \xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ is an entire function of order 1. Its density decays like $\exp(-\pi e^{2|t|})$, so X has Gaussian moments of all orders, and RH is the statement that X is ferromagnetic; by Theorem 13.1.2, with $b = 0$, it is the product formula $L_X(s) = \prod_k (1 + s^2/\gamma_k^2)$ over the positive ordinates. The law is symmetric and unimodal, since $\Phi'(t) < 0$ for $t > 0$ [68, Theorem A].

13.3.1 Necessary conditions

If X is ferromagnetic, Corollary 13.1.3 with $c_k = \gamma_k$ gives $\kappa_2 = 2\sum_k \gamma_k^{-2}$, $\kappa_4 = -12\sum_k \gamma_k^{-4}$ and $\kappa_6 = 240\sum_k \gamma_k^{-6}$. Table 13.3 computes the cumulants. The agreement of the columns is not evidence for RH. The Hadamard product $\xi(\frac12 + s)/\xi(\frac12) = \prod_k (1 + s^2/\gamma_k^2)$, with one factor for each pair of zeros $\frac12 \pm i\gamma_k$, holds with no hypothesis if γ_k is allowed to be complex, and it converges absolutely because $\sum_k |\gamma_k|^{-2} < \infty$; it gives the same formulas for the cumulants.

	moments of Φ	Taylor coefficients of $\log\xi$	zero sums
κ_2	0.0462099862308	0.0462099862308	0.046209986105
κ_4	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$
κ_6	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$

Table 13.3: Even cumulants of the Pólya law, by quadrature of the moments of Φ, from the Taylor coefficients of $\log\xi(\frac12 + s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k \gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference 1.3×10^{-10} in κ_2 is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed by `fm_polya.py`.

The moment test of Section 13.2.4 is sharper. For the Pólya law the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ are the Taylor coefficients of Ξ, up to sign and the factor $\Xi(0)$:

$$\frac{\Xi(t)}{\Xi(0)} = \sum_{j\ge0} (-1)^j b_j t^{2j}.$$

In this form (13.2.2), $b_j^2 \ge (1 + \frac1j) b_{j-1} b_{j+1}$, is the system of Turán inequalities for the moments $\int_0^\infty t^{2j}\Phi(t)\,dt$, which Csordas, Norfolk and Varga proved for every $j \ge 1$, answering a question of Pólya [68]. Table 13.4 shows the margins. They decrease slowly; the Gaussian law, for which $b_j = (\sigma^2/2)^j/j!$, satisfies every inequality with equality.

The Turán inequalities are the first of a hierarchy. For $d, n \ge 0$ let

$$J_{d,n}(x) = \sum_{i=0}^{d} \binom{d}{i} (n+i)!\, b_{n+i}\, x^i,$$

the Jensen polynomial of degree d and shift n of the sequence $j!\,b_j$. Pólya proved that RH holds if and only if every $J_{d,n}$ has only real zeros, and for $d = 2$ this is (13.2.2) with $j = n + 1$. Griffin, Ono, Rolen and Zagier proved that for each d all but finitely many of the $J_{d,n}$ have only real zeros, and that all of them do when $d \le 8$ [99]. Each degree is a necessary condition for the Pólya law to be ferromagnetic, and none of them is known to fail. With $F(x) = \sum_{j\ge0} b_j x^j$, so that $F(-t^2) = \varphi_X(t)$, the polynomials $J_{d,n}$ with n fixed are the Jensen polynomials

j	b_j	$b_j^2/(b_{j-1}b_{j+1})$	$1+1/j$
1	2.3104993×10^{-2}	2.1496879	2
2	2.4833405×10^{-4}	1.5941155	1.5
3	1.6743526×10^{-6}	1.4057378	1.3333333
4	8.0306974×10^{-9}	1.3100631	1.25
5	2.9401358×10^{-11}	1.2518550	1.2
10	1.1310848×10^{-24}	1.1322525	1.1
15	1.7510030×10^{-39}	1.0907154	1.0666667
20	2.9919818×10^{-55}	1.0693835	1.05
24	2.1176410×10^{-68}	1.0585214	1.0416667

Table 13.4: The moment test (13.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1+\frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed by `fm_polya.py`.

of the derivative $F^{(n)}$, and all of them have only real zeros exactly when $F^{(n)}$ is in the Laguerre Pólya class. For $n \geq 1$ that says nothing about the zeros of φ_X. For the spin one law of Table 13.2 with $q > \frac{1}{2}$, $F(x) = q + (1-q)\cosh\sqrt{x}$ and $F'(x) = (1-q)\sinh\sqrt{x}/(2\sqrt{x})$, whose zeros $-k^2\pi^2$ are real, so every $J_{d,n}$ with $n \geq 1$ has only real zeros, while $\varphi_X(t) = q + (1-q)\cos t$ has no real zero at all. In the other direction, if ξ had only finitely many zeros off the critical line, then F, which has order $\frac{1}{2}$, would have only finitely many nonreal zeros, and $F^{(n)}$ would be in the Laguerre Pólya class for every large n, by the theorem of Craven, Csordas and Smith on real entire functions of order less than two [67].

A necessary condition of a different kind comes from the correlation inequalities of Section 13.1.1. Write $\Phi = e^{-F}$. Newman proved that F' is convex on $[0,\infty)$ [211]; in his normalisation $\Xi(x/2) = 4\int e^{ixt - V(t)}\,dt$, and $V(t) = F(2t) + \log 2$. So the Pólya law belongs to the class of Ellis, Monroe and Newman, and every ferromagnet whose single spins are independent copies of X satisfies the Griffiths, Hurst and Sherman inequality [83].

13.3.2 A mixture of uniform laws

Since p is symmetric and unimodal, Khinchine's theorem of Section 13.2.1 writes $X \overset{d}{=} UR$, with R independent of U and with the density $h(r) = -2rp'(r) = -2r\Phi'(r)/\Xi(0)$ on $(0,\infty)$. Since $\mathbb{E}|X|^m = \mathbb{E}R^m/(m+1)$, the mixing variable has $\mathbb{E}R = 2\mathbb{E}|X| = 0.3464543599$ and $\mathbb{E}R^2 = 3\operatorname{Var}X = 0.1386299587$. Integrating by parts as in (13.2.1), now on the half line,

$$z\,\varphi_X(z) = 2\int_0^\infty \big(-p'(u)\big)\sin(zu)\,du,$$

so RH says that the sine transform of the positive function $-\Phi'$ on $(0,\infty)$ has only real zeros. The shape of this slope is that of neither part of Theorem 13.2.3. Since $p'' = p\,(F'^2 - F'')$, the density is concave near the origin, where $p''(0) = -F''(0)p(0) = -33.65$, and convex for large u, where $F'^2/F'' \to \infty$. On a bounded support, a slope that rises gives only real zeros, by Pólya's theorem, and one that falls, and is not constant on the support, gives zeros off the real line, by Theorem 13.2.3(ii); the slope of the Pólya law does both. The damped densities, proportional to $e^{tu^2/4}\Phi(u)$ with $t < 0$, are symmetric and unimodal too, hence mixtures of uniform laws, and they are mixtures of arcsine laws as well (Proposition 11.5.13), while their transforms have nonreal zeros, so neither mixture can decide RH by itself (Section 11.5.5).

The mixture does not discretise well. Replacing R by J equally weighted atoms at its quantiles r_j of order $(j - \frac{1}{2})/J$ gives the characteristic function $\varphi_J(z) = J^{-1}\sum_j \sin(r_j z)/(r_j z)$. For $J = 5$ it has 17 zeros in the rectangle $(0.25, 100) \times (-30, 30)$, of which only 7 are real, and for $J = 20$ only 3 of 19 are real, while φ_X has 29 zeros in $(0, 100)$, all real (computed by `fm_polya.py`). As Theorem 13.2.1 suggests, mixing in the R variable destroys the property easily.

13.3.3 No component with finitely many zeros

A sum of independent ferromagnetic variables is ferromagnetic (Section 13.1.5), and one might hope to reach the Pólya law, or to approximate it, as such a sum of simple pieces. The growth of L_X rules out most pieces.

Theorem 13.3.1. *Let X be a symmetric random variable with $\mathbb{E}e^{sX} < \infty$ for every real s and $\log \mathbb{E}e^{sX} = o(s^2)$ as $s \to \infty$, and let $X \stackrel{d}{=} D + R$ with D and R independent. Then $\log \mathbb{E}e^{sD} = o(s^2)$ as $|s| \to \infty$. In particular, if D is ferromagnetic and not 0, then $b = 0$ in its Hadamard form* (13.1.1)*, and its characteristic function has infinitely many zeros. The Pólya law satisfies the hypothesis, so it has no Gaussian component, and it is not a finite or convergent infinite sum of independent ferromagnetic variables of which one is not 0 and has finitely many zeros.*

Proof. By independence $\mathbb{E}e^{sX} = \mathbb{E}e^{sD}\,\mathbb{E}e^{sR}$, and both factors are finite because the product is. Jensen's inequality gives $\mathbb{E}e^{sR} \geq e^{s\mathbb{E}R}$, and $\mathbb{E}R = -\mathbb{E}D$ because $\mathbb{E}X = 0$. So $\log \mathbb{E}e^{sD} \leq \log \mathbb{E}e^{sX} + s\mathbb{E}D$, and the right side is $o(s^2)$ as $s \to \pm\infty$, by the symmetry of X. If D is ferromagnetic, $\mathbb{E}e^{sD} = e^{bs^2}\prod_k (1 + s^2/c_k^2) \geq e^{bs^2}$, so $b = 0$. If moreover there are finitely many zeros, then $\varphi_D(t) = \prod_{k\leq K}(1 - t^2/c_k^2)$ is a polynomial bounded on the real line, hence constant, so $K = 0$ and $D = 0$. For the Pólya law, $\log \mathbb{E}e^{sX} = \log \xi(\frac{1}{2} + s) - \log \xi(\frac{1}{2}) = O(s \log s)$, since ξ has order 1. A summand of an almost surely convergent sum of independent variables is an independent component of the sum. □

The laws of Section 13.4 that keep the first K zeros of φ_X exactly carry a

Gaussian factor, so none of them is a component of the Pólya law. If RH holds, any decomposition of X into independent ferromagnetic pieces splits the ordinates into infinite sets, each the zero set of a characteristic function $\prod_k (1 - t^2/c_k^2)$ with no Gaussian factor, like those of the uniform law, the fair sign and the symmetric beta laws in Table 13.1.

The simplest such splitting would be a Viète product $L_X(s) = \prod_j \cosh(a_j s)$, which would write $X = \sum_j a_j \varepsilon_j$ as a sum of independent scaled fair signs, a member of the Griffiths Simon class with uncoupled spins. The ordinates would then split into arithmetic progressions, and three consecutive terms of a progression satisfy $\gamma - 2\gamma' + \gamma'' = 0$. The hypothesis that the positive ordinates are linearly independent over the rationals, which Ingham used in his work on the Mertens conjecture [125], forbids such a relation. Under that hypothesis the Pólya law is not a sum of independent scaled fair signs, and if it belongs to the Griffiths Simon class at all, the spins must be coupled. Section 13.6 returns to this question.

13.4 Ferromagnetic approximations of the Pólya law

By Corollary 13.1.7 and the remark after it, RH holds exactly when the total variation distance δ from the Pólya law to the class of ferromagnetic laws is 0. This section bounds δ from above with no hypothesis, by the distances to laws that are proved to be ferromagnetic. The distances are enclosed in ball arithmetic, with explicit bounds for the discretisation and the tails, so the bounds on δ they give are proved.

13.4.1 Pólya's approximations

The Gaussian law $N(0, \operatorname{Var} X)$ is ferromagnetic and lies at total variation distance 1.48×10^{-2} from the Pólya law. Pólya kept the leading part of the first term of Φ and made it even:

$$\Phi^*(t) = 8\pi^2 \cosh \tfrac{9t}{2}\, e^{-2\pi \cosh 2t}, \qquad \int \Phi^*(t) e^{i\theta t}\, dt = 4\pi^2 \Big[K_{\frac{9}{4} + \frac{i\theta}{2}}(2\pi) + K_{\frac{9}{4} - \frac{i\theta}{2}}(2\pi) \Big],$$

by $\int e^{-a \cosh u + \nu u}\, du = 2K_\nu(a)$, and he proved that this transform has only real zeros [273]. So the law with density proportional to Φ^* is ferromagnetic. It is the law of $\frac{\varepsilon}{2} \log G$, where G is generalised inverse Gaussian with density proportional to $g^{5/4} e^{-\pi(g + 1/g)}$, since the substitution $g = e^{2t}$ turns $e^{9t/2} e^{-2\pi \cosh 2t}$ into that density; so it can be sampled exactly. Keeping the whole first term gives the positive kernel

$$\Psi(t) = \Big(8\pi^2 \cosh \tfrac{9t}{2} - 12\pi \cosh \tfrac{5t}{2} \Big) e^{-2\pi \cosh 2t},$$

and the same theorem of Pólya covers it.

Proposition 13.4.1. *The law with density proportional to Ψ is ferromagnetic.*

Proof. Write $\Psi(t) = \psi(it)f(t)$ with $f(t) = e^{-2\pi\cosh 2t}$ and $\psi(z) = 8\pi^2\cos\frac{9z}{2} - 12\pi\cos\frac{5z}{2}$. In the variable $w = z/2$, ψ is the trigonometric polynomial $8\pi^2\cos 9w - 12\pi\cos 5w$ of degree 9. At $w_j = j\pi/9$, $0 \le j < 18$, the first term is $8\pi^2(-1)^j$ and the second is at most $12\pi < 8\pi^2$ in absolute value, so ψ changes sign at least 18 times in a period, and since a trigonometric polynomial of degree 9 has at most 18 zeros in a period strip, all its zeros are real. So ψ is a real entire function of order 1 with only real zeros. The transform of f is $K_{i\theta/2}(2\pi)$, which has only real zeros, by Pólya's theorem on $\int e^{-\lambda\cosh t}e^{izt}dt$ [73, (1.2)], and f decays faster than $e^{-|t|^b}$ for every b. Pólya's theorem on universal factors [73, p. 197], [54, Proposition 1] then says that the transform of $\psi(it)f(t)$ has only real zeros. The density decays like f, so it has Gaussian moments of all orders. □

As densities both approximations are poor: their total variation distances to p are 0.099 and 0.161, their variances 0.0624 and 0.0726 against $\operatorname{Var} X = 0.0462$, and their values at the origin 1.404 and 1.163 against $p(0) = 1.797$, all enclosed in ball arithmetic (`fm_tvcert.py`). Their transforms have 14 zeros each in $(0.25, 62.5)$, all real, and an argument principle count in $(0.25, 62.5) \times (-2, 2)$ finds no others, the same number as the ordinates below 62.5. The positions differ: the first five are 8.99, 19.07, 24.47, 28.97, 33.01 for Φ^* and 7.60, 18.72, 24.28, 28.83, 32.89 for Ψ, against 14.13, 21.02, 25.01, 30.42, 32.94 (computed by `fm_polya.py`).

13.4.2 Tilts that shrink the strip

Every zero γ of $\varphi_X = \Xi/\Xi(0)$ satisfies $|\operatorname{Im}\gamma| < \frac{1}{2}$, because $0 < \operatorname{Re}\rho < 1$. Multipliers of the density move the zeros towards the real axis, and de Bruijn measured by how much. For $\lambda_1, \dots, \lambda_N > 0$ and $\mu \ge 0$ put

$$m(t) = e^{\mu t^2}\prod_{k\le N}\cosh(\lambda_k t), \qquad c(m) = 2\mu + \sum_{k\le N}\lambda_k^2.$$

Then $m(t) = \mathbb{E}e^{tY} = \mathbb{E}\cosh(tY)$ for $Y = \sqrt{2\mu}\,Z + \sum_k \lambda_k\varepsilon_k$, and $c(m) = \operatorname{Var} Y$.

Proposition 13.4.2. *The zeros of $\int \Phi(t)m(t)e^{izt}\,dt$ lie in the strip $|\operatorname{Im} z| \le (\max(\frac{1}{4} - c(m), 0))^{1/2}$. The same holds for $N = \infty$ when $\sum_k \lambda_k^2 < \infty$. In particular, with no hypothesis, the law with density proportional to $m\,p$ is ferromagnetic whenever $\operatorname{Var} Y \ge \frac{1}{4}$; for example $\cosh(at)\,p$ for $a \ge \frac{1}{2}$, $\frac{\sinh(ct)}{ct}\,p$ for $c \ge \frac{\sqrt{3}}{2}$, and $e^{\mu t^2}p$ for $\mu \ge \frac{1}{8}$. If no zero of ζ has real part greater than $\frac{7}{8}$, as announced in October 2026 (Section 12.1.2), all of this holds with $\frac{9}{64}$ in place of $\frac{1}{4}$; then $\cosh(at)\,p$ is ferromagnetic for $a \ge \frac{3}{8}$ and $\frac{\sinh(ct)}{ct}\,p$ for $c \ge \frac{3\sqrt{3}}{8}$.*

Proof. For finite N this is de Bruijn's Theorem 13, for the Gaussian factor $e^{\frac{1}{2}(2\mu)t^2}$, followed by his Theorem 12, for the product of cosines [73]; both apply to an even kernel F with $F(t) = O(e^{-|t|^b})$ for some $b > 2$, which Φ and $\Phi e^{\mu t^2}$ are, and both take as input a closed strip containing the zeros, here $|\operatorname{Im} z| \le \frac{1}{2}$. Under the

hypothesis of the last sentence every zero has $\frac{1}{8} \le \operatorname{Re}\rho \le \frac{7}{8}$, by $\xi(s) = \xi(1-s)$, so the zeros of φ_X lie in $|\operatorname{Im} z| \le \frac{3}{8}$, and the same two theorems give the strip with $\frac{9}{64}$ in place of $\frac{1}{4}$. For $N = \infty$ the partial products m_N have $c(m_N) \uparrow c(m)$, and their transforms converge to that of Φm locally uniformly in z, by dominated convergence with $m_N \le m$. By Hurwitz's theorem the zeros of the limit lie in the intersection of the strips. The tilted densities decay like Φ, so they have Gaussian moments of all orders. The three examples have $\operatorname{Var} Y = a^2$, $c^2/3$ (by Viète's product $\frac{\sinh ct}{ct} = \prod_k \cosh \frac{ct}{2^k}$) and 2μ. □

The single cosine is elementary, and the proof shows where the number $\frac{1}{2}$ comes from. The characteristic function of $\cosh(at)p(t)$ is proportional to $\varphi_X(z+ia) + \varphi_X(z-ia)$. Take $z = x + iy$ with $y > 0$ and a pair of zeros $\beta \pm i\eta$ of φ_X, with $c = (x-\beta)^2$. De Bruijn's identity [73, Lemma 1]

$$\big(c + (y+a-\eta)^2\big)\big(c + (y+a+\eta)^2\big) - \big(c + (y-a-\eta)^2\big)\big(c + (y-a+\eta)^2\big) = 8ay\,(c + y^2 + a^2 - \eta^2)$$

shows that the pair contributes a factor larger than 1 to $|\varphi_X(z+ia)|^2/|\varphi_X(z-ia)|^2$ as soon as $a > |\eta|$. With $a \ge \frac{1}{2}$ this holds for every zero, by the Hadamard product, so the two terms cannot cancel above the real axis, nor, by symmetry, below it. In the variable $s = \frac{1}{2} + iz$ the transform is $\frac{1}{2}(\xi(s+a) + \xi(s-a))$, and Lagarias proved, in addition, that its zeros are simple [161]. For the de Bruijn Newman flow the multiplier is $e^{\lambda t^2/4}$, with $c(m) = \lambda/2$; the proposition gives de Bruijn's bound $\Lambda \le \frac{1}{2}$, and the bound $\Lambda \le 0.2$ of Platt and Trudgian [247] makes $e^{\lambda t^2/4}p$ ferromagnetic for every $\lambda \ge 0.2$, again with no hypothesis.

The three multipliers of the proposition with $\operatorname{Var} Y = \frac{1}{4}$ give total variation distances 2.731×10^{-3}, 2.739×10^{-3} and 2.751×10^{-3} from p, in the order listed. The flow at $\lambda = 0.2$, with $c(m) = 0.1$, gives 1.097×10^{-3}. For comparison, $\cosh(\sqrt{0.1}\,t)\,p$, with the same contraction, gives 1.094×10^{-3}, but whether it is ferromagnetic is not known: the bound of Platt and Trudgian is a statement about the Gaussian flow, and Proposition 13.4.2 needs $\operatorname{Var} Y \ge \frac{1}{4}$, or $\frac{9}{64}$ if the announced half plane stands, both above 0.1 (the distances are enclosed in ball arithmetic by `fm_tvcert.py`).

The cosine tilts define a constant in the spirit of Λ. Let Ω be the infimum of the $a_0 \ge 0$ such that $\cosh(at)\,p$ is ferromagnetic for every $a \ge a_0$. Proposition 13.4.2 gives $\Omega \le \frac{1}{2}$, and $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros. If RH holds, Proposition 13.1.6(i) gives $\Omega = 0$. Conversely, if $\Omega = 0$, the tilted laws converge to p in total variation as $a \downarrow 0$, and p is ferromagnetic by Theorem 13.1.5. So RH holds if and only if $\Omega = 0$. Unlike the flow, the operators $\varphi \mapsto \varphi(\cdot + ia) + \varphi(\cdot - ia)$ do not form a semigroup, and we do not know whether the set of good values of a is an interval.

13.4.3 Laws that keep the first zeros

Let $0 < \gamma_1 \le \gamma_2 \le \dots$ be the ordinates of the zeros on the critical line, with multiplicity, and put $s_K = \sum_{k\le K} \gamma_k^{-2}$ and $\beta_K = \frac{1}{2}\operatorname{Var} X - s_K$. The idea is to keep the first K factors of the Hadamard product exactly and to replace the others by the Gaussian factor with the same effect on the variance:

$$\varphi_K(\theta) = e^{-\beta_K \theta^2} \prod_{k\le K} \Big(1 - \frac{\theta^2}{\gamma_k^2}\Big).$$

A symmetric Maxwell variable M, with density $x^2 e^{-x^2/2}/\sqrt{2\pi}$, has $\mathbb{E}e^{i\theta M} = (1-\theta^2)e^{-\theta^2/2}$ and $\operatorname{Var} M = 3$; it is ferromagnetic, with $L_M(s) = (1+s^2)e^{s^2/2}$. Let $M_1, M_2, \dots$ be independent copies.

Theorem 13.4.3. *(i) $\beta_K > 0$ for every K. If $3s_K \le \operatorname{Var} X$, then φ_K is the characteristic function of*

$$X_K = \big(\operatorname{Var} X - 3s_K\big)^{1/2} Z + \sum_{k\le K} \frac{M_k}{\gamma_k},$$

whose law f_K is ferromagnetic, with $\operatorname{Var} X_K = \operatorname{Var} X$ and $\kappa_4(X_K) = -12\sum_{k\le K}\gamma_k^{-4}$. The condition holds for $K \le 16$ and fails for $K = 17$.

(ii) For every K, if the inverse Fourier transform f_K of φ_K is nonnegative, then f_K is a ferromagnetic probability density.

(iii) RH holds if and only if $\varphi_K \to \varphi_X$ pointwise as $K \to \infty$. Under RH, $f_K \to p$ in law along every sequence of K for which $f_K \ge 0$.

Proof. With no hypothesis, $\frac{1}{2}\operatorname{Var} X = \sum_k \gamma_k^{-2}$, where the sum runs over one zero of each pair $\pm\gamma$ and complex γ are allowed (Section 13.3.1). A zero off the line contributes, with its conjugate, $2\operatorname{Re}\gamma^{-2} > 0$, since $|\operatorname{Re}\gamma| > 14 > |\operatorname{Im}\gamma|$. So β_K is the sum of γ_k^{-2} over the line zeros with $k > K$ and of these positive terms, and it is positive.

(i) M_k/γ_k has characteristic function $(1-\theta^2/\gamma_k^2)e^{-\theta^2/2\gamma_k^2}$, so the sum over $k \le K$ accounts for every factor of φ_K except $e^{-(\beta_K - s_K/2)\theta^2}$, a Gaussian factor of variance $2\beta_K - s_K = \operatorname{Var} X - 3s_K$. A sum of independent ferromagnetic variables is ferromagnetic. The cumulants come from $\log\varphi_K$. Since s_K increases with K, the last statement is $3s_{16} < \operatorname{Var} X < 3s_{17}$. In ball arithmetic, with the certified zeros of Arb [130] and $\operatorname{Var} X = (\log\xi)''(\frac{1}{2})$ from the Taylor series of $\log\xi$, $3s_{16} = 0.0456803$, $\operatorname{Var} X = 0.0462100$ and $3s_{17} = 0.0463006$ (`fm_cutoff.py`).

(ii) f_K has total mass $\varphi_K(0) = 1$, and its moment generating function $e^{\beta_K s^2}\prod_{k\le K}(1 + s^2/\gamma_k^2)$ vanishes only on the imaginary axis. By Proposition 13.4.4 below, f_K is a Gaussian density times a polynomial, so it has Gaussian moments.

(iii) As $K \to \infty$, $\beta_K \downarrow \beta_\infty$, the sum of the terms $2\operatorname{Re}\gamma^{-2}$ over the zeros off the line, and $\varphi_K \to \varphi_\infty = e^{-\beta_\infty \theta^2} \prod_k (1 - \theta^2/\gamma_k^2)$, a product over the line zeros, whose zeros are all real. If RH holds, $\beta_\infty = 0$ and $\varphi_\infty = \varphi_X$ by the Hadamard product. If RH fails, φ_X vanishes at a nonreal point and φ_∞ does not, so $\varphi_\infty \neq \varphi_X$. Along a sequence with $f_K \geq 0$ the φ_K are characteristic functions, and Lévy's continuity theorem gives convergence in law. □

The law f_{16} is proved ferromagnetic with no hypothesis, and its distance from the Pólya law is $\mathrm{TV}(f_{16}, p) = 1.27 \times 10^{-4}$, ten times smaller than that of the tilt by the flow at $\lambda = 0.2$. Beyond $K = 16$ the Maxwell construction runs out of Gaussian variance, and positivity must be checked directly. The exact condition comes from a closed form, in which He_n is the Hermite polynomial with $\mathrm{He}_n(y) = \mathbb{E}(y + iZ)^n$, g_v is the $N(0, v)$ density, and e_j is the jth elementary symmetric function of $\gamma_1^{-2}, \dots, \gamma_K^{-2}$.

Proposition 13.4.4. *For $v > 0$ the function $e^{-v\theta^2/2} \prod_{k \leq K} (1 - \theta^2/\gamma_k^2)$ is the Fourier transform of*

$$f_{K,v}(x) = g_v(x)\, Q_{K,v}\big(x/\sqrt{v}\big), \qquad Q_{K,v}(y) = \sum_{j=0}^{K} \frac{e_j}{v^j}\, \mathrm{He}_{2j}(y),$$

and $f_K = f_{K,2\beta_K}$. There is a threshold $v^*(K)$ such that $f_{K,v} \geq 0$ if and only if $v \geq v^*(K)$, and $v^*(K) \leq s_K$. Moreover $Q_{K,v}(y) > 0$ for $|y| \geq \sqrt{8K+2}$, so the sign of $f_{K,v}$ is decided on $|x| < \sqrt{(8K+2)v}$.

Proof. Differentiating g_v gives $g_v^{(n)}(x) = (-1)^n v^{-n/2} \mathrm{He}_n(x/\sqrt{v}) g_v(x)$, and the transform of $g_v^{(2j)}$ is $(-1)^j \theta^{2j} e^{-v\theta^2/2}$. Expanding $\prod_k (1 - \theta^2/\gamma_k^2) = \sum_j (-1)^j e_j \theta^{2j}$ gives the formula. For $v < v'$, $f_{K,v'} = f_{K,v} * g_{v'-v}$, so positivity persists as v increases, and the set of good v is closed. At $v = s_K$ the transform is that of $\sum_{k \leq K} M_k/\gamma_k$, so $v^*(K) \leq s_K$. The zeros of He_n lie in $|y| < \sqrt{4n+2}$, by a Sturm comparison for the Hermite equation, so for $|y| \geq \sqrt{8K+2}$ every He_{2j} with $j \leq K$ is positive, and so are the coefficients $e_j v^{-j}$. □

For $K = 1$, $Q_{1,v}(y) = 1 + (y^2 - 1)/(v\gamma_1^2)$, so $v^*(1) = \gamma_1^{-2} = s_1$, the Maxwell value. For larger K the threshold is far smaller, and Table 13.5 brackets it for $K \leq 800$. The Hermite sum cancels catastrophically, so the computation runs in ball arithmetic [130] at thousands of bits (8210 at $K = 800$), from the certified ordinates $\gamma_1, \dots, \gamma_{800}$ and $\operatorname{Var} X = (\log \xi)''(\frac{1}{2})$. The recurrence $\mathrm{He}_{n+1} = y\mathrm{He}_n - n\mathrm{He}_{n-1}$ gives $Q_{K,v}$ in the monomial basis. On an interval $[c - r, c + r]$ its Taylor coefficients T_m at c come exactly from one polynomial product, and $Q_{K,v}(c + h) \geq T_0 - \sum_{m \geq 1} |T_m| r^m$ for $|h| \leq r$, with no remainder; intervals on which this bound is not positive are bisected. A cover of $[0, \sqrt{8K+2}]$ by such intervals, with Proposition 13.4.4 beyond it and the evenness of $Q_{K,v}$, proves $Q_{K,v} > 0$ on the line.

In this way $f_K > 0$ is proved for every $1 \le K \le 200$ and for $K = 300$, 400, 600 and 800. For all these K the minimum of $Q = Q_{K,2\beta_K}$ is attained only at $y = 0$: near 0 because $Q(y) - Q(0) \ge y^2\big(b_2 - \sum_{m\ge 2} |b_{2m}| y^{2m-2}\big) > 0$ for $0 < |y| \le 2^{-5}$, the b_n being the coefficients of Q, and elsewhere by covers on which the lower bound exceeds $Q(0)$. For the column $v^*(K)$, a complete cover at $v = t_+ \cdot 2\beta_K$ gives $v^*(K) \le t_+ \cdot 2\beta_K$, and a certified negative value of $Q_{K,v}$ at $v = t_- \cdot 2\beta_K$ gives $v^*(K) > t_- \cdot 2\beta_K$, for dyadic rationals $t_- < t_+$ at most 2^{-18} apart.

K	$2\beta_K$	s_K	$v^*(K)$	$v^*(K)/2\beta_K$	$\min Q_{K,2\beta_K}$	$\mathrm{TV}(f_K, p)$
0	0.046210	0	0	0	1	1.48×10^{-2}
1	0.036199	0.005005	0.005005	0.138	0.862	3.81×10^{-3}
2	0.031674	0.007268	0.004955	0.156	0.804	2.09×10^{-3}
3	0.028477	0.008867	0.005416	0.190	0.762	1.30×10^{-3}
5	0.024472	0.010869	0.005527	0.226	0.705	6.96×10^{-4}
10	0.019121	0.013544	0.005441	0.285	0.623	2.64×10^{-4}
16	0.015756	0.015227	0.005222	0.331	0.566	1.27×10^{-4}
17	0.015343	0.015434	0.005190	0.338	0.558	1.14×10^{-4}
20	0.014273	0.015969	0.005065	0.355	0.538	8.75×10^{-5}
50	0.009108	0.018551	0.004146	0.455	0.430	1.70×10^{-5}
100	0.006222	0.019994	0.003303	0.531	0.355	4.42×10^{-6}
150	0.004907	0.020652	0.002816	0.574	0.316	1.93×10^{-6}
200	0.004122	0.021044	0.002487	0.603	0.289	
300	0.003199	0.021505	0.002059	0.644	0.255	
400	0.002659	0.021775	0.001784	0.671	0.232	
600	0.002036	0.022087	0.001441	0.708	0.203	
800	0.001677	0.022267	0.001229	0.733	0.184	

Table 13.5: The zero completed laws f_K. For $K \le 16$, f_K is a ferromagnetic probability density by Theorem 13.4.3(i) ($s_K \le 2\beta_K$). For every $K \ge 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so f_K is a ferromagnetic probability density by Theorem 13.4.3(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 13.4.4 lies in a certified interval of width at most 1.2×10^{-7}, within 6×10^{-7} of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. Computed by `fm_qcert.py` and `fm_tvcert.py` from the certified zeros and variance, except $v^*(1) = s_1$.

The question is the trend of the ratio $v^*(K)/2\beta_K$. While it stays below 1, f_K is a probability density. It increases slowly, to 0.733 at $K = 800$, and its slope in $\log K$ falls from 0.110 between $K = 30$ and 50 to 0.086 between $K = 600$ and 800. A straight line in $\log K$ through the last two points reaches 1 near $K = 1.8 \times 10^4$, while the power law $1 - v^*(K)/2\beta_K \approx 1.65 K^{-0.27}$, fitted for $K \ge 100$, never does. The data do not decide between them. Either outcome is consistent with RH, since Theorem 13.4.3(iii) asks only for the convergence of φ_K, not for positivity.

13.4.4 The Maxwell series

For $K \le 16$, Theorem 13.4.3(i) reads as a law the factorisation

$$\varphi_K(\theta) = e^{-(\operatorname{Var} X - 3s_K)\theta^2/2}\, \mathbb{E} e^{i\theta \sum_{k \le K} M_k/\gamma_k}.$$

It holds for every K, though from $K = 17$ on its Gaussian factor has negative variance, and letting $K \to \infty$ in it, with that factor moved to the other side, gives an identity in law. Let $\mathcal{M} = \sum_k M_k/\gamma_k$, summed over the zeros on the critical line. The series converges almost surely, because its terms are independent and centred with summable variances $3\gamma_k^{-2}$.

Theorem 13.4.5. *Let Z be independent of X. Then RH holds if and only if*

$$X + \left(\tfrac{1}{2} \operatorname{Var} X\right)^{1/2} Z \overset{d}{=} \mathcal{M}.$$

Proof. $\mathbb{E} e^{i\theta\mathcal{M}} = \prod_k (1 - \theta^2/\gamma_k^2) e^{-\theta^2/2\gamma_k^2}$, an entire function of θ with only real zeros. Under RH, $\sum_k \gamma_k^{-2} = \frac{1}{2} \operatorname{Var} X$, and the product is $\varphi_X(\theta) e^{-\operatorname{Var} X\, \theta^2/4}$, the characteristic function of the left side. If RH fails, the characteristic function of the left side vanishes at a nonreal point, and that of $\mathcal{M}$ does not. □

A simulation checks the identity numerically (computed by `fm_maxwell.py`, with 10^7 draws). Like Table 13.3, this checks the computation, not RH.

13.4.5 How close

Proposition 13.4.6. *With no hypothesis, the distance δ of Section 13.1.5 is at most $\mathrm{TV}(f_{150}, p) < 1.932 \times 10^{-6}$, since f_{150} is a ferromagnetic law by Theorem 13.4.3(ii) and the positivity proof of Section 13.4.3; without that computer assisted proof, f_{16} gives $\delta < 1.266 \times 10^{-4}$.*

Every density here is even, and $\mathrm{TV}(q, p) = \int_0^\infty |q - p|$ is enclosed in ball arithmetic. On $[0, 3]$, cut into intervals $[c - r, c + r]$ with $r = 2^{-12}$, the integral of $|d|$, $d = q - p$, differs from the exact integral of $|d(c) + d'(c)h|$ over $|h| \le r$ by at most $|d''(c)|r^3/3 + M_3 r^4/12$, where M_3 bounds $|d'''|$ by interval evaluation (for the laws f_K, by $\frac{1}{\pi} \int_0^\infty \theta^3 |\varphi_K(\theta)|\, d\theta$); beyond 3 each density is bounded by an explicit function of known logarithmic decay rate. The laws f_K are built from the certified zeros and variance of Section 13.4.3, so the distances are those of the true laws (`fm_tvcert.py`).

13.4.6 A cusped weight never works

The multipliers of Proposition 13.4.2 are entire. A weight with a cusp at the origin behaves differently: no tilt of p by $e^{-a|t|}$, $a > 0$, is ferromagnetic, however small a is. We write the tilts in the variable $u = t/2$ of the de Bruijn Newman flow (Section 8.4). An even weight $w(u)$ gives the multiplier $m(t) = w(t/2)$ in the notation

of Proposition 13.4.2, and $8\int_0^\infty w(u)\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du = \int_{\mathbb{R}} \Phi(t)m(t)\cos(\theta t)\,dt$ with $\theta = z/2$, so the zeros of the first function are twice those of the transform of the tilt $m\,p$. Thus $e^{-\mu|u|}$ is $e^{-a|t|}$ with $a = \mu/2$, and $\cosh(\mu u)$ is $\cosh(at)$.

THEOREM 13.4.7. *Let $\mu > 0$ and $H^{(\mu)}(z) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$. With C a standard Cauchy variable,*

$$H^{(\mu)}(z) = \mathbb{E}\, H_0(z + \mu C),$$
$$z^2 H^{(\mu)}(z) \to \mu\,\Phi_{\mathrm{dBN}}(0) = 0.4467\mu \quad (z \to \infty,\ z \in \mathbb{R}).$$

Hence $H^{(\mu)}$ is positive beyond some height, has finitely many real zeros and infinitely many non real ones, the law with density proportional to $e^{-\mu|t|/2}p(t)$ is not ferromagnetic, and the tilted clock $R^{(\mu)}(s) = H^{(\mu)}(0)/H^{(\mu)}(2i\sqrt{s})$ is not the Laplace transform of a GGC, for any $\mu > 0$.

Proof. Since $e^{-\mu|u|} = \mathbb{E}\cos(\mu C u)$ and $\mathbb{E}\sin(\mu C u) = 0$, Fubini gives $H^{(\mu)}(z) = \mathbb{E}H_0(z + \mu C) = \int H_0(y)P_\mu(z - y)\,dy$ with $P_\mu(y) = \mu/(\pi(\mu^2 + y^2))$. As $H_0(y) = \frac{1}{8}\Xi(y/2)$ decays exponentially and $\int_{\mathbb{R}} H_0(y)\,dy = \pi\Phi_{\mathrm{dBN}}(0)$ by Fourier inversion at $u = 0$, dominated convergence, with $z^2 P_\mu(z - y) \le 2(\mu + y^2/\mu)/\pi$, gives $z^2 H^{(\mu)}(z) \to \mu\Phi_{\mathrm{dBN}}(0)$. The function $H^{(\mu)}$ is even and entire of order one; with finitely many zeros it would be a polynomial times $e^{\alpha z}$, hence a polynomial by evenness, contradicting $H^{(\mu)}(z) \to 0$. So it has infinitely many zeros, almost all non real. It is also positive on the imaginary axis, $H^{(\mu)}(iy) = \int_0^\infty e^{-\mu u}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, so the argument of Theorem 4.2.4 applies to $R^{(\mu)}$ verbatim, as it does to R_t in Corollary 4.2.5, and $R^{(\mu)}$ is the Laplace transform of a GGC only if all zeros of $H^{(\mu)}$ are real. By the relation between the variables above, the characteristic function of the tilt $e^{-\mu|t|/2}p(t)$ is $H^{(\mu)}(2\theta)/H^{(\mu)}(0)$, whose non real zeros show that the tilt is not ferromagnetic. □

Numerically the real zeros are lost at once (`xg_cauchy.py`). For $\mu = 1$ the function $H^{(1)}$ has no sign change on $[0, 80]$, where H_0 has six, and the Laguerre inequality $(H')^2 - HH'' \ge 0$, satisfied by every real rooted even entire function of order one, fails for $24 \le z \le 38$ and for $43 \le z \le 80$. Even for $\mu = 0.01$ only two sign changes survive on $[0, 400]$, against the six of H_0 on $[0, 80]$. The mechanism is the cusp of $e^{-\mu|u|}$ at $u = 0$, which adds to $H^{(\mu)}$ a positive term decaying like z^{-2} and swamps the exponentially decaying oscillation that carries the zeros of Ξ. The table gives $H^{(\mu)}(z)$ against the tail value $\mu\Phi_{\mathrm{dBN}}(0)/z^2$ of the theorem.

μ	z	$H^{(\mu)}(z)$	$\mu\Phi_{\mathrm{dBN}}(0)/z^2$
1	80	7.237×10^{-5}	6.980×10^{-5}
1	400	2.796×10^{-6}	2.792×10^{-6}
3	400	8.387×10^{-6}	8.376×10^{-6}
0.01	400	2.796×10^{-8}	2.792×10^{-8}

13.5 The exponential of the Pólya variable

The positive variable e^{2X} lives on the Mellin side of Section 9.1.1: it is the variable $\frac{\pi}{2}S_2$ of Biane, Pitman and Yor, size biased by its power $\frac{1}{4}$. Along the family $e^{\beta X}$ the picture is complete except on one interval: $e^{\beta X}$ is not infinitely divisible for $0 < |\beta| < 2$, it is self decomposable and not a GGC at $|\beta| = 2$, and it has a hyperbolically completely monotone density, so it is a GGC, for $|\beta| \geq 5.5791422$, a bound certified in ball arithmetic. None of this involves a hypothesis on the zeros. GGC, HCM and self decomposable laws are as in Section 1.3.2.

13.5.1 Size biased laws and their phase

Write $B_0 = \frac{\pi}{2}S_2 = \sum_{m\geq 1} \Gamma_m/(\pi m^2)$, with independent Gamma$(2,1)$ variables Γ_m as in Section 9.1.1. It is a GGC with Thorin atoms of mass 2 at the points πm^2 and Laplace transform $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$, and by (9.1.1) $\mathbb{E}B_0^\sigma = 2\xi(2\sigma)$ for every complex σ. With Riemann's function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ of Section 1.2, its density is

$$f_0(y) = \sum_{m\geq 1} \bigl(4\pi^2 m^4 y - 6\pi m^2\bigr) e^{-\pi m^2 y} = 4y\,\omega''(y) + 6\,\omega'(y)\,:$$

two integrations by parts give $\int_0^\infty y^\sigma (4y\omega'' + 6\omega')\,dy = 2\sigma(2\sigma-1)\pi^{-\sigma}\Gamma(\sigma)\zeta(2\sigma) = 2\xi(2\sigma)$. For real q let B_q be B_0 size biased by B_0^q, with density $y^q f_0(y)/(2\xi(2q))$, so that

$$\mathbb{E}B_q^\sigma = \frac{\xi(2\sigma+2q)}{\xi(2q)}, \qquad e^{2X} \stackrel{d}{=} B_{1/4},$$

the second because $\mathbb{E}e^{2\sigma X} = \xi(\frac{1}{2}+2\sigma)/\xi(\frac{1}{2})$. In the notation of Section 9.1.1, where V_α is the Esscher tilt of the Pólya density at the basepoint α, $B_q = e^{2V_{2q}}$; so the B_q with $0 < q < 1$ correspond to the basepoints $0 < \alpha < 2$, and $e^{2X} = e^{2V_{1/2}}$ to the centre. Since X is symmetric, $e^{-\beta X} \stackrel{d}{=} e^{\beta X}$, and for $\beta > 0$, $e^{\beta X} = B_{1/4}^{\beta/2}$.

Proposition 13.5.1. *For $0 < |\beta| < 2$ the variable $e^{\beta X}$ is not infinitely divisible.*

Proof. Take $\beta > 0$. The first term of Φ dominates for large t, so $\log P(X > x) = -\pi e^{2x} + O(x)$ and, with $Y = e^{\beta X}$, $\log P(Y > y) = -\pi y^{2/\beta} + O(\log y)$. Since $2/\beta > 1$, $\mathbb{E}\exp(cY\log Y) < \infty$ for every $c > 0$. By Sato's theorem on the tails of infinitely divisible laws [297, Theorem 26.1], an infinitely divisible law with this property has no Lévy measure, so it is Gaussian or degenerate, and Y is neither. □

The other results of the section come from the continuation of the Laplace transform

$$\Lambda_q(\lambda) = \mathbb{E}e^{-\lambda B_q} = \frac{1}{2\xi(2q)} \int_0^\infty y^q f_0(y)\, e^{-\lambda y}\, dy,$$

which is finite for $\lambda > -\pi$. Let $0 < q < 1$, the range of the results below, fix $M \geq 1$ and let R_M be the tail $\sum_{m>M}$ of the series for f_0. It is bounded near 0, because $f_0(y) = y^{-5/2} f_0(1/y)$ by the evenness of Φ, so that $f_0(y) \to 0$ as $y \downarrow 0$, and it is $O(y\, e^{-\pi(M+1)^2 y})$ at infinity. Integrating the first M terms gives, for $\operatorname{Re}\lambda > -\pi$,

$$2\xi(2q)\,\Lambda_q(\lambda) = \sum_{m\leq M} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(\pi m^2+\lambda)^{q+2}} - \frac{6\pi m^2 \Gamma(q+1)}{(\pi m^2+\lambda)^{q+1}}\Big] + \int_0^\infty y^q R_M(y)\, e^{-\lambda y}\, dy, \tag{13.5.1}$$

with principal powers. The last integral is analytic for $\operatorname{Re}\lambda > -\pi(M+1)^2$ and real on the real axis. So (13.5.1) continues Λ_q analytically to the plane cut along $(-\infty, -\pi]$, and Λ_q has boundary values $\Lambda_q(-r+i0)$ that are continuous in $r \neq \pi m^2$, while near each $-\pi m^2$ it behaves like a multiple of $(\pi m^2+\lambda)^{-q-2}$.

The boundary values carry the Thorin measure. If G is a GGC with drift b and Thorin measure μ, its Laplace transform $H(\lambda) = \exp\{-b\lambda - \int \log(1+\lambda/t)\,\mu(dt)\}$ continues analytically to $\mathbb{C} \setminus (-\infty, 0]$, and at a point r that is not an atom of μ and where $H(-r+i0)$ is finite and not zero, the argument of $H(-r+i0)$ is $-\pi\mu((0,r))$, modulo 2π, as the proof of the lemma below shows. We call a continuous determination of $-\frac{1}{\pi}\arg H(-r+i0)$ on an interval a phase of H there. Fourati used the argument of such boundary values to show that certain powers of positive stable variables are not GGC [87].

Lemma 13.5.2. *Let G be a GGC with drift b, Thorin measure μ and Laplace transform H, and let $I \subset (0,\infty)$ be an open interval such that for each $r \in I$, $H(-r+i\epsilon)$ tends as $\epsilon \downarrow 0$ to a nonzero limit $H(-r+i0)$, continuous in r. Then μ has no atoms in I, and $\mu((0,r)) - \phi(r)$ is constant on I for every phase ϕ of H on I.*

Proof. Let $L(\lambda) = -b\lambda - \int \log(1+\lambda/t)\,\mu(dt)$, so that $H = e^L$. For $r \in I$ and $0 < \epsilon \leq 1$, $\operatorname{Im}\log(1+(-r+i\epsilon)/t) = \arg(t-r+i\epsilon)$ lies in $(0,\pi)$, tends to π, $\frac{\pi}{2}$ or 0 according as $t < r$, $t = r$ or $t > r$, and is at most $2\epsilon/t$ for $t \geq 2r$. Since $\mu((0,2r)) < \infty$ and $\int_{[2r,\infty)} \mu(dt)/t < \infty$, dominated convergence gives $\operatorname{Im} L(-r+i\epsilon) \to -\pi\big(\mu((0,r)) + \frac{1}{2}\mu(\{r\})\big)$. If $\mu(\{r\}) = a > 0$ then, since $|t-r+i\epsilon| \leq t+r+1$,

$$\operatorname{Re} L(-r+i\epsilon) \;\geq\; br - \int_{t\neq r} \log\Big(1+\frac{r+1}{t}\Big)\,\mu(dt) - a\log\frac{\epsilon}{r} \;\longrightarrow\; \infty,$$

so $|H(-r+i\epsilon)| \to \infty$, which is excluded. Hence μ has no atoms in I, and $e^{i\operatorname{Im} L(-r+i\epsilon)}$, the argument factor of $H(-r+i\epsilon)$, tends to that of $H(-r+i0)$. So $\mu((0,r)) - \phi(r)$ is an even integer for each $r \in I$, and being continuous on I it is constant. □

The same lower bound for $\operatorname{Re} L$ shows that the boundary values of a GGC transform never vanish.

THEOREM 13.5.3. *For $0 < q < 1$ the variable B_q is not a GGC. In particular e^{2X} is not a GGC.*

Proof. Let $r > \pi$, $r \neq \pi m^2$, and take M with $\pi(M+1)^2 > r$ in (13.5.1). At $\lambda = -r + i0$ the terms with $\pi m^2 > r$ and the last integral are real, and $(\pi m^2 - r + i0)^{-p} = (r - \pi m^2)^{-p} e^{-i\pi p}$ for $\pi m^2 < r$, so

$$2\xi(2q)\ \mathrm{Im}\,\Lambda_q(-r+i0) = -\sin(\pi q) \sum_{\pi m^2 < r} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(r-\pi m^2)^{q+2}} + \frac{6\pi m^2 \Gamma(q+1)}{(r-\pi m^2)^{q+1}} \Big] < 0. \tag{13.5.2}$$

Suppose that B_q is a GGC, and apply Lemma 13.5.2 on $I = (\pi, 4\pi)$. By (13.5.2) a phase ϕ of Λ_q on I takes its values in $(2k, 2k+1)$ for a single integer k. As $r \downarrow \pi$ the first term of (13.5.1) dominates, so that $\Lambda_q(-r+i0) \sim c\,(r-\pi)^{-q-2} e^{-i\pi(q+2)}$ with $c > 0$, and $\phi(r) \to 2k+q$. As $r \uparrow 4\pi$ the term with $m = 2$ dominates and is positive, and $\phi(r) \to 2k$. So ϕ falls by q across I, while $\mu((0,r)) = \phi(r) + \mathrm{const}$ is nondecreasing, a contradiction. □

Size biasing a GGC by a negative power gives a GGC: Fourati's Lemma 6.3 [87] states this for every real exponent, and its proof, which multiplies the variable by one plus an independent gamma variable of shape $\beta > 0$ and lets the rate of the gamma variable tend to 0, covers the size bias by $x^{-\beta}$, once slips in the printed normalising constants are corrected. For positive exponents the statement fails. By Theorem 13.5.3 it fails for every exponent in $(0,1)$, and it fails at 1 as well: the size bias of order one of the GGC $e_1 + \frac{1}{2}e_2$ has Laplace transform $\frac{4}{3}(2\lambda+3)(1+\lambda)^{-2}(2+\lambda)^{-2}$, which vanishes at the point $-\frac{3}{2}$ of the cut. With the negative orders, B_q is a GGC for every $q \leq 0$ and for no $q \in (0,1)$.

13.5.2 The size bias principle

A gamma variable of shape t size biased by y^q is a gamma variable of shape $t+q$ with the same rate. For a sum of gamma variables with different rates the same happens at every rate, up to an independent GGC that lives between the rates. The proof uses divided differences. For a polynomial Q with real zeros $s_1 \leq \cdots \leq s_n$, listed with multiplicity, and a function h analytic near $[s_1, s_n]$, put

$$h[Q] = \frac{1}{2\pi i} \oint \frac{h(\zeta)}{Q(\zeta)}\, d\zeta = \sum_{\zeta_0} \mathrm{Res}_{\zeta=\zeta_0} \frac{h(\zeta)}{Q(\zeta)},$$

the contour enclosing the zeros of Q inside the domain of h and the sum running over the distinct zeros ζ_0 of Q, so that $h[Q]$ is the divided difference of h at $s_1, \ldots, s_n$ when Q is monic. By the formula of Hermite and Genocchi, $h[Q]$ is the integral of $h^{(n-1)}(\sum_j w_j s_j)$ over the simplex of weights $w_j \geq 0$, $\sum_j w_j = 1$, with its uniform measure of total mass $1/(n-1)!$; so $h[Q]$ has the sign of $h^{(n-1)}$ when that sign is constant on $[s_1, s_n]$.

LEMMA 13.5.4. *Let* $Q(\zeta) = \prod_{n=1}^{K}(\zeta - \tau_n)^{t_n}$ *with* $\tau_1 < \cdots < \tau_K$ *and integers* $t_n \geq 1$, *let* $1 \leq k < K$ *and* $\tau_k < x < \tau_{k+1}$, *and write* $Q = Q_-Q_+$ *with* $Q_-(\zeta) = \prod_{n\leq k}(\zeta - \tau_n)^{t_n}$, *of degree* N_-, *and* Q_+ *of degree* N_+. *For* $p > 0$ *put*

$$D_+(x) = h_+[Q_+], \qquad h_+(\zeta) = \frac{(\zeta - x)^{-p}}{Q_-(\zeta)},$$
$$D_-(x) = h_-[Q_-], \qquad h_-(\zeta) = \frac{(x - \zeta)^{-p}}{Q_+(\zeta)}.$$

Then $(-1)^{N_+-1}D_+(x) > 0$ *and* $(-1)^{N_+}D_-(x) > 0$, *and as functions of* $x \in (\tau_k, \tau_{k+1})$, $|D_+|$ *increases and* $|D_-|$ *decreases.*

Proof. On (x, ∞) the function h_+ is a product of the completely monotone functions $(\zeta - x)^{-p}$ and $(\zeta - \tau_n)^{-1}$, $n \leq k$, so $(-1)^j h_+^{(j)} > 0$ there for every j, and the formula of Hermite and Genocchi with $j = N_+ - 1$ gives the first sign. On $(-\infty, x)$ the function $(-1)^{N_+}h_-(\zeta) = (x - \zeta)^{-p}\prod_{n>k}(\tau_n - \zeta)^{-t_n}$ is a product of functions all of whose derivatives are positive, which gives the second. Finally $\partial_x(\zeta - x)^{-p} = p(\zeta - x)^{-p-1}$ and $\partial_x(x - \zeta)^{-p} = -p(x - \zeta)^{-p-1}$, so D'_+ and D'_- are p and $-p$ times the same expressions with $p + 1$ in place of p: D'_+ has the sign of D_+, and D'_- the sign opposite to that of D_-. □

THEOREM 13.5.5 (size bias principle). *Let* $Y = \sum_{n=1}^{K}\Gamma_n/\tau_n$ *with* $0 < \tau_1 < \cdots < \tau_K$ *and independent gamma variables* Γ_n *with integer shapes* $t_n \geq 1$ *and unit scale, let* $0 < q < 1$, *and let* Y_q *be* Y *size biased by* Y^q. *Then*

$$Y_q + G \stackrel{d}{=} \sum_{n=1}^{K}\frac{\Gamma'_n}{\tau_n},$$

where the Γ'_n *are independent gamma variables with shapes* $t_n + q$ *and unit scale, and* G, *independent of* Y_q, *is a GGC without drift whose Thorin measure has a continuous distribution function* θ_G *that vanishes on* $(0, \tau_1]$, *increases by exactly* q *on each interval* $[\tau_k, \tau_{k+1}]$ *and is constant on* $[\tau_K, \infty)$.

Proof. Let $N = \sum_n t_n$ and $Q(\zeta) = \prod_n(\zeta - \tau_n)^{t_n}$. The density of Y is $c_0(-1)^{N-1}(e^{-\zeta y})[Q]$ with $c_0 = \prod_n \tau_n^{t_n}$, as taking Laplace transforms under the contour integral shows: $(1/(\zeta + \lambda))[Q] = -1/Q(-\lambda)$, by the residue at $\zeta = -\lambda$. Multiplying by y^q before integrating gives, for $\lambda \geq 0$,

$$\Lambda(\lambda) = \mathbb{E}e^{-\lambda Y_q} = c\,(-1)^{N-1}g_\lambda[Q], \qquad g_\lambda(\zeta) = (\zeta + \lambda)^{-1-q},$$

with a constant $c > 0$, and since $g_\lambda^{(N-1)} = (-1)^{N-1}(1+q)\cdots(N-1+q)\,(\zeta+\lambda)^{-N-q}$, the formula of Hermite and Genocchi gives

$$\Lambda(\lambda) = \frac{\mathbb{E}(\bar{s} + \lambda)^{-N-q}}{\mathbb{E}\bar{s}^{\,-N-q}}, \qquad \bar{s} = \sum_{j=1}^{N} w_j s_j,$$

where $s_1, \dots, s_N$ are the rates listed with multiplicity and (w_j) is uniform on the simplex. This continues Λ analytically to $\mathbb{C} \setminus (-\infty, -\tau_1]$, where the residue form $c(-1)^{N-1} g_\lambda[Q]$ remains valid, and shows that $|\Lambda(\lambda)|$ is of exact order $|\lambda|^{-N-q}$ as $|\lambda| \to \infty$, uniformly up to the cut.

Boundary values. Let $\tau_k < r < \tau_{k+1}$. At points $\zeta < r$ the derivatives of $(\zeta - r + i0)^{-1-q}$ are $e^{-i\pi(1+q)}$ times those of $(r-\zeta)^{-1-q}$. Splitting the residues of g_λ/Q into those at τ_n with $n > k$ and those with $n \le k$ gives, with $D_\pm$ of Lemma 13.5.4 at $p = 1+q$ and the signs found there,

$$\Lambda(-r+i0) = c\,(-1)^{N-1}\big(D_+(r) + e^{-i\pi(1+q)} D_-(r)\big) = c\,(-1)^{N_-} z(r),$$
$$z(r) = |D_+(r)| + e^{-i\pi q} |D_-(r)|.$$

The point $z(r)$ is the sum of two nonzero vectors at the angle $\pi q < \pi$, so its argument lies in $(-\pi q, 0)$. The ratio $|D_-|/|D_+|$ decreases, from ∞ as $r \downarrow \tau_k$, where D_- has the singular term $(r-\tau_k)^{-t_k-q}$, to 0 as $r \uparrow \tau_{k+1}$; so $\arg z(r)$ increases from $-\pi q$ to 0. For $r > \tau_K$ the same computation with $Q_+ = 1$ gives $\Lambda(-r+i0) = c(-1)^{N-1} e^{-i\pi(1+q)} D_-(r)$ with $D_- > 0$.

The factor G. Put $\theta_G = 0$ on $(0, \tau_1]$, $\theta_G(r) = kq + \frac{1}{\pi} \arg z(r)$ for $\tau_k < r < \tau_{k+1}$, and $\theta_G = (K-1)q$ on $[\tau_K, \infty)$. This function is continuous and nondecreasing. Near τ_k, $|D_+/D_-|$ is $(r-\tau_k)^{t_k+q}$ times a function that is continuously differentiable up to τ_k, and similarly near τ_{k+1}; since $t_k + q > 1$, θ_G is continuously differentiable, with derivative 0 at the points τ_n. Let G be the GGC without drift and with Thorin measure $d\theta_G$, with Laplace transform $H_G(\lambda) = \exp\{-\int \log(1+\lambda/t)\, d\theta_G(t)\}$, and let $\Lambda_+(\lambda) = \prod_n (1+\lambda/\tau_n)^{-t_n-q}$, the Laplace transform of the right side. The function $J = \Lambda H_G/\Lambda_+$ is analytic on $\mathbb{C} \setminus (-\infty, -\tau_1]$ and positive on $(-\tau_1, \infty)$. For $\tau_k < r < \tau_{k+1}$ the arguments of Λ, H_G and $1/\Lambda_+$ at $-r+i0$ are, modulo 2π,

$$\pi N_- + \arg z(r), \qquad -\pi k q - \arg z(r), \qquad \pi(N_- + kq),$$

and for $r > \tau_K$ they are $\pi N - \pi q$, $-\pi(K-1)q$ and $\pi(N+Kq)$. In both cases they add up to a multiple of 2π, so J has real boundary values from above, by symmetry the same boundary values from below, and it continues analytically across the cut between the points $-\tau_n$. Near $-\tau_n$, $\Lambda = O(|\lambda + \tau_n|^{-t_n-q})$ by the residue form, and $1/\Lambda_+$ cancels this, while H_G is bounded because θ_G has a bounded density; so these singularities are removable, and J is entire. As $|\lambda| \to \infty$, $|\Lambda|$, $|1/\Lambda_+|$ and $|H_G|$ are of the exact orders $|\lambda|^{-N-q}$, $|\lambda|^{N+Kq}$ and $|\lambda|^{-(K-1)q}$, so J is bounded. By Liouville's theorem $J \equiv J(0) = 1$, that is $\Lambda H_G = \Lambda_+$, which is the identity in law. □

THEOREM 13.5.6. *For $0 < q < 1$,*

$$B_q + G_q \stackrel{d}{=} \frac{\pi}{2} S_{2+q},$$

where G_q is independent of B_q and is a GGC without drift whose Thorin measure has a continuous distribution function θ_q with $\theta_q = 0$ on $(0,\pi]$ and $\theta_q(\pi m^2) = (m-1)q$ for every $m \geq 1$. In particular $e^{2X} + G_{1/4} \stackrel{d}{=} \frac{\pi}{2} S_{9/4}$.

Proof. Let $B^{(M)} = \sum_{m \leq M} \Gamma_m/(\pi m^2)$ and let $B_q^{(M)}$ be its size bias of order q. By Theorem 13.5.5, $B_q^{(M)} + G_M \stackrel{d}{=} \sum_{m \leq M} \Gamma'_m/(\pi m^2)$, with shapes $2+q$ on the right. As $M \to \infty$, $\mathbb{E}[(B^{(M)})^q e^{-\lambda B^{(M)}}] \to \mathbb{E}[B_0^q e^{-\lambda B_0}]$ for $\lambda \geq 0$ by dominated convergence, since $B^{(M)} \leq B_0$; so $B_q^{(M)} \to B_q$ in law, and the right sides converge in law to $\frac{\pi}{2} S_{2+q}$, whose Laplace transform is $\Lambda_+(\lambda) = (\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^{2+q}$. Hence the Laplace transforms of the G_M converge to $H = \Lambda_+/\Lambda_q$, which is continuous at 0. So G_M converges in law to a variable G_q with Laplace transform H, G_q is a GGC by the closure theorem of Section 4.1.1, and $B_q + G_q \stackrel{d}{=} \frac{\pi}{2} S_{2+q}$. A GGC with drift b is at least b, while $\frac{\pi}{2} S_{2+q}$ is smaller than any given positive number with positive probability, so G_q has no drift.

It remains to find the Thorin measure μ of G_q. By (13.5.1), near each $-\pi m^2$ both Λ_q and Λ_+ are $(\pi m^2 + \lambda)^{-2-q}$ times functions with continuous nonzero boundary values, and between these points $\Lambda_q(-r+i0) \neq 0$ by (13.5.2). So H has continuous nonzero boundary values on all of $(0,\infty)$, and by Lemma 13.5.2 μ has no atoms and $\mu((0,r)) = \phi(r)$, where ϕ is the phase of H that vanishes on $(0,\pi)$, where $H > 0$. On $(\pi m^2, \pi(m+1)^2)$ the argument of $\Lambda_+(-r+i0)$ is $-\pi(2+q)m$, and $\Lambda_q(-r+i0)$ lies in the lower half plane by (13.5.2), with argument tending to $-\pi(2+q)$ modulo 2π at the left end and to 0 at the right end, as in the proof of Theorem 13.5.3. So, modulo 2, ϕ takes its values in $(mq-1, mq)$ on this interval and tends to $(m-1)q$ at its left end and to mq at its right end. Since ϕ is continuous and $\phi(\pi) = 0$, induction on m gives $\phi(\pi m^2) = (m-1)q$. □

The theorem gives B_q a signed Thorin measure, $(2+q)\sum_m \delta_{\pi m^2} - d\theta_q$, with a negative continuous part of mass q on each gap, which is what Theorem 13.5.3 detects.

13.5.3 Self decomposability and the theta Lévy density

Theorem 13.5.7. *For $0 < q < 1$ the variable B_q is self decomposable, with no drift and Lévy density $k_q(x)/x$, where*

$$k_q(x) = (2+q)\,\omega(x) - \int_0^\infty e^{-tx}\, d\theta_q(t), \qquad 2\,\omega(x) \leq k_q(x) \leq (2+q)\,\omega(x).$$

In particular e^{2X} is self decomposable, hence infinitely divisible and unimodal, its Lévy density lies between $2\omega(x)/x$ and $\frac{9}{4}\omega(x)/x$, and it is not a GGC.

Proof. By Theorem 13.5.6, $-\log\Lambda_q = -\log\Lambda_+ + \log H$, and Frullani's integral $\log(1+\lambda/t) = \int_0^\infty (1-e^{-\lambda x})e^{-tx}\,dx/x$, applied to the atoms of $\frac{\pi}{2} S_{2+q}$ and to the

Thorin measure of G_q, gives

$$-\log \Lambda_q(\lambda) = \int_0^\infty (1 - e^{-\lambda x})\, k_q(x)\, \frac{dx}{x}.$$

On $[\pi m^2, \pi(m+1)^2]$ the measure $d\theta_q$ has mass q and $e^{-tx} \le e^{-\pi m^2 x}$, so $0 \le \int e^{-tx}\, d\theta_q(t) \le q\,\omega(x)$, which gives the bounds. In particular $k_q > 0$, so B_q is infinitely divisible with Lévy density $k_q(x)/x$, and it is self decomposable if k_q is nonincreasing (Section 1.3.2), that is, if

$$\int_0^\infty t e^{-tx}\, d\theta_q(t) \;\le\; (2+q) \sum_{m\ge 1} \pi m^2 e^{-\pi m^2 x}, \qquad x > 0.$$

Bound the contribution of $[\pi m^2, \pi(m+1)^2]$ to the left side by $q\, s_m(x)$, with $s_m(x) = \max\{te^{-tx} : \pi m^2 \le t \le \pi(m+1)^2\}$. For $m \ge 2$, $s_m(x) \le \pi(m+1)^2 e^{-\pi m^2 x} \le \frac{9}{4}\pi m^2 e^{-\pi m^2 x}$, and $\frac{9}{4}q < 2+q$, which leaves $(2 - \frac{5}{4}q)\pi m^2 e^{-\pi m^2 x}$ of the mth term on the right. It remains to show $q\, s_1(x) \le (2+q)\pi e^{-\pi x} + (2 - \frac{5}{4}q) 4\pi e^{-4\pi x}$. If $x \ge 1/\pi$, then $s_1(x) = \pi e^{-\pi x}$. If $x \le 1/(4\pi)$, then $s_1(x) = 4\pi e^{-4\pi x}$, and $4q \le 4(2 - \frac{5}{4}q) + (2+q)$ because $8q \le 10$. Otherwise $s_1(x) = 1/(ex)$, and $(2+q)\pi x\, e^{1-\pi x} \ge (2+q)\frac{1}{4} e^{3/4} > 1 > q$, because ue^{1-u} increases on $(\frac{1}{4}, 1)$. Self decomposable laws are unimodal [347], and $e^{2X} = B_{1/4}$ is not a GGC by Theorem 13.5.3. □

Near 0, $\omega(x) \sim 1/(2\sqrt{x})$, and $0 \le q\sqrt{t/\pi} - \theta_q(t) \le 2q$, since $\theta_q = 0$ on $(0, \pi]$ and, on $[\pi m^2, \pi(m+1)^2]$, the first lies between mq and $(m+1)q$ and the second between $(m-1)q$ and mq; so the subtracted fraction $1 - k_q/((2+q)\omega)$ tends to $q/(2+q)$ as $x \to 0$.

The script `fm_sizebias.py` checks the three theorems numerically. It continues Λ_q by (13.5.1) with $M = 5$, at $30 + r/4$ digits, on 1800 points of $r \in [0.05, 90]$, for $q = 0.1$, 0.25, 0.5, 0.75 and 0.9. In every case the phase of Λ_q decreases along the grid on each of the four gaps $(\pi m^2, \pi(m+1)^2)$, $m \le 4$, so that θ_q increases, and at the last grid point below $\pi(m+1)^2$, θ_q is within 10^{-7} of mq, as Theorem 13.5.6 requires. Extending θ_q beyond the scan by $q(\sqrt{t/\pi} - 1)$, the value it takes at the points πm^2, the function k_q is positive and decreasing on 300 points of $[10^{-3}, 3]$, and the subtracted fraction at $x = 10^{-3}$ is 0.046, 0.108, 0.194, 0.264 and 0.300, against the limits 0.048, 0.111, 0.200, 0.273 and 0.310.

The factorization can also be tested away from the cut, through the characteristic function on the real line. For a positive infinitely divisible variable B with no drift and Lévy density $k(x)/x$, $-\log|\mathbb{E}e^{iyB}|^2 = \int_0^\infty 4\sin^2(\frac{yx}{2})\, k(x)\, dx/x$, and Frullani's integral $\int_0^\infty 4\sin^2(\frac{yx}{2}) e^{-tx}\, dx/x = \log(1 + y^2/t^2)$ turns this into the Thorin form $\int \log(1+y^2/t^2)$ against the Thorin measure, here the signed one. For B_q the atoms sum in closed form, by $\prod_m (1 + u^4/m^4) = (\cosh\sqrt{2}\pi u - \cos\sqrt{2}\pi u)/(2\pi^2 u^2)$:

$$-\log\left|\mathbb{E}e^{iyB_q}\right|^2 = (2+q)\log\frac{\cosh\sqrt{2\pi y} - \cos\sqrt{2\pi y}}{2\pi y} - \int_0^\infty \log\left(1 + \frac{y^2}{t^2}\right) d\theta_q(t).$$

Table 13.6 compares, for e^{2X}, the direct Fourier integral of the density with the Thorin form and the theta Lévy form, both built from the computed $\theta_{1/4}$. The two forms agree with each other to a relative 2×10^{-7}, and with the direct integral to a relative 1.2×10^{-5}. As a last check, $\mathbb{E}G_{1/4} = \mathbb{E}\frac{\pi}{2}S_{9/4} - \mathbb{E}e^{2X} = \frac{3\pi}{8} - \xi(\frac{5}{2})/\xi(\frac{1}{2}) = 0.0815947$, and $\int \theta_{1/4}(t)\,t^{-2}dt = 0.0815351$ from the computed Thorin measure; the extension beyond $t = 90$ differs from $\theta_{1/4}$ by at most q, which can move the integral by at most $q/90 < 0.003$.

y	direct	Thorin form	theta Lévy form	atoms alone
0.5	0.0588838	0.0588840	0.0588840	0.0609722
1	0.227615	0.227616	0.227616	0.235869
2	0.808286	0.808290	0.808290	0.839868
4	2.347416	2.347431	2.347431	2.457076
8	5.261493	5.261552	5.261553	5.575941

Table 13.6: $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points πm^2 summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed by `fm_sizebias.py`.

On the dual side the roles are reversed. Section 10.1.4 puts the characteristic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ of $Y_{1/2}$ in the same form, with no hypothesis: $\log(\xi(\frac{1}{2}+u)/\xi(\frac{1}{2})) = 2\int_0^\infty(1-\cos ux)\,K_{1/2}(x)\,dx/x$ for real u. So the logarithm of the moment generating function of X has the form of $-\log|\mathbb{E}e^{iyB_q}|^2$, with the Poisson trace $K_{1/2}$, a sum over the zeros, in place of the theta kernel k_q. The Poisson trace is completely monotone exactly when RH holds (Corollary 10.1.4), while $k_{1/4}$ is positive and decreasing with no hypothesis.

13.5.4 Hyperbolically completely monotone powers

The kernel Φ is analytic in the strip $|\operatorname{Im}t| < \pi/4$, where $\operatorname{Re}e^{2t} > 0$, and since it is real and even, $F(v) = \Phi(iv)$ is real for real v. We have $F(0) = 0.8933938$; F has a local maximum $F(v_m) = 7.0315837$ at $v_m = 0.5630962646$, and it vanishes at $v = 0.6388301$, which is the zero α_1 of the continuation in Proposition 7.1.18(iii).

Proposition 13.5.8. *The variable $e^{\beta X}$ has an HCM density, and so is a GGC, for $|\beta| \geq 5.5791422$, and it does not have an HCM density for $|\beta| \leq 5.5791395$. The proof of the first statement is computer assisted.*

Proof. Let $\beta > 0$. The density $p(\beta^{-1}\log y)/(\beta y)$ of $e^{\beta X}$ has elasticity $1 - \beta^{-1}(\Phi'/\Phi)(\beta^{-1}\log y)$, and the upper half plane in y corresponds to the strip $0 < \operatorname{Im}t < \pi/\beta$. By Theorem 4.3.2, as in Proposition 4.3.26, $e^{\beta X}$ has an HCM

density if and only if Φ continues to a zero free analytic function on $|\operatorname{Im} t| < \pi/\beta$ and

$$h(x,v) = \operatorname{Im}\Big(-\frac{\Phi'}{\Phi}\Big)(x+iv) \ge 0 \qquad \text{for } 0 < v < \pi/\beta,$$

and HCM laws are GGC by (F5) of Section 4.2.1. On the imaginary axis $h(0,v) = F'(v)/F(v)$. Ball arithmetic [130] encloses v_m in an interval of width 2×10^{-17} with certified signs $F' > 0$ at its left end and $F > 0 > F'$ at its right end v_b. If $\pi/\beta > v_b$, the strip contains iv_b, where Φ is analytic and $h < 0$, so $e^{\beta X}$ is not HCM. Since $\pi/v_b > 5.5791395$, this proves the second statement.

For the first, let $v_* = 0.563096$, so that $\pi/v_* < 5.5791422$, and let φ_1 be the term $n = 1$ of Φ. Three estimates, in ball arithmetic with rigorous bounds for the tail of the series, show that $h \ge 0$ on $0 < v \le v_*$. (Z) For $\operatorname{Re} t \ge 0$ and $|\operatorname{Im} t| \le 0.5634$, $|\Phi/\varphi_1 - 1| \le 0.6015$, so $\Phi \ne 0$ there, and also for $\operatorname{Re} t \le 0$, because $\Phi(-\bar{t}) = \overline{\Phi(t)}$. (T) On the segment $v = v_*$, $0 \le x \le 2$, adaptive bisection into 2896 intervals encloses h in positive balls. The smallest lower bound, 1.8×10^{-8}, occurs next to $x = 0$, where $h(0,v)$ tends to 0 as $v \uparrow v_m$. Since $h(-x,v) = h(x,v)$, $h > 0$ on the whole segment $|x| \le 2$. (S) For $x \ge 2$ and $0 < v \le v_*$, the first term gives $\operatorname{Im}(-\varphi_1'/\varphi_1) \ge 2\pi e^{2x} \sin 2v \ge 8e^4 v$, and Cauchy's estimate on discs of radius 0.1 bounds the imaginary part of the derivative of $\log(\Phi/\varphi_1)$ by $10^{-40}v$; so $h > 0$ there, and by symmetry also for $x \le -2$. By (Z), h is harmonic on the strip $|\operatorname{Im} t| < 0.5634$. It vanishes on the real axis and, by (T) and (S), is positive on the top and the sides of the rectangle $[-2,2] \times [0, v_*]$, so $h \ge 0$ inside by the minimum principle, and with (S), $h \ge 0$ on $0 < v \le v_*$. So $e^{\beta X}$ is HCM for $\beta \ge \pi/v_*$. □

The certificate stops 3×10^{-6} short of $\pi/v_m = 5.5791396$, because h vanishes at iv_m and the strip of height v_m cannot be covered by balls with positive lower bounds; presumably the threshold is exactly π/v_m. The computation is `fm_hcm.py`, which works in the variable $t/2$ of Rodgers and Tao [289], where the heights are halved and the thresholds doubled.

The HCM threshold, the least $|\beta|$ for which $e^{\beta X}$ has an HCM density (these powers form a closed half line, by (F2) and (F4) of Section 4.2.1), lies between 5.5791395 and π/v_*, and for $2 < |\beta|$ below it infinite divisibility and the GGC property are open (Section 13.6). The bracket comes from the kernel, not from the zeros, and involves no hypothesis. It is the Pólya side counterpart of Theorem 4.3.27, where, under RH and two further hypotheses on the zeros, the HCM powers of the Mellin variable W are limited by the first gap $\gamma_2 - \gamma_1$. For the clock, whose Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the moment generating function of X inverted at $\sqrt{u}$, the GGC property is RH itself (Proposition 1.2.1).

13.6 Open problems

The last two problems concern the variable e^{2X} of Section 13.5, where no hypothesis on the zeros enters.

PROBLEM 13.1. Mixed shoulders. Theorem 13.2.3 decides the symmetric laws on $[-1,1]$ whose density is concave on $[0,1]$ and those whose density is convex there and not linear on its support: by (13.2.1) the question is whether the sine transform of the slope $g = -p'$ has only real zeros, and the theorem answers it when g is monotone. Decide the slopes that first rise and then fall. One class is settled. If g is symmetric about $\frac{1}{2}$, then $p(0) = \int_0^1 g = 1$, because $1 = 2\int_0^1 p = 2\int_0^1 xg(x)\,dx = \int_0^1 g$ by the symmetry of g, so $u \mapsto g(\frac{1}{2}+u)$ is a probability density on $[-\frac{1}{2}, \frac{1}{2}]$; and $X \stackrel{d}{=} \frac{1}{2}U + B$, with B independent of U and distributed with that density, since the density of $\frac{1}{2}U + B$ has slope g on $(0,1)$. As $\sin(z/2)/(z/2)$ has only real zeros, X is then ferromagnetic exactly when B is. For the beta slopes $g(y) \propto y^a(1-y)^b$, $a, b \geq 0$, this decides the diagonal $a = b$, where B is a symmetric beta law and X is ferromagnetic (Section 13.1.4), and Theorem 13.2.3 decides the edges: X is ferromagnetic for $b = 0$, where p is concave, and not for $a = 0 < b$, where p is convex and not linear. The chapter does not decide the rest of the quadrant. On the half line the slope of the Pólya law rises and then falls (Section 13.3.2), and RH is the statement that its sine transform has only real zeros.

PROBLEM 13.2. The Griffiths Simon class. Newman's theorem places the Pólya law in the class of Ellis, Monroe and Newman (Section 13.3.1). Does it belong to the Griffiths Simon class of Section 13.1.1? The laws of that class are ferromagnetic, so a positive answer would imply RH, and under Ingham's hypothesis the spins of any approximating system would have to be coupled (Section 13.3.3).

PROBLEM 13.3. Splitting the zeros. Assume RH, so that $\varphi_X(t) = \prod_\gamma (1 - t^2/\gamma^2)$ over the positive ordinates, with no Gaussian factor by Theorem 13.3.1. Can the ordinates be split into two nonempty sets A and A' such that $\prod_{\gamma\in A}(1 - t^2/\gamma^2)$ and $\prod_{\gamma\in A'}(1 - t^2/\gamma^2)$ are both characteristic functions? This is the same as writing $X = D + R$ with D and R independent, ferromagnetic and not 0, and by Theorem 13.3.1 both sets are then infinite. Under Ingham's hypothesis no three ordinates are in arithmetic progression, so no piece is a scaled fair sign or a scaled uniform variable, whose zeros form arithmetic progressions.

PROBLEM 13.4. Positivity of the zero completed laws. The laws f_K of Section 13.4.3 are proved to be ferromagnetic probability densities for $K \leq 16$ by the Maxwell construction, and for every $K \leq 200$ and for $K = 300$, 400, 600 and 800 by a computer assisted proof in ball arithmetic. Compute the threshold $v^*(K)$ near $K = 5000$, where the linear and the power law fits of the ratio $v^*(K)/2\beta_K$ part; the cancellation there calls for a working precision of tens of thousands of bits. Is $v^*(K) < 2\beta_K$ for every K? If so, every f_K is a ferromagnetic probability density by Theorem 13.4.3(ii), and with no hypothesis their limit in law, with characteristic function $e^{-\beta_\infty\theta^2}\prod_k(1 - \theta^2/\gamma_k^2)$ over the zeros on the critical line, is a ferromagnetic law by Theorem 13.1.5; by Theorem 13.4.3(iii) it is the Pólya law

exactly when RH holds.

PROBLEM 13.5. The cosine constant. By Section 13.4.2, $\Omega \le \frac{1}{2}$ with no hypothesis, $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros (Section 12.1.2), and RH holds if and only if $\Omega = 0$. Is the set of $a \ge 0$ for which $\cosh(at)\, p$ is ferromagnetic an interval? Improve these bounds by computation, as computations with the zeros have improved de Bruijn's bound $\Lambda \le \frac{1}{2}$ on the de Bruijn Newman constant, to $\Lambda \le 0.2$ [247]. A first case is $a = \sqrt{0.1}$, below both bounds: the law $\cosh(\sqrt{0.1}\, t)\, p$ is as close to p as the flow at $\lambda = 0.2$, which is ferromagnetic, and deciding it is a computation with the zeros of $\frac{1}{2}(\xi(s+a)+\xi(s-a))$, of the kind that gave the bound 0.2 for the flow.

PROBLEM 13.6. The size bias principle for real shapes. Theorem 13.5.5 is proved for integer shapes. Does it hold for all real shapes $t_n > 0$? For $K = 1$ it is the gamma identity, with $G = 0$. Integer shapes enter the proof twice, through the divided difference formula for the density of Y and through Lemma 13.5.4, and together they show that on each gap (τ_k, τ_{k+1}) the phase of the Laplace transform of Y_q is nonincreasing and falls by exactly q. Is this still true for real shapes? With it, the rest of the proof goes through, provided the estimates of the transform near the points $-\tau_n$ and at infinity can be checked as well.

PROBLEM 13.7. The powers of e^X. For $2 < |\beta|$ below the HCM threshold, which lies between 5.5791395 and $\pi/0.563096 = 5.57914219\ldots$ (Proposition 13.5.8), is $e^{\beta X}$ infinitely divisible, self decomposable or a GGC? Below the threshold it has no HCM density, which leaves the GGC property open, and Theorem 13.5.7 gives self decomposability only at $|\beta| = 2$. Is the HCM threshold exactly $\pi/v_m = 5.5791396$? That requires $h \ge 0$ on the open strip $0 < v < v_m$, where $h(0, v) \to 0$ as $v \uparrow v_m$, so a certificate must replace the balls with positive lower bounds near iv_m by a local argument.

Chapter 14

The Pólya density as a maximum order statistic and the Thorin measure of the clock

This chapter follows the first author's manuscript [257], which starts from one observation: the law behind the Pólya kernel of the Riemann Xi function is built from the maximum M_4 of independent exponential random variables of Section 8.5.4.2, in the sense that e^{2Y} is a sum of two independent copies of M_4 (Proposition 14.2.1). Everything else follows from asking which parts of that observation survive the passage to the Thorin measure of the clock.

The Thorin programme of this book asks whether the clock, with Laplace transform $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{\lambda})$, is a generalised gamma convolution. The unconditional statements about the clock one level below the GGC property are those of Section 5.1, and this chapter uses them as they stand.

What is not claimed. Nothing here proves RH. Every statement equivalent to RH is labelled as such.

The dictionary. Table 14.1 sets the trigonometric side against the Riemann side.

14.1 Notation

The Laplace variable of this chapter is λ, the variable u of Section 1.2, so that the Laplace transform of the clock is $1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{\lambda})$, written $\varphi_{1/2}$ when the basepoint family is in view.

14.2 The Pólya density as a maximum order statistic

Section 8.5.4.2 shows that $\vartheta_4(t) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-\pi n^2 t}$ is the distribution function of the maximum M_4 of independent exponential variables, one of each rate πn

	Trigonometric side	Riemann side
entire function	$\sin(\sqrt{\pi}\theta)/(\sqrt{\pi}\theta)$	$\Xi(\theta)/\Xi(0)$
real zeros	$\theta = \pm\sqrt{\pi}\, n$	$\theta = \pm\gamma_k$ (all real iff RH)
space component	$\sqrt{\pi/2}\, v$, with v uniform on $(-1,1)$	$X/\sqrt{2}$, with X the Pólya variable
time component	M_4	T, the clock
Laplace transform	$\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda}$	$1/G(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda})$
exists as a law	yes	yes (Theorem 5.1.1)
ID and SD	yes	yes (Theorem 5.1.1)
x times the Lévy density	$\sum_{n\geq 1} e^{-\pi n^2 x}$	$W(x) = \sum_k e^{-a_k x}$
GGC, Thorin measure	$\sum_{n\geq 1} \delta_{\pi n^2}$	$\sum_k \delta_{\gamma_k^2}$ if and only if RH (Proposition 1.2.1)
maximum of exponentials	yes, with rates πn and $\pi(2m-1)$	no (Proposition 14.9.1)
Gnedenko norming	$a_n = 1/\pi$	$a_n = 1/\gamma_1^2$

Table 14.1: The dictionary between the trigonometric side, where everything is classical, and the Riemann side developed in this chapter. The variable θ is that of Section 1.2; ID and SD abbreviate infinitely divisible and self decomposable, and the Wald couple of the third and fourth rows is that of Theorem 14.4.2 and Remark 14.5.1.

for $n \geq 1$ and one of each rate $\pi(2m-1)$ for $m \geq 1$: by independence and Jacobi's triple product, $\Pr(M_4 \leq t) = \prod_{n\geq 1}(1-e^{-\pi n t})\prod_{m\geq 1}(1-e^{-\pi(2m-1)t}) = \vartheta_4(t)$. Since the Kolmogorov distribution function is $k \mapsto \vartheta_4(2k^2/\pi)$, πM_4 is twice the square of a variable with the Kolmogorov law. The same section has $\mathbb{E}e^{-\lambda M_4} = \sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda} = \prod_{n\geq 1}(1+\lambda/(\pi n^2))^{-1}$, so that $M_4 \stackrel{d}{=} \sum_{n\geq 1} E_n/(\pi n^2)$ with independent standard exponential variables E_n, a GGC with Thorin measure $\sum_n \delta_{\pi n^2}$; and $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\xi(s)/(s-1)$ for every complex s, with the right side continued at $s=1$ by $\log 2$. The Mellin transform is a case of the dictionary of Biane, Pitman and Yor [32]. By the Mellin transform, inside the open critical strip the zeros of $s \mapsto \mathbb{E}M_4^{s/2}$ are exactly the nontrivial zeros of ζ, since $1-2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The same law is thus a maximum of exponentials with linear rates πn and a sum of exponentials with quadratic rates πn^2; this coincidence is the Jacobi triple product, and Section 14.9 shows that it has no counterpart for the clock.

Up to a constant factor, e^Y is the maximum of a normalised Brownian excursion [32].

Proposition 14.2.1. *Let M_4 and M_4' be independent, and let Y be the variable of Section 1.2, $\mathbb{E}e^{sY} = 2\xi(s)$. Then:*

(i) $e^{2Y} = \frac{\pi}{2} S_2 \stackrel{d}{=} M_4 + M_4'$.

(ii) If f is the density of $e^Y = \sqrt{M_4 + M_4'}$, then the Pólya kernel (1.2.3) *is* $\Phi(u) = \frac{1}{2}\, e^{3u/2} f(e^u)$, *and*

$$f(y) = 4 \sum_{n \geq 1} \bigl(2\pi^2 n^4 y^3 - 3\pi n^2 y\bigr) e^{-\pi n^2 y^2}.$$

Proof. (i) By definition $e^{2Y} = \frac{\pi}{2} S_2$, and $\frac{\pi}{2} S_2 = \sum_{k \geq 1} (e_k + e_k')/(\pi k^2)$ by the definition of S_2. Since $M_4 \stackrel{d}{=} \sum_{n \geq 1} E_n/(\pi n^2)$, a sum of two independent copies of M_4 has the same law.

(ii) Let μ be the law of Y. By Section 1.2, $2\Xi(\theta) = 2\xi(\frac{1}{2} + i\theta) = \mathbb{E} e^{(\frac{1}{2} + i\theta)Y} = \int_{\mathbb{R}} e^{u/2} e^{i\theta u} \mu(du)$, and the measure $e^{u/2}\mu(du)$ is finite because $\mathbb{E} e^{Y/2} = 2\xi(\frac{1}{2})$. On the other hand $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u) e^{i\theta u}\, du$ by (1.2.3), since Φ is even. The finite measures $\frac{1}{2} e^{u/2}\mu(du)$ and $\Phi(u)\, du$ have the same Fourier transform, so they are equal: $\mu(du) = 2e^{-u/2}\Phi(u)\, du$. The density of e^Y is $f(y) = y^{-1} \cdot 2y^{-1/2}\Phi(\log y)$, which is the first claim, and substituting (1.2.3) gives the series. □

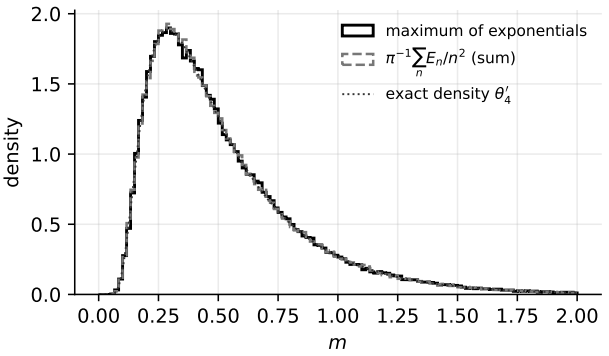


Figure 14.1: Histograms of M_4 simulated as a maximum of exponentials and as a weighted sum of exponentials.

14.3 Khinchin: the zeta distribution inside the tilted maximum

Khinchin [148] proved that for $\sigma > 1$ the function $\zeta(\sigma + it)/\zeta(\sigma)$ is an infinitely divisible characteristic function; Gnedenko and Kolmogorov [94] reproduce it as a worked example, and Lin and Hu [180] name the law the Riemann zeta distribution. If $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, the zeta distribution of Sections 6.1.6 and 7.1.2.3, then $\mathbb{E} N_\sigma^{-it} = \zeta(\sigma + it)/\zeta(\sigma)$, and the Euler product gives $N_\sigma = \prod_p p^{G_p}$ with independent geometric variables G_p, $\Pr(G_p = k) = (1 - p^{-\sigma})p^{-k\sigma}$. Neither Khinchin nor Gnedenko worked on the zeros or on Ξ; the work that continues their line stays at $\sigma > 1$ [180, 103, 207]. The completed ratio $\xi(\sigma + it)/\xi(\sigma)$ is the

characteristic function of the variable V_σ of Section 9.1.1, whose reflection $-V_\sigma$ is Nakamura's complete Riemann zeta distribution [206], and V_σ is quasi infinitely divisible but not infinitely divisible for $\sigma > 1$ (Proposition 11.1.3). The Pólya density, on the other hand, is not infinitely divisible, because its characteristic function $\Xi(\theta)/\Xi(0)$ has real zeros.

THEOREM 14.3.1. *Fix $\sigma > 1$ and let $\widetilde{M}_4$ have the tilted law $\Pr(\widetilde{M}_4 \in dm) \propto m^{\sigma/2} \Pr(M_4 \in dm)$. Let K, N_σ and $\Gamma_{1+\sigma/2}$ be independent of each other and of $\widetilde{M}_4$, with $\Pr(K = k) = (1-a)a^k$ for $k \geq 0$ and $a = 2^{1-\sigma}$, with $N_\sigma = \prod_p p^{G_p}$ zeta distributed with parameter σ, and with $\Gamma_{1+\sigma/2}$ gamma distributed with shape $1 + \sigma/2$ and scale 1. Then*

$$\widetilde{M}_4\, 4^{-K} \;\stackrel{d}{=}\; \frac{\Gamma_{1+\sigma/2}}{\pi N_\sigma^2} \;\stackrel{d}{=}\; \frac{\Gamma_{1+\sigma/2}}{\pi} \prod_p p^{-2G_p}.$$

Proof. By Section 8.5.4.2, $\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\,\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ for every complex s, because $2\xi(s)/(s-1) = 2\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$. With $s = \sigma + it$, $\mathbb{E}\widetilde{M}_4^{it/2}$ is the ratio of this at s and at σ, that is

$$\mathbb{E}\widetilde{M}_4^{it/2} = \frac{1 - ae^{-it\log 2}}{1-a} \cdot \pi^{-it/2} \cdot \frac{\Gamma(1+\frac{\sigma+it}{2})}{\Gamma(1+\frac{\sigma}{2})} \cdot \frac{\zeta(\sigma+it)}{\zeta(\sigma)}.$$

The first factor is the reciprocal of $\mathbb{E}e^{-itK\log 2} = \mathbb{E}(4^{-K})^{it/2}$. The remaining factors are $\pi^{-it/2}$, $\mathbb{E}\Gamma_{1+\sigma/2}^{it/2}$ and $\mathbb{E}N_\sigma^{-it}$. Hence $\mathbb{E}(\widetilde{M}_4\, 4^{-K})^{it/2} = \mathbb{E}\big(\Gamma_{1+\sigma/2}/(\pi N_\sigma^2)\big)^{it/2}$ for every real t, which is the first identity in law; the second is the Euler product. □

14.4 The Wald couple of the maximum

DEFINITION 14.4.1. A pair (τ, ς) of independent variables with $\tau \geq 0$ is a Wald couple if $\mathbb{E}e^{s\varsigma} < \infty$ and $\mathbb{E}e^{s\varsigma}\,\mathbb{E}e^{-s^2\tau/2} = 1$ for all real s. Equivalently, with Z standard normal and independent of τ, the characteristic functions of ς and of $\sqrt{\tau}Z$ form a van Dantzig pair in the sense of Section 1.2: $\mathbb{E}e^{-\theta^2\tau/2} = 1/\mathbb{E}e^{-\theta\varsigma}$ for real θ.

The name comes from Wald's identity $\mathbb{E}\exp(sB_\tau - s^2\tau/2) = 1$: if a stopping time τ of Brownian motion B has B_τ independent of τ and the stopped martingales $\exp(sB_{t\wedge\tau} - s^2(t\wedge\tau)/2)$ are uniformly integrable, then (τ, B_τ) is a Wald couple.

THEOREM 14.4.2. *Let $\upsilon, \upsilon_1, \upsilon_2$ be uniform on $(-1,1)$, with υ_1 and υ_2 independent, let M_4' be an independent copy of M_4, and let $c = \sqrt{\pi/2}$. Then $(M_4, c\upsilon)$ and $(M_4 + M_4', c(\upsilon_1 + \upsilon_2))$ are Wald couples. The associated van Dantzig pairs are $(\sin ct/(ct),\ ct/\sinh ct)$ and their squares, and $\sqrt{M_4}\,Z$ has the logistic law with characteristic function $ct/\sinh ct$.*

Proof. $\mathbb{E}e^{-s^2M_4/2} = cs/\sinh(cs)$ by Section 8.5.4.2, with $\lambda = s^2/2$, and $\mathbb{E}e^{scv} = \sinh(cs)/(cs)$. Squaring gives the second couple. □

The pair $(\sin\theta/\theta, \theta/\sinh\theta)$ is the classical van Dantzig pair of Section 1.2 [195], at $\theta = ct$, and the language of Wald couples is that of the first author's [251]. The logistic statement is the representation of the logistic law as a normal scale mixture with Kolmogorov mixing of Andrews and Mallows [3] and Stefanski [311], read through the maximum.

14.5 The clock exists and is infinitely divisible

Theorem 5.1.1 proves unconditionally, using only the verification of RH below height 1000, that $W > 0$ and $W' < 0$ on $(0, \infty)$, so that the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density $W(x)/x$; the law is unimodal by Yamazato's theorem [347] (Section 11.1), and its mean is $\mathbb{E}T = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ (Section 1.2). The ladder of signs $(-1)^k W^{(k)} > 0$ is Theorem 5.1.6 for $0 \le k \le 10^{17}$, Theorem 5.1.9 for every order on $(0, 1.29]$ and Theorem 5.1.13 for every $t > 0$ and every $k \le 1.16 \times 10^{25}$.

Remark 14.5.1. The clock gives a Wald couple on the Riemann side: $(T, X/\sqrt{2})$ is a Wald couple. Indeed T exists by Theorem 5.1.1, $\mathbb{E}e^{sX/\sqrt{2}} = \xi(\frac{1}{2} + s/\sqrt{2})/\xi(\frac{1}{2})$ and $\mathbb{E}e^{-s^2T/2} = \xi(\frac{1}{2})/\xi(\frac{1}{2} + |s|/\sqrt{2})$, and $\xi(\frac{1}{2} + w)$ is even in w. This is the van Dantzig pair $(\Xi(\theta)/\Xi(0),\ \xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta))$ of Theorem 5.1.1 read at $\theta = s/\sqrt{2}$.

14.6 A Stieltjes criterion

Section 1.3 reads the GGC property through the Stieltjes transform of the Thorin measure. For the clock, $G'/G = -\frac{d}{d\lambda}\log(1/G)$ takes the place of the function f_H there, whether or not RH holds, and by Section 1.2 $(G'/G)(\lambda) = \sum_k (\lambda + a_k)^{-1} = \int_0^\infty e^{-\lambda x} W(x)\, dx$. The quantity of this section is

$$r_\eta(x) = \frac{1}{\pi}\ \mathrm{Im}\,\frac{G'}{G}(-x - i\eta), \qquad x \in \mathbb{R},\ \eta > 0,$$

defined wherever $-x - i\eta$ is not a pole of G'/G. If the clock is a GGC with Thorin measure U, then r_η is the Poisson smoothing of U at distance η below the cut, as in (1.3.2).

Theorem 14.6.1. *The following are equivalent.*

(a) The Riemann Hypothesis.

(b) T is a GGC; then $T \overset{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ and its Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, the ordinates counted with multiplicity.

(c) G'/G *is a Stieltjes function.*

(d) $r_\eta(x) \geq 0$ *for all* $x \in \mathbb{R}$ *and* $\eta > 0$ *for which* $-x - i\eta$ *is not a pole of* G'/G.

(e) W *is completely monotone.*

In that case r_η *is the Poisson smoothing of* U_{RH} *at scale* η, *and* $U_{\mathrm{RH}}((0,x]) = N(\sqrt{x})$.

Items (a), (b), (c) and (e) are in the book: (a) if and only if (b) is Proposition 1.2.1, (b) implies (c) by the Stieltjes form of the GGC property in Section 1.3, and (b) if and only if (e) is the characterisation of a GGC by the complete monotonicity of k, x times its Lévy density [39], with $k = W$ for the clock (Sections 1.2 and 1.3).

Proof. It remains to prove that (c) implies (d) and that (d) implies (a). (c) implies (d): a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$ and maps the upper half plane into the closed lower half plane, and $(G'/G)(\bar\lambda) = \overline{(G'/G)(\lambda)}$, so $\mathrm{Im}(G'/G)(-x - i\eta) \geq 0$ for $\eta > 0$. (d) implies (a): if RH fails, then some zero has $\beta \neq \frac12$, and since the zeros are symmetric under $\rho \mapsto \bar\rho$ and $\rho \mapsto 1-\rho$ there is a zero $\rho = \beta + i\gamma$ with $\gamma > 0$ and $\beta > \frac12$; Lemma 14.6.2 gives $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$, so $r_\eta(x_\rho) < 0$ for $\eta < b$ close to b and (d) fails. The last statement is (1.3.2) and the counting formula of Section 1.3 for U_{RH}. □

Lemma 14.6.2 (Detection). *Let* $\rho = \beta + i\gamma$ *be a zero of multiplicity* m *with* $\beta > \frac12$ *and* $\gamma > 0$, *put* $\delta = \beta - \frac12$, *and write* $a_\rho = x_\rho - ib$ *with* $x_\rho = \gamma^2 - \delta^2$ *and* $b = 2\delta\gamma$, *so that* $\lambda_\rho = -a_\rho = -x_\rho + ib$. *Then* $r_\eta(x_\rho) \to -\infty$ *as* $\eta \uparrow b$. *Moreover, for* $0 < \eta < b$ *such that* $-x_\rho - i\eta$ *is not a pole of* G'/G,

$$\pi r_\eta(x_\rho) = \eta\Big(A_\eta - \frac{2m}{b^2 - \eta^2}\Big),$$

where

$$A_\eta = \frac{1}{\eta} \sum_{\lambda_{\rho'} \neq \lambda_\rho, \bar\lambda_\rho} \mathrm{Im}\,\big(-x_\rho - i\eta - \lambda_{\rho'}\big)^{-1},$$

the sum running over the zeros ρ' *with* $\mathrm{Im}\,\rho' > 0$, *counted with multiplicity, whose* $\lambda_{\rho'}$ *is neither* λ_ρ *nor* $\bar\lambda_\rho$. *Hence* $r_\eta(x_\rho) < 0$ *at every* $\eta \in (0,b)$ *with* $A_\eta < 2m/b^2$. *If all the other zeros lie on the line,* $A_\eta \to A_0 = \sum_{\gamma'} (\gamma'^2 - x_\rho)^{-2}$ *as* $\eta \downarrow 0$.

Proof. At $x = x_\rho$ we have $-x_\rho - i\eta - \lambda_\rho = -i(\eta + b)$ and $-x_\rho - i\eta - \bar\lambda_\rho = -i(\eta - b)$. Since $\mathrm{Im}(-iy)^{-1} = 1/y$, each of the m copies of the pair $\lambda_\rho, \bar\lambda_\rho$ contributes $1/(\eta + b) + 1/(\eta - b) = -2\eta/(b^2 - \eta^2)$, which gives the display. This tends to $-\infty$ as $\eta \uparrow b$, while all other terms stay bounded, because the other poles of G'/G lie at a positive distance from $-x_\rho - ib$. If $A_\eta < 2m/b^2$ then $A_\eta < 2m/(b^2 - \eta^2)$ for every $\eta \in (0,b)$. For a zero ρ' on the line, $\lambda_{\rho'} = -\gamma'^2$ and $\mathrm{Im}(-x_\rho - i\eta + \gamma'^2)^{-1} = \eta/((\gamma'^2 - x_\rho)^2 + \eta^2)$, which gives the limit A_0. □

For $x \neq x_\rho$ the same computation shows that the m copies of the pair contribute $m\,[P_{b+\eta}(x-x_\rho) - P_{b-\eta}(x-x_\rho)]$ to $r_\eta(x)$ for $0 < \eta < b$, with P_h the Poisson kernel of width h: a positive kernel of width $b+\eta$ and a negative kernel of width $b-\eta$, which becomes an atom of mass $-m$ at x_ρ as $\eta \uparrow b$. A zero with $b < \sqrt{x_\rho}$ is therefore seen by a search over the window $\{0 < \eta < \sqrt{x},\ x \le X\}$ as soon as $x_\rho \le X$. The condition $b < \sqrt{x_\rho}$ is $\delta < \gamma/\sqrt{4\gamma^2+1}$, that is $\beta < 1 - 1/(16\gamma^2)$ up to terms of higher order, and it follows from the zero free region $\beta < 1 - c/\log\gamma$ for γ large [324, Theorem 3.8]; it does not follow from $b < \gamma$. A search that finds $r_\eta \ge 0$ on the whole window therefore certifies that no such zero has $x_\rho \le X$, while a search on a grid is evidence only. Figure 14.2 tests the detection on a model in which G'/G is built from the first 60 zeros: with all zeros on the line, r_η is positive on the grid of the figure; after moving γ_5 to $\beta = 0.6$ (so $b = 6.59$) a negative region appears at $x \approx 1084.7$ for η up to 6.5, and $r_\eta(x_\rho) \to -\infty$ as $\eta \uparrow b$, as the lemma predicts.

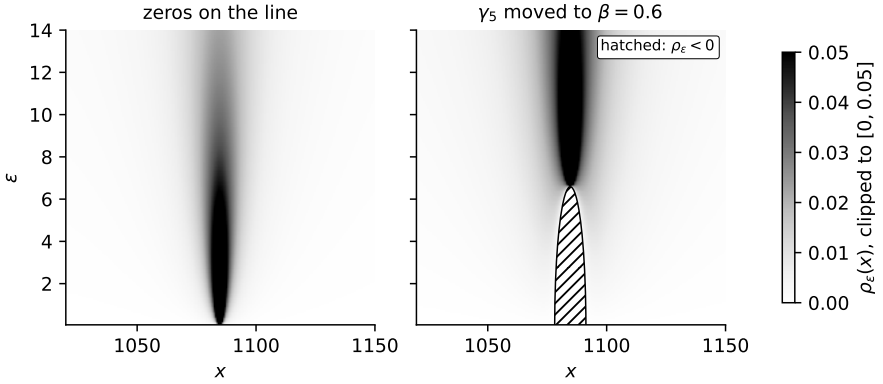


Figure 14.2: The smoothed Thorin density $r_\eta(x)$ in the (x, η) plane near $\gamma_5^2 = 1084.7$. Left: zeros on the line. Right: γ_5 moved off the line; the off line pair is an atom of negative mass at (x_ρ, b).

14.7 Is the clock a weak limit of GGC laws?

The class of GGC laws is closed under weak limits [39, Theorem 3.1.5] (Section 4.1.1), and with Proposition 1.2.1 this gives a sharp answer: T is a weak limit of GGC laws if and only if RH holds (Corollary 4.1.2).

The de Bruijn Newman flow of Section 8.4 is the second family of candidates, and Theorem 7.2.15 places RH at the end of a path of GGC laws. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ be the kernel in the rescaled variable of Rodgers and Tao [289], as in Theorem 7.2.15, let $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with Λ the de Bruijn Newman constant in this normalisation, and let V_t be the variable with density proportional to $e^{tu^2}\Phi_{\mathrm{dBN}}(u)$,

the Gaussian tilt of the Pólya variable $V_0 = X/2$, so that $\mathbb{E}e^{sV_t} = H_t(is)/H_t(0)$ for real s. Whenever a law T_t makes (T_t, V_t) a Wald couple (Definition 14.4.1), $\mathbb{E}e^{-\lambda T_t} = H_t(0)/H_t(i\sqrt{2\lambda}) = R_t(\lambda/2)$, where $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ is the transform of the flow clock of Section 8.4 and $R_0 = 1/G$. Thus T_t is half the flow clock, and $T_0 = T/2$; at $t = 0$ the couple $(T/2, X/2)$ is the one of Remark 14.5.1 rescaled by $\sqrt{2}$.

By Theorem 7.2.15, for every $t \geq \Lambda$ the transform R_t is that of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the positive zeros of the even function H_t (each pair $\pm z$ gives one atom). Hence for $t \geq \Lambda$ the law T_t exists and is a GGC with Thorin measure $\sum_k \delta_{z_k(t)^2/2}$, and since $\Lambda \leq 0.2$ [247, 282] this holds unconditionally for $t \geq 0.2$. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t(\lambda/2) \to 1/G(\lambda/2)$ pointwise, which is the transform of $T/2$, and $T_t \to T/2$ in law along any sequence $t_n \downarrow 0$ for which T_{t_n} exists. With the bound $\Lambda \geq 0$ of Rodgers and Tao [289], Theorem 7.2.15 gives that RH holds if and only if T_t is a GGC for every $t > 0$, if and only if $\Lambda = 0$.

Two cautions. Unconditionally T_t is known to exist only for $t \geq 0.2$ and at $t = 0$ (Theorem 5.1.1), so the convergence $T_t \to T/2$ is a statement about the range of t on which T_t exists. Under RH this is every $t \geq 0$; without RH, whether T_t exists as a probability law for $0 < t < 0.2$ is open (Problem 14.3 in Section 14.15). And the Thorin atoms $z_k(t)^2/2$ of T_t move with the zeros of H_t only on the range $t \geq \Lambda$ where T_t is a GGC; Table 8.8 gives the zeros of $H_t(2\theta)$, in the variable $\theta = z/2$.

Definition 14.7.1 (Laplace distance). For laws P and Q on $[0, \infty)$ with Laplace transforms L_P and L_Q, the Laplace distance is $d_L(P, Q) = \sup_{\lambda \geq 0} |L_P(\lambda) - L_Q(\lambda)|$.

Proposition 14.7.2. *Let $H = 3 \times 10^{12}$, let $N = N(H)$ be the number of zeros ρ_k with $0 < \gamma_k \leq H$, and let $T^{[N]} = \sum_{k \leq N} e_k/\gamma_k^2$ with independent standard exponential variables e_k, the GGC with transform $1/G_N$ of Section 4.1.1. With a_k as in Section 1.2, put $\varepsilon_H = \sum_{k>N} 1/|a_k|$ and, for $K \leq N$, $S_K = \sup_{\lambda \geq 0} \lambda/G_K(\lambda)$. Then*

$$d_L\big(T, T^{[N]}\big) \leq S_K\, \varepsilon_H, \qquad \varepsilon_H < 1.4798 \times 10^{-12}, \qquad S_1 = \gamma_1^2, \qquad S_{1700} < 16.85, \tag{14.7.1}$$

so that $d_L(T, T^{[N]}) < 2.5 \times 10^{-11}$, while the first factor alone gives $\gamma_1^2 \varepsilon_H < 2.96 \times 10^{-10}$. The proof uses only the verification of RH up to height H [247], and makes no assumption on the zeros above H.

Proof. By (1.2.1), $1/G = \prod_k f_k$ with $f_k(\lambda) = (1 + \lambda/a_k)^{-1}$, and $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > 0$ for every zero, so for $\lambda \geq 0$ we have $|1 + \lambda/a_k| \geq \operatorname{Re}(1 + \lambda/a_k) \geq 1$, hence $|f_k| \leq 1$ and $|f_k - 1| = |\lambda/a_k|\,|f_k| \leq \lambda/|a_k|$. For complex numbers of modulus at most one, $|\prod_j f_j - 1| \leq \sum_j |f_j - 1|$. The zeros with $k \leq N$ lie on the line, so $\prod_{k \leq N} f_k = 1/G_N$ is the transform of $T^{[N]}$, and its factors lie in $(0, 1]$; leaving out all but the first K of them can only increase the product, so $1/G_N \leq 1/G_K$.

Therefore

$$\Big|\frac{1}{G(\lambda)}-\frac{1}{G_N(\lambda)}\Big|=\frac{1}{G_N(\lambda)}\Big|\prod_{k>N}f_k(\lambda)-1\Big|\le\frac{\lambda}{G_K(\lambda)}\,\varepsilon_H\le S_K\,\varepsilon_H.$$

For $K=1$, $\lambda/G_1(\lambda)=\lambda/(1+\lambda/\gamma_1^2)$ increases to γ_1^2, a supremum that is not attained. For $K\ge 2$, $\frac{d}{d\lambda}\log(\lambda/G_K(\lambda))=-\phi_K(\lambda)/\lambda$ with $\phi_K(\lambda)=\sum_{k\le K}\lambda/(\lambda+\gamma_k^2)-1$, which increases strictly from -1 to $K-1$, so λ/G_K increases up to the root λ^* of ϕ_K and decreases after it. For $K=1700$ the root is $\lambda^*=47.362$, and evaluating ϕ_{1700} and λ/G_{1700} in ball arithmetic with the first 1700 zeros, on both sides of λ^*, gives $S_{1700}<16.8497$ (code/maxwald_weak.py).

For ε_H, $|a_k|=|\rho_k-\frac12|^2\ge\gamma_k^2$, and partial summation gives $\varepsilon_H\le\sum_{\gamma_k>H}\gamma_k^{-2}=2\int_H^\infty(N(t)-N(H))\,t^{-3}\,dt$. Write $N(t)=N_0(t)+E(t)$ with $N_0(t)=\frac{t}{2\pi}\log\frac{t}{2\pi e}$; by [109, Corollary 1.2], $|E(t)|\le 0.1038\log t+0.2573\log\log t+9.3675$ for $t\ge e$. The main term integrates to exactly $(\log\frac{H}{2\pi}+1)/(2\pi H)=1.47970\times10^{-12}$, and the bound for E, used at t and at H, adds less than 10^{-23}. □

14.8 The sine squared lemma and the basepoint family in the λ chart

The sine squared lemma (1.2.9) [249, Lemma 8.1] turns the Lévy measure of $\log\xi$ at a basepoint α into the Thorin measure of the reciprocal at the same basepoint. For $\alpha>1$ its input is the signed canonical measure of [252, Theorem 8.1]; for $\frac12<\alpha\le1$ there is no such Lévy form, and the family is defined by the Hadamard product, not by the lemma. The family is that of the shifted clocks T_α with transform $\varphi_\alpha(\lambda)=\xi(\alpha)/\xi(\alpha+\sqrt{\lambda})$, $T_{1/2}=T$. For every $\alpha>\frac12$ put $u_\alpha(\lambda)=\mathrm{Re}(\xi'/\xi)(\alpha+i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$, the Thorin density of T_α when every zero has $\beta<\alpha$ (Theorem 1.2.2) and in every case the density of U_α of (1.2.6) in the chart $\lambda=\theta^2$, a sum over the zeros of Poisson kernels (Proposition 1.2.3), and for $\alpha>1$ it has the arithmetic form (3.1.1) with $\theta=\sqrt{\lambda}$ (drift, archimedean, pole and prime pieces), which Table 1.5 compares with the Pólya route. Since $\varphi_\alpha\to1/G$ pointwise as $\alpha\downarrow\frac12$ and T_α exists for every $\alpha>\frac12$, $T_\alpha\to T$ in law, with no hypothesis (Section 4.1.1).

In the chart θ a zero $\rho=\beta+i\gamma$ with $\gamma>0$ contributes, paired with $\bar\rho$, the sum $P_\delta(\theta-\gamma)+P_\delta(\theta+\gamma)$ of two Poisson kernels with $\delta=\alpha-\beta$ (Proposition 1.2.3; $P_{-a}=-P_a$ as there). In the chart $\lambda=\theta^2$ the map $\theta\mapsto\theta^2$ folds the kernel centred at $-\gamma$ onto the one centred at γ and brings in the Jacobian $1/(2\sqrt{\lambda})$.

Proposition 14.8.1. *Let $\alpha>\frac12$, let $\rho=\beta+i\gamma$ be a zero with $\gamma>0$, and put $\delta=\alpha-\beta\ne0$ and $c=\alpha-\rho=\delta-i\gamma$. The term of ρ in $u_\alpha=\sum_{\gamma>0}u_\rho$, where the zeros ρ and $\bar\rho$ are paired, is*

$$u_\rho(\lambda)=\frac{P_\delta(\sqrt{\lambda}-\gamma)+P_\delta(\sqrt{\lambda}+\gamma)}{2\sqrt{\lambda}},\qquad\lambda>0,\tag{14.8.1}$$

and it satisfies the following.

(a) $u_\rho(\lambda) = \kappa_\rho(\lambda)\, C_\rho(\lambda) = \delta\,(\lambda + |c|^2)/(\pi\sqrt{\lambda}\,|c^2+\lambda|^2)$, *where*

$$\kappa_\rho(\lambda) = \frac{\lambda + \gamma^2 + \delta^2}{2\gamma\sqrt{\lambda}},$$
$$C_\rho(\lambda) = \frac{1}{\pi}\,\frac{2\delta\gamma}{(\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2}.$$

For $\delta > 0$, C_ρ is the Cauchy density centred at $\gamma^2 - \delta^2$ with half width $2\delta\gamma$.

(b) $\int_0^\infty u_\rho(\lambda)\, d\lambda = \operatorname{sign}\delta$, *so for $\delta > 0$ the term has unit mass.*

(c) $\kappa_\rho(\lambda) \ge \sqrt{1 + \delta^2/\gamma^2} > 1$ *for every* $\lambda > 0$, $\kappa_\rho(\gamma^2 - \delta^2) = (1 - \delta^2/\gamma^2)^{-1/2}$ *when* $|\delta| < \gamma$, *and for* $K \ge 1$ *and* $|\delta| \le \gamma/(2K)$,

$$|\kappa_\rho(\lambda) - 1| \le \Big(\frac{K^2}{2} + 1\Big)\frac{\delta^2}{\gamma^2} \qquad \text{for } |\lambda - \gamma^2| \le K|\delta|\gamma.$$

Hence near its peak the term is a Lorentzian of full width $4|\delta|\gamma$ at half height, with the sign of δ, up to a relative error $O(\delta^2/\gamma^2)$; away from the peak it is not: $u_\rho(\lambda) \sim \delta/(\pi(\gamma^2+\delta^2)\sqrt{\lambda})$ as $\lambda \downarrow 0$ and $u_\rho(\lambda) \sim \delta/(\pi\lambda^{3/2})$ as $\lambda \to \infty$, while C_ρ is bounded at 0 and $O(\lambda^{-2})$ at infinity.

Proof. Write $\theta = \sqrt{\lambda}$ and $A = \lambda + \gamma^2 + \delta^2$. Since $(\theta \mp \gamma)^2 + \delta^2 = A \mp 2\gamma\theta$,

$$P_\delta(\theta - \gamma) + P_\delta(\theta + \gamma) = \frac{\delta}{\pi}\Big[\frac{1}{A - 2\gamma\theta} + \frac{1}{A + 2\gamma\theta}\Big] = \frac{\delta}{\pi}\,\frac{2A}{A^2 - 4\gamma^2\lambda},$$

and $A^2 - 4\gamma^2\lambda = (\lambda - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2$, since both sides equal $\lambda^2 + 2(\delta^2 - \gamma^2)\lambda + (\gamma^2 + \delta^2)^2$. For $c = \delta - i\gamma$ the right side is also $|c^2 + \lambda|^2$, because $c^2 + \lambda = (\lambda + \delta^2 - \gamma^2) - 2i\delta\gamma$, and $|c|^2 = \gamma^2 + \delta^2$. Dividing by 2θ gives $u_\rho = \delta A/(\pi\sqrt{\lambda}\,|c^2+\lambda|^2)$, which is the second form in (a), and writing $\delta A/\sqrt{\lambda} = \frac{A}{2\gamma\sqrt{\lambda}} \cdot 2\delta\gamma$ gives $\kappa_\rho C_\rho$. For (b) substitute $\lambda = \theta^2$: $\int_0^\infty u_\rho\, d\lambda = \int_0^\infty [P_\delta(\theta-\gamma) + P_\delta(\theta+\gamma)]\, d\theta = \int_{-\gamma}^\infty P_\delta + \int_\gamma^\infty P_\delta$, which is 1 for $\delta > 0$ because P_δ is an even probability density, and -1 for $\delta < 0$ because $P_{-a} = -P_a$. For (c), $\lambda + (\gamma^2 + \delta^2) \ge 2\sqrt{\lambda}\sqrt{\gamma^2 + \delta^2}$ gives the lower bound, and $A = 2\gamma^2$ at $\lambda = \gamma^2 - \delta^2$ gives the value there. For the window put $\lambda = \gamma^2(1 + \varepsilon)$, so that $|\varepsilon| \le K|\delta|/\gamma \le \frac{1}{2}$ and

$$\kappa_\rho(\lambda) = \frac{1 + \varepsilon/2}{\sqrt{1+\varepsilon}} + \frac{\delta^2}{2\gamma^2}\,\frac{1}{\sqrt{1+\varepsilon}}.$$

The first term is 1 at $\varepsilon = 0$ and has derivative $\varepsilon/(4(1+\varepsilon)^{3/2})$, so it differs from 1 by at most $\varepsilon^2/2$ for $|\varepsilon| \le \frac{1}{2}$, and the second term is at most δ^2/γ^2. Take $\delta > 0$, the case $\delta < 0$ being the same with a change of sign. The half height points of

C_ρ are at $\gamma^2 - \delta^2 \pm 2\delta\gamma$, inside the window with $K = 3$ (for $\delta \le \gamma/6$), where $|(\log C_\rho)'| = 1/(2\delta\gamma)$; a relative perturbation η of the density therefore moves them by $O(\eta\,\delta\gamma)$, which gives the width $4\delta\gamma\,(1 + O(\delta^2/\gamma^2))$. The behaviour at 0 and at infinity is read off (a). □

Remark 14.8.2. The leading correction to the width is the factor $1 + \eta$ with $\eta = \delta^2/(2\gamma^2)$: with $x = (\lambda - \gamma^2 + \delta^2)/(2\delta\gamma)$ the expansion above gives $\kappa_\rho = 1 + \eta(1 + x^2) + O(\delta^3/\gamma^3)$, so near the peak $u_\rho \approx C_\rho + \eta/(2\pi\delta\gamma)$, with a relative error $O(\delta^3/\gamma^3)$, is a Lorentzian on a flat background of relative height η, whose half height points are at $x = \pm(1 + \eta)$ to first order.

Numerically the three expressions for u_α agree, up to their truncations: the line form $\mathrm{Re}(\xi'/\xi)(\alpha + i\sqrt{\lambda})/(2\pi\sqrt{\lambda})$ computed through the Pólya density, the zero form $\sum_\rho u_\rho$ of Proposition 14.8.1, and the prime form (3.1.1). At $\alpha = 2$ the three forms agree to 4 to 5 significant digits at $\lambda = 100, 250, 500, 1000$; at $\alpha = 1.5$ the prime form with $n \le N = 2 \times 10^5$ agrees to 1 to 2 digits, the gap being the truncated prime tail $\sum_{n>N} \Lambda(n) n^{-\alpha} \approx N^{1-\alpha}/(\alpha - 1)$ as for Table 1.5. The minimum of u_α on $[1, 2500]$ is positive for $\alpha = 0.55, 0.6, 0.75, 1, 1.5$ and lies near $\lambda = 59$ for each of them, which on a window where every zero is known to lie on the line checks the computation and does not test RH.

14.9 The maximum representation does not transfer

The distribution function of T can be computed by residues, as ϑ_4 is for M_4 (Section 8.5.4). Along a vertical line $\mathrm{Re}\,\lambda = -L$ that contains no pole of $1/G$, the point $\frac{1}{2} + \sqrt{\lambda}$ runs off to the right as $|\,\mathrm{Im}\,\lambda| \to \infty$, where $|\xi|$ grows, so $1/(\lambda G(\lambda))$ is integrable on the line and the Bromwich contour for $F_T(t) = \Pr(T \le t)$, whose transform is $1/(\lambda G(\lambda))$, can be shifted from $\mathrm{Re}\,\lambda > 0$ to $\mathrm{Re}\,\lambda = -L$ past the finitely many poles in between, because $\mathrm{Re}(-a_k) \le \frac{1}{4} - \gamma_k^2$. Write R_k for the residue of $1/G$ at $-a_k$, which for a simple zero $\rho_k = \frac{1}{2} + i\gamma_k$ is $R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ (Section 8.1.1). When the poles with $\mathrm{Re}\,a_k < L$ are simple,

$$\Pr(T > t) = \sum_{\mathrm{Re}\,a_k < L} \frac{R_k}{a_k}\, e^{-a_k t} + O(e^{-Lt}), \qquad t \to \infty. \tag{14.9.1}$$

Only this finite shift is used below: the full residue series needs contours between the poles on which $1/|\xi|$ is controlled, and under RH with simple zeros it is the spectral expansion of Section 8.1.1.

The coefficients R_k/a_k alternate in sign, because Ξ' changes sign at consecutive simple zeros, and they grow like $e^{\pi\gamma_k/4}$ up to a power of γ_k and the factor $1/|\zeta'(\rho_k)|$ (Section 8.1.1). The factor $e^{-\gamma_k^2 t}$ still wins for every $t > 0$.

A law F on $(0, \infty)$ that is the distribution function of a maximum of independent exponential variables E_i/r_i with deterministic rates r_i, where $\sum_i e^{-r_i t} < \infty$

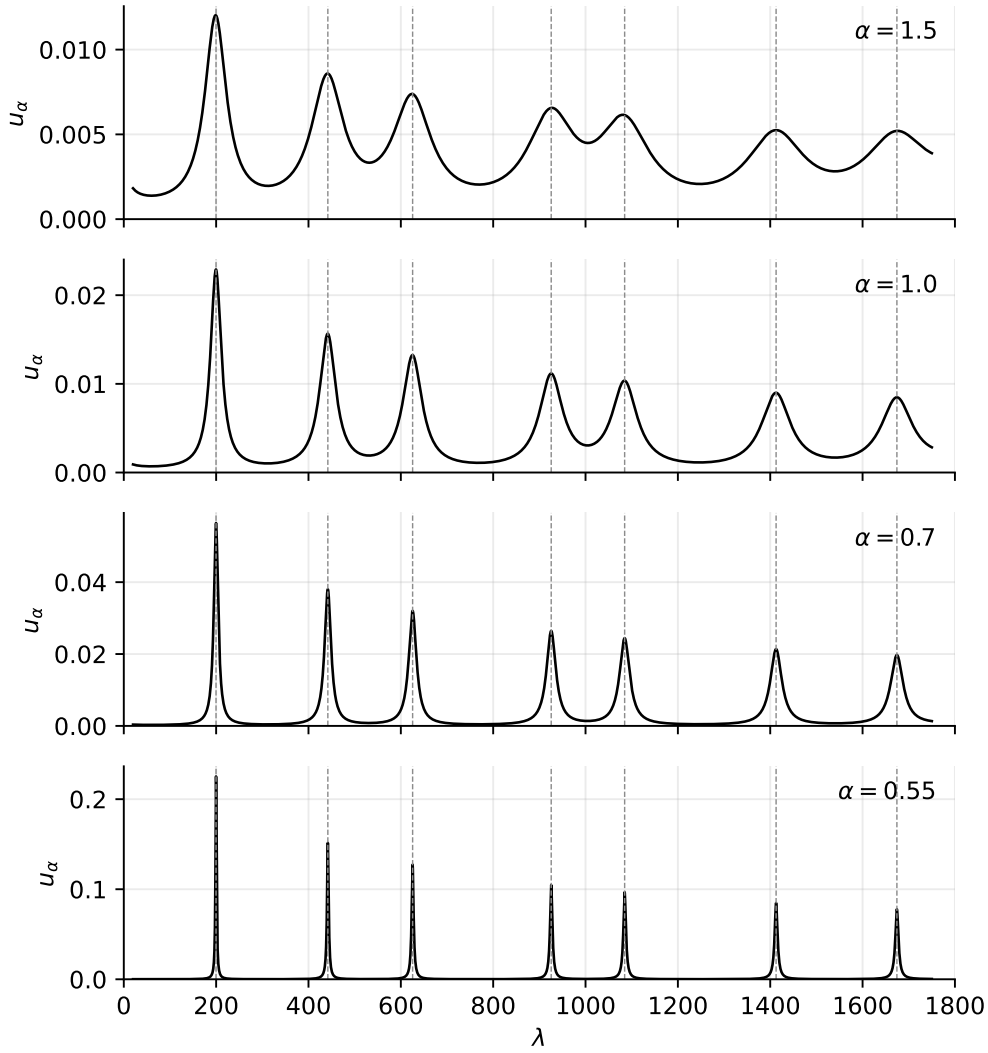


Figure 14.3: The density u_α of (1.2.6), computed from $\operatorname{Re}\,\xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ with no zero as input, for $\alpha = 1.5, 1, 0.7, 0.55$; for $\alpha = 0.7$ and 0.55 it is the Thorin density of T_α only if no zero lies to the right of the line, as under RH. Dashed lines mark γ_k^2. The term of a zero on the line has unit mass and is close to a Lorentzian of width $4(\alpha - \frac{1}{2})\gamma_k$ near its peak (Proposition 14.8.1); as $\alpha \downarrow \frac{1}{2}$ these terms converge to unit atoms at γ_k^2.

for $t > 0$, has $-\log F(t) = \sum_i \sum_{m\ge1} e^{-mr_i t}/m$, a sum of exponentials with positive coefficients. The same holds, with $-\log F(t) = \int e^{-xt}\,\nu(dx)$, for the maximum of a Poisson cloud of exponential clocks with a rate measure ν, that is, of the clocks E_i/x_i attached to the points x_i of a Poisson process of intensity ν, since the number of clocks above t is Poisson with mean $\int e^{-xt}\nu(dx)$.

PROPOSITION 14.9.1. *$-\log F_T$ is not completely monotone. Hence T is not distributed as a maximum of independent exponential variables, for any choice of rates, nor as the maximum of a Poisson cloud of exponential clocks.*

Proof. Put $q_k = R_k/a_k$. For $k \le 3$ the zero ρ_k lies on the line, so the $a_k = -(\rho_k - \frac12)^2$ of Section 1.2 equals γ_k^2. The zeros with $\operatorname{Re} a < 3a_1 = 599.37$ are ρ_1 and ρ_2 only: $a_1 = 199.79$, $a_2 = 441.93$ and $a_3 = 625.54$, the zeros below height $H = 3\times10^{12}$ lie on the line [247], and a zero above H has $\operatorname{Re} a \ge \gamma^2 - \frac14 > H^2 - \frac14$ wherever it lies. The zeros below height 200 are simple (Section 3.1.1). Equation (14.9.1) with $L = 3a_1$ gives $\Pr(T > t) = q_1 e^{-a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$, and since $-\log(1-y) = y + \frac12 y^2 + O(y^3)$ and $2a_1 < a_2 < 3a_1 < a_1 + a_2$,

$$-\log F_T(t) = q_1 e^{-a_1 t} + \tfrac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t}).$$

Suppose that $-\log F_T(t) = \int_{[0,\infty)} e^{-xt}\nu(dx)$ for $t > 0$ with $\nu \ge 0$ (Bernstein's theorem). First $\nu([0,a_1)) = 0$: for $x_0 < a_1$, $\nu([0,x_0])e^{-x_0 t} \le -\log F_T(t) = O(e^{-a_1 t})$. Then $e^{a_1 t}(-\log F_T(t)) = \nu(\{a_1\}) + \int_{(a_1,\infty)} e^{-(x-a_1)t}\nu(dx)$, whose last term tends to 0 by dominated convergence, so $\nu(\{a_1\}) = q_1$, and $f_1(t) = \int_{(a_1,\infty)} e^{-xt}\nu(dx) = \frac12 q_1^2 e^{-2a_1 t} + q_2 e^{-a_2 t} + O(e^{-3a_1 t})$. The same two steps give $\nu((a_1, 2a_1)) = 0$ and $\nu(\{2a_1\}) = \frac12 q_1^2$, and then $f_2(t) = \int_{(2a_1,\infty)} e^{-xt}\nu(dx) \ge 0$ satisfies $e^{a_2 t} f_2(t) \to q_2$ because $a_2 < 3a_1$. So $q_2 \ge 0$. But $R_2 = -2\gamma_2 \Xi(0)/\Xi'(\gamma_2)$ and $\Xi'(\gamma_2) > 0$, since Ξ is positive on $[0,\gamma_1)$, negative on (γ_1,γ_2) and changes sign at the simple zero γ_2; so $q_2 = R_2/a_2 < 0$, a contradiction. □

For M_4 the coefficient at $a_2 = 4\pi$ also has the wrong sign: $\Pr(M_4 > t) = \sum_{n\ge1} 2(-1)^{n+1} e^{-\pi n^2 t}$ has $q_1 = 2$ and $q_2 = -2$. But $a_2 = 4\pi = 4a_1$ coincides with a term of the expansion in q_1, and the coefficient of $e^{-4\pi t}$ in $-\log\vartheta_4$ becomes $q_2 + \frac14 q_1^4 = -2 + 4 = 2 > 0$. Commensurability of the exponents πn^2 rescues complete monotonicity (the Jacobi triple product writes $-\log\vartheta_4$ as a sum of exponentials with positive coefficients), while $\gamma_2^2/\gamma_1^2 = 2.2119$ is not an integer, so nothing rescues it for T.

14.10 Gnedenko: the first zero as a norming constant

Gnedenko's theorem [93] classifies the limits of normalised maxima of independent identically distributed variables, and a law with a tail $\Pr(X > t) \sim Ce^{-at}$ lies in the Gumbel domain with norming constant $1/a$ and centring $\log(Cn)/a$. The

proposition reads the norming constant of T as the first atom γ_1^2 of its Thorin measure, with the constant given by the first residue.

PROPOSITION 14.10.1. *(i)* $\Pr(M_4 > t) \sim 2e^{-\pi t}$, *so for independent copies* $M_4^{(j)}$, $j \le n$,

$$\Pr\left(\pi \max_{j\le n} M_4^{(j)} - \log(2n) \le x\right) \to \exp(-e^{-x}).$$

(ii) $\Pr(T > t) \sim (R_1/\gamma_1^2)\, e^{-\gamma_1^2 t}$, *so for independent copies* $T^{(j)}$, $j \le n$, *of* T,

$$\Pr\left(\gamma_1^2 \max_{j\le n} T^{(j)} - \log \frac{R_1 n}{\gamma_1^2} \le x\right) \to \exp(-e^{-x}),$$

$$\max_{j\le n} T^{(j)} \Big/ \log n \to \frac{1}{\gamma_1^2} \quad \textit{in probability.}$$

Proof. (i) is $1-\vartheta_4(t) = 2e^{-\pi t} + O(e^{-4\pi t})$. For (ii), T exists by Theorem 5.1.1, and (14.9.1) with $a_1 < L < a_2$ gives $\Pr(T > t) = (R_1/a_1)e^{-a_1 t} + O(e^{-Lt})$, because the simple pole at $-\gamma_1^2$ is the rightmost singularity of $1/(\lambda G(\lambda))$ after 0, every other pole having $\operatorname{Re}\lambda \le -\gamma_2^2$. In both cases, if $\Pr(X > t) \sim Ce^{-at}$ and $y_n = (\log(Cn) + x)/a$, then $n\Pr(X > y_n) \to e^{-x}$ and $\Pr(\max_{j\le n} X^{(j)} \le y_n) = (1 - \Pr(X > y_n))^n \to \exp(-e^{-x})$. The last statement follows because $a\max_{j\le n} X^{(j)} - \log n$ is then tight. □

This gives a way to find γ_1 that uses no zeros at all: invert the Laplace transform $(1 - 1/G(\lambda))/\lambda$ numerically and read off the exponential rate of the tail. It is the experiment of Section 8.1.1, where the hazard of T tends to γ_1^2 by Talbot inversion with no zero as input. The local logarithmic slopes of $\Pr(T > t)$ on $[0.04, 0.05]$, $[0.05, 0.06]$, $[0.06, 0.07]$ and $[0.07, 0.08]$ are 199.496, 199.764, 199.788 and 199.790, converging to $\gamma_1^2 = 199.7905$ at the rate $e^{-(\gamma_2^2-\gamma_1^2)t}$ predicted by the second term of (14.9.1), and the intercept $e^{\gamma_1^2 t}\Pr(T > t)$ at $t = 0.08$ gives $R_1/\gamma_1^2 = 50.87103$ against the residue value 50.87104.

14.11 Finding the zeros from the Thorin density in the z chart

Theorem 14.6.1 gives a procedure in the chart $z = \lambda$: compute the smoothed Thorin density r_η of that theorem on a grid from ξ'/ξ, with no zero used, locate its local maxima, and take square roots. Under RH $r_\eta = P_\eta * U_{\mathrm{RH}}$ is the Poisson smoothing of the Thorin measure $U_{\mathrm{RH}} = \sum_k \delta_{\gamma_k^2}$ at distance η below the cut (Theorem 14.6.1 and Section 1.3), so each atom appears as a spike of unit mass and height near $1/(\pi\eta)$ at $x \approx \gamma_k^2$. The book carries out the same recovery in the chart $\theta = \sqrt{z}$ (Sections 3.1.1 and 8.6). The table below is a cross check in the other chart, at $\eta = 3$.

The peak of r_η near γ_k^2 is displaced from it by the pull of the neighbouring spikes, to leading order by $\eta^4 \sum_{j\ne k}(\gamma_j^2 - \gamma_k^2)^{-3}$, which is of modulus below 2×10^{-5}

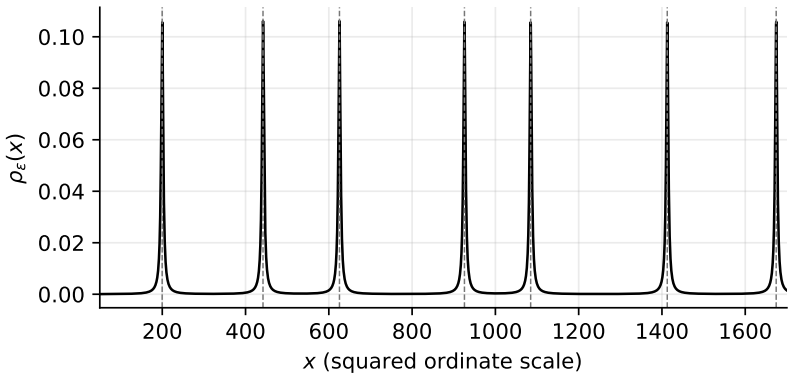


Figure 14.4: The smoothed Thorin density r_η of T, $\eta = 3$, computed from ξ'/ξ with no zero as input. Dashed lines mark γ_k^2. Each spike has height near $1/(\pi\eta)$ and unit mass.

k	peak of r_η	$\sqrt{\text{peak}}$	γ_k
1	199.790462	14.1347254	14.1347251
2	441.926159	21.0220398	21.0220396
3	625.542987	25.0108574	25.0108576
4	925.673105	30.4248764	30.4248761
5	1084.718263	32.9350613	32.9350616
6	1412.720791	37.5861782	37.5861782
7	1674.341571	40.9187191	40.9187190

Table 14.2: Zeros recovered as atoms of the smoothed Thorin measure, $\eta = 3$. The integral of r_η over $[50, 1700]$ is 6.95, against 7 zeros.

in the table. On the window $[50, 1700]$ with $\eta = 3$ the minimum of r_η is $5.6\times10^{-5} > 0$, attained at the endpoint $x = 50$. This is a consistency check of the computation with Theorem 14.6.1(d) and Lemma 14.6.2, not a test of RH: the window contains only the squares of the zeros below height $\sqrt{1700} = 41.2$, which are known to lie on the line.

14.12 Finite maxima and a limit argument

The maximum interpretation of Section 14.2 suggests a direct strategy: truncate the maximum at N terms, show that the finite object has no zeros off the critical line, and pass to the limit. For the sum representation instead of the maximum the book has the truncations of the Biane, Pitman and Yor lattice (Subsection 7.2.1).

14.12.1 Finite maxima

For $N \geq 1$ let

$$M_4^{(N)} = \max\Big\{ \frac{E_n}{\pi n} \ (n \leq N), \quad \frac{E'_m}{\pi(2m-1)} \ (m \leq N) \Big\},$$

$$\mathcal{P}_N(q) = \prod_{n \leq N} (1 - q^n) \prod_{m \leq N} (1 - q^{2m-1}) = \sum_{m \geq 0} p_m^{(N)} q^m,$$

where $(E_n)_{n \geq 1}$ and $(E'_m)_{m \geq 1}$ are independent standard exponential variables, and $M_4 = \lim_N M_4^{(N)}$ is the maximum of Subsection 8.5.4, written with the rates πn and $\pi(2m-1)$. Then $\Pr(M_4^{(N)} \leq t) = \mathcal{P}_N(e^{-\pi t})$ exactly, a polynomial in $e^{-\pi t}$ with integer coefficients. Expanding the product and integrating term by term, for $\operatorname{Re} s > 0$,

$$\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\Gamma(1 + \tfrac{s}{2})\, \pi^{-s/2} D_N(s), \qquad D_N(s) = \sum_{m \geq 1} p_m^{(N)}\, m^{-s/2},$$

a finite Dirichlet polynomial D_N with integer coefficients. Since the factors left out of $\mathcal{P}_N$ are $1 - q^a$ with $a > N$, $p_m^{(N)}$ equals the coefficient of ϑ_4 for $m \leq N$: it is $2(-1)^n$ at $m = n^2$ and 0 otherwise. So the first block of D_N is exactly minus the partial sum $2\sum_{n^2 \leq N} (-1)^{n+1} n^{-s}$ of $2(1 - 2^{1-s})\zeta(s)$, followed by a tail of integer coefficients at the frequencies $\frac{1}{2}\log m$ with $N < m \leq N(3N+1)/2$, the degree of $\mathcal{P}_N$.

By Subsection 8.5.4, $(1 - 2^{1-s})\zeta(s) = \mathbb{E}[(2K^2)^{s/2}]/(2\Gamma(1 + \frac{s}{2}))$ for every complex s, with $\pi M_4 = 2K^2$ and K the Kolmogorov variable, and since $2\xi(s)/(s-1) = 2\pi^{-s/2}\Gamma(1 + \frac{s}{2})\zeta(s)$ this reads

$$\mathbb{E} M_4^{s/2} = \frac{2(1 - 2^{1-s})\, \xi(s)}{s - 1} \qquad \text{for every complex } s.$$

Define the normalised finite stage and its symmetrisation

$$\xi_N(s) = \frac{(s-1)\, \mathbb{E}(M_4^{(N)})^{s/2}}{2(1 - 2^{1-s})}, \qquad \xi_N^{\mathrm{sym}}(s) = \tfrac{1}{2}\big(\xi_N(s) + \xi_N(1-s)\big),$$

so that the same expression with M_4 in place of $M_4^{(N)}$ is ξ. In the open strip $0 < \operatorname{Re} s < 1$ the factor $1 - 2^{1-s}$ does not vanish and $\mathbb{E}(M_4^{(N)})^{s/2}$ is holomorphic (it is holomorphic for $\operatorname{Re} s > -4N$, since $\Pr(M_4^{(N)} \leq t) \sim C t^{2N}$ as $t \to 0$), so ξ_N and ξ_N^{sym} are holomorphic there. The function ξ_N^{sym} is invariant under $s \mapsto 1 - s$ and, because ξ_N is real on the real axis, $\xi_N^{\mathrm{sym}}(\frac{1}{2} + it) = \operatorname{Re}\, \xi_N(\frac{1}{2} + it)$ is real for real t.

THEOREM 14.12.1. *(i)* $M_4^{(N)} \uparrow M_4$ *almost surely, and* $\Pr(M_4^{(N)} \neq M_4) \leq C\sqrt{N}\, e^{-\pi\sqrt{N}/2}$.

(ii) On every compact K *in the open strip* $0 < \operatorname{Re} s < 1$,

$$\sup_K |\xi_N - \xi| \leq C_K\, N^{1/4} e^{-\pi\sqrt{N}/4}, \qquad \sup_K |\xi_N^{\mathrm{sym}} - \xi| \leq C_K\, N^{1/4} e^{-\pi\sqrt{N}/4}.$$

Proof. (i) The variables of the stages are the same for all N, so adding variables can only raise the maximum, and $M_4^{(N)} \neq M_4$ requires a left out variable to exceed $M_4^{(N)}$. For any $\tau > 0$,

$$\Pr(M_4^{(N)} \neq M_4) \leq \Pr(M_4^{(N)} \leq \tau) + \sum_{n>N} e^{-\pi n\tau} + \sum_{m>N} e^{-\pi(2m-1)\tau}.$$

The last two sums are at most $2e^{-\pi N\tau}/(1-e^{-\pi\tau})$. For the first, $\Pr(M_4^{(N)} \leq \tau) = \vartheta_4(\tau)/\Pi_N(\tau)$, where $\Pi_N(\tau) = \prod_{n>N}(1-e^{-\pi n\tau}) \prod_{m>N}(1-e^{-\pi(2m-1)\tau})$ is the product of the left out factors. Since $1 - x \geq e^{-x/(1-x)}$, $\Pi_N(\tau) \geq \exp\bigl(-2e^{-\pi N\tau}/((1-e^{-\pi\tau})(1-e^{-\pi N\tau}))\bigr)$, while Jacobi's transformation $\vartheta_4(\tau) = \tau^{-1/2} \sum_n e^{-\pi(n+1/2)^2/\tau}$ gives $\vartheta_4(\tau) \leq 3\tau^{-1/2} e^{-\pi/(4\tau)}$ for $\tau \leq 1$. The choice $\tau = 1/(2\sqrt{N})$ balances $\pi/(4\tau)$ against $\pi N\tau$, both equal to $\pi\sqrt{N}/2$, and gives the bound.

(ii) For $\sigma = \operatorname{Re} s > 0$, $M_4^{(N)} \leq M_4$ gives $|\mathbb{E}M_4^{s/2} - \mathbb{E}(M_4^{(N)})^{s/2}| \leq 2\mathbb{E}[M_4^{\sigma/2}\mathbf{1}\{M_4 \neq M_4^{(N)}\}] \leq 2(\mathbb{E}M_4^{\sigma})^{1/2} \Pr(M_4^{(N)} \neq M_4)^{1/2}$ by Cauchy and Schwarz. The prefactor $(s-1)/(2(1-2^{1-s}))$ is bounded on K because $1 - 2^{1-s}$ vanishes only on $\operatorname{Re} s = 1$. The symmetrised bound follows since K and $1 - K$ both lie in the strip and $\xi(1-s) = \xi(s)$. □

The exponent in (ii) is not sharp. A heuristic count of the event $M_4^{(N)} \neq M_4$, in which the jth left out variable wins with probability $\mathbb{E}e^{-\pi j M_4} = \pi\sqrt{j}/\sinh(\pi\sqrt{j})$, of order $\sqrt{j}\, e^{-\pi\sqrt{j}}$, gives a rate of order $Ne^{-\pi\sqrt{N}}$.

COROLLARY 14.12.2 (Hurwitz). *(a) Every zero of* ξ *in the open strip is a limit of zeros of* ξ_N*, and of zeros of* ξ_N^{sym}*, with multiplicity; every limit of zeros of* ξ_N *inside the open strip is a zero of* ξ*.*

(b) If $\rho = \frac{1}{2} + i\gamma$ *is a simple zero of* ξ*, then for all large* N*,* ξ_N^{sym} *has exactly one zero in a small disc about* ρ*, and that zero lies exactly on the critical line.*

(c) Suppose that there are $N_j \to \infty$ *and heights* $h_j \to \infty$ *such that* $\xi_{N_j}^{\mathrm{sym}}$ *has no zeros in* $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$ *for every* j*. Then the Riemann Hypothesis holds. In particular this is so if there are* $c > 0$ *and* N_0 *such that for every* $N \geq N_0$*,* ξ_N^{sym} *has no zeros in* $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < c\sqrt{N}\}$*.*

Proof. (a) is Hurwitz's theorem with Theorem 14.12.1(ii). For (b), Hurwitz gives exactly one zero z_N of ξ_N^{sym} in a disc D centred at ρ for large N. The symmetries $s \mapsto 1 - s$ and $s \mapsto \bar{s}$ of ξ_N^{sym} make its zero set invariant under $s \mapsto 1 - \bar{s}$, which

maps D to itself; uniqueness forces $1-\bar{z}_N = z_N$, that is $\operatorname{Re} z_N = \frac{1}{2}$. For (c), fix Γ and let $U = \{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < \Gamma\}$. For j so large that $h_j > \Gamma$ the functions $\xi^{\mathrm{sym}}_{N_j}$ have no zeros in U, and they converge to $\xi \not\equiv 0$ uniformly on compact subsets of U, so by Hurwitz ξ has no zeros in U. Letting $\Gamma \to \infty$ and using $\xi(s) = \xi(1-s)$ gives RH. □

Part (b) is pointwise in ρ and gives no height up to which the stages reproduce all the zeros of ξ, since that would need an error bound uniform up to the height together with lower bounds for $|\xi|$ on circles around every zero below it, that is, bounds for $1/\zeta'(\rho)$, and none is proved here.

14.12.2 What the finite stages look like

The unsymmetrised stage ξ_N is not zero free off the line at any of the N from 4 to 128 that we tested, so the naive version of the strategy fails. Its zeros near ρ_k approach ρ_k as N grows, and more slowly the larger γ_k is (Table 14.3), and it carries spurious zeros elsewhere, such as the zero of ξ_{16} at $1.055 + 17.984i$, just to the right of the strip.

N	zero near ρ_1	zero near ρ_2	zero near ρ_5
16	$0.51577 + 14.09915i$	$0.95703 + 20.91901i$	$1.56809 + 34.68128i$
32	$0.49948 + 14.13483i$	$0.48159 + 21.02790i$	$1.02797 + 32.22350i$
64	$0.50000052 + 14.1347256i$	$0.50003418 + 21.0220191i$	$0.51254 + 32.92764i$

Table 14.3: Zeros of the unsymmetrised finite stage ξ_N. For $N = 16$ and $N = 32$ the zero listed near ρ_5 does not track ρ_5, since ρ_5 lies above the height $4\sqrt{N}$ discussed below.

N	zeros	$\lvert t_1 - \gamma_1\rvert$	$\lvert t_2 - \gamma_2\rvert$	$\lvert t_3 - \gamma_3\rvert$	$\lvert t_5 - \gamma_5\rvert$
16	7	3.6×10^{-2}	9.9×10^{-2}	1.7×10^{-1}	1.3×10^{-3}
32	10	1.0×10^{-4}	5.9×10^{-3}	6.8×10^{-2}	1.1
64	8	4.1×10^{-7}	2.1×10^{-5}	3.8×10^{-4}	7.5×10^{-3}

Table 14.4: Real zeros t_k of the symmetrised stage $\operatorname{Re} \xi_N(\frac{1}{2} + it)$: the column "zeros" is the number of sign changes on $[10, 44]$, where ξ has 8 zeros ($10 \le \gamma \le 44$). At $N = 16$ the agreement at γ_5 is accidental.

From $N = 32$ to $N = 64$ the error at γ_1 falls by a factor of about 250, against $e^{\pi(8-\sqrt{32})} = 1574$ for the exponential factor of the heuristic rate (about 800 with its prefactor N) and $e^{\pi(8-\sqrt{32})/4} = 6.3$ for that of the proven one; this is a rate of about $e^{-0.75\pi\sqrt{N}}$ (about $e^{-0.60\pi\sqrt{N}}$ between $N = 8$ and $N = 16$). The supremum of $|\xi_N - \xi|$ over the grid $\{0.3, 0.5, 0.7\} + i\{0, 2, \dots, 10\}$ decays like $e^{-c\pi\sqrt{N}}$ with

$c \approx 0.93$ between $N = 32$ and $N = 64$ and $c \approx 0.80$ between $N = 8$ and $N = 16$, and that of $|\xi_N^{\mathrm{sym}} - \xi|$ with $c \approx 0.81$ and $c \approx 0.80$, all between the exponent $\frac{1}{4}$ of the proof and the exponent 1 of the heuristic. Between consecutive stages the exponents for ξ_N^{sym} and for the error at γ_1 oscillate, between 0.73 and 1.10 and between 0.49 and 1.37, so the exponent 1 of the heuristic is not a bound. Since $|\xi(\frac{1}{2} + i\gamma)|$ decays like $e^{-\pi\gamma/4}$, the heuristic predicts that zeros up to height γ are captured once $\pi\sqrt{N}$ exceeds $\pi\gamma/4$ by a margin, that is $N \gtrsim \gamma^2/16$. The tables agree: the symmetrised stage $N = 64$ captures every zero up to height 33 to within 10^{-2} (and ξ_{64} itself to within 1.5×10^{-2}), and the accuracy degrades beyond 37. This suggests that the stage of index N tracks the zeros of ξ up to a height of about $4\sqrt{N}$; this tracking height is not the zero free height of Corollary 14.12.2(c), which may grow with N in any way.

14.12.3 The symmetric stages F_N^{sym} and how far they stay on the line

The normalisation ξ_N divides by $1 - 2^{1-s}$, which has zeros on $\operatorname{Re} s = 1$ and, after symmetrisation, on $\operatorname{Re} s = 0$. For counting zeros it is better to keep the factor. Put

$$F_N(s) = \frac{s-1}{2}\,\mathbb{E}\big(M_4^{(N)}\big)^{s/2} = -\frac{s-1}{2}\,\Gamma\big(1+\tfrac{s}{2}\big)\pi^{-s/2}D_N(s),$$

$$F_N^{\mathrm{sym}}(s) = \tfrac{1}{2}\big(F_N(s) + F_N(1-s)\big),$$

so that $F_N = (1 - 2^{1-s})\xi_N$. These functions are holomorphic on $-4N < \operatorname{Re} s < 4N + 1$, but they are not entire: F_N has a pole at $s = -4N$ and F_N^{sym} also at $s = 4N + 1$, far from every region used below. As $N \to \infty$, $F_N \to (1 - 2^{1-s})\xi(s)$ and $F_N^{\mathrm{sym}} \to \mathfrak{q}(s)\xi(s)$, with $\mathfrak{q}(s) = 1 - 2^{-s} - 2^{s-1}$. The zeros of $\mathfrak{q}$ are simple and are the solutions of $2^s = 1 \pm i$, all on $\operatorname{Re} s = \frac{1}{2}$, at $t = (2\pi k \pm \pi/4)/\log 2$.

Theorem 14.12.1 and Corollary 14.12.2 hold for F_N^{sym} with $\mathfrak{q}\xi$ in place of ξ, on compact subsets of the open strip, since $1 - 2^{1-s}$ is bounded there. Part (c) becomes: if there are $N_j \to \infty$ and $h_j \to \infty$ such that $F_{N_j}^{\mathrm{sym}}$ has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < h_j\}$, then RH holds, by the same argument, since a zero of ξ in that region is a zero of $\mathfrak{q}\xi$. We call (c) for F_N^{sym} the F_N^{sym} form of Corollary 14.12.2(c): it is a different statement from (c) for ξ_N^{sym}, since F_N^{sym} is not ξ_N^{sym} times a fixed factor.

Zeros of F_N^{sym} in a rectangle $[-1, 2] \times [t_1, t_2]$ can be counted by the argument principle, and zeros on the line are sign changes of the real function $F_N^{\mathrm{sym}}(\frac{1}{2} + it)$. When the two counts agree, every zero in the rectangle is on the line. Write $T^*(N)$ for the height below which the counts agree in every window of height 10 and above which off line zeros appear in the rectangle. Since the rectangle contains the strip, F_N^{sym} has no zeros in $\{\frac{1}{2} < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < T^*(N)\}$, which is the hypothesis of the F_N^{sym} form of Corollary 14.12.2(c) at the height $T^*(N)$. Table 14.5 reports the counts.

N	terms	height	on the line	first off line window	$T^*(N)$
8	72	400	70 of 192	$(110, 120]$	≈ 110
16	350	1000	146 of 510	$(250, 260]$	≈ 250
32	1478	800	312 of 360	$(530, 540]$	≈ 530
48	3372	1000	494 of 496	$(870, 880]$	≈ 870
64	6034	1500	791 of 797	$(1060, 1070]$	≈ 1060
96	13670	2500	1463 of 1465	$(1590, 1600]$	≈ 1590
128	24354	130	64 of 66	$(60, 70]$	≈ 60

Table 14.5: Zeros of the symmetric stage F_N^{sym} in $[-1, 2] \times [0, T]$ by the argument principle, against sign changes on the critical line: "terms" is the number of nonzero coefficients $p_m^{(N)}$, $m \geq 1$, "height" the height T searched, and "on the line" the number of sign changes out of the number of zeros counted. The windows have height 10. Above $T^*(N)$ the off line zeros arrive in bulk for $N \leq 32$: for $N = 16$ the two windows of $(300, 320]$ have 10 of their 13 zeros off the line. The sum D_N, whose terms are much larger than the sum itself, is evaluated in double precision, with a bound on its rounding error on the critical line in every window, and in multiprecision in the windows below 100 for $N = 96$ and below 130 for $N = 128$; above 130 the double precision margin of $N = 128$ is lost, so that stage is reported only up to 130. Off line zeros come in pairs $\sigma + it$, $1 - \sigma + it$, so an off line pair adds two to the count of its window, and a difference of one between the two counts is an artefact, such as a zero at the window boundary.

The first off line pairs of F_N^{sym} are $1.5643 + 111.7906i$ for $N = 8$, $1.1950 + 254.3196i$ for $N = 16$, $1.2079 + 537.2287i$ for $N = 32$, $1.0434 + 870.3807i$ for $N = 48$, $1.1025 + 1063.9243i$ for $N = 64$, $1.1177 + 1593.1234i$ for $N = 96$ and $1.5311 + 60.7314i$ for $N = 128$, each with its mirror image under $s \mapsto 1 - \bar{s}$. The first off line zero inside the strip appears for $N = 8$ in $(120, 130]$, for $N = 16$ in $(330, 340]$ and for $N = 32$ in $(550, 560]$, and none was found there for $N = 48$, 64, 96 and 128 up to the heights 1000, 1500, 2500 and 130 searched, so the height up to which F_N^{sym} has no off line zeros in the strip is at least $T^*(N)$ and may be larger. The counts depend on the rectangle: for $N = 16$ the rectangle $[-8, 9] \times [240, 250]$ contains 7 zeros where $[-1, 2] \times [240, 250]$ contains 3, all on the line, so zeros of F_N^{sym} far from the strip exist below $T^*(N)$, and $T^*(N)$ refers to $[-1, 2]$.

Three things follow. First, the symmetric stages are not real rooted: off line zeros exist in $[-1, 2] \times [0, T]$ for every N we tested, but only above a height $T^*(N)$, and the first ones found lie outside the critical strip. Second, up to $N = 96$ the height $T^*(N)$ grows roughly linearly, while the height up to which the stage approximates $\mathfrak{q}\xi$ well enough to capture its zeros is only about $4\sqrt{N}$ (Table 14.4 is for ξ_N^{sym}; the same height is expected for F_N^{sym}, whose limit $\mathfrak{q}\xi$ has the same zeros besides those of $\mathfrak{q}$). At $N = 128$ the growth breaks: $T^*(128)/128 = 0.47$, although the strip has no off line zero of that stage below 130, the height to which

it was counted. Below $4\sqrt{N}$ the zeros of F_N^{sym} are the zeros of $\mathfrak{q}\xi$ moved along the line; between $4\sqrt{N}$ and $T^*(N)$ the stage has its own zeros, fewer than $\mathfrak{q}\xi$ has, but still all on the line; above $T^*(N)$ it has zeros off the line. The linear growth is what a heuristic based on the approximate functional equation suggests for a symmetrised Dirichlet polynomial whose longest frequency is $\frac{1}{2}\log m_{\max}$ with $m_{\max} = N(3N+1)/2$: the two halves of F_N^{sym} can balance off the line only when $(t/2\pi)^{\sigma-1/2}$ is matched by a ratio of the Dirichlet polynomial at σ and $1-\sigma$, which first becomes possible at t of order N; the pair of $N = 128$, far from the strip, falls outside this heuristic. Third, the hypothesis of the F_N^{sym} form of Corollary 14.12.2(c) holds up to height $T^*(N)$ for the N tested, but finitely many N cannot verify an asymptotic hypothesis, and (c) needs no particular height: it suffices that $T^*(N_j) \to \infty$ along some sequence $N_j \to \infty$. The finite N question becomes the growth of $T^*(N)$: the conjecture $T^*(N) \geq \kappa N$ for some $\kappa > 0$ and all N would imply RH, as would any $T^*(N_j) \to \infty$, in particular the far weaker $T^*(N) \geq c\sqrt{N}$.

We do not have a mechanism for $T^*(N)$. The classical argument that puts zeros of a symmetrised Dirichlet polynomial on the line, dominance of the first term of D_N, that is $|D_N(s) - p_1| < |p_1|$ on $\mathrm{Re}\, s \geq \frac{1}{2}$ (with $|p_1| = 2$), fails here because $\sum_m |p_m^{(N)}| m^{-1/4}$ is large: about 50 for $N = 8$ and 2.4×10^9 for $N = 64$. The observed $T^*(N)$ is nevertheless what the balance heuristic predicts, so the right statement to prove may be about the ratio $|D_N(\sigma + it)/D_N(1 - \sigma + it)|$ rather than about D_N alone.

14.12.4 Comparison with the symmetrised lattice truncations

Subsection 7.2.1 carries out the same strategy for the symmetrised lattice truncations K_n, with Proposition 7.2.6 and Theorem 7.2.7 as the analogues of Corollary 14.12.2(b) and (c); the off line zeros of F_N^{sym} above $T^*(N)$ do not bear on Conjecture S of that subsection, which concerns K_n.

14.13 Bondesson classes containing GGC

The classes BO, GGC and SD (Subsection 1.3.2) have finite order versions. This section establishes, for the clock T, the finite order versions up to the orders 1.16×10^{25} on the Lévy side and 9×10^{12} on the transform side from finite verification, and shows that every infinite order version is equivalent to RH.

Definition 14.13.1. A function f on an interval is k monotone if $(-1)^j f^{(j)} \geq 0$ for $0 \leq j \leq k$, and completely monotone if it is k monotone for every k. For $k \geq 1$ let GGC_k be the laws on $[0,\infty)$ with a Lévy density ℓ such that $x\ell(x)$ is k monotone, and BO_k those with ℓ itself k monotone. Thus $\mathrm{GGC}_1 = \mathrm{SD}$ is the self decomposable class, $\bigcap_k \mathrm{GGC}_k = \mathrm{GGC}$ and $\bigcap_k \mathrm{BO}_k = \mathrm{BO}$. A positive function φ on $(0,\infty)$ is HCM of order k if for every $u > 0$ the function $w \mapsto \varphi(uv)\varphi(u/v)$,

$w = v + 1/v \geq 2$, is k monotone on $[2,\infty)$; HCM means HCM of every order (Subsection 4.1.2).

The finite order versions are introduced here. They are not Urbanik's nested classes, and other finite order notions in the literature (Williamson's k monotone functions, Bondesson's classes of finite order for densities) use other conventions. By the Leibniz identity $(-1)^j(fg)^{(j)} = \sum_i \binom{j}{i}[(-1)^i f^{(i)}][(-1)^{j-i} g^{(j-i)}]$ a product of two nonnegative k monotone functions is k monotone; as x^{-1} is completely monotone and $\ell(x) = x\ell(x)\cdot x^{-1}$, this gives $\mathrm{GGC}_k \subset \mathrm{BO}_k$ for every k.

THEOREM 14.13.2. *Let T be the clock, with Lévy density $W(x)/x$ and Laplace transform $\varphi_{1/2}(\lambda) = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{\lambda}) = 1/G(\lambda)$, and let $H = 3\times 10^{12}$. Statements (i) and (ii) are unconditional: they use the verification of RH to height H [247].*

(i) $T \in \mathrm{GGC}_k$, and hence $T \in \mathrm{BO}_k$, for every $k \leq 1.16\times 10^{25}$.

(ii) $\varphi_{1/2}$ is HCM of order k for every $k \leq 9\times 10^{12}$.

(iii) The following are equivalent: RH; T is a GGC; $T \in \mathrm{BO}$; $\varphi_{1/2}$ is HCM; $\varphi_{1/2}$ is HCM of every order.

Proof. (i) By Theorem 5.1.1 the clock T exists and is infinitely divisible with Lévy density $\ell(x) = W(x)/x$, and Theorem 5.1.13 gives $(-1)^j W^{(j)}(x) > 0$ for every $x > 0$ and every integer $0 \leq j \leq 1.29(H-23)^2$, which exceeds 1.16×10^{25}. Since $x\ell(x) = W(x)$, this says $T \in \mathrm{GGC}_k$ for every $k \leq 1.16\times 10^{25}$, and the Leibniz step above gives $T \in \mathrm{BO}_k$.

(ii) Zeros are counted with multiplicity, and $\rho = \frac{1}{2}+b+i\gamma$ with $\gamma > 0$ runs over the zeros of positive ordinate. Put $a_\rho = -(\rho-\frac{1}{2})^2 = \gamma^2 - b^2 - 2ib\gamma$, as in Section 1.2, so that $\operatorname{Re} a_\rho > 0$ and $\sum_\rho |a_\rho|^{-1} < \infty$, and by the Hadamard product (1.2.1), $\varphi_{1/2}(\lambda) = \prod_\rho (1+\lambda/a_\rho)^{-1}$, absolutely convergent. Since $(1+uv/a)(1+u/(va)) = 1 + uw/a + u^2/a^2$ with $w = v + 1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v) = \prod_{b=0,\ \gamma>0} \Big(1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big)^{-1} \times \prod_{b>0,\ \gamma>0} \Big|1 + \frac{uw}{a_\rho} + \frac{u^2}{a_\rho^2}\Big|^{-2}, \qquad w = v + \frac{1}{v},$$

where in the second product ρ stands for the pair $\{\rho, 1-\bar\rho\}$ of zeros with $\gamma > 0$ and $b \neq 0$, whose values a_ρ and $a_{1-\bar\rho} = \bar a_\rho$ are conjugate, so that their two factors multiply to a squared modulus. Put $c_\rho(u) = a_\rho/u + u/a_\rho$, so that $1 + uw/a + u^2/a^2 = (u/a)(w + c)$. For a zero on the line $a_\rho = \gamma^2$ and $c_\rho(u) \geq 2$ is real, and the factor $(a_\rho/u)/(w + c_\rho(u))$ is completely monotone in w on $[2,\infty)$. For a pair the factor is $(|a_\rho|^2/u^2)/((w+c)(w+\bar c))$ with $c = c_\rho(u)$ complex. Write $a_\rho = re^{-i\phi}$ with $0 < \phi < \pi/2$. Then $\tan\phi = 2b\gamma/(\gamma^2 - b^2) \leq \gamma/(\gamma^2 - \frac{1}{4}) < 1.01/\gamma$

for $\gamma \geq 10$, because $0 < b < \frac{1}{2}$. Moreover $\operatorname{Re} c = (r/u + u/r)\cos\phi$ and $|\operatorname{Im} c| = |r/u - u/r|\sin\phi \leq (r/u + u/r)\sin\phi$, so for every $w \geq 0$ the angle $\vartheta = |\arg(w+c)|$ satisfies $\tan\vartheta = |\operatorname{Im} c|/(w + \operatorname{Re} c) \leq \tan\phi$, that is $\vartheta \leq \phi$. Partial fractions give $1/((w+c)(w+\bar{c})) = -\operatorname{Im}[(w+c)^{-1}]/\operatorname{Im} c$, whence

$$(-1)^j \frac{d^j}{dw^j} \frac{1}{(w+c)(w+\bar{c})} = \frac{j!\,\sin((j+1)\vartheta)}{|\operatorname{Im} c|\,|w+c|^{j+1}},$$

which is positive as long as $(j+1)\vartheta < \pi$, hence for all $j < \pi/\phi - 1$ (when $\operatorname{Im} c = 0$, that is $u = r$, the factor is $(w+c)^{-2}$, whose signed derivatives $(j+1)!\,(w+c)^{-j-2}$ are positive). By the verification of RH to height H [247] every zero with $\gamma \leq H$ is on the line, so every pair has $\gamma > H$ and $\phi < 1.01/H$, and its factor is k monotone for every $k \leq \pi H/1.01 - 1$, which is 9.33×10^{12} and exceeds 9×10^{12}. Products of nonnegative k monotone functions are k monotone by the Leibniz identity, and the partial products converge, with all their w derivatives, locally uniformly on $\operatorname{Re} w > 0$: the factors have no zeros there, since the poles $w = -c$ lie in $\operatorname{Re} w < 0$, and $\sum_\rho |uw/a_\rho + u^2/a_\rho^2| < \infty$ locally uniformly. So the limit is k monotone.

(iii) RH holds if and only if T is a GGC by Proposition 1.2.1, and if and only if $\varphi_{1/2}$ is HCM by Corollary 4.1.3(ii), that is by Bondesson's theorem [39, Theorem 6.1.1] (a function φ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM; see also [129]). A GGC is in BO by the Leibniz step. Conversely suppose that $\ell(x) = W(x)/x$ is completely monotone, $\ell(x) = \int_{[0,\infty)} e^{-xy} m(dy)$ with $m \geq 0$. Taking Laplace transforms of $W(x) = x\ell(x)$ for $\operatorname{Re}\lambda > 0$,

$$\int_0^\infty e^{-\lambda x} W(x)\,dx = \int_{[0,\infty)} \frac{m(dy)}{(\lambda+y)^2} = \sum_\rho \frac{1}{\lambda + a_\rho} = \frac{G'}{G}(\lambda).$$

The middle expression is analytic on $\mathbb{C} \setminus (-\infty, 0]$, while G'/G has a pole at every $-a_\rho$, and $-a_\rho \notin (-\infty, 0]$ exactly when $a_\rho \notin (0, \infty)$, that is when ρ is off the line. So BO forces RH. □

For a model pair factor with a zero at $\beta = 0.6$, $\gamma = 100$, the first order at which the closed form fails is 1571 at $u = 1$, $w = 2$, and 1570 as $u \to 0$, against the bound $\pi\gamma/(2(\beta - \frac{1}{2})) = 1570.8$.

14.14 Is there a new path to the Riemann Hypothesis?

The honest answer is no.

14.14.1 Three reformulations and what each costs

The derivative ladder: RH holds if and only if W is completely monotone (Theorem 14.6.1(e)), and order k and height H trade quadratically (Subsection 5.1.6), so climbing the ladder to infinity is verifying RH to infinite height; the ladder

is a reformulation, not a path. The basepoint family: the basepoint clock T_α (Section 14.8) is GGC for every $\alpha > \frac{1}{2}$ if and only if RH (Theorem 10.2.7), and this is Lagarias's criterion [160] (Remark 1.2.4), so the Thorin route at basepoints and the Lagarias route are one route. The finite maxima: Corollary 14.12.2(c) reduces RH to zero freeness of the symmetrised stages ξ_N^{sym} off the line up to heights $h_j \to \infty$, for infinitely many N, a hypothesis that is not weaker than RH (whether RH implies it is open). Unsymmetrised sections of ζ are known to have zeros off the line and even to the right of $\operatorname{Re} s = 1$ [325, 204], which is why the symmetrisation is essential; for the symmetrised sections there are results of Spira and of Gonek and Montgomery [309, 96] (the paragraph after Table 7.2); we know of no result of this kind for the symmetrised stages of the maximum.

14.14.2 The arithmetic form below $\alpha = 1$, and how far it sees

The sine squared prime form of Section 1.2 (the paragraph after (1.2.9), with Table 1.5) writes $v_\alpha(\theta) = \operatorname{Re} \xi'/\xi(\alpha + i\theta)$ for $\alpha > 1$ through the signed canonical measure with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density: the prime and gamma terms are nonnegative and the pole terms negative, so the positivity of v_α for $\alpha > 1$, which is Lagarias positivity there, comes from the absence of zeros with $\beta \geq 1$ and not from the terms one by one. Theorem 10.2.7 already continues it below $\alpha = 1$ by compensation: its last condition, that the compensated sine squared sum converges for every $\alpha' > \alpha$, is equivalent to zero freeness above α. A smoothed form of the continuation, with explicit pole terms, is the following.

Proposition 14.14.1. *Let $\alpha = \frac{1}{2} + \delta$ with $\delta > 0$, $t \in \mathbb{R}$ and $X > 0$. Put $g_X(x) = e^{-\delta|x|}\cos(tx)\,(1 - |x|/X)_+$ and $h_X(r) = \int_{\mathbb{R}} g_X(x)e^{irx}dx$, so that $h_X \geq 0$ on $\mathbb{R}$: it is $(2\pi)^{-1}$ times the convolution of the two sided Poisson kernel $h_\infty(r) = \delta/(\delta^2 + (t-r)^2) + \delta/(\delta^2 + (t+r)^2)$ with the Fejér kernel $4\sin^2(Xr/2)/(Xr^2)$. Define*

$$A_X(\delta,t) = \frac{1}{2}\Big[h_X(\tfrac{i}{2}) + h_X(-\tfrac{i}{2}) - \log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h_X(r)\,\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\,dr - 2\sum_{n<e^X}\frac{\Lambda(n)}{n^\alpha}\cos(t\log n)\Big(1 - \frac{\log n}{X}\Big)\Big],$$

where ψ is the digamma function and $h_X(\frac{i}{2}) + h_X(-\frac{i}{2}) = 2\operatorname{Re}[J(-\delta + \frac{1}{2} + it) + J(-\delta - \frac{1}{2} + it)]$ with $J(c) = -1/c + (e^{cX} - 1)/(Xc^2)$. Then:

(a) $A_X(\delta,t) = \frac{1}{2}\sum_\rho h_X(\gamma_\rho)$, with $\gamma_\rho = (\rho - \frac{1}{2})/i$ and the sum over all nontrivial zeros, unconditionally.

(b) Under RH, $A_X(\delta,t) \geq 0$ for every X, δ and t. In particular a single triple (δ,t,X) with $A_X(\delta,t) < 0$ would disprove RH.

(c) If every zero has $|\beta - \frac12| < \delta$ (under RH, or unconditionally when $\delta \ge \frac12$), then $A_X(\delta, t) \to \operatorname{Re} \xi'/\xi(\alpha + it)$ as $X \to \infty$.

Proof. (a) This is Weil's explicit formula $\sum_\rho h(\gamma_\rho) = h(\frac{i}{2}) + h(-\frac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\operatorname{Re}\psi(\frac14 + \frac{ir}{2})\,dr - 2\sum_n \Lambda(n)n^{-1/2}g(\log n)$, in the form of [128], which evaluates the distribution D of Proposition 7.2.13 on a test function g with Fourier transform h, applied to the even test function g_X, which is continuous, compactly supported and piecewise smooth, with Fourier transform h_X, so that $g_X(0) = 1$ and $n^{-1/2}g_X(\log n) = n^{-\alpha}\cos(t\log n)(1 - \log n/X)_+$; the sum over zeros converges absolutely because $h_X(r) = O(r^{-2})$ uniformly in $|\operatorname{Im} r| \le \frac12$. Here $J(c) = \int_0^X e^{cx}(1 - x/X)\,dx$, and $h_X(r) = 2\int_0^X e^{-\delta x}\cos(tx)\cos(rx)(1 - x/X)\,dx = \frac12\sum_{s_1,s_2=\pm1} J(-\delta + i(s_1 t + s_2 r))$. At $r = \pm\frac{i}{2}$ two of the four terms are the complex conjugates of the other two, so $h_X(\pm\frac{i}{2}) = \operatorname{Re}[J(-\delta + \frac12 + it) + J(-\delta - \frac12 + it)]$ and the two pole terms add up to twice that, which is the stated value.

(b) Under RH every γ_ρ is real and $h_X \ge 0$ on $\mathbb{R}$, so every term of (a) is nonnegative.

(c) For a zero $\rho = \frac12 + b + i\gamma$ the four exponentials of $h_X(\gamma_\rho) = h_X(\gamma - ib)$ have $\operatorname{Re} c = -\delta \pm b$, which is negative when $|b| < \delta$. Then $J(c) \to -1/c$ as $X \to \infty$, and $|J(c) + 1/c| \le 2/(X|c|^2)$, so $h_X(\gamma_\rho) \to h_\infty(\gamma_\rho) = \delta/(\delta^2 + (t - \gamma_\rho)^2) + \delta/(\delta^2 + (t + \gamma_\rho)^2)$, with $|h_X(\gamma_\rho)| \le C(1 + |\gamma_\rho|)^{-2}$ uniformly in $X \ge 1$, where C depends on δ, t and the finitely many zeros with $||\gamma_\rho| - t| \le 1$, because the four terms $-1/c$ combine in pairs to $O(\gamma^{-2})$ and $|c| \ge 1$ for the other zeros. By dominated convergence $A_X \to \frac12\sum_\rho h_\infty(\gamma_\rho)$. Finally $h_\infty(\gamma_\rho)$ is the sum of the four terms $\frac12/(\alpha \pm it - \rho)$ and $\frac12/(\rho - (1 - \alpha) \pm it)$, and since the zeros are symmetric under $\rho \mapsto 1 - \rho$ and $\rho \mapsto \bar\rho$, $\frac12\sum_\rho h_\infty(\gamma_\rho) = \operatorname{Re}\sum_\rho(\alpha + it - \rho)^{-1} = \operatorname{Re}\,\xi'/\xi(\alpha + it)$ by the symmetric Hadamard product. □

Remark 14.14.2. The hypothesis of (c) is needed. If a zero has $b = \beta - \frac12 > \delta$, the terms of $h_X(\gamma - ib)$ with $\operatorname{Re} c = b - \delta > 0$ contain $e^{(b-\delta)X}/(Xc^2)$, which grows with X, and $h_\infty(\gamma - ib)$ is only the analytic continuation of a divergent integral. At $t = \gamma$ the four zeros $\frac12 \pm b \pm i\gamma$ contribute $e^{(b-\delta)X}/(X(b-\delta)^2)\,(1 + O((b-\delta)^2/\gamma^2)) + O(1)$ to $A_X(\delta, \gamma)$, which is positive and unbounded in X. No converse is claimed: whether an off line zero forces $A_X(\delta, t) < 0$ somewhere for large X is not shown here.

The proposition is Lagarias's criterion restricted to a family, and it is also a form of Weil's positivity criterion (compare Proposition 7.2.13) for the positive definite test functions g_X, which have corners. Table 14.6 gives the minimum over $t \in [0, 60]$, positive at every X and every α tested.

The positivity in the table is not evidence of anything new, and it is important to say why. The contribution of zeros above height H to A_X is at most about $(e^{X/2}/X + \delta)\log H/H$, because $|h_X(\gamma_\rho)| \lesssim e^{|\operatorname{Im}\gamma_\rho| X}/(X\gamma^2)$ and $|\operatorname{Im}\gamma_\rho| \le \frac12$. With $H = 3 \times 10^{12}$ and $X = 13$ that is below 10^{-8}, about 4.9×10^{-10}, far under the minima in the table. So the verified zeros force $A_X \ge 0$ for t in the range of the

X	$\alpha = 0.75$	$\alpha = 0.6$	$\alpha = 0.55$
2	0.03275	0.02495	0.02230
4	0.02208	0.01432	0.01160
8	0.01707	0.00954	0.00681
13	0.01510	0.00812	0.00576

Table 14.6: $\min_{0\le t\le 60} A_X(\alpha - \frac{1}{2}, t)$ from primes below e^X (up to 4.4×10^5 at $X = 13$). The values decrease towards $\xi'/\xi(\alpha) = 0.011551$, 0.0046209, 0.0023105 for the three α.

table as long as this bound stays below the minima, that is up to X of about 46 to 50, where it reaches the values from 0.0058 down to the limit 0.0023; the bound itself stays small up to about $2\log H \approx 57$. The arithmetic test at length e^X sees zeros up to a height of order $e^{X/2}$, the usual duality between primes and zeros, and to see past the present verification one would need primes up to about 10^{25}. As a computation the arithmetic form is strictly weaker than zero verification. As a target for proof it is sharper than before: RH needs $A_X \ge 0$ uniformly in X (whether this suffices is open, Remark 14.14.2), with the prime sum, the gamma integral and the pole terms each explicit, and the whole difficulty is that the pole terms, of size $e^{(1/2-\delta)X}/X$, are cancelled by the prime sum only through the prime number theorem with an error term, which is to say through the zeros.

14.15 Open problems

Problem 14.1. Prove the hypothesis of the F_N^{sym} form of Corollary 14.12.2(c): find heights $h_N \to \infty$ such that for infinitely many N the symmetric stage F_N^{sym} has no zeros with $\frac{1}{2} < \operatorname{Re} s < 1$ and $|\operatorname{Im} s| < h_N$, for instance $h_N = c\sqrt{N}$ with some $c > 0$. Subsection 14.12.3 finds, for the N tested, that every zero of the stages with $-1 \le \operatorname{Re} s \le 2$ and $0 < \operatorname{Im} s \le T^*(N)$ lies on the line, with $T^*(N)$ of order N for $N \le 96$, far more than needed (at $N = 128$ a pair outside the strip comes down to height 60.7, while the strip has none below 130), and the sharper conjecture is $T^*(N) \ge \kappa N$. Either statement is about explicit functions built from the Dirichlet polynomials D_N with integer coefficients, and either implies RH. A mechanism for $T^*(N)$, presumably through the ratio $|D_N(\sigma+it)/D_N(1-\sigma+it)|$, is the first thing to find. The corresponding statement for the symmetrised lattice truncations K_n is Conjecture S of Subsection 7.2.1 (Theorem 7.2.7).

Problem 14.2. Theorem 10.2.7 continues the sine squared prime form below $\alpha = 1$ by compensation, and shows that the compensated sum converges for every $\alpha' > \alpha$ exactly when ζ has no zero with $\operatorname{Re} s > \alpha$; Proposition 14.14.1 gives a smoothed form of the continuation with explicit pole terms, which is not positive term by term. Is there a continuation of the prime form below $\alpha = 1$ that is positive term by term? The prime sum diverges there, but smoothed versions

of it, with $n^{-\alpha}$ replaced by a weight that cuts off at $n \approx e^{1/\varepsilon}$, remain positive termwise, and the question is whether the loss from the cutoff can be controlled by $(\xi'/\xi)(\alpha)$, the gamma sum and the pole terms. A positive answer on $\alpha \in (\alpha_0, 1]$ would prove the zero free region $\beta \le \alpha_0$.

Problem 14.3. For $0 < t < 0.2$, does the law T_t of Section 14.7, half of the flow clock of Section 8.4, with transform $R_t(\lambda/2)$, exist as a probability law, and is it infinitely divisible? Unconditionally it is known to be a GGC only for $t \ge 0.2$, and to exist only for $t \ge 0.2$ and at $t = 0$, where $T_0 = T/2$ (Theorem 5.1.1); under RH it exists and is a GGC for every $t \ge 0$. This is the analogue along the de Bruijn Newman flow of the existence of T (Theorem 5.1.1); it would need an explicit formula for the heat trace of the flow, which the Gaussian tilt destroys.

Remark 14.15.1. No published result gives an explicit height below which the zeros of H_t are real for $0 < t < 0.2$: [282] and [247] give $\Lambda \le 0.2$ through a barrier at $|\operatorname{Im} z| \ge y_0 > 0$, and [150] show that all but finitely many zeros are real, without a height. Real zeros do not collide forward in time (a multiple real zero splits into real zeros, as in Lemma 8.4.23, after the statement of [73] cited in Section 8.4), so the verification of RH to 3×10^{12} passes to H_t except for zeros entering from above, which a box barrier touching the real axis would exclude. That route has not been carried out. The forward statement is Lemma 8.4.23, and the barrier hypothesis and its consequence are in Proposition 8.4.24. With a certified height H', an explicit count of the zeros of H_t and the low zeros of H_t located, the certificate of Section 5.1.8 would give about $0.76H'$ rungs, and rungs 0 and 1 (infinite divisibility and self decomposability of T_t for every $t \in [0, 0.2]$) would need only an explicit zero density bound for H_t and a small computation.

Problem 14.4. Is there an analogue of Theorem 14.3.1 at $\sigma = \frac{1}{2}$, that is, a factorisation of the tilted law of T or of the Pólya variable through a law built from the primes, with the explicit formula replacing the Euler product?

14.16 Numerical checks

The numerical results of this chapter are recomputed by the programs `maxwald_*.py` of the repository (`code/README.md`). The zero counts of Table 14.5 are the most delicate, and the other checks use multiprecision arithmetic, except the grids of Figure 14.2, the grid values and the gamma factors of the finite stages, and the sums and integrals of Table 14.6, which are in double precision; those of Table 14.6 are checked against the sum over the zeros, to 5×10^{-7}. Table 14.7 lists the checks of the results kept in this chapter.

result	test points	agreement
kernel identity of Proposition 14.2.1	six values of u	24 digits
$\mathbb{E}M_4^{s/2} = 2(1-2^{1-s})\xi(s)/(s-1)$	$s = 0.5,\ 2,\ 3+2i$	8 to 30 digits
Theorem 14.3.1, Mellin form	$\sigma = 1.5,\ 2,\ 3.5,\ t = 1,\ 5,\ 20$	30 digits
low coefficients of $\mathcal{P}_N$ equal those of ϑ_4	$N = 4, 8, 16$	exact
zero of ξ_{64} near ρ_1	$N = 64$	$\lvert \operatorname{Re} - \frac{1}{2}\rvert = 5.2 \times 10^{-7}$
zero counts of F_N^{sym} in $[-1,2] \times [0,T]$ against sign changes on the line (Table 14.5)	$N = 8,\ 16,\ 32,\ 48,\ 64,\ 96,\ 128$	windows of height 10
unit mass of the term of one zero, Proposition 14.8.1(b)	$\alpha = 0.55$	10 digits
width at half height of the term of ρ_1, Proposition 14.8.1	$\alpha = 0.55$	6 digits, up to the factor $1 + \delta^2/(2\gamma_1^2)$
γ_1^2 by Laplace inversion, Proposition 14.10.1	$t \in [0.04, 0.08]$	199.790

Table 14.7: Numerical verification of results of this chapter that are not covered by the book's tables.

Chapter 15

The Thorin phase diagram

This chapter sets out the most promising computational path to RH that the book has found, and says exactly where it stops.

The sine squared lemma and the Thorin side of Chapter 3 offer no computational alternative to the de Bruijn Newman flow, only the right language for it. The Thorin side is the object to compute, prime free and well defined at every basepoint, and the flow is the only known source of an analytic tail below the basepoint $\frac{7}{8}$. What separates the theorems $\Lambda \leq 0.22$ of Polymath15 [282] and $\Lambda \leq 0.2$ of Platt and Trudgian [247] from RH is not anything a computed density shows; it is the existence of the analytic tail at $t = 0.2$ and its absence at $t = 0$. Sections 15.2, 15.5 and the two experiments of Section 15.6 were first written as a working report by the authors [264], which this chapter supersedes. We use the normalisation of Section 8.4: $H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cos(zu)\, du$, $\xi_t(s) = 8H_t(-2i(s - \frac{1}{2}))$, a zero $x + iy$ of H_t is the zero $\frac{1}{2} - \frac{y}{2} + \frac{ix}{2}$ of ξ_t, and $P_a(x) = a/(\pi(a^2 + x^2))$ is the Cauchy density of scale $a > 0$.

15.1 Which computations see past the verified height

Every route to RH in this book replaces the hypothesis by an infinite family of statements, each of which a finite computation can certify. The question that separates the routes is what happens above the height the computation reaches. If nothing does, the route is finite information in the sense of Section 5.2: by Theorem 5.2.1 a perturbation with the same zeros below that height and zeros off the line above it passes the same test, so the test certifies no more than a verification of the zeros up to a height determined by its size.

Definition 15.1.1. A route *reaches every height* if there are a finite computation C and an analytic statement A valid above a barrier X such that C and A together imply a statement about all zeros of ξ, or of a fixed approximant or deformation of ξ, that is not implied by the zeros below X alone.

15.1.1 The finite information routes

Most of the book's tests are finite information, and the book measures exactly how much height each one buys. The heat trace has its derivative signs to order 1.16×10^{25} from the verification to 3×10^{12} (Theorem 5.1.13), and the order is a height in disguise: an off line zero at height γ first breaks the alternation at an order of about $t\gamma^2$, with t the parameter of the heat trace and not the time of the flow. The Bosch function $\varphi(x^p)$ of Section 4.3 is HCM for $|p| \leq 1 - 1/(\pi T_0)$, and $1 - p \approx 1/(\pi T)$ is again a height (Corollary 4.3.7). Li coefficients see an off line zero at height γ and distance d from the line only from an order of about γ^2/d, while the Hankel test of Section 8.4.3 often registers a pair at an order close to $2k + 2$, set by its index k, at a cost in precision that grows linearly with the order (Section 11.5). The Cauchy flow ρ_ε is positive on the verified window for every ε, but its minimum is attained at the origin and tends to zero with ε (Theorem 9.1.18 and Remark 9.1.19), so positivity is never won by a fixed margin, and positivity on one line is not known to exclude zeros to its right (Remark 1.2.4). The ferromagnetic distance of Section 13.4 is below 1.932×10^{-6}, and it is zero exactly under RH, but the approximating laws keep only the first K zeros, so it sees nothing above them. The Weil Gram matrices and the arithmetic forms of Section 15.6.1 below have the same property, with a resolution in height set by the support of the test functions.

15.1.2 The three routes that reach every height

Three routes in the book reach every height in the sense of Definition 15.1.1.

The Euler product: basepoints $\alpha \geq 1$. At the basepoint one, the Thorin density of the shifted clock is positive at every height, and the zeros below 2000 in ball arithmetic and explicit bounds for $S(T)$ [109, 11] certify that it is smallest only at the origin (Propositions 9.1.11 and 9.1.12). Above the barrier, the analytic input is the explicit bound for $S(T)$, and positivity itself comes from the Euler product, through the absence of zeros in $\operatorname{Re} s \geq 1$. Below $\alpha = 1$ the same structure gives a statement at every height only together with a zero free half plane (Proposition 9.1.13); the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [226] would extend the route to $\alpha \geq \frac{7}{8}$ (Section 12.1.2), and below $\frac{7}{8}$ nothing is known.

The functional equation: symmetrised truncations. The symmetrised truncations of the Biane, Pitman and Yor lattice of Section 7.2 have all their zeros in the strip on the line, at every height, for $n \leq 24$ and $n = 32, 36$, proved above explicit thresholds and computed below them (Theorem 7.2.7 and Corollaries 7.2.9 and 7.2.11). Above the threshold the analytic input is the symmetry of the functional equation. RH would follow from infinitely many n (Problem 7.1), and no argument uniform in n is known.

The heat flow: positive time. At time $t > 0$ and for a suitable $y_0 > 0$, the function H_t has no zeros with $\operatorname{Im} z \geq y_0$ above a barrier, by an analytic argument, the effective Riemann Siegel approximation of Polymath15 [282]; a computation of winding numbers keeps zeros from entering through a strip of width one at the barrier for the times up to t, and the verification of RH covers the zeros below it. Together they give an upper bound for Λ, since then $\Lambda \leq t + y_0^2/2$ by de Bruijn's strip theorem [73, Theorem 13] (Polymath15 take $t = y_0 = 0.2$): $\Lambda \leq 0.22$ [282] and, from the verification to 3×10^{12}, $\Lambda \leq 0.2$ [247].

route	analytic input above the barrier	what is certified at every height	what is missing
basepoint $\alpha \geq 1$	Euler product, explicit bounds for $S(T)$	the shifted clock is a GGC, its density minimal at the origin	any basepoint below $\frac{7}{8}$
symmetrised truncations	functional equation symmetry	zeros of K_n in the strip on the line, $n \leq 24$ and $n = 32, 36$ (floating point counts below the thresholds)	infinitely many n
heat flow	Riemann Siegel at $t > 0$	$\Lambda \leq 0.2$; GGC at points (t, α)	a tail at $t = 0$

Table 15.1: The three routes of the book that reach every height. All other routes are finite information.

The first and the third routes are the two coordinates of the phase diagram of Section 15.3. The second route has no coordinate on the diagram and stands apart; Problem 15.3 asks whether it can be given one.

15.2 Clocks along the flow at every basepoint

This section extends Theorem 1.2.2, which decides when the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$ is a generalised gamma convolution, and Theorem 7.2.15 with Corollary 4.2.5, which decide it for the centre clock R_t of the flow, to every time $t \geq 0$ and every basepoint $\alpha \geq \frac{1}{2}$.

Lemma 15.2.1 (Structure of ξ_t). *For every $t \geq 0$: (i) ξ_t is entire of order at most one, real on the real axis, and $\xi_t(1 - s) = \xi_t(s)$; (ii) $\xi_t(\frac{1}{2} + x) > 0$ for real x; (iii) with $w = s - \frac{1}{2}$ and the zeros written $\rho = \frac{1}{2} + \tilde{\rho}$,*

$$\xi_t(\tfrac{1}{2} + w) = \xi_t(\tfrac{1}{2}) \prod \Big(1 - \frac{w^2}{\tilde{\rho}^2}\Big),$$

the product running over one representative of each pair $\{\tilde{\rho}, -\tilde{\rho}\}$, with multiplicity, and $\sum_\rho |\tilde{\rho}|^{-2} < \infty$.

Proof. Since $\Phi_{\mathrm{dBN}}(u) \le Ce^{9u-\pi e^{4u}}$ for $u \ge 0$ and $|\cos(zu)| \le e^{|z|u}$,

$$|H_t(z)| \le C\int_0^\infty e^{tu^2+|z|u+9u-\pi e^{4u}}\,du \le \exp(C'|z|\log|z|)$$

for large $|z|$; so H_t and ξ_t are entire of order at most one. Evenness of H_t gives the reflection symmetry, and reality of the integrand on the imaginary axis gives ξ_t real on the real axis, with

$$\xi_t(\tfrac12+x) = 8\int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(2xu)\,du > 0,$$

since $\Phi_{\mathrm{dBN}} > 0$. Hadamard's theorem for order at most one gives $H_t(z) = e^{a+bz}\prod_k E_1(z/z_k)$ over the zeros, none of which is zero since $H_t(0) > 0$. Pairing z_k with $-z_k$, a zero by evenness, gives $E_1(z/z_k)E_1(-z/z_k) = 1 - z^2/z_k^2$, and $\sum |z_k|^{-2} < \infty$ since the convergence exponent is at most one. Evenness forces $b = 0$, and the substitution $z = -2iw$, $\tilde\rho = iz_k/2$ gives (iii). □

Lemma 15.2.2 (One conjugate pair). *Let $d, \gamma \in \mathbb{R}$, not both zero, and $\varphi(u) = (d^2+\gamma^2)/((\sqrt{u}+d)^2+\gamma^2)$ for $u \ge 0$.*

(a) If $d > 0$, then φ is the Laplace transform of a GGC whose Thorin measure, in the chart $\theta = \sqrt{z}$, is $[P_d(\theta-\gamma) + P_d(\theta+\gamma)]\,d\theta$ on $\theta > 0$ (compare Proposition 1.2.3):

$$-\log\varphi(u) = \int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big[P_d(\theta-\gamma)+P_d(\theta+\gamma)\big]\,d\theta.$$

(b) If $d = 0$, then $\varphi(u) = (1+u/\gamma^2)^{-1}$, an exponential law, with Thorin measure δ_{γ^2} in the variable z, that is, δ_γ in the chart $\theta = \sqrt{z}$.

(c) If $d < 0$, then φ has a pole at $u_0 = (-d+i\gamma)^2 \in \mathbb{C}\setminus(-\infty,0]$ and is not the Laplace transform of a GGC.

Proof. (a) Both sides vanish at $u = 0$, so it suffices to compare derivatives in $u > 0$. On the left, with $F(u) = (\sqrt{u}+d)^2+\gamma^2$, $F'/F = (\sqrt{u}+d)/(\sqrt{u}\,F)$. On the right, differentiating under the integral and using the symmetry $\theta \mapsto -\theta$ of the second kernel,

$$\begin{aligned}\int_{-\infty}^\infty \frac{P_d(\theta-\gamma)}{\theta^2+u}\,d\theta &= \frac{\pi}{\sqrt{u}}\int_{-\infty}^\infty P_d(\theta-\gamma)P_{\sqrt{u}}(\theta)\,d\theta\\ &= \frac{\pi}{\sqrt{u}}P_{d+\sqrt{u}}(\gamma) = \frac{d+\sqrt{u}}{\sqrt{u}\,((d+\sqrt{u})^2+\gamma^2)},\end{aligned}$$

by the semigroup property $P_a * P_b = P_{a+b}$. The integral converges, since $\log(1+u/\theta^2)$ is integrable at 0 and $O(\theta^{-2})$ at infinity, and the measure is positive since

$d > 0$. (b) is immediate. (c) The equation $\sqrt{u} + d - i\gamma = 0$ has the solution $\sqrt{u} = -d + i\gamma$, whose real part is positive, so it is the principal root and u_0 lies off $(-\infty, 0]$ (if $\gamma = 0$ then $u_0 = d^2 > 0$); the Laplace transform of a GGC is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$ [39]. □

THEOREM 15.2.3 (Threshold along the flow). *Let $t \geq 0$ and $\alpha \geq \frac{1}{2}$. The shifted time t clock*

$$\varphi_{t,\alpha}(u) = \frac{\xi_t(\alpha)}{\xi_t(\alpha + \sqrt{u})}$$

is the Laplace transform of a GGC if and only if $\alpha \geq \sup_\rho \operatorname{Re}\rho$ over the zeros ρ of ξ_t. In that case its Thorin measure, in the chart $\theta = \sqrt{z}$, is

$$\sum_{\substack{\rho=\beta+i\gamma \\ \gamma>0,\ \beta<\alpha}} \left[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\right] d\theta \;+\; \sum_{\substack{\rho=\alpha+i\gamma \\ \gamma>0}} m_\rho\, \delta_\gamma,$$

where the first sum counts each zero with its multiplicity and m_ρ is the multiplicity of ρ; this is the Poisson form of Proposition 1.2.3 for ξ_t.

Proof. By Lemma 15.2.1, with $a = \alpha - \frac{1}{2} \geq 0$,

$$\varphi_{t,\alpha}(u) = \prod \frac{1 - a^2/\tilde{\rho}^2}{1 - (a + \sqrt{u})^2/\tilde{\rho}^2} = \prod_\rho \Big(1 - \frac{\sqrt{u}}{\rho - \alpha}\Big)^{-1},$$

where the last product runs over all zeros and is grouped so that each finite partial product contains, with every zero, its partner under $\tilde{\rho} \mapsto -\tilde{\rho}$; the denominators do not vanish since $\xi_t(\alpha) > 0$. Because ξ_t is real on the real axis, the zeros are also closed under conjugation, and regrouping into conjugate pairs is legitimate: both groupings give absolutely convergent products, the conjugate pairing because $(1-\sqrt{u}/(\rho-\alpha))(1-\sqrt{u}/(\bar{\rho}-\alpha)) = 1+(2d\sqrt{u}+u)/(d^2+\gamma^2)$ with $d = \alpha-\beta$ bounded (the zeros of H_t lie in $|\operatorname{Im} z| \leq 1$ by de Bruijn's strip theorem [73, Theorem 13], as in Section 8.4.1; and $0 \leq d \leq 2\alpha - 1$ directly when $\alpha \geq \sup \operatorname{Re}\rho$, the only case in which the pairing is used, by the symmetry $\rho \mapsto 1 - \rho$) and $\sum |\rho - \alpha|^{-2} < \infty$, and the partial products over $|\gamma| \leq T$ agree. Hence

$$\varphi_{t,\alpha}(u) = \prod_{\{\rho,\bar{\rho}\}} \frac{d^2 + \gamma^2}{(\sqrt{u} + d)^2 + \gamma^2},$$

a product of the functions of Lemma 15.2.2, one for each conjugate pair, every pair being genuine because ξ_t has no real zeros by Lemma 15.2.1(ii).

If $\alpha \geq \sup \operatorname{Re}\rho$, every factor is a GGC transform by Lemma 15.2.2(a) and (b). Finite products of GGC transforms are GGC transforms with Thorin measures adding, the partial products converge pointwise to $\varphi_{t,\alpha}$, which is continuous with $\varphi_{t,\alpha}(0) = 1$, so by the continuity theorem and the closure of the class under

weak limits ([39, Theorem 3.1.5], Section 4.1.1) the limit is a GGC transform. Its Thorin measure is the full sum μ displayed in the theorem, and there is no drift, by the following monotone convergence argument. Write μ_T for the sum of the kernels and atoms over the zeros with $0 < \gamma \le T$, so that $\mu_T \uparrow \mu$ as $T \to \infty$, and φ_T for the partial product over the same zeros. By Lemma 15.2.2, $-\log \varphi_T(u) = \int \log(1 + u/\theta^2)\, \mu_T(d\theta)$ for $u \ge 0$, and the integrand is nonnegative, so by monotone convergence $-\log \varphi_T(u)$ increases to $\int \log(1 + u/\theta^2)\, \mu(d\theta)$. Since $\varphi_T(u) \to \varphi_{t,\alpha}(u) > 0$, this gives $-\log \varphi_{t,\alpha}(u) = \int \log(1 + u/\theta^2)\, \mu(d\theta) < \infty$ for every $u \ge 0$. If some zero has $\beta > \alpha$, then $\xi_t(\alpha + \sqrt{u_0}) = 0$ at $u_0 = (\rho - \alpha)^2$, whose principal root is $\rho - \alpha$ because $\operatorname{Re}(\rho - \alpha) > 0$, so $\varphi_{t,\alpha}$ has a pole in $\mathbb{C} \setminus (-\infty, 0]$ and is not a GGC transform. □

Remark 15.2.4. The proof uses nothing about ξ_t beyond Lemma 15.2.1, so it applies to any even real entire function of order at most one with a genus zero product in w^2, positive on the real axis, in particular to the perturbation of Theorem 5.2.1, whose shifted clocks are therefore not GGC transforms at any basepoint below the real part of the displaced zeros. The Thorin density of $\varphi_{t,\alpha}$ in the chart θ is $\pi^{-1} \operatorname{Re}(\xi_t'/\xi_t)(\alpha + i\theta)$ away from atoms, which is $\rho_{\alpha - 1/2, t}(\theta)$ of Section 8.4.1. The shifted time t clock is a function of u only: where the threshold condition fails the theorem claims no probability law behind it, and for $\alpha = \frac{1}{2}$ and $0 < t < 0.2$ it is open, without RH, whether R_t is the transform of a probability law at all (Problem 14.3); the basepoint family at $t = 0$ and the flow at $\alpha = \frac{1}{2}$ are the two one parameter families of laws compared in Section 14.7.

15.3 The Thorin phase diagram

Theorem 15.2.3 gives every time t its own threshold basepoint, $\alpha^*(t) = \max(\frac{1}{2}, \sup \operatorname{Re} \rho)$ over the zeros of ξ_t.

Definition 15.3.1. The *GGC region* $\mathcal{G}$ is the set of points (t, α) with $t \ge 0$ and $\alpha \ge \frac{1}{2}$ at which the shifted clock $\varphi_{t,\alpha}$ is the Laplace transform of a GGC, where at $\alpha = \frac{1}{2}$ the clock is R_t of Section 8.4.

By Theorem 15.2.3, $(t, \alpha) \in \mathcal{G}$ exactly when every zero $x + iy$ of H_t has $|y| \le 2(\alpha - \frac{1}{2})$; at $\alpha = \frac{1}{2}$ this is the statement that the zeros are real, which is the GGC property of R_t. The slice $t = 0$ of $\mathcal{G}$ concerns the basepoint family of Section 14.8, a GGC for $\alpha \ge \sup \operatorname{Re} \rho$ and unconditionally for $\alpha \ge 1$, and the slice $\alpha = \frac{1}{2}$ the flow family R_t of Section 8.4, known to be a GGC unconditionally only for $t \ge 0.2$ (Problem 14.3); the diagram joins these two one parameter families. The region is closed upward in α, since the threshold condition is monotone.

Theorem 15.3.2 (de Bruijn's strip theorem in Thorin form). *(i) If $(t_0, \alpha_0) \in \mathcal{G}$, then $(t, \alpha) \in \mathcal{G}$ for every $t \ge t_0$ and every $\alpha \ge \frac{1}{2}$ with*

$$(\alpha - \tfrac{1}{2})^2 + \tfrac{t}{2} \;\ge\; (\alpha_0 - \tfrac{1}{2})^2 + \tfrac{t_0}{2}.$$

(ii) $\Lambda = \min\{\, t + 2(\alpha - \frac{1}{2})^2 : (t,\alpha) \in \mathcal{G} \,\}$.

(iii) RH holds if and only if $(0, \frac{1}{2}) \in \mathcal{G}$*, and if and only if* $(t, \frac{1}{2} + \sqrt{\delta - t}/\sqrt{2}) \in \mathcal{G}$ *for every* $\delta > 0$ *and some* $t \in [0, \delta]$.

Proof. De Bruijn's theorem on strips [73, Theorem 13], in the form used by Polymath15 [282], says that if the zeros of H_{t_0} lie in $|\operatorname{Im} z| \le \Delta$, then the zeros of H_t lie in $(\operatorname{Im} z)^2 \le \max(\Delta^2 - 2(t - t_0), 0)$ for $t \ge t_0$. With $\Delta = 2(\alpha_0 - \frac{1}{2})$ the zeros of H_t then satisfy $|\operatorname{Im} z| \le 2(\alpha - \frac{1}{2})$ whenever $4(\alpha - \frac{1}{2})^2 \ge 4(\alpha_0 - \frac{1}{2})^2 - 2(t - t_0)$, which is (i). For (ii), a point $(t,\alpha) \in \mathcal{G}$ puts the zeros of H_t in $|\operatorname{Im} z| \le 2(\alpha - \frac{1}{2})$, so by (i) those of $H_{t'}$ are real for $t' = t + 2(\alpha - \frac{1}{2})^2$, and $\Lambda \le t + 2(\alpha - \frac{1}{2})^2$. Conversely $\Lambda \ge 0$ [289], the zeros of H_Λ are real, and so $(\Lambda, \frac{1}{2}) \in \mathcal{G}$, where the quantity equals Λ. (iii) The first equivalence is (ii) with $\Lambda \ge 0$. For the second, the points named lie on the level set $t + 2(\alpha - \frac{1}{2})^2 = \delta$, so their membership gives $\Lambda \le \delta$ for every $\delta > 0$, and conversely RH puts $(0, \frac{1}{2})$ in $\mathcal{G}$, from which (i) gives every point of every level set. □

Corollary 15.3.3 (The known region). *The following points lie in* $\mathcal{G}$*, and with them every point that Theorem 15.3.2(i) carries them to; the bounds on* Λ *in (a) to (c) are those of Table 8.14.*

(a) $(0, 1)$*, by the critical strip; it gives de Bruijn's* $\Lambda \le \frac{1}{2}$.

(b) $(0.2, 0.6)$*, by the certificate of Polymath15 [282]; it gives* $\Lambda \le 0.22$.

(c) A point on the level set $t + 2(\alpha - \frac{1}{2})^2 = 0.2$*, by the certificate of Polymath15 with the verification of Platt and Trudgian [247]; it gives* $\Lambda \le 0.2$.

(d) Conditionally on the half plane announced in October 2026 [226], $(0, \frac{7}{8})$*; it gives* $\Lambda \le \frac{9}{32}$.

Proof. (a) is the absence of zeros of ζ in $\operatorname{Re} s \ge 1$ and the functional equation. (b) is the statement proved in [282]: $H_{0.2}$ has no zeros with $\operatorname{Im} z > 0.2$, which is $(0.2, 0.6) \in \mathcal{G}$. (c) is [247], which takes a row of the table of [282] whose hypothesis their verification satisfies. (d) is immediate from [226]. The bounds on Λ are Theorem 15.3.2(ii). □

Figure 15.1 shows the region. A half plane at time zero improves the current bound only below $\alpha = \frac{1}{2} + \sqrt{0.1} \approx 0.816$, as Section 8.4 notes.

Remark 15.3.4 (Two semigroups). The two coordinates of the diagram carry two monotonicities, and they run in compatible directions. In α the Cauchy densities form a semigroup, so when no zero of ξ_t lies to the right of α the Thorin density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the one at α, and positivity propagates to the right; going left is deconvolution, which does not preserve positivity (Section 11.4.7). In t the heat flow carries real zeros to real

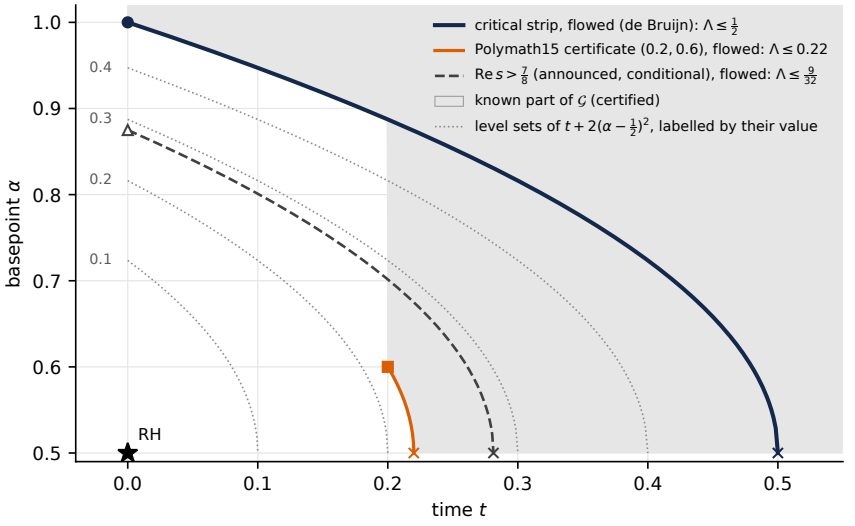


Figure 15.1: The Thorin phase diagram. Solid curves are the boundaries of certified GGC regions carried along the flow by Theorem 15.3.2(i), and the dashed curve is that of the region conditional on the half plane of Corollary 15.3.3(d); the dotted curves are the level sets of $t + 2(\alpha - \frac{1}{2})^2$, labelled by the bound on Λ that a point on them gives. RH is the point $(0, \frac{1}{2})$. Script `phase_thorin.py`.

zeros and narrows strips, so membership propagates forward in time. RH asks for the corner where both coordinates are smallest, and neither semigroup can be run backward.

15.4 What a Thorin computation sees at positive time

The Pólya route computes ξ_t at the cost of ξ, since the flow only reweights the kernel (Section 8.4.1), and, when $(t, \alpha) \in \mathcal{G}$, the Thorin density of the shifted time t clock is $\pi^{-1} v_t(\alpha, \theta)$ with

$$v_t(\alpha, \theta) = \operatorname{Re} \frac{\xi_t'}{\xi_t}(\alpha + i\theta),$$

$$\xi_t(s) = 8 \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u) \cosh(2wu)\, du,$$

$$\xi_t'(s) = 8 \int_0^\infty e^{tu^2} \Phi_{\mathrm{dBN}}(u)\, 2u \sinh(2wu)\, du,$$

$w = s - \frac{1}{2}$, by Remark 15.2.4; where membership in $\mathcal{G}$ is not known, as at $t = 0$ with $\alpha = 0.6$, the function v_t/π is still defined. We computed $v_t(\alpha, \theta)$ at the two times $t = 0$ and $t = 0.2$ and the two basepoints $\alpha = 0.6$ and 0.75, for $0 \le \theta \le 100$ in

steps of 0.25, with no primes and no list of zeros. The integrals were evaluated by the trapezoid rule of Section 2.1.1 with step $1/160$ on $[0, 1.6]$ (the tail beyond 1.6 is below 10^{-812}) at $\lfloor 0.36\,\theta \rfloor + 35$ digits, the term $0.36\,\theta$ absorbing the cancellation $|\xi_t(\alpha + i\theta)| \approx e^{-\pi\theta/4}$ that the Pólya route always carries (Section 2.1). A second discretisation agrees with the first in v_t at all 1604 points of the four grids to a relative 2×10^{-35}; at $t = 0$ the integrals, evaluated with 10 more digits, agree with the closed forms of ξ and ξ'/ξ through Γ, ψ, ζ and ζ' to at least 43 digits at both basepoints and at $\theta = 0$, 17.5, 50 and 100, and at $t = 0.2$ the integral for ξ_t' agrees with a central difference of ξ_t to 30 digits at two points.

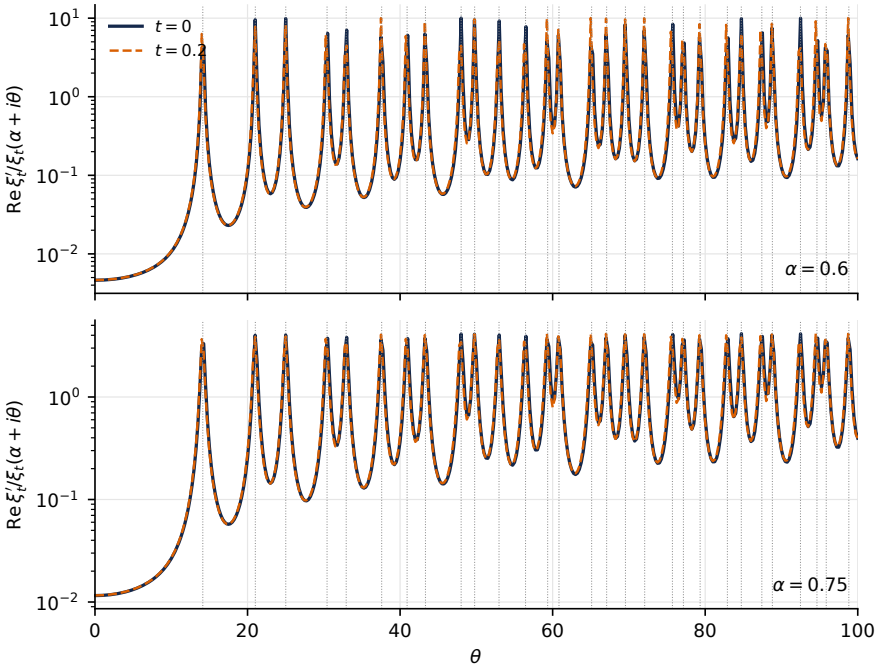


Figure 15.2: $v_t(\alpha, \theta) = \mathrm{Re}\ \xi_t'/\xi_t(\alpha + i\theta)$, which is π times the Thorin density of the shifted time t clock when $(t, \alpha) \in \mathcal{G}$, on the lines $\mathrm{Re}\, s = 0.6$ and 0.75, at $t = 0$ (solid) and $t = 0.2$ (dashed), from the Pólya integral. Dotted lines mark the ordinates of the zeros of ζ. Script `phase_thorin.py`.

Proposition 15.4.1 (Computed). *On $0 \le \theta \le 100$:*

(i) At both times and both basepoints the minimum of $v_t(\alpha, \cdot)$ is at the origin, $v_t(\alpha, 0) = \xi_t'/\xi_t(\alpha)$, in agreement with Theorem 9.1.18 at $t = 0$, and $v_t(\alpha, \cdot)$ is positive.

(ii) Between consecutive zeros the valleys are set by the gaps relative to the kernel width $\alpha - \frac{1}{2}$, and they scale roughly in proportion to it: at $t = 0$ and $\alpha = 0.6$

	$\alpha = 0.6$			$\alpha = 0.75$		
θ	$t = 0$	$t = 0.2$	change	$t = 0$	$t = 0.2$	change
0	0.004621	0.004640	+0.4%	0.011551	0.011599	+0.4%
17.50	0.0230	0.0231	+0.4%	0.0573	0.0575	+0.4%
23.01	0.0582	0.0578	−0.7%	0.1439	0.1429	−0.7%
27.67	0.0389	0.0393	+0.9%	0.0968	0.0976	+0.9%
35.27	0.0520	0.0526	+1.1%	0.1291	0.1304	+1.1%
42.12	0.1539	0.1496	−2.8%	0.3729	0.3627	−2.7%
45.63	0.0569	0.0579	+1.8%	0.1412	0.1438	+1.8%
95.26	0.5625	0.4829	−14.2%	1.2474	1.0906	−12.6%

Table 15.2: The value at the origin and at the minimum of a selection of the valleys of $v_t(\alpha, \cdot)$ between consecutive zeros. The ordinate θ is that of the minimum at $t = 0$ and $\alpha = 0.6$; at $t = 0.2$ the minimum lies 0.06 to 0.08 to the left, and at $\alpha = 0.75$ it moves by less than 0.002. Script `phase_thorin.py`.

they range from 0.023 *near* $\theta = 17.5$ *to* 0.56 *near* $\theta = 95$*, and at* $\alpha = 0.75$ *they are* 2.2 *to* 2.5 *times larger.*

(iii) The 28 *valleys above* $\theta = 10$*, each taken at its minimum, change between* $t = 0$ *and* $t = 0.2$ *by a median of* 2.3% *and at most* 14.2% *at* $\alpha = 0.6$ *(*2.2% *and* 12.6% *at* $\alpha = 0.75$*), the largest change being that of the valley near* $\theta = 95$*; the zeros below* 60 *move by* 0.044 *to* 0.118*, with mean* 0.075*, the first from* 14.135 *to* 14.077 *(Section 15.5.2).*

Part (ii) corrects a natural expectation. Averaged over a window wider than the spacing of the zeros, $v_t(\alpha, \theta)$ is close to $\frac{1}{2}\log(\theta/2\pi)$, so one might expect the certificate to enjoy a margin of that size. At $t = 0$ Remark 9.1.17 proves a margin that does grow with the height, $v_\alpha(\theta)/\pi \geq c \log\theta$ for every $c < (\alpha - \frac{1}{2})/(\pi(2\alpha+1))$ and all large θ outside a set of density zero, but its constant is proportional to $\alpha - \frac{1}{2}$: at $\alpha = 0.6$ it is below $0.015 \log\theta$, against the average $(2\pi)^{-1}\log(\theta/2\pi)$ of v_α/π, and its exceptional set contains the window of every zero to the right of $\operatorname{Re} s = \alpha$. Below height 100 the computed margin is smaller still, set by the largest gaps between consecutive zeros, because the kernel width $\alpha - \frac{1}{2}$ is small against the mean spacing $2\pi/\log(\theta/2\pi)$.

On the windows computed here, below height 100, the clock of $\xi_{0.2}$, which is a GGC at the basepoint 0.6 by Corollary 15.3.3(b), and the clock of ξ, which is a GGC at the basepoint 0.6 only if there is no zero with $\beta > 0.6$ anywhere, are described by the functions $v_{0.2}/\pi$ and v_0/π, whose valleys differ by a median of 2.3% and at most 14.2%, and both are positive. Nothing in the computed density distinguishes the theorem from the conjecture. The Thorin side therefore does not offer a cheaper finite computation than the flow; it offers the same finite computation in the language of this book, and the whole question is where the tail comes from. The same comparison at $t = 0.2$ and $t = 0$ above the barrier

of Polymath15, at heights near 3×10^{10}, would require a Riemann Siegel type evaluator of $H_{0.2}$ such as the effective approximation of [282] (Section 2.2 gives the one for ζ); Problem 15.1 asks for it.

15.5 A certificate for one point of the diagram

This section writes out the finite part of the statement $(0.2, 0.6) \in \mathcal{G}$ as a certified computation with explicit constants, and runs it in ball arithmetic below height 160. The statement is that $\xi_{0.2}$ has no zero with real part above 0.6. Given a barrier X (in the variable of $H_{0.2}$, so that it sits at height $X/2$ for $\xi_{0.2}$; see Theorem 15.5.6) beyond which this is known [282], it is the vanishing of the winding number of $\xi_{0.2}$ around the rectangle $Q = [0.6, 0.9] \times [0, X/2]$. Throughout, $t = 0.2$.

15.5.1 The lemmas

Lemma 15.5.1 (Bottom edge). *$\xi_t(\sigma) > 0$ for every real σ and every $t \geq 0$.*

Proof. Lemma 15.2.1(ii). □

Lemma 15.5.2 (Right edge). *For $0 \leq t \leq \frac{1}{2}$ every zero of ξ_t satisfies $|\beta - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1-2t}$. In particular for $t = 0.2$ every zero has $\beta \leq 0.8873$, so $\xi_{0.2}$ has no zero on $\operatorname{Re} s = 0.9$.*

Proof. De Bruijn's theorem on strips, as in the proof of Theorem 15.3.2, with $\Delta = 1$ from the critical strip and $\beta - \frac{1}{2} = \frac{1}{2}\operatorname{Im} z$ up to sign. □

Lemma 15.5.3 (Tilted bound). *Let $0 < \eta < \pi/8$, $t \geq 0$ and $s = \frac{1}{2} + x + i\theta$ with $\theta \geq 0$. Then*

$$|\xi_t(s)| \leq K_\eta(t,x)\, e^{-2\eta\theta}, \qquad K_\eta(t,x) = 4\int_{-\infty}^{\infty} \left|e^{t(u+i\eta)^2}\right| |\Phi_{\mathrm{dBN}}(u+i\eta)|\, e^{2xu}\, du,$$

and $K_\eta(t,x)$ is finite and even in x. Write $\bar{K}_\eta(t,R) = \max_{|x|\leq R} K_\eta(t,x)$.

Proof. Since Φ_{dBN} is even, $\xi_t(s) = 4\int_{\mathbb{R}} e^{tu^2}\Phi_{\mathrm{dBN}}(u)e^{2(x+i\theta)u}\, du$. The kernel extends analytically to $|\operatorname{Im} u| < \pi/8$ with the same doubly exponential decay, which is Proposition 2.1.1 in the variable $v = 2u$, so the contour may be moved to $\operatorname{Im} u = \eta$, and $|e^{2i\theta(u+i\eta)}| = e^{-2\eta\theta}$. □

Lemma 15.5.4 (Derivative bound). *With the notation of Lemma 15.5.3, for $r > 0$, $|\xi_t'(s)| \leq r^{-1}\bar{K}_\eta(t, |x| + r)\, e^{-2\eta(\theta - r)}$, and the factor $e^{-2\eta(\theta-r)}$ may be replaced by 1 when $\theta \leq r$.*

Proof. Cauchy's estimate on the disc of radius r about s. A point z of the disc has $|\operatorname{Re} z - \frac{1}{2}| \le |x| + r$ and $\operatorname{Im} z \ge \theta - r$. If $\operatorname{Im} z \ge 0$, Lemma 15.5.3 bounds $|\xi_t(z)|$ by $\bar{K}_\eta(t, |x|+r)e^{-2\eta \operatorname{Im} z}$, which is at most $\bar{K}_\eta(t, |x|+r)e^{-2\eta(\theta-r)}$ and at most $\bar{K}_\eta(t, |x| + r)$. If $\operatorname{Im} z < 0$, which occurs only when $\theta < r$, then $\xi_t(\bar{z}) = \overline{\xi_t(z)}$ gives the bound $\bar{K}_\eta(t, |x| + r)$ for $|\xi_t(z)|$. So on the disc $|\xi_t| \le \bar{K}_\eta(t, |x| + r)e^{-2\eta(\theta-r)}$, and $|\xi_t| \le \bar{K}_\eta(t, |x| + r)$ when $\theta \le r$. □

Lemma 15.5.5 (Grid). *Let s_0, s_1 be points of a segment on which ξ_t has no zero, and suppose $|s_1 - s_0| \cdot \sup_{[s_0,s_1]} |\xi_t'| < |\xi_t(s_0)|$. Then the change of $\arg \xi_t$ along the segment equals the principal value $\operatorname{Arg}(\xi_t(s_1)/\xi_t(s_0)) \in (-\pi, \pi)$.*

Proof. The image of the segment lies in the open disc of radius $|\xi_t(s_0)|$ about $\xi_t(s_0)$, which does not contain 0, and on that disc a continuous branch of the argument varies by less than π from its value at the centre. □

Theorem 15.5.6 (The certificate). *Let X be a barrier for $(0.2, 0.6)$ in the variable of $H_{0.2}$, as Polymath15 provide one [282] (no script of this book checks it), that is, a number such that $H_{0.2}$ has no zero z with $\operatorname{Re} z \ge X$ and $\operatorname{Im} z > 0.2$, so that $\xi_{0.2}$ has no zero with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \ge X/2$. Let $s_0, \dots, s_M$ be points of the left, top and right edges of Q, in order along ∂Q from one bottom corner, s_0, to the other, s_M, and including the two top corners, and let B_j be ball enclosures of $\xi_{0.2}(s_j)$ with radii ε_j. Suppose that for every j there are $r > 0$ and $\eta \in (0, \frac{\pi}{8})$ such that*

$$|s_{j+1} - s_j| \, \frac{\bar{K}_\eta(0.2,\, 0.4 + r)}{r} \, e^{-2\eta(\theta_j - r)} + \varepsilon_j + \varepsilon_{j+1} < \min |B_j|,$$
$$\theta_j = \min(\operatorname{Im} s_j, \operatorname{Im} s_{j+1}),$$

where the factor $e^{-2\eta(\theta_j - r)}$ may be replaced by 1 when $\theta_j \le r$, and that $\sum_j \operatorname{Arg}(B_{j+1}/B_j)$, evaluated in ball arithmetic, lies within $(-\pi, \pi)$. Then $\xi_{0.2}$ has no zero in Q, so $(0.2, 0.6) \in \mathcal{G}$ and $\Lambda \le 0.22$.

Proof. By Lemma 15.5.1, $\xi_{0.2}$ is real and positive on the bottom edge, so $\arg \xi_{0.2}$ does not change along it, and by Lemma 15.5.2 the right edge is zero free. The hypothesis gives $|\xi_{0.2}(s_j)| \ge \min |B_j|$ at the grid points, and Lemma 15.5.4 with $|x| \le 0.4$ and $\operatorname{Im} z \ge \theta_j$ on the segment from s_j to s_{j+1} bounds the derivative between them, so Lemma 15.5.5 applies to every step, and the winding number is the integer nearest to $\frac{1}{2\pi} \sum_j \operatorname{Arg}(B_{j+1}/B_j)$, which is 0. So $\xi_{0.2}$ has no zero in Q; it has none with $\operatorname{Re} s > 0.9$ by Lemma 15.5.2, none with $\operatorname{Re} s > 0.6$ and $\operatorname{Im} s \ge X/2$ by the barrier, and its zeros in $\operatorname{Im} s < 0$ are the conjugates of those in $\operatorname{Im} s > 0$. Hence it has no zero with $\operatorname{Re} s > 0.6$, which is $(0.2, 0.6) \in \mathcal{G}$, and the last clause is Theorem 15.3.2. □

The number of grid points is of order $(X/2) \sup_j K_\eta e^{2\eta r}/(r\,|\xi_t(s_j)|e^{2\eta\theta_j})$. The decay of ξ_t, of order $e^{-\pi\theta/4}$ away from its zeros, cancels against the decay $e^{-2\eta\theta}$

of the derivative bound only at $\eta = \pi/8$, which Lemma 15.5.3 excludes, and K_η grows without bound as $\eta \uparrow \pi/8$ (Table 15.3). With a tilt $\eta < \pi/8$ the count grows like $e^{(\pi/4-2\eta)\theta}$. We have no proof that the count is $O(X \log X)$, or polynomial in X, when the tilt is chosen at each height. In the runs below the left edge needs on average about 45 steps per unit height up to height 60, 73 up to 120 and 94 up to 160 (Theorem 15.5.8).

PROPOSITION 15.5.7 (The certificate in Thorin form). *Let $t \geq 0$ and $\frac{1}{2} < \alpha < \sigma_1$, and suppose $v_t(\sigma, \theta) > 0$ for every $(\sigma, \theta) \in [\alpha, \sigma_1] \times [0, T]$ at which $\xi_t(\sigma + i\theta) \neq 0$. Then ξ_t has no zero $\beta + i\gamma$ with $\alpha < \beta \leq \sigma_1$ and $0 < \gamma < T$.*

Proof. Near a zero ρ of multiplicity m, $\xi_t'/\xi_t(s) = m/(s-\rho) + O(1)$, so at $s = \rho - \delta$ with small $\delta > 0$ the real part is $-m/\delta + O(1) < 0$. If $\alpha < \beta \leq \sigma_1$, these points lie in the rectangle for small δ. This is the argument of Remark 1.2.4, restricted to a rectangle of finite height. □

Positivity on the single line $\operatorname{Re} s = \alpha$ is not enough (Remark 1.2.4), which is why the proposition asks for the whole rectangle. It is the pointwise counterpart of Theorem 15.5.6: the same evaluations of ξ_t and ξ_t' certify it, with a two dimensional grid and a bound on the derivative of v_t in place of the argument increments, and the margin is the one measured in Proposition 15.4.1.

15.5.2 Constants and the march

Table 15.3 gives the constants of Lemma 15.5.3 at $t = 0.2$ and the values of $\xi_{0.2}$ on the real axis that fix the scale.

η	decay $e^{-2\eta\theta}$	$K_\eta(0.2, 0.4)$	$K_\eta(0.2, 0.9)$
0.30	0.60	23.67	25.31
0.35	0.70	199.2	231.1
0.38	0.76	4847	6776

σ	$\xi_{0.2}(\sigma)$
0.5	0.49827
0.6	0.49839
0.9	0.50013
1.0	0.50117
1.5	0.50996

Table 15.3: Constants for the certificate at $t = 0.2$, in floating point. Script `phase_cert.py`.

A floating point march of Lemma 15.5.5 around $Q_{60} = [0.6, 0.9] \times [0, 60]$, with the Pólya route evaluator (the trapezoid rule of Section 2.1.1 with the weight e^{tu^2}) at 40 digits (working precision 72 digits) and an adaptive step keeping $|\xi_t(s_{j+1}) - \xi_t(s_j)| < \frac{1}{2}|\xi_t(s_j)|$, returned the winding -2.9×10^{-72} in 910 evaluations. The same evaluator located the zeros of $\xi_{0.2}$ on the critical line below 60,

$$14.077,\ 20.947,\ 24.952,\ 30.330,\ 32.888,\ 37.503,$$
$$40.833,\ 43.282,\ 47.887,\ 49.730,\ 52.905,\ 56.358,\ 59.237,$$

thirteen zeros, as for ξ below 60, each moved slightly by the flow (five of them are in Table 8.8). For the control function that moves the pair at 30.330 and 32.888 to the quadruple $\frac12 \pm \frac14 + 31.609i$ and its conjugates, as in Theorem 5.2.1 (there for ξ and ordinates $1000 < A < B \le 2000$), the computed winding was 1 to within 10^{-35} in 905 evaluations: exactly one of the four new zeros, $0.75 + 31.609i$, lies in Q_{60}.

The finite part of Theorem 15.5.6 was then run in ball arithmetic, adding every truncation error to the radius of the enclosure. The evaluator is the same trapezoid rule, with $w = s - \frac12 = x + i\theta$ and $u_j = jh$: $\xi_t(s) = 8h \sum_{j=0}^{J} \omega_j e^{tu_j^2} \Phi_{\mathrm{dBN}}(u_j) \cosh(2wu_j) + E$, where $Jh \ge 3$, $\omega_0 = \frac12$, $\omega_j = 1$ for $j \ge 1$, and $|E| \le E_{\mathrm{cut}} + E_{\mathrm{trap}}$. For $u \ge 3$ the kernel satisfies $|\Phi_{\mathrm{dBN}}(u)| \le 60\, e^{9u} e^{-\pi e^{4u}}$, so that for $|x| \le 1$ the summand is at most $8h\, G(u_j)$ with $G(u) = 60 \exp(tu^2 + 11u - \pi e^{4u})$, which decreases with $-G'/G \ge \lambda = 4\pi e^{4Jh} - 2tJh - 11$ for $u \ge Jh$, and the nodes beyond Jh contribute at most $E_{\mathrm{cut}} = 8G(Jh)/\lambda < 10^{-229000}$. The error E_{trap} of the rule is bounded by Poisson summation on the strip $|\operatorname{Im} u| < a$, $a = 0.30 < \pi/8$, in which the integrand of Lemma 15.5.3 is analytic: $E_{\mathrm{trap}} \le 2K_{0.30}(0.2, x) \cosh(2a\theta)/(e^{2\pi a/h} - 1)$, with $K_{0.30}(0.2, x) \le 23.72$ for $|x| \le 0.4$, bounded in the way described below, and h is taken so that this is below $5 \times 10^{-4} e^{-\pi T/4}$ at the top of the rectangle of height T: $h = 0.0199$, 0.0106 and 0.008 at $T = 60$, 120 and 160, at the working precisions 219, 288 and 334 bits. The series for the kernel is summed until its tail is below $2^{-\mathrm{prec}-30}$, the tail after N terms being enclosed by

$$\sum_{n>N} \left| (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}} \right| \le |e^{9u}| \Big(2\pi^2 + \frac{3\pi}{\kappa}\Big) \Big(\frac{4}{\pi\kappa}\Big)^2 e^{-2}\, \frac{e^{-\pi(N+1)^2\kappa/2}}{1 - e^{-\pi(N+1)\kappa}}, \qquad \kappa = \operatorname{Re}(e^{4u}),$$

using $|e^{5u}| \le |e^{9u}|/\kappa$, $n^4 e^{-\pi n^2 \kappa/2} \le (4/\pi\kappa)^2 e^{-2}$ and $n^2 \ge (N+1)^2 + 2(N+1)(n - N - 1)$ for the geometric sum, for every N and every $\kappa > 0$. At a real node $\kappa = e^{4u}$; at a complex u κ is a lower bound of $\operatorname{Re}(e^{4u})$ over the input ball. The constants $\bar K_\eta(0.2, 0.4 + r)$ were bounded above by interval subdivision of $[0, 3]$ into $3 \cdot 2^{14}$ cells, plus the tail $u \ge 3$, below e^{-25000} since $\operatorname{Re} e^{4(u+i\eta)} \ge e^{12} \cos(4\eta) > 8000$ there, for the three tilts $\eta = 0.30$, 0.35 and 0.38 and the three radii $r = \frac14$, $\frac38$ and $\frac12$: 24.37, 24.82 and 25.36, then 212.33, 221.11 and 231.71, then 5633.6, 6170.8 and 6841.2, each rounded up. The ordinates of the grid points are exact dyadic numbers, and the abscissae 0.6 and 0.9 are balls of radius $2^{-\mathrm{prec}}$ enclosing them, so that the radius of $\xi_t(s_j)$ is that of the evaluator. At each step the tilt η and the radius r giving the smallest derivative bound are used, the step is $0.8\, |\xi_t(s_j)|_{\mathrm{low}}$ divided by that bound, capped at 0.5, and the inequality of Theorem 15.5.6, with the exact length of the step and its exact lowest height, is then verified in ball arithmetic, the step being halved when it fails.

THEOREM 15.5.8. *$\xi_{0.2}$ has no zero s with $0.6 \le \operatorname{Re} s \le 0.9$ and $0 \le \operatorname{Im} s \le 160$.*

Proof. For $T = 60$, 120 and 160 the script walks the boundary of $Q_T = [0.6, 0.9] \times [0, T]$ counterclockwise, up the right edge, along the top and down the left edge, with the steps of Theorem 15.5.6. The bottom edge is not walked, since $\xi_{0.2}$ is real and positive on it (Lemma 15.5.1). The run took 4483 steps for $T = 60$ (1737 up the right edge, 31 along the top and 2715 down the left edge), 13,679 for $T = 120$ (4955, 11 and 8713) and 23,219 for $T = 160$ (8142, 20 and 15,057), and returned the winding number enclosures $[\pm 3.94 \times 10^{-5}]$, $[\pm 1.24 \times 10^{-6}]$ and $[\pm 3.47 \times 10^{-7}]$. In every run the inequality of Theorem 15.5.6 was verified in ball arithmetic at every step and no step failed, and the sum of the arguments lies within π of 0, so by Lemmas 15.5.1 to 15.5.5 the winding number is 0 and $\xi_{0.2}$ has no zero in Q_T, which for $T = 160$ is the theorem. Script `phase_rig.py`, data `phase_rig.json`. □

With Lemma 15.5.2 this says that every zero of $\xi_{0.2}$ below height 160 has $|\beta - \frac{1}{2}| < 0.1$; as a control, the same script run around $[0.4, 0.6] \times [0, 60]$, which contains the thirteen zeros on the line, returns the winding number enclosure $[13.0000 \pm 5.04 \times 10^{-5}]$ in 4841 steps. The barrier of Polymath15 is near $X = 6 \times 10^{10}$, height 3×10^{10} for $\xi_{0.2}$, and reaching it in this framework requires the evaluator of the programme below.

15.6 Where the uniform step could come from

Section 12.1.2 names three mechanisms that could supply the missing uniform step: a positive approximating family whose positivity is proved independently, a Hilbert Pólya operator, or a direct Weil inequality.

15.6.1 The Weil Gram matrix on compactly supported test functions

For h smooth enough, even or not (each term depends on h only through its even part h_e, defined below), with $\hat{h}(r) = \int h(x) e^{irx}\, dx$, the explicit formula reads

$$\sum_{\rho} \hat{h}(\gamma_\rho) = \hat{h}(\tfrac{i}{2}) + \hat{h}(-\tfrac{i}{2}) - \sum_{n \ge 2} \frac{\Lambda(n)}{\sqrt{n}} \big(h(\log n) + h(-\log n) \big) + \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(r)\, \operatorname{Re} \psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\, dr - h(0) \log \pi,$$

with $\rho = \frac{1}{2} + i\gamma_\rho$ and $\psi = \Gamma'/\Gamma$. For $h = g * \tilde{g}$ with $\tilde{g}(x) = \overline{g(-x)}$ and g supported in $[-L/2, L/2]$, the left side is $\sum_\rho |\hat{g}(\gamma_\rho)|^2 \ge 0$ under RH, which is Weil's criterion [338], h is supported in $[-L, L]$, and the prime sum runs over prime powers $n \le e^L$.

The archimedean term was evaluated in x space,

$$\frac{1}{2\pi}\int \hat{h}(r)\ \mathrm{Re}\,\psi\big(\tfrac{1}{4}+\tfrac{ir}{2}\big)\,dr = \int_0^\infty \Big[\frac{h(0)e^{-2x}}{x} - \frac{2h_e(x)e^{-x/2}}{1-e^{-2x}}\Big]dx,$$
$$h_e(x) = \tfrac{1}{2}\big(h(x)+h(-x)\big),$$

which is far better conditioned than the r integral for compactly supported h: truncated at $r = 400$, the r integral carries an error of 6×10^{-6} in the archimedean term of the diagonal entry at $w = 0.1$, more than two thousand times the smallest eigenvalue 2.5×10^{-9} of Table 15.4. With hat functions $g_j(x) = \max(0, 1-|x-c_j|/w)$ at nodes spaced w apart inside $[-L/2, L/2]$, $N = L/w - 1$ functions in all, $h_{jk} = g_j * \tilde{g}_k$ is an explicit piecewise cubic, and the Gram matrix $G_{jk} = W(h_{jk})$ is Hermitian, with $W(g*\tilde{g}) \ge 0$ on the span exactly when G is positive semidefinite.

w	L	N	$\lambda_{\min}$	$\lambda_{\max}$	pole, −prime, archimedean
0.25	1	3	4.22×10^{-4}	0.034	0.3468, −0.0141, −0.3322
0.25	2	7	2.22×10^{-5}	0.050	0.8410, −0.2472, −0.5938
0.25	4	15	4.70×10^{-7}	0.096	1.6494, −0.8550, −0.7944
0.25	6	23	3.56×10^{-8}	0.144	2.7255, −1.8067, −0.9187
0.1	1	9	1.84×10^{-5}	0.083	0.1287, −0.0019, −0.1268
0.1	2	19	1.27×10^{-6}	0.149	0.2900, −0.0676, −0.2224
0.1	4	39	5.70×10^{-8}	0.290	0.7281, −0.3981, −0.3300
0.1	6	59	2.48×10^{-9}	0.431	1.1050, −0.7353, −0.3697

Table 15.4: The Weil Gram matrix on hat functions. The last column splits $\lambda_{\min} = v^*Gv$ for the minimising unit vector into the pole, prime (with its sign) and archimedean pieces. Script `phase_weil.py`, output `phase_weil.json`.

w	L	$\lambda_{\min}$ (arithmetic)	$\sum_\rho \lvert\hat{g}(\gamma_\rho)\rvert^2$ (zeros)	zeros carrying most of the sum
0.25	1	4.2184×10^{-4}	4.2184×10^{-4}	#2 (56%), #4 (20%), #5 (11%)
0.25	3	1.8238×10^{-6}	1.8237×10^{-6}	#9 (44%), #11 (24%), #20 (5%)
0.25	6	3.5562×10^{-8}	3.5560×10^{-8}	#10 (40%), #20 (13%), #30 (9%)
0.1	2	1.2691×10^{-6}	1.2691×10^{-6}	#13 (37%), #15 (15%), #16 (15%)
0.1	4	5.7008×10^{-8}	5.7004×10^{-8}	#14 (36%), #13 (23%), #16 (7%)
0.1	6	2.4846×10^{-9}	2.4841×10^{-9}	#41 (32%), #15 (17%), #14 (8%)

Table 15.5: The smallest eigenvalue from the arithmetic side against the sum of squares of $\hat{g}_{\min}$ over the first 1500 zeros, with the zeros (by index) that carry the largest shares. Script `phase_weil.py`, output `phase_weil.json`.

The experiment says three things (Tables 15.4 and 15.5, Figure 15.3). First, $\lambda_{\min} > 0$ throughout and it falls roughly geometrically in L, by a factor between about 2.5 and 20 per unit of L up to $L = 6$, as it must, since the nodes are nested

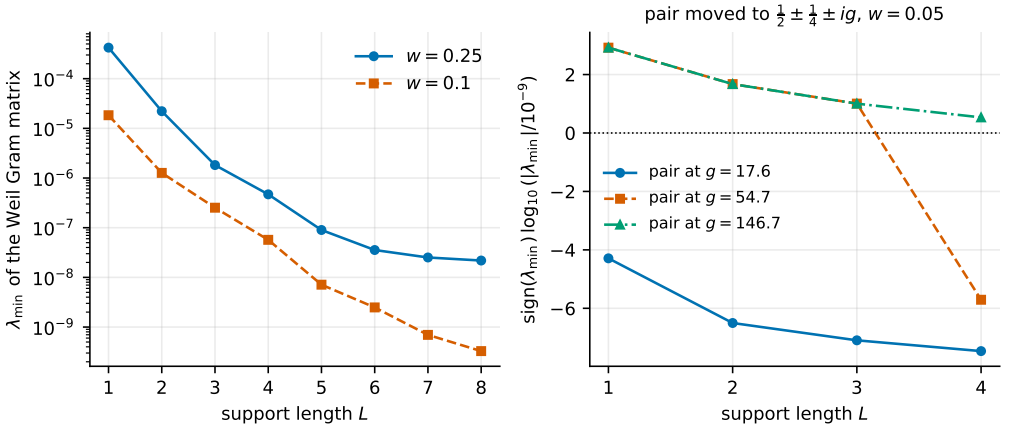


Figure 15.3: Left: $\lambda_{\min}$ of the Weil Gram matrix against the support length L for two resolutions. Right: a pair of zeros at height g moved to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$, signed logarithm of $\lambda_{\min}$ for $w = 0.05$. The pair at $g \approx 17.6$ is detected at every L, the pair at $g \approx 54.7$ only at $L = 4$, and the pair at $g \approx 146.7$ not at all within $L \leq 4$. Script `phase_weil.py`, output `phase_weil.json`.

(for the values of L used), so that the matrix for a shorter support is a principal submatrix of the matrix for a longer one, and the smallest eigenvalue of the Weil form on $[-L/2, L/2]$ tends to zero as L grows: a long enough support can make $\hat{g}$ small at every low ordinate. Second, the two sides of Weil's identity agree to a relative 2×10^{-4} at every (w, L), down to $\lambda_{\min} = 2.5 \times 10^{-9}$; and the margin is carried by a handful of zeros, zero 41 at height 124.3 carrying 32% at $w = 0.1$, $L = 6$. Third, at $w = 0.05$ a pair displaced at height $g \approx 17.6$ makes $\lambda_{\min}$ negative at every $L \leq 4$, one at $g \approx 54.7$ only at $L = 4$ and one at $g \approx 146.7$ at no $L \leq 4$, an instance of the finite information principle of Section 15.1.1. How fine w must be to see a pair at height g is not fixed by one resolution: at $w = 0.2$ the pair at $g \approx 17.6$ is already seen at $L = 2$, with $wg \approx 3.5$.

The sign structure is the reverse of the Lévy picture of Section 9.1, where the pole of ζ is a negative input to $\log \xi$; in the Weil form the pole, on the minimising vectors of Table 15.4, contributes positively. A certificate for the Weil form would therefore have to show, for every L, that the three terms together are nonnegative on the whole span, although the pole term alone is not (for real odd g it is $-2G(\frac{1}{2})^2$, with $G(a) = \int g(x)e^{ax}\,dx$), and on the minimising vectors the pole term outweighs the prime and archimedean terms only by a margin that tends to zero. No uniformly positive quadratic form can do that. What could is a factorisation $W(g * \tilde{g}) = \|Ag\|^2$ with A built from the arithmetic and exact positivity, the shape of the Hilbert Pólya mechanism, and Problem 15.4 asks for the first experiment in that direction.

15.6.2 The zero gas along the flow

In the variable $z = 2\gamma$ the zeros of H_t move, while simple, by $dz_k/dt = 2\sum_{j\neq k}(z_k - z_j)^{-1}$ (Section 8.4.1, with five of the zeros at four times in Table 8.8). We integrated this system for the first 600 zeros, heights up to 939.02, with a frozen tail for the omitted zeros (the zeros 601 to 1700 held at their positions at $t = 0$ and a continuum of density $\frac{1}{4\pi}\log(z/4\pi)$ beyond them, whose drift on z_k is $-4z_k\sum_{j>600} z_j^{-2}$ to leading order), backward from $t = 0$ until the first gap fell below 0.02 and forward to $t = 0.3$. Along the trajectory we recorded the minimum normalised gap $s_{\min}(t) = \min_k (z_{k+1} - z_k)\frac{1}{4\pi}\log(z_{k+1}/4\pi)$ and the Csordas, Smith and Varga quantity $\min_k \Delta_k^2 g_k$ over the first 400 pairs, with $\Delta_k = z_{k+1} - z_k$ and $g_k = \sum_{j\neq k,k+1}[(z_k - z_j)^{-2} + (z_{k+1} - z_j)^{-2}]$, where j runs over all the other zeros $\pm z_j$ of the model, the frozen tail and the continuum included; a pair with $\Delta_k^2 g_k < \frac{4}{5}$ is a Lehmer pair in the sense of [70] and Section 8.4.6.

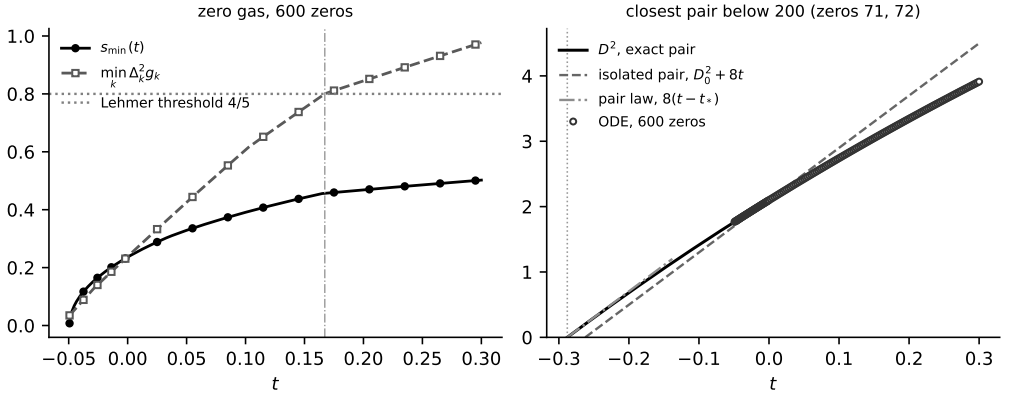


Figure 15.4: Left: the minimum normalised gap and $\min_k \Delta_k^2 g_k$ along the flow, both monotone on the computed range; the second crosses the Lehmer threshold $\frac{4}{5}$ near $t = 0.17$. Right: D^2 for the closest pair below height 200, at ordinates 184.87 and 185.60, against the isolated pair law $D_0^2 + 8t$.

Proposition 15.6.1 (Computed; the upper end of (iii) certified). *(i) The backward integration stops at* $t = -0.0493$*, when the pair at ordinates* 750.656 *and* 750.966*, the* 453*rd and* 454*th zeros, collides; the isolated pair estimate is* $-D_0^2/8 = -0.048$*. (ii) At the* 171 *times sampled in* $[-0.049, 0.30]$ *both* $s_{\min}(t)$ *and* $\min_k \Delta_k^2 g_k$ *are strictly increasing, from* 0.0076 *to* 0.502 *and from* 0.035 *to* 0.977*; at* $t = 0$ *the minimiser of* $\Delta_k^2 g_k$ *is the pair at* 564.16 *and* 564.51*, with value* 0.238*. (iii) For the pair near* 185*, the pair* $(\gamma_{71}, \gamma_{72})$ *of Section 8.4.2, certified signs of* $\xi_t(\frac{1}{2} + i\theta)$ *in Arb at* 640 *bits (the trapezoid sum of Section 2.1.1 with an enclosure of its discretisation error) at* 131 *points of* $[184.6, 185.9]$*, with bisection in* t *on the number of sign changes, show that the pair is still real at* $t = -0.287979$*, and with the pair*

law below place the collision in $[-0.28803, -0.287979]$*, in agreement with the value* -0.2880 *of Section 8.4 and inside the bracket of Section 8.4.5. Newton's method on the same sum puts the double zero at* $z_* = 370.78087$*,* $t_* = -0.287979837009$*; a Krawczyk test certifies a real double zero of the trapezoid sum itself in a box of half width* 2^{-400} *about this point (a statement about the sum, not about* H_t*), and at* $t = t_* + 1.2 \times 10^{-12}$ *the signs* $+, -, +$ *of* H_t *at* $z_* - 4 \times 10^{-6}$*,* z_* *and* $z_* + 4 \times 10^{-6}$ *show that the pair is still real, which certifies that the collision lies below* -0.287979837008*: the upper end of the bracket is certified and the lower end is not. The extrapolation of the integrated* D^2 *was* 7% *early and the isolated pair estimate* -0.262 *was* 9% *late. Script* `phase_gas.py` *for (i) and (ii), output* `phase_gas.json`*; script* `phase_collide.py` *for (iii), output* `phase_collide.json`*.*

The bracket in (iii) uses that a pair at distance D in z obeys $D^2 \approx 8(t - t_*)$ near the collision (Section 8.4.2), so sign changes on a grid of step 0.01 in θ vanish at most $0.01^2/2 = 5 \times 10^{-5}$ above t_*. At $t = -0.287981$ the grid finds none, which does not exclude two real zeros less than 0.01 apart, and the lower end rests on the pair law, an asymptotic statement and not an enclosure.

The question is whether a gap functional, monotone as in (ii), can carry a positive lower bound from $t > 0$ down to $t = 0$ uniformly in the height, and the data suggest that gap functionals cannot. Both are minimised at Lehmer pairs, whose normalised gaps are expected, on the GUE conjecture, to be arbitrarily small at large height, so their infimum over all heights would be zero at $t = 0$. A Lyapunov functional for RH must be global, insensitive to individual close pairs, in the manner of the pair correlation that Rodgers and Tao used to prove the reverse inequality [289]; we did not find a candidate. Two rigorous companions of the data are in Sections 8.4.7 and 8.4.10: an exact formula for the drift of a gap, and a proof that no bound through the absolute value of the integrand gives a step uniform in the height [213].

15.6.3 The cost of the flow route

The published points of the phase diagram give the cost of descending along it (Table 8.14 lists the bounds on Λ). Polymath15 proved $\Lambda \le 0.22$ with the barrier $X \approx 6 \times 10^{10}$, and Table 1 of their paper lists, for each bound from $\Lambda \le 0.21$ down to $\Lambda \le 0.1$, the barrier X that the same method needs, with RH verified to height $X/2$ [282]: $X \approx 2 \times 10^{12}$ for 0.21, 5×10^{12} for 0.2, 2×10^{13} for 0.19, 7×10^{15} for 0.15 and 9×10^{21} for 0.1. Platt and Trudgian obtained $\Lambda \le 0.2$ from the second row, which by their account needs verification to 2.51×10^{12}, and the third row, 0.19, requires verification slightly above 10^{13} [247]. The product $\Lambda \log X$ is 5.46 at 0.22 and falls along the table from 5.95 at 0.21 to 5.06 at 0.1.

Proposition 15.6.2 (Computed). *The least squares fit of* $\Lambda = c/\log X$ *in* $1/\log X$ *to the bound* $\Lambda \le 0.22$ *and the twelve rows of the table gives* $c = 5.59$*, with residuals below* 0.013 *in* Λ*, which along the table are errors of up to a factor of about* 200 *in*

X; on this law a bound $\Lambda \le \delta$ requires a barrier X of order $e^{c/\delta}$, that is verification to height $X/2$: $X = 10^{48.6}$ for $\delta = 0.05$ and $10^{242.9}$ for $\delta = 0.01$. Since $\Lambda \log X$ falls along the table, the law $\log X = a/\Lambda + b$ fits its twelve rows better, with $a = 4.24$, $b = 8.28$ and residuals below 0.003 in Λ, within a factor 1.6 in X, and gives $X = 10^{40.4}$ for $\delta = 0.05$ and $10^{187.7}$ for $\delta = 0.01$. Script `phase_scaling.py`*, output* `phase_scaling.json`*.*

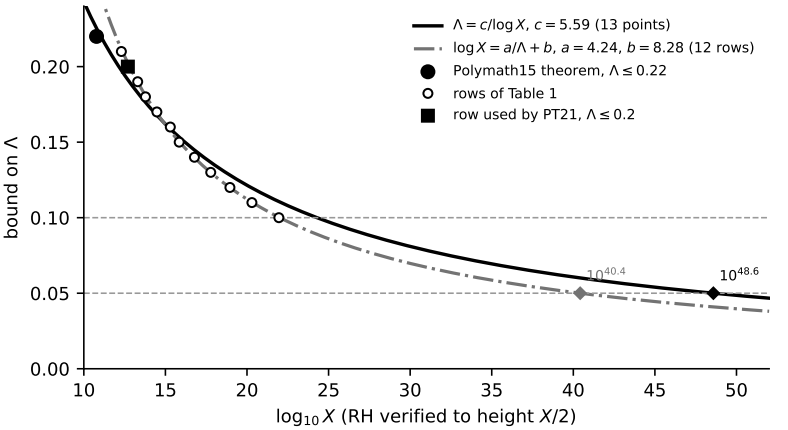


Figure 15.5: The bound $\Lambda \le 0.22$ of Polymath15 and the twelve rows of their table of conditional bounds [282] against the barrier X (verification height $X/2$), with the fits $\Lambda = c/\log X$ and $\log X = a/\Lambda + b$. Dashed lines mark $\Lambda = 0.1$ and 0.05.

Beyond the table the two laws should be read as orders of magnitude. Their form agrees with the heuristic that the flow at time t makes the zeros near height x real once $t \log x$ exceeds a constant, which is how the Riemann Siegel type approximation of H_t enters [282]. Along the table, halving the bound from 0.2 to 0.1 raises the required barrier from 5×10^{12} to 9×10^{21}, to about its power 1.7, and the power is 2 at every bound on the law $c/\log X$ and tends to 2 as the bound decreases on the law $\log X = a/\Lambda + b$. By the table itself the bound $\Lambda \le 0.15$ needs verification to a height of about 3.5×10^{15}, some 1200 times the record 3×10^{12} [247], and $\Lambda \le 0.1$ about 4.5×10^{21}; no conceivable computation reaches 0.05. The flow route produces a theorem at every stage, a monotone sequence of upper bounds for Λ, which is 0 exactly when RH holds [289], as no other route does, and it cannot reach the corner of the diagram by computation.

15.6.4 A programme

The phase diagram suggests five steps, of which the first three are engineering.

1. An evaluator: a Riemann Siegel type form of ξ_t and ξ_t', such as the effective

approximation of H_t in [282], in ball arithmetic, at a cost of about $\sqrt{\theta/2\pi}$ terms per point (Table 2.4), replacing the Pólya integral, whose required precision grows like $0.341\,\theta$ digits (Section 2.1.1); see Section 12.2.2 for $t = 0$.

2. Certificates in Thorin form: Theorem 15.5.6 or Proposition 15.5.7 at a target (t_0, α_0) up to the barrier.

3. The tail in the same framework: the barrier and the analytic region beyond it [282], so that the points of Corollary 15.3.3(b) and (c) are certified end to end and stated as GGC theorems.

4. A measurement: the optimal (t_0, α_0) on each level set as a function of the verification height, and the constant c of Proposition 15.6.2 as a function of the height.

5. The structural step: a tail at $t = 0$ below the basepoint $\frac{7}{8}$. Every certified point is a theorem, and by Proposition 15.4.1 no computed window distinguishes the corner from them; the step is the existence of the tail, and it must come from one of the mechanisms of Section 12.1.2.

15.6.5 Open problems

Problem 15.1. Compute $v_t(\alpha, \theta)$ at $t = 0$ and $t = 0.2$ near the Polymath barrier, heights near 3×10^{10}, with a Riemann Siegel type evaluator of $H_{0.2}$ such as the effective approximation of [282] (Section 2.2 gives the one for ζ), and measure how the difference between the functions $v_{0.2}/\pi$ and v_0/π grows with the height. Proposition 15.4.1(iii) measures it below 100. For $t = 0$ see work package 2 of Section 12.2.2 and Section 12.1.3.

Problem 15.2. Describe the boundary $\alpha^*(t)$ of $\mathcal{G}$ for $0 \le t \le 0.2$ as far as computation and the known zero free regions allow. Is α^* continuous, and is $(\alpha^*(t) - \frac{1}{2})^2 + \frac{t}{2}$ strictly decreasing in t where $\alpha^*(t) > \frac{1}{2}$? Theorem 15.3.2(i) shows that it is nonincreasing there. Problem 14.3 asks the neighbouring question for the flow clock T_t, $0 < t < 0.2$.

Problem 15.3. Give the symmetrised truncations of Section 7.2 a coordinate on the phase diagram: for some (t, α) outside the region of Corollary 15.3.3, is there a family of explicit truncations of ξ_t, built from finitely many terms of the Pólya integral or of a Dirichlet series and making no use of the zeros, as the symmetrised lattice truncations K_n of Subsection 7.2.1 are, that converges to ξ_t on compact sets, together with heights $h_n \to \infty$ such that for infinitely many n the truncation has no zero with $\operatorname{Re} s > \alpha$ and $|\operatorname{Im} s| < h_n$? Hurwitz's theorem then makes (t, α) a new point of $\mathcal{G}$, as in Corollary 14.12.2(c); at $t = 0$ and $\alpha = \frac{1}{2}$ the question for the K_n is Conjecture S (Theorem 7.2.7), and for the stages of Chapter 14 it is Problem 14.1.

Problem 15.4. Compute the Cholesky factor of the Weil Gram matrix of Section 15.6.1 for decreasing w and increasing L, and decide whether its entries converge to an object built from the arithmetic, the first test of a factorisation $W(g * \tilde{g}) = \|Ag\|^2$.

Problem 15.5. Find a global functional of the zeros of H_t, insensitive to individual close pairs, that is monotone along the flow and whose positivity at $t = 0$ excludes collisions at positive time. Biane's gap functional of Section 8.4.4 is local, the minimum over one gap of Laguerre's quantity, and Problems 8.1 and 12.5 ask for the control of the nonreal zeros that would make its brackets unconditional; the functional asked for here should instead be one quantity for all the zeros.

Notes

The phase diagram theorem is de Bruijn's strip theorem [73, Theorem 13] read through Theorem 15.2.3; its case $t = 0$ is in Section 8.4. Theorem 15.2.3, the certificate of Section 15.5 with Theorem 15.5.8, and the experiments of Sections 15.6.1 and 15.6.2 are from the authors' working report [264]. The bounds on Λ are those of de Bruijn [73], Ki, Kim and Lee [150], Polymath15 [282] and Platt and Trudgian [247], and the lower bound is Rodgers and Tao [289]. The half plane $\operatorname{Re} s > \frac{7}{8}$ is the announcement [226], discussed in Section 12.1.2. The description of the Thorin measure as one object with a prime face and a zero face, joined by the sine squared lemma, is developed for automorphic L functions in [265].

15.7 Exercises

On Sections 15.2 and 15.3.

15.1. Prove the semigroup identity $P_a * P_b = P_{a+b}$ from the characteristic function $e^{-a|\xi|}$ of P_a, and deduce the identity $\int_{\mathbb{R}} P_d(\theta - \gamma)(\theta^2 + u)^{-1} d\theta = \pi P_{d+\sqrt{u}}(\gamma)/\sqrt{u}$ used in Lemma 15.2.2.

15.2. Show that the Thorin density of Lemma 15.2.2(a), in the chart $z = \theta^2$, has total mass one, and that it converges vaguely to δ_{γ^2} as $d \downarrow 0$.

15.3. Show that the level sets of $t + 2(\alpha - \frac{1}{2})^2$ are the parabolas along which Theorem 15.3.2(i) carries the boundary of a certified region, and that the parabola through (t_0, α_0) meets the axis $\alpha = \frac{1}{2}$ at $t_0 + 2(\alpha_0 - \frac{1}{2})^2$.

15.4. Using Theorem 15.3.2(ii) and Corollary 15.3.3(d), find the α for which a zero free half plane $\operatorname{Re} s > \alpha$ at time zero would improve the bound $\Lambda \le 0.2$, and the smallest t for which the flowed half plane $\operatorname{Re} s > \frac{7}{8}$ gives a point of $\mathcal{G}$ at the basepoint 0.6.

15.5. Let F be the perturbation of Theorem 5.2.1, with a quadruple of zeros at $\frac{1}{2} \pm \frac{1}{4} \pm ig$. Assuming RH, show that the shifted clock of F at the basepoint α is a GGC exactly for $\alpha \geq \frac{3}{4}$, and compute its Thorin density near $\theta = g$ for $\alpha = 0.8$.

On Sections 15.4 and 15.5.

15.6. Assuming RH, show that the average of $v_0(\alpha, \theta)$ over a window of length H about θ is $\frac{1}{2}\log(\theta/2\pi) + O(\log\theta/H)$ when H is large compared with $\alpha - \frac{1}{2}$, and explain why the minimum over the window can be much smaller than the average when $\alpha - \frac{1}{2}$ is small compared with the mean spacing.

15.7. Show that $v_t(\alpha, 0) = \xi_t'/\xi_t(\alpha)$ is positive for every $t \geq 0$ and $\alpha > \frac{1}{2}$, using Lemma 15.2.1(ii), and compute it at $t = 0.2$, $\alpha = 0.6$ from the Pólya integral.

15.8. Prove Lemma 15.5.5 and show that it remains true with $\leq$ in place of $<$, the change of argument then lying in $(-\frac{\pi}{2}, \frac{\pi}{2})$.

15.9. Explain why the number of grid points in Theorem 15.5.6 grows like $e^{(\pi/4 - 2\eta)\theta}$ for a tilt $\eta < \pi/8$.

15.10. Show that positivity of v_t on the boundary of a rectangle does not by itself exclude a zero inside it, and explain why Proposition 15.5.7 asks for the whole rectangle.

On Section 15.6.

15.11. Show that for g supported in $[-L/2, L/2]$ the function $h = g * \tilde{g}$ is supported in $[-L, L]$, and that the prime sum in Weil's explicit formula runs over $n \leq e^L$.

15.12. Derive the x space form of the archimedean term of Section 15.6.1 from the integral representation $\psi(z) = \int_0^\infty (e^{-t}/t - e^{-zt}/(1 - e^{-t}))\, dt$.

15.13. For an isolated pair of zeros of H_t at distance D, show from the equation of motion that $D^2 \approx D_0^2 + 8t$ (compare Exercises 8.15 and 8.16). Explain why two certified sign changes of $\xi_t(\frac{1}{2} + i\theta)$ on a grid of step 0.01 show that the pair of Proposition 15.6.1(iii) is still real at t, while their absence does not show that it has left the line, and why the lower end of the bracket there rests on the law $D^2 \approx 8(t - t_*)$.

15.14. Fit $\Lambda = c/\log X$ to the thirteen points of Proposition 15.6.2 by least squares in $1/\log X$ and check the value $c = 5.59$; then fit $\Lambda = c' \log\log X/\log X$ to the same points and compare the barrier each law gives for $\Lambda = 0.1$ with the last row of the table, $X \approx 9 \times 10^{21}$.

Chapter 16

Pólya and the zeros of entire functions

This chapter reads Pólya's work through the reciprocal Xi law $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, its Thorin measure, the basepoint laws $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ and the $2\sin^2$ lemma. The statement that carries the chapter is Proposition 16.6.3: the atoms of the even part laws produced by Pólya's universal factor $\cosh(au)$ sit exactly where the argument clock of the basepoint law crosses half integers.

What is not claimed. Nothing here proves RH. Section 16.12 is conditional on a half plane that has not been refereed. A number is computed in floating point at the precision stated, and a certificate that says so is a floating point certificate with explicit analytic bounds, not an interval arithmetic proof.

16.1 Pólya's question

Pólya spent a decade, from 1918 to 1927, on a single question: when does the Fourier transform of a positive even kernel have only real zeros? The motivation was Riemann's formula

$$\begin{aligned} \Xi(t) &= \xi\big(\tfrac{1}{2}+it\big) = 2\int_0^\infty \Phi(u)\cos(tu)\,du, \\ \Phi(u) &= 2\sum_{n\geq 1}\Big(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\Big)\,e^{-\pi n^2 e^{2u}}, \end{aligned} \tag{16.1.1}$$

in which the Riemann Hypothesis is the statement that Ξ has only real zeros. The Thorin programme of this book reads the same function from the other side, through the reciprocal $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$, which is unconditionally the Laplace transform of a self decomposable law μ on $[0,\infty)$ (Theorem 5.1.1), and is the Laplace transform of a generalised gamma convolution (GGC) exactly when RH holds (Proposition 1.2.1). Pólya works on the kernel Φ, multiplying it by factors that preserve real zeros, while the Thorin programme works on the logarithm of the transform, asking for positivity of a measure.

This chapter asks three questions. Which of Pólya's results have not yet been brought into the Thorin programme? Which of them would help to show RH? Is the $2\sin^2$ lemma useful with regard to his work?

A remark on sources. Pólya's papers on zeros are collected in Volume II of his Collected Papers, edited by Boas [280]. We did not have the full text of every paper discussed below. The Acta paper of 1926 [273] was read in part, and what is said of it comes from its first page and from the careful secondary accounts of Newman and Wu [212], Lagarias [161], Gasper [91] and Haglund [105]; every formula used below for Pólya's Ξ^* was rederived and checked numerically. Whether a result is already part of the programme is judged by comparison with the earlier chapters, the first author's paper [249] (version 24 of arXiv 1708.02653) and monograph [252].

16.2 Notation and the Thorin dictionary

We use the Titchmarsh normalisation (16.1.1). Since Φ is even and positive,

$$\xi\big(\tfrac12+w\big)=2\int_0^\infty \Phi(u)\cosh(wu)\,du=\sum_{k\ge0}c_k w^{2k},\qquad c_k=\frac{2}{(2k)!}\int_0^\infty \Phi(u)u^{2k}\,du>0. \tag{16.2.1}$$

Put $F(s)=\xi(1/2+\sqrt{s})/\xi(1/2)=\sum_{k\ge0}a_k s^k$ with $a_k=c_k/c_0>0$. Then F is entire of order one half with $F(s)=\prod_\rho(1-s/s_\rho)$, $s_\rho=(\rho-1/2)^2$, the product taken over zeros in the upper half plane (one factor for each pair $\rho,1-\rho$), and $\varphi=1/F$ is the Laplace transform of the reciprocal Xi law μ; we write $Y\sim\mu$. Its exponent $\psi=\log F$ satisfies $\psi'(s)=\int_0^\infty e^{-sx}W(x)\,dx$ with the heat trace $W(x)=\sum_{\gamma>0}e^{(\rho-1/2)^2x}$, and the dictionary of Section 5.1 reads: $W\ge0$ (infinite divisibility, unconditional), $-W'\ge0$ (self decomposability, unconditional), W completely monotone (GGC, equivalent to RH). Under RH the Thorin measure is $U=\sum_{\gamma>0}\delta_{\gamma^2}$ and $Y\overset{d}{=}\sum_\gamma E_\gamma/\gamma^2$.

At a basepoint α the law $\mu^{(\alpha)}$ has Laplace transform $\xi(\alpha)/\xi(\alpha+\sqrt{s})$; it is a GGC if and only if $\alpha\ge\sigma_{\max}=\sup\operatorname{Re}\rho$, unconditionally for $\alpha\ge1$ (Theorem 1.2.2), and its Thorin distribution in the ordinate variable is the clock

$$U_\alpha(\tau)=\frac{1}{\pi}\,\Delta_0^\tau\arg\xi(\alpha+i\tau),\qquad \frac{d}{d\tau}U_\alpha(\tau)=\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau). \tag{16.2.2}$$

With τ_α the inverse of the clock and g a standard gamma subordinator, $\mu^{(\alpha)}$ is the law of $\int_0^\infty dg_v/\tau_\alpha(v)^2$.

The $2\sin^2$ lemma (Lemma 6.2.4) rests on the identity, for $x,w>0$ and $u=$

$w^2/2$,

$$e^{-wx}+wx-1=\int_0^\infty \log\Bigl(1+\frac{u}{z}\Bigr)\,\varpi_x(z)\,dz,$$
$$\varpi_x(z)=\frac{x}{\pi\sqrt{2z}}\,2\sin^2\Bigl(x\sqrt{z/2}\Bigr)\ge 0. \tag{16.2.3}$$

Consequently, if $\log f(\alpha+w)/f(\alpha)-wf'(\alpha)/f(\alpha)=\int(e^{-wx}-1+wx)e^{-\alpha x}\mu_f(dx)/x$ with $\mu_f\ge 0$, the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)}f(\alpha)/f(\alpha+w)$, $w=\sqrt{2u}$, is a GGC in u with Thorin density $\int\varpi_x(z)e^{-\alpha x}\mu_f(dx)/x$. Differentiating twice in w, the hypothesis $\mu_f\ge 0$ is the statement that $(\log f)''$ is completely monotone on (α,∞).

Notation. The function F is the function G of Section 1.2, and $Y\sim\mu$ is the clock variable T, so that $\varphi=1/F$ is its Laplace transform. The coefficients a_k are the scaled moments b_k of Section 13.2.4 (Table 13.4), not the exponents a_j of Section 5.1. The word clock in this chapter means the argument clock U_α, the cumulative Thorin mass $M_\varepsilon(\theta)=\pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$ of Section 3.4 with $\varepsilon=\alpha-\frac12$ and $\theta=\tau$, and not the Laplace transform $1/G$ of the earlier chapters. We write $b_\alpha=\frac{\xi'}{\xi}(\alpha)$ for the drift of Section 9.1. Finally $X=Z\sqrt{2Y}$ is the variable $Y_{1/2}$ of Section 10.1, with density f_X, and not the Pólya variable X of Section 1.2, whose density is $\Phi/\Xi(0)$.

16.3 Pólya's papers on zeros, one by one

P1. Approximation by polynomials with real zeros (1913)

Über Annäherung durch Polynome mit lauter reellen Wurzeln, Rend. Circ. Mat. Palermo 36 [266]. Completing Laguerre, Pólya characterised the entire functions that are locally uniform limits of real polynomials with only real zeros: they are exactly

$$f(z)=cz^me^{-\lambda z^2+\kappa z}\prod_k\left(1-\frac{z}{b_k}\right)e^{z/b_k},$$
$$m\in\{0,1,2,\dots\},\ \lambda\ge 0,\ c,\kappa\in\mathbb{R},\ b_k\in\mathbb{R}\setminus\{0\},\ \sum_k b_k^{-2}<\infty, \tag{16.3.1}$$

the Laguerre Pólya class $\mathcal{LP}$. The functions of type I, $cz^me^{\sigma z}\prod_k(1+z/b_k)$ with $c\ge 0$, $\sigma\ge 0$, $b_k>0$ and $\sum_k b_k^{-1}<\infty$, are exactly the locally uniform limits of polynomials with only negative zeros and positive coefficients. RH is the statement $\Xi\in\mathcal{LP}$.

P2. Pólya and Schur, two kinds of factor sequences (1914)

Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen, J. reine angew. Math. 144 [281]. A sequence (g_k) is a multiplier sequence if $\sum p_k x^k$ with only real zeros implies $\sum g_k p_k x^k$ with only real zeros. Pólya and Schur proved that a positive sequence is a multiplier sequence if and only if $\sum g_k z^k/k!$ is in $\mathcal{LP}$ of type I, if and only if every Jensen polynomial $\sum_{k=0}^{d} \binom{d}{k} g_k x^k$ has only real zeros.

P3. Genus zero and one (1915)

Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins, J. reine angew. Math. 145 [267]. Operations on entire functions of genus at most one that preserve reality of zeros. The statement used later is the shift lemma, quoted as Hilfssatz II of the 1926 paper P8 in Section 7.1.3: if G is real, of genus 0 or 1, with only real zeros, at least one of them, then $G(z+ia)+G(z-ia)$ has only real zeros for real a.

P4. Zeros of certain entire functions (1918)

Über die Nullstellen gewisser ganzer Funktionen, Math. Z. 2 [268]. The continuous Eneström Kakeya theorem: if f is positive and nondecreasing on $(0,1)$, then $\int_0^1 f(t)\cos(zt)\,dt$ and $\int_0^1 f(t)\sin(zt)\,dt$ have only real zeros. The Riemann kernel is positive but decreasing on $(0,\infty)$, so the theorem does not apply.

P5. Number theoretic remarks (1919)

Verschiedene Bemerkungen zur Zahlentheorie, Jahresber. DMV 28 [269]. Among other things, the Pólya conjecture that the summatory Liouville function $L(x) = \sum_{n\le x} \lambda(n)$ satisfies $L(x) \le 0$ for $x \ge 2$. It implies RH and is false: Haselgrove (1958) [111] disproved it and Tanaka (1980) [317] found the first sign change at $x = 906{,}150{,}257$.

P6. Zeros of successive derivatives (1922)

Über die Nullstellen sukzessiver Derivierten, Math. Z. 12 [270]. Pólya introduced the final set of a function, the set of limit points of the zeros of $f^{(n)}$ as $n \to \infty$, and determined it in several classes. The statements about real entire functions that we use are recorded under P13.

P7. A Fourier integral (1923)

On the zeros of an integral function represented by Fourier's integral, Messenger of Math. 52 [272]. A short note on the zeros of cosine transforms of positive kernels, a precursor of the 1926 and 1927 papers.

P8. The integral representation of the Riemann xi function (1926)

Bemerkung über die Integraldarstellung der Riemannschen ξ Funktion, Acta Math. 48 [273]. As $u \to +\infty$ the kernel is dominated by its $n = 1$ term, and Pólya replaced it by the even function

$$\Phi^*(u) = 4\pi^2 \left(e^{9u/2} + e^{-9u/2}\right) e^{-\pi(e^{2u}+e^{-2u})} = 8\pi^2 \cosh\left(\tfrac{9u}{2}\right) e^{-2\pi \cosh 2u}, \quad (16.3.2)$$

and proved that $\Xi^*(t) = 2\int_0^\infty \Phi^*(u)\cos(tu)\,du$ has only real zeros, with a zero count asymptotic to that of Ξ. The proof combines the reality of the zeros of $z \mapsto K_{iz}(a)$, $a > 0$, with the shift lemma P3. A refinement keeping the second exponential term is also real rooted.

P9. Zeros of certain trigonometric integrals (1926)

J. London Math. Soc. 1 [274]. Following Hardy's proof of infinitely many zeros on the critical line, Pólya explained why infinitely many real zeros is much easier than no complex zeros, and gave a general criterion for infinitely many real zeros. In Thorin language the infinitude of real zeros is the statement that the on line part $\varphi_{\text{on}} = \prod_{\text{on line}}(1 + s/\gamma^2)^{-1}$, a GGC whatever the zero set, has a Thorin measure with infinitely many atoms.

P10. Trigonometric integrals with only real zeros (1927)

Über trigonometrische Integrale mit nur reellen Nullstellen, J. reine angew. Math. 158 [275]. The central paper. A real analytic ϕ is a universal factor if $\int \mathcal{F}(t)e^{izt}dt$ with only real zeros implies the same for $\int \phi(t)\mathcal{F}(t)e^{izt}dt$, for every admissible kernel $\mathcal{F}$, that is, every real kernel that decays fast enough for these integrals to define entire functions of z. Pólya proved that ϕ is a universal factor if and only if $z \mapsto \phi(iz)$ belongs to $\mathcal{LP}$. Hence $e^{\lambda t^2}$ ($\lambda > 0$) and $\cosh(bt)$ ($b \in \mathbb{R}$) are universal factors. He also proved a Mellin criterion: if $\mathcal{F}$ is admissible, holomorphic near the origin, and $H(z) = \int_0^\infty \mathcal{F}(t)t^{z-1}dt$ has no zeros in $\mathbb{C} \setminus (-\infty, 0]$, then $\int \mathcal{F}(t^{2q})e^{izt}dt$ has only real zeros for every positive integer q. Consequences: $\int e^{-t^{2q}}e^{izt}dt$ and $\int \cosh(bt)e^{-a\cosh t}e^{izt}dt$ with $a > 0$ and $b \in \mathbb{R}$ have only real zeros. The factor $e^{\lambda t^2}$ is the starting point of de Bruijn (1950) and Newman (1976), hence of the de Bruijn Newman constant Λ (Table 8.14).

P11. Jensen's investigations (1927)

Über die algebraisch funktionentheoretischen Untersuchungen von J. L. W. V. Jensen, Kgl. Danske Vid. Selsk. Math. Fys. Medd. 7 [276]. Pólya edited and proved Jensen's criteria: a real entire function of genus at most one is in $\mathcal{LP}$ if and only if all its Jensen polynomials have only real zeros, together with Jensen's

integral criteria. This is the source of the Jensen polynomial approach to RH (Section 13.3.1); Farmer (2022) [85] argued that this direction cannot reach RH.

P12. Fourier's work on transcendental equations (1930)

Quart. J. Math. Oxford 1 [278]. The Fourier Pólya conjectures on the number of nonreal zeros of successive derivatives.

P13. Derivatives and analytic character (1943)

Bull. AMS 49 [279]. The Pólya Wiman conjecture: if f is real entire of order below two with finitely many nonreal zeros, then $f^{(n)}$ has only real zeros for all large n; proved by Craven, Csordas and Smith (1987) [67] and Kim (1990) [151], with the Fourier Pólya conjecture settled by Ki and Kim (2000) [149]. For Ξ, whose zeros lie in a strip, the theorem of Ki and Kim gives the local statement that for every R the zeros of $\Xi^{(n)}$ in $|z| < R$ are real for all large n; Conrey showed that the proportion of real zeros of $\Xi^{(n)}$ tends to one [61]. Global real rootedness of $\Xi^{(n)}$ for some n is not known.

P14. The Hilbert Pólya suggestion

In a 1982 letter to Odlyzko, Pólya recalled suggesting to Landau, around 1914, that the zeros might be the eigenvalues of a self adjoint operator. The Krein string route of Corollary 9.2.3, which assumes RH and simple zeros, is a probabilistic form of this.

16.4 The Laguerre Pólya class in the Thorin variable

Proposition 16.4.1. *The following are equivalent.*

(i) RH.

(ii) $F(s) = \xi(1/2 + \sqrt{s})/\xi(1/2)$ *belongs to the Laguerre Pólya class of type I, that is, it is a locally uniform limit of polynomials with only negative zeros.*

(iii) The sequence $g_k = k!\, a_k$*, where* $F(s) = \sum a_k s^k$*, is a Pólya Schur multiplier sequence.*

(iv) For every d *the Jensen polynomial* $J_d(x) = \sum_{k=0}^{d} \binom{d}{k} g_k x^k$ *has only real, negative zeros.*

(v) The reciprocal Xi law is a GGC.

Proof. Under RH every $s_\rho = -\gamma^2$ is negative and $F(s) = \prod_\gamma (1 + s/\gamma^2)$ with $\sum \gamma^{-2} < \infty$, which is of type I; conversely a function of type I has only real

nonpositive zeros, so every $s_\rho \le 0$, and $s_\rho = (\rho - 1/2)^2 \le 0$ forces $\rho - 1/2 \in i\mathbb{R}$. The equivalence of (ii), (iii) and (iv) is the theorem of Pólya and Schur, since $\sum g_k s^k/k! = F(s)$. The equivalence with (v) is Proposition 1.2.1. □

The ladder (iv) is the coefficient counterpart of the monotonicity ladder $T_k = (-1)^k W^{(k)} \ge 0$ of Theorem 5.1.6. Its first rung, for F and for every derivative of F, is the Turán inequality $g_k^2 \ge g_{k-1} g_{k+1}$ for $k \ge 1$, the condition that the Jensen polynomials $J_{2,n}$, $n \ge 0$, are real rooted; for F alone it is the case $k = 1$. With $m_k = (2k)!\, c_k = 2\int_0^\infty \Phi(u) u^{2k}\, du$ it reads $m_k^2 \ge \frac{2k-1}{2k+1} m_{k-1} m_{k+1}$, which is the Turán inequality proved unconditionally by Csordas, Norfolk and Varga (1986) [68]; in the scaled moments a_k, which are the b_j of Section 13.2.4, it is (13.2.2), whose margins are in Table 13.4.

Numerics. Computing the c_k from (16.2.1) by quadrature on the kernel reproduces $\Xi(0) = \xi(1/2) = 0.497120778188314$ to fifteen digits and gives $a_1 = 0.0231049931$ and the values a_2, a_3, a_4 of Table 13.4. The Turán ratios $g_k^2/(g_{k-1} g_{k+1})$ decrease slowly from 1.0748 at $k = 1$ and stay above 1.0109 for all $k \le 40$. The Jensen polynomials J_d of every degree $2 \le d \le 30$, computed with 120 digits, have only real negative zeros.

Remark 16.4.2. The ladder (iv) at fixed degree d is a finite set of polynomial inequalities in $a_1, \dots, a_d$, and so is checkable to any degree; but for each fixed d it is implied by information about finitely many zeros, which is why it cannot prove RH by itself.

16.5 Pólya frequency functions: the two sided criterion

Schoenberg (1951) [300] proved that an integrable function f on $\mathbb{R}$ is a Pólya frequency function of every order (PF_∞) if and only if its bilateral Laplace transform is $1/\Psi(s)$ on a strip around the origin, with Ψ a function of the form (16.3.1) with $m = 0$ and $\Psi(0) > 0$ that is not of the form $Ce^{\kappa s}$. The name honours Pólya, and the theorem is the two sided counterpart of the Thorin theory.

Proposition 16.5.1. *Let $Y \sim \mu$ be the reciprocal Xi law and Z a standard normal variable independent of Y. Unconditionally, $X = Z\sqrt{2Y}$ has a smooth symmetric density f_X with characteristic function and bilateral Laplace transform*

$$\mathbb{E}e^{itX} = \frac{\xi(1/2)}{\xi(1/2+t)}, \qquad \mathbb{E}e^{zX} = \frac{\Xi(0)}{\Xi(z)} \quad (|\operatorname{Re} z| < \gamma_1). \tag{16.5.1}$$

Moreover RH holds if and only if f_X is PF_∞.

This is the equivalence of Section 1.2, between RH and the law of $Z\sqrt{2Y}$ as a sum of Laplace variables, read through Schoenberg's theorem; the same equivalence is stated for the Laguerre Pólya class in [153, Theorem 2.3 and Corollary 2.4], and its forward half is item (iv) of Theorem 13.1.4.

Proof. $\mathbb{E}e^{itX} = \mathbb{E}e^{-t^2Y} = \varphi(t^2) = \xi(1/2)/\xi(1/2+t)$, which is integrable in t since $\xi(1/2+t)$ grows like $\exp(\frac{t}{2}\log t)$, so f_X exists and is smooth. For real z, $\mathbb{E}e^{zX} = \mathbb{E}e^{z^2Y} = \varphi(-z^2)$, and $\varphi(-z^2) = \xi(1/2)/\xi(1/2+iz) = \Xi(0)/\Xi(z)$ by evenness; the Laplace transform of the positive measure μ converges up to its first singularity on the negative real axis (Landau's theorem), which is at $-\gamma_1^2$, because a zero off the line gives a nonreal s_ρ and the zero of lowest ordinate is on the line. Under RH, $\Xi(z)/\Xi(0) = \prod_\gamma(1 - z^2/\gamma^2) = \prod_\gamma(1 - z/\gamma)e^{z/\gamma}(1 + z/\gamma)e^{-z/\gamma}$ is of Schoenberg's form, so f_X is PF_∞. Conversely if f_X is PF_∞ then $\Xi(z)/\Xi(0)$ agrees on a strip with an entire function having only real zeros, hence everywhere, and RH follows. □

The Thorin atoms γ^2 become Schoenberg's $\pm 1/\gamma$. Under RH, $X \overset{d}{=} \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables L_γ, because a Laplace variable is $Z\sqrt{2E}$. The PF ladder $\mathrm{PF}_2 \supset \mathrm{PF}_3 \supset \cdots \supset \mathrm{PF}_\infty$ is a third ladder, after T_k and the Jensen degrees. PF_2 is log concavity of f_X.

Numerics. Fourier inversion of (16.5.1) gives f_X. On $[0, 1.6]$ with step 0.02 every second difference of $\log f_X$ is negative (the largest is -2.6×10^{-5}), so PF_2 holds on the grid; all 3000 random pairs of ordered triples $x_1 < x_2 < x_3$, $y_1 < y_2 < y_3$ in $[-0.8, 0.8]$ have a positive normalised determinant $\det[f_X(x_i - y_j)]_{i,j\le 3} / \prod_i f_X(x_i - y_i)$, the smallest being 6.7×10^{-9}.

Problem 16.1. Prove unconditionally, in exact or interval arithmetic, that f_X is log concave, that is PF_2 (Theorem 16.11.5 gives a floating point certificate), and then that it is PF_n for $n \ge 3$ (Section 16.11). The case of PF_2 would be a two sided analogue of the unconditional self decomposability of μ, one rung up the Schoenberg ladder.

16.6 The universal factor $\cosh(au)$ and the clock

Multiplying the kernel by e^{au} is an Esscher tilt of the Pólya density and moves the basepoint: $\int_{-\infty}^{\infty} \Phi(u)e^{au}\cosh(wu)\,du = \frac{1}{2}\left(\xi(\frac{1}{2}+a+w) + \xi(\frac{1}{2}+a-w)\right)$. The tilt alone destroys evenness. Pólya's universal factor $\cosh(au)$ is the even average of the two tilts. Define, for $a \ge 0$ and $\alpha = 1/2 + a$,

$$A_\alpha(w) = 2\int_0^\infty \Phi(u)\cosh(au)\cosh(wu)\,du = \tfrac{1}{2}\left(\xi(\alpha+w) + \xi(\alpha-w)\right), \quad (16.6.1)$$

the even part in w of the basepoint function $w \mapsto \xi(\alpha + w)$. For real t, $A_\alpha(it) = \mathrm{Re}\,\xi(\alpha + it)$, and in the variable $s = 1/2 + w$ this is Lagarias's function $A_h(s) = \frac{1}{2}(\xi(s+h) + \xi(s-h))$ with $h = a$.

Theorem 16.6.1 (Pólya, de Bruijn, Lagarias). *(i) Under RH, A_α has only purely imaginary zeros for every $\alpha > 1/2$.*

(ii) If every zero of Ξ satisfies $|\operatorname{Im} t| \le \Delta$, then every zero of $t \mapsto A_\alpha(it)$ satisfies $|\operatorname{Im} t| \le \max(\Delta^2 - a^2, 0)^{1/2}$. In particular, unconditionally, A_α has only purely imaginary zeros for $\alpha \ge 1$, and they are simple.

(iii) If A_{α_n} has only imaginary zeros along a sequence $\alpha_n \downarrow 1/2$, then RH holds.

Proof. (i) $\cosh(au)$ is a universal factor since $\cos(az) \in \mathcal{LP}$. (ii) is de Bruijn's strip theorem for the factor cosh, and $\Delta \le 1/2$ because ζ has no zeros with $\operatorname{Re} s > 1$; simplicity is Lagarias's theorem [161]. (iii) $A_\alpha \to \xi(1/2 + w)$ locally uniformly as $\alpha \downarrow 1/2$, and Hurwitz's theorem applies. □

Corollary 16.6.2 (Even part basepoint laws). *Whenever A_α has only imaginary zeros $\pm i\tau_k$, the function $\varphi_\alpha^P(s) = \xi(\alpha)/A_\alpha(\sqrt{s})$ is the Laplace transform of a GGC law μ_α^P with Thorin measure $\sum_k \delta_{\tau_k^2}$, so μ_α^P is the law of $\sum_k E_k/\tau_k^2$. This holds unconditionally for $\alpha \ge 1$, for all $\alpha > 1/2$ under RH, and RH follows if it holds along a sequence $\alpha_n \downarrow 1/2$.*

Proof. $A_\alpha(\sqrt{s})$ is entire of order one half in s with positive Taylor coefficients by (16.6.1), value $\xi(\alpha) > 0$ at $s = 0$, and zeros $s = -\tau_k^2$; the genus zero product gives $\varphi_\alpha^P(s) = \prod_k (1 + s/\tau_k^2)^{-1}$. □

Proposition 16.6.3 (Half integer crossings). *Let α be such that ξ has no zeros in $\operatorname{Re} s \ge \alpha$ (this holds for $\alpha > \sigma_{\max}$, and unconditionally for every $\alpha \ge 1$, since ζ has no zeros on $\operatorname{Re} s = 1$). Then the clock U_α is continuous and strictly increasing from $U_\alpha(0) = 0$, and the positive zeros of $t \mapsto \operatorname{Re} \xi(\alpha + it)$ are exactly*

$$\tau_k = \tau_\alpha\big(k - \tfrac{1}{2}\big), \qquad k = 1, 2, \dots, \tag{16.6.2}$$

where τ_α is the inverse of the clock. Consequently the Thorin distribution of the even part law is the rounding of the Thorin distribution of the basepoint law,

$$U_\alpha^P\big((0, T^2]\big) = \big\lfloor U_\alpha(T) + \tfrac{1}{2} \big\rfloor, \tag{16.6.3}$$

and in law $\mu_\alpha^P = \mathcal{L}\big(\sum_k E_k/\tau_\alpha(k - \frac{1}{2})^2\big)$ while $\mu^{(\alpha)} = \mathcal{L}\big(\int_0^\infty dg_v/\tau_\alpha(v)^2\big)$: the even part law is the midpoint rule for the gamma clock integral.

Proof. Since no zero has $\beta_\rho \ge \alpha$, the Hadamard sum $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) = \sum_\rho \frac{\alpha - \beta_\rho}{|\alpha + i\tau - \rho|^2}$ has every term positive, so the clock is strictly increasing, and $\xi(\alpha + i\tau) \ne 0$. Write $\xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| e^{i\pi U_\alpha(\tau)}$, which is legitimate since $\xi(\alpha) > 0$. Then $\operatorname{Re} \xi(\alpha + i\tau) = |\xi(\alpha + i\tau)| \cos(\pi U_\alpha(\tau))$ vanishes exactly when $U_\alpha(\tau) \in \mathbb{Z} + \frac{1}{2}$, and a strictly increasing clock starting at 0 and tending to infinity (by Stirling, $U_\alpha(\tau) \sim \frac{\tau}{2\pi} \log \tau$) crosses each level $k - \frac{1}{2}$ exactly once, transversally. This gives (16.6.2) and simple zeros. Every zero ρ has $|\beta_\rho - \frac{1}{2}| < \alpha - \frac{1}{2}$, by the hypothesis and the functional equation, so Theorem 16.6.1(ii) with $\Delta = \alpha - \frac{1}{2}$ shows that A_α has no zeros off the imaginary axis, and the corollary on even part laws applies;

counting levels below $U_\alpha(T)$ then gives (16.6.3). The two laws follow from the gamma process representation: the increments of g over $[k-1,k]$ are independent standard exponentials, and replacing $\tau_\alpha(v)$ on $[k-1,k]$ by its midpoint value gives the sum. □

Under the hypothesis of Proposition 16.6.3, which holds unconditionally for $h \geq \frac{1}{2}$ and under RH for every $h > 0$, the proposition explains why the zeros of A_h are asymptotically equally spaced, as in Lagarias's theorem [161]: they are level crossings of a smooth increasing clock whose derivative $\pi^{-1} \operatorname{Re} \xi'/\xi(\alpha + i\tau)$ is a Cauchy smoothing of the zero density, with fluctuations of lower order than its mean $\frac{1}{2\pi} \log \frac{\tau}{2\pi}$.

Numerics. Table 16.1 lists, for four basepoints, the number of zeros of $\operatorname{Re} \xi(\alpha+it)$ on $(0, 100]$, the clock at 100, its rounding, and the largest distance of the clock from a half integer at the computed zeros (on a grid of step 0.01 with linear interpolation). As $\alpha \downarrow 1/2$ the atoms converge to the ordinates.

α	zeros on $(0, 100]$	$U_\alpha(100)$	$\lfloor U_\alpha(100) + \frac{1}{2} \rfloor$	max distance to $\mathbb{Z} + \frac{1}{2}$
1.00	29	29.1112	29	2.7×10^{-5}
0.75	29	29.0548	29	5.6×10^{-5}
0.60	29	29.0218	29	1.7×10^{-4}
0.52	29	29.0044	29	1.6×10^{-3}

Table 16.1: Atoms of the even part law against the rounded clock of the basepoint law.

Remark 16.6.4 (Is the Pólya family easier?). Real rootedness of A_α is a counting condition: it can be certified to a given height by sign changes of $\operatorname{Re} \xi(\alpha + it)$ together with an argument principle count, exactly as in Turing's method. So the Pólya family is not easier in principle, but it converts the analytic positivity of $\operatorname{Re} \xi'/\xi$ into a discrete, verifiable statement about atoms. The zero trajectories $\alpha \mapsto \tau_k(\alpha)$ satisfy $d\tau_k/d\alpha = \operatorname{Re} \xi'(s)/\operatorname{Im} \xi'(s)$ at $s = \alpha + i\tau_k$. Under RH no two trajectories collide for $\alpha > \frac{1}{2}$, because the zeros are simple (Proposition 16.6.3); conversely a nonimaginary zero of A_α can appear only through a collision of two imaginary zeros, provided no zeros enter from infinity as α decreases, which needs uniform control of the zero counts in α. In this heuristic sense RH is the statement that no two trajectories collide before $\alpha = \frac{1}{2}$, a discrete analogue of the de Bruijn Newman dynamics.

16.7 Pólya's Ξ^* as the archimedean model

Proposition 16.7.1. *With* (16.3.2),

$$\Xi^*(t) = 4\pi^2\left[K_{\frac{9}{4}+\frac{it}{2}}(2\pi) + K_{\frac{9}{4}-\frac{it}{2}}(2\pi)\right] = 8\pi^2 \operatorname{Re} K_{\frac{9}{4}+\frac{it}{2}}(2\pi). \tag{16.7.1}$$

Proof. Substitute $v = 2u$ in $2\int_0^\infty \Phi^*(u)\cos(tu)\,du$, write $\cosh(9v/4)\cos(tv/2)$ as the average of $\cos((t/2 \mp 9i/4)v)$, and use $\int_0^\infty e^{-a\cosh v}\cos(\omega v)\,dv = K_{i\omega}(a)$ and the evenness of K_ν in ν. □

This is the identity of Section 8.7, where $P(\theta) = K_{i\theta/2}(2\pi)$ and $\Xi^*(\theta) = 4\pi^2[P(\theta - \frac{9i}{2}) + P(\theta + \frac{9i}{2})]$.

Formula (16.7.1) says that Ξ^* is exactly the construction (16.6.1) applied to the Bessel function: it is the even part, at basepoint $9/4$ in the order variable, of $\nu \mapsto K_\nu(2\pi)$. Pólya's theorem that $K_{iz}(a)$ has only real zeros is the Riemann Hypothesis for this function: its zeros in ν lie on $\operatorname{Re}\nu = 0$. The basepoint $9/4$ lies strictly to the right of that line, so by the argument of Theorem 16.6.1 and Proposition 16.6.3 the even part has only real zeros (the argument needs $\operatorname{Re}\frac{d}{d\nu}\log K_\nu(2\pi) > 0$ on $\operatorname{Re}\nu > 0$, and this holds because $\nu \mapsto K_\nu(2\pi)$ is even, of order one, with only the zeros $\pm iz_n$, $z_n > 0$, so that $K_\nu(2\pi) = K_0(2\pi)\prod_n(1+\nu^2/z_n^2)$ and each term $\frac{d}{d\nu}\log(1+\nu^2/z_n^2) = (\nu+iz_n)^{-1} + (\nu - iz_n)^{-1}$ has positive real part there), located where the Bessel clock $\pi^{-1}\arg K_{9/4+it/2}(2\pi)$ crosses half integers.

Proposition 16.7.2 (Half Gram points). *Let t_k^* be the positive zeros of Ξ^* in increasing order and ϑ the Riemann Siegel theta function. Then*

$$\frac{\vartheta(t_k^*)}{\pi} = k - \frac{3}{2} - \frac{2\pi^2 - 3}{\pi\, t_k^*} + O\big((t_k^*)^{-2}\big). \tag{16.7.2}$$

Hence, with $N^*(T) = \#\{k : t_k^* \le T\}$, $N^*(T) = \lfloor \vartheta(T)/\pi + \frac{3}{2}\rfloor$ for all large T outside the intervals $[t_k^*, t_k^* + O(1/(t_k^*\log t_k^*)))$, on which $N^*(T)$ is one larger; in particular $N^*(T) - \vartheta(T)/\pi$ is bounded, whereas $N(T) = \vartheta(T)/\pi + 1 + S(T)$.

Sketch. For $\nu = \frac{9}{4} + \frac{it}{2}$, the ascending series of $I_{-\nu}$ and the reflection formula give $K_\nu(2\pi) = \frac{1}{2}\Gamma(\nu)\pi^{-\nu}\big(1 + \frac{\pi^2}{1-\nu} + O(t^{-2})\big)$, the companion term $\frac{1}{2}\Gamma(-\nu)\pi^{\nu}$ being smaller by a factor of order $t^{-9/2}$. With $s = \frac{1}{2} + it$ we have $\Gamma(\nu) = (\frac{s}{2}+1)\frac{s}{2}\,\Gamma(\frac{s}{2})$ and $\Gamma(\frac{s}{2})\pi^{-it/2} = |\cdot|e^{i\vartheta(t)}$, so the phase of $K_\nu(2\pi)$ is $\vartheta(t) + \arg s + \arg(s+2) + \operatorname{Im}\frac{\pi^2}{1-\nu} + O(t^{-2})$. Here $\arg s + \arg(s+2) = \pi - \frac{3}{t} + O(t^{-3})$ and $\operatorname{Im}\frac{\pi^2}{1-\nu} = \frac{2\pi^2 t}{t^2+25/4} = \frac{2\pi^2}{t} + O(t^{-3})$, so the phase is $\vartheta(t) + \pi + \frac{2\pi^2-3}{t} + O(t^{-2})$. The zeros of the real part are where this phase lies in $\frac{\pi}{2} + \pi\mathbb{Z}$, which is (16.7.2); the indexing is fixed by the first zero. □

Compare the Thorin form of the zero count, $U((0,t]) = \vartheta(\sqrt{t})/\pi + 1 + S(\sqrt{t})$ under RH. Pólya's Ξ^* replaces the arithmetic fluctuation S by a bounded rounding

error: by Proposition 16.7.2 its Thorin distribution $N^*(\sqrt{t})$ is the rounding of the smooth archimedean distribution $\vartheta(\sqrt{t})/\pi + 1$. This is the distribution that the $2\sin^2$ lemma computes for the gamma factor in Theorem 6.2.12, up to an affine function of $\sqrt{t}$: that theorem gives $(\vartheta(\sqrt{t}) - \vartheta'(0)\sqrt{t})/\pi$ for the compensated factor $e^{-c\sqrt{s}}/G(s)$, with $c = -\vartheta'(0)$. In this sense Ξ^* is the zeta function with ζ replaced by its first Dirichlet term and the result symmetrised, and its reciprocal $\Xi^*(0)/\Xi^*(\sqrt{-s})$ is, by Pólya's theorem, an unconditional GGC with Thorin measure $\sum_k \delta_{(t_k^*)^2}$.

Numerics. Ξ^* has 29 zeros in $(0, 100]$, as does Ξ, but they are displaced: the first eight are 8.9928, 19.0654, 24.4748, 28.9750, 33.0069, 36.7367, 40.2491, 43.5947. There are 138 zeros up to height 300; at the last, $t^*_{138} = 299.627$, the deviation from $k - \frac{3}{2}$ times t_k^* is -5.329, against the constant $-(2\pi^2 - 3)/\pi = -5.328$ of (16.7.2) (Figure 16.1, right).

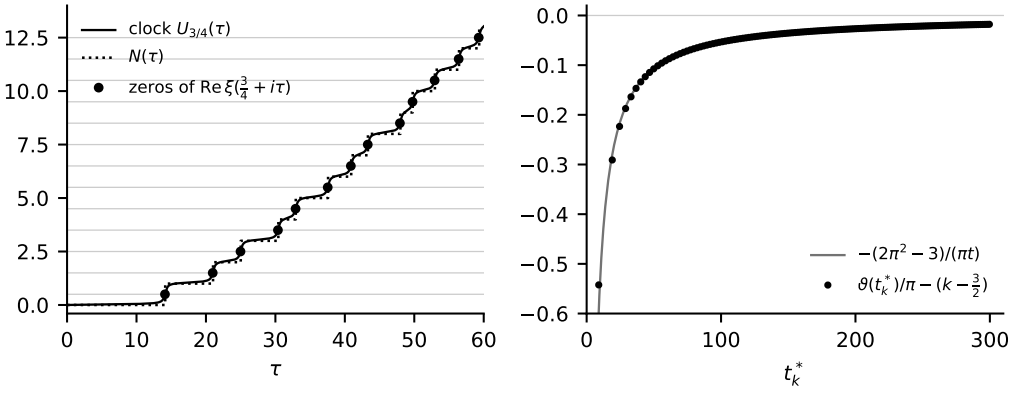


Figure 16.1: Left: the clock $U_{3/4}(\tau)$ of the basepoint law at $\alpha = 3/4$ for $0 \le \tau \le 60$, the zero count $N(\tau)$ (dotted), and the zeros of $\operatorname{Re}\xi(3/4 + i\tau)$, which sit exactly on the half integer levels (grey lines). Right: $\vartheta(t_k^*)/\pi - (k - \frac{3}{2})$ for the 138 zeros of Pólya's Ξ^* below height 300, with the asymptote $-(2\pi^2 - 3)/(\pi t)$ of Proposition 16.7.2.

16.8 Pólya's Mellin criterion applied to the Riemann kernel

Take $q = 1$ and $\mathcal{F}(v) = \Phi(\sqrt{v})$ in P10, so that $\mathcal{F}(t^2) = \Phi(t)$ and the conclusion would be RH. The criterion asks that

$$M(z) = \int_0^\infty \Phi(u)\, u^{2z-1}\, du = \tfrac{1}{2} \int_0^\infty \mathcal{F}(v) v^{z-1} dv \tag{16.8.1}$$

have no zeros in $\mathbb{C}\setminus(-\infty,0]$. The integral converges for $\operatorname{Re} z>0$, and $M(k+\frac12)=\frac12(2k)!\,c_k>0$ and $2M(\frac12)=\xi(\frac12)$. Near the origin $\Phi(u)=\sum_{j\ge0}p_ju^{2j}$, with radius of convergence $\pi/4$, $p_0=\Phi(0)=0.893394$ and $p_1=-8.36525$; integrating termwise over $(0,\delta)$, $\delta<\pi/4$, gives $\int_0^\delta\Phi(u)u^{2z-1}du=\sum_j p_j\delta^{2z+2j}/(2z+2j)$, which continues M to a meromorphic function with simple poles at $z=0,-1,-2,\dots$, and this continuation is the function that the criterion refers to.

The criterion does not apply to Φ. If X_Φ has the Pólya density $\Phi/\Xi(0)$ (the variable X of Theorem 5.2.3, and not the X of this chapter), then $\mathbb{E}|X_\Phi|^{2s}=2M(s+\frac12)/\Xi(0)$ with the M of (16.8.1), so that theorem says that M has a simple zero at $z_1=s_1+\frac12=-7.99418226+2.93776707\,i$, which lies in $\mathbb{C}\setminus(-\infty,0]$. We checked this independently (`polya_mellin_z1.py`, at 60 digits, with two quadrature layouts): $|M|=8.0\times10^{-16}$ at the point $s_1+\frac12$ printed in that theorem, and on the circle $|z-z_1|=\frac12$ we have $|M|\ge0.224$ and the winding number of M is 1, so that z_1 is the only zero in the disc. So the hypothesis of the criterion fails for $q=1$. For $q\ge2$ the function $\mathcal{F}(t^{2q})$ has no t^2 term whenever $\mathcal{F}$ is holomorphic near the origin, whereas $\Phi(t)=0.893394-8.36525\,t^2+\cdots$, so there is no factorisation $\Phi(t)=\mathcal{F}(t^{2q})$ with $q\ge2$. The Mellin route of the 1927 paper is therefore closed for Φ itself, by the zero that closes the Bernstein Pick route in Theorem 5.2.3.

The zero z_1 is the only obstruction that we found. An argument principle scan of the continuation (`polya_mellin.py`, floating point) finds no zero with $0.05\le|\operatorname{Im} z|\le14.05$ in $-3.95\le\operatorname{Re} z\le8.05$. A plain Gauss Legendre rule for (16.8.1) produces spurious zeros with $\operatorname{Re} z<\frac12$, because the integrand has an integrable singularity $u^{2\operatorname{Re} z-1}$ at the origin that oscillates like $\cos(2\operatorname{Im} z\log u)$; the termwise integration on $(0,\frac14)$ removes the difficulty.

16.9 Is the $2\sin^2$ lemma useful with regard to Pólya's work?

16.9.1 Where it helps

The lemma is applied at a basepoint α through the weight $e^{-\alpha x}$ in its hypothesis, which on the kernel side is the tilt e^{au}, the non even half of Pólya's universal factor $\cosh(au)$; it computes the continuous Thorin densities of which Pólya's approximations are the roundings, up to the affine term of its compensation (Theorem 6.2.12, Proposition 16.7.2); and positivity of the Thorin density at a basepoint makes the clock increasing along $\operatorname{Re} s=\alpha$ and places the real zeros of $\operatorname{Re}\xi(\alpha+it)$ at its half integer crossings (Proposition 16.6.3).

16.9.2 Where it cannot help

The Bessel test. The Bessel model $f(\nu)=K_\nu(2\pi)$ is even in ν, of order one, with zeros $\pm iz_n$ on $\operatorname{Re}\nu=0$ (Section 16.7) and $\sum z_n^{-2}<\infty$, so the product formula and the basepoint theorem (Theorem 1.2.2) apply verbatim, as in Theorem 8.7.10,

which is the case $\alpha = 0$: $K_\alpha(2\pi)/K_{\alpha+\sqrt{s}}(2\pi)$ is a GGC for every $\alpha \geq 0$, with Lorentzian Thorin density when $\alpha > 0$. The hypothesis of the lemma at basepoint α is that $(\log f)''$ is completely monotone on (α, ∞). Table 16.2 shows the signs of $(-1)^j d^{2+j} \log K_\nu(2\pi)/d\nu^{2+j}$: the third entry is negative at every $\nu \leq 5$ and a later entry fails at $\nu = 8$. So the primal measure of the lemma is signed at every basepoint tested, although the reciprocal law is a GGC at every basepoint. The lemma's success for ξ at $\alpha > 1$ is a cancellation, as Chapter 9 already observes.

ν	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
0.25	0.1482	6.7×10^{-4}	-2.7×10^{-3}	-9.8×10^{-5}	3.9×10^{-4}	3.6×10^{-5}
1	0.1469	2.6×10^{-3}	-2.5×10^{-3}	-3.7×10^{-4}	3.2×10^{-4}	1.3×10^{-4}
3	0.1373	6.6×10^{-3}	-1.3×10^{-3}	-6.8×10^{-4}	-1.4×10^{-5}	1.5×10^{-4}
5	0.1223	8.0×10^{-3}	-1.5×10^{-4}	-4.6×10^{-4}	-1.6×10^{-4}	6.9×10^{-6}
8	0.0991	7.1×10^{-3}	5.6×10^{-4}	-6.9×10^{-5}	-7.7×10^{-5}	-3.5×10^{-5}

Table 16.2: $(-1)^j \frac{d^{2+j}}{d\nu^{2+j}} \log K_\nu(2\pi)$.

Kernel multipliers. Pólya's universal factors $e^{\lambda u^2}$ and $\cosh(au)$ multiply the kernel, but the lemma acts on $\log f$ and has no composition rule that carries a lemma statement at one value of λ or a to another, which is why the heat flow row of Table 15.1 stands apart from the basepoint row.

16.9.3 Verdict

The lemma is not a mechanism for producing real zeros: in the Bessel model, where Pólya's theorem gives real zeros and every basepoint law is a GGC, the positivity the lemma needs is false. Any route to RH through the lemma must therefore control a cancellation between signed primal measures, and Pólya's own tools (universal factors, difference equations for K_ν, Hermite Biehler) are the natural candidates for that control.

16.10 What would help show RH

None of Pólya's theorems crosses into the critical strip. Each stops at the same place as the basepoint programme.

1. The cosh factor and de Bruijn's strip theorem give real zeros of the even part for $\alpha \geq 1/2 + \Delta$, the same threshold $\sigma_{\max}$ as the basepoint laws. Proposition 16.6.3 identifies the two families, so no gain in the strip.

2. The heat flow factor $e^{\lambda u^2}$ reaches $\Lambda \leq 0.2$ (Platt and Trudgian, sharpening the bound 0.22 of Polymath 15), inside the strip in the Gaussian sense, and $\Lambda \geq 0$ by Rodgers and Tao; the remaining gap is the whole problem.

3. The Pólya Schur and Schoenberg ladders are equivalent to RH; their first rungs hold: the Turán inequality for the first is a theorem, and log concavity for the second is settled by a floating point certificate in Section 16.11.

4. The Mellin criterion does not apply to Φ (Section 16.8): for $q = 1$ the Mellin transform has the zero of Theorem 5.2.3, and for $q \geq 2$ there is no factorisation $\Phi(t) = \mathcal{F}(t^{2q})$. A different criterion, applied to a different factorisation of the Pólya density, such as its signed mixture of Gamma(2) and Gamma(3) densities (Section 8.5.4.3) or its representation through maxima (Chapter 14), would be needed.

5. The Bessel model is a laboratory. Since its analogue of RH is a theorem of Pólya, any proposed mechanism can be tested on it before it is tried on ξ; the $2\sin^2$ positivity mechanism already fails this test.

Problem 16.2. Find a sign condition on the primal measure of the lemma that holds for the Bessel model at every basepoint $\alpha > 0$, is preserved by the Euler product, that is, by adding the primal measures of the Euler factors, and fails for the Davenport Heilbronn function.

Problem 16.3. Prove real rootedness of $A_\alpha = \frac{1}{2}(\xi(\alpha + w) + \xi(\alpha - w))$ for one fixed $\alpha \in (\frac{1}{2}, 1)$ without proving the quasi Riemann hypothesis at α, by a Pólya type argument that uses the Euler product (the odd part B_α need not be real rooted).

Problem 16.4. Determine whether the derivative laws $\xi^{(2n)}(\frac{1}{2})/\xi^{(2n)}(\frac{1}{2} + \sqrt{s})$, or their counterparts $F^{(n)}(0)/F^{(n)}(s)$ with derivatives taken in the Thorin variable s, are GGCs for some n. For the first family, Ki and Kim give a local statement and Conrey the proportion of zeros on the line, for large n; for the second, Griffin, Ono, Rolen and Zagier give the first d rungs of the Pólya Schur ladder for $n \geq N(d)$ (Section 16.11). For neither family is a global statement known for any n, and one for a single n would be the first unconditional GGC in its family.

Problem 16.5. Under RH, which is what makes the midpoint rule of Proposition 16.6.3 available for every $\alpha > \frac{1}{2}$, prove that it converges, $\mu_\alpha^P - \mu^{(\alpha)} \to 0$ weakly as $\alpha \downarrow \frac{1}{2}$, with an explicit rate, so that the atomic and continuous descents can be run together.

16.11 Progress on the open problems

16.11.1 Problem 16.5: the even part and basepoint laws differ by an explicit factor

Theorem 16.11.1. *For every $\alpha > 1/2$ and real $s \geq 0$ put $R_\alpha(s) = \xi(\alpha - \sqrt{s})/\xi(\alpha + \sqrt{s})$. Then, unconditionally, $R_\alpha(0) = 1$, $0 < R_\alpha(s) < 1$ for $s > 0$, and*

$$\begin{aligned} \frac{\varphi_\alpha^P(s)}{\varphi^{(\alpha)}(s)} &= \frac{2}{1 + R_\alpha(s)}, \\ \psi_\alpha^P(s) - \psi_\alpha(s) &= \log \frac{1 + R_\alpha(s)}{2}, \end{aligned} \tag{16.11.1}$$

where $\psi = -\log \varphi$ denotes the Laplace exponent, and for $s > 0$ the difference of the exponents lies in $(-\log 2,\, 0)$. Hence $\varphi^{(\alpha)} \leq \varphi_\alpha^P \leq 2\varphi^{(\alpha)}$ on $[0, \infty)$, and as $\alpha = \frac{1}{2} + a \downarrow \frac{1}{2}$,

$$\psi_\alpha^P(s) - \psi_\alpha(s) = -a\, \frac{\xi'}{\xi}\big(\tfrac{1}{2} + \sqrt{s}\big) + O(a^2) \tag{16.11.2}$$

locally uniformly in s. In particular $\varphi_\alpha^P \to \varphi$ and $\varphi^{(\alpha)} \to \varphi$ pointwise, so whenever the two families are laws they converge weakly to the reciprocal Xi law.

Proof. By definition $A_\alpha(\sqrt{s}) = \frac{1}{2}\xi(\alpha + \sqrt{s})\,(1 + R_\alpha(s))$, which gives (16.11.1). For real σ, $\xi(\sigma) > 0$, ξ is symmetric about $\frac{1}{2}$ and increasing on $[\frac{1}{2}, \infty)$, because $\xi(\frac{1}{2} + w) = \sum c_k w^{2k}$ with $c_k > 0$. Since $\xi(\alpha - \sqrt{s}) = \xi(1 - \alpha + \sqrt{s})$ and $|\alpha + \sqrt{s} - \frac{1}{2}| > |\frac{1}{2} - \alpha + \sqrt{s}|$ when $\alpha > \frac{1}{2}$ and $s > 0$, we get $0 < R_\alpha < 1$ for $s > 0$, while $R_\alpha(0) = 1$. For (16.11.2), $\log R_\alpha(s) = \log \xi(\frac{1}{2} + \sqrt{s} - a) - \log \xi(\frac{1}{2} + \sqrt{s} + a) = -2a\frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) + O(a^3)$, and $\log \frac{1+R}{2} = \frac{1}{2}\log R + O((\log R)^2)$. □

Combined with Proposition 16.6.3, Theorem 16.11.1 evaluates the error of the midpoint rule exactly. For every α as in Proposition 16.6.3, that is for $\alpha > \sigma_{\max}$ and unconditionally for $\alpha \geq 1$, and every $s \geq 0$,

$$\sum_{k \geq 1} \log\left(1 + \frac{s}{\tau_\alpha(k - \frac{1}{2})^2}\right) = \int_0^\infty \log\left(1 + \frac{s}{\tau_\alpha(v)^2}\right) dv + \log \frac{1 + R_\alpha(s)}{2}. \tag{16.11.3}$$

The defect of the midpoint sum is a ratio of two values of ξ on the real axis, and no modulus of continuity of the clock is needed. Numerically, for $\alpha = 1, 0.75, 0.6, 0.52$ the defect divided by $a = \alpha - \frac{1}{2}$ varies by 1.2% at $s = 1$ and by 10% at $s = 100$, and approaches $-\frac{\xi'}{\xi}(\frac{1}{2} + \sqrt{s}) = -0.0461$, -0.408, as (16.11.2) predicts.

16.11.2 Problem 16.1: f_X is log concave

Write $N = f_X f_X'' - (f_X')^2$, so that PF_2 is $N < 0$, and recall $\varphi(t^2) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + t) > 0$ for real t.

LEMMA 16.11.2 (Origin). *$f_X'(0) = 0$ and $f_X''(0) = -\frac{1}{\pi}\int_0^\infty t^2\varphi(t^2)\,dt < 0$, so f_X is log concave on a neighbourhood of 0, unconditionally.*

LEMMA 16.11.3 (Residue expansion). *Let $H = 3\times 10^{12}$ be the height up to which the zeros of ξ are verified to be simple and on the critical line [247], let $0 < T < H$ be different from every ordinate, and put $\eta_k = -\Xi(0)/\Xi'(\gamma_k)$. Then, unconditionally, for $x > 0$,*

$$f_X(x) = \sum_{\gamma_k<T} \eta_k\, e^{-\gamma_k x} + E_T(x), \qquad \bigl|E_T^{(j)}(x)\bigr| \le C_j(T)\, e^{-Tx},$$
$$C_j(T) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{|u-iT|^j\,\xi(\frac12)}{|\xi(\frac12+u-iT)|}\,du. \tag{16.11.4}$$

The residues alternate in sign, $\eta_1 > 0 > \eta_2$, and so on.

Proof. $f_X(x) = \frac{1}{2\pi}\int \varphi(t^2)e^{-itx}dt$. Move the line of integration to $\operatorname{Im} t = -T$. The poles crossed are the zeros of $\xi(\frac12+t)$ with $-T < \operatorname{Im} t < 0$, which below H are $t = -i\gamma_k$, simple; the vertical sides vanish because $|\xi(\sigma+i\tau)|$ grows like a gamma function as $|\sigma|\to\infty$, uniformly for bounded τ. The residue at $-i\gamma_k$ is computed from $\xi'(\frac12 - i\gamma) = i\,\Xi'(\gamma)$. The signs alternate because Ξ changes sign at each simple real zero, starting from $\Xi(0) > 0$. □

Numerically $\log|\eta_k| - \pi\gamma_k/4$ equals $-5.2, -8.0, -8.8, -9.7, -11.1$ for $k = 1, 10, 20, 40, 60$, so $|\eta_k|$ grows like $e^{\pi\gamma_k/4}$ up to a power of γ_k, and the expansion is useful only for $x > \pi/4$.

LEMMA 16.11.4 (Two exponentials). *If $f = Ae^{-ax} + Be^{-bx}$, then $ff'' - (f')^2 = AB\,(b-a)^2e^{-(a+b)x}$. With $A = \eta_1 > 0 > B = \eta_2$ this is negative.*

The proof is a one line expansion. With $T = 23$, between γ_2 and γ_3, Lemmas 16.11.3 and 16.11.4 give $Ne^{(\gamma_1+\gamma_2)x} = \eta_1\eta_2(\gamma_2-\gamma_1)^2 + \varepsilon(x)$, where $|\varepsilon(x)|$ is bounded by a sum of terms each decreasing in x, built from C_0, C_1, C_2 and $|\eta_1|, |\eta_2|$. Numerically $\eta_1 = 359.524$, $\eta_2 = -28007.2$, $\eta_1\eta_2(\gamma_2-\gamma_1)^2 = -4.776\times 10^8$, and $C_0, C_1, C_2 = 5.99\times 10^4,\ 1.38\times 10^6,\ 3.19\times 10^7$.

THEOREM 16.11.5 (PF_2, floating point certificate). *The density f_X of $X = Z\sqrt{2Y}$ is log concave on $\mathbb{R}$. The certificate has three parts, for $x \ge 0$ (the density is even).*

(A) On $[0, 0.95]$, direct quadrature of $f^{(j)}(x) = \frac{1}{\pi}\int_0^\infty \varphi(t^2)\frac{d^j}{dx^j}\cos(tx)\,dt$ at the left end x_i of each of 770 cells, with $\sup|N'|$ on a cell of length h bounded by $B(h)$ through $|f^{(j)}(y)| \le |f^{(j)}(x_i)| + h\,m_{j+1}$ and $m_j = \frac{1}{\pi}\int_0^\infty t^j\varphi(t^2)dt$ ($m_0,\dots,m_4 = 1.899,\ 7.321,\ 44.94,\ 355.8,\ 3361$), proves $N < 0$. The cells are chosen from the left, each as long as possible with $h\,B(h) \le 0.99\,|N(x_i)|$, so that $N \le N(x_i) + h\,B(h) < 0$ on the cell; any factor in $(0,1)$ in place of 0.99 works (814 cells at 0.9, 1165 at 0.5).

(B) On $[0.95, 2.1]$*, the expansion* (16.11.4) *with the first twelve zeros and* $T = \frac{1}{2}(\gamma_{12} + \gamma_{13}) = 57.90$ *gives* $Ne^{(\gamma_1+\gamma_2)x}$ *as an explicit exponential sum plus a remainder with decreasing majorant;* 26 *equal cells certify negativity (*22 *equal cells are the least that do, and two adaptive cells suffice). At* $x = 0.95$ *the sum is* -2.18×10^8 *and the remainder at most* 7.03×10^6*.*

(C) For $x \geq 2.1$ *the two term form with* $T = 23$ *has* $|\varepsilon(x)| \leq 4.687 \times 10^8 < 4.776 \times 10^8$*: the majorant decreases and equals* 4.687×10^8 *at* $x = 2.1$*.*

Figure 16.2 (left) shows $-(\log f_X)''$. It is 23.66 at the origin and then decays like $\frac{|\eta_2|}{\eta_1}(\gamma_2 - \gamma_1)^2 e^{-(\gamma_2-\gamma_1)x}$, which is 4.0×10^{-9} at $x = 4$: the log concavity of the tail is exponentially thin, and it is held up by the sign of η_2 alone.

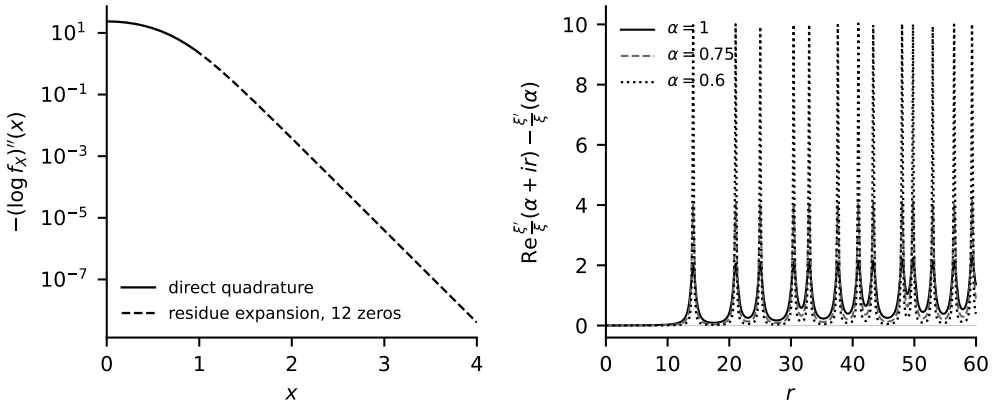


Figure 16.2: Left: $-(\log f_X)''(x)$ on a log scale, from direct quadrature on $[0, 0.95]$ (solid) and from the residue expansion with twelve zeros on $[0.95, 4]$ (dashed); it is positive throughout, which is Theorem 16.11.5. Right: the compensated clock condition $D_\alpha(r) = \mathrm{Re}\,\frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha) \geq 0$ of Theorem 16.11.7 for $\alpha = 1$ (solid), 0.75 (dashed) and 0.6 (dotted), computed from the Hadamard sum over the first 1700 zeros at step 0.02; D_α vanishes at $r = 0$ and is positive after, with a peak of height about $1/(\alpha - \frac{1}{2})$ at each ordinate. Drawn by `polya_fig.py`.

Numerics. The certificate is reproduced by `code/polya_density.py` (`code/README.md`) in mpmath at 80 digits. The values $f_X^{(j)}$ on $[0, 1.6]$ come from composite Gauss Legendre quadrature and beyond 1.6 from the residue expansion; the two agree to a relative 2×10^{-20} at $x = 1.6$ with fourteen zeros and to 8×10^{-42} with forty (for the derivatives of orders one to three, to 2×10^{-18} and 6×10^{-39}). The quadrature errors are many orders of magnitude below the margins of Parts (A) and (B). Part (C) is thin: the majorant falls below 4.776×10^8 from $x = 2.0904$ on, so that at $x = 2.1$ the margin is 1.9% of the

bound, and a relative error of 2% in C_0, C_1, C_2 would break it; the constants are stable to many digits under two quadrature layouts. The only arithmetic input is that the first thirteen zeros are simple and on the line.

The mechanism of Lemma 16.11.4 extends, as a sketch, to every order n with $\gamma_n < H$ in the far tail. If $x_1 < \cdots < x_n$, $y_1 < \cdots < y_n$ and every $x_i - y_j$ is large, the Cauchy Binet expansion of $\det[f_X(x_i - y_j)]$ is dominated by the term from $\gamma_1, \dots, \gamma_n$, whose sign is $\operatorname{sgn}(\eta_1 \cdots \eta_n)\,(-1)^{n(n-1)/2} = (-1)^{\lfloor n/2 \rfloor + n(n-1)/2} = +1$. So the PF_n minors with $\gamma_n < H$ are positive far in the tail, unconditionally, provided the remainder E_T of (16.11.4), which is not a sum of exponentials, can be dominated inside the determinant; Proposition 16.11.14 does this for coalescing points.

THEOREM 16.11.6 (f_X is uniformly close to a PF_∞ density). *Let $X_{\mathrm{on}} = \sum L_\gamma/\gamma$, the sum over the zeros on the critical line with independent standard Laplace variables. Unconditionally X_{on} has a PF_∞ density f_{on}, and for $j = 0, 1, 2$*

$$\sup_x \left| f_X^{(j)}(x) - f_{\mathrm{on}}^{(j)}(x) \right| \le \varepsilon_j := \frac{1}{\pi} \int_0^\infty t^j \, \Pi_K(t) \left(2S\, t^2 + S^2 t^4 \right) dt, \tag{16.11.5}$$

where $\Pi_K(t) = \prod_{k \le K} (1 + t^2/\gamma_k^2)^{-1}$ for any K with $\gamma_K < H$, and $S = \sum_{\gamma > H} \gamma^{-2} \le (\log \frac{H}{2\pi} + 2)/(2\pi H) = 1.533 \times 10^{-12}$ for $H = 3 \times 10^{12}$. With $K = 300$, $\varepsilon_0 \le 1.57 \times 10^{-10}$, $\varepsilon_1 \le 1.3 \times 10^{-9}$ and $\varepsilon_2 \le 1.3 \times 10^{-8}$. Consequently every $n \times n$ minor $\det[f_X(x_i - y_j)]$ with ordered x and y is at least $-n \cdot n!\,(2\varepsilon_0 + f_{\mathrm{on}}(0))^{n-1} \varepsilon_0$, which is -1.1×10^{-8} for $n = 3$ and -1.2×10^{-7} for $n = 4$, since $f_{\mathrm{on}}(0) \le \frac{1}{\pi} \int_0^\infty \Pi_K = 1.976$.

Proof. The characteristic function of X_{on} is $\varphi_{\mathrm{on}}(t^2) = \prod_{\text{on line}} (1 + t^2/\gamma^2)^{-1}$, the reciprocal of a function of type (16.3.1), so f_{on} is PF_∞ by Schoenberg's theorem whatever the zeros off the line do. Write $\varphi(t^2) = \varphi_{\mathrm{on}}(t^2) R(t^2)$, where R collects the off line zeros. Each conjugate pair $s_\rho, \bar{s}_\rho$ with $s_\rho = -a^2$, $a = \gamma - i\delta$, contributes $1/P_\rho(t^2)$ with $P_\rho(v) = 1 + 2v \operatorname{Re}(a^{-2}) + v^2 |a|^{-4}$. Since $\gamma > H > \frac{1}{2} > |\delta|$, $\operatorname{Re} a^{-2} = (\gamma^2 - \delta^2)/|a|^4 > 0$, so $P_\rho \ge 1$ on $[0, \infty)$, $0 < R \le 1$ and $1 - R \le \sum_\rho (P_\rho - 1) \le 2St^2 + S^2 t^4$. Fourier inversion gives $|f_X^{(j)} - f_{\mathrm{on}}^{(j)}| \le \frac{1}{\pi} \int_0^\infty t^j \varphi_{\mathrm{on}}(t^2)(1 - R(t^2))\, dt$, and $\varphi_{\mathrm{on}} \le \Pi_K$ because the first K zeros are on the line. The tail sum S follows by partial summation from the zero counting formula with an explicit error term [109]; Proposition 14.7.2 gives the sharper value 1.4798×10^{-12} for the same sum, and the constants of the theorem were computed with the larger one. For the minors, expand the determinant multilinearly in the differences $f_X - f_{\mathrm{on}}$ and use $\det[f_{\mathrm{on}}(x_i - y_j)] \ge 0$ and $|f_{\mathrm{on}}| \le f_{\mathrm{on}}(0)$. □

RH is the statement that the uniform distance of f_X to a PF_∞ density is zero. The bound on the minors grows quickly with n (it exceeds 1 at $n = 10$), so for large orders the theorem gives no information, and a failure of PF_3 would be invisible at the level 10^{-8}. Numerically there is none: over 6050 Toeplitz type minors for each of $n = 3$ and $n = 4$, with spacings $0.02d$ and $0.02(d + 1)$,

$1 \le d \le 25$, and offsets up to 1.2, all are positive and the smallest normalised minor $\det[f_X(x_i - y_j)]/\prod_i f_X(x_i - y_i)$ is 1.6×10^{-10} for $n = 3$ and 3.5×10^{-19} for $n = 4$. Both occur at the smallest spacing; most of the smallness comes from nearly coinciding points, since for $n = 3$ the smallest value over the offsets grows with d from 1.6×10^{-10} at $d = 1$ to 6.0×10^{-2} at $d = 25$.

16.11.3 Problem 16.2: the exact form of the $2\sin^2$ lemma

Stieltjes inversion gives the Thorin density itself, with no sign assumption on the primal measure.

Theorem 16.11.7. *Let f be a real entire function with no zeros in $\operatorname{Re} z > \alpha - \delta$ for some $\delta > 0$ and with $\frac{f'}{f}(z) = O(\log|z|)$ uniformly on $\operatorname{Re} z \ge \alpha$. This holds for ξ with $\alpha > \sigma_{\max}$, with $\delta = \alpha - \sigma_{\max}$, the bound following from the Hadamard product because $O(\log T)$ zeros have ordinate in $[T, T+1]$, and for $\nu \mapsto K_\nu(2\pi)$ with $\alpha > 0$, with $\delta = \alpha$, the zeros being purely imaginary with spacing about $\pi/\log y$ at height y. Let $\psi(u) = \log f(\alpha + w) - \log f(\alpha) - w\frac{f'}{f}(\alpha)$ with $w = \sqrt{2u}$ the principal root. Then ψ' is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, $\psi'(u) = O(|u|^{-1/2}\log|u|)$, and*

$$\begin{aligned} \psi'(u) &= \int_0^\infty \frac{\omega_\alpha(z)}{z+u}\,dz, \\ \omega_\alpha(z) &= \frac{1}{\pi\sqrt{2z}}\left[\operatorname{Re}\frac{f'}{f}\Big(\alpha + i\sqrt{2z}\Big) - \frac{f'}{f}(\alpha)\right]. \end{aligned} \tag{16.11.6}$$

Hence the compensated reciprocal $e^{wf'(\alpha)/f(\alpha)}f(\alpha)/f(\alpha+w)$ is the Laplace transform in u of a GGC if and only if

$$\operatorname{Re}\frac{f'}{f}(\alpha + ir) \ge \frac{f'}{f}(\alpha) \qquad \text{for all } r > 0, \tag{16.11.7}$$

that is, if and only if the clock at basepoint α runs at least as fast at every height as at height zero.

Proof. For $u \notin (-\infty, 0]$ the root w has positive real part, so $\alpha + w$ stays in the zero free half plane and $\psi'(u) = \big[\frac{f'}{f}(\alpha+w) - \frac{f'}{f}(\alpha)\big]/w$ is holomorphic there. By the hypothesis on $\frac{f'}{f}$ the numerator of ψ' is $O(\log|w|)$, whence the bound on ψ'; the zero free strip of width δ makes ψ' continuous up to the cut. Cauchy's formula on a keyhole contour around the cut, with the bound to discard the large circle and the integrability of ω_α at 0 (where $\omega_\alpha(z) = O(\sqrt{z})$ by the evenness of $r \mapsto \operatorname{Re}\frac{f'}{f}(\alpha+ir)$), gives the Stieltjes representation with density $-\frac{1}{\pi}\operatorname{Im}\psi'(-z+i0)$. At $u = -z+i0$ the root is $w = ir$, $r = \sqrt{2z}$, and $-\frac{1}{\pi}\operatorname{Im}\psi' = \frac{1}{\pi r}\operatorname{Re}\big[\frac{f'}{f}(\alpha+ir) - \frac{f'}{f}(\alpha)\big]$, which is ω_α. A Stieltjes function with zero constant term is the derivative of a Thorin Bernstein function exactly when its representing density is nonnegative,

which is (16.11.7). Finally, if $\log f$ has the representation of the lemma with $\nu(dx) = e^{-\alpha x}\mu_f(dx)$, then $\frac{f'}{f}(\alpha+w) - \frac{f'}{f}(\alpha) = \int(1-e^{-wx})\nu(dx)$ and $\omega_\alpha(z) = \frac{1}{\pi r}\int 2\sin^2(rx/2)\,\nu(dx)$, which is the $2\sin^2$ lemma. □

The right sign condition on the primal measure is not $\nu \geq 0$ but

$$\int \big(1-\cos(rx)\big)\,\nu(dx) \geq 0 \qquad \text{for all } r, \tag{16.11.8}$$

that the cosine transform of ν is maximal at the origin, read in this way because ν has infinite mass for ξ and for the Bessel model. Its local form at $r=0$ is $-(\log f)'''(\alpha) \geq 0$, the $j=1$ column of Table 16.2, which is positive at every basepoint; the failures in that table are in higher columns, which (16.11.8) does not require. Numerically, (16.11.7) holds for the Bessel model at $\alpha = 0.25, 1, 3, 8$ on $0 < r \leq 60$ (grid of step 0.02), and for ξ at $\alpha = 1, 0.75, 0.6, 0.55$ on $0 < r \leq 100$ (grid of step 0.01). That the condition holds for the Bessel model at every $\alpha > 0$ rests on these numerics, and that it fails for the Davenport Heilbronn function (Problem 16.2) is not discussed here. For ξ, condition (16.11.7) reads $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) \geq b_\alpha = \frac{\xi'}{\xi}(\alpha)$, the compensated companion of Lagarias's criterion $\operatorname{Re}\frac{\xi'}{\xi} > 0$; in the variable $u = w^2/2$ the Thorin densities of the basepoint law and of the compensated reciprocal differ by the stable term $b_\alpha/(\pi\sqrt{2z})$ of Proposition 9.1.12, where the stable factor is $e^{-b_\alpha\sqrt{u}}$ in the variable u of that proposition. To the right of 1 the primes enter exactly as the lemma predicts. Write $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$ with the archimedean part $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})$.

Proposition 16.11.8 (Primes and the compensated clock). *For $\alpha > 1$ and real r, write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha)$, which is $v_\alpha(r) - b_\alpha$ in the notation of Section 9.1. Then*

$$D_\alpha(r) = \big[\operatorname{Re}\mathcal{A}(\alpha+ir) - \mathcal{A}(\alpha)\big] + \sum_{n\geq 2}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{r\log n}{2}\Big). \tag{16.11.9}$$

The prime sum is nonnegative; it is the $2\sin^2$ transform of the positive measure $\sum\Lambda(n)n^{-\alpha}\delta_{\log n}$. The archimedean bracket equals $\sum_{k\geq 1} q_{\alpha+2k}(r) - q_{\alpha-1}(r)$ with $q_c(r) = r^2/(c(c^2+r^2))$, so that only the pole term is negative, and it is nonnegative for every r if and only if $\zeta(3, 1+\frac{\alpha}{2}) \geq 8(\alpha-1)^{-3}$, that is, if and only if $\alpha \geq \alpha_1 = 7.78918\ldots$; the compensated condition then holds from the identity alone. For $1 < \alpha < \alpha_1$ the bracket dips below zero (its minimum is -4.83 at $\alpha = 1.2$, $r = 1.57$; -0.74 at $\alpha = 2$; -0.011 at $\alpha = 6$), and the primes must make up the deficit. They do, for every $\alpha \geq 1$, by Proposition 9.1.12, whose proof combines the zeros below height 2000, the bounds of [109] for the zeros above, and a computation in ball arithmetic; (16.11.9) *is* (9.1.7).

Proof. For $\alpha > 1$, $\operatorname{Re}\frac{\zeta'}{\zeta}(\alpha+ir) - \frac{\zeta'}{\zeta}(\alpha) = \sum \Lambda(n)n^{-\alpha}(1-\cos(r\log n))$ by the Dirichlet series, and $1-\cos x = 2\sin^2(x/2)$. For the bracket, $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\ge0} y^2/((x+k)((x+k)^2+y^2))$ for the digamma function ψ, so with $x=\alpha/2$, $y=r/2$ the term $\frac12\Gamma'/\Gamma(s/2)$ contributes $\sum_{k\ge0} q_{\alpha+2k}(r)$; the term $k=0$ cancels the contribution $-q_\alpha(r)$ of $1/s$, and $1/(s-1)$ contributes $-q_{\alpha-1}(r)$. Each ratio $((\alpha-1)^2+r^2)/((\alpha+2k)^2+r^2)$ increases with r, so the bracket $r^2\big[\sum_{k\ge1}\frac{1}{(\alpha+2k)((\alpha+2k)^2+r^2)} - \frac{1}{(\alpha-1)((\alpha-1)^2+r^2)}\big]$ is nonnegative for every r if and only if it is nonnegative as $r\to0$, that is, if and only if $\sum_{k\ge1}(\alpha+2k)^{-3} \ge (\alpha-1)^{-3}$, which is $\zeta(3,1+\frac{\alpha}{2}) \ge 8(\alpha-1)^{-3}$. The quantity $(\alpha-1)^3\sum_{k\ge1}(\alpha+2k)^{-3} = \sum_{k\ge1}\big(\frac{\alpha-1}{\alpha+2k}\big)^3$ increases with α, so the condition holds exactly for $\alpha\ge\alpha_1$, where this quantity equals 1. □

As $\alpha\downarrow1$ the deficit of the bracket comes from the pole term, $\operatorname{Re}\frac{1}{\alpha-1+ir} - \frac{1}{\alpha-1} \sim -\frac{1}{\alpha-1}$ for fixed $r\neq0$, and the prime sum compensates it because $-\frac{\zeta'}{\zeta}(\alpha)\sim\frac{1}{\alpha-1}$. Grouping $\frac{1}{s-1}$ with $\frac{\zeta'}{\zeta}(s)$, which is holomorphic on $\operatorname{Re}s\ge1$ by the prime number theorem, both pieces stay finite at $\alpha=1$. Below 1 the Dirichlet series is gone and (16.11.9) has no direct analogue; this is the wall that Proposition 9.1.12 meets at $\alpha=1$.

Proposition 16.11.9. *If* $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)\ge b_\alpha$ *for all real* r *and every* $\alpha>\alpha_0\ge\frac12$, *then no zero of* ζ *has real part greater than* α_0. *For* $\alpha_0=\frac12$ *it implies RH.*

Proof. $b_\alpha = \frac{\xi'}{\xi}(\alpha) > 0$ for $\alpha>\frac12$, because ξ is increasing on $(\frac12,\infty)$. The hypothesis therefore gives $\operatorname{Re}\frac{\xi'}{\xi}>0$ on the half plane $\operatorname{Re}s>\alpha_0$. Directly, near a zero ρ with $\operatorname{Re}\rho>\alpha_0$ and multiplicity $m\ge1$ the function $\frac{\xi'}{\xi}(s) - m/(s-\rho)$ is holomorphic, hence bounded, so $\operatorname{Re}\frac{\xi'}{\xi}(\rho-\varepsilon) = -m/\varepsilon + O(1)$ as $\varepsilon\downarrow0$, while $b_{\operatorname{Re}\rho-\varepsilon}$ stays bounded; the hypothesis at $\alpha=\operatorname{Re}\rho-\varepsilon>\alpha_0$ and $r=\operatorname{Im}\rho$ fails for small $\varepsilon>0$, so there is no such zero; this is condition (iv) of Theorem 10.2.7. □

Under RH the compensated condition holds at every $\alpha>\frac12$, and it holds unconditionally for $\alpha\ge1$ (Theorem 9.1.18 and Proposition 9.1.12), so Proposition 16.11.9 has a converse for $\alpha_0=\frac12$. The delicate range is below the first zero: there the numerics show $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)$ rising from its value at $r=0$ like r^2 (Figure 16.2, right). The refined problem is to prove $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir)\ge b_\alpha$ at one $\alpha<1$ for all heights. By Proposition 16.11.9, doing so for all α above some $\alpha_0<1$ implies a quasi Riemann hypothesis at α_0, and by Corollary 16.12.9 it is equivalent to one, whereas on a single line it is not known to imply anything about the zeros to its right (compare Remark 1.2.4).

16.11.4 Problem 16.3: local reality and certificates to height 300

Proposition 16.11.10 (Local reality, unconditional). *Let K be such that the first K zeros of ξ in the upper half plane, ordered by ordinate, are simple and on the line, with ordinates $\gamma_1 < \cdots < \gamma_K$, and fix $\Gamma \in (\gamma_K, \gamma_{K+1})$. There is $\alpha_K > \frac{1}{2}$ such that for $\frac{1}{2} < \alpha < \alpha_K$ the zeros of A_α in $|w| < \Gamma$ are exactly $\pm i\tau_k(\alpha)$, $k \le K$, simple and purely imaginary, with $\tau_k(\alpha) \to \gamma_k$.*

Proof. $A_\alpha \to \xi(\frac{1}{2} + w)$ uniformly on $|w| \le \Gamma$, so by Hurwitz's theorem A_α has exactly one zero in a small disc around each $\pm i\gamma_k$ and none elsewhere in $|w| < \Gamma$. A_α is even and real on the real axis, so its zero set is invariant under $w \mapsto -\bar{w}$; a unique zero in a disc centred on the imaginary axis is therefore on that axis. □

This is the even part analogue of the local theorem of Ki and Kim for the derivatives of ξ (item P13 of Section 16.3). For $\alpha = 0.6$ and 0.75 the argument principle on the rectangle $|\operatorname{Re} w| \le 1$, $0 \le \operatorname{Im} w \le 100$, which contains every zero of A_α up to that height by de Bruijn's strip theorem, counts 29.000 zeros, and Table 16.1 lists 29 sign changes of $\operatorname{Re}\xi(\alpha + it)$. So every zero of $A_{0.6}$ and $A_{0.75}$ up to height 100 is purely imaginary. At $\alpha = 0.55$ the same count to height 300 gives 138.000 zeros against 138 sign changes, so every zero of $A_{0.55}$ up to height 300 is purely imaginary as well. Together with Proposition 16.6.3 this is a Turing type method for Thorin atoms in the strip: it verifies the even part laws to a given height without any statement about $\operatorname{Re}\xi'/\xi$.

16.11.5 Problem 16.4: derivatives in the Thorin variable are the shifted Jensen polynomials

Since $\xi(\frac{1}{2} + z) = \frac{1}{8}(4z^2 - 1)\widehat{\zeta}(\frac{1}{2} + z)$ with $\widehat{\zeta}(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, the sequence of Griffin, Ono, Rolen and Zagier [99], defined by $(4z^2 - 1)\widehat{\zeta}(\frac{1}{2} + z) = \sum_n \gamma_{\mathrm{GORZ}}(n) z^{2n}/n!$, is $\gamma_{\mathrm{GORZ}}(n) = 8\xi(\frac{1}{2})\, n!\, a_n$, a constant multiple of our $g_n = n!\, a_n$. Their shifted Jensen polynomials are therefore, up to a constant factor, the polynomials $J_{d,n}(x) = \sum_{j=0}^{d} \binom{d}{j} g_{n+j} x^j$ of Section 13.2.4, the Jensen polynomials of the nth derivative $F^{(n)}(s)$ in the Thorin variable. By Pólya and Schur, the law with Laplace transform $F^{(n)}(0)/F^{(n)}(s)$ is a GGC if and only if $J_{d,n}$ is real rooted for every d.

Corollary 16.11.11. *For every d there is $N(d)$ such that for all $n \ge N(d)$ the derivative laws $F^{(n)}(0)/F^{(n)}(s)$ satisfy the first d rungs of the Pólya Schur ladder [99]. Under RH they are GGCs for every n, by Laguerre's theorem that the Laguerre Pólya class is closed under differentiation.*

The first case is classical. Since $F'(s) = \xi'(\frac{1}{2} + \sqrt{s})/(2\sqrt{s}\,\xi(\frac{1}{2}))$, the zeros of F' are the squares $s = w^2$ of the zeros $w \ne 0$ of $\xi'(\frac{1}{2} + w)$, so the law $F'(0)/F'(s)$ is a GGC if and only if every zero of ξ' other than $\frac{1}{2}$ lies on the critical line. That is the

Riemann Hypothesis for ξ', which RH implies and which is not known to imply RH; Conrey's theorem [61] gives a positive proportion of its zeros on the line. In general $F^{(n)}(w^2) = (D^n\xi)(\tfrac12+w)/\xi(\tfrac12)$ with $D = \frac{1}{2w}\frac{d}{dw}$, so the derivative laws in the Thorin variable correspond to powers of this Bessel type operator rather than of $\frac{d}{dw}$. Numerically $\xi'(\tfrac12+w)$ has 29 zeros with $1 < \operatorname{Im} w < 100$ in the box $|\operatorname{Re} w| \le 0.6$, all on the imaginary axis.

Numerically every $J_{d,n}$ with $2 \le d \le 15$ and $0 \le n \le 25$, 364 polynomials in all, has only real negative zeros, computed at 150 digits from the coefficients of Section 16.4. The refined problem is a uniform statement: one n for which all rungs hold. As in Section 13.2.4, and following Farmer [85], large n carries little information about RH, so the useful direction is the opposite one, small n with all d, which is RH itself at $n = 0$.

16.11.6 Problem 16.2 again: the compensated condition at almost all heights in the strip

Write $D_\alpha(r) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+ir) - \frac{\xi'}{\xi}(\alpha)$ and let $N(\sigma,T)$ count zeros with $\beta > \sigma$ and $0 < \gamma \le T$.

Lemma 16.11.12. *For $\alpha > \frac12$, unconditionally, $0 < b_\alpha = \frac{\xi'}{\xi}(\alpha) \le 2a_1(\alpha-\frac12) < 0.04621\,(\alpha-\frac12)$.*

Proof. With $a = \alpha - \frac12$, $b_\alpha = \frac{d}{da}\log F(a^2) = 2a\,\psi'(a^2)$, where $\psi = \log F$. By Theorem 5.1.1 ψ is a Bernstein function, so ψ' is completely monotone and in particular $\psi'(a^2) \le \psi'(0) = a_1 = 0.0231050$. This is part (i) of Theorem 9.1.18, where $V_f = 2a_1 = 0.04620998623$ (Table 9.4) and the bound is proved for every $\alpha > \frac12$. □

Theorem 16.11.13 (Almost all heights). *Fix $\frac12 < \alpha \le 1$ and put $\sigma = \frac12(\alpha+\frac12)$. There is T_0 such that for $T \ge T_0$ the exceptional set $E_\alpha(T) = \{r \in [T,2T] : D_\alpha(r) \le 0\}$ has measure*

$$|E_\alpha(T)| \le C\Big(1 + \frac{1}{\alpha-\frac12}\Big)\big(N(\sigma,3T)+1\big) \tag{16.11.10}$$

with an absolute constant C. By Ingham's density theorem [124]

$$|E_\alpha(T)| \ll_\alpha T^{3(1-\sigma)/(2-\sigma)}(\log T)^5 = o(T),$$

and under the density hypothesis $|E_\alpha(T)| \ll T^{2(1-\sigma)+\epsilon}$. In particular the compensated clock condition holds at every basepoint in the strip on a set of heights of full density.

Proof. By the Hadamard product, summed symmetrically, $\operatorname{Re}\frac{\xi'}{\xi}(s)=\sum_\rho \operatorname{Re}\frac{1}{s-\rho}$ over all nontrivial zeros, so with $c_\rho=\alpha-\beta_\rho$,

$$D_\alpha(r)=\sum_\rho\left[\frac{c_\rho}{c_\rho^2+(r-\gamma_\rho)^2}-\frac{c_\rho}{c_\rho^2+\gamma_\rho^2}\right],$$

the sum running over zeros of both signs of γ. Call a zero harmful if $\beta>\alpha$, so $c<0$, and good if $|\beta-\frac12|\le\sigma-\frac12$, so $c\in[c_{\min},c_{\max}]$ with $c_{\min}=\alpha-\sigma=\frac12(\alpha-\frac12)$ and $c_{\max}=\alpha-1+\sigma\le\frac34$. Every zero that is not harmful contributes a nonnegative first term. Hence

$$D_\alpha(r)\ge L(r)-b_\alpha-N_h(r),$$
$$N_h(r)=\sum_{\text{harmful}}\frac{|c|}{c^2+(r-\gamma)^2},$$

where $L(r)$ is the first term of any one good zero. The second terms of all zeros, harmful ones included, sum to b_α, so no separate bound for the harmful second terms is needed.

The local term. By Littlewood's theorem [185] that gaps between consecutive ordinates tend to zero, for $r\ge T_0$ the window $[r-1,r+1]$ contains an ordinate. If one such zero is good, $L(r)\ge\frac{c}{c^2+1}\ge\kappa:=\frac{c_{\min}}{c_{\min}^2+1}\ge\frac45 c_{\min}$, since $c\mapsto c/(c^2+1)$ increases on $(0,1)$. Otherwise r lies within 1 of a zero that is not good, that is one with $\beta>\sigma$ or its mirror with $\beta<1-\sigma$; the measure of such r in $[T,2T]$ is at most $4N(\sigma,2T+1)$.

The harmful sum. Since $\int_{\mathbb R}\frac{|c|}{c^2+(r-\gamma)^2}dr=\pi$, the harmful zeros with $\frac T2\le\gamma\le 3T$ contribute at most $2\pi N(\alpha,3T)\le 2\pi N(\sigma,3T)$ to $\int_T^{2T}N_h$, and the remaining harmful zeros contribute $o(1)$ by the trivial bound $|c|/(r-\gamma)^2$ and Ingham's bound for $N(\alpha,t)$. By Markov's inequality, $N_h(r)>\kappa/2$ on a set of measure at most $\frac{2}{\kappa}(2\pi N(\sigma,3T)+o(1))$.

Outside these two sets, Lemma 16.11.12 gives $b_\alpha\le 0.04621(\alpha-\frac12)=0.09242\,c_{\min}$, hence $D_\alpha(r)\ge\kappa-\frac{\kappa}{2}-0.0925\,c_{\min}\ge(0.4-0.0925)c_{\min}>0$. Adding the two exceptional measures and using $\kappa\ge\frac45 c_{\min}$ gives (16.11.10). □

Theorem 16.11.13 has the same content as Remark 9.1.17, which obtains the comparison from Theorem 9.1.15 with $\sigma_0=\frac12+(\alpha-\frac12)/\sqrt2$ in place of σ and a lower bound growing like $\log\theta$; the proof here is direct, and Theorem 11.4.2 gives a statement of the same kind for the mass of the negative part of the density.

To the right of the strip the inequality holds on the grid numerically (Section 16.12), and at all heights for every $\alpha\ge1$ by Proposition 9.1.12. Below 1 it holds at all heights, on every line to the right of α_0, under a quasi Riemann hypothesis at α_0 (Theorem 16.12.8).

16.11.7 Problem 16.1 again: coalescing points and the PF_3, PF_4 conditions

Theorem 16.11.6 controls PF_n minors up to an additive error of 1.1×10^{-8} for $n = 3$ and 1.2×10^{-7} for $n = 4$, which is enough wherever the minors of f_{on} are larger than that, but not where they vanish. They vanish in two places: at coalescing points and far in the tail. When all x_i and all y_j coalesce at points whose difference is x, the minor divided by the two Vandermonde products tends to a constant multiple of the Hankel determinant $\mathcal{H}_n(x) = \det[f_X^{(i+j)}(x)]_{i,j<n}$, and the PF_n condition becomes $(-1)^{n(n-1)/2}\mathcal{H}_n(x) \geq 0$; for $n = 2$ this is log concavity.

Proposition 16.11.14. *In the tail the coalescing conditions hold, unconditionally, for every n with $\gamma_n < H$: for an exponential sum $f = \sum_k \eta_k e^{-\gamma_k x}$,*

$$\mathcal{H}_n(x) = \sum_{|S|=n} \Big(\prod_{k\in S} \eta_k \Big) \prod_{k<l\in S} (\gamma_l - \gamma_k)^2 \, e^{-x\sum_{k\in S}\gamma_k},$$

and the term of $S = \{1, \dots, n\}$ dominates as $x \to \infty$, with sign $(-1)^{\lfloor n/2 \rfloor} = (-1)^{n(n-1)/2}$ by the alternation of the residues.

Proof. $[f^{(i+j)}(x)]_{i,j<n} = V \operatorname{diag}(\eta_k e^{-\gamma_k x}) V^{\top}$ with $V = [(-\gamma_k)^i]_{i<n,\,k}$, so the Cauchy Binet formula gives the sum over n element subsets S, each weighted by $\det(V_S)^2 = \prod_{k<l\in S}(\gamma_l - \gamma_k)^2$. The residue expansion of Lemma 16.11.3 with T between γ_n and $\min(\gamma_{n+1}, H)$ shows that the remaining terms, including those that contain the remainder E_T and its derivatives, are smaller by a factor of order $e^{-(T-\gamma_n)x}$. □

Numerically, combining quadrature on $[0, 1.6]$ with the residue expansion with fourteen zeros beyond, $-\mathcal{H}_3/f_X^3$ and $\mathcal{H}_4/f_X^4$ are positive at every point of a grid of step 0.02 on $[0, 4]$; their smallest values, 1.05×10^{-21} and 2.08×10^{-39}, occur at $x = 4$, where Proposition 16.11.14 predicts exponential decay. So PF_3 and PF_4 hold at coalescing points on the grid and, by Proposition 16.11.14, as $x \to \infty$, and the scan of separated minors in Section 16.11 finds none that is negative. A complete proof of PF_3 would need three things that are not proved here: an explicit threshold in x in Proposition 16.11.14, a bound for nearly coalescing points, and a bound for separated points in the tail.

16.11.8 Problem 16.3 again: almost all zeros of the even and odd parts are on the line

Let $B_\alpha(w) = \frac{1}{2i}(\xi(\alpha+w) - \xi(\alpha-w))$ be the odd part, so that $A_\alpha(it) = \operatorname{Re}\xi(\alpha+it)$ and $B_\alpha(it) = \operatorname{Im}\xi(\alpha+it)$, and let $N_A(T)$, $N_B(T)$ count the zeros with $0 < \operatorname{Im} w \leq T$ and $N_A^0(T)$, $N_B^0(T)$ those on the imaginary axis.

THEOREM 16.11.15 (Zero accounting for the even and odd parts). *Let $\frac{1}{2} < \alpha \le 1$ be such that no zero of ξ lies on the line $\operatorname{Re} s = \alpha$, which excludes at most countably many α. Then, with no other hypothesis, as $T \to \infty$ through values that are not ordinates of zeros of ξ, A_α or B_α,*

$$\begin{aligned} N_A(T) &= U_\alpha(T) + 2N(\alpha, T) + O(\log T), \\ N_A^0(T) &\ge \left\lfloor U_\alpha(T) + \tfrac{1}{2} \right\rfloor, \end{aligned} \tag{16.11.11}$$

and the same holds for B_α with $\lfloor U_\alpha(T) \rfloor$ in place of $\lfloor U_\alpha(T) + \frac{1}{2} \rfloor$. Consequently

$$\begin{aligned} N_A(T) - N_A^0(T) &\le 2N(\alpha, T) + O(\log T), \\ N_B(T) - N_B^0(T) &\le 2N(\alpha, T) + O(\log T), \end{aligned} \tag{16.11.12}$$

so by Ingham's density theorem all but $O(T^{3(1-\alpha)/(2-\alpha)} \log^5 T)$ of the zeros of A_α and of B_α up to height T are purely imaginary, a proportion tending to one.

Proof. For $\operatorname{Re} w \ge \alpha$ both $\alpha + w$ and $\alpha - w$, reflected by the functional equation to $1 - \alpha + w$, have real part at least 1, where $\log|\xi(\sigma + it)|$ is strictly increasing in σ because every zero has $\beta < 1$; as $\alpha + \operatorname{Re} w > 1 - \alpha + \operatorname{Re} w$, $|\xi(\alpha + w)| > |\xi(\alpha - w)|$ and $A_\alpha(w) \ne 0$. So all zeros lie in $|\operatorname{Re} w| < \alpha$. Apply the argument principle to A_α on the rectangle $|\operatorname{Re} w| \le X$, $0 \le \operatorname{Im} w \le T$ with $X = 2$. On the real axis $A_\alpha > 0$. On the right side $A_\alpha = \frac{1}{2}\xi(\alpha + w)(1 + q)$ with $|q| < 1$, so its argument changes by that of ξ along $\operatorname{Re} s = \alpha + X$ up to $O(1)$, and the left side contributes the same by the symmetries $A_\alpha(-w) = A_\alpha(w)$ and $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. The argument principle for ξ on $[\alpha, \alpha + X] \times [0, T]$ converts the change along $\operatorname{Re} s = \alpha + X$ into $\pi U_\alpha(T) + 2\pi N(\alpha, T)$ plus a horizontal variation at height T. The two horizontal variations, of ξ and of A_α, are $O(\log T)$ by Backlund's argument [11] with Jensen's formula, since $\log|\xi|$ and $\log|A_\alpha|$ vary by $O(\log T)$ on discs of bounded radius at height T, by Stirling's formula and polynomial bounds for ζ. This gives the first identity in (16.11.11). For the second, $A_\alpha(it) = |\xi(\alpha + it)| \cos(\pi U_\alpha(t))$ with U_α continuous from $U_\alpha(0) = 0$, so every level $k - \frac{1}{2} \le U_\alpha(T)$ is crossed at least once. The odd part is treated in the same way with sin, except that B_α is odd, with a simple zero at $w = 0$ on the bottom edge, so the contour is indented above 0, which changes the count by $\frac{1}{2}$, inside $O(\log T)$, and $B_\alpha(\bar{w}) = -\overline{B_\alpha(w)}$ replaces $A_\alpha(\bar{w}) = \overline{A_\alpha(w)}$. Subtracting gives (16.11.12). □

Theorem 16.11.15 sharpens the half integer crossing picture into a balance sheet. Numerically, at $\alpha = 0.55$ the clock reaches $U_{0.55}(300) = 137.921$; $\operatorname{Re}\xi(0.55 + it)$ has $138 = \lfloor 137.921 + \frac{1}{2} \rfloor$ zeros on $(0, 300]$ and $\operatorname{Im}\xi(0.55 + it)$ has $137 = \lfloor 137.921 \rfloor$, and the argument principle counts 138.000 zeros of $A_{0.55}$ and 137.0 zeros of $B_{0.55}$ in the box, so the balance is exact there with no off axis zeros.

The odd part family also clarifies Problem 16.4. Since $B_{\frac{1}{2}+a}(w)/a \to -i\,\xi'(\frac{1}{2} + w)$ locally uniformly as $a \downarrow 0$, the derivative ξ' is the endpoint of the odd part

family, just as ξ is the endpoint of the even part family. Under RH every B_α has only imaginary zeros, and Hurwitz's theorem then gives the Riemann Hypothesis for ξ'. Theorem 16.11.15 cannot be pushed to $a \to 0$ directly, because the known bounds for $N(\alpha, T)$ grow to a positive proportion of all zeros as $\alpha \downarrow \frac{1}{2}$, and Hurwitz's theorem needs every zero of B_α on the axis, not almost every one; this is why the Riemann Hypothesis for ξ' remains open.

16.12 Consequences of a quasi Riemann hypothesis at 7/8

16.12.1 The claim and its status

On 6 October 2026 OpenAI released a manuscript dated 30 September 2026 [226], with a formal proof in Lean, that every Dirichlet L function, $\zeta(s)$ included, has no zeros in the half plane $\operatorname{Re} s > \frac{7}{8}$, and a companion manuscript dated 5 October 2026 [227] with a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$. The work has not been refereed; Section 12.1.2 describes it. Everything in this section is conditional on

$(\mathrm{Q}_{7/8})$ $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{7}{8}$, so that $\sigma_{\max} \leq \frac{7}{8}$ and every zero lies in $\frac{1}{8} \leq \beta \leq \frac{7}{8}$.

Write $\Delta = \sigma_{\max} - \frac{1}{2} \leq \frac{3}{8}$ for the half width of the strip containing the zeros of Ξ.

16.12.2 What moves from 1 to 7/8

Theorem 16.12.1 (Consequences of $\mathrm{Q}_{7/8}$). *Assume* $(\mathrm{Q}_{7/8})$.

(i) Basepoint laws. For every $\alpha \geq \frac{7}{8}$ the law $\mu^{(\alpha)}$ with Laplace transform $\xi(\alpha)/\xi(\alpha + \sqrt{s})$ is a GGC. For $\alpha > \frac{7}{8}$ its Thorin measure is absolutely continuous, with density $\pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau)$ in the ordinate variable, a sum of Lorentzians of widths at least $\alpha - \frac{7}{8}$; it has a stable tail $\mu^{(\alpha)}((y, \infty)) \sim b_\alpha / \sqrt{\pi y}$.

(ii) The clock and Lagarias positivity. For $\alpha > \frac{7}{8}$ the clock U_α is strictly increasing and $\operatorname{Re} \frac{\xi'}{\xi} > 0$ on $\operatorname{Re} s > \frac{7}{8}$. In terms of ζ this is the one sided bound $-\operatorname{Re} \frac{\zeta'}{\zeta}(\alpha + i\tau) < \operatorname{Re} \mathcal{A}(\alpha + i\tau) = \frac{1}{2} \log \frac{\tau}{2\pi} + O(\tau^{-2})$, with $\mathcal{A}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma}(\frac{s}{2})$, at every height τ and for every $\alpha > \frac{7}{8}$.

(iii) Even and odd parts. For $\alpha \geq \frac{7}{8}$ the functions A_α and B_α have only purely imaginary zeros, simple for $\alpha > \frac{7}{8}$. Proposition 16.6.3, the exact midpoint formula (16.11.3) *and, for $\alpha \leq 1$, the accounting theorem with $N(\alpha, T) = 0$ hold for every $\alpha > \frac{7}{8}$, and the even part laws μ_α^P are GGCs there.*

(iv) Heat flow. The de Bruijn Newman constant satisfies $\Lambda \le 2\Delta^2 \le \frac{9}{32} = 0.28125$ *by de Bruijn's theorem alone.*

(v) Densities. $N(\sigma, T) = 0$ *for* $\sigma > \frac{7}{8}$*, so in Theorem 16.11.13 the exceptional set at a basepoint* $\alpha \le \frac{7}{8}$ *is generated only by zeros with* $\frac{1}{2}(\alpha + \frac{1}{2}) < \beta \le \frac{7}{8}$*.*

Proof. (i) By Theorem 1.2.2 and Theorem 7.1.14, $\mu^{(\alpha)}$ is a GGC when no zero has real part greater than α, and its Thorin measure is the Poisson smoothed zero density, in which a zero of real part $\beta < \alpha$ contributes a Lorentzian of width $\alpha - \beta \ge \alpha - \frac{7}{8}$; a zero produces an atom only when $\beta = \alpha$, which is why absolute continuity is claimed for $\alpha > \frac{7}{8}$ only. For the tail, $1 - \xi(\alpha)/\xi(\alpha + \sqrt{s}) \sim b_\alpha \sqrt{s}$ as $s \downarrow 0$, and Karamata's Tauberian theorem for the survival function, with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, gives $\mu^{(\alpha)}((y, \infty)) \sim b_\alpha / \sqrt{\pi y}$. (ii) For $\alpha > \sigma_{\max}$ every term $\frac{\alpha - \beta}{|\alpha + i\tau - \rho|^2}$ of the Hadamard sum is positive, so $\operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) > 0$ and the clock, whose derivative is $\pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau)$, runs forward. Since $\frac{\xi'}{\xi} = \mathcal{A} + \frac{\zeta'}{\zeta}$, positivity is the stated bound, and $\operatorname{Re} \mathcal{A}(\alpha + i\tau) = \frac{1}{2} \log \frac{\tau}{2\pi} + O(\tau^{-2})$ by Stirling's formula for $\frac{\Gamma'}{\Gamma}$. (iii) For A_α this is Theorem 16.6.1(ii) with $\Delta \le \frac{3}{8}$ and $a = \alpha - \frac{1}{2} \ge \frac{3}{8}$. For B_α it is the Hermite Biehler property of $z \mapsto \xi(\alpha - iz)$, which holds whenever $\sigma_{\max} \le \alpha$ (Remark 16.6.4; the pairing of ρ with $1 - \bar{\rho}$ proves it). Simplicity follows from the strict monotonicity of the clock in (ii), since the zeros of $A_\alpha(it)$ are transversal crossings of half integer levels. The remaining statements are the propositions cited with $\alpha > \sigma_{\max}$. (iv) In the normalisation $H_0(z) = \frac{1}{8}\Xi(\frac{z}{2})$ the zeros of H_0 lie in $|\operatorname{Im} z| \le 2\Delta$, and de Bruijn's theorem [73] makes H_t real rooted for $t \ge \frac{1}{2}(2\Delta)^2$. (v) is immediate. □

The bound $\Lambda \le \frac{9}{32}$ is weaker than the bound 0.2 of Platt and Trudgian [247], which sharpens the bound 0.22 of the Polymath project [282] (Table 8.14), obtained by their barrier method and extensive computation; a smaller a priori Δ might lower those bounds, but that requires redoing the computation. And the stable tails of (i) show what the GGC laws inside the strip look like: the laws at basepoints between $\frac{7}{8}$ and 1 have heavy tails of nearly the same weight as the law at 1, and only the law at $\frac{1}{2}$ has a finite mean. At the edge the density of (i) can be computed from ξ with no zeros: on $0 \le \tau \le 60$, $\operatorname{Re} \frac{\xi'}{\xi}(\frac{7}{8} + i\tau)$ is smallest at $\tau = 0$, where it equals $\frac{\xi'}{\xi}(\frac{7}{8}) = 0.0173248$, and near each ordinate it has a peak of height at least about $\frac{8}{3}$ and width $\frac{3}{4}$, narrower than the smallest gap 1.77 between ordinates there, so the Thorin measure at $\alpha = \frac{7}{8}$ already separates the zeros (Figure 16.3).

Elsewhere in the book the half plane acts as Table 16.3 lists; every entry is conditional on [226] and is proved where it is cited, except the heuristic count in the last row.

PROPOSITION 16.12.2 (Poles of the reciprocal under $\mathrm{Q}_{7/8}$). *Assume* $(\mathrm{Q}_{7/8})$*. Every pole* $(\rho - \frac{1}{2})^2$ *of* $1/\xi(\frac{1}{2} + \sqrt{w})$ *lies in the closed region* $(\operatorname{Im} w)^2 \le 4\Delta^2(\Delta^2 - \operatorname{Re} w) \le$

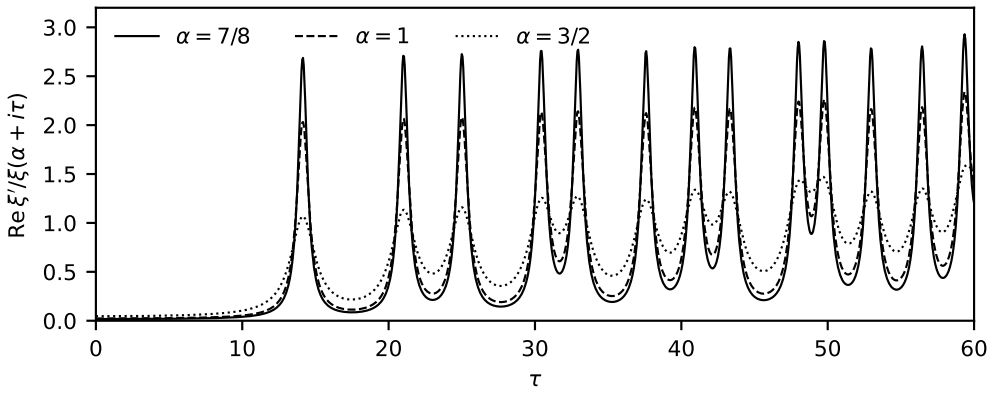


Figure 16.3: The function $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$, which is π times the Thorin density of Theorem 16.12.1(i) in the ordinate variable, at $\alpha=\frac{7}{8}$ (solid), 1 (dashed) and $\frac{3}{2}$ (dotted) for $0\le\tau\le 60$, computed from ξ with no zeros (`pol_qrh_edge.py`). The zeros in this range lie on the critical line, and each contributes a Lorentzian of width $\alpha-\frac{1}{2}$ and height $1/(\alpha-\frac{1}{2})$, that is $\frac{8}{3}$, 2 and 1; at $\alpha=\frac{7}{8}$ the computed peaks are 2.69 to 2.93. On this range the minimum is at $\tau=0$, where the value is $\frac{\xi'}{\xi}(\alpha)=0.0173$, 0.0231 and 0.0461. Positivity at every height for $\alpha>\frac{7}{8}$ is Theorem 16.12.1(ii), under $(\mathrm{Q}_{7/8})$, and not a consequence of the figure.

Chapter	What $(\mathrm{Q}_{7/8})$ gives there
9	the compensated Lévy form for $\frac{7}{8}<\alpha<1$, with error $O(X^{7/8-\alpha+\delta})$ (Proposition 9.1.13)
10	$R(x)=O(x^{7/16+\delta})$ (Theorem 10.2.7); the range $0<y<\frac{4\pi}{3}$ of Remark 10.1.7
12	step 7 of the path, with $\alpha=\frac{7}{8}$ (Section 12.1.2)
13	$\cosh(at)\,p$ ferromagnetic for $a\ge\frac{3}{8}$; $\Omega\le\frac{3}{8}$ (Section 13.4.2)
15, 19	$\Lambda\le\frac{9}{32}$, weaker than 0.2 (Corollary 15.3.3)
18	the bound on ζ'/ζ behind $v_\alpha>0$ is Theorem 16.12.1(ii) (Section 18.5)
20	equivalent to the damped form being nonnegative; $\lambda_{\min}(\frac{7}{8},a)\ge 0$ (Theorem 20.1.2)
21	with the half plane for every Dirichlet L function, Theorem 21.6.5 for every $q\ge 3$; the heuristic count of Section 21.1.2 does not lower K

Table 16.3: Where the half plane $\operatorname{Re} s>\frac{7}{8}$ of [226], which has not been refereed, acts outside this section.

$\frac{9}{16}(\frac{9}{64}-\operatorname{Re} w)$, *bounded by a parabola with focus* 0 *and vertex* $\Delta^2\le\frac{9}{64}$, *against the vertex* $\frac{1}{4}$ *with no hypothesis. Theorem 4.3.9 holds with the sliver* $\mathcal{E}_{T_0}$ *replaced by*

the thinner set

$$\mathcal{E}'_{T_0} = \Big\{ x+iy : \ x < -T_0^2 + \tfrac{3}{4}T_0 + \tfrac{9}{64},\ y^2 \le \tfrac{9}{16}\big(\tfrac{9}{32} - x\big) \Big\},$$

whose half width at the abscissa $-X^2$ is about $\frac{3}{4}X$, against about X for $\mathcal{E}_{T_0}$.

Proof. In the notation of the proof of Theorem 4.3.9 a zero is $\gamma = a + ib$ with $0 \le b \le \Delta \le \frac{3}{8}$, and its pole $w = -\gamma^2$ has $\operatorname{Re} w = b^2 - a^2$ and $(\operatorname{Im} w)^2 = 4a^2b^2 = 4b^2(b^2 - \operatorname{Re} w)$, which increases in b. In that proof, $b \le \frac{3}{8}$ in place of $b < \frac{1}{2}$ gives $x \le 2b^2 - (a-b)^2 < -T_0^2 + \frac{3}{4}T_0 + \frac{9}{64}$ and $y^2 \le 4b^2(2b^2 - x) \le \frac{9}{16}(\frac{9}{32} - x)$ on D_γ, and the rest is unchanged. □

16.12.3 The Gaussian smoothed Möbius sum

The proof of [226] bounds smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$ (Section 12.1.2), and its last step is a continuation: a bound on the growth of such a sum makes the reciprocal L function analytic in a half plane. Liu [192] claims an optimisation of its exponents to the half plane $\operatorname{Re} s > 0.874957\ldots$, and in a second preprint [193], the half plane $\operatorname{Re} s > B_\flat = 0.8748866470\ldots$, with $B_\flat$ the root near 0.8749 of $133488B^4 - 376596B^3 + 384534B^2 - 167541B + 26231$, for every Hecke L function of finite order over every number field K, obtained over $K(\zeta_{12})$ and carried down to K. Both claims rest on the argument of [226]; neither is refereed, and the Lean proof covers $\frac{7}{8}$ only. We checked only that $B_\flat$ is a root of the quartic. Nothing below uses them: $B_\flat$ lies 1.13×10^{-4} below $\frac{7}{8}$ and would change the bound of Theorem 16.12.1(iv) from 0.28125 to $2(B_\flat - \frac{1}{2})^2 = 0.28108$. For ζ alone, with a Gaussian weight in the Mellin variable, the sum and its explicit formula can be written down and computed, and nothing in this subsection depends on $(\mathrm{Q}_{7/8})$. For $x > 0$ put

$$M_{\mathrm{g}}(x) = \frac{1}{2\pi i}\int_{(2)} x^{v-2/3}\,\frac{e^{(v-5/6)^2}}{\zeta(v)}\,dv = \frac{x^{-2/3}}{2\sqrt{\pi}}\sum_{n\ge 1}\mu(n)\Big(\frac{x}{n}\Big)^{5/6}\exp\Big(-\frac{1}{4}\log^2\frac{x}{n}\Big), \tag{16.12.1}$$

where the second form follows from $1/\zeta(v) = \sum_n \mu(n) n^{-v}$ and

$$\frac{1}{2\pi i}\int_{(c)} e^{vw}e^{(v-c)^2}dv = \frac{e^{cw - w^2/4}}{2\sqrt{\pi}},$$

a Möbius sum smoothed by a Gaussian of standard deviation $\sqrt{2}$ in $\log n$ around $\log x$.

Proposition 16.12.3. *Let $\sigma_{\max} = \sup \operatorname{Re}\rho$.*

(i) For every $K \ge 1$ there are heights $T_m \in [m, m+1]$ such that for every $x > 0$

$$M_{\mathrm{g}}(x) = \lim_{m\to\infty}\sum_{|\gamma|<T_m}\operatorname*{Res}_{v=\rho}\frac{x^{v-2/3}e^{(v-5/6)^2}}{\zeta(v)} + \sum_{k=1}^{K}\frac{e^{(2k+5/6)^2}}{\zeta'(-2k)}\,x^{-2k-2/3} + I_K(x),$$

with

$$I_K(x) = \frac{1}{2\pi i}\int_{(-2K-1)} x^{v-2/3}e^{(v-5/6)^2}\zeta(v)^{-1}dv = O(x^{-2K-5/3}).$$

At a simple zero the residue is $e^{(\rho-5/6)^2}x^{\rho-2/3}/\zeta'(\rho)$. *As* $K\to\infty$ *the trivial zero series diverges: its kth term has modulus*

$$\frac{2(2\pi)^{2k}e^{(2k+5/6)^2}x^{-2k-2/3}}{(2k)!\,\zeta(2k+1)}.$$

(ii) $\sigma_{\max}\le\sigma$ *if and only if* $M_{\mathrm g}(x)=O(x^{\sigma-2/3+\varepsilon})$ *as* $x\to\infty$ *for every* $\varepsilon>0$. *RH is the exponent* $-\frac16$ *and* $(\mathrm{Q}_{7/8})$ *the exponent* $\frac{5}{24}$.

(iii) $M_{\mathrm g}(x)=c_1x^{-8/3}+E(x)+O(x^{-11/3})$, *with* $c_1=e^{(17/6)^2}/\zeta'(-2)=-1.00659\times10^5$ *and* E *the sum over the zeros in (i), and* $E(x)$ *is not* $O(x^{\sigma-2/3})$ *for any* $\sigma<\sigma_{\max}$. *So* $c_1x^{-8/3}$ *is the leading term of* $M_{\mathrm g}$ *on a finite range of* x *only, not its asymptotic.*

Proof. (i) Move the line of integration from $\operatorname{Re}v=2$ to $\operatorname{Re}v=-2K-1$ inside the rectangles $|\operatorname{Im}v|<T_m$. On the horizontal sides $|e^{(v-5/6)^2}|=e^{(\operatorname{Re}v-5/6)^2-T_m^2}$, and $T_m\in[m,m+1]$ can be chosen with $|\zeta(\sigma\pm iT_m)|^{-1}\le e^{C(\log T_m)^2}$ for $-2K-1\le\sigma\le2$ ([324, Chapter 9] for $-1\le\sigma\le2$, the functional equation to the left of -1), so the horizontal integrals tend to zero. The residues at $v=-2k$ use $\zeta'(-2k)=(-1)^k(2k)!\,\zeta(2k+1)/(2(2\pi)^{2k})$; at $v=1$ the integrand vanishes. (ii) If $M_{\mathrm g}(x)=O(x^{\sigma-2/3+\varepsilon})$, then $\int_0^\infty M_{\mathrm g}(x)x^{2/3-v}\,dx/x$ converges absolutely for $\operatorname{Re}v>\sigma+\varepsilon$, since near $x=0$ the series in (16.12.1) gives $M_{\mathrm g}(x)x^{2/3}=O(x^{5/6}e^{-\log^2x/4})$. By Mellin inversion it equals $e^{(v-5/6)^2}/\zeta(v)$ for $\operatorname{Re}v>1$, hence for $\operatorname{Re}v>\sigma+\varepsilon$ by continuation, so ζ has no zero there. Conversely, if $\sigma_{\max}\le\sigma$, then $1/\zeta(v)=O(|\operatorname{Im}v|^{\varepsilon})$ on $\operatorname{Re}v\ge\sigma+\varepsilon$, $|\operatorname{Im}v|\ge1$, by the Borel and Carathéodory argument of [324, Section 14.2], which uses only the absence of zeros to the right of σ; moving the line to $\operatorname{Re}v=\sigma+\varepsilon$ gives the bound. (iii) Take $K=1$ in (i). If $E(x)=O(x^{\sigma-2/3})$ with $\sigma<\sigma_{\max}$, then $M_{\mathrm g}(x)=O(x^{\sigma'-2/3})$ with $\sigma'=\max(\sigma,-2)<\sigma_{\max}$, and (ii) gives $\sigma_{\max}\le\sigma'$, a contradiction. □

Computed three ways, from the Möbius sum over $n\le4\times10^7$ and from the integral on $\operatorname{Re}v=2$ and on $\operatorname{Re}v=0.6$, between which no zero lies below height 3×10^{12} [247], far beyond the reach of the Gaussian factor, $M_{\mathrm g}(10^3)=-5.53094\times10^{-4}$; the ratio $M_{\mathrm g}(x)/(c_1x^{-8/3})$ is 0.5495, 0.9470, 0.99856 and 0.999983 at $x=10^3$, 10^4, 10^5 and 10^6 (`qrh.py`; the Möbius sum is accurate only while its window $n\lesssim xe^8$ fits below 4×10^7). The first pair of zeros has coefficient $|e^{(\rho_1-5/6)^2}/\zeta'(\rho_1)|=2.404\times10^{-87}$ and overtakes $c_1x^{-8/3}$ only at $x\approx10^{36.5}$. Below that the sum is numerically its trivial zero term, and whether its exponent is $-\frac16$ or larger cannot be seen at any

computable x. The continuation step does not need to see it: it reads the exponent, which the rightmost zero sets however small its coefficient. Since $\sigma_{\max} = \frac{1}{2}$ is RH, reading an exponent is no obstruction in itself; what stops a method of this kind is the size of the exponents it can prove.

M_{g} is the Mellin counterpart of the density f of the clock T (Section 8.1.1). Both invert a reciprocal, $e^{(v-5/6)^2}/\zeta(v)$ on a vertical line and $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ on a Bromwich line, and both have two expansions. On the prime side each is a Möbius series with no hypothesis, (16.12.1) here and Proposition 8.1.1 there. On the zero side each is a residue series, Proposition 16.12.3(i) here and, under RH with simple zeros, $f(t) = \sum_k R_k e^{-\gamma_k^2 t}$ there. They differ in the archimedean factor. The Gaussian is chosen, and it puts the factor $e^{-\gamma^2}$ into the weight of each zero. The clock carries the gamma factor of ξ, which puts $e^{-\pi\gamma_k/4}$ into $|\xi'(\rho_k)|$, so that the residues R_k grow like $e^{\pi\gamma_k/4}$ up to powers of γ_k and the factor $1/|\zeta'(\rho_k)|$. That is why the spectral expansion of $P(T > t)$ at $t = 0.008$ is far off with 20 zeros and within 10^{-8} with 50 (`bn_tail.py`, part b), while M_{g} shows no zero below $x = 10^{36}$.

Proposition 16.12.4 (Mertens function under $\mathrm{Q}_{7/8}$). *Assume* ($\mathrm{Q}_{7/8}$). *Then* $M(x) = \sum_{n\le x}\mu(n) = O(x^{7/8+\varepsilon})$ *for every* $\varepsilon > 0$, *and* $\sum_n \mu(n)n^{-s}$ *converges to* $1/\zeta(s)$ *on* $\operatorname{Re} s > \frac{7}{8}$. *So the Möbius series of* $1/\zeta(\frac{1}{2}+\sqrt{s})$ *converges on* $H_{3/8} = \{\operatorname{Re}\sqrt{s} > \frac{3}{8}\} = \{u+iy:\ u > \frac{9}{64} - \frac{16}{9}y^2\}$, *which contains the region* $H_{1/2}$ *of absolute convergence of Section 6.1.*

Proof. With no zero in $\operatorname{Re} s > \frac{7}{8}$, $\log\zeta$ is analytic there, and the argument of [324, Section 14.2], with $\frac{7}{8}$ in place of $\frac{1}{2}$, gives $1/\zeta(s) = O(|t|^{\varepsilon})$ uniformly on $\operatorname{Re} s \ge \frac{7}{8}+\varepsilon$, $|t| \ge 2$. Perron's formula with the contour on $\operatorname{Re} s = \frac{7}{8}+\varepsilon$, as in [324, Section 14.25], gives the bound on M. Partial summation makes the series converge locally uniformly on $\operatorname{Re} s > \frac{7}{8}$, and its sum, analytic there, equals $1/\zeta$ on $\operatorname{Re} s > 1$. With $\sqrt{s} = c+id$, $s = c^2-d^2+2icd$, and $c = \frac{3}{8}$ traces $x = \frac{9}{64} - \frac{16}{9}y^2$. □

This is the unsmoothed form of Proposition 16.12.3(ii), whose exponent $\frac{5}{24}$ is $\frac{7}{8} - \frac{2}{3}$.

Remark 16.12.5 (Least prime in a progression). Assume, as [226] claims, that no Dirichlet L function vanishes in $\operatorname{Re} s > \frac{7}{8}$, so that every nontrivial zero has $\frac{1}{8} \le \beta \le \frac{7}{8}$ and $|\rho| \ge \frac{1}{8}$. The explicit formula for $\psi(x,\chi)$ [71, Chapters 19 and 20] with $T = x$, and $\sum_{|\gamma|\le x}|\rho|^{-1} = O(\log^2(qx))$, give $\psi(x;q,a) = x/\varphi(q) + O(x^{7/8}\log^2(qx))$ uniformly in $(a,q) = 1$, so the least prime $p \equiv a \pmod q$ is $O(q^{8+\varepsilon})$ (7.9928 with the unrefereed $B_{\flat}$). This is weaker than the unconditional exponent 5 of Xylouris [194], and it is not the Linnik constant K of Chapter 21.

16.12.4 The compensated clock condition at every height

Lemma 16.12.6 (Window count). *For $d > 0$ and $r > d + e$, the number of zeros, counted with multiplicity, with ordinates in $[r - d, r + d]$ is at least*

$$\nu(r,d) = \frac{\vartheta(r+d) - \vartheta(r-d)}{\pi} - s(r+d) - s(r-d),$$

where ϑ is the Riemann Siegel theta function, $s(t) = 2.5167$ for $e \le t \le X_0 = 30\,610\,046\,000$, and $s(t) = 0.1095 \log t + 0.2042 \log\log t + 3.0305$ for $t > X_0$.

Proof. $N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t)$ (Section 2.2.1), and $|S(t)| \le s(t)$ for $t \ge e$ by (1.7) and Corollary 1.4 of Hasanalizade, Shen and Wong [109], as in the proof of Proposition 9.1.11. The bound of Trudgian [329], which is often quoted, has an incomplete proof [23, 109] and is not used. □

Theorem 16.12.7 (Compensated clock under $\mathrm{Q}_{7/8}$, counting form). *Assume $(\mathrm{Q}_{7/8})$ and let $\alpha > \frac{7}{8}$. Put $c_{\min} = \alpha - \frac{7}{8}$, $c_{\max} = \alpha - \frac{1}{8}$, $f(c,x) = c/(c^2 + x^2)$ and $m(d) = \min\{f(c_{\min}, d), f(c_{\max}, d)\}$. If for some $d > 0$ and some $T_0 > d + e$ one has $\nu(r,d)\, m(d) > b_\alpha$ for all $r \ge T_0$, then*

$$D_\alpha(r) = \operatorname{Re} \frac{\xi'}{\xi}(\alpha + ir) - \frac{\xi'}{\xi}(\alpha) > 0 \qquad \text{for all } r \ge T_0.$$

For $r + d \le H = 3 \times 10^{12}$ the conclusion also holds if $m(d)$ is replaced by $f(\alpha - \frac{1}{2}, d)$.

Proof. By the Hadamard product, $D_\alpha(r) = \sum_\rho f(\alpha - \beta_\rho, r - \gamma_\rho) - b_\alpha$, summed symmetrically over all zeros $\rho = \beta_\rho + i\gamma_\rho$ counted with multiplicity. Under $(\mathrm{Q}_{7/8})$ every width $c_\rho = \alpha - \beta_\rho$ lies in $[c_{\min}, c_{\max}]$, because $\frac{1}{8} \le \beta_\rho \le \frac{7}{8}$, so every term is positive. For $|r - \gamma_\rho| \le d$ we have $f(c_\rho, r - \gamma_\rho) \ge f(c_\rho, d)$, and $f(\cdot, d)$ increases on $(0, d)$ and decreases on (d, ∞), so its minimum on an interval is attained at an endpoint and $f(c_\rho, d) \ge m(d)$. Lemma 16.12.6 counts the terms, whence $D_\alpha(r) \ge \nu(r,d)\, m(d) - b_\alpha$. For $r + d \le H$ every zero with ordinate in $[r - d, r + d]$ lies on the critical line by the verification of [247], so $c_\rho = \alpha - \frac{1}{2}$ and $f(c_\rho, d) = f(\alpha - \frac{1}{2}, d)$. □

Table 16.4 gives the heights obtained. For each α and each half width d put $k = \lfloor b_\alpha / m(d) \rfloor + 1$; if the window holds at least k zeros then $k\, m(d) > b_\alpha$, and by Lemma 16.12.6 it does as soon as $\nu(r,d) \ge k$. The table lists, for the d that makes the last height smallest, the height T_0 beyond which $\nu(r,d) \ge k$. Under $(\mathrm{Q}_{7/8})$ the off line threshold is finite for every $\alpha > \frac{7}{8}$, because the window count grows like $(\frac{d}{\pi} - 0.219) \log r$ for $d > 0.69$ while $m(d)$ is fixed. For α close to $\frac{7}{8}$, however, it exceeds the verified height H, and then the off line range $r \ge T_0$ and the on line range $r + d \le H$ leave the heights between H and T_0 uncovered. The two ranges overlap from $\alpha_* = 0.8888$ on: at $\alpha = 0.8888$ the count gives $T_0 = 2.30 \times 10^{12}$ with $d = 3.5$ and sixteen zeros, whereas at $\alpha = 0.8887$ it gives $3.19 \times 10^{12} > H$, and

the smallest α for which the off line threshold T_0 is at most H is 0.88875. For the eight basepoints of the table the ranges overlap, and the count covers every height beyond the on line threshold. The scan in α is made by the script on a grid; we do not prove that the thresholds are monotone in α, and Theorem 16.12.8 makes this unnecessary.

α	b_α	T_0 off line	(d,k)	T_0 on line	(d,k)
0.900	0.01848	1.9×10^6	$(3,7)$	147	$(13,8)$
0.925	0.01963	5.2×10^4	$(3.5,5)$	147	$(13,8)$
0.950	0.02079	1.3×10^4	$(5,7)$	147	$(13,8)$
1.000	0.02310	3.5×10^3	$(6.5,8)$	147	$(13,8)$
1.500	0.04614	324	$(8,5)$	147	$(13,8)$
2.000	0.06907	231	$(10.5,7)$	147	$(13,8)$
5.000	0.20166	191	$(12,8)$	167	$(12.5,8)$
7.800	0.31322	238	$(13,10)$	214	$(12.5,9)$

Table 16.4: Heights beyond which Theorem 16.12.7 guarantees the compensated clock inequality under $(\mathrm{Q}_{7/8})$, rounded up, with the half width d of the window and the number k of zeros that the proof asks for in it. "Off line" uses only $\frac{1}{8}\le\beta\le\frac{7}{8}$; "on line" uses $\beta=\frac{1}{2}$, valid for $r+d\le H$. Computed by `polya_qrh.py` with the bound of Lemma 16.12.6.

The count loses information in one place. A zero at distance $c_{\min}=\alpha-\frac{7}{8}$ from the line $\operatorname{Re}s=\alpha$ contributes only about $c_{\min}/d^2$ to D_α, which tends to 0 with $\alpha-\frac{7}{8}$, whereas b_α stays near 0.0173; this is why the method stops at $\alpha_*=0.8888$ rather than at $\frac{7}{8}$, and why any extension of the verified height only lowers α_* towards $\frac{7}{8}$.

THEOREM 16.12.8 (Compensated clock under a quasi Riemann hypothesis). *Let $\frac{1}{2}\le\alpha_0<1$ and assume that ζ has no zero with $\operatorname{Re}s>\alpha_0$. Then $D_\alpha(r)>0$ for every α with $\alpha_0<\alpha\le1$ and every real $r\neq0$. Together with Proposition 9.1.12, which has no hypothesis, $D_\alpha(r)>0$ for every $\alpha>\alpha_0$ and every real $r\neq0$.*

Proof. Fix α and put $a=\alpha-\frac{1}{2}$ and $c_{\max}=\alpha+\alpha_0-1$, which is less than 1 because $\alpha\le1$ and $\alpha_0<1$. As D_α is even we take $r>0$. By the Hadamard product, as in the proof of Theorem 16.12.7, $D_\alpha(r)=\sum_\rho f(\alpha-\beta_\rho,r-\gamma_\rho)-b_\alpha$ with $f(c,x)=c/(c^2+x^2)$. By hypothesis and the functional equation, $1-\alpha_0\le\beta_\rho\le\alpha_0<\alpha$ for every zero, so every width $c_\rho=\alpha-\beta_\rho$ lies in $(0,c_{\max}]$ and every term is positive.

Window estimate. Fix $d>0$ and let W be the multiset of zeros with ordinate in $[r-d,r+d]$, with n elements. The map $\rho\mapsto1-\bar\rho$ preserves the ordinate and the multiplicity and fixes exactly the zeros on the critical line, so W consists of zeros on the line, of width a, and of pairs $\{\rho,1-\bar\rho\}$ of the same ordinate whose widths c,c' satisfy $c+c'=2a$ however far from the line the zero lies. For $0<c\le c_{\max}$ and $|x|\le d$ we have $c^2+x^2\le c_{\max}^2+d^2$, hence $f(c,x)\ge c/(c_{\max}^2+d^2)$; summing

over W, in which the widths add up to na,

$$D_\alpha(r) \;\ge\; \frac{na}{c_{\max}^2+d^2} - b_\alpha \;\ge\; a\Big(\frac{n}{c_{\max}^2+d^2} - 0.04621\Big),$$

by Lemma 16.11.12. With $d=4$ and $c_{\max}<1$ we have $c_{\max}^2+d^2<17$ and $0.04621\cdot 17 = 0.7856 < 1$, so $n\ge 1$ suffices: if the window $[r-4,r+4]$ contains the ordinate of at least one zero, then $D_\alpha(r) > a\big(\frac{1}{17}-0.04621\big) > 0.0126\,a$.

It remains to show that the window contains an ordinate, or that $D_\alpha(r)>0$ for another reason, at every $r>0$. There are three ranges.

(a) $0<r\le\theta_0=\gamma_1-\alpha/\sqrt{3}$, where $\theta_0\ge\gamma_1-1/\sqrt{3}=13.557$. Every zero has $|r-\gamma_\rho|\ge\gamma_1-r\ge\alpha/\sqrt{3}>c_\rho/\sqrt{3}$, because $c_\rho<\alpha$, and $x\mapsto f(c,x)$ is strictly convex where $|x|>c/\sqrt{3}$, since $\partial_x^2 f = 2c(3x^2-c^2)/(c^2+x^2)^3$. So D_α is strictly convex on $[-\theta_0,\theta_0]$ (the series of second derivatives converges uniformly there), even, and zero at the origin, hence positive for $0<|r|\le\theta_0$; this is the argument of Proposition 9.1.11.

(b) $\theta_0<r\le\gamma_{1700}+4=2201.26$. The first 1700 ordinates, computed in ball arithmetic and stored in `data/zeros_1700.txt`, have consecutive differences at most 6.8873 (the largest, between $\gamma_1=14.1347$ and $\gamma_2=21.0220$; the next is 5.4140), so the windows $[\gamma_k-4,\gamma_k+4]$ cover $[\gamma_1-4,\gamma_{1700}+4]$, and $\gamma_1-4=10.13<\theta_0$. Every r of the range lies in one of them, that is, its window contains an ordinate.

(c) $r\ge 2200$. By Lemma 16.12.6 with $d=4$ the window holds at least $\nu(r,4)$ zeros, and we show $\nu(r,4)\ge 2.42$. Put $I(r)=(\vartheta(r+4)-\vartheta(r-4))/\pi$. Since $\vartheta''(t)=\frac14\sum_{n\ge0}(n+\frac14)t\,\big((n+\frac14)^2+\frac{t^2}{4}\big)^{-2}>0$, I increases, and on $[2200,X_0-4]$ we have $\nu(r,4)=I(r)-2\cdot 2.5167\ge\nu(2200,4)=7.4591-5.0334=2.4257$. For $r\ge X_0-4$ Stirling's formula gives $I(r)=\frac{4}{\pi}\log\frac{r}{2\pi}+O(r^{-2})$ with an error below 10^{-9}, and s is nondecreasing and, at $t=r+4\ge X_0$, at most $s_{\log}(t)=0.1095\log t+0.2042\log\log t+3.0305$, so $\nu(r,4)\ge\frac{4}{\pi}\log\frac{r}{2\pi}-2s_{\log}(r+4)-10^{-9}$. The derivative of the right side is at least $\big(\frac{4}{\pi}-2\cdot0.1095-\frac{2\cdot0.2042}{\log X_0}\big)/(r+4)>1/(r+4)$, so it increases, and at $r=X_0-4$ it equals $28.4018-12.6490-10^{-9}>15.75$. Hence $\nu(r,4)\ge2.42>0$ for all $r\ge2200$, and the window contains a zero. Ranges (b) and (c) overlap in $[2200,2201.26]$. □

For $\alpha_0=\frac12$ the theorem is the direction RH ⇒ compensated positivity of Theorem 9.1.18(vi), and the window argument is that of the proof of part (v) of Theorem 9.1.18.

Corollary 16.12.9. *For $\frac12\le\alpha_0<1$ the following are equivalent: (a) ζ has no zero with $\operatorname{Re}s>\alpha_0$; (b) $D_\alpha(r)\ge0$ for every $\alpha\in(\alpha_0,1)$ and every real r at which D_α is defined. When they hold, $D_\alpha(r)>0$ for every $\alpha\in(\alpha_0,1)$ and every real $r\ne0$.*

Proof. (a) implies (b) by Theorem 16.12.8, and (b) implies (a) by Proposition 16.11.9, since $D_\alpha\ge0$ for $\alpha\ge1$ by Proposition 9.1.12. □

For $\alpha_0 = \frac{1}{2}$ this is part (vi) of Theorem 9.1.18.

COROLLARY 16.12.10. *Assume* $(\mathrm{Q}_{7/8})$ *and let* $\alpha > \frac{7}{8}$. *Then the compensated reciprocal* $e^{wb_\alpha}\xi(\alpha)/\xi(\alpha+w)$ *is the Laplace transform in* $u = w^2/2$ *of a GGC, with the Thorin density* (16.11.6) *of the* $2\sin^2$ *lemma, positive at every height. For* $\alpha \geq 1$ *this holds with no hypothesis, by the factorisation of Proposition 9.1.12.*

Proof. The hypotheses of Theorem 16.11.7 hold for ξ with $\delta = \alpha - \frac{7}{8}$, and its condition (16.11.7) is $D_\alpha \geq 0$, which holds by Theorem 16.12.8 for $\frac{7}{8} < \alpha \leq 1$ and by Proposition 9.1.12 for $\alpha \geq 1$. □

So under the claimed theorem the conclusion of the $2\sin^2$ lemma, positivity of the Thorin density of the compensated reciprocal, holds on every line to the right of $\frac{7}{8}$, inside the critical strip for $\frac{7}{8} < \alpha < 1$, even though the hypothesis of the lemma, positivity of the primal measure, cannot even be stated there, because the Euler product diverges.

Numerics. The script `polya_qrhpair.py` (`code/README.md`) computes the constants of the proof of Theorem 16.12.8 for $\alpha_0 = \frac{7}{8}$ and writes them to `data/polya_qrhpair.json`. The margin $a/(c_{\max}^2 + 16) - b_\alpha$ for one zero in the window, with b_α computed by mpmath at sixty values of α in $(\frac{7}{8}, 1]$, is at least 0.0053, with the minimum at $\alpha = 0.8751$, and the ratio b_α/a is at most 0.04620; the inequality $f(c,x) + f(2a-c,x) \geq 2a/(c_{\max}^2 + d^2)$ for a pair holds on grids of α, c and x with smallest ratio 1.0000047. The quantity $\nu(r,4)$ was recomputed with the Stirling series for ϑ and with mpmath at ten heights up to 10^{16}, agreeing to 7.1×10^{-15}; it never exceeds the number of ordinates in $[r-4, r+4]$ for r from 40 to 2190.

The direct computation of D_α agrees. The script `polya_primes.py` finds $D_\alpha(r) > 0$ on $(0, 600]$ at step 0.01 (60 000 heights) at $\alpha = 0.9$, 0.95, 1, 1.001, 2, 4 and 7.8, and on $(0, 200]$ at the fifteen equally spaced basepoints from 1 to 7.8 and at twelve further ones (1.001, 1.01, 1.1, 1.2, 1.5, 2, 3, 4, 5, 6, 7 and 7.789), with the smallest value of D_α at the first height of the grid, where D_α grows like r^2 and $D_\alpha(r)/r^2$ is 1.1×10^{-4} at $\alpha = 1$ and 9.0×10^{-4} at $\alpha = 7.8$. At heights above 2 the minimum over $(2, 600]$ is 3.7×10^{-4}, 4.2×10^{-4} and 4.6×10^{-4} at $\alpha = 0.9$, 0.95 and 1, attained just above $r = 2$. These computations use no hypothesis on the zeros, and for $\alpha < 1$ they are floating point evidence, not a proof.

16.12.5 What it does not give, and what to look for in the proof

The claimed theorem is a fixed half plane. The routes of this chapter and of the book need more.

1. Bondesson's closure gives RH from GGC basepoint laws along a sequence $\alpha_n \downarrow \frac{1}{2}$ (Corollary 4.1.2). $(\mathrm{Q}_{7/8})$ supplies them for $\alpha \geq \frac{7}{8}$ only; the open strip is $\frac{1}{2} < \alpha < \frac{7}{8}$.

2. The even part laws and the accounting theorem need no zero to the right of α; below $\frac{7}{8}$ the term $2N(\alpha, T)$ of Theorem 16.11.15 remains and is controlled only by density estimates.

The structural point is Corollary 16.12.9: a quasi Riemann hypothesis at α_0 is equivalent to the positivity of the compensated Thorin densities of the $2\sin^2$ lemma on every line to the right of α_0. The proof announced in [226] does not go through the Thorin picture (Section 12.1.2), but the equivalence says that whatever it proves has a Thorin reading. If it contains an explicit lower bound for $\operatorname{Re}\frac{\zeta'}{\zeta}$ or for a mollified Dirichlet polynomial on $\operatorname{Re} s > \frac{7}{8}$, then a bound at least as strong as $-\operatorname{Re}\mathcal{A}$ is, by part (ii) of Theorem 16.12.1, positivity of the clock, a Thorin statement in disguise, and the natural next step towards $\alpha_0 < \frac{7}{8}$ is to transfer it directly to the clock: the identity (16.11.9) at $\alpha < 1$, with the prime sum replaced by whatever the proof controls.

Chapter 17

Yoshida's Hermitian forms, Weil positivity and the Thorin programme

In 1992 Hiroyuki Yoshida studied Weil's explicit formula as a Hermitian form on functions supported in a window $[-a, a]$ [349]. As the first author reads the paper, Yoshida proved the form positive definite for $a \leq \frac{1}{2}\log 2$, the range in which no prime power enters, and reduced the Riemann Hypothesis to its nondegeneracy on a completed space for every a (Section 17.1). This chapter reads Yoshida against the Thorin programme of this book (Section 1.1) and the sine squared lemma (1.2.9), and works the open problems that come out of the comparison.

Yoshida's functional $\mathcal{Y}(F) = \sum_\rho \Phi(\rho)$ is Weil's distribution, and the heat trace W of the reciprocal Xi law is half its value on a Gaussian. The sine squared lemma is the hinge: its Brownian level x is Yoshida's support length $2a$. There is an uncertainty barrier for one coefficient: Proposition 17.9.1 bounds the off line part of the tent coefficient $D(x) = x\langle \chi_0, \chi_0 \rangle$ at $a = x/2$ by an explicit $\varepsilon_H(x)$; against the lower bound for the on line part that the first 1700 zeros give (a double precision computation), this keeps D positive at levels below about 47 (Proposition 17.9.2) whatever the zeros above the verified height 3×10^{12} [247] do, so that no computation of D at those levels can detect an off line zero.

What is not claimed. Nothing here proves RH. A number is computed in floating point at the precision stated, or in ball arithmetic where the text says so. The barrier concerns the tent coefficient only. The account of Yoshida's paper in Section 17.1 is the first author's reading of it, used here as context; it was not checked against the paper.

17.1 Yoshida's paper

For $F \in C_c^\infty(\mathbb{R})$ put

$$\Phi(s) = \int_{\mathbb{R}} F(y) e^{(s-1/2)y}\, dy, \qquad \widehat{F}(t) = \Phi(\tfrac{1}{2} + it) = \int_{\mathbb{R}} F(y) e^{ity}\, dy.$$

The letter Φ is local to this chapter and is not the Pólya kernel of Section 1.2. As there, write each nontrivial zero as $\rho = \frac{1}{2} + i\gamma_\rho$, that is $\gamma_\rho = -i(\rho - \frac{1}{2})$. The number γ_ρ is complex for a zero off the line: if $\rho = \frac{1}{2} + \delta + i\gamma$ with γ and δ real, then $\gamma_\rho = \gamma - i\delta$, whereas $\gamma = \operatorname{Im}\rho$ is the ordinate and $\beta_\rho = \operatorname{Re}\rho$ the real part. Then $\Phi(\rho) = \widehat{F}(\gamma_\rho)$, the entire extension of $\widehat{F}$, and Weil's explicit formula reads

$$\begin{aligned}\mathcal{Y}(F) := \sum_\rho \widehat{F}(\gamma_\rho) = \int_{\mathbb{R}} F(y)\big(e^{y/2} + e^{-y/2}\big)dy - (\log\pi)F(0) \\ - \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\big(F(\log n) + F(-\log n)\big) + V_1(F),\end{aligned} \tag{17.1.1}$$

summed over all nontrivial zeros with multiplicity, with the archimedean term in two forms,

$$\begin{aligned}V_1(F) &= \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{F}(t)\operatorname{Re}\psi\Big(\frac{1}{4} + \frac{it}{2}\Big)dt \\ &= -\tfrac{1}{2}\operatorname{Pf}\Big(\frac{1}{|y|}\Big)(F) - (\gamma_E + \log 2)F(0) - \int_{\mathbb{R}}\Big(\frac{e^{|y|/2}}{2\sinh|y|} - \frac{1}{2|y|}\Big)F(y)\,dy,\end{aligned} \tag{17.1.2}$$

where $\psi = \Gamma'/\Gamma$ and $\operatorname{Pf}(1/|y|)(F) = \lim_{\epsilon\downarrow 0}\big(\int_{|y|>\epsilon} F(y)\,|y|^{-1}dy + 2F(0)\log\epsilon\big)$ is the finite part of $1/|y|$, the integral cut at $|y| = 1$. Each term of (17.1.1) depends on F only through its even part, because the zeros are symmetric under $\rho \mapsto 1-\rho$, which negates γ_ρ. The formula holds for the Gaussians used below, whose transforms are entire and decrease rapidly in every strip (the class of Section 1.2), and, in Weil's original form [338, 38], for continuous, piecewise C^1 functions of compact support, the class of the tent and of the products $\chi_n * \widetilde{\chi_m}$ of Sections 17.6 and 17.7.

Remark 17.1.1 (The archimedean term in the real variable). The second line of (17.1.2) is the explicit formula of Section 1.2 with the archimedean integral carried to the variable y and its divergence at $y = 0$ cut at $|y| = 1$; the forms of Theorem 20.1.1 and Section 15.6.1 differ from it only in where γ_E is placed.

The Hermitian form is $\langle\varphi_1, \varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ with $\widetilde{\varphi}(y) = \overline{\varphi(-y)}$. Since $\widehat{\varphi * \widetilde{\varphi}} = |\widehat{\varphi}|^2$ on $\mathbb{R}$, Weil's criterion [338, 38] says that RH holds if and only if $\langle\varphi,\varphi\rangle \geq 0$ for all $\varphi \in C_c^\infty(\mathbb{R})$ (Proposition 7.2.13 and Section 3.5.2). Let $C(a)$ be the space of the functions of $C_c^\infty(\mathbb{R})$ supported in $[-a, a]$, let $\chi_n(y) = (2a)^{-1/2}e^{\pi i n y/a}1_{[-a,a]}(y)$ for $n \in \mathbb{Z}$, an orthonormal basis of $L^2[-a, a]$, and let $K(a)$ be the space of the restrictions to $[-a, a]$ of smooth $2a$ periodic functions, that is the series $\sum c_n\chi_n$ with rapidly decreasing coefficients. Yoshida's results, as the first author reads the paper, are the following.

(Y1) Positivity on odd φ alone is equivalent to RH; on even φ alone, to RH up to

real zeros. The proof is a localisation lemma that concentrates a test vector on one zero.

(Y2) The prime sum in $\langle \varphi_1, \varphi_2 \rangle$ for $\varphi_1, \varphi_2 \in C(a)$ runs over $\log n \le 2a$ only. The form is positive definite on $C(a)$ for small a, and on the complement of the low Fourier modes for any a.

(Y3) The form is nondegenerate on $C(a)$ for every a, unconditionally.

(Y4) There are closed formulas for the coefficients $\langle \chi_n, \chi_m \rangle$ on the basis of $K(a)$.

(Y5) Theorem 1: the form is positive definite on $K(a)$ for $a = \frac{1}{2}\log 2$.

(Y6) Theorem 2: RH holds if and only if the form is nondegenerate on a completion of $K(a)$ for every a. If RH fails there is a first a_0 beyond which positivity is lost, and continuity in a forces degeneracy there.

(Y7) For $\mathbb{Q}$ the form is nondegenerate on $K(a)$ for every a.

The statements (Y1) to (Y7) were not checked against the paper, and the proofs of Sections 17.2, 17.3, 17.4 and 17.5 do not use them.

Throughout, $W(x) = \sum_{\gamma>0} e^{(\rho-1/2)^2 x}$ is the heat trace of Section 1.2, the sum over one zero from each pair $\{\rho, 1-\rho\}$; $T_k = (-1)^k W^{(k)}$ is the ladder, the function written c_k in Section 5.1; U is the Thorin functional of Section 5.3.1, which under RH is the measure $\sum_{\gamma>0} \delta_{\gamma^2}$; and $p_1 = \sum_{\gamma>0} \gamma_\rho^{-2} = \frac{1}{2}(\log \xi)''(\frac{1}{2}) = 0.0231049931$, so that $2p_1$ is the number V_ξ of Remark 9.1.19. The sine squared lemma, in the chart of this chapter, is

$$k_x(w) := e^{-wx} + wx - 1 = \int_0^\infty \log\Big(1 + \frac{u}{z}\Big)\varpi_x(z)\,dz, \qquad u = \frac{w^2}{2},$$
$$\varpi_x(z) = \frac{x}{\pi\sqrt{2z}}\, 2\sin^2\Big(x\sqrt{\tfrac{z}{2}}\Big). \tag{17.1.3}$$

It is (1.2.9) after the substitution $z \mapsto 2z$, and (16.2.3) of Chapter 16.

17.2 The ladder is the diagonal of Weil's form

Let $h_x(y) = (4\pi x)^{-1/2} e^{-y^2/(4x)}$, so $\widehat{h_x}(t) = e^{-t^2 x}$ and $h_x = h_{x/2} * \widetilde{h_{x/2}}$.

Proposition 17.2.1. *For $k \ge 0$ and $x > 0$ let $g_{k,x} = (i\,d/dy)^k h_{x/2}$, so that $\widehat{g_{k,x}}(t) = t^k e^{-t^2 x/2}$. Then*

$$\langle g_{k,x}, g_{k,x} \rangle = \sum_\rho \gamma_\rho^{2k} e^{-\gamma_\rho^2 x} = 2\,T_k(x), \qquad W(x) = \tfrac{1}{2}\langle h_{x/2}, h_{x/2} \rangle.$$

Proof. $F = g_{k,x} * \widetilde{g_{k,x}}$ has $\widehat{F}(t) = t^{2k}e^{-t^2x}$, so $\Phi(\rho) = \gamma_\rho^{2k}e^{-\gamma_\rho^2 x}$. The summand is even in γ_ρ and $\rho \mapsto 1-\rho$ negates γ_ρ, so the sum over all zeros is twice the sum over the zeros with $\gamma > 0$, which is $2T_k(x)$. □

Dividing the prime side of $\langle h_{x/2}, h_{x/2}\rangle$ by two gives the explicit formula (1.2.5) for W. Both sides give $W(0.003) = 1.09522012145284$ (the first 1700 zeros against the primes, agreeing to thirty digits; script `yo_explicit.py`). Since W is completely monotone exactly under RH (Proposition 1.2.1), Proposition 17.2.1 says that RH is positivity of Weil's form on the diagonal of a single two parameter family (k, x), with no cross terms.

17.3 Every large rung fails

The rungs $T_k(x) = \sum_{\gamma>0} w_\rho^k e^{-w_\rho x}$, with $w_\rho = \gamma_\rho^2$, satisfy $T'_{k-1} = -T_k$, and $T_{k-1}(x) \to 0$ as $x \to \infty$ because every exponent has real part $\gamma^2 - \delta^2 > \gamma_1^2 - \frac{1}{4} > 0$, where $\gamma_1 = 14.1347\ldots$ is the least ordinate. Hence

$$T_{k-1}(x) = \int_x^\infty T_k(s)\, ds, \qquad k \geq 1,\ x > 0. \tag{17.3.1}$$

So $T_k \geq 0$ on $(0,\infty)$ forces $T_{k-1} \geq 0$ there: the orders for which $T_k \geq 0$ form an initial segment. This is the argument of Section 5.1.4, where it is stated for the Williamson measures U_k, which are positive exactly when $T_k \geq 0$ on $(0,\infty)$.

Theorem 17.3.1. *(i) If $T_k \geq 0$ on $(0,\infty)$ then $T_j \geq 0$ on $(0,\infty)$ for every $j \leq k$, so the set of such orders is an initial segment of $\mathbb{N}$, and it is all of $\mathbb{N}$ if and only if RH holds. (ii) If RH fails there is K_0 such that for every $k \geq K_0$ there is $x > 0$ with $T_k(x) < 0$. Consequently, for any infinite set $\mathcal{K}$ of orders, RH holds if and only if $T_k \geq 0$ on $(0,\infty)$ for every $k \in \mathcal{K}$. In particular RH is equivalent to $\langle g_{k,x}, g_{k,x}\rangle \geq 0$ for all odd k and all $x > 0$, a positivity statement on odd Hermite Gaussians only.*

Proof. (i) The first claim is (17.3.1). All T_k are nonnegative on $(0,\infty)$ if and only if W is completely monotone, which holds if and only if RH holds (Proposition 1.2.1). (ii) If RH fails, some T_{k_0} takes a negative value, and then by (i) so does every T_k with $k \geq k_0$; let K_0 be the least such k_0. If $T_k \geq 0$ for every k in an infinite set $\mathcal{K}$, then by (i) it holds for every k, since $\mathcal{K}$ is unbounded, and RH holds; the converse is clear. The last statement is Proposition 17.2.1 for odd k. □

For infinite sets $\mathcal{K}$ this is the equivalence of Section 5.1.4 between RH and $U_k \geq 0$ for infinitely many k. What is added here is the reading of T_k as the diagonal of Yoshida's form (Proposition 17.2.1), which turns the odd orders into the Hermite counterpart of Yoshida's parity reduction (Y1), and the location of the failure in the next proposition, which is not needed for the theorem. With

$y = u^2$ and $x = k/y$ the computation is the ray $t = k/u^2$ of Proposition 5.1.10: $e^{-k(\log y - 1)} T_k(k/y)$ is the quantity $e^{-f_0} c_k(t)$ there.

PROPOSITION 17.3.2 (Where the rungs fail). *Suppose RH fails. For a zero* $\rho = \frac{1}{2} + \delta + i\gamma$ *with* $\gamma > 0$ *write* $w_\rho = \gamma_\rho^2 = r_\rho e^{i\theta_\rho}$, *so that* $\theta_\rho = -2\arctan(\delta/\gamma)$ *and* $\sec\theta_\rho = (\gamma^2 + \delta^2)/(\gamma^2 - \delta^2)$. *Then* $S = \max \log\sec\theta_\rho > 0$ *is attained, by finitely many zeros. Let* $w_* = r_* e^{i\theta_*}$ *be one maximiser, of multiplicity* m, *and* $y_* = r_* \cos\theta_* = \gamma_*^2 - \delta_*^2$. *There are* $\eta > 0$ *and* K_1 *and, for every* $k \geq K_1$, *a point* y_k *with* $|y_k - y_*| = O(1/k)$ *such that*

$$e^{-k(\log y_k - 1)} T_k(k/y_k) = -2m\, e^{kS}\big(1 + O(1/k)\big) + O\big(e^{k(S-\eta)}\big) < 0.$$

Proof. Since $|\delta| < \frac{1}{2} < \gamma$, $\operatorname{Re} w_\rho = \gamma^2 - \delta^2 > 0$. The zeros ρ and $1 - \overline{\rho}$ give conjugate values of w, so T_k is real. Evaluate at $x = k/y$ and multiply by the positive factor $e^{-k(\log y - 1)}$:

$$e^{-k(\log y - 1)} T_k(k/y) = \sum_{\gamma > 0} e^{k g(w_\rho, y)}, \qquad g(w, y) = \log\frac{w}{y} - \frac{w}{y} + 1.$$

For fixed $w = re^{i\theta}$, $\operatorname{Re} g(w, y) = \log(r/y) - r\cos\theta/y + 1$ has a unique maximum in y, equal to $\log\sec\theta$, at $y = r\cos\theta$. On line zeros $\theta = 0$ and $\operatorname{Re} g \leq 0$. Suppose an off line zero exists. Since $|\theta_\rho| < 1/\gamma \to 0$, only finitely many zeros have $|\theta_\rho|$ above any given positive bound, so $S = \max\log\sec\theta_\rho$ is attained, by finitely many zeros, and $S > 0$. Every zero other than w_* and $\overline{w_*}$ has, for y in a small neighbourhood J of y_*, $\operatorname{Re} g(w_\rho, y) \leq S - \eta$ for some $\eta > 0$: on line zeros because $\operatorname{Re} g \leq 0$, for zeros with $|\theta_\rho| < |\theta_*|$ because only finitely many have $\sec\theta_\rho$ near the maximum, and for the other maximisers because y_* is not their optimal point. Since $e^{ku} \leq e^u$ for $u \leq 0$ and $k \geq 1$, the remainder satisfies

$$\Big|\sum_{\rho \neq w_*, \overline{w_*}} e^{k g(w_\rho, y)}\Big| \leq e^{k(S-\eta)} \sum_\rho e^{\operatorname{Re} g(w_\rho, y) - S + \eta} \leq C\, e^{k(S-\eta)}, \qquad y \in J,$$

with C independent of k, because $\sum_\rho (r_\rho/y) e^{1 - r_\rho \cos\theta_\rho / y}$ converges uniformly on J. The main term is $2m\, e^{k \operatorname{Re} g(w_*, y)} \cos\big(k \operatorname{Im} g(w_*, y)\big)$ with $\operatorname{Im} g(w_*, y) = \theta_* - r_* \sin\theta_*/y$, whose derivative in y is $r_* \sin\theta_*/y^2 \neq 0$. Choose y_k with $k \operatorname{Im} g(w_*, y_k) \equiv \pi \pmod{2\pi}$ and $|y_k - y_*| = O(1/k)$. Then $\operatorname{Re} g(w_*, y_k) = S - O(1/k^2)$, the main term is $-2m\, e^{kS}(1 + O(1/k))$, and it dominates the remainder for $k \geq K_1$. Hence $T_k(k/y_k) < 0$. □

The proposition gives no explicit K_1, and so no explicit bound for K_0, because C depends on the zeros, and two constraints enter. The first is the phase condition: y_k lies at distance about $\pi y_*/(k\tan\theta_*)$ from y_*, which costs a factor $\exp(-\pi^2/(2k\tan^2\theta_*))$ in the main term, so the asymptotics start only once $k\theta_*^2$ is of order π. The second is the competition of the main term e^{kS}, with $S = \log\sec\theta_*$ and $\sec\theta_* - 1 = 2\delta_*^2/(\gamma_*^2 - \delta_*^2)$, with the nearest true zeros.

Remark 17.3.3 (A stress test). Add a hypothetical pair at $\gamma = 60.5$, $\delta = 0.45$ to the true zeros (only those between heights 40 and 85 matter at these orders: every excluded term is below $e^{-0.23k}$ and, relative to the main terms, below 10^{-420} for $k \geq 3349$), so that $\sec\theta_* - 1 = 1.1065 \times 10^{-4}$ and $y_* = 3660.05$, and test the sign of $T_k(k/y)$ on 3001 points of $|y/y_* - 1| \leq 0.06$. The first failing order is $K_0 = 6349$. Every one of the 3000 orders in $[6349, 9348]$ fails, every one of 395 sampled orders $6349 + 997j$ up to 399167 fails, and none of the orders 1 to 6348 fails on this grid. For the order 6348 the sign is positive at every point of two further grids (200001 points near y_*, with both local minima of the scale free margin polished, where the margin is at least 2.9×10^{-4}, and 18421 points of $y \in [1, 10^8]$ with all 1700 zeros), while at $y = 3584.65$, that is $x = 1.771$, arithmetic with 60 digits gives $T_{6348}(6348/y) = +1.289 \times 10^{19803}$ and $T_{6349}(6349/y) = -5.700 \times 10^{19805}$. By Theorem 17.3.1(i) positivity of T_{6348} gives that of every lower order, so 6349 is the first failing order of this configuration; all of this is computed, not proved (script `yo_ladder.py`, `code/README.md`). For this pair $K_0(\sec\theta_* - 1) = 0.7025$.

The first failing order found on a grid is an upper bound for K_0. For the pair $(\gamma, \delta) = (30.5, 0.45)$ the grid of the stress test gives 1064, while a grid of ± 30 percent gives 396, the negative lobe lying 27 percent from y_*; the other thirteen pairs of the next sentence give the same order on both grids. Over fourteen pairs, with γ from 30.5 to 250.5 and δ from 0.02 to 0.45 (`data/yo_ladder.json`), $K_0(\sec\theta_* - 1)$ lies between 0.03 and 0.82, and at fixed γ it is proportional to δ: the logarithmic slope of $K_0(\sec\theta_* - 1)$ against δ is 1.04 for $\gamma = 60.5$ and 1.05 for $\gamma = 120.5$, and the ratio $K_0(\sec\theta_* - 1)/\delta$ lies between 0.97 and 2.01 for thirteen of the fourteen pairs. So K_0 grows with γ^2 and, at fixed γ over the range tested, like $1/\delta$, not like $(\sec\theta_* - 1)^{-1} \approx \gamma^2/(2\delta^2)$: for $\gamma = 60.5$ and $\delta = 0.02$ the second quantity is 4.6×10^6 and the computed order is 128610. This is an observation from computation, not a theorem; a scale argument (the pair is a Gaussian envelope times a cosine in the variable $\sqrt{k}\log(y/y_*)$, and its negative lobe has to beat the true zeros that sit near it) suggests that K_0 is of order $\gamma^2/(\delta d)$, where d is the distance from γ to the nearest true ordinate, and that γ^2/δ^2 appears only when δ is of the size of d.

17.4 Compact support, the signed Thorin formula, and the first windows

Section 5.3.1 defines the Thorin functional U on the class $\mathcal{T}$ of the functions $\varphi(t) = h(\sqrt{t})$ by (5.3.1), and identifies it with $\varphi \mapsto \sum_k \varphi(a_k)$ through Weil's formula, with no hypothesis on the zeros. In the variable $t = r^2$ the same functional

is the formal signed measure

$$U^{\pm}(dt) = \delta_{-1/4}(dt) + \frac{\operatorname{Re}\psi(\frac14 + \frac{i\sqrt{t}}{2}) - \log\pi}{4\pi\sqrt{t}}\,dt - \frac{1}{2\pi\sqrt{t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(\sqrt{t}\log n)\,dt, \tag{17.4.1}$$

that is $\langle U,\varphi\rangle = \int \varphi\, dU^{\pm}$: the atom is the first term of (5.3.1), the density is its integral term with $t = r^2$, and the third term is the prime sum. The prime series in (17.4.1) is not a tempered distribution in the ordinate $r = \sqrt{t}$. Let Z be the space of the functions $\widehat{F}$ with $F \in C_c^\infty(\mathbb{R})$, and Z' its dual.

Proposition 17.4.1. *In the ordinate $r = \sqrt{t}$ the prime series of* (17.4.1) *defines an element of Z', and for every even $F \in C_c^\infty(\mathbb{R})$ with support in $[-L,L]$,*

$$\mathcal{Y}(F) = 2\int \widehat{F}(\sqrt{t})\,U^{\pm}(dt), \qquad \text{with } \widehat{F}(\sqrt{-1/4}) = \widehat{F}(i/2);$$

only the prime powers $n \le e^L$ contribute.

Proof. Pairing $\cos(r\log n)$ with $\widehat{F}(r)$ gives $\pi\big(F(\log n) + F(-\log n)\big)$, which vanishes for $\log n > L$; hence the series is a finite sum on every element of Z and defines an element of Z'. With $t = r^2$ the three parts of (17.4.1) give, after the factor 2, which is the passage from one zero of each pair $\{\rho, 1-\rho\}$ in (5.3.1) to all zeros in (17.1.1), the pole term $\widehat{F}(i/2) + \widehat{F}(-i/2)$, the term $V_1(F) - (\log\pi)F(0)$, and the prime sum of (17.1.1). The function $\varphi(t) = \widehat{F}(\sqrt{t})$ lies in $\mathcal{T}$, because $\widehat{F}$ is even, entire and $O((1+|r|)^{-N})$ in every strip. □

For $\varphi \in K(a)$ with $a = \frac12\log 2$, $F = \varphi * \widetilde{\varphi}$ is continuous, vanishes at $\pm\log 2$, and the only prime power level in its support is that endpoint. On Yoshida's window, then, the form is the pole and archimedean part alone: his Theorem 1, as the first author reads it, uses only the gamma factor, the pole and the fact that Dirichlet series begin at $n = 2$. Connes and Consani reproved this window from the semilocal trace formula [59]. Beyond it the primes enter: the window of half length 0.8, of autocorrelation support 1.6, contains the prime powers 2, 3 and 4, and Zhu [350] has certified positivity of the form on it (Chapter 18). Windows are indexed by their half length a in the ladder of Section 20.2, each of them a finite problem in the prime powers up to e^{2a}.

17.5 Localisation: a modulated heat trace

Yoshida's localisation lemma (Y1) concentrates a test vector on one zero. Its Gaussian form is already in the book as the Gaussian smoothing $\mathcal{G}(u,t) = \sum_{\gamma>0} e^{-2t(\gamma_\rho - u)^2}$ of the zeros of Proposition 5.1.10, the sum over one zero of each pair $\{\rho, 1-\rho\}$ (there $\widetilde{\gamma}$ is the γ_ρ of this chapter). For real u and $x > 0$ let g be the vector with $\widehat{g}(t) = e^{-(t-u)^2x/2}$, so that $g * \widetilde{g} = h_x e^{-iuy}$ and $\widehat{g * \widetilde{g}}(t) = e^{-(t-u)^2x}$.

PROPOSITION 17.5.1. *(i)* $\langle g,g\rangle = \mathcal{Y}(h_x e^{-iuy}) = \sum_\rho e^{-(\gamma_\rho - u)^2 x} = \mathcal{G}(u,\frac{x}{2}) + \mathcal{G}(-u,\frac{x}{2})$, *which is* $2W(x)$ *at* $u=0$. *(ii) Unconditionally,*

$$\mathcal{G}(u,\tfrac{x}{2}) + \mathcal{G}(-u,\tfrac{x}{2}) = 2e^{-(u^2-1/4)x}\cos(ux) - \frac{\log\pi}{\sqrt{4\pi x}} + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-(t-u)^2 x}\,\operatorname{Re}\psi\Big(\frac{1}{4}+\frac{it}{2}\Big)dt - \frac{1}{\sqrt{\pi x}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\cos(u\log n)\,e^{-(\log n)^2/(4x)}.$$

(iii) Under RH it equals $\int \big[e^{-(\sqrt{t}-u)^2 x} + e^{-(\sqrt{t}+u)^2 x}\big]U(dt)$. *(iv) RH holds if and only if it is nonnegative for all real* u *and all* $x>0$.

Proof. (i) The transform $e^{-(t-u)^2x}$ is entire, so (17.1.1) applies; the zero $1-\rho$ of the pair $\{\rho,1-\rho\}$ has $\gamma_{1-\rho} = -\gamma_\rho$ and gives $\mathcal{G}(-u,\frac{x}{2})$, and at $u=0$ the sum is $2W(x)$ by Proposition 17.2.1. (ii) Apply (17.1.1) to $F = h_x e^{-iuy}$: the pole term is $\widehat{F}(i/2)+\widehat{F}(-i/2) = e^{-(u+i/2)^2x} + e^{-(u-i/2)^2x} = 2e^{-(u^2-1/4)x}\cos(ux)$, the term $-(\log\pi)F(0)$ is $-(\log\pi)/\sqrt{4\pi x}$, $V_1(F)$ is the integral, and the prime sum is $-\sum\Lambda(n)n^{-1/2}h_x(\log n)\big(e^{-iu\log n}+e^{iu\log n}\big)$. This is (5.1.3) at $t=x/2$, term by term: its polar term $2e^{t/2-2tu^2}\cos 2tu$, its integral $A(u,t)$ (with $m(r) = \operatorname{Re}\psi(\frac14+\frac{ir}{2}) - \log\pi$) and its prime term $P(u,t)$ are the four terms above, because $2t=x$. (iii) Group $\pm\gamma$ in $\sum_{\gamma>0}\big[e^{-x(\gamma-u)^2}+e^{-x(\gamma+u)^2}\big]$ and use $U=\sum_{\gamma>0}\delta_{\gamma^2}$. (iv) This is Proposition 5.1.11, in the form of the remark after its proof: the term $\mathcal{G}(-u,t)$, which comes from the zeros $1-\rho$, is too small there to change the sign. □

The formula (ii) agrees with (5.1.3); Table 5.3 computes the same identity. A vector modulated to a centre u within $\frac{\delta}{2}$ of the ordinate γ of an off line zero $\rho = \frac12+\delta+i\gamma$, with finitely many exceptional centres, makes the form negative at arbitrarily large x (Proposition 5.1.11), and u is free: modulation reaches a zero above any verified height, which is why the barrier of Proposition 17.9.1 is stated for the tent coefficient only.

The ladder reaches the height u only at order $k \approx u^2 x$, because the weight $r^{2k}e^{-r^2x}$ of $T_k(x)$ peaks at $r=\sqrt{k/x}$ and the power r^{2k} is its only way to move the window; Theorem 5.1.13 states the consequence, that verification to a height H reaches the orders up to about the square of H. Modulation moves the window at no cost. Figure 17.1 shows the modulated heat trace resolving isolated zeros into bumps of height one at $x=0.5$; nearby zeros overlap, and zeros less than 2 apart merge into one bump (computed, script `yo_ladder.py`).

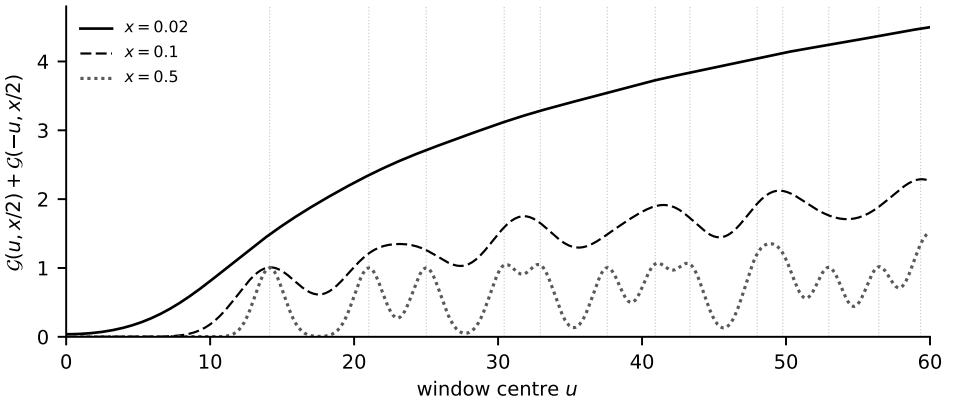


Figure 17.1: The modulated heat trace $\mathcal{G}(u, \frac{x}{2}) + \mathcal{G}(-u, \frac{x}{2})$ against the window centre u at $x = 0.02$ (solid), 0.1 (dashed) and 0.5 (dotted), from the first 1700 zeros; dotted verticals mark the ordinates of the zeros.

17.6 The tent and the sine squared lemma

Yoshida's basic vector is $\chi_0 = (2a)^{-1/2} 1_{[-a,a]}$, and $\chi_0 * \widetilde{\chi_0} = (2a)^{-1}(2a - |y|)_+$. Put $x = 2a$,

$$\Delta_x(y) = (x - |y|)_+, \qquad D(x) = \mathcal{Y}(\Delta_x) = x\,\langle \chi_0, \chi_0 \rangle,$$

where χ_0 is taken for $a = x/2$. So Δ_x is the tent of height x on $[-x, x]$, and $D(x)$ is the single coefficient $\langle \chi_0, \chi_0 \rangle$ of Yoshida's matrix multiplied by the length x of the window.

Proposition 17.6.1. *Let $k_x(w) = e^{-wx} + wx - 1$ be the compensated exponential on the left of the sine squared lemma* (1.2.9)*, and let $\gamma_\rho = -i(\rho - \frac{1}{2})$, which is complex for an off line zero.*

(a) For complex w and real t,

$$\int_{\mathbb{R}} \Delta_x(y) e^{wy}\, dy = \frac{k_x(w) + k_x(-w)}{w^2} = \frac{4\sinh^2(wx/2)}{w^2},$$

$$\widehat{\Delta_x}(t) = \frac{4\sin^2(xt/2)}{t^2} \geq 0.$$

(b) Unconditionally $D(x) = \sum_\rho 4\sin^2(\gamma_\rho x/2)/\gamma_\rho^2$, the sum over all nontrivial zeros. Under RH, with the Thorin measure $U = \sum_{\gamma>0} \delta_{\gamma^2}$ of Proposition 1.2.1,

$$D(x) = 8 \int \frac{\sin^2(x\sqrt{t}/2)}{t}\, U(dt)$$

$$= \frac{8\pi}{x} \int \frac{x \sin^2(x\sqrt{t}/2)}{\pi\sqrt{t}}\, \frac{U(dt)}{\sqrt{t}},$$

so D samples the Thorin density of the sine squared lemma (1.2.9) *at level x on the Thorin atoms $t = \gamma^2$, with the weight $t^{-1/2}$.*

(c) Unconditionally,

$$D(x) = 32\sinh^2\frac{x}{4} - x\log\pi - 2\sum_{n\le e^x}\frac{\Lambda(n)}{\sqrt{n}}\log\frac{e^x}{n} + V_1(\Delta_x),$$

$$V_1(\Delta_x) = x - x\log x - (\gamma_E + \log 2)x \\ - 2\int_0^x \Big(\frac{e^{y/2}}{2\sinh y} - \frac{1}{2y}\Big)(x-y)\,dy,$$

and the prime term is a Riesz mean of order one of $\Lambda(n)n^{-1/2}$.

(d) With ψ the digamma function, $V_1(\Delta_x)$ is also the residue series

$$V_1(\Delta_x) = x\,\psi(\tfrac14) + \tfrac12\psi'(\tfrac14) - 2\sum_{k\ge0}\frac{e^{-(2k+1/2)x}}{(2k+\frac12)^2}.$$

Proof. (a) Taylor's formula with integral remainder gives $k_x(w) = w^2\int_0^x (x-y)e^{-wy}dy$, so $\int \Delta_x e^{wy}dy = (k_x(w)+k_x(-w))/w^2$, and $k_x(w)+k_x(-w) = e^{wx} + e^{-wx} - 2 = 4\sinh^2(wx/2)$; put $w = it$.

(b) Evaluate $\widehat{\Delta_x}$ at γ_ρ and sum over the zeros; the series converges absolutely, since $|\sin^2(x\gamma_\rho/2)| \le \cosh^2(x/4)$ for $|\operatorname{Im}\gamma_\rho| < \frac12$ and $\sum|\gamma_\rho|^{-2} < \infty$. Under RH group $\pm\gamma$, which gives $D(x) = 2\sum_{\gamma>0} 4\sin^2(\gamma x/2)/\gamma^2$.

(c) Insert Δ_x in (17.1.1). By (a) at $w = \pm\frac12$ the first term is $2\cdot 4\sinh^2(x/4)/(1/4) = 32\sinh^2(x/4)$, the second is $-(\log\pi)\Delta_x(0) = -x\log\pi$, and $\Delta_x(\log n) + \Delta_x(-\log n) = 2(x - \log n)_+$. For V_1 use the second form of (17.1.2), with $\operatorname{Pf}(1/|y|)(\Delta_x) = 2(x\log x - x)$ for all $x > 0$, because $\int_{\epsilon<|y|<x}(x-|y|)\,dy/|y| = 2x\log(x/\epsilon) - 2x + 2\epsilon$. To see that (17.1.1) holds for the tent, apply it to $\Delta_x * h_\eta$, with $h_\eta(y) = (4\pi\eta)^{-1/2}e^{-y^2/(4\eta)}$, whose transform $\widehat{\Delta_x}(t)e^{-\eta t^2}$ decays rapidly in every strip, and let $\eta\downarrow 0$. The zero sum converges by dominated convergence, since $|e^{-\eta\gamma_\rho^2}| \le e^{\eta/4}$ and the bound of (b) holds; the Fourier form of V_1 converges because $\widehat{\Delta_x}(t) = O(t^{-2})$; and the prime sum converges because $\Delta_x * h_\eta \to \Delta_x$ uniformly with Gaussian tails.

(d) By the Fourier form of (17.1.2), $V_1(\Delta_x) = \frac{1}{2\pi}\int \widehat{\Delta_x}(t)\psi(\frac14 + \frac{it}{2})\,dt$, because the imaginary part of $\psi(\frac14+\frac{it}{2})$ is odd in t and $\widehat{\Delta_x}$ is even. With $w = it$ and (a) this is

$$V_1(\Delta_x) = \frac{1}{2\pi i}\int_{\operatorname{Re} w=0}\frac{e^{wx}+e^{-wx}-2}{w^2}\,\psi\big(\tfrac14+\tfrac{w}{2}\big)\,dw.$$

The integrand is regular at $w = 0$, so the line may be moved to $\operatorname{Re} w = -\frac14$. Split the numerator into its three terms, and close the contour to the left for e^{wx}

and to the right for e^{-wx} and for -2; the arcs, which pass between the poles of ψ, contribute nothing in the limit, because the integrands are $O(|w|^{-2}\log|w|)$ on them. On the left, $\psi(\frac14+\frac{w}{2})$ has simple poles at $w=-(2k+\frac12)$ with residue -2, which gives $-2\sum_k e^{-(2k+1/2)x}/(2k+\frac12)^2$. On the right the double pole at $w=0$ gives, with the clockwise orientation, $x\psi(\frac14)-\frac12\psi'(\frac14)$ from e^{-wx} and $\psi'(\frac14)$ from -2. □

The kernel $K(w,x)=\int\Delta_x(y)e^{wy}dy=(k_x(w)+k_x(-w))/w^2$ of (a) has two readings. The sine squared lemma fixes w and sums over the level x against the canonical measure of Section 5.3, built from $\Lambda(n)n^{-1/2}$ at $x=\log n$, the gamma factor and the pole; Yoshida fixes the level x and sums over the zeros.

The tent in the variogram form. Put $\varphi_x(z)=\widehat{\Delta_x}(\sqrt z)=4\sin^2(x\sqrt z/2)/z$. By (b) and (c), $D(x)=2\langle U,\varphi_x\rangle$, where $\langle U,\cdot\rangle$ is the Thorin functional of Section 5.3.1, and (c) is twice (5.3.1) for φ_x. The tent lies just outside the class $\mathcal T$ of that section, because $\widehat{\Delta_x}(r)=O(r^{-2})$, and the limiting argument in the proof of (c) takes the place of the decay condition. In the variogram form of Weil's criterion (Section 3.5.2) the tent is the function $g=\Delta_x$ of (3.5.2): its transform $h=\widehat{\Delta_x}$ is nonnegative, and its variogram is $g(0)-g(u)=\min(u,x)$ for $u\ge0$,

$$\int_0^\infty\varphi_x(z)\,\frac{2\sin^2(u\sqrt z/2)}{2\pi\sqrt z}\,dz=\min(u,x),$$

since both sides are $\pi^{-1}\int_0^\infty\widehat{\Delta_x}(r)(1-\cos ur)\,dr$, and (3.5.2) needs only $h\in L^1$. Theorem 17.6.2 below is therefore Weil's criterion restricted to the one parameter family of variograms $\min(u,x)$, $x>0$, of tents.

THEOREM 17.6.2. *RH holds if and only if $D(x)\ge0$ for all $x>0$, if and only if D is bounded below. Equivalently, RH holds if and only if the single coefficient $\langle\chi_0,\chi_0\rangle$ is nonnegative for every $a>0$.*

Proof. Under RH every term in (b) is nonnegative. Conversely put $\omega(x)=\sum_\rho e^{(\rho-1/2)x}/(\rho-\frac12)^2$, which converges absolutely. Using $\rho\mapsto1-\rho$ and $(\rho-\frac12)^2=-\gamma_\rho^2$ we get $D=2\omega+4p_1$, where $p_1=\sum_k a_k^{-1}=\frac12\sum_\rho\gamma_\rho^{-2}$ is the mean of the clock (Section 1.2). For $\operatorname{Re}s>\frac12$,

$$\int_0^\infty\omega(x)e^{-sx}dx=R(s):=\sum_\rho\frac{1}{(\rho-\frac12)^2\,(s-\rho+\frac12)},$$

which is meromorphic on $\mathbb C$ with poles only at the points $\rho-\frac12$, and with the nonzero residues $m/(\rho-\frac12)^2$, m the multiplicity. If $D\ge-C$, then $f=\omega+2p_1+C/2\ge0$ has the Laplace transform $R(s)+(2p_1+C/2)/s$. By Landau's theorem [340, Chapter II, Theorem 5b] this transform is singular at the real point of its abscissa of convergence σ_c. No $\rho-\frac12$ is real, because ξ has no real zero, so the

only real singularity is 0, $\sigma_c \le 0$, and the transform is holomorphic in $\operatorname{Re} s > 0$; hence no zero has $\operatorname{Re}(\rho - \frac{1}{2}) > 0$, and by the symmetry $\rho \mapsto 1-\rho$ none has $\operatorname{Re}(\rho - \frac{1}{2}) < 0$. □

The equivalence is classical in substance (Landau's theorem applied to a Riesz mean, as in Ingham [123]); its interest is its place in Yoshida's matrix and in the lemma. Yoshida's Theorem 1 gives $D > 0$ on $(0, \log 2]$ (Section 17.1), where no prime power contributes to (c), and $\langle \chi_0, \chi_0 \rangle = D(\log 2)/\log 2 = 0.18338$ at $a = \frac{1}{2}\log 2$. Computed from the prime side (c), which is explicit between consecutive prime powers, on a grid of step 10^{-4}, $D(x) \ge 0.0550$ on $[0.05, 16]$, with the minimum at $x = 8.04$, and the mean of D on $[2, 16]$ is 0.092409 against $4p_1 = 0.092420$. The difference -1.07×10^{-5} is the mean of the oscillating part 2ω: it equals $-\frac{4}{14}\sum_{\gamma>0}(\sin 16\gamma - \sin 2\gamma)/\gamma^3$, which the first 1700 zeros give to the same digits (script `yo_tent.py`, `code/README.md`).

Remark 17.6.3 (Detection scale). A hypothetical off line zero $\rho = \frac{1}{2} + \delta + i\gamma$ with its three companions adds to D the term $4\operatorname{Re}\big[4\sin^2\big((\gamma - i\delta)x/2\big)/(\gamma - i\delta)^2\big]$, whose amplitude is about $4e^{\delta x}/\gamma^2$ with oscillating sign. Against the mean level $4p_1$ of D this makes D first turn negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$. For $(\gamma, \delta) = (100, 0.3)$, $(100, 0.2)$, $(1000, 0.3)$ the first negative values of D plus the pair term occur at $x = 17.3$, 25.4, 32.4, against the predictions 18.1, 27.2, 33.5 (Figure 17.2); the predictions are upper estimates. On a grid of step 0.01 the third value is 32.83 and not 32.4, because at $\gamma = 1000$ the pair term has period $2\pi/\gamma = 0.0063$, shorter than the step. A negative value of the single coefficient $D(2a)$ shows that the form is not positive on $C(a)$, so Yoshida's threshold a_0 of Section 17.1 is at most about $x_*/2$, logarithmic in the height. This is a bound and not an estimate, because the tent is one vector of the window: a vector that vanishes at the on line ordinates near γ sees the pair earlier, and Table 18.3 of Chapter 18 gives, for the full form on a window, detection at half lengths far below $x_*/2$ in the cases computed there (a heuristic experiment of that chapter, with its own cutoff).

Table 17.1 compares the two sides of Proposition 17.6.1. The prime side is (c) with the exact $\Lambda(n)$. The zero side is $\sum_{\gamma>0} 8\sin^2(\gamma x/2)/\gamma^2$ over the first 1500 zeros (to height 1980.91) plus the mean tail $2(\log(T/2\pi)+1)/(\pi T)$ at $T = \gamma_{1500}$. The two sides agree to 3.6×10^{-6} at every level of the table except $x = \log 2$ and $x = \log 5$, where they differ by 1.6×10^{-4} and 2.3×10^{-4}. These are the corners of the prime side at $x = \log n$, where its slope drops by $2\Lambda(n)/\sqrt{n}$, which a zero sum cut at T and completed by a mean tail cannot follow: by Landau's formula [165] for $\sum_{0<\gamma\le T} n^{i\gamma}$ the zeros above T contribute $2\Lambda(n)/(\pi\sqrt{n}\,T)$ at $x = \log n$, which is 1.575×10^{-4} and 2.313×10^{-4} at $T = \gamma_{1500}$, and with it the two corner rows agree to 1.3×10^{-6}. The last two columns agree to 10^{-29} in every row (eight decimals are printed). The values come from the script `yo_explicit.py` (`code/README.md`).

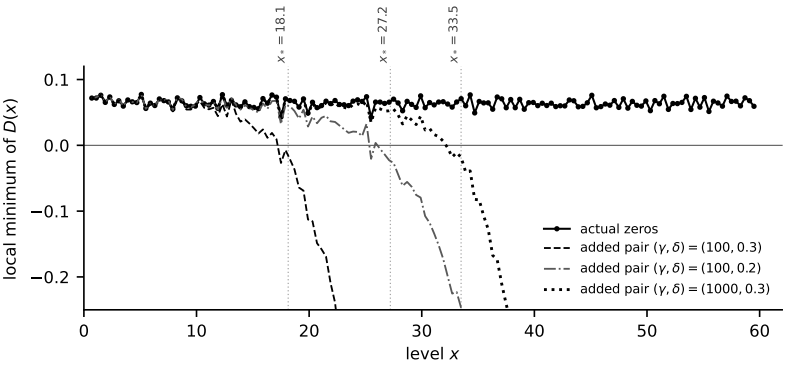


Figure 17.2: The least value of D over windows of length 0.4 on $[0, 60]$, for the actual zeros (solid, the first 1700 zeros with the smooth tail; the least window minimum is 0.0427, in the window centred at $x = 25.5$) and with one hypothetical off line pair added at height γ and displacement δ, $(\gamma, \delta) = (100, 0.3)$ (dashed), $(100, 0.2)$ (dash dotted) and $(1000, 0.3)$ (dotted). Vertical dotted lines mark the predicted levels $x_* = 18.1$, 27.2 and 33.5 of Remark 17.6.3. Script `yo_tent.py`.

x	prime side	zero side	$V_1(\Delta_x)$ by (c)	by (d)
0.2	0.1089996	0.1089981	0.25787887	0.25787887
0.5	0.0803252	0.0803270	0.15008051	0.15008051
$\log 2$	0.1271112	0.1269547	−0.04998528	−0.04998528
1.0	0.0880146	0.0880147	−0.50847597	−0.50847597
1.5	0.0898189	0.0898186	−1.52909231	−1.52909231
$\log 5$	0.1343264	0.1340964	−1.78866457	−1.78866457
2.0	0.1066822	0.1066850	−2.80144646	−2.80144646
3.0	0.0794186	0.0794176	−5.86891443	−5.86891443
4.0	0.0702819	0.0702855	−9.39384635	−9.39384635
6.0	0.0962464	0.0962437	−17.16435327	−17.16435327

Table 17.1: The tent value $D(x)$ from the primes (Proposition 17.6.1(c)) and from the zeros, and $V_1(\Delta_x)$ by the closed form (c) and by the residue series (d). The rows $x = \log 2$ and $x = \log 5$ differ by Landau's term, see the text.

17.7 The sine squared lemma off the diagonal: Yoshida's matrix is a Löwner matrix

Recall Yoshida's basis $\chi_n = (2a)^{-1/2} e^{\pi i n y/a} 1_{[-a,a]}$ of $K(a)$ (Section 17.1, [349]), and write $\nu_n = \pi n/a$. The Fourier transform of the basis vector is

$$\widehat{\chi_n}(t) = (2a)^{-1/2} \int_{-a}^{a} e^{i(t+\nu_n)y}\, dy = (2a)^{-1/2}\, \frac{2(-1)^n \sin(at)}{t + \nu_n},$$

because $\sin(a(t+\nu_n)) = (-1)^n \sin(at)$. Every basis vector carries the same factor $\sin(at)$; only the pole moves. The Löwner matrix of a function g on the nodes ν_n has the entries $(g(\nu_n) - g(\nu_m))/(\nu_n - \nu_m)$ off the diagonal and $g'(\nu_n)$ on it, and a Pick function is a function holomorphic on the upper half plane with nonnegative imaginary part there.

THEOREM 17.7.1. *(i) Unconditionally,*

$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \sum_\rho \frac{\sin^2(a\gamma_\rho)}{(\gamma_\rho + \nu_n)(\gamma_\rho + \nu_m)},$$

an absolutely convergent series.

(ii) Let $g_a(\nu) = \sum_\rho \sin^2(a\gamma_\rho)/(\gamma_\rho - \nu)$, the sum over all zeros taken in symmetric pairs; unconditionally it is an odd meromorphic function of ν. Then for $n \neq m$

$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \frac{g_a(\nu_n) - g_a(\nu_m)}{\nu_n - \nu_m}, \qquad \langle \chi_n, \chi_n \rangle = \frac{2}{a} g_a'(\nu_n),$$

so Yoshida's matrix is, up to the sign change $\operatorname{diag}((-1)^n)$, *$\frac{2}{a}$ times the Löwner matrix of g_a on the lattice $\frac{\pi}{a}\mathbb{Z}$.*

(iii) Under RH, with the weighted Thorin measure $U_a^\sharp(dt) = 2\sin^2(a\sqrt{t})\, U(dt) \geq 0$ and its Stieltjes transform $\mathcal{S}_a(s) = \int U_a^\sharp(dt)/(t+s)$,

$$g_a(\nu) = \nu\, \mathcal{S}_a(-\nu^2),$$
$$\langle \chi_n, \chi_m \rangle = (-1)^{n+m} \frac{2}{a} \int \frac{2\sin^2(a\sqrt{t})\,(t + \nu_n\nu_m)}{(t-\nu_n^2)(t-\nu_m^2)}\, U(dt),$$

and g_a is an odd Pick function. By the elementary half of Löwner's theorem every finite Yoshida matrix is then positive semidefinite, for every a.

(iv) For each fixed $a > 0$, RH holds if and only if g_a is a Pick function.

Proof. (i) $F = \chi_n * \widetilde{\chi_m}$ has $\widehat{F}(t) = \widehat{\chi_n}(t)\overline{\widehat{\chi_m}(\bar{t})} = (-1)^{n+m}\frac{2}{a}\sin^2(at)/((t+\nu_n)(t+\nu_m))$, entire because $\sin^2(at)$ vanishes at $-\nu_n$ and $-\nu_m$. It is $O(t^{-2})$ in every strip, and F is continuous, piecewise smooth and compactly supported, so (17.1.1) applies to it as it does to the tent (proof of Proposition 17.6.1(c)). Sum $\widehat{F}(\gamma_\rho)$.

(ii) Replace γ_ρ by $-\gamma_\rho$ (the zero set is symmetric), then use $\frac{1}{(\gamma-\nu_n)(\gamma-\nu_m)} = \frac{1}{\nu_n-\nu_m}\big(\frac{1}{\gamma-\nu_n} - \frac{1}{\gamma-\nu_m}\big)$; the diagonal is the limit. The series of g_a converges in symmetric pairs, since the pair $\pm\gamma_\rho$ contributes $2\nu\sin^2(a\gamma_\rho)/(\gamma_\rho^2 - \nu^2) = O(|\nu|\,|\gamma_\rho|^{-2})$, and $g_a(-\nu) = -g_a(\nu)$ because the zero set is symmetric. A zero with $\gamma_\rho = \nu_n$ contributes nothing at the lattice, since $\sin^2(a\gamma_\rho) = 0$.

(iii) Under RH group $\pm\gamma$: $\sin^2(a\gamma)\big(\frac{1}{\gamma-\nu}-\frac{1}{\gamma+\nu}\big)=\frac{2\nu\sin^2(a\gamma)}{\gamma^2-\nu^2}$, and with $t=\gamma^2$ this is $\nu\int U_a^\sharp(dt)/(t-\nu^2)$. In the same way the pair $\pm\gamma$ in (i) gives $2\sin^2(a\gamma)(\gamma^2+\nu_n\nu_m)/((\gamma^2-\nu_n^2)(\gamma^2-\nu_m^2))$, which is the second formula. Each atom $\nu/(t-\nu^2)$ has imaginary part $\operatorname{Im}\nu\,(t+|\nu|^2)/|t-\nu^2|^2>0$ for $\operatorname{Im}\nu>0$, so g_a is a Pick function. For a Pick function g and real nodes $\nu_1,\dots,\nu_N$, the matrix $[(g(\nu_i)-g(\nu_j))/(\nu_i-\nu_j)]$ is positive semidefinite, which is Löwner's easy direction [189, 78].

(iv) If RH fails, an off line pair gives the nonreal ordinates $\pm\gamma\pm i\delta$, two of them in the upper half plane, where g_a has a pole with residue $-m\sin^2(a\gamma_\rho)$, m the multiplicity. This residue is nonzero because sin vanishes only on the real axis, and a Pick function is holomorphic in the upper half plane. The converse is (iii). □

The mechanism of the theorem is that of the rigidity of the clock, Proposition 1.2.1: there G'/G is the Stieltjes transform of U, holomorphic off the negative real axis exactly when every zero is on the line, and $\mathcal{S}_a$ is the same transform with the atoms weighted by $2\sin^2(a\gamma)$, sampled on the cut at the squared modal frequencies ν_n^2 and differenced.

Numerically, the prime side of $\langle\chi_n,\chi_m\rangle$, computed from (17.1.1) and (17.1.2) by Gauss Legendre quadrature, gives the coefficient $\langle\chi_0,\chi_0\rangle=0.18338$ of Yoshida's matrix at $a=\frac12\log 2$, and at $a=1$ it matches part (i) summed over the first 1500 zeros, entry by entry, up to the zero tail. The tail of the smooth count from $T=1982.0$ accounts for the difference up to 9×10^{-7} (`yo_explicit.py`, `code/README.md`).

17.8 Basepoints: damped forms, multipliers and sharp coercivity

The Thorin programme of this book leaves the critical line for a basepoint $\alpha>\frac12$, where the Thorin density of the shifted clock is $\rho_{\alpha-1/2}=\pi^{-1}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$ (equation (1.2.7)). On Yoshida's side the same move damps the autocorrelation $\varphi*\widetilde{\varphi}$ by $e^{-(\alpha-1/2)|y|}$.

17.8.1 The multiplier identity

For a basepoint $\alpha\ge\frac12$ put $\varepsilon=\alpha-\frac12$ and define the damped Yoshida form

$$\langle\varphi_1,\varphi_2\rangle_\alpha=\mathcal{Y}\big(e^{-\varepsilon|y|}\,\varphi_1*\widetilde{\varphi_2}\big),$$

where $\mathcal{Y}$ is the functional of (17.1.1) and $\widetilde{\varphi}(y)=\overline{\varphi(-y)}$; at $\alpha=\frac12$ it is the form of Section 17.1, and $C(a)\subset K(a)$ are the spaces of that section. On the prime side the damping replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$, and in the pole term $e^{y/2}+e^{-y/2}$ becomes $e^{(1-\alpha)|y|}+e^{-\alpha|y|}$; on $K(a)$ the prime sum is still finite, since only $\log n\le 2a$ occurs. Let $\sigma_{\max}=\sup_\rho\operatorname{Re}\rho\in[\frac12,1]$, let $v_\alpha(\tau)=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)$

and $b_\alpha = v_\alpha(0) = \xi'(\alpha)/\xi(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (1.2.7), and write $P_w(\tau) = w/(\pi(w^2+\tau^2))$ for the Poisson kernel of width w, with $P_{-w} = -P_w$, as in Proposition 1.2.3.

PROPOSITION 17.8.1. *Let $\alpha \geq \sigma_{\max}$, and suppose that no zero has real part α. Then ρ_ε is real analytic and positive, $(1+\tau^2)^{-1}\rho_\varepsilon$ is integrable over $\mathbb{R}$, and for every $a > 0$ and every $\varphi \in K(a)$*

$$\langle \varphi, \varphi \rangle_\alpha = \int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2 \, \rho_\varepsilon(\tau)\, d\tau = \frac{1}{\pi} \int_{\mathbb{R}} |\widehat{\varphi}(\tau)|^2 \, v_\alpha(\tau)\, d\tau.$$

Proof. For $w > 0$ the kernel P_w has Fourier transform $\int P_w(\tau) e^{i\tau y}\, d\tau = e^{-w|y|}$, so for a bounded F with compact support and $\gamma \in \mathbb{R}$, $\int \widehat{F}(\tau) P_w(\tau - \gamma)\, d\tau = \int F(y) e^{-w|y|} e^{i\gamma y}\, dy$. Let $F = \varphi * \widetilde{\varphi}$, which is continuous, piecewise smooth and supported in $[-2a, 2a]$, so that (17.1.1) applies to $e^{-\varepsilon|y|}F$ and

$$\mathcal{Y}(e^{-\varepsilon|y|}F) = \sum_\rho \int F(y) e^{-\varepsilon|y|} e^{i\gamma_\rho y}\, dy,$$

the series converging absolutely. Group the zeros in fours $\{\rho, \bar\rho, 1-\rho, 1-\bar\rho\}$, counted with multiplicity; for a zero on the critical line the group is $\{\rho, \bar\rho\}$. For the four zeros $\frac12 \pm \delta \pm i\gamma$ with $\gamma > 0$ and $\delta > 0$ we have $e^{i\gamma_\rho y} = e^{\pm i\gamma y} e^{\pm \delta y}$, so their contributions add up to

$$\begin{aligned} &\int F(y) e^{-\varepsilon|y|} \big(e^{i\gamma y} + e^{-i\gamma y}\big)\big(e^{\delta y} + e^{-\delta y}\big) dy \\ &\quad = \int F(y) \big(e^{-(\varepsilon-\delta)|y|} + e^{-(\varepsilon+\delta)|y|}\big)\big(e^{i\gamma y} + e^{-i\gamma y}\big) dy, \end{aligned}$$

because $e^{\delta y} + e^{-\delta y}$ is even. The widths $\varepsilon - \delta = \alpha - \beta$ and $\varepsilon + \delta = \alpha - (1-\beta)$, with $\beta = \frac12 + \delta$, are the differences between α and the real parts of the zeros of the group, and they are positive, since every zero has real part less than α. By the identity for P_w the group contributes $\int \widehat{F}(\tau) \sum P_{\alpha-\beta_{\rho'}}(\tau - \operatorname{Im}\rho')\, d\tau$, the sum over its four members ρ' with real part $\beta_{\rho'}$; the same holds for a zero on the line, with the single width ε. Hence $\mathcal{Y}(e^{-\varepsilon|y|}F) = \sum_\rho \int \widehat{F}(\tau) P_{\alpha-\beta_\rho}(\tau - \operatorname{Im}\rho)\, d\tau$ over all zeros, where $\beta_\rho = \operatorname{Re}\rho$. Now $\widehat{F} = |\widehat{\varphi}|^2 \geq 0$ on $\mathbb{R}$, so every term is nonnegative and the sum and the integral may be interchanged. By Proposition 1.2.3, which is Hadamard's product, $\sum_\rho P_{\alpha-\beta_\rho}(\tau - \operatorname{Im}\rho) = \pi^{-1} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\tau) = \rho_\varepsilon(\tau)$, a sum of positive kernels; it is real analytic because no zero lies on the line $\operatorname{Re} s = \alpha$. Finally $\int P_w(\tau - t)(1+\tau^2)^{-1} d\tau = (1+w)/((1+w)^2 + t^2) \leq (1+\alpha)/(1+t^2)$ for $0 < w \leq \alpha$, so by Tonelli $\int (1+\tau^2)^{-1} \rho_\varepsilon\, d\tau \leq (1+\alpha) \sum_\rho (1 + (\operatorname{Im}\rho)^2)^{-1} < \infty$, because the number of zeros with $|\operatorname{Im}\rho| \leq T$ is $O(T \log T)$. The proof used only that $F = \varphi * \widetilde{\varphi}$ is continuous, piecewise C^1 and supported in $[-2a, 2a]$, so the identity holds as well for every bounded piecewise C^1 function φ supported in $[-a, a]$. □

For the Fejér taper $\varphi = e^{ity}\chi_0$ on the window $a = X/2$ one has $\varphi * \widetilde{\varphi} = e^{ity}(1-|y|/X)_+$, and since $\mathcal{Y}(F)$ depends only on the even part of F, $\langle\varphi,\varphi\rangle_\alpha = \mathcal{Y}\big(e^{-\varepsilon|y|}\cos(ty)(1-|y|/X)_+\big) = 2A_X(\varepsilon,t)$ in the notation of Proposition 14.14.1, whose part (c), $A_X(\varepsilon,t)\to v_\alpha(t)$ as $X\to\infty$ when $\alpha>\sigma_{\max}$, so that every zero has $|\beta-\frac12|<\varepsilon$, is the concentration argument in the proof of Theorem 17.8.5(i) below for this family of test functions.

Remark 17.8.2. For $\alpha<\sigma_{\max}$ the two sides of the identity differ: a zero with $\beta>\alpha$ contributes the growing factor $e^{(\beta-\alpha)|y|}$ to the damped Weil form, but the negative kernel $P_{\alpha-\beta}=-P_{\beta-\alpha}$ to the symbol. The script `yo_damped.py` checks the identity at $\alpha=1$, $a=0.8$ and $\varphi=\chi_0$ (data in `yo_damped.json`, `code/README.md`).

Theorem 17.8.3. *The damped forms have the following properties.*

(i) For $\alpha\ge\sigma_{\max}$ the damped form is positive semidefinite on $C_c^\infty(\mathbb{R})$; unconditionally this holds for every $\alpha\ge1$.

(ii) RH holds if and only if the damped form is positive semidefinite for every $\alpha>\frac12$.

(iii) If the damped form at α is positive semidefinite on $C(a)$, so is the form at every $\alpha'>\alpha$; thus $a_0(\alpha)=\sup\{a:\langle\cdot,\cdot\rangle_\alpha\ge0 \text{ on } C(a)\}$ is nondecreasing in α.

Proof. (i) If no zero has real part α this is Proposition 17.8.1, because $\rho_\varepsilon\ge0$ and every $\varphi\in C_c^\infty(\mathbb{R})$ lies in some $K(a)$. If a zero has real part α, then $\alpha=\sigma_{\max}$ and $\langle\varphi,\varphi\rangle_\alpha=\lim_{\alpha'\downarrow\alpha}\langle\varphi,\varphi\rangle_{\alpha'}\ge0$, since for fixed φ each part of the form is continuous in ε: the pole term and the finite prime sum obviously, and $V_1(e^{-\varepsilon|y|}F)$ because $\widehat{e^{-\varepsilon|y|}F}=P_\varepsilon*\widehat F$ with $|\widehat F(t)|\le C(1+t^2)^{-1}$, which gives a bound $C'(1+t^2)^{-1}$ uniform in $\varepsilon\le1$, while $\operatorname{Re}\psi(\frac14+\frac{it}{2})=O(\log(2+|t|))$, and dominated convergence applies to the Fourier form of V_1 in (17.1.2). Unconditionally $\sigma_{\max}\le1$ and no zero has real part 1, so every $\alpha\ge1$ is covered.

(ii) Under RH, $\sigma_{\max}=\frac12$ and (i) applies to every $\alpha>\frac12$. Conversely, positivity for a sequence $\alpha_n\downarrow\frac12$ gives $\langle\varphi,\varphi\rangle\ge0$ for every $\varphi\in C_c^\infty(\mathbb{R})$ by the same continuity, which is Weil's criterion (Section 17.1).

(iii) Write $e^{-\varepsilon'|y|}=\int P_{\varepsilon'}(\lambda)e^{i\lambda y}\,d\lambda$ with $\varepsilon'=\alpha'-\alpha$. Since $(\varphi_\lambda*\widetilde{\varphi_\lambda})(y)=e^{i\lambda y}(\varphi*\widetilde\varphi)(y)$ for $\varphi_\lambda=e^{i\lambda y}\varphi\in C(a)$, we get $\langle\varphi,\varphi\rangle_{\alpha'}=\int P_{\varepsilon'}(\lambda)\langle\varphi_\lambda,\varphi_\lambda\rangle_\alpha\,d\lambda$ (Fubini, as in (i)), which is nonnegative when the form at α is positive semidefinite on $C(a)$. □

Remark 17.8.4 (One basepoint). Part (ii) uses every $\alpha>\frac12$. The converse of (i) holds at a single basepoint: if $\frac12\le\alpha\le1$ and $\langle\varphi,\varphi\rangle_\alpha\ge0$ for every $\varphi\in C_c^\infty(\mathbb{R})$, then no zero has real part greater than α. This is Theorem 20.1.2 of Chapter 20. With (i), the damped form at $\alpha\in[\frac12,1]$ is positive semidefinite if and only if

$\sigma_{\max} \le \alpha$. A statement about the symbol on one line is weaker: positivity of v_α on the line $\operatorname{Re} s = \alpha$ is not known to exclude zeros to its right (Remark 1.2.4).

17.8.2 Sharp coercivity

For $a > 0$ let

$$\lambda_{\min}(\alpha, a) = \inf\Big\{ \frac{\langle \varphi, \varphi \rangle_\alpha}{\|\varphi\|^2} : \ 0 \ne \varphi \in K(a) \Big\},$$

the least eigenvalue of the damped form on the window $[-a, a]$, where $\|\cdot\|$ is the norm of $L^2(\mathbb{R})$ and the test functions are complex valued. The matrix of the form on the basis χ_n, $|n| \le N$, is the form restricted to a subspace of $K(a)$, so its least eigenvalue is an upper bound for $\lambda_{\min}(\alpha, a)$: these are the finite sections of Table 17.2. Under the hypotheses of Proposition 17.8.1 the infimum over the smaller space $C(a)$ is the same. Indeed, let $\varphi \in K(a)$ and let $\psi_\delta \in C_c^\infty(-a, a)$ satisfy $0 \le \psi_\delta \le 1$ and $\psi_\delta = 1$ on $[-a + \delta, a - \delta]$, with ψ_δ monotone on each of the two end intervals. Then $\eta_\delta = \varphi(1 - \psi_\delta)$ lies in $K(a)$, has total variation bounded independently of δ and L^1 norm at most $2\delta\|\varphi\|_\infty$, so that $|\widehat{\eta_\delta}(\tau)| \le C \min(\delta, |\tau|^{-1})$, and $\langle \eta_\delta, \eta_\delta \rangle_\alpha = \int |\widehat{\eta_\delta}|^2 \rho_\varepsilon \to 0$ as $\delta \downarrow 0$ by dominated convergence, since $(1 + \tau^2)^{-1} \rho_\varepsilon$ is integrable. The form being positive semidefinite on $K(a)$, the Rayleigh quotients of $\varphi\psi_\delta \in C(a)$ tend to that of φ. The argument applies verbatim to the functions supported in $[-a, a]$ that are smooth on $[-a, a]$, to which Proposition 17.8.1 extends, so the infimum is the same over that larger class; it contains the Legendre polynomials on the window used in Chapter 20, whose finite sections are therefore upper bounds as well. Put $m_\alpha = \inf_{\tau \in \mathbb{R}} v_\alpha(\tau)$.

Theorem 17.8.5. *Let $V_\xi = (\log \xi)''(\frac{1}{2})$ and $p_1 = \frac{1}{2} V_\xi = 0.0231049931\ldots$, which is $\sum_{\gamma > 0} \gamma^{-2}$ under RH, and, under RH, let $p_2 = \sum_{\gamma > 0} \gamma^{-4} = 3.71726\ldots \times 10^{-5}$.*

(i) If $\alpha \ge \sigma_{\max}$ and no zero has real part α, then $\lambda_{\min}(\alpha, a) \ge 2m_\alpha$ for every $a > 0$, $\lambda_{\min}(\alpha, a)$ is nonincreasing in a, and $\lambda_{\min}(\alpha, a) \to 2m_\alpha$ as $a \to \infty$.

(ii) Under RH, $m_\alpha = b_\alpha = \xi'(\alpha)/\xi(\alpha)$ for every $\alpha > \frac{1}{2}$, and $v_\alpha(\tau) > b_\alpha$ for $\tau \ne 0$: the minimum of $\operatorname{Re} \xi'/\xi$ on each vertical line lies on the real axis, and only there. For $\alpha \ge 1$ this holds with no hypothesis (Proposition 9.1.12, and Proposition 9.1.11 for $\alpha = 1$); for $\frac{1}{2} < \alpha < 1$ it holds with no hypothesis for $0 < |\tau| \le 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)); and RH is equivalent to it on every line $\frac{1}{2} < \alpha < 1$ (Theorem 9.1.18(vi), which sharpens Lagarias's criterion [160]).

(iii) Hence RH holds if and only if

$$\langle \varphi, \varphi \rangle_\alpha \ \ge \ 2b_\alpha \, \|\varphi\|^2 \qquad \text{for all } \alpha \in (\tfrac{1}{2}, 1],\ a > 0,\ \varphi \in K(a),$$

and under RH the constant is sharp: $\lambda_{\min}(\alpha, a) \downarrow 2b_\alpha$ as $a \to \infty$.

(iv) For every $\alpha > \frac12$, with no hypothesis, $2b_\alpha \le 4p_1(\alpha-\frac12)$; under RH also $4p_1(\alpha-\frac12) - 4p_2(\alpha-\frac12)^3 \le 2b_\alpha$. The coercivity constant vanishes linearly at $\alpha = \frac12$.

Proof. (i) By Proposition 17.8.1 and Plancherel, $\int|\widehat{\varphi}|^2 = 2\pi\|\varphi\|^2$, we have, for $\varphi \in K(a)$,

$$\langle\varphi,\varphi\rangle_\alpha = \pi^{-1}\int|\widehat{\varphi}|^2 v_\alpha \ge \pi^{-1}m_\alpha\int|\widehat{\varphi}|^2 = 2m_\alpha\|\varphi\|^2.$$

Monotonicity holds because $\lambda_{\min}(\alpha,a)$ is also the infimum over $C(a)$ and $C(a)\subset C(a')$ for $a<a'$. For the limit, fix τ_0 and $\psi\in C_c^\infty(-1,1)$ with $\|\psi\|=1$, and let $\varphi_a(y) = a^{-1/2}e^{-i\tau_0 y}\psi(y/a)\in C(a)$. Then $\|\varphi_a\|=1$, $|\widehat{\varphi_a}(\tau)|^2 = a\,|\widehat{\psi}(a(\tau-\tau_0))|^2$, and $|\widehat{\varphi_a}|^2/2\pi$ is a probability density that concentrates at τ_0 as $a\to\infty$. The function v_α is continuous and $(1+\tau^2)^{-1}v_\alpha$ is integrable (Proposition 17.8.1), and $\widehat{\psi}$ is a Schwartz function, so $\langle\varphi_a,\varphi_a\rangle_\alpha = \pi^{-1}\int|\widehat{\varphi_a}|^2 v_\alpha \to 2v_\alpha(\tau_0)$. Hence $\limsup_a \lambda_{\min}(\alpha,a)\le 2v_\alpha(\tau_0)$ for every τ_0, that is, $\le 2m_\alpha$.

(ii) Under RH the whole of (ii) is in the book, at the places cited; for $\alpha\ge1$ no hypothesis is needed.

(iii) Under RH, (i) and (ii) give the inequality and its sharpness. Conversely, $b_\alpha>0$ for every $\alpha>\frac12$ with no hypothesis (Theorem 9.1.18(i)), so the inequality gives positive semidefiniteness for $\alpha\in(\frac12,1]$, Theorem 17.8.3(i) covers $\alpha>1$, and part (ii) of that theorem gives RH.

(iv) The upper bound is Theorem 9.1.18(i), $b_\alpha \le (\alpha-\frac12)V_\xi$, with $V_\xi = 2p_1$. Under RH, $b_\alpha = \sum_{\gamma>0} 2\varepsilon/(\varepsilon^2+\gamma^2)$ and $\gamma^{-2} - \varepsilon^2\gamma^{-4} \le (\varepsilon^2+\gamma^2)^{-1} \le \gamma^{-2}$. □

Remark 17.8.6. At a single $\alpha\in(\frac12,1)$ with no zero on the line, the inequality $\langle\varphi,\varphi\rangle_\alpha \ge 2b_\alpha\|\varphi\|^2$ for all a and φ holds if and only if $\sigma_{\max}\le\alpha$ and $v_\alpha\ge b_\alpha$ on the line: if it holds, the form is positive semidefinite (as $b_\alpha>0$), so $\sigma_{\max}\le\alpha$ by Remark 17.8.4, and then (i) gives $2m_\alpha\ge 2b_\alpha$, while $m_\alpha\le v_\alpha(0)=b_\alpha$; the converse is (i).

Lemma 17.8.7. *Consecutive ordinates of the zeros of ζ differ by at most* 6.8874. *Hence every real τ with $|\tau|\ge\gamma_1$ lies within* 3.4437 *of an ordinate.*

Proof. The lemma needs no hypothesis on the zeros. The first 1800 zeros of Proposition 9.1.11, computed in ball arithmetic, reach height 2304.3: consecutive ordinates among them differ by at most $\gamma_2-\gamma_1 = 6.88731\ldots$, and the later gaps by at most 5.4141 (between $\gamma_3=25.0109$ and $\gamma_4=30.4249$). It remains to show that $\gamma'-\gamma\le h := 6.8874$ for consecutive ordinates $\gamma<\gamma'$ with $\gamma\ge 2304.3$. Take T with $\gamma<T<\gamma'$ and write $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2.1). The second derivative $\vartheta''(T) = \frac{T}{4}\sum_{k\ge0}(k+\frac14)\big((k+\frac14)^2+\frac{T^2}{4}\big)^{-2}$ is positive, so ϑ is convex, $\vartheta(T+h)-\vartheta(T)\ge h\vartheta'(T)$, and

$$N(T+h)-N(T) \;\ge\; \frac{h\,\vartheta'(T)}{\pi} - |S(T)| - |S(T+h)|.$$

We use the bounds of [109] as they are quoted in the proof of Proposition 9.1.11: $|S(t)| \le 2.5167$ for $0 \le t \le X_0 = 30\,610\,046\,000$, and $|S(t)| \le B(t) = 0.1095 \log t + 0.2042 \log\log t + 3.0305$ for $t \ge e$, with B increasing. For $2304.3 \le T \le X_0 - h$ the first bound applies to both terms, and $\vartheta'(T) \ge \vartheta'(2304.3) = 2.9523$ gives $N(T+h) - N(T) \ge 6.4725 - 5.0334 > 1.4$. For $T > X_0 - h$ both $|S(T)|$ and $|S(T+h)|$ are at most $B(T+h)$, because $B(X_0) = 6.32 > 2.5167$, so $N(T+h) - N(T) \ge D(T) = h\vartheta'(T)/\pi - 2B(T+h)$. At $T = X_0 - h$, $h\vartheta'(T)/\pi = 24.452$ and $2B(X_0) = 12.649$, so $D(X_0 - h) = 11.80 > 0$; and D increases, because a comparison of the sum for ϑ'' with an integral gives $\vartheta''(T) \ge \frac{1}{2T} - \frac{1}{T^2}$, while $2T\,B'(T+h) \le 0.219 + 0.4084/\log T$, so that $T\,D'(T) \ge \frac{h}{2\pi}(1 - \frac{2}{T}) - 0.219 - \frac{0.4084}{\log T} > 0.8$ for $T \ge X_0 - h$. In every case $N(T+h) - N(T) > 0$, so some ordinate lies in $(T, T+h]$, and letting $T \downarrow \gamma$ gives $\gamma' - \gamma \le h$. Finally, a real τ with $|\tau| \ge \gamma_1$ lies between two consecutive ordinates, hence within half a gap of one of them. □

Corollary 17.8.8. *For every $a > 0$ and every $\varphi \in K(a)$, with no hypothesis on the zeros,*

$$\langle \varphi, \varphi \rangle_1 \;\ge\; 2b_1 \, \|\varphi\|^2 = \big(2 + \gamma_E - \log 4\pi\big) \|\varphi\|^2 = 0.0461914\ldots \, \|\varphi\|^2,$$

and no larger constant works for every a: $\lambda_{\min}(1, a) \downarrow 2 + \gamma_E - \log 4\pi$ as $a \to \infty$.

Proof. At $\alpha = 1$ no zero has real part 1 and $\sigma_{\max} \le 1$, so Theorem 17.8.5(i) applies, and $m_1 = b_1$ by Proposition 9.1.11. Finally $b_1 = \xi'(1)/\xi(1) = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, twice which is $2 + \gamma_E - \log 4\pi = 0.0461914179\ldots$. □

Remark 17.8.9 (A Laguerre type inequality). Laguerre's quantity is

$$\mathcal{L}_\Xi(\tau) = -(\log \Xi)''(\tau) = \sum_\rho (\tau - \gamma_\rho)^{-2}, \qquad \Xi(\tau) = \xi(\tfrac{1}{2} + i\tau),$$

and it is the limit of $v_{1/2+\varepsilon}(\tau)/\varepsilon$ as $\varepsilon \downarrow 0$, taken termwise: a zero on the line gives $(\tau - \gamma)^{-2}$, and a pair $\frac{1}{2} \pm \delta + i\gamma$ gives $2\,\mathrm{Re}(\tau - \gamma + i\delta)^{-2}$. Dividing $v_\alpha(\tau) \ge b_\alpha$ by ε, part (ii) of Theorem 17.8.5 becomes, as $\varepsilon \downarrow 0$,

$$\mathcal{L}_\Xi(\tau) \;\ge\; \mathcal{L}_\Xi(0) = 2p_1,$$

a sharpening of Laguerre's inequality $\Xi'^2 - \Xi\Xi'' \ge 0$, which says $\mathcal{L}_\Xi \ge 0$. The sharpened inequality holds under RH for every τ, and with no hypothesis for $|\tau| \le 3 \times 10^{12} - 1.21$ (Theorem 9.1.18(ii)). Numerically (`yo_symbol.py`) $\mathcal{L}_\Xi(0) = 0.0462100$ is the global minimum of $\mathcal{L}_\Xi$ on $[0, 1800]$, and the next local minimum, 0.2301087 at $\tau = 17.498$ between the first two ordinates, is 4.98 times larger.

Table 17.2 lists $\lambda_{\min}(\alpha, a)$ for five windows and the four levels $\alpha = 0.55$, 0.6, 0.75, 1, with the floor $2b_\alpha$ of Theorem 17.8.5 in the last row, and the finite section values at $\alpha = \frac{1}{2}$. The matrices of the form are assembled from the prime side of (17.1.1) with the damping.

a	$\alpha=\frac12$	$\alpha=0.55$	$\alpha=0.6$	$\alpha=0.75$	$\alpha=1$
0.35	1.20×10^{-3}	0.0066141	0.0120197	0.0281496	0.0547526
0.8	2.28×10^{-17}	0.0047355	0.0094659	0.0236477	0.0472506
1.5	2.43×10^{-64}	0.0046489	0.0092974	0.0232400	0.0464620
2.5	1.95×10^{-145}	0.0046305	0.0092608	0.0231497	0.0462846
4.0	1.59×10^{-286}	0.0046246	0.0092490	0.0231206	0.0462270
$2b_\alpha$	0	0.0046210	0.0092418	0.0231027	0.0461914

Table 17.2: The least eigenvalue $\lambda_{\min}(\alpha,a)$ of the damped Yoshida form on the window $[-a,a]$, from the matrices of the form on the modes χ_n, $|n|\le N$. For $\alpha>\frac12$ the entries are converged values, from Richardson extrapolation of $N=96$, 192, 384, with an uncertainty of at most 3×10^{-7} at $a=0.35$ and 1.1×10^{-8} for $a\ge0.8$ (an independent Legendre basis agrees to 1.5×10^{-7} at most). For $\alpha=\frac12$ the entries are the finite section values at $N=\max(24,24a)$, upper bounds computed in extended precision, which are not converged; the entry for $a=4$ ($N=96$) was computed with 640 digits. The last row is the floor $2b_\alpha=2\xi'(\alpha)/\xi(\alpha)$. Computed by `yo_damped.py` with `yo_matrix.py`, data in `yo_damped.json`.

Every entry for $\alpha>\frac12$ exceeds the floor and decreases with a, as Theorem 17.8.5 says (under RH for $\alpha<1$). The excess over the floor falls roughly like a^{-2}, as in Theorem 20.2.1(b) of Chapter 20.

The column $\alpha=\frac12$ is of another kind. The entries depend strongly on N (at $a=0.8$ the value is 6.9×10^{-17} at $N=12$, 2.3×10^{-17} at $N=24$ and 1.9×10^{-17} at $N=48$). Two other tables of the book compute the same eigenvalue differently: Table 18.1 of Chapter 18 at $\alpha=\frac12$, to 60 digits, for even and for odd test functions separately, and Table 20.1 of Chapter 20, the rungs at more levels $\alpha>\frac12$, as converged values.

Figure 17.4 shows the symbol on four lines.

At $\alpha=\frac12$ Theorem 17.8.5 does not apply. Under RH the symbol there is the atomic measure on the ordinates, the vague limit of ρ_ε as $\varepsilon\downarrow0$; it vanishes between the atoms, which is why $\lambda_{\min}(\frac12,a)\to0$, and under RH the decay is faster than any power of a (Theorem 20.2.1(c) of Chapter 20). The speed of the collapse in Table 17.2 is a numerical observation. The finite section values at $\alpha=\frac12$ follow $\exp(-2c\,ae^{2a})$ with $c=-\log\lambda/(2ae^{2a})=4.12$, 4.77, 5.10, 5.01, 4.86 at $a=0.25$, 0.35, 0.5, 0.65, 0.8 (so $4.1\le c\le5.1$, at $N=48$). Zhu proves, under RH and for all large a, only the bound $\exp(-ae^a)$ for the least eigenvalue over real even test functions (Theorem 1.3 of [350], Chapter 18): it has e^a where the observed decay has e^{2a}. For $a\ge1.5$ the finite sections saturate and c falls, so those values say nothing about the infimum. A heuristic for the form of the decay: $\widehat{\varphi}$ is an entire function of exponential type a and has at most about $2aH/\pi$ real zeros in $[-H,H]$, against about $(H/\pi)\log(H/2\pi)$ ordinates there, so a vector of type a can vanish at every ordinate up to a height of about $2\pi e^{2a}$ and no further, which

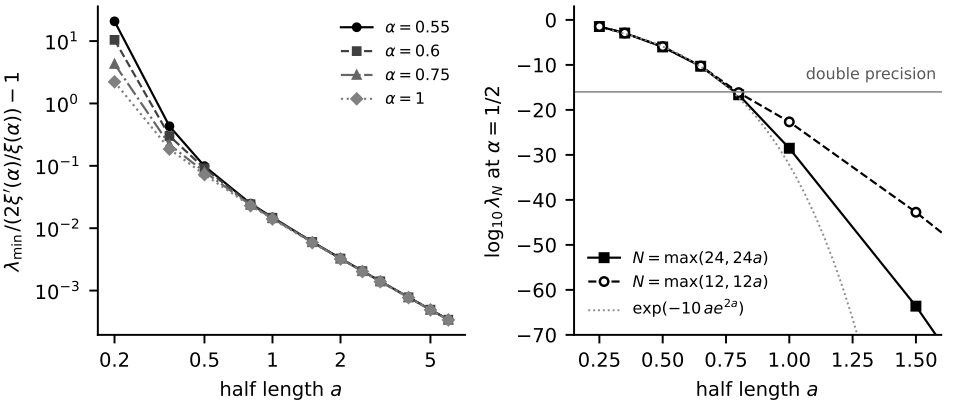


Figure 17.3: Left: the excess of the least eigenvalue of the damped Yoshida form over its floor, $\lambda_{\min}(\alpha, a)/(2b_\alpha) - 1$, against the window half length a, for $\alpha = 0.55$, 0.6, 0.75, 1 (finite sections of $\max(96, \lceil 24a \rceil)$ modes, logarithmic scales); it tends to zero, roughly like a^{-2}. Right: $\log_{10}$ of the finite section value at $\alpha = \frac{1}{2}$ for $N = \max(24, 24a)$ and $N = \max(12, 12a)$ modes, with the level of double precision and the curve $\exp(-10ae^{2a})$.

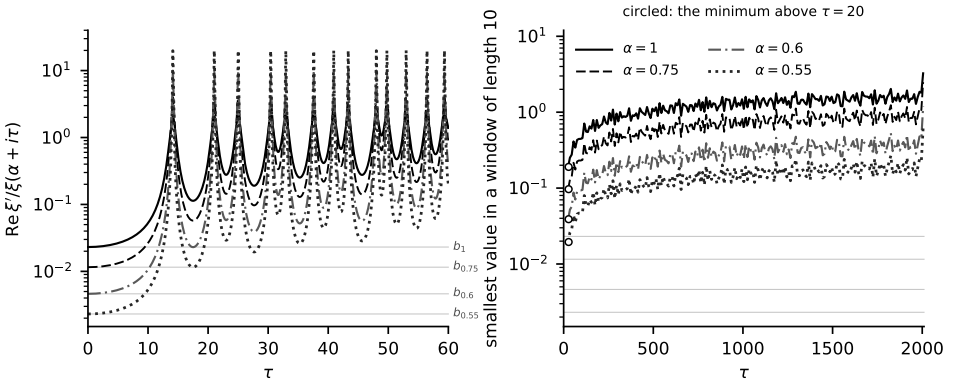


Figure 17.4: Left: the symbol $v_\alpha(\tau) = \mathrm{Re}\,\xi'/\xi(\alpha + i\tau)$ on $[0, 60]$ for $\alpha = 1$, 0.75, 0.6, 0.55 (logarithmic scale), with b_α as thin horizontal lines. Right: the smallest value of v_α in windows of length 10 on $[0, 2000]$, with the minimum above $\tau = 20$ circled. Above $\tau = 20$ the minima are 0.1901, 0.0968, 0.0389 and 0.0195 for $\alpha = 1$, 0.75, 0.6, 0.55, which are 8.2, 8.4, 8.4 and 8.4 times b_α, all at $\tau = 27.67$, between the third and the fourth ordinate (the largest gap above the second); the lowest local minimum other than the one at the origin is lower, $4.90\,b_1$ at $\tau = 17.49$ for $\alpha = 1$ and $4.98\,b_{0.55}$ for $\alpha = 0.55$, in the gap between the first two ordinates.

suggests a decay like $\exp(-c\,ae^{2a})$. This is not a proof.

17.9 An uncertainty barrier

Every prime side computation is limited in one of two ways. Compact support in the level variable keeps the prime sum finite but limits the level to $\log n \le X$. A finite basis limits the frequency, the ordinate $|\tau| \le T$. For the tent coefficient D of Proposition 17.6.1 the limit in level runs into zero verification, which is what the next proposition makes precise.

Proposition 17.9.1. *Suppose all zeros with $0 < \gamma \le H$ lie on the critical line, and let $D_{\mathrm{on}} \ge 0$ be the part of D from on line zeros. Then for every $x > 0$*

$$|D(x) - D_{\mathrm{on}}(x)| \le \varepsilon_H(x) := 8\cosh^2\frac{x}{4}\sum_{\gamma>H}\frac{1}{\gamma^2},$$

the sum being over all zeros above H, counted with multiplicity. With $H = 3\times10^{12}$ (Platt and Trudgian [247], the height H_0 of Theorem 9.1.18) the sum differs from $(\log(H/2\pi)+1)/(2\pi H) = 1.4797\times10^{-12}$ by less than 2×10^{-24} (Theorem 11.4.6), so that $\varepsilon_H(x) < 1.184\times10^{-11}\cosh^2(x/4)$.

Proof. Off line zeros have $|\gamma| > H$ and $|\delta| < \frac12$, so $|\sin(\gamma_\rho x/2)|^2 \le \cosh^2(\delta x/2) \le \cosh^2(x/4)$ and $|\gamma_\rho|^2 \ge \gamma^2$. Each off line zero therefore contributes at most $4\cosh^2(x/4)/\gamma^2$ to D, and summing over all such zeros, two (a zero and its conjugate) for each ordinate $\gamma > H$ of the upper half plane, gives the bound. □

The proposition is of use only against a lower bound for D_{on}. Every term of D_{on} is nonnegative, so the first N zeros, which are on the line, give one: $D_{\mathrm{on}}(x) \ge L_N(x) := \sum_{n\le N} 8\sin^2(\gamma_n x/2)/\gamma_n^2$. For the $N = 1700$ certified zeros of the book the least value of L_N on $[0.05, 47.1]$ is 0.040704, at $x = 25.383$, and 0.04070 is a lower bound for it computed by branch and bound (`yo_tent.py`, `code/README.md`); on $[0.05, 16]$ the least value is 0.0531.

Proposition 17.9.2 (Computed positivity of the tent coefficient). *For the $N = 1700$ certified zeros of the book let $L_N(x) = \sum_{n\le N} 8\sin^2(\gamma_n x/2)/\gamma_n^2$, and let $H = 3\times10^{12}$. Since all zeros with $0 < \gamma \le H$ lie on the critical line [247], $D(x) \ge L_N(x) - \varepsilon_H(x)$ for every $x > 0$. Moreover $L_N(x) > \varepsilon_H(x)$ on $[0.05, 47.48]$, so $D > 0$ on $[0.05, 47.48]$ from the verification of RH up to H and the first 1700 zeros alone, with no prime sum. The first inequality is proved. The second is computed in double precision and not in ball arithmetic: for $x < 46.688$ it follows from the floor 0.04070 of L_N on $[0.05, 47.1]$, found by branch and bound, and for $46.688 \le x \le 47.48$ from a grid of step 5×10^{-4} and the Lipschitz constant 10.85 of L_N.*

The level 47.10 at which ε_H reaches 0.05 ($e^{47.10} = 2.85 \times 10^{20}$) is arithmetically right, but it is the barrier only if D_{on} stays above 0.05, which it does not; with the floor 0.0407 the bound reaches it at $x = 46.69$ ($e^{46.69} = 1.9 \times 10^{20}$), and pointwise at $x = 47.48$ ($e^{47.48} = 4.2 \times 10^{20}$). The barrier is at about 47 either way, because ε_H grows exponentially. A prime side computation of D at a level below 47.48, which needs all prime powers up to e^x, can therefore only confirm a positive sign that the zeros already give; it cannot produce a negative value of D, which is how an off line zero would show. To see one through D a computation must reach levels beyond 47.48, with all prime powers up to 4×10^{20}, or heights beyond H, and the second is the zero verification itself.

In frequency, Proposition 17.8.1 makes the same point for a basepoint $\alpha \geq \sigma_{\max}$: a test vector with spectrum concentrated in $|\tau| \leq T < H$ sees only the symbol there, and a zero $\beta + i\gamma$ with $\gamma > H$ contributes to the Hadamard sum for the symbol the term $(\alpha - \beta)/((\alpha - \beta)^2 + (\tau - \gamma)^2)$, of modulus at most $|\alpha - \beta|/(\gamma - T)^2$.

What the barrier does not say. Proposition 17.9.1 is proved for the single coefficient D and for no other vector. A normalised function supported in the window $[-a, a]$ and modulated to a height above H (Proposition 17.5.1) sees an off line zero there with a weight that does not decay with the height, and ε_H, which carries the factor γ^{-2} of the tent, does not bound the contribution of the zeros above H to the form of such a vector. Nor could a perturbation bound of size 10^{-11} settle the sign of a windowed form at $\alpha = \frac{1}{2}$, whose least eigenvalue is below 10^{-15} from $a = 0.8$ on (Table 17.2). The tent is one vector of the window, so the level x_* of Remark 17.6.3 bounds Yoshida's threshold only from above, $a_0 \leq x_*/2$, and the window experiment of Chapter 18 (Table 18.3) finds a hypothetical off line pair at half lengths far below $x_*/2$.

17.10 The question of a path to RH

The chapter does not give a path to RH. Every criterion proved here has the same structure. On the RH side it is a positivity statement: $D \geq 0$, $T_k \geq 0$ for infinitely many k, the modulated heat trace of Proposition 17.5.1 nonnegative, a Pick function, a coercive damped form. On the other side it is the presence of a pole off the real axis, or a negative Cauchy kernel, at an off line ordinate. The barrier of Section 17.9 then says, for the tent coefficient D, that a computation at levels below 47 cannot go beyond what zero verification already knows; for the other quantities no such statement is proved there. A proof needs an inequality valid at all levels or all heights at once, and it must use the Euler product: the function f_{DH} of Davenport and Heilbronn [72] (Section 11.5.5.4) shares the functional equation and has zeros off the line, so it violates the analogue of every criterion in this chapter.

What the chapter does change is the shape of the missing inequality. Theo-

rem 17.8.5 puts it in its sharpest form:

$$\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\tau)\ \geq\ \frac{\xi'}{\xi}(\alpha)\qquad\text{for all }\tau,\ \alpha\in(\tfrac12,1],\tag{17.10.1}$$

which we write $v_\alpha(\tau)\geq b_\alpha$ as in Theorem 9.1.18. Taken as a family over $\alpha\in(\frac12,1]$, and not line by line, this is equivalent to the coercivity of every damped Yoshida form with the constant it has on functions of low frequency. Three facts about it:

1. It is true with no hypothesis for every $\alpha\geq 1$ and every τ (Propositions 9.1.11 and 9.1.12), and for $\frac12<\alpha\leq\alpha_{\max}=7.35$ and $0<|\tau|\leq H-\frac{1+\sqrt2}{2}$ (Theorem 9.1.18(ii)); for $\frac12<\alpha<1$ and all τ it is equivalent to RH (Theorem 9.1.18(vi)). On the line $\alpha=1$ the minimum is at $\tau=0$, the next lowest local minimum of v_1, at $\tau=17.49$, is 4.90 times b_1, and on $[20,2000]$ the least value is $8.23\,b_1$, at $\tau=27.67$ (computed by `yo_symbol.py`, `code/README.md`).

2. If it holds, even in the weak form $v_{\alpha'}(\tau)>0$, on every line $\operatorname{Re}s=\alpha'$ with $\alpha<\alpha'\leq 1$, then $\sigma_{\max}\leq\alpha$, since a zero $\rho_0=\beta_0+i\gamma_0$ with $\beta_0>\alpha$ sends $v_{\alpha'}(\gamma_0)$ to $-\infty$ as $\alpha'\uparrow\beta_0$; on all of $(\frac12,1]$ it is RH. On a single line it is not known to exclude a zero to its right (Remark 1.2.4). Its $\alpha\downarrow\frac12$ limit, under RH, is the sharpened Laguerre inequality $-(\log\Xi)''(\tau)\geq 2p_1$.

3. Its failure mode is local. A zero with $\beta>\alpha$ would create a negative Cauchy kernel of height $1/(\pi(\beta-\alpha))$ at its ordinate, competing against a symbol of size $\frac{1}{2\pi}\log\tau$ there. So the inequality at α asks for control of $-\operatorname{Re}\zeta'/\zeta(\alpha+i\tau)$ by $\frac12\log(\tau/2\pi)$, a one sided bound on ζ'/ζ; the Fejér tapered form of the question is Proposition 14.14.1. No method known today proves it at any fixed $\alpha<1$ for all τ; the zero free half plane $\operatorname{Re}s>\frac78$ announced in October 2026 (Section 12.1.2) has not been refereed and is not counted.

17.11 Open problems

What remains:

Problem 17.1 (The symbol inequality below the line $\alpha=1$). The inequality $v_\alpha(\tau)\geq b_\alpha$ is a theorem of the book for every $\alpha\geq 1$ and every τ (Propositions 9.1.11 and 9.1.12), and for $\frac12<\alpha\leq\alpha_{\max}$ up to height $3\times10^{12}-1.21$ (Theorem 9.1.18(ii)); for $\frac12<\alpha<1$ and all τ it is equivalent to RH (Theorem 9.1.18(vi)). What is open is the region $\alpha<1$ above the verified height. Determine the part of it in which $v_\alpha(\tau)>b_\alpha$ can be proved with no hypothesis and with explicit constants; the Vinogradov Korobov region $\alpha\geq 1-c(\log|\tau|)^{-2/3}(\log\log|\tau|)^{-1/3}$ [335, 155] is the natural target. A proof on every line of a strip $\alpha_0<\alpha\leq 1$ and for all τ would give $\sigma_{\max}\leq\alpha_0$ (Section 17.10), which is why the target is a region that narrows with the height.

Problem 17.2 (A prime side proof at $\alpha = 1$). The book proves $v_1(\tau) \geq b_1$ through the zeros (Proposition 9.1.11), by the Cauchy kernels of the ordinates nearest to τ and a bound on the gaps between them (compare Lemma 17.8.7). At $\alpha = 1$ the statement can be written entirely in primes. Since $\xi'/\xi(1) = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$ and $\operatorname{Re}\frac{1}{i\tau} = 0$, it reads

$$\sum_{n\geq 2} \frac{\Lambda(n)}{n}\cos(\tau \log n) \;\leq\; \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\tau}{2}\Big) - \frac{1}{2}\log\pi + \frac{1}{1+\tau^2} - \frac{\xi'}{\xi}(1) \qquad (\tau \neq 0),$$

the left side being the conditionally convergent series for $-\operatorname{Re}\zeta'/\zeta(1+i\tau)$. Find a proof using only Mertens type information, in the spirit of the Fejér taper of Proposition 14.14.1; the right side grows like $\frac{1}{2}\log(\tau/2\pi)$ and the left side is $O(\log\log\tau)$ under RH.

Problem 17.3 (Löwner at a single window). By Theorem 17.7.1, positivity on $K(a)$ is positivity of the Löwner matrices of g_a on the lattice $\frac{\pi}{a}\mathbb{Z}$. Löwner's hard theorem [189, 78] recovers the Pick property from Löwner matrices on all finite subsets of an interval. What does positivity on the lattice alone, for one fixed a, imply about the location of the zeros? A positive answer that grew with a would quantify Yoshida's threshold a_0.

Problem 17.4 (The constants of the ladder and of the window). Determine the order of K_0 for the ladder and of a_0 for Yoshida's window, for a single off line pair, and compare with the displacement $\sigma_{\max} - \frac{1}{2}$ and with the de Bruijn Newman constant $\Lambda \leq 2(\sigma_{\max} - \frac{1}{2})^2$ (de Bruijn [73], in the normalisation of Rodgers and Tao [289] used in Section 8.4). For the ladder the computations of Section 17.3 give $K_0(\sec\theta_* - 1) \approx 0.70$ for the pair of the stress test (Remark 17.3.3) and, over fourteen pairs, a first failing order that grows like γ^2 and, at fixed γ, like $1/\delta$ and not like the heuristic γ^2/δ^2; its dependence on δ and on the spacing of the true ordinates is what is to be determined. For the window only $a_0 \leq x_*/2$ is known (Remark 17.6.3), and the window experiment of Chapter 18 (Table 18.3) finds the threshold far below it.

Problem 17.5 (Zero density and the barrier). Proposition 17.9.1 allows every zero above H to have δ arbitrarily close to $\frac{1}{2}$. Zero density estimates make zeros with large δ rare. Sharpen the barrier level using an explicit $N(\sigma, T)$ bound, and determine whether any prime side computation of D at feasible levels can, even in principle, exclude off line zeros beyond the verified height. For windows no barrier is proved: decide whether a window of half length a below about 23.7 (level 47.48) can see an off line zero above H, for which Section 18.4 of Chapter 18 gives numerical evidence and a heuristic horizon, not a proof.

Notes

The forms of this chapter are Yoshida's [349], and the explicit formula is Weil's [338]. Lagarias's criterion [160] asks only for $v_\alpha > 0$; the minimum at the origin of Theorem 9.1.18(vi) sharpens it. In integrated form, under RH, the minimum at the origin says that $\sigma \mapsto |\xi(\sigma+i\tau)|/\xi(\sigma)$ is nondecreasing on $(\frac{1}{2}, \infty)$, a normalised version of the reformulation of Sondow and Dumitrescu [308] that $|\xi(\sigma + i\tau)|$ increases in σ. Theorem 17.6.2 is classical in substance.

Chapter 18

Zhu's compact windows and the Landau Widom horizon

Xuefeng Zhu's preprint on Weil positivity in compact windows [350] continues Yoshida's programme of Chapter 17 by certified computation.

What is not claimed. The account of Zhu's preprint is the first author's reading of version 2 of 2 September 2026, checked statement by statement: his theorems are quoted with the status he gives them and are not reproved, and his certificates are not rechecked. A number is computed in floating point at the precision stated; the least eigenvalues of Table 18.1 are consistent with Zhu's certified intervals and carry no interval certificate of their own. The passage from a statement about a window to a statement about the height of the zeros (the horizon of Section 18.4) is Zhu's heuristic and is marked as one wherever it is used. The detection experiment of that section rests on a hypothetical configuration of zeros, a quadruple added to the actual zeros of ξ; it is a computation about that configuration and not a statement about ξ.

18.1 What Zhu proves

Zhu [350] writes L for the half length of the window. Here it is a, as in Chapter 17, where L is the support of an autocorrelation, so that his $\lambda^*(L)$, A_L and $T_1(L)$ appear below as $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)$, A_a and $T_1(a)$. For real even continuous f supported in $[-a,a]$, with $F=\widehat{f}$, $\widehat{f}(t)=\int_{\mathbb{R}} f(y)e^{ity}\,dy$, Zhu writes Weil's quadratic form as

$$Q(f)=2F\big(\tfrac{i}{2}\big)^2+\frac{1}{2\pi}\int_{\mathbb{R}}|F(t)|^2\,\Psi_a(t)\,dt,\qquad \Psi_a(t)=\operatorname{Re}\psi\Big(\frac{1}{4}+\frac{it}{2}\Big)-\log\pi-\mathcal{P}_a(t), \tag{18.1.1}$$

where $\mathcal{P}_a(t)=\sum_{\log n<2a}2\Lambda(n)n^{-1/2}\cos(t\log n)$ is the prime comb. He sets $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a)=\inf Q(f)/\|f\|^2$ over the real even $f\neq 0$ supported in $[-a,a]$. His results, with the status he gives them, are the following.

(Z1) One stroke reduction (Theorem 1.1, unconditional). Let $A_a = \sup_t \mathcal{P}_a(t) = \sum_{\log n<2a} 2\Lambda(n)n^{-1/2}$: the supremum is attained at $t = 0$, and by Weyl equidistribution of the phases $t \log p$ values arbitrarily close to it occur at arbitrarily large t (his Lemma 3.2). The envelope $\operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2}) - \log\pi \geq \log\frac{t}{2\pi} - \frac{1}{t}$ for $t \geq \frac{15}{4}$ (Lemma 3.1) gives $\Psi_a(t) \geq \beta^* > 0$ for $t \geq T_\sharp$, where $\beta^* = \log\frac{T_\sharp}{2\pi} - \frac{1}{T_\sharp} - A_a$ and $\beta^* > 0$ holds for every $T_\sharp$ above a threshold that exceeds $T_1 := 2\pi e^{A_a}$ by the factor $1 + O(T_1^{-1})$ (120.1 against 119.1 at $a = 0.8$). Positivity on the window then reduces to the positive semidefiniteness of a finite matrix in an even Legendre basis, with explicit tails.

(Z2) Certificate (Theorem 1.2, a computer assisted theorem). At $a = 0.8$, with $T_\sharp = 200$, 200 even Legendre modes (orders up to 398), working precision 50 digits and a verified Cholesky factorisation, $Q(f) \geq 8.9\times10^{-18}\|f\|^2$ for every real even f supported in $[-0.8, 0.8]$. The autocorrelation then has support $[-1.6, 1.6]$, which is 2.3 times the range $2a \leq \log 2$ that Zhu attributes to Yoshida [349] and, by a different method, to Connes and Consani [59]. His variational upper bound, computed on the geometric side in interval arithmetic, is $\lambda_{\min}^{\text{even}}(\frac{1}{2}, 0.8) \leq 2.27 \times 10^{-17}$, which encloses the infimum within a factor 2.6.

(Z3) Under RH, $\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) \leq \exp(-ae^a)$ for a large (Theorem 1.3, conditional on RH), by an explicit test function: a canonical product over the ordinates up to $T = 2\pi e^a$ times a power of the sinc function. The construction works at half saturation, and with $T = 2\pi e^{2\theta a}$ it closes only for $\theta \lesssim 0.65$ (his Remark 13.6).

(Z4) Decay law (Conjecture 12.1, fitted to computed bounds). The certified variational upper bounds for a from 0.5 to 2, unconditional and computed on the geometric side, fall from 10^{-6} to 3.2×10^{-283} and follow $-\log\lambda_{\min}^{\text{even}}(\frac{1}{2}, a) = C\,N(T_*)/\log N(T_*)\,(1 + o(1))$, where $T_* = 2\pi e^{2a}$, $N(T)$ is the number of ordinates up to T and $C = 20.13\ldots \approx 2\pi^2$ (the fit of his equation (5)). The constant is fitted and not derived: Zhu calibrates the law on $a \leq 1.2$ and tests it on $a = 1.4$ to 2, where it underpredicts by about 5 percent with no trend. That the law describes the infimum itself, with $2\pi^2$ in place of the fitted C, is his Conjecture 12.1.

(Z5) Barrier (Theorem 1.4, unconditional). Since $A_a = (4 + o(1))e^a$ by the prime number theorem, any application of (Z1) needs $T_\sharp > T_1(a) = 2\pi e^{A_a}$, about 1.0×10^7 at $a = 1.6$ and 2.4×10^{11} at $a = 2$ ($A_{1.6} = 14.323$, $A_2 = 24.383$), so that the size of the matrix grows doubly exponentially in a; and since $\sup_t \mathcal{P}_a(t) = A_a$ exactly, no pointwise bound on the comb lowers T_1. To improve on pointwise envelopes one would have to prove that the phases $t \log p$ cannot align, which Zhu calls an arithmetic statement about simultaneous Diophantine approximation of $\{\log p\}$ (his Remark 1.6).

(Z6) Parity (Lemma 6.1, Theorem 6.2 and Corollary 6.3). For real $f = f_e + f_o$, $Q(f) = Q(f_e) + Q(f_o)$, with the pole term entering as $+2c^2$ on the even part and $-2s^2$ on the odd part, where $c = \int f_e(x)\cosh\frac{x}{2}\,dx$ and $s = \int f_o(x)\sinh\frac{x}{2}\,dx$. At $a = 0.8$ the even ground state is simple, with the certified bounds $8.9\times10^{-18} \le \lambda_1^{\mathrm{even}} \le 2.523\times10^{-16}$ (the upper bound 2.27×10^{-17} of (Z2) is the sharper), $\lambda_2^{\mathrm{even}} \ge 2.085\times10^{-12}$ and $8.206\times10^{-15} \le \lambda_1^{\mathrm{odd}} \le 2.347\times10^{-14}$, where $\lambda_1^{\mathrm{even}} \le \lambda_2^{\mathrm{even}}$ are the first two min max values of $Q(f)/\|f\|^2$ over real even f supported in $[-0.8, 0.8]$ and λ_1^{odd} is the infimum over real odd f. Hence $Q(f) \ge 8.9\times10^{-18}\|f\|^2$ for complex f as well. Zhu connects the simple even ground state with one of the two spectral steps that the programme of Connes, Consani and Moscovici [60] isolates as missing.

Zhu also retracts a claim of an earlier draft at autocorrelation support 2.38 ($a = 1.19$), keeping a computation that is consistent with positivity there and no certificate (his Section 7), and he warns that evaluating the geometric side of Q in double precision produces spurious negative eigenvalues (his Remark 2.1 and Section 10).

For complex test functions the least eigenvalue of Chapter 17 is $\lambda_{\min}(\frac12, a)$, the infimum of $\langle\varphi,\varphi\rangle/\|\varphi\|^2$ over $0 \ne \varphi \in C(a)$ (Section 17.8.2). Write $\lambda_{\min}^{\mathrm{odd}}(\frac12, a)$ for the infimum over real odd test functions. The form has no cross terms between the real and the imaginary part of φ, nor between its even and its odd part (Zhu's Lemma 6.1), so that

$$\lambda_{\min}(\tfrac12, a) = \min\bigl(\lambda_{\min}^{\mathrm{even}}(\tfrac12, a),\, \lambda_{\min}^{\mathrm{odd}}(\tfrac12, a)\bigr),$$

and the two sides $\lambda_{\min}(\frac12, a)$ and $\lambda_{\min}^{\mathrm{even}}(\frac12, a)$ are equal when the ground state is even, which Zhu's Theorem 6.2 shows at $a = 0.8$. Yoshida's parity reduction (Y1) of Section 17.1 concerns the same two sectors. The splitting holds for the damped forms of Section 17.8.1 at every α as well, because the damping $e^{-\varepsilon|y|}$ is even.

18.2 The Thorin dictionary

The functional $\mathcal{Y}$, the form $\langle\varphi_1,\varphi_2\rangle = \mathcal{Y}(\varphi_1 * \widetilde{\varphi}_2)$ and the spaces $C(a)$ and $K(a)$ are those of Section 17.1. Call a real even f supported in $[-a, a]$ admissible if the explicit formula applies to $f * f$, as it does for every $f \in C_c^\infty(\mathbb{R})$. Let PW_a be the Paley Wiener space of the entire functions of exponential type at most a that are square integrable on $\mathbb{R}$, and PW_a^{even} its real even members, which contain the transforms $F = \widehat{f}$ of the admissible f. The measure $U = \sum_{\gamma>0}\delta_{\gamma^2}$ is the Thorin measure of the clock under RH (Section 1.2), each ordinate counted with its multiplicity.

Proposition 18.2.1. *For admissible f, with $F = \widehat{f}$, Zhu's form is Yoshida's form*

on the window $C(a)$: $Q(f)=\langle f,f\rangle=\mathcal{Y}(f*f)$. Under RH

$$Q(f)=\sum_{\rho}|F(\gamma_\rho)|^2=2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt),$$

$$\lambda^{\mathrm{even}}_{\min}(\tfrac12,a)=\inf_f\frac{2\int_{(0,\infty)}|F(\sqrt{t})|^2\,U(dt)}{\frac{1}{2\pi}\int_{\mathbb{R}}|F|^2},$$

so that $\lambda^{\mathrm{even}}_{\min}(\frac12,a)$ is the lower sampling constant of the Thorin measure U for the transforms $F\in PW^{\mathrm{even}}_a$ of the admissible f.

Proof. The autocorrelation $g=f*f$ is even, supported in $[-2a,2a]$ and has $\widehat{g}=F^2=|F|^2\geq 0$ on $\mathbb{R}$. By Proposition 17.4.1 (there $L=2a$; the proposition is stated for smooth f, and for an admissible f the identity is the term by term explicit formula that follows), with no hypothesis on the zeros, $\langle f,f\rangle=\mathcal{Y}(g)=2\int\widehat{g}(\sqrt{t})\,U^{\pm}(dt)$, and only the prime powers $n\leq e^{2a}$ enter; under RH the functional $\varphi\mapsto\int\varphi\,dU^{\pm}$ is the sum of φ over the atoms of U (Section 5.3.1). Term by term, the explicit formula (17.1.1) for g has the pole term $\widehat{g}(\frac{i}{2})+\widehat{g}(-\frac{i}{2})=2F(\frac{i}{2})^2$. Its prime term $-\sum_{n\geq2}\Lambda(n)n^{-1/2}\big(g(\log n)+g(-\log n)\big)$ equals $-\frac{1}{2\pi}\int|F|^2\mathcal{P}_a$, because $\widehat{g}=|F|^2$ and g vanishes beyond $2a$. Its archimedean term is $V_1(g)-g(0)\log\pi=\frac{1}{2\pi}\int|F|^2\big(\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi\big)dt$, by the first line of (17.1.2) and $g(0)=\frac{1}{2\pi}\int|F|^2$. The three terms add up to (18.1.1). Under RH every γ_ρ is real, and the zero side of (17.1.1) is $\sum_\rho\widehat{g}(\gamma_\rho)=\sum_\rho F(\gamma_\rho)^2=2\sum_{\gamma>0}F(\gamma)^2$, because F is even and the ordinates come in pairs $\pm\gamma$; this is $2\int F(\sqrt{t})^2\,U(dt)$. Finally $\|f\|^2=\frac{1}{2\pi}\int|F|^2$ by Plancherel. □

Two consequences follow. First, under RH the positivity of the infimum for every a is a sampling statement, and it is Zhu's Proposition 2.3. Littlewood's theorem on the gaps between consecutive ordinates [324, Section 9.12], which holds without RH although Zhu states it as a consequence of RH, says that the gaps tend to zero, so under RH the set $\{\pm\gamma\}$ contains, above some height, a separated subset Λ of lower uniform density larger than a/π (a maximal subset of points at least $\pi/(4a)$ apart), and Beurling's sampling theorem [302, Chapters 2 and 3] gives $c>0$ with $\sum_{\lambda\in\Lambda}|F(\lambda)|^2\geq c\int_{\mathbb{R}}|F|^2$ for all $F\in PW_a$. Hence $Q(f)\geq 2\pi c\|f\|^2$ and $\lambda^{\mathrm{even}}_{\min}(\frac12,a)>0$, with a constant that is not explicit (Zhu's Remark 2.4). Without RH the argument gives nothing (Zhu's Remark 2.4 counts the gap bound as a second use of RH and says that unconditionally the gaps are only known to be bounded). Second, in this picture the smallness of the infimum is the poor conditioning of sampling by a set that is sparse at low frequency: there is no ordinate in $(-14.13,14.13)$.

Zhu's parity lemma (Z6) has a Thorin reading. The pole is the atom of mass one of the signed Thorin measure $U^{\pm}$ at $t=-\frac14$ (equation (17.4.1)), that is at the imaginary ordinates $\pm\frac{i}{2}$, and it is paired with $F(\frac{i}{2})F(-\frac{i}{2})$. For even f this is

a	modes N	$\lambda_{\min}^{\text{even}}$	$\lambda_{\min}^{\text{odd}}$	Zhu, even
0.5	8	1.336×10^{-6}	2.51×10^{-4}	
0.5	14	1.082×10^{-6}	2.39×10^{-4}	
0.5	20	1.005×10^{-6}	2.30×10^{-4}	
0.5	26	9.743×10^{-7}	2.24×10^{-4}	$\leq 1.05 \times 10^{-6}$
0.5	160	9.344×10^{-7}	1.997×10^{-4}	
0.8	8	5.49×10^{-15}	2.67×10^{-12}	
0.8	14	4.127×10^{-17}	3.86×10^{-14}	
0.8	20	2.651×10^{-17}	2.333×10^{-14}	
0.8	26	2.144×10^{-17}	2.157×10^{-14}	$[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$
0.8	160	1.673×10^{-17}	1.644×10^{-14}	

Table 18.1: Least eigenvalues of Weil's form on the window $[-a, a]$ in the even and the odd sector, from the prime side Yoshida matrices in 60 digit arithmetic with cosine and sine modes up to the order N shown (`code/zh_window_eigs.py`). The last column gives Zhu's certified bounds for the even sector. The values at 26 modes and below are not converged, and the odd values at 160 modes are still falling.

$c^2 \geq 0$ and for odd f it is $-s^2 \leq 0$ (Proposition 17.4.1): it helps the even sector and hurts the odd one. That the even ground state is nevertheless the smaller is explained by Zhu (Section 6.3) through the constraint $F_o(0) = 0$ on the odd sector, which denies odd test functions the zero free interval $[0, \gamma_1)$ in which even minimisers park most of their mass: the low frequency gap, and not the pole, controls the conditioning. At $a = 0.8$ the odd sector lies about three orders of magnitude above the even one and at $a = 0.5$ two orders (Table 18.1), and Zhu reports that the ratio grows almost geometrically in a over $[0.5, 1.2]$, as $10^{2.07a+1.32}$.

The least eigenvalues of the two sectors are computed from the prime side matrices of Yoshida's form. The even sector has the basis $(2a)^{-1/2}$ and $a^{-1/2}\cos(\pi jy/a)$, the odd sector the basis $a^{-1/2}\sin(\pi jy/a)$, for $1 \leq j \leq N$, and N is the number of modes in Table 18.1.

The values of Table 18.1 are consistent with Zhu's numbers, which come from another basis and another code: Fourier modes on $[-a, a]$ against his Legendre polynomials, and quadrature of the explicit formula in the variable y against his spherical Bessel integrals. At $a = 0.8$ both sectors lie inside his certified intervals at 26 modes. At 160 modes the even values 9.344×10^{-7} and 1.673×10^{-17} lie 0.04 and 1.4 percent above the reference floors 9.34×10^{-7} and 1.65×10^{-17} of Zhu's Table 1 (a converged exact assembly, which he gives as reference values and not as certificates), and the odd values 1.997×10^{-4} and 1.644×10^{-14} lie 2.9 and 4.7 percent above his 1.94×10^{-4} and 1.57×10^{-14} and are still falling. At $a = 0.8$ the second even eigenvalue is 8.42×10^{-12} at 160 modes (Zhu: 8.38×10^{-12} in his

Table 1, and the certified bound $\lambda_2^{\mathrm{even}} \geq 2.085 \times 10^{-12}$), about 5×10^5 times the first, in agreement with the simple ground state of (Z6). Raising the precision to 90 digits, or the number of quadrature nodes by half, changes the least eigenvalues by a relative 2×10^{-58} at most (checked at 26 and 40 modes), so every value is an upper bound for the infimum over its sector.

As a check of the normalisation of Proposition 18.2.1, take the unit bottom eigenvector f of the matrices of 26 and 40 modes at $a = 0.5$ and 0.8, in both sectors. The partial sums $2\sum_{\gamma \leq \gamma_K} F(\gamma)^2$ over the first K ordinates of `data/zeros_1700.txt` increase with K and stay below the eigenvalue, and with the mean tail $2\int_{\gamma_K}^{\infty} F(t)^2 \, \frac{1}{2\pi} \log \frac{t}{2\pi} \, dt$ of the missing ordinates they reproduce it to within 0.16 percent for $K = 300$, 400 and 1700 in all eight cases.

Three other tables of the book list the same eigenvalue at other levels: Table 17.2 of Chapter 17 for the windows $a = 0.35$ to 4 (its entry 2.28×10^{-17} at $a = 0.8$, from 24 modes, lies between our values at 20 and 26 modes), and Tables 20.1 and 20.2 of Chapter 20. Table 18.1 is at $\alpha = \frac{1}{2}$ only, with the even and the odd sector apart, in 60 digits, and it shows the convergence in the number of modes without extrapolating; it reaches $a = 0.8$.

18.3 Zhu's test function through a truncated reciprocal Xi transform

Zhu's proof of (Z3) takes $\widehat{f_0}(r) = B(r)h(r)$, with $B(r) = \prod_{0<\gamma_j \leq T}(1 - r^2/\gamma_j^2)$ a finite product over the ordinates up to a truncation height T, and a window kernel h of type a. In the notation of Section 1.2, $G(u) = \prod_{\gamma>0}(1 + u/\gamma^2)$ is the reciprocal of the Laplace transform of the clock, which under RH is $\sum_{\gamma>0} e_\gamma/\gamma^2$. Truncate the product, $G_T(u) = \prod_{0<\gamma_j \leq T}(1 + u/\gamma_j^2)$. Then $1/G_T$ is the Laplace transform of the finite gamma convolution $\sum_{\gamma_j \leq T} e_{\gamma_j}/\gamma_j^2$ and

$$B(r) = G_T(-r^2),$$

the reciprocal of that Laplace transform continued to the cut $s = -r^2$. Zhu's test function is the window kernel divided by the truncated reciprocal Xi transform on the cut. In the variable $t = r^2$ it vanishes at the Thorin atoms $\gamma_j^2 \leq T^2$ (and at the zeros of h), and under RH

$$Q(f_0) = 2\int_{t>T^2} h(\sqrt{t})^2 \, G_T(-t)^2 \, U(dt).$$

The optimisation is a trade: each atom killed costs a factor r^2/γ_j^2 of growth, which h must absorb with its limited type.

We evaluated this family numerically (script `zh_product_bound.py`), with $h(r) = \big(\sin(ar/m)/(ar/m)\big)^m$, which has type a, optimising over the number K of atoms killed and over the even exponent $m \geq 2K+2$. Table 18.2 is a computation

a	best bound	K	m	γ_K	$N(T_*)$	$T_* = 2\pi e^{2a}$
0.5	3.88×10^{-4}	2	6	21.0	1	17.1
0.8	4.03×10^{-9}	6	18	37.6	4	31.1
1.0	7.04×10^{-14}	9	26	48.0	8	46.4

Table 18.2: The best Rayleigh quotient $Q(f_0)/\|f_0\|^2$ of Zhu's product family over the number K of atoms killed and the even exponent m, with the optimal K and m, the last ordinate γ_K killed, the number $N(T_*)$ of ordinates below the resolution height $T_* = 2\pi e^{2a}$ of Section 18.4, and T_* itself.

of this chapter and not of Zhu. The quotient $Q(f_0)/\|f_0\|^2$ is evaluated on the zero side under RH, with the 1700 ordinates of `data/zeros_1700.txt` and an explicit Riemann von Mangoldt tail [109], and on the geometric side by Zhu's formula (18.1.1), which needs no hypothesis. The two agree to 13 digits or more at $a = 0.8$ and 1.0 and to 3.5×10^{-7} relative at $a = 0.5$, where the tail beyond 1700 ordinates limits the comparison, so every entry is an unconditional upper bound for $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2}, a)$.

The product construction is crude. At $a = 0.8$ it gives 4.03×10^{-9} against about 1.7×10^{-17} for the window infimum (the entry at 160 modes of Table 18.1, an upper bound), and a general element of PW_a does much better. The number of atoms worth killing grows with a and exceeds the number $N(T_*)$ of ordinates below the horizon T_* of the next section: in each row the last atom killed lies above T_* and below eT_*. When K is restricted to $N(T_*)$ the bound is worse by the factors 1.27, 1.40 and 1.05. At $a = 0.35$ and $a = 1.2$ (not tabulated) the optimum is $K = 0$ and $K = 16$, equal to $N(T_*)$ in both cases. So atoms above the horizon are worth killing at the three tabulated half lengths, and three rows with $a \le 1$ cannot settle the asymptotics (Zhu's Remark 13.6 in (Z3)).

18.4 The Landau Widom horizon

Zhu's resolution height comes from a comparison of densities (his Section 9). By Jensen's formula a real entire function of exponential type a has real zeros of average density at most a/π on long intervals $[-R, R]$ about the origin, which Zhu takes as a bound for the local density, while the ordinates have density $\nu(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$ at height t. The two meet at

$$\frac{a}{\pi} = \frac{1}{2\pi} \log \frac{T}{2\pi} \iff T = T_*(a) := 2\pi e^{2a}, \tag{18.4.1}$$

which we call the Landau Widom horizon, or the horizon. Zhu's argument is that below T_* a test function of type a can in principle interpolate the ordinates, placing a zero of F on each of them, and that above T_* the ordinates outnumber

the available oscillations, so that $|F|^2$ must leak onto them. A count of degrees of freedom gives the same height up to a factor. An even function of PW_a concentrated on the frequencies $[0,T]$ has about aT/π degrees of freedom, since the time frequency rectangle $[-a,a]\times[-T,T]$ has about $2aT/\pi$ [307, 166], halved for evenness, while $N(T)\approx\frac{T}{2\pi}\log\frac{T}{2\pi e}$ ordinates lie below T. Equating the two counts gives $2\pi e^{2a+1}=eT_*$.

Zhu's decay law (Z4) uses T_*. He models the cost of killing $N(T_*)$ ordinates at saturated budget as an excursion of order $N(T_*)$ eigenvalues beyond the Shannon number of the window, at the plunge rate $\lambda_{K+j}\approx\exp(-\pi^2 j/\log K)$ of the eigenvalues of the concentration operator of time and band limiting [167]. This gives $-\log\lambda_{\min}^{\mathrm{even}}(\frac12,a)\asymp\pi^2N(T_*)/\log N(T_*)$ up to a constant, and the fit supplies the constant (his Section 9 and Conjecture 12.1). It is his model, fitted and not derived.

The passage from a window statement to a statement about the height of the zeros is a heuristic and not a theorem. Nothing in this book proves that positivity of Weil's form on every window of half length a is implied by, or implies, the verification of the zeros up to a height of order $T_*(a)$, and Problem 18.1 asks for the theorem that the heuristic would need.

A quadruple of off line zeros $\frac12\pm\delta\pm i\gamma$ contributes $4\operatorname{Re}F(\gamma-i\delta)^2$ to $Q(f)$ for real even f, and a negative value of Q needs a test function that nearly vanishes at the on line ordinates near γ while keeping its energy near $\gamma\pm i\delta$. Heuristically this is possible only when the window has more degrees of freedom near γ than the ordinates have, $a/\pi>\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, which predicts a detection threshold

$$a_{\mathrm{det}}(\gamma,\delta)\approx\tfrac12\log\frac{\gamma}{2\pi}+O(1),$$

with a weak dependence on δ for small δ. Once the test function vanishes at the neighbours and at γ itself, the contribution of the quadruple is $4\operatorname{Re}F(\gamma-i\delta)^2=-4\delta^2F'(\gamma)^2+O(\delta^4)$, which is negative for small $\delta>0$, and it must exceed the residual of the on line zeros.

We tested this on a hypothetical configuration (scripts `zh_detect.py` and `zh_zeros.py`). Add to the on line zeros one quadruple $\frac12\pm\delta\pm i\gamma$, where γ is the midpoint of the gap between two consecutive ordinates (numbers 10 and 11, 79 and 80, 649 and 650, 1517 and 1518). The matrix is the zero side formula of Theorem 17.7.1 for Yoshida's form on $C(a)$, with the first 2100 ordinates (the last at 2620.28) and the modes $|n|\le\lceil a(\gamma+45)/\pi\rceil+2$, whose frequencies $\pi n/a$ reach 45 beyond the height of the quadruple. The on line zeros contribute a positive semidefinite Gram matrix and the quadruple a real symmetric matrix of rank at most four, so any negativity comes from the quadruple. Two criteria are used. In criterion (i) the entry is the smallest a on a grid of step 0.05 at which the least eigenvalue of the float64 matrix is below -10^{-11}. In criterion (ii) it is the smallest a at which the exact least eigenvalue is negative, which is decided in ball arithmetic: in each parity sector the quadruple is a perturbation of rank two of

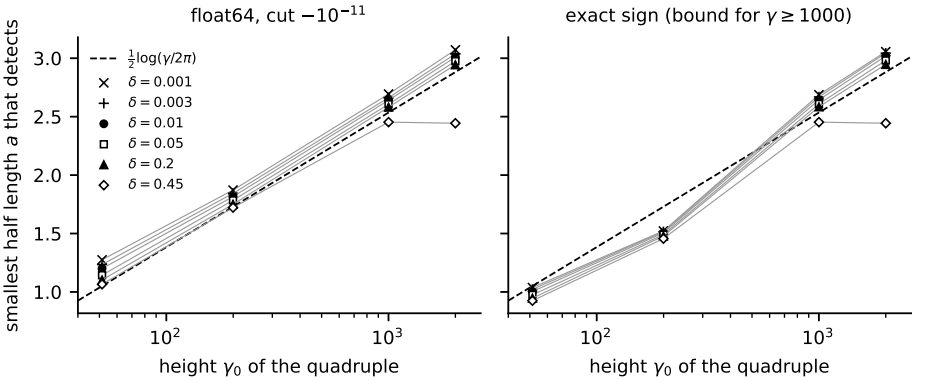


Figure 18.1: Detection thresholds of Table 18.3 and of the displacements $\delta = 0.001$ and 0.003 of the text against the height γ of the hypothetical quadruple (logarithmic axis), for $\delta = 0.001$, 0.003, 0.01, 0.05, 0.2, 0.45 (points of equal δ joined), with the horizon $\frac{1}{2}\log(\gamma/2\pi)$ dashed; the vertical axis is the half length a of the window. Left: criterion (i). Right: criterion (ii), with upper bounds at the heights 1000 and 2000. The points are the thresholds refined by bisection to 0.005, so that they lie up to 0.05 below the entries of the table. Drawn by `code/zh_detect.py` (option `--figure`) from `data/zh_detect.json`.

a positive definite matrix, so the sign is decided by a two by two determinant. At the heights 1000.57 and 1999.99 criterion (ii) is only an upper bound, from the principal submatrix on the modes with $|\pi n/a - \gamma| \leq 150$: by interlacing a detection there is a detection for the full matrix, and a failure to detect proves nothing. The entry of the table is the smaller of this band value and the entry of criterion (i).

The rounding error of the float64 eigenvalue is bounded a priori, from the rounding of the Gram matrix and a textbook bound for the eigensolver, by 4×10^{-12} to 5×10^{-12} at height 51, 9×10^{-12} at 200, about 2.6×10^{-11} at 1000 and 4×10^{-11} at 2000. The cut -10^{-11} of criterion (i) is therefore safe in double precision at the heights 51 and 200, whereas at 1000 and 2000 the bound exceeds the cut, so that there a detection is certified only when the float64 eigenvalue lies below minus the bound. At the printed value of a every one of the sixteen events is certified: in fourteen cells by the float64 eigenvalue and this bound, in the other two (at $(51.37, 0.01)$ and $(199.64, 0.2)$) by the exact test of the cut. The grid point below the printed value is certified as a non detection at the heights 51 and 200 only. Each entry of criterion (i) is an upper bound, for the model of the table, for the sign change in exact arithmetic.

For $\delta \leq 0.2$ every threshold of criterion (i) lies 0.02 to 0.17 above $\frac{1}{2}\log(\gamma/2\pi)$. For $\delta = 0.45$ it lies above it by 0.05 and 0.02 at the heights 51 and 200 and below it

γ	$\frac{1}{2}\log\frac{\gamma}{2\pi}$	$\delta=0.01$	$\delta=0.05$	$\delta=0.2$	$\delta=0.45$
Criterion (i): the float64 least eigenvalue falls below -10^{-11}					
51.37	1.051	1.20	1.15	1.15	1.10
199.64	1.729	1.85	1.80	1.75	1.75
1000.57	2.535	2.65	2.65	2.60	2.50
1999.99	2.882	3.05	3.00	2.95	2.45
Criterion (ii): the exact least eigenvalue is negative					
51.37	1.051	1.05	1.00	0.95	0.95
199.64	1.729	1.55	1.50	1.50	1.50
1000.57	2.535	≤ 2.65	≤ 2.65	≤ 2.60	≤ 2.50
1999.99	2.882	≤ 3.05	≤ 3.00	≤ 2.95	≤ 2.45

Table 18.3: Smallest window half length a at which Yoshida's form on $[-a,a]$ detects a hypothetical off line quadruple $\frac{1}{2}\pm\delta\pm i\gamma$ added to the first 2100 zeros, on a grid of step 0.05 (script `code/zh_detect.py`, record `data/zh_detect.json`). The entries of the lower block are exact on the full matrix at the heights 51.37 and 199.64; at 1000.57 and 1999.99 the entries marked $\le$ are upper bounds, the smaller of the value from a band of modes and the entry of criterion (i) above. At the cell $(1999.99, 0.2)$ the band computation of 150 modes does not detect the quadruple at 2.90 and 2.95 (it does at 3.00, with 2100, 4000 and 8000 zeros and with the margin 90 in place of 45 in the number of modes), so that the entry 2.95 there is the entry of criterion (i); the exact test detects the quadruple in a band of 300 modes at 2.95 and in a band of 600 modes at 2.90.

by 0.04 and 0.43 at the heights 1000 and 2000 (Figure 18.1). Refined by bisection to 0.005 the thresholds of criterion (ii) are 1.003, 0.975, 0.947, 0.925 at $\gamma=51.37$ and 1.506, 1.491, 1.478, 1.456 at $\gamma=199.64$, for $\delta=0.01$, 0.05, 0.2, 0.45, which lie below the horizon by 0.05 to 0.13 and by 0.22 to 0.27. So the horizon is neither an upper bound nor a lower bound for the detection threshold in these data. The table is set by the cut and by the model as much as by the horizon: near the cut the negative eigenvalue grows by a factor 2×10^2 to 2×10^4 per step 0.05 in a, so that any cut between 10^{-16} and 10^{-9} moves the thresholds by less than 0.15, but double precision cannot see the sign change itself, because between the exact sign change and the cut the float64 eigenvalue is rounding noise (between -2×10^{-15} and -5×10^{-15} for every a from 1.3 to 1.7 at height 200).

The dependence on δ is weak for small δ. At $\delta=0.001$ the thresholds of criterion (i) are 1.30, 1.90, 2.70, 3.10 at the four heights, and at $\delta=0.003$ they are 1.25, 1.85, 2.70, 3.05. From $\delta=0.001$ to 0.2 they fall by 0.10 to 0.15, and they are nonincreasing in δ in every row. So a displacement of 0.01 is detected almost as early as one of 0.2, but a displacement of 0.45 is detected at height 2000 much earlier than the horizon suggests. The weak dependence is a property of

these computations and not a consequence of the expansion above. What lets a large displacement be seen early is the growth of a test function off the real axis: for $F \in PW_a$ bounded on $\mathbb{R}$, the Phragmén Lindelöf bound gives $|F(\gamma - i\delta)|^2 \le e^{2a\delta} \sup_{\mathbb{R}} |F|^2$, which is 9.1 at $a = 2.45$, $\delta = 0.45$. It is an upper bound and not a measured gain, and the data are consistent with the reading that at height 2000 the term of the quadruple, amplified by up to that factor, can beat on line neighbours spaced about 0.9 apart (the gap that contains the quadruple is 0.889, the mean spacing 1.09) before the window resolves them individually. The experiment does not test this reading.

Three features of the model limit what the table says. First, the configuration is not count preserving. The variant that removes the two on line zeros flanking the gap is detected earlier, by 0.15 to 0.30 in each of the eight cells computed ($\delta = 0.01$ and 0.2 at the four heights), and every one of its thresholds lies at or below the horizon, by 0.03 to 0.24. The thresholds of the table are therefore those of a configuration with two extra zeros and not a statement about a configuration with the true zero count. Second, the truncation to 2100 ordinates can only advance detection, because the omitted zeros add positive semidefinite terms. In the eight cells that were also run with 4000 and with 8000 ordinates the float64 thresholds are unchanged, but the rounding bound grows with the number of zeros, and the entry $(1999.99, 0.2)$ is no longer certified at either size (the certified value is 3.00), nor is $(1000.57, 0.01)$ at 8000. Third, with the margin 90 in place of 45 in the number of modes the entry $(1999.99, 0.01)$ is 3.00 and not 3.05.

Compare the tent coefficient $D(x) = x\langle \chi_0, \chi_0 \rangle$ of Chapter 17 at the level $x = 2a$. By Remark 17.6.3 it turns negative near $x_* \approx (2\log\gamma + \log p_1)/\delta$, with $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231049931$. The tent is one vector of the window $C(a)$, so $x_*/2$ is only an upper bound for the half length at which the window detects the quadruple. For $\delta = 0.2$ the level at which the quadruple can first push D negative for some phase, computed from the first 8000 ordinates, gives $x_*/2 = 8.68$, 16.2, 24.2, 27.8 at the four heights, against the thresholds 1.15, 1.75, 2.60, 2.95 of Table 18.3. Over the 16 cells of the table the window thresholds are 4.0 to 177 times smaller than $x_*/2$, the ratio being roughly $2/\delta$ (144 to 177 at $\delta = 0.01$, 29 to 37 at 0.05, 7.5 to 9.4 at 0.2 and 4.0 to 5.1 at 0.45). The formula for x_* overestimates the computed level by 2 to 22 percent. The window data do not support an order $\delta^{-1}\log\gamma$ for the window threshold itself. For the height 3×10^{12} the formula gives $x_*/2 = 26.85/\delta$, that is 59.7 at $\delta = 0.45$ and 134 at $\delta = 0.2$, against the half length 13.45 of the horizon, with the same ratio $2/\delta$. Proposition 17.9.1 (level 47) is proved for D only, and Table 18.3 shows that the window is far more sensitive than the coefficient. The level 47 is therefore not the level at which windows begin to see beyond the verification of the zeros, and the conversion from half lengths to levels, here and in Chapter 17, is a heuristic.

The horizon turns a window statement into a height statement only heuristically, and only for small displacements. Large displacements are seen below the

window statement	half length a	height $2\pi e^{2a}$
Yoshida's Theorem 1, as Zhu reports it	0.347	12.6
Zhu's certificate (Z2)	0.8	31.1
practical limit of envelope certificates, Zhu's Remark 1.6	≈ 1.6	154
verification of the zeros to 3×10^{12} [247]	≈ 13.4	3×10^{12}
RH	∞	∞

Table 18.4: The horizon $2\pi e^{2a}$ at the half lengths of some window statements, and the half length $\frac{1}{2}\log(H/2\pi) = 13.45$ at which it equals the height $H = 3 \times 10^{12}$ of the verification of the zeros. The conversion is a heuristic and not a theorem.

horizon at the heights 1000 and 2000 (Table 18.3), and if the margin below the horizon keeps growing beyond the range tested, a window of half length near 13 could see a large displacement far above 3×10^{12}. Table 18.4 lists the heights $2\pi e^{2a}$ of some window statements.

Yoshida's window lies below the first ordinate: its height $4\pi = 12.57$ is less than $\gamma_1 = 14.13$. Zhu's certificate reaches about the fourth ordinate ($\gamma_4 = 30.42 < 31.12 < \gamma_5 = 32.94$). Under the heuristic, a statement about these windows speaks about the zeros below the height shown, which the verification of the zeros covers far beyond.

18.5 The comb barrier, basepoints and damping

Damping to a basepoint $\alpha \geq \frac{1}{2}$ (Section 17.8) replaces $\Lambda(n)n^{-1/2}$ by $\Lambda(n)n^{-\alpha}$ in the comb. Put

$$A_a(\alpha) = \sum_{\log n < 2a} \frac{2\Lambda(n)}{n^{\alpha}}, \qquad T_1(\alpha, a) = 2\pi e^{A_a(\alpha)}, \tag{18.5.1}$$

so that $A_a(\frac{1}{2}) = A_a$ and $T_1(\frac{1}{2}, a) = T_1$. For $\alpha > \frac{1}{2}$ the height $T_1(\alpha, a)$ is a definition by analogy and not the cutoff of a theorem: Zhu's Theorem 1.1 is a statement about the undamped form, no damped analogue is proved here, and for $\frac{1}{2} < \alpha < 1$ the pole part of the damped form, with kernel $2e^{-(\alpha-1/2)|y|}\cosh(y/2)$, grows like $e^{(1-\alpha)|y|}$ and is not a positive rank one term. By the prime number theorem, as $a \to \infty$,

$$\begin{gathered} A_a(\alpha) \sim \frac{2e^{2(1-\alpha)a}}{1-\alpha} \quad (\tfrac{1}{2} \leq \alpha < 1), \qquad A_a(1) = 4a - 2\gamma_E + o(1), \\ A_a(\alpha) \text{ increases to } -2\frac{\zeta'}{\zeta}(\alpha) \quad (\alpha > 1). \end{gathered} \tag{18.5.2}$$

Table 18.5 lists the values, computed by `zh_comb.py` (`code/README.md`).

	$\alpha=\frac12$		$\alpha=0.75$		$\alpha=1$	
a	$A_a(\alpha)$	$\log_{10}T_1$	$A_a(\alpha)$	$\log_{10}T_1$	$A_a(\alpha)$	$\log_{10}T_1$
0.35	0.980	1.22	0.824	1.16	0.693	1.10
0.80	2.942	2.08	2.278	1.79	1.772	1.57
1.60	14.323	7.02	8.497	4.49	5.222	3.07
2.00	24.383	11.39	12.566	6.26	6.874	3.78
3.00	75.190	33.45	26.647	12.37	10.856	5.51
4.00	213.548	93.54	49.930	22.48	14.855	7.25
5.00	588.024	256.17	88.183	39.10	18.844	8.98

Table 18.5: The comb constant $A_a(\alpha)$ of (18.5.1) and the barrier height $T_1(\alpha,a)=2\pi e^{A_a(\alpha)}$ at three basepoints. The column $\alpha=\frac12$ is Zhu's [350], who prints $A_{0.8}(\frac12)=2.9420$ and $T_1=119$ at $a=0.8$, 1.0×10^7 at $a=1.6$ and 2.4×10^{11} at $a=2$; the digits here are 2.94197, 119, 1.04×10^7 and 2.44×10^{11}. For $\alpha>1$ the constant is bounded by $-2\zeta'(\alpha)/\zeta(\alpha)=18.88$, 3.01, 1.14 at $\alpha=1.1$, 1.5, 2.

Corollary 18.5.1. *Let $\alpha\ge1$, $a>0$ and $f\in C(a)$, and write $F=\widehat{f}$ and $b_\alpha=\xi'(\alpha)/\xi(\alpha)$. With no hypothesis on the zeros,*

$$\langle f,f\rangle_\alpha=\frac{1}{\pi}\int_{\mathbb{R}}|F(t)|^2\,\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt\;\ge\;2b_\alpha\|f\|^2,\qquad b_\alpha=\inf_{t\in\mathbb{R}}\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)>0,$$

and the constant $2b_\alpha$ does not depend on a.

First proof, $\alpha\ge1$. Let $\sigma_{\max}=\sup_\rho\operatorname{Re}\rho$ be the largest real part of a zero (Section 17.8.1). Then $\sigma_{\max}\le1\le\alpha$ and no zero has real part α, so Proposition 17.8.1 gives the identity, and Theorem 17.8.5(i) gives the inequality with $m_\alpha=\inf_t\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$. By part (ii) of that theorem, which for $\alpha\ge1$ needs no hypothesis (Propositions 9.1.11 and 9.1.12), $m_\alpha=b_\alpha$, and $b_\alpha>0$ because the density $v_\alpha=\operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\cdot)$ is positive (Theorem 1.2.2). □

Second proof, $\alpha>1$. Let $g=f*\widetilde{f}$, which is supported in $[-2a,2a]$ and has $\widehat{g}=|F|^2$. By (17.1.1) with the damping $e^{-(\alpha-1/2)|y|}$, the form is the sum of a pole part, an archimedean part and the prime part $-\sum_{n\ge2}\Lambda(n)n^{-\alpha}\big(g(\log n)+g(-\log n)\big)$. Since $\int_{\mathbb{R}}|F(t)|^2\cos(t\log n)\,dt=\pi\big(g(\log n)+g(-\log n)\big)$, which vanishes for $\log n>2a$, and since for $\alpha>1$ the full comb $\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)$ converges absolutely and uniformly in t, the prime part equals $-\frac{1}{2\pi}\int|F|^2\sum_{n\ge2}2\Lambda(n)n^{-\alpha}\cos(t\log n)\,dt$ with the full comb, inserted term by term. The pole part is $\int g(y)\big(e^{-(\alpha-1)|y|}+e^{-\alpha|y|}\big)dy$, which is $\frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\big(\frac1s+\frac{1}{s-1}\big)dt$ at $s=\alpha+it$, because $e^{-w|y|}$ has Fourier transform $2w/(w^2+t^2)$. The archimedean part is the Poisson smoothing by $P_{\alpha-1/2}$ of the multiplier $\operatorname{Re}\psi(\frac14+\frac{it}{2})-\log\pi$, and this is $\operatorname{Re}\psi(\frac s2)-\log\pi$ at $s=\alpha+it$, because $z\mapsto\psi(\frac14+\frac{iz}{2})$ is holomorphic for $\operatorname{Im}z<\frac12$.

Now $\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)$ and $-\frac{\zeta'}{\zeta}(s) = \sum\Lambda(n)n^{-s}$ for $\operatorname{Re} s > 1$, so the three multipliers add up to $2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$, and $\langle f,f\rangle_\alpha = \frac{1}{2\pi}\int|F|^2\,2\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)\,dt$. This symbol is positive (Theorem 1.2.2) and tends to infinity like $\frac{1}{2}\log|t|$, because $\operatorname{Re}\frac{\zeta'}{\zeta}$ is bounded on the line $\operatorname{Re} s=\alpha$; its infimum is therefore positive and equals $2b_\alpha$ by Theorem 17.8.5(ii), and Plancherel, $\int|F|^2 = 2\pi\|f\|^2$, gives the bound. □

For $\alpha \geq 1$ no certificate is needed: the damped form is positive on every window (Theorem 17.8.3(i)), although the envelope height $T_1(1,a)$ still grows like e^{4a} (Table 18.5). The line $\operatorname{Re} s = 1$ is the same boundary as in the basepoint picture: the shifted clock $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is a GGC with Thorin density $v_\alpha > 0$ for every $\alpha \geq 1$ with no hypothesis, and for $\alpha < 1$ exactly when $\alpha \geq \sigma_{\max}$ (Theorem 1.2.2). The one sided bound on ζ'/ζ that corresponds to $v_\alpha > 0$ is the bound of Theorem 16.12.1(ii), which for $\alpha < 1$ holds under the hypothesis of that theorem.

For $\frac{1}{2} \leq \alpha < 1$ the full comb diverges, since $\sum\Lambda(n)n^{-\alpha}$ diverges, and the second proof is not available. The symbol $\operatorname{Re}\frac{\xi'}{\xi}(\alpha+it)$ still exists by analytic continuation, and the damped form is its multiplier whenever $\alpha \geq \sigma_{\max}$ and no zero has real part α (Proposition 17.8.1). Positivity of the symbol on this one line follows from the absence of zeros with real part at least α and is not known to imply it (Remark 1.2.4), whereas positivity of the damped form on every window is equivalent to the absence of zeros with real part greater than α (Theorem 20.1.2, Remark 17.8.4). What the comb would have to supply there is the one sided bound for $-\operatorname{Re}\zeta'/\zeta(\alpha+it)$ of Section 17.10, that is cancellation in $\sum\Lambda(n)n^{-\alpha-it}$ beyond what absolute convergence gives, which is Problem 17.1 of Chapter 17.

18.5.1 Damping removes the collapse

Write $\lambda_{\min}(\alpha,a)$ for the least eigenvalue of the damped form of Section 17.8 on the window $[-a,a]$, over complex test functions in $C(a)$ (Section 17.8.2), and $\lambda_{\min}^{\text{even}}(\alpha,a)$ for the infimum of $\langle f,f\rangle_\alpha/\|f\|^2$ over real even $f \in C(a)$. By the splitting of Section 18.1, $\lambda_{\min}(\alpha,a) \leq \lambda_{\min}^{\text{even}}(\alpha,a)$.

Let $p_1 = \frac{1}{2}(\log\xi)''(\frac{1}{2}) = 0.0231049931\ldots$, which is $\sum_{\gamma>0}\gamma^{-2}$ under RH, and $p_2 = 3.7172599\ldots\times10^{-5}$, which is $\sum_{\gamma>0}\gamma^{-4}$ under RH, as in Theorem 17.8.5.

Theorem 18.5.2 (Two parameter bound). *Let $a > 0$ and $\alpha > \frac{1}{2}$, and write $b_\alpha = \xi'(\alpha)/\xi(\alpha)$.*

(i) With no hypothesis, $\lambda_{\min}(\alpha,a) \geq \lambda_{\min}(\frac{1}{2},a)$.

(ii) If RH holds, or if $\alpha \geq 1$, then $\lambda_{\min}(\alpha,a) \geq 2b_\alpha$. Here $2b_\alpha \leq 4p_1(\alpha-\frac{1}{2})$ with no hypothesis, and under RH $2b_\alpha \geq 4p_1(\alpha-\frac{1}{2}) - 4p_2(\alpha-\frac{1}{2})^3$.

Hence, under RH, $\lambda_{\min}(\alpha,a) \geq \max\bigl(\lambda_{\min}(\tfrac12,a),\, 2b_\alpha\bigr)$ *for every* $\alpha > \tfrac12$ *and every* $a > 0$.

Proof. (i) Put $\varepsilon = \alpha - \tfrac12$ and recall $e^{-\varepsilon|y|} = \int_{\mathbb{R}} P_\varepsilon(\lambda) e^{i\lambda y} d\lambda$ with the Poisson kernel $P_\varepsilon(\lambda) = \varepsilon/(\pi(\varepsilon^2+\lambda^2))$. For $f \in C(a)$ put $f_\lambda(y) = e^{i\lambda y} f(y) \in C(a)$, so that $f_\lambda * \widetilde{f_\lambda} = e^{i\lambda y}\,(f * \widetilde{f})$. The interchange of $\mathcal{Y}$ and the integral over λ is that of the proof of Theorem 17.8.3(iii), taken from the basepoint $\frac12$, and gives

$$\begin{aligned}\langle f,f\rangle_\alpha = \mathcal{Y}\bigl(e^{-\varepsilon|y|} f * \widetilde{f}\bigr) &= \int_{\mathbb{R}} P_\varepsilon(\lambda)\langle f_\lambda, f_\lambda\rangle_{1/2}\, d\lambda \\ &\geq \lambda_{\min}(\tfrac12,a)\int_{\mathbb{R}} P_\varepsilon(\lambda)\|f_\lambda\|^2 d\lambda = \lambda_{\min}(\tfrac12,a)\|f\|^2,\end{aligned}$$

because $\|f_\lambda\| = \|f\|$ and $\int P_\varepsilon = 1$.

(ii) Under RH, $\sigma_{\max} = \frac12$ and no zero has real part $\alpha > \frac12$; for $\alpha \geq 1$, $\sigma_{\max} \leq 1 \leq \alpha$ and no zero has real part α. In both cases Theorem 17.8.5(i) gives $\lambda_{\min}(\alpha,a) \geq 2m_\alpha$, and part (ii) of that theorem gives $m_\alpha = b_\alpha$, under RH and with no hypothesis for $\alpha \geq 1$. The bounds on $2b_\alpha$ are Theorem 17.8.5(iv). $\square$

At $a = 0.8$ the first bound is certified: Corollary 6.3 of [350] gives $\lambda_{\min}(\frac12, 0.8) \geq 8.9 \times 10^{-18}$ for complex test functions, so by (i) $\langle f,f\rangle_\alpha \geq 8.9\times10^{-18}\|f\|^2$ for every $\alpha \geq \frac12$ and every $f \in C(0.8)$, with no hypothesis. The second bound is not a multiple of $4p_1(\alpha-\frac12)$ but only comparable to it: $2b_\alpha$ lies between $4p_1(\alpha-\frac12) - 4p_2(\alpha-\frac12)^3$ and $4p_1(\alpha-\frac12)$, so under RH a bound $\lambda_{\min}(\alpha,a) \geq 4p_1(\alpha-\frac12)$ would fail for large a, where $\lambda_{\min}(\alpha,a) \to 2b_\alpha$ (Theorem 17.8.5(i) and (ii)). Unconditionally the second bound is available for $\alpha \geq 1$ only (Corollary 18.5.1); at a single $\alpha < 1$ with no zero on the line the inequality $\lambda_{\min}(\alpha,a) \geq 2b_\alpha$ for all a is equivalent to $\sigma_{\max} \leq \alpha$ together with $v_\alpha \geq b_\alpha$ on the line (Remark 17.8.6).

The theorem has a numerical counterpart for the windows of Table 18.1. The matrices of that table, with the damping, give upper bounds for $\lambda^{\mathrm{even}}_{\min}(\alpha,a)$ (computed at 60 digits, for 26 and 40 modes, `code/README.md`). In all 44 values, for $a = 0.5$ and 0.8 and $\alpha - \frac12$ between 10^{-17} and 10^{-3}, the finite section value lies above both bounds of the theorem: the least ratio to the floor $2b_\alpha$ is 1.030 at $a = 0.8$, $\alpha - \frac12 = 10^{-4}$ and 1.121 at $a = 0.5$, $\alpha - \frac12 = 10^{-3}$, and the least ratio to the undamped value of the same size is 1.011 at $a = 0.5$, $\alpha - \frac12 = 10^{-7}$ and 1.047 at $a = 0.8$, $\alpha - \frac12 = 10^{-17}$.

The damped value is the sum of the two terms and not their maximum. For $\alpha - \frac12 = \varepsilon$ the difference $\lambda^{\mathrm{even}}_{\min}(\frac12+\varepsilon,a) - \lambda^{\mathrm{even}}_{\min}(\frac12,a)$ equals $s \cdot 4p_1\varepsilon$ with $s = 1.111$ at $a = 0.5$ for ε from 10^{-7} to 10^{-4} and $s = 1.082$ at $a = 0.8$ for ε from 10^{-17} to 10^{-14}, the same at 26 and 40 modes to the digits quoted. For larger ε at $a = 0.8$ the factor s falls, to 1.067, 1.043, 1.040 and 1.030 at $\varepsilon = 10^{-10}$, 10^{-8}, 10^{-6} and 10^{-4}, so that from $\varepsilon = 10^{-8}$ on the finite section value lies within 4.5 percent of the floor $2b_\alpha$; the fall begins at ε of order 10^{-10}, where ε times the slope is of the size of the gap of about 10^{-11} to the second even eigenvalue.

Far above the crossover the least eigenvalue is close to the floor, as the converged entries of Table 17.2 show.

The two terms of the theorem are equal, to first order in the damping, at the crossover $\varepsilon_\times = \lambda^{\text{even}}_{\min}(\frac{1}{2}, a)/(4p_1)$, at which the least eigenvalue is about twice $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$ (the damping at which it exactly doubles is 0.90 times $\varepsilon_\times$ at $a = 0.5$ and 0.92 times $\varepsilon_\times$ at $a = 0.8$, for 26 modes). Its value depends on which number is used for $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$, and Table 18.6 gives the choices. The 26 mode values are not converged.

a	value used for $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$	λ	$\lambda/(4p_1)$
0.5	Zhu, certified upper bound	1.05×10^{-6}	1.14×10^{-5}
0.5	26 modes, Table 18.1	9.743×10^{-7}	1.05×10^{-5}
0.5	extrapolated from 256 to 400 modes	9.33×10^{-7}	1.01×10^{-5}
0.8	Zhu, certified lower bound	8.9×10^{-18}	9.6×10^{-17}
0.8	extrapolated from 256 to 400 modes	1.63×10^{-17}	1.77×10^{-16}
0.8	26 modes, Table 18.1	2.144×10^{-17}	2.32×10^{-16}
0.8	Zhu, certified upper bound	2.27×10^{-17}	2.46×10^{-16}

Table 18.6: The crossover $\varepsilon_\times = \lambda/(4p_1)$ of Theorem 18.5.2 for several values of $\lambda = \lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$, with $4p_1 = 0.0924200$. The extrapolated values are three point power law fits to the finite sections of 60 digit matrices (script `code/zh_window_eigs.py`, record `data/zh_window_eigs.json`), not bounds; Zhu's reference values of his Table 1, 9.34×10^{-7} and 1.65×10^{-17}, give 1.01×10^{-5} and 1.79×10^{-16}.

The explanation of the collapse is the one given at the end of Section 17.8. A damping of the size of the crossover already doubles the finite section values (computed); for larger damping the least eigenvalue is, under RH, at least $2b_\alpha$, a margin of order $\alpha - \frac{1}{2}$ in place of $\lambda^{\text{even}}_{\min}(\frac{1}{2}, a)$. Whether certificates for the damped forms say more or less about the zeros than the certificates at $\alpha = \frac{1}{2}$ is not settled here (Problem 18.3).

18.6 What window certificates reach

A window of half length a resolves the zeros up to the horizon $2\pi e^{2a}$ (a heuristic, Section 18.4, after Section 9 of [350]), and whether the verification of the zeros up to a height H implies positivity on the window of half length $\frac{1}{2}\log(H/2\pi) - C$ is Problem 18.1; for large displacements it may fail.

18.7 Open problems

Problem 18.1 (The horizon as a theorem). Determine the largest half length $a(H, \delta_0)$ such that, if all zeros with $0 < \gamma \le H$ lie on the line and every zero above H has $|\beta - \frac{1}{2}| \le \delta_0$, Weil's form is positive on the window $[-a, a]$ with $a = a(H, \delta_0)$.

Is $a(H,\delta_0) \geq \frac{1}{2}\log\frac{H}{2\pi} - C(\delta_0)$, with a constant $C(\delta_0)$ that depends only on δ_0? Status: open. The heuristic of Section 18.4 suggests the bound for small δ_0. A window gains up to a factor $e^{2a\delta}$ on a zero at distance δ from the line, and in Table 18.3 the largest displacement is detected below the horizon at the largest heights, so a constant C that does not depend on δ_0 may not exist. A proof would need a quantitative form of the lower sampling inequality of Proposition 18.2.1 for the ordinates, with constants that depend only on the local density of the zeros. Compare Proposition 17.9.1, which is proved for the tent coefficient only.

Problem 18.2 (Decay law from the Thorin side). Derive the exponent of Zhu's decay law, $-\log\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a) = C\,N(T_*)/\log N(T_*)\,(1+o(1))$ with $T_* = 2\pi e^{2a}$ and the fitted constant $C = 20.13\ldots$ of his equation (5) (his Conjecture 12.1 [350] states the law with $C = 2\pi^2$), from Zhu's test function $\widehat{f} = h\,G_T(-r^2)$ of Section 18.3, optimising T and h jointly, and explain the constant, which Zhu finds close to $2\pi^2 = 19.74\ldots$, through the Landau Widom plunge. Status: open. Zhu fits the law to certified upper bounds for $a \leq 1.2$, tests it on $a = 1.4$ to 2, and states that the constant is fitted and not derived. The only proved decay beyond the bound $O(a^{-N})$ for every N of Theorem 20.2.1(c) of Chapter 20 is his bound $\lambda_{\min}^{\mathrm{even}}(\frac{1}{2},a) \leq \exp(-ae^a)$ for large a, under RH (Theorem 1.3), which kills the zeros only up to $2\pi e^a$, half the exponent of T_*, and is weaker than the law.

Problem 18.3 (Certified damped windows). Extend the certification of Zhu's Theorem 1.2 to the damped forms at $\alpha \in (\frac{1}{2},1)$, where under RH the margin is at least $2\xi'(\alpha)/\xi(\alpha)$, of order $\alpha - \frac{1}{2}$, in place of $\lambda_{\min}(\frac{1}{2},a)$ (Theorem 18.5.2), and measure how far a can be pushed. Status: open. No damped analogue of Zhu's Theorem 1.1 is proved here, and below $\alpha = 1$ the damped pole part is not a positive rank one term (Section 18.5); for $\alpha \geq 1$ no certificate is needed (Corollary 18.5.1). The question for the rungs of Chapter 20 is Problem 20.1.

Problem 18.4 (Comb cancellation on average). Zhu's barrier concerns the supremum of the comb $\mathcal{P}_a(t) = \sum_{\log n<2a} 2\Lambda(n)n^{-1/2}\cos(t\log n)$. The form sees the comb only through $\int |F|^2\mathcal{P}_a$ with F in the Paley Wiener space PW_a of Proposition 18.2.1, an average over a frequency band of width about $1/a$. Determine the supremum of $\int_{|t|\geq T}|F|^2\mathcal{P}_a \big/ \int_{|t|\geq T}|F|^2$ over $F \in PW_a$, as a function of T and a. Status: open. Zhu's Lemma 3.2 shows that the pointwise supremum of $\mathcal{P}_a$ is A_a, which bounds the ratio from above. The average over the whole line is smaller than A_a for every F, because $g = f * \widetilde{f}$ vanishes at $\pm 2a$ and damps the terms with $\log n$ near $2a$; how much smaller the restricted ratio is, uniformly in T, is not known, and a bound of smaller order than $A_a = (4+o(1))e^a$ would be a statement about short sums of $\Lambda(n)n^{-1/2-it}$, the object of the Fejér tapered sums of Proposition 14.14.1.

Notes

The theorems of Zhu quoted in this chapter are from [350] (version 2 of 2 September 2026), which the chapter reads in the language of the book and does not reprove; they and his heuristics (the horizon and the plunge rate of his Section 9, the parity of the ground state) are credited where they are stated. Yoshida's form is [349], Weil's criterion is [338, 38], and the verification of the zeros to 3×10^{12} is [247]. The computations of the chapter are its own: the table of least eigenvalues in 60 digit arithmetic (Table 18.1), the optimisation of Zhu's product test function (Section 18.3), the comb table (Table 18.5), the detection experiment (Table 18.3 and Figure 18.1) and the damping numbers (Table 18.6). `code/README.md` lists the scripts.

Chapter 19

Three families of generalised gamma convolutions and the deconvolution obstruction

Three one parameter families of generalised gamma convolutions (GGCs) contain the clock of Riemann's ξ as a boundary case: the Esscher tilt, the basepoint family, which smooths the zero counting measure by Cauchy kernels, and the de Bruijn Newman heat flow; in each the Thorin property propagates away from the centre, and RH asks for it at the centre.

What is not claimed. Nothing here proves RH, and the word obstruction in the title names what the theorems on the three families have in common, not a theorem of its own. A number of the chapter's own is computed in floating point at the precision stated. Where a sentence reads a theorem or an identity as saying what a route needs, it is a heuristic and is marked as one.

19.1 Three families, and the tilt

Let ξ be Riemann's function, with zeros $\rho = \beta + i\gamma$, and $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$. The centre clock is the variable $T \geq 0$ with

$$\mathbb{E}\big[e^{-uT}\big] = \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})}, \qquad u \geq 0,$$

which exists with no hypothesis (Theorem 5.1.1) and is a GGC exactly under RH, with Thorin measure $\sum_{\gamma>0} \delta_{\gamma^2}$ (Proposition 1.2.1). Heights are written h in this chapter. A basepoint $\alpha = \frac{1}{2} + \varepsilon$ is also called the level ε, the resolution of Section 1.2. Put

$$\sigma_{\max} = \sup_{\rho} \operatorname{Re} \rho \in [\tfrac{1}{2}, 1],$$

so that RH is $\sigma_{\max} = \frac{1}{2}$. Since $\beta < 1$ for every zero, every level $\varepsilon \geq \frac{1}{2}$ lies strictly to the right of every zero, $\varepsilon > \beta - \frac{1}{2}$, whether or not $\sigma_{\max} < 1$; the latter is the

quasi Riemann hypothesis, and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [226] (Section 12.1.2), not yet refereed, would give $\sigma_{\max} \le \frac{7}{8}$.

The book meets the clock inside three one parameter families, each containing it as the boundary case $p = 0$:

(A) the *tilt* $u_0 \ge 0$, the Esscher transform $\xi(\frac{1}{2} + \sqrt{u_0})/\xi(\frac{1}{2} + \sqrt{u_0 + u})$ (Theorem 9.2.2 and Corollary 9.2.3 at $u_0 = \frac{1}{4}$);

(B) the *basepoint* $\alpha = \frac{1}{2} + \varepsilon$, the shifted clock $\xi(\alpha)/\xi(\alpha + \sqrt{u})$, whose Thorin density is a sum of Cauchy kernels over the zeros (Theorem 1.2.2 and Proposition 1.2.3);

(C) the *heat time* $t \ge 0$, the clock of the de Bruijn Newman function H_t (Section 8.4.1 and Theorem 7.2.15).

This chapter adds the converse half of the Poisson semigroup and the deconvolution form of RH (Theorems 19.2.1(b) and 19.2.2), the measured exponents of Table 19.1 for the smoothed prime sums, and, in Section 19.3, the visibility curve at every level with its closed forms in three lattice models and its level count.

Proposition 19.1.1 (Invertible families do not localise the difficulty). *Let $\{S_p\}_{p \ge 0}$ be maps on a set $\mathcal{D}$ of Laplace transforms, with S_0 the identity. If each S_p is injective on $\mathcal{D}$, maps GGC Laplace transforms in $\mathcal{D}$ to GGC Laplace transforms, and has an inverse on $S_p(\mathcal{D})$ that does the same, then for every $\varphi \in \mathcal{D}$ and every p, $S_p\varphi$ is a GGC Laplace transform exactly when φ is.*

Proof. If φ is a GGC transform so is $S_p\varphi$, and if $S_p\varphi$ is one so is $S_p^{-1}S_p\varphi = \varphi$. □

In such a family the Thorin property is equally hard at every parameter: knowing it at some $p > 0$ would already prove RH. The tilt is a family of this kind, and no unconditional point is known in it. The basepoint and heat families are not of this kind: the property is known at $\varepsilon \ge \frac{1}{2}$ and at $t \ge 0.2$, and their inverses do not preserve it. The two fail in opposite ways. The inverse of the Cauchy smoothing is an unbounded operator, with Fourier multiplier $e^{a|x|}$, so the step towards RH is an ill posed computation. The inverse of the heat flow is bounded, a Gaussian average, so the step towards RH is a stable computation; but by Corollary 19.4.2 a property that Gaussian averaging preserves and that implies real zeros holds for no H_t.

19.1.1 The tilt: invertible and uninformative

Write $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.2, an entire function of u, and for $u_0 \ge 0$

$$\widetilde{\varphi}_{u_0}(u) = \frac{G(u_0)}{G(u_0 + u)}, \qquad u \ge 0,$$

the Laplace transform of the clock tilted by $e^{-u_0 T}$, that is of the variable $\widetilde{T}_{u_0}$ with law $e^{-u_0 t}P(T \in dt)/\mathbb{E}e^{-u_0 T}$ (the $\widetilde{T}$ of Theorem 9.2.2 for $u_0 = \frac{1}{4}$). Each pair $\{\rho, 1-\rho\}$ of zeros gives the number $a_\rho = -(\rho - \frac{1}{2})^2 = \gamma_\rho^2 = \gamma^2 - (\beta - \frac{1}{2})^2 - 2i\gamma(\beta - \frac{1}{2})$, the a_k of (1.2.1), where $\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ is the complex ordinate of ρ, and Hadamard's product (1.2.1) gives $G(u) = \prod(1 + u/a_\rho)$, absolutely convergent since $\sum |a_\rho|^{-1} < \infty$. On the line $a_\rho = \gamma^2$.

THEOREM 19.1.2 (The tilt family). *(a) For $u_1 \geq 0$,*

$$\widetilde{\varphi}_{u_0+u_1}(u) = \frac{\widetilde{\varphi}_{u_0}(u_1 + u)}{\widetilde{\varphi}_{u_0}(u_1)},$$

so the forward tilt is the Esscher transform by $e^{-u_1 \widetilde{T}_{u_0}}$, and it maps GGCs to GGCs, moving the Thorin measure up by u_1. Backward, if $\widetilde{\varphi}_{u_0+u_1}$ is a GGC transform with Thorin measure in (u_1, ∞), then

$$\widetilde{\varphi}_{u_0}(u) = \frac{\widetilde{\varphi}_{u_0+u_1}(u - u_1)}{\widetilde{\varphi}_{u_0+u_1}(-u_1)}$$

is a GGC transform, with the Thorin measure moved down by u_1; the denominator is $G(u_0 + u_1)/G(u_0)$, finite and positive.

(b) $\widetilde{\varphi}_{u_0}$ is the Laplace transform of a GGC exactly when every $u_0 + a_\rho$ is real and positive. In particular $\mathrm{Im}(u_0 + a_\rho) = -2\gamma(\beta - \frac{1}{2})$ *does not depend on u_0.*

(c) For ξ, the Thorin property at one tilt is equivalent to the property at every tilt and to RH.

Proof. (a) The identity is the definition. If $\varphi(u) = \exp(-du - \int \log(1 + u/z)\, U(dz))$, then $\varphi(u_1 + u)/\varphi(u_1) = \exp(-du - \int \log(1 + u/(z + u_1))\, U(dz))$, a GGC with the same drift and the Thorin measure moved up by u_1. Backward, if U lies in (u_1, ∞) and $\varphi(-u_1)$ is finite, as it is here, then $\varphi(u - u_1)/\varphi(-u_1) = \exp(-du - \int \log(1 + u/(z - u_1))\, U(dz))$, a GGC with the measure moved down. (b) If every $u_0 + a_\rho$ is real and positive, $\widetilde{\varphi}_{u_0}(u) = \prod(u_0 + a_\rho)/(u_0 + a_\rho + u)$ is the transform of $\sum_\rho e_\rho/(u_0 + a_\rho)$ with independent unit exponentials e_ρ, a GGC. Conversely a GGC transform extends to an analytic zero free function on $\mathbb{C} \setminus (-\infty, 0]$, while $\widetilde{\varphi}_{u_0}$ has poles at $u = -(u_0 + a_\rho)$, so each of these lies in $(-\infty, 0)$. (c) The zeta function has no real zeros in $(0, 1)$, so $\gamma \neq 0$ for every zero, and $\xi(\frac{1}{2}) \neq 0$. By (b) the property at any u_0 says that every a_ρ is real, that is $\beta = \frac{1}{2}$; conversely, under RH every $a_\rho = \gamma^2 > 0$, and (b) gives the property at every $u_0 \geq 0$. □

Theorem 21.2.3 states the equivalence for the Dedekind zeta function of every number field and every $u_0 \geq 0$, and Theorem 19.1.2 is its case of $\mathbb{Q}$.

In the chart $\theta = \sqrt{z}$ of the Thorin measure the tilt is the deformation $\theta \mapsto \sqrt{\theta^2 + u_0}$ of the atoms, not a smoothing. What the tilt can move is real zeros: for

the Dedekind zeta function ζ_K of a quadratic field K with $\zeta_K(\frac12)\neq 0$, a real zero $\beta\in(\frac12,1)$ of ζ_K, which is a real zero of $L(s,\chi)$ for the quadratic character χ of K, gives a real $a_\rho=-(\beta-\frac12)^2<0$, admissible exactly when $u_0>(\beta-\frac12)^2$.

19.2 The basepoint family: a Poisson semigroup to the right of the zeros

For real ε and θ such that $\frac12+\varepsilon+i\theta$ is not a zero of ξ, let

$$\rho_\varepsilon(\theta)=\frac{1}{\pi}\,\mathrm{Re}\,\frac{\xi'}{\xi}\big(\tfrac12+\varepsilon+i\theta\big)$$

be the density of (1.2.7), now for every real ε and not only for $\varepsilon>0$, and let $P_a(x)=a/(\pi(a^2+x^2))$ for real $a\neq 0$ be the Poisson kernel of Section 1.2, with $P_a*P_b=P_{a+b}$ for $a,b>0$. For $\varepsilon>0$ the shifted clock at the basepoint $\frac12+\varepsilon$ has Thorin density ρ_ε in the chart $\theta=\sqrt{z}$ whenever it is a GGC and no zero lies on the line $\mathrm{Re}\,s=\frac12+\varepsilon$ (Theorem 1.2.2). The function $(\theta,\varepsilon)\mapsto\rho_\varepsilon(\theta)$ is harmonic off the zeros, odd in ε by the functional equation, and Hadamard's product gives the absolutely convergent expansion

$$\rho_\varepsilon(\theta)=\sum_\rho P_{\varepsilon-(\beta-\frac12)}(\theta-\gamma),\qquad \varepsilon\neq\beta-\tfrac12, \tag{19.2.1}$$

over all zeros with multiplicity (Proposition 1.2.3). A zero with $\beta-\frac12<\varepsilon$ contributes a positive Cauchy density and one with $\beta-\frac12>\varepsilon$ a negative one.

Theorem 19.2.1 (Where the Poisson semigroup holds). *(a) If $\varepsilon_0>\beta-\frac12$ for every zero, then $\rho_{\varepsilon_0+a}=P_a*\rho_{\varepsilon_0}$ for every $a>0$, and $\rho_\varepsilon>0$ for every $\varepsilon\geq\varepsilon_0$. This holds for every $\varepsilon_0\geq\frac12$, with no hypothesis.*

(b) If $\varepsilon_0\geq 0$ lies at positive distance from the set $\{\beta-\frac12\}$ and some zero has $\beta-\frac12>\varepsilon_0$, then $\rho_{\varepsilon_0+a}\neq P_a*\rho_{\varepsilon_0}$ for some $a>0$.

Proof. (a) Every term of (19.2.1) at ε_0 is a Cauchy density $P_c(\theta-\gamma)$ with $c=\varepsilon_0-(\beta-\frac12)>0$, and $P_a*P_c=P_{a+c}$. The terms are positive, so the convolution passes through the sum by monotone convergence. For $\varepsilon_0\geq\frac12$, $c\geq 1-\beta>0$. (b) Now $|c|\geq c_0>0$ for every term, and $|c|<1$ because $0\leq\varepsilon_0<\frac12$, so $|P_c(\theta-\gamma)|\leq\min\big((\pi c_0)^{-1},(\pi(\theta-\gamma)^2)^{-1}\big)$, and by the Riemann von Mangoldt formula the sum is $O(\log(2+|\theta|))$. Hence $w(\theta,a)=(P_a*\rho_{\varepsilon_0})(\theta)$ is defined and harmonic in $a>0$. If $w(\cdot,a)=\rho_{\varepsilon_0+a}$ for every $a>0$, then off the zeros $\rho_\varepsilon(\theta)$ would agree with a function harmonic on the whole half plane $\{\varepsilon>\varepsilon_0\}$, and in particular bounded near every point $(\gamma,\beta-\frac12)$ with $\beta-\frac12>\varepsilon_0$. But near a zero, $\xi'/\xi(s)=m_\rho/(s-\rho)+O(1)$, so $\rho_\varepsilon(\theta)$ is unbounded there. □

Part (a) is the semigroup of Remark 15.3.4. Part (b) is its converse (a rightmost zero need not exist).

THEOREM 19.2.2 (Deconvolution form of RH). *Let $\varepsilon_0 > \beta - \frac{1}{2}$ for every zero (for instance $\varepsilon_0 \geq \frac{1}{2}$), and let $0 \leq \varepsilon_1 < \varepsilon_0$.*

(a) There is a positive measure ν on $\mathbb{R}$ with $\int (1+x^2)^{-1}\nu(dx) < \infty$ and $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon_1} * \nu$ if and only if $\beta - \frac{1}{2} \leq \varepsilon_1$ for every zero. When it exists, ν is unique: it is $\rho_{\varepsilon_1}\, d\theta$ if $\beta - \frac{1}{2} < \varepsilon_1$ for every zero, and otherwise the vague limit of $\rho_\varepsilon\, d\theta$ as $\varepsilon \downarrow \varepsilon_1$, which carries point masses $m_\rho \delta_\gamma$ at the zeros with $\beta - \frac{1}{2} = \varepsilon_1$.

(b) In particular RH holds if and only if $\rho_{\varepsilon_0} = P_{\varepsilon_0} * \mu$ for a positive measure μ (with $\int (1+x^2)^{-1}\mu(dx) < \infty$, which is automatic since $P_{\varepsilon_0} * \mu = \rho_{\varepsilon_0}$ is finite), and then $\mu = \sum_\gamma \delta_\gamma$, the sum over the ordinates of all zeros, of both signs, with multiplicity.

Proof. If every zero has $\beta - \frac{1}{2} < \varepsilon_1$, Theorem 19.2.1(a) gives $\nu = \rho_{\varepsilon_1}\, d\theta$. If every zero has $\beta - \frac{1}{2} \leq \varepsilon_1$ with equality for some, write $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon} * \rho_\varepsilon$ for $\varepsilon_1 < \varepsilon < \varepsilon_0$ and let $\varepsilon \downarrow \varepsilon_1$: by (19.2.1) the terms of the zeros with $\beta - \frac{1}{2} < \varepsilon_1$ converge to Cauchy densities and those with $\beta - \frac{1}{2} = \varepsilon_1$ to point masses at their ordinates, and the sum over the zeros passes to the limit by the domination of the discussion after Remark 1.2.4, so $\rho_\varepsilon\, d\theta$ converges vaguely to a positive measure ν, the sum of these limits. Then $P_{\varepsilon_0-\varepsilon_1} * \nu = \rho_{\varepsilon_0}$ term by term, because $P_{\varepsilon_0-\varepsilon_1} * P_{\varepsilon_1-(\beta-\frac{1}{2})} = P_{\varepsilon_0-(\beta-\frac{1}{2})}$ and $P_{\varepsilon_0-\varepsilon_1} * \delta_\gamma = P_{\varepsilon_0-(\beta-\frac{1}{2})}(\cdot - \gamma)$ when $\beta - \frac{1}{2} = \varepsilon_1$, all terms being positive; in particular $\int (1+x^2)^{-1}\nu(dx) < \infty$. Conversely, suppose ν exists and some zero has $\beta - \frac{1}{2} > \varepsilon_1$. The Poisson integral $u(\theta,\varepsilon) = (P_{\varepsilon-\varepsilon_1} * \nu)(\theta)$ is harmonic on $\{\varepsilon > \varepsilon_1\}$, and for $\varepsilon > \varepsilon_0$ it equals $P_{\varepsilon-\varepsilon_0} * \rho_{\varepsilon_0} = \rho_\varepsilon(\theta)$ by Theorem 19.2.1(a). The set $\{\varepsilon > \varepsilon_1\}$ minus the zeros is connected and $u(\theta,\varepsilon) - \rho_\varepsilon(\theta)$ is harmonic on it and vanishes on the open set $\{\varepsilon > \varepsilon_0\}$, so $u = \rho_\varepsilon$ there. Then $\rho_\varepsilon(\theta)$ is bounded near a zero with $\beta - \frac{1}{2} > \varepsilon_1$, which contradicts the pole of ξ'/ξ. For uniqueness, let ν and ν' both qualify and put $\sigma = \nu - \nu'$, a signed measure with $\int (1+x^2)^{-1}|\sigma|(dx) < \infty$. For $\varepsilon > \varepsilon_0$ the Poisson integrals of ν and ν' at height $\varepsilon - \varepsilon_1$ both equal $P_{\varepsilon-\varepsilon_0} * \rho_{\varepsilon_0}$, so the harmonic function $(P_{\varepsilon-\varepsilon_1} * \sigma)(\theta)$ on $\{\varepsilon > \varepsilon_1\}$ vanishes for $\varepsilon > \varepsilon_0$, hence everywhere by real analyticity, and letting $\varepsilon \downarrow \varepsilon_1$ gives $\sigma = 0$, since the Poisson integral of a signed measure σ with $\int (1+x^2)^{-1}|\sigma|(dx) < \infty$ converges vaguely to σ. Part (b) is the case $\varepsilon_1 = 0$: $\beta - \frac{1}{2} \leq 0$ for every zero means $\beta = \frac{1}{2}$ for every zero by the symmetry $\rho \mapsto 1 - \rho$, and (19.2.1) at $\beta = \frac{1}{2}$ gives μ. □

The *if* direction of (b), and the atoms at a boundary line in (a), are Theorem 1.2.2, Proposition 1.2.3 and Theorem 15.2.3; what this chapter adds is the *only if* direction and the uniqueness. Theorem 19.2.2 turns the obstruction of Section 11.4.7 into an exact statement: $\sigma_{\max} - \frac{1}{2}$ is the depth to which the Thorin density at a basepoint beyond the zeros can be deconvolved into a positive measure. Two remarks qualify it. First, positivity of ρ_{ε_1} is necessary for deconvolvability to ε_1 but not sufficient: below $\sigma_{\max} - \frac{1}{2}$ the function ρ_{ε_1} may well be positive (Remark 1.2.4 and Section 19.3), and still $P_{\varepsilon_0-\varepsilon_1} * \rho_{\varepsilon_1} \neq \rho_{\varepsilon_0}$. Second,

the deconvolution is unbounded. On the Fourier side $\widehat{P_a}(x) = e^{-a|x|}$, and in the explicit formula the frequency $x = \log n$ carries the weight $\Lambda(n)n^{-\alpha}$ at basepoint α. Deconvolving by P_a multiplies that weight by n^a, so the prime sums converge absolutely only for $\alpha > 1$, and below the basepoint one they need cancellation, which is a zero free region (Proposition 9.1.13).

PROPOSITION 19.2.3 (The smoothed prime sums at a basepoint). *Let χ be a primitive nonprincipal character, $N > 0$, and $s = \alpha + ib$ such that $s - k$ is not a zero of $L(s,\chi)$ for any integer $k \geq 0$. Put*

$$E(N,s) = \sum_{n\geq 1} \chi(n)\Lambda(n)n^{-s}e^{-n/N} + \frac{L'}{L}(s,\chi),$$

the difference between the smoothed prime sum and $-L'/L(s,\chi)$. Then

$$\begin{aligned} E(N,s) = & -\sum_{\rho} m_\rho\, \Gamma(\rho - s)N^{\rho - s} \\ & + \sum_{k\geq 1} \frac{(-1)^k}{k!}\Big(-\frac{L'}{L}\Big)(s-k,\chi)N^{-k}, \end{aligned} \tag{19.2.2}$$

where ρ runs over all zeros of $L(s,\chi)$, trivial ones included, with multiplicity m_ρ. Hence, for fixed s, $E(N,s) = O(N^{\sigma_{\max}(\chi)-\alpha}) + O(N^{-1})$ as $N \to \infty$, where $\sigma_{\max}(\chi)$ is the supremum of β over the nontrivial zeros $\beta + i\gamma$ of $L(s,\chi)$. Under GRH the sum over the nontrivial zeros is $N^{1/2-\alpha}F_s(\log N)$, where $F_s(x) = -\sum m_\rho \Gamma(\rho - s)e^{i(\gamma-b)x}$, summed over the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$, is an almost periodic function none of whose coefficients vanishes; so $F_s(\log N)$ does not tend to zero, and for $\alpha < \frac{3}{2}$ the error $E(N,s)$ is not $o(N^{1/2-\alpha})$.

Proof. This is the explicit formula for the smoothed prime sum (compare [71, Chapter 19]); the derivation is given for the order of the error. Insert $e^{-x} = \frac{1}{2\pi i}\int_{(c)} \Gamma(w)x^{-w}dw$ with $c > \max(0, 1-\alpha)$ and sum: the smoothed prime sum is $\frac{1}{2\pi i}\int_{(c)}(-L'/L)(s+w,\chi)\Gamma(w)N^w\,dw$. Move the contour to $\operatorname{Re} w = -R$ with $R \to \infty$ along a sequence avoiding the poles; $\Gamma(w)$ decays like $e^{-\pi|\operatorname{Im} w|/2}$ and L'/L grows at most polynomially on suitable horizontal segments and vertical lines [71, Chapters 16 and 19], so the integrals vanish in the limit. The poles are simple by the hypothesis on s. The residues are $(-L'/L)(s,\chi)$ at $w = 0$, which is moved to the left side, $-m_\rho\Gamma(\rho - s)N^{\rho-s}$ at $w = \rho - s$, since $-L'/L$ has residue $-m_\rho$ at ρ, and $(-1)^k(k!)^{-1}(-L'/L)(s-k,\chi)N^{-k}$ at $w = -k$. The sum over ρ converges absolutely and is dominated by the zeros with $|\gamma - b|$ small, because $|\Gamma(\rho - s)|$ decays exponentially in $|\gamma - b|$; the nontrivial zeros contribute $O(N^{\sigma_{\max}(\chi)-\alpha})$, the trivial zeros $O(N^{-\alpha})$, and $O(N^{-1-\alpha})$ for odd χ, and the last sum is $O(N^{-1})$. Under GRH each nontrivial term is $|\Gamma(\rho - s)|N^{1/2-\alpha}$ times a phase, Γ has no zeros, and the series defining F_s converges absolutely, so F_s is almost periodic in

the sense of Bohr with the nonzero coefficients $-m_\rho\Gamma(\rho-s)$, and its mean square is the sum of the squares of their moduli, which is positive. □

So recovering the Thorin density at basepoint α to accuracy η from the primes with this smoothing needs primes up to about $N \approx \eta^{-1/(\alpha-\sigma_{\max}(\chi))}$, which is unbounded as $\alpha \downarrow \sigma_{\max}(\chi)$. This is a heuristic reading of the proposition as the deconvolution of Theorem 19.2.2 paid for in primes: the identity is for the weight $e^{-n/N}$, and it does not bound every method that uses the primes up to N. Proposition 9.1.13 gives the same exponent for the compensated prime sums of ξ (the book writes Θ for $\sigma_{\max}$ there), and Section 11.3.2 treats $L(s,\chi_{12})$, where no pole needs compensating. Table 19.1 shows the exponent for χ_4, for which GRH is verified to height 2.5×10^7 [245]: the measured exponent matches $\frac12-\alpha$ to within 0.006 for $0.6\le\alpha\le1.2$, while at $\alpha=1.5$ the $O(N^{-1})$ term is of the same size as the zero sum, between 1.1 and 2.3 times it.

α	$N=10^3$	10^4	10^5	2.5×10^5	slope	$\frac12-\alpha$
1.5	3.83×10^{-3}	3.97×10^{-4}	4.72×10^{-5}	2.00×10^{-5}	−0.949	−1.00
1.2	1.56×10^{-2}	3.03×10^{-3}	6.24×10^{-4}	3.36×10^{-4}	−0.695	−0.70
1.0	5.86×10^{-2}	1.83×10^{-2}	5.73×10^{-3}	3.78×10^{-3}	−0.499	−0.50
0.9	1.21×10^{-1}	4.78×10^{-2}	1.87×10^{-2}	1.35×10^{-2}	−0.400	−0.40
0.8	2.60×10^{-1}	1.29×10^{-1}	6.36×10^{-2}	4.95×10^{-2}	−0.302	−0.30
0.7	5.66×10^{-1}	3.57×10^{-1}	2.22×10^{-1}	1.86×10^{-1}	−0.203	−0.20
0.6	1.23×10^{0}	9.78×10^{-1}	7.69×10^{-1}	6.93×10^{-1}	−0.103	−0.10

Table 19.1: Mean modulus of $E(N,s)$ of (19.2.2) for χ_4 over $s=\alpha+10i$ and $s=\alpha+25i$, with prime powers summed up to $40N$, and the slope of $\log_{10}$ of the mean against $\log_{10}N$ fitted by least squares on the four values of N. Four values of N and two values of b illustrate the GRH statement of the proposition and do not verify it. Script `tf_primes.py`.

The identity (19.2.2) itself was checked in the 56 cases of the table: $E(N,s)$ and the right side, built from the zeros of $L(s,\chi_4)$ of height below 62, the trivial zeros $-1,-3,\dots,-15$ and the terms $m\le9$, agree to 5.4×10^{-14} in absolute value (script `tf_primes.py`).

The book's route through basepoints, positivity at every height for $\alpha\ge1$ (Section 15.1.2), stops at $\alpha=1$, the edge of the known zero free half plane; the cost exponent and the zero free half plane are two faces of one fact (Proposition 9.1.13), not cause and effect.

19.3 What a single basepoint sees

Theorem 19.2.2, like Lagarias's criterion [160] (Remark 1.2.4), needs every level. A test at one level ε sees less, and Section 5.2.2, with its Table 5.5 and Exercise 5.14, sets out what it sees in a lattice model. Two facts frame it. A zero at distance

$\delta = \beta - \frac{1}{2} > 0$ from the line contributes positively at every level $\varepsilon > \delta$, so those levels cannot see it. As $\varepsilon \uparrow \delta$ its contribution at $\theta = \gamma$ is $-1/(\pi(\delta - \varepsilon)) \to -\infty$, so the levels just below δ always see it. Near the line the test reduces to the sign of Laguerre's quantity $\mathcal{L}_\Xi = -(\log|\Xi|)''$.

A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ with $\delta > 0$ is added by hand to quantities computed from the actual zeros, to ξ'/ξ and to $\mathcal{L}_\Xi$, in the way that Section 11.4.4 inserts quadruples into the computed density, and nothing below says that ξ has such zeros. The tables, the figures and Corollary 19.3.3 describe what a test would see if it did, and are statements about a model. A quadruple puts two zeros in the upper half plane, and it is placed among the actual zeros in one of three ways, which differ in the number of zeros up to a given height:

(i) the quadruple replaces one on line zero, at the ordinate γ^* of that zero, as in the lattice of Section 5.2.2; the count of zeros up to a given height above γ^* grows by one;

(ii) the quadruple is inserted at the midpoint γ^* of a gap between consecutive on line zeros; the count grows by two;

(iii) the quadruple replaces the two on line zeros that bound a gap and sits at the midpoint γ^* of the gap, as in the perturbation of Theorem 5.2.1; the count is unchanged.

Proposition 19.3.1 (The test near the line). *Let θ be real and not the ordinate of a zero on the line, and let $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$, defined by the same formula for every real ε. Then*

$$\rho_\varepsilon(\theta) = \frac{\varepsilon}{\pi}\,\mathcal{L}_\Xi(\theta) + O(\varepsilon^3) \quad (\varepsilon \to 0),$$

where

$$\mathcal{L}_\Xi(\theta) = -\frac{d^2}{d\theta^2}\log|\Xi(\theta)| = \sum_\rho \frac{1}{(\theta - \gamma_\rho)^2},$$

$\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ are the zeros of Ξ, and the sum is real, since the zeros of Ξ come in conjugate pairs. Under RH, $\mathcal{L}_\Xi(\theta) = \sum_\gamma (\theta - \gamma)^{-2} > 0$. A quadruple of zeros $\frac{1}{2} \pm \delta \pm i\gamma^*$ contributes

$$2\operatorname{Re}\frac{1}{(\theta - \gamma^* + i\delta)^2} + 2\operatorname{Re}\frac{1}{(\theta + \gamma^* + i\delta)^2},$$

which equals $-2/\delta^2 + O(\gamma^{*-2})$ at $\theta = \gamma^*$.

Proof. As a function of (θ, ε), $\rho_\varepsilon(\theta)$ is harmonic off the zeros and odd in ε by the functional equation, so near $(\theta, 0)$ it equals $\varepsilon\,\partial_\varepsilon \rho_\varepsilon(\theta)|_{\varepsilon=0} + O(\varepsilon^3)$. With $F = \xi'/\xi$ and $\Xi'(\theta) = iF(\frac{1}{2} + i\theta)\Xi(\theta)$, differentiating gives $F'(\frac{1}{2} + i\theta) = -(\Xi'/\Xi)'(\theta)$, which

is real, and $\partial_\varepsilon \rho_\varepsilon = \pi^{-1} \operatorname{Re} F'$. Hadamard's product for Ξ, with the zeros $\pm\gamma_\rho$ paired, gives $\Xi'/\Xi = \sum (\theta - \gamma_\rho)^{-1}$, and differentiating once more gives the sum. For the quadruple the zeros of Ξ are $\gamma^* \pm i\delta$ and $-\gamma^* \pm i\delta$, and each conjugate pair contributes $2\operatorname{Re}(\theta \mp \gamma^* + i\delta)^{-2}$; at $\theta = \gamma^*$ the first is $2\operatorname{Re}(i\delta)^{-2} = -2/\delta^2$. □

At first order the test at γ^* sees the quadruple exactly when

$$\delta < \delta_c(\gamma^*) = \sqrt{2/\mathcal{L}_0(\gamma^*)},$$

with $\mathcal{L}_0(\gamma^*)$ the contribution of the other zeros to $\mathcal{L}_\Xi(\gamma^*)$; this is the local visibility condition of Section 5.2.2, written with Laguerre's quantity. In (i) $\mathcal{L}_0$ omits the replaced zero; with neighbours at the mean spacing $\Delta = 2\pi/\log(h/2\pi)$ this gives $\mathcal{L}_0 = \pi^2/(3\Delta^2)$ and $\delta_c = (\sqrt{6}/\pi)\Delta = 0.780\,\Delta$. In (ii), midway between two on line zeros, $\mathcal{L}_0 = \pi^2/\Delta^2$ and $\delta_c = (\sqrt{2}/\pi)\Delta = 0.450\,\Delta$. In (iii) the two zeros at $\gamma^* \pm \frac{1}{2}\Delta$ are removed from the lattice of (ii), $\mathcal{L}_0 = (\pi^2 - 8)/\Delta^2$ and $\delta_c = \sqrt{2/(\pi^2 - 8)}\,\Delta = 1.034\,\Delta$. A distance $\delta_c = c\Delta$ is below $\frac{1}{2}$ exactly when $h > 2\pi e^{4\pi c}$, that is above $2\pi e^{4\sqrt{6}} = 1.131 \times 10^5$ in (i), above $2\pi e^{4\sqrt{2}} = 1799$ in (ii) and above 2.8×10^6 in (iii). Table 19.2 replaces the lattice by the actual zeros.

		(i) replacing a zero			(ii) at a midpoint			(iii) count preserving		
height	mean gap	median	least	ratio	median	least	ratio	median	least	ratio
14	3.177	2.065	1.503	0.716	1.339	0.692	0.450	2.987	2.364	1.035
50	2.557	1.824	1.335	0.719	1.128	0.589	0.447	2.551	2.187	1.000
237	1.679	1.201	0.951	0.733	0.674	0.447	0.454	1.691	1.508	1.093
1419	1.178	0.799	0.655	0.711	0.469	0.297	0.452	1.167	0.975	1.062
9878	0.848	0.579	0.419	0.701	0.375	0.164	0.444	0.856	0.715	0.973
74921	0.661	0.446	0.235	0.662	0.296	0.088	0.437	0.651	0.527	0.902

Table 19.2: The first order visibility distance $\delta_c = \sqrt{2/\mathcal{L}_0}$ at the actual zeros, from $\mathcal{L}_\Xi = -(\log|\Xi|)''$ evaluated with `mpmath` at 30 digits, in the three configurations (i), (ii) and (iii) of the start of this section, at six heights $n_0 = 1, 10, 10^2, \ldots, 10^5$; the height is the ordinate γ_{n_0} of the n_0th zero. Replacing a zero: the 20 zeros γ_n, $n_0 < n \le n_0 + 20$, each removed from $\mathcal{L}_0$, with the local gap $(\gamma_{n+1} - \gamma_{n-1})/2$; the mean gap is the mean of the local gap over these 20 zeros. At a midpoint and count preserving: the 20 midpoints of γ_n and γ_{n+1}, $n_0 \le n < n_0 + 20$, with the local gap $\gamma_{n+1} - \gamma_n$, and in the count preserving block the two zeros γ_n and γ_{n+1} are removed as well. Median and least are over the 20 values of δ_c, and the ratio is the median over the 20 points of δ_c divided by the local gap. Script `tf_deltac.py`, data `tf_deltac.json`.

The ratio δ_c/gap is stable across heights in each configuration: about 0.45 at midpoints, as in the lattice; somewhat below the lattice value 0.78 when a zero is replaced; around the lattice value 1.034 when the count is preserved. The median of δ_c falls below $\frac{1}{2}$, the largest possible distance of a zero from the line, already by

height 1419 at midpoints and by height 74921 when replacing a zero, below the lattice estimate 1.1×10^5 above. Above these heights an off line zero at distance between δ_c and $\frac{1}{2}$ passes the first order test at its own ordinate wherever $\delta_c < \frac{1}{2}$, which is so at half or more of the sampled points once the median is below $\frac{1}{2}$, that is, the test does not see it.

Table 19.3 and Figure 19.1 show what happens beyond δ_c. We insert a hypothetical quadruple at the midpoint γ^* of a gap, configuration (ii), add its Cauchy terms exactly to ρ_ε computed from ξ'/ξ, and record the levels $\varepsilon \in (0, \delta)$, on the grid $\varepsilon = \delta k/40$, $k = 1, \dots, 39$, with $0.999\,\delta$, at which ρ_ε is negative at some of 41 equally spaced points of the gap. At height 9878, where $\delta_c = 0.381$, the quadruple with $\delta = 0.45$ is seen only for $0.19125 \le \varepsilon < 0.45$; at every level of the grid below 0.19, the first order test included, ρ_ε stays positive at the 41 points of the gap. The level at which the sign changes at the ordinate is $\varepsilon = 0.1895$ (Proposition 19.3.2 below), and the minimum over the gap, which lies about 0.004 to the left of the midpoint, changes sign at $\varepsilon = 0.1893$; the grid level 0.19125 is the first above both.

γ^*	δ_c	$\delta = 0.05$	$\delta = 0.2$	$\delta = 0.3$	$\delta = 0.45$
100.07	1.109	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$
9878.22	0.381	all $\varepsilon < \delta$	all $\varepsilon < \delta$	all $\varepsilon < \delta$	$0.19125 \le \varepsilon < 0.45$

Table 19.3: Levels ε at which ρ_ε is negative (grid and points as in the text), near a hypothetical quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$ inserted at the midpoint of the gap between the 29th and 30th zeros and between the 10000th and 10001st (configuration (ii)). Script `tf_window.py`, data `tf_window.json`.

19.3.1 The visibility curve

Proposition 19.3.2 (The visibility curve). *Let ρ_ε^0 be the Thorin density of all zeros other than the two zeros $\frac{1}{2} \pm \delta + i\gamma^*$ of a quadruple $\frac{1}{2} \pm \delta \pm i\gamma^*$, so that the terms of the quadruple's images at $-\gamma^*$ are included in it (and δ_{det} below depends on δ through them, by 10^{-9} in the computations of this section), and suppose $\rho_\varepsilon^0(\gamma^*) > 0$. For $0 < \varepsilon < \delta$, $\rho_\varepsilon(\gamma^*) < 0$ if and only if*

$$\delta < \delta_{\mathrm{det}}(\varepsilon), \qquad \delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + \frac{2\varepsilon}{\pi \rho_\varepsilon^0(\gamma^*)}.$$

In the lattice models the other zeros are those of the lattice alone, and the images at $-\gamma^*$ and the distant zeros are neglected, as in Section 5.2.2. Let the other zeros lie on the line at the points $\gamma^* \pm (k + \frac{1}{2})\Delta$ with $k \ge 0$ in configuration (ii), $\gamma^* \pm k\Delta$

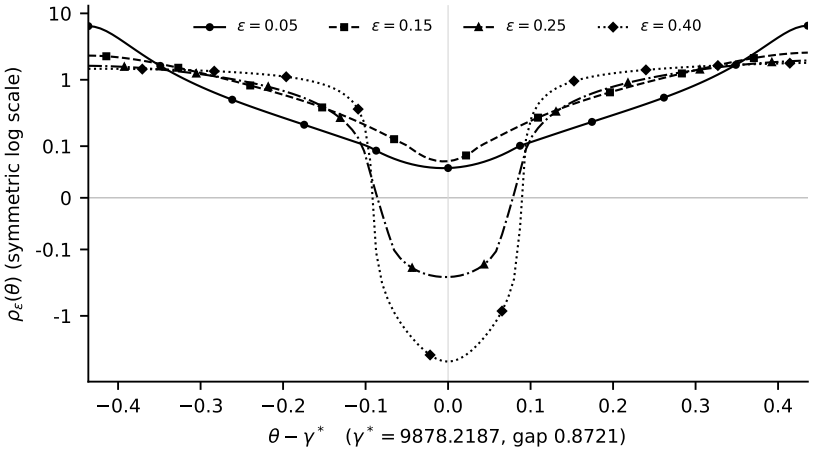


Figure 19.1: The Thorin density ρ_ε across the gap at height 9878 (gap 0.872) with a hypothetical quadruple at distance $\delta = 0.45$ inserted at its midpoint, at four levels $\varepsilon = 0.05$, 0.15, 0.25 and 0.4. It is positive at $\varepsilon = 0.05$ and $\varepsilon = 0.15$ and negative near the midpoint at $\varepsilon = 0.25$ and $\varepsilon = 0.4$. The vertical scale is symmetric logarithmic, linear within ± 0.1. Script `tf_window.py` with the option `--plot`, data `tf_window.json`.

with $k \geq 1$ in (i), and $\gamma^* \pm (k + \frac{1}{2})\Delta$ with $k \geq 1$ in (iii). Then

$$\delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + \begin{cases} \dfrac{2\varepsilon\Delta}{\pi \tanh(\pi\varepsilon/\Delta)} & \text{in (ii),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\coth\frac{\pi\varepsilon}{\Delta} - \frac{1}{\varepsilon}} & \text{in (i),} \\[2ex] \dfrac{2\varepsilon}{\frac{\pi}{\Delta}\tanh\frac{\pi\varepsilon}{\Delta} - \frac{2\varepsilon}{\varepsilon^2+\Delta^2/4}} & \text{in (iii).} \end{cases}$$

As $\varepsilon \to 0$ these tend to $\delta_c^2 = 2\Delta^2/\pi^2$, $6\Delta^2/\pi^2$ and $2\Delta^2/(\pi^2 - 8)$, and for $\varepsilon \geq \Delta$ all three satisfy $\delta_{\mathrm{det}}(\varepsilon)^2 - \varepsilon^2 = 2\varepsilon\Delta/\pi + O(\Delta^2)$, that is $\delta_{\mathrm{det}}(\varepsilon) - \varepsilon = (\Delta/\pi)\big(1 + O(\Delta/\varepsilon)\big)$. In all three, δ_{det} is increasing.

Proof. The two zeros at height γ^* contribute $\pi^{-1}\big((\varepsilon - \delta)^{-1} + (\varepsilon + \delta)^{-1}\big) = -2\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ at $\theta = \gamma^*$, so $\rho_\varepsilon(\gamma^*) < 0$ exactly when $\pi\rho_\varepsilon^0(\gamma^*) < 2\varepsilon/(\delta^2 - \varepsilon^2)$, which rearranges to the stated inequality. A zero on the line at distance δ_j from γ^* contributes $\varepsilon/(\pi(\varepsilon^2 + \delta_j^2))$ to $\rho_\varepsilon^0(\gamma^*)$ (Proposition 1.2.3), so in the lattice models $\pi\rho_\varepsilon^0(\gamma^*) = \varepsilon\sum_j(\varepsilon^2 + \delta_j^2)^{-1}$, which is positive. The sum decreases with ε, and $\delta_{\mathrm{det}}(\varepsilon)^2 = \varepsilon^2 + 2\big/\sum_j(\varepsilon^2 + \delta_j^2)^{-1}$ increases. For the lattices, $\sum_{k\in\mathbb{Z}} \varepsilon/(\varepsilon^2 + (k + \frac{1}{2})^2\Delta^2) = (\pi/\Delta)\tanh(\pi\varepsilon/\Delta)$ and

$\sum_{k\neq 0}\varepsilon/(\varepsilon^2+k^2\Delta^2)=(\pi/\Delta)\coth(\pi\varepsilon/\Delta)-1/\varepsilon$, by the partial fractions of tanh and coth, and removing the two zeros at $\gamma^*\pm\frac12\Delta$ subtracts $2\varepsilon/(\varepsilon^2+\Delta^2/4)$ from the first. The limits follow from $\tanh x=x+O(x^3)$ and $\coth x=1/x+x/3+O(x^3)$, which give $\pi\rho_\varepsilon^0(\gamma^*)/\varepsilon\to\pi^2/\Delta^2$, $\pi^2/(3\Delta^2)$ and $(\pi^2-8)/\Delta^2$. For $\varepsilon\geq\Delta$, $\tanh(\pi\varepsilon/\Delta)$ and $\coth(\pi\varepsilon/\Delta)$ equal 1 up to $O(e^{-2\pi\varepsilon/\Delta})$, so that $\pi\rho_\varepsilon^0(\gamma^*)=\pi/\Delta+O(1/\varepsilon)$ in all three, whence $\delta_{\det}^2-\varepsilon^2=2\varepsilon\Delta/\pi+O(\Delta^2)$; dividing by $\delta_{\det}+\varepsilon=2\varepsilon+O(\Delta)$ gives $\delta_{\det}-\varepsilon=(\Delta/\pi)(1+O(\Delta/\varepsilon))$. □

The first display of the proposition is the formula for $\pi\rho_\varepsilon(\gamma)$ at the ordinate in Section 5.2.2, solved for the distance, with all the other zeros and not only the neighbours in ρ_ε^0. So a level ε sees, at the ordinate, exactly the zeros at distances in $(\varepsilon,\delta_{\det}(\varepsilon))$: a band that starts at width δ_c near the line and has width close to $\Delta/\pi=2/\log(h/2\pi)$ well above it, the scale $1/\log\gamma$ of Section 5.2.2 with its constant. The width $\delta_{\det}(\varepsilon)-\varepsilon$ is $\Delta/\pi+c\,\Delta^2/\varepsilon+\cdots$ with $c=\frac{1}{2\pi^2}$, $-\frac{1}{2\pi^2}$ and $\frac{3}{2\pi^2}$ in (i), (ii) and (iii), so that for large ε it is above Δ/π in (i) and (iii) and below it in (ii); in (i) and (iii) it falls monotonically to Δ/π (computed), while in (ii) it falls from $0.450\,\Delta$ to a least value $0.272\,\Delta$ at $\varepsilon=0.581\,\Delta$ and then rises to Δ/π from below (computed by `tf_visibility.py`). The curve explains Table 19.3. At the midpoint of the 10000th gap, $\pi\rho_\varepsilon^0(\gamma^*)$ computed from ξ'/ξ gives $\delta_{\det}(0.19)=0.4503$, and solving $\delta_{\det}(\varepsilon)=0.45$ gives $\varepsilon=0.1895$. Figure 19.2 draws the region for the actual zeros and for the lattice with the same gap, which lies within 0.015 of it.

Corollary 19.3.3 (How many levels, in the lattice model). *Take any of the three lattice models, with gap $\Delta=2\pi/\log(h/2\pi)$, and put $\varepsilon_1=\delta_c$ and $\varepsilon_{k+1}=\delta_{\det}(\varepsilon_k)$. Let K be the least k with $\varepsilon_{k+1}>\frac12$, so that $K=0$ if $\delta_c>\frac12$.*

(a) The sign of $\mathcal{L}_\Xi(\gamma^*)$, together with positivity of $\rho_\varepsilon(\gamma^*)$ at the levels $\varepsilon_1,\dots,\varepsilon_K$, each moved down slightly so that consecutive bands overlap, detects a single off line quadruple at every distance in $(0,\frac12]$.

(b) No test made at the ordinate, that is the sign of $\mathcal{L}_\Xi(\gamma^*)$ together with the sign of $\rho_\varepsilon(\gamma^*)$ at finitely many levels, that detects a single off line quadruple at every distance in $(0,\frac12]$ uses fewer than K levels.

(c) $K=\frac14\log(h/2\pi)+O(\log\log h)$.

Proof. By Proposition 19.3.1 the first order test detects the distances in $(0,\delta_c)$, and by Proposition 19.3.2 the level ε detects exactly those in $(\varepsilon,\delta_{\det}(\varepsilon))$, since a zero at distance $\delta\leq\varepsilon$ contributes positively and $\rho_\varepsilon^0(\gamma^*)>0$. Write $f=\delta_{\det}$, so that $\varepsilon_{k+1}=f(\varepsilon_k)$; f is continuous and increasing with $f(\varepsilon)>\varepsilon$.

(a) The bands $(0,\varepsilon_1)$ and $(\varepsilon_k,\varepsilon_{k+1})$, $1\leq k\leq K$, cover $(0,\varepsilon_{K+1})$ except the points $\varepsilon_1,\dots,\varepsilon_K$. Moving the levels down by suitably small amounts, which is possible because f is continuous and $\varepsilon_{K+1}=f(\varepsilon_K)>\frac12$, makes consecutive bands overlap and still reach beyond $\frac12$.

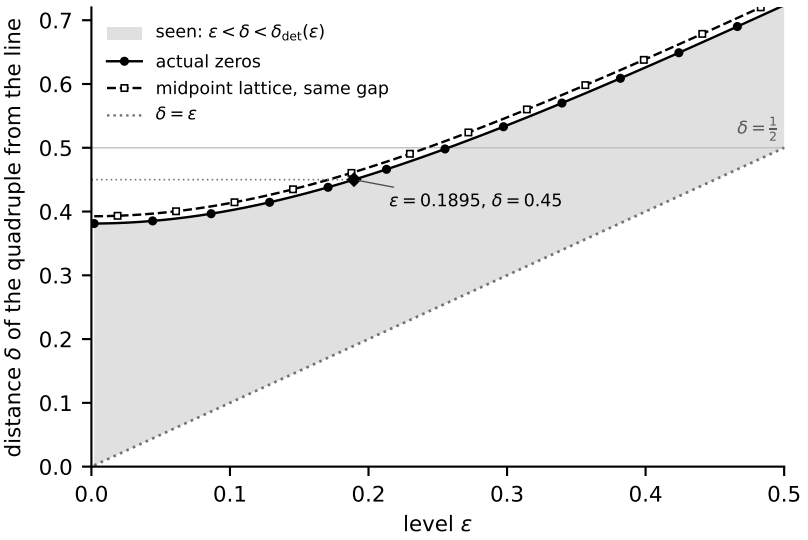


Figure 19.2: The visibility region $\varepsilon < \delta < \delta_{\det}(\varepsilon)$ (shaded band) at the midpoint of the 10000th gap. Solid: $\delta_{\det}$ from the actual zeros through ξ'/ξ; dashed: the midpoint lattice of Proposition 19.3.2 with the same gap; dotted: $\delta = \varepsilon$, the lower edge of the band. The marked point is $\varepsilon = 0.1895$, $\delta = 0.45$. Script `tf_visibility.py`.

(b) We claim that if $j \geq 1$ bands $(\varepsilon', f(\varepsilon'))$ cover $[\ell, x)$ with $\ell > 0$, then $x < f^{(j)}(\ell)$, the jth iterate. A band containing ℓ has $\varepsilon' < \ell$, so its right end is $r = f(\varepsilon') < f(\ell)$. For $j = 1$ this gives $x \leq r < f(\ell)$. For $j > 1$, either $x \leq r < f(\ell) \leq f^{(j)}(\ell)$, or the other $j-1$ bands cover $[r, x)$ and the claim for $j-1$ gives $x < f^{(j-1)}(r) < f^{(j-1)}(f(\ell)) = f^{(j)}(\ell)$. Now suppose that m levels, with the first order test, detect every distance in $(0, \frac{1}{2}]$. The point δ_c is not in $(0, \delta_c)$, so unless $\delta_c > \frac{1}{2}$ (and then $K = 0$) some bands cover $[\delta_c, \frac{1}{2}]$, and the claim with $\ell = \delta_c$ and $x = \frac{1}{2}$ gives $\frac{1}{2} < f^{(m)}(\delta_c) = \varepsilon_{m+1}$. By the definition of K, $m \geq K$.

(c) The bands advance by $a(\varepsilon) = \delta_{\det}(\varepsilon) - \varepsilon = \big(\delta_{\det}(\varepsilon)^2 - \varepsilon^2\big)/\big(\delta_{\det}(\varepsilon) + \varepsilon\big)$, which by the formulas of Proposition 19.3.2 is $\frac{\Delta}{\pi}\big(1 + O(\Delta/\varepsilon)\big)$ for $\varepsilon \geq \Delta$ and, for $\varepsilon \leq \Delta$, lies between two positive constants times Δ, since $a(\varepsilon)/\Delta$ is a positive continuous function of ε/Δ. The number of steps from δ_c to $\frac{1}{2}$ is therefore $\int_\Delta^{1/2} \frac{d\varepsilon}{a(\varepsilon)} + O(\log(1/\Delta)) = \frac{\pi}{\Delta}\big(\frac{1}{2} + O(\Delta \log(1/\Delta))\big)$, and $\pi/(2\Delta) = \frac{1}{4}\log(h/2\pi)$, while $\log(1/\Delta) = O(\log\log h)$. □

Table 19.4 gives the numbers of levels. The count preserving lattice has the larger $\delta_c = 1.034\,\Delta$, which stays above $\frac{1}{2}$ up to $h = 2.8 \times 10^6$, so it uses fewer levels, and at 10^5 the $\delta_c = 0.506$ of the replacement lattice already covers every distance. At the height 3×10^{12} to which RH has been verified [247] the three

h	$\frac{1}{4}\log(h/2\pi)$	(i) replacing	(ii) midpoint	(iii) count preserving
10^5	2.4	0	2	0
10^8	4.1	2	4	1
3×10^{12}	6.7	4	6	3
10^{22}	12.2	9	12	7

Table 19.4: The number K of greedy levels of Corollary 19.3.3, counted without the downward moves, in the three lattice models with $\Delta = 2\pi/\log(h/2\pi)$, against $\frac{1}{4}\log(h/2\pi)$. Script `tf_visibility.py`.

models use between three and six levels. The corollary is a statement about a model with a single quadruple and regularly spaced neighbours, and about tests at the ordinate: a test that looks for negative values anywhere in the window can see more (Section 5.2.2, Exercise 5.14), and for it no lower bound is proved here. Turning it into a certificate for the actual zeros is Problem 19.1.

19.4 The heat flow: a stable step that cannot carry real zeros

With the de Bruijn Newman kernel Φ_{dBN} of Section 8.4.1,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\,\Xi(z/2),$$

the function H_t is defined for every real t, since the kernel decays like $e^{9u}\exp(-\pi e^{4u})$, and the zeros of H_0 are the points $2\gamma_\rho$, with imaginary parts $-2(\beta-\frac{1}{2})$. The de Bruijn Newman constant Λ satisfies $0 \le \Lambda \le 0.2$ [289, 247].

Theorem 19.4.1 (The heat family). *(a) $\partial_t H_t = -\partial_z^2 H_t$, so $H_t = e^{-tD^2}H_0$ with $D = d/dz$.*

(b) For real t, $s>0$ and complex z, with Z a standard normal variable,

$$H_{t-s}(z) = \mathbb{E}\big[H_t\big(z+\sqrt{2s}\,Z\big)\big].$$

(c) Forward in t real zeros persist, and if the zeros of H_{t_0} lie in $|\operatorname{Im} z|\le r$, then those of H_t lie in $(\operatorname{Im} z)^2 \le \max(r^2-2(t-t_0),0)$ for $t\ge t_0$ [73, Theorem 13].

(d) Backward in t real zeros are not preserved: $f(z)=z^2-c$, $c>0$, has real zeros, and $\mathbb{E}f(z+\sqrt{2s}Z) = z^2-c+2s$ has none once $s>c/2$. For H itself, H_t has nonreal zeros for every $t<0$ [289].

Proof. (a) Differentiate under the integral: ∂_t brings down u^2 and ∂_z^2 brings down $-u^2$. (b) $\mathbb{E}\cos((z+\sigma Z)u) = \cos(zu)e^{-\sigma^2u^2/2}$ for complex z, and with $\sigma^2=2s$ the factor e^{-su^2} turns the kernel $e^{tu^2}\Phi_{\mathrm{dBN}}$ into $e^{(t-s)u^2}\Phi_{\mathrm{dBN}}$; Fubini applies because

Φ_{dBN} decays like $e^{9u}\exp(-\pi e^{4u})$, for every real t. (c) is de Bruijn's theorem, in the form used by Polymath15 [282] and in the proof of Theorem 15.3.2. (d) is the computation shown and the theorem of Rodgers and Tao. □

COROLLARY 19.4.2 (No averaging stable property carries real zeros). *Let $\mathcal{P}$ be a property of entire functions that is preserved by $f \mapsto \mathbb{E}f(\cdot + \sqrt{2s}Z)$ for some $s > 0$. If $\mathcal{P}$ implies that all zeros are real, then H_t has $\mathcal{P}$ for no real t.*

Proof. If H_{t_0} has $\mathcal{P}$, then by (b), applied k times, so does H_{t_0-ks} for every integer $k \geq 1$, and for $t_0 - ks < 0$ it has only real zeros, against (d). The proof uses only that some H_t has a nonreal zero, which is Newman's theorem $\Lambda > -\infty$ [210]. □

The corollary is the heat flow form of the criticality filter of Section 11.5.5: Proposition 11.5.12 shows that no property of the kernel Φ_t that holds at some $t < 0$ can force real zeros. The step towards RH, from $t = 0.2$ down to 0, is an average of $H_{0.2}$ against a Gaussian of variance 0.4: a contraction, stable to compute.

A Gauss Legendre rule with 200 nodes in u on $[0, 1.5]$, beyond which $\Phi_{\mathrm{dBN}} < 10^{-540}$, and a Gauss Hermite rule with 80 nodes reproduce $H_0(z)$ from $H_{0.2}$ at $z = 0$, 10 and 28.27 to a relative error below 10^{-35} in 40 digit arithmetic. The step away from RH, e^{-tD^2}, preserves real zeros, as the operator of the Laguerre Pólya theory, and is a deconvolution in z. In the basepoint family the roles are reversed: the positivity preserving step is a convolution and the step towards RH an unbounded deconvolution.

The two informative families are tied by the conserved quantity of Theorem 15.3.2: a basepoint offset ε is worth heat time $2\varepsilon^2$, and $\Lambda \leq 2(\sigma_{\max} - \frac{1}{2})^2$. The critical strip, $\sigma_{\max} \leq 1$, gives de Bruijn's $\Lambda \leq \frac{1}{2}$; the half plane announced in October 2026 [226], not yet refereed, would give $\sigma_{\max} \leq \frac{7}{8}$ and, conditionally on it, $\Lambda \leq \frac{9}{32}$, weaker than the known 0.2; and $\Lambda \leq 0.2$ corresponds to the basepoint $\frac{1}{2} + \sqrt{0.1} = 0.816$ (Corollary 15.3.3). In the language of Theorem 19.2.2, whose *if* direction holds for ξ_t by the product of Lemma 15.2.1, the bound $\Lambda \leq 0.2$ of Platt and Trudgian, obtained through the Polymath15 framework, says that the Thorin density of $\xi_{0.2}$ deconvolves to a positive measure all the way to the line; RH asks for the same at $t = 0$, and by Corollary 19.4.2 no property preserved by the Gaussian average joining the two times can carry it.

What would get past the obstruction. This paragraph is a heuristic reading of the theorems above. In the basepoint family a proof needs a bound, uniform in the height, for the deconvolution of ρ_{ε_0}; from the primes that means cancellation in $\sum \Lambda(n)n^{-\alpha-ib}$ below $\alpha = 1$, which is a zero free region by Proposition 9.1.13, and the half plane announced in October 2026 [226], not yet refereed, would move the end of the range to $\alpha = \frac{7}{8}$. In the heat family a proof needs a property of H_0 that implies real zeros and that fails for every H_t with $t < 0$, since those have nonreal

zeros (Theorem 19.4.1(d)): a property that sees the boundary $t = \Lambda$ itself. In the tilt family nothing can be gained. And since $\Lambda \geq 0$, any such statement must hold at the boundary with no slack (Remark 15.3.4).

19.4.1 Open problems

Problem 19.1 (How many levels, for the actual zeros). Corollary 19.3.3 counts levels in a model with one quadruple among regularly spaced zeros, and only at its ordinate. Turn it into a certificate: find levels $\varepsilon_1 < \cdots < \varepsilon_K$ such that positivity of every ρ_{ε_k} on $[0, h]$, together with positivity of $\mathcal{L}_\Xi$, excludes all off line zeros up to height h, allowing several off line zeros, irregular spacing, and minima off the ordinate (Exercise 5.14). Is K still of order $\log h$? Such a certificate is finite information in the sense of Section 5.2 and Theorem 5.2.1, so its interest is as a different verification of RH up to height h, from ξ on lines to the right of the critical line rather than from sign changes on it.

Problem 19.2 (Stability with an a priori bound). On a strip $\frac{1}{2}+\varepsilon_1 \leq \operatorname{Re} s \leq \frac{1}{2}+\varepsilon_0$ free of zeros, Hadamard's three lines theorem bounds the error of an approximation g to ξ'/ξ at level ε by $\eta^\lambda M^{1-\lambda}$, $\lambda = (\varepsilon - \varepsilon_1)/(\varepsilon_0 - \varepsilon_1)$, where η is the error at ε_0 and M a bound at ε_1. The bound M presupposes the zero free strip. Is there a formulation in which the a priori bound comes from the Thorin structure itself, for instance from positivity of ρ_ε at the levels already certified?

Problem 19.3 (Seeing the boundary). By Theorem 19.4.1(d), a property of H_0 that implies real zeros fails for H_t at every $t < 0$. Which of the positivity statements of the book, for instance the GGC property of the clock, whose Laplace transform is R_0, the case $t = 0$ of $R_t(s) = H_t(0)/H_t(2i\sqrt{s})$ in Theorem 7.2.15, or the Pólya frequency property of the density of $Z\sqrt{2T}$ (Proposition 16.5.1, where the clock is written Y), can be shown to fail for $t < 0$ by a mechanism visible at finite height, as Lehmer pairs make Λ close to 0 visible [70]?

Problem 19.4 (The window below δ). For a quadruple at distance $\delta > \delta_c$ the level test detects it on a set of levels contained in $(0, \delta)$, an interval (ε_c, δ) in our examples. At the ordinate ε_c is the inverse of the visibility curve of Proposition 19.3.2. Describe the distribution of δ_{det}, and of the onset $\varepsilon_c(\delta, \gamma^*)$, over heights in terms of the local zero statistics, and compare it with the prediction of the sine kernel; Table 19.2 gives its lower end, δ_c.

Notes

The ordinate formula for $\pi\rho_\varepsilon(\gamma)$, the first order visibility condition and the lattice threshold $(\sqrt{6}/\pi)\Delta$ of Section 19.3 are those of Section 5.2.2, and inserting hypothetical quadruples into the density computed from the zeros is the device of Section 11.4.4; the closed forms of Proposition 19.3.2, the level count of Corollary 19.3.3 and the computations at the actual zeros are in this chapter. The

bound $\Lambda \leq 0.2$ is that of Platt and Trudgian [247], and the visibility of a Lehmer pair through Λ is the work of Csordas, Smith and Varga [70].

Chapter 20

Deconvolution as a damped Weil criterion

What is not claimed. The half plane $\operatorname{Re} s > \frac{7}{8}$ has not been refereed. The eigenvalues of the ladder are variational upper bounds, the converged values are extrapolations of such bounds with an error estimate, and none of them is a certificate.

The question. Let ξ have zeros $\rho = \beta + i\gamma$, let $\gamma_\rho = -i(\rho - \frac{1}{2}) = \gamma - i(\beta - \frac{1}{2})$ be the zeros of $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$, and put $\sigma_{\max} = \sup_\rho \operatorname{Re}\rho \in [\frac{1}{2}, 1]$. We say that a level $\varepsilon > 0$ is *to the right of the zeros* if $\beta < \frac{1}{2} + \varepsilon$ for every zero. Every $\varepsilon \geq \frac{1}{2}$ is, since $\beta < 1$, whether or not $\sigma_{\max} < 1$; the latter is the quasi Riemann hypothesis. For real α write $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha + i\theta)$ and $b_\alpha = v_\alpha(0) = \frac{\xi'}{\xi}(\alpha)$, so that $\rho_\varepsilon = \pi^{-1} v_{1/2+\varepsilon}$ is the density (1.2.7): for $\varepsilon > 0$ the Poisson smoothing of the zeros (Proposition 1.2.3) and, when ε is to the right of the zeros, the Thorin density of the shifted clock at basepoint $\frac{1}{2}+\varepsilon$ (Theorem 1.2.2). Write $P_a(x) = a/(\pi(a^2+x^2))$ for the Cauchy density of scale $a > 0$. Theorem 19.2.2 of Chapter 19 says that, for ε_0 to the right of the zeros and $0 \leq \varepsilon < \varepsilon_0$, there is a positive measure ν with $\int (1+\theta^2)^{-1}\nu(d\theta) < \infty$ and $\rho_{\varepsilon_0} = P_{\varepsilon_0-\varepsilon} * \nu$ if and only if no zero has $\beta > \frac{1}{2}+\varepsilon$. The proof runs a Poisson integral of ν down from ε_0 and uses harmonic continuation: a positive ν would make $\rho_\varepsilon(\theta)$ harmonic in (θ, ε) near a zero with $\beta - \frac{1}{2} > \varepsilon$, where ξ'/ξ has a pole.

Asking whether the deconvolution *can be done* has two meanings. One is constructive: can $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ be produced, from data we can compute, without assuming where the zeros are? The other is the theorem itself: can it be shown to be a positive measure? We answer the first completely (Section 20.1.1) and reduce the second to a two parameter family of finite problems (Sections 20.1.2 and 20.2), whose behaviour we partly prove and compute.

20.1 Weil's distribution and the damped criterion

20.1.1 The deconvolved object exists: it is Weil's distribution

With $\hat f(t)=\int f(x)e^{itx}dx$ the Fourier transform of P_a is $e^{-a|x|}$, so deconvolution by P_a multiplies Fourier transforms by $e^{a|x|}$. Let $\mathcal B$ be the space of even functions $\varphi(\theta)=\frac{1}{2\pi}\int\hat\varphi(x)e^{-i\theta x}dx$ with $\hat\varphi$ of compact support, Lipschitz, and with $\hat\varphi'$ of bounded variation. Then φ is entire of exponential type and $\varphi(\theta+is)=O(\theta^{-2})$ uniformly for $|s|\le 1$, since $\hat\varphi(x)e^{sx}$ has the same properties. For $a\ge 0$ let $P_a^{-1}\varphi\in\mathcal B$ have Fourier transform $\hat\varphi(x)e^{a|x|}$; the corner of $e^{a|x|}$ at 0 keeps it in $\mathcal B$.

Theorem 20.1.1 (The deconvolved functional). *Let ε_0 be to the right of the zeros. For $\varphi\in\mathcal B$ put $D(\varphi)=\int\rho_{\varepsilon_0}(\theta)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$. Then D does not depend on ε_0, and with $g(x)=\hat\varphi(x)/2\pi$, so that $\varphi(\theta)=\int g(x)e^{i\theta x}dx=\hat g(\theta)$,*

$$D(\varphi)=\sum_\rho\varphi(\gamma_\rho)=\hat g\big(\tfrac{i}{2}\big)+\hat g\big(-\tfrac{i}{2}\big)-(\gamma_E+\log\pi)\,g(0)\\ +\int_0^\infty\frac{2\big(e^{-2x}g(0)-e^{-x/2}g(x)\big)}{1-e^{-2x}}\,dx\\ -2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt n}\,g(\log n),$$

where the sum over ρ runs over all zeros with multiplicity, in conjugate pairs, and converges absolutely. Through the x form of its archimedean term, the Thorin functional U of Section 5.3.1 extends to $\mathcal B$, and then $D(\varphi)=2\langle U,\varphi(\sqrt{\cdot})\rangle$.

Proof. By the expansion of ρ_{ε_0} over the zeros (Proposition 1.2.3, summed over all zeros $\rho=\beta+i\gamma$, with γ of both signs), $D(\varphi)=\sum_\rho\int P_{c_\rho}(\theta-\gamma)\,(P_{\varepsilon_0}^{-1}\varphi)(\theta)\,d\theta$ with $c_\rho=\varepsilon_0-(\beta-\frac12)>0$; the interchange is justified because $\rho_{\varepsilon_0}=O(\log(2+|\theta|))$ and $P_{\varepsilon_0}^{-1}\varphi=O(\theta^{-2})$. By Parseval each term is $\frac{1}{2\pi}\int\hat\varphi(x)e^{(\varepsilon_0-c_\rho)|x|}e^{-i\gamma x}\,dx$. A zero ρ off the line and its partner $1-\bar\rho$ have the same γ and opposite $\beta-\frac12=\pm\delta$, so the pair gives $\frac{1}{2\pi}\int\hat\varphi(x)e^{-i\gamma x}\,2\cosh(\delta x)\,dx=\varphi(\gamma-i\delta)+\varphi(\gamma+i\delta)$, because $e^{\delta|x|}+e^{-\delta|x|}=e^{\delta x}+e^{-\delta x}$; a zero on the line is its own partner and gives the single term $\varphi(\gamma)$. This is $\sum_\rho\varphi(\gamma_\rho)$, absolutely convergent by the uniform $O(\theta^{-2})$ bound, and it does not involve ε_0. The second expression is the explicit formula of Weil [338], [128, Theorem 5.12], with the archimedean term written in the variable x by $\operatorname{Re}\psi(\frac14+\frac{it}{2})=-\gamma_E+\int_0^\infty\frac{e^{-u}-e^{-u/4}\cos(tu/2)}{1-e^{-u}}\,du$; this integral converges because g is Lipschitz at 0, and the formula extends from the usual class to $\mathcal B$ by approximation: mollifying g gives functions of the usual class supported in a fixed interval, with uniformly bounded Lipschitz constants, and both sides are continuous along such sequences, the zero side by the uniform $O(\theta^{-2})$ bound. The comparison with (5.3.1), whose functional sums once over each pair $\{\rho,1-\rho\}$, gives the factor 2. □

For g even, Lipschitz and of compact support, with g' of bounded variation, so that $\hat{g} \in \mathcal{B}$, write $\langle D, g\rangle = D(\hat{g})$, and for g not even apply it to the even part of g. By the theorem $\langle D, g\rangle = \sum_\rho \hat{g}(\gamma_\rho)$, which is the action on g of Weil's distribution $D(x) = \sum_\rho e^{(\rho-\frac{1}{2})x}$ of Proposition 7.2.13.

The distribution $P_{\varepsilon_0}^{-1}\rho_{\varepsilon_0}$ on $\mathcal{B}$, which exists with no hypothesis, is a positive measure exactly when RH holds: under RH it is $\sum_\gamma \delta_\gamma$, and if $D(\varphi) = \int \varphi\, d\mu$ on $\mathcal{B}$ with $\mu \geq 0$, then $\int \rho_{\varepsilon_0}\psi = D(P_{\varepsilon_0} * \psi) = \int (P_{\varepsilon_0} * \psi)\, d\mu$ for every $\psi \in \mathcal{B}$, which gives $\int (1+\theta^2)^{-1} d\mu < \infty$ (take $\psi \geq 0$ with $P_{\varepsilon_0} * \psi \geq c(1+\theta^2)^{-1}$) and $\rho_{\varepsilon_0} = P_{\varepsilon_0} * \mu$, so that RH follows from Theorem 19.2.2. On the functions $\varphi = |\hat{h}|^2$ the statement $D(\varphi) \geq 0$ is Weil's criterion [338], in the form the book uses in Section 3.5.2. We checked the second expression against $2\sum_{\gamma>0} \hat{g}(\gamma)$ for the Gaussian $g(x) = e^{-(x/c)^2}$ (`code/dc_weil.py`): at $c = \frac{1}{2}$ both sides are $6.69239857682 \times 10^{-6}$ and they differ by less than 10^{-69} at seventy digits of working precision.

20.1.2 The damped Weil criterion

Partial deconvolution, to depth ε instead of to the line, corresponds to damping. For bounded h of bounded variation and compact support put $\tilde{h}(x) = \overline{h(-x)}$, so that $h * \tilde{h}$ is Lipschitz and Hermitian, and

$$Q_\varepsilon(h) = \big\langle D,\, e^{-\varepsilon|x|}\,(h * \tilde{h})\big\rangle.$$

The Fourier transform of $e^{-\varepsilon|x|}$ is $2\pi P_\varepsilon$ and that of $h * \tilde{h}$ is $|\hat{h}|^2$, so

$$\widehat{e^{-\varepsilon|\cdot|}(h * \tilde{h})} = P_\varepsilon * |\hat{h}|^2,$$

a nonnegative function. Real h suffice in what follows, since $Q_\varepsilon(h) = Q_\varepsilon(\operatorname{Re} h) + Q_\varepsilon(\operatorname{Im} h)$.

Theorem 20.1.2 (Damped Weil criterion). *Let $0 \leq \varepsilon \leq \frac{1}{2}$. Then $Q_\varepsilon(h) \geq 0$ for every $h \in C_c^\infty(\mathbb{R})$ if and only if no nontrivial zero of ζ has $\beta > \frac{1}{2} + \varepsilon$. When ε is to the right of the zeros,*

$$Q_\varepsilon(h) = \int_{\mathbb{R}} |\hat{h}(\theta)|^2\, \rho_\varepsilon(\theta)\, d\theta \tag{20.1.1}$$

for every bounded h of bounded variation and compact support. In particular: at $\varepsilon = 0$ this is Weil's criterion for RH; at $\varepsilon = \frac{1}{2}$ it holds with no hypothesis; and the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 [226], which has not been refereed, is equivalent to $Q_{3/8} \geq 0$.

Proof. Formula (20.1.1) follows from Theorem 20.1.1. Let h be real, bounded, of bounded variation and of compact support. Then $F = h * \tilde{h}$ is Lipschitz and of compact support, and its derivative $h * d\tilde{h}$, the convolution of a bounded

function of bounded variation with a finite measure, is of bounded variation; so $e^{-\varepsilon|x|}F$ is Lipschitz, of compact support, with a derivative of bounded variation, and $\varphi = P_\varepsilon * |\hat{h}|^2$, the Fourier transform of $e^{-\varepsilon|x|}F$, and $|\hat{h}|^2$, that of F, lie in $\mathcal{B}$. As $P_\varepsilon^{-1}\varphi = |\hat{h}|^2$, the first identity of Theorem 20.1.1 with $\varepsilon_0 = \varepsilon$ gives $Q_\varepsilon(h) = D(\varphi) = \int |\hat{h}|^2 \rho_\varepsilon \, d\theta$; for complex h add the two parts, the cross terms of $|\hat{h}|^2$ being odd and ρ_ε even. For bounded piecewise C^1 functions h this is Proposition 17.8.1 of Chapter 17 at $\alpha = \frac{1}{2} + \varepsilon$, where the form $\langle \varphi, \varphi \rangle_\alpha$ is $Q_\varepsilon(\varphi)$. The *if* direction is Theorem 17.8.3(i): if no zero has $\beta > \frac{1}{2} + \varepsilon$, then either ε is to the right of the zeros and $\rho_\varepsilon > 0$, so that $Q_\varepsilon \geq 0$ by (20.1.1), or some zero has $\beta = \frac{1}{2} + \varepsilon$, and then $Q_\varepsilon(h) = \lim_{\varepsilon' \downarrow \varepsilon} Q_{\varepsilon'}(h) \geq 0$, the limit holding because $Q_\varepsilon(h)$ is a finite combination of integrals continuous in ε.

Conversely, suppose $Q_\varepsilon \geq 0$. The map $T_\varepsilon : g \mapsto \langle D, e^{-\varepsilon|x|} g \rangle$ is a distribution of order at most one, the corner of $e^{-\varepsilon|x|}$ at 0 being harmless in the archimedean integral, and $Q_\varepsilon(h) = T_\varepsilon(h * \tilde{h}) \geq 0$ says that it is positive definite. By the Bochner Schwartz theorem [92, Chapter II] it is tempered and $T_\varepsilon(g) = \int \hat{g} \, d\nu_\varepsilon$ for a positive tempered measure ν_ε and every Schwartz g. Fix $\varepsilon_0 > \varepsilon$ with $\varepsilon_0 \geq \frac{1}{2}$, put $c = \varepsilon_0 - \varepsilon$, and let $k_\eta(x) = e^{-c\sqrt{x^2+\eta^2}}$, smooth, positive definite and of exponential decay, with $\hat{k}_\eta \geq 0$, $\hat{k}_\eta \to 2\pi P_c$ pointwise and $\hat{k}_\eta \leq C(1+\theta^2)^{-1}$ uniformly in $\eta \leq 1$. For $h \in C_c^\infty$ and $g = h * \tilde{h}$, $T_\varepsilon(k_\eta g) = \frac{1}{2\pi} \int (\hat{k}_\eta * |\hat{h}|^2) \, d\nu_\varepsilon$, and as $\eta \to 0$ the left side tends to $T_\varepsilon(e^{-c|x|} g) = Q_{\varepsilon_0}(h) = \int |\hat{h}|^2 \rho_{\varepsilon_0}$, because $k_\eta g \to e^{-c|x|} g$ uniformly, with supports in a fixed interval and uniformly bounded Lipschitz constants, which is enough for dominated convergence in the archimedean integral and in the other terms of $\langle D, \cdot \rangle$. By Fatou, $\int (P_c * |\hat{h}|^2) \, d\nu_\varepsilon \leq \int |\hat{h}|^2 \rho_{\varepsilon_0} < \infty$; choosing h with $|\hat{h}|^2 > 0$ near 0 gives $P_c * |\hat{h}|^2 \geq c'(1+\theta^2)^{-1}$, so $\int (1+\theta^2)^{-1} d\nu_\varepsilon < \infty$, and dominated convergence then gives $\int (P_c * |\hat{h}|^2) \, d\nu_\varepsilon = \int |\hat{h}|^2 \rho_{\varepsilon_0}$, that is $\int |\hat{h}|^2 \, (P_c * \nu_\varepsilon) = \int |\hat{h}|^2 \rho_{\varepsilon_0}$. Polarising, the same holds with $|\hat{h}|^2$ replaced by $\hat{h}_1 \overline{\hat{h}_2}$, and these products span a dense set, so $\rho_{\varepsilon_0} = P_{\varepsilon_0 - \varepsilon} * \nu_\varepsilon$. Theorem 19.2.2 gives $\beta \leq \frac{1}{2} + \varepsilon$ for every zero. □

The theorem is the Fourier dual of the deconvolution theorem; at $\varepsilon = 0$ it is a proof of Weil's criterion along the lines of the harmonic continuation of Theorem 19.2.2. The damping also explains why $\varepsilon = \frac{1}{2}$ is free. The primes enter with weight $\Lambda(n) n^{-1/2-\varepsilon}$, absolutely summable exactly for $\varepsilon > \frac{1}{2}$: the damped criterion at $\varepsilon = \frac{1}{2}$ sits at the edge of the Euler product, the basepoint one of Chapter 9.

20.2 The ladder

Restrict the support of h. For $a > 0$ and $\varepsilon \geq 0$ let

$$\lambda_{\min}\big(\tfrac12+\varepsilon,a\big) = \inf\Big\{ \frac{Q_\varepsilon(h)}{\|h\|_2^2} : h \text{ bounded, real, of bounded variation, supported in } [-a,a],\ h \neq 0 \Big\}.$$

This is the least eigenvalue $\lambda_{\min}(\alpha,a)$, $\alpha = \frac12+\varepsilon$, of Chapter 17, with the same normalisation. The function $e^{-\varepsilon|x|}(h * \tilde h)$ is supported in $[-2a,2a]$, so Q_ε involves only the prime powers $n \leq e^{2a}$: each $\lambda_{\min}(\frac12+\varepsilon,a)$ is the bottom of the spectrum of an explicit form. RH is $\lambda_{\min}(\frac12,a) \geq 0$ for every a. Yoshida [349, Theorem 1] proved positivity of Weil's form for $a \leq \frac12\log 2$, the range in which no prime enters, Bombieri [38] treated other test functions of small support, and Connes and Consani reproved Yoshida's range from the semilocal trace formula [59]. Zhu certifies the lower bound 8.9×10^{-18} for $\lambda_{\min}(\frac12,a)$ at $a = 0.8$ (Corollary 6.3 of [350]), and reports an empirical law for the decay of $\lambda_{\min}(\frac12,a)$ [350] (Chapter 18), and Connes, Consani and Moscovici find that the Fourier transform of the lowest eigenvector of a truncated form reproduces the low zeros to high accuracy [60]. The last two papers are cited as preprints.

Theorem 20.2.1 (The ladder at depth $\varepsilon > 0$). *Let $\varepsilon > 0$ be to the right of the zeros, and suppose that $v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$ for $\theta \neq 0$, with $v''_{1/2+\varepsilon}(0) > 0$ and $\liminf_{|\theta|\to\infty} v_{1/2+\varepsilon}(\theta) > v_{1/2+\varepsilon}(0)$.*

(a) $\lambda_{\min}(\frac12+\varepsilon,a)$ is nonincreasing in a, and $\lambda_{\min}(\frac12+\varepsilon,a) \geq 2b_{1/2+\varepsilon} > 0$ for every a.

(b) $\lambda_{\min}(\frac12+\varepsilon,a) \to 2b_{1/2+\varepsilon}$ as $a\to\infty$, and there are constants $c, a_0 > 0$ with

$$\frac{c}{a^2} \leq \lambda_{\min}\big(\tfrac12+\varepsilon,a\big) - 2b_{1/2+\varepsilon} \leq \frac{\pi^2 v''_{1/2+\varepsilon}(0)/4 + o(1)}{a^2} \qquad (a \geq a_0).$$

(c) At $\varepsilon = 0$, with none of the hypotheses above, under RH, $\lambda_{\min}(\frac12,a) \geq 0$ and $\lambda_{\min}(\frac12,a) = O(a^{-N})$ for every N, so $\lambda_{\min}(\frac12,a) \to 0$; if RH fails, $\lambda_{\min}(\frac12,a) < 0$ for all large a.

Proof. (a) is Theorem 17.8.5(i) of Chapter 17 at $\alpha = \frac12+\varepsilon$, whose $m_\alpha = \inf_\theta v_\alpha(\theta)$ is b_α here, the minimum being at the origin; and $b_{1/2+\varepsilon} = \sum_\rho \mathrm{Re}(\frac12+\varepsilon-\rho)^{-1} > 0$ because every term is positive. We use the first step of that proof: by (20.1.1) and Parseval, $\|\hat h\|_2^2 = 2\pi\|h\|_2^2$, so $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$, the average of $v_{1/2+\varepsilon}$ against the probability density $|\hat h|^2/\|\hat h\|_2^2$, which is at least $2\min v_{1/2+\varepsilon} = 2b_{1/2+\varepsilon}$.

(b) Write $v = v_{1/2+\varepsilon}$. Upper bound: take $h(u) = \cos(\pi u/2a)$ on $[-a,a]$. Then $\langle\theta^2\rangle_h = \|h'\|_2^2/\|h\|_2^2 = \pi^2/4a^2$, and the normalised density satisfies $|\hat h(\theta)|^2/\|\hat h\|_2^2 =$

$O(a^{-3}\theta^{-4})$ for $|\theta| \geq 1/a$. Write $v(\theta) = v(0) + \frac{1}{2}v''(0)\theta^2 + R(\theta)$, with $|R(\theta)| \leq C\theta^4$ for $|\theta| \leq \eta$ and $|R(\theta)| \leq C\theta^2 \log(2 + |\theta|)$ for all θ. Splitting at $|\theta| = \eta$ gives $\langle R \rangle_h = O(\eta a^{-3}) + O(a^{-3}\eta^{-1})$, and with $\eta = a^{-1/2}$ this is $o(a^{-2})$. Hence $\lambda_{\min}(\frac{1}{2} + \varepsilon, a) \leq 2v(0) + v''(0)\pi^2/4a^2 + o(a^{-2})$. Lower bound: by the hypotheses there is $c_1 > 0$ with $v(\theta) - v(0) \geq c_1 \min(\theta^2, 1)$. For h supported in $[-a, a]$ the fraction of $\|\hat{h}\|_2^2$ carried by $|\theta| \leq \kappa/a$ is at most the largest eigenvalue $\ell_0(\kappa)$ of the time and band limiting operator with product κ, which is less than 1 for every finite κ and tends to 0 with κ [307]. So $\langle \min(\theta^2, 1) \rangle_h \geq (\kappa/a)^2(1 - \ell_0(\kappa))$ once $\kappa \leq a$, which is the lower bound with $c = 2c_1\kappa^2(1 - \ell_0(\kappa))$. Convergence $\lambda_{\min}(\frac{1}{2} + \varepsilon, a) \to 2b_{1/2+\varepsilon}$ follows from the two bounds.

(c) Under RH, $Q_0(h) = \sum_\gamma |\hat{h}(\gamma)|^2 \geq 0$ by Theorem 20.1.1. Fix a real $h_1 \in C_c^\infty[-1, 1]$, $h_1 \neq 0$, and put $h_a(u) = h_1(u/a)$. Then $\hat{h}_a(\theta) = a\hat{h}_1(a\theta)$ with $|\hat{h}_1(t)| \leq C_N|t|^{-N}$, so $Q_0(h_a)/\|h_a\|_2^2 \leq C'_N a^{1-2N} \sum_\gamma \gamma^{-2N}$, every $|\gamma|$ being larger than 14. If RH fails, Weil's criterion gives $h \in C_c^\infty$, which may be taken real, with $Q_0(h) < 0$; it is supported in some $[-a_0, a_0]$, and by monotonicity $\lambda_{\min}(\frac{1}{2}, a) \leq \lambda_{\min}(\frac{1}{2}, a_0) < 0$ for $a \geq a_0$. □

At $\varepsilon = \frac{1}{2}$ the hypotheses of the theorem hold with no assumption on the zeros: the level is to the right of the zeros because ζ has no zeros on $\operatorname{Re} s = 1$, and Proposition 9.1.11 gives the minimum at the origin, with $v_1''(0) > 0$, and in its proof $v_1 > 0.05 > b_1$ for $|\theta| \geq 2000$, which is the third condition. For $0 < \varepsilon < \frac{1}{2}$ the level is to the right of the zeros exactly when the closed half plane $\operatorname{Re} s \geq \frac{1}{2} + \varepsilon$ is free of zeros, which the announced half plane gives only for $\varepsilon > \frac{3}{8}$; the minimum at the origin is certified for $0 < |\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, $H_0 = 3 \times 10^{12}$, by Theorem 9.1.18(ii) (Table 9.4), and its holding on every line with $\frac{1}{2} < \alpha < 1$ jointly is equivalent to RH (part (vi) there). The third condition is not known for $0 < \varepsilon < \frac{1}{2}$ without RH; under RH it holds, because $v_{1/2+\varepsilon}$ is then a sum of Poisson kernels of the zeros and tends to infinity.

Remark 20.2.2. At $\varepsilon = \frac{1}{2}$ the hypotheses of Theorem 20.2.1(a) and (b) hold, as just said, and $2b_1 = 2\frac{\xi'}{\xi}(1) = 2\big(1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi\big) = 2 + \gamma_E - \log 4\pi = 0.0461914\ldots$. So for every $a > 0$ and every bounded real h of bounded variation supported in $[-a, a]$,

$$Q_{1/2}(h) \;\geq\; \big(2 + \gamma_E - \log 4\pi\big)\,\|h\|_2^2 = 0.0461914\ldots\,\|h\|_2^2,$$

and by (20.1.1) this is Proposition 9.1.11 read through Plancherel, and Theorem 17.8.5(i) at $\alpha = 1$, which is Corollary 17.8.8 of Chapter 17; what the ladder adds is the rate in a of Theorem 20.2.1(b).

20.2.1 Numerical results

We computed $\lambda_{\min}(\frac{1}{2} + \varepsilon, a)$ variationally, with h in the span of the first M Legendre polynomials $q_0, \ldots, q_{M-1}$ on $[-a, a]$, normalised in L^2: the correlations $q_j * \tilde{q}_k$ by

Gauss Legendre quadrature, which is exact for polynomials; the archimedean and pole integrals by Gauss Legendre quadrature on $[0, 2a]$, plus the exact tail of the $g(0)$ term; and the primes exactly. The script `dc_ladder.py` (`code/README.md`) repeats this for $M = 24$ to 128 and computes the same eigenvalue on a second basis, the normalised Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, for $N = 24$ to 1536, from the matrices of the module `yo_matrix.py` of Chapter 17. Every computed eigenvalue is an upper bound for $\lambda_{\min}$, since a finite basis is a variational class, and needs no hypothesis on the zeros, because Q_ε is an explicit finite form; the hypotheses of Theorem 20.2.1 enter only in what the numbers are compared with. Rounding is not the limit: double precision agrees with thirty digit arithmetic to 1.4×10^{-14} at most, and 1.5 times the quadrature nodes changes the eigenvalue by at most 2×10^{-15}. The limit is the size of the basis.

Between $M = 24$ and $M = 32$ the Legendre bounds change by at most 2.6×10^{-6}, but they are not converged at that level: for $\varepsilon > 0$ the $M = 32$ values lie above the converged values defined next by 0.6×10^{-7} to 1.8×10^{-6}, so the change between two sizes understates the error. The Fourier sections are nested, and their eigenvalues converge like a power, $\lambda_N - \lambda_\infty \propto N^{-p}$ with p between 1.18 and 1.69 over the whole grid. The *converged value* is the power law extrapolation from the three largest sizes $N = 384$, 768, 1536, and its uncertainty is the shift of that value when the largest size is dropped (sizes 192, 384, 768): at most 2.7×10^{-8} at $a = 0.25$, 3.0×10^{-9} at $a = 0.5$ and 1.7×10^{-9} for $a \ge 0.75$. This is an error estimate and not a bound. Two checks support it. The Legendre bound with $M = 128$ lies above the converged value in all 40 entries with $\varepsilon > 0$, by 1.8×10^{-8} to 1.5×10^{-7}, as an upper bound must. And the converged values of Table 17.2 of Chapter 17, obtained there by Richardson extrapolation of the sizes $N = 96$, 192, 384, agree with the values computed here for the same cells to 2.0×10^{-8} at most over its twenty entries with $\alpha > \frac{1}{2}$ and to 4.1×10^{-9} for $a \ge 0.8$.

Table 20.1 shows the converged values for $\varepsilon > 0$ and, at $\varepsilon = 0$, for $a = 0.25$ and 0.5, and Table 20.2 the rung at $\varepsilon = 0$ for $0.25 \le a \le 0.5$. Table 17.2 of Chapter 17 and Table 18.1 of Chapter 18 list eigenvalues of the same kind for other windows and levels; the level 0.375 of Table 20.1 is that of the announced half plane. Figure 20.1 plots the eigenvalue on a logarithmic scale.

For every level in Table 20.1 the converged values decrease to $2b_{1/2+\varepsilon}$ and never cross it (nor does any computed upper bound, in either basis), as Theorem 20.2.1(a) predicts when its hypotheses hold; at $\varepsilon = \frac{1}{2}$ this is Remark 20.2.2, and at $\varepsilon = \frac{3}{8}$ the announced half plane gives $\lambda_{\min}(\frac{7}{8}, a) \ge 0$ (Theorem 20.1.2), while $\lambda_{\min} \ge 2b_{7/8}$ needs in addition the minimum of $v_{7/8}$ at the origin, which Theorem 9.1.18(ii) certifies only up to height 3×10^{12}. The margin is linear in ε. With no hypothesis $2b_{1/2+\varepsilon} \le 2V_\xi\, \varepsilon$, where $V_\xi = (\log \xi)''(\frac{1}{2}) = 0.04620998623$ is the constant of Theorem 9.1.18(i), and $2b_{1/2+\varepsilon}/\varepsilon \to 2V_\xi = 0.0924200$ as $\varepsilon \downarrow 0$; under RH $V_\xi = 2p_1$ with $p_1 = \sum_{\gamma>0} \gamma^{-2} = 0.0231050$.

The rate of approach changes character at the line. Table 20.3 compares the

a	$\varepsilon = 0.5$	0.375	0.2	0.1	0.05	0
0.25	0.0894856	0.0756372	0.0560458	0.0447429	0.0390618	0.0333609
0.5	0.0495266	0.0372606	0.0200130	0.0100860	0.0050779	9.33×10^{-7}
0.75	0.0474318	0.0355980	0.0190070	0.0095123	0.0047593	
1	0.0468396	0.0351417	0.0187506	0.0093779	0.0046899	
1.5	0.0464620	0.0348541	0.0185931	0.0092974	0.0046489	
2	0.0463394	0.0347613	0.0185428	0.0092720	0.0046361	
3	0.0462555	0.0346979	0.0185087	0.0092548	0.0046275	
4	0.0462270	0.0346764	0.0184971	0.0092490	0.0046246	
$2b_{1/2+\varepsilon}$	0.0461914	0.0346497	0.0184828	0.0092418	0.0046210	0

Table 20.1: The ladder $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$, converged values: the limit of the variational upper bounds on the Fourier modes $e^{i\pi nu/a}$, $|n| \le N$, by power law extrapolation from $N = 384$, 768, 1536 (`code/dc_ladder.py`, parts `grid` for $\varepsilon > 0$ and `y0` for $\varepsilon = 0$, `data/dc_ladder.json`). The uncertainty of an entry is the one defined in the text; it is an error estimate and not a bound. The last row is the limit $2b_{1/2+\varepsilon} = 2\,\xi'/\xi(\frac{1}{2}+\varepsilon)$ of Theorem 20.2.1 where its hypotheses hold (for $\varepsilon < \frac{1}{2}$ they are conditional, and at $\varepsilon = 0$ the limit is 0 under RH), computed by `code/dc_weil.py`, part W2 (`data/dc_weil.json`). At $\varepsilon = 0$ the entries for $a \ge 0.75$ are omitted: in double precision they are below the rounding level of about 10^{-13}, and the upper bounds from extended precision are given in the text.

a	$M = 32$	converged	uncertainty
0.25	3.3363×10^{-2}	3.3361×10^{-2}	2.7×10^{-8}
0.3	7.5722×10^{-3}	7.5715×10^{-3}	1.4×10^{-8}
$\frac{1}{2}\log 2$	1.3294×10^{-3}	1.3293×10^{-3}	3.4×10^{-9}
0.35	1.1938×10^{-3}	1.1935×10^{-3}	1.3×10^{-9}
0.4	1.8173×10^{-4}	1.8126×10^{-4}	1.1×10^{-9}
0.45	1.6232×10^{-5}	1.6149×10^{-5}	1.7×10^{-10}
0.5	9.4039×10^{-7}	9.3317×10^{-7}	1.4×10^{-11}

Table 20.2: The rung at $\varepsilon = 0$ for $a \le 0.5$: the Legendre bound with $M = 32$, the converged value (Fourier modes to 1536, power law extrapolation) and its uncertainty, the shift of the extrapolation when $N = 1536$ is dropped, an error estimate and not a bound. At $a = \frac{1}{2}\log 2 = 0.3466$, the largest window in which no prime enters, the value is 1.3293×10^{-3}. Script `code/dc_ladder.py`, part `y0`, data `data/dc_ladder.json`.

scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ of the converged values with the upper constant $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): the ratio falls towards 1 as a grows. An upper bound for $\lambda_{\min}$ that approaches the constant from above says nothing about whether the true limit is that constant or a smaller one, and the

converged values are extrapolations, so we fit them: a least squares fit on $2 \le a \le 7$ (prime powers to e^{14} at $a = 7$) of $a^2(\lambda_{\min} - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$ gives $C_0 = 1.0002$, 1.0003, 1.0005 at $\varepsilon = \frac{1}{2}$, 0.2, 0.1, with an uncertainty from the errors of the data of 0.00004, 0.0001, 0.0002, and over the choice of the window ($a \ge 1.5$, 2 or 3) and of the degree (2 or 3) C_0 ranges over 0.9999 to 1.0004, 0.9995 to 1.0010 and 0.9991 to 1.0016. The data therefore support the limit $\bar{c}_\varepsilon$ to within 0.04 to 0.2 percent, with a positive correction $C_1 = 0.139$, 0.159, 0.166 and $C_2 = 0.037$, 0.045, 0.051 and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$; the term in a^{-2} is needed, a fit by $C_0 + C_1/a$ alone being rejected by its residuals. This is numerical evidence and not a proof (Problem 20.2). It also indicates, numerically, that $\bar{c}_\varepsilon$ is not a bound at finite a: the scaled excess exceeds it at every computed a ($1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$).

At $\varepsilon = 0$ and $a = 0.35$ the 1.20×10^{-3} of Table 17.2 is the finite section with $N = 24$, an upper bound for the value 1.19×10^{-3} of Table 20.2. From $a = 0.75$ on double precision returns only rounding noise, below about 10^{-13}, and extended precision gives the upper bounds $\lambda_{\min}(\frac{1}{2}, a) < 4.4 \times 10^{-15}$, 1.92×10^{-17} and 7.3×10^{-30} at $a = 0.75$, 0.8 and 1 (Fourier modes $|n| \le 48$, 48 and 60, at 60, 60 and 80 digits). They fall steeply with the size of the section and depend on the basis (the Legendre bounds with $M = 24$ are 2.7×10^{-13}, 2.7×10^{-14} and 2.1×10^{-16}, and with $M = 96$ they are 3.7×10^{-15}, 1.66×10^{-17} and 6.3×10^{-30}, each of these three below the Fourier bound above), so they are upper bounds and not the order of the eigenvalue. Zhu's certified lower bound at $a = 0.8$ is 8.9×10^{-18} (Corollary 6.3 of [350], Chapter 18), and the bounds 1.92×10^{-17} and 1.66×10^{-17} lie above it, the second within a factor 1.9.

The decay of the rung at $\varepsilon = 0$ is stated four ways in this book: Section 15.6.1 observes that the least eigenvalue on hat functions falls roughly geometrically in the support length $L = 2a$, up to $L = 8$; Chapter 17 observes that the finite sections follow $\exp(-2c\,ae^{2a})$ with $4.1 \le c \le 5.1$ for $0.25 \le a \le 0.8$, and does not prove it; Zhu proves, under RH and for large a, that the least eigenvalue over real even test functions, and so $\lambda_{\min}(\frac{1}{2}, a)$, is at most $\exp(-ae^a)$ (Theorem 1.3 of [350], Chapter 18, a preprint); and Theorem 20.2.1(c) proves, under RH, that the rung is $O(a^{-N})$ for every N. The four do not conflict: each of these falls is faster than any power of a, and Theorem 20.2.1(c) is the weakest of the conditional bounds and the only one whose proof is in the text.

The proof of Theorem 20.2.1(b) can be run with numbers, and Table 20.4 does so (`dc_rate.py`, parts `lower` and `trial`). Write $\psi(\theta) = v_{1/2+\varepsilon}(\theta) - v_{1/2+\varepsilon}(0)$. The infimum c_1 of $\psi(\theta)/\min(\theta^2, 1)$ is the limit $v''_{1/2+\varepsilon}(0)/2$ at the origin, on a grid of 600 points to height 40 and with the assumption that ψ stays above its minimum on $[20, 40]$ beyond 40, and $\max_\kappa \kappa^2(1 - \ell_0(\kappa)) = 0.5348$ at $\kappa = 1.54$, where $\ell_0(1) = 0.5726$ and $\ell_0(2) = 0.8806$. So the lower constant of the theorem is $c = 2c_1 \cdot 0.5348 = 0.2167\,\bar{c}_\varepsilon$ for $a \ge 1.54$. Using at every radius at once the bound $P(|\theta| \le r) \le \ell_0(ar)$ for the probability density $|\hat{h}|^2/\|\hat{h}\|_2^2$ gives the layer

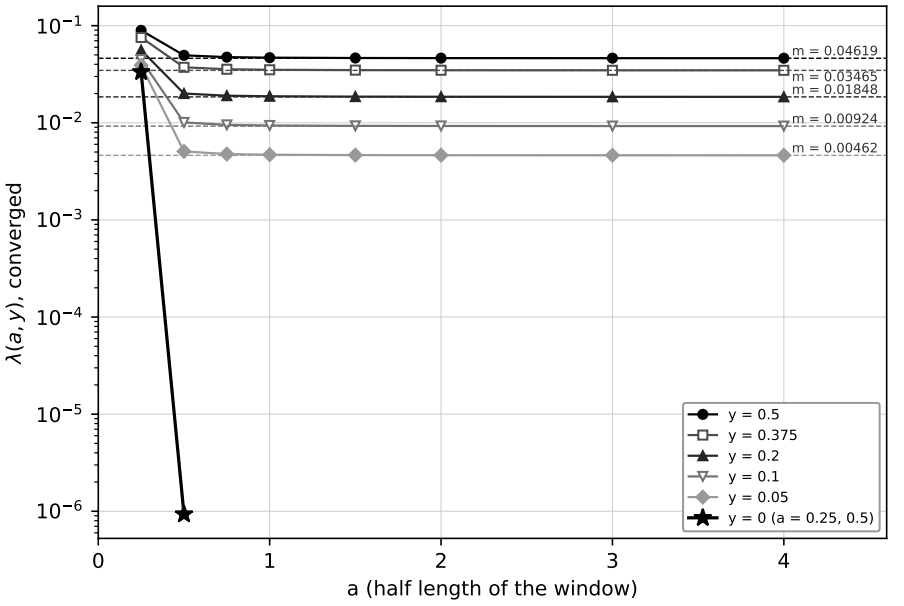


Figure 20.1: The converged ladder on a logarithmic scale: $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ against the window half length a at $a = 0.25$, 0.5, 0.75, 1, 1.5, 2, 3, 4, for $\varepsilon = 0.5$, 0.375, 0.2, 0.1, 0.05 (five marker styles in black and grey), each with the value $2b_{1/2+\varepsilon}$, the limit where the hypotheses of Theorem 20.2.1 hold, as a dashed line labelled m, and for $\varepsilon = 0$ at $a = 0.25$ and 0.5 only (heavy line with stars). The values are those of Table 20.1; the legend and the axis write y and $\lambda(a, y)$ for ε and $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$. Drawn by `code/dc_ladder.py` (part `fig`) from `data/dc_ladder.json`.

cake bound $\lambda_{\min} - 2b_{1/2+\varepsilon} \geq 2\int_0^\infty \tilde{\psi}'(r)\,(1-\ell_0(ar))\,dr$, with $\tilde{\psi}$ the nondecreasing minorant of ψ in $|\theta|$; it is $0.666\,\bar{c}_\varepsilon$ at $a = 4$, $\varepsilon = \frac{1}{2}$ (3.66×10^{-4}), and tends to $8I_1/\pi^2\,\bar{c}_\varepsilon = 0.683\,\bar{c}_\varepsilon$ as $a \to \infty$, with $I_1 = \int_0^\infty \kappa\,(1-\ell_0(\kappa))\,d\kappa = 0.8425$. These lower bounds are floating point computations on a grid, numerical and not certificates, and for $\varepsilon < \frac{1}{2}$ the identity $Q_\varepsilon(h)/\|h\|_2^2 = 2\langle v_{1/2+\varepsilon}\rangle_h$ behind them presupposes that the level is to the right of the zeros.

The cosine trial function of the proof gives upper bounds, computed from the prime side (Table 20.4, last column). At $\varepsilon = \frac{1}{2}$ the ratio of its scaled excess to $\bar{c}_\varepsilon$ is 1.52, 1.37, 1.17 and 1.11 at $a = 1$, 2, 4 and 6, and a times its remainder over $\bar{c}_\varepsilon$ is 0.65 to 0.68 for $3 \leq a \leq 8$, a positive correction of order a^{-1}. At $\varepsilon = 0.2$ and 0.1 the remainder oscillates in a (a times it over $\bar{c}_\varepsilon$ lies between 1.4 and 2.2 at $\varepsilon = 0.2$ and between 2.3 and 4.8 at $\varepsilon = 0.1$ for $1.5 \leq a \leq 8$), because the tail of $|\hat{h}|^2$, of order $\cos^2(a\theta)/\theta^4$, meets the peaks of $v_{1/2+\varepsilon}$ at the zeros, of height $1/\varepsilon$, and the cosine is a poor trial function for small ε.

ε	$\pi^2 v''_{1/2+\varepsilon}(0)/4$	$a=1$	$a=2$	$a=3$	$a=4$
0.5	5.485	6.48	5.92	5.76	5.69
0.2	2.200	2.68	2.40	2.33	2.29
0.1	1.100	1.36	1.21	1.17	1.15

Table 20.3: The scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$ in units of 10^{-4}, against the constant $\pi^2 v''_{1/2+\varepsilon}(0)/4$ of Theorem 20.2.1(b): converged $\lambda_{\min}$ (Fourier modes to 1536, power law extrapolation, as in Table 20.1), rounded to three digits. The uncertainty of an entry, the shift of the extrapolation times a^2, is at most 4×10^{-5} in these units, an error estimate and not a bound. Script `code/dc_rate.py`, parts `const` and `table`, data `data/dc_rate.json`.

ε	a	lower (κ)	lower (layer cake)	converged	upper (trial)
0.5	1	0.950 (1.00)	3.834	6.482	8.353
0.5	2	1.189 (1.54)	3.692	5.919	7.493
0.5	4	1.189 (1.54)	3.656	5.690	6.413
0.5	6	1.189 (1.54)	3.661	5.619	6.095
0.2	1	0.381 (1.00)	1.538	2.678	4.291
0.2	2	0.477 (1.54)	1.481	2.401	4.584
0.2	4	0.477 (1.54)	1.466	2.295	3.258
0.2	6	0.477 (1.54)	1.468	2.262	2.857
0.1	1	0.191 (1.00)	0.769	1.361	2.813
0.1	2	0.239 (1.54)	0.741	1.206	3.745
0.1	4	0.239 (1.54)	0.733	1.150	2.331
0.1	6	0.239 (1.54)	0.734	1.133	1.863

Table 20.4: Bounds on the scaled excess $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon})$, in units of 10^{-4}: the lower bound of the proof of Theorem 20.2.1(b) with the single radius $\kappa \le a$ that maximises $\kappa^2(1-\ell_0(\kappa))$ (the radius in brackets), the layer cake lower bound, the converged value (as in Table 20.3, extended to $a=6$ in `data/dc_rate.json`), and the value of the trial function $\cos(\pi u/2a)$ computed from the prime side, which is an upper bound. Script `code/dc_rate.py`, parts `lower` and `trial`.

20.3 What can be done and what is open

20.3.1 What can be done

Partly, and the parts can be stated exactly: the deconvolution of the Thorin density always exists, as Weil's distribution computed from the primes with no hypothesis (Theorem 20.1.1), and its positivity to depth ε is a zero free half plane (Theorem 20.1.2), known for $\varepsilon = \frac{1}{2}$ and announced for $\varepsilon = \frac{3}{8}$, each rung $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) \ge 0$ being a finite problem in the primes up to e^{2a} whose limit, under the

hypotheses of Theorem 20.2.1, is a margin $2b_{1/2+\varepsilon}$ of about $0.092\,\varepsilon$, approached at a rate of order a^{-2}.

A rung is finite information in the sense of Section 5.2.2 (Theorem 5.2.1 is the rigorous form of this for the heat trace), so heuristically no finite set of rungs sees every height, and at $\varepsilon = 0$ the margin is 0 and, under RH, the approach to it is faster than any power of a (Theorem 20.2.1(c)): this is the deconvolution obstruction, consistent with the theorem of Rodgers and Tao that RH, if true, holds with no slack [289].

20.3.2 Open problems

Problem 20.1 (Certified rungs). Certify $\lambda_{\min}(\frac{1}{2}+\varepsilon, a) > 0$ in interval arithmetic for $\varepsilon \in \{0.05, 0.1, 0.2, 0.375\}$ and $a = 4$, which by monotonicity in a covers every $a \le 4$, and determine how far in a a certificate can be pushed at fixed ε. Each certificate is unconditional, and at $\varepsilon = \frac{3}{8}$ each agrees with what the announced half plane gives, namely $\lambda_{\min} \ge 0$ (Theorem 20.1.2), and can be checked independently of it.

Problem 20.2 (The exact rate). Prove that, under the hypotheses of Theorem 20.2.1, $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) \to \pi^2 v''_{1/2+\varepsilon}(0)/4$, and find the next term. The converged values of Table 20.3 exceed the constant by a relative amount that falls like a^{-1}: a least squares fit on $2 \le a \le 7$ gives $a^2(\lambda_{\min}(\frac{1}{2}+\varepsilon, a) - 2b_{1/2+\varepsilon}) = \bar{c}_\varepsilon(C_0 + C_1/a + C_2/a^2)$, with $\bar{c}_\varepsilon = \pi^2 v''_{1/2+\varepsilon}(0)/4$, $C_0 = 1.0002,\ 1.0003,\ 1.0005$, $C_1 = 0.139,\ 0.159,\ 0.166$ and $C_2 = 0.037,\ 0.045,\ 0.051$ at $\varepsilon = \frac{1}{2},\ 0.2,\ 0.1$, and residuals below $1.1 \times 10^{-4}\,\bar{c}_\varepsilon$. So, numerically, the correction is positive and $\bar{c}_\varepsilon$ is not a bound at finite a (the scaled excess is $1.024\,\bar{c}_\varepsilon$ at $a = 6$, $\varepsilon = \frac{1}{2}$); the fit is numerical evidence for the limit, to within 0.04 to 0.2 percent, and for the order a^{-1} of the correction, and neither is proved.

Problem 20.3 (The transition at the line). Describe $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ jointly as $a \to \infty$ and $\varepsilon \to 0$. For fixed $\varepsilon > 0$, under the hypotheses of Theorem 20.2.1, the limit is $2b_{1/2+\varepsilon}$, which is at most $2V_\xi\,\varepsilon$ and asymptotic to it as $\varepsilon \downarrow 0$, and the rate is algebraic; at $\varepsilon = 0$, under RH, the decay is faster than any power. On which curves $\varepsilon = \varepsilon(a) \to 0$ does $\lambda_{\min}(\frac{1}{2}+\varepsilon, a)$ stay comparable to $2b_{1/2+\varepsilon}$, and does the crossover occur at $\varepsilon \approx$ the mean spacing of the zeros at height e^{2a}?

Problem 20.4 (Uniformity from structure). Find a structural reason, valid for all a, why $Q_\varepsilon \ge 0$ on $L^2[-a, a]$ for some fixed $\varepsilon < \frac{1}{2}$, using properties of the archimedean and pole terms that dominate the primes up to e^{2a} uniformly. By Theorem 20.1.2 any such argument proves a zero free half plane $\operatorname{Re} s > \frac{1}{2} + \varepsilon$.

Chapter 21

Thorin representations, Dirichlet L functions and the Linnik constant

For a number field F, the reciprocal of the completed Dedekind zeta function, tilted by $u_0 \geq 0$ and so read at the point $\frac{1}{2} + \sqrt{u_0}$, is a generalised gamma convolution (GGC) exactly when every nontrivial zero lies on the critical line or is real and at distance less than $\sqrt{u_0}$ from the centre, with no zero at the centre when $u_0 = 0$, and its Thorin measure is then the zero counting measure shifted by u_0 (Theorem 21.2.3). For cyclotomic fields the Thorin property at the centre, for every modulus, is equivalent to GRH for all Dirichlet L functions together with $L(\frac{1}{2}, \chi) \neq 0$ for every primitive character χ, so it implies the GRH value $K = 7$ of the Linnik constant (Corollary 21.7.1); this is weaker than the known conditional result, because the nonvanishing at the centre is an extra hypothesis, and with the present methods it gives nothing beyond what GRH gives (Section 21.1.2).

What is not claimed. Nothing here proves RH, and nothing here lowers the Linnik constant. A statement that rests on RH, on GRH or on the zero free half plane of October 2026, which has not been refereed, says so. A number not attributed to a source is computed in floating point at the precision stated.

21.1 Linnik's problem and where zeros enter

21.1.1 The theorem and the records

Linnik [182, 183] proved that there is an absolute constant K such that every sufficiently large even integer N can be written

$$N = p_1 + p_2 + 2^{k_1} + \cdots + 2^{k_K}, \qquad p_1, p_2 \text{ prime}, \quad k_i \in \mathbb{Z}_{\geq 0}. \tag{21.1.1}$$

The 1951 paper assumes GRH and the 1953 paper does not; neither gives a value of K. Gallagher [90] recast the argument so that explicit constants became possible, and the values of K that followed were lowered in a series of papers [186, 187, 177, 337, 178], from 54000 (Liu, Liu and Wang) to 1906 (Li), and under GRH to 200

(Liu, Liu and Wang), until Heath-Brown and Puchta [112] reached the values in the first line of the table below.

The best values on record are the following.

Authors	K unconditionally	K under GRH
Heath-Brown and Puchta [112]	13	7
Pintz and Ruzsa [241]		7
Pintz and Ruzsa [242]	8	

Platt and Trudgian [246] sharpened the numerical constants in the method of [112].

21.1.2 Where K is determined

Fix a small constant $\varpi > 0$, write $e(x) = \exp(2\pi i x)$ and $L = \lfloor \log_2(N/2K) \rfloor$, and put

$$\mathcal{S}(\alpha) = \sum_{\varpi N < p \le N} e(p\alpha), \qquad \mathcal{D}(\alpha) = \sum_{1 \le \nu \le L} e(2^\nu \alpha).$$

The number of representations (21.1.1) with both primes in $(\varpi N, N]$ and the exponents restricted to $1 \le k_i \le L$, so that the powers of two add up to at most $N/2$, is

$$R(N) = \int_0^1 \mathcal{S}(\alpha)^2 \, \mathcal{D}(\alpha)^K \, e(-N\alpha)\, d\alpha. \tag{21.1.2}$$

Let $\mathfrak{M} \subset [0,1]$ be the set of major arcs, the neighbourhoods of rationals with small denominator, and $\mathfrak{m} = [0,1] \setminus \mathfrak{M}$ the minor arcs.

We follow the argument of Heath-Brown and Puchta [112], whose constants we quote with their pinpoints; the proof has five steps.

(1) *Major arcs.* Gallagher's method, with the major arcs $q \le P$, $|\alpha - a/q| \le P/(qN)$ and $P \le N^{45/154 - 4\varpi}$, gives on $\mathfrak{M}$ a main term of at least

$$2.7895\,(1 - 3\varpi)\, C_0\, N (\log 2)^{-2} L^{K-2}$$

(Lemma 7 of [112], stated there for $K \ge 9$), where $C_0 = \prod_{p>2} \big(1 - (p-1)^{-2}\big) = 0.6601\ldots$ is the twin prime constant, and 2.7895 is $2\big(1 + \frac{1}{3}(1 - 2^{-6}) + \frac{1}{15}(1 - 4^{-6})\big) = 2.78955\ldots$ rounded down, the contribution of the divisors 1, 3 and 5 to the singular series of Goldbach's problem averaged over the powers of two (equation (25) there). No hypothesis on zeros is used, and no explicit numerical zero free region. [112] control the error terms, character sums over primes, by the zero density estimates of Huxley and Jutila, $\sum_{r \le R} N(\sigma, r, T) \ll (R^2 T)^{\kappa(\sigma)(1-\sigma)}$ with $\kappa(\sigma) = \frac{3}{2-\sigma} + \varpi$ for $\frac{1}{2} \le \sigma \le \frac{3}{4}$ and $\frac{12}{5} + \varpi$ for $\frac{3}{4} \le \sigma \le 1$, which also fix the restriction on P, and, for the few characters with a zero near 1 and for the principal character,

by the zero free regions of Iwaniec (for nonreal characters, with an implied constant that is not explicit) and of Vinogradov. Zeros close to $\sigma = 1$ play no role.

(2) *Minor arc mean square.* $\int_0^1 |\mathcal{S}(\alpha)\mathcal{D}(\alpha)|^2 d\alpha$ counts the solutions of $p_1 - p_2 = 2^\mu - 2^\nu$ with $\mu, \nu \le L$. Chen's sieve bound for the number of prime differences, $r(n) \le C_0 C_1 h(n) N (\log N)^{-2}$ with $C_1 = 7.8342$ (Theorem 3 of Chen's 1978 paper on Goldbach's problem and the sieve methods, as used in [112]), bounds the integral over $[0,1]$ from above, and a lower bound for the part over $\mathfrak{M}$ is subtracted, which gives $\int_{\mathfrak{m}} |\mathcal{S}\mathcal{D}|^2 d\alpha \le (13.968 + o(1))\, C_0 N (\log 2)^{-2}$ (Lemmas 8 to 10 of [112]).

(3) *Minor arc sup bound for the primes.* $\mathcal{S}(\alpha) \ll N^{\theta + o(1)}$ on $\mathfrak{m}$, with $\theta = 263/308$ unconditionally, from Theorem 3.1 of Vaughan's book on the Hardy Littlewood method with $P = N^{45/154 - 4\varpi}$, and $\theta = 3/4$ under GRH, from Lemma 12 of Baker and Harman (Section 6 of [112]).

(4) *Large deviations for the powers of two.* The set $\mathcal{A}_\lambda = \{\alpha : |\mathcal{D}(\alpha)| \ge \lambda L\}$ has measure $\ll N^{-E(\lambda)}$, with $E(0.863665) > \frac{109}{154}$ and $E(0.722428) > \frac{1}{2}$ (Lemma 1 of [112], proved there by a martingale argument suggested by Keith Ball). The contribution of $\mathfrak{m} \cap \mathcal{A}_\lambda$ is $\ll N^{-E(\lambda) + 2\theta + o(1)} L^K$, negligible once $E(\lambda) > 2\theta - 1$, that is $\frac{109}{154}$ unconditionally and $\frac{1}{2}$ under GRH.

(5) *Comparison.* Off $\mathcal{A}_\lambda$, $|\mathcal{D}| < \lambda L$, so the contribution of $\mathfrak{m} \setminus \mathcal{A}_\lambda$ is at most $(\lambda L)^{K-2} \cdot 13.968\, C_0 N (\log 2)^{-2}$, and the count is positive, for N large and ϖ small, when $13.968\, \lambda^{K-2} < 2.7895$ (end of Section 6 of [112]). This needs $K - 2 > 10.99$ at $\lambda = 0.863665$ and $K - 2 > 4.95$ at $\lambda = 0.722428$, which gives $K = 13$ unconditionally and $K = 7$ under GRH.

In this summary a hypothesis on zeros enters only at step (3): GRH lowers θ to $\frac{3}{4}$ and hence lowers λ. In the scheme of [112] the remaining gains come from the constants of steps (1), (2) and (4), which do not depend on a hypothesis on zeros; Elsholtz is reported there to have obtained $K = 12$ unconditionally, in unpublished work, by improving the constants of the first two steps. The refinements of Pintz and Ruzsa [241, 242] differ from this argument in a number of respects, as [112] remark, and we do not describe them. In the scheme of [112] an input that delivered GRH would be worth the passage from $K = 13$ to $K = 7$, and against the records it is the passage from $K = 8$ to $K = 7$.

The zero free half plane of October 2026. In October 2026 OpenAI released a proof, produced with one of its AI models, that $\zeta(s)$ and every Dirichlet L function have no zeros in $\operatorname{Re} s > \frac{7}{8}$ [226], and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [227]. The claims have not been refereed (Section 12.1.2). A zero free region can act in step (3), through the error term of the exponential sum, and in

step (1), through the zero density estimates. For step (3), the exponent $\frac{3}{4}$ under GRH comes from the lemma of Baker and Harman used in [112]: for $\alpha = a/q + \beta$ with $|\beta| \le N^{-1/2}$, GRH gives $\mathcal{S}(\alpha) \ll (\log N)\{N/q + N^{1/2}q^{1/2} + N(q|\beta|)^{1/2}\}$. By Dirichlet's theorem every α has such an approximation with $q \le N^{1/2}$ and $|\beta| \le 1/(qN^{1/2})$, which makes each of the three terms at most $N^{3/4}$ once $q > N^{1/4}$, and $q \le N^{1/4}$ does not occur on the minor arcs because $P = N^{45/154-4\varpi} \ge N^{1/4}$. In the second term GRH enters as the factor $N^{1/2}$, the size of the error term in the explicit formula for $\psi(x, \chi)$. Replacing GRH by a zero free half plane $\operatorname{Re} s > \theta_0$ for all Dirichlet L functions would replace that factor by N^{θ_0}, and the same choice of approximation, with $q \le N^{1-\theta_0}$, would give $\theta = \frac{1}{2}(1 + \theta_0)$. This is a heuristic count: the lemma is proved under GRH and its analogue for a half plane is not proved here. It beats the unconditional exponent $\frac{263}{308}$ only if $\theta_0 < \frac{109}{154} = 0.7078$. The half plane $\operatorname{Re} s > \frac{7}{8}$ gives $\frac{15}{16}$, worse than the unconditional exponent, so used this way it does not lower K. For step (1), the constant 2.7895 is arithmetic, and zeros close to $\sigma = 1$ play no role in [112], so excluding exceptional zeros is not what a half plane would add. Zeros enter the major arcs through the zero density estimates, which fix the restriction $P \le N^{45/154-4\varpi}$ by way of the bound $(\frac{45}{77}\kappa(\sigma) - 1)(1 - \sigma) \le \frac{31}{77}(1 - \sigma) \le \frac{31}{154}$ for the exponent of the estimate (Section 3 there), taken at the left end $\sigma = \frac{1}{2}$; a half plane $\operatorname{Re} s > \theta_0$ removes only the zeros to the right of θ_0, so it does not relax this restriction. Whether a finer combination of the half plane with Vaughan's identity does better, on the major or the minor arcs, is Problem 21.9. In the Thorin language of this chapter, the half plane would give the Thorin property of the shifted clock at every basepoint $\alpha > \frac{7}{8}$ for ξ_F with F abelian over $\mathbb{Q}$ (quadratic and cyclotomic fields in particular, by (21.5.4)) and for every real character (Theorem 21.5.8; for ξ this is the remark after Theorem 1.2.2). Liu [193] claims the half plane $\operatorname{Re} s > B_\flat = 0.87489$ for every Hecke L function of finite order over every number field (Section 16.12.3); if that unrefereed and unformalised claim stands, the same holds at every basepoint $\alpha > B_\flat$ for ξ_F with F any number field, abelian or not.

21.2 The Thorin property at every tilt

21.2.1 Conventions

The centre clock is the book's T (Section 1.2), the variables e_γ are independent standard exponentials as in the book, and $Y = \sqrt{2T}\,Z$, with Z a standard normal variable independent of T, is the variable $Y_{1/2}$ of Section 10.1. The letter T also stands for a height, but only in $N(T)$ and $S(T)$, as in the book, and as the cut height of the anchored tail of Section 21.3.3. The transfer function from the real axis to the critical line is written $\mathfrak{T}(t)$ (Proposition 21.3.6), and $\mathfrak{B}(z)$ is its Blaschke product in $z = t^2$, so that $\mathfrak{T}(t) = \mathfrak{B}(t^2)$ (Proposition 21.3.8). Completed Dirichlet L functions are written $\xi(s, \chi)$, as in the book, and Λ is the von Mangoldt function. The tilt u_0 and the basepoint α are different parameters: the clock tilted by u_0

reads ξ at $\frac{1}{2}+\sqrt{u_0+u}$, the shifted clock at basepoint α reads it at $\alpha+\sqrt{u}$, and the two agree only at the centre. In Section 21.1.2 only, α is also the circle method variable, and N, L, λ, θ, c and ϖ have their circle method meanings there.

21.2.2 Generalised gamma convolutions

A probability law on $[0,\infty)$ is a generalised gamma convolution (GGC) if its Laplace transform has the form

$$\varphi(s)=\exp\Big(-bs-\int_{(0,\infty)}\log\Big(1+\frac{s}{t}\Big)\,U(dt)\Big),\qquad s\ge 0, \tag{21.2.1}$$

with $b\ge 0$ and a nonnegative Thorin measure U satisfying $\int_{(0,1]}|\log t|\,U(dt)<\infty$ and $\int_{(1,\infty)}t^{-1}U(dt)<\infty$; see Bondesson [39] and equation (1.3.1), where the drift is d and the Thorin variable z. We use three standard facts.

(G1) The right side of (21.2.1) extends to an analytic, zero free function on $\mathbb{C}\setminus(-\infty,-t_0]$, where $t_0=\inf\operatorname{supp}U$.

(G2) If $t_j>0$ and $\sum_j 1/t_j<\infty$, then $\prod_j t_j/(t_j+s)$ is the Laplace transform of $\sum_j e_j/t_j$ with e_j independent unit exponentials, a GGC with $b=0$ and $U=\sum_j\delta_{t_j}$.

(G3) If $\inf\operatorname{supp}U>c$ with $c>0$, then $\varphi(-c)<\infty$ and $\varphi(s-c)/\varphi(-c)$ is again of the form (21.2.1), with Thorin measure $U(\cdot+c)$. This is closure under exponential tilting.

The class is also closed under independent sums, positive scaling and weak limits (equation (1.3.1)), and, by a theorem of Bondesson [40], under products of independent GGC variables. It is not closed under products with arbitrary independent positive variables (an exponential times an independent two point variable is a mixture of two exponentials with rates $a\neq b$, whose Thorin measure would be the signed measure $\delta_a+\delta_b-\delta_c$ for a rate c between a and b), under harmonic means (Proposition 7.1.7), or under mixtures (Proposition 21.6.3).

21.2.3 Completed Dedekind zeta functions

Let F be a number field of degree $n=r_1+2r_2$ and discriminant d_F. Put $\Gamma_{\mathbb{R}}(s)=\pi^{-s/2}\Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s)=2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_F(s)=s(s-1)\,|d_F|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\,\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_F(s). \tag{21.2.2}$$

Then ξ_F is entire of order one, real on the real axis, and $\xi_F(s)=\xi_F(1-s)$. For $F=\mathbb{Q}$ it is $2\xi(s)$, with Riemann's $\xi(s)=\frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$. Its zeros are the nontrivial zeros of ζ_F, written $\rho=\frac{1}{2}+w_\rho$ with $|\operatorname{Re}w_\rho|<\frac{1}{2}$.

LEMMA 21.2.1 (Evenness). *$E_F(w) := \xi_F(\frac{1}{2}+w)$ is even and entire, so $E_F(w) = G_F(w^2)$ with G_F entire. The zeros of G_F are the points $z_\rho = w_\rho^2$, one for each pair $\{\rho, 1-\rho\}$.*

Proof. The functional equation gives $E_F(-w) = \xi_F(\frac{1}{2}-w) = \xi_F(\frac{1}{2}+w) = E_F(w)$, and an even entire function is an entire function of w^2. □

LEMMA 21.2.2 (Product formula). *Let Z_F be the multiset of the values $z_\rho \neq 0$, one for each pair of zeros, and let $2m_F$ be the order of vanishing of ξ_F at $\frac{1}{2}$. Then, with absolute convergence since $\sum_{z\in Z_F} |z|^{-1} < \infty$,*

$$G_F(u) = c_F\, u^{m_F} \prod_{z\in Z_F} \Big(1 - \frac{u}{z}\Big), \qquad c_F \neq 0. \tag{21.2.3}$$

Proof. G_F is entire of order $\frac{1}{2}$ because E_F has order one, so Hadamard's theorem gives the product with no exponential factor. Convergence of $\sum |z_\rho|^{-1} = \sum |\rho - \frac{1}{2}|^{-2}$ follows from the Riemann von Mangoldt formula $N_F(T) \asymp T\log T$. □

The function G_F is not normalised, unlike the function G of Section 1.2: for $F = \mathbb{Q}$ one has $G_{\mathbb{Q}} = 2\xi(\frac{1}{2})\, G$.

Under GRH every $z \in Z_F$ equals $-\gamma^2$ for an ordinate $\gamma > 0$, while a real zero $\beta \in (\frac{1}{2}, 1)$, which GRH excludes, would give $z = (\beta - \frac{1}{2})^2 \in (0, \frac{1}{4})$.

21.2.4 The equivalence theorem

For $u_0 \geq 0$ with $\xi_F(\frac{1}{2} + \sqrt{u_0}) \neq 0$, tilt the centre clock by u_0:

$$\varphi_{F,u_0}(s) := \frac{\xi_F(\frac{1}{2}+\sqrt{u_0})}{\xi_F(\frac{1}{2}+\sqrt{u_0+s})} = \frac{G_F(u_0)}{G_F(u_0+s)}, \qquad s \geq 0. \tag{21.2.4}$$

By Lemma 21.2.1 the branch of the square root does not matter. For $F = \mathbb{Q}$ this is the function $\widetilde{\varphi}_{u_0}$ of Section 19.1.1.

THEOREM 21.2.3 (Thorin property of the tilted clock). *Let $u_0 \geq 0$. The following are equivalent.*

(i) $\xi_F(\frac{1}{2}+\sqrt{u_0}) \neq 0$ and φ_{F,u_0} is the Laplace transform of a GGC.

(ii) Every z_ρ is real with $z_\rho < u_0$, and $m_F = 0$ if $u_0 = 0$. Equivalently, every nontrivial zero of ζ_F lies on the critical line or is a real zero β with $|\beta - \frac{1}{2}| < \sqrt{u_0}$, and $\zeta_F(\frac{1}{2}) \neq 0$ when $u_0 = 0$.

When they hold, $b = 0$ and the Thorin measure is

$$U_{F,u_0} = \sum_{\gamma>0} \delta_{u_0+\gamma^2} + m_F\, \delta_{u_0} + \sum_{\beta} \delta_{u_0-(\beta-\frac{1}{2})^2}, \tag{21.2.5}$$

where $\frac{1}{2}+i\gamma$ runs over critical line zeros with $\gamma > 0$ and β over real zeros in $(\frac{1}{2}, 1)$, all with multiplicity.

Proof. (i) implies (ii). By (G1) the Laplace transform extends to an analytic zero free function on $\mathbb{C} \setminus (-\infty, 0]$, and it agrees with $G_F(u_0)/G_F(u_0+s)$, which is meromorphic, on $s > 0$. Hence $G_F(u_0+s)$ has no zeros off $(-\infty, 0]$, so every zero z of G_F satisfies $z - u_0 \le 0$, and $z \ne u_0$ because $G_F(u_0) \ne 0$. Imaginary w_ρ is a critical line zero, real w_ρ a real zero with $(\beta - \frac{1}{2})^2 < u_0$. At $u_0 = 0$ a central zero would make $G_F(0) = 0$.

(ii) implies (i). Under (ii), Lemma 21.2.2 gives

$$\varphi_{F,u_0}(s) = \Big(\frac{u_0}{u_0+s}\Big)^{m_F} \prod_{z \in Z_F} \frac{u_0 - z}{u_0 - z + s},$$

where every rate $u_0 - z$ is positive, equal to $u_0 + \gamma^2$ or to $u_0 - (\beta - \frac{1}{2})^2$, and the reciprocals are summable. Fact (G2) gives the GGC with Thorin measure (21.2.5). □

For a general field at $u_0 = 0$ the theorem is Proposition 10.3.6, which assumes $\zeta_F(\frac{1}{2}) \ne 0$. That assumption is not automatic, since Armitage found fields with $\zeta_F(\frac{1}{2}) = 0$ [8], and here a central zero appears in the conclusion as the atom $m_F \delta_{u_0}$.

COROLLARY 21.2.4 (The tilts). *(a) At $u_0 = 0$: $\xi_F(\frac{1}{2})/\xi_F(\frac{1}{2} + \sqrt{s})$ is a GGC Laplace transform if and only if all nontrivial zeros of ζ_F lie on the critical line and $\zeta_F(\frac{1}{2}) \ne 0$.*

(b) For every $u_0 \ge \frac{1}{4}$, in particular at $u_0 = \frac{1}{4}$ and $u_0 = 1$, where ξ_F is read from 1 and from $\frac{3}{2}$: φ_{F,u_0} is a GGC Laplace transform if and only if every nontrivial zero lies on the critical line or on the real axis.

(c) Define the Thorin threshold $u_F^* = \max\{(\beta - \frac{1}{2})^2\}$ over real zeros $\beta \in [\frac{1}{2}, 1)$, with $u_F^* = 0$ if there are none. If all nonreal zeros are on the critical line, the Thorin property holds at tilt u_0 exactly for $u_0 > u_F^*$, and also at $u_0 = 0$ when there are no real zeros at all.

(d) When the property holds at two tilts $u_1 < u_0$, the measures are shifts of one another, $U_{F,u_0} = U_{F,u_1}(\cdot - (u_0 - u_1))$, and $\varphi_{F,u_1}(s) = \varphi_{F,u_0}(s - (u_0 - u_1))/\varphi_{F,u_0}(-(u_0 - u_1))$, the tilt (G3).

Proof. (a) and (b) are Theorem 21.2.3 at $u_0 = 0$ and for $u_0 \ge \frac{1}{4}$, since every real zero in $(0,1)$ has $(\beta - \frac{1}{2})^2 < \frac{1}{4}$. (c) restates the condition $z_\rho < u_0$ for the real values z_ρ. (d) follows from (21.2.5) and (21.2.4); for ξ it is the forward and backward tilt of Theorem 19.1.2(a). □

REMARK 21.2.5 (The tilt and Landau's theorem). Suppose the Thorin property holds at the tilt 1. Descending from tilt 1 to tilt 0 is the tilt (G3) with $c = 1$, admissible exactly when $\varphi_{F,1}(-1) < \infty$. Since $\varphi_{F,1}$ is the Laplace transform of a

positive measure, Landau's theorem [340, Ch. II, Theorem 5b] puts its abscissa of convergence at a real singularity. So $\varphi_{F,1}(-1) < \infty$ if and only if G_F has no zero in $[0,1]$, that is, ξ_F has no real zero in $[\frac{1}{2}, \frac{3}{2}]$. For ξ there is no real zero, so under RH the descent to tilt 0 is possible, and it is Theorem 19.1.2(a) (Section 19.1.1). What is specific to number fields is that a central zero or a real zero in $(\frac{1}{2}, 1)$ blocks the descent to tilt 0.

Remark 21.2.6 (The Thorin measure is the zero counting measure). For $F = \mathbb{Q}$, $\zeta(\frac{1}{2}) \neq 0$ and $\zeta(\sigma) < 0$ on $(0,1)$, so there are no central or real zeros. By Corollary 21.2.4, the Thorin property of ξ at any tilt u_0 is equivalent to RH, with Thorin measure $\sum_{\gamma>0} \delta_{u_0+\gamma^2}$; at tilt $u_0 = 1$ the smallest atom is $1+\gamma_1^2 = 200.7905$.

21.3 The centre clock: moments, transfer function and negative moments

21.3.1 The zeta function at $\sigma > 1$

Lemma 21.3.1. *For $\sigma > 1$ and $\mathrm{Re}(s) \geq 0$,*

$$\log \frac{\zeta(\sigma + s)}{\zeta(\sigma)} = \int_0^\infty (e^{-sx} - 1)\, \mu^\sigma(dx), \tag{21.3.1}$$

where $\mu^\sigma = \sum_{p,\, r \geq 1} r^{-1} p^{-r\sigma}\, \delta_{r \log p}$, the sum over primes p and integers $r \geq 1$, has total mass $\log \zeta(\sigma)$.

Proof. Take logarithms of the Euler product, expand $\log(1-z)^{-1} = \sum_{r \geq 1} z^r / r$, subtract $\log \zeta(\sigma)$ and write $p^{-rs} = e^{-rs \log p}$. □

Lemma 21.3.2. *Let N follow the Riemann zeta distribution $\mathrm{RZ}(\sigma)$, $\Pr(\mathsf{N} = n) = n^{-\sigma}/\zeta(\sigma)$, and $J^\sigma := \log \mathsf{N}$. Then J^σ is compound Poisson with Lévy measure μ^σ and $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma - it)/\zeta(\sigma)$.*

Proof. $\mathbb{E}[e^{itJ^\sigma}] = \zeta(\sigma)^{-1} \sum_n n^{-\sigma} n^{it}$, and (21.3.1) at $s = -it$ identifies the exponent. □

The Riemann zeta distribution and the infinite divisibility of $\log \mathsf{N}$ go back to Khinchin [148]; see Lin and Hu [180], Theorem 14.3.1 and Section 6.1.6.

Theorem 21.3.3 (Wiener spectral identity). *For $\sigma > 1$ and integer $k \geq 1$,*

$$\lim_{X \to \infty} \frac{1}{2X} \int_{-X}^{X} \left|\zeta(\sigma + it)\right|^{2k} dt = \sum_{n \geq 1} \frac{d_k(n)^2}{n^{2\sigma}}, \tag{21.3.2}$$

where $d_k(n)$ counts ordered factorisations $n = n_1 \cdots n_k$.

Proof. By Lemma 21.3.2, $|\zeta(\sigma+it)|^{2k} = \zeta(\sigma)^{2k}\mathbb{E}[e^{itV}]$ with $V = \sum_{i\le k} J_i - \sum_{i\le k} J_i'$ for independent copies of J^σ. Wiener's theorem on the atom of a measure at 0 [342] gives the limit $\Pr(V=0) = \Pr(\mathsf{N}_1\cdots\mathsf{N}_k = \mathsf{N}_1'\cdots\mathsf{N}_k')$. The product $\mathsf{N}_1\cdots\mathsf{N}_k$ has law $d_k(m)m^{-\sigma}/\zeta(\sigma)^k$, so the collision probability is $\zeta(\sigma)^{-2k}\sum_m d_k(m)^2 m^{-2\sigma}$. □

Corollary 21.3.4. *The right side of* (21.3.2) *is* $\zeta(2\sigma)$ *for* $k=1$, $\zeta(2\sigma)^4/\zeta(4\sigma)$ *for* $k=2$ *(through Ramanujan's identity), and in general the Euler product*

$$\prod_p \sum_{r\ge 0} \binom{r+k-1}{k-1}^2 p^{-2r\sigma}.$$

These are the classical mean value theorems of Carlson ($k=1$) and Ingham ($k=2$), which hold for $\sigma > \frac12$ (see [324, Chapter VII]); for $k\ge 3$ they are known only on a smaller range of σ, and on the whole range $\sigma > \frac12$ they follow from the Lindelöf hypothesis [324, Chapter XIII].

21.3.2 The Riemann case in detail

Take $F=\mathbb{Q}$ and $u_0 = 0$, and assume RH. By Corollary 21.2.4,

$$\frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u})} = \mathbb{E}\big[e^{-uT}\big], \qquad T \overset{d}{=} \sum_{\gamma>0}\frac{e_\gamma}{\gamma^2}, \qquad u\ge 0, \tag{21.3.3}$$

with independent standard exponential variables e_γ. The measure $\sum_{\gamma>0}\delta_{\gamma^2}$ is the measure U_{RH} of Proposition 1.2.1, which Chapters 3 to 5 compute at finite resolution. Without RH the clock still exists, as a self decomposable variable (Theorem 5.1.1), with the same Laplace transform, and the moment identities, the Mellin transform and the interpolation below hold for it with γ^2 replaced by $a_k = -(\rho_k - \frac12)^2$, the numbers of the product $G(u) = \prod_k (1+u/a_k)$ of equation (1.2.1).

21.3.2.1 The normal scale mixture and the moments

Define $Y := \sqrt{2T}\,Z$ with $Z\sim N(0,1)$ independent of T. Conditioning on T,

$$\mathbb{E}\big[e^{itY}\big] = \mathbb{E}\big[e^{-t^2T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+|t|)}, \tag{21.3.4}$$

which evaluates ξ on the real axis, where it is positive, even and super exponentially growing. This is the variable $Z\sqrt{2T}$ of Section 1.2. It is the variable $X = Z\sqrt{2Y}$ of Proposition 16.5.1, where Chapter 16 writes Y for the clock. The moments of T are the Taylor coefficients of (21.3.3): with $p_j := \sum_{\gamma>0}\gamma^{-2j}$,

$$\mathbb{E}[T^k] = k!\,h_k\big(\{\gamma^{-2}\}\big), \qquad \mathbb{E}[Y^{2k}] = (2k-1)!!\,2^k\,\mathbb{E}[T^k], \tag{21.3.5}$$

where h_k is the complete homogeneous symmetric polynomialpolynomial. In particular $\mathbb{E}[T] = p_1$ and $\mathrm{Var}(T) = p_2$.

21.3.2.2 The Hadamard product and the transfer function

The same factorisation gives the critical line value

$$\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \prod_{\gamma>0}\Big(1-\frac{t^2}{\gamma^2}\Big), \tag{21.3.6}$$

which alternates in sign; read in $u = t^2$ it is the Laplace transform of no measure, while in t it is the characteristic function of the Pólya density (Section 1.2). Multiplying (21.3.4) by (21.3.6) gives

$$\mathbb{E}[e^{itY}]\cdot\frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2})} = \frac{\xi(\frac{1}{2}+it)}{\xi(\frac{1}{2}+|t|)} = \prod_{\gamma>0}\frac{\gamma^2-t^2}{\gamma^2+t^2}, \tag{21.3.7}$$

with zeros at $\pm\gamma$ and, continued in t, poles at $\pm i\gamma$, where the Laplace transform of T stops being analytic. The left side is the characteristic function of the sum of Y and an independent variable with the Pólya density, so without any hypothesis the middle term is a characteristic function; the product on the right is its form under RH. The critical line moments $\int_0^X |\zeta(\frac{1}{2}+it)|^{2k}dt$ are governed by (21.3.6), not by Y; the transfer function links the two and has a closed form (Proposition 21.3.6).

21.3.3 Closed forms and their verification

21.3.3.1 The moment identity

We computed $\mathbb{E}[T^k]$ two ways. The zero side uses (21.3.5), with p_j summed over the first 800 ordinates plus the Riemann von Mangoldt tail $\int_{\gamma_{800}}^{\infty} t^{-2j}\,\frac{1}{2\pi}\log\frac{t}{2\pi}\,dt$, started at the ordinate $\gamma_{800} = 1183.7128$. The function side uses the Taylor expansion of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+w)$ in powers of w^2 at $w=0$, from the Taylor coefficients of $\xi(\frac{1}{2}+s)$ at $s=0$. Table 21.1 gives the result. The residual is mostly the error

Table 21.1: The moments $\mathbb{E}[T^k]$ from the zeros (800 ordinates and the smooth tail from γ_{800}) and from the function (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`).

k	zero side	function side	relative error
1	0.0231054	0.0231050	1.9×10^{-5}
2	5.71033×10^{-4}	5.71013×10^{-4}	3.5×10^{-5}
3	1.52001×10^{-5}	1.51994×10^{-5}	4.9×10^{-5}
4	4.38851×10^{-7}	4.38824×10^{-7}	6.0×10^{-5}

of starting the smooth tail at an ordinate: with the first 800 ordinates and the tail started at the midpoint of γ_{800} and γ_{801} the relative errors for $k=1,\dots,4$ are 3.1×10^{-7}, 5.8×10^{-7}, 8.1×10^{-7} and 1.0×10^{-6}, and with the anchored

tail below they are 8.8×10^{-10}, 1.6×10^{-9}, 2.3×10^{-9} and 2.8×10^{-9}. The function side computes $\sum_k a_k^{-j}$ with no hypothesis (the cumulants of T from $G(u) = \prod_k (1 + u/a_k)$, Section 11.2.3), so the agreement checks the zero list and the tail. One missing zero at γ_{800} would change $\mathbb{E}[T]$ by 3.1×10^{-5} in relative terms, the order of the residuals in Table 21.1, so the table alone cannot show that none is missing; the anchored residual quoted above is smaller than that change by a factor of 3.5×10^4, and with it the agreement shows that no zero below the cut is missing from the list. It does not test that the zeros lie on the line: moving a pair of zeros at height γ_{800} off the line by 0.25 changes p_1 by 8.3×10^{-12} in relative terms, below every residual. Nor is it a check on the Thorin property itself, which no finite computation can establish (Theorem 5.2.1).

Anchoring the tail. Most of the residual comes from the fluctuation $S(T)$ of the zero count at the cut. Write $N(t) = N_0(t) + 1 + S(t)$ with $N_0 = \vartheta/\pi$. Integration by parts gives, exactly,

$$\begin{aligned} \sum_{\gamma > T} \gamma^{-2j} = {} & \int_T^\infty t^{-2j}\, \frac{\vartheta'(t)}{\pi}\, dt + \big(N_0(T) + 1 - N(T)\big) T^{-2j} \\ & + 2j \int_T^\infty S(t)\, t^{-2j-1}\, dt. \end{aligned} \tag{21.3.8}$$

The smooth tail keeps only the first term, with the error $-S(T)T^{-2j}$ of the second, of order $T^{-2j} \log T$ at most. Since $\int_T^X S(t)\, dt = O(\log X)$ [324, Section 9.9], the third term is $O(jT^{-2j-1} \log T)$, so the anchored tail, which keeps the first two terms, typically gains over the smooth tail a factor of the order of the height of the cut. Table 21.2 gives the relative errors of p_1, p_2, p_3 against the values of the function side, for the first N ordinates and the cut at the midpoint of γ_N and γ_{N+1}, so that $N(T) = N$. The gain is typical, not guaranteed: the smooth error

Table 21.2: Relative errors of p_1, p_2, p_3 for the smooth and the anchored tail with the first N ordinates; the cuts are $T = 1183.1475$ ($S(T) = 0.083$) for $N = 799$, $T = 397.1503$ ($S(T) = 0.244$) for $N = 200$ and $T = 680.8186$ ($S(T) = -0.211$) for $N = 400$ (`gl_zeta_moments.py`, 100 digits, `data/gl_zeta_moments.json`).

N	tail	p_1	p_2	p_3
799	smooth	2.6×10^{-6}	1.1×10^{-9}	2.1×10^{-13}
	anchored	6.3×10^{-9}	5.6×10^{-12}	1.6×10^{-15}
200	smooth	6.7×10^{-5}	2.6×10^{-7}	4.3×10^{-10}
	anchored	6.8×10^{-8}	5.1×10^{-10}	1.2×10^{-12}
400	smooth	2.0×10^{-5}	2.6×10^{-8}	1.4×10^{-11}
	anchored	1.4×10^{-7}	3.7×10^{-10}	3.1×10^{-13}

is $|S(T)|T^{-2j}$ to 0.1%, so the gain is large only when $|S(T)|$ is not small. Over

the 1650 cuts $N = 50$ to 1699 the ratio of the smooth to the anchored error of p_1 has median 354, about a third of the height of the cut, and deciles 42 and 2700, and it is below 10 at 3.2% of the cuts. This is the first half of Problem 21.3.

21.3.3.2 The tilt by one quarter and a classical constant

The classical evaluation

$$\sum_{\rho} \frac{1}{\rho} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi = 0.0230957089661\ldots,$$

with γ_E Euler's constant, which under RH is $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1}$, differs from $\mathbb{E}[T] = 0.0231049931\ldots$ by about 9.3×10^{-6}. The shift by $\frac14$ is the tilt $u_0 = \frac14$. The book identifies the classical constant as the mean of this tilted clock in Section 9.2 (after Corollary 9.2.3), and as π times the Thorin density of the shifted clock at basepoint one at the origin (Proposition 9.1.2); what we add is the expansion (21.3.11) in the cumulants.

Proposition 21.3.5 (Esscher tilt). *Assume* (21.3.3) *and let* $u_0 > 0$. *The tilted law* $\Pr_{u_0}(dh) = e^{-u_0 h}\Pr(T \in dh)/\mathbb{E}[e^{-u_0 T}]$ *is the GGC with Thorin measure* $U_{\mathbb{Q},u_0} = \sum_{\gamma>0}\delta_{u_0+\gamma^2}$ *of Theorem 21.2.3, which is the forward tilt of Theorem 19.1.2(a), and*

$$\mathbb{E}\big[e^{-u_0 T}\big] = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u_0})}, \tag{21.3.9}$$

$$\mathbb{E}_{u_0}[T] = \sum_{\gamma>0}\frac{1}{\gamma^2+u_0} = \frac{1}{2\sqrt{u_0}}\frac{\xi'}{\xi}\big(\tfrac12+\sqrt{u_0}\big). \tag{21.3.10}$$

At $u_0 = \frac14$, *since* $\xi(1) = \frac12$,

$$\mathbb{E}\big[e^{-T/4}\big] = 2\,\xi(\tfrac12) = 0.99424155637\ldots,$$
$$\mathbb{E}_{1/4}[T] = \frac{\xi'(1)}{\xi(1)} = 1 + \frac{\gamma_E}{2} - \frac12\log 4\pi.$$

Proof. $\mathbb{E}_{u_0}[e^{-vT}] = \mathbb{E}[e^{-(u_0+v)T}]/\mathbb{E}[e^{-u_0T}] = \prod_{\gamma>0}(\gamma^2+u_0)/(\gamma^2+u_0+v)$, which is $\varphi_{\mathbb{Q},u_0}(v)$. Equation (21.3.9) is (21.3.3) at $u = u_0$, and $\mathbb{E}_{u_0}[T] = -\frac{d}{du}\log\mathbb{E}[e^{-uT}]$ at u_0, computed either from the product or from $\log\xi(\frac12+\sqrt{u})$. At $u_0 = \frac14$ the evaluation point is $\frac12+\sqrt{u_0} = 1$, where $\xi(1) = \xi(0) = \frac12$ and $\xi'(1)/\xi(1) = \sum_\rho \rho^{-1}$. □

From the first 800 zeros with the anchored tail, $\prod_{\gamma>0}\gamma^2/(\gamma^2+\frac14) = 0.99424155637$ and $\sum_{\gamma>0}(\gamma^2+\frac14)^{-1} = 0.02309570899$, within 5×10^{-12} and

2×10^{-11} of $2\xi(\frac{1}{2})$ and of the constant. Expanding $(\gamma^2 + \frac{1}{4})^{-1}$ in powers of γ^{-2}, valid since $\gamma_1 > \frac{1}{2}$,

$$1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi \;=\; \sum_{j\ge 1} (-1)^{j+1} 4^{1-j}\, p_j, \tag{21.3.11}$$

ties it to the whole cumulant sequence of T. With the p_j of the function side the series agrees with the constant to a relative 2×10^{-30} after ten terms, the size of the first omitted term.

21.3.3.3 The transfer function

Proposition 21.3.6. *Assume RH. For all real t,*

$$\mathfrak{T}(t) \;:=\; \frac{\xi(\frac{1}{2} + it)}{\xi(\frac{1}{2} + |t|)} \;=\; \prod_{\gamma > 0} \frac{\gamma^2 - t^2}{\gamma^2 + t^2}, \tag{21.3.12}$$

and $\mathfrak{T}$ is real, even, bounded by 1 in modulus, and vanishes exactly at $t = \pm\gamma$. For $|t| < \gamma_1$,

$$\log \mathfrak{T}(t) \;=\; -2 \sum_{\gamma > 0} \operatorname{artanh}\Big(\frac{t^2}{\gamma^2}\Big) \;=\; -2 \sum_{m \ge 1} \frac{p_{2m-1}}{2m-1}\, t^{4m-2}, \tag{21.3.13}$$

which involves only the odd index power sums $p_1, p_3, p_5, \dots$.

Proof. Lemma 21.2.2 gives $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2}) = \prod_{\gamma>0}(1 + s^2/\gamma^2)$ for all $s \in \mathbb{C}$. Setting $s = it$ and $s = |t|$ and dividing gives (21.3.12), each factor real with modulus at most 1. For (21.3.13), $\log \frac{\gamma^2 - t^2}{\gamma^2 + t^2} = -2\operatorname{artanh}(t^2/\gamma^2)$; expand and interchange the absolutely convergent double sum. □

Proposition 21.3.7. *Assume RH. The odd index power sums determine the zeros: the series $\sum_{m\ge1} p_{2m-1} z^{2m-1}$ converges for $|z| < \gamma_1^2$ and continues to the meromorphic function*

$$\sum_{m \ge 1} p_{2m-1} z^{2m-1} = \frac{1}{2} \sum_{\gamma > 0} \Big(\frac{1}{1 - z/\gamma^2} - \frac{1}{1 + z/\gamma^2} \Big),$$

whose poles are the points $\pm\gamma^2$, each with residue $-\frac{1}{2}\gamma^2$ times the multiplicity of γ. Hence $\{p_{2m-1}\}$ determines $\{\gamma\}$ with multiplicities, and with it every p_j.

Proof. For $|z| < \gamma_1^2$, $\sum_m (z/\gamma^2)^{2m-1} = (z/\gamma^2)/(1 - z^2/\gamma^4)$, which is half the bracket; summing over γ and interchanging is justified by $\sum \gamma^{-2} < \infty$. The sum over γ converges locally uniformly off $\{\pm\gamma^2\}$, since the bracket is $O(|z|\gamma^{-2})$ for $\gamma^2 > 2|z|$, so it defines a meromorphic function with the stated poles. A meromorphic function is determined by its Taylor coefficients at 0, and its poles and residues give back the multiset $\{\gamma^2\}$. □

Numerically, $\mathfrak{T}(2) = 0.8312$, $\mathfrak{T}(5) = 0.3145$ and $\mathfrak{T}(13) = 1.7242 \times 10^{-4}$, the last close to the first zero $\gamma_1 \approx 14.13$. The Hadamard product over the first 1699 ordinates with the anchored tail reproduces these three values, computed from ξ at the two points $\frac{1}{2} + it$ and $\frac{1}{2} + t$, to relative errors 3.8×10^{-10}, 2.4×10^{-9} and 1.6×10^{-8}.

PROPOSITION 21.3.8. *Let* $\mathfrak{B}(z) = \prod_{\gamma>0}(\gamma^2 - z)/(\gamma^2 + z)$, *so* $\mathfrak{T}(t) = \mathfrak{B}(t^2)$. *Under RH,* $\mathfrak{B}$ *is a Blaschke product for the right half plane: it converges there,* $|\mathfrak{B}| < 1$ *for* $\mathrm{Re}(z) > 0$, $|\mathfrak{B}| = 1$ *for* $\mathrm{Re}(z) = 0$, *its zeros are the points* γ^2, *and it has no singular inner factor.*

Proof. Each factor $b_\gamma(z) = (\gamma^2 - z)/(\gamma^2 + z)$ is a Cayley map taking $\mathrm{Re}(z) > 0$ into the unit disc with $|b_\gamma| = 1$ on $\mathrm{Re}(z) = 0$. Since $1 - b_\gamma(z) = 2z/(\gamma^2 + z)$ and $\sum \gamma^{-2} < \infty$, the product converges locally uniformly away from $\{-\gamma^2\}$; this is the half plane Blaschke condition. A convergent product of Blaschke factors is a Blaschke product, so there is no singular inner factor, and by Lemma 21.2.2 its value is $\mathfrak{B}(z) = \xi(\frac{1}{2} + i\sqrt{z})/\xi(\frac{1}{2} + \sqrt{z})$, the product formula of the lemma having no exponential factor. □

REMARK 21.3.9 (Without RH). Let $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2}) = \prod_k (1 + u/a_k)$ be the function of Section 1.2 (equation (1.2.1)), with $a_k = -(\rho_k - \frac{1}{2})^2$ and one factor for each pair $\{\rho_k, 1 - \rho_k\}$ of zeros. Then $\mathfrak{T}(t) = G(-t^2)/G(t^2)$ for real t with no hypothesis, and $\mathfrak{B}(z) := G(-z)/G(z) = \prod_k (a_k - z)/(a_k + z)$ is still a Blaschke product for the right half plane, with zeros at the points $a_k = -(\rho_k - \frac{1}{2})^2$. They lie in the open right half plane, since $\mathrm{Re}\, a_k = \gamma_k^2 - (\beta_k - \frac{1}{2})^2 > 0$ for $\rho_k = \beta_k + i\gamma_k$ ($|\beta_k - \frac{1}{2}| < \frac{1}{2}$ and $|\gamma_k| \geq \gamma_1$). The product converges because $1 - (a_k - z)/(a_k + z) = 2z/(a_k + z)$ and $\sum_k |a_k|^{-1} < \infty$ (Lemma 21.2.2). RH is the statement that these zeros are real, hence equal to the γ^2, not that $\mathfrak{B}$ is inner. In particular $|\mathfrak{B}(iy)| = 1$ is an identity for every real entire G, and the reality of $\mathfrak{T}(t)$ for real t is the functional equation; a numerical check of either tests rounding only.

21.3.3.4 Negative moments as a Mellin transform

PROPOSITION 21.3.10. *For* $\mathrm{Re}(s) > 0$, *with no hypothesis on the zeros,*

$$\begin{aligned} \mathbb{E}\big[T^{-s}\big] &= \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \, \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + \sqrt{u})} \, du \\ &= \frac{2\,\xi(\frac{1}{2})}{\Gamma(s)} \int_0^\infty \frac{w^{2s-1}}{\xi(\frac{1}{2} + w)} \, dw. \end{aligned} \tag{21.3.14}$$

Proof. Since $T > 0$ almost surely, $T^{-s} = \Gamma(s)^{-1} \int_0^\infty u^{s-1} e^{-uT} \, du$. Take expectations, with Fubini justified by positivity, substitute the Laplace transform (21.3.3),

which holds without RH, and put $u = w^2$. The integral converges at 0 for $\mathrm{Re}(s) > 0$ and at ∞ for every s. □

In a computation at 30 digits the two forms agree to 10^{-18} at $s = \frac{1}{2}$ and to 1.4×10^{-27} at $s = \frac{3}{4}$; at $s = \frac{1}{4}$ the integrand $u^{-3/4}$ of the first form is singular at the origin, and the value below is that of the second form. The values are

$$\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots,$$
$$\mathbb{E}[T^{-1/4}] = 2.590049154\ldots, \qquad \mathbb{E}[T^{-3/4}] = 17.567694990\ldots.$$

The value $\mathbb{E}[T^{-1/2}]$ is a spectral average weighted toward the low lying zeros; Problem 21.4 compares it with random matrix theory.

PROPOSITION 21.3.11. *With no hypothesis on the zeros, $s \mapsto \mathbb{E}[T^{-s}]$ is an entire function, equal for $\mathrm{Re}\, s > 0$ to the Mellin integral* (21.3.14)*, and at the nonpositive integers it takes the values of the positive moments:*

$$\mathbb{E}[T^{-s}]\big|_{s=-n} \;=\; \mathbb{E}[T^n], \qquad n = 0, 1, 2, \ldots \tag{21.3.15}$$

Proof. Since $T > 0$ almost surely and T has all positive and all negative moments, $\mathbb{E}|T^{-s}| = \mathbb{E}[T^{-\mathrm{Re}\, s}] < \infty$ for every s, so $\mathbb{E}[T^{-s}]$ is entire as an expectation and (21.3.15) holds by definition. What the residues add is that the continuation of the Mellin formula is this function. Write $M(s) = \int_0^\infty u^{s-1}\varphi(u)\,du$ with $\varphi(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u}) = \sum_{n\ge 0} c_n u^n$ near 0, $c_n = (-1)^n \mathbb{E}[T^n]/n!$. Subtracting Taylor polynomials continues M to a meromorphic function with simple poles at $s = -n$ and $\mathrm{Res}_{s=-n} M = c_n$. Since $1/\Gamma$ is entire with simple zeros exactly there, M/Γ is entire and

$$\mathbb{E}[T^{-s}]\big|_{s=-n} = \frac{\mathrm{Res}_{s=-n} M(s)}{\mathrm{Res}_{s=-n} \Gamma(s)} = \frac{(-1)^n \mathbb{E}[T^n]/n!}{(-1)^n/n!} = \mathbb{E}[T^n].$$

□

The entire function $\mathbb{E}[T^{-s}]$ is not of exponential type, because the radius of convergence of $\sum_n \mathbb{E}[T^n]\lambda^n/n!$ is γ_1^2 (Theorem 11.2.3(a)), so $\mathbb{E}[T^n]$ is not bounded by any geometric sequence, whereas the values of a function of exponential type along the integers grow at most geometrically. Its modulus is nevertheless bounded on every vertical line by $|\mathbb{E}[T^{-s}]| \le \mathbb{E}[T^{-\mathrm{Re}\, s}]$. We claim no uniqueness theorem of Carlson type.

21.3.3.5 Total positivity of the density of Y

The density of $Y = \sqrt{2T}\, Z$ is

$$f(y) = \frac{1}{\pi} \int_0^\infty \cos(ty)\, \frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2} + t)}\, dt,$$

computed from ξ on the real axis alone. By Proposition 16.5.1, f is unconditionally a smooth symmetric density, and RH holds if and only if f is a Pólya frequency function of infinite order. One direction is Theorem 13.1.4(iv); the equivalence in the language of the Laguerre Pólya class is in [153, Theorem 2.3 and Corollary 2.4].

The low orders are decided by the low zeros (Chapter 16). Order two holds without RH, that is, f is log concave (Theorem 16.11.5, a floating point certificate), and, given the verification of RH to height 3×10^{12} [247], f lies within 1.6×10^{-10} of a Pólya frequency function of every order (Theorem 16.11.6), so that every minor of order 3 and 4 is at least -1.1×10^{-8} and -1.2×10^{-7}. What follows tests f at the orders two to eight on other grids. The computation is by `gl_density_pf.py` at 70 digits, with the trapezoid rule in t; $f(2)$ agrees to 3×10^{-62} with the residue series $\sum_k r_k e^{-\gamma_k y}$, $r_k = -\xi(\frac12)/\Xi'(\gamma_k)$, over the first 60 zeros.

At the origin $f(0) = 1.8994210510239\ldots$, which equals $\mathbb{E}[T^{-1/2}]/(2\sqrt{\pi})$ with $\mathbb{E}[T^{-1/2}] = 6.7332723127566\ldots$ (Proposition 21.3.10 at $s=\frac12$). On $[0,2]$ with step 0.05, $ff''-f'^2<0$ at all 41 grid points; the largest value, -1.37×10^{-22}, is at $y=2$, where $f(2) = 1.89857001286\times10^{-10}$. The 121 minors $\det[f(x_r-y_c)]_{r,c<3}$ of order three with $x_r=(i+r)/10$ and $y_c=(j+c)/10$, $0\le i,j\le10$, depend only on $i-j$ and are all positive; the smallest, 6.8275×10^{-16} at $|i-j|=10$, is far smaller than the 1.1×10^{-8} by which Theorem 16.11.6 allows a minor of order three to be negative. The Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ are positive for $n=3$ to 8 at $h=0.05$, 0.2 and 0.4 (Table 21.3), the smallest being 1.4971×10^{-21}. These are

Table 21.3: Toeplitz minors $\det[f((r-c)h)]_{r,c<n}$ of the density of Y (`gl_density_pf.py`, 70 digits).

n	$h=0.05$	$h=0.2$	$h=0.4$
3	2.7009×10^{-3}	2.1714	6.4789
4	6.4860×10^{-6}	2.0489	11.961
5	3.8720×10^{-9}	1.9020	22.083
6	7.2088×10^{-13}	1.7560	40.770
7	5.0043×10^{-17}	1.6184	75.271
8	1.4971×10^{-21}	1.4906	138.97

finite order tests: a negative minor would refute RH, and positive minors cannot prove it (Problem 21.10).

21.4 The master theorem route

By Theorem 21.2.3 and Remark 21.2.6, the Thorin property of the centre clock is equivalent to RH, so a proof of it that avoids the zeros (Problem 21.1) cannot take RH as an input. The route of [249] expands $\xi(\frac12)/\xi(\frac12+\sqrt{u})$ term by term and

passes to an integral representation of the GGC through the Ramanujan master theorem. We isolate what the extension of Chaudhry and Qadir [55] settles in that step and what it leaves open.

21.4.1 The growth hypothesis

Chaudhry and Qadir [55, §4] enlarge Hardy's class to $Q_\infty(\kappa,\delta)$ (their ε is called κ here, because $\varepsilon=\alpha-\frac12$ in Section 21.5): meromorphic $F(s;0)$, analytic for $\operatorname{Re}(s)<\delta$, not identically zero at the nonpositive integers, with

$$|\Gamma(s)F(s;0)|\le C\,|\tau|^{\kappa(\sigma-\delta)},\qquad s=\sigma+i\tau,\ |\tau|\to\infty, \tag{21.4.1}$$

uniformly for $\sigma_1\le\sigma\le\sigma_2<\delta$. Their Theorem 4.1 grants the master theorem to the class, and their Examples 5.5 and 5.6 place $\zeta(s)/[\Gamma(1-s)\zeta(1-s)]$ and $\zeta(1-s)/[\Gamma(s)\zeta(s)]$ in $Q_\infty(1,\frac12)$. The function to which the route applies the theorem is $F(s;0)=\xi(\frac12)/\xi(\frac12+s)$.

Lemma 21.4.1. *Assume RH, and let $0<\sigma_1\le\sigma_2<\frac12$. Then $F(s;0)=\xi(\frac12)/\xi(\frac12+s)$ satisfies*

$$|\Gamma(s)\,\xi(\tfrac12)/\xi(\tfrac12+s)|\le C_\eta\,e^{-\pi|\tau|/4}\,|\tau|^{\sigma/2-9/4+\eta},\qquad |\tau|\ge1,$$

uniformly for $\sigma_1\le\sigma\le\sigma_2$, for every $\eta>0$. In particular the growth condition (21.4.1) *of $Q_\infty(1,\frac12)$ holds on such strips, with exponential room. By the functional equation the same argument gives, for $-\sigma_2\le\sigma\le-\sigma_1$, the exponent $\frac32\sigma-\frac94+\eta$, which is smaller.*

Proof. Write $\xi(\frac12+s)=\frac12 w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$ with $w=\frac12+s$, so $\frac12<\operatorname{Re}w<1$. Stirling's formula $|\Gamma(x+iy)|\asymp|y|^{x-1/2}e^{-\pi|y|/2}$ gives $|\Gamma(w/2)|\asymp|\tau|^{\sigma/2-1/4}e^{-\pi|\tau|/4}$, and $|w(w-1)|\asymp\tau^2$. Under RH, $1/\zeta(w)=O(|\tau|^{\eta})$ uniformly for $\operatorname{Re}w\ge\frac12+\sigma_1$ [324, Section 14.2]. So $|1/\xi(\frac12+s)|\ll e^{\pi|\tau|/4}|\tau|^{-\sigma/2-7/4+\eta}$, and multiplying by $|\Gamma(s)|\asymp|\tau|^{\sigma-1/2}e^{-\pi|\tau|/2}$ gives the bound. For $-\sigma_2\le\sigma\le-\sigma_1$ use $F(s;0)=F(-s;0)$, which is the functional equation $\xi(\frac12+s)=\xi(\frac12-s)$, and $|\Gamma(s)/\Gamma(-s)|\asymp|\tau|^{2\sigma}$: the bound at $-s$ has the exponent $-\sigma/2-\frac94+\eta$, and the factor $|\tau|^{2\sigma}$ turns it into $\frac32\sigma-\frac94+\eta$. □

The strip must avoid $\sigma=0$, where $F(s;0)$ has its poles at $s=i\gamma$. And without RH no polynomial bound for $1/\zeta$ is known on any fixed vertical strip $\sigma_0<\operatorname{Re}w<1$ of the critical strip, so unconditionally the growth condition is only known where $\operatorname{Re}(\frac12+s)\ge1$, outside the range of the class.

21.4.2 The absolute convergence hypothesis

With g the density of the clock T (it exists, since T is self decomposable and not degenerate), the interchange of sum and integral requires

$$\int_0^\infty g(y)\,\mathfrak{R}(z/y)\,dy < \infty, \qquad \mathfrak{R}(w) := \sum_{k\ge 1} b_k w^k, \quad b_k := \frac{1}{\Gamma(k)}\,\frac{\xi(\frac12)}{\xi(\frac12+k)}. \tag{21.4.2}$$

Lemma 21.4.2. *With no hypothesis on the zeros,* $\mathfrak{R}$ *is entire of order* $\frac23$*, and* (21.4.2) *holds for every* $z > 0$*.*

Proof. Stirling applied to $\Gamma(k/2 + 1/4)$ gives $\log \xi(\frac12 + k) = \frac{k}{2}\log k + O(k)$, so the coefficients satisfy $\log(1/b_k) = \frac{3k}{2}\log k + O(k)$ and $\mathfrak{R}$ has order $\frac23$. In particular $\mathfrak{R}(w) \le e^{w^{3/4}}$ for $w \ge w_0$, so $\mathfrak{R}(z/y)$ grows at most like $e^{(z/y)^{3/4}}$ as $y \to 0^+$, and convergence is a condition on the left tail of g. Stirling's formula for $\xi(\frac12 + x)$ gives $\log \mathbb{E}[e^{-uT}] = -\log\big(\xi(\frac12 + \sqrt{u})/\xi(\frac12)\big) \sim -\frac14\sqrt{u}\log u$ as $u \to \infty$, and the exponential Tauberian theorem of Kasahara [139] converts this into $-\log \Pr(T \le y) \sim \frac{1}{16}\,y^{-1}\log^2(1/y)$ as $y \to 0^+$. The clock is self decomposable (Theorem 5.1.1) with the infinite Lévy measure $W(x)\,dx/x$, so it has no atom, and it is unimodal [347] (Section 11.1.1), with a mode $m > 0$ and a density g that is nondecreasing on $(0, m)$. Hence $g(y) \le y^{-1}\Pr(T \le 2y)$ for $2y \le m$, that is, $g(y) \le y^{-1}\exp\{-\frac{1}{32}\,y^{-1}\log^2(1/y)\,(1 + o(1))\}$, which beats $e^{(z/y)^{3/4}}$ since $1 > \frac34$. For $y \ge y_0$ the argument z/y stays in $[0, z/y_0]$, where $\mathfrak{R}$ is bounded, and $\int g = 1$. □

21.4.3 What remains open

Two obstructions remain.

(i) *The poles at the zeros.* The class requires analyticity on $\operatorname{Re} s < \frac12$, and the residue argument of Chaudhry and Qadir assumes that the only poles are those of $\Gamma(s)$. But $\xi(\frac12)/\xi(\frac12 + s)$ has further poles at $s = \rho - \frac12$, which lie on $\operatorname{Re} s = 0$ under RH and in the strip $|\operatorname{Re} s| < \frac12$ in general, so it is not in $Q_\infty(1, \frac12)$ with or without RH. A contour argument that handles them is available only once their location is known, and assuming it is assuming RH.

(ii) *The* $\sqrt{u}$ *distortion.* The theorem of Chaudhry and Qadir is stated on vertical strips in the variable s, while the Laplace variable of the clock is $u = s^2$, and the squaring map carries the strip $|\operatorname{Re} s| < 1$ onto the parabolic region $\{|\operatorname{Im} u| < 2\sqrt{1 - \operatorname{Re} u}\}$. Whether the master theorem survives this change of variable needs a separate argument; Proposition 21.4.3 gives it in the shifted variable $w = u - 1$, where it holds and is empty.

21.4.4 The master theorem in other variables

Hardy's class consists of the φ analytic on $\operatorname{Re} w \ge -\delta$ $(0<\delta<1)$ with $|\varphi(v+i\tau)| \le Ce^{Pv+A|\tau|}$, $A<\pi$; for them $\int_0^\infty x^{w-1}\sum_{k\ge0}(-x)^k\varphi(k)/k!\,dx = \Gamma(w)\varphi(-w)$ on $0<\operatorname{Re} w<\delta$.

Proposition 21.4.3. *The function $\varphi(w)=\xi(\frac32)/\xi(\frac12+\sqrt{1+w})$, with the principal root, is analytic and bounded on $\operatorname{Re} w\ge-\delta$. Hence it is in Hardy's class with $P=0$ and any A, with no hypothesis on the zeros. The image of $\operatorname{Re} w\ge-\delta$ under $w\mapsto\frac12+\sqrt{1+w}$ lies in $\{\sigma+i\tau:\sigma-\frac12\ge\sqrt{1-\delta+\tau^2}\}$, which contains no zero, so the identity carries no information on them.*

Proof. Let $z=x+iy=\sqrt{1+w}$. Then $x^2-y^2=\operatorname{Re}(1+w)\ge1-\delta>0$, which gives the region $(\sigma-\frac12=x,\ \tau=y)$ and shows that $|\arg z|<\frac\pi4$ and $|z|^2\ge1-\delta$. A zero $\rho=\beta+i\gamma$ with $\frac12+z=\rho$ would need $|\gamma|<\beta-\frac12<\frac12$, but $|\gamma|>14$, so φ is analytic. For large $|z|$, $\operatorname{Re}(\frac12+z)\ge\frac12+|z|/\sqrt2>1$ and $\zeta(\frac12+z)\to1$, so Stirling gives $\log|\xi(\frac12+z)|=\frac12|z|(\cos\theta\log|z|-\theta\sin\theta)(1+o(1))$ with $\theta=\arg z$, and $\cos\theta\ge1/\sqrt2$. Thus $\xi(\frac12+z)$ grows faster than any $e^{c|z|}$ and φ is bounded. □

By contrast, the Thorin property is a statement on the cut $w\in(-\infty,-1]$, which maps onto the critical line. This does not contradict Lemma 21.4.1, which concerns the variable s and needs RH. The other two variables fail for exact reasons.

The variable α. Fix $s>0$ and $\delta<\min(1,\sqrt s)$. The function $a\mapsto\xi(1+a)/\xi(1+a+\sqrt s)$ is analytic on $\operatorname{Re} a\ge-\delta$, since $\operatorname{Re}(1+a+\sqrt s)>1$ there, and of polynomial growth by Stirling and the convexity bound for ζ. The master theorem evaluates $\xi(\alpha)/\xi(\alpha+\sqrt s)$ at $\alpha=1-w\in(1-\delta,1)$, but only while $\alpha+\sqrt s>1$, in the region of the Euler product. Letting $s\to0$, which the Thorin property at α requires, needs $\delta\to0$ unless there is a zero free strip $\beta\le1-\delta$, that is $\Theta\le1-\delta$ for $\Theta=\sup\beta$, which is not known for any $\delta>0$. So the route is exactly as strong as the hypothesis on Θ it needs.

The Thorin variable. Unconditionally $\xi(\frac12)/\xi(\frac12+\sqrt u)=\exp\{-\int_0^\infty(1-e^{-ut})W(t)\,dt/t\}$ with $W(t)=\sum_{\operatorname{Im}\rho>0}e^{-\lambda_\rho t}$, $\lambda_\rho=-(\rho-\frac12)^2$, as in Theorem 5.1.1. The master theorem needs W to be given by a Taylor series at 0. But $W(t)\sim\log(1/t)/(8\sqrt{\pi t})$ (Section 5.1), so W has none: its Mellin transform is $\Gamma(w)Z(w)$ with $Z(w)=\sum_{\operatorname{Im}\rho>0}\lambda_\rho^{-w}$, convergent for $\operatorname{Re} w>\frac12$, and the singular term gives the double pole $\int_0^1 t^{w-3/2}\log(1/t)\,dt=(w-\frac12)^{-2}$, whereas a Taylor series would give only simple poles at $w=0,-1,\dots$.

21.4.5 Why the square root stays

Let L be the class of the functions $s\mapsto\mathbb{E}e^{-sX}$, X a real random variable, finite on an interval.

THEOREM 21.4.4. *(a) For every real α, $s \mapsto \xi(\alpha)/\xi(\alpha+s)$ is not in L, whether or not RH holds.*

(b) For every real α, $t \mapsto \xi(\alpha+it)/\xi(\alpha)$ is, unconditionally, the characteristic function of the density $\Phi(u)e^{(\alpha-\frac{1}{2})u}/\xi(\alpha)$, with Φ of (1.2.3), but it is never infinitely divisible, hence never an EGGC. At $\alpha=\frac{1}{2}$ it is the member $\Xi(t)/\Xi(0)$ of the van Dantzig pair of Theorem 5.1.1.

(c) For every real α, $t \mapsto \xi(\alpha)/\xi(\alpha+it)$ is unbounded, so it is not a characteristic function.

Proof. (a) By Stirling, $\log\{\xi(\alpha)/\xi(\alpha+s)\} = -\frac{1}{2}s\log s\,(1+o(1))$, while, if $\mathbb{E}e^{-sX}$ agreed with it on an interval, it would be finite and agree with it on all of $\mathbb{R}$, since the transform of a positive measure is singular at the real end points of its interval of convergence [340] and $\xi(\alpha)/\xi(\alpha+s)$ is analytic near the real axis; then $\mathbb{E}e^{-sX} \geq \Pr(X \leq M)e^{-sM}$ decays at most exponentially as $s \to \infty$. (b) Φ is even and positive, since $2\pi n^2 e^{2u} > 3$ for $u \geq 0$, and $\xi(\frac{1}{2}+iz) = \int_{\mathbb{R}} \Phi(u)e^{izu}du$ continues to $z = t - i(\alpha-\frac{1}{2})$, which gives the density, integrable because $\Phi(u) = O(e^{\frac{9}{2}|u|-\pi e^{2|u|}})$. The function is entire in t and vanishes at $t = \gamma - i(\beta-\alpha)$ for every zero ρ, whereas an infinitely divisible law with $\mathbb{E}e^{cX} < \infty$ for every real c has $\mathbb{E}e^{zX} = e^{\psi(z)}$ for every complex z, with ψ its Lévy Khintchine exponent [297], so its characteristic function has no zero in $\mathbb{C}$. (c) $|\xi(\alpha+it)| \to 0$ faster than any power. □

The EGGC reading of the reciprocal that survives is therefore the symmetric one, $\xi(\alpha)/\xi(\alpha+|t|) = \mathbb{E}e^{-t^2H}$, with H a GGC when $\alpha \geq \sup_\rho \operatorname{Re}\rho$ (Theorem 1.2.2, Section 1.3.2), which contains the square root. Without it, the structure that exists lives on the ξ side and stops at $\alpha = 1$. For $\alpha > 1$, with $G \sim \text{Gamma}(1+\frac{\alpha}{2}, 1)$ and N independent, $\Pr(N=n) = n^{-\alpha}/\zeta(\alpha)$, the identity $\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+\frac{s}{2})\zeta(s)$ gives $\xi(\alpha+s)/\xi(\alpha) = (1+\frac{s}{\alpha-1})\mathbb{E}e^{-sL}$ with $L = \frac{1}{2}\log\pi - \frac{1}{2}\log G + \log N$ infinitely divisible, and $\xi(\alpha)/\xi(\alpha+s)$ is the quotient of a gamma transform by an infinitely divisible one, whose Lévy measure enters with its sign reversed, $1/\mathbb{E}N^{-s} = \zeta(\alpha)\sum_n \mu(n)n^{-\alpha-s}$ being Möbius signed.

21.5 Dirichlet L functions and number fields

This section carries the construction of Section 21.2 from ξ to Dirichlet L functions and to Dedekind zeta functions. The numbers are computed by the scripts `gl_dirichlet.py` and `gl_sinsq_chi4.py` (`code/README.md`).

Let χ be a primitive character modulo $q > 1$ of parity $a \in \{0,1\}$, $\chi(-1) =$

$(-1)^a$, and

$$\xi(s,\chi) = \Big(\frac{q}{\pi}\Big)^{(s+a)/2}\Gamma\Big(\frac{s+a}{2}\Big)L(s,\chi),$$
$$\xi(s,\chi) = W_\chi\,\xi(1-s,\bar\chi), \qquad |W_\chi| = 1,$$

an entire function of order one [71].

21.5.1 The positivity obstacle

For $\mathrm{Re}(s) > 1$,

$$\log L(s,\chi) = \sum_{n\ge 2} \frac{\chi(n)\Lambda(n)}{\log n}\, n^{-s}, \tag{21.5.1}$$

so the analogue of μ^σ (Lemma 21.3.1) carries the weights $\chi(p)^r$: complex for complex χ, signed for real nonprincipal χ. Then $L(\sigma+it,\chi)/L(\sigma,\chi)$ is the characteristic function of no probability law: by uniqueness its only candidate has the weights $\chi(n)n^{-\sigma}/L(\sigma,\chi)$ at $-\log n$, and these are not all nonnegative.

On the critical line the Thorin construction needs no Euler product at all, only evenness and the Hadamard product of the completed function.

21.5.2 Real characters

For a real primitive character, $\bar\chi = \chi$ and $W_\chi = 1$, so $\xi(s,\chi) = \xi(1-s,\chi)$ and $\xi(\cdot,\chi)$ is real on the real axis.

Proposition 21.5.1. *Let χ be a real primitive character, and assume GRH for $L(s,\chi)$ and $L(\frac12,\chi)\neq 0$. Write the zeros as $\frac12 \pm i\gamma$ with $\gamma>0$, counted with multiplicity. Then $T_\chi \overset{d}{=} \sum_{\gamma>0} e_\gamma/\gamma^2$ is a GGC with*

$$\frac{\xi(\frac12,\chi)}{\xi(\frac12+\sqrt{u},\chi)} = \mathbb{E}\big[e^{-uT_\chi}\big], \qquad u\ge 0, \tag{21.5.2}$$

with cumulants $(j-1)!\,p_j(\chi)$, $p_j(\chi) = \sum_{\gamma>0}\gamma^{-2j}$. Propositions 21.3.5 to 21.3.11 hold with ξ replaced by $\xi(\cdot,\chi)$, with GRH for $L(s,\chi)$ and $L(\frac12,\chi)\neq 0$ in the place of RH and $\xi(\frac12)\neq 0$, and with the values at $u_0 = \frac14$, which use $\xi(1) = \frac12$, replaced by those below; in particular

$$\sum_{\gamma>0}\frac{1}{\gamma^2+\frac14} = \frac{\xi'}{\xi}(1,\chi) = \frac12\log\frac{q}{\pi} + \frac12\,\psi\Big(\frac{1+a}{2}\Big) + \frac{L'}{L}(1,\chi), \tag{21.5.3}$$

$$\mathbb{E}\big[e^{-T_\chi/4}\big] = \frac{\xi(\frac12,\chi)}{\xi(1,\chi)}.$$

Proof. $\xi(\frac12+s,\chi)$ is even, entire of order one and nonzero at $s=0$, so the argument of Lemmas 21.2.1 and 21.2.2 gives $\xi(\frac12+s,\chi)/\xi(\frac12,\chi)=\prod_\gamma(1+s^2/\gamma^2)$, and (G2) gives (21.5.2). The later propositions used only this product, evenness and growth on the real axis. For (21.5.3), take the logarithmic derivative of the product at $s=\frac12$ and of the definition of $\xi(\cdot,\chi)$ at $s=1$. □

The hypothesis $L(\frac12,\chi)\neq0$ is Chowla's conjecture for real characters, still open; a central zero would put an atom at u_0 in the tilted Thorin measure (Theorem 21.2.3) and block the untilted clock (Corollary 21.2.4(a)). Lemma 21.4.1 holds for $\xi(\cdot,\chi)$ under GRH, with the same decay $e^{-\pi|\tau|/4}$ and the power $|\tau|^{\sigma/2-a/2-1/4+\eta}$ in place of $|\tau|^{\sigma/2-9/4+\eta}$, so the master theorem analysis of Section 21.4 carries over with the same two obstructions.

21.5.3 Complex characters

For complex χ the functional equation links $\xi(s,\chi)$ to a different function, $\xi(1-s,\bar\chi)$. So $\xi(\frac12+s,\chi)$ is not even, its ordinates are not symmetric under $\gamma\mapsto-\gamma$, and the pairing behind (21.5.2) fails. Rotating by the root number makes the function real on the critical line, but not even, and it stays complex on the real axis, where a Laplace transform must be positive.

Proposition 21.5.2. *Let χ be a complex primitive character and $\xi_{\chi\bar\chi}(s):=\xi(s,\chi)\xi(s,\bar\chi)$. Then $\xi_{\chi\bar\chi}(s)=\xi_{\chi\bar\chi}(1-s)$ and $\xi_{\chi\bar\chi}(\sigma)=|\xi(\sigma,\chi)|^2$ for real σ. If GRH holds for $L(s,\chi)$ and $L(\frac12,\chi)\neq0$, then*

$$\frac{\xi_{\chi\bar\chi}(\frac12)}{\xi_{\chi\bar\chi}(\frac12+\sqrt{u})}=\mathbb{E}\big[e^{-uT_{\chi\bar\chi}}\big],\qquad T_{\chi\bar\chi}\overset{d}{=}\sum_{\gamma\in\Gamma_\chi}\frac{e_\gamma}{\gamma^2},$$

where Γ_χ is the multiset of all ordinates of zeros of $L(s,\chi)$, of both signs.

Proof. The root numbers multiply to $W_\chi W_{\bar\chi}=|W_\chi|^2=1$, and $\xi(\sigma,\bar\chi)=\overline{\xi(\sigma,\chi)}$ for real σ. The zeros of $L(s,\bar\chi)$ are the conjugates $\frac12-i\gamma$, so $\xi_{\chi\bar\chi}$ has zeros at $\frac12\pm i\gamma$ for every $\gamma\in\Gamma_\chi$, and the even product argument applies: each ordinate of $L(s,\chi)$ gives one factor $1+u/\gamma^2$, its partner $\frac12-i\gamma$ among the zeros of $L(s,\bar\chi)$ being the other zero of the pair, so no ordinate is counted twice. □

21.5.4 The Dedekind route

Let χ be a real primitive character of conductor q and $F=\mathbb{Q}(\sqrt{D})$, with $D=\chi(-1)q$ a fundamental discriminant. Then $\zeta_F(s)=\zeta(s)L(s,\chi)$, and F is real quadratic, $(r_1,r_2)=(2,0)$, for even χ and imaginary quadratic, $(r_1,r_2)=(0,1)$, for odd χ. More generally, for $q\geq3$ the cyclotomic field $F_q=\mathbb{Q}(\zeta_q)$ is totally complex of degree $\varphi(q)$ and

$$\zeta_{F_q}(s)=\prod_{\chi \bmod q}L(s,\chi^*),\tag{21.5.4}$$

with χ^* the primitive character inducing χ. Every primitive character of conductor dividing q occurs, complex ones together with their conjugates. And Dedekind zeta functions have nonnegative coefficients,

$$\log \zeta_F(s) = \sum_{n\geq 2} \frac{\Lambda_F(n)}{\log n}\, n^{-s}, \qquad \Lambda_F(n) = \sum_{N\mathfrak{p}^r = n} \log N\mathfrak{p} \;\geq 0,$$

which is why one passes to fields.

PROPOSITION 21.5.3. *For a quadratic field, $\xi_F(s) = 2\,\xi(s)\xi(s,\chi)$ when χ is even and $\xi_F(s) = 2q^{-1/2}\xi(s)\xi(s,\chi)$ when χ is odd. Consequently, at any tilt $u_0 \geq 0$ at which ξ and $\xi(\cdot,\chi)$ both have the Thorin property in the sense of Theorem 21.2.3, so does ξ_F, its tilted clock is the independent sum of the tilted clocks of ξ and of $\xi(\cdot,\chi)$, and*

$$U_{F,u_0} = U_{\mathbb{Q},u_0} + U_{\chi,u_0},$$

where U_{χ,u_0} is the Thorin measure of the tilted clock of $\xi(\cdot,\chi)$, defined as U_{F,u_0} is in Theorem 21.2.3; at $u_0 = 0$ this reads $T_F \overset{d}{=} T_\zeta + T_\chi$, with $T_\zeta = T$ the centre clock of ξ. For a cyclotomic field, under the hypotheses of Propositions 21.5.1 and 21.5.2, T_{F_q} is the independent sum of T_ζ, of T_χ over the real primitive characters of conductor $f \mid q$ with $f > 1$, and of $T_{\chi\bar\chi}$ over the conjugate pairs of complex primitive characters of conductor dividing q. In particular the cumulants add: $p_j(F) = p_j(\zeta) + p_j(\chi)$ for a quadratic field.

For $\mathbb{Q}(i)$ at basepoints $\alpha \geq 1$, where no hypothesis is needed, this is Proposition 8.5.27 and Corollary 8.5.28, with the constant $2 \cdot 4^{-1/2} = 1$.

Proof. For even χ the gamma factors agree, $\Gamma_{\mathbb{R}}(s)^2$ on both sides, and $|d_F| = q$. For odd χ the duplication formula gives $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$, while $\xi(s,\chi) = q^{(s+1)/2}\Gamma_{\mathbb{R}}(s+1)L(s,\chi)$ carries an extra $q^{1/2}$. The constants cancel in the ratios (21.2.4), so the Laplace transforms multiply and the variables add. The cyclotomic case follows from (21.5.4) in the same way: the factors there are primitive, half of the characters modulo $q \geq 3$ are odd, so the gamma factors multiply to $\Gamma_{\mathbb{C}}(s)^{\varphi(q)/2}$, and the discriminant is the product of the conductors. □

21.5.5 The Lévy decomposition of ξ_F, with signs

For $\alpha > 1$ and $s \geq 0$ put $h_s(x) = e^{-sx} - 1 + sx \geq 0$. Binet's formula gives

$$\log \frac{\Gamma_{\mathbb{R}}(\alpha+s)}{\Gamma_{\mathbb{R}}(\alpha)} - s\frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-2x}}\frac{dx}{x},$$
$$\log \frac{\Gamma_{\mathbb{C}}(\alpha+s)}{\Gamma_{\mathbb{C}}(\alpha)} - s\frac{\Gamma'_{\mathbb{C}}}{\Gamma_{\mathbb{C}}}(\alpha) = \int_0^\infty h_s(x)\frac{e^{-\alpha x}}{1-e^{-x}}\frac{dx}{x},$$

a linear factor gives $\log\frac{z+s}{z}-\frac{s}{z}=-\int_0^\infty h_s(x)e^{-zx}\frac{dx}{x}$ for $z>0$, and

$$\log\frac{\zeta_F(\alpha+s)}{\zeta_F(\alpha)}-s\frac{\zeta_F'}{\zeta_F}(\alpha)=\sum_{n\geq 2}\frac{\Lambda_F(n)}{\log n}\,n^{-\alpha}h_s(\log n).$$

Proposition 21.5.4. *For $\alpha>1$ and $s\geq 0$,*

$$\log\frac{\xi_F(\alpha+s)}{\xi_F(\alpha)}-s\,\frac{\xi_F'}{\xi_F}(\alpha)=\int_0^\infty h_s(x)\,e^{-\alpha x}\,\frac{\nu_F(dx)}{x} \tag{21.5.5}$$

with the signed measure

$$\nu_F(dx)=\Big(\frac{r_1}{1-e^{-2x}}+\frac{r_2}{1-e^{-x}}-1-e^x\Big)dx+\sum_{n\geq 2}\Lambda_F(n)\,\delta_{\log n}(dx). \tag{21.5.6}$$

Proof. Add the decompositions above. The density $-(1+e^x)$ comes from the factors $\alpha+s$ and $\alpha-1+s$ of $s(s-1)$, both with positive z since $\alpha>1$, and $|d_F|^{s/2}$ is linear in s and only changes the drift. □

For $F=\mathbb{Q}$, where $\xi_{\mathbb{Q}}=2\xi$, this is Proposition 9.1.10; for χ_4 and χ_{12} it is Propositions 8.5.23 and 11.3.2.

Remark 21.5.5 (Where positivity fails, and why it is the same question). The polynomial factor contributes the density $-(1+e^x)$, which dominates the positive continuous part for large x. By the prime ideal theorem Λ_F has mean value one, so on average the atoms $\sum\Lambda_F(n)\delta_{\log n}$ cancel $-e^x\,dx$, and by the explicit formula what remains is a sum over the zeros of ζ_F. Positivity of any smoothed ν_F is therefore again a statement about zeros: the Dedekind route does not give the Thorin property for free.

21.5.6 The sine squared lemma: from primes to the Thorin measure

Proposition 21.5.4 writes $\log\xi_F$ against a signed Lévy measure. The sine squared lemma [249, Lemma 8.1], first in [250, Lemma 1], converts any such decomposition into a Thorin representation of the reciprocal [249, Theorem 8.2]; in the book it is equation (1.2.9) of Section 1.2, applied to ξ in Proposition 9.1.3 and to $\xi(\cdot,\chi_4)$ after the Lévy form of Proposition 8.5.23 (Section 8.5.3). In the chart $z=b^2$ the kernel identity (1.2.9) reads, for $w,x>0$,

$$e^{-wx}-1+wx=\frac{x}{\pi}\int_0^\infty\log\Big(1+\frac{w^2}{b^2}\Big)\,2\sin^2\Big(\frac{bx}{2}\Big)\,db, \tag{21.5.7}$$

and the book gives its proof. For a function f below write $v_\alpha(b)=\operatorname{Re}\frac{f'}{f}(\alpha+ib)$. For $f=\xi$ this is the function v_α of Section 1.2, v_α/π is the density $\rho_{\alpha-1/2}$ of (1.2.7), and the measure $U_\alpha(dt)=v_\alpha(\sqrt{t})\,dt/(2\pi\sqrt{t})$ is that of (1.2.6).

PROPOSITION 21.5.6 (The lemma on the prime side). *Let f be ξ_F, or $\xi(\cdot,\chi)$ for a real primitive χ, and let $\alpha > 1$. Then*

$$\frac{f(\alpha)}{f(\alpha+\sqrt{u})} = \exp\Big(-\int_0^\infty \log\Big(1+\frac{u}{b^2}\Big)\,\frac{v_\alpha(b)}{\pi}\,db\Big), \qquad u \ge 0, \tag{21.5.8}$$

where

$$v_\alpha(b) = \frac{f'}{f}(\alpha) + \int_0^\infty 2\sin^2\Big(\frac{bx}{2}\Big)e^{-\alpha x}\,\nu(dx) = \operatorname{Re}\frac{f'}{f}(\alpha+ib), \tag{21.5.9}$$

and ν is the Lévy measure of f: ν_F of (21.5.6) *for ξ_F, and for $\xi(\cdot,\chi)$*

$$\nu_\chi(dx) = \sum_{n\ge 2}\chi(n)\Lambda(n)\,\delta_{\log n}(dx) + \frac{e^{(1-a)x}}{2\sinh x}\,dx,$$

that is $(2\sinh x)^{-1}dx$ for odd χ and $(1-e^{-2x})^{-1}dx$ for even χ, from the gamma factor $\Gamma_{\mathbb{R}}(s+a)$.

Proof. For ξ_F the decomposition

$$\log\frac{f(\alpha+w)}{f(\alpha)} - w\,\frac{f'}{f}(\alpha) = \int_0^\infty h_w(x)\,e^{-\alpha x}\,\frac{\nu(dx)}{x}$$

is Proposition 21.5.4, and for $\xi(\cdot,\chi)$ it follows in the same way from Binet's formula for $\Gamma_{\mathbb{R}}(s+a)$ and (21.5.1). Insert (21.5.7) and interchange the integrals; this is justified for $\alpha > 1$ because the kernels are nonnegative and $\int \min(1,x)\,e^{-\alpha x}|\nu|(dx) < \infty$ (near 0 the density of $|\nu|$ is of order $1/x$, and at infinity $e^{-\alpha x}|\nu|(dx)$ is dominated by $e^{-(\alpha-1)x}dx$ and by the prime sum $\sum \Lambda_F(n)n^{-\alpha}$). The linear term is absorbed using $\int_0^\infty \log(1+w^2/b^2)\,db = \pi w$, which gives the first form of v_α with $u = w^2$. For the second, differentiate the decomposition in w: $\int(1-e^{-wx})e^{-\alpha x}\nu(dx) = (f'/f)(\alpha+w) - (f'/f)(\alpha)$. The left side is analytic for $\operatorname{Re} w > 1-\alpha$ by the same bound, the right side is meromorphic with poles at the zeros, and they agree for $w > 0$, so both continue to $w = ib$; taking real parts gives $\int 2\sin^2(bx/2)e^{-\alpha x}\nu(dx) = \operatorname{Re}(f'/f)(\alpha+ib) - (f'/f)(\alpha)$. □

For a real character the arithmetic part of $v_\alpha(b)$ is

$$-\sum_{n\ge 2}\chi(n)\Lambda(n)\,n^{-\alpha}\cos(b\log n),$$

absolutely convergent for $\alpha > 1$. At $\alpha = 1$ it converges conditionally for nonprincipal χ, and for ξ_F the prime sum diverges and converges only together with the density $-e^x$ of (21.5.6), as in Proposition 9.1.3.

LEMMA 21.5.7. *For $c \in \mathbb{C}$ with $\operatorname{Re} c > 0$ and $w > 0$,*

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \log\Big(1+\frac{w^2}{b^2}\Big)\,\operatorname{Re}\frac{1}{c+ib}\,db = 2\log\Big|1+\frac{w}{c}\Big|.$$

For $\operatorname{Re} c < 0$ the right side is replaced by $-2\log|1-w/c|$.

Proof. Write $\log(1+w^2/b^2) = \int_0^w 2w'(b^2+w'^2)^{-1}dw'$. For $w' > 0$ the function $b \mapsto (c+ib)^{-1}(b^2+w'^2)^{-1}$ is rational and $O(|b|^{-3})$, with poles at ic, iw' and $-iw'$. If $\operatorname{Re} c > 0$ the pole ic lies in the upper half plane, and closing the contour in the lower half plane picks up only $-iw'$, so that $\int_{-\infty}^{\infty} db/((c+ib)(b^2+w'^2)) = -2\pi i/((c+w')(-2iw')) = \pi/(w'(c+w'))$. Since $|\operatorname{Re}(c+ib)^{-1}| \le |\operatorname{Re} c|^{-1}$ and $\int_{-\infty}^{\infty} 2w'(b^2+w'^2)^{-1}db = 2\pi$, Fubini applies, and taking real parts,

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{c+ib}\,db = \operatorname{Re}\int_0^w \frac{2\,dw'}{c+w'} = 2\log\Big|\frac{c+w}{c}\Big|.$$

If $\operatorname{Re} c < 0$ the same computation, closing in the upper half plane, gives $-2\log|(c-w)/c|$. □

Zeros $\rho = \beta + i\gamma$ are counted with multiplicity. Say that f has the *Thorin property at the basepoint* $\alpha > \frac12$ if $f(\alpha) \neq 0$ and $u \mapsto f(\alpha)/f(\alpha+\sqrt{u})$ is the Laplace transform of a GGC.

THEOREM 21.5.8 (The basepoint family of the lemma). *Let f be as in Proposition 21.5.6, with zeros $\rho = \beta + i\gamma$.*

(a) If every zero has $\beta < \alpha$, then (21.5.8) *holds with*

$$\frac{v_\alpha(b)}{\pi} = \frac{1}{\pi}\sum_\rho \operatorname{Re}\frac{1}{\alpha+ib-\rho} = \frac{1}{\pi}\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(b-\gamma)^2} > 0,$$

so f has the Thorin property at the basepoint α, with no drift and Thorin measure $U_\alpha(dt) = v_\alpha(\sqrt{t})\,dt/(2\pi\sqrt{t})$. Since every zero has $\beta < 1$, this holds unconditionally for every $\alpha \ge 1$.

(b) If some zero has $\beta > \alpha$ and $f(\alpha) \neq 0$, then f does not have the Thorin property at the basepoint α. Hence f has it at every basepoint $\alpha > \frac12$ if and only if every zero lies on the critical line.

(c) Under GRH, and if $f(\frac12) \neq 0$, with $\varepsilon = \alpha - \frac12$,

$$\frac{v_\alpha(b)}{\pi} = \sum_{\gamma>0}\big(P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)\big), \qquad P_\varepsilon(x) = \frac{\varepsilon}{\pi(\varepsilon^2+x^2)},$$

the Poisson smoothing at height ε of the zero counting measure. As $\alpha \downarrow \frac12$, $(v_\alpha(b)/\pi)\,db \to \sum_{\gamma>0}\delta_\gamma$ vaguely on $[0,\infty)$, and U_α tends to the untilted Thorin measure $\sum_\gamma \delta_{\gamma^2}$ of Theorem 21.2.3.

Proof. (a) By Hadamard's factorisation $f(s) = e^{A+Bs}\prod_\rho (1-s/\rho)e^{s/\rho}$ and the functional equation, $\operatorname{Re}(f'/f)(s) = \sum_\rho \operatorname{Re}(s-\rho)^{-1}$, absolutely convergent, for all s off the zeros, and $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$ [71, Ch. 12]. Each term of the sum is positive when $\operatorname{Re} s > \beta$, which gives the formula for v_α and its positivity. For a zero ρ put $\omega_\rho = \alpha - \rho$, so that $\operatorname{Re}\omega_\rho = \alpha - \beta > 0$ and the term of ρ in $v_\alpha(b)$ is $\operatorname{Re}(\omega_\rho + ib)^{-1}$, and for $w > 0$ put

$$I_\rho = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\operatorname{Re}\frac{1}{\omega_\rho + ib}\,db \;\ge 0.$$

Since $\operatorname{Re}(\omega_{\bar\rho}+ib)^{-1} = \operatorname{Re}(\omega_\rho - ib)^{-1}$, the substitution $b \mapsto -b$ and Lemma 21.5.7 give $I_\rho + I_{\bar\rho} = 2\log|1+w/\omega_\rho|$, and $|1+w/\omega_{\bar\rho}| = |1+w/\omega_\rho|$. The function f is real on the real axis, so $\rho \mapsto \bar\rho$ permutes the zeros, and since all terms are nonnegative we may sum:

$$\int_0^\infty \log\Big(1+\frac{w^2}{b^2}\Big)\frac{v_\alpha(b)}{\pi}\,db = \sum_\rho I_\rho = \sum_\rho \log\Big|1+\frac{w}{\omega_\rho}\Big|.$$

On the other side the Hadamard product gives $f(\alpha+w)/f(\alpha) = e^{Bw}\prod_\rho(1+w/\omega_\rho)e^{w/\rho}$, an absolutely convergent product, and the left side is positive because f is real and has no zero on $[\alpha,\infty)$. Taking logarithms and using $\operatorname{Re} B = -\sum_\rho \operatorname{Re}\rho^{-1}$, we get $\log f(\alpha+w)/f(\alpha) = \sum_\rho \log|1+w/\omega_\rho|$, the series converging absolutely. With $u = w^2$ this is (21.5.8) with density v_α/π. In the variable $t = b^2$ the exponent is $\int_0^\infty \log(1+u/t)\,U_\alpha(dt)$ and it is finite; this is equivalent to the integrability conditions on a Thorin measure stated with (21.2.1), so U_α is a Thorin measure with no drift.

(b) Let ρ be a zero with $\beta > \alpha$ and put $u_\rho = (\rho-\alpha)^2$. Since $\operatorname{Re}(\rho-\alpha) = \beta-\alpha > 0$, the principal square root of u_ρ is $\rho-\alpha$, and $u_\rho \notin (-\infty,0]$, because otherwise $\rho-\alpha$ would be purely imaginary. The function $f(\alpha+\sqrt{u})$ is analytic on $\mathbb{C}\setminus(-\infty,0]$ and vanishes at u_ρ, so $f(\alpha)/f(\alpha+\sqrt{u})$ has a pole there. If f had the Thorin property at α, this function would coincide on $u>0$, hence on $\mathbb{C}\setminus(-\infty,0]$, with a GGC transform, which is analytic there by (G1): a contradiction. If every zero lies on the critical line, then $\beta = \frac12 < \alpha$ for every $\alpha > \frac12$ and (a) gives the Thorin property at every basepoint $\alpha > \frac12$. If some zero is off the line, the functional equation gives one with $\beta > \frac12$, and the property fails at every $\alpha \in (\frac12,\beta)$, by the argument above where $f(\alpha)\ne 0$ and by definition where $f(\alpha) = 0$.

(c) Under GRH every zero is $\frac12 \pm i\gamma$ with $\gamma > 0$, since $f(\frac12) \ne 0$, and (a) applies for every $\alpha > \frac12$; with $\alpha-\beta = \varepsilon$ the two zeros $\frac12 \pm i\gamma$ contribute $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$. Let h be continuous with compact support in $[0,R]$ and let $0 < \varepsilon \le 1$. For $\gamma \le R+1$ the integral of h against $P_\varepsilon(b-\gamma)+P_\varepsilon(b+\gamma)$ over $[0,\infty)$ tends to $h(\gamma)$, because $\gamma > 0$ is a point of continuity of h extended by zero and $\int_0^\infty P_\varepsilon(b+\gamma)\,db \le \varepsilon/(\pi\gamma)$. For $\gamma > R+1$ and $b \in [0,R]$ one has $|b\mp\gamma| \ge \gamma - R \ge \gamma/(R+1)$, so the integral is at most $2R\|h\|_\infty(R+1)^2\varepsilon/(\pi\gamma^2)$. The zeros with $\gamma > R+1$ therefore contribute

at most a constant times $\varepsilon \sum_\gamma \gamma^{-2}$, which tends to 0, and the finitely many others give $\sum_{\gamma>0} h(\gamma)$ in the limit, so $\int h\,(v_\alpha/\pi)\,db \to \sum_{\gamma>0} h(\gamma)$. Under $b \mapsto b^2$ this is the convergence of U_α to $\sum_\gamma \delta_{\gamma^2}$. □

If the half plane of [226], which has not been refereed (Section 12.1.2), stands, then every zero of every Dirichlet L function has $\beta \le \frac{7}{8}$, and part (a) gives the Thorin property of $\xi(\cdot,\chi)$ for every real primitive character χ, and of ξ_F for every quadratic field F, at every basepoint $\alpha > \frac{7}{8}$, with Thorin density v_α/π a sum of Lorentzians of widths at least $\alpha - \frac{7}{8}$. This is Theorem 16.12.1(i) carried from ζ to the family, and it lowers the unconditional end $\alpha \ge 1$ of part (a) to $\alpha > \frac{7}{8}$ under that hypothesis. At $\alpha = \frac{7}{8}$ itself part (a) does not apply, since the half plane does not exclude a zero on the line $\operatorname{Re} s = \frac{7}{8}$.

For ξ these statements are in the book: part (a) is the first sentence of Theorem 1.2.2; parts (a) and (b) together, with the borderline case in which a zero on the line $\operatorname{Re} s = \alpha$ contributes an atom, are Theorem 7.1.14, and Theorem 10.2.7 treats the basepoints $\frac{1}{2} \le \alpha < 1$; and the Poisson form of part (c) is Proposition 1.2.3. For χ_4 the corresponding result is Proposition 8.5.24, and for χ_{12} at the basepoint one Proposition 11.3.1. In part (c) the vague convergence holds without RH, as shown after Remark 1.2.4 and in [249, Proposition 9.5]: the limit is then the counting measure of the ordinates of the zeros on the line, the pairs off the line contributing signed mass whose total variation tends to zero. Without $f(\frac{1}{2}) \ne 0$ a central zero of order $2m$ adds the term $2m\,P_\varepsilon(b)$ to v_α/π, of mass m on $(0,\infty)$, which concentrates at $b = 0$ as $\varepsilon \downarrow 0$; the limit is then not a Thorin measure (Corollary 21.2.4(a)).

The tilt family and the basepoint family meet at the centre. Theorem 21.2.3 tilts by u_0, reads f at $\frac{1}{2} + \sqrt{u_0 + u}$, and shifts each atom from γ^2 to $u_0 + \gamma^2$. The lemma's family reads f at $\alpha + \sqrt{u}$ and smooths each atom by a Poisson kernel of width $\alpha - \frac{1}{2}$. One consequence differs. At the centre $f(\frac{1}{2} + w)$ is even in w and $v_{1/2}(0) = 0$; for $\alpha > \frac{1}{2}$ the function $f(\alpha + w)$ is not even, and under the hypothesis of part (a) $v_\alpha(0) = (f'/f)(\alpha) > 0$ is a sum of positive terms. The Thorin measure then has a component like that of a stable law of index $\frac{1}{2}$, its density behaving like $v_\alpha(0)/(2\pi\sqrt{t})$ as $t \downarrow 0$, and the shifted clock T_α has $\mathbb{E}[T_\alpha] = \infty$. At every basepoint $\alpha \ge 1$ the book proves the exact split into a positive stable variable of index $\frac{1}{2}$ and a GGC, for ξ in Propositions 9.1.11 and 9.1.12 and for χ_4 and χ_{12} in Theorem 9.1.18(v) with the paragraph after it. The moment dictionary of Section 21.3.3 lives only at the centre.

Remark 21.5.9 (What the lemma buys, and what it cannot). Reaching the centre means running the Poisson smoothing backwards from α to $\frac{1}{2}$. In the variable $x = \log n$ the smoothing multiplies the weight of n by $n^{-(\alpha - 1/2)}$, so undoing it replaces $n^{-\alpha}$ by the divergent $n^{-1/2}$: the backward step is the explicit formula, and it is ill posed. Chapter 19 measures its cost. By Theorem 19.2.2 the Thorin density at a basepoint beyond the zeros is the Poisson smoothing of a positive

measure down to the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon_1$ exactly when no zero has $\beta - \frac{1}{2} > \varepsilon_1$. Under GRH, by Proposition 19.2.3 and Table 19.1 (for χ_4) the number of primes needed for accuracy δ is of order $\delta^{-1/(\alpha-1/2)}$, unbounded as $\alpha \downarrow \frac{1}{2}$. Section 20.1.1 of Chapter 20 shows that the deconvolved object is Weil's distribution of the explicit formula. Positivity on a single line is not known to exclude zeros to its right (Remark 1.2.4 and Section 12.1.2). For ξ, Proposition 9.1.12 proves that the minimum of v_α on the line is at $b = 0$ for every $\alpha \geq 1$, and for χ_4 and χ_{12} Theorem 9.1.18(v) does; the theorem also certifies it for every $\alpha \in (\frac{1}{2}, \alpha_{\max}]$ up to the height to which GRH is verified (Table 9.4), and shows that the statement on the whole lines for all $\alpha \in (\frac{1}{2}, 1)$ together is equivalent to GRH. Our computations for χ_4 below agree. A real zero $\beta > \alpha$ adds the term $-(\beta-\alpha)/((\beta-\alpha)^2 + b^2)$ to $v_\alpha(b)$, whose modulus is largest at $b = 0$, so the origin is where a real exceptional zero acts first: a Siegel zero is a failure of the lemma's positivity at the origin. The classical zero free region arguments live on the Lévy side of the lemma, which restates their conclusions as positivity of a Thorin density. That is a translation, and this chapter obtains no new estimate from it, for Linnik's constant K or for the least prime in a progression.

Numerical check for χ_4. The script `code/gl_sinsq_chi4.py` checks the lemma for χ_4 at 30 digits. At $\alpha = 2$, with the prime powers up to N and the integral over b done term by term (each prime term is the kernel identity), the integral in (21.5.8) differs from $\log \xi(\alpha + w, \chi_4)/\xi(\alpha, \chi_4)$ by -4.4×10^{-8}, -1.2×10^{-9} and 6.2×10^{-11} for $N = 10^4$, 10^5 and 10^6 at $w = 1$ and $w = 3$; this difference is the tail of the prime sum beyond N. The prime side of (21.5.9), over the prime powers up to 2×10^6, agrees with $v_\alpha(b)/\pi$ computed directly from ξ'/ξ to 9.4×10^{-11} at $\alpha = 2$ and to 1.9×10^{-7} at $\alpha = \frac{3}{2}$. The zero side, the sum of part (a) over the 203 zeros of $L(s, \chi_4)$ below height 300 found by the script, with the tail beyond height 300 taken from the smooth zero density, differs from the direct value by 1.1×10^{-6} to 1.7×10^{-6} at $\alpha = \frac{3}{2}$ and 2 for $0 \leq b \leq 50$; once the tail is anchored to the exact zero count at the cut, as in (21.3.8), the difference falls to between 9.7×10^{-9} and 1.7×10^{-8}, far inside the enclosure of 1.6×10^{-5} to 2.7×10^{-5} that the explicit bound for the zero count used in the proof of Theorem 9.1.18(v) guarantees.

On the lines $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$ the minimum of $v_\alpha(b)$ over $0 \leq b \leq 150$, on grids of step 0.1 and 0.005, is at $b = 0$ (Table 21.4), where $v_\alpha(0) = b_\alpha = (\xi'/\xi)(\alpha, \chi_4)$.

These checks test different things. For $\alpha = 2$ and $\frac{3}{2}$ every zero of $L(s, \chi_4)$ has real part below $1 < \alpha$, so Proposition 21.5.6 and part (a) of Theorem 21.5.8 hold with no hypothesis, and the comparisons above are identities up to truncation: they test the quadrature, the truncation of the prime sums, the zero list and its tail, and the code, not the theorem and not GRH. The grid of the table is a finite test of the function v_α, consistent with GRH; part (ii) of Theorem 9.1.18 turns it into a theorem for $0 < |b| \leq 2.5 \times 10^7 - 1.21$, using the verification of GRH for χ_4 to height 2.5×10^7 [245], and for $\alpha < 1$ the function v_α/π is the Thorin density of $\xi(\alpha, \chi_4)/\xi(\alpha + \sqrt{u}, \chi_4)$ only if no zero lies to the right of α, which is known below

α	$v_\alpha(0) = b_\alpha$	$2(\alpha - \frac{1}{2})\, p_1(\chi_4)$	ratio
1	0.0777839900	0.0780148250	0.99704
$\frac{3}{4}$	0.0389784293	0.0390074125	0.99926
$\frac{3}{5}$	0.0156011077	0.0156029650	0.99988
$\frac{11}{20}$	0.0078012503	0.0078014825	0.99997

Table 21.4: The minimum of $v_\alpha(b)$ over $0 \le b \le 150$ for χ_4, always at $b = 0$, against the bound $2(\alpha - \frac{1}{2})\, p_1(\chi_4) = (\alpha - \frac{1}{2})V_f$ of Theorem 9.1.18(i), with $V_f = 2p_1(\chi_4) = 0.1560296499$. Computed by `code/gl_sinsq_chi4.py` (`data/gl_sinsq_chi4.json`).

that height and open above it. The lemma's output is not computable from the primes alone in practice for $\alpha \le 1$: the plain prime sum at $b = 0$ over the prime powers up to 2×10^6 differs from $v_\alpha(0)$ by 0.17, 12, 260 and 1100 percent of the value at $\alpha = 1, \frac{3}{4}, \frac{3}{5}, \frac{11}{20}$, so the minima of the table come from the zeros and from the direct values of ξ'/ξ, not from the primes.

Finally, with the smoothing run backwards by a smooth cutoff, the peaks of $-\sum_{n \le N} \chi_4(n)\Lambda(n)n^{-1/2}\cos(b\log n)\,\cos^2\left(\frac{\pi}{2}\frac{\log n}{\log N}\right)$, $N = 2 \times 10^6$, lie within 0.003 of all fourteen zeros of $L(s, \chi_4)$ below height 40. Up to height 45 the seventeen zeros are matched by seventeen peaks and there is no other peak above half the largest value. This tests the explicit formula with the smoothing run backwards, and it locates the ordinates from the primes; it does not test the lemma.

21.6 GRH from the Thorin property, and families of characters

Theorem 21.6.1. *Let χ be a real primitive character with quadratic field F.*

(a) If $\varphi_{F,1}$ is the Laplace transform of a GGC, then RH holds and every non-trivial zero of $L(s, \chi)$ lies on the critical line or is real.

(b) If $\varphi_{F,0}$ is the Laplace transform of a GGC, equivalently $\varphi_{F,1}$ is one with Thorin measure in $(1, \infty)$, then $L(s, \chi)$ satisfies GRH and $L(\frac{1}{2}, \chi) \ne 0$.

If $\varphi_{F_q,0}$ is a GGC Laplace transform for every $q \ge 3$, with $F_q = \mathbb{Q}(\zeta_q)$, then GRH holds for all Dirichlet L functions, and $L(\frac{1}{2}, \chi) \ne 0$ for every primitive character χ.

Proof. Apply Corollary 21.2.4 to $\zeta_F = \zeta\, L(s, \chi)$, part (b) of the corollary for the tilt $u_0 = 1$ and part (a) for $u_0 = 0$; for ζ itself real zeros in $(0, 1)$ are excluded because $\zeta(\sigma) < 0$ there, and $\zeta(\frac{1}{2}) \ne 0$. The last statement uses equation (21.5.4): the zeros of ζ_{F_q} are those of the functions $L(s, \chi)$ for the primitive characters χ of conductor dividing q, and every primitive character of conductor $f \ge 3$ occurs

for $q = f$. Part (a) of the corollary then puts all of them on the critical line and excludes $L(\frac{1}{2}, \chi) = 0$. □

REMARK 21.6.2 (The Siegel zero is a threshold crossing). A real zero β contributes the atom $u_0 - (\beta - \frac{1}{2})^2$, positive while $u_0 > (\beta - \frac{1}{2})^2$ and crossing zero as the tilt descends to $(\beta - \frac{1}{2})^2$. A Siegel zero near 1 pushes the threshold u_F^* of Corollary 21.2.4(c) towards $\frac{1}{4}$, leaving only tilts above a value near $\frac{1}{4}$.

21.6.1 Numerical verification for the characters modulo 4 and 5

We take χ_4 (odd, so $a = 1$) and the real character χ_5 modulo 5, the Legendre symbol (even, so $a = 0$). The numbers are computed by `gl_dirichlet.py`, in ball arithmetic where a count or a sign is certified and otherwise in mpmath, except those of the last paragraph, by `gl_zeta_moments.py` (`code/README.md`). Write $N(T, \chi)$ for the number of zeros of $\xi(s, \chi)$ with $0 < \operatorname{Im} s < T$, $\theta_\chi(T)/\pi = \frac{T}{2\pi} \log \frac{qT}{2\pi e} + \frac{a}{4} - \frac{1}{8}$ for the smooth count and $S(T, \chi) = N(T, \chi) - \theta_\chi(T)/\pi$ for its fluctuation.

Zeros. Sign changes of the real function $\xi(\frac{1}{2} + it, \chi)$ on $0 < t < 300$ give the first ordinates 6.0209489 and 6.6484533. The counts, 203 and 214, are certified: the argument principle in ball arithmetic gives $N(300, \chi) = 203$ and 214, equal to the numbers of sign changes, so no zero below 300 was missed and all are simple and on the line. The smooth counts $\theta_\chi(T)/\pi$ at $T = 300$ are 203.15 and 213.56, so $S(300, \chi_4) = -0.153$ and $S(300, \chi_5) = +0.443$.

No real or central zeros. On a grid of step 0.005 in $[\frac{1}{2}, 1)$ the smallest values of $L(\sigma, \chi)$ are the central ones, $L(\frac{1}{2}, \chi_4) = 0.66769$ and $L(\frac{1}{2}, \chi_5) = 0.23175$, and ball arithmetic certifies $L(\sigma, \chi) > 0$ on $[\frac{1}{2}, 2]$, with 32 pieces and smallest lower bound 0.033 for χ_4 and 101 pieces and 0.0031 for χ_5. Since $\xi(\sigma, \chi) = \xi(1 - \sigma, \chi)$ and the gamma factor is positive, $\xi(\sigma, \chi) > 0$ on $[0, 1]$, so neither function has a real zero in the critical strip, and neither has a central zero. So both Thorin thresholds are 0, and the untilted clock is available under GRH. For χ_4 the book also certifies the absence of real zeros with a positive kernel (Section 8.5.7.1).

Power sums. Let $p_j(\chi) = \sum_{\gamma > 0} \gamma^{-2j}$ be the power sums of the ordinates. The zero side is

$$\sum_{0 < \gamma < X} \gamma^{-2j} + \int_X^\infty t^{-2j} \, \frac{1}{2\pi} \log \frac{qt}{2\pi} \, dt, \qquad X = 300,$$

the sum over the ordinates below X plus the tail started at X. The function side is read from the Taylor coefficients of $\log \xi(\frac{1}{2} + s, \chi)$, in which s^{2j} has the coefficient $(-1)^{j+1} p_j / j$; it uses no zeros. Table 21.5 gives the two sides.

With $N(t, \chi) = \theta_\chi(t)/\pi + S(t, \chi)$, integration by parts gives the error of a tail started at the height X as $S(X, \chi) X^{-2j} - 2j \int_X^\infty S(t, \chi) \, t^{-2j-1} \, dt$, up to the difference between the density of the tail and θ_χ'/π, which is of order X^{-2j-1}. So the error falls with j like X^{-2j}, and it is not of the size $X^{1-2j} \log X$ of the tail

Table 21.5: The power sums $p_j(\chi)$ of the ordinates for the characters modulo 4 and 5: the zero side, the function side, the relative error, and the difference of the two sides times 300^{2j}.

q	j	zero side	function side	relative error	difference $\times 300^{2j}$
4	1	0.0780131	0.0780148	2.2×10^{-5}	-0.1512
	2	9.288458×10^{-4}	9.288458×10^{-4}	2.0×10^{-8}	-0.1498
	3	2.216962×10^{-5}	2.216962×10^{-5}	9.2×10^{-12}	-0.1484
5	1	0.0784620	0.0784571	6.3×10^{-5}	0.4423
	2	7.178001×10^{-4}	7.178001×10^{-4}	7.6×10^{-8}	0.4414
	3	1.316152×10^{-5}	1.316152×10^{-5}	4.6×10^{-11}	0.4405

itself. In Table 21.5 the last column is within 3 percent of $S(300, \chi_4) = -0.1526$ and $S(300, \chi_5) = +0.4431$ for every j.

Transfer function and products. The transfer function $\mathfrak{T}_\chi(t)$, the ratio $\xi(\frac12 + it, \chi)/\xi(\frac12 + t, \chi)$, has the values $\mathfrak{T}_{\chi_4}(2) = 0.53522$ and $\mathfrak{T}_{\chi_4}(5) = 0.014573$. The Hadamard product $\prod_{\gamma>0}(1 + s^2/\gamma^2)$ at $s = 3$, over the 203 and 214 ordinates with the tail started at 300, is 1.9524380 and 1.9736682, against $\xi(\frac72, \chi)/\xi(\frac12, \chi) =$ 1.9524675 and 1.9735809 for χ_4 and χ_5. The relative differences, -1.5×10^{-5} and $+4.4 \times 10^{-5}$, are the size $S(300, \chi)\, s^2/300^2$ of the fluctuation term of the tail. A match at this level does not by itself show that the product has no exponential factor; that is Lemma 21.2.2, whose argument covers $\xi(s, \chi)$ for a real character, because $\xi(\frac12 + s, \chi)$ is even in s (Proposition 21.5.1).

Negative moments and the tilt. The negative moments, computed as the Mellin integral of equation (21.3.14) for $\xi(s, \chi)$, are

$$\mathbb{E}\big[T_{\chi_4}^{-1/2}\big] = 3.7574677\ldots, \qquad \mathbb{E}\big[T_{\chi_5}^{-1/2}\big] = 3.7097072\ldots,$$

smaller than 6.7333 for ζ because the first zeros are lower. In equation (21.5.3), $(\xi'/\xi)(1, \chi_4) = 0.0777840$ and $(\xi'/\xi)(1, \chi_5) = 0.0782785$. The difference from $p_1(\chi)$ is 2.30835×10^{-4} for χ_4 and 1.78632×10^{-4} for χ_5; the series of equation (21.3.11) through $j = 12$ agrees to 3×10^{-28}. Since $\xi(1, \chi_4) = \frac{4}{\pi} \cdot \frac{\pi}{4} = 1$, $\mathbb{E}[e^{-T\chi_4/4}] = \xi(\frac12, \chi_4) = 0.9807136$, and the product over the zeros with the tail gives 0.9807140, a relative difference of 4.2×10^{-7}.

The tilt $u_0 = 1$. For ζ, the product $\prod_\gamma(1 + \gamma^2)/(1 + \gamma^2 + s)$ over the first 400 ordinates with the Riemann von Mangoldt tail started at γ_{400} matches $\xi(\frac32)/\xi(\frac12 + \sqrt{1+s})$ to the tail error: the zero side is 0.9332865 against 0.9332901 at $s = 3$, a relative difference of 3.8×10^{-6}, and 0.3287603 against 0.3287814 at $s = 50$, a relative difference of 6.4×10^{-5}. With the anchored tail of equation (21.3.8) the differences are 9.5×10^{-9} and 1.6×10^{-7}.

21.6.2 The family: sum and mixture

The cyclotomic field settles one family question. By Proposition 21.5.3, the independent sum of the character variables of conductor dividing q is the Thorin variable of $\mathbb{Q}(\zeta_q)$, a GGC whenever the Thorin property holds there. Averages are another matter. Let $\mathcal{F}_q$ be the real primitive characters modulo q together with one representative of each conjugate pair of complex ones, and let $T^{(q)}$ be the variable of a uniformly chosen member, under GRH for the members and with their central values nonzero, so that their clocks exist (Propositions 21.5.1 and 21.5.2). Its moments are the averages

$$\mathbb{E}\big[(T^{(q)})^k\big] = \frac{1}{|\mathcal{F}_q|}\sum_{\mathcal{F}_q}\mathbb{E}[T_\bullet^k],$$

but the next proposition shows that $T^{(q)}$ is in general not a GGC. Here the Thorin measures of the members are sums of unit point masses, repeated according to multiplicity, so their Laplace transforms $\varphi_i(s) = \prod_t t/(t+s)$ are meromorphic on $\mathbb{C}$ with poles at the points $s = -t$.

Proposition 21.6.3 (Mixtures of clocks). *Let φ_1, φ_2 be Laplace transforms of GGCs with zero drift and Thorin measures U_1, U_2 that are sums of unit point masses, repeated according to multiplicity, let $0 < \lambda < 1$, and put $\varphi = \lambda\varphi_1 + (1-\lambda)\varphi_2$.*

(a) If $\varphi(-x) = 0$ for some $x > 0$ that is not an atom of U_1 or U_2, then φ is not the Laplace transform of a GGC.

(b) Let $t_1 < t_2$ be consecutive points of $\operatorname{supp} U_1 \cup \operatorname{supp} U_2$ with t_1 an atom of only one of the measures and t_2 an atom of only the other. If $U_1 + U_2$ has an odd number of atoms in $(0, t_1]$, counted with multiplicity, then φ has a zero in $(-t_2, -t_1)$, so φ is not the Laplace transform of a GGC, for any λ.

Proof. (a) Suppose $\varphi = g$ on $(0,\infty)$ with $g(s) = \exp(-ds - \int \log(1 + s/t)\, U(dt))$. Both are analytic on $\mathbb{C} \setminus (-\infty, 0]$, so they agree there. For $|s| \le x+1$ off the negative axis, $|1 + s/t| \le 1 + (x+1)/t$, so $|g(s)| \ge \exp(-d(x+1) - \int \log(1 + (x+1)/t)\, U(dt)) > 0$, the integral being finite by the conditions on a Thorin measure. Letting $s \to -x$ from the upper half plane, $\varphi(s) \to \varphi(-x) = 0$ because $-x$ is not a pole, a contradiction. (b) For $y > 0$ off the atoms, $\varphi_i(-y) = \prod_t t/(t-y)$ has the sign $(-1)^{N_i(y)}$, with $N_i(y)$ the number of atoms of U_i in $(0, y)$. Say t_1 is an atom of U_1 and t_2 of U_2. As $y \downarrow t_1$, $|\varphi_1(-y)| \to \infty$ while $\varphi_2(-y)$ stays bounded, so $\varphi(-y)$ has the sign $(-1)^{n_1}$, where n_i is the number of atoms of U_i in $(0, t_1]$. As $y \uparrow t_2$, φ_2 dominates and the sign is $(-1)^{n_2}$, since U_2 has no atom in (t_1, t_2). These signs differ when $n_1 + n_2$ is odd, and the intermediate value theorem gives the zero; (a) applies. □

Part (b) concerns two clocks. If no atom is common to the two measures, its hypothesis fails only when every maximal run of consecutive atoms of one measure, except the last run, has an even number of atoms (counted with multiplicity), because the numbers of atoms at the changes from one measure to the other are the partial sums of the run lengths; the odd configuration of (b) is the case in which some such partial sum is odd.

Modulo 5 the family has two members, χ_5 and the conjugate pair $\{\psi, \bar{\psi}\}$ with $\psi(2) = i$, so that $\lambda = \frac{1}{2}$ is the uniform average. Under GRH and with $L(\frac{1}{2}, \chi_5)L(\frac{1}{2}, \psi) \neq 0$ the clocks of the two members are the Thorin variables of Propositions 21.5.1 and 21.5.2, the second for the product $\xi_{\psi\bar{\psi}}(s) = \xi(s, \psi)\,\xi(s, \bar{\psi})$; numerically $L(\frac{1}{2}, \chi_5) = 0.23175$ and $L(\frac{1}{2}, \psi) = 0.76375 + 0.21696\,i$, with enclosures that exclude 0 (the part `central` of `gl_mixture_mod5.py`, at 128 bits). In the variable $y = \sqrt{-s}$ the atoms are the ordinates: for the pair 4.1329037, 6.1835782, 8.4572292, 9.4429311 (the absolute values of the ordinates of $L(s, \psi)$), and for χ_5 6.6484533, 9.8314444. These are all the atoms below 9.84: the argument principle certifies exactly 3 zeros of $\xi(\frac{1}{2} + it, \chi_5)$ and 5 zeros of $L(s, \psi)$, two with positive and three with negative ordinate, below height 12. Three atoms lie in $(0, 6.6484533]$, and the next atom belongs to the pair, so by (b) the mixture vanishes between $y = 6.6484533$ and $y = 8.4572292$ for every λ; the same happens between 9.4429311 and 9.8314444, with five atoms below.

Numerically, on the negative axis the mixture is $M_\lambda(y) = \lambda A(y) + (1-\lambda)B(y)$, with $A(y) = \xi(\frac{1}{2}, \chi_5)/\xi(\frac{1}{2} + iy, \chi_5)$ and $B(y) = \xi_{\psi\bar{\psi}}(\frac{1}{2})/\xi_{\psi\bar{\psi}}(\frac{1}{2} + iy)$. We computed it in ball arithmetic at 128 bits for $\lambda = 0.01$, 0.1, 0.5, 0.9 and 0.99. In all ten cases the signs at the two ends of the interval are opposite, and each interval contains exactly one zero (the argument principle on a thin rectangle about the interval gives the count); Table 21.6 lists the zeros.

Table 21.6: The zero y of the mixture $M_\lambda = \lambda A + (1-\lambda)B$ modulo 5 in each of the two intervals between a pole of A and a pole of B, for five weights; each interval contains exactly one zero.

λ	zero in (6.6484533, 8.4572292)	zero in (9.4429311, 9.8314444)
0.01	6.64874689	9.83141547
0.1	6.65169528	9.83112598
0.5	6.67865792	9.82859297
0.9	6.98112355	9.80692785
0.99	8.03540132	9.65957988

Proposition 21.6.4 (The family modulo 5). *Under GRH for $L(s, \chi_5)$ and $L(s, \psi)$, for every $\lambda \in (0, 1)$ the function $\lambda\,\xi(\frac{1}{2}, \chi_5)/\xi(\frac{1}{2} + \sqrt{u}, \chi_5) + (1-\lambda)\,\xi_{\psi\bar{\psi}}(\frac{1}{2})/\xi_{\psi\bar{\psi}}(\frac{1}{2} + \sqrt{u})$ of $u \geq 0$ is not the Laplace transform of a GGC. In particular the variable $T^{(5)}$, the case $\lambda = \frac{1}{2}$, is not a GGC.*

Proof. The central values are nonzero, as computed, so under GRH the clocks of the two members are the Thorin variables of Propositions 21.5.1 and 21.5.2, whose Thorin measures are sums of unit point masses at the squared ordinates. The certified atoms above put the two members in the odd configuration of Proposition 21.6.3(b). The location of the zeros uses no hypothesis; the hypothesis enters only through the identification of the Thorin measures of the two members. □

21.6.3 Landau Siegel zeros and the twisted heat trace

Let χ be a primitive real character of conductor $q \geq 3$. Siegel's theorem gives $1-\beta \gg_\varepsilon q^{-\varepsilon}$ for a real zero β of $L(s,\chi)$, with an ineffective constant, and the classical effective bound is only $1-\beta \gg q^{-1/2}\log^{-2} q$ [71].

Theorem 21.6.5 (Banks, Kerr; unrefereed). *There are absolute constants q_0 and $\delta_0 > 0$ such that for every primitive real character χ of conductor $q \geq q_0$, every real zero $\beta \in (0,1)$ of $L(s,\chi)$ satisfies $(1-\beta)\log q \geq \delta_0$. Banks gives $q_0 = 10^{10}$ and $\delta_0 = 3.0 \times 10^{-3}$.*

The preprints [14, 142] are dated October 2026 and have not been refereed (Section 21.1). Banks describes the proof as an arithmetic comparison of the archimedean and the p adic size of an interpolation determinant in the field $\mathbb{Q}(\sqrt{2}, \sqrt{\chi(-1)q})$, and Kerr gives a simplification; we have not checked either. The half plane $\operatorname{Re} s > \frac{7}{8}$ for every Dirichlet L function reported in Section 21.1 [226], if it stands, would give $1-\beta \geq \frac{1}{8}$ for every real zero and so imply Theorem 21.6.5 for every $q \geq 3$. Banks refers to a proof by OpenAI; whether that is the release of [226] we cannot say.

The twisted heat trace. Let $z_\rho = (\rho - \frac{1}{2})^2$, the variable of this chapter (the atom of a zero is $u_0 - z_\rho$, Theorem 21.2.3), so that $z_\rho = -\gamma^2$ for $\rho = \frac{1}{2} + i\gamma$ and $z_\rho = (\beta - \frac{1}{2})^2 > 0$ for a real zero β. As in Section 5.1, with the zeros counted with multiplicity and $t > 0$ the heat time, the twisted heat trace is

$$W(t;\chi) = \sum_{\operatorname{Im}\rho > 0} e^{t z_\rho} + \tfrac{1}{2} \sum_{\rho \text{ real}} e^{t z_\rho}.$$

This symbol $W(t;\chi)$ is not the root number W_χ of Section 21.5, which equals 1 for real χ. If $L(\frac{1}{2},\chi) \neq 0$ and $|\gamma| > |\beta - \frac{1}{2}|$ for every zero $\beta + i\gamma$ (so that $\operatorname{Re}(-z_\rho) > 0$ and the integral converges), then for $u \geq 0$

$$\log \frac{\xi(\frac{1}{2} + \sqrt{u}, \chi)}{\xi(\frac{1}{2}, \chi)} = \int_0^\infty (1 - e^{-ut}) \, \frac{W(t;\chi)}{t} \, dt,$$

and the ratio $\xi(\frac{1}{2},\chi)/\xi(\frac{1}{2} + \sqrt{u},\chi)$ is the Laplace transform of a GGC if and only if $W(\cdot\,;\chi)$ is completely monotone, which as in Section 5.1 is the ladder

$(-1)^k W^{(k)}(t;\chi) > 0$. The Gaussian explicit formula, with $a \in \{0,1\}$ the parity of χ and ψ the digamma function, reads

$$W(t;\chi) = \frac{\log(q/\pi)}{4\sqrt{\pi t}} + \frac{1}{4\pi}\int_{-\infty}^{\infty} e^{-tr^2}\,\mathrm{Re}\,\psi\Big(\frac{1+2a}{4}+\frac{ir}{2}\Big)dr \\ - \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)\chi(n)}{\sqrt{n}}\,e^{-(\log n)^2/4t}. \tag{21.6.1}$$

Compared with the formula for ξ in equation (1.2.5) the polar term $e^{t/4}$ is absent, because $L(s,\chi)$ is entire, $-\log\pi$ in $m(r)$ becomes $-\log(\pi/q)$, and the prime sum carries the factor $\chi(n)$.

A real zero as a failure of rung one. A real zero $\beta \in (\frac12, 1)$ and its mirror $1-\beta$ contribute together the single term e^{ct} with $c = (\beta - \frac12)^2 = \frac14 - \beta(1-\beta)$, which grows with t. This is the defect that a complex off line pair cannot model in Theorem 5.1.14, whose defect needs a nonzero imaginary part.

Proposition 21.6.6. *Let $\beta \in (\frac12, 1)$ be the largest real zero of $L(s,\chi)$, suppose every nonreal zero of $L(s,\chi)$ lies on the critical line, and put $c = (\beta - \frac12)^2$. Then $W(t;\chi) \to \infty$ and $-W'(t;\chi) < 0$ for all $t > t_1 = \max\{1, c^{-1}\log(C'\log q/c)\}$, where C' is an absolute constant. If moreover $q \ge q_0$, put $\eta = \delta_0/\log q$; then $c \le \frac14 - \eta(1-\eta)$, so by Theorem 21.6.5 the growth rate c of such a term is at least $\eta(1-\eta)$ below the rate $\frac14$ of the pole term $e^{t/4}$ of ζ.*

Proof. Under the hypotheses,

$$W(t;\chi) = \sum_{\gamma>0} e^{-t\gamma^2} + me^{ct} + (\text{terms } e^{c't} \text{ from the other real zeros, } c' \le c),$$

and $-W'(t;\chi) \le \sum_{\gamma>0}\gamma^2 e^{-t\gamma^2} - ce^{ct}$. By the bound $N(T,\chi) \ll (T+1)\log(q(T+2))$ for the zero counting function, the sum is at most $C'\log q$ at $t=1$ and decreases in t, so $-W' < 0$ as soon as $ce^{ct} > C'\log q$, which holds for $t \ge t_1$; and $W \ge e^{ct} \to \infty$. For the last statement, the identity $c = \frac14 - \beta(1-\beta)$ and $\beta \le 1-\eta$ (Theorem 21.6.5), with $\beta(1-\beta)$ decreasing on $(\frac12,1)$, give $\beta(1-\beta) \ge \eta(1-\eta)$. □

Thus a real zero with $\beta > 1-\eta$, which Theorem 21.6.5 excludes for $q \ge q_0$, would, under the hypotheses of Proposition 21.6.6, be seen on rung one at the scale t_1. The difficulty is uniformity in q: for a given χ of small conductor the verification of GRH by Platt [245] covers a real zero, and for general q equation (21.6.1) is the only handle. For complex pairs the book's detection of a pair at height a is Proposition 5.1.10, and nothing in Theorem 21.6.5 touches complex zeros.

The pole lock, a heuristic. The prime sum of equation (21.6.1) suggests why a zero near 1 is tied to the primes with $\chi(p) = -1$. Suppose, as a model and not as a claim that such a character exists, that $\chi(p) = -1$ for every prime p. The prime term of equation (21.6.1) is then

$$\frac{1}{2\sqrt{\pi t}} \sum_{p} \frac{\log p}{\sqrt{p}}\, e^{-(\log p)^2/4t} + O(1) \;\approx\; \frac{1}{2\sqrt{\pi t}} \int_0^\infty e^{u/2 - u^2/4t}\, du = e^{t/4}\, \Phi\big(\sqrt{t/2}\big),$$

where Φ is the standard normal distribution function and the $O(1)$, which comes from the prime powers (mainly the squares, where $\chi(p^2) = 1$) and tends to $-\frac{1}{4}$ as $t \to \infty$, is not negligible for $t \le 3$. As $t \to \infty$ this is $e^{t/4}(1 + o(1))$: on the prime side, such a character produces the growth that the pole of ζ produces in equation (1.2.5). A real zero β is the same picture with the rate reduced from $\frac{1}{4}$ to $(\beta - \frac{1}{2})^2$, and the reduction $\beta(1-\beta)$ is what Theorem 21.6.5 bounds below. Numerically (the script `gl_pole_lock.py`, primes up to 6×10^6) the ratio of the prime sum to $e^{t/4}$ is 0.402, 0.510, 0.645, 0.735 This is a heuristic about a hypothetical character.

What does not transfer. The proof of Theorem 21.6.5 as Banks describes it is not a positivity statement and not a complete monotonicity statement, and it concerns the single point $s = 1$ through the pole of $\zeta_K(s) = \zeta(s)L(s,\chi)$ of the quadratic field $K = \mathbb{Q}(\sqrt{\chi(-1)q})$ of χ. The Thorin measure is a statement about the whole zero set in the variable z_ρ, and we see no way to recast the determinant argument as a rung of the ladder.

21.7 Consequences for the Linnik constant and for RH

21.7.1 The Linnik constant

Corollary 21.7.1. *Suppose that for every $q \ge 3$ the function $\varphi_{\mathbb{Q}(\zeta_q),0}$ is the Laplace transform of a GGC. Then every sufficiently large even integer is a sum of two primes and at most seven powers of two.*

Proof. Theorem 21.6.1 gives GRH for all Dirichlet L functions, and under GRH every sufficiently large even integer is a sum of two primes and at most seven powers of two, by the theorem of Heath-Brown and Puchta [112] (see also Pintz and Ruzsa [241]). □

The corollary is weaker than the known conditional result. By Corollary 21.2.4(a) and equation (21.5.4), its hypothesis is GRH for every Dirichlet L function together with $L(\frac{1}{2}, \chi) \neq 0$ for every primitive character χ. It contains GRH and adds a statement that is not known to follow from it, because the nonvanishing at the centre is open (for real characters it is Chowla's conjecture), whereas $K = 7$ already follows from GRH alone. A direct proof of the

Thorin property for all cyclotomic fields would prove GRH, and with it $K = 7$ unconditionally, one better than the record $K = 8$ of Pintz and Ruzsa [242].

For K itself the Thorin property gives what GRH gives. Nothing in the proof of Corollary 21.7.1 uses the nonvanishing at the centre, and the inputs that decide K below seven, the large deviation estimate for the sum $\mathcal{D}(\alpha)$ over the powers of two (step (4) of Section 21.1.2), the sieve constant for prime differences (step (2)) and the refinements of Pintz and Ruzsa, are additive questions about the powers of two and not statements about zeros. With the present methods the framework therefore does not reach below seven.

A weaker hypothesis may suffice for seven. The Thorin property at tilt $u_0 = 1$ for cyclotomic fields (Corollary 21.2.4(b) with equation (21.5.4)) is GRH with possible real zeros. Where could a real zero hurt? Write $\mathcal{S}(a/q+\eta)$ as a combination of twisted prime sums over the characters modulo q. A real zero β of $L(s,\chi^*)$, with χ^* of conductor $q^* \mid q$, contributes a term of size about $q^{1/2}\varphi(q)^{-1}N^{\beta}(1+|\eta|N)^{-1}$, which exceeds $N^{3/4}$ only if $q < N^{2\beta-3/2}$ and $|\eta| < N^{\beta-7/4}$. Those arcs lie close to the major arcs, and moving them there, where a real zero changes the main term by a factor of the shape $1-\chi^*(\cdot)N^{\beta-1}$ that Gallagher's method [90] already handles for one exceptional zero, would reduce the question to major arc estimates with real zeros (Problem 21.2). If it works, real zeros, and with them Siegel zeros, become irrelevant to $K = 7$.

21.7.2 What would help towards RH

The three smoothings of the centre clock are compared in Chapter 19.

A proof has to establish a closed positivity condition at the boundary, from input that sees the arithmetic beyond positivity. Zero free regions of the kind announced in October 2026 [226] would be such input, if confirmed, and each one would move the unconditional end of the basepoint family inwards (Section 21.1.2); averages over families of L functions, where the low lying zeros are conjectured to obey the symmetry laws of Katz and Sarnak [140], are another (Problem 21.4). Section 19.4 says what would get past each of the three obstructions of Chapter 19.

21.8 Open problems

Problem 21.1 (The Thorin property without the zeros). Establish condition (i) of Theorem 21.2.3 for $\mathbb{Q}$, or for cyclotomic fields, without first knowing (ii). In the master theorem route, settle the parabolic distortion of Section 21.4.3, uniformly in the field, and find a substitute for the residue bookkeeping at the zeros. *Status.* Open. For $\mathbb{Q}$ at $u_0 = 0$ the statement to be proved is RH, so a growth bound that is itself proved under RH, such as Lemma 21.4.1, cannot be an input to its proof.

PROBLEM 21.2 (Seven with real zeros). Show that $K = 7$ follows from the Thorin property at tilt $u_0 = 1$ for cyclotomic fields, that is from GRH with possible real zeros. Section 21.7 locates the arcs where a real zero can exceed the bound $N^{3/4}$; what remains is to move them into the major arcs and to redo the main term with several real zeros, checking that the constant 2.7895 survives. In the paper of Heath-Brown and Puchta [112] the major arc estimate that gives this constant (their Lemma 7) is stated for $K \geq 9$, with zero density estimates in place of GRH, and the paper does not say how it is used for $K = 7$; the computation of the constant itself holds for $K \geq 7$. *Status.* Open. Only the location of the arcs is done, and it is a heuristic estimate.

PROBLEM 21.3 (The moment identity at scale). Anchoring the tail to the exact zero count, equation (21.3.8), already brings the check of p_1 from 2.6×10^{-6} to 6.3×10^{-9} with 799 zeros. Carry it to large height with Odlyzko's tables [224], where the remaining term of (21.3.8), the integral of $S(t)\,t^{-2j-1}$ over the tail, sets the limit and calls for an explicit formula for $\int S$, and to $L(s,\chi)$ with the zeros of Platt [245]. *Status.* Partly done. The script `gl_zeta_moments.py` (`code/README.md`) reproduces the anchoring with the first 1700 ordinates at the cuts $N = 200$ and $N = 799$ and at every cut $N = 50, \dots, 1699$, at 100 digits (`data/gl_zeta_moments.json`). For p_1 the anchored tail typically gains about a third of the height of the cut over the smooth tail (the first and ninth deciles of the gain are 42 and 2700), but it gains nothing where $|S|$ is small at the cut: the gain is below a factor 10 at 3.2% of the cuts. Large heights and the characters are open.

PROBLEM 21.4 (The negative moment as a family statistic). The value $\mathbb{E}[T^{-1/2}] = 6.7333$ (and 3.7575, 3.7097 for T_{χ_4}, T_{χ_5}) is computed from ξ alone: $\mathbb{E}[T^{-1/2}] = \Gamma(\frac12)^{-1}\int_0^\infty u^{-1/2}\prod_\gamma(1+u/\gamma^2)^{-1}\,du$, the product being over the ordinates $\gamma > 0$. It is mostly a smooth statistic. Let E_m be this integral with the first m ordinates exact and the factors of the others replaced by the smooth model $\exp\big(-\frac{1}{\pi}\int_{t_m}^\infty \log(1+u/t^2)\,\vartheta'(t)\,dt - a_m\log(1+u/t_m^2)\big)$, where ϑ is the Riemann Siegel theta function, t_m is the midpoint of γ_m and γ_{m+1}, and $a_m = N_0(t_m) + 1 - m$, with $N_0 = \vartheta/\pi$, is the mass of the atom at t_m that the anchored tail (21.3.8) adds. For $m = 0$ the cut is $t_0 = 9.6669$, the point where $N_0 + 1 = 0$, and there is no atom. The relative error $E_m/\mathbb{E}[T^{-1/2}] - 1$ is -7.0×10^{-3} for $m = 0$, -2.5×10^{-3} for $m = 1$, 1.0×10^{-4} for $m = 2$, -3.9×10^{-4} for $m = 5$ and 1.4×10^{-5} for $m = 20$. The atom matters: cutting at the midpoint without it gives -5.5×10^{-3}, -1.2×10^{-3}, -1.4×10^{-3} and -6.5×10^{-4} for $m = 1$, 2, 5, 20, and cutting at the point where $N_0 + 1 = m$ without it gives -2.5×10^{-3}, 9.1×10^{-5}, -4.0×10^{-4} and 1.0×10^{-5}. The elasticity $(\gamma_k/E)\,\partial E/\partial\gamma_k$ of $E = \mathbb{E}[T^{-1/2}]$ with respect to one ordinate, the others fixed in the integral above, is 0.1799 for γ_1 and falls to 0.0319 for γ_6 (over all ordinates the elasticities sum to 1, by scaling). A single function therefore says little, and the natural comparison with random matrices is in families. Compute the average of $\mathbb{E}[T_\chi^{-1/2}]$ over real primitive characters of conductor in $[Q, 2Q]$ and

compare it with the prediction of the symplectic ensemble. Since $\mathbb{E}[T^{-1/2}]$ is not a linear statistic of the zeros, the one level density of Katz and Sarnak [140] alone does not give this prediction, and all the correlation functions enter. The circular unitary comparisons of Section 11.4.5 and the moments of Keating and Snaith [141] (Section 11.4.6.1) are the models for a single function. *Status.* Partly done: the single function is computed, with the model above, by `gl_density_pf.py`; the family average is open.

Problem 21.5 (The family mixture). Let $T^{(q)}$ be the variable of a uniformly chosen member of $\mathcal{F}_q$ (Section 21.6.2, under GRH for the members and with their central values nonzero). Is $T^{(q)}$ infinitely divisible, and does any q with two or more members avoid the odd configuration of Proposition 21.6.3(b), which is stated for two clocks, for every pair of its members? *Status.* Open. That $T^{(q)}$ is not a GGC is settled for $q = 5$ and every weight, under GRH for the two members (their values at $\frac{1}{2}$ are nonzero numerically; Proposition 21.6.4), and holds, under the same hypotheses, for every q whose family has exactly two members, when these are in the odd configuration (Proposition 21.6.3(b)). It does not decide whether $T^{(5)}$ is infinitely divisible.

Problem 21.6 (Sign definite explicit formulae). Under the hypothesis of Corollary 21.7.1 the explicit formula for $\psi(x; q, 1)$, which up to the factor $\varphi(q)$ counts the prime ideals of $\mathbb{Q}(\zeta_q)$, becomes an integral against the nonnegative measure $U_{F_q,0}$, and Theorem 21.5.8 supplies the positive family $\pi^{-1}\operatorname{Re}(\xi'_{F_q}/\xi_{F_q})(\alpha + ib)$ joining it to the primes. For $a \not\equiv 1 \pmod q$ the characters enter the formula for $\psi(x; q, a)$ with the weights $\bar\chi(a)$, which are complex or negative, and no sign is available. These are zero counting measures and their Poisson smoothings, so any gain must come from structure beyond counting, such as monotonicity or log concavity of the associated GGC densities [39, Ch. 4], transported to bounds on L'/L near the line. Is there a sign definite explicit formula for $\psi(x; q, a)$ with $a \not\equiv 1 \pmod q$, in terms of such measures? *Status.* Open.

Problem 21.7 (Certified backward continuation). The Thorin density $\pi^{-1}\operatorname{Re}(f'/f)(\alpha + ib)$ of Theorem 21.5.8, which is $\rho_{\alpha-1/2}(b)$ for $f = \xi$, is computable from the primes, with rigorous error bounds, for $\alpha > 1$, and its positivity on every line $\operatorname{Re} s = \alpha'$ with $\alpha' > \alpha$ is a zero free half plane $\operatorname{Re} s > \alpha$. The book certifies the minimum at the origin for ξ, χ_4 and χ_{12} for $\alpha < 1$ up to a large height, from the zeros (Theorem 9.1.18). How far into the strip can positivity be certified from the primes alone, using smoothed prime sums of the kind that located the zeros of $L(s, \chi_4)$ in Section 21.5.6, before the ill posedness of the backward step takes over? Chapter 19 measures the cost of the step: by Theorem 19.2.2 it is a deconvolution, which is unbounded for the basepoint family, and Proposition 19.2.3 counts the primes it needs, up to about $\eta^{-1/(\alpha-\sigma_{\max})}$ for accuracy η; for the heat flow the corresponding step is stable but cannot carry real zeros (Section 19.4). Section 20.1.1 of Chapter 20 reads the same step as

a deconvolution of the explicit formula. *Status.* Partly done. For χ_4 the prime side (21.5.9) of π times the density at $b = 0$, summed over the prime powers up to 2×10^6 without smoothing, differs from $(\xi'/\xi)(\alpha, \chi_4)$ by 1.3×10^{-4}, 4.8×10^{-3}, 4.1×10^{-2} and 8.3×10^{-2} at $\alpha = 1$, $\frac{3}{4}$, $\frac{3}{5}$ and $\frac{11}{20}$, that is by 0.2%, 12%, 260% and 1100% of the value, so that for $\alpha \leq 1$ the minima computed in Section 21.5.6 come from the zeros and not from the primes. What smoothed prime sums can certify below $\alpha = 1$ is open.

Problem 21.8 (The arithmetic Lévy measure and the minor arcs). For $F = \mathbb{Q}$ the atoms of ν_F in (21.5.6) at the primes p have mass $\log p$, the weights that enter $\mathcal{S}(\alpha)$; the prime powers of (21.5.6) do not enter $\mathcal{S}(\alpha)$. Can $\mathcal{S}(\alpha)$ on the minor arcs be written as an expectation against the GGC laws above? Such a representation would be a different route to the sup bound in step (3); nothing in the chapter shows that it exists. *Status.* Open.

Problem 21.9 (Zero free half planes in the circle method). Assume $L(s, \chi) \neq 0$ for $\operatorname{Re} s > \theta_0$ and every Dirichlet character. What is the best minor arc exponent $\theta(\theta_0)$ obtainable by combining the explicit formula with Vaughan's identity? The one line balance of Section 21.1.2 gives $\frac{1}{2}(1+\theta_0)$, which is better than the unconditional exponent $\frac{263}{308}$ of step (3) exactly when $\theta_0 < \frac{109}{154}$, so it does nothing at $\theta_0 = \frac{7}{8}$; it is not shown to be optimal. Does a hybrid bound, using the half plane only for the characters of small conductor, improve λ in step (4)? And does the half plane, which excludes exceptional zeros of large conductor, raise the constant 2.7895 of step (1)? The paper of Heath-Brown and Puchta [112] suggests that it does not by itself: they state that zeros close to $\sigma = 1$ play no role and that no explicit numerical zero free region is needed, the constant is a sum over the divisors $d \leq 5$ (the improvement of Elsholtz to 2.96169 that they report takes $d \leq 21$ and a better sieve constant), and the restriction $q \leq P = N^{45/154 - 4\varpi}$ of their major arcs comes from zero density estimates, whose critical zeros have $\sigma \leq \frac{3}{4}$ and are not removed by a half plane at $\frac{7}{8}$. That restriction is not sharp for those estimates: the exponent $(\frac{45}{77}\kappa(\sigma) - 1)(1 - \sigma)$ of their Section 3 has its maximum 0.1050 at $\sigma = 0.676$, against the bound $\frac{31}{154} = 0.2013$ that they use. *Status.* Open.

Problem 21.10 (Finite order total positivity). RH holds if and only if the density of $Y = \sqrt{2T}\, Z$ is a Pólya frequency function of infinite order (Proposition 16.5.1, where the variable is called X; Section 21.7.2). For this one density the question whether a finite order already implies RH has a trivial answer. The orders decrease, so if RH holds every such implication is true, and if RH fails the property fails at some finite order, where the implication is vacuous. The question has content for a class of functions. Let $\mathcal{E}$ be the class of even entire functions F with $F(0) = 1$ and the order, type and reflection symmetry of $\xi(\frac{1}{2} + s)/\xi(\frac{1}{2})$ for which $1/F$ is the characteristic function of a probability density f_F; the density of Y is f_F for $F = \xi(\frac{1}{2} + \cdot)/\xi(\frac{1}{2})$ (Proposition 16.5.1). Is there a finite k such that, for every $F \in \mathcal{E}$, the property that f_F is a Pólya frequency function of order k implies that

every zero of F lies on the imaginary axis? *Status.* Open. Chapter 16 shows that the low orders are decided by the low zeros: order two holds without RH (Theorem 16.11.5), and the density lies within 1.6×10^{-10} of a Pólya frequency function of every order (Theorem 16.11.6), so that, given the verification of RH to height 3×10^{12} [247], every minor of order 3 and 4 is at least -1.1×10^{-8} and -1.2×10^{-7}. Problem 16.1 asks for proofs of the low orders, not for the converse asked here. Section 21.3.3.5 reports the computations of this chapter. The perturbation of Theorem 5.2.1, made at a height that is large compared with k, is the natural candidate for a negative answer in the class $\mathcal{E}$; whether it keeps the order k has not been computed here.

Notes

The Thorin representation and the van Dantzig pair are from [249, 251]; the GGC background is Bondesson's monograph [39]; the master theorem class is from Chaudhry and Qadir [55], and the exponential Tauberian theorem from Kasahara [139]. Completed Dirichlet L functions and Dedekind zeta functions are in Davenport [71], mean values of ζ in Titchmarsh [324]. The account of the circle method in Section 21.1.2 is a summary of the paper of Heath-Brown and Puchta [112].

Appendix A

Code and reproducibility

The scripts, the data files they write and the reruns made for the book are not printed here: they are released with the PDF, in the book's repository, `https://github.com/VadimSokolov/computing_rh`, beside the LaTeX source of the book. Most scripts are in the folder `code/` and the data files they write in `data/`, the reruns and reconstructions made for the book in `repro/`, and the scripts of some chapters in other folders. The file `code/README.md` lists every script with its purpose, the data files it reads and writes, the results of the book that it produces and its usage, and describes the reruns of September 2026, the computations that have no script in the archive and the gaps that remain.

The scripts run with Python 3.11 and the libraries of `requirements.txt`: `python-flint` 0.9.0, the Python interface to FLINT and Arb [130], for ball arithmetic and the certificates, and `mpmath` [205] for reference values. The long computations ran on the Hopper cluster of George Mason University. The scripts of the later chapters keep a log of each run beside the data file, those of Chapters 17 to 21 take a `--quick` option that runs a reduced case in seconds, and where a chapter printed a value that was recomputed for the book, the data file or the log records the value computed beside it and says whether the two agree. Where the book says that a statement is proved in ball arithmetic, the script named beside it is the certificate, and its output records the enclosures.

A.1 The Pólya evaluators

The evaluator of Section 2.1 on the real u axis is `code/polya_flint.py`. In ball arithmetic it computes the node values $\Phi_j = \Phi(jh)$, $0 \le j \le J$, from the theta series (2.1.1), and from them $P(E) + P(E^{-1})$ and $Q(E) - Q(E^{-1})$ with $E = e^{sh}$ and the polynomials P and Q of Section 2.1.1, the trapezoid approximations to $\xi(\frac{1}{2} + s)$ and $\xi'(\frac{1}{2} + s)$. The defaults are 240 bits, $h = 1/200$ and $|u| \le 3$, so that $J = 600$. Most runs use 420 bits, the parameters of Section 2.1.1; the Keiper Li coefficients of Section 8.3 use 600 bits, and the Selberg computation of Section 3.5 uses 300.

The tilted evaluator `code/tilted.py` implements Proposition 2.1.1. With the nodes $u_j = x_j + i\eta$, $x_j = jh$, $|j| \le J$, where $J = \lceil x_{\max}/h \rceil$ for the cut $|x| \le x_{\max}$, it returns the trapezoid sums of the two integrals of the proposition. The production scripts evaluate each exponential directly, not by Horner's rule, whose ball radii grow as described in Section 2.1.5. The Lehmer pair and its de Bruijn Newman flow use $\delta = 0.01$, 256 bits, $h = 6 \times 10^{-4}$ and $|x| \le 3.9$; the count at height 1000 in Section 2.2.3.2 uses $\delta = 0.1$, 200 bits, $h = 0.004$ and $|x| \le 4.2$; the Laplace transform of the clock for Sections 8.1 and 8.2 switches to the tilted evaluator with $\delta = 0.1$, 320 bits, $h = 0.003$ and $|x| \le 4.5$ when $|\operatorname{Im} s| \ge 120$; and the runs of Table 2.2 take δ, the precision, h, the cut and the heights from the command line, which `code/README.md` records.

In both evaluators the radii of the returned balls bound the rounding errors of the finite sums, and nothing else. Three errors lie outside them: the discretisation error of the trapezoid rule, bounded in Section 2.1.1; the cut of the axis at $|u| \le 3$, where $\Phi < 10^{-500}$, or at $|x| \le x_{\max}$ on the tilted line; and the truncation of the theta series, whose loop stops at the first term with midpoint below 2^{-b-20} at working precision b and does not add the tail to the radius. The stopping test compares double precision numbers, so for $b \ge 1055$ its threshold underflows to zero and the test is never met. The untilted route needs more than 1054 bits for thirty digits above height about 840 (Table 2.3), and there the test has to be made in ball arithmetic. The runs in this book use at most 600 bits and are not affected.

Appendix B

A short history of computations on the zeros

Every verification in Table B.1 from Backlund's on rests on an evaluator of ζ on or near the critical line and a certificate that no zero has been missed (Section 2.2). The window on which Chapters 3 to 5 measure distances, the 79 zeros below height 200, is the range that Backlund verified by hand in 1914 [10].

year	authors	result
1859	Riemann	the first zeros, in unpublished notes later found by Siegel [304]
1903	Gram	the first 15 zeros; Gram points [98]
1914	Backlund	all 79 zeros below height 200 [10, 11]
1925	Hutchinson	all 138 zeros below height 300 [118]
1936	Titchmarsh	the first 1041 zeros, all below height 1468, by the Riemann Siegel formula, computed with the help of Comrie [322]
1953	Turing	all 1104 zeros below height 1540, on the Manchester computer; Turing's method [332]
1956	Lehmer	the first 25000 zeros; the close pair near height 7005 [173, 174]
1969	Rosser, Yohe, Schoenfeld	the first 3.5×10^6 zeros; Rosser's rule [292]
1986	van de Lune, te Riele, Winter	the first 1.5×10^9 zeros [334]
1987	Odlyzko	10^5 zeros near the 10^{12}th zero, and the statistics of their spacings [219]
1988	Odlyzko, Schönhage	fast evaluation of ζ at many points [220]
2004	Gourdon	the first 10^{13} zeros, not interval certified [97]
2021	Platt, Trudgian	RH verified rigorously below height 3×10^{12} [247]

Table B.1: Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [81, Chapters 6 to 8].

Where Turing bounds an average of S to certify a count (Section 2.2.1), the zero density identity of Section 3.4 is exact, and its value is an integer at every resolution; at the resolution $\frac{1}{2}$, the basepoint one of Section 9.1, it is Backlund's method with the vertical side moved from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = 1$, and it gives the count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$ with no hypothesis, which Section 2.2.3.2 computes at height 1000. Lehmer's close pair near height 7005 [173] is computed from the Pólya kernel in Section 2.2.3.1 and followed under the de Bruijn Newman flow in Section 8.4. The Odlyzko Schönhage algorithm, with Odlyzko's zeros as the reference measure, is one of the evaluators that Section 2.2.2 proposes for the Thorin comparison at large height.

Appendix C

Notation

The tables list the notation used in more than one section; the first five cover the book and record the second meanings that some letters carry in particular chapters and sections. Sums and products over k run over the pairs $\{\rho, 1-\rho\}$.

symbol	meaning	defined in
ξ, Ξ	the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$	Section 1.2
ρ, γ_k	a zero $\rho = \beta + i\gamma$ of ξ, that is a nontrivial zero of ζ; the ordinates $0 < \gamma_1 \le \gamma_2 \le \cdots$ of the zeros with $\gamma > 0$	Section 1.2
a_k	$-(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$, equal to γ_k^2 on the line (in Section 8.3, the Taylor coefficients of $\log \xi(1+w)$)	(1.2.1)
Φ, p, X	the Pólya kernel in Titchmarsh's normalisation; the Pólya density $p = \Phi/\Xi(0)$; a random variable with density p (in Section 6.2, p is the Meixner density; in Section 4.3, X is the clock variable; in Sections 9.1 and 11.3, X is the prime cutoff)	(1.2.3), (1.2.4)
S_2, Y, J	$S_2 = \sum_{k\ge1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponential variables, the sum of two independent copies of the first passage time J of the level 1 by a three dimensional Bessel process started at 0; $Y = \frac{1}{2}\log(\pi S_2/2)$, with $\mathbb{E}e^{sY} = 2\xi(s)$; X is Y tilted by $e^{Y/2}$	Section 1.2, Section 9.1.1
u, z, θ	the variable $u = s^2$ of the clock (also the variable of Φ); the variable of Thorin measures; the chart $\theta = \sqrt{z}$, which is also the ordinate (the Kent approximants of Section 8.1 write w for u)	Section 1.1
$G(u)$	$\xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$; the clock is $1/G$ (in Section 7.1.3, G is a Laguerre Pólya function)	Section 1.1

symbol	meaning	defined in
$\mathcal{G}(u,t)$	the Gaussian smoothing $\sum e^{-2t(\tilde{\gamma}-u)^2}$ of the zeros at resolution $1/(2\sqrt{t})$, with $\tilde{\gamma}=\gamma-i\delta$ for a zero $\frac{1}{2}+\delta+i\gamma$ with $\gamma>0$	Section 5.1.5
T, e_γ	the clock variable, with $\mathbb{E}e^{-uT}=1/G(u)$, which exists with no hypothesis and is the centre clock of Chapters 9 to 11; under RH, $T=\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (T is also a height, as in $N(T)$)	Section 1.2, Proposition 1.2.1
U, U_α	the Thorin measure of a GGC H, with $\mathbb{E}e^{-uH}=\exp\{-du-\int\log(1+u/z)\,U(dz)\}$ and drift d; the signed measure $U_\alpha(dz)=v_\alpha(\sqrt{z})\,dz/(2\pi\sqrt{z})$ at a basepoint α; for the clock, the Thorin functional $\sum_k \delta_{a_k}$	Section 1.2, (1.2.6), Section 5.3.1
k	the canonical function of an infinitely divisible law on $[0,\infty)$ with Lévy density $k(t)/t$ ($k(y)$ in Sections 5.1 and 6.2)	Section 1.3

symbol	meaning	defined in
U_{RH}, U_{line}, $U^{\pm}_{\mathrm{RH}}$	$\sum_{\gamma>0}\delta_{\gamma^2}$; the same sum over the zeros on the line; $\sum_{\gamma>0}(\delta_\gamma+\delta_{-\gamma})$	(1.1.1), Proposition 1.2.3
$\mathcal{W}$, $T_{\mathcal{W}}$, $U_{\mathcal{W}}$, U	the principal branch of Lambert's function, $\mathcal{W}(u)e^{\mathcal{W}(u)}=u$ (not the heat trace W); the Lambert law, with Laplace transform $\mathcal{W}(u)/u=e^{-\mathcal{W}(u)}$; its Thorin measure; a variable uniform on $(0,1)$ (not the Thorin functional U)	Sections 11.5.1, 11.5.3
φ_α, T_α	$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, the clock at a basepoint $\alpha\geq\frac{1}{2}$, with $\varphi_{1/2}=1/G$; the variable with $\mathbb{E}e^{-uT_\alpha}=\varphi_\alpha(u)$, which exists for every $\alpha>\frac{1}{2}$ and is a GGC if and only if no zero has $\beta>\alpha$, in particular for every $\alpha\geq 1$; $T_{1/2}=T$	Theorems 1.2.2, 9.2.5
$\widetilde{T}$, $\widetilde{T}_\alpha$, $\sigma^{(\alpha)}$	the centre clock tilted by $e^{-T/4}$, and by $e^{-(\alpha-\frac{1}{2})^2T}$, so that $\widetilde{T}=\widetilde{T}_1$; the subordinator with Laplace exponent $u+2(\alpha-\frac{1}{2})\sqrt{u}$, with $T_\alpha\stackrel{d}{=}\sigma^{(\alpha)}_{\widetilde{T}_\alpha}$	Theorems 9.2.2, 9.2.5
H_a, $H_a^{(c)}$	the first passage time of B_t+ct to the level $a\geq 0$, for a standard Brownian motion B, and $H_a=H_a^{(0)}$; unrelated to H_t	Section 9.2
Y_α, Z	the normal scale mixture $Y_\alpha=\sqrt{2T_\alpha}\,Z$, with Z standard normal and independent of T_α, whose characteristic function is $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ (in Section 11.1, Z is the subordinator with $T\stackrel{d}{=}\int_0^\infty e^{-t}\,dZ_t$)	Proposition 10.1.1

symbol	meaning	defined in
$W(t)$, $c_k(t)$	the heat trace $\sum_k e^{-a_k t}$, also written $\int e^{-tz} U(dz)$; the signed derivatives $(-1)^k W^{(k)}(t)$ (in Chapters 6 and 7 the time is x, or y where x is a Lévy variable, as in the proof of Lemma 6.2.5 and from Proposition 6.2.16 to the end of Section 6.2; the numbers $c_k(u)$ of Section 4.1 are local to it)	Sections 1.2, 5.1
m_n, $\mathcal{H}_N$, $\mathcal{H}'_N$, ι_N, ι'_N	the Stieltjes moments $\sum_k a_k^{-n-1}$, with $m_0 = \mathbb{E}T$; the Hankel matrices $[m_{i+j}]_{i,j<N}$ and $[m_{i+j+1}]_{i,j<N}$; the numbers of their negative eigenvalues, which count the zeros off the line (Section 8.4 uses the same symbols for the moments $m_n(t)$ of H_t)	Section 5.1, Theorem 5.1.7
$W^{[j]}$, β_j, B_j	the ladder functions $(-t\frac{d}{dt})^j W$, with $W^{[0]} = W$ (along the de Bruijn Newman flow $W^{[j]}(x;t)$, with x the time of the heat trace and t that of the flow); the polynomials with $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)e^{-at}$; the Touchard polynomials, $\beta_j(u) = (-1)^j B_j(-u)$	Section 11.5.4
L_n, L_∞	Urbanik's classes of multiply self decomposable laws, with L_0 the self decomposable laws; unrelated to the Post Widder functionals $L_n[W]$ of Section 7.2.2	Section 11.5.4.1
$Z^{[j]}$	the jth driving process of the centre clock, $Z^{[1]} = Z$, with Laplace exponent $(u\frac{d}{du})^j \log G$	Lemma 11.5.7
g, g_k, H, λ^*	the candidate generating exponent $\Psi_Z \circ H$ and its Taylor coefficients, with H the inverse of $\log G$; the branch point $\log\lvert G(u^*)\rvert + i\pi$ of H	Section 11.5.4.4
Φ_t, $\widehat{g}$, R_g	the damped Pólya kernel $e^{tv^2/4}\Phi(v)$; the cosine transform $\int g(v)\cos(\theta v)\,dv$ of an even kernel; $\widehat{g}(0)/\widehat{g}(i\sqrt{s})$, with $R_{\Phi_t} = R_t$	Section 11.5.5.1
X_t, p_t, Q_t, ϱ_t, h_t	the variable with density $p_t = \Phi_t/\Xi_t(0)$, $X_0 = X$; the radius of Khinchine's mixture, $X_t = UQ_t$; the radius of the arcsine mixture, $X_t = \varrho_t \cos\Theta$, and its density	Section 11.5.5.2
$\mathcal{B}$, f, $\mathcal{B}_g$, $\mathcal{B}_t$	$\frac{\pi}{2}S_2$, with $\mathbb{E}\mathcal{B}^{s/2} = 2\xi(s)$; its density $b^{-5/4}\Phi(\frac{1}{2}\log b)$; the variable with density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; $\mathcal{B}_{\Phi_t}$	Section 11.5.5.3
ℓ, $\kappa_k(\lambda)$, $c_k(\lambda)$, σ_λ	a Laplace transform $\mathbb{E}e^{-\lambda\mathcal{V}}$; the cumulants $(-1)^k(\log\ell)^{(k)}(\lambda)$; $\kappa_{k+1}(\lambda)/k!$, the moments of σ_λ for a GGC (not the $c_k(t)$ of the heat trace)	(11.5.9)
$\mathcal{K}_n(\lambda)$, $\mathcal{K}'_n(\lambda)$	the Hankel matrices $[c_{i+j}(\lambda)]_{i,j<n}$ and $[c_{i+j+1}(\lambda)]_{i,j<n}$, positive semidefinite for a GGC	Section 11.5.5.3

symbol	meaning	defined in
ℓ_t, $\dot{\ell}$, $\mathcal{J}$	$\mathbb{E}e^{-\lambda\mathcal{B}_t}$; its derivative in t at $t=0$, up to a term in ℓ_0; the first order change of the Thorin measure in a gap, in the proof of Proposition 11.5.15	Section 11.5.5.3
ϑ_a, $\xi(s,a)$, K_a, $\mathcal{B}_a$	the theta series $\sum n^p a_n e^{-\pi n^2 x/q}$ of a periodic sequence; its completed Dirichlet series; the kernel $2e^{(p+1/2)v}\vartheta_a(e^{2v})$; $\mathcal{B}_{K_a}$ (for $a=\chi_{-4}$, $1/\mathcal{B}_a$ is the variable Z of Remark 8.5.22)	Section 11.5.5.4
χ_d, S_a, Σ_a	the real primitive character of discriminant d ($\chi_{-4}=\chi_4$); the positivity sum of Lemma 11.5.17; the numerator of the reflected transform (11.5.10)	Section 11.5.5.4
f_{DH}, $\varkappa$, f_r	the function of Davenport and Heilbronn; its coefficient 0.28407904…; the self dual family modulo 7	Section 11.5.5.4
k_n	the inverse Laplace transform of Li's φ_n, $\lambda_n = \int k_n W$	Problem 11.4

symbol	meaning	defined in
ε	the resolution; the basepoint is $\frac{1}{2}+\varepsilon$	Section 1.1
$v_\alpha(r)$	$\mathrm{Re}(\xi'/\xi)(\alpha+ir)$, which is π times the density of U_α in the chart $\theta=\sqrt{z}$ (Chapters 9 to 11 write $v_\alpha(\theta)$)	Section 1.2
b_α, Π_α	$b_\alpha = \frac{\xi'}{\xi}(\alpha) = v_\alpha(0)$, the drift of the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx}-1+sx)\,\Pi_\alpha(dx)$ (for $\alpha\geq 1$, and for $\frac{1}{2}<\alpha<1$ when $\Theta<\alpha$), with $b_1 = \sum_\rho \rho^{-1} = \lambda_1$; its signed Lévy measure, with atoms $\Lambda(n)/(n^\alpha \log n)$ at $\log n$	Proposition 9.1.10, Section 9.1.5
A, A_α, Q, ν_1	the gamma piece $A(\theta) = \frac{1}{2}[\mathrm{Re}\,\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})]$ and the compensated prime piece $Q(\theta) = \lim_{X\to\infty}\int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ of $v_1 = b_1 + A + Q$; the gamma piece A_α at α; the prime measure $\nu_1 = \sum_n \Lambda(n) n^{-1}\delta_{\log n} - dx$ on $(0,\infty)$, compensated by Lebesgue measure and unrelated to $\nu(\sigma,T)$ (Section 11.3 uses A, A_α and Q for $L(s,\chi_{12})$)	(9.1.6), Proposition 9.1.3
S, T_1°; S_α, T_α°	the positive stable variable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$; the GGC with no drift, independent of S, whose Thorin density in the chart $\theta=\sqrt{z}$ is $(A+Q)/\pi$, so that $T_1 \overset{d}{=} S + T_1^\circ$ (unrelated to $S(t)$ and S_2); the same at a basepoint $\alpha\geq 1$, with b_α and $(v_\alpha - b_\alpha)/\pi$, so that $T_\alpha \overset{d}{=} S_\alpha + T_\alpha^\circ$	Propositions 9.1.11, 9.1.12

symbol	meaning	defined in
V_α, V	the variable with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, whose density $e^{(\alpha-\frac12)y}p(y)\,\xi(\frac12)/\xi(\alpha)$ is a tilt of the Pólya density, with no hypothesis; $V_{1/2} = X$ and $V = V_1$	Section 1.2, Section 9.1.1
$\rho_\varepsilon(\theta)$	the Thorin density at basepoint $\frac12+\varepsilon$ in the chart $\theta = \sqrt{z}$, $\pi^{-1}v_{1/2+\varepsilon}(\theta)$	(1.2.7)
μ_ε	the signed measure $U_{1/2+\varepsilon}$, with density $\rho_\varepsilon(\theta)\,d\theta$ in the chart θ; the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac12+\varepsilon > \operatorname{Re}\rho$ for every zero, and called the Thorin measure at resolution ε in general ($\mu(n)$ is the Möbius function)	Section 1.2

symbol	meaning	defined in
P_a	the Poisson (Cauchy) kernel $a/(\pi(a^2+x^2))$, with $P_{-a} = -P_a$	Section 1.2
$d_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\|_{\mathrm{TV}}$, the distance to the RH measure smoothed at the same resolution, which vanishes under RH	Section 3.2
$\tilde d_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, the matched distance computed against the model $\widetilde{U}$ of the zero list (the first 1700 ordinates and the density $\pi^{-1}\vartheta'$ beyond them, each with its reflection)	Section 3.2
κ_{cell}	the cell constant $\pi^{-1}\sum_k \lvert\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\rvert$ of the 79 cells $(b_{k-1}, b_k]$ below height 200, so that $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$	Section 3.2
$M_\varepsilon(\theta)$	the cumulative mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the branch of the argument that vanishes at $\theta = 0$ (the Mertens function $M(t) = \sum_{d\le t}\mu(d)$ in Section 10.3)	Section 2.1.2, Section 3.4
$\tau_\varepsilon(T)$	π^{-1} times the change of $\arg\xi$ along the segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$, so that $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac12+\varepsilon, T)$ ($\tau(n)$ is Ramanujan's function in Section 8.5)	Proposition 3.4.2
$N(T)$, $\nu(\sigma,T)$	the number of zeros with $0 < \operatorname{Im}\rho \le T$; the number of those with $\operatorname{Re}\rho > \sigma$ (Sections 3.4 and 11.4 write $N(\sigma,T)$ for the number with $\operatorname{Re}\rho \ge \sigma$)	Section 1.1
$S(t)$, $S_\varepsilon(t)$	$\pi^{-1}\arg\zeta(\frac12+it)$, with the argument continued along $2 \to 2+it \to \frac12+it$; $\pi^{-1}\arg\zeta(\frac12+\varepsilon+it)$, continued along the line $\operatorname{Re}s = \frac12+\varepsilon$ from $S_\varepsilon(0+) = -1$ if $0 < \varepsilon < \frac12$ and from 0 if $\varepsilon > \frac12$	Sections 2.2.1, 3.4.1
$\vartheta(t)$	the Riemann Siegel theta function $\operatorname{Im}\log\Gamma(\frac14+\frac{it}{2}) - \frac{t}{2}\log\pi$	Section 2.2
$Z(t)$	Hardy's function $e^{i\vartheta(t)}\zeta(\frac12+it)$, real for real t	Section 2.2

symbol	meaning	defined in
Θ	a height, the top of a range of ordinates, with $\Theta = 200$ for the window of Chapters 2 to 5; in Chapters 6 and 7 and Sections 9.1 and 10.2, $\sup \operatorname{Re} \rho$	Section 2.1.2, Section 6.1
T_0	the height 3×10^{12} below which RH has been verified [247]	Section 1.1
λ_n	the Keiper Li coefficients $\sum_\rho [1 - (1 - 1/\rho)^n]$	Section 8.3
H_t, Λ, Φ_{dBN}	the de Bruijn Newman flow, normalised by $H_0(z) = \frac{1}{8}\Xi(z/2)$; the de Bruijn Newman constant; the kernel $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u)$ of the flow ($\Lambda(n)$ is von Mangoldt's function)	Section 8.4
$P(\theta)$, ν_n, A_P, K^P	Pólya's Bessel function $P(\theta) = K_{i\theta/2}(2\pi)$; the zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta = \pm 2\nu_n$; the operator $-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ with a Dirichlet condition, whose eigenvalues are the ν_n^2; the trace $K^P(y) = \sum_n e^{-2\nu_n y}$	Section 8.7

symbol	meaning	defined in
δ	the displacement of an off line zero $\frac{1}{2} \pm \delta + i\gamma$, written d in Sections 5.2, 5.3, 5.4, 8.3, 8.4 and 12.2; in Chapter 2 and where its parameters are quoted, the tilt margin, with $0 < \delta < \pi/4$; in Section 4.3, the angle of a zero	Propositions 1.2.3, 2.1.1
η	the tilted height $\eta = \pi/4 - \delta$ (Dedekind's eta function in Sections 8.5, 4.3 and 11.3; Dirichlet's eta function $\eta(s) = (1 - 2^{1-s})\zeta(s)$ in Section 10.1 and, as $\eta(1 + v) = (1 - 2^{-v})\zeta(1 + v)$, in Chapters 6 and 7)	Proposition 2.1.1
$K_\eta(\varepsilon)$	the kernel constant $\int_{\mathbb{R}} \lvert\Phi(x \pm i\eta)\rvert\, e^{\varepsilon\lvert x\rvert}\, dx$ of the trapezoid bound, with $K_\eta = K_\eta(0)$	Section 2.1.1
ψ	the digamma function Γ'/Γ (Chebyshev's function $\sum_{n \le y} \Lambda(n)$ at real $y \ge 1$ in Sections 7.1, 9.1, 10.2 and 11.4, and $\psi(y, \chi_{12})$ in Section 11.3)	Section 1.2
γ_E	Euler's constant	Section 5.1
$\mathcal{N}$	the standard normal distribution function, with $\overline{\mathcal{N}} = 1 - \mathcal{N}$	Section 8.2
$\xi(s, \chi_{12})$, K	the completed L function $(12/\pi)^{s/2}\Gamma(s/2)L(s, \chi_{12}) = \int_0^\infty \eta(ix) x^{s/2}\, dx/x$; its kernel $K(u) = 2e^{u/2}\eta(ie^{2u})$, with $\xi(\frac{1}{2} + s, \chi_{12}) = \int_{\mathbb{R}} K(u) e^{su}\, du$ (a variable Y with density $K/\xi(\frac{1}{2}, \chi_{12})$ plays the role of X)	Section 8.5.1

symbol	meaning	defined in
$\xi(s,\chi_4)$, K_4, Y_4	the completed Dirichlet beta function $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_4)$, equal to $\int_0^\infty \eta(ix)^3 x^{(s+1)/2}\,dx/x$; its kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$, with $\xi(\frac{1}{2}+s,\chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\,du$; the variable with density $K_4/\xi(\frac{1}{2},\chi_4)$	Section 8.5.7.1
$R(x)$, $R_h(x)$	Riesz's function $x\sum_{n\geq 1}\mu(n)n^{-2}e^{-x/n^2}$; its twist $x\sum_{n\geq 1}\mu(n)n^{-2}(x/n^2)^{ih}e^{-x/n^2}$, with $R_0 = R$	Sections 10.1.8, 10.2.1
$\xi_K(s)$, C_K	the completed Dedekind zeta function $s(s-1)\lvert d_K\rvert^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ of a number field K, with $\xi_{\mathbb{Q}} = 2\xi$; the constant of $\sum_{N(\mathfrak{a})\leq x}\mu(\mathfrak{a})^2/N(\mathfrak{a})$, with $C = C_{\mathbb{Q}} = 1.7171377$	Sections 10.3.1, 10.3.3
$\lVert\cdot\rVert_{\mathrm{TV}}$	the total variation $\lvert\cdot\rvert(E)$ of a signed measure on the window E	Section 1.1
GGC, HCM, ID, SD, BO, EGGC	generalised gamma convolution; hyperbolically completely monotone; infinitely divisible; self decomposable; the Bondesson class; the extended class of generalised gamma convolutions, the weak limits of convolutions of gamma laws and their reflections	Section 1.3, Section 1.3.2

symbol	meaning	defined in

Sections 4.2 and 4.3 use the following notation.

symbol	meaning	defined in
s, $\varphi(s)$	the variable u of the clock; the clock $\varphi_{1/2}(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$	Section 4.2
W	the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace	Section 4.2
T_α	Bosch's variable $(Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with independent positive α stable $Z_\alpha, \tilde{Z}_\alpha$; here α is a stability index, not a basepoint	Section 4.2
X, p, ε_h, δ, θ, ϑ	under RH, the clock variable T; an exponent, not the Pólya density; an elasticity, not a resolution; the angle of a zero; angles, not the ordinate and the Riemann Siegel function	Section 4.3
ψ, S_{on}, R_{off}	$-\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$, the function of the Stieltjes form; its sums over the real and over the nonreal γ (in Section 11.4.7, the parts of G'/G in the variable u)	Theorems 4.2.4, 4.3.9
D_γ, $\mathcal{E}_{T_0}$, θ_0	for a nonreal zero $\gamma = a + ib$, the closed disc $\lvert s - (b^2 - a^2)\rvert \le 2ab$ with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$; the parabolic sliver $\{x + iy :\ x < -T_0^2 + T_0 + \frac{1}{4},\ y^2 < \frac{1}{2} - x\}$ that contains every D_γ; its angle at the tip, seen from the origin	Theorem 4.3.9

The numerical checks behind Section 4.3 come from one script in the archive (`code/README.md`) and, for Theorem 4.3.9, Remark 4.3.10 and Figure 4.2, from `pakes_disc.py`.

Chapters 6 and 7 use the following notation.

symbol	meaning	defined in
s, t, x	the variable u of the clock; the variable z of Thorin measures; the time of the heat trace $W(x)$	Section 6.1
τ, z	the ordinate; the argument of ζ and ξ	Section 6.1
G, R, φ	the gamma factor $\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$; the reciprocal Xi function $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, the clock $1/G$ of Section 1.2; the normalised reciprocal zeta function $\varphi(s) = \zeta(\frac{1}{2})/\{(1-4s)\zeta(\frac{1}{2} + \sqrt{s})\}$, with $\varphi = RG$	(6.1.4)

symbol	meaning	defined in
R_σ, φ_σ, R_t	$\xi(\sigma)/\xi(\sigma+\sqrt{s})$, which the rest of the book writes φ_σ; the normalised reciprocal zeta function $\zeta(\sigma)/\{(1-s/(1-\sigma)^2)\zeta(\sigma+\sqrt{s})\}$ at σ, with $\varphi_{1/2}=\varphi$, not the φ_σ of the rest of the book; $H_t(0)/H_t(2i\sqrt{s})$, with $R_0=R$ (also in Section 8.4, which writes u for s)	Sections 6.1, 7.2
θ, c, c_n	the Riemann Siegel theta function (ϑ elsewhere); $c=-\theta'(0)=\zeta'(\frac12)/\zeta(\frac12)=2.6860917$ (also quoted in Sections 5.1, 8.5 and in `code/README.md`); $c_n=2n+\frac12$	Section 6.1, (6.1.6)
$W_0(x)$, $w(x)$, $M(t)$	the archimedean part $\frac{1}{\pi}\int_0^\infty e^{-x\tau^2}\theta'(\tau)\,d\tau$ of $W(x)$; the canonical function $w=W-W_0$ of φ; $N(\sqrt{t})-\theta(\sqrt{t})/\pi=1+S(\sqrt{t})$	Section 6.1
X_φ, X_R	random variables with Laplace transforms φ and R, so that X_R has the law of the clock variable T	Section 6.2
κ, Y_κ	a Poisson width; the variable with $\mathbb{E}e^{-sY_\kappa}=\kappa/(\kappa+\sqrt{s})$, whose Thorin density is a Poisson kernel of width κ in $\sqrt{t}$	Section 6.2
$\eta(1+v)$, χ_4, $\mathcal{S}_n$	Dirichlet's eta function; the nontrivial character modulo 4; the truncated lattice variables $\mathcal{S}_n=\sum_{k\le n}k^{-2}(e_k+e'_k)$, with $\mathcal{S}=\frac{\pi^2}{2}S_2$ and tails $\mathcal{T}_n=\mathcal{S}-\mathcal{S}_n$ (also in Section 11.1)	Sections 6.1, 7.2
B_N, h_λ	the weights $B_N=\sum_{n^m=N}d_m a_n$ of the Vere-Jones representation, with $a_n=\Lambda(n)/(\sqrt{n}\log n)$ and $\sum_n d_n n^{-v}=1/\eta(1+v)$, which vanish unless N is a prime power p^n and are then sums of necklace polynomials at $p^{-1/2}$; the heat trace of the sawtooth measure U_λ, a Jacobi theta function	Section 6.1, Theorem 7.1.8

Bibliography

[1] J. Abate and P. P. Valkó. Multi-precision Laplace transform inversion. *International Journal for Numerical Methods in Engineering* 60 (2004), 979–993.

[2] L. Alpöge and R. Furman. More than two thirds of the zeta zeros are simple and on the critical line. arXiv:2608.13637, 2026.

[3] D. F. Andrews and C. L. Mallows. Scale mixtures of normal distributions. *Journal of the Royal Statistical Society Series B (Methodological)* 36 (1974), 99–102.

[4] S. V. Anulova. On Markov stopping times with a given distribution for a Wiener process. *Theory of Probability and its Applications* 25 (1980), 362–366.

[5] J. Arias de Reyna. High precision computation of Riemann's zeta function by the Riemann-Siegel formula, I. *Mathematics of Computation* 80 (2011), 995–1009.

[6] O. Arizmendi, T. Hasebe and N. Sakuma. On the law of free subordinators. *ALEA Latin American Journal of Probability and Mathematical Statistics* 10 (2013), 271–291.

[7] O. Arizmendi and D. Perales. Cumulants for finite free convolution. *Journal of Combinatorial Theory, Series A* 155 (2018), 244–266.

[8] J. V. Armitage. Zeta functions with a zero at $s = \frac{1}{2}$. *Inventiones Mathematicae* 15 (1971), 199–205.

[9] R. Arratia, L. Goldstein and F. Kochman. Size bias for one and all. *Probability Surveys* 16 (2019), 1–61.

[10] R. J. Backlund. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences* 158 (1914), 1979–1981.

[11] R. J. Backlund. Über die Nullstellen der Riemannschen Zetafunktion. *Acta Mathematica* 41 (1918), 345–375.

[12] L. Báez-Duarte. A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis. *Atti della Accademia Nazionale dei Lincei, Rendiconti Lincei, Matematica e Applicazioni* 14 (2003), 5–11.

[13] R. C. Baker and J. Pintz. The distribution of square-free numbers. *Acta Arithmetica* 46 (1985), 73–79.

[14] W. D. Banks. On the OpenAI proof that Landau Siegel zeros do not exist. arXiv:2610.15483, 2026.

[15] E. Balanzario and J. Sánchez-Ortiz. Zeros of the Davenport–Heilbronn counterexample. *Mathematics of Computation* 76 (2007), 2045–2049.

[16] Á. Baricz and F. Štampach. The Hurwitz-type theorem for the regular Coulomb wave function via Hankel determinants. *Linear Algebra and its Applications* 548 (2018), 259–272.

[17] O. E. Barndorff-Nielsen and D. R. Cox. Edgeworth and saddle-point approximations with statistical applications. *Journal of the Royal Statistical Society B* 41 (1979),

279 ff.

[18] O. E. Barndorff-Nielsen, J. Kent and M. Sørensen. Normal variance-mean mixtures and z distributions. *International Statistical Review* 50 (1982), 145 ff.

[19] O. E. Barndorff-Nielsen, M. Maejima and K. Sato. Some classes of multivariate infinitely divisible distributions admitting stochastic integral representations. *Bernoulli* 12 (2006), 1–33.

[20] O. E. Barndorff-Nielsen and S. Thorbjørnsen. Self-decomposability and Lévy processes in free probability. *Bernoulli* 8 (2002), 323–366.

[21] O. E. Barndorff-Nielsen and S. Thorbjørnsen. A connection between free and classical infinite divisibility. *Infinite Dimensional Analysis, Quantum Probability and Related Topics* 7 (2004), 573–590.

[22] T. Barnet-Lamb, D. Geraghty, M. Harris and R. Taylor. A family of Calabi–Yau varieties and potential automorphy II. *Publications of the Research Institute for Mathematical Sciences* 47 (2011), 29–98.

[23] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer. Counting zeros of Dirichlet L-functions. *Mathematics of Computation* 90 (2021), 1455–1482.

[24] H. Bercovici and V. Pata. Stable laws and domains of attraction in free probability theory (with an appendix by P. Biane). *Annals of Mathematics* 149 (1999), 1023–1060.

[25] H. Bercovici and D. Voiculescu. Free convolution of measures with unbounded support. *Indiana University Mathematics Journal* 42 (1993), 733–773.

[26] C. Berg. Correction to a paper by A. G. Pakes. *Journal of the Australian Mathematical Society* 76 (2004), 67–74.

[27] C. Berg and G. Forst. Multiply self-decomposable probability measures on $\mathbb{R}_+$ and $\mathbb{Z}_+$. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 62 (1983), 147–163.

[28] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41 (1999), 236–266.

[29] A. Beurling. A closure problem related to the Riemann zeta-function. *Proceedings of the National Academy of Sciences* 41 (1955), 312–314.

[30] P. Biane. On the free convolution with a semi-circular distribution. *Indiana University Mathematics Journal* 46 (1997), 705–718.

[31] P. Biane. Matrix valued Brownian motion and a paper by Pólya. In *Séminaire de Probabilités XLII*, Lecture Notes in Mathematics, Springer, 2009, 171–185.

[32] P. Biane, J. Pitman and M. Yor. Probability laws related to the Jacobi theta and Riemann zeta functions, and Brownian excursions. *Bulletin of the American Mathematical Society* 38 (2001), 435–465.

[33] N. H. Bingham, C. M. Goldie and J. L. Teugels. *Regular Variation*. Encyclopedia of Mathematics and its Applications 27, Cambridge University Press, 1987.

[34] L. I. Bluestein. A linear filtering approach to the computation of discrete Fourier transform. *IEEE Transactions on Audio and Electroacoustics* 18 (1970), 451–455.

[35] E. Bogomolny, O. Bohigas, P. Leboeuf and A. G. Monastra. On the spacing distribution of the Riemann zeros: corrections to the asymptotic result. *Journal of Physics A: Mathematical and General* 39 (2006), 10743–10754.

[36] H. Bohr and E. Landau. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences* 158 (1914), 106–110.

[37] E. Bombieri and J. C. Lagarias. Complements to Li's criterion for the Riemann hypothesis. *Journal of Number Theory* 77 (1999), 274–287.

[38] E. Bombieri. Remarks on Weil's quadratic functional in the theory of prime numbers, I. *Rendiconti Lincei. Matematica e Applicazioni* (9) 11 (2000), 183–233.
[39] L. Bondesson. *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*. Lecture Notes in Statistics 76, Springer, 1992.
[40] L. Bondesson. A class of probability distributions that is closed with respect to addition as well as multiplication of independent random variables. *Journal of Theoretical Probability* 28 (2015), 1063–1081.
[41] P. Bosch. HCM property and the half-Cauchy distribution. *Probability and Mathematical Statistics* 35 (2015), 191–200.
[42] P. Bosch and T. Simon. A proof of Bondesson's conjecture on stable densities. *Arkiv för Matematik* 54 (2016), 31–38.
[43] P. Bourgade, T. Fujita and M. Yor. Euler's formulae for $\zeta(2n)$ and products of Cauchy variables. *Electronic Communications in Probability* 12 (2007), 73–80.
[44] P. Bourgade, C. P. Hughes, A. Nikeghbali and M. Yor. The characteristic polynomial of a random unitary matrix: a probabilistic approach. *Duke Mathematical Journal* 145 (2008), 45–69.
[45] J. Bourgain. On large values estimates for Dirichlet polynomials and the density hypothesis for the Riemann zeta function. *International Mathematics Research Notices* 2000 (2000), 133–146.
[46] C. J. Bouwkamp. A conjectured definite integral (A. H. Nuttall), solution. *SIAM Review* 28 (1986), 568–569.
[47] R. P. Brent. On the zeros of the Riemann zeta function in the critical strip. *Mathematics of Computation* 33 (1979), 1361–1372.
[48] S. Broadbent, H. Kadiri, A. Lumley, N. Ng and K. Wilk. Sharper bounds for the Chebyshev function $\theta(x)$. *Mathematics of Computation* 90 (2021), 2281–2315.
[49] M. Broadie, P. Glasserman and S. Kou. A continuity correction for discrete barrier options. *Mathematical Finance* 7 (1997), 325–349.
[50] D. Bump, K.-K. Choi, P. Kurlberg and J. Vaaler. A local Riemann hypothesis, I. *Mathematische Zeitschrift* 233 (2000), 1–18.
[51] J. Büthe. An analytic method for bounding $\psi(x)$. *Mathematics of Computation* 87 (2018), 1991–2009.
[52] B. Candelpergher. A new expansion of the Riemann zeta function. arXiv:2512.11405, 2025.
[53] D. A. Cardon. Sums of exponential functions having only real zeros. *Manuscripta Mathematica* 113 (2004), 307–317.
[54] D. A. Cardon. Fourier transforms having only real zeros. *Proceedings of the American Mathematical Society* 133 (2005), 1349–1356.
[55] M. A. Chaudhry and A. Qadir. Extension of Hardy's class for Ramanujan's interpolation formula and master theorem with applications. *Journal of Inequalities and Applications* 2012 (2012), paper 52.
[56] X. Chen, L. Cheng, J. Chadam and D. Saunders. Existence and uniqueness of solutions to the inverse boundary crossing problem for diffusions. *Annals of Applied Probability* 21 (2011), 1663–1693.
[57] Z. Ciesielski and S. J. Taylor. First passage times and sojourn times for Brownian motion in space and the exact Hausdorff measure of the sample path. *Transactions of the American Mathematical Society* 103 (1962), 434–450.
[58] M. W. Coffey and G. Csordas. On the log-concavity of a Jacobi theta function. *Mathematics of Computation* 82 (2013), 2265–2272.

[59] A. Connes and C. Consani. Weil positivity and trace formula, the archimedean place. *Selecta Mathematica (N.S.)* 27 (2021), Art. 77, arXiv:2006.13771.

[60] A. Connes, C. Consani and H. Moscovici. Zeta spectral triples. arXiv:2511.22755, 2025.

[61] J. B. Conrey. Zeros of derivatives of Riemann's ξ-function on the critical line. *Journal of Number Theory* 16 (1983), 49–74.

[62] J. B. Conrey and X.-J. Li. A note on some positivity conditions related to zeta and L-functions. *International Mathematics Research Notices* 2000 (2000), 929–940.

[63] J. B. Conrey, D. W. Farmer, J. P. Keating, M. O. Rubinstein and N. C. Snaith. Integral moments of L-functions. *Proceedings of the London Mathematical Society* 91 (2005), 33–104.

[64] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey and D. E. Knuth. On the Lambert W function. *Advances in Computational Mathematics* 5 (1996), 329–359.

[65] H. Cramér. Studien über die Nullstellen der Riemannschen Zetafunktion. *Mathematische Zeitschrift* 4 (1919), 104–130.

[66] H. Cramér. Ein Mittelwertsatz in der Primzahltheorie. *Mathematische Zeitschrift* 12 (1922), 147–153.

[67] T. Craven, G. Csordas and W. Smith. The zeros of derivatives of entire functions and the Pólya-Wiman conjecture. *Annals of Mathematics* 125 (1987), 405–431.

[68] G. Csordas, T. S. Norfolk and R. S. Varga. The Riemann hypothesis and the Turán inequalities. *Transactions of the American Mathematical Society* 296 (1986), 521–541.

[69] G. Csordas, A. M. Odlyzko, W. Smith and R. S. Varga. A new Lehmer pair of zeros and a new lower bound for the de Bruijn-Newman constant Λ. *Electronic Transactions on Numerical Analysis* 1 (1993), 104–111.

[70] G. Csordas, W. Smith and R. S. Varga. Lehmer pairs of zeros, the de Bruijn-Newman constant Λ, and the Riemann hypothesis. *Constructive Approximation* 10 (1994), 107–129.

[71] H. Davenport. *Multiplicative Number Theory*, third edition, revised by H. L. Montgomery. Graduate Texts in Mathematics 74, Springer, 2000.

[72] H. Davenport and H. Heilbronn. On the zeros of certain Dirichlet series I, II. *Journal of the London Mathematical Society* 11 (1936), 181–185 and 307–312.

[73] N. G. de Bruijn. The roots of trigonometric integrals. *Duke Mathematical Journal* 17 (1950), 197–226.

[74] H. Delange. Une remarque sur la dérivée logarithmique de la fonction zêta de Riemann. *Colloquium Mathematicum* 53 (1987), 333–335.

[75] P. Deligne. La conjecture de Weil. I. *Publications Mathématiques de l'IHÉS* 43 (1974), 273–307.

[76] L. Devroye. *Non-Uniform Random Variate Generation*. Springer, 1986.

[77] A. Dobner. A proof of Newman's conjecture for the extended Selberg class. *Acta Arithmetica* 201 (2021), 29–62.

[78] W. F. Donoghue, Jr. *Monotone Matrix Functions and Analytic Continuation*. Grundlehren der mathematischen Wissenschaften 207, Springer, 1974.

[79] H. Dym and H. P. McKean. *Gaussian Processes, Function Theory, and the Inverse Spectral Problem*. Probability and Mathematical Statistics 31, Academic Press, 1976.

[80] F. J. Dyson. A Brownian-motion model for the eigenvalues of a random matrix. *Journal of Mathematical Physics* 3 (1962), 1191–1198.

[81] H. M. Edwards. *Riemann's Zeta Function*. Academic Press, 1974.
[82] E. Ekström and S. Janson. The inverse first-passage problem and optimal stopping. *Annals of Applied Probability* 26 (2016), 3154–3177.
[83] R. S. Ellis, J. L. Monroe and C. M. Newman. The GHS and other correlation inequalities for a class of even ferromagnets. *Communications in Mathematical Physics* 46 (1976), 167–182.
[84] P. Embrechts, C. M. Goldie and N. Veraverbeke. Subexponentiality and infinite divisibility. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 49 (1979), 335–347.
[85] D. W. Farmer. Jensen polynomials are not a plausible route to proving the Riemann Hypothesis. *Advances in Mathematics* 411 (2022), 108781; arXiv:2008.07206.
[86] W. Feller. *An Introduction to Probability Theory and Its Applications*, volume II, second edition. Wiley, 1971.
[87] S. Fourati. On some analytical properties of stable densities. arXiv:1604.07705, 2016.
[88] S. A. Fulling. Nonuniqueness of canonical field quantization in Riemannian spacetime. *Physical Review D* 7 (1973), 2850–2862.
[89] W. Gabcke. *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*. Dissertation, Georg-August-Universität Göttingen, 1979.
[90] P. X. Gallagher. Primes and powers of 2. *Inventiones Mathematicae* 29 (1975), 125–142.
[91] G. Gasper. Using integrals of squares of certain real-valued special functions to prove that the Pólya $\Xi^*(z)$ function, the functions $K_{iz}(a)$, $a > 0$, and some other entire functions have only real zeros. arXiv:0801.2996, 2008.
[92] I. M. Gel'fand and N. Ya. Vilenkin. *Generalized Functions, Volume 4: Applications of Harmonic Analysis*. Academic Press, 1964.
[93] B. V. Gnedenko. Sur la distribution limite du terme maximum d'une série aléatoire. *Annals of Mathematics* 44 (1943), 423–453.
[94] B. V. Gnedenko and A. N. Kolmogorov. *Limit Distributions for Sums of Independent Random Variables*. Translated by K. L. Chung. Addison-Wesley, Cambridge, MA, 1954; revised edition 1968.
[95] C. Goldie. A class of infinitely divisible random variables. *Proceedings of the Cambridge Philosophical Society* 63 (1967), 1141–1143.
[96] S. M. Gonek and H. L. Montgomery. Zeros of a family of approximations of the Riemann zeta-function. *International Mathematics Research Notices* 2013 (2013), 4712–4733.
[97] X. Gourdon. The 10^{13} first zeros of the Riemann Zeta function, and zeros computation at very large height. Manuscript, 2004, `http://numbers.computation.free.fr/Constants/Miscellaneous/zetazeros1e13-1e24.pdf`, accessed 27 September 2026.
[98] J.-P. Gram. Note sur les zéros de la fonction $\zeta(s)$ de Riemann. *Acta Mathematica* 27 (1903), 289–304.
[99] M. Griffin, K. Ono, L. Rolen and D. Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences* 116 (2019), 11103–11110.
[100] R. B. Griffiths, C. A. Hurst and S. Sherman. Concavity of magnetization of an Ising ferromagnet in a positive external field. *Journal of Mathematical Physics* 11 (1970), 790–795.

[101] B. Grigelionis. Processes of Meixner type. *Lithuanian Mathematical Journal* 39 (1999), 33–41.
[102] J. Grommer. Ganze transzendente Funktionen mit lauter reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 144 (1914), 114–166.
[103] A. Gut. Some remarks on the Riemann zeta distribution. *Revue Roumaine de Mathématiques Pures et Appliquées* 51 (2006).
[104] L. Guth and J. Maynard. New large value estimates for Dirichlet polynomials. *Annals of Mathematics* 203 (2026), 623–675.
[105] J. Haglund. Some conjectures on the zeros of approximates to the Riemann Ξ-function and incomplete gamma functions. arXiv:0910.5228, 2009.
[106] G. H. Hardy and J. E. Littlewood. Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes. *Acta Mathematica* 41 (1916), 119–196.
[107] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*, sixth edition, revised by D. R. Heath-Brown and J. H. Silverman. Oxford University Press, 2008.
[108] P. Hartman and G. S. Watson. "Normal" distribution functions on spheres and the modified Bessel functions. *Annals of Probability* 2 (1974), 593–607.
[109] E. Hasanalizade, Q. Shen and P.-J. Wong. Counting zeros of the Riemann zeta function. *Journal of Number Theory* 235 (2022), 219–241.
[110] T. Hasebe and S. Thorbjørnsen. Unimodality of the freely selfdecomposable probability laws. *Journal of Theoretical Probability* 29 (2016), 922–940.
[111] C. B. Haselgrove. A disproof of a conjecture of Pólya. *Mathematika* 5 (1958), 141–145.
[112] D. R. Heath-Brown and J. C. Puchta. Integers represented as a sum of primes and powers of two. *Asian Journal of Mathematics* 6 (2002), 535–566.
[113] E. Hecke. Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung. *Mathematische Annalen* 112 (1936), 664–699.
[114] H. Hedenmalm, P. Lindqvist and K. Seip. A Hilbert space of Dirichlet series and systems of dilated functions in $L^2(0,1)$. *Duke Mathematical Journal* 86 (1997), 1–37.
[115] C. Hermite. Extrait d'une lettre de Mr. Ch. Hermite de Paris à Mr. Borchardt de Berlin sur le nombre des racines d'une équation algébrique comprises entre des limites données. *Journal für die reine und angewandte Mathematik* 52 (1856), 39–51.
[116] G. 't Hooft. On the quantum structure of a black hole. *Nuclear Physics B* 256 (1985), 727–745.
[117] G. Hurst. Computations of the Mertens function and improved bounds on the Mertens conjecture. *Mathematics of Computation* 87 (2018), 1013–1028.
[118] J. I. Hutchinson. On the roots of the Riemann zeta function. *Transactions of the American Mathematical Society* 27 (1925), 49–60.
[119] M. N. Huxley. On the difference between consecutive primes. *Inventiones Mathematicae* 15 (1972), 164–170.
[120] Y. Ihara. On the Euler–Kronecker constants of global fields and primes with small norms. In *Algebraic Geometry and Number Theory*, V. Ginzburg, ed., Progress in Mathematics 253, Birkhäuser, 2006, 407–451.
[121] A. M. Iksanov, Z. J. Jurek and B. M. Schreiber. A new factorization property of the selfdecomposable probability measures. *Annals of Probability* 32 (2004), 1356–1369.

[122] A. E. Ingham. Mean-value theorems in the theory of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (2) 27 (1928), 273–300.
[123] A. E. Ingham. *The Distribution of Prime Numbers*. Cambridge Tracts in Mathematics and Mathematical Physics 30, Cambridge University Press, 1932.
[124] A. E. Ingham. On the estimation of $N(\sigma, T)$. *Quarterly Journal of Mathematics* 11 (1940), 291–292.
[125] A. E. Ingham. On two conjectures in the theory of numbers. *American Journal of Mathematics* 64 (1942), 313–319.
[126] H. Inoue. Expansion of Dirichlet L-function on the critical line in Meixner-Pollaczek polynomials. arXiv:1412.1220, 2014.
[127] K. Itô and H. P. McKean. *Diffusion Processes and Their Sample Paths*. Springer, 1965.
[128] H. Iwaniec and E. Kowalski. *Analytic Number Theory*. American Mathematical Society Colloquium Publications 53, American Mathematical Society, 2004.
[129] L. F. James, B. Roynette and M. Yor. Generalized gamma convolutions, Dirichlet means, Thorin measures, with explicit examples. *Probability Surveys* 5 (2008), 346–415.
[130] F. Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers* 66 (2017), 1281–1292.
[131] D. R. Johnston and A. Yang. Some explicit estimates for the error term in the prime number theorem. *Journal of Mathematical Analysis and Applications* 527 (2023), 127460.
[132] L. Jones and N. O'Connell. Weyl chambers, symmetric spaces and number variance saturation. *ALEA Latin American Journal of Probability and Mathematical Statistics* 2 (2006), 91–118.
[133] Z. J. Jurek and W. Vervaat. An integral representation for selfdecomposable Banach space valued random variables. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 62 (1983), 247–262.
[134] M. Kac. A remark on Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 1155–1157.
[135] G. A. Kalugin and D. J. Jeffrey. Unimodal sequences show Lambert W is Bernstein. arXiv:1011.5940, 2010.
[136] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Stieltjes, Poisson and other integral representations for functions of Lambert W. Preprint, arXiv:1103.5640, 2011; published as [137, 138].
[137] G. A. Kalugin, D. J. Jeffrey, R. M. Corless and P. B. Borwein. Stieltjes and other integral representations for functions of Lambert W. *Integral Transforms and Special Functions* 23 (2012), 581–593.
[138] G. A. Kalugin, D. J. Jeffrey and R. M. Corless. Bernstein, Pick, Poisson and related integral expressions for Lambert W. *Integral Transforms and Special Functions* 23 (2012), 817–829.
[139] Y. Kasahara. Tauberian theorems of exponential type. *Journal of Mathematics of Kyoto University* 18 (1978), 209–219.
[140] N. M. Katz and P. Sarnak. Zeroes of zeta functions and symmetry. *Bulletin of the American Mathematical Society* 36 (1999), 1–26.
[141] J. P. Keating and N. C. Snaith. Random matrix theory and $\zeta(1/2 + it)$. *Communications in Mathematical Physics* 214 (2000), 57–89.
[142] B. Kerr. A simpler proof that there are no Landau Siegel zeros. arXiv:2610.21099,

2026.

[143] J. B. Keiper. Power series expansions of Riemann's ξ function. *Mathematics of Computation* 58 (1992), 765–773.

[144] J. T. Kent. Some probabilistic properties of Bessel functions. *Annals of Probability* 6 (1978), 760–770.

[145] J. T. Kent. Eigenvalue expansions for diffusion hitting times. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 52 (1980), 309–319.

[146] J. T. Kent. The spectral decomposition of a diffusion hitting time. *Annals of Probability* 10 (1982), 207–219.

[147] D. Kershaw. Some extensions of W. Gautschi's inequalities for the gamma function. *Mathematics of Computation* 41 (1983), 607–611.

[148] A. Ya. Khinchin. *Limit Laws for Sums of Independent Random Variables* (in Russian). ONTI, Moscow, 1938.

[149] H. Ki and Y. O. Kim. On the number of nonreal zeros of real entire functions and the Fourier-Pólya conjecture. *Duke Mathematical Journal* 104 (2000), 45–73.

[150] H. Ki, Y.-O. Kim and J. Lee. On the de Bruijn–Newman constant. *Advances in Mathematics* 222 (2009), 281–306.

[151] Y. O. Kim. A proof of the Pólya-Wiman conjecture. *Proceedings of the American Mathematical Society* 109 (1990), 1045–1052.

[152] R. Koekoek, P. A. Lesky and R. F. Swarttouw. *Hypergeometric Orthogonal Polynomials and Their q-Analogues*. Springer, 2010.

[153] T. Konstantopoulos, P. Patie and R. Sarkar. A new class of solutions to the van Dantzig problem, the Lee–Yang property, and the Riemann hypothesis. *Annales de l'Institut Fourier* 74 (2024), 377–421.

[154] J. Korevaar. Distribution proof of Wiener's Tauberian theorem. *Proceedings of the American Mathematical Society* 16 (1965), 353–355.

[155] N. M. Korobov. Estimates of trigonometric sums and their applications. *Uspekhi Matematicheskikh Nauk* 13 (1958), no. 4, 185–192.

[156] G. K. Kristiansen. A proof of Steutel's conjecture. *Annals of Probability* 22 (1994), 442–452.

[157] A. B. J. Kuijlaars and W. Van Assche. The asymptotic zero distribution of orthogonal polynomials with varying recurrence coefficients. *Journal of Approximation Theory* 99 (1999), 167–197.

[158] A. Kuznetsov. Integral representations for the Dirichlet L-functions and their expansions in Meixner–Pollaczek polynomials and rising factorials. *Integral Transforms and Special Functions* 18 (2007), 827–835.

[159] A. Kuznetsov. Expansion of the Riemann Ξ function in Meixner–Pollaczek polynomials. *Canadian Mathematical Bulletin* 51 (2008), 561–569.

[160] J. C. Lagarias. On a positivity property of the Riemann ξ-function. *Acta Arithmetica* 89 (1999), 217–234.

[161] J. C. Lagarias. Zero spacing distributions for differenced L-functions. *Acta Arithmetica* 120 (2005), 159–184.

[162] C. D. Lai and D. Vere-Jones. Odd man out: the Meixner hypergeometric distribution. *Australian Journal of Statistics* 21 (1979), 256–265.

[163] J. Lamperti. An occupation time theorem for a class of stochastic processes. *Transactions of the American Mathematical Society* 88 (1958), 380–387.

[164] Y. Lamzouri. A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line. arXiv:2609.02882, 2026.

[165] E. Landau. Über die Nullstellen der Zetafunktion. *Mathematische Annalen* 71 (1912), 548–564.
[166] H. J. Landau and H. O. Pollak. Prolate spheroidal wave functions, Fourier analysis and uncertainty, III: the dimension of the space of essentially time and band limited signals. *Bell System Technical Journal* 41 (1962), 1295–1336.
[167] H. J. Landau and H. Widom. Eigenvalue distribution of time and frequency limiting. *Journal of Mathematical Analysis and Applications* 77 (1980), 469–481.
[168] S. Lang. *Algebraic Number Theory*, second edition. Graduate Texts in Mathematics 110, Springer, 1994.
[169] C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice Hall, 1974.
[170] I. Lazarević. Certaines inégalités où interviennent les fonctions hyperboliques. *Univerzitet u Beogradu, Publikacije Elektrotehničkog Fakulteta, Serija Matematika i Fizika* 159–170 (1966), 41–48.
[171] T. D. Lee and C. N. Yang. Statistical theory of equations of state and phase transitions. II. Lattice gas and Ising model. *Physical Review* 87 (1952), 410–419.
[172] R. S. Lehman. On the distribution of zeros of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (3) 20 (1970), 303–320.
[173] D. H. Lehmer. On the roots of the Riemann zeta-function. *Acta Mathematica* 95 (1956), 291–298.
[174] D. H. Lehmer. Extended computation of the Riemann zeta-function. *Mathematika* 3 (1956), 102–108.
[175] B. Ja. Levin. *Distribution of Zeros of Entire Functions*. Translations of Mathematical Monographs 5, American Mathematical Society, 1964.
[176] X.-J. Li. The positivity of a sequence of numbers and the Riemann hypothesis. *Journal of Number Theory* 65 (1997), 325–333.
[177] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes. *Acta Arithmetica* 92 (2000), 229–237.
[178] H. Z. Li. The number of powers of 2 in a representation of large even integers by sums of such powers and of two primes, II. *Acta Arithmetica* 96 (2001), 369–379.
[179] E. H. Lieb and A. D. Sokal. A general Lee-Yang theorem for one-component and multicomponent ferromagnets. *Communications in Mathematical Physics* 80 (1981), 153–179.
[180] G. D. Lin and C. Y. Hu. The Riemann zeta distribution. *Bernoulli* 7 (2001), 817–828.
[181] A. Lindner, L. Pan and K. Sato. On quasi-infinitely divisible distributions. *Transactions of the American Mathematical Society* 370 (2018), 8483–8520.
[182] Yu. V. Linnik. Prime numbers and powers of two. *Trudy Matematicheskogo Instituta imeni V. A. Steklova* 38 (1951), 152–169.
[183] Yu. V. Linnik. Addition of prime numbers with powers of one and the same number. *Matematicheskii Sbornik* 32 (1953), 3–60.
[184] Yu. V. Linnik. *Decomposition of Probability Distributions*. Oliver and Boyd, 1964.
[185] J. E. Littlewood. On the zeros of the Riemann zeta-function. *Proceedings of the Cambridge Philosophical Society* 22 (1924), 295–318.
[186] J. Liu, M. C. Liu and T. Wang. The number of powers of 2 in a representation of large even integers, I. *Science in China Series A* 41 (1998), 386–398.
[187] J. Liu, M. C. Liu and T. Wang. On the almost Goldbach problem of Linnik. *Journal de Théorie des Nombres de Bordeaux* 11 (1999), 133–147.

[188] The LMFDB Collaboration. *The L-functions and modular forms database*, `https://www.lmfdb.org`, accessed 27 September 2026.

[189] K. Löwner. Über monotone Matrixfunktionen. *Mathematische Zeitschrift* 38 (1934), 177–216.

[190] R. Lugannani and S. Rice. Saddle point approximation for the distribution of the sum of independent random variables. *Advances in Applied Probability* 12 (1980).

[191] D. Ostrovsky. Theory of Barnes beta distributions. *Electronic Communications in Probability* 18 (2013).

[192] Liu. Improved zero free half planes for the quasi Riemann hypothesis. arXiv:2610.12234, 2026.

[193] B. Liu. Zero free half planes for Hecke L functions over number fields. Preprint, 2026.

[194] T. Xylouris. On the least prime in an arithmetic progression and estimates for the zeros of Dirichlet L-functions. *Acta Arithmetica* 150 (2011).

[195] E. Lukacs. Contributions to a problem of D. van Dantzig. *Theory of Probability and its Applications* 13 (1968), 116–127.

[196] S. Mandelbrojt. *Séries de Dirichlet. Principes et méthodes*. Gauthier-Villars, 1969.

[197] V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR, Sbornik* 1 (1967), 457–483.

[198] A. W. Marcus, D. A. Spielman and N. Srivastava. Interlacing families II: mixed characteristic polynomials and the Kadison Singer problem. *Annals of Mathematics* 182 (2015), 327–350.

[199] A. W. Marcus, D. A. Spielman and N. Srivastava. Finite free convolutions of polynomials. *Probability Theory and Related Fields* 182 (2022), 807–848.

[200] S. Mase. Decomposition of infinitely divisible characteristic functions with absolutely continuous Poisson spectral measure. *Annals of the Institute of Statistical Mathematics* 27 (1975), 289–298.

[201] Math StackExchange question 2063166. Dirichlet series associated to squared Möbius. December 2016, `https://math.stackexchange.com/q/2063166`, accessed 9 October 2026.

[202] N. Metropolis and G.-C. Rota. Witt vectors and the algebra of necklaces. *Advances in Mathematics* 50 (1983), 95–125.

[203] H. L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, H. G. Diamond, ed., Proceedings of Symposia in Pure Mathematics 24, American Mathematical Society, 1973, 181–193.

[204] H. L. Montgomery. Zeros of approximations to the zeta function. In *Studies in Pure Mathematics: To the Memory of Paul Turán*, Birkhäuser, Basel, 1983, 497–506.

[205] The mpmath development team. *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, version 1.3.0, 2023.

[206] T. Nakamura. A complete Riemann zeta distribution and the Riemann hypothesis. *Bernoulli* 21 (2015), 604–617.

[207] T. Nakamura. Zeta distributions generated by Dirichlet series and their (quasi) infinite divisibility. arXiv:2209.13257, 2022.

[208] C. M. Newman. Zeros of the partition function for generalized Ising systems. *Communications on Pure and Applied Mathematics* 27 (1974), 143–159.

[209] C. M. Newman. Inequalities for Ising models and field theories which obey the Lee-Yang theorem. *Communications in Mathematical Physics* 41 (1975), 1–9.

[210] C. M. Newman. Fourier transforms with only real zeros. *Proceedings of the American*

Mathematical Society 61 (1976), 245–251.

[211] C. M. Newman. The GHS inequality and the Riemann hypothesis. *Constructive Approximation* 7 (1991), 389–399.

[212] C. M. Newman and W. Wu. Constants of de Bruijn–Newman type in analytic number theory and statistical physics. *Bulletin of the American Mathematical Society* 57 (2020), 595–614.

[213] T. Newton. Kernel-concavity obstructions and Fourier-side Laguerre bounds for the Riemann Ξ programme. Zenodo, 2026, `https://doi.org/10.5281/zenodo.23461586`, accessed 20 October 2026.

[214] A. Nica and R. Speicher. *Lectures on the Combinatorics of Free Probability*. London Mathematical Society Lecture Note Series 335, Cambridge University Press, 2006.

[215] A. Nikeghbali and M. Yor. The Barnes G function and its relations with sums and products of generalized gamma convolution variables. *Electronic Communications in Probability* 14 (2009), 396–411.

[216] A. A. Novikov. Martingales, Tauberian theorem, and strategies of gambling. *Theory of Probability and its Applications* 41 (1997), 716–729.

[217] A. H. Nuttall. A conjectured definite integral, problem. *SIAM Review* 27 (1985), 573.

[218] B. Nyman. *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*. Thesis, University of Uppsala, 1950.

[219] A. M. Odlyzko. On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation* 48 (1987), 273–308.

[220] A. M. Odlyzko and A. Schönhage. Fast algorithms for multiple evaluations of the Riemann zeta function. *Transactions of the American Mathematical Society* 309 (1988), 797–809.

[221] A. M. Odlyzko. *The 10^{20}-th zero of the Riemann zeta function and 175 million of its neighbors*. Unpublished manuscript, 1992, `https://www-users.cse.umn.edu/~odlyzko/unpublished/zeta.10to20.1992.pdf`, accessed 27 September 2026.

[222] A. M. Odlyzko. An improved bound for the de Bruijn–Newman constant. *Numerical Algorithms* 25 (2000), 293–303.

[223] A. M. Odlyzko. The 10^{22}-nd zero of the Riemann zeta function. In *Dynamical, Spectral, and Arithmetic Zeta Functions*, M. L. Lapidus and M. van Frankenhuysen, eds., Contemporary Mathematics 290, American Mathematical Society, 2001, 139–144.

[224] A. M. Odlyzko. *Tables of zeros of the Riemann zeta function*. Online, `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`, accessed 27 September 2026.

[225] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain, eds. *NIST Digital Library of Mathematical Functions*. `https://dlmf.nist.gov/`, Release 1.2.8 of 2026-09-15.

[226] OpenAI. The quasi-Riemann hypothesis: a zero-free half-plane $\mathrm{Re}(s) > 7/8$. Preprint dated 30 September 2026, released 6 October 2026 with a formalisation in Lean, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[227] OpenAI. The quasi-Riemann hypothesis. Preprint dated 5 October 2026, released 6 October 2026, `https://github.com/openai/math` (family 003), accessed 7 October 2026.

[228] A. G. Pakes. Length biasing and laws equivalent to the log-normal. *Journal of Mathematical Analysis and Applications* 197 (1996), 825–854.

[229] A. G. Pakes. Remarks on converse Carleman and Krein criteria for the classical moment problem. *Journal of the Australian Mathematical Society* 71 (2001), 81–104.
[230] A. G. Pakes. Structure of Stieltjes classes of moment-equivalent probability laws. *Journal of Mathematical Analysis and Applications* 326 (2007), 1268–1290.
[231] A. G. Pakes. Lambert's W, infinite divisibility and Poisson mixtures. *Journal of Mathematical Analysis and Applications* 378 (2011), 480–492.
[232] A. G. Pakes. On the preservation of infinite divisibility under length-biasing. *Journal of Probability* 2014 (2014), 703697.
[233] A. G. Pakes. The Lambert W function, Nuttall's integral, and the Lambert law. *Statistics and Probability Letters* 139 (2018), 53–60.
[234] A. G. Pakes. Self-decomposable laws from continuous branching processes. *Journal of Theoretical Probability* 33 (2020), 361–395.
[235] A. G. Pakes. Self-decomposability and the Lambert law. *Axioms* 15 (2026), 574.
[236] A. G. Pakes, W.-L. Hung and J.-W. Wu. Criteria for the unique determination of probability distributions by moments. *Australian and New Zealand Journal of Statistics* 43 (2001), 101–111.
[237] A. G. Pakes and R. Khattree. Length-biasing, characterizations of laws and the moment problem. *Australian Journal of Statistics* 34 (1992), 307–322.
[238] A. G. Pakes, T. Sapatinas and E. B. Fosam. Characterizations, length-biasing, and infinite divisibility. *Statistical Papers* 37 (1996), 53–69.
[239] V. Pérez-Abreu and N. Sakuma. Free generalized gamma convolutions. *Electronic Communications in Probability* 13 (2008), 526–539.
[240] G. Peskir. On integral equations arising in the first-passage problem for Brownian motion. *Journal of Integral Equations and Applications* 14 (2002), 397–423.
[241] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, I. *Acta Arithmetica* 109 (2003), 169–194.
[242] J. Pintz and I. Z. Ruzsa. On Linnik's approximation to Goldbach's problem, II. *Acta Mathematica Hungarica* 161 (2020), 569–582.
[243] J. Pitman and M. Yor. Bessel processes and infinitely divisible laws. In *Stochastic Integrals*, Lecture Notes in Mathematics 851, Springer, 1981, 285–370.
[244] J. Pitman and M. Yor. Infinitely divisible laws associated with hyperbolic functions. *Canadian Journal of Mathematics* 55 (2003), 292–330.
[245] D. J. Platt. Numerical computations concerning the GRH. *Mathematics of Computation* 85 (2016), 3009–3027.
[246] D. J. Platt and T. S. Trudgian. Linnik's approximation to Goldbach's conjecture, and other problems. *Journal of Number Theory* 153 (2015), 54–62.
[247] D. J. Platt and T. S. Trudgian. The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bulletin of the London Mathematical Society* 53 (2021), 792–797.
[248] N. G. Polson, J. G. Scott and J. Windle. The Bayesian bridge. *Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 76 (2014), 713; sections and theorems are cited by the numbering of arXiv:1109.2279v2.
[249] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v24, 2026.
[250] N. G. Polson. On Hilbert's 8th problem. arXiv:1708.02653v5, 2017.
[251] N. G. Polson. Riemann, Thorin, van Dantzig pairs, Wald couples and Hadamard factorisation. arXiv:1804.10043, 2018.
[252] N. G. Polson. *On Hilbert's 8th Problem: the reciprocal Xi clock, Thorin measures, van Dantzig pairs, zero free approximations, hitting times and the reciprocal zeta*

function, first edition. Independently published, 2026.
[253] N. G. Polson. The Dirichlet beta function in Thorin form. Manuscript, October 2026.
[254] N. G. Polson. Biane on Pólya: hitting times, Thorin measures and a Wick rotation. Manuscript, October 2026.
[255] N. G. Polson. Thorin measures of the Dedekind eta function. Manuscript, September 2026.
[256] N. G. Polson. Maxima of exponentials, incomplete gamma continuations, and the Gamma(2) structure of L functions. Manuscript, October 2026.
[257] N. G. Polson. The Pólya density as a maximum order statistic and the Thorin measure of the reciprocal Xi function. Manuscript, October 2026.
[258] N. G. Polson. Thorin type representations of the reciprocal zeta function on the critical line. Manuscript, September 2026, now Part IX of the first edition of *On Hilbert's 8th Problem* (2026).
[259] N. G. Polson. The squarefree Dirichlet series $\zeta(s)/\zeta(2s)$: a summary of Math StackExchange 2063166, Pólya densities, Gamma(2) smoothing and Thorin measures of Ξ. Manuscript, October 2026.
[260] N. G. Polson. Gamma, reciprocal Gamma, η^3 and Δ: Thorin forms and the Riemann Ξ function. Manuscript, October 2026.
[261] N. G. Polson. Wigner's semicircle, free probability and the Thorin framework for the reciprocal Xi function. Manuscript, version 7, October 2026.
[262] N. G. Polson. Pakes, the Lambert law and the Thorin condition for the Riemann Xi function. Manuscript, October 2026.
[263] N. G. Polson and V. Sokolov. Computing the Thorin measure of the reciprocal Xi function: a Pólya route and a total variation comparison with the Riemann atoms. Manuscript, September 2026.
[264] N. G. Polson and V. Sokolov. Four attacks on the computational path: a working report. Manuscript, 7 October 2026.
[265] N. G. Polson and V. Sokolov. *On Langlands Conjectures: Probabilistic Methods*, Chapter 21. Draft, 2026.
[266] G. Pólya. Über Annäherung durch Polynome mit lauter reellen Wurzeln. *Rendiconti del Circolo Matematico di Palermo* 36 (1913), 279–295.
[267] G. Pólya. Algebraische Untersuchungen über ganze Funktionen vom Geschlechte Null und Eins. *Journal für die reine und angewandte Mathematik* 145 (1915), 224–249.
[268] G. Pólya. Über die Nullstellen gewisser ganzer Funktionen. *Mathematische Zeitschrift* 2 (1918), 352–383.
[269] G. Pólya. Verschiedene Bemerkungen zur Zahlentheorie. *Jahresbericht der Deutschen Mathematiker-Vereinigung* 28 (1919), 31–40.
[270] G. Pólya. Über die Nullstellen sukzessiver Derivierten. *Mathematische Zeitschrift* 12 (1922), 36–60.
[271] G. Pólya. Über die Existenz unendlich vieler singulärer Punkte auf der Konvergenzgeraden gewisser Dirichletscher Reihen. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse* (1923), 45–50.
[272] G. Pólya. On the zeros of an integral function represented by Fourier's integral. *Messenger of Mathematics* 52 (1923), 185–188.
[273] G. Pólya. Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion. *Acta Mathematica* 48 (1926), 305–317.

[274] G. Pólya. On the zeros of certain trigonometric integrals. *Journal of the London Mathematical Society* 1 (1926), 98–99.
[275] G. Pólya. Über trigonometrische Integrale mit nur reellen Nullstellen. *Journal für die reine und angewandte Mathematik* 158 (1927), 6–18.
[276] G. Pólya. Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen. *Det Kongelige Danske Videnskabernes Selskab, Mathematisk-fysiske Meddelelser* 7 (1927), no. 17, 3–33.
[277] G. Pólya. Untersuchungen über Lücken und Singularitäten von Potenzreihen. *Mathematische Zeitschrift* 29 (1929), 549–640.
[278] G. Pólya. Some problems connected with Fourier's work on transcendental equations. *The Quarterly Journal of Mathematics* (Oxford) 1 (1930), 21–34.
[279] G. Pólya. On the zeros of the derivatives of a function and its analytic character. *Bulletin of the American Mathematical Society* 49 (1943), 178–191.
[280] G. Pólya. *Collected Papers, Vol. II: Location of Zeros*, edited by R. P. Boas. Mathematicians of Our Time 8, MIT Press, 1974.
[281] G. Pólya and J. Schur. Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen. *Journal für die reine und angewandte Mathematik* 144 (1914), 89–113.
[282] D. H. J. Polymath. Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant. *Research in the Mathematical Sciences* 6 (2019), paper 31.
[283] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler. More than five-twelfths of the zeros of ζ are on the critical line. *Research in the Mathematical Sciences* 7 (2020), article 2.
[284] R. Remmert. *Classical Topics in Complex Function Theory*, translated by L. Kay. Graduate Texts in Mathematics 172, Springer, 1998.
[285] M. Riesz. Sur l'hypothèse de Riemann. *Acta Mathematica* 40 (1916), 185–190.
[286] H. Robbins. A remark on Stirling's formula. *American Mathematical Monthly* 62 (1955), 26–29.
[287] G. O. Roberts. Asymptotic approximations for Brownian motion boundary hitting times. *Annals of Probability* 19 (1991), 1689–1731.
[288] G. O. Roberts and C. F. Shortland. The hazard rate tangent approximation for boundary hitting times. *Annals of Applied Probability* 5 (1995), 446–460.
[289] B. Rodgers and T. Tao. The de Bruijn–Newman constant is non-negative. *Forum of Mathematics, Pi* 8 (2020), e6.
[290] D. Romik. Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases. *Acta Arithmetica* 200 (2021), 259–329.
[291] J. B. Rosser and L. Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois Journal of Mathematics* 6 (1962), 64–94.
[292] J. B. Rosser, J. M. Yohe and L. Schoenfeld. Rigorous computation and the zeros of the Riemann zeta-function. In *Information Processing 68: Proceedings of IFIP Congress 1968*, volume 1, A. J. H. Morrell, ed., North-Holland, 1969, 70–76.
[293] R. Rumely. Numerical computations concerning the ERH. *Mathematics of Computation* 61 (1993), 415–440.
[294] R. Salem. Sur une proposition équivalente à l'hypothèse de Riemann. *Comptes Rendus de l'Académie des Sciences* 236 (1953), 1127–1128.
[295] P. Salminen and P. Vallois. On maximum increase and decrease of Brownian motion.

Annales de l'Institut Henri Poincaré, Probabilités et Statistiques 43 (2007), 655–676.

[296] K. Sato. Class L of multivariate distributions and its subclasses. *Journal of Multivariate Analysis* 10 (1980), 207–232.

[297] K. Sato. *Lévy Processes and Infinitely Divisible Distributions*. Cambridge Studies in Advanced Mathematics 68, Cambridge University Press, 1999.

[298] A. Saumard and J. A. Wellner. Log-concavity and strong log-concavity: a review. *Statistics Surveys* 8 (2014), 45–114.

[299] R. L. Schilling, R. Song and Z. Vondraček. *Bernstein Functions: Theory and Applications*, second edition. Walter de Gruyter, 2012.

[300] I. J. Schoenberg. On Pólya frequency functions. I. The totally positive functions and their Laplace transforms. *Journal d'Analyse Mathématique* 1 (1951), 331–374.

[301] W. Schoutens and J. L. Teugels. Lévy processes, polynomials and martingales. *Communications in Statistics. Stochastic Models* 14 (1998), 335–349.

[302] K. Seip. *Interpolation and Sampling in Spaces of Analytic Functions*. University Lecture Series 33, American Mathematical Society, 2004.

[303] A. Selberg. Contributions to the theory of the Riemann zeta-function. *Archiv for Mathematik og Naturvidenskab* 48 (1946), 89–155.

[304] C. L. Siegel. Über Riemanns Nachlaß zur analytischen Zahlentheorie. *Quellen und Studien zur Geschichte der Mathematik, Astronomie und Physik, Abteilung B: Studien* 2 (1932), 45–80.

[305] G. Sierra. The Riemann zeros as energy levels of a Dirac fermion in a potential built from the prime numbers in Rindler spacetime. *Journal of Physics A: Mathematical and Theoretical* 47 (2014), 325204.

[306] B. Simon and R. B. Griffiths. The $(\phi^4)_2$ field theory as a classical Ising model. *Communications in Mathematical Physics* 33 (1973), 145–164.

[307] D. Slepian and H. O. Pollak. Prolate spheroidal wave functions, Fourier analysis and uncertainty, I. *Bell System Technical Journal* 40 (1961), 43–63.

[308] J. Sondow and C. Dumitrescu. A monotonicity property of Riemann's xi function and a reformulation of the Riemann hypothesis. *Periodica Mathematica Hungarica* 60 (2010), 37–40.

[309] R. Spira. Zeros of approximate functional approximations. *Mathematics of Computation* 21 (1967), 41–48.

[310] R. Spira. Some zeros of the Titchmarsh counterexample. *Mathematics of Computation* 63 (1994), 747–748.

[311] L. A. Stefanski. A normal scale mixture representation of the logistic distribution. *Statistics & Probability Letters* 11 (1991), 69–70.

[312] F. W. Steutel. Note on the infinite divisibility of exponential mixtures. *Annals of Mathematical Statistics* 38 (1967), 1303–1305.

[313] F. W. Steutel. Some recent results in infinite divisibility. *Stochastic Processes and their Applications* 1 (1973), 125–143.

[314] F. W. Steutel and J. G. F. Thiemann. On the independence of integer and fractional parts. *Statistica Neerlandica* 43 (1989), 53–59.

[315] F. W. Steutel and K. van Harn. *Infinite Divisibility of Probability Distributions on the Real Line*. Marcel Dekker, 2004.

[316] A. Talbot. The accurate numerical inversion of Laplace transforms. *Journal of the Institute of Mathematics and its Applications* 23 (1979), 97–120.

[317] M. Tanaka. A numerical investigation on cumulative sum of the Liouville function.

Tokyo Journal of Mathematics 3 (1980), 187–189.

[318] J. T. Tate. Fourier analysis in number fields and Hecke's zeta-functions. In *Algebraic Number Theory*, J. W. S. Cassels and A. Fröhlich, eds., Academic Press, 1967, 305–347.

[319] O. Thorin. On the infinite divisibility of the Pareto distribution. *Scandinavian Actuarial Journal* (1977), 31–40.

[320] O. Thorin. On the infinite divisibility of the lognormal distribution. *Scandinavian Actuarial Journal* (1977), 121–148.

[321] O. Thorin. An extension of the notion of a generalized Γ-convolution. *Scandinavian Actuarial Journal* (1978), 141–149.

[322] E. C. Titchmarsh. The zeros of the Riemann zeta-function. *Proceedings of the Royal Society of London, Series A* 157 (1936), 261–263.

[323] E. C. Titchmarsh. *Eigenfunction Expansions Associated with Second-Order Differential Equations*. Clarendon Press, Oxford, 1946.

[324] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*, second edition, revised by D. R. Heath-Brown. Oxford University Press, 1986.

[325] P. Turán. On some approximative Dirichlet polynomials in the theory of the zeta function of Riemann. *Danske Vid. Selsk. Mat.-Fys. Medd.* 24 (1948).

[326] L. N. Trefethen and J. A. C. Weideman. The exponentially convergent trapezoidal rule. *SIAM Review* 56 (2014), 385–458.

[327] T. S. Trudgian. Improvements to Turing's method. *Mathematics of Computation* 80 (2011), 2259–2279.

[328] T. S. Trudgian. On the success and failure of Gram's Law and the Rosser Rule. *Acta Arithmetica* 148 (2011), 225–256.

[329] T. S. Trudgian. An improved upper bound for the argument of the Riemann zeta-function on the critical line II. *Journal of Number Theory* 134 (2014), 280–292.

[330] K.-M. Tsang. Some Ω-theorems for the Riemann zeta-function. *Acta Arithmetica* 46 (1986), 369–395.

[331] N. Tschebotareff. Über die Realität von Nullstellen ganzer transzendenter Funktionen. *Mathematische Annalen* 99 (1928), 660–686.

[332] A. M. Turing. Some calculations of the Riemann zeta-function. *Proceedings of the London Mathematical Society* (3) 3 (1953), 99–117.

[333] K. Urbanik. Limit laws for sequences of normed sums satisfying some stability conditions. In *Multivariate Analysis III*, P. R. Krishnaiah, ed., Academic Press, 1973, 225–237.

[334] J. van de Lune, H. J. J. te Riele and D. T. Winter. On the zeros of the Riemann zeta function in the critical strip. IV. *Mathematics of Computation* 46 (1986), 667–681.

[335] I. M. Vinogradov. A new estimate of the function $\zeta(1+it)$. *Izvestiya Akademii Nauk SSSR, Seriya Matematicheskaya* 22 (1958), 161–164.

[336] D. Voiculescu. Addition of certain non-commuting random variables. *Journal of Functional Analysis* 66 (1986), 323–346.

[337] T. Wang. On Linnik's almost Goldbach theorem. *Science in China Series A* 42 (1999), 1155–1172.

[338] A. Weil. Sur les "formules explicites" de la théorie des nombres premiers. *Communications du Séminaire Mathématique de l'Université de Lund*, tome supplémentaire (1952), 252–265.

[339] G. C. Wick. Properties of Bethe-Salpeter wave functions. *Physical Review* 96 (1954), 1124–1134.

[340] D. V. Widder. *The Laplace Transform*. Princeton Mathematical Series 6, Princeton University Press, 1941.

[341] N. Wiener. Tauberian theorems. *Annals of Mathematics* 33 (1932), 1–100.

[342] N. Wiener. *The Fourier Integral and Certain of Its Applications*. Cambridge University Press, 1933.

[343] E. P. Wigner. On the distribution of the roots of certain symmetric matrices. *Annals of Mathematics* 67 (1958), 325–327.

[344] D. Williams. Brownian motion and the Riemann zeta-function. In *Disorder in Physical Systems*, G. R. Grimmett and D. J. A. Welsh, eds., Oxford University Press, 1990, 361–372.

[345] R. E. Williamson. Multiply monotone functions and their Laplace transforms. *Duke Mathematical Journal* 23 (1956), 189–207.

[346] S. J. Wolfe. On a continuous analogue of the stochastic difference equation $X_n = \rho X_{n-1} + B_n$. *Stochastic Processes and their Applications* 12 (1982), 301–312.

[347] M. Yamazato. Unimodality of infinitely divisible distribution functions of class L. *Annals of Probability* 6 (1978), 523–531.

[348] M. Yor. Loi de l'indice du lacet brownien, et distribution de Hartman-Watson. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 53 (1980), 71–95.

[349] H. Yoshida. On Hermitian forms attached to zeta functions. In *Zeta Functions in Geometry (Tokyo, 1990)*, N. Kurokawa and T. Sunada, eds., Advanced Studies in Pure Mathematics 21, Kinokuniya, 1992, 281–325.

[350] X. Zhu. Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law. arXiv:2608.24827v2, 2 September 2026.